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Road Design Manual

Volume 4: Bridge and Retaining Structures Design

Part 2: Bridge and Culvert Design

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Foreword

This manual was developed by the Ministry pursuant to The Fourth Schedule of the Constitution which assigns to the National Government the functions and powers of setting standards for the construction and maintenance of all public roads including those under the County Governments.

It is part of a series of manuals that replace the first generation of road manuals developed in the first and second decades after independence. This second generation of the road manuals were developed to cover the entire road project cycle covering planning, appraisal, design, contracts, construction, maintenance, operations and monitoring. The series incorporates best practices, climate change considerations, and recent technologies to enable the provision of road infrastructure that is safe, secure, and efficient.

Under the Kenya Vision 2030 long term plan, infrastructure expansion and modernisation are some of the foundations for the realisation of economic, social and political transformation of Kenya into a rapidly industrialising middle-income country. The plan envisages an integrated, safe and efficient transport and communication infrastructure network consisting of roads, railways, ports, airports, waterways, and telecommunications infrastructure.

The strategies to be pursued under the Vision 2030 plan to improve infrastructure services and to maximise the economic and social impacts of infrastructure development and management include: Strengthening of the institutional framework for infrastructure development and maintenance; Raising efficiency and quality of infrastructure projects; Enhancing local content of identified infrastructure projects to minimise import content; Benchmarking infrastructure facilities and services provision with globally acceptable performance standards; and, Implementing infrastructure projects that will stimulate demand in hitherto marginalised areas.

The first three 5-year Medium Term Plans (MTP) under the Vision 2030 from 2008 to 2022 targeted construction of 1,950 km, 5,500 km and 10,000 km of new paved roads under MTP I, II and III, respectively, totalling 17,450 km. This was a massive infrastructure development program intended to double the paved road network in 10 years compared to 8,600 km developed from independence in 1963 to 2008.

Implementation of MTP I to III resulted in the construction of 14,000 km of paved roads, which extended the paved road coverage to Arid and Semi-Arid regions, that had been previously neglected. However, some key milestones of the Vision 2030 goals have not been realised. This has been due to internal and external challenges. External challenges included: climate change – prolonged droughts; the emergence of COVID-19 pandemic; global supply chain disruptions; exchange rate volatility; and rising interest rates in the leading economies.

The internal challenges included: inadequate road maintenance equipment; pavement overloading by heavy goods vehicles; huge maintenance backlog of the road network; low contracting and supervision capacity particularly in the Counties; poor quality control and assurance of works; congestion in urban areas; encroachment on road reserves; high costs and delays in payments of land acquisition; lack of harmonisation of cross-border transport regulation and operational procedures; rapid urbanisation; increased traffic volume with exponential growth of motorcycle traffic; high cost/delays in relocation of utilities and services along and across road reserves; inadequate funding of projects and programs; and, delay or default in payments for goods, services and works.

The implementation of MTP III came to an end on 30th June 2023, ushering in the implementation of the Fourth Medium Term Plan (MTP IV), which has been aligned to the aspirations of the Kenya Vision 2030 and the Kenya Kwanza Government's Bottom-Up Economic Transformation Agenda (BETA) planning approach and its key priorities.

BETA is the Government's transformation agenda geared towards economic turnaround through a value chain approach. BETA has targeted sectors with the highest impact to drive economic recovery and growth. This will be achieved through bringing down the cost of living; eradicating hunger; creating jobs; expanding the tax base; improving foreign exchange balances; and inclusive growth. BETA ensures rational resource allocation by eliminating wastage of resources occasioned by duplication, overlaps, fragmentation and ineffective coordination in the implementation of programmes and projects.

The Fourth Medium Term Plan key priorities are clustered under five key sectors, namely: Finance and Production; Infrastructure; Social; Environment and Natural Resources; and Governance and Public Administration. The infrastructure sector seeks to: enhance transport connectivity by constructing 6,000 km of new roads, maintaining rural and urban roads, rail, air and seaport facilities and services; expand communication and broadcasting systems; and promote the development of energy generation and distribution by increasing investments in green energy (geothermal, wind, solar and hydro). The infrastructure gap is expected to be bridged by promoting economic participation of the private sector through public private partnerships in the financing, construction, development, operation, and maintenance of infrastructure.

The plan entails a shift of focus to fundamentals in project planning and implementation which include: respect for technical input, regulations and standard practices; adherence to project life cycle i.e., planning, feasibility studies and design before procurement of works; public and stakeholder consultation; procurement within budgetary ceilings; shifting focus during project implementation from the finished product 'black top' to the construction of the foundation; building local capacity particularly MSMEs by ensuring prompt payments; and capacity building at all levels to enable internalisation of policies and processes.

The first generation of the road manuals were used for 35 to 45 years. It is my sincere hope that the second generation of the road standards which have been developed in alignment with the Government's strategy will provide guidance in solving most of the above challenges and those expected to emerge in the next 50 years. Implementation of the manuals will enable achievement of the Government aspirations which include inclusive growth; creation of sustainable employment; building of MSMEs; climate change adaptation and realisation of the UN SDGs; enhanced efficiency in management of infrastructure and transport system; and, laying the foundation for the next national long-term plan at the end of the Vision 2030.

On behalf of the Government of Kenya, I would wish to thank the European Union for financing the development of the first drafts of the manuals in 2009 and the African Development Bank for the financial support in the review and updating of the manuals. I would also like to thank the members of the National Steering Committee and the Technical Task Force for their input. The Technical Administrators, and the Kenya National Highways Authority (KeNHA) for the procurement and able administration of the consultancy Contract. I also thank the Consultant, TRL Limited for their role in providing technical expertise that was essential for the success of the manuals updating exercise. I also wish to express my deepest appreciation to our stakeholders and all those who have contributed to this process and the staff of the Ministry for their continued input.

Hon. Davis K. Chirchir, E.G.H

Cabinet Secretary, Ministry of Roads and Transport

Preface

The earlier Bridge Design Manual (Draft), November-1989 broadly covered the design loads, combination of loads, materials, bridge deck cross-sections, site investigation & preliminary design, types of bridges, foundations, bearings, expansion joints, construction aspects, etc. The details provided in the Manual are of preliminary level. This Manual was revised in August 1993, which was in very concise form. In this revised Manual, BS 5400 is referred for majority of provisions except few deviations like applicability of HA & HB loadings to different class of roads (Class A to Class E), values of mean hourly wind speed, bridge cross sections, etc. Later, certain Eurocodes were adopted as design standards in Kenya through the Kenya Gazette Notice No. 13048 of 14th September 2012.

This Manual is intended to establish common procedures for the design of road bridges and culverts in Kenya. With the adoption of Eurocodes and the trainings imparted by KEBS, the engineers and bridge design practitioners have familiarity with these design standards. This current Manual provides design guidelines for concrete, steel and composite bridges broadly guided by Eurocodes, based on limit state philosophy and partial factor method for verification. Use of prestress concrete is gaining prominence in the region for increased spans. Guidance has also been provided on design of prestress concrete bridges.

Guidance on classification of bridges, width of bridge deck elements, various design requirements such Geometric, hydraulic, geotechnical, constructability has also been covered in the Manual. Climatic conditions affect the bridge designs. Provisions has been made to take effect of wind, temperatures, and seismic actions.

Bridges are important elements of any road infrastructure project. The design and construction of road bridges require an extensive and thorough knowledge of the engineering involved. It is hoped that this Manual will serve as a guide to the engineers in design and construction of road bridges in Kenya.

Eng. Joseph M. Mbugua

Principal Secretary, Ministry of Roads and Transport

Document Management

Document Status

This document has the status of a Manual. Users shall apply the contents there-in to fully satisfy the requirements set out. The content of the manual is based on current practice in Kenya and latest practices in the road sector, both regionally and internationally.

Sources of the Document

Copies of the document can be obtained from:

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Notification of Errors and Requests for Amendments

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Requests for edits and corrections can be freely sent to the following address:

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Amendments Request Form

Request No.	Name	Organisation	Chapter	Page	Section/ Clause	Ref. to: Figure/ Table/	Type of Request	Request

Type of request: General – G; Editorial – E; Technical - T

Amendments to Date

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A National Steering Committee was set up and chaired by the Permanent Secretary, Ministry of Roads and Transport, with the following membership: Principal Secretary for Devolution, Office of the Deputy President; Chief Executive Officer, Inter-Governmental Relations Technical Committee; Chief Executive Officer, Council of Governors; Managing Director and Council Secretary, Kenya Bureau of Standards; Director, National Transport and Safety Authority; Director General, Kenya Roads Board; Director General, Kenya Wildlife Services; Chief Executive Officer, Engineers Board of Kenya; Director General, Kenya Rural Roads Authority; Director General, Kenya Urban Roads Authority; President, Institution of Engineers Kenya; Director Policy, Strategy and Compliance; Kenya National Highways Authority; Chief Engineer, Roads Division, State Department for Roads; Chief Engineer, Materials Testing and Research Division, State Department for Roads.

The technical work was undertaken under the guidance of a Technical Task Force, chaired by Eng. David Maganda, with the following gazetted members: Francis Gichaga (Prof.) (Eng.), Andrew Gitonga (Eng.), Timothy Nyomboi (Dr.) (Eng.), Rosemary Kungu (Eng.), Charles Obuon (Eng.), Sylvester Abuodha (Prof.) (Eng.), Samuel Kathindai (Eng.), Nicholas Musuni (Eng.), Charles Muriuki (Eng.), Tom Opiyo (Eng.), John Maina (Eng.), Fidelis Sakwa (Eng.), Daniel Cherono (Eng.), Maurice Ndeda (Eng.), Theo Uwamba (Eng.).

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Abbreviations

DAF	Dynamic Amplification Factor (DAF)
EQU	Equilibrium Ultimate Limit State
GEO	Geotechnical Ultimate Limit State
LM1	Load Model-1
LM2	Load Model-2
LM3	Load Model-3
LM4	Load Model-4
LOS	Level of Service
NDP	Nationally Determined Parameter
PSC	Prestressed Cement Concrete
RCC	Reinforced Cement Concrete
SLS	Serviceability Limit State
SOV	Special Order Vehicle
SV	Special Vehicle
STR	Structural Ultimate Limit State
TS	Tandem System
UDL	Uniformly Distributed Load
ULS	Ultimate Limit State

Definitions and Glossary of Terms

Accidental design situation	The design situation that refers to exceptional condition of the structure including vehicle impact with pier or superstructure or local failure.
Accidental Action	The action which usually has short duration but of significant magnitude and that is unlikely to occur on a given structure during the design working life.
Accompanying value of a variable action (ψQ_k)	A value of a variable action that accompanies the leading action in a combination. The accompanying value of a variable action may be the combination value ($\psi_0 Q_k$), the frequent value ($\psi_1 Q_k$) or the quasi-permanent value ($\psi_2 Q_k$).
Action (F)	A set of forces (loads) applied to the structure or imposed deformations or accelerations caused such as by temperature changes, moisture variation, uneven settlement, or earthquakes.
Basic wind velocity	The fundamental basic wind velocity modified to account for the direction of the wind being considered and the season (if required).
Behaviour factor	A factor used for design purposes to reduce the forces obtained from linear analysis of seismic actions in order to account for the non-linear response of a structure, associated with the material, the structural system and the design procedures.
Buckling length	System length of an otherwise similar member with pinned ends, which has the same critical buckling load as a given member or segment of member.
Bulk weight density	The bulk weight density is the overall weight per unit volume of a material including a normal distribution of voids and pores.
Characteristic value of an action	A value that may be derived statistically with a probability of not being exceeded during a reference period.
Combination of actions	Set of design values used for the verification of the structure for a limit state under simultaneous influence of different actions.
Combination value of a variable action ($\psi_0 Q_k$)	A value chosen so that the probability of the effects caused by the combination will be exceeded is approximately the same as by the characteristic value of an individual action. It may be expressed as a part of the characteristic value by using a factor ($\psi_0 \leq 1$).
Construction load	The load that can be present due to execution activities, but is not present when the execution activities are complete.
Deck	Deck is that part of a bridge which carry the traffic loading.
Design situations	Sets of physical conditions representing the real conditions occurring during a certain time interval for which the design will demonstrate that relevant limit states are not exceeded.
Design value of an action (F_d)	The value obtained by multiplying the representative value of an action by the partial factor γ_f .
Design working life	Assumed period for which a structure or part of a structure is to be used for its intended purpose with an anticipated maintenance but without major repair being necessary.
Ductile member	A member able to dissipate energy through the formation of plastic hinge.

Ductile structure	Structure that under strong seismic motions can dissipate significant amounts of input energy through the formation of an intended configuration of plastic hinges or by other mechanisms.
Effect of action (E)	This refers to the effect of actions on structural members such as internal force, moment, stress, strain or on the whole structure such as deflection, rotation
Footbridge	The bridge mainly used to carry pedestrian and/or cycle track loads. The road traffic is not permitted on footbridge except permitted vehicles such maintenance vehicle.
Frequent value of a variable action ($\psi_1 Q_k$)	A value determined so that the total time for during which it exceeds in reference period is only a small given part, or the frequency of it being exceeded is limited to a given value. It may be expressed as a part of the characteristic value by using a factor ($\psi_1 \leq 1$).
Fundamental basic wind velocity	The 10-minute mean wind velocity with an annual risk of being exceeded of 0.02 i.e. 50 years return period, irrespective of wind direction, at a height of 10m above flat open country terrain and accounting for altitude effects (if required).
Geotechnical action	The action transmitted to the structure by the ground, fill, standing water or ground-water.
Gross cross-section	The total cross-sectional area of a member but excluding discontinuous longitudinal stiffeners.
Initial temperature	The temperature of a structural element at the relevant stage of the restraint.
Limit states	The states beyond which the structure no longer fulfils the relevant design criteria.
Limited ductile behaviour	Seismic behaviour of bridges without significant dissipation of energy in plastic hinges under the design seismic action.
Load case	Compatible load arrangement, set of deformations and imperfections considered simultaneously with fixed variable actions and permanent actions for a particular verification.
Mean wind velocity	The basic wind velocity modified to account for the effect of terrain roughness and orography.
Nominal Value	A value fixed on non-statistical bases, for instance on acquired experience or on physical conditions.
Non-structural elements	The non-structural elements are those that include completion and finishing elements connected with the structure, comprising such as road surfacing, non-structural parapets.
Permanent action (G)	The action that is likely to act throughout a given reference period and for which the variation in magnitude with time is negligible, or for which the variation is always in the same direction (monotonic) until the action attains a certain limiting value.
Persistent design situation	The design situation that is relevant during a period of the same order as the design working life of the structure. It refers to conditions of normal use.
Prestress	The process of prestressing consists in applying forces to the concrete structure by stressing tendons relative to the concrete member.
Quasi-permanent value of a variable action ($\psi_2 Q_k$)	A value determined so that the total period of time for which it will be exceeded is a large fraction of the reference period. It may be expressed as a part of the characteristic value by using a factor ($\psi_2 \leq 1$).
Quasi-static action	Dynamic action represented by an equivalent static action in a static model.

Reference period	A chosen period of time that is used as a basis for assessing statistically variable actions, and possibly for accidental actions.
Resistance	The capacity of a member or component, or a cross-section of a member to withstand actions without mechanical failure such as bending resistance, buckling resistance, tension resistance.
Representative value of an action (F_{rep})	A value used for the verification of a limit state, which may be the characteristic value, combination value, frequent value or quasi-permanent value of the action.
Seismic Action (A_E)	The action that arises due to earthquake ground motions.
Seismic design situation	The design situation involving exceptional condition of the structure when subjected to seismic event.
Serviceability limit states	The states that correspond to conditions beyond which specified service requirements for a structure or structural member are no longer met.
Special vehicle / Abnormal vehicle	Vehicle used to carry heavy equipment, machinery. Such vehicle not to be carried on a route without permission from relevant authority.
Stiffness	Material resistance against deformation
Strength	Mechanical property of a material indicating its ability to resist actions, usually given in units of stress.
Stress range	The algebraic difference between the two extremes of a particular stress cycle derived from a stress history.
Structural analysis	Procedure for determination of action effects in every point of a structure.
Structural elements	The structural elements comprise the primary structural frame and supporting structures, comprising such as girders, structural slabs, piers.
Structure	Organised combination of connected parts designed to carry loads and provide adequate rigidity
Substructure	Substructure is that part of a structure which supports the superstructure. The foundations, abutments, piers and columns etc. are referred to as substructure for bridges.
Superstructure	Superstructure is that part of a structure which is supported by the substructure. Bridge deck is referred to as superstructure.
Temperature Difference component	The part of a temperature profile in a structural element representing the temperature difference between the outer face of the element and any in-depth point.
Thermal actions	Thermal actions on a structure or a structural element are those actions that arise from the changes of temperature.
Transient design situation	The design situation that is relevant during a period much shorter than the design working life of the structure. It refers to temporary conditions of the structure such as during execution or maintenance.
Uniform Bridge Temperature component	The temperature, constant over the cross section, which governs the expansion or contraction of an element or structure.
Ultimate limit states	The states associated with collapse or with other similar forms of structural failure. It generally corresponds to the maximum load-carrying resistance of a structure or structural member.
Variable action (Q)	The action for which the magnitude varies with time.

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1 Introduction

1.1 General

This Manual was prepared by the Ministry of Roads and Transport as part of a series of manuals that cover the entire project cycle. The series incorporate best practices, climate change considerations, and recent technologies thereby enabling the provision of road infrastructure that is safe, secure, and resilient.

The Kenya road manual series is as follows:

Project Cycle Stage	Manual: Volume or Part/Chapter	Code
A. General	Procedures and Standards Manual	PSM
	1. General	
	2. Policies	
	3. Procedures Guidance	
	4. Codes of Practice	
	5. Guidelines	
B. Planning	Network and Project Planning Manual	NPM
	1. Road Classification	
	2. Route/Corridor Planning	
	3. Route/Corridor Planning	
	4. Highway Capacity	
C. Appraisal	Project Appraisal Manual	PAM
	1. Environmental Impact Assessment and Audit	
	2. Social Impact Assessment	
	3. Traffic Impact Assessment	
	4. Road Safety Audits	
	5. Project Appraisal	
D. Design	Road Design Manual	RDM
	1. Geometric Design	
	2. Hydrology and Drainage Design	
	3. Materials and Pavement Design for New Roads	
	4. Bridges and Retaining Structures Design	
	5. Pavement Maintenance, Rehabilitation and Overlay Design	
	6. Traffic Control Facilities and Communication Systems Design	
E. Contracts	Works and Services Contracts Manual	WSCM
	1. Forms of contracts	
	2. Standard Specification for Road and Bridge Construction	
	3. Bills of Quantities	
F. Construction	Road Construction Manual	RCM
	1. Construction Management	
	2. Project Management	
	3. Site Supervision	
	4. Quality Assurance	
	5. Quality Control	

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Project Cycle Stage	Manual: Volume or Part/Chapter	Code
G. Maintenance	Road Asset Management Manual	RAAM
	1. Maintenance Management	
	2. General Maintenance	
	3. Pavement Maintenance	
	4. Bridges and Structures Maintenance	
H. Operations	Road Operation Manual	ROM
	1. Traffic Management	
	2. Vehicle Load Control	
	3. Emergency Services	
	4. Tolling	
I. Monitoring & Evaluation	Road Design Manual	MEM
	1. Performance Monitoring Manual	
	2. Technical Audits	
	3. Poverty, Gender Equality and Social Inclusion Monitoring	

This manual, as indicated below Volume 4, Part 2 –Bridge and Culvert Design and is part of the Road Design Manual (RDM) whose volumes, parts and coding system are shown in Table 1.1.

Table 1.1 Road Design Manual (RDM) Coding Structure

Vol.	Manual Title	Part Name	Code
1	Road Design Manual: Vol. 1 Geometric Design	Part 1 - Topographic Survey	RDM 1.1
		Part 2 – Traffic Surveys	RDM 1.2
		Part 3 – Geometric Design of Highways, Rural and Urban Roads	RDM 1.3
2	Road Design Manual: Vol. 2 Hydrology & Drainage Design	Part 1 – Hydrological Surveys	RDM 2.1
		Part 2 – Drainage Design	RDM 2.2
3	Road Design Manual: Vol. 3 Materials & Pavement Design for New Roads	Part 1 – Ground Investigations and Material Prospecting	RDM 3.1
		Part 2 – Materials Field and Laboratory Testing	RDM 3.2
		Part 3 – Pavement Foundation and Materials Design	RDM 3.3
		Part 4 – Flexible Pavement Design	RDM 3.4
		Part 5 – Rigid Pavement Design	RDM 3.5
4	Road Design Manual: Vol. 4 Bridges & Retaining Structures Design	Part 1 – Geotechnical Investigation and Design	RDM 4.1
		Part 2 – Bridge and Culvert Design	RDM 4.2
		Part 3 – Retaining Structures Design	RDM 4.3
		Part 4 – Reinforced Fill Structures Design	RDM 4.4
		Part 5 – Bridges and Structures Condition Survey	RDM 4.5
		Part 6 – Bridge Maintenance Design	RDM 4.6
5	Road Design Manual: Vol. 5 Pavement Maintenance, Rehabilitation & Overlay Design	Part 1 – Pavement Condition Survey	RDM 5.1
		Part 2 – Pavement Maintenance, Rehabilitation and Overlay Design	RDM 5.2
6	Road Design Manual: Vol. 6 Traffic Control Facilities & Communication Systems Design	Part 1 – Road Marking	RDM 6.1
		Part 2 – Traffic Signs	RDM 6.2
		Part 3 – Traffic Signals and Communication System	RDM 6.3
		Part 4 – Other Traffic Control Devices	RDM 6.4
7	Road Design Manual: Vol. 7 Road Lighting Design	Part 1 – Grid-connected Road Lighting	RDM 7.1
		Part 2 – Solar Road Lighting	RDM 7.2

This Manual must be applied sensibly and flexibly in conjunction with the skill and judgement of the designer. Compliance with the guidance given in the Manual does not relieve designers of the responsibility for establishing that their design is suitable, appropriate, safe, and adequate for the purpose stated in the project requirements.

1.2 Objectives of this Part

Bridge and Culvert Design Manual has been prepared to assist practitioners in guiding on design of different components of road bridges and culverts. The objective of this Manual is to establish a common procedure for road bridges and culverts design which are appropriate to the conditions found in Kenya. The requirements specified in the Manual aim at achieving design of safe, serviceable, durable, and economical bridges and culverts.

1.3 Scope of this Part

The scope covers determination of loads and forces to which bridges and culverts are subjected, combinations of loads and partial factors to be applied for limit state design, structural design principles of reinforced concrete, prestressed concrete, steel and composite bridges, joints, and bearings.

Loads on bridges are caused by several actions including permanent actions, variable actions, accidental and seismic actions. These loads are used for the design of bridge structural elements including super structure, piers, abutments, upstand walls, wing walls and bridge foundations.

This Manual provides the common bridge and culvert design guidelines for design of whole structure and its components. Special bridges such as cable supported bridges, arch bridges and other bridges using innovative structural arrangements and materials, which require extensive study, are beyond the scope of this Manual.

While applying the provisions of this Manual, it is assumed that the choice of structural system and design of structures are made by well qualified and experienced personnel. Execution is carried out by qualified, skilled, and experienced personnel. Proper supervision and quality control are maintained during all stages of design and construction. The structures are used as intended in design and are maintained adequately.

Note 1: Kenya adopted the Eurocode standards EN 1990 to EN 1999 vide Gazette Notice no. 13048 of 14th Sep 2012. Eurocodes have many Nationally Determined Parameters (NDPs) that are yet to be determined. KEBS is the regulatory authority and adopted the UK National Annex in the intervening period. KEBS is organising trainings to Engineers on Eurocodes using UK National Annex.

Note 2: This Manual is prepared primarily based on Eurocodes using UK National Annex for NDPs. After availability of Kenya national parameters, the UK NDPs may be replaced with the Kenya NDPs.

1.4 Organisation of the Manual

The document is arranged into following eleven Chapters.

- **Chapter 1:** Introduction
- **Chapter 2:** General Requirements
- **Chapter 3:** Actions and Combination of Actions
- **Chapter 4:** Concrete Structures
- **Chapter 5:** Steel Structures
- **Chapter 6:** Composite Structures
- **Chapter 7:** Masonry and Timber Structures
- **Chapter 8:** Joints and Bearings
- **Chapter 9:** Footbridges
- **Chapter 10:** Culverts
- **Chapter 11:** Foundations

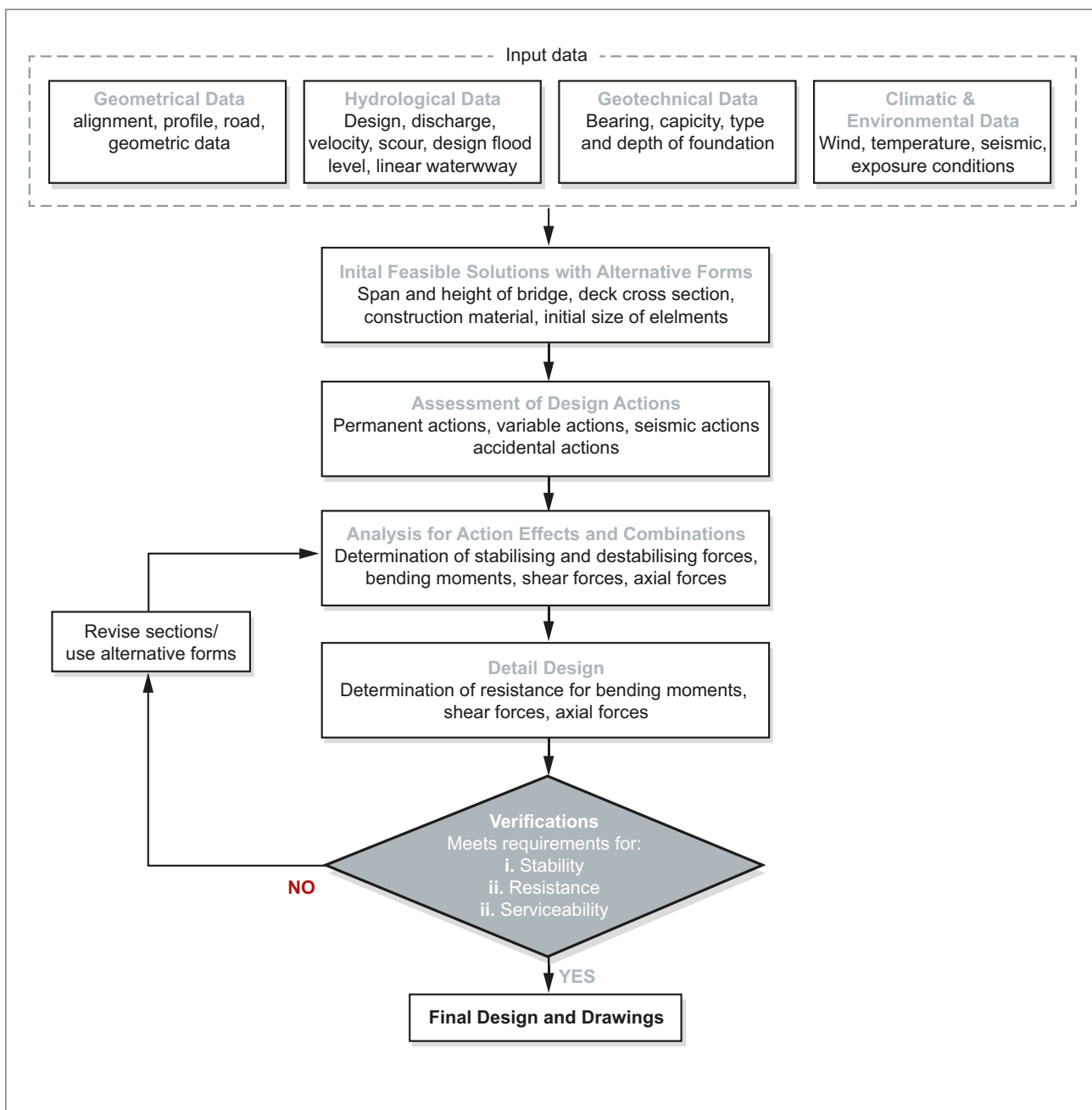
This Manual has references from following other volumes of Road Design Manual.

- Vol. 1 – Geometric Design
- Vol. 2 – Hydrology and Drainage Design
- Vol 3 Part 1 – Ground Investigations and Materials Prospecting
- Vol 4 Part 1 – Geotechnical Investigation and Design
- Vol 4 Part 3 – Retaining Structures Design

1.5 Design Process

The main objective of this Manual is to arrive at the design solutions to various types of bridges. Before carrying out the designs, the designer should have obtained various inputs data. The various steps involved in the design of structures can broadly be represented as shown in Figure 1.1.

Figure 1.2 Design Process



2 General Requirements

2.1 General

This Chapter deals with the design philosophy, limit states of design and the planning considerations specifying various requirements for geometry, hydraulic, general design and construction of bridges and culverts.

2.2 Design Philosophy

This section provides the principles and requirements for safety, serviceability and durability of structures and describes the basis of design. It is based on the limit state concept and used in conjunction with a partial factor method.

2.2.1 Basic Requirements

A structure shall be designed and executed in such a way that during its intended life with appropriate degrees of reliability and in an economical way, it will sustain all actions and influences likely to occur during execution and use and meet the specified serviceability requirements.

The reliability required for the structures shall be achieved by designing in accordance with the design guidelines provided in this Manual.

The structure shall be designed such that deterioration over its design working life does not impair the performance of the structure below that intended, having due regard to its environment and the anticipated level of maintenance. The environmental conditions shall be identified at the design stage so that their significance can be assessed in relation to durability and adequate provisions can be made for protection of the materials used in the structure. Appropriate quality management measures should be adopted during design, execution, use and maintenance.

2.2.2 Limit States of Design

The limit states are those states beyond which the structure no longer fulfils the relevant design criteria. The design for limit states shall be based on the use of structural and load models with appropriate partial factors for relevant limit states. It shall be verified that no limit state is exceeded when relevant design values for actions, material properties, or product properties and geometrical data are used in these models. The verifications shall be carried out for all relevant design situations and load cases.

Ultimate Limit State (ULS)

The ultimate limit state is associated with collapse or with other similar forms of structural failure and generally correspond to the maximum load carrying resistance of the structure or structural member. It concerns with the safety of people and structure. The state prior to structural collapse, considered in place of the collapse itself, may be treated as ultimate limit state. The various ultimate limit states that need to verify may constitute loss of equilibrium of structure, failure by excessive deformations, failure caused by fatigue or other time dependent effects.

Serviceability Limit State (SLS)

The serviceability limit state corresponds to the conditions beyond which specified service requirements for a structure or structural member are no longer met. The limit state that concerns the functioning of the structure or structural members under normal use, comfort of people or the appearance of the construction works, shall be classified as serviceability limit states. The various service state limits that need to verify may constitute:

- i. Deformations that affect appearance, comfort of user, functioning of the structure or that cause damage to non-structural members.

- ii. Vibrations that cause discomfort to people or limit the functional effectiveness of the structure.
- iii. Damage that is likely to adversely affect the appearance, durability, or the functioning of the structure.

Limit states shall be related to design situations which can be classified as persistent, transient or accidental design situations. The persistent design situation refers to the conditions of normal use. The transient design situation refers to temporary conditions applicable to the structure, e.g. during execution or repair. The accidental design situation refers to exceptional conditions applicable to the structure, e.g. vehicle impact, explosion or the consequences of localised failure. The seismic design situation refers to conditions applicable to the structure when subjected to seismic events. The selected design situations shall be sufficiently severe and varied so as to encompass all conditions that can reasonably be foreseen to occur during the execution and use of the structure.

2.2.3 Actions

A set of forces applied to the structure, imposed deformations or accelerations are called actions. The actions can be classified by their variation in time as follows:

Permanent actions (G), Actions for which, the variation in magnitude with time is negligible or the variation is always in the same direction until the action attains a certain limiting value, such as self-weight of structures, fixed equipment and road surfacing, and indirect actions caused by shrinkage and uneven settlements.

Variable actions (Q), Action for which the variation in magnitude with time is not negligible such as imposed loads on floors, beams and roofs, wind actions.

Accidental actions (A), Action for which duration is short but magnitude is significant such as explosions or impact from vehicles.

The characteristic value F_k of an action is its representative value which can be a mean value, upper value, lower value or nominal value.

The characteristic value of a permanent action can be used as a single value (G_k) equal to the mean value if variability is considered as small. If the variability of permanent action cannot be considered as small, two values shall be used: an upper value $G_{k,sup}$ and a lower value $G_{k,inf}$. The $G_{k,inf}$ is the 5% fractile and $G_{k,sup}$ is the 95% fractile of the statistical distribution for permanent action G .

The characteristic value for variable actions (Q_k), shall correspond to either an upper value with an intended probability of not being exceeded or a lower value with an intended probability of being achieved during some specific reference period or a nominal value in cases where a statistical distribution is not known.

The combination value of a variable action, represented as a product $\psi_0 Q_k$ is used for verification of ultimate limit states, irreversible serviceability limit states.

The frequent value of a variable action, represented as a product $\psi_1 Q_k$, is used for the verification of ultimate limit states involving accidental actions and for verifications of reversible serviceability limit states.

The infrequent value of a variable action, represented as a product $\psi_{1,infq} Q_k$ may be used only for the verification of certain serviceability limit states for concrete bridges.

The quasi-permanent value of a variable action, represented as a product $\psi_2 Q_k$, is used for the verification of ultimate limit states involving accidental actions and for the verification of reversible serviceability limit states. Quasi-permanent values are also used for the calculation of long-term effects.

2.2.4 Verification

Verification is based on the partial factor method. In this method, it is verified that no relevant limit states are exceeded when design values and resistances are used in the design models in all the relevant design situations.

2.2.5 Design Working Life

The design working life of a highway structure should be determined, taking into account:

1. Functional requirement.
2. Safety.
3. Consequence.
4. Traffic disruption.
5. Durability.
6. Sustainability.
7. Financial.
8. Practical feasibility if it were to be replaced in the future.

The recommended design working life of highway structures can be taken from Table 2.1.

Table 2.1 Recommended Design Working Life

Recommended Design Working Life	Type of Structures
10 years	Temporary structures
10 to 30 years	Replaceable structural parts (bearings, expansion joints)
50 years	Culverts
100 years	Bridges, retaining walls
120 years	Special bridges and other structures
<i>Note: A structure with a total length up to 6.0 m will be treated as culvert and exceeding 6.0 m will be treated as bridge.</i>	

2.3 Planning Considerations

Bridges and culverts are the structures which provide transportation route over the physical obstructions, usually over river, lake, road, rail or valleys. Planning of a bridge is important aspect, and it should comply various technical, administrative, environmental and social requirements. The required information and data shall be collected for proper planning.

2.3.1 Classification of Bridges

The bridges can be classified based on the criteria considered e.g. principal function, material of construction, structural form, inter-span system of support, position of deck level and type of physical obstruction. The broad classifications are given below:

1. Classification of bridges according to their function:
 - i. Road bridge.
 - ii. Pedestrian bridge.
 - iii. Railway bridge.
 - iv. Aqueduct bridge.
 - v. Combined bridge (more than one function).

2. Classification of bridges according to the material of construction:
 - i. Reinforced concrete bridge.
 - ii. Prestressed concrete bridge.
 - iii. Steel bridge.
 - iv. Composite bridge.
 - v. Masonry bridge.
 - vi. Timber bridge.
3. Classification of bridges according to the structural form of bridge:
 - i. Slab bridge.
 - ii. Girder and slab bridge.
 - iii. Box girder bridge.
 - iv. Truss bridge.
 - v. Arch bridge.
 - vi. Suspension bridge.
 - vii. Cable-stayed bridge.
 - viii. Rigid frame bridge.
4. Classification of bridges according to the inter-span system of support:
 - i. Simply supported bridge.
 - ii. Continuous bridge.
 - iii. Integral bridge.
 - iv. Cantilever bridge.
5. Classification of bridges according to position of deck level:
 - i. Deck bridge.
 - ii. Through bridge.
 - iii. Semi-through bridge.
6. Classification of bridges according to type of physical obstruction crossing:
 - i. River bridge.
 - ii. Over bridge.
 - iii. Viaduct.
7. Based on the total length of bridge/culvert, though not common, may be classified as below:
 - i. Culvert (length ≤ 6 m).
 - ii. Small bridge ($6 \text{ m} < \text{length} \leq 15 \text{ m}$).
 - iii. Medium bridge ($15 \text{ m} < \text{length} \leq 50 \text{ m}$).
 - iv. Large bridge (length $> 50 \text{ m}$).

Length of a bridge structure shall be taken as the total length measured along the centre line of the bridge between inner faces of abutment upstand walls or between the end deck joints or outer edge of deck where there is no upstand wall.

The cross-drainage structure with pipes will be termed as culvert irrespective of the length.

2.3.2 Siting of a Bridge

The first step in the planning of a bridge is to ascertain its location in the link of the road network and the functions thereof in a seamless manner. A bridge shall ideally be sited across the narrowest width of the river with straight stretches of the channel in considerable length on upstream and downstream of the proposed bridge site and the channel have stable banks. The meandering stretches of the rivers

shall be avoided for bridge siting. Where the banks of the rivers are not well defined or unstable, river training works/ guide bunds can also be considered. The preliminary site selection can be based on the satellite images supplemented by preliminary survey. A model study may be required for selection of bridge site on rivers having undefined cross section and large flood plains. Safety and economy shall also be considered while assessing the suitability of bridge site.

Location of bridges with a length up to 60 m are generally governed by road alignment unless there are specific problems at the bridge site. Site for bridges of length ranging from 60 m to 300 m are selected based on suitability of bridge and the alignment of approaches. For bridge with length more than 300m, the requirement of suitability of bridge shall generally have the precedence over the alignment of approaches and the approach alignment has to be modified to integrate with the selected bridge site.

The location and configuration of the flyover/ grade separator/ underpass shall be determined according to the alignment, geometry, category of intersecting roads, traffic and degree of segregation required.

The minimum length of 100m in approaches to bridge on either side or the defined length up to end of approach gradients, shall be considered as part of the planning aspects. The approaches of the flyovers can be provided with viaducts in inhabited areas or wall type structure in remote areas.

Initial environmental examination shall be carried out to assess the likely adverse impacts of the proposed structure. Likely impacts to be considered and assessed are:

- i. Increase in submergence of surrounding area.
- ii. Likelihood of river bank erosion.
- iii. Possibility of silting.
- iv. Relocation and rehabilitation of the people.
- v. Effect on monuments or heritage structures.
- vi. Impact on wildlife, flora and fauna.
- vii. Impact on surroundings.

Economic evaluation of the different alternatives of a bridge project shall be carried out to determine the relative viability together with the environmental and social requirements.

2.3.3 Collection of Data

Generally, the following data shall be collected to have a detailed information and appreciation of the bridge project and shall be included in the project documents.

- i. Site map, plans and topographical features
- ii. Hydraulic data.
- iii. Geological data.
- iv. Environmental data.
- v. Traffic data.
- vi. Construction materials.
- vii. Existing services.
- viii. Details of existing bridges.

2.3.4 Width of Bridge Deck

The width of the bridge should correspond to the carriageway width as determined from “RDM 1.3”. The paved width as provided for both side approaches shall be continued through the bridge. The design should include an assessment of the need to provide additional widening at structures to allow for future widening of the cross-section.

For road bridges, the width of carriageway shall not be less than 3.5 m for a single lane bridge, 7.0 m for a two-lane bridge. A hard strip of minimum of 0.50 m both sides of carriageway shall be provided on bridges. A minimum clear width of 5.50 m is desirable for single lane bridge for vehicles to cross at slow speed. The width of walkway shall be extra on both sides of the carriageway, may be separated by safety barrier. For a bridge on horizontal curve, the carriageway width shall be increased to conform the geometric requirements provided in relevant Clause of “RDM 1.3”.

The width of central verge on bridges for dual carriageway shall generally be maintained same as that on approach road. The provision for walkway and cycle track including horizontal separation from the carriageway, at a structure shall be at least equivalent to any provision immediately upstream and downstream of the structure. All bridges shall be provided with walkway on both sides unless a separate footbridge is provided for pedestrians. The minimum width of walkway on a road bridge shall not be less than 1.8 m. For bridges in urban areas, the width of walkway shall be increased based on pedestrian traffic as per the provisions in “RDM 1.3”.

Table 2.2 Minimum Width of Bridges

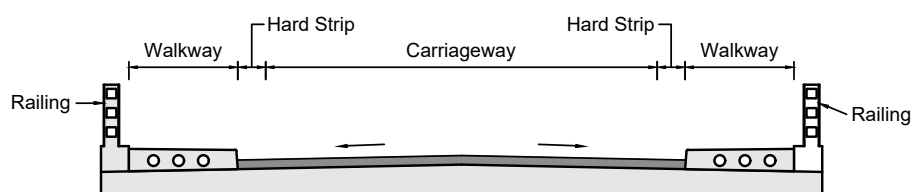
Bridge Elements		Minimum Width (m)
Single Lane Bridge	Carriageway	$1 \times 3.5 = 3.5$
	Hard strip	$2 \times 0.5 = 1$
	Walkway	$2 \times 1.8 = 3.6$
	Railing kerb	$2 \times 0.5 = 1$
	Total	9.1
Two Lane Bridge	Carriageway	$2 \times 3.5 = 7$
	Hard strip	$2 \times 0.5 = 1$
	Walkway	$2 \times 1.8 = 3.6$
	Railing kerb	$2 \times 0.5 = 1$
	Total	12.6

Notes:

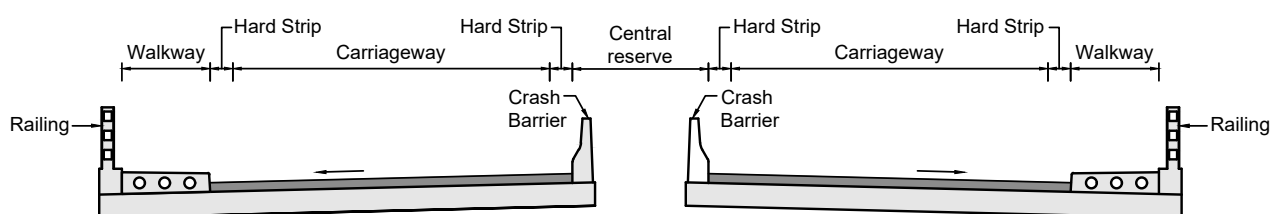
1. The clear width on bridges shall not be less than paved width on approaches.
2. The desirable clear width on single lane bridge is 5.5 m. The total width in that case would be $5.5 + 2 \times 1.8$ (walkway) + 2×0.5 (railing kerb) = 10.1 m.
3. If the lane widths on approaches to bridge are higher than the widths indicated above, the higher lane widths shall continue on bridge also.
4. It is desirable to keep the width of hard strip/ shoulders on bridge not less than that on approaches.

To account for future widening, the authority may decide to keep additional width of carriageway on bridge more than the approach road carriageway. Some typical sections showing geometric elements of bridge are shown in Figure 2.1.

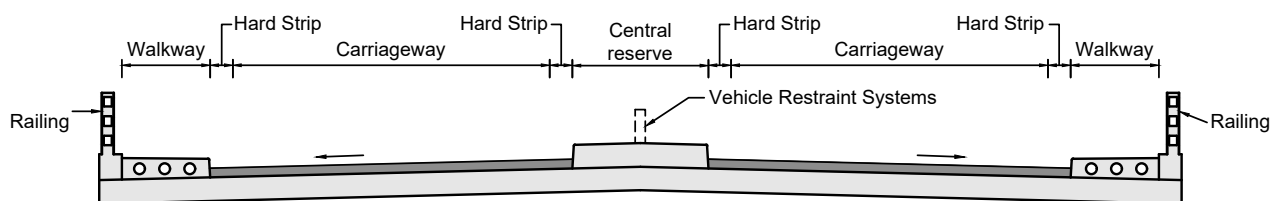
Figure 2.1 Typical Cross-sections for Bridge Deck Width



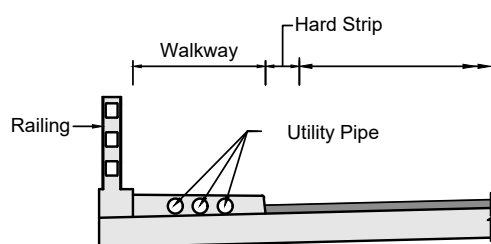
(a) Bridge deck at single carriageway



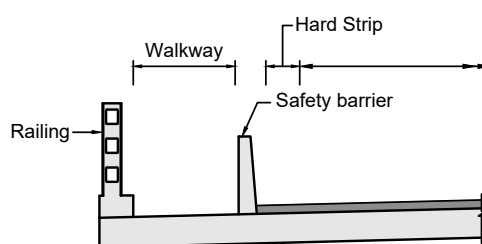
(b) Bridge deck at dual carriageway divided structure



(c) Bridge deck at dual carriageway single structure



(d) Raised walkway



(e) Walkway with safety barrier

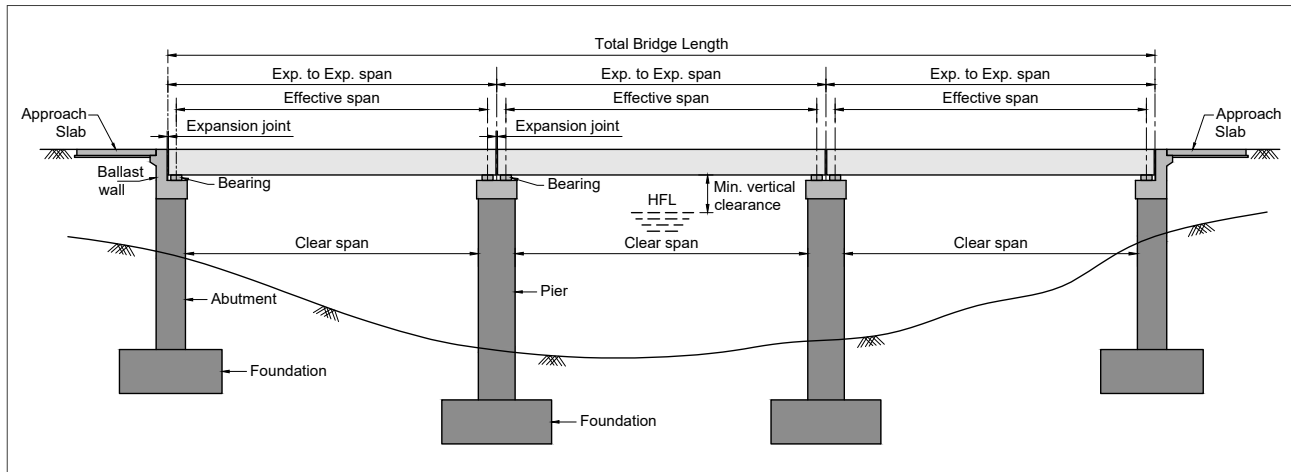
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For minor utility services to pass at bridge locations or for the purpose of illuminations on bridges, the services pipes can be provided in the raised walkway to accommodate cables.

2

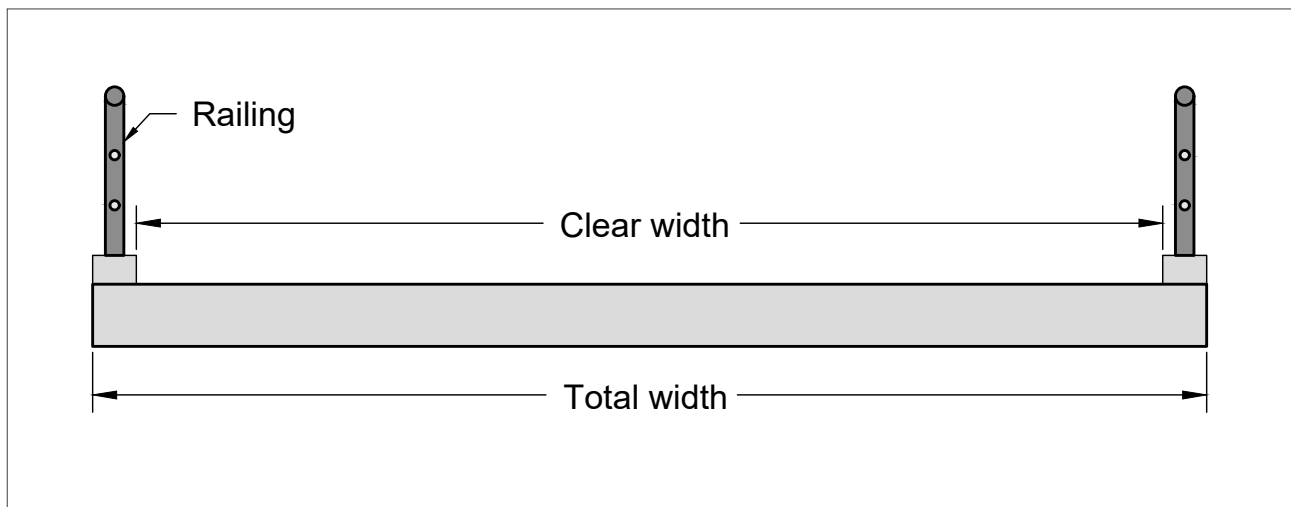
General Requirements

Figure 2.2 Typical Long Section of a Bridge



Where required, footbridges shall be provided to cross the physical obstacle such as over rivers or to avoid pedestrian – vehicle conflicts such as over roads to cross. Pedestrian facility should provide enough space for safe and comfortable walking. The width should be provided based on the current and expected pedestrians. Footbridge may have segregated provision for pedestrians and cyclists, in which case it will need to be wider. Parapets are provided for the safety of the pedestrians. Continuity of pedestrian facilities shall be maintained on approaches to footbridge.

Figure 2.3 Typical Cross Section for Footbridge



The layouts and design of footbridges are discussed in detail under Chapter 9.

2.3.5 Minimum Dimensions of Bridge Elements

The dimensions of various elements of a bridge will depend upon the design parameters and the detail design. Minimum dimensions of the different elements of reinforced concrete bridge shall be adopted as given Table 2.3.

Table 2.3 Minimum Dimensions of Structural Elements for Bridges

Element	Dimension (mm)
Concrete deck slab thickness for road bridge	≥ 200 mm
Concrete deck slab thickness for footbridge/ pedestrian's bridge	≥ 150 mm
Stem and Head wall/ballast wall of abutment thickness	≥ 250 mm
Bearing shelf of abutment or pier	≥ 400 mm
Wingwall thickness	≥ 200 mm
Pier columns thickness	≥ 500 mm
Pier walls thickness	≥ 300 mm
Footing thickness	≥ 300 mm

2.3.6 Selection of Type of Superstructure and Span Lengths

The selection of type of bridge and span lengths depends upon various factors. Some important factors are:

- Estimated length of bridge.
- Subsoil characteristics.
- Hydraulic characteristics of crossing stream.
- Available construction materials, plants and machinery.
- Available technology.
- Constructability factors.
- Operation and maintenance requirements.
- Overall economics of the bridge.

As a general guidance, the general range of economic span length for different type of superstructure may be taken as suggested in Table 2.4.

Table 2.4 Type of Superstructure and Suggestive Span Lengths

Type of Super Structure	Span Range (m)
Simply supported RCC slab bridge	up to 10
Simply supported RCC girder bridge	10 – 20
Simply supported prestress girder bridge	25 – 40
Simply supported RCC box girder	20 – 30
Simply supported prestress box girder	40 – 60
Integral RCC slab bridge	10 – 20
Continuous RCC girder bridge	15 – 25
Continuous prestress girder bridge	25 – 40
Rolled steel girder bridge	up to 10
Steel plate girder bridge	25-50
Steel truss bridge	40-100

2.4 General Design Requirements

The general structural scheme, span arrangement, structure type, size and shape of the bridge elements shall be decided based on following aspects:

- i. Functional requirements.
- ii. Structural requirements.
- iii. Safety considerations.
- iv. Construction requirements.
- v. Operation and maintenance requirements.
- vi. Environmental requirements.
- vii. Durability requirement for concrete structures.
- viii. Durability requirement for steel structures.
- ix. Aesthetic and durability requirements.

2.4.1 Functional Requirements

Bridges based on their basic functions they perform can be classified into road bridges, pedestrian bridges, railway bridges, aqueduct bridges or combined bridges which caters for more than one function. The requirements of bridge shall be assessed properly to meet out the intended functions. The width of bridge deck, carriageway, walkways, horizontal and vertical clearances, length and span arrangement shall be provided as discussed in relevant Clauses of this Chapter. The length of river bridge is usually based on width of river to be crossed and assessed from the hydrological studies. Alignment of the bridge shall be decided carefully keeping the geometric requirements in accordance with Clause 2.6. In general, the bridges shall be on straight alignment. Curved bridges require extra design and detailing and costs more for construction and maintenance. Provision for utility ducts of 75 mm diameter PVC pipes for communication or electric cables may be kept through raised walkways on both sides.

2.4.2 Structural Requirements

A structure shall be designed such that it will sustain all actions and influences likely to occur during execution and use and meet the specified serviceability requirements. The deterioration over its design working life does not impair the intended performance of the structure. The details on different actions such as traffic loads, vehicle collision loads, wind actions, thermal actions, seismic actions and other actions to be considered for structural design are provided in Chapter 3. The details on structural designs of different components of bridges are provided in Chapter 4 for concrete structures and Chapter 5 for steel structures.

Bridge foundations can be shallow or deep (pile) foundations depending upon the type of geotechnical properties of sub-strata, hydrological and other considerations. The foundation shall rest at a depth where the bearing capacity of the strata is adequate to sustain the foundation loads without excessive settlement or failure. It is also important to take the foundation at a depth which provides adequate safety against maximum scour, where applicable.

The bridge substructure can be circular piers, wall type, portal or any other shape with solid or hollow sections. The bridge abutment can be of various configurations such as U-type, wall type, spill through type, bank seat type.

The super structure can be reinforced concrete, pre-stress concrete, steel or any other type depending upon the span length, functional and techno-economic considerations.

The Retaining/ Wing walls can be at an angle to bridge axis (splayed type) or parallel to the bridge axis (return wall type) or cantilever type projecting from abutment. Reinforced earth or gabion walls may also be provided depending upon the site conditions and availability of materials.

Approach slab of reinforced concrete shall be provided on all bridges covering full width of carriageway including walkways but excluding support on wing walls. A support of minimum 300 mm shall be provided on abutment. Minimum length of approach slab perpendicular to abutment shall be 3.5 m and minimum thickness 300 mm. The minimum strength class of concrete for approach slab shall be C25/30 laid over 150 mm thick blinding concrete of C12/15 strength class. The maximum angular rotation (Δ/L) of approach slab (length L) due to settlement (Δ) of approach embankment shall not exceed 0.004 radians. The steel reinforcement shall be provided in two layers (near top and bottom surface of approach slab) and shall not be less than 12 mm at 150 mm spacing in each direction each layer.

Bridges shall be designed to prevent the accumulation of water and allow for surface water on bridge decks to be effectively removed. A minimum 0.3% longitudinal gradient is desirable for drainage from bridge decks. Drainage spouts shall be provided at adequate spacing to intercept and drain off the rainwater from deck. For urban areas, the drainage spouts shall be connected with ground drainage system through down-take pipes. Continuous drip groove shall be provided at underside edge of the bridge.

Appropriate bearings shall be provided to transfer the vertical/ lateral loads from superstructure depending upon the type of structure, magnitude and direction of loads, movements and rotations. Bearings shall be easy to install, inspect and replace when required. Provision shall be made for replacement of bearings, since the design life of bearings is less than the design life of bridge. Location of jacks shall be marked on the structure for lifting and replacement of bearings. The structure shall be checked for jacked up condition. Bearings may not be required for spans ≤ 10 m. In such structures, separation layer of paper impregnated with bitumen or equivalent may be provided between superstructure and substructure at supports.

Expansion joints shall be provided to have designed movements and rotations of the structure and shall have smooth riding while vehicle passing on joints. Expansion joints shall be easy to install, inspect and replace when required.

All bridges shall be provided with wearing surface to protect the structural concrete of the bridge deck from the damage caused by the moving vehicles. Wearing coat/surfacing over the deck can be either asphalt wearing coat or concrete wearing coat depending upon the approach road surfacing. Concrete wearing coat may also be provided irrespective of approach road surfacing for bridges located in remote areas and in situations where water accumulation is expected. For new concrete decks, the designed total minimum thickness of surfacing material shall be 120 mm including the waterproofing system and any protective layer. On flexible pavement roads, the surface course on the bridge deck should be the same as for the road and carried through over the bridge at the same thickness. The surfacing material laid over the waterproofing and additional protective layer (if present) shall consist of binder and surface courses.

2.4.3 Safety Consideration

Vehicle restraint system/ Safety barriers/ parapets shall be provided on both side of deck for all high-level bridges and grade separated structures. For the bridges with walkway, the safety barriers shall be provided between carriageway and walkway and pedestrian railing at edge of the deck i.e., outside edge of the walkway. The safety barriers shall also be provided to bridge approaches for embankment heights above 3m. The safety barriers/ railings shall be provided with expansion gaps consistent with the deck expansion joints. The height of safety barrier shall not be less than 0.90m above the adjacent roadway surface and railings shall not be less than 1.20m high above the adjacent walkway surface. The detail provisions shall be in accordance with "RDM 1.3".

Adequate illumination shall be provided for all bridges and interchanges in urban areas and near to habitations. Provision of cabling, mounting of poles and installation of luminaries shall be considered in bridge design and detailing. The bridge lighting and its support system, mountings and other fixtures required for illumination of the bridge shall not impair the functionality of other road furniture.

The road markings, signage, delineators and signals (where required) shall be planned and integrated with overall bridge planning to facilitate smooth and safe movement/ operation of traffic, in accordance with “RDM Vol 6”. Where the number of lanes on bridge and the approach road are different, a transition taper shall be provided with appropriate markings and signs.

2.4.4 Construction Requirements

Construction requirements aim at optimum use of construction knowledge and experience in planning, design and construction. Requirements to have construction technology that leads to fast-track construction shall be considered at design stage. Various factors that may influence the construction of structure shall be considered at design stage. Some of the construction aspects are:

- Availability of construction materials.
- Availability of plant and machinery.
- Construction and erection methodology.
- Availability of land for the proposed structure and construction camps.
- Access to the work area.
- Weather conditions and construction time.
- Interference with existing traffic.
- Interference with existing utilities.
- Existing ground and sub-surface conditions.
- Simplicity of design and clarity in detailing.
- Ease of fabrication of steel members.
- Standardisation of design elements to extent possible.
- Availability of skilled manpower.

All the construction stage actions depending upon the process of construction, erection methodology and the stages of construction and shall be considered in design of the structure. Alternative design and construction methodology shall be compared and considered in the design including inspection, maintenance and repair/ replacement aspects.

Construction shall be carried out by appropriate skilled personnel to ensure quality as envisaged. Adequate supervision and quality control shall be provided during all stages of construction.

The design process shall include the necessary detailing for the proposed construction methods, available construction materials, equipment and man power. The choice of structural system and the design and detailing of the structure shall be made appropriately for the proposed construction methodology and be reflected in design and drawings. If the proposed methodology of construction is required to be changes during implementation, its impacts on the design shall be investigated and considered before carrying out construction.

2.4.5 Operation and Maintenance Requirements

Periodic inspections of bridges and culverts are essential to assess the structural health and maintenance requirements. Suitable access arrangements for inspection and routine maintenance of bearings, expansion joints, anchorages, inside of box type decks should be planned and integrated

at design stage. Bearing replacement scheme shall be planned at design stage and corresponding design forces shall be considered during design. Ventilation shall be provided to all accessible closed box sections. The access facilities provided for inspection and maintenance shall be secure, preventing access by unauthorised personal. Access points can be at each end of the structure at points that are safe and easily accessible and do not require traffic control.

The different components of the structure shall be easy to inspect and maintain in intended operational conditions. Exposed parts of the steel structures shall be accessible for inspection and periodic maintenance such as protective coatings/ painting of surfaces.

Information and guidance on various requirements and procedures for maintenance of the structure shall be in the maintenance manual.

2.4.6 Environmental Requirements

The environmental studies shall be conducted in accordance with environmental regulations to evaluate the adverse impacts of proposed infrastructure on environment. The adverse impacts due to the structures may include increase in submergence, riverbank erosion, relocation and rehabilitation of affected people, impacts on wildlife, flora and fauna in ecologically sensitive areas may be subject to environmental regulations. Assessment shall be made for potential impacts on wildlife, vegetation, and water bodies, and incorporate mitigation measures to minimise disruption to the ecosystem.

Environmental impact assessment shall also be carried out where surface protection systems are proposed for structures subject to the following situations:

- i. Structures located within water bodies.
- ii. Structures located near water bodies.
- iii. Where the surface protection system products could find their way into the water environment.

2.4.7 Durability Requirements for Concrete Structures

A durable structure shall meet the requirements of serviceability, strength and stability throughout its design working life for its intended purpose with anticipated maintenance but without major repair. Corrosion protection of steel reinforcement depends on density, quality and thickness of concrete cover and cracking. The different environmental conditions of exposure shall be considered in the design. The cover density and quality are achieved by controlling the maximum water/cement ratio and minimum cement content. The details on durability aspects of concrete structures specifying exposure classes related to environmental conditions, minimum cover requirements, concrete strength class, maximum water cement ratio and minimum cement content are provided in Clause 4.4. The details on crack control measures are provided in Clause 4.7.

Application of waterproofing to the upper surface of concrete deck have an important role in controlling the corrosion of reinforcement. The waterproofing shall be robust to resist transient vehicular loading, maintain good adhesion to the deck and surfacing and have long term durability (minimum working life of 25 years). The waterproofing systems for installation on concrete bridge decks and top slabs of buried structures may be sheet applied systems or liquid applied systems and shall be installed in accordance with the manufacturer's specified procedures. Waterproofing shall be continuous and cover the entire deck between parapet kerbs including walkways, central reserves and verges. The sheet applied waterproofing systems shall comply with BS EN 14695.

Additional protective layer shall not be used unless it is a specific requirement of the waterproofing system. The nominal thickness of additional protective layer, if required, shall not exceed 20mm. Where the structure is covered with fill, such as in the case of underpasses and culverts, a waterproofing system shall be applied for the buried top slab. Where the top slab of the buried structure is continuous with the abutments, waterproofing on the top surface shall be continued down the outside of the abutment to a level 200 mm below the soffit.

Surface protection systems when used in combination with other surface protection systems shall be compatible with each other. Surface protection systems shall be transparent and colourless for application to the concrete substrate. The need for the application of anti-carbonation coatings on concrete structures shall be assessed. New concrete structures subject to the following conditions can be vulnerable to carbonation:

- i. High carbon dioxide concentrations.
- ii. Relative humidity between 50 % and 70 %.

The performance requirements for anti-carbonation coatings shall be in accordance with BS EN 1504-2.

2.4.8 Durability Requirements for Steel Structures

Steel components should either be designed to function adequately for the full design life of the bridge, with appropriate levels of inspection and maintenance or should be designed to be replaceable (such as bearings, expansion joints). Access shall be available for in-service inspection and maintenance for different parts of the structure. The inaccessible surfaces that have not been effectively sealed against the effects of air, moisture and contaminants should be provided with an additional sacrificial thickness appropriate to the environment. The details on durability aspects are provided in Clause 5.4.

Steel bridge structures should be fabricated to EN 1090-2 in which the fabrication procedures are designed to ensure durable steel components. Corrosion protection is an important issue to consider when designing and detailing steel bridges. Corrosion protection is usually achieved by the application of coatings which may generally be paint coatings, thermally sprayed metallic coatings or hot-dip galvanised coating. It should be ensured that the particular type of protective coating system is carefully and accurately specified including the initial surface preparation works. The corrosion protection system shall be replaceable safely at the end of its design life. Following standards may be referred for execution and corrosion protection measures for steel structures.

EN 1090: Execution of steel structures - Technical requirements.

EN ISO 12944: Paints and varnishes - Corrosion protection of steel structures by protective paint systems.

EN ISO 1461: Hot dip galvanised coatings on fabricated iron and steel articles - specifications and test methods.

2.4.9 Aesthetic Requirements

Bridges have visual impacts on their surroundings and the structure's appearance shall be considered as an important consideration. The structural form of bridges and culvert shall be such as to have aesthetic appearance in harmony with its surroundings. The well-proportioned structural form, texture, colour, and other details of visible component are important in aesthetics of bridges. The aesthetic and architectural considerations shall be taken into account in bridge design in harmony with general landscape and local architecture. The advice of an Architect may be sought for good aesthetics of the bridges and culverts.

The potential risks arising from wilful misuse, vandalism, graffiti, and other types of anti-social behaviour shall be taken into account in the design. Use of enclosures shall be avoided at such locations, wherever possible, unless they are required for other reasons. Space shall be maintained clear off the visual barriers which reduce visibility.

Anti-graffiti coatings may be applied to concrete structures which have anticipated problem with offensive graffiti. Anti-graffiti coatings shall be specified to ensure that graffiti on treated areas of a concrete structure can be removed without damage to the concrete substrate. The anti-graffiti coating shall be applied in accordance with the manufacturer's specified procedures. The performance requirements for anti-graffiti coatings shall be in accordance with BS EN 1504-2.

2.5 Vertical and Lateral Clearances

Vertical clearance for an overpass is the height above any highest point on roadway or rail top level to the corresponding lowest point of the super structure. Minimum vertical clearance for any structure over road shall be in accordance with the provisions in “RDM 1.3” but not less than 5.5m. For the structures on a route carrying over-dimension vehicles, increased vertical clearance can be considered.

The lateral dimension between the abutment faces for an overpass shall not be less than the width of approach roadway section including shoulder, kerbs, gutters and walkways, utility corridor, any future widening, etc. For a bridge constructed over a road on horizontal curve super elevated sections, the lateral dimension shall be increased on the side of inner kerb by an amount equal to 5.5 metres multiplied by super elevation. This extra horizontal clearance is in addition to the increase in width required on curves.

For underpass carrying only pedestrian traffic or cycle tracks, a minimum vertical clearance of 2.4 m shall be provided. For cattle and wildlife, underpasses must be designed as the normal height of the actual animal plus 0.5 m. Bridges above railways must have a clearance height of at least 7.2 m - if not otherwise stated - to facilitate possible future electrification and shall be agreed with the Approving Authority.

2.6 Geometric Design Requirements

The geometric design of bridges and the bridge approaches should include the integration of vertical and horizontal alignment of the bridges with the highway geometry. This should cover the following.

Alignment Plan: Shall provide the details on horizontal alignment, curve detail, width of bridge deck elements and other details such as carriageway, walkway, median, crash barrier, railing, shoulders, side drains, right of way, etc.

Vertical Profile: Shall provide the vertical alignment, curves, grade line, high flood level, ground profile, etc.

Cross Sections: Shall provide the carriageway, camber/ super elevation, side slopes, median, walkway, hydraulic details, structural arrangement, type of super structure, sub structure, foundations, abutment, protection works, retaining walls, etc.

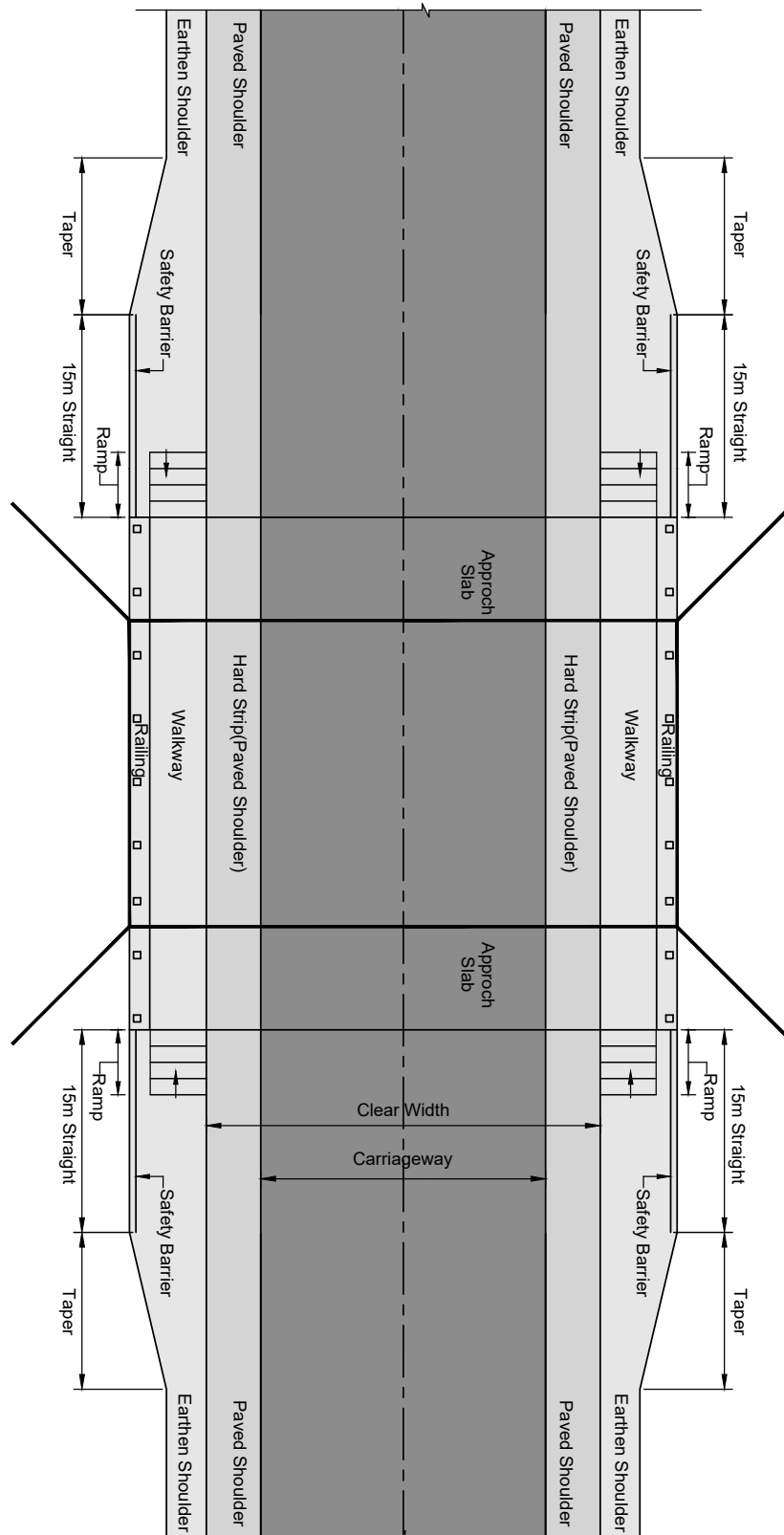
Railings/ safety barriers shall be provided along the edges of the bridge deck as per the requirements of Safety Standards. The walkway/ cycle tracks adjacent to the carriageway shall be separated by safety crash barrier. For the bridges on navigational streams, the bridge structure shall be protected against vessel collision by fenders, protection islands or any other means.

The horizontal and vertical alignments of the bridges shall meet the requirements of “RDM 1.3”. Following aspects shall also be considered.

- i. The alignment of the roadway shall be selected such as to make the bridge crossing nearly square as possible. If not possible, the skew angle shall preferably not exceed 30°.
- ii. The longitudinal gradient and cross slope on bridge shall be consistent with the approach roads. A minimum 0.3% longitudinal gradient is desirable for drainage of deck.
- iii. For the bridges in curve, constant curvature is desirable within bridge portion. Sudden change in curvature in bridge portion shall be avoided. Vertical sag curve shall be avoided as far as possible.
- iv. Design speed for the bridge shall be consistent with the design speed on approach roads and shall preferably be maintained uniform for entire length of the bridge.
- v. The bridge approaches on either side for a length of at least 15 m shall continue to have the same horizontal profile as that of bridge. Transition to be provided after that to match with the roadway section as per requirements in “Geometric Design Manual”.

- vi. For divided carriageways, the finished road levels at the median for both carriageways shall preferably be same at any cross-section.
- vii. When a new bridge is constructed adjacent to the existing bridge, necessary space shall be maintained to have clearance between the foundations. A minimum of 2.5 m clear gap shall be provided between the two bridges to allow for working space.

Figure 2.4 Typical Plan of Bridge and Merging with Approach Road



2.7 Hydrological and Hydraulic Design Requirements

2.7.1 General

All the hydrological and hydraulic data and parameters required for the design of bridges shall be collected and analysed in accordance with the provisions in “RDM Vol 2”. Design high flood level, design discharge, waterway, afflux, freeboard, and clearances are the important hydraulic parameters for planning and design of the bridges.

The design flood level shall be decided based on past flood records, site visits, flood marks on the banks, survey maps, hydrology and hydraulic studies. Highest flood levels shall also be obtained from local enquiries from aged persons that have lived for long periods proximity to the crossing at 2-3 different locations on site.

2.7.2 Hydrological Design Requirements

Hydrological investigations, survey, data collection, method, analysis and optimal flow return period shall be carried out in accordance with the guidelines provided in “RDM 2.1”. The data on hydrological parameters such as rainfall, river flow, river gauge levels, known highest flood marks, key hydrological year data and river cross sectional profiles shall be collected to determine the peak design flood discharge. The catchment area characteristics shall be established carefully to arrive at realistic flood discharges. The peak design discharge shall be based on the maximum flood discharge for a return period recommended in “RDM 2.1”. The recommendations are reproduced in Table 2.5 for ready reference of users.

Table 2.5 Proposed Applicable Design Flow Return Periods (Years)

S/No	Structure Type	Design Frequency	Check Frequency
1	Pipe Culverts	10	25
2	Box Culverts (total area of opening/s $\leq 6.0 \text{ m}^2$)	25	50
3	Box Culverts (total area of opening/s $> 6.0 \text{ m}^2$)	50	100
4	Small ($6\text{m} < \text{length} \leq 15\text{m}$) and Medium ($15\text{m} < \text{length} \leq 50\text{m}$) Bridges	50	100
5	Large ($\text{length} > 50\text{m}$) Bridges	100	200

2.7.3 Hydraulic Design Requirements

The detailed hydraulic design for bridges and culverts shall be carried out in accordance with “RDM 2.2”. The length of bridge, peak design flood discharge and velocity, peak flood levels, afflux, vertical clearance above design peak flood levels and scour are the important aspect of bridge design and to be assessed in accordance with the principles of hydraulics and the design guidelines provided in “RDM 2.2”.

Linear waterway to be used for deciding the length of bridge for the channels with defined banks shall be the distance between the banks at high flood level for the design discharge. For undefined channels with erodible bed/ banks, the linear waterway shall be determined with the principles of hydraulics provided in “RDM 2.2”. For meandering rivers, adequate river training works shall be provided. Hydraulic or Mathematical model studies may be conducted for such rivers to ascertain the design parameters. If a new bridge is constructed close to an existing bridge and if the existing bridge is functioning well, the linear waterway for new bridge shall generally be the same as in existing bridge.

Restricted waterway can be considered for the rivers flowing over large widths with shallow depths. The effect of such restriction shall be analysed on flow velocities, scouring, afflux and other site conditions. Adequate protection measures shall be provided for safety against erosion and to guide the flow through bridge section.

1

The effective linear waterway is the net distance between the abutment faces perpendicular to the direction of flow calculated after deducting the total width of obstructions due to piers or other structures obstructing the flow.

2

Vertical clearance (freeboard) for river bridges is the height from the design flood level with afflux or high tide level to the lowest point of the super structure at the position of bridge where clearance is being denoted. The minimum vertical clearance for river bridges shall be taken as per the “RDM 2.2”. The recommendations are reproduced in Table 2.6 for ease of reference to users.

Table 2.6 Minimum Vertical Clearance for River Bridges

Structure Type	Minimum Vertical Clearance (m)
Small and medium bridges (6 m < length ≤ 50 m)	1.0*
Large bridges (length > 50 m)	1.5
Notes: 1. * Recommended up to 1.5m if vertical alignment permits. 2. For the bridges on navigational rivers or over canals, the vertical clearance shall be provided as per the requirements of concerned Authorities.	

The bearings shall not be less than 0.50m above the design high flood level. The minimum free board for approach embankments shall not be less than 1.50m.

The maximum depth of scour shall be adopted for the design of foundations for piers, abutments, protection and river training works. The slopes in front of abutments and bridge approaches in contact with water at peak design flood level shall be provided with pitching/ rip-rap and launching apron.

The hydraulic and mathematical modelling shall be carried out under complex situations such as for big rivers having uncertain and complex flow conditions, linear waterway and scour cannot be assessed fairly, or the protection and river training works cannot be designed fairly due to uncertain or complex conditions.

2.8 Geotechnical Design Requirements

Geotechnical aspects are important in design of foundations, retaining and protection works. The geotechnical investigations and studies shall be carried out in accordance with the “RDM 4.1” and other relevant standards which plays an important role in deciding the type of foundations. The study may include geology, geomorphology, seismicity and history of the site. Ground investigations should consist of field investigations, laboratory testing and additional desk studies where appropriate.

The geotechnical investigations shall be carried out to determine the essential properties of sub-strata and assessment of the geotechnical parameters used in design. The location and depth of investigations shall be based on the anticipated type and design of the foundations e.g., shallow or pile foundations, dimensions of the structure, variability and complexity in sub-soil conditions. The depth of investigations shall extend to the strata that will affect the project.

The detail geotechnical investigations shall be carried out by a qualified Geotechnical Engineer. The detail requirements are to be assessed by the geotechnical engineer. Exploratory drilling shall be carried out to assess the sub-soil properties. All the measurements and related information shall be recorded for the tests conducted at site and samples collected for onward testing in laboratory to determine the geotechnical properties. As a general guidance for normal structures and non-complex ground conditions, the spacing and depth of investigation points for foundations may be selected as per the guidelines provided below. Detailed information is provided in “RDM 3.1”.

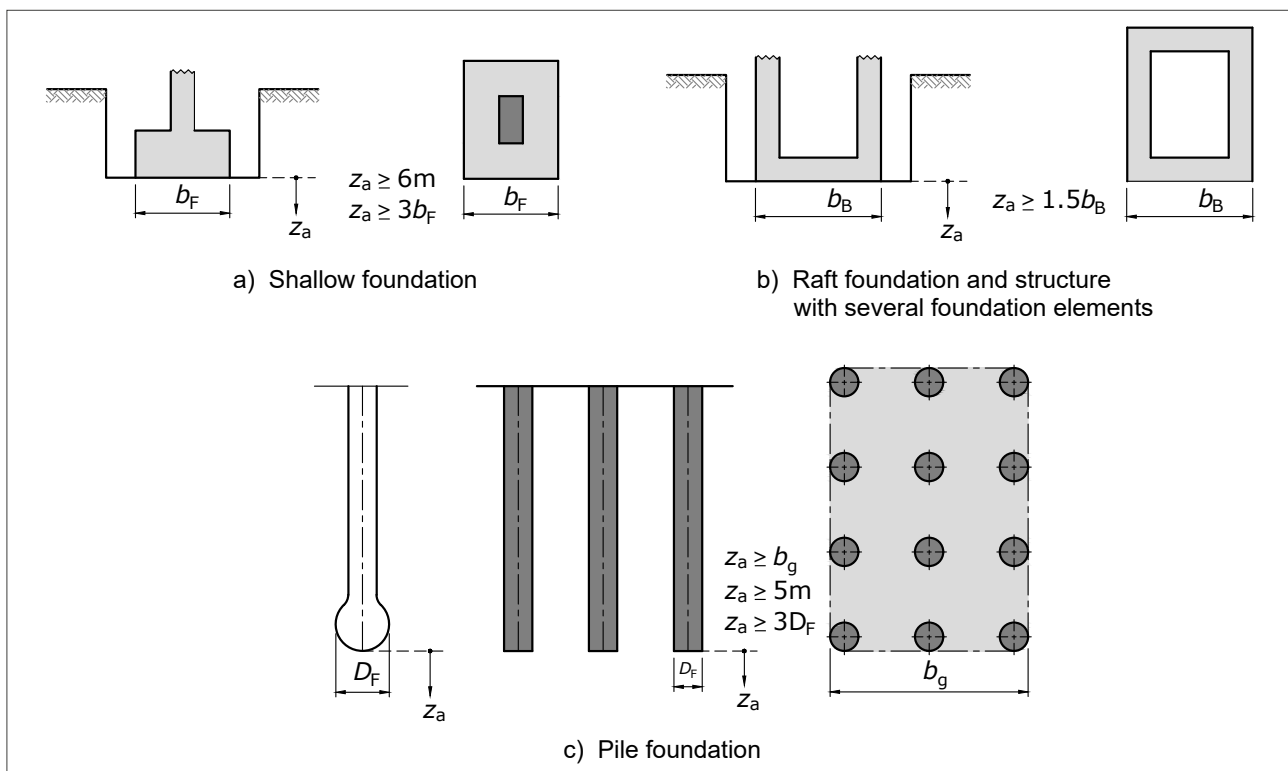
- a. A minimum of two boreholes per substructure shall be required for bridge piers or abutments ≤ 30 m wide and a minimum of four borings per substructure for bridge piers or abutments greater than 30 m wide. Additional exploration points may be provided where erratic subsurface conditions are encountered.

- b. For drilled shaft foundations, at least one borehole should be drilled for each shaft where the diameter of the shaft is greater than 1.5m, especially for shafts socketed into bedrock.
- c. Depth of preliminary investigations below ground level shall be at least ≥ 10 m for box culverts, ≥ 15 m for bridge structures and ≥ 20 m for hardstone investigations.
- d. Depth of investigations for shallow foundations shall be greater of the following (refer Figure 2.5 a):
 - $z_a \geq 6$ m
 - $z_a \geq 3 b_F$ where b_F is the smaller side length of the foundation
- e. For raft foundations, and structures with several foundation elements whose effects in deeper strata are superimposed on each other, the depth of investigation shall be greater than the following (refer Figure 2.5 b):
 - $z_a \geq 1.5 b_B$ where b_B is the smaller side length of the structure.
- f. Depth of investigations for pile foundations shall be greater of the following (refer Figure 2.5 c):
 - $z_a \geq 5$ m
 - $z_a \geq b_g$ where b_g is the smaller side length of the rectangle circumscribing the group of piles forming the foundation at the level of the pile base.
 - $z_a \geq 3 D_F$ where D_F is the pile base diameter.
- g. The depth of investigation (z_a), as shown in Figure 2.5, shall be counted from the lowest point of foundation. If a bedrock formation is encountered at the proposed base of the structure, this level shall be taken as the reference level for z_a . Otherwise, z_a shall refer to the bedrock formation.

Greater investigation depths should always be selected, where unfavourable geological conditions, such as weak or compressible strata below strata of higher bearing capacity, are presumed.

Where the structure is built on competent strata, the depth of investigation may be reduced to $z_a = 2$ m. If the geology is indistinct, at least one borehole shall be taken down to a minimum depth of $z_a = 5$ m.

Figure 2.5 Minimum Depth of Investigations for Structure Foundations



1

2

General Requirements

The aims of a geotechnical investigations are to establish the soil, rock and groundwater conditions, to determine the properties of the soil and rock, and to gather additional relevant knowledge about the site. Geotechnical investigation assists in determining the foundation design by considering factors like soil/rock-bearing capacity, settlement, and lateral stability.

Following parameters, in general, may be considered as part of geotechnical investigations.

Characterisation of soil and rock types in accordance with relevant test under RDM 3.1:

- a. Weight density.
- b. Density index.
- c. Degree of compaction of natural ground.
- d. Shear strength.
- e. Soil stiffness.
- f. Quality and properties of rocks and rock masses.
- g. Permeability and consolidation parameters of soil and rock.

All the results of field and laboratory investigations shall be presented in Ground Investigation Report. The geotechnical investigations shall provide sufficient data for ground and ground water conditions at and around the site to assess the essential ground properties and geotechnical properties to be used in design calculations. The safe bearing capacities of the ground stratum at different depths shall be assessed and presented in the report along with the characteristic values of geotechnical parameters that are to be used in design calculations.

The test results shall be evaluated and interpreted by a qualified Geotechnical Engineer. The geotechnical parameters and coefficients shall be derived from the test results using appropriate theoretical relationships, correlation and experience. The report shall normally consist of presentation of all available geotechnical information including geological features and relevant data and geotechnical evaluation of the information, stating the assumptions made in the interpretation of the test results.

2.9 Foundation Design Requirements

2.9.1 General

Foundations are one of the essential parts of the bridges and culverts and transfers the load from the structure to the ground in such a way that the bearing capacity of the founding strata is not exceeded and the settlement of the structure remains within permissible limits. The maximum permissible limits of settlement for cohesive soils shall be 25 mm and 65 mm for cohesionless soils unless otherwise specified in relevant clause of this Manual.

Foundation types depend primarily on the depth and safe bearing pressures of the bearing stratum and also restrictions placed on differential settlement due to the type of bridge deck.

Bridge foundations are classified into following two main categories:

- Shallow foundations such as spread footing, strip footing, raft footing
- Deep foundations such as pile foundations

Design of foundations comprises of the following:

- To select the stratum that has the adequate bearing capacity in accordance with the site investigations.
- To select the type of foundation comparing the suitability of different types based on technical, economic and feasibility considerations.
- Design of the foundation to transfer and distribute the loads from structure to the ground by verification for ULS and SLS conditions with applying appropriate partial factors to actions and the geotechnical parameters.

Size and depth of the footing is determined by considering the effects of vertical loads and rotational loads. The combination of these two must neither exceed the safe bearing capacity of the stratum nor produce uplift. The thickness of the footings shall be capable of withstanding moments and shears produced by the loads and the ground resistances.

The minimum depth of shallow foundations shall not be less than 2.0 m below the maximum scour level or the protected bed level when foundations are resting in soil strata. For foundations resting on rock, the minimum embedment of the foundations shall not be less than 1.5 m when embedded in weak rock (unconfined compressive strength 5 to 25 MPa) and 0.6m when embedded in medium strong rock or higher (unconfined compressive strength > 25 MPa).

For pile foundations of river bridges, the minimum diameter of pile shall not be less than 1.0 m. The pile diameter may be reduced up to 0.75 m for bridges beyond water zone and for bridges on land. For friction piles, the spacing at centre to centre of piles should not be less than three times the diameter of pile. The spacing may be reduced to two times the diameter if the piles are deriving their resistance mainly from end bearing.

2.9.2 Shallow Foundations

Shallow foundations distribute the loads from the structure to a relatively large area and do not extend much deeper into the ground. These foundations are suitable when the ground strata with higher bearing capacities are available at shallow depths to support the loads from the structure.

Spread footings may be provided individually for each pier/ abutment. Strip foundations are used to support the piers in a row when the individual footings either overlap or are close to each other.

Raft foundations are suitable where the soil has a low bearing capacity and the bridge loads are higher such that the spread footings for individual piers/ abutments overlap or close to each other. The raft foundations are also suitable when the soil strata have some compressible soft pockets or when there is a risk of unequal settlements.

2.9.3 Deep Foundations

Deep foundations extend much deeper into the ground compared to shallow foundations. Pile foundations support the loads using friction along their depth in soil strata (friction piles) or using end bearing when the piles extend up to bedrock/hard soil (end bearing piles) or a combination of both. Piles may be bored cast-in-situ piles that are formed by drilling a hole and filling it with reinforced concrete. Driven piles may be of concrete or steel and that are driven deep into the ground by blows of power hammer or jacked into the ground.

Deep foundations are suitable when the bearing capacity at shallow depths are relatively low and the ground strata conditions are such that it is desirable to transmit the load to a stratum that are deeper than the reach of shallow foundations. It is also favourable where heavy scouring of river bed is expected. Pile foundations offer greater resistance against uplift and provide structural stability against lateral and overturning forces.

2.9.4 Selection of Foundation

Bridge foundations are the most critical design consideration for all bridge projects. Proper selection of the foundation type is necessary to maintain the structural integrity and stability of the entire structure. Some of the important points to consider while selecting foundation are:

- Location of structure, e.g. on land, river or sea.
- Scour depth for river bridges.
- Geological conditions.
- Type of superstructure – continuous or simply supported.
- Availability of time for dry river bed for construction of foundations.
- Cost of temporary works, e.g. cofferdams, stream diversion, etc.
- Available equipment and construction technology.
- Magnitude and type of applied loads (dead loads, imposed loads, wind loads, etc.).
- Permissible bearing pressures.
- Acceptable level of settlement.
- Location and proximity of adjacent structures.
- Economic considerations.

As a guidance, following site conditions may be considered in selection of foundations.

Table 2.7 Site Conditions for Foundation Selection

Type of Foundation	Suitable Site Conditions	Unsuitable Site Conditions
Shallow foundations	• Presence of hard soil and rock formations at shallow depths • Ground water surface elevation is below bottom of foundation level	• Soft compressible soil • Collapsible / expansive soils • Excavation below ground water level • Rock dipping in adverse direction
Deep foundations	• Lower bearing capacity at ground surface and near surface • Good tip resistance of piles on dense / hard soil and rock	• Presence of hard/ rock at shallow depths • Deep soft soils • Potential caving in loose soils

2.10 Software for Bridge Design

There are numerous computer software programs used in analysis and design of bridges. They provide necessary design, analysis, and reporting tools. The software shall have a user-friendly interface for modelling, analysis, and generation of reports with checking for geometrical and other data errors. It should be able to create load models for different load positions of moving traffic loads to analyses the critical load position and the effect of the loads together with the appropriate load combinations with other load models. The analysis may be 2D or 3D as per the structural analysis requirements of the bridge.

The software to be used for a particular bridge analysis and design shall be agreed with the approving authority.

3 Actions and Combination of Actions

3.1 General

3.1.1 Scope

The bridges are subjected to various actions such as self-weight of construction material, traffic loads, wind actions, seismic actions, etc. during its service life and some of these during construction stage. An assessment of these actions is essential to enable structural designs. This Chapter provides guidance on assessment of the actions to be considered in the design of structures.

3.1.2 Definitions

For general definitions, refer initial sections of this Manual. Other definitions are as under.

Basic wind velocity	The fundamental basic wind velocity modified to account for the direction of the wind being considered and the season (if required).
Behaviour factor	A factor used for design purposes to reduce the forces obtained from linear analysis of seismic actions in order to account for the non-linear response of a structure, associated with the material, the structural system and the design procedures.
Bulk weight density	The bulk weight density is the overall weight per unit volume of a material including a normal distribution of voids and pores.
Construction load	The load that can be present due to execution activities, but is not present when the execution activities are complete.
Deck	Deck is that part of a bridge which carry the traffic loading.
Ductile member	A member able to dissipate energy through the formation of plastic hinge.
Ductile structure	Structure that under strong seismic motions can dissipate significant amounts of input energy through the formation of an intended configuration of plastic hinges or by other mechanisms.
Uniform Bridge Temperature component	The temperature, constant over the cross section, which governs the expansion or contraction of an element or structure.
Footbridge	The bridge mainly used to carry pedestrian and/or cycle track loads. The road traffic is not permitted on footbridge except permitted vehicles such maintenance vehicle.
Fundamental basic wind velocity	The 10-minute mean wind velocity with an annual risk of being exceeded of 0.02 i.e. 50 years return period, irrespective of wind direction, at a height of 10m above flat open country terrain and accounting for altitude effects (if required).
Initial temperature	The temperature of a structural element at the relevant stage of the restraint.
	The load that can be present due to execution activities, but is not present when the execution activities are complete.
Limited ductile behaviour	Seismic behaviour of bridges without significant dissipation of energy in plastic hinges under the design seismic action.
Mean wind velocity	The basic wind velocity modified to account for the effect of terrain roughness and orography.

1

Non-structural elements

The non-structural elements are those that include completion and finishing elements connected with the structure, comprising such as road surfacing, non-structural parapets.

2

Special vehicle / Abnormal vehicle

Vehicle used to carry heavy equipment, machinery. Such vehicle not to be carried on a route without permission from relevant authority.

3

Stress range

The algebraic difference between the two extremes of a particular stress cycle derived from a stress history.

Structural elements

The structural elements comprise the primary structural frame and supporting structures, comprising such as girders, structural slabs, piers.

Substructure

Substructure is that part of a structure which supports the superstructure. The foundations, abutments, piers and columns etc. are referred to as substructure for bridges.

Superstructure

Superstructure is that part of a structure which is supported by the substructure. Bridge deck is referred to as superstructure.

Temperature Difference component

The part of a temperature profile in a structural element representing the temperature difference between the outer face of the element and any in-depth point.

3.1.3 Symbols

The general symbols used in this Chapter are as under. Additional symbols are also defined where they appear. Symbols may have various meanings.

Q_{ik}	Characteristic value of vertical axle load of Load Model 1 on notional lane number i of the road bridge.
q_{ik}	Characteristic value of vertical uniformly distributed load of Load Model 1 on notional lane number i of the road bridge.
α_{Qi}, α_{qi}	Adjustment factors for some load models for lane i .
Q_{ak}	Characteristic value of vertical single axle load of Load Model 2 of a road bridge.
β_Q	Adjustment factor for Load Model 2.
q_{fk}	Characteristic value of vertical uniformly distributed load on walkways or footbridges.
Q_{fwk}	Characteristic value of vertical concentrated load (wheel load) on walkways or footbridges.
Q_{tk}	Characteristic value of transverse or centrifugal force on road bridge.
Q_{lk}	Characteristic value of longitudinal forces (braking and acceleration forces) on a road bridge.
Q_{trk}	Transverse breaking force on road bridges.
γ	Unit weight of soil (in kN/m^3)
F_{wa}	Forces on structural component due to water
v_b	Basic wind speed
A	Area
F_w	Wind force
R_e	Reynolds number
T_{max}	Maximum shade air temperature
T_{min}	Minimum shade air temperature

$T_{e\ max}$	Maximum effective bridge temperature
$T_{e\ min}$	Minimum effective bridge temperature
T_0	Initial temperature when structural element is restrained
S_e	Elastic response spectrum
S_d	Design response spectrum
a_g	Design ground acceleration
S	Soil factor
q	Seismic behaviour
Q_c	Construction loads
G_k	Characteristic value of permanent action
P	Representative value of pre-stressing action
Q_{kl}	Characteristic value of leading variable action
Q_k	Characteristic value of accompanying variable actions
A_d	Design value of an accidental action
A_{Ed}	Design value of a seismic action
γ_G	Partial factor for permanent action
γ_Q	Partial factor for variable action
ψ_0	Factor for combination value of a variable action
ψ_1	Factor for frequent value of a variable action
ψ_2	Factor for quasi-permanent value of a variable action
$G_{k\ sup}$	Upper characteristic value of permanent action
$G_{k\ inf}$	Lower characteristic value of permanent action

3.2 Self-weight of Construction Works

The self-weight of the construction works, in most cases shall be represented by a single characteristic value and be calculated on the basis of characteristic values of the densities of the materials of the components under consideration and the nominal dimensions. The total self-weight of structural and non-structural fixed elements are classified as permanent actions and shall be taken into account in combination of actions as a single action.

The self-weight of the construction works shall consist of weight of the materials and parts of the structure that are structural elements such as girders and deck slabs and non-structural elements such as road surfacing, parapets, ducts, miscellaneous furniture, etc. including fixed services as well as the weight of earth. The characteristic value for densities of some construction materials given in Table 3.1 shall be used for initial calculations unless they have been established by actual weighing of representative samples. For materials which are not covered in this Table, the characteristic value of the density shall be assessed and agreed for individual project.

The upper and lower characteristic values of densities for non-structural parts, such as fill above culverts, should be taken into account if the material is expected to consolidate, become saturated or change its properties, during use. Self-weight of fill should be defined for the individual project depending on the estimated change with time due to consolidation, saturation, etc. and considering local conditions, type of soil, ground water variation and degree of consolidation.

To determine the upper and lower characteristic values of self-weight of waterproofing, surfacing and other coatings for bridges, where the variability of their thickness may be high, a deviation of the total thickness from the nominal or other specified values should be taken into account. This deviation may be taken equal to $\pm 40\%$ if a post-execution coating is included in the nominal value, and to $+55\%$ and -40% if such a coating is not included.

Table 3.1 Densities of Construction Materials

Material	Unit Weight (kN/m ³)
Concrete and Steel	
Normal weight concrete	24
Reinforced Concrete	25
Pre-stressed Concrete	25
Steel	78.5
Pavement of road bridges	
Asphalt concrete	24 to 25
Mastic asphalt	18 to 22
Hot rolled asphalt	23
Infills for bridges	
Sand (dry)	15 to 16
Ballast, gravel (loose)	15 to 16
Hardcore	18.5 to 19.5
Crushed slag	13.5 to 14.5
Packed stone rubble	20.5 to 21.5
Note: The unit weight may be increased by 1 kN/m ³ for unhardened/ fresh concrete	

For the self-weight of cables, pipes and service ducts, the upper and lower characteristic values should be taken into account. Deviation from the mean value of the self-weight of $\pm 20\%$ should be taken into account.

For the self-weight of other non-structural elements such as handrails, safety barriers, parapets, kerbs, joints/fasteners, void formers, the characteristic values should be taken equal to the nominal values.

3.3 Traffic Loads on Bridges

The characteristic design traffic loads shall consist of that produced due to standard vehicles defined in this Manual. The standard vehicle shall be assumed to travel within kerb-to-kerb width of roadway parallel to the length of the bridge and to occupy any position which will produce maximum forces provided that minimum clearances from the roadway face of the kerb or between the vehicles are not encroached upon. For each standard vehicle, all the axles of a unit of vehicle shall be considered to act simultaneously in a position causing maximum stresses covering all normally foreseeable traffic situations (i.e. traffic conditions in either direction on any lane due to the road traffic). The load models defined in this Chapter shall be used for design of road bridges with loaded lengths less than 200 m.

Note: UK National Annex specifies that Load Model-1 may be used for loaded lengths up to 1500 m.

The traffic and pedestrian loads under normal use shall be considered as variable action. The actions due to collision of road vehicles or their accidental presence shall be considered as actions under accidental design situation.

3.3.1 Notional Lanes

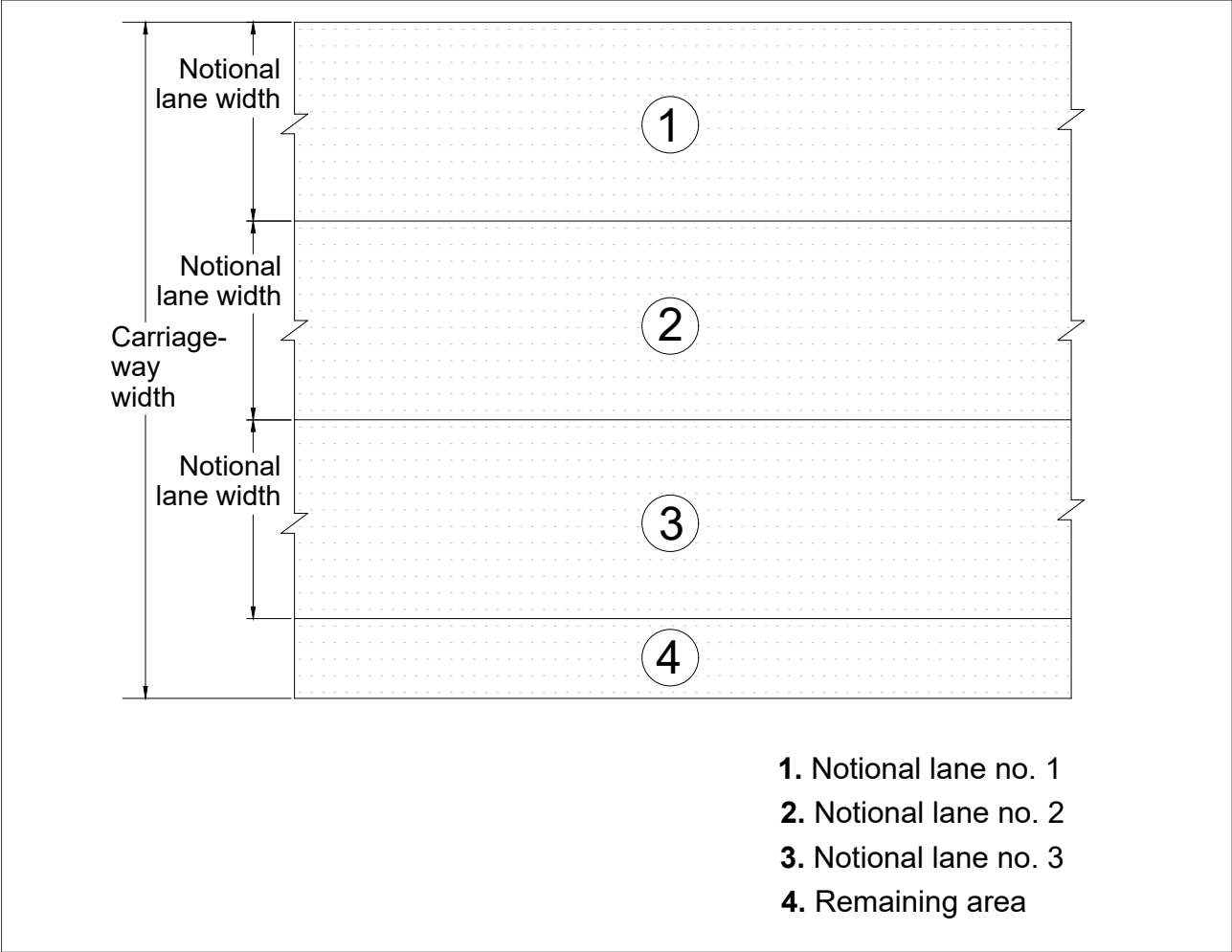
The carriageway of the bridge to be divided into number of notional lanes and the width of notional lanes for the purpose of design. The number of notional lanes shall be the integer part of clear carriageway width (in meters) between the kerbs divided by 3. Any future changes in clear carriageway width of the bridge shall be considered. The number of notional lanes shall be taken as per Table 3.2.

Where the carriageway on a bridge deck is physically divided into two parts separated by a permanent central reservation, then each part including hard shoulder or strip, shall be separately divided into notional lanes. If it is divided by a temporary central reservation, then the whole carriageway including central reservation shall be divided into notional lanes.

Table 3.2 Notional Lanes

Carriageway width (w)	Number of notional lanes (n_l)	Width of notional lane (w_l)	Width of remaining area
$w < 5.4$ m	1	3 m	$w - 3$
$5.4 \leq w < 6$	2	$w/2$	0
$w \geq 6$	$n = \text{Int}(w/3)$	3 m	$w - 3 n_l$

Figure 3.1 Example of Lane Numbering



The location of notional lanes shall not be necessarily related to their numbering. The numbering, their location on the carriageway and number of lanes to be loaded for each individual verification should be chosen such that the effects from different load models are the most adverse. The lane giving the most unfavourable effect shall be numbered as Lane Number 1.

Where the carriageway on a single deck consists of two separate parts, only one numbering system shall be used for the whole carriageway. Where the carriageway consists of two separate parts on two independent decks, separate numbering shall be used for design of each deck. If both decks are supported on a single pier or abutments, one numbering shall be used for both parts of the carriageway for the design of pier or abutments.

3.3.2 Design Load Models

Traffic loading consists of the following:

- i. Load Model-1
- ii. Load Model-2
- iii. Load Model-3
- iv. Load Model-4

All permanent structures shall be designed to carry above loadings independently. The load models on each notional lane shall be applied on such a length and located longitudinally so that most adverse effect is obtained. On the remaining area, the associated load model shall be applied on such lengths and widths so that most adverse effect is obtained.

3.3.3 Vertical Loads

The vertical load due to different load models are defined as follow-

3.3.3.1 Load Model-1 (LM1)

The Load Model-1 consists of a combination of concentrated load and uniformly distributed load as shown in Figure 3.2-A. The concentrated load has double axles (Tandem System - TS) with each axle having a weight of $\alpha_Q Q_k$. The uniformly distributed load (UDL) has a weight of $\alpha_q q_k$ per square meter of notional lane. The α_Q and α_q are adjustment factors. This model shall be used for general and local verification.

For notional lane i , the load magnitudes are referred as $\alpha_{Qi} Q_{ik}$ and $\alpha_{qi} q_{ik}$. On remaining area, the load magnitude is referred as $\alpha_{qr} q_{rk}$. The adjustment factors α_{Qi} , α_{qi} and α_{qr} can be taken from Table 3.3.

Table 3.3 Adjustment factors α_{Qi} , α_{qi} and α_{qr} for Load Model-1

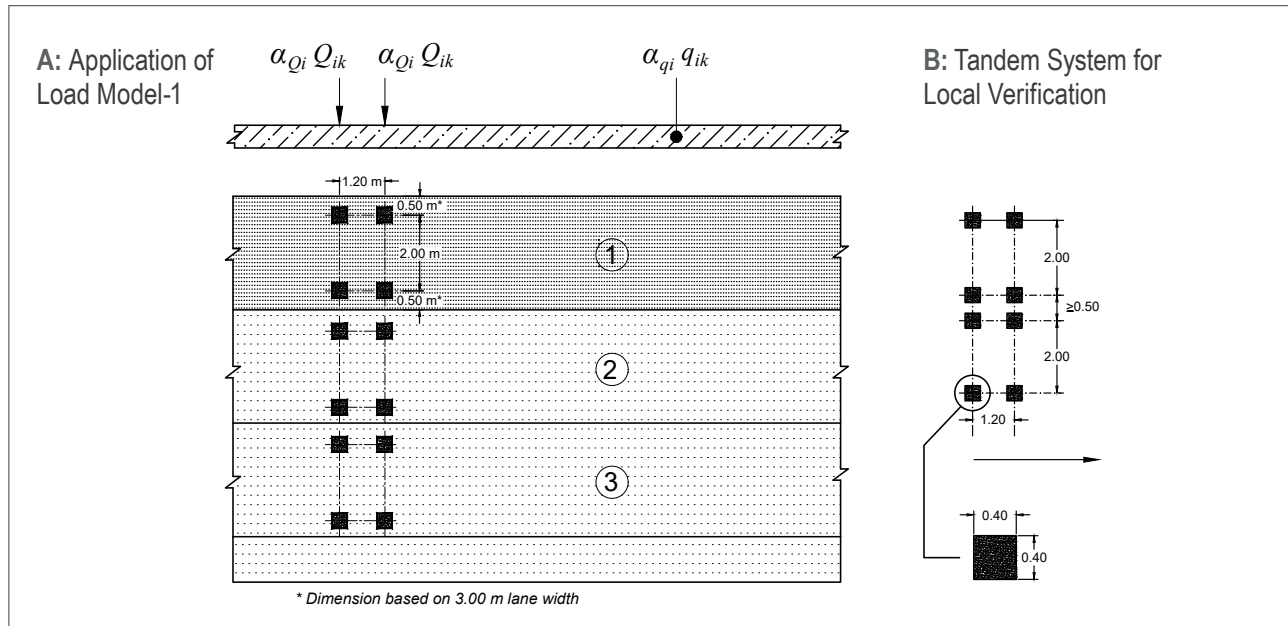
Location	α_{Qi} for tandem axle loads	α_{qi} or α_{qr} for UDL loading
Lane 1	$\alpha_{Q1} = 1.0$	$\alpha_{q1} = 0.61$
Lane 2	$\alpha_{Q2} = 1.0$	$\alpha_{q2} = 2.2$
Lane 3	$\alpha_{Q3} = 1.0$	$\alpha_{q3} = 2.2$
Other lanes	-	$\alpha_{qn} = 2.2$
Remaining area	-	$\alpha_{qr} = 2.2$
<i>The value of α_{Q1} shall be taken as 1.0 for Clause 3.3.5.1.</i>		
Note: The adjustment factors α_{Qi} , α_{qi} and α_{qr} for Load Model-1 are NDPs and the values given in Table 3.3 are based on UK National Annex.		

The characteristic values of Q_{ik} and q_{ik} including dynamic amplification to be considered for different lanes are given in Table 3.4.

Table 3.4 Load Model-1

Location	Tandem System	UDL System
	Axle load Q_{ik} (kN)	q_{ik} or q_{rk} (kN/m ²)
Lane 1	300	9
Lane 2	200	2.5
Lane 3	100	2.5
Other lanes	0	2.5
Remaining area	0	2.5

Figure 3.2 Load Model-1



For lane number 1, $Q_{1k} = 300 \text{ kN}$, $q_{1k} = 9 \text{ kN/m}^2$. For lane number 2, $Q_{2k} = 200 \text{ kN}$, $q_{2k} = 2.5 \text{ kN/m}^2$. For lane number 3, $Q_{3k} = 100 \text{ kN}$, $q_{3k} = 2.5 \text{ kN/m}^2$. For remaining area, $q_{rk} = 2.5 \text{ kN/m}^2$. Tandem axle spacing = 1.2 m

While applying LM-1, following guidelines shall be used.

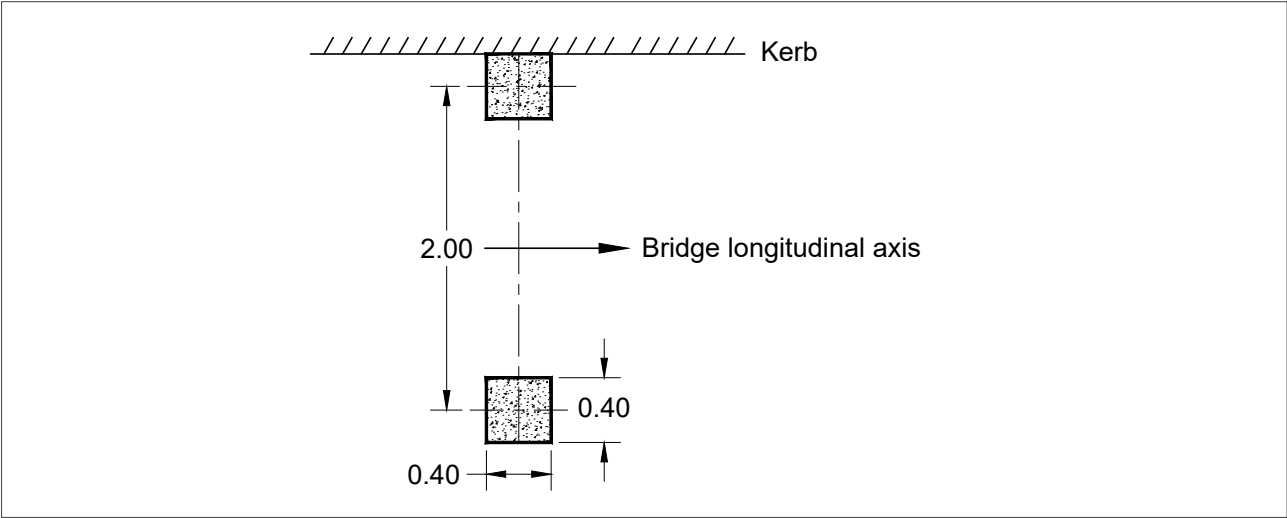
- i. Not more than one tandem system shall be considered per notional lane.
- ii. Complete tandem systems shall be taken into account.
- iii. Each tandem shall be assumed to travel centrally along the notional lanes for general effects.
- iv. For local effects, a tandem shall be applied at the most unfavourable location. Where two tandem systems are considered on two adjacent lanes, the distance between wheel axes shall not be less than 0.50 m as shown in Figure 3.2-B.
- v. Each axle of a tandem system comprises two identical wheels, load on each wheel equal to $0.5 \alpha_Q Q_k$.
- vi. Contact surface area for each wheel shall be a square of side 0.40 m.
- vii. The uniformly distributed loads shall be applied on relevant adverse parts of the influence surfaces, longitudinally and transversally.
- viii. Load model-1 shall be applied on each notional lane and on the remaining area.

3.3.3.2 Load Model-2 (LM2)

The Load Model-2 consists of one axle load of $\beta_Q Q_{ak}$ where Q_{ak} equal to 400 kN including dynamic amplification. The load can be applied on any location on the carriageway. Only one wheel of $200 \beta_Q$ kN may also be considered if relevant. The contact surface for each wheel shall be taken as a square of sides 0.40 m (See Figure 3.3).

The adjustment factor β_0 may be taken equal to α_{01} used for LM-1

Figure 3.3 Load Model-2



An additional dynamic load amplification factor as defined below shall be applied in the vicinity of expansion joints.

Additional amplification factor = $1.30 \left(1 - \frac{D}{26}\right)$, shall be ≥ 1 .

Where, D is the distance (m) of the cross-section under consideration from the expansion joint.

3.3.3.3 Load Model-3 (LM3)

The Load Model-3 represents Special Vehicles comprising a set of assemblies of axle loads. These vehicles can travel on routes permitted for abnormal loads and used to carry heavy equipment and machinery such as stator units, turbines, boilers, etc. occasionally. This Load Model is to be adopted for the design of new bridges on selected routes where passage of such vehicles may occur. This model shall be used for general and local verification.

Following are the different SV and SOV model vehicles for Load Model-3. The use of particular SV or SOV model vehicle for the design of bridges shall be approved by the authority for the individual project.

Note 1: The SV and SOV models are NDPs and the models presented are based on UK National Annex. The DMRB CD 350 recommends following guidelines for the use of SV and SOV models.

Table 3.5 SV Models to be Used in the Design

Class or road carried by highway structure	SV models
Motorway and all-purpose trunk road, principal road extensions of trunk roads	SV80, SV100, SV196
Principal roads as agreed by Approving Authority	SV80 , SV100
Other public roads as agreed by Approving Authority	SV80

Note 2: Where a highway structure carries a traffic on a route designated as a heavy load route, the structure shall be designed for the appropriate SOV models and/ or specified individual vehicles as required. The appropriate SOV model shall be agreed by the Approving Authority.

3.3.3.3.1 SV Model Vehicles

Following are three types of SV models.

- a. **SV80:** Maximum gross weight of 80 tonnes and a maximum basic axle load of 12.5 tonnes.
- b. **SV100:** Maximum gross weight of 100 tonnes and a maximum basic axle load of 16.5 tonnes.
- c. **SV196:** Represents the effect of a single locomotive pulling a special vehicle load with a maximum gross weight of 150 tonnes and a maximum basic axle load of 16.5 tonnes with the gross weight of the vehicle train not exceeding 196 tonnes.

The plan and axle load configuration are shown in Figure 3.4. The wheel loads of all the three SV model vehicles shall be uniformly distributed over a square contact area of sides 0.35m.

3.3.3.3.2 SOV Model Vehicles

Following are four types of SOV models:

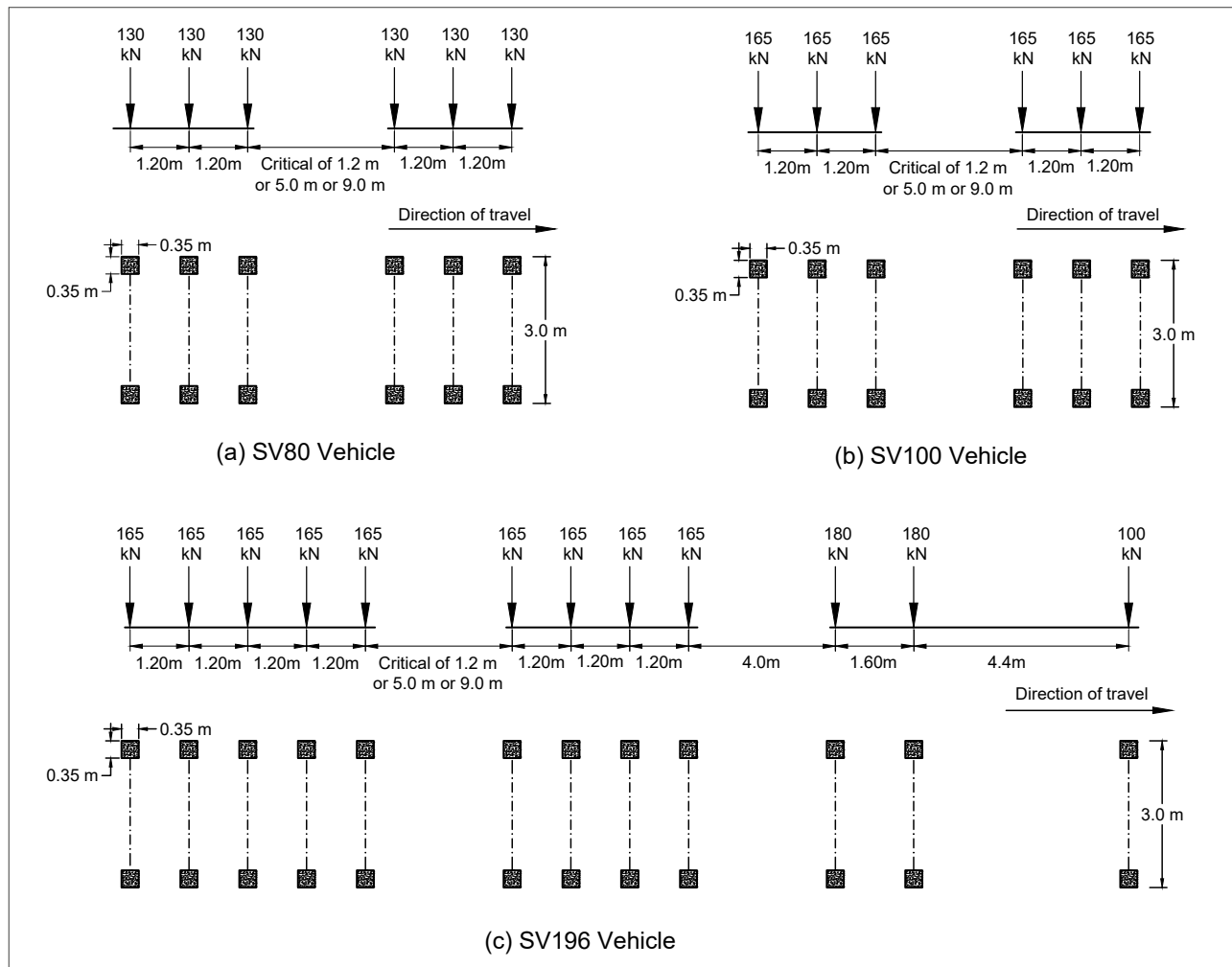
- a. **SOV-250:** Maximum total weight of SO trailer units up to 250 tonnes.
- b. **SOV-350:** Maximum total weight of SO trailer units up to 350 tonnes.
- c. **SOV-450:** Maximum total weight of SO trailer units up to 450 tonnes.
- d. **SOV-600:** Maximum total weight of SO trailer units up to 600 tonnes.

The longitudinal configuration of the four SOV load model vehicles is shown in Figure 3.5. The standard configuration has a trailer with two bogies and two tractors: one pulling and one pushing. However, on structures located on a stretch of road with a gradient steeper than 1 in 25, six tractor units in any combination of pulling and pushing that produces the worst effect, should be used for design.

The lateral wheel arrangement for the trailer axles of all the SOV model vehicles is shown in Figure 3.5 (e). All the wheels are of equal weight. The contact surface of each wheel should be taken as a square of sides 0.35 m.

The tractor axles of the model vehicles have two wheels, each of equal weight and with square contact areas of side 0.35 m. The outside track and overall width of the vehicle is 3.0 m.

Figure 3.4 Load Model 3 - SV Model Vehicles



3.3.3.3.3 Dynamic Amplification Factor

To determine the load effects of SV and SOV vehicles, the basic axle loads given in Figure 3.4 and Figure 3.5 shall be multiplied by the appropriate Dynamic Amplification Factor (DAF) for each axle as given in Table 3.6.

Table 3.6 Dynamic Amplification Factor for SV and SOV Vehicles

Basic Axle Load	Dynamic Amplification Factor
100 kN	1.20
130 kN	1.16
165 kN	1.12
180 kN	1.10
225 kN	1.07

3.3.3.3.4 Application of Special Vehicle Models

The SV or SOV vehicle loads shall be combined with Load Model 1 together with the load adjustment factors. Only one SV or SOV model vehicle should be considered on any one superstructure. The Load Model 1 should be considered to be at the "frequent" values. The loads shall be applied to each notional lane and the remaining area of the bridge deck.

The SV or SOV vehicle should be placed at the most unfavourable position transversely, either wholly within one notional lane or straddling two adjacent lanes and longitudinally over the loaded length, in order to produce the most severe load effect at the section being considered.

Figure 3.5 Load Model 3 - SOV Model Vehicles

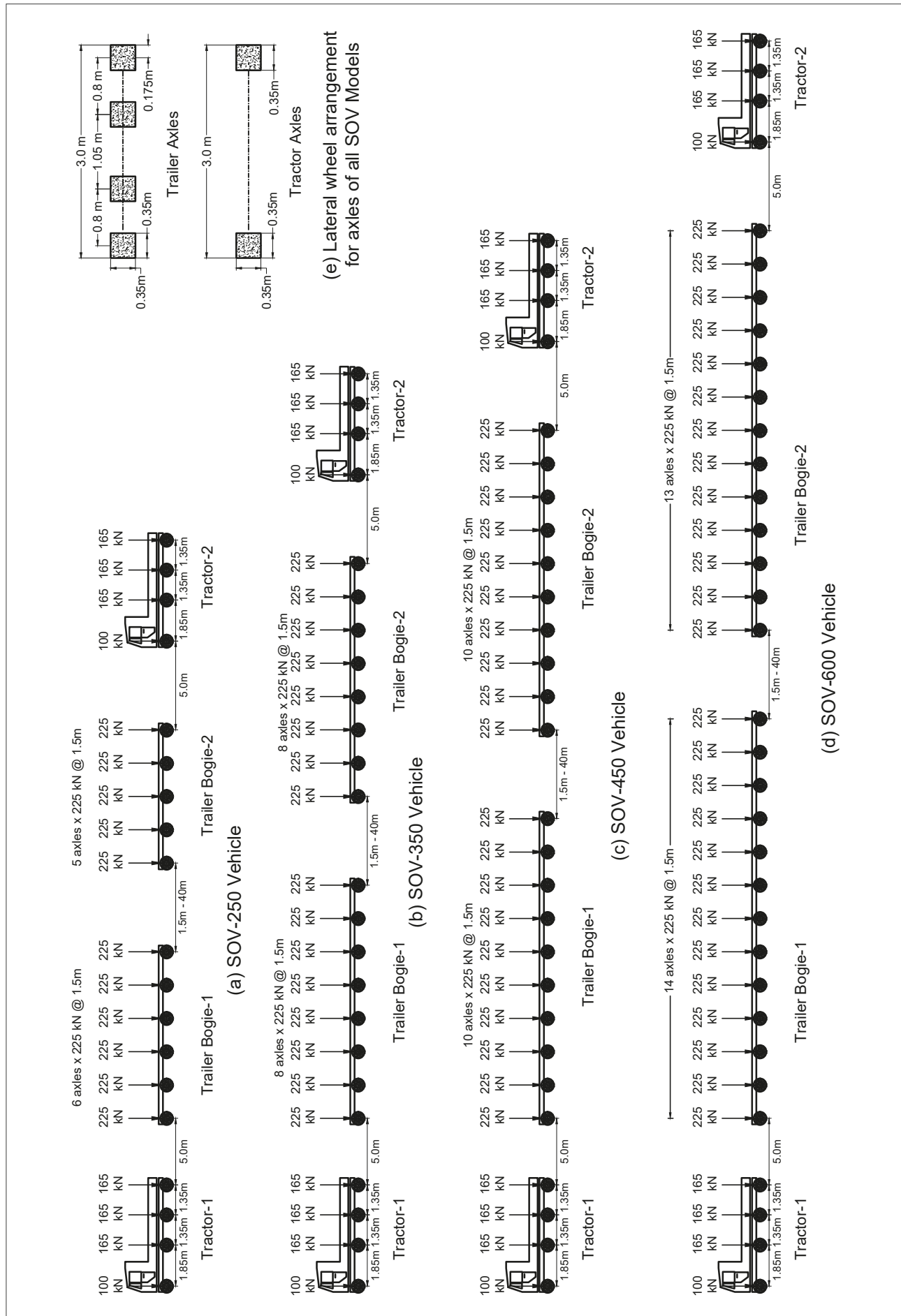
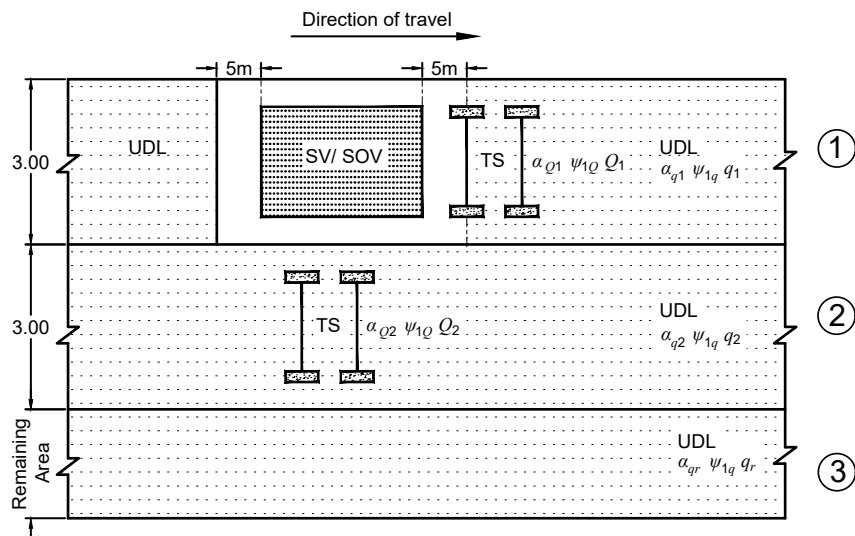
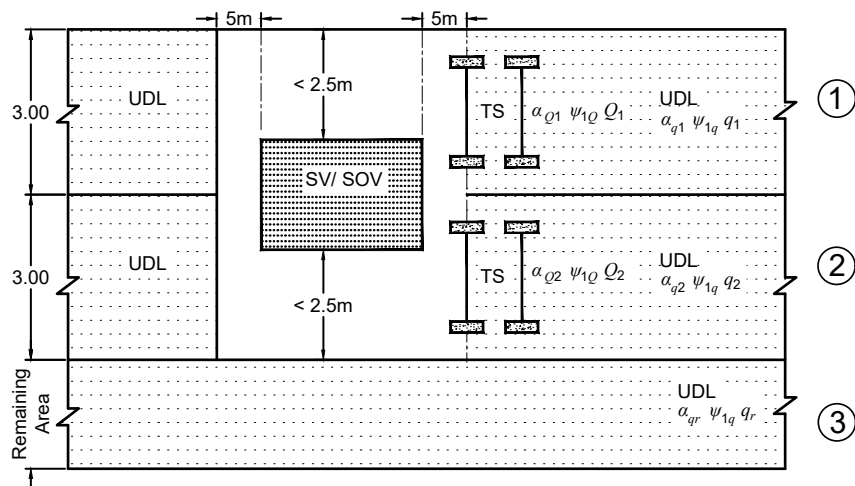


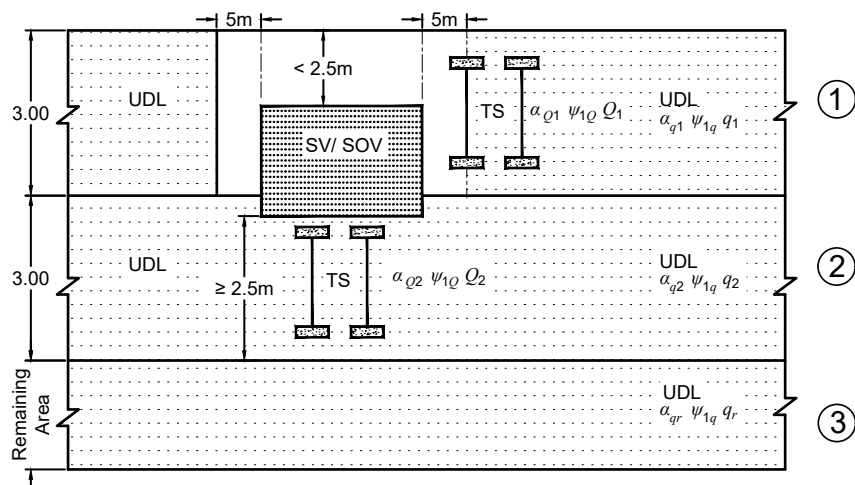
Figure 3.6 Typical Application of SV or SOV and Load Model 1



(a) SV or SOV lies fully within notional lane



(b) SV or SOV straddles two adjacent lane and distance to lane edge < 2.5 m

(c) SV or SOV straddles two adjacent lane and distance to lane edge ≥ 2.5 m

Where the SV or SOV vehicle lies fully within a notional lane, the associated Load Model 1 loads shall not be applied within 5 m from the centre of outermost axles (front and rear) of the SV or SOV vehicle in that lane as shown in Figure 3.6 (a).

Where the SV or SOV vehicle lies partially within a notional lane and the remaining width of the lane is less than 2.5 m, as shown in Figure 3.6 (b), the associated Load Model 1 loads shall not be applied within 5 m of the centre of the outermost axles (front and rear) of the SV or SOV vehicle in that lane. If the remaining width of lane is greater than or equal to 2.5 m, as shown in Figure 3.6 (c), the "frequent" value of the uniformly distributed load of the Load Model 1 may be applied over the remaining width of the notional lane. The "frequent" value of the tandem system for that notional lane may be applied anywhere along its length.

On the remaining lanes not occupied by the SV or SOV vehicle, the Load Model 1 at its "frequent" value should be applied. The notional lanes are located so as to give maximum load effect at the part of the structure considered.

3.3.3.4 Load Model-4 (LM4)

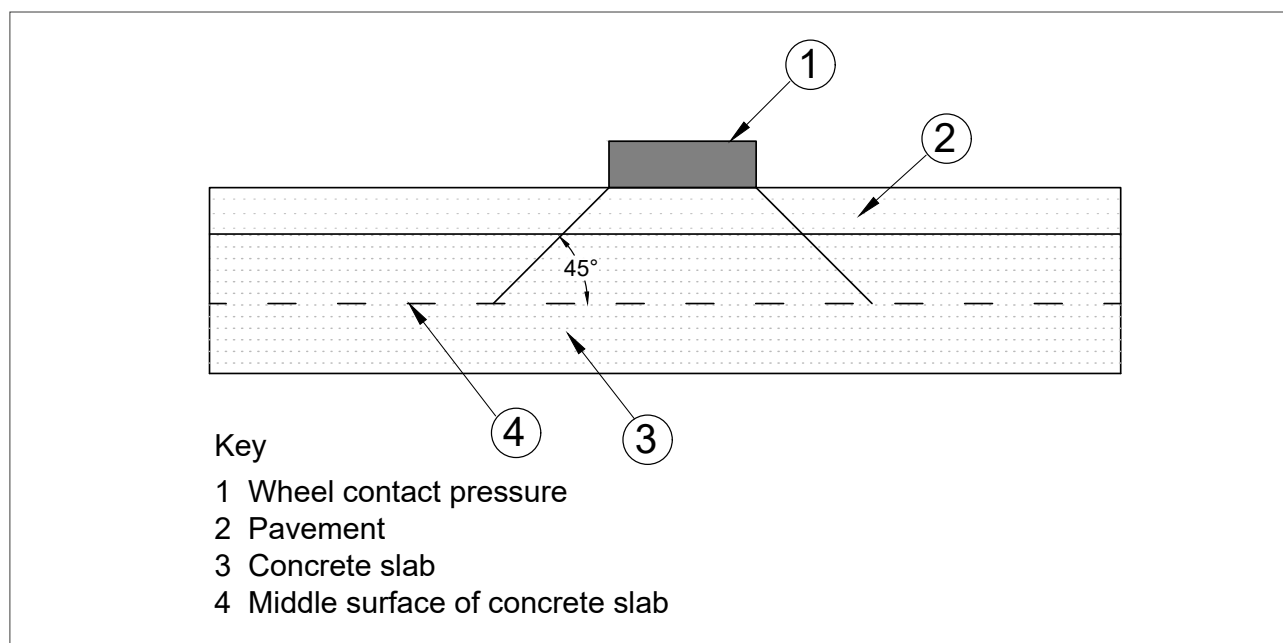
The load model-4 consists of a uniformly distributed load for crowd loading. This loading is more relevant for bridges located in or near towns. It is represented by a uniformly distributed load of 5 kN/m² which includes dynamic amplification.

Load Model 4 should be applied on the relevant parts of the length and width of the road bridge deck, the central reservation being included where relevant. This loading system, intended for general verifications, should associated only with a transient design situation.

3.3.3.5 Dispersal of Wheel Loads

The various wheel loads associated with Load Model 1 and 2 shall be taken as uniformly distributed on the respective wheel contact area. The dispersal through pavement (wearing coat) and concrete slabs should be taken at a spread to depth ratio of 1 horizontally to 1 vertically down to the level of the centroid of the slab (Figure 3.7).

Figure 3.7 Dispersal of Wheel Loads Through Pavement and Concrete Slab



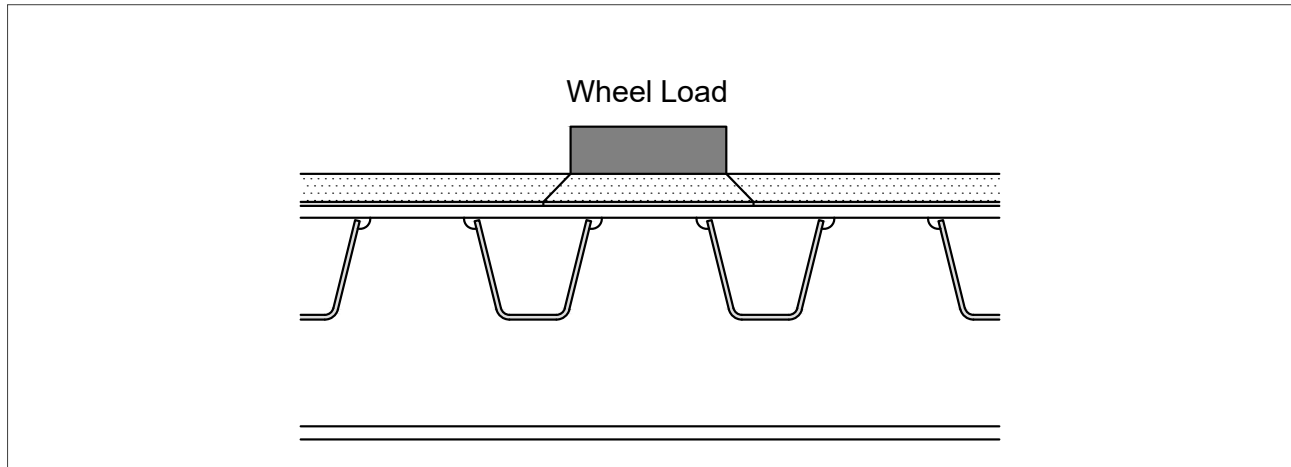
1

2

3

The dispersal through the pavement and orthotropic decks should be taken at a spread-to-depth ratio of 1 horizontally to 1 vertically down to the level of the middle plane of the structural top plate.

Figure 3.8 Dispersal of Wheel Loads Through Pavement and Orthotropic Decks



In case of dispersal of wheel loads through backfills or earth properly consolidated (such as buried structures, box culverts), the dispersal angle, transversely and longitudinally, from vertical equal to 30° may be considered.

When the wheel is near to a vertical wall, such as wing wall or head wall, the dispersal zone is curtailed by the wall, the wheel load is dispersed over the reduced length as given in Clause 10.4.7. Where the dispersion zone is curtailed by the wall or where the dispersion zone of one or more-wheel overlaps, reference shall be made to Clause 10.4.7.

3.3.4 Actions on Walkways and Cycle Tracks

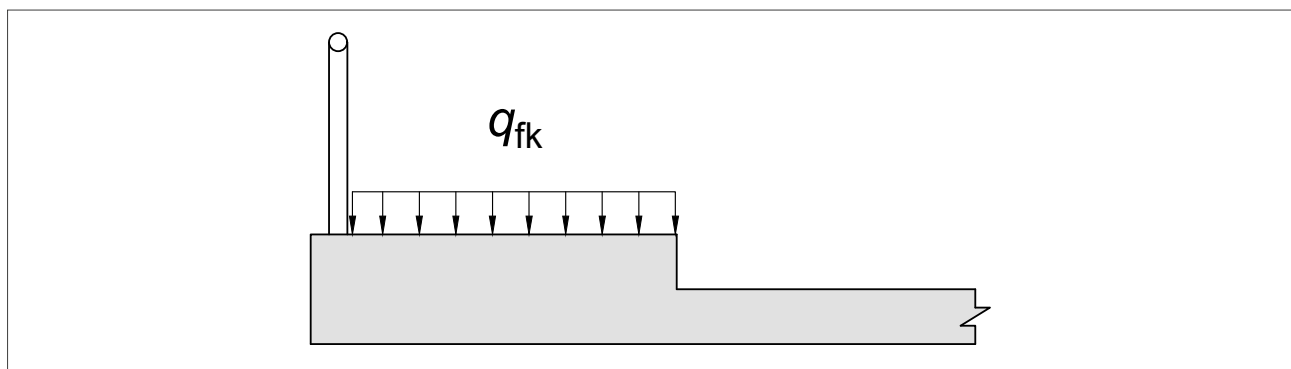
For road bridges supporting walkways or cycle tracks (Figure 3.9), the following loads shall be considered.

1. Where the risk of continuous dense crowd exists (e.g. footbridges serving sports stadiums), the load model 4 defined in Clause 3.3.3.4, corresponding to uniformly distributed load of $q_{fk} = 5 \text{ kN/m}^2$ shall be considered over the area used for pedestrians or cycle traffic which is not part of the carriageway for local and general effects. For other cases, the uniformly distributed load q_{fk} shall be taken as under.

$$q_{fk} = 2.0 + 120 / (L+10) \quad \text{kN/m}^2, \text{ where, } L \text{ is loaded length}$$

$$2.5 \text{ kN/m}^2 \leq q_{fk} \leq 5.0 \text{ kN/m}^2$$

Figure 3.9 Live Load on Walkways or Cycle tracks



2. A concentrated load of Q_{fwk} equal to 10 kN acting on a square surface of sides 0.10 m shall be considered for local effects.

3.3.5 Horizontal Forces

3.3.5.1 Braking and Acceleration Forces

The characteristic value of braking force Q_{lk} is calculated as a fraction of total maximum vertical loads corresponding to Load Model-1 likely to be applied on Lane Number 1, as follows:

$$Q_{lk} = 0.6 \alpha_{Q1} (2Q_{1k}) + 0.1 \alpha_{q1} q_{1k} w_l L$$

Equation 3.1

Where, L is the length of the deck or part of it under consideration (loaded length).

The maximum value of braking force (Q_{lk}) shall be limited to 900 kN for the total width of the bridge. The minimum value of braking force shall not be less than $180 \alpha_{q1}$ in kN.

The braking force is acting at the surface level of the carriageway and located along the axis of any lane. If the eccentricity effects are not significant, it may be considered to apply along the carriageway axis and uniformly distributed over the loaded length.

The acceleration forces shall be of the same magnitude as braking force but acting in the opposite direction.

The horizontal force transmitted to expansion joints or applied to structural members that can be loaded by one axle only, shall be $Q_{lk} = 0.6 \alpha_{Q1} (Q_{1k})$.

For buried structures with earth fill total height 0.6 m or less, the full value of longitudinal forces (Q_{lk}) due to the braking or acceleration force as determined above, should be applied. When the total fill height is more than the distance between outer faces of walls of the box structure, the braking or acceleration forces may be neglected. For intermediate values of fill heights, the braking or acceleration forces may be interpolated.

The longitudinal force due to load model 3 should be taken as the more severe of either the braking or the acceleration force calculated as below.

The characteristic value of longitudinal braking force on individual axles, Q_{lk} , S expressed in kN, of special vehicles (both SV and SOV) shall be taken as δw where δ is the deceleration factor and w is the basic axle load of the relevant SV or SOV vehicle in kN. The value of δ should be taken as 0.5 for SV80, 0.40 for SV100, 0.25 for the SV196 and 0.20 for all of the SOV model vehicles.

The acceleration force should be taken as 10 % of the gross weight of the SV or SOV vehicle and distributed between the axles and wheels in the same proportion as the vertical loads.

3.3.5.2 Centrifugal and Other Transverse Forces

When a road bridge is situated on a curve, the structural elements that are affected by centrifugal action of the moving vehicle shall be designed to carry this action in addition to all other actions which they may carry.

The characteristic value of centrifugal force Q_{tk} shall be taken as a point load acting at the finished level of carriageway and radially in a direction transverse to the axis of carriageway. The value of centrifugal force Q_{tk} shall be taken as given in Table 3.7.

Table 3.7 Centrifugal Force

Centrifugal Force	Carriageway Centreline Radius
$Q_{tk} = 0.20 Q_v$	If $r < 200$ m
$Q_{tk} = 40 Q_v / r$	If $200 \leq r \leq 1500$ m
$Q_{tk} = 0$	If $r > 1500$ m

1

Where,

Q_v = Total maximum weight of vertical concentrated loads of the tandem systems of Load Model-1 (in kN) i.e. $\sum \alpha_{Q_i} (2Q_{ik})$.

2

r = Horizontal radius of carriageway centre line (in m).

Q_{tk} = Centrifugal force (in kN), assumed to act as a point load at any deck cross section.

3

Where relevant, lateral forces from skew braking or skidding should be taken into account. A minimum transverse force due to skew braking or skidding, Q_{trk} equal to 50% of the longitudinal braking force Q_{lk} for loaded lengths up to 120 m should be considered to act simultaneously with Q_{lk} at the finished carriageway level. For loaded length greater than or equal to 120m, a transverse force of 280 kN shall be used.

The characteristic value of centrifugal force from load model 3 vehicles, $Q_{tk,S}$ shall be calculated as follows.

$$V = \rho \left\{ \text{greater of } 30 \text{ or } \sqrt{\frac{100 g r}{r + 150}} \right\} \leq V_{Limit}$$

Equation 3.2

$$Q_{tk,S} = \frac{W V^2}{g r}$$

Equation 3.3

Where,

V = velocity of the SV or SOV vehicle in m/sec.

V_{Limit} = speed limit on the road in m/sec.

W = weight of the SV or SOV vehicle in kN.

r = radius of curvature in m.

g = acceleration due to gravity = 9.8 m/sec².

ρ = 0.86 for SV80, 0.77 for SV100, 0.55 for SV196, 0.41 for SOV 250, 0.36 for SOV 350, 0.33 for SOV 450 and 0.30 for SOV 600.

The centrifugal force should be distributed between axles and wheels in the same proportion as the vertical loads.

3.3.6 Group of Traffic Loads on Road Bridges

3.3.6.1 Characteristic Values of the Multi-component Action

The simultaneity of the loading systems as defined earlier for Load Mode 1, Load Model 2, Load Model 3, Load Model 4, braking force, centrifugal force and the loads for walkways should be taken into account by considering the groups of loads Table 3.8. Each of these groups, which are mutually exclusive, should be considered as defining a characteristic action for combination with other non-traffic loads.

3.3.6.2 Frequent Values of the Multi-component Action

The frequent action should consist only of either the frequent values of LM1 or the frequent value of LM2, or the frequent values of loads on walkways and cycle-tracks (taking the more unfavourable), without any accompanying component.

Table 3.8 Assessment of Groups of Traffic Loads

Load Type		Carriageway						Walkways & Cycle Tracks
		Vertical Forces				Horizontal Forces		Vertical Forces Only
Load System		LM1 (TS and UDL)	LM2 (Single axle)	LM3 (Special vehicles)	LM4 (Crowd loading)	Braking and acceleration forces	Centrifugal Force	Uniformly distributed load
Group of Loads	gr1a	Characteristic value						0.6 times characteristic value
	gr1b		Characteristic value					
	gr2	Frequent value				Characteristic value	Characteristic value	
	gr3							Characteristic value
	gr4				Characteristic value			Characteristic value
	gr5	Frequent value		Characteristic value				
	gr6			Characteristic value		Characteristic value	Characteristic value	

Notes: The shaded cells indicate the dominant component action

1. The group gr3 is irrelevant if gr4 is considered. One walkway shall be considered to be loaded if the effect is more unfavourable than the effect of two loaded walkway.

2. In group gr1a, the reduced value of walkway and cycle track load is accompanying the characteristic value of LM1 and shall not be factored by ψ_1 . However, when gr1a is combined with leading non-traffic actions, this value should be factored by ψ_0 .

Table 3.9 Assessment of Groups of Traffic Loads

Load Type		Carriageway		Walkways & Cycle Tracks
		Vertical Forces		
Load System		LM1 (TS and UDL)	LM2 (Single axle)	Uniformly distributed load
Group of Loads	gr1a	Frequent values		0.6 times characteristic value
	gr1b		Frequent values	
	gr2	Frequent value		Frequent values

Notes: The shaded cells indicate the dominant component action

For group gr3 one walkway shall be considered to be loaded if the effect is more unfavourable than the effect of two loaded walkway.

3.3.6.3 Groups of Loads in Transient Design Situations

For transient design situations, the rules given in Clause 3.3.6.1 and 3.3.6.2 are applicable with following modifications.

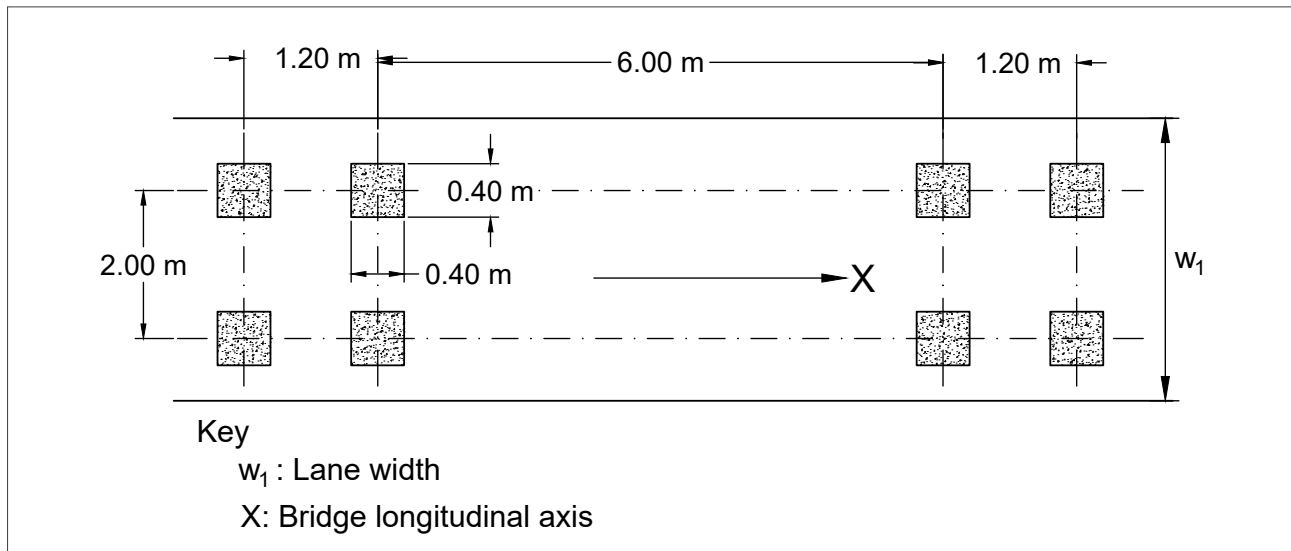
For verifications in transient design situations, the characteristic values associated with the tandem system should be taken equal to $0.8 \alpha_{Qi} Q_{ik}$. All other characteristic, frequent and quasi-permanent values and the horizontal forces are as specified for persistent design situations without any modification (i.e. they are not reduced proportionally to the weight of the tandems).

3.4 Fatigue Loads

Traffic running on bridges produces a stress spectrum which may cause fatigue. The stress spectrum depends on the geometry of the vehicles, the axle loads, the vehicle spacing, the composition of the traffic and its dynamic effects.

Load model as shown in Figure 3.10 may be used for fatigue life assessment of steel, concrete and composite bridges. The model consists of four axles, each axle weighing 120 kN and have two identical wheels. The contact area of each wheel is a square of side 0.40 m.

Figure 3.10 Fatigue Load Model



The maximum and minimum stresses and the stress ranges resulting from the single transit of the model along the longitudinal direction of bridge should be calculated for fatigue assessment.

Where relevant, two vehicles in the same lane should be considered. In such case, the first vehicle shall be as defined in Figure 3.10. Second vehicle shall have the same geometry, but weight of each axle is 36 kN instead 120 kN. The distance between two vehicles measured from centre to centre of vehicles is not less than 40 m.

Traffic category on a bridge should be defined based on the number of slow lanes (traffic lane predominantly used by lorries) and the number N_{obs} of heavy vehicles (gross vehicle weight more than 100 kN) per year per slow lane. The indicative vales for N_{obs} for heavy goods vehicle numbers for use in fatigue design may be adopted as given in Table 3.10. Heavy goods vehicle counts may be obtained from site surveys by doubling the observed number of lorries with three or more axles.

Table 3.10 Indicative Heavy Vehicles Per Year Per Lane

Type	Traffic Categories		N _{obs} per Lane (millions per year)	
	Carriageway Layout	No. of Lanes per Carriageway	Each Slow Lane	Each Fast Lane
Motorway	Dual	3	2.0	1.5
Motorway	Dual	2	1.5	1
All purpose	Dual	3		n/a
All purpose	Dual	2		
Slip road	Single	2		
All purpose	Single	3	1.0	0
All purpose	Single	2		0
Slip road	Single	1		n/a
All purpose	Single	2	0.5	0
Local (low lorry flow)	Single	2	0.05	0

Notes:

1. Intermediate values of N_{obs} are not excluded but are unlikely to have significant effect on the fatigue life.

2. The values presented in this table are design values that are intended to reflect approximate road capacities, and they may not match observations of current usage. Traffic flows at a small number of sites may exceed these values, but the differences are unlikely to have a very significant influence on designs.

For the assessment of general action effects (e.g. in main girders), the fatigue load model should be placed centrally on the notional lanes.

For the assessment of local action effects (e.g. in slabs) the model should be cantered on notional lanes assumed to be located anywhere on the carriageway.

The load model includes dynamic load amplification. An additional amplification factor $\Delta\phi_{fat}$ should be taken into account near expansion joints and applied to all loads.

$$\Delta\phi_{fat} = 1.30 \left(1 - \frac{D}{26} \right), \text{ shall be } \geq 1$$

Equation 3.4

Where, D is the distance (m) of the cross-section under consideration from the expansion joint.

3.5 Actions for Accidental Design Situations

The forces due to road vehicles in accidental design situations shall be considered, if relevant, resulting from:

- Vehicle collision with bridge piers or other supporting system, soffit of bridge decks.
- Presence of vehicle wheels on walkways or cycle tracks (for bridges where walkways are not protected by effective rigid restraint).
- Vehicle collision with kerb, vehicle parapets and other structural components.

3.5.1 Collision Forces on Piers and Other Supporting Structures

The force due to vehicular impact on members supporting bridges over or adjacent to roads will depend upon the impact velocity of the impacting object and the mass distribution, deformation behaviour and damping characteristics of both the impacting object and the structure. Other factors such as the angle of impact, the construction of the impacting object and movement of the impacting object after collision may also be relevant. For simplified structural design the actions due to impact may be represented by an equivalent static force giving the equivalent action effects in the structure shown in Figure 3.11. Detail analysis to be carried out for important structures using refined methods such as dynamic analyses, non-linear models and interaction between the load and the structure.

The bridge piers and abutments shall be protected by rigid restraint system from vehicular collision. However, the piers which are susceptible to vehicle collision, an equivalent static design force as given in Table 3.11 may be considered to act horizontally. The main and residual component shall be applied simultaneously.

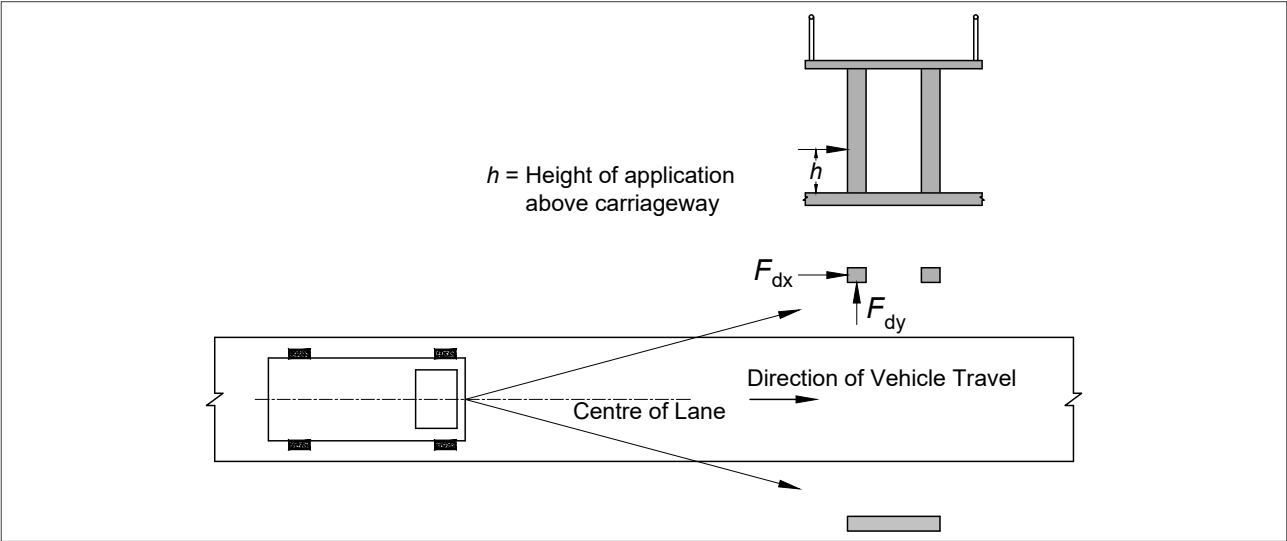
Table 3.11 Equivalent Static Design Force Due to Vehicular Impact on Bridge Supports

SN	Class of road under bridge	Force F_{dx} In the direction of normal travel kN	Force F_{dy} Perpendicular to the direction of normal travel kN	Point of application on bridge support
1	Bridges over Motorways, Trunk and Principal Roads			
a)	Main component	1650	825	At the most severe point between 0.75m and 1.5m above carriageway level
b)	Residual component	825	415	At the most severe point between 1.0m and 3.0m above carriageway level
2	Bridges over other roads where speed limit ≥ 72 km per hour, e.g. other rural roads			
a)	Main component	1240	620	At the most severe point between 0.75m and 1.5m above carriageway level
b)	Residual component	620	290	At the most severe point between 1.0m and 3.0m above carriageway level
3	Bridges over other roads where speed limit < 72 km per hour, e.g. other urban roads			
a)	Main component	825	415	At the most severe point between 0.75m and 1.5m above carriageway level
b)	Residual component	415	205	At the most severe point between 1.0m and 3.0m above carriageway level
4	Bridges over roads: minimum forces for robustness			
a)	Main component	250	250	At the most severe point between 0.75m and 1.5m above carriageway level
b)	Residual component	165	165	At the most severe point between 1.0m and 3.0m above carriageway level

The forces F_{dx} and F_{dy} shall not act simultaneously. For individual piers, the impact force may be considered to act as point load. For wall type piers and abutments, it may be considered to act over an area of 0.5 m (height) by 1.50 m (width) or the member width, whichever is the smaller.

The reference shall be made to BS EN 1991-1-7 and it's UK National Annex for more details and adjustment factor to be applied on values given in Table 3.11.

Figure 3.11 Collision Force on Bridge Supports



3.5.2 Collision Forces on Bridge Decks

Adequate vertical clearance between the ground surface and the soffit of the deck above ground shall be maintained in the design. Suitable protection measures shall be provided to avoid impact for inadequate vertical clearance.

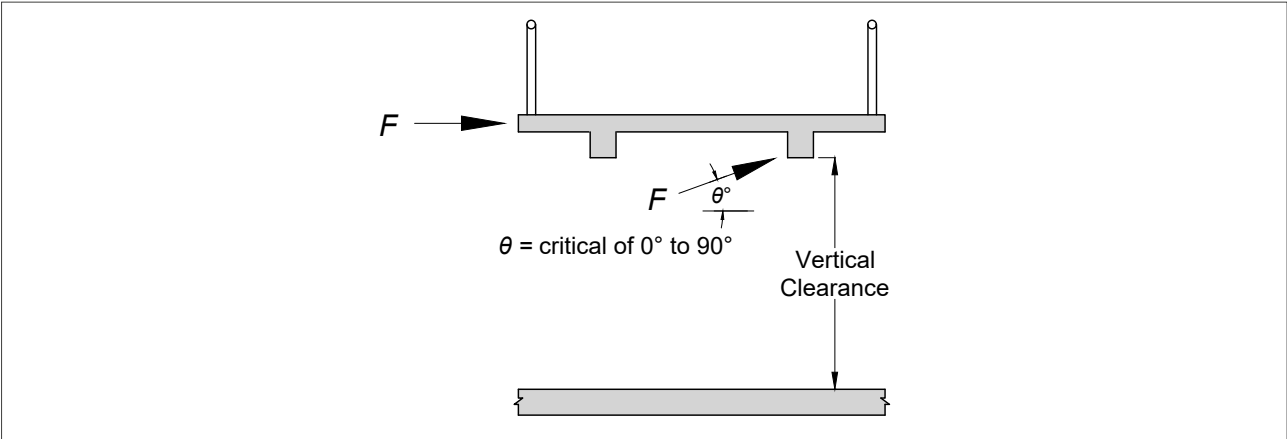
To account for accidental situation, the equivalent static design force F_{dx} in the direction of travel and F_{dy} perpendicular to the direction of travel due to impact on superstructure may be taken as given in Table 3.11 for all vehicular bridges with vertical clearance < 5.7 m. This force need not be considered if a vertical clearance ≥ 5.7 m is provided.

It is recommended that F_{dx} and F_{dy} shall not act simultaneously. The impact force may be assumed to act on a square area with sides of 0.25 m.

Table 3.12 Equivalent Static Design Force Due to Vehicular Impact on Superstructures

SN	Class of Road Under Bridge	Force F_{dx} In the direction of normal travel kN	Force F_{dy} Perpendicular to the direction of normal travel kN
1	Motorways, Trunk and Principal Roads	825	415
2	Other rural roads where speed limit > 72 km per hour	620	310
3	Other urban roads where speed limit ≤ 72 km per hour	415	205
4	Courtyards and parking garages	75	Not required

Figure 3.12 Collision Force on Bridge Decks



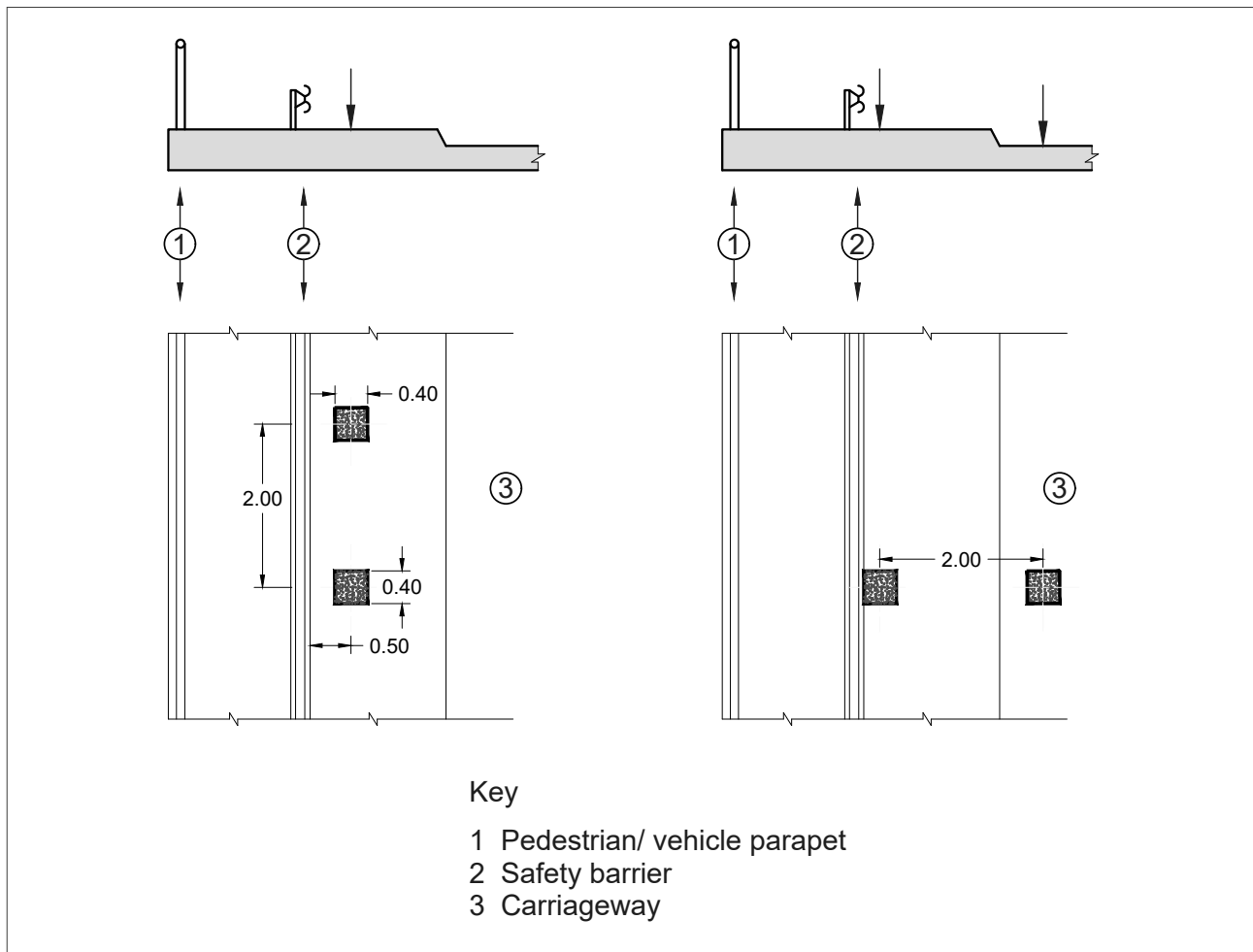
The loads F_{dx} and F_{dy} on underside of the structure shall be applied in any direction between horizontal and vertically upwards which ever gives most unfavourable effects in the element being considered.

3.5.3 Vehicle on Walkways and Cycle Tracks on Road Bridges

If a non-mountable safety barrier is provided which separates the carriageway and walkway/cycle track, the wheel or vehicle loading beyond this barrier need not be taken into account.

Adjacent to such safety barrier, one accidental axle load of $\alpha_{Q2} Q_{2k}$ ($=200$ kN) shall be placed on the unprotected part of the deck to give most adverse effect (Figure 3.13). No other live load on the deck shall be considered under this situation. A single wheel alone may be considered if geometrical constraints make a two-wheel arrangement impossible.

Figure 3.13 Accidental Axle Load on Walkway or Cycle Track



3.5.4 Collision Forces on Kerbs

The lateral force by vehicle collision with kerb, illustrated in Figure 3.14, shall be taken as 100 kN acting at a depth of 0.05 m below top of kerb in a length of 0.50 m. In rigid structural members, the load should be assumed to have an angle of dispersal of 45° . Under unfavourable conditions, a vertical load of $0.75 \alpha_{Q1} Q_{1k}$ ($= 225$ kN) shall also be taken simultaneously.

3.5.5 Collision Forces on Vehicle Restraint Systems

The vehicle restraint system shall be designed for an appropriate class of forces given in Table 3.13. The force shall be applied uniformly over a length of 3 m and at top of the traffic face of the vehicle restraint system and in a position along the line of vehicle restraint system that gives maximum effects on the part of structure under consideration.

Figure 3.14 Vehicle Collision on Kerbs

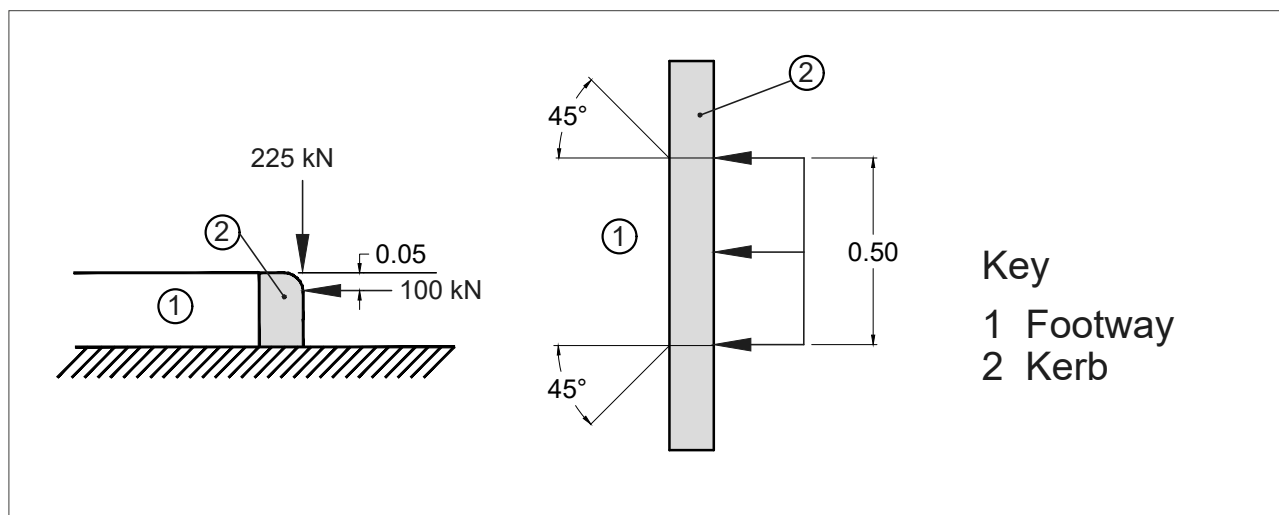


Table 3.13 Forces Due to Collision With Vehicle Restraint System

Class	Transverse force (kN)	Longitudinal force (kN)	Vertical force (kN)	Examples of application
A	100	-	-	Normal containment flexible parapets (e.g. metal post and rail parapets)
B	200	-	-	Normal containment rigid parapets (e.g. reinforced concrete parapets)
C	400	100	175	Very high containment flexible parapets (e.g. metal post and rail parapets)
D	600	100	175	Very high containment rigid parapets (e.g. reinforced concrete parapets)

The vertical forces acting simultaneously with the collision forces shall be taken as 0.75 times the loading given by Load Model 1 given in Clause 3.3.3.1 and the full accidental wheel / vehicle loading given in Clause 3.5.3. The three sets of forces shall be applied in such a way that will have the most severe effect on the part under consideration. The structure supporting the vehicle parapet shall be designed for an accidental load effect of 1.25 times the characteristic load resistance of vehicle parapet and need not be combined with any other variable load.

3.5.6 Collision Forces on Structural Members on Bridge Deck

The vehicle collision forces on unprotected structural members above or beside the carriageway level shall be designed considering following options.

- Design for vehicle collision forces as provided in Clause 3.5.1.
- Design for nominal vehicle collision forces for the provision of minimum robustness and for situation where damage or failure to the structural member will not cause collapse of the structure. These forces shall not be considered to act simultaneously with any other variable load.

3.5.7 Actions on Pedestrian Parapets

For structural design, forces that are transferred to the bridge deck by pedestrian parapets should be taken into account as variable loads and defined, depending on the selected loading class of the parapet. For bridges, class C is the recommended minimum class. The characteristic value of forces transferred to the structure should be taken as the design loads given as below for the relevant class of pedestrian parapet.

- a. For the design of pedestrian parapets, the loads are defined in horizontal and vertical directions and can be applied as distributed or as point loads.
- b. The characteristic value of the horizontal uniformly distributed load q_{hk} on the top rail and other horizontal rails can be specified from the loads given in Table 3.14 (reference from EN 1317-6) for the relevant class of pedestrian parapet.

Table 3.14 Minimal Loads for Parapets

Load	A	B	C	D	E	F	G	H	J
q_{hk} kN/m	0.4	0.8	1.0	1.2	1.6	2.0	2.4	2.8	3.0

- c. A pedestrian parapet rail shall withstand a horizontal concentrated load $Q_{hk} = 1.0$ kN applied to any point of the rail.
- d. All horizontal and inclined elements up to 60° of a pedestrian parapet shall withstand a vertical uniformly distributed load $q_{vk} = 1.0$ kN/m.
- e. A vertical concentrated load $Q_{hk} = 1.0$ kN shall be applied to any point of the horizontal and inclined elements.

For the design of the supporting structure, the minimum horizontal load should be taken as 1.6 kN/m, corresponding to Class E, for normal situations, and 3.0 kN/m, corresponding to Class J, for exceptional situations where crowding can occur. The horizontal load should be applied at the top of the pedestrian parapet.

For design of supporting structure, if the pedestrian parapets are adequately protected against vehicle collision, the horizontal actions shall be considered as simultaneously with uniformly distributed vertical loads on walkways or footbridges.

If the pedestrian parapets are not adequately protected against vehicle collision, the supporting structure shall be designed to sustain an accidental load corresponding to 1.25 times the load resistance of the parapet, exclusive of any other variable load.

3.5.8 Load Models for Abutments and Walls Adjacent to Bridges

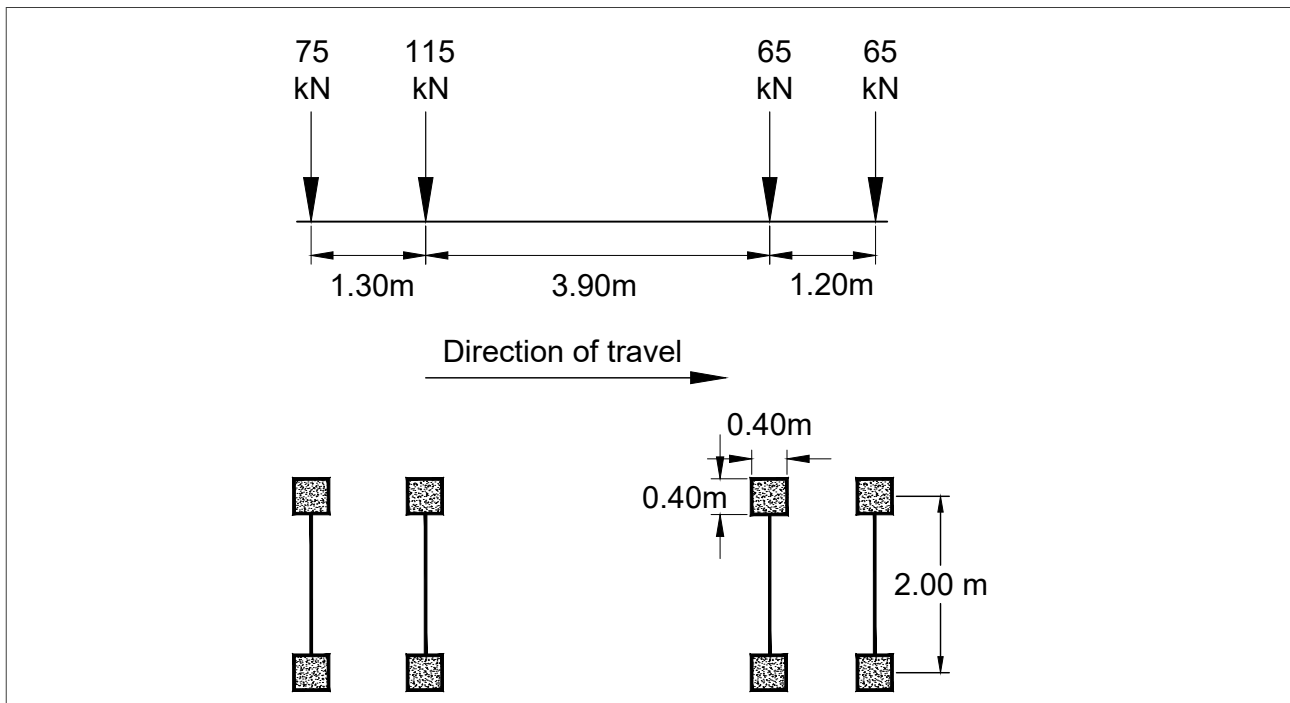
3.5.8.1 Vertical Loads from Traffic

For determining the vertical and horizontal pressures in the backfill from traffic, the carriageway located behind abutments, wing walls, side walls and other parts of the bridge in contact with earth, should be loaded with appropriate models corresponding to normal traffic and special vehicles where required.

3.5.8.1.1 Loading from Normal Traffic

For determining the vertical and horizontal pressures in the backfill behind an abutment or wing wall, the carriageway located behind the abutments is loaded with the model vehicle loads and configuration given in Figure 3.15. These vehicle loads should be considered as characteristic loads. Each axle consists of two wheels of equal weight and contact surface of each wheel should be taken as a square of sides 0.40 m.

Figure 3.15 Vehicle Model for Abutment and Wing Walls



All the axle loads given in Figure 3.15 shall be multiplied by an Overload Factor of 1.5 and a Dynamic Amplification Factor of 1.4. The Dynamic Amplification Factor on vertical and horizontal earth pressure may be reduced linearly from 1.4 at surface level to 1.0 at a depth of 7.0 m. Vehicles shall be positioned in a maximum of three adjacent notional lanes. The axle loads for the vehicle in the third lane should be factored by a lane factor of 0.5.

The maximum load effect from the following two cases should be used for design:

- i. A single vehicle in each notional lane.
- ii. Convoy of vehicles in each notional lane with the Dynamic Amplification Factor set to 1.0 (represents a traffic jam situation).

The lateral and longitudinal position of vehicles within each lane shall be positioned so as to maximise the load effects at the part of the structure under consideration. A minimum lateral spacing of 1.0 m is maintained between the centrelines of wheels from two adjacent vehicles. In case of a convoy of vehicles, a minimum longitudinal spacing of 3.0 m should be kept between the last axle of the leading vehicle and the first axle of the trailing vehicle.

1

Where the load model behind the abutment is applied in conjunction with either Load Model 1 or Load Model 2 on the deck, the two load models should be applied simultaneously without modification to their rules of application.

2

The load model should be considered to form part of gr1a and gr1b, in which it should be applied at its characteristic value, and gr2, in which it should be applied at its frequent value. The combination factors for the load model should be taken equal to those for the tandem axle system of Load Model 1.

3

3.5.8.1.2 Loading From Special Vehicles

For the determination of vertical and horizontal pressures, due to special vehicle loading behind the abutment, only one SV or SOV vehicle model, appropriate to the road class, and in one notional lane, should be considered. The vehicle load model given in Clause 3.5.8.1.1 for normal traffic may be applied in two adjacent lanes but with all the axle loads multiplied by a factor of 0.75.

The effect of the Dynamic Amplification Factor on vertical and horizontal earth pressure may be considered to reduce linearly from the values given in Table 3.5 for SV and SOV vehicles at the surface to 1.0 at a depth of 7.0 m.

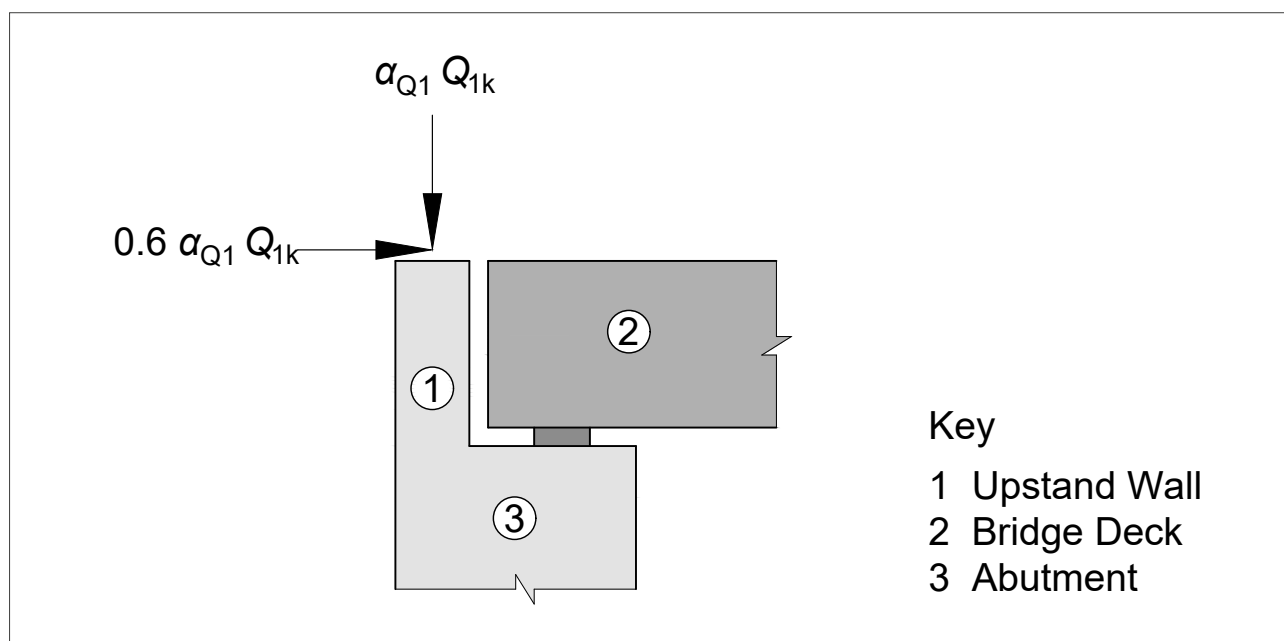
3.5.8.2 Horizontal Earth Pressure From Traffic Loads

The traffic load models to determine the horizontal earth pressure on abutment walls and retaining/wing walls shall be in accordance with the Clause 3.5.8.1. The horizontal earth pressure from the traffic load models may be determined as per geotechnical design guidelines. The Appendix A.1 may be referred for simplified traffic surcharge models and detailed analysis of horizontal pressures on abutments and walls from traffic loads.

3.5.8.3 Horizontal Loads on Upright Walls

The abutment upstand walls shall be designed for a longitudinal braking force with a characteristic value of $0.6 \alpha_{Q1} Q_{1k}$, acting simultaneously with a vertical axle load $\alpha_{Q1} Q_{1k}$ of Load Model-1 and the earth pressure from backfill without considering live load surcharge. The load configuration is shown in Figure 3.16.

Figure 3.16 Loads on Upright Walls



3.6 Actions on Walkways

The load models and other actions that are applicable for footbridges used for pedestrians and cycle tracks shall be considered in accordance with Chapter 9.

3.7 Earth Pressure on Abutments and Retaining Structures

The design provisions applicable for geotechnical design of abutments and retaining structures shall be in accordance with “RDM 4.3”.

The structural design of concrete abutments and retaining structures shall be carried in accordance with the design provisions given in Chapter 4.

3.7.1 General Design Considerations

The determination of the magnitudes of earth pressures and directions of forces resulting from earth pressures shall take account of –

- The surcharge on and slope of the ground surface.
- The inclination of the wall to the vertical.
- The water tables in the ground.
- The amount and direction of the movement of the wall relative to the ground.
- The horizontal as well as vertical equilibrium for the entire retaining structure.
- The shear strength and weight density of the ground.
- The wall roughness.

The forces on retaining walls and abutments due to retained earth to be worked out in accordance with the “Part-3: Retaining Structures Design”. If the retained earth is not allowed to drain or ponding expected behind the wall, the wall shall be designed to resist the hydrostatic pressure plus earth pressure. The static water pressure at any depth h below water level is given by the product of depth h and the unit weight of water. This pressure will act in a direction perpendicular to the surface of wall face in contact with water. Submerged unit weight of retained earth shall be used for calculating lateral earth pressure below the water table.

3.7.2 Lateral Earth Pressure in Complex Situations

There could be a lot of complex situations which could be due to variation in properties of retained earth or the type of retaining structure. The lateral earth pressure in complex situations shall be treated differently as per principles of Soil Mechanics. A few of the situations could be due to one of more of the following reasons:

- Stratified soil deposits.
- Presence of ground water table and its variation.
- Embedded walls and sheet piles.
- Anchored walls.

3.8 Actions Caused by Water

In general, actions due to water, including ground water, (Q_{wa}) shall be represented as static pressures and/or hydrodynamic effects, whichever gives the most unfavourable effects. The hydrodynamic forces are due to water currents on immersed obstacles. Actions caused by water may be considered in combinations as permanent and/or variable actions taking into account of the specific environmental conditions.

3.8.1 Static Water Pressure

The static water pressure at any depth h below water level is given by the product of depth h and the unit weight of water. This pressure will act in a direction perpendicular to the surface of object in contact with water. Design value of the static water pressure for ultimate limit state shall represent the unfavourable values that could occur during the design lifetime of the structure. Design value of the static water pressure for serviceability limit state shall represent the unfavourable values that could occur in normal circumstances.

3.8.2 Water Current Forces

Any part of the bridge which may be submerged in running water for the design floods shall be designed to sustain the horizontal forces due to the water current which are given below. The design flood parameters shall be determined in accordance with the "Hydrology and Drainage Design Manual".

3.8.2.1 Water Current Pressure

The intensity of water pressure parallel to the direction of stream flow exerted on vertical surface of any object shall be calculated from the Equation:

$$p = \frac{1}{2} k \rho_w v^2$$

Equation 3.5

Maximum pressure intensity will be at water surface level, where velocity of flow is $\sqrt{2}$ times the average velocity of flow, given by:

$$p_{sur} = k \rho_w v_{av}^2$$

Equation 3.6

Where,

p = Intensity of pressure in N/m^2 .

ρ_w = Unit weight of water in kg/m^3 .

v = Velocity of flow in m/s , at the point where pressure intensity is being calculated.

v_{av} = Average velocity of flow in m/s for the design flood, (averaged over the depth of flow).

k = A coefficient having values for different shapes of piers as in Table 3.15.

Table 3.15 Coefficient k for Pier Shapes

Pier Shape		Coefficient k
	Piers with square ends	1.44
	Circular piers or piers with semi circular ends	0.70
	Piers with angle ends where:	
	Angle = $< 30^\circ$	0.50
	$30^\circ < \text{angle} < 60^\circ$	0.50 – 0.70
	$60^\circ = \text{angle} = < 90^\circ$	0.70 – 0.90
	Piers with equilateral arc end	0.50

3.8.2.2 Flow Velocity

The velocity of flow at any point can be determined from the Figure 3.17. The value of square of velocity (v^2) shall be assumed to vary linearly from zero at the point of deepest scour to the square of maximum velocity at free surface of water. The maximum velocity shall be assumed to be $\sqrt{2}$ times the average velocity of flow.

V_{av} = Average velocity of flow in m/s to be determined for design flood condition.

v_{max} = Maximum velocity at free surface in m/s = $\sqrt{2} v_{av}$.

v^2 = Square of velocity at any height x above scour level = $2 v_{av}^2 \times x / H$.

3.8.2.3 Water Current Force

The water current force (F_{wa}) parallel to the direction of stream flow on vertical surface shall be calculated from the Equation:

$$F_{wa} = p_{wa} A$$

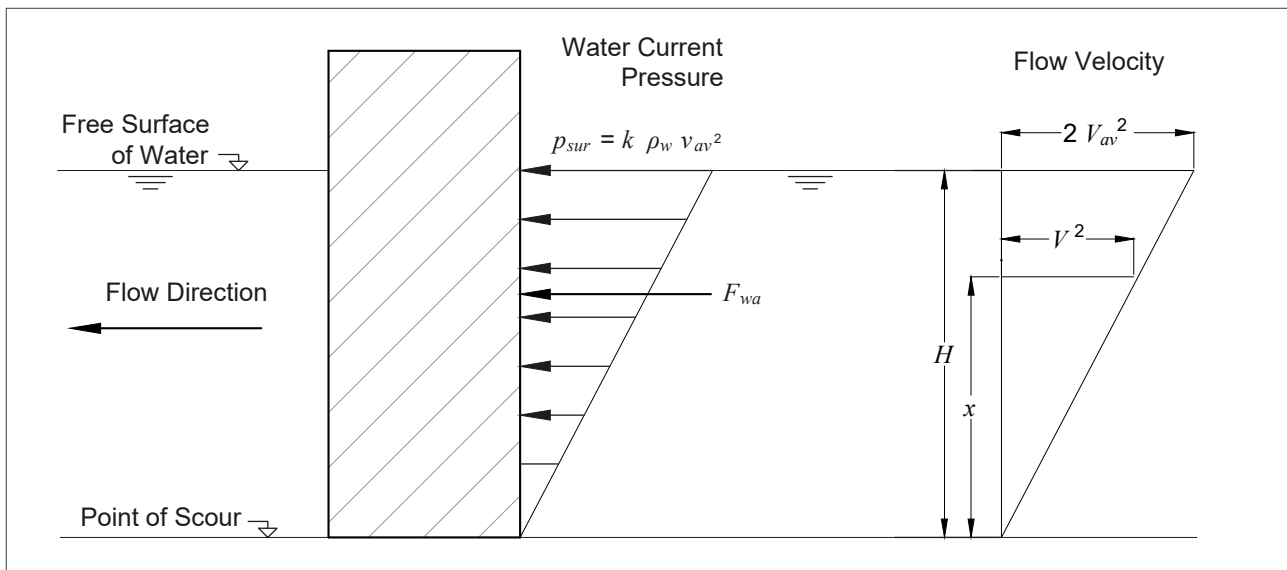
Equation 3.7

Where,

p_{wa} = Average of the intensity of pressure (averaged over the area under consideration).

A = Project area of the object in m^2 , on a vertical plane normal to stream flow.

Figure 3.17 Pressure and Velocity Due to Water Currents



While making combination of water current forces with self-weight of structure, due allowance shall be made for buoyancy effect on the parts of structure that are submerged in water. While calculating the area and the pressure thereon, maximum depth of scour under the design flood conditions shall be considered.

When water current strikes the pier at an angle, the velocity of flow can be resolved into two components – one parallel and other normal to the pier. The pressure parallel to pier shall be determined taking the velocity component parallel to pier. The pressure normal to pier and acting on the area of the side elevation of the pier, shall be calculated taking velocity component normal to pier. However, the value of k for normal component shall be taken as 0.70 for circular piers and 1.44 for other shapes.

1

Allowance shall be made in the design of piers for a variation of $\pm 20^\circ$ from the normal direction of flow to account for possible variation in the normal direction of flow.

2

When supports are made with two or more piles or trestle columns, spaced less than three times the diameter of pile/ width of column across the direction of flow, the group shall be treated as a solid rectangle of the same overall length and width. The value of k shall be taken as 1.25 for calculating pressure.

3

3.8.3 Effect of Buoyancy

In the design of abutments, especially those for submersible bridges, the effects of buoyancy shall also be considered assuming that the fill behind the abutments has been removed by scour. To account for full buoyancy effect, a reduction shall be made in the gross weight of the member affected by reducing its density by the density of the displaced water. The density of water may be taken as 10 kN/m^3 . High flood level or actual water head, whichever is higher, shall be considered for calculating the uplift forces due to Buoyancy. In case of submersible bridges, the full buoyancy effect on the superstructure shall be taken into consideration.

3.9 Wind Actions

The wind forces on a bridge depends upon the geographic location, terrain of the surrounding area, height of the bridge above ground, meteorological data, dimensions of the bridge, etc.

The provisions of this Clause are not applicable for special bridges such as cable stayed bridges, suspension bridges, arch bridges, bridges with multiple or significantly curved decks etc. Bridge deck vibrations from transverse wind turbulence are also not covered here. Provisions of this Manual are applicable for the bridges that are not susceptible to dynamic effects and no span greater than 200 m.

The wind effect is represented by forces whose effects are equal to the extreme effects of the wind.

3.9.1 Fundamental Basic Wind Velocity

Fundamental Basic Wind Velocity (v_{b0}) is the characteristic 10 minutes mean wind velocity with an annual risk of exceeding 0.02 (i.e. a return period of 50 years), irrespective of wind direction, at a height of 10 m above flat open country terrain and shall be determined from following Equation.

$$v_{b,0} = c_{alt} v_{b,map}$$

Equation 3.8

Where,

$v_{b,map}$ = is the value of the fundamental basic wind velocity before the altitude correction is applied.

$v_{b,map}$ = given in Figure 3.18. (Refer note below this figure).

c_{alt} = the altitude factor given as:

$$c_{alt} = 1 + 0.001 A \text{ for } z \leq 10 \text{ m}$$

Equation 3.9

$$c_{alt} = 1 + 0.001 A (10/z)^{0.2} \text{ for } z > 10 \text{ m}$$

Equation 3.10

Where,

A = the altitude of the site in meters above mean sea level.

z = height of the part above ground.

3.9.2 Basic Wind Velocity

Basic Wind Velocity (v_b) is the fundamental basic wind velocity modified to account for the direction being considered and the season.

$$v_b = c_{dir} c_{season} v_{b,0}$$

Equation 3.11

Where,

c_{dir} = The directional factor.

c_{season} = The seasonal factor.

Considering the directional factor and seasonal factor as 1.0 conservatively, the basic wind velocity (v_b) may be taken equal to the fundamental basic wind velocity ($v_{b,0}$).

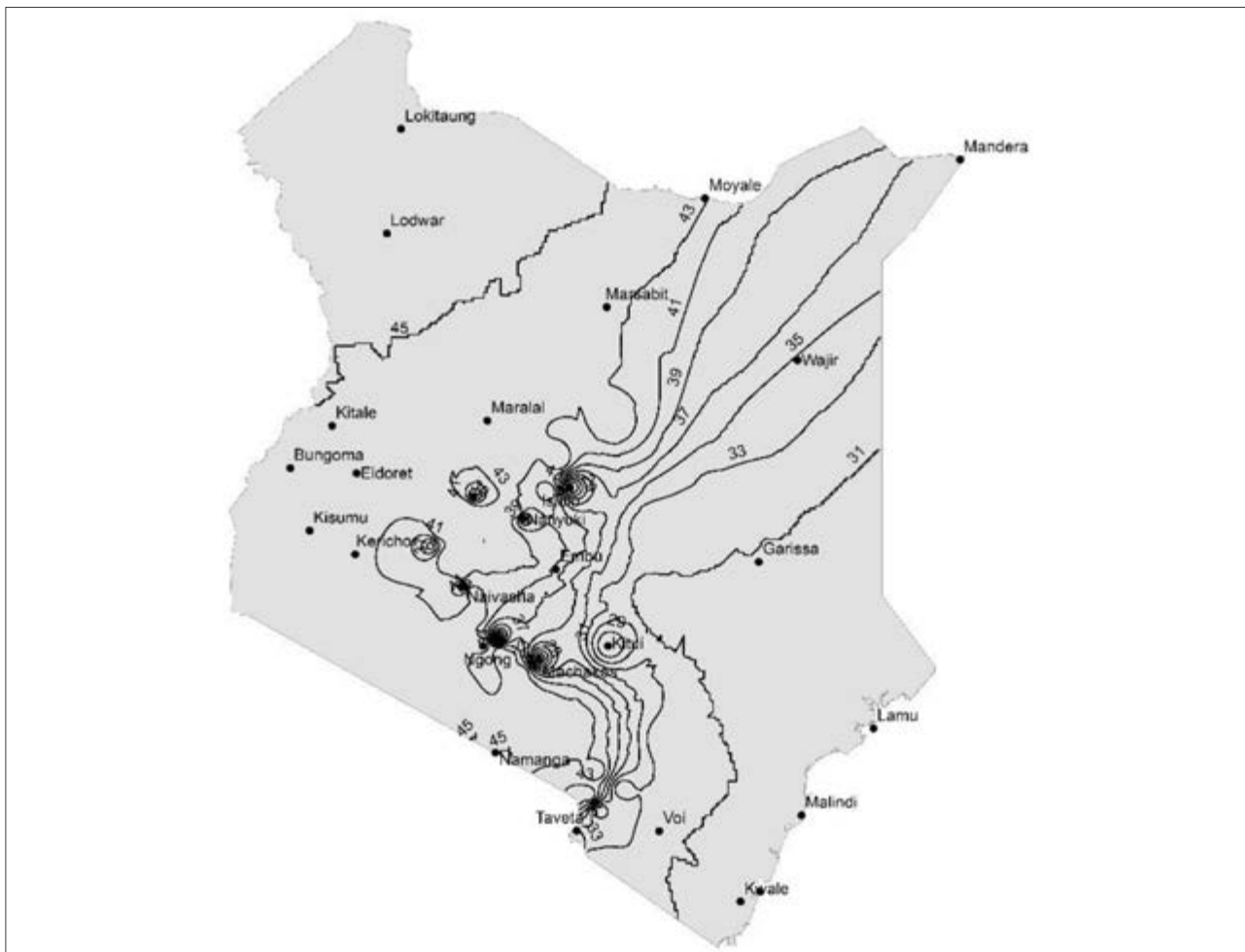
The 10 minutes mean wind velocity having the probability p for an annual exceedance is determined by multiplying the basic wind velocity v_b by the probability factor, c_{prob} given by following Expression:

$$c_{prob} = \left(\frac{1 - K \cdot \ln(-\ln(1 - p))}{1 - K \cdot \ln(-\ln(0.98))} \right)^n$$

Equation 3.12

The recommended value of shape factor $K = 0.2$ and exponent $n = 0.50$ are to be used.

Figure 3.18 Fundamental Basic Wind Velocity $v_{b,map}$ (m/s) (before altitude correction is applied)



Source: Paper by Muthomi Munyua, Kenya, (generated from Table 1 of KS 02-26:1977)

Note: The wind map shown in Figure 3.18 is based on old data and is provided as an indicative map. The actual values of fundamental basic wind velocity for the bridge specific site are to be obtained from the Meteorology Department, Kenya. The $v_{b,map}$ corresponds to a 10 minutes mean wind velocity with an annual risk of exceeding 0.02 (i.e. a return period of 50 years) before altitude correction.

3.9.3 Mean Wind Velocity

3.9.3.1 Variation With Height

The mean wind velocity (v_m) at a height z above the terrain depends upon the terrain roughness, orography (hills, cliffs, etc.) and basic wind velocity.

$$v_m(z) = c_r(z) c_o(z) v_b$$

Equation 3.13

Where,

$c_r(z)$ = Roughness factor

$c_o(z)$ = Orography factor

3.9.3.2 Terrain Roughness

The roughness factor, $c_r(z)$, accounts for the variability of the mean wind velocity at the site of the structure due to the height above ground level and the ground roughness of the terrain upwind of the structure in the wind direction considered.

For determining roughness factor $c_r(z)$, the terrain is classified into one of three categories Sea, Country terrain or Town terrain as per the Table 3.16.

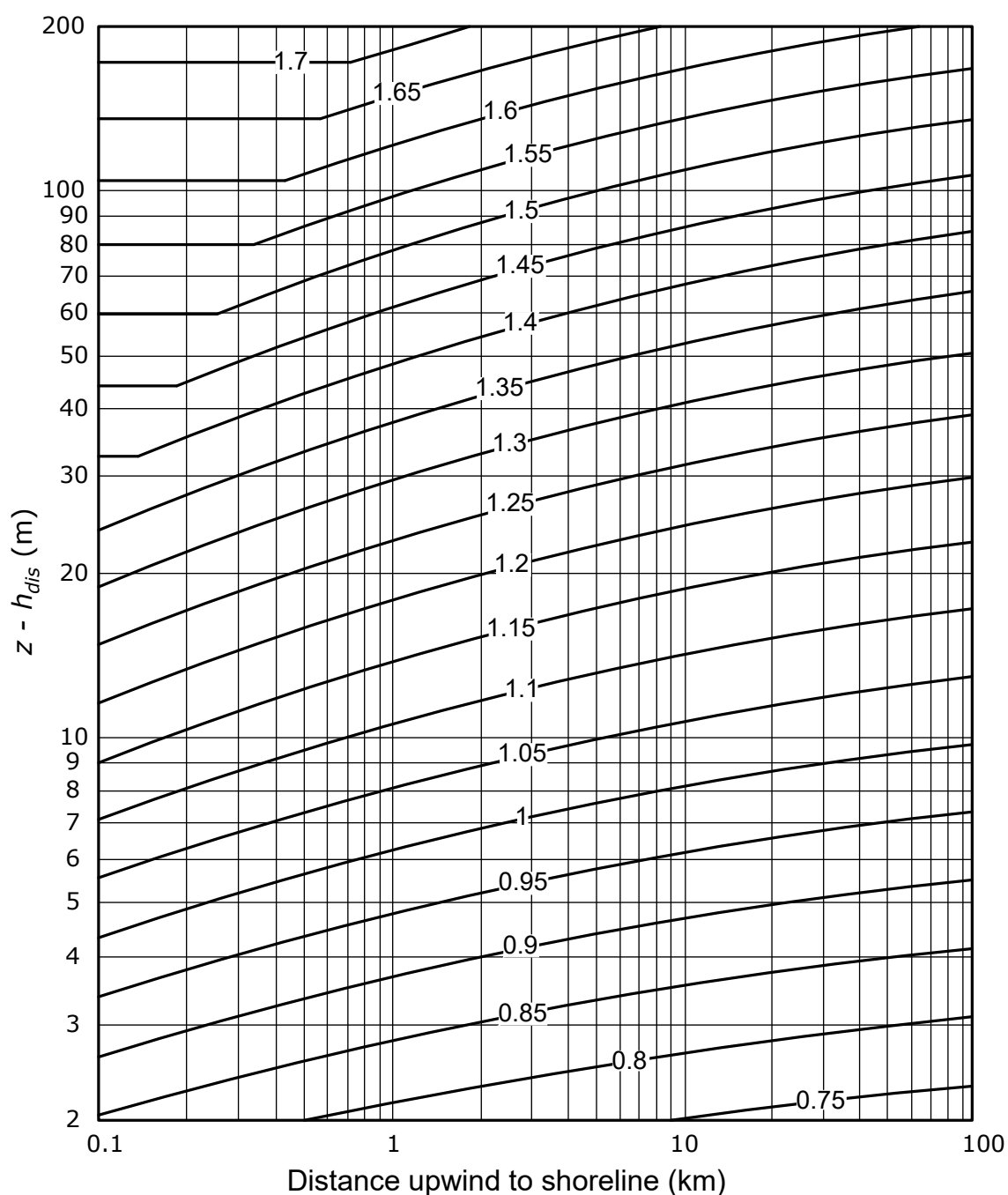
Table 3.16 Terrain Category and the Parameters

Terrain Category	
Sea	
0	Sea or costal area exposed to the open sea.
Country Terrain	
I	Lakes or flat and horizontal area with negligible vegetation and without obstacles.
II	Area with low vegetation such as grass and isolated obstacles (trees, buildings) with separation of at least 20 obstacle heights.
Town Terrain	
III	Area with regular cover of vegetation or buildings or with isolated obstacles with separations of maximum 20 obstacle heights (such as villages, suburban terrain, permanent forest).
IV	Area in which at least 15% of the surface is covered with buildings and their average height exceeds 15m.

All inland lakes extending more than 1 km in the direction of wind and closer than 1 km upwind of the site should be treated as Sea. The roughness factor $c_r(z)$ depends on upwind distance to sea and additionally on the distance upwind to the edge of the urban area for sites in Town terrain.

For sites in Country terrain, the roughness factor $c_r(z)$ given in Figure 3.19 should be used. For sites adjacent to Sea terrain (sea or large inland lakes), the distance upwind from the shoreline should be taken as 0.1 km. For sites in Town terrain, the roughness factor $c_r(z)$ given in Figure 3.19 should be multiplied by the roughness correction factor $c_{r,T}$ for Town terrain given in Figure 3.20.

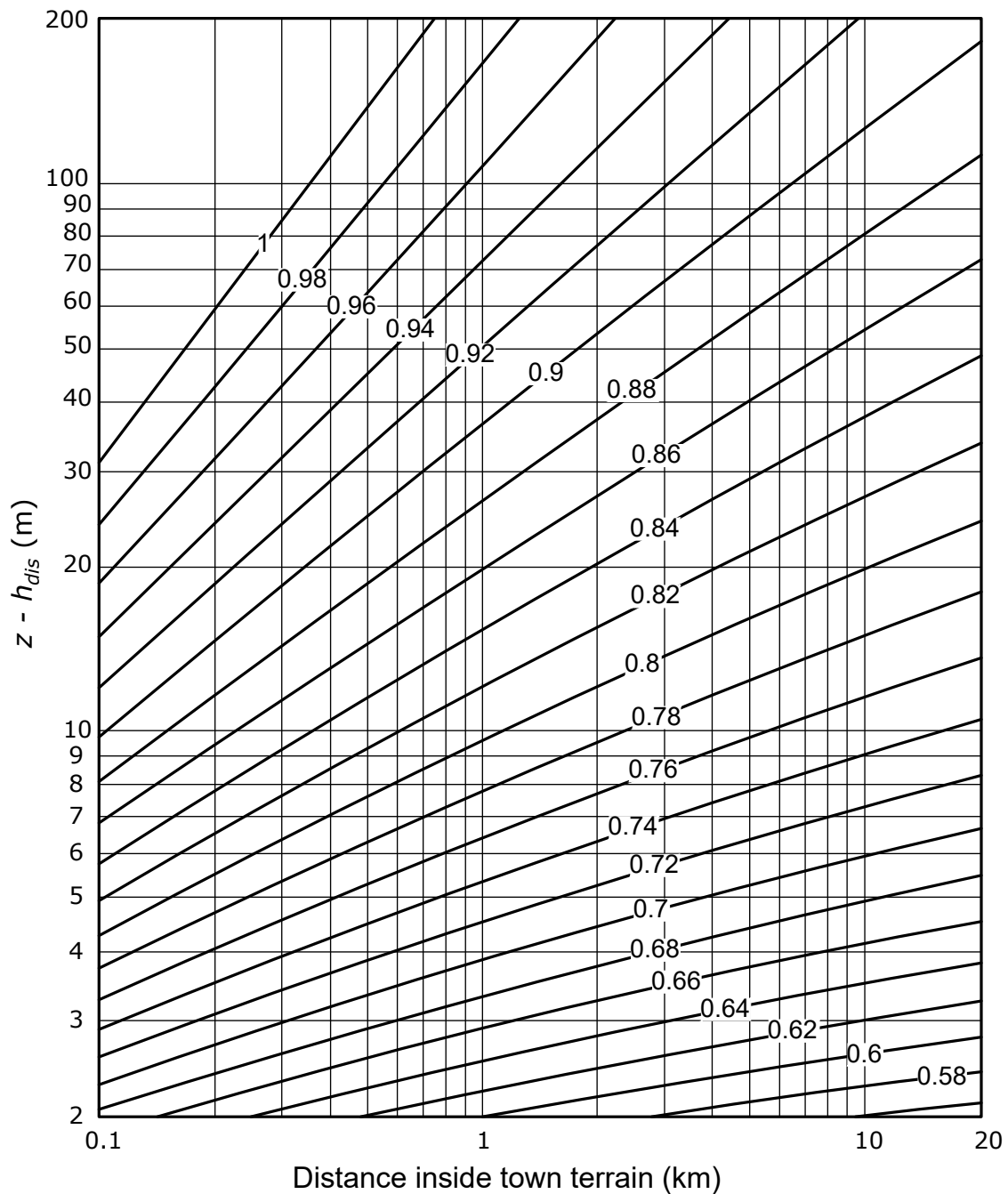
Note: The appropriate value of h_{dis} is to be used in Figure 3.19 and Figure 3.20. For sites in Country terrain, $h_{dis} = 0$. For sites in Town terrain, h_{dis} is given by BS EN 1991-1-4 Annex A.5.

Figure 3.19 Values of Terrain Roughness Factor $c_r(z)$ 

Note 1: The height z is the height at which v_m is sought

3.9.3.3 Terrain Orography

The orography factor $c_o(z)$ can be considered as 1 if the increase in wind velocity due to orography is not more than 5%. The orography may be neglected when the average slope of the upwind terrain is within 3%. The procedure given in BS EN 1991-1-4: Annex A-3 may be referred if orography effects are to be considered. When the orography factor is taken as 1.0, the altitude A shall be taken as the site altitude.

Figure 3.20 Values of Correction Factor c_{rT} for Sites in Town Terrain

Note 1: The height z is the height at which v_m is sought.

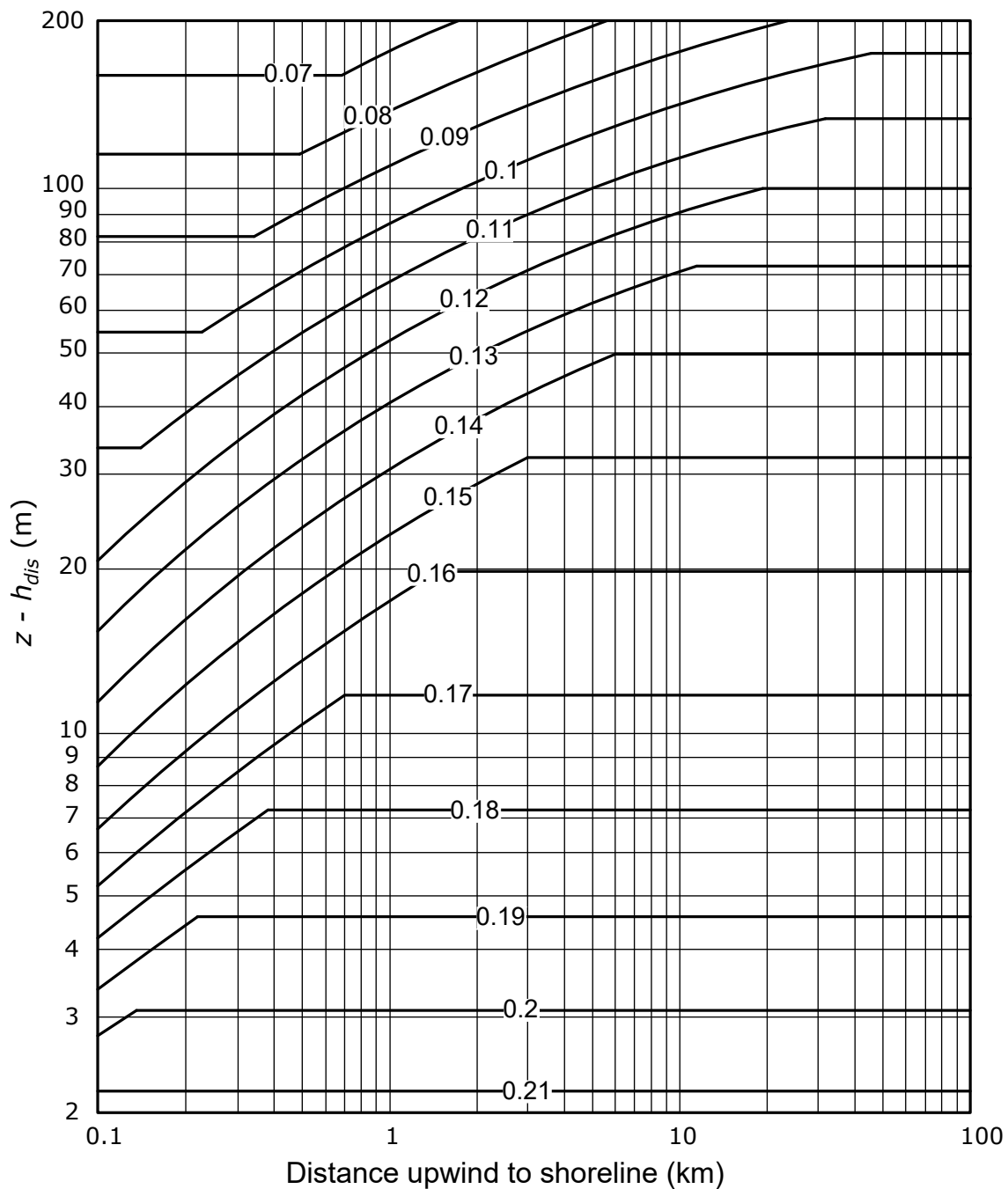
Note 2: Sites less than 0.1 km inside town boundary should be treated as being in country terrain.

3.9.4 Wind Turbulence

The turbulence intensity $I_v(z)$ at height z is defined as the standard deviation of the turbulence by mean wind velocity and can be determined by following Equations:

$$I_v(z)_{flat} = \left(\frac{k_I}{\ln((z - h_{dis})/z_0)} \right)$$

Equation 3.14

Figure 3.21 Values of Turbulence Intensity $I_v(z)_{flat}$ 

Note: The height z is the height at which q_n is sought using Equation 3.18.

The values of $I_v(z)_{flat}$ are given in Figure 3.21

When orography is not significant,

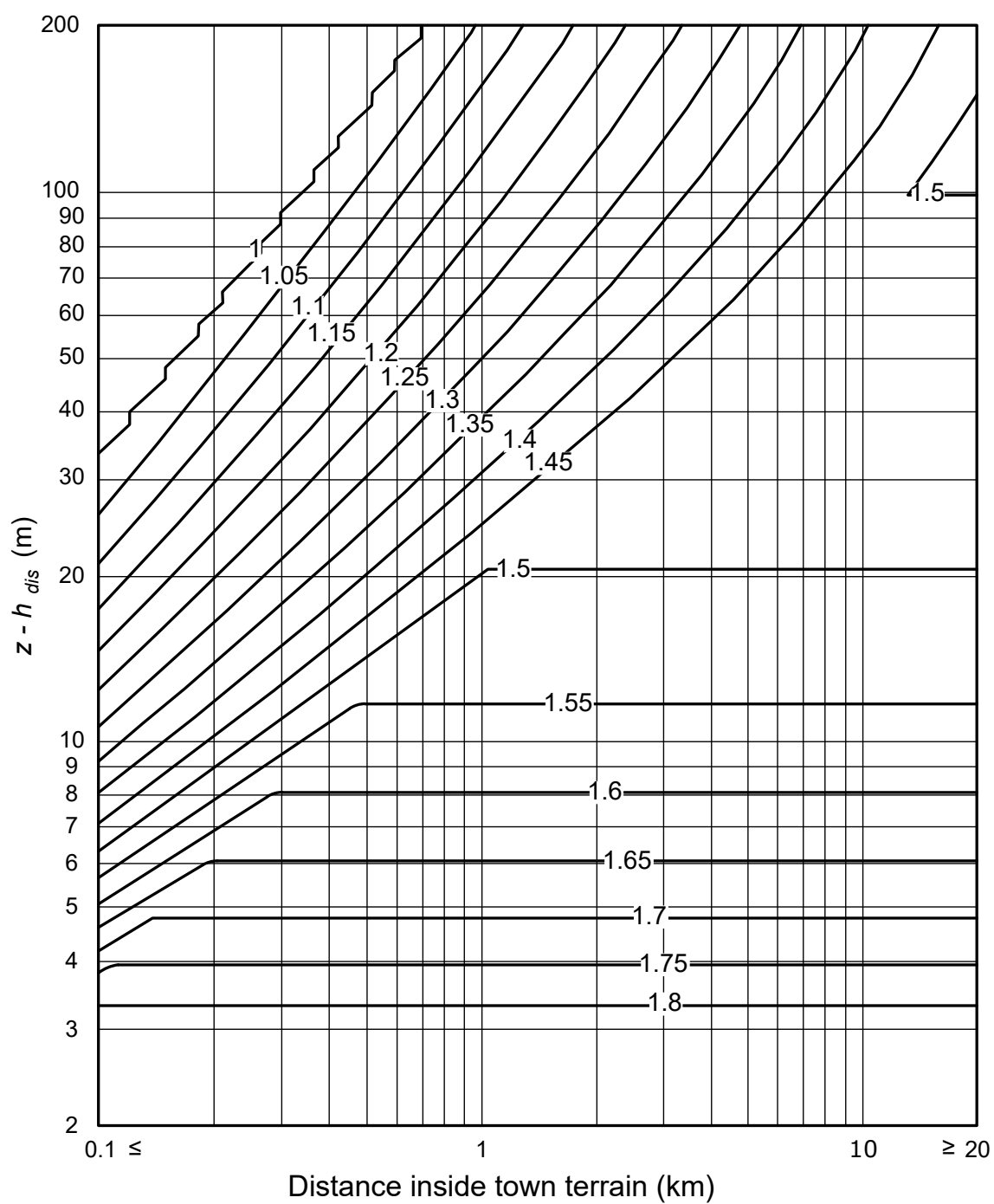
$$I_v(z) = I_v(z)_{flat} \text{ for sites in country terrain.}$$

$$I_v(z) = I_v(z)_{flat} k_{I,T} \text{ for sites in town terrain.}$$

$k_{I,T}$ = The turbulence correction factor for Town terrain, values as given in Figure 3.22.

When orography is significant, the values obtained from Figure 3.21 shall be divided by the orography factor c_0 .

Note: The appropriate value of h_{dis} to be used in Figure 3.21 and Figure 3.22. The h_{dis} is defined in BS EN 1991-1-4 Annex A.5

Figure 3.22 Values of Turbulence Intensity K_{it} for Sites in Town Terrain

Note 1: The height z is the height at which v_m is sought using Expression 3.18.

Note 2: Sites less than 0.1 km inside town boundary should be treated as being in country terrain.

3.9.5 Peak Velocity Pressure

The peak velocity pressure $q_p(z)$ at a height z is given by following Expressions.

When orography is not significant ($c_0 = 1.0$):

$$q_p(z) = c_e(z) q_b \text{ for sites in country terrain}$$

Equation 3.15

$$q_p(z) = c_e(z) c_{e,T} q_b \text{ for sites in town terrain}$$

Equation 3.16

The values of exposure factor $c_e(z)$ are given in Figure 3.23 and the values of exposure correction factor for Town terrain $c_{e,T}$ are given in Figure 3.24.

When orography is significant:

$$q_p(z) = \{q_p(z) \text{ from Equation 3.15 or 3.16}\} \{(c_e(z) + 0.6)/1.6\}^2$$

Equation 3.17

Where, $z \leq 50$ m

or

$$qp(z) = \{1 + 3.0 I_v(z)\}^2 (1/2) \rho v_m^2$$

Equation 3.18

Where, $z > 50$ m

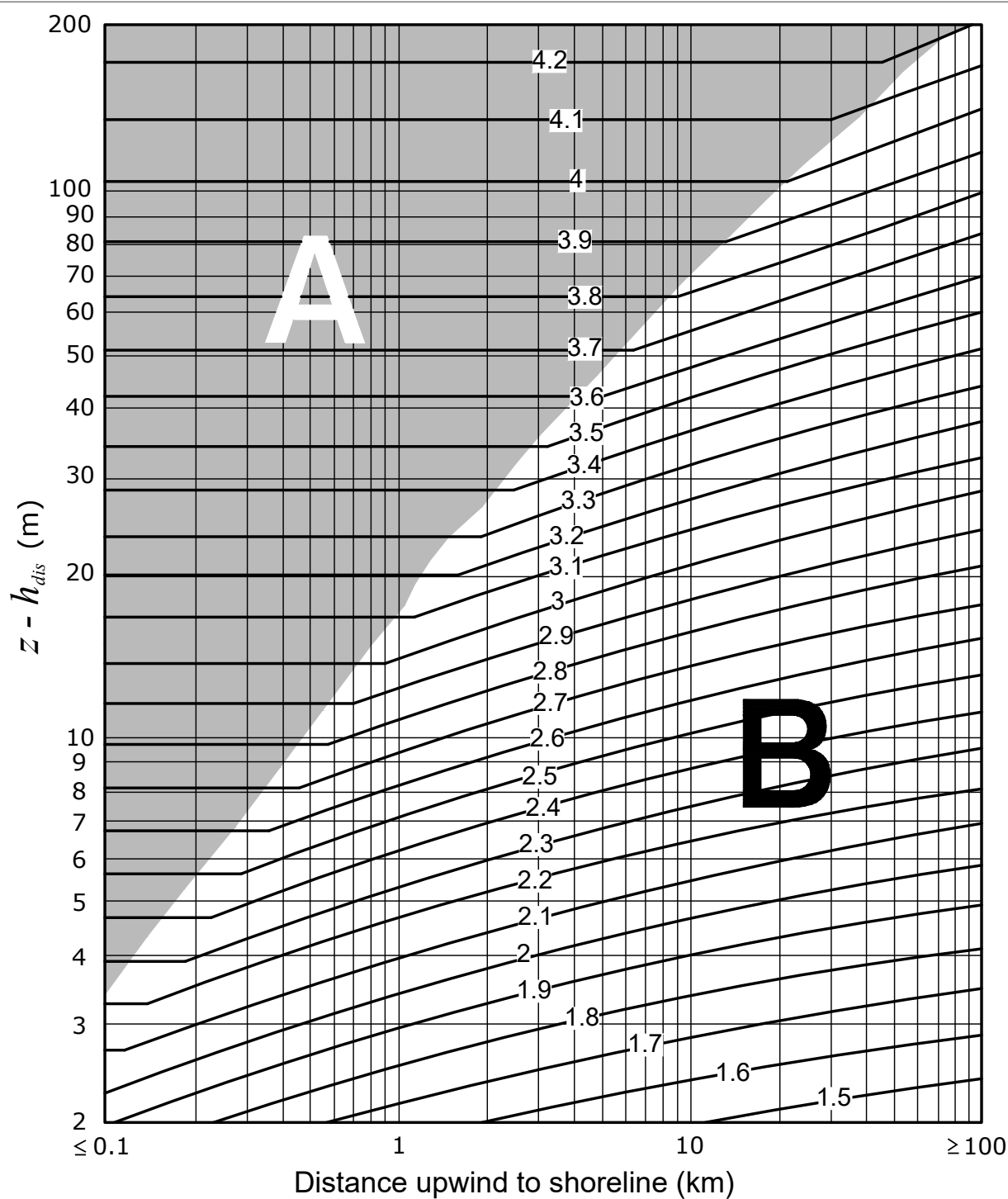
ρ = Air density, to be taken as 1.226 kg/m^3 .

$c_e(z)$ = Exposure factor, values are given in Figure 3.23.

$c_{e,T}$ = Exposure correction factor, values are given in Figure 3.24.

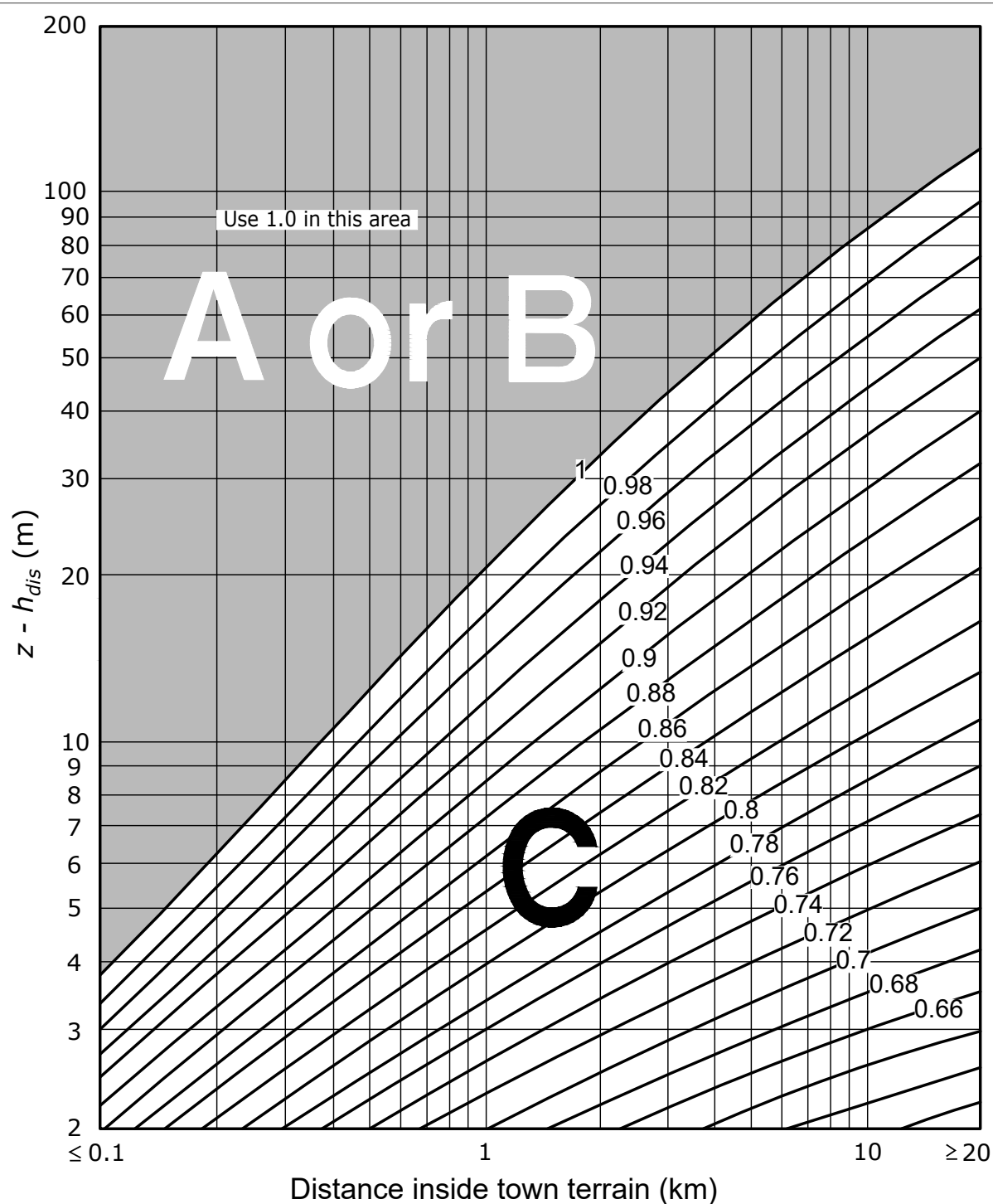
q_b = Basic velocity pressure = $(1/2) \rho v_b^2$.

v_m = Mean wind velocity at a height z .

Figure 3.23 Values of Exposure Factor $c_e(z)$ 

Note 1: The height z is the height at which q_p is sought using Equations 3.15 and 3.16.

Note 2: Zone A and B are indicated for use in Table 3.17.

Figure 3.24 Values of Exposure Correction Factor $c_{e,T}$ for Sites in Town Terrain

Note 1: The height z is the height at which q_p is sought using Equations 3.15 or 3.16.

Note 2: Zone A and B are indicated for use in Table 3.17.

Note 3: Sites less than 0.1 km inside town boundary should be treated as being in country terrain.

1

2

3

Actions and Combinations of Actions

Table 3.17 Size Factor c_s for Zones A, B and C Indicated in Figure 3.23 and Figure 3.24.

$b + h$ m	$z - h_{dis} \leq 6 \text{ m}$			$z - h_{dis} = 10 \text{ m}$			$z - h_{dis} = 30 \text{ m}$			$z - h_{dis} = 50 \text{ m}$			$z - h_{dis} = 200 \text{ m}$		
	A	B	C	A	B	C	A	B	C	A	B	C	A	B	C
≤ 5	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
10	0.95	0.94	0.88	0.95	0.95	0.90	0.96	0.96	0.93	0.97	0.96	0.94	0.98	0.97	0.97
20	0.93	0.91	0.84	0.93	0.92	0.87	0.95	0.94	0.90	0.95	0.95	0.92	0.96	0.96	0.95
30	0.91	0.89	0.81	0.92	0.91	0.84	0.94	0.93	0.88	0.94	0.93	0.90	0.96	0.95	0.93
40	0.90	0.88	0.79	0.91	0.89	0.82	0.93	0.91	0.86	0.93	0.92	0.88	0.95	0.94	0.92
50	0.89	0.86	0.77	0.90	0.88	0.80	0.92	0.90	0.85	0.92	0.91	0.87	0.94	0.94	0.91
70	0.87	0.84	0.74	0.88	0.86	0.77	0.90	0.89	0.83	0.91	0.90	0.85	0.93	0.92	0.90
100	0.85	0.82	0.71	0.86	0.84	0.74	0.89	0.87	0.80	0.90	0.88	0.82	0.92	0.91	0.88
150	0.83	0.80	0.67	0.84	0.82	0.71	0.87	0.85	0.77	0.88	0.86	0.79	0.90	0.89	0.85
200	0.81	0.78	0.65	0.83	0.80	0.69	0.85	0.83	0.74	0.86	0.84	0.77	0.89	0.88	0.83
300	0.79	0.75	0.62	0.80	0.77	0.65	0.83	0.80	0.71	0.84	0.82	0.73	0.87	0.85	0.80

Notes: b = cross wind breadth of building or building part or width of element h = height of building or building part or length of element z = height of building or height to top of element

interpolation may be used.

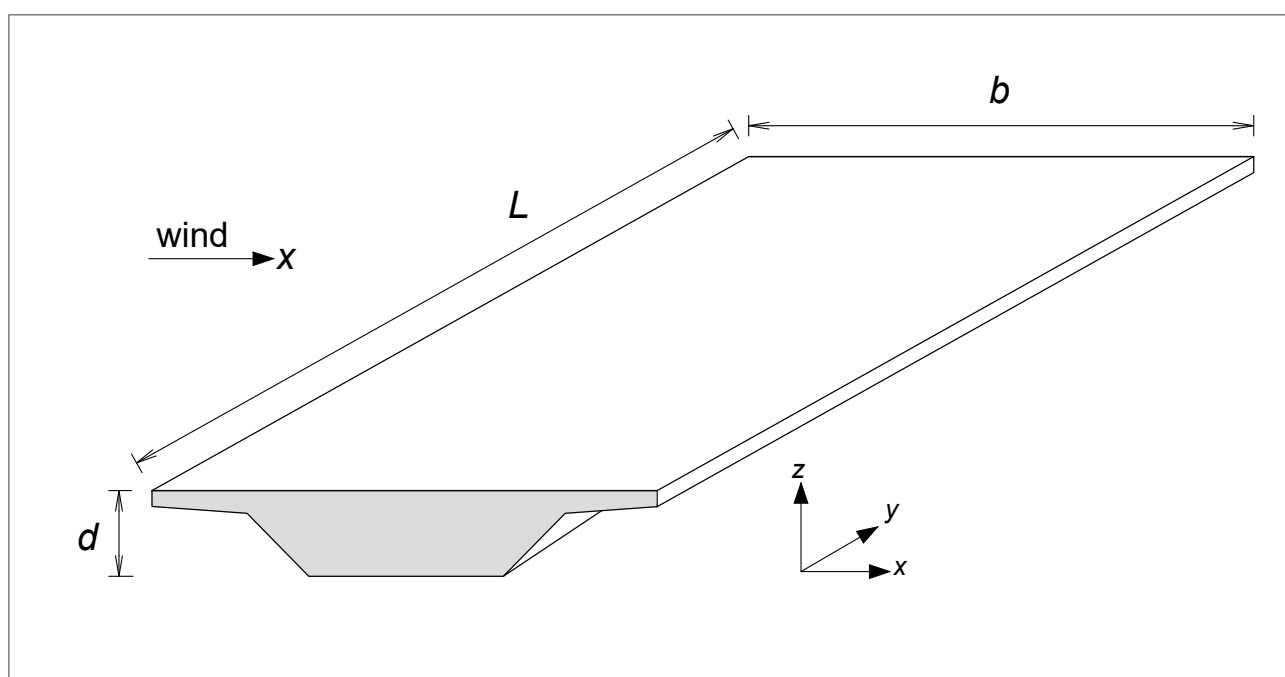
The zone A, B or C to be used for a building can be determined as follows:

For sites in country terrain, it is determined with respect to distance from shore and $(z - h_{dis})$ using Figure 3.23.For sites in town terrain, using the distance into town and $(z - h_{dis})$ in Figure 3.24, it is first determined whether zone C applies. If not, zone A or B will apply depending on the distance of the site from shore and $(z - h_{dis})$ as shown in Figure 3.23.

3.9.6 Wind Actions on Bridge Decks

The wind actions on bridges produces forces in the x , y and z directions representing transverse (i.e. perpendicular to the span), longitudinal (i.e. along the span) and vertical direction as shown in Figure 3.25.

Figure 3.25 Directions of Wind Actions on Bridges



When road traffic is considered to be simultaneous with the wind, the combination value $\psi_0 F_{wk}$ of the wind action on bridge and on the vehicles should be limited to a value F_w^* determined by using the fundamental value of the basic wind velocity v_{b0} but limiting the maximum value of peak velocity pressure $q_p(z)$ to 750 Pa, where z is the height of the deck above ground level.

It should be assessed whether a dynamic response procedure is needed for bridges in accordance with the guidelines provided in BS EN 1991-1-4 and its UK National Annex. Highway bridges of less than 200 m span do not normally require explicit allowance for dynamic response in the along wind direction, in service condition. Similarly, if the conditions specified therein for vertical wind response and aerodynamic stability are met, the dynamic effects may be neglected.

3.9.6.1 Force Coefficient

The force coefficient for wind actions in x-direction ($c_{f,x}$) on bridge decks can be taken as:

$$c_{f,x} = c_{f,x,0}$$

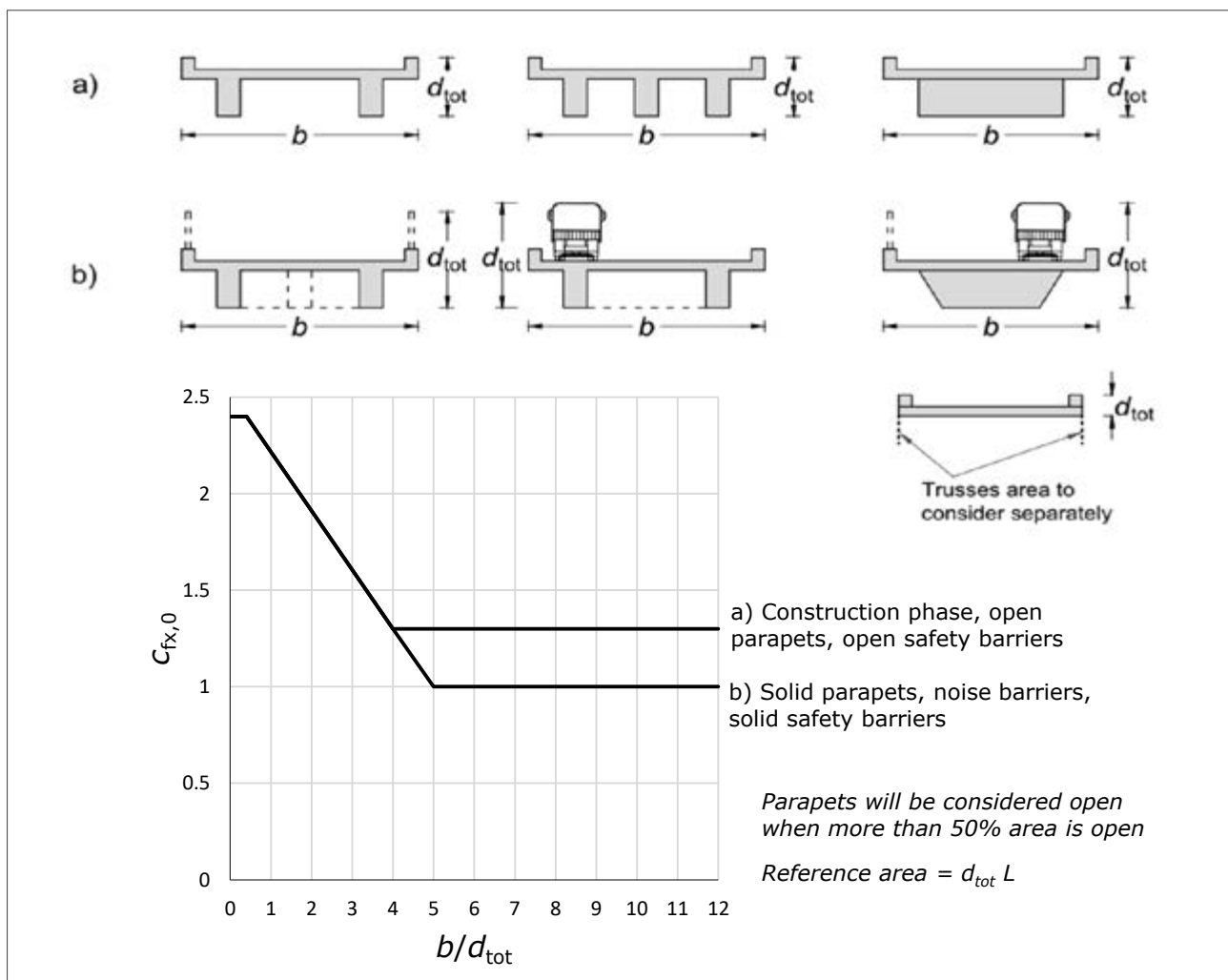
Equation 3.19

Where,

$c_{f,x,0}$ = force coefficient without free end flow, may be taken from Figure 3.26.

A bridge has usually no free-end flow because the flow is deviated only along two sides (over and under the bridge deck).

Figure 3.26 Force Coefficient for Bridge Decks ($c_{f,x,0}$)



1

Where two similar decks are at the same level, separated transversely by a gap not more than 1m, the wind on the windward structure may be calculated considering it a single structure.

2

Where a bridge deck is slopped transversely, the force coefficient $c_{fx,0}$ shall be increased by 3 % per degree of inclination, limiting maximum increase up to 25 %.

3

Where the angle of inclination of the wind exceeds 10°, the force coefficient may be derived from special studies.

3.9.6.2 Reference Area

The reference area in transverse direction ($A_{ref,x}$) for the purpose of calculating wind forces shall be the projected area on a plane perpendicular to the transverse direction. The reference area for load combination without traffic shall be calculated considering total equivalent depth d_{tot} as shown in Figure 3.27 and Table 3.18.

- For decks with plain beams, the projected area shall be calculated using total depth of structure. It is the sum of face area of the front main girder, face area of those parts of other girders projecting under this first one, face area of part of one cornice or walkway projecting over the front main girder and the face area of solid restraints or noise barriers where relevant. In case of open parapets or open safety barriers, a height of 0.30 m may be considered for each open parapet or barrier.
- In case of deck with trussed girders, it is the sum of face area of one cornice or walkway, the projected face area of solid parts of all main truss girders, face area of solid restraints or noise barriers where relevant or 0.3 m for each open parapet or barrier. The total reference area thus calculated shall not exceed that obtained by considering an equivalent plain beam with same overall depth including all projected parts.
- In case of decks with several main girders during construction phase, before placement of the carriageway slab, the face area of two main girders shall be considered.

Figure 3.27 Depth to be Used for Reference Area $A_{ref,x}$

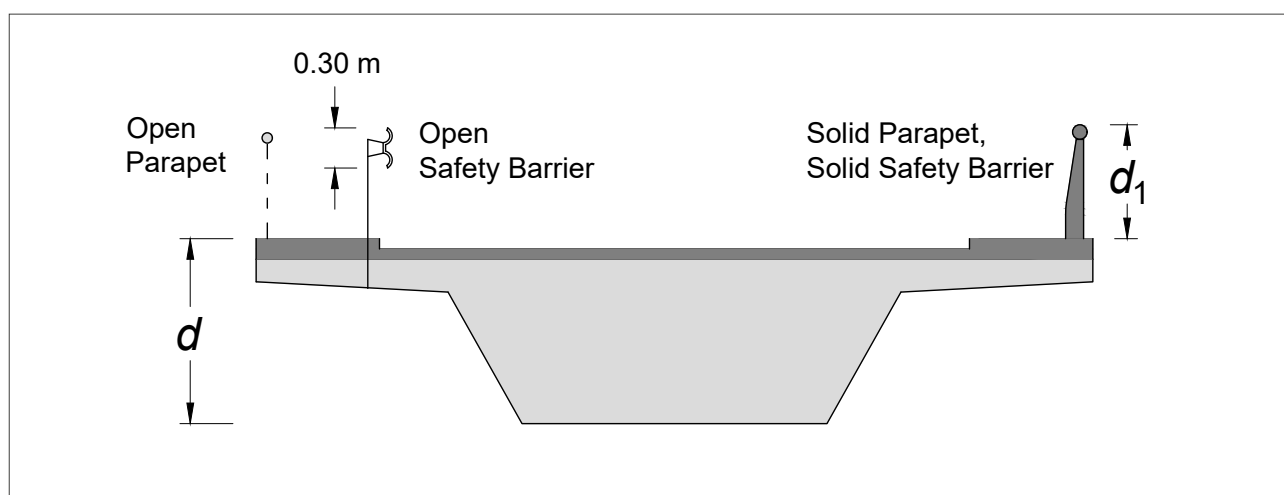


Table 3.18 Depth d_{tot} to be used for Calculating Reference Area ($A_{ref,x}$).

Road Restraint System	On One Side	On Both Sides
Open parapet or open safety barrier	$d + 0.3 \text{ m}$	$d + 0.6 \text{ m}$
Solid parapet or solid safety barrier	$d + d_1$	$d + 2 d_1$
Open parapet and open safety barrier	$d + 0.6 \text{ m}$	$d + 1.2 \text{ m}$

The reference area for load combination with traffic load shall be as determined above excluding the area of walkway and parapets but adding the area of live load. The area for live load height of 2m from the level of carriageway, on most unfavourable loaded length independent of the location of the vertical traffic loads, shall be considered. For calculating wind forces, the reference area with restraint system (parapet/ barrier) or with live load, whichever is critical, shall be considered.

The reference height z_e of structure may be taken as the distance from the lowest ground level to the centre of the bridge deck.

3.9.6.3 Forces in Transverse Direction

The wind force in a horizontal x-direction, transverse to the traffic movement, shall be considered to act horizontally at the centroid of the area and determined using simplified method as follows:

$$F_{wx} = (1/2) \rho v_b^2 C A_{ref,x} \quad \text{Equation 3.20}$$

Where,

- ρ = Air density, which depends on the altitude, temperature and atmospheric pressure. Normally, it can be taken as 1.226 kg/m³
- v_b = Basic wind speed (m/s)
- C = Wind load factor = $c_e c_{f,x}$
- c_e = Exposure factor
- $c_{f,x}$ = Force coefficient
- $A_{ref,x}$ = Reference area as defined above of the structure under consideration projected on a plane normal to the transverse direction (= $L d_{tot}$)

The values of C given in Table 3.19 shall be used.

Table 3.19 Values of Wind Load Factor C

b/d_{tot}	$z_e \leq 20 \text{ m}$	$z_e = 50 \text{ m}$
≤ 0.5	7.4	9.1
≥ 4.0	4.0	4.9

Note 1: The above values of C are a simplification for the product of c_e and $c_{f,x}$. In general, the individual values of c_e and $c_{f,x}$ are to be used.

Note 2: Linear interpolation may be used for intermediate values of b/d_{tot} and z_e

3.9.6.4 Forces in Vertical Direction

The wind force in a vertical i.e. z-direction shall be determined as follows:

$$F_{wz} = (1/2) \rho v_b^2 C A_{ref,z} \quad \text{Equation 3.21}$$

Where,

- ρ = Air density, which depends on the altitude, temperature and atmospheric pressure. Normally, it can be taken as 1.226 kg/m³.
- v_b = Basic wind speed (m/s).
- C = Force factor = $c_e c_{f,z}$.
- $c_{f,z}$ = May be taken as ± 0.9 . This value takes into account the possible transverse slope of the deck, slope of the terrain and fluctuations of the angle of wind directions with the deck due to turbulence.
- $A_{ref,z}$ = Reference area of the structure in plan = $b L$.

Eccentricity of the vertical force in transverse direction may be taken as one fourth of width ($e = b/4$).

3.9.6.5 Forces in Longitudinal Direction

Longitudinal forces (F_{wy}) due to wind on bridge superstructure shall be taken as acting at the centroids of the appropriate areas and be more severe of either:

- a. The longitudinal wind load on the superstructure, F_{sy} alone; or
- b. The sum of the nominal longitudinal wind load on the superstructure, F_{sy} and the nominal longitudinal wind load on the live load, F_{Ly} derived separately, as given as appropriate in 1) to 6);

1) All super structures with solid elevation

Longitudinal wind force shall be 25% of the wind force in x-direction i.e.

$$F_{sy} = 0.25 q_p(z) C_{fx} A_{ref}$$

Equation 3.22

Where,

$q_p(z)$ = Peak velocity pressure, as defined in Clause 3.9.5, appropriate value for superstructures with or without live load.

A_{ref} = Reference area for superstructure alone, as defined in Clause 3.9.6.2.

C_{fx} = Force coefficient for the superstructure as defined in Clause 3.9.6.1, but not less than 1.3.

2) All truss girder superstructure

Longitudinal wind force shall be 50% of the wind force in x-direction i.e.

$$F_{sy} = 0.50 q_p(z) C_{fx} A_{ref}$$

Equation 3.23

Where,

$q_p(z)$ = Peak velocity pressure, as defined in Clause 3.9.5, appropriate value for superstructures with or without live load.

A_{ref} = Reference area for superstructure alone, as defined in Clause 3.9.6.2.

C_{fx} = Force coefficient for the superstructure as defined in Clause 3.9.6.1.

3) Live load on all superstructures

Longitudinal wind force shall be 50% of the wind force on live load in x-direction i.e.

$$F_{Ly} = 0.50 q_p(z) C_{fx} A_{ref}$$

Equation 3.24

Where,

$q_p(z)$ = Peak velocity pressure, as defined in Clause 3.9.5, appropriate value for superstructures with live load.

A_{ref} = Reference area for the live load, as defined in Clause 3.9.6.2 and the appropriate horizontal wind loaded length giving the most severe effect.

C_{fx} = 1.45.

4) Parapet and safety fences

- i. With vertical infill members, the longitudinal force should be taken as 0.8 times the force in the x-direction on the element.
- ii. With two or three horizontal rails only, the longitudinal force should be taken as 0.4 times the force in the x-direction on the element.
- iii. With mesh panels, the longitudinal force should be taken as the force in the x-direction on the element.

5) Cantilever brackets extending outside main girders or trusses

The longitudinal force is the force derived from a horizontal wind acting at 45° to the longitudinal axis on the areas of each bracket not shielded by a fascia girder or adjacent bracket. C_{fx} is the drag coefficient, should be taken as appropriate.

6) Piers

The force derived from a horizontal wind acting along the longitudinal axis of the bridge should be taken as:

$$F_{Ly} = q_p(z) C_{fy} A_{ref}$$

Equation 3.25

Where,

$q_p(z)$ = Peak velocity pressure, as defined in Clause 3.9.5, appropriate value for superstructures with or without live load.

A_{ref} = Reference area for piers, which is the solid area in projected elevation normal to the longitudinal wind direction (in m²).

C_{fy} = Force coefficient, taken from Table 3.19 with values of b and t interchanged.

3.9.7 Wind Forces on Piers

Wind forces on bridge decks and the supporting piers shall be calculated considering most unfavourable direction of the wind. Structure shall also be checked for design situations during construction phase if critical.

The wind force for any structural component of pier shall be determined by:

$$F_w = C_{fp} q_p A_{ref}$$

Equation 3.26

Where,

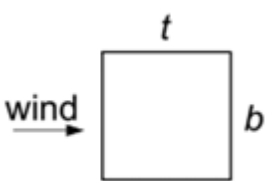
C_{fp} = Force coefficient defined for different sections.

q_p = Peak Velocity Pressure at reference height.

A_{ref} = Reference area of the structural component under consideration.

For simplicity, upper bound values of the force coefficient C_{fp} for piers may be taken from Table 3.20.

Table 3.20 Force Coefficient C_{fp} for Piers

Plan Shape	t/b	C_{fp} for Pier With Height/Breadth Ratios						
		1	2	4	6	10	20	40
 b is the dimension normal to the direction of wind	≤ 0.25	1.3	1.4	1.4	1.5	1.6	1.8	1.9
	0.333	1.4	1.4	1.5	1.5	1.7	1.8	2.0
	0.50	1.4	1.5	1.6	1.6	1.7	1.9	2.1
	0.667	1.5	1.6	1.6	1.7	1.8	2.0	2.2
	1.0	1.4	1.4	1.5	1.5	1.6	1.8	2.0
	1.5	1.2	1.2	1.3	1.4	1.4	1.6	1.7
	2.0	1.0	1.1	1.2	1.2	1.3	1.4	1.6
	3.0	0.9	0.9	1.0	1.0	1.1	1.2	1.3
	≥ 4.0	0.8	0.8	0.8	0.9	1.0	1.0	1.1
Square on diagonal (Note: dimension b in this case is the dimension of the diagonal)		1.0	1.1	1.1	1.2	1.2	1.4	1.5
Octagonal		0.83	0.87	.91	0.94	1.00	1.10	1.21
12-sided polygon		0.72	0.75	0.79	0.81	0.87	0.96	1.05
Circle with smooth surface where $1.5 t V_m(z) \geq 6 \text{ m}^2/\text{s}$		0.48	0.50	0.63	0.56	0.60	0.66	0.72
Circle with smooth surface where $1.5 t V_m(z) < 6 \text{ m}^2/\text{s}$. Also circle with rough surface or with projections		0.75	0.78	0.82	0.84	0.91	1.00	1.09
Notes: 1. After erection of the superstructure, C_{fp} is to be derived for a height/breadth ratio of 40. 2. For a rectangular pier with radiused corners, the value of C_{fp} derived from this Table is to be multiplied by $(1-2.5 r/b)$ or 0.5, whichever is greater. 3. For a pier with triangular nosing, C_{fp} is to be derived as for the rectangle encompassing the outer edges of the pier. 4. For a pier tapering with height, C_{fp} is to be derived for each of the unit heights into which the support has been subdivided. Mean values of t and b for each unit height is to be used to evaluate t/b . The overall pier height and the mean breadth of each unit height is to be used to evaluate height/breadth. 5. On relieving areas, use F_m instead of F_w .								

3.10 Thermal Actions

3.10.1 General

The shade air temperature depends upon the climatic conditions and the geographic location. Daily and seasonal variations in shade air temperature, solar radiation, etc. cause the following-

- Changes in the overall temperature of the bridge, referred to as the uniform bridge temperature. Over a prescribed period, there will be a minimum and a maximum, together with a range of uniform bridge temperature, resulting in actions and/or action effects within the bridge due to:
 - Restraint offered to the associated expansion/contraction by the form of construction (e.g., portal frame, arch, flexible pier, elastomeric bearings) referred to as temperature restraint; and
 - Friction at roller or sliding bearings referred to as frictional bearing restraint.
- Differences in temperature between the top surface and other levels through the depth of the superstructure, referred to as temperature difference and resulting in associated actions and/or action effects within the structure.

Representative values of thermal actions shall be determined for each relevant design situation considering uniform bridge temperature component and the temperature difference components. Provisions shall be made for thermal stresses and/or movements resulting from variations in the temperature. Thermal actions are considered as variable actions.

For the purpose of determining temperature effect, the bridge decks are grouped as follows:

Type-1: Steel deck (steel box girder, steel truss or plate girder).

Type-2: Composite deck (concrete deck on steel box, truss or plate girder).

Type-3: Concrete deck (concrete slab, concrete beam, concrete box girder).

The temperature effect for other type of bridges not covered here shall be agreed for individual project with the approving authority.

The effect of temperature changes may be ignored on following structural components:

- a. Walls and the base slab of buried concrete box structures and the walls of buried concrete frame structures.
- b. In-situ buried concrete structures (box culverts, portals frame structures, etc.) which have more than 0.60 m total fill height (fill including surfacing) and which have length (in transverse direction) more than five times of clear span (for single span structure) or five times of largest clear span (for multi-span structures).
- c. For buried concrete segments which have more than 0.60 m fill height (fill including surfacing) and which are located more than 1.25 times the clear span from the edge of the structures.

Buried concrete box culverts and the portal frame structures shall be classified as Type-3 structures.

3.10.2 Uniform Temperature Component

3.10.2.1 General

The uniform bridge temperature in a bridge is an evenly distributed temperature over the cross section which governs the expansion or contraction of the bridge structure. The effects of restraint of associated expansion or contraction due to type of construction and friction at roller or sliding bearings shall be taken into account where relevant.

The uniform bridge temperatures depend upon the maximum and minimum shade air temperatures where the bridge is erected. For the maximum and minimum shade air temperatures (T_{max} and T_{min}) for the location of the bridge, refer Clause 3.10.2.3. The maximum and minimum uniform bridge temperatures ($T_{e,max}$ and $T_{e,min}$) for different type of construction may be derived from Figure 3.29 for specific values of maximum and minimum shade air temperatures subject to adjustment for deck surfacing given in Table 3.21.

The uniform bridge components are dependent on the depth of surfacing on bridge deck. The values given in Figure 3.29 are based on surfacing depth of 40 mm for Type-1 and 100 mm for Type-2 and Type-3 decks. Adjustment shall be made by the amounts given in Table 3.21 when depth of surfacing differs.

3.10.2.2 Uniform Temperature Component for Buried Concrete Structures

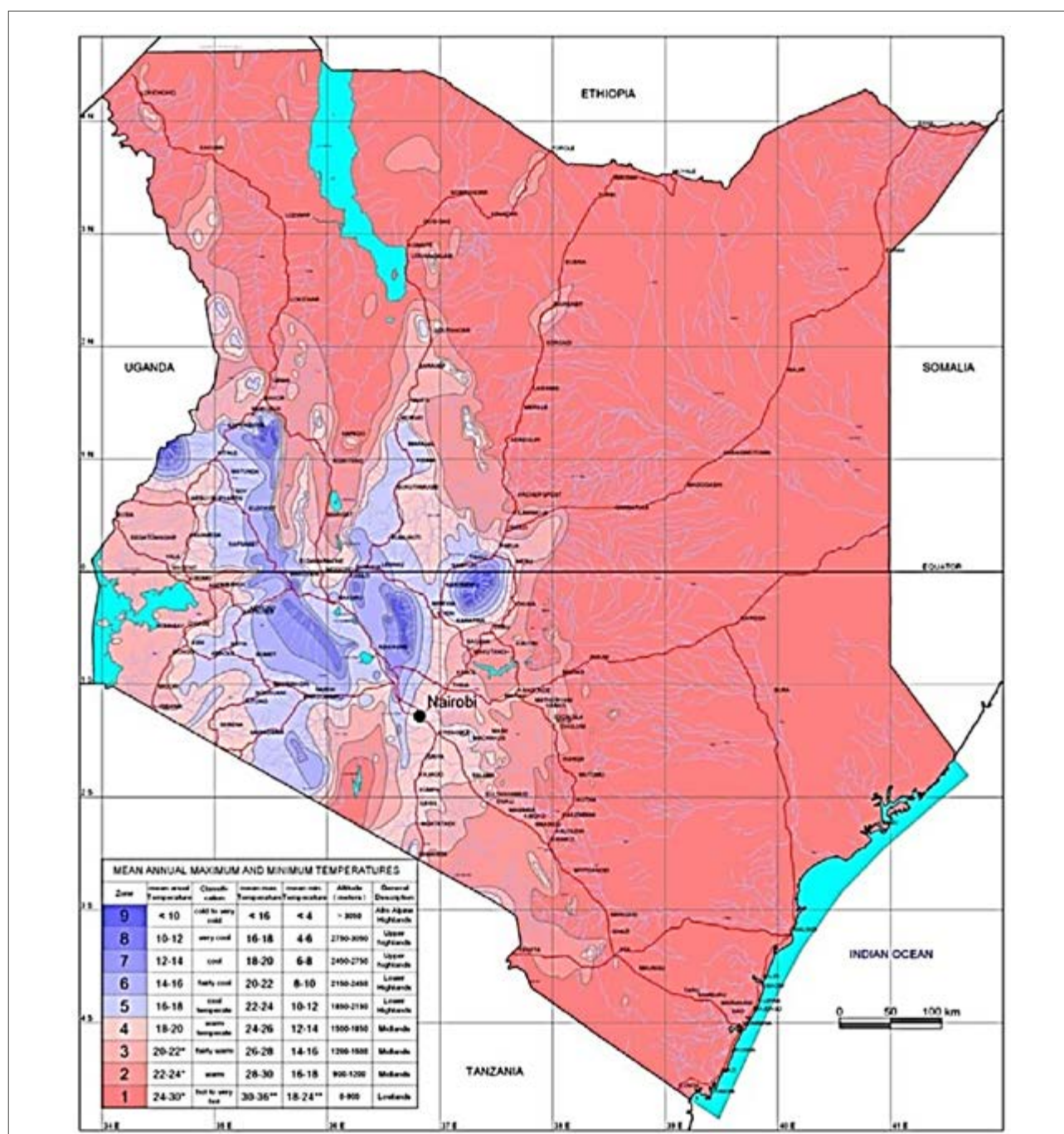
For buried concrete box culverts and portal structures where the total fill height from top of the roof slab to top of surfacing is more than 200 mm, the minimum and maximum uniform bridge temperature derived from Figure 3.29 and adjusted for Table 3.21, may further be modified. For every additional 100 mm total fill height in excess of 200 mm, the minimum uniform bridge temperature may be increased by 1°C and maximum uniform bridge temperature may be reduced by 2 °C. However, the difference between the maximum and minimum uniform bridge temperature shall not be taken less than 15°C. When the total fill height from top of roof slab to top of surfacing is 1.5 m or more, the changes in uniform bridge temperature may be ignored.

3.10.2.3 Shade Air Temperature

The characteristic values of maximum (T_{max}) and minimum (T_{min}) shade air temperatures represent shade air temperatures for mean sea level in open country with an annual probability of being exceeded of 0.02 (i.e., 1 in 50 years return period). The temperature values shall be adjusted for other height above mean sea level. The values of shade air temperature may be adjusted for height above sea level by subtracting 0.5°C per 100 m height for minimum shade air temperatures and 1.0°C per 100 m height for maximum shade air temperatures. If the values represent temperatures at actual ground level, this correction is not applied.

Note: The characteristic values of maximum (T_{max}) and minimum (T_{min}) shade air temperatures with an annual probability of being exceeded of 0.02 (i.e., 1 in 50 years return period) based on maximum and minimum hourly records shall be obtained from the Meteorology Department, Kenya. As a preliminary design, in absence of any alternative data, the minimum and maximum temperatures values determined from map provided at Figure 3.28 may be used which is borrowed from "RDM 3.3".

Figure 3.28 Map Showing Temperature Zones



3.10.2.4 Range of Uniform Bridge Temperature Component

In determining action effects due to temperature, the initial uniform bridge temperature (T_0) when the structure is effectively restrained shall be taken as datum in calculating the expansion up to the maximum uniform bridge temperature and contraction down to the minimum uniform bridge temperature component.

The characteristic value of maximum expansion range of uniform bridge temperature shall be taken as:

$$\Delta T_{N,exp} = T_{e,max} - T_0 \quad \text{Equation 3.27}$$

The characteristic value of maximum contraction range of uniform bridge temperature shall be taken as:

$$\Delta T_{N,con} = T_0 - T_{e,min} \quad \text{Equation 3.28}$$

Overall range of effective bridge temperature component is given by:

$$\Delta T_N = T_{e,max} - T_{e,min} \quad \text{Equation 3.29}$$

For bearings and expansion joints, the recommended value of maximum expansion range of uniform bridge temperature component and maximum contraction range of the uniform temperature component are ($\Delta T_{N,exp} + 20^\circ\text{C}$) and ($\Delta T_{N,con} + 20^\circ\text{C}$) respectively. The maximum expansion and contraction for this temperature range can be determined as follows:

Maximum expansion $\Delta L_{exp} = L \alpha_T (T_{e,max} - T_0 + 20^\circ\text{C})$

Maximum contraction $\Delta L_{con} = L \alpha_T (T_0 - T_{e,min} + 20^\circ\text{C})$

Where,

L = Effective expansion/contraction length of the bridge deck under consideration.

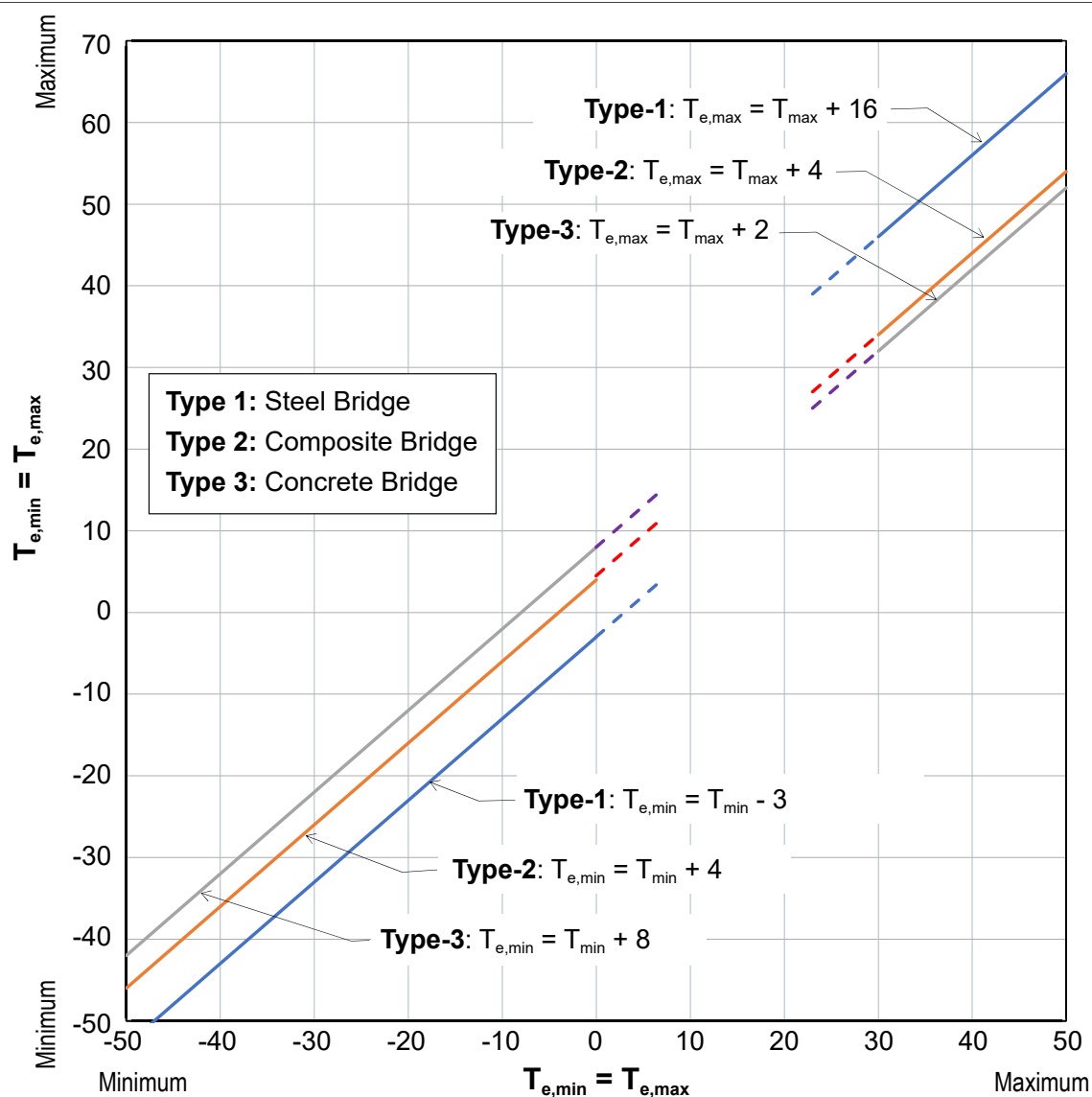
α_T = Coefficient of thermal expansion.

T_0 = Initial effective bridge temperature.

If the temperature at which the bearings and expansion joints are set is specified, then the maximum expansion and contraction uniform temperature range may be taken as ($\Delta T_{N,exp} + 10^\circ\text{C}$) and ($\Delta T_{N,con} + 10^\circ\text{C}$) respectively.

The initial temperature T_0 should be taken as the temperature of a structural element at the relevant stage of its restraint (completion). If it is not predictable the average temperature during the construction period should be taken. In case of uncertainty concerning sensitivity of the bridge to T_0 , it is recommended that a lower and upper bound of an interval expected for T_0 is considered.

Figure 3.29 Correlation Between Shade Air Temperature and Uniform Bridge Temperature



Note: For steel truss and plate girders, the maximum values given for type-1 may be reduced by 3°C.

Table 3.21 Adjustment to Uniform Bridge Temperature for Deck Surfacing

Deck Surfacing	Addition to minimum uniform bridge temperature component, °C			Addition to maximum uniform bridge temperature component, °C		
	Type-1	Type-2	Type-3	Type-1	Type-2	Type-3
Unsurfaced	0	-3	-1	+4**	0	0
Water-proofed	0	-3	-1	+4**	+4	+2
40mm surfacing*	0	-2	-1	0	+2	+1
100 mm surfacing*	N/A	0	0	N/A	0	0
200 mm surfacing*	N/A	+3	+1	N/A	-4	-2

* Surfacing depths include waterproofing.

** For steel truss and plate girders, the values for unsurfaced and waterproofed deck surfaces may be reduced to +2 °C.

3.10.3 Temperature Difference Component

Due to variations in temperature over a time period, heating and cooling of bridge deck will result in maximum heating and maximum cooling temperature variations. Top surface will become warmer during heating and bottom surface remains warmer during cooling of the bridge deck. The vertical temperature difference may produce effects within a structure due to restraint of free curvature and friction at rotational bearings.

Note: Eurocode provide details on Approach 1 (vertical linear component) and Approach 2 (vertical temperature components with non-linear effects) for determining the temperature difference components. Approach 1 is provided in this section as a simplified procedure. UK National Annex recommends to use Approach 2. The Approach 2 should be used unless the use of Approach 1 is agreed for individual project with the approving authority. The details for Approach 2 are provided at Appendix A.2.

3.10.3.1 Vertical Linear Component – Approach 1

The effect of vertical temperature gradient may be considered by using an equivalent linear temperature gradient $\Delta T_{m,heat}$ and $\Delta T_{m,cool}$ to be applied between top and bottom of the bridge deck. The values of $\Delta T_{m,heat}$ and $\Delta T_{m,cool}$ may be taken from Table 3.22. The values given in this table are based on 50 mm surfacing depth.

Table 3.22 Linear Temperature Difference for Bridge Decks

Type of Deck		Top Warmer than Bottom	Bottom Warmer than Top
		$\Delta T_{m,heat} (^{\circ}\text{C})$	$\Delta T_{m,cool} (^{\circ}\text{C})$
Type 1: Steel deck		18	13
Type 2: Composite deck		15	18
Type 3: Concrete deck	Concrete box girder	10	5
	Concrete beam	15	8
	Concrete slab	15	8

Note: The values given in this table are based on 50mm surfacing depth.

For surface thickness other than 50 mm, the values should be multiplied by a factor k_{sur} given in Table 3.23.

Table 3.23 Values of k_{sur} for Different Surfacing Depths

Surface thickness (mm)	Deck Type 1		Deck Type 2		Deck Type 3	
	Top warmer than bottom	Bottom warmer than top	Top warmer than bottom	Bottom warmer than top	Top warmer than bottom	Bottom warmer than top
Unsurfaced	0.7	0.9	0.9	1.0	0.8	1.1
Water-proofed	1.6	0.6	1.1	0.9	1.5	1.0
50	1.0	1.0	1.0	1.0	1.0	1.0
100	0.7	1.2	1.0	1.0	0.7	1.0
150	0.7	1.2	1.0	1.0	0.5	1.0

3.10.3.2 Vertical Linear Component – Approach 2

The effect of vertical temperature difference under this approach is considered by a non-linear temperature difference component. The details of this approach is provided at Appendix A.2.

3.10.3.3 Temperature Difference Component for Buried Concrete Structures

The temperature difference component for buried concrete box and portal frame structures shall be established using Appendix A.2. In Table A.5, the value of h for buried concrete structure shall be taken as the distance from underside of the surfacing to the soffit of the roof slab of the structure. The ΔT_1 in Figure A.3, Figure A.4 and Figure A.5 should be taken as occurring at the underside of the surfacing and the dimensions h and h_1 should be measured from that level so that the temperature profiles shown in Figure A.5 are applied through the fill as well as through the roof slab.

The heating and cooling temperature difference may be ignored for buried concrete box and portal frame structures if the total filling height from top of roof slab to top of the surfacing is more than 0.50 m.

3.10.4 Coefficient of Thermal Expansion

For the purpose of calculating temperature effects, the coefficient of thermal expansion for concrete and steel structures may be taken as $10 \times 10^{-6} / ^\circ\text{C}$ and $12.0 \times 10^{-6} / ^\circ\text{C}$ respectively, except where more accurate values have been experimentally determined.

3.11 Seismic Actions

3.11.1 General

The provision herein covers the seismic actions for conventional bridges with slab, beam girder, box girder supported on piers in which the horizontal seismic actions are mainly resisted by bending of piers or at the abutments. The structures in seismic regions shall be designed and constructed to withstand the design seismic forces.

Suspension bridges, cable stayed bridges, masonry bridges, moveable bridges and floating bridges are not covered in the scope of this Section.

The design seismic action (A_{Ed}) is expressed in terms of:

- The reference seismic action (A_{Ek}) associated with a reference probability of exceedance P_{NCR} of 10% in 50 years or a reference return period T_{NCR} of 475 years.
- The importance factor γ_I (refer Table 3.24).

$$A_{Ed} = \gamma_I A_{Ek}$$

Equation 3.30

Note: The Eurocode specifies $P_{NCR} = 10\%$ and $T_{NCR} = 475$ years. The UK National Annex specifies $T_{NCR} = 2500$ years. UK is considered an area of very low seismicity and using the same return period in Kenya will be on higher side. It may be more appropriate to use the Eurocode recommended return period of 475 years in Kenya till the local parameters are determined. Authority may consider any alternative values or the UK National Annex values for individual project.

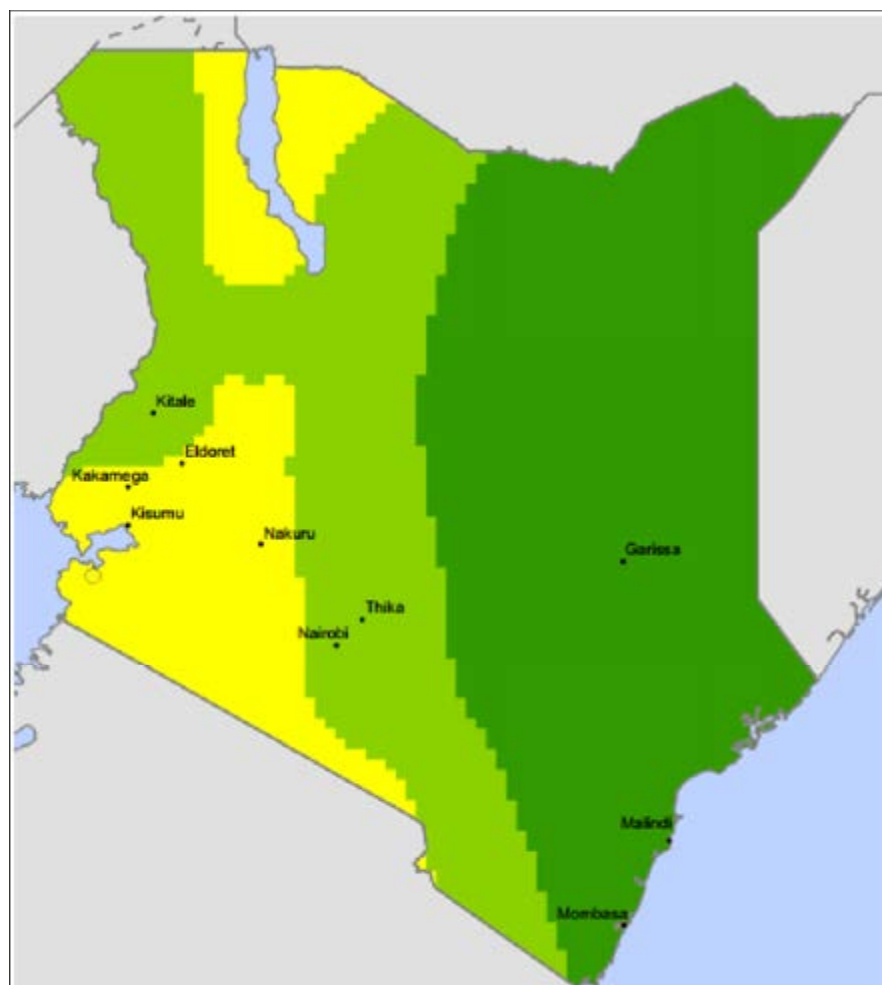
The elastic spectrum of the design seismic action in accordance with Clause 3.11.4.2, 3.11.4.3 and 3.11.4.4 shall apply. For application of the equivalent linear method of Clause 3.11.5.6 (using the behaviour factor q), the spectrum shall be the design spectrum in accordance with Clause 3.11.4.5.

3.11.2 Seismic Zones

The seismicity in Kenya is categorised into three zones for the purpose of determining seismic actions as shown in Figure 3.30. The reference peak ground acceleration (a_{gR}) may be taken as shown in Figure 3.30. The bridge shall be assigned to one of the seismic zones based on the location of the bridge.

Note: Seismic zones and the peak ground acceleration shown in Figure 3.30 are based on the seismic map developed by WHO and provided as indicative values. The actual values for bridge specific site for a return period of 475 years or a period as agreed by the Authority are to be obtained from Mines and Geological Department, Kenya.

Figure 3.30 Seismic Zone Map



Seismic Zone	Peak Ground Acceleration (a_{gR}) m/sec ²
I	0 - 0.2
II	0.2 - 0.8
III	0.8 - 2.4

1

2

3

Actions and Combinations of Actions

Source: WHO 2010

For the purpose of calculating seismic effect, the bridges shall be classified in importance classes as shown Table 3.24 depending on the consequences of their damage or failure from seismic events. The various factors which decide the importance category may include extent of disturbance to traffic affecting human life, importance of maintaining communications especially in immediate post-earthquake period, cost of repairs or replacement and time involved, indirect economic loss, social, security or defence requirements. An importance factor γ_I is assigned to each importance class.

Table 3.24 Seismic Importance Factors

Category	Description	Importance Factor (γ_I)
I	Bridges which are not critical for maintaining communication and have less than average importance.	0.85
II	General road bridges of average importance.	1.0
III	Bridges of critical importance for maintaining communications, especially in the immediate post-earthquake period, bridges the failure of which is associated with a large number of probable fatalities and major bridges where a design life greater than normal is required.	1.3

Note: Where a reference return period $T_{NCR} = 2500$ years is adopted for Category III bridges, the $\gamma_I = 1.0$ may be assumed.

The design ground acceleration (a_g) = $\gamma_I a_{gR}$

In case of low seismicity, a reduced or simplified seismic design procedures may be used. Low seismicity is considered if the design ground acceleration on type A ground (a_g) is not greater than 0.78 m/s^2 (for $T_{NCR} = 475$ years). If $T_{NCR} = 2500$ years as per UK National Annex, a_g not greater than 2 m/s^2 shall be considered as low seismicity.

In case of very low seismicity, the seismic action effects may be neglected. Very low seismicity is considered if the design ground acceleration on type A ground (a_g) is not greater than 0.39 m/s^2 (for $T_{NCR} = 475$ years). If $T_{NCR} = 2500$ years as per UK National Annex, a_g not greater than 1.8 m/s^2 shall be considered as very low seismicity.

3.11.3 Ground Conditions

The local ground conditions have influence on the seismic action and to account for this, the ground is classified into different types from A to E, described by stratigraphic profiles and parameters as given in Table 3.25.

Table 3.25 Ground Types

Ground Type	Description of Stratigraphic Profile	Parameters		
		v_{s30} (m/s)	N_{SPT} (blows/30cm)	c_u (N/mm ²)
A	Rock or other rock-like geological formation, including at most 5 m of weaker material at the surface.	>800	–	–
B	Deposits of very dense sand, gravel, or very stiff clay, at least several tens of metres in thickness, characterised by a gradual increase of mechanical properties with depth.	360-800	> 50	> 0.25
C	Deep deposits of dense or medium-dense sand, gravel or stiff clay with thickness from several tens to many hundreds of metres.	180-360	15-50	0.07-0.25
D	Deposits of loose-to-medium cohesionless soil (with or without some soft cohesive layers), or of predominantly soft-to-firm cohesive soil.	<180	<15	<0.07
E	A soil profile consisting of a surface alluvium layer with v_s values of type C or D and thickness varying between about 5 m and 20 m, underlain by stiffer material with $v_s > 800 \text{ m/s}$.			
S1	Deposits consisting, or containing a layer at least 10 m thick, of soft clays/silts with a high plasticity index ($PI > 40$) and high-water content.	<100 (indicative)	–	0.01 – 0.02
S2	Deposits of liquefiable soils, of sensitive clays, or any other soil profile not included in types A – E or S1.			

v_{s30} = Average shear wave velocity in top 30m of the soil at shear strain of 10^{-5} or less. N_{SPT} = Standard penetration test blow counts.
 c_u = Undrained shear strength of soil (N/mm²).

The ground type shall be classified according to the average shear wave velocity ($v_{s\ 30}$), if available, otherwise use value of N_{SPT} . The average shear wave velocity shall be calculated as:

$$v_{s\ 30} = \frac{30}{\sum \frac{h_i}{v_i}} \quad \text{Equation 3.31}$$

The h_i and v_i denote the thickness in meters and the shear wave velocity at a shear strain level of 10^{-5} or less for the i -th layer existing in the top 30 m.

The ground type S1 and S2 represents the two special ground conditions which needs special studies for the definition of the seismic action.

3.11.4 Basic Representation of the Seismic Action

3.11.4.1 General

The earthquake motion at a given point of the surface is represented by an elastic ground acceleration response spectrum called “elastic response spectrum”. The horizontal seismic action is described by two orthogonal components considered as independent and represented by the same response spectrum.

3.11.4.2 Horizontal Elastic Response Spectrum

For the horizontal components of the seismic action, elastic response spectrum $S_e(T)$ is given by the following Equations:

$$S_e(T) = a_g S \left[1 + \frac{T}{T_B} (\eta \ 2.5-1) \right] \text{ for } 0 \leq T \leq T_B \quad \text{Equation 3.32}$$

$$S_e(T) = a_g S \eta \ 2.5 \text{ for } T_B \leq T \leq T_C \quad \text{Equation 3.33}$$

$$S_e(T) = a_g S \eta \ 2.5 \left[\frac{T_C}{T} \right] \text{ for } T_C \leq T \leq T_D \quad \text{Equation 3.34}$$

$$S_e(T) = a_g S \eta \ 2.5 \left[\frac{T_C T_D}{T^2} \right] \text{ for } T_D \leq T \leq 4 \text{ seconds} \quad \text{Equation 3.35}$$

Where,

$S_e(T)$ = Elastic response spectrum.

T = Vibration period of a linear single-degree-of-freedom system.

a_g = Design ground acceleration. ($a_g = \gamma_I a_{gR}$)

T_B = Lower limit of the period of the constant spectral acceleration branch.

T_C = Upper limit of the period of the constant spectral acceleration.

T_D = Value at beginning of the constant displacement response range of the spectrum.

S = Soil factor.

η = Damping correction factor with a reference value of $\eta = 1$ for 5% damping.

The values of the time periods T_B , T_C and T_D and the soil factor S depends upon the ground type. Depending on the characteristics of the most significant earthquake contributing to the local hazard, two types of elastic response spectrum are:

Type 1: surface-wave magnitude, M_s , greater than 5.5

Type 2: surface-wave magnitude, M_s , less than or equal to 5.5

Table 3.26 Values of the Parameters for Elastic Response Spectra

Spectrum	Ground Type	S	T_B (s)	T_C (s)	T_D (s)
Type 1	A	1.0	0.15	0.4	2.0
	B	1.2	0.15	0.5	2.0
	C	1.15	0.20	0.6	2.0
	D	1.35	0.20	0.8	2.0
	E	1.4	0.15	0.5	2.0
Type 2	A	1.0	0.05	0.25	1.2
	B	1.35	0.05	0.25	1.2
	C	1.5	0.1	0.25	1.2
	D	1.8	0.1	0.30	1.2
	E	1.6	0.05	0.25	1.2

Note: UK National Annex specifies that in the absence of site-specific information, the recommended values for Type 2 earthquake may be used.

The parameters T_B , T_C and T_D and S for two types of spectra are given in Table 3.26.

The shape of Type 1 and Type 2 normalised elastic response spectrum, $(S_e(T) / a_g)$ corresponding to the type of ground conditions are given in Figure 3.31 for 5% damping.

For any other value of damping, the damping correction factor η may be determined as:

$$\eta = \sqrt{\frac{10}{5 + \zeta}} \geq 0.55$$

Equation 3.36

Where, ζ is the damping ratio of the structure in percent.

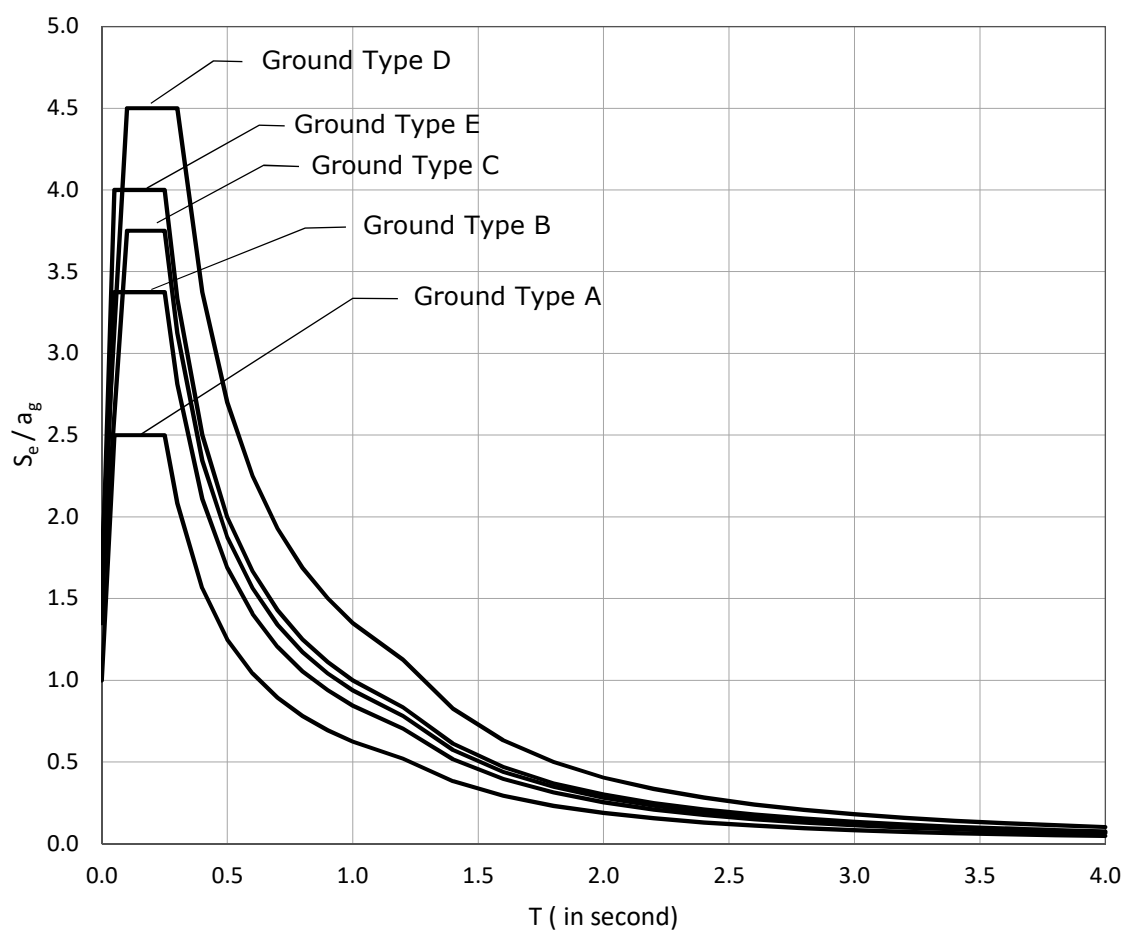
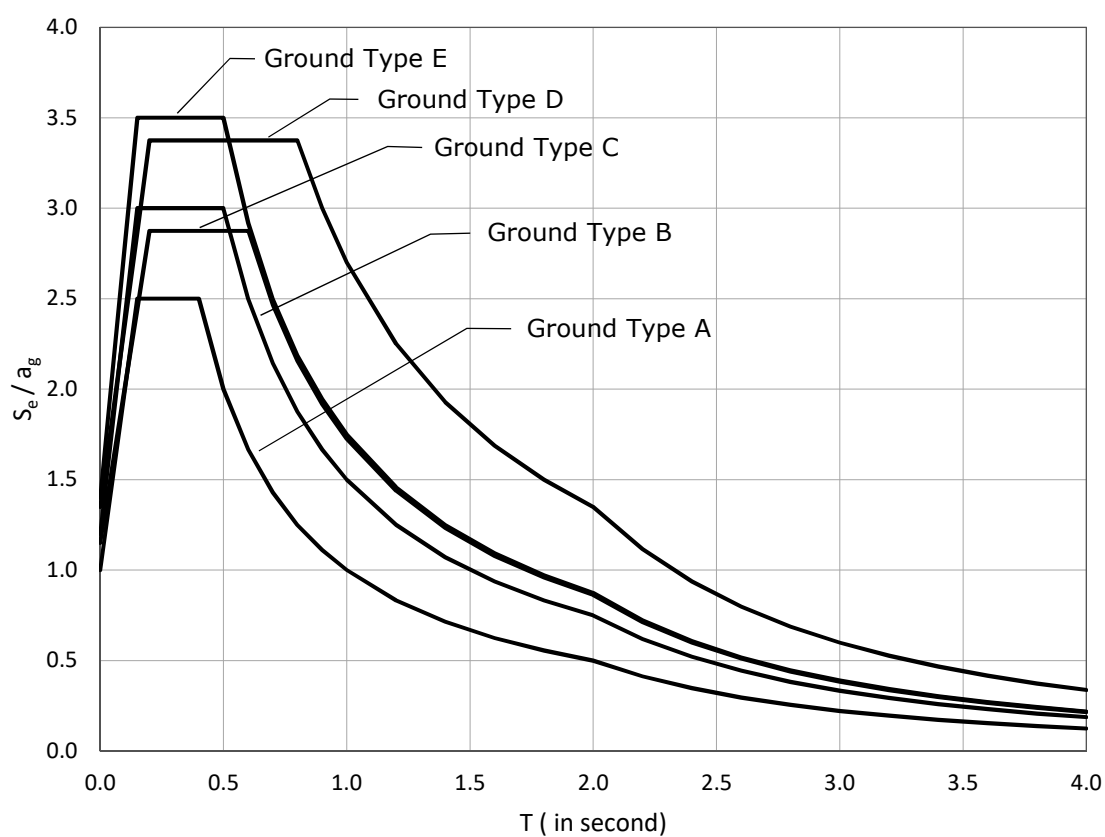
The elastic displacement response spectrum $S_{De}(T)$ shall be obtained by direct transformation of elastic acceleration response spectra $S_e(T)$ as follows:

$$S_{De}(T) = S_e \left[\frac{T}{2\pi} \right]^2$$

Equation 3.37

The Equation 3.37 shall normally be applied for vibration periods not exceeding 4.0 seconds.

Figure 3.31 Horizontal Elastic Seismic Response Spectra (5% Damping)



3.11.4.3 Vertical Elastic Response Spectrum

The vertical component of the seismic action shall be represented by an elastic response spectrum (S_{ve}) given as:

$$S_{ve}(T) = a_{vg} S \left[1 + \frac{T}{T_B} (\eta \cdot 3.0 - 1) \right] \text{ for } 0 \leq T \leq T_B \quad \text{Equation 3.38}$$

$$S_{ve}(T) = a_{vg} S \eta \cdot 3.0 \text{ for } T_B \leq T \leq T_C \quad \text{Equation 3.39}$$

$$S_{ve}(T) = a_{vg} S \eta \cdot 3.0 \left[\frac{T_C}{T} \right] \text{ for } T_C \leq T \leq T_D \quad \text{Equation 3.40}$$

$$S_{ve}(T) = a_{vg} S \eta \cdot 3.0 \left[\frac{T_C T_D}{T^2} \right] \text{ for } T_D \leq T \leq 4 \text{ seconds} \quad \text{Equation 3.41}$$

The values of the time periods T_B , T_C , T_D and a_{vg} for ground types A, B, C, D and E for vertical spectra are given in Table 3.27.

Table 3.27 Values of the Parameters for Vertical Elastic Response Spectra

Spectrum	a_{vg}	T_B (s)	T_C (s)	T_D (s)
Type 1	0.90 a_g	0.05	0.15	1.0
Type 2	0.45 a_g	0.05	0.15	1.0
<i>Note: UK National Annex specifies that in the absence of site-specific information, the recommended values for Type 2 earthquake may be used.</i>				

3.11.4.4 Design Ground Displacement

The design ground displacement (dg) corresponding to the design ground acceleration may be estimated as:

$$d_g = 0.025 a_g S T_C T_D \quad \text{Equation 3.42}$$

3.11.4.5 Design Spectrum for Elastic Analysis

3.11.4.5.1 General

In order to avoid explicit inelastic analysis in design, the capacity of the structure to dissipate energy through ductile behaviour is taken into account by performing an elastic analysis based on a response spectrum reduced with respect to the elastic response spectrum by introducing a behaviour factor q . Both ductile and limited ductile structures shall be designed by performing linear analysis.

The behaviour factor q is an approximation of the ratio of the seismic forces that the structure would experience if its response was completely elastic with 5% damping, to the seismic forces that may be used in the design, with a conventional elastic analysis model, still ensuring a satisfactory response of the structure.

3.11.4.5.2 Horizontal Design Spectrum for Elastic Analysis

The horizontal components of the seismic action, the design spectrum $S_d(T)$ are given by following Equations:

$$S_d(T) = a_g S \left[\frac{2}{3} + \frac{T}{T_B} \left(\frac{2.5}{q} - \frac{2}{3} \right) \right] \text{ for } 0 \leq T \leq T_B \quad \text{Equation 3.43}$$

$$S_d(T) = a_g S \frac{2.5}{q} \text{ for } T_B \leq T \leq T_C \quad \text{Equation 3.44}$$

$$S_d(T) = a_g S \frac{2.5}{q} \left[\frac{T_C}{T} \right] \text{ but } \geq \beta a_g \text{ for } T_C \leq T \leq T_D$$

Equation 3.45

$$S_d(T) = a_g S \frac{2.5}{q} \left[\frac{T_C T_D}{T^2} \right] \text{ but } \geq \beta a_g \text{ for } T_D \leq T \text{ seconds}$$

Equation 3.46

Where,

a_g , S , T_B , T_C and T_D = defined earlier.

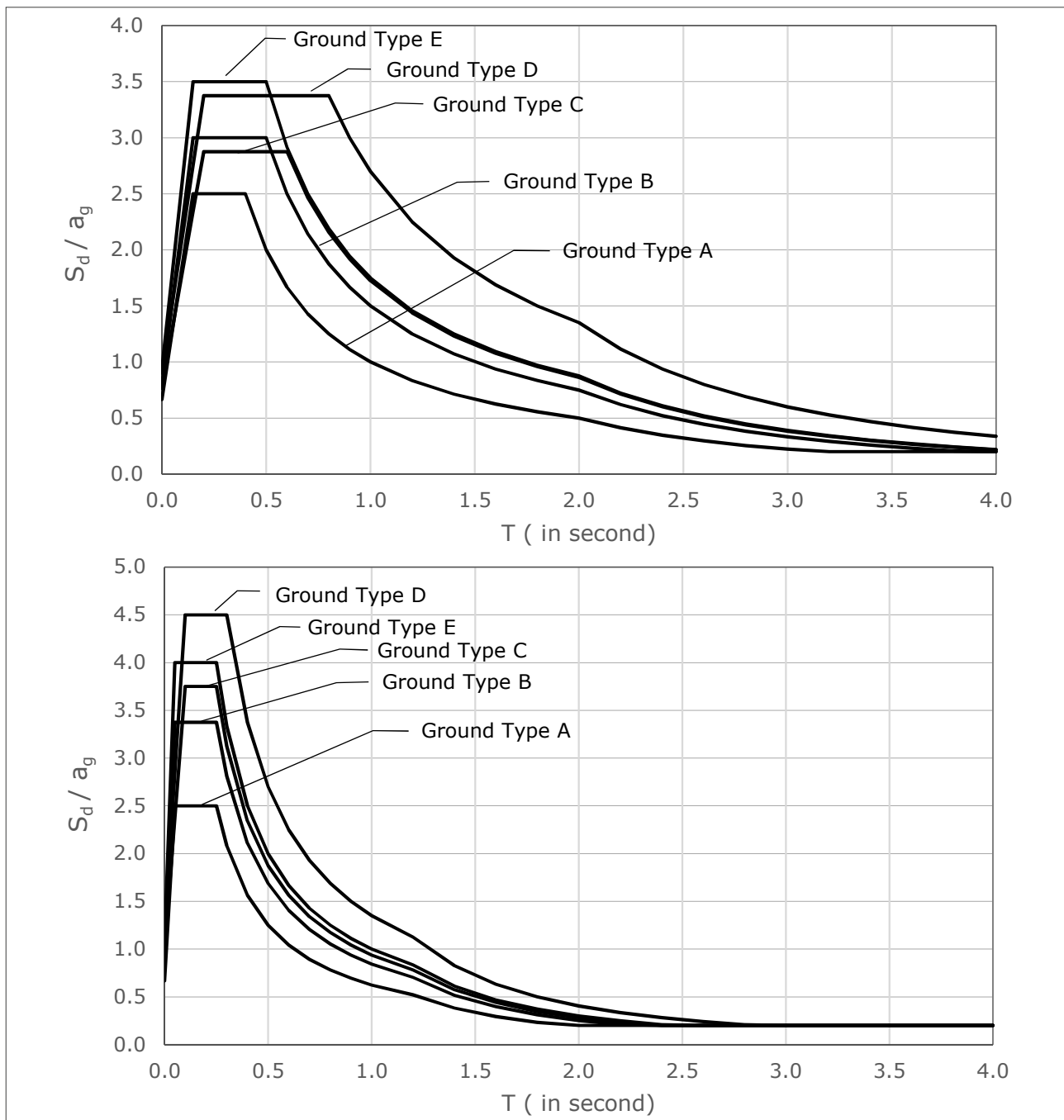
$S_d(T)$ = Design spectrum.

q = Behaviour factor.

β = Lower bound factor for the horizontal design spectrum = 0.2.

The normalised design response spectrum $S_d(T)/a_g$ corresponding to the type of ground conditions are given in Figure 3.32 for 5% damping and for a behaviour factor of $q = 1.0$.

Figure 3.32 Design Response Spectra Normalised (5% Damping, $q = 1$)



3.11.4.5.3 Vertical Design Spectrum for Elastic Analysis

For vertical component of the seismic action, the design spectrum is given by the Equation 3.43 to 3.46, for the design ground acceleration in the vertical direction, replace a_g by a_{vg} , take value of S as 1.0 and other parameters as provided in Table 3.27. For vertical component of seismic action, a behaviour factor q up to 1.50 shall generally be adopted for all materials and structural systems.

3.11.5 Analysis of Seismic Actions

3.11.5.1 General

The response of the random ground motion can be analysed by modelling the structure in three mutually perpendicular directions. The components are considered to act simultaneously but independently. In certain cases, it is sufficient to analyse the response in two separate models in the analysis, one for modelling the response in longitudinal direction and other for the transverse direction. The cases when it is necessary to analyse in vertical direction are described later.

3.11.5.2 Seismic Masses

Mean value of the permanent masses equal to their characteristic values and the quasi-permanent values of the masses corresponding to variable actions shall be considered. The quasi-permanent value of the variable action due to traffic shall be taken as $\psi_{2,l} Q_{k,l}$ where $Q_{k,l}$ is the characteristic value of traffic load. The value of factor $\psi_{2,l} = 0$ for road bridges with normal traffic and footbridges. For bridges with severe traffic and for the UDL system of Load Model 1 (LM1), the value of factor $\psi_{2,l} = 0.2$ may be considered. For the purpose of severe traffic condition, all routes of international and national importance may be considered. When piers are immersed in water, hydrodynamic effect shall also be considered. The hydrodynamic effect may be omitted on the vertical seismic action.

3.11.5.3 Damping of the Structure

When response spectrum analysis is used, following values of the damping ratio may be assumed depending upon the material of the member where large part of the deformation energy is dissipated during the seismic response. In general, it will occur in piers.

Reinforced concrete	5%
Prestressed concrete	2%
Welded steel	2%
Bolted steel	4%

When the structure comprises several components i with different damping ratios ζ_i , the effective damping ratio ζ_{eff} of the structure may be estimated by weighted average considering damping ratio and the deformation energy E_{di} induced in different components by the seismic action.

$$\zeta_{eff} = \frac{\sum \zeta_i E_{di}}{\sum E_{di}}$$

Equation 3.47

3.11.5.4 Soil-structure Interaction

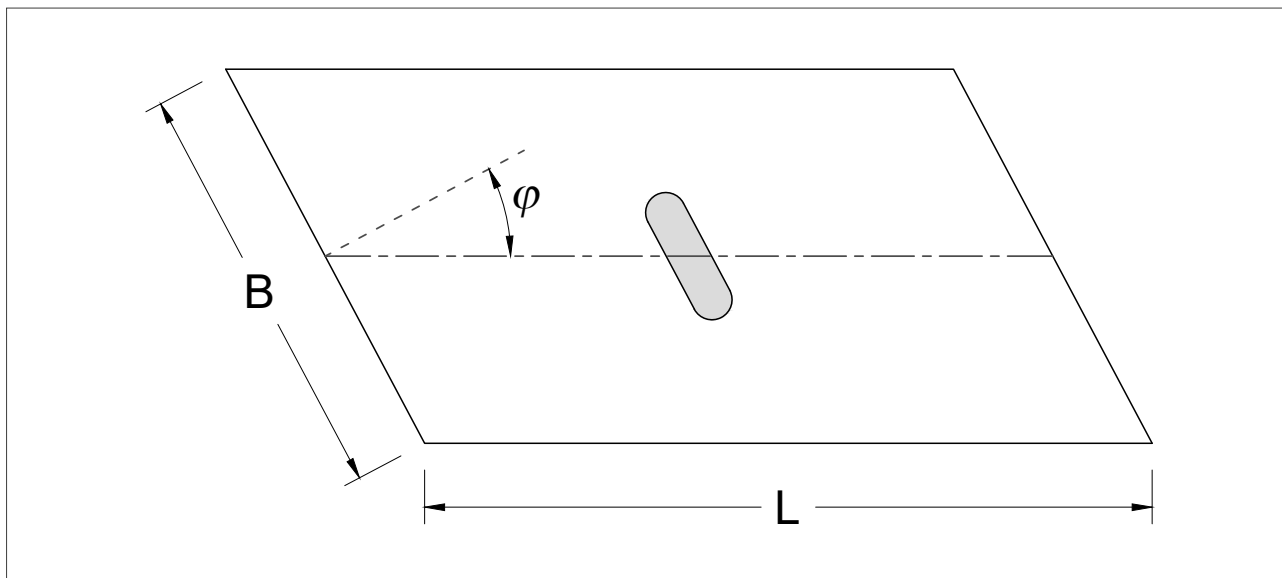
For the seismic analysis of the global system, the supporting members which transmit the seismic action from the soil to the deck shall, in general, be assumed as fixed relative to the foundation soil. The soil-structure interaction effect may be considered using appropriate impedances or the appropriately defined soil springs.

Soil-structure interaction effects should always be accounted for in piers where, under the action of a unit horizontal load in a given direction at the top of the pier, the soil flexibility contributes more than 20% of the total displacement at the top of the pier.

3.11.5.5 Torsional Effects

Torsional effect of a bridge about the vertical axis shall be considered only in skewed bridges (skew angle $> 20^\circ$) and bridges with a ratio $B/L > 2$. The skew bridges tend to rotate about the vertical axis even when the centre of mass theoretically coincides with the centre of stiffness. Skew bridges with skew angle greater than 45° shall in general be avoided in high seismic regions.

Figure 3.33 Skew Bridge



3.11.5.6 Behaviour Factor for Linear Analysis

The linear analysis using behaviour factor q (global force reduction factor) is the normal analysis method. The behaviour factor is defined globally for the entire structure and reflects its ductility capacity. Two types of intended seismic behaviour are foreseen for bridges.

- Ductile behaviour: Corresponding values of behaviour factor, $1.50 < q \leq 3.5$
- Limited ductile behaviour: corresponding values of behaviour factor, $q \leq 1.50$

The maximum values of behaviour factor q which may be used for the two horizontal components shall be as given in Table 3.28, depending on the post elastic behaviour of the ductile member. The capability of ductile members to develop flexural plastic hinges is an essential requirement for the values of behaviour factor specified in the Table for ductile behaviour. If a bridge has various types of ductile members, the behaviour factor shall be used corresponding to the type-group with major contribution to the seismic resistance. Different values may be used in two horizontal directions.

Table 3.28 Maximum Values of the Behaviour Factor q

Type of Ductile Members		Seismic Behaviour	
		Limited Ductile	Ductile
Reinforced concrete piers:	Vertical piers in bending	1.5	3.5λ
	Inclined struts in bending	1.2	2.1λ
Steel Piers:	Vertical piers in bending	1.5	3.5
	Inclined struts in bending	1.2	2.0
	Piers with normal bracing	1.5	2.5
	Piers with eccentric bracing	–	3.5
Abutments rigidly connected to the deck:	In general	1.5	1.5
	Locked-in structures	1.0	1.0
Arches		1.2	2.0

$$\lambda = 1.0 \quad \text{for } \alpha_s \geq 3$$

$$\lambda = \sqrt{\frac{\alpha_s}{3}} \quad \text{for } 3 > \alpha_s \geq 1.0$$

$\alpha_s = L_s/h$ is the shear span ratio of the pier, where L_s is the distance from the plastic hinge to the point of zero moment and h is the depth of the cross section in the direction of flexure of the plastic hinge.

For reinforced concrete ductile members, the values of q -factors specified in the Table are applicable when the normalised axial force η_k does not exceed 0.30. When $0.30 < \eta_k \leq 0.60$, even in a single ductile member, the value of the q factor should be reduced to:

$$q_r = q - \frac{\eta_k - 0.3}{0.3} (q - 1) \text{ but } \geq 1.0 \quad \text{Equation 3.48}$$

The normalised axial force is given by:

$$\eta_k = \frac{N_{Ed}}{A_c f_{ck}} \quad \text{Equation 3.49}$$

Where,

N_{Ed} = Value of axial force at the plastic hinge corresponding to the design seismic combination, positive if compressive.

A_c = Cross sectional area of the section.

f_{ck} = Characteristic concrete compressive strength.

A value for reduced behaviour factor $q_r = 1.0$ shall be used for the bridges in which the seismic force resisting system contains members with $\eta_k \geq 0.6$.

The values of the q -factor for ductile behaviour specified in the Table, may be used only if the locations of all the relevant plastic hinges are accessible for inspection and repair. Otherwise, these values are multiplied by 0.6. However final q -values less than 1.0 need not be used.

The inertial response of bridge structures whose mass follows essentially the horizontal seismic motion of the ground ("locked-in" structures), may be assessed using the design value of the seismic ground acceleration and $q = 1$. Such structures are characterised by a very low value of the natural period in the horizontal directions ($T \leq 0.03$ s). Abutments flexibly connected to the deck belong to this category.

When the main part of the design seismic action is resisted by elastomeric bearings, the flexibility of the bearings imposes a practically elastic behaviour of the system. Such bridges are designed in accordance with the rules of seismic isolation.

The behaviour factor for the analysis in the vertical direction shall be taken as 1.0.

3.11.5.7 Vertical Component of the Seismic Action

The effects of the vertical seismic component on the piers may be omitted in cases of low and moderate seismicity. In zones of high seismicity, these effects need only be taken into account if the piers are subjected to high bending stresses due to vertical permanent actions of the deck, or when the bridge is located within 5 km of an active seismotectonic fault.

The effects of the vertical seismic component acting in the upward direction on prestressed concrete decks, bearings and links shall be taken into account.

The estimation of the effects of the vertical component may be carried out using the Fundamental Mode Method and the Flexible Deck Model.

3.11.5.8 Regular and Irregular Seismic Behaviour of Ductile Bridges

Depending upon the maximum value of the design moment ($M_{Ed,i}$) at the intended plastic hinge location of a ductile member derived from the analysis for the seismic design situation and the design flexural resistance ($M_{Rd,i}$) of the same section with its actual reinforcement under the concurrent action of the non-seismic action effects in the seismic design situation, then a local force reduction factor r_i associated with that member, under the specific seismic action is defined as:

$$r_i = q \frac{M_{Ed,i}}{M_{Rd,i}} \quad \text{Equation 3.50}$$

The value of $r_i \leq q$, since $M_{Ed} \leq M_{Rd}$

When in a regular bridge, the maximum value of r_i among all ductile members $r_{i \max} = 1.0$, the bridge responds elastically to the earthquake considered.

A bridge shall be considered to have regular seismic behaviour in the considered horizontal direction, when the following condition is satisfied.

$$\rho_r = \frac{r_{i \max}}{r_{i \min}} \leq \rho_0 \quad \text{Equation 3.51}$$

Where,

$r_{i \max}$ = Maximum value of r_i among all ductile members

$r_{i \min}$ = Minimum value of r_i among all ductile members

ρ_0 = 2.0, limiting value selected so as to ensure that sequential yielding of the ductile members will not cause unacceptably high ductility demands on one member

One or more ductile members (piers) may be exempted from the above calculation of $r_{i \max}$ and $r_{i \min}$, if their total shear contribution does not exceed 20% of the total Seismic shear in the considered horizontal direction.

Bridges that do not conform to above condition of ρ_{ri} , shall be considered to have irregular seismic behaviour, in the considered horizontal direction. Such bridges shall either be designed using a reduced q-value

$$q_{ri} = q \frac{\rho_0}{\rho_{ri}} \geq 1.0 \quad \text{Equation 3.52}$$

or shall be designed based on results of non-linear analysis.

3.11.5.9 Non-linear Analysis of Irregular Bridges

In bridges of irregular seismic behaviour, the sequential yielding of the ductile members (piers) may cause substantial deviations of the results of the equivalent linear analysis performed with the assumption of a global behaviour factor q from those of the non-linear response of the bridge structure.

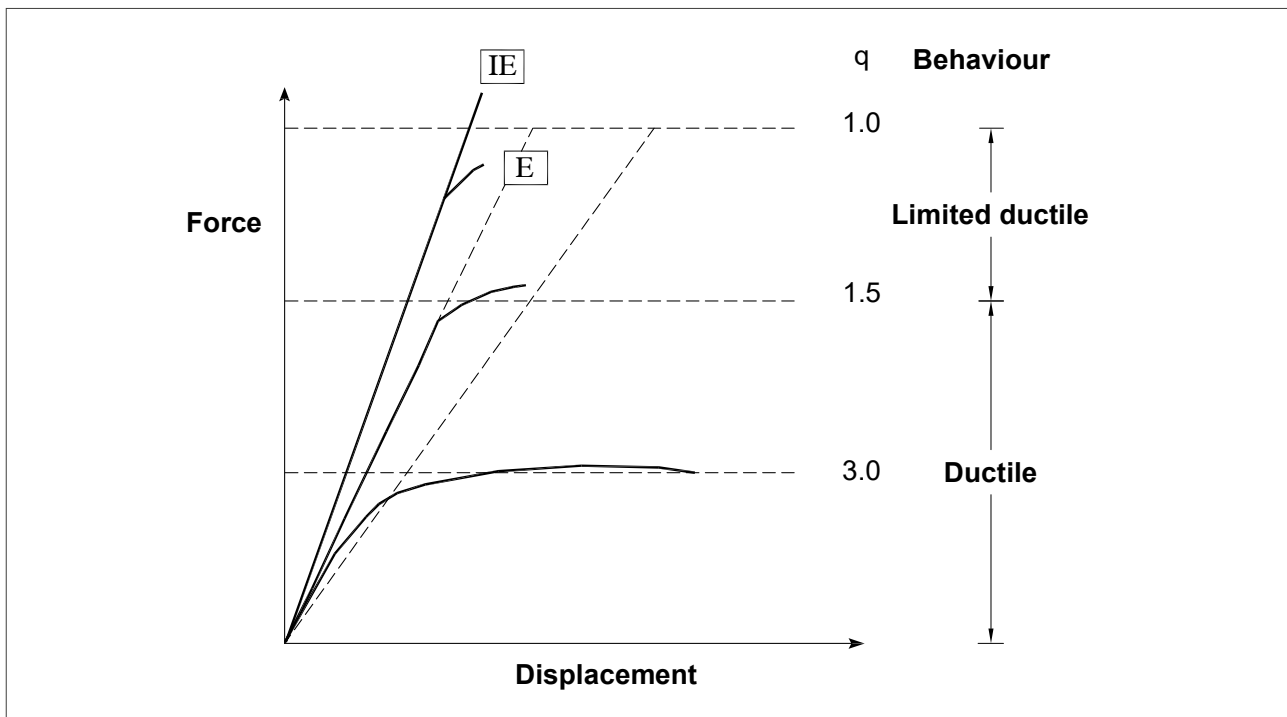
In general, the realistic response of irregular bridges under the design seismic action may be estimated by means of a dynamic non-linear time-history analysis. An approximation of the non-linear response may also be obtained by a combination of an equivalent linear analysis with a non-linear static analysis (pushover analysis).

3.11.5.10 Seismic Behaviour of Structure

3.11.5.10.1 General

The bridge shall be designed so that its behaviour under the design seismic action is either ductile, or limited ductile/essentially elastic, depending on the seismicity of the site, on whether seismic isolation is adopted, or any other constraints prevails. This behaviour (ductile or limited ductile) is characterised by the global force-displacement relationship of the structure, shown in Figure 3.34.

Figure 3.34 Seismic Behaviour of Structure



3.11.5.10.2 Ductile Behaviour

In moderate to high seismic regions, it is preferable to design a bridge for ductile behaviour. This is achieved by providing for the formation of an intended configuration of flexural plastic hinges or by using isolating devices. The flexural hinges normally form in the piers and act as the primary energy dissipating components. The location of plastic hinges should be selected at points which are accessible for inspection and repair. The bridge deck shall remain within the elastic range. Ductile behaviour corresponds to a behaviour factor of $1.50 < q \leq 3.5$.

Plastic hinges shall not be formed in reinforced concrete sections where the normalised axial force ηk given as below exceeds 0.6.

$$\eta_k = \frac{N_{Ed}}{A_c f_{cK}}$$

Equation 3.53

The pre-stress members should be protected from formation of plastic hinges under the design seismic action. Supporting piers or abutments connected to the deck through sliding or flexible elastomeric bearings should, in general, remain within the elastic range.

3.11.5.10.3 Limited Ductile Behaviour

In structures with limited ductile behaviour, a yielding region with significant reduction in secant stiffness need not appear under the design seismic action, but deviation from the ideal elastic behaviour provides some hysteretic energy dissipation. The limited ductile behaviour corresponds to a behaviour factor of $q \leq 1.5$.

For bridges where the detailing of plastic hinges for ductility may not be reliable (e.g. due to a high axial force), a behaviour factor of $q = 1$ is recommended, corresponding to elastic behaviour.

3.11.5.11 Effective Stiffness

When equivalent linear analysis methods are used, the stiffness of each member shall be chosen corresponding to its secant stiffness under the maximum calculated stresses under the design seismic action. For members containing plastic hinges this corresponds to the secant stiffness at the theoretical yield point.

For reinforced concrete members in bridges designed for ductile behaviour, unless a more accurate method is used for its estimation, the effective flexural stiffness to be used in linear analysis (static or dynamic) for the design seismic action may be estimated as follows.

- For reinforced concrete piers, a value calculated on the basis of the secant stiffness at the theoretical yield point.
- For prestressed or reinforced concrete decks, the stiffness of the uncracked gross concrete sections.

In bridges designed for limited ductile behaviour, the flexural stiffness of the uncracked gross concrete sections may be used for the entire structure.

For both ductile and limited ductile bridges, the significant reduction of the torsional stiffness of concrete decks, in relation to the torsional stiffness of the uncracked deck, should be accounted for. Unless a more accurate calculation is made, the following fractions of the torsional stiffness of the uncracked gross section may be used:

- For open sections or slabs, the torsional stiffness may be ignored.
- For prestressed box sections, 50% of the uncracked gross section stiffness.
- For reinforced concrete box sections, 30% of the uncracked gross section stiffness.

3.11.6 Linear Methods of Analysis

In case of low to medium seismicity, simplified seismic design methods may be used for normal bridges. The simplified design methods are based on the limited ductile/ essentially elastic behaviour of the bridge for which no ductility requirements are necessary. In case of high seismicity, the selection of ductile behaviour is generally advisable.

1

3.11.6.1 Fundamental Mode Method

2

3.11.6.1.1 General

The equivalent static seismic forces are derived from the inertia forces corresponding to the fundamental mode and natural period of the structure in the direction under consideration. The site dependent design spectrum shall be used in deriving seismic forces. The method also includes simplifications regarding the shape of the first mode and the estimation of the fundamental period.

3

This method may be applied using three different approaches for the model depending on the particular characteristics of the bridge.

- The Rigid Deck Model
- The Flexible Deck Model
- The Individual Pier Model

3.11.6.1.2 Field of Application

The fundamental mode method may be applied when the dynamic behaviour of the structure can be sufficiently approximated by a single dynamic degree of freedom model. This may be considered in the following cases.

- In the longitudinal direction of approximately straight bridges with continuous deck, when the seismic forces are carried by piers and the total mass of which is less than 20% of the mass of the deck.
- In the transverse direction for straight bridges as defined above (a), and the structural system is approximately symmetric about the centre of the deck.
- In the case of piers with simply-supported spans, if no significant interaction between piers is expected and the total mass of each pier is less than 20% of the contributory mass of the deck.

3.11.6.1.3 Rigid Deck Model

This model may only be applied if, under the seismic action, the deformation of the deck in a horizontal plane is negligible compared to the horizontal displacements of the pier tops. This condition is always met in the longitudinal direction of approximately straight bridges with continuous deck. In the transverse direction the deck may be assumed rigid either if $L/B \leq 4,0$, or if the maximum difference of the transverse displacements at pier tops is not exceeding 20% of the average displacements of all pier tops under transverse seismic action or under similar transverse load distribution.

The seismic effect shall be determined by applying a horizontal equivalent static force (F) at the deck given as:

$$F = MS_d(T)$$

Equation 3.54

Where,

M = Total effective mass of the structure, equal to the mass of the deck plus the mass of the upper half of the piers

$S_d(T)$ = Design Spectral acceleration (Clause 3.11.4.5) corresponding to the fundamental period T of the bridge

The fundamental period T of the bridge is given by:

$$T = 2\pi \sqrt{\frac{M}{\sum K_i}}$$

Equation 3.55

Where,

$\sum K_i$ = Stiffness of the system, equal to the sum of the stiffness's of the resisting members

The force F in transverse direction may be distributed along the desk proportionally to the distribution of the effective masses.

3.11.6.1.4 Flexible Deck Model

The flexible model shall be applied if the maximum difference of the transverse displacements at pier tops is exceeding 20% of the average displacements of all pier tops under transverse seismic action or under similar transverse load distribution.

The fundamental period of the structure in the horizontal direction considered, may be estimated via the Rayleigh quotient, using a generalised single-degree-of-freedom system, as follows:

$$T = 2\pi \sqrt{\frac{\sum M_i d_i^2}{g \sum M_i d_i}} \quad \text{Equation 3.56}$$

Where,

M_i = Mass at the i^{th} nodal point

d_i = Displacement in the direction under consideration when the structure is acted upon by forces gM_i acting at all nodal points in the horizontal direction considered

The seismic effects shall be determined by applying horizontal forces F_i at all nodal points given as:

$$F_i = \frac{4\pi^2}{gT^2} S_d(T) d_i M_i \quad \text{Equation 3.57}$$

Where,

T = Period of the fundamental mode of vibration for the horizontal direction considered

M_i = Mass concentrated at the i^{th} point

d_i = Displacement of i^{th} nodal point in an approximation of the shape of the first mode

$S_d(T)$ = Design spectral acceleration

g = Acceleration of gravity

3.11.6.1.5 Torsional Effects

When the seismic effects are determined in transverse direction using Rigid or the Flexible Deck Model, torsional effects in the transverse direction i.e. rotation about the vertical axis may be estimated by applying a static torsional moment M_t :

$$M_t = \pm F e \quad \text{Equation 3.58}$$

Where,

F = Horizontal force determined for Rigid Deck Model (F) or the Flexible Deck Model ($\sum F_i$) which is under consideration

e = $e_0 + e_a$

e_0 = Theoretical eccentricity

e_a = Additional eccentricity to account for accidental and dynamic amplification effects
= 0.05L

The torsional moment M_t may be distributed to the supporting members using Rigid Deck Model.

3.11.6.1.1 Individual Pier Model

When seismic action in the transverse direction of the bridge is resisted mainly by the piers, without significant interaction between adjacent piers, the seismic effects in the i^{th} pier may be determined by applying on it an equivalent static force given as:

$$F = M_i S_d(T_i) \quad \text{Equation 3.59}$$

1

Where,

M_i = Effective mass attributed to i^{th} pier.

$S_d(T_i)$ = Design spectral acceleration corresponding to the fundamental time period T_i for the i^{th} pier considered independently.

2

$$T_i = 2\pi \sqrt{\frac{M_i}{K_i}}$$

3

K_i = Stiffness of the i^{th} pier.

This simplified approach may be applied as a first approximation for preliminary analyses, if the fundamental period of all adjoining piers meet the following condition.

$$0.90 \leq \frac{T_i}{T_{i+1}} \leq 1.10$$

If above condition is not met, a redistribution of the effective masses attributed to each pier is required, leading to the satisfaction of the above condition.

3.11.6.2 Response Spectrum Method – Linear Dynamic Analysis

3.11.6.2.1 General

The Response Spectrum Analysis is an elastic calculation of the peak dynamic response of all significant modes of the structure using site dependent design spectrum. The overall response shall be obtained by statistical combination of the maximum model contributions. This type of analysis may be applied in cases where linear analysis is permitted. The earthquake action effects shall be determined from an appropriate discrete linear model (Full Dynamic Model).

3.11.6.2.2 Significant Modes

All modes making significant contribution to the total structural response shall be taken into account. For bridges in which the total mass M can be considered as a sum of "effective modal masses" M_i , this criterion is deemed to be satisfied if the sum of the effective modal masses for the modes considered, $(\sum M_i)_c$, amounts to at least 90% of the total mass of the bridge.

3.11.6.2.3 Combination of Model Responses

The bridge may be analysed separately considering different modes of vibration. To have maximum value of the seismic action effect, the model responses of the different modes shall be combined either by SRSS (square root of the sum of the squares) rule.

$$E_E = \sqrt{\sum E_{Ei}^2}$$

Equation 3.60

Where,

E_E = Seismic action effect under consideration (force, displacement, etc.).

E_{Ei} = Value of the seismic action effect due to vibration mode i .

3.11.6.2.4 Combination of Components of the Seismic Action

The three translational components of the seismic action need to be taken into account for the design of bridges. The bridge may be analysed separately for the three components of the seismic action in the longitudinal, transverse and vertical directions. The probable maximum seismic action effects E_{Ed} due to simultaneous occurrence of the three components of seismic action along axes X, Y and Z should be derived through the application of SRSS rule to the maximum action effects E_x , E_y and E_z due to independent seismic actions along each axis:

$$E_{Ed} = \sqrt{E_x^2 + E_y^2 + E_z^2}$$

Equation 3.61

The probable maximum action effect E may be taken as the most adverse of the effects calculated from following Equations.

$$E_{Edx} + " 0.30 E_{Edy} + "0.30E_{Edz}$$

Equation 3.62

$$0.30E_{Edx} + E_{Edy} + "0.30E_{Edz}$$

Equation 3.63

$$0.30E_{Edx} + "0.30 E_{Edy} + "E_{Edz}$$

Equation 3.64

E_{Edx} , E_{Edy} and E_{Edz} are the action effects due to the application of design seismic actions in each direction X (longitudinal), Y (transverse) and Z (vertical) respectively and "+" implies "to be combined with".

3.11.7 Non-linear Methods of Analysis

The non-linear methods are intended to be used for bridges with complex and irregular behaviour. These methods require special studies and are beyond the scope of this Manual. Outline of the Time-History and Pushover Analysis are given for general guidance.

3.11.7.1 Non-linear Dynamic Time-history Analysis

The time dependent response of the structure shall be obtained through direct numerical integration of its non-linear differential equations of motion. The seismic input shall consist of ground motion time-histories. The effect of gravity loads and of the other quasi-permanent actions in the seismic design situation, as well as second order effects, shall be taken into account.

In general, this method is used in combination with a normal response spectrum analysis to provide insight into the post - elastic response and comparison between required and available local ductility. Generally, the results of the non-linear analysis are not intended to be used to relax requirements resulting from the response spectrum analysis. However, in the case of bridges with isolating devices and irregular bridges, lower results from a rigorous time-history analysis may be substituted for the results of the response spectrum analysis.

When a non-linear time-history analysis is carried-out, at least three pairs of horizontal ground motion time-history components shall be used. The pairs should be selected from recorded events with magnitudes, source distances, and mechanisms consistent with those that define the design seismic action. The components of each pair of time histories shall be applied simultaneously.

When non-linear dynamic analysis is performed for at least seven independent pairs of horizontal ground motions, the average of the individual responses may be used as the design value of the action effects. When less than seven non-linear dynamic analyses are performed for the corresponding independent pairs of input motions, the maximum responses of the ensemble should be used as design action effects.

3.11.7.1 Non-linear Static Pushover Analysis

Pushover analysis is a static non-linear analysis of the structure under constant vertical (gravity) loads and monotonically increased horizontal loads, representing the effect of a horizontal seismic component. Second order effects should be accounted for. The horizontal loads are increased until the target displacement is reached at the reference point.

The analysis should be carried out in the two horizontal directions (the longitudinal and the transverse direction). The target displacement is the maximum of the displacements in the relevant direction, at the centre of mass of the deformed deck, resulting from equivalent linear multi-mode spectrum analysis, assuming $q = 1.0$, for the following combinations of seismic components:

$$E_x \text{ "+" } 0.3E_y \text{ and } E_y \text{ "+" } 0.3E_x.$$

The spectrum analysis should be carried out using effective stiffness of ductile members. The reference point should be the centre of mass of the deformed deck.

3.12 Construction Loads

Construction loads are those which are incident upon a structure or its components during construction stage of the structures. The effects of construction/ erection as per actual loads based on the construction programme shall be considered in the design. This shall also include the condition of one span being completed in all respects and the adjacent span not in position, which causes eccentric loading.

A detailed construction procedure associated with a method statement shall be drawn up during design and the corresponding construction loads shall be considered in the design to ensure stability and strength of the structure.

Construction loads (Q_c) may be represented as one single variable action, or where appropriate different types of construction loads may be grouped and applied as a single variable action. Single and/or a grouping of construction loads should be considered to act simultaneously with non-construction loads as appropriate. Characteristic values of construction loads, including vertical and horizontal components where relevant, shall be determined according to the technical requirements for the execution of the works. The construction loads shall be considered as variable actions.

The construction loads which should be accounted in the design are given below.

1. **Personnel and Hand Tools (Q_{ca}):** This includes the load of working personnel, staff and visitors including with hand tools or other site equipment. This load can be represented as a uniformly distributed load q_{ca} . The characteristic value of UDL, may be taken as 1.0 kN/m².
2. **Storage of moveable items (Q_{cb}):** This includes the load of construction materials, pre-cast elements, equipment, etc. or other site equipment. This load can be represented as a uniformly distributed load q_{cb} and a concentrated load F_{cb} and shall be defined for individual project. The characteristic value of q_{cb} and F_{cb} may be taken 0.2 kN/m² and 100 kN respectively as a minimum value.
3. **Non-Permanent Equipment (Q_{cc}):** This includes the non-permanent equipment for use during execution such as formwork panels, scaffolding, falsework, machinery, travelling forms, launching girders and nose, counterweights. The load can be represented as a uniformly distributed load q_{cc} . The load may be defined for the individual project based on the information provided by equipment supplier. However, a minimum of 0.5 kN/m² may be taken as characteristic value of q_{cc} .
4. **Moveable heavy machinery and Equipment (Q_{cd}):** This includes the load due to moveable heavy machinery and equipment e.g. cranes, lifts, vehicles, trucks, lifting devices, etc. The load, designated as Q_{cd} , may be defined for the individual project depending upon the type of equipment used based on the information provided by equipment supplier. The action due to vehicles may be taken as defined in the Section for Traffic Loads.
5. **Accumulation of waste material (Q_{ce}):** This includes the load due to accumulation of waste material such as demolition material, surplus construction material, soil, etc. The load, designated as Q_{ce} , can be taken into account by considering the possible weight effect of accumulated material. These loads may vary significantly depending upon the type of material and the quantity of accumulation.
6. **Loads from parts of a structure in a temporary state (Q_{cf}):** This includes the load from parts of a structure in temporary state under construction such as loads from lifting operations, weight of fresh concrete. The load, designated as Q_{cf} , can be taken into account by considering the load effect of planned construction sequence.

The construction loads can be at fixed location on the structure, in that case, the tolerances for possible deviation from the theoretical position shall be considered in the designs. When construction loads are free to move, then the limits of the area of their movement shall be defined and to consider in designs.

Simultaneity of loads during casting of concrete shall be considered as appropriate. This may include loads due to working personnel with small site equipment (Q_{ca}), formwork and load bearing members (Q_{cc}) and weight of fresh concrete (a component of Q_{cf}). For the purpose of assessment of actions due to construction loads during casting of concrete, the actual area is divided into two parts, the working area of size 3 m x 3 m or 3 m x span length whichever is less and the remaining area (outside working area) in such a way to maximise the load effects. The characteristic value of construction loads during casting of concrete may be taken as under:

- **On actual area:** Self weight of the formwork, load bearing elements (Q_{cc}) and weight of fresh concrete (Q_{cf}).
- **On working area:** 10% of the self-weight of the concrete but not less than 0.75 kN/m² and not more than 1.5 kN/m². Includes Q_{ca} and Q_{cf} .
- **Outside working area:** 0.75 kN/m² covering Q_{ca} .

Transient, accidental and seismic design situations shall be identified in accordance with the execution process and conditions from stage to stage during execution anticipated in design and taken into account as appropriate for designs for execution.

The duration of different stages of execution is much shorter than the design return period of the variable actions (e.g. climatic conditions) adopted for in-service design. The recommended value of return period for determining the characteristic value of the climatic actions during execution stage may be taken as under.

Duration of Execution Stage	Return period
≤ 3 months	5 years
≤ 1 year, but >3 months	10 years
> 1 year	50 years

The basic value of the wind velocity during execution for stages up to 3 months duration may be taken as 20 m/s. The various processes such as movement of cranes, launching/ pushing of girders, etc. shall not be carried out if the wind velocities exceed this limit.

For the incremental launching of bridge superstructure, the horizontal friction forces shall be determined and should be applied between bridge superstructure, the bearings and the supporting structure with dynamic action effects where appropriate. The total horizontal friction forces should not be less than 10% of the vertical loads. The design value of horizontal friction forces at every pier shall be determined with appropriate friction coefficients μ_{min} as 0 and μ_{max} as 0.04.

3.13 Combination of Actions and Partial Factors

3.13.1 General

A structure shall be designed such that it will sustain all actions and influences likely to occur during execution and use and meet the specified serviceability requirements for a structure or a structural element during its intended life. The verification shall be carried out under Ultimate Limit State (ULS) and Serviceability Limit state (SLS) for all relevant design situations and load cases using the partial factor method. Design situations are circumstances / sets of physical conditions that the structure might experience during its life. The design situations may be classified as:

- **Persistent design situations:** Which refer to the conditions of normal use.
- **Transient design situations:** Which refer to temporary conditions applicable to the structure, e.g. during execution or maintenance.

1

- **Accidental design situations:** Which refer to exceptional conditions applicable to the structure or to its exposure involving accidental action, e.g. vehicle impact.
- **Seismic design situations:** Which refer to conditions applicable to the structure when subjected to seismic events.

2

The structure or a structural element may be subjected to following actions.

3

- **Permanent actions (denoted G),** e.g. self-weight of structures, road surfacing and indirect actions such as uneven settlements.
- **Variable actions (denoted Q),** e.g. traffic load, wind and thermal actions.
- **Accidental actions (denoted A),** e.g. impact from vehicles.

The characteristic value of a permanent action is its main representative value and is denoted by G_k . In case when limit state is very sensitive to variation in magnitude of permanent actions, the upper ($G_{k\ sup}$) and lower characteristic value ($G_{k\ inf}$) of these actions should be taken into account.

Variable Action has four different representative values. They are called characteristic, combination, frequent and quasi-permanent value. The characteristic value is a statistically extreme value and is the main representative value. The other representative values are determined by multiplying the characteristic value by ψ_0 , ψ_1 and ψ_2 respectively. The factors ψ_0 , ψ_1 and ψ_2 are always less than 1. For design of road bridges, recommended ψ -factors are given Table 3.29 and for footbridges in Table 3.30.

For multi-component actions the characteristic action should be represented by groups of values each to be considered separately in design calculations.

Table 3.29 Value of ψ -factors for Road Bridges

Action	Group	Load Description	ψ_0	ψ_1	ψ_2
Traffic loads	gr1a	LM1 Tandem System	0.75	0.75	0
		LM1 UDL System	0.75	0.75	0
		Walkway & cycle-track loads	0.40	0.40	0
	gr1b	LM2 Single axle	0	0.75	0
	gr2	Horizontal forces	0	0	0
	gr3	Pedestrian loads	0	0.40	0
	gr4	LM4 Crowd loading	0	-	0
	gr5	LM3 - Vertical forces from SV & SOV vehicles	0	-	0
Wind forces	F_{wk}	Persistent design situations	0.5	0.2	0
		During Execution	0.8	-	0
	F_w^*	During Execution	1.0	-	0
Thermal actions	T_k		0.6	0.6	0.5
Construction loads	Q_c		1.0	-	1.0

Notes:

1. The values of ψ_0 , ψ_1 and ψ_2 for gr1a and gr1b are for roads with traffic corresponding to α_{Qi} , α_{qi} , α_{qr} and β_Q defined in this Manual.
2. The value of the pedestrian and cycle-track load group gr1a is already a 'reduced' value accompanying the characteristic value of LM1 and should not be factored again by ψ_1 . However, when gr1a is combined with leading non-traffic actions, this value should be factored by ψ_0 .
3. The frequent values of load groups gr4 and gr5 do not need to be considered.
4. The ψ_0 factors specified for a group of loads apply to all the component actions in that group, except for gr1a where they are individually specified. The ψ_1 and ψ_2 factors always apply to individual components of loading and the values for a given component are the same in all load groups in which the component occurs.
5. Where relevant, representative values of water actions F_{wa} should be determined for the individual project.
6. The combination of special vehicles (SV and SOV) with Load Model 1 shall be as given in Clause 3.3.3.3.4.

Table 3.30 Value of ψ -factors for Foot Bridges

Action	Load Group Description	ψ_0	ψ_1	ψ_2
Traffic Loads	gr1	0.40	0.40	0
	Q_{fwh}	0	0	0
	gr2	0	0	0
Wind forces	F_{wk}	0.3	0.2	0
Thermal actions	T_k	0.6	0.6	0.5
Construction loads	Q_c	1.0	-	1.0
Note: The recommended ψ_0 value for thermal actions may in most cases be reduced to 0 for ultimate limit states verification.				

3.13.2 Limit State Design for Bridges

The partial factor method is applied in limit state verifications. In this method it is verified that, in all relevant design situations, no relevant limit state is exceeded when design values for actions and resistances are used in the design models. The design values of action effects are determined accounting for uncertainties in the actions themselves and also uncertainties in the evaluation of effects of actions. This is done by using two partial factors in determining action effects (with one applied to the action and the other to the effect of the action). These factors are:

- γ_f partial factor for the action which takes account of the possibility of unfavourable deviations of the action values from the representative values
- γ_{sd} partial (model) factor taking account of uncertainties in modelling the effects of actions

The partial factor is generally simplified by combining γ_f and γ_{sd} into a single partial factor denoted γ_F (or more specifically γ_Q for variable actions and γ_G for permanent actions). Where a distinction has to be made between favourable and unfavourable permanent actions, two different partial factors shall be used ($\gamma_{G,inf}$ and $\gamma_{G,sup}$).

Design value of action: As a simplification, the design value F_d of an action F can be represented as:

$$F_d = \gamma_F F_{rep} \text{ and } F_{rep} = \psi F_k$$

Where,

- F_k = Characteristic value of the action.
- F_{rep} = Relevant representative value of the action.
- γ_F = A single partial factor combining γ_f and γ_{sd} .
- ψ = A factor, may be 1.0, ψ_0 , ψ_1 , or ψ_2 .

Design value of the effects of actions: As a simplification, the design values of the effects of actions (E_d) can be represented as:

$$E_d = E \{ \gamma_{F,i} F_{rep,i} ; a_d \}, i \geq 1$$

Where,

- $\gamma_{F,i}$ = A single partial factor combining γ_f and γ_{sd} .
- = $\gamma_{sd} \gamma_{fi}$.
- a_d = The design value of the geometrical data.

Design value of material or product properties: As a simplification, the design value X_d of a material or product property can be expressed as:

$$X_d = X_k / \gamma_M$$

1

Design value of geometrical data: The design value of geometrical data such as dimensions of members that are used to determine the action effects and/or resistances, may be represented by nominal values:

$$a_d = a_{nom}$$

2

Where, the effects of deviations in geometrical data are significant, the design values of geometrical data shall be defined as:

3

$$a_d = a_{nom} \pm \Delta a$$

Where, Δa takes account of possibility of unfavourable deviations from the characteristic or nominal value.

Design resistance: As a simplification, the design resistance can be represented as:

$$R_d = R \{ X_{k,i} / \gamma_{M,i} ; a_d \}, i \geq 1$$

Or, alternatively, may be obtained directly from characteristic value of a material or product resistance

$$R_d = R_k / \gamma_M$$

3.13.2.1 Ultimate Limit State

Under ultimate limit state, loads are to be combined to check the equilibrium and the structural strength. The equilibrium of the structure shall be checked against overturning, sliding and uplift. It shall be ensured that the destabilising loads (overturning, sliding and uplifting) shall always be less than the stabilising or restoring actions. The structural strength under ultimate limit state shall be estimated in order to avoid internal failure or excessive deformation. Geotechnical verification shall be carried out for excessive deformation of the structure where strength of the ground is significant in providing resistance.

Following verifications shall be done for ultimate limit states:

Verification of Static Equilibrium (EQU): For static equilibrium of the structure, it shall be verified that:

$$\begin{array}{ccc} \text{Design value of effect of destabilising actions} & \leq & \text{Design value of effect of stabilising actions} \\ (E_{d,dst}) & & (E_{d,stb}) \end{array}$$

Verification of Resistance (STR and or GEO): For a limit state of rupture or excessive deformation of a section or member or connection, it shall be verified that:

$$\begin{array}{ccc} \text{Design value of the effect of actions } (E_d) & \leq & \text{Design value of the corresponding resistance} \\ \text{e.g. internal forces, moments} & & (R_d) \end{array}$$

3.13.2.2 Serviceability Limit States

Under service limit state, loads are to be combined to check the serviceability requirements. The serviceability limit state verification shall be carried out in order to have control on deformations (affecting appearance, comfort or functioning of structure), vibrations (causing discomfort or limiting effectiveness of the structure) and damage (adversely affecting appearance, durability or function). It shall be ensured that the design value obtained by using the appropriate combination shall be less than the limiting value of serviceability criterion as per the relevant design standards.

Following verification shall be made for Serviceability Limit State.

$$\begin{array}{ccc} \text{Design value of the effects of actions} & \leq & \text{Limiting design value of the relevant} \\ \text{specified in the serviceability criterion,} & & \text{serviceability criterion } (C_d) \\ \text{determined on the basis of the relevant} & & \\ \text{combination } (E_d) & & \end{array}$$

3.13.3 Combination of Actions

The design values of the effects of actions shall be determined by combining the values of actions that are considered to occur simultaneously. When verifications are sensitive to variation of the magnitude of a permanent action from place to place in the structure, the unfavourable and favourable parts of this action shall be considered as individual action.

While working the combinations, only one variable action shall be considered as the leading action at a time. All other variable actions shall be considered as accompanying actions. In case if the variable actions produce favourable effect (relieving effect) the same shall be ignored.

Effects of actions that cannot occur simultaneously due to physical or functional reasons need not be considered together in combinations of actions.

Traffic Load Model 2 and the concentrated load on walkways need not be combined with any other variable non traffic action.

The wind actions need be combined with braking and acceleration forces or the centrifugal forces, loads on walkways and cycle tracks.

Wind actions and thermal actions need not be taken into account simultaneously.

3.13.3.1 Combination for Ultimate Limit States

Three combinations of actions are used for ultimate limit state verifications: one is used for persistent and transient design situations, one for accidental design situations and one for seismic design situations.

- Combination of the effects of actions for persistent or transient design situations (fundamental combination) may be represented as:

$$\Sigma \gamma_G G_k \text{ "+" } \gamma_P P \text{ "+" } \gamma_{Ql} Q_{kl} \text{ "+" } \Sigma \gamma_Q \psi_0 Q_k$$

Equation 3.65

- Combination of the effects of actions for accidental design situations may be represented as:

$$\Sigma G_k \text{ "+" } P \text{ "+" } A_d \text{ "+" } \psi_{1l} Q_{kl} \text{ "+" } \Sigma \psi_2 Q_k$$

Equation 3.66

- Combination of the effects of actions for seismic design situations may be represented as:

$$\Sigma G_k \text{ "+" } P \text{ "+" } A_{Ed} \text{ "+" } \Sigma \psi_2 Q_k$$

Equation 3.67

Where,

"+" = Implies to be combined with.

G_k = Characteristic value of permanent action.

P = Representative value of prestressing action.

Q_{kl} = Characteristic value of leading variable action.

Q_k = Characteristic value of accompanying variable actions.

A_d = Design value of an accidental action.

A_{Ed} = Design value of a seismic action.

γ_G = Partial factor for permanent action.

γ_{Ql} = Partial factor for permanent action.

γ_Q = Partial factor for accompanying variable action.

ψ_0 = Factor for combination value of a variable action.

ψ_{1l} = Factor for frequent value of a leading variable action.

ψ_2 = Factor for quasi-permanent value of a variable action

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3

Actions and Combinations of Actions

In case when limit state is very sensitive to variation in magnitude of permanent actions, the upper ($G_{k\ sup}$) and lower characteristic value ($G_{k\ inf}$) shall be used in above Equations and accordingly, the associated partial factor as $\gamma_{G\ sup}$ and $\gamma_{G\ inf}$ with upper and lower values. The characteristic values of all permanent actions from one source are multiplied by $\gamma_{G\ sup}$ if the total resulting action effect is unfavourable and $\gamma_{G\ inf}$ if the total resulting action effect is favourable.

3.13.3.1.1 Design Value of Actions in Persistent and Transient Design Situations

The design value of actions for the persistent and transient design situations are given in Table 3.31, Table 3.32 and Table 3.33. When the limit state is very sensitive to variations in the magnitude of permanent actions, the upper and lower characteristic values of these actions should be taken. For verification of static equilibrium (EQU) of bridges, use design value of actions as given in Table 3.31. For verification of resistance of structural members (STR) not involving geotechnical actions, use design values of actions as given in Table 3.32. For verification of structural members e.g. footings, abutments, wing walls, etc. (STR) involving geotechnical actions and resistance of the ground (GEO), use separate calculation design values from Table 3.33 and Table 3.32 to the geotechnical actions as well as actions from the structure.

Table 3.31 Design Value of Actions for Verification of Equilibrium (EQU) (Set A)

Design Situation	Permanent Actions		Prestress	Leading Variable Actions	Accompanying variable actions	
	Unfavourable	Favourable			Main (if any)	Others
Persistent and transient design situation Equation 3.65	$\gamma_{G\ sup} G_{k\ sup}$	$\gamma_{G\ inf} G_{k\ inf}$	$\gamma_p P$	$\gamma_{Ql} Q_{kl}$		$\gamma_Q \psi_0 Q_k$
Notes: 1. For persistent design situations, following values of partial factor (γ) may be taken: Permanent actions (following components shall be combined as appropriate) Concrete self-weight $\gamma_{G\ sup} = 1.05$ $\gamma_{G\ inf} = 0.95$ Steel self-weight $\gamma_{G\ sup} = 1.05$ $\gamma_{G\ inf} = 0.95$ Superimposed dead weight $\gamma_{G\ sup} = 1.05$ $\gamma_{G\ inf} = 0.95$ Road surfacing $\gamma_{G\ sup} = 1.05$ $\gamma_{G\ inf} = 0.95$ Weight of soil $\gamma_{G\ sup} = 1.05$ $\gamma_{G\ inf} = 0.95$ Self-weight of other contributing materials $\gamma_{G\ sup} = 1.05$ $\gamma_{G\ inf} = 0.95$ Prestressing γ_p as defined in the relevant Chapter or for the individual project Variable actions Road traffic actions: gr1a, gr1b, gr2, gr5, gr6 $\gamma_Q = 1.35$ = 0 where favourable Pedestrian actions: gr3, gr4 $\gamma_Q = 1.35$ = 0 where favourable Wind actions $\gamma_Q = 1.70$ = 0 where favourable Thermal actions $\gamma_Q = 1.55$ = 0 where favourable 2. For transient design situations, during which there is a risk of loss of static equilibrium, Q_{kl} represents dominant (leading) destabilising variable action and Q_k represents relevant accompanying destabilising variable actions. 3. The characteristic values of all unfavourable permanent actions are multiplied by $\gamma_{G\ sup}$ and the characteristic values of all favourable permanent actions are multiplied by $\gamma_{G\ inf}$ irrespective of whether they arise from a single source. 4. During execution, if the construction process is properly controlled, the values of partial factors for persistent design situation given above may be used with the exceptions set out as below: a. Where a counterweight is used, the variability of its characteristics may be taken into account. for example, by one or both of the following rules: • applying a partial factor $\gamma_{G\ inf}$ as 0.8 where the self-weight is not well defined (e.g. containers). • by considering a variation in defined location, with a value to be specified proportionately to the dimensions of the bridge, where the magnitude of the counterweight is well defined. For steel bridges during launching, the variation of the counterweight location is often taken equal to ± 1 m. b. Where loss of equilibrium could result in multiple fatalities (for example bridges constructed over railways or motorways), partial factors for permanent actions affecting stability ($\gamma_{G\ sup}$ and $\gamma_{G\ inf}$), should be increased to 1.15 and decreased to 0.85 respectively. 5. The design value of earth pressures should be based on the design value of the actions giving rise to the earth pressure. In some cases, model factors may be required when evaluating horizontal earth pressures as given in Chapter 11. 6. For self-weight of water, ground-water pressure and other actions dependent on the level of water, no partial factor is specified. The design value of such actions may be directly assessed considering most unfavourable values that could occur during design lifetime of the structure. Partial factors such actions may be determined for individual project.						

Table 3.32 Design Value of Actions for Verification of Equilibrium (STR/EQU) (Set B)

Design Situation	Permanent Actions		Prestress	Leading Variable Actions	Accompanying variable actions	
	Unfavourable	Favourable			Main (if any)	Others
Persistent and transient design situation Equation 3.65	$\gamma_{G\ sup} G_{k\ sup}$	$\gamma_{G\ inf} G_{k\ inf}$	$\gamma_p P$	$\gamma_{Ql} Q_{kl}$		$\gamma_Q \psi_0 Q_k$

Notes:

1. For persistent design situations, following values of partial factor (γ) may be taken:

Permanent actions (following components shall be combined as appropriate)

Concrete self-weight $\gamma_{G\ sup} = 1.35$ $\gamma_{G\ inf} = 0.95$

Steel self-weight $\gamma_{G\ sup} = 1.20$ $\gamma_{G\ inf} = 0.95$

Superimposed dead weight $\gamma_{G\ sup} = 1.20$ $\gamma_{G\ inf} = 0.95$

Road surfacing $\gamma_{G\ sup} = 1.20$ $\gamma_{G\ inf} = 0.95$

Weight of soil $\gamma_{G\ sup} = 1.35$ $\gamma_{G\ inf} = 0.95$

Self-weight of other contributing materials $\gamma_{G\ sup} = 1.35$ $\gamma_{G\ inf} = 0.95$

Settlement (linear structural analysis) $\gamma_{G\ set\ sup} = 1.20$ $\gamma_{G\ set\ inf} = 0$

Settlement (nonlinear structural analysis) $\gamma_{G\ set\ sup} = 1.35$ $\gamma_{G\ set\ inf} = 0$

Prestressing γ_p as defined in the relevant Chapter or for the individual project

Variable actions

Road traffic actions

gr1a, gr1b, gr2, gr5, gr6 $\gamma_Q = 1.35$ = 0 where favourable

Pedestrian actions

gr3, gr4 $\gamma_Q = 1.35$ = 0 where favourable

Variable horizontal earth pressure from soil, ground water, free water, traffic load

surcharge earth pressure $\gamma_Q = 1.50$ = 0 where favourable

Wind actions $\gamma_Q = 1.70$ = 0 where favourable

Thermal actions $\gamma_Q = 1.55$ = 0 where favourable

2. The characteristic values of all permanent actions from one source are multiplied by $\gamma_{G\ sup}$ if total resulting action effect from this source is unfavourable and by $\gamma_{G\ inf}$ if the total resulting action effect from this source is favourable.

3. The design value of earth pressures should be based on the design value of the actions giving rise to the earth pressure. In some cases, model factors may be required when evaluating horizontal earth pressures as given in Chapter 11.

4. For self-weight of water, ground-water pressure and other actions dependent on the level of water, no partial factor is specified. The design value of such actions may be directly assessed considering most unfavourable values that could occur during design lifetime of the structure. Partial factors such actions may be determined for individual project.

Table 3.33 Design Value of Actions for Verification of Equilibrium (STR/EQU) (Set C)

Design Situation	Permanent Actions		Prestress	Leading Variable Actions	Accompanying variable actions	
	Unfavourable	Favourable			Main (if any)	Others
Persistent and transient design situation Equation 3.65	$\gamma_{G\ sup} G_{k\ sup}$	$\gamma_{G\ inf} G_{k\ inf}$	$\gamma_p P$	$\gamma_{Ql} Q_{kl}$		$\gamma_Q \psi_0 Q_k$

Notes:

1. For persistent design situations, following values of partial factor (γ) may be taken:

Permanent actions (following components shall be combined as appropriate)

Concrete self-weight	$\gamma_{G\ sup}$	= 1.00	$\gamma_{G\ inf}$	= 1.0
Steel self-weight	$\gamma_{G\ sup}$	= 1.00	$\gamma_{G\ inf}$	= 1.0
Superimposed dead weight	$\gamma_{G\ sup}$	= 1.00	$\gamma_{G\ inf}$	= 1.0
Road surfacing	$\gamma_{G\ sup}$	= 1.00	$\gamma_{G\ inf}$	= 1.0
Weight of soil	$\gamma_{G\ sup}$	= 1.00	$\gamma_{G\ inf}$	= 1.0
Self-weight of other contributing materials	$\gamma_{G\ sup}$	= 1.00	$\gamma_{G\ inf}$	= 1.0
Settlement (linear structural analysis)	$\gamma_{G\ set\ sup}$	= 1.00	$\gamma_{G\ set\ inf}$	= 0
Settlement (nonlinear structural analysis)	$\gamma_{G\ set\ sup}$	= 1.00	$\gamma_{G\ set\ inf}$	= 0

Prestressing
 γ_p as defined in the relevant Chapter or for the individual project

Variable actions

Road traffic actions				
gr1a, gr1b, gr2, gr5, gr6	γ_Q	= 1.15	= 0 where favourable	
Pedestrian actions				
gr3, gr4	γ_Q	= 1.15	= 0 where favourable	
Variable horizontal earth pressure from soil, ground water, free water, traffic load				
surcharge earth pressure	γ_Q	= 1.30	= 0 where favourable	
Wind actions	γ_Q	= 1.45	= 0 where favourable	
Thermal actions	γ_Q	= 1.30	= 0 where favourable	

2. The characteristic values of all permanent actions from one source are multiplied by $\gamma_{G\ sup}$ if total resulting action effect from this source is unfavourable and by $\gamma_{G\ inf}$ if the total resulting action effect from this source is favourable.

3. The design value of earth pressures should be based on the design value of the actions giving rise to the earth pressure. In some cases, model factors may be required when evaluating horizontal earth pressures as given in Chapter 11.

4. For self-weight of water, ground-water pressure and other actions dependent on the level of water, no partial factor is specified. The design value of such actions may be directly assessed considering most unfavourable values that could occur during design lifetime of the structure. Partial factors such actions may be determined for individual project.

3.13.3.1.2 Design Value of Actions in Accidental and Seismic Design Situations

The design value for actions for the ultimate limit states in the accidental and seismic design situations are given in Table 3.34.

For execution phase, during which there is a risk of loss of static equilibrium, the combination of actions shall be as follows:

$$\Sigma G_k \text{ "+" } P \text{ "+" } A_d \text{ "+" } \psi_2 Q_{ck}$$

Equation 3.68

Where Q_{ck} is the characteristic value of construction loads (i.e. the characteristic value of the relevant combination of groups Q_{ca} , Q_{cb} , Q_{cc} , Q_{cd} , Q_{ce} and Q_{cf}).

Table 3.34 Design Value of Actions for Accidental and Seismic Combinations

Ultimate Limit State						
Design Situation	Permanent Actions		Prestress	Accidental or Seismic Action	Accompanying variable actions	
	Unfavourable	Favourable			Main	Others
Accidental Equation 3.66	$G_{k\ sup}$	$G_{k\ inf}$	P	A_d	$\psi_{1l}\ Q_{k,l}$	$\psi_2\ Q_k$
Seismic Equation 3.67	$G_{k\ sup}$	$G_{k\ inf}$	P	$A_{ED} = \gamma_l A_{Ek}$	$\psi_2\ Q_k$	
Notes: 1. All the partial factors γ shall be taken as 1.0 using design values in above table for all non-seismic actions. 2. The values of ψ_1 and ψ_2 factors shall be taken from Table 3.29 or Table 3.30 as appropriate. 3. In the case of accidental design situations, the main variable action may be taken with its frequent value with the combination factor ψ_1 . 4. The seismic design situation should be used when specified for the individual project.						

3.13.3.2 Combination for Serviceability Limit State

Three combinations of actions are used for serviceability limit state verifications. These are called the characteristic combination, the frequent combination and the quasi-permanent combination.

- Characteristic combination of the effects of actions may be represented as:

$$\Sigma G_k \text{ "+" } P \text{ "+" } Q_{kl} \text{ "+" } \Sigma \psi_0 Q_k \quad \text{Equation 3.69}$$

- Frequent combination of the effects of actions may be represented as:

$$\Sigma G_k \text{ "+" } P \text{ "+" } \psi_{1l} Q_{kl} \text{ "+" } \Sigma \psi_2 Q_k \quad \text{Equation 3.70}$$

- Quasi-permanent combination of the effects of actions may be represented as:

$$\Sigma G_k \text{ "+" } P \text{ "+" } \psi_{2l} Q_{kl} \text{ "+" } \Sigma \psi_2 Q_k \quad \text{Equation 3.71}$$

Table 3.35 Design Value of Actions for Combination

Design Situation	Permanent Actions		Prestress	Accompanying variable actions	
	Unfavourable	Favourable		Main	Others
Characteristic	$G_{k\ sup}$	$G_{k\ inf}$	P	Q_{kl}	$\psi_0 Q_k$
Frequent	$G_{k\ sup}$	$G_{k\ inf}$	P	$\psi_{1l} Q_{kl}$	$\psi_1 Q_k$
Quasi-permanent	$G_{k\ sup}$	$G_{k\ inf}$	P	$\psi_{2l} Q_{kl}$	$\psi_2 Q_k$

All the partial factors shall be taken as 1.0 and the design values of actions in above table shall be used for verification of serviceability limit state. The value of ψ factors shall be as given in Table 3.29.

3.13.4 Serviceability Regarding Deflection of Bridges

Serviceability criteria for concrete and steel bridges are given in Chapter 4 and Chapter 5 respectively. Appropriate limiting values of deflection taking into account the nature of the structure, of the finishes, partitions and fixings and upon the function of the structure should be established for individual project.

Note: In the absence of other criteria, following deflection limits may be considered for concrete and steel bridges.

Vehicular load, general	Span/800
Vehicular and pedestrian loads	Span/1000
Vehicular load on cantilever arms	Span/300
Vehicular and pedestrian load on cantilever arms	Span/375

1

3.13.5 Verifications Concerning Vibration of Footbridges due to Pedestrian Traffic

2

3.13.5.1 Design Situation

The design situation should be selected depending on the pedestrian traffic to be admitted on the individual footbridge during its design working life. For common pedestrian traffic classes and densities for footbridges, refer Clause 9.8.7.2.

3

Depending on the deck area or the part of the deck area under consideration, the traffic categories, associated with design situations which may be persistent, transient or accidental, should be specified when relevant, including:

- The presence of streams of pedestrians.
- Occasional festive events.

The traffic categories and the relevant design situations may have to be agreed for the individual project.

3.13.5.2 Pedestrian Comfort Criteria

Pedestrian comforts are commonly represented as a limiting acceleration for the footbridge. The recommended comfort class are given in Clause 9.8.7.3. The comfort class should be defined as appropriate for the individual project.

A verification of the comfort criteria should be performed if the fundamental frequency of the deck is less than:

- 5 Hz for vertical vibrations.
- 2.5 Hz for horizontal (lateral) and torsional vibrations.

4 Concrete Structures

4.1 General

4.1.1 Scope

This Chapter deals with the design of concrete road bridges and culverts. It covers design principles, detailed design criteria and material properties for reinforced concrete and prestressed concrete bridges and culverts. It provides general basis for the design of bridges and culverts in plain, reinforced and pre-stressed concrete made with normal weight aggregates.

A structure shall be designed and executed in such a way that it will sustain all actions and influences likely to occur during execution and use and meet the specified serviceability requirements for a structure or a structural element during its intended life, with appropriate degrees of reliability and in an economical way.

4.1.2 References Documents

Following documents may be referred while using this Manual:

EN 197-1: Cement: Composition, specification and conformity criteria for common cements

EN 206-1: Concrete: Specification, performance, production and conformity

EN 12390: Testing hardened concrete

EN 10080: Steel for the reinforcement of concrete

EN 10138: Prestressing steels

EN ISO 17660: Welding- Welding of reinforcing steel

EN 1536: Execution of special geotechnical work- Bored Piles

EN ISO 15630: Steel for the reinforcement and prestressing of concrete

KS EAS 412: Steel for reinforcement of concrete

KS ISO 6934: Steel for prestressing of concrete

4.1.3 Definitions

For general definitions, refer initial sections of this Manual. Other definitions are as under.

Precast structures	Precast structures are characterised by structural elements manufactured elsewhere than in the final position in the structure.
Unbonded and external tendons	Unbonded tendons for post-tensioned members having ducts which are permanently un-grouted, and tendons external to the concrete cross-section.

4.1.4 Symbols

The general symbols used in this Chapter are as under. Additional symbols are also defined where they appear. Symbols may have various meanings.

A	Accidental action.
A	Cross sectional area.
A_c	Cross sectional area of concrete.
A_{ct}	Area of concrete in tensile zone.
A_p	Area of a prestressing tendon or tendons.
A_s	Cross sectional area of reinforcement.
$A_{s,min}$	minimum cross-sectional area of reinforcement.
A_{sw}	Cross sectional area of shear reinforcement.
D	Diameter of mandrel.
E	Effect of action.
$E_c, E_{c(28)}$	Tangent modulus of elasticity of normal weight concrete at a stress of $\sigma_c = 0$ and at 28 days.
$E_{c,eff}$	Effective modulus of elasticity of concrete.
E_{cd}	Design value of modulus of elasticity of concrete.
E_{cm}	Secant modulus of elasticity of concrete.
$E_c(t)$	Tangent modulus of elasticity of normal weight concrete at a stress of $\sigma_c = 0$ and at time t .
E_p	Design value of modulus of elasticity of prestressing steel.
E_s	Design value of modulus of elasticity of reinforcing steel.
EI	Bending stiffness.
F	Action.
F_d	Design value of an action.
F_k	Characteristic value of an action.
G_k	Characteristic permanent action.
I	Second moment of area of concrete section.
K_c	Factor for cracking and creep effects.
K_s	Factor for reinforcement contribution.
L	Length.
M	Bending moment.
M_{Ed}	Design value of the applied internal bending moment.
M_{rep}	Cracking bending moment.
N	Axial force or number of cyclic loads in fatigue.
N_{Ed}	Design value of the applied axial force (tension or compression).
P	Prestressing force.
P_0	Initial force at the active end of the tendon immediately after stressing.
Q_k	Characteristic variable action.
R	Resistance or relaxation function.
S	Internal forces and moments.
S	First moment of area.
SLS	Serviceability Limit State.
T	Torsional Moment.

T_{Ed}	Design value of the applied torsional moment.
ULS	Ultimate limit state.
V	Shear force.
V_{Ed}	Design value of the applied shear force.
Vol	Volume of traffic.
a	Distance.
a	Geometrical data.
Δa	Deviation for geometrical data.
b	Overall width of a cross-section, or actual flange width in a T or L beam.
b_w	Width of the web on T, I or L beams.
c_{min}	Minimum cover.
d	Diameter; Depth.
d	Effective depth of a cross-section.
d_g	Largest nominal maximum aggregate size.
e	Eccentricity.
f_c	Compressive strength of concrete.
f_{cd}	Design value of concrete compressive strength.
f_{ck}	Characteristic compressive cylinder strength of concrete at 28 days.
f_{cm}	Mean value of concrete cylinder compressive strength.
f_{ctb}	Tensile strength prior to cracking in biaxial state of stress.
f_{ctk}	Characteristic axial tensile strength of concrete.
f_{ctm}	Mean value of axial tensile strength of concrete.
f_{ctx}	Appropriate tensile strength for evaluation of cracking bending moment.
f_p	Tensile strength of prestressing steel.
f_{pk}	Characteristic tensile strength of prestressing steel.
$f_{p0.1}$	0.1 % proof-stress of prestressing steel.
$f_{p0.1k}$	Characteristic 0.1 % proof-stress of prestressing steel.
$f_{0.2k}$	Characteristic 0.2 % proof-stress of reinforcement.
f_t	Tensile strength of reinforcement.
f_{tk}	Characteristic tensile strength of reinforcement.
f_y	Yield strength of reinforcement.
f_{yd}	Design yield strength of reinforcement.
f_{yk}	Characteristic yield strength of reinforcement.
f_{ywd}	Design yield of shear reinforcement.
h	Height.
h	Overall depth of a cross-section.
i	Radius of gyration.
k	Coefficient; Factor.
l	Length, span or height.
$1/r$	Curvature at a particular section.
s	Spacing between cracks.
t	Thickness.
t	Time being considered..
t_0	The age of concrete at the time of loading.

1

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4

Concrete Structures

u	Perimeter of concrete cross-section, having area A_c .
w	Component of the displacement of a point or crack width.
x	Neutral axis depth.
x, y, z	Coordinates.
x_u	Neutral axis depth at ULS after redistribution.
z	Lever arm of internal forces.
α	Angle; ratio; Long term effects coefficient or ratio between principal stresses.
α_e	E_s/E_{cm} ratio.
α_h	Reduction factor for θ_1 .
β	Angle; ratio; coefficient.
γ	Partial factor.
γ_A	Partial factor for accidental actions A.
γ_C	Partial factor for concrete.
γ_F	Partial factor for actions, F.
$\gamma_{F,fat}$	Partial factor for fatigue actions.
$\gamma_{C,fat}$	Partial factor for fatigue of concrete.
γ_G	Partial factor for permanent actions, G.
γ_M	Partial factor for a material property, taking account of uncertainties in the material property itself, in geometric deviation and in the design model used.
γ_P	Partial factor for actions associated with prestressing, P.
γ_Q	Partial factor for variable actions, Q.
γ_S	Partial factor for reinforcing or prestressing steel.
$\gamma_{S,fat}$	Partial factor for reinforcing or prestressing steel under fatigue loading.
γ_f	Partial factor for actions without taking account of model uncertainties.
γ_g	Partial factor for permanent actions without taking account of model uncertainties.
γ_m	Partial factors for a material property, taking account only of uncertainties in the material property.
δ	Increment/redistribution ratio.
ξ	Creep redistribution function or bond strength ratio.
ζ	Reduction factor/distribution coefficient.
ϵ_c	Compressive strain in the concrete.
ϵ_{ca}	Autogenous shrinkage.
ϵ_{cc}	Creep strain.
ϵ_{cd}	Dry shrinkage.
ϵ_{cl}	Compressive strain in the concrete at the peak stress f_c .
ϵ_{cu}	Ultimate compressive strain in the concrete.
ϵ_u	Strain of reinforcement or prestressing steel at maximum load.
ϵ_{uk}	Characteristic strain of reinforcement or prestressing steel at maximum load.
θ	Angle.
θ_i	Inclination for geometric imperfections.
λ	Slenderness ratio or damage equivalent factors in fatigue.
μ	Coefficient of friction between the tendons and their ducts.
ν	Poisson's ratio.
ν	Strength reduction factor for concrete cracked in shear.

ζ	Ratio of bond strength of prestressing and reinforcing steel.
ρ	Oven-dry density of concrete in kg/m ³ .
ρ_{1000}	Value of relaxation loss (in %), at 1000 hours after tensioning and at a mean temperature of 20°C.
ρ_l	Reinforcement ratio for longitudinal reinforcement.
ρ_w	Reinforcement ratio for shear reinforcement.
σ_c	Compressive stress in the concrete.
σ_{cp}	Compressive stress in the concrete from axial load or prestressing.
σ_{cu}	Compressive stress in the concrete at the ultimate compressive strain ε_{cu} .
τ	Torsional shear stress.
ϕ	Diameter of a reinforcing bar or of a prestressing duct.
ϕ_n	Equivalent diameter of a bundle of reinforcing bars.
$\varphi(t, t_0)$	Creep coefficient, defining creep between times t and t_0 , related to elastic deformation at 28 days.
$\varphi(\infty, t_0)$	Final value of creep coefficient.
ψ	Factors defining representative values of variable actions.
ψ_0	for combination values.
ψ_1	for frequent values.
ψ_2	for quasi-permanent values.

4.2 Basis of Design

4.2.1 Requirements

The design shall be carried out in accordance with the guidelines provided in this Manual. The limit state design in conjunction with the partial factor method, actions and combination of actions shall be applied in accordance with the guidelines provided in this Manual.

4.2.2 Principles of Limit State Design

The principles of limit state and design situations are given in Chapter 2. The structures are designed for ultimate limit state and serviceability limit state. The limit state verification shall be carried out for persistent, transient, accidental or seismic design situations. The ultimate limit state verification shall be carried, where they are relevant, for loss of equilibrium, failure by excessive deformation or failure caused by fatigue. Design for limit states shall be based on the use of structural and load models for relevant limit states. Verifications shall be carried out for all relevant design situations and load case by the partial factor method.

4.2.3 Basic Variables

4.2.3.1 Actions

General

The actions / forces (permanent actions, traffic loads, wind actions, thermal actions, seismic actions, construction loads, accidental actions, water pressure, earth pressure, etc.) to be used in design may be obtained in accordance with the Chapter 3.

Thermal Effects

1

Thermal effects should be taken into account when checking serviceability limit states.

2

Thermal effects should be considered for ultimate limit states only where they are significant (e.g. integral bridges, long span bridges). In other cases, they need not be considered where the ductility and rotation capacity of the elements are sufficient.

3

Where thermal effects are taken into account, they should be considered as variable actions and applied with a partial factor and ψ factor.

4

Differential Settlements

Differential settlements/movements of the structure due to soil subsidence should be classified as a permanent action, G_{set} which is introduced as such in combinations of actions.

The effects of differential settlements should generally be considered for the verification for serviceability limit states. For ultimate limit states they should be considered only where they are significant. Where differential settlements are considered, a partial safety factor for settlement effects should be applied.

Prestress

The prestress is applied by tendons made of high-strength steel (wires, strands or bars). Tendons may be embedded in the concrete. They may be pre-tensioned and bonded or post-tensioned and bonded or unbonded. Tendons may also be external to the structure with points of contact occurring at deviators and anchorages.

4.2.3.2 Material and Product Properties

General

The material and product properties for concrete, reinforcement and prestressing steel are given in Section 4.3 or the relevant Product Standards may be referred.

Shrinkage and Creep

Shrinkage and creep are time-dependent properties of concrete. Their effects should generally be considered for the verification of serviceability limit states. The effects of shrinkage and creep should be considered at ultimate limit states only where their effects are significant. When creep is taken into account its design effects should be evaluated under the quasi-permanent combination of actions irrespective of the design situation considered.

4.2.3.3 Deformations of Concrete

The consequences of deformation due to temperature, creep and shrinkage shall be considered in design.

4.2.3.4 Geometric Data

Geometrical data shall be represented by their characteristic values and the dimensions specified in the design may be taken as characteristic values. The imperfections that should be taken into account in the design of structural members are also given in the relevant sections of this Manual.

Uncertainties related to the cross-section of cast in place piles and concreting procedures shall be allowed for in design. The diameter used in design calculations, of cast in place piles without permanent casing should be taken as:

- if $d_{nom} < 400$ mm, $d = d_{nom} - 20$ mm
- if $400 \leq d_{nom} \leq 1000$ mm, $d = 0.95 d_{nom}$
- if $d_{nom} > 1000$ mm, $d = d_{nom} - 50$ mm

Where, d_{nom} is the nominal diameter of the pile.

4.2.4 Verification by the Partial Factor Method

4.2.4.1 General

The partial factor method is used to verify that no relevant limit state is exceeded when design values for actions or effects of actions and resistances are used in the design models in all relevant design situations.

The design values shall be obtained by using the characteristic or other representative values in combination with partial and other factors as provided in relevant sections of this Manual.

4.2.4.2 Design Values

The design values of actions can be obtained by applying partial factors (γ_F) and combination factors (Ψ_0 or Ψ_1 or Ψ_2) from the characteristic values in accordance with the details given in Chapter 3. The values of partial factors and combination factors for self-weight, traffic actions, wind actions, thermal actions are given in Chapter 3. Following partial factors shall be used to obtain the design values for other actions.

4.2.4.2.1 Partial Factor for Shrinkage Action

Where consideration of shrinkage actions is required for ultimate limit state a partial factor, γ_{SH} , should be used. The recommended value of γ_{SH} is 1.0.

4.2.4.2.2 Partial Factors for Prestress Loads

Prestress in most situations is intended to be favourable and for the ultimate limit state verification the value of $\gamma_{P,fav}$ should be used. The design value of prestress may be based on the mean value of the prestressing force. The recommended value for persistent and transient design situations is 0.9. This value may also be used for fatigue verification.

In the verification of the limit state for stability with external prestress, where an increase of the value of prestress can be unfavourable, $\gamma_{P,unfav}$ should be used. The recommended value $\gamma_{P,unfav}$ for global analysis is 1.1.

In the verification of local effects (for anchorage zones), $\gamma_{P,unfav}$ should also be used. The recommended value $\gamma_{P,unfav}$ for local effects is 1.2.

4.2.4.2.3 Partial Factor for Fatigue Loads

The partial factor for fatigue loads is $\gamma_{F,fat}$. The recommended value $\gamma_{F,fat}$ is 1.0.

4.2.4.2.4 Partial Factors for Materials

Partial factors for materials for ultimate limit states, γ_C and γ_S should be used. The recommended values for persistent & transient and accidental design situations are given in Table 4.1.

For fatigue verification, the partial factors for persistent design situations given in Table 4.1 are recommended for the values of $\gamma_{C,fat}$ and $\gamma_{S,fat}$. The values for partial factors for materials for serviceability limit state verification should be taken as those given in the particular Clauses of this Manual. For situations not covered by particular Clause, a value of 1.0 may be taken.

4.2.4.2.5 Partial Factors for Materials for Foundations

Table 4.1 Partial Factors for Materials for ULS and SLS

Design Situations	γ_c for Concrete	γ_s for reinforcing steel	γ_s for prestressing steel
ULS Persistent and Transient	1.5	1.15	1.15
Accidental	1.2	1.0	1.0
Fatigue	1.5	1.15	1.15
SLS	Refer relevant Clause in the Manual 1.0 may be taken if not covered in particular Clause		

The partial factor γ_c for concrete for ultimate limit state given in above Clause should be multiplied by a factor 1.1 for calculation of design resistance of cast in place piles without permanent casing.

4.2.4.3 Combinations of Actions

The details for combinations of actions for the serviceability and ultimate limit states for different design situations are given in Chapter 3.

4.3 Material Properties

The knowledge of the physical, chemical, mechanical, load-dependent, time-dependant and process dependent properties of materials are important in analysis and design of the structure and its elements.

4.3.1 Concrete

4.3.1.1 Strength

Concrete strength is denoted by concrete strength class (C) which relate to characteristic (5%) cylindrical strength f_{ck} or the cube strength $f_{ck,cube}$ determined at 28 days. The strength classes in this Manual are based on cylindrical strength f_{ck} . The characteristic strengths f_{ck} and the corresponding mechanical characteristics necessary for design, are given in Table 4.2. The notation ' C ' refers to normal weight concrete, first number refers to cylindrical strength f_{ck} and second number to cube strength $f_{ck,cube}$. The recommended minimum and maximum strength class to be used in road bridges are C_{min} as C25/30 for reinforced and C35/45 for prestress concrete and C_{max} as C70/85 respectively. However, the shear strength of concrete strength class higher than C50/60 may be limited to that of C50/60.

In certain situations (e.g. demoulding, pre-stressing) it may be required to assess the concrete compressive strength before or after 28 days on the basis of test specimens. If the concrete strength is determined at an age of $t > 28$ days, the values of α_{cc} and α_{ct} used in calculating design compressive and design tensile strength is reduced by a multiplication factor k_t , refer Clause 4.3.1.5. The recommended value of k_t is 1.0.

The characteristic compressive strength $f_{ck}(t)$ of concrete at time t may be specified as below.

$$f_{ck}(t) = f_{cm}(t) - 8 \text{ (MPa) for } 3 < t < 28 \text{ days}$$

$$f_{ck}(t) = f_{ck} \text{ for } t \geq 28 \text{ days}$$

The compressive strength of concrete at an age t depends on the type of cement, temperature and curing conditions. For a mean temperature of 20°C and standard curing, the mean compressive strength of concrete at various ages $f_{cm}(t)$ may be estimated from following Expressions.

$$f_{cm}(t) = \beta_{cc}(t) f_{cm}$$

Equation 4.1

The factor $\beta_{cc}(t)$ is given as:

$$\beta_{cc}(t) = \exp \left\{ s \left[1 - \left(\frac{28}{t} \right)^{1/2} \right] \right\}$$

Equation 4.2

Where,

- $f_{cm}(t)$ = Mean concrete compressive strength at an age of t days.
 f_{cm} = Mean concrete compressive strength at an age of 28 days.
 $\beta_{cc}(t)$ = A coefficient which depends on the age of concrete t .
 t = age of concrete in days.
 s = A coefficient which depends on type of cement.
 = 0.20 for cement of strength class CEM 42.5R, CEM 52.5N, CEM 52.5R (class R)
 = 0.25 for cement of strength class CEM 32.5R, CEM 42.5N (class N).
 = 0.38 for cement of strength class CEM 32.5N (class S).

Where, Class R = high early strength, Class N = normal early strength, Class S = slow early strength.

For situations where heat curing is applied to concrete, refer separately.

The tensile strength refers to the highest stress reached under concentric tensile loading. Where the tensile strength is determined as the splitting tensile strength $f_{ct,sp}$, the approximate value of axial tensile strength may be taken as:

$$f_{ct} = 0.9 f_{ct,sp}$$

Equation 4.3

The development of tensile strength with time depends upon the curing and drying conditions as well as by the dimensions of the structural members. As a first approximation, it can be taken as:

$$f_{ctm}(t) = (\beta_{cc}(t))^\alpha f_{ctm}$$

Equation 4.4

Where,

- $\beta_{cc}(t)$ = as defined earlier
 α = 1 for $t < 28$ days
 α = 2/3 for $t \geq 28$ days

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Table 4.2 Strength and Deformation Characteristics for Normal Concrete

Symbol	Description	C12/15	C16/20	C20/25	C25/30	C28/35	C30/37	C32/40	C35/45	C40/50	C45/55	C50/60	C55/67	C60/75	C70/85	C80/95	C90/105	Analytical Relation
f_{ck} (MPa)	Characteristic cylinder compressive strength	12	16	20	25	28	30	32	35	40	45	50	55	60	70	80	90	
$f_{ck,cube}$ (MPa)	Characteristic cube compressive strength	15	20	25	30	35	37	40	45	50	55	60	67	75	85	95	105	
f_{cm} (MPa)	Mean cylinder compressive strength	20	24	28	33	36	38	40	43	48	53	58	63	68	78	88	98	$f_{cm} = f_{ck} + 8$ (MPa)
f_{ctm} (MPa)	Mean tensile strength	1.6	1.9	2.2	2.6	2.8	2.9	3.0	3.2	3.5	3.8	4.1	4.2	4.4	4.6	4.8	5.0	$f_{ctm} = 0.30 f_{ck}^{(2/3)} \leq C50/60$ $f_{ctm} = 2.2 \ln(1 + f_{cm}/10) > C50/60$
$f_{ctk,0.05}$ (MPa)	5% fractile tensile strength	1.1	1.3	1.5	1.8	1.9	2.0	2.1	2.2	2.5	2.7	2.9	3.0	3.1	3.2	3.4	3.5	$f_{ctk,0.05} = 0.7 f_{ctm}$
$f_{ctk,0.95}$ (MPa)	95% fractile tensile strength	2.0	2.5	2.9	3.3	3.6	3.8	3.9	4.2	4.6	4.9	5.3	5.5	5.7	6.0	6.3	6.6	$f_{ctk,0.95} = 1.3 f_{ctm}$
E_{cm} (GPa)	Elastic modulus	27	29	30	31	32	33	33	34	35	36	37	38	39	41	42	44	$E_{cm} = 22 (f_{cm}/10)^{0.3}$ (f_{cm} in MPa)
ε_{cl} (‰)	Compressive strain at peak stress f_c (non-linear diagram)	1.8	1.9	2.0	2.1	2.1	2.2	2.2	2.25	2.3	2.4	2.45	2.5	2.6	2.7	2.8	2.8	$\varepsilon_{cl}(\text{‰}) = 0.7 f_{cm}^{-0.31} \leq 2.8$
ε_{cu1} (‰)	Ultimate compressive strain (non-linear diagram)	3.5										3.2	3.0	2.8	2.8	2.8	2.8	$\varepsilon_{cu1}(\text{‰}) = 2.8 + 27 \{(98 - f_{cm})/100\}^4$ for $f_{ck} \geq 50$ MPa
ε_{c2} (‰)	Compressive strain at maximum strength (parabola-rectangular diagram)	2.0										2.2	2.3	2.4	2.5	2.6	2.6	$\varepsilon_{c2}(\text{‰}) = 2.0 + 0.085 \{(f_{ck} - 50)^{0.53}\}$ for $f_{ck} \geq 50$ MPa
ε_{cu2} (‰)	Ultimate compressive strain (parabola-rectangular diagram)	3.5										3.1	2.9	2.7	2.6	2.6	2.6	$\varepsilon_{cu2}(\text{‰}) = 2.6 + 35 \{(90 - f_{ck})/100\}^4$ for $f_{ck} \geq 50$ MPa
n	Exponent (Refer Equation 4.17)	2.0										1.75	1.6	1.45	1.4	1.4	1.4	$n = 1.4 + 23.4 \{(90 - f_{ck})/100\}^4$ for $f_{ck} \geq 50$ MPa
ε_{c3} (‰)	Compressive strain at maximum strength (bi-linear diagram)	1.75										1.8	1.9	2.0	2.2	2.3	2.3	$\varepsilon_{c3}(\text{‰}) = 1.75 + 0.55 \{(f_{ck} - 50)/40\}$ for $f_{ck} \geq 50$ MPa
ε_{cu3} (‰)	Ultimate compressive strain (bi-linear diagram)	3.5										3.1	2.9	2.7	2.6	2.6	2.6	$\varepsilon_{cu3}(\text{‰}) = 2.6 + 35 \{(90 - f_{ck})/100\}^4$ for $f_{ck} \geq 50$ MPa

4.3.1.2 Elastic Deformation

The elastic deformations of concrete broadly depend on its composition, especially the aggregates. The values given in the Table 4.2 shall be regarded as indicative for general applications. The elastic deformations should be specifically assessed if the structure is likely to be sensitive to deviations from these general values.

The modulus of elasticity of a concrete is controlled by the moduli of elasticity of its components. Approximate values for the modulus of elasticity E_{cm} , secant value between $\sigma_c = 0$ and $0.4f_{cm}$, for concretes with quartzite aggregates, are given in Table 4.2. For limestone and sandstone aggregates the value should be reduced by 10 % and 30 % respectively. For basalt aggregates the value should be increased by 20 %.

Variation of the modulus of elasticity with time can be estimated by:

$$E_{cm}(t) = (f_{cm}(t)/f_{cm})^{0.3} E_{cm} \quad \text{Equation 4.5}$$

Where, $E_{cm}(t)$ and $f_{cm}(t)$ are the values at an age of t days; E_{cm} and f_{cm} are the values determined at an age of 28 days. The relation between $f_{cm}(t)$ and f_{cm} follows from Expression defined earlier.

Poisson's ratio may be taken equal to 0.2 for uncracked concrete and 0 for cracked concrete.

Unless more accurate information is available, the linear coefficient of thermal expansion may be taken equal to $10 \times 10^{-6} \text{ K}^{-1}$.

4.3.1.3 Creep and Shrinkage

Creep is the effect due to which concrete undergoes continuous deformation under sustained loading applied for a considerable time. Shrinkage is drying of concrete and is independent of applied loads.

The Creep and shrinkage of the concrete depend on the ambient humidity, the dimensions of the element and the composition of the concrete. Creep is also influenced by the maturity of the concrete when the load is first applied and depends on the duration and magnitude of the loading.

4.3.1.3.1 Creep

The creep coefficient, $\varphi(t, t_0)$ is related to E_c , the tangent modulus, which may be taken as $1.05 E_{cm}$. Where great accuracy is not required, the value found from Figure 4.1 may be considered as the creep coefficient, provided that the concrete is not subjected to a compressive stress greater than $0.45 f_{ck}(t_0)$ at an age t_0 , the age of concrete at the time of loading.

The creep deformation of concrete $\varepsilon_{cc}(\infty, t_0)$ at time $t = \infty$ for a constant compressive stress σ_c applied at the concrete age t_0 , is given by:

$$\varepsilon_{cc}(\infty, t_0) = \varphi(\infty, t_0)(\sigma_c/E_c) \quad \text{Equation 4.6}$$

When the compressive stress of concrete at an age t_0 exceeds the value $0.45 f_{ck}(t_0)$ then creep non-linearity should be considered. Such a high stress can occur as a result of pre-tensioning, e.g. in precast concrete members at tendon level. In such cases the non-linear notional creep coefficient should be obtained as follows:

$$\varphi_{nl}(\infty, t_0) = \varphi(\infty, t_0) \exp(1.5(k_\sigma - 0.45)) \quad \text{Equation 4.7}$$

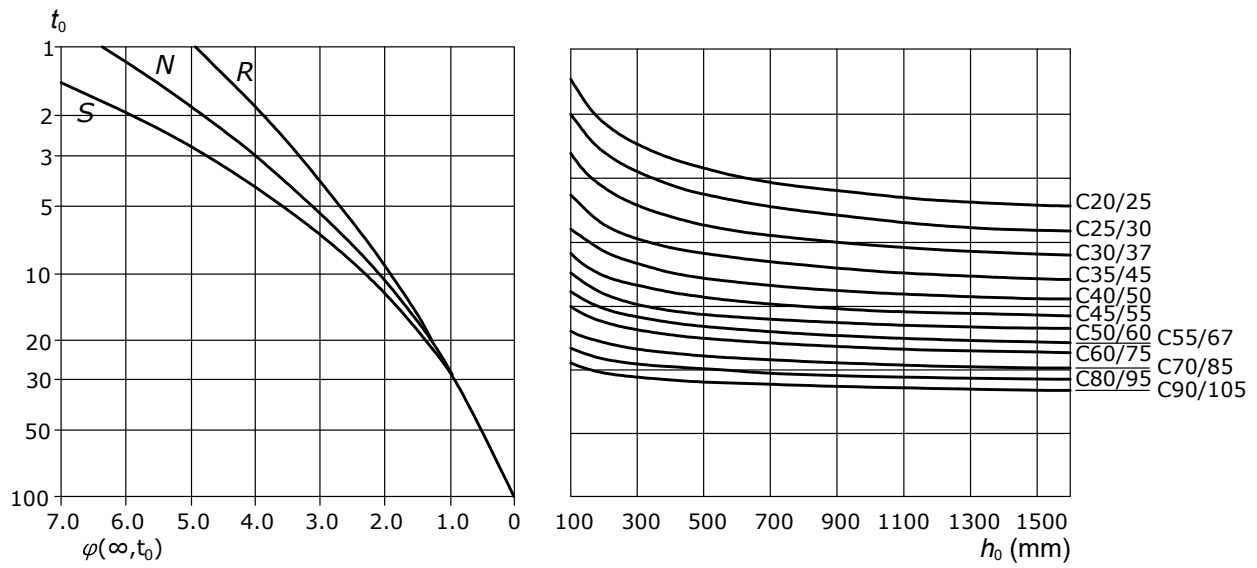
Where,

$\varphi_{nl}(\infty, t_0)$ = non-linear notional creep coefficient which replaces $\varphi(\infty, t_0)$

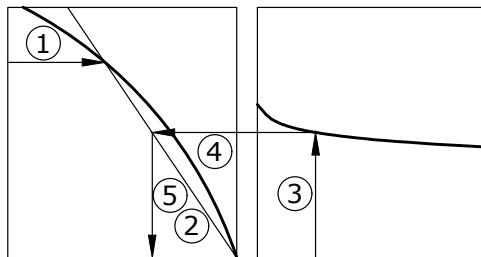
k_σ = stress-strength ratio $\sigma_c / f_{ck}(t_0)$

σ_c = compressive stress

$f_{ck}(t_0)$ = characteristic concrete compressive strength at the time of loading

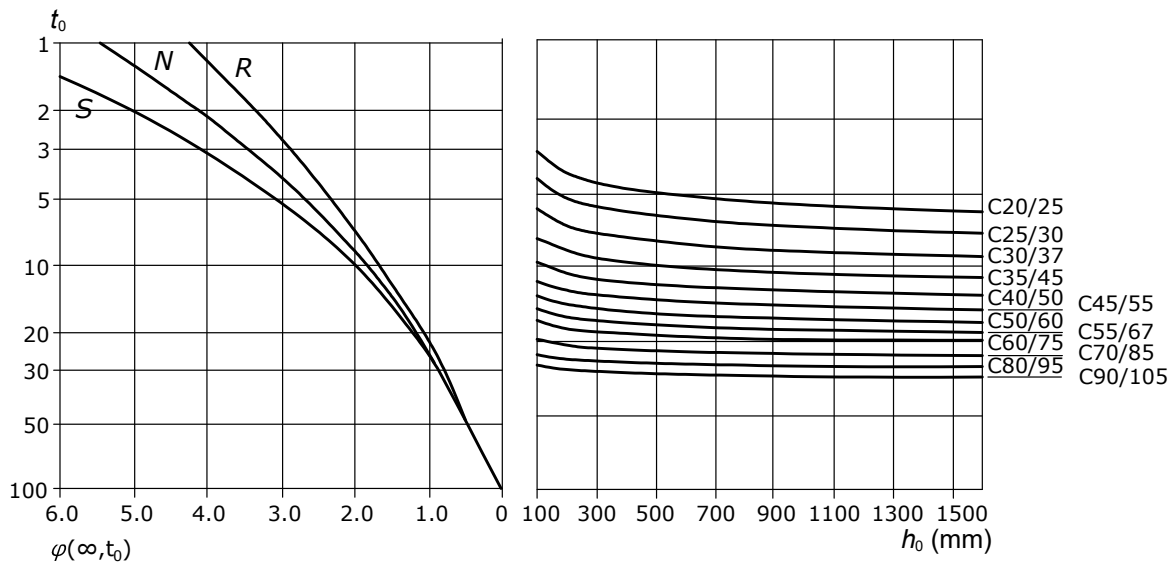
Figure 4.1 Method for Determining the Creep Coefficient $\varphi(\infty, t_0)$ for Concrete Under Normal Environmental Conditions

a) inside conditions - RH = 50%



Note:

- intersection point between lines 4 and 5 can also be above point 1
- for $t_0 > 100$ it is sufficiently accurate to assume $t_0 = 100$ (and use the tangent line)



b) outside conditions - RH = 80%

The values given in Figure 4.1 are valid for ambient temperatures between -40°C and +40°C and a mean relative humidity between RH = 40% and RH = 100%. The following symbols are used:

$\phi_{nl}(\infty, t_0)$ = Final creep coefficient

t_0 = Age of concrete at the time of loading in days

h_0 = Notional size = $2A_c/u$

A_c = Concrete cross-sectional area

u = Perimeter of that part which is exposed to drying

S = Class S of Cement (strength class CEM 32.5 N)

N = Class N of Cement (strength class CEM 32.5 R, CEM 42.5 N)

R = Class R of Cement (strength class CEM 42.5 R, CEM 52.5 N and CEM 52.5 R)

4.3.1.3.2 Shrinkage

The total shrinkage strain is composed of two components, the drying shrinkage strain and the autogenous shrinkage strain. The drying shrinkage strain develops slowly, since it is a function of the migration of the water through the hardened concrete. The autogenous shrinkage strain develops during hardening of the concrete. The major part therefore develops in the early days after casting. Autogenous shrinkage is a linear function of the concrete strength. It should be considered specifically when new concrete is cast against hardened concrete. The values of the total shrinkage strain ε_{cs} is given by:

$$\varepsilon_{cs} = \varepsilon_{cd} + \varepsilon_{ca}$$

Equation 4.8

Where,

ε_{cs} = Total shrinkage strain

ε_{cd} = Drying shrinkage strain

ε_{ca} = Autogenous shrinkage strain

The final value of the drying shrinkage strain, $\varepsilon_{cd,\infty}$ is equal $k_h \varepsilon_{cd,0}$. The value of $\varepsilon_{cd,0}$ may be taken from Table 4.3. The formula for $\varepsilon_{cd,0}$ is given in Appendix B.2.

Table 4.3 Nominal Unrestrained Drying Shrinkage Values $\varepsilon_{cd,0}$ (in ‰) for Concrete with Cement CEM Class N

$f_{ck} / f_{ck,cube}$ (MPa)	Relative Humidity (in ‰)					
	20	40	60	80	90	100
20/25	0.62	0.58	0.49	0.3	0.17	0.00
40/50	0.48	0.46	0.38	0.24	0.13	0.00
60/75	0.38	0.36	0.30	0.19	0.10	0.00
80/95	0.30	0.28	0.24	0.15	0.08	0.00
90/105	0.27	0.25	0.21	0.13	0.07	0.00

The development of the drying shrinkage strain in time is as follows:

$$\varepsilon_{cd}(t) = \beta_{ds}(t, t_s) k_h \varepsilon_{cd,0}$$

Equation 4.9

Where, k_h is a coefficient depends upon nominal size h_0 , as in Table 4.4.

Table 4.4 Values for k_h in Equation 4.9

h_0	k_h
100	1.00
200	0.85
300	0.75
≥500	0.70

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$$\beta_{ds}(t, t_s) = \frac{(t - t_s)}{(t - t_s) + 0.04\sqrt{h_0^3}}$$

Where,

t = The age of the concrete at the moment considered, in days

t_s = The age of concrete (days) at the beginning of drying shrinkage (or swelling). Normally this is at the end of curing

h_0 = The notional size (mm) of the cross section = $2 A_c / u$

Where,

A_c = The concrete cross section area

u = The perimeter of that part of the cross section which is exposed to drying.

The autogenous shrinkage strain follows from:

$$\varepsilon_{ca}(t) = \beta_{as}(t) \varepsilon_{ca}(\infty)$$

Equation 4.11

Where,

$$\varepsilon_{ca}(\infty) = 2.5(f_{ck} - 10)10^{-6}$$

Equation 4.12

and

$$\beta_{as}(t) = 1 - \exp(-0.2t^{0.5})$$

Equation 4.13

t is given in days

4.3.1.4 Stress-strain Relation for Non-linear Structural Analysis

The relation between compressive stress (σ_c) and strain (ε_c) shown in Figure 4.2 (compressive stress and shortening strain shown as absolute values) for short term uniaxial loading is described as:

$$\frac{\sigma_c}{f_{cm}} = \frac{k\eta - \eta^2}{1 + (k - 2)\eta}$$

Equation 4.14

Where,

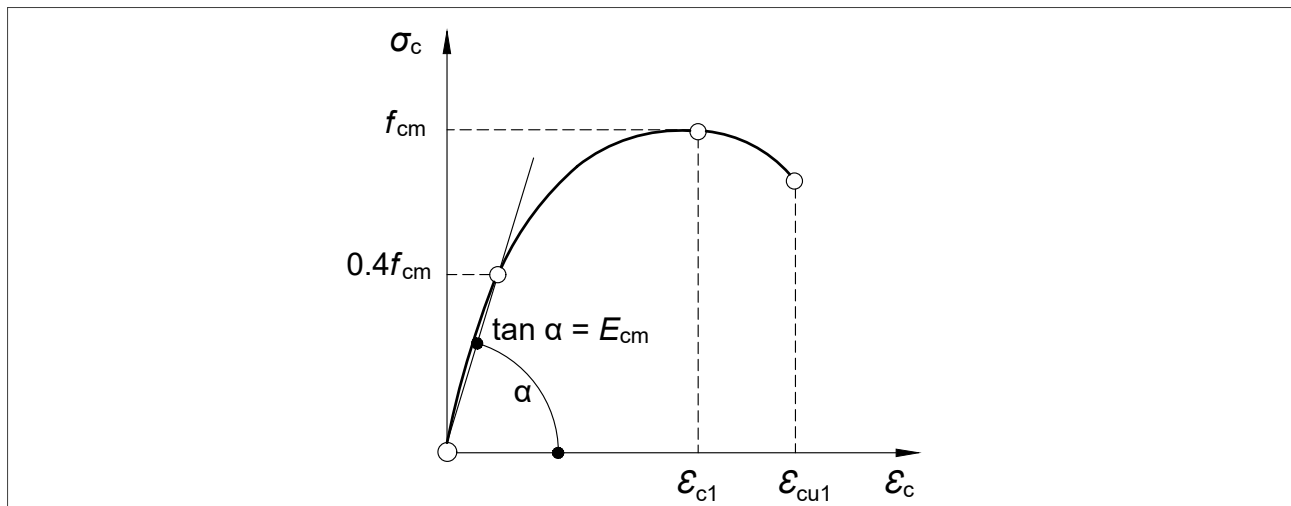
$\eta = \varepsilon_c / (\varepsilon_{c1})$

ε_{c1} = The strain at the peak stress according to Table 4.2.

$k = 1.05 E_{cm} | \varepsilon_{c1} | / f_{cm}$ (the f_{cm} according to Table 4.2).

The Equation 4.14 is valid for $0 < |\varepsilon_c| < |\varepsilon_{cu1}|$ where ε_{cu1} is the nominal ultimate strain.

Figure 4.2 Schematic Representation of the Stress-strain Relation for Structural Analysis



4.3.1.5 Design Compressive and Tensile Strengths

The design compressive strength of concrete is defined as:

$$f_{cd} = \alpha_{cc} \frac{f_{ck}}{\gamma_c}$$

Equation 4.15

Where,

f_{ck} = Characteristic compressive cylinder strength of concrete at 28 days

γ_c = Partial safety factor for concrete

α_{cc} = Coefficient accounting for long term effect on compressive strength of concrete and of unfavourable effects = 0.85

The design tensile strength of concrete is defined as:

$$f_{ctd} = \frac{\alpha_{ct} f_{ctk,0.05}}{\gamma_c}$$

Equation 4.16

Where,

$f_{ctk,0.05}$ = Characteristic axial tensile strength of concrete, 5% fractile value

γ_c = partial safety factor for concrete

α_{ct} = coefficient accounting for long term effect on tensile strength of concrete and of unfavourable effects = 1.0

4.3.1.6 Stress-strain Relations for the Design of Cross-sections

4.3.1.6.1 Parabolic - Rectangular

For the design of cross-sections, the following stress-strain relationship may be used:

$$\sigma_c = f_{cd} \left[1 - \left(1 - \frac{\epsilon_c}{\epsilon_{c2}} \right)^n \right] \quad \text{for } 0 \leq \epsilon_c \leq \epsilon_{c2}$$

Equation 4.17

$$\sigma_c = f_{cd} \quad \text{for } \epsilon_{c2} \leq \epsilon_c \leq \epsilon_{cu2}$$

Equation 4.18

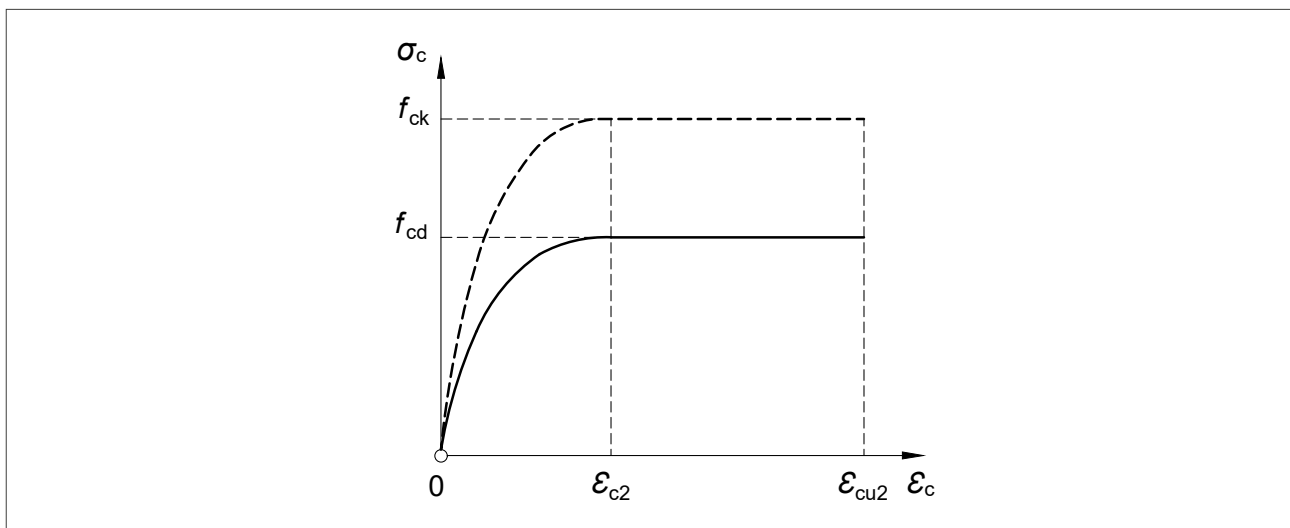
Where,

n = The exponent according to Table 4.2.

ϵ_{c2} = The strain at the reaching the maximum strength according to Table 4.2.

ϵ_{cu2} = The ultimate strain according to Table 4.2.

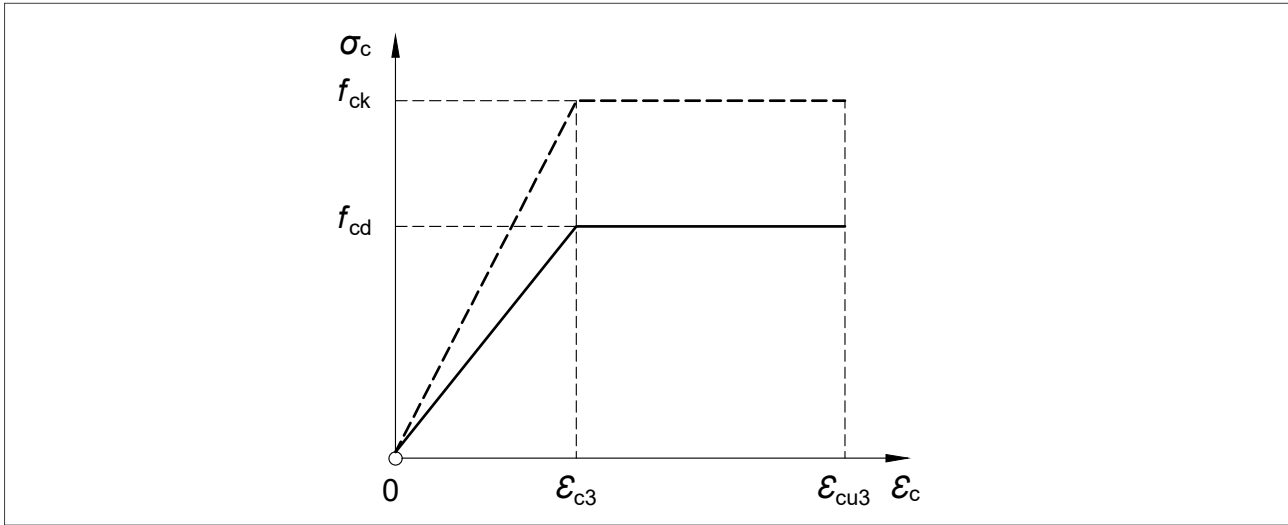
Figure 4.3 Parabola-rectangle Diagram for Concrete Under Compression



4.3.1.6.2 Bi-linear

Other simplified stress-strain relationships may also be used which is more conservative.

Figure 4.4 Bi-linear Stress-strain Relation



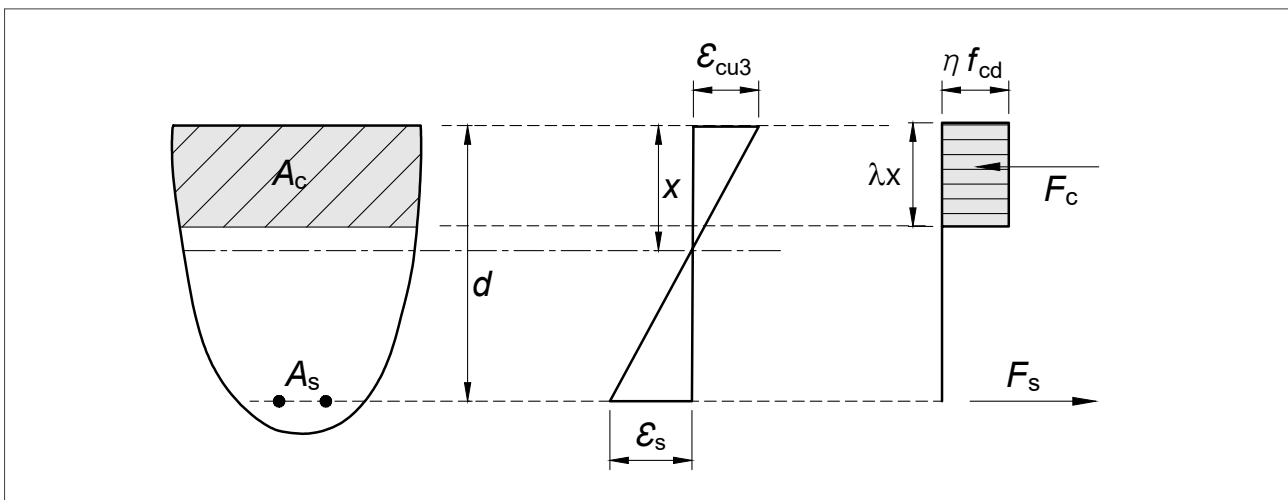
4.3.1.6.3 Rectangular

A rectangular stress distribution may be assumed for more simplification. The factor λ defining the effective height of compression zone and factor η defining the effective strength can be given as:

$$\begin{aligned} \lambda &= 0.8 && \text{for } f_{ck} \leq 50 \text{ MPa} \\ \lambda &= 0.8 - (f_{ck} - 50)/400 && \text{for } 50 < f_{ck} \leq 90 \text{ MPa} \\ \eta &= 1.0 && \text{for } f_{ck} \leq 50 \text{ MPa} \\ \eta &= 1.0 - (f_{ck} - 50)/200 && \text{for } 50 < f_{ck} \leq 90 \text{ MPa} \end{aligned}$$

If the width of the compression zone decreases in the direction of the extreme compression fibre, the value of ηf_{cd} should be reduced by 10 %.

Figure 4.5 Rectangular Stress Distribution



The parabolic rectangular stress-strain block described above is of general validity for all design situations. However, simplified equivalent stress blocks such as bilinear or rectangle may be used for design purposes where the net results are sufficiently accurate.

4.3.1.7 Flexural Tensile Strength

The mean flexural tensile strength of reinforced concrete members depends on the mean axial tensile strength and the depth of the cross-section. The following relationship may be used:

$$f_{ctm,fl} = \max \left\{ \frac{1.6 - h}{1000} f_{ctm} ; f_{ctm} \right\}$$

Equation 4.19

Where,

h = Total depth of member in mm.

f_{ctm} = Mean axial tensile strength following from Table 4.2.

The relation given above also applies for the characteristic tensile strength values.

4.3.1.8 Confined Concrete

Confinement of concrete results in a modification of the effective stress-strain relationship, higher strength and higher critical strains are achieved. The other basic material characteristics may be considered as unaffected for design.

The stress-strain relation shown in Figure 4.6 (compressive strain shown positive) may be used, with increased characteristic strength and strains according to:

$$f_{ck,c} = f_{ck} \left(1.000 + \frac{5.0 \sigma_2}{f_{ck}} \right) \quad \text{for } \sigma_2 \leq 0.05 f_{ck}$$

Equation 4.20

$$f_{ck,c} = f_{ck} \left(1.125 + \frac{2.5 \sigma_2}{f_{ck}} \right) \quad \text{for } \sigma_2 > 0.05 f_{ck}$$

Equation 4.21

$$\varepsilon_{c2,c} = \varepsilon_{c2} \left(\frac{f_{ck,c}}{f_{ck}} \right)^2$$

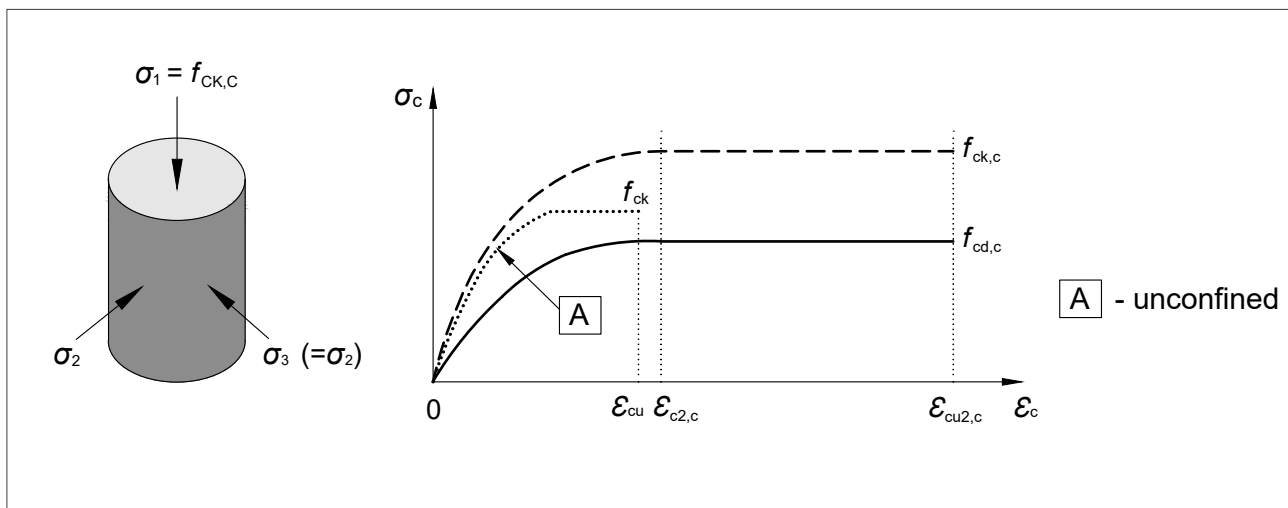
Equation 4.22

$$\varepsilon_{cu2,c} = \varepsilon_{cu2} + \left(\frac{0.2 \sigma_2}{f_{ck}} \right)$$

Equation 4.23

Where, σ_2 ($= \sigma_3$) is the effective lateral compressive stress at the ULS due to confinement and ε_{c2} and ε_{cu2} follow from Table 4.2. Confinement can be generated by adequately closed links or cross-ties, which can reach the plastic condition due to lateral extension of the concrete.

Figure 4.6 Stress-strain Relationship for Confined Concrete Distribution



4.3.2 Steel Reinforcement

4.3.2.1 General

The following Clauses are applicable for reinforcement which is in the form of bars and de-coiled rods. They do not apply to specially coated bars.

The requirements for the properties of the reinforcement are for the material as placed in the hardened concrete. If site operations can affect the properties of the reinforcement, then those properties shall be verified after such operations.

4.3.2.2 Properties

The behaviour of reinforcing steel is specified by the following properties:

- Yield strength (f_{yk} or $f_{0.2k}$).
- Maximum actual yield strength ($f_{y,max}$).
- Tensile strength (f_t).
- Ductility (ε_{uk} and f_t/f_{yk}).
- Bendability.
- Bond characteristics (f_R).
- Section sizes and tolerances.
- Fatigue strength.
- Weldability.

The application rules for design and detailing in this Manual are valid for a specified yield strength range, $f_{yk} = 400$ to 600 MPa.

Table 4.5 gives the properties of reinforcement suitable for use with this Manual. The properties are valid for temperatures between -40°C and 100°C for the reinforcement in the finished structure.

Table 4.5 Properties of Reinforcement

Product Form	Bars and De-coiled Rods			Wire Fabrics		
Class	A	B	C	A	B	C
Characteristic yield strength f_{yk} or $f_{0.2k}$ (MPa)	400 to 600					
Minimum value $k = (f_t/f_y)_k$	≥ 1.05	≥ 1.08	≥ 1.15 < 1.35	≥ 1.05	≥ 1.08	≥ 1.15 < 1.35
Characteristic strain at maximum force, ε_{uk} (%)	≥ 2.5	≥ 5.0	≥ 7.5	≥ 2.5	≥ 5.0	≥ 7.5

The properties of reinforcing steel are given in EN 10080 and KS EAS 412.

4.3.2.3 Strength

The yield strength f_{yk} (or the 0.2% proof stress, $f_{0.2k}$) and the tensile strength f_{tk} are defined respectively as the characteristic value of the yield load, and the characteristic maximum load in direct axial tension, each divided by the nominal cross-sectional area.

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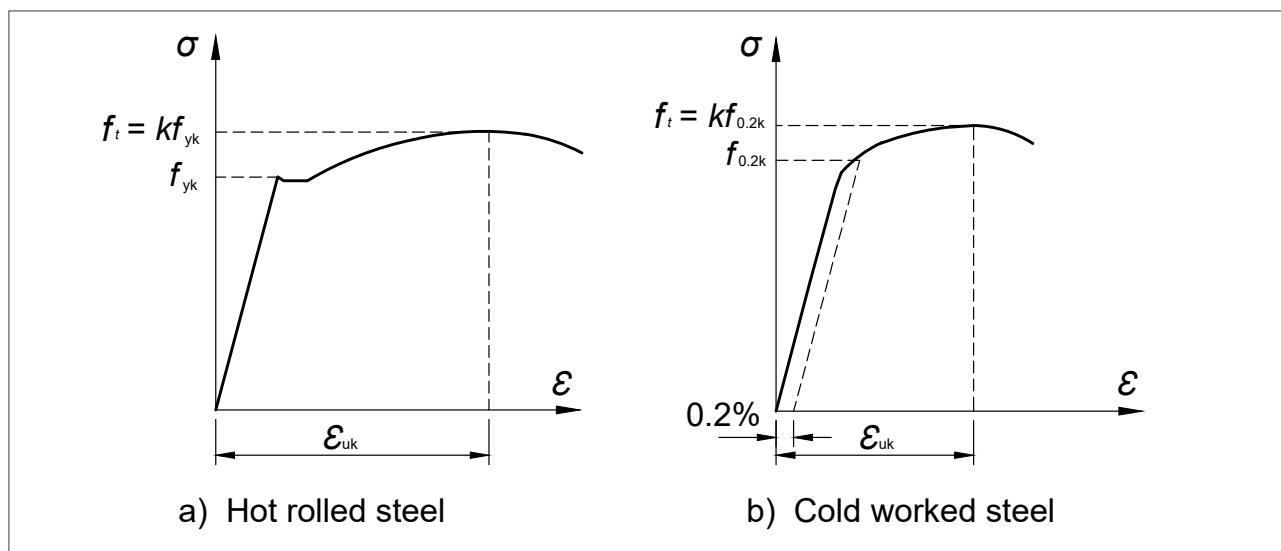
4.3.2.4 Ductility Characteristics

The reinforcement shall have adequate ductility as defined by the ratio of tensile strength to the yield stress, $(f_t/f_y)_k$ and the elongation at maximum force, ϵ_{uk} .

The reinforcement Class B and Class C shall be used in Bridges.

Figure 4.7 shows stress-strain curves for typical hot rolled and cold worked steel.

Figure 4.7 Stress-strain Diagrams of Typical Reinforcing Steel



4.3.2.5 Welding

Welding processes for reinforcing bars shall be in accordance with Table 4.6 and the weldability shall be in accordance with EN 10080.

Table 4.6 Permitted Welding Processes and Examples of Application

Loading Case	Welding Method	Bars in Tension ¹	Bars in Compression ²
Predominantly Static	Flash-welding	Butt joint	
	Manual metal arc welding and metal arc welding with filling electrode	Butt joint with $\phi \geq 20\text{mm}$, splice, lap, cruciform joints ³ , joint with other steel members	
	Metal arc active welding	Splice, lap, cruciform ³ joints & joint with other Steel members	
		-	Butt joint with $\phi \geq 20\text{mm}$
	Friction welding	Butt joint, joint with other steels	
	Resistance spot welding	Lap joint ⁴ Cruciform joint ^{2,4}	
Not Predominantly Static	Flash-welding	Butt welding	
	Manual metal arc welding	-	Butt joint with $\phi \geq 14\text{mm}$
	Metal arc active welding	-	Butt joint with $\phi \geq 14\text{mm}$
	Resistance spot welding	Lap joint ⁴ Cruciform joint ^{2,4}	

Notes:

1. Only bars with approximately the same nominal diameter may be welded together.
2. Permitted ratio of mixed diameter bars ≥ 0.57 .
3. For bearing joints $\phi \leq 16\text{mm}$.
4. For bearing joints $\phi \leq 28\text{mm}$.

1

The strength of the welded joints along the anchorage length of welded fabric shall be sufficient to resist the design forces.

2

The strength of the welded joints of welded fabric may be assumed to be adequate if each welded joint can withstand a shearing force not less than 25% of a force equivalent to the specified characteristic yield stress times the nominal cross-sectional area. This force should be based on the area of the thicker wire if the two are different.

3

4.3.2.6 Fatigue

Recommended values of fatigue stress range with an upper limit of βf_{yk} and for the minimum relative rib area are given below. The value of β may be taken as 0.6.

4

Table 4.7 Properties of Reinforcement

Product Form		Bars and De-coiled Rods			Wire Fabrics			Requirement or Quantile Value (%)
Class		A	B	C	A	B	C	
Fatigue stress range (MPa) (for $N \geq 2 \times 10^6$ cycles) with an upper limit of βf_{yk}		≥ 150			≥ 100			10.0
Bond: Minimum relative rib area, $f_{R,min}$	Nominal bar size (mm)							5.0
	5 - 6	0.035						
	6.5 to 12	0.040						
	>12	0.056						

4.3.2.7 Design Assumptions

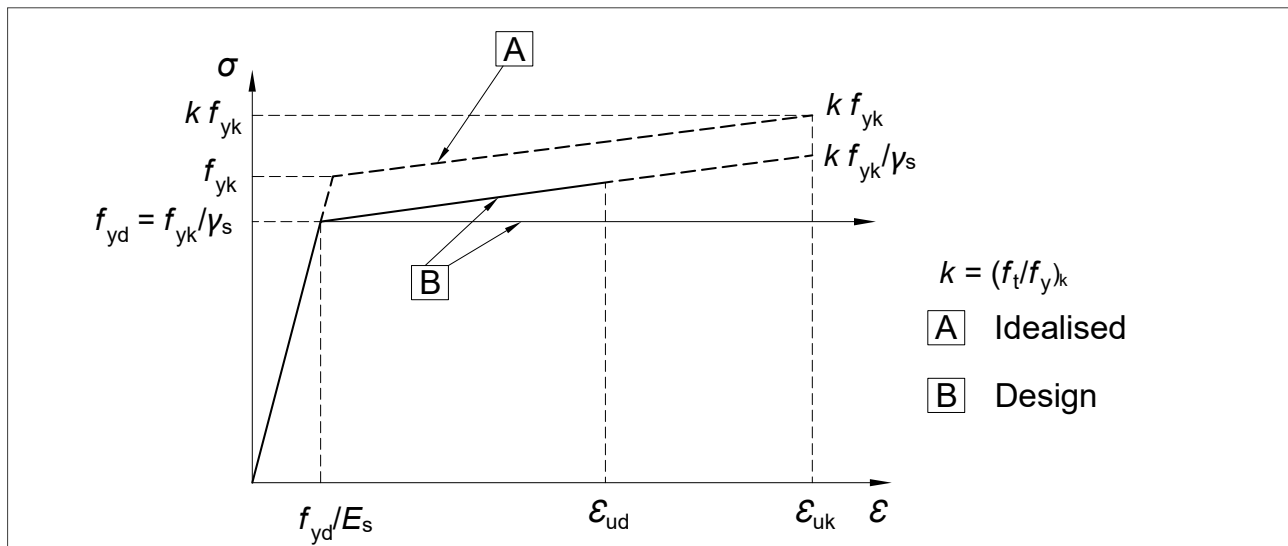
Design should be based on the nominal cross-section area of the reinforcement and the design values derived from the characteristic values. For normal design, either of the following assumptions may be made (shown in Figure 4.8).

- an inclined top branch with a strain limit of ϵ_{ud} and a maximum stress of $k f_{yk} / \gamma_s$ at ϵ_{uk} , where $k = (f_t/f_y)_k$.
- a horizontal top branch without the need to check the strain limit.

The value of ϵ_{ud} may be taken as $0.9 \epsilon_{uk}$.

The value of $(f_t/f_y)_k$ is given in Table 4.5.

Figure 4.8 Idealised and Design Stress-strain Diagrams for Reinforcing Steel (for tension and compression)



For $\varepsilon_s \leq f_{yd} / E_s$ $\sigma_s = E_s \varepsilon_s$

For $\varepsilon_s > f_{yd} / E_s$ $\sigma_s = f_{yd} + (k f_{yd} - f_{yd}) (\varepsilon_s - f_{yd} / E_s) / (\varepsilon_{uk} - f_{yd} / E_s)$

The mean value of density may be assumed to be 7850 kg/m³.

The design value of the modulus of elasticity, E_s may be assumed to be 200 GPa.

4.3.3 Prestressing Steel

4.3.3.1 General

The prestressing steel can be in the form of wires, bars and strands used as prestressing tendons in concrete structures. The tensile strength, 0.1 % proof stress, and elongation at maximum load are specified in terms of characteristic values and these values are designated as f_{pk} , $f_{p0.1k}$ and ε_{uk} respectively.

4.3.3.2 Properties

The properties of prestressing steel are given in EN 10138 and KS ISO 6934. Manufacturer's technical data sheet shall be referred for detail information. The prestressing tendons (wires, strands and bars) shall be classified according to:

- i. Strength, denoting the value of the 0.1 % proof stress ($f_{p0.1k}$) and the value of the ratio of tensile strength to proof strength ($f_p / f_{p0.1k}$) k and elongation at maximum load (ε_{uk})
- ii. Class, indicating the relaxation behaviour
- iii. Size
- iv. Surface characteristics

The three classes of relaxation are defined:

- Class 1: wire or strand - ordinary relaxation
- Class 2: wire or strand - low relaxation
- Class 3: hot rolled and processed bars

The design calculations for the losses due to relaxation of the prestressing steel should be based on the value of ρ_{1000} , the relaxation loss (in %) at 1000 hours after tensioning and at a mean temperature of 20 °C. The value of ρ_{1000} is expressed as a percentage ratio of the initial stress and is obtained for an initial stress equal to $0.7f_{pk}$, where f_{pk} is the characteristic tensile strength. The values for ρ_{1000} can be either assumed equal to 8 % for Class 1, 2.5 % for Class 2, and 4 % for Class 3, or taken from the certificate.

The relaxation loss may be obtained from the manufacturers test certificates or defined as the percentage ratio of the variation of the Prestressing stress over the initial prestressing stress, should be determined by applying one of the Equations below.

For wires or strands for ordinary relaxation:

$$\text{Class 1 } \frac{\Delta\sigma_{pr}}{\sigma_{pi}} = 5.39 \rho_{1000} e^{6.7\mu} \left(\frac{t}{1000} \right)^{0.75(1-\mu)} 10^{-5}$$

Equation 4.24

For wires or strands for low relaxation

$$\text{Class 2 } \frac{\Delta\sigma_{pr}}{\sigma_{pi}} = 0.66 \rho_{1000} e^{9.1\mu} \left(\frac{t}{1000} \right)^{0.75(1-\mu)} 10^{-5}$$

Equation 4.25

For hot rolled and processed bars:

$$\text{Class 3 } \frac{\Delta\sigma_{pr}}{\sigma_{pi}} = 1.98 \rho_{1000} e^{8\mu} \left(\frac{t}{1000} \right)^{0.75(1-\mu)} 10^{-5}$$

Equation 4.26

Where,

$\Delta\sigma_{pr}$ = Absolute value of the relaxation losses of the prestress.

σ_{pi} = Absolute value of the initial prestress.

= σ_{pm0} for post-tensioning.

= Maximum tensile stress applied to the tendon minus the immediate losses occurred during the stressing process for pre-tensioning.

t = The time after tensioning (in hours).

μ = σ_{pi}/f_{pk} , where f_{pk} is the characteristic value of the tensile strength of the prestressing steel.

ρ_{1000} = The value of relaxation loss (in %), at 1000 hours after tensioning and at a mean temperature of 20 °C.

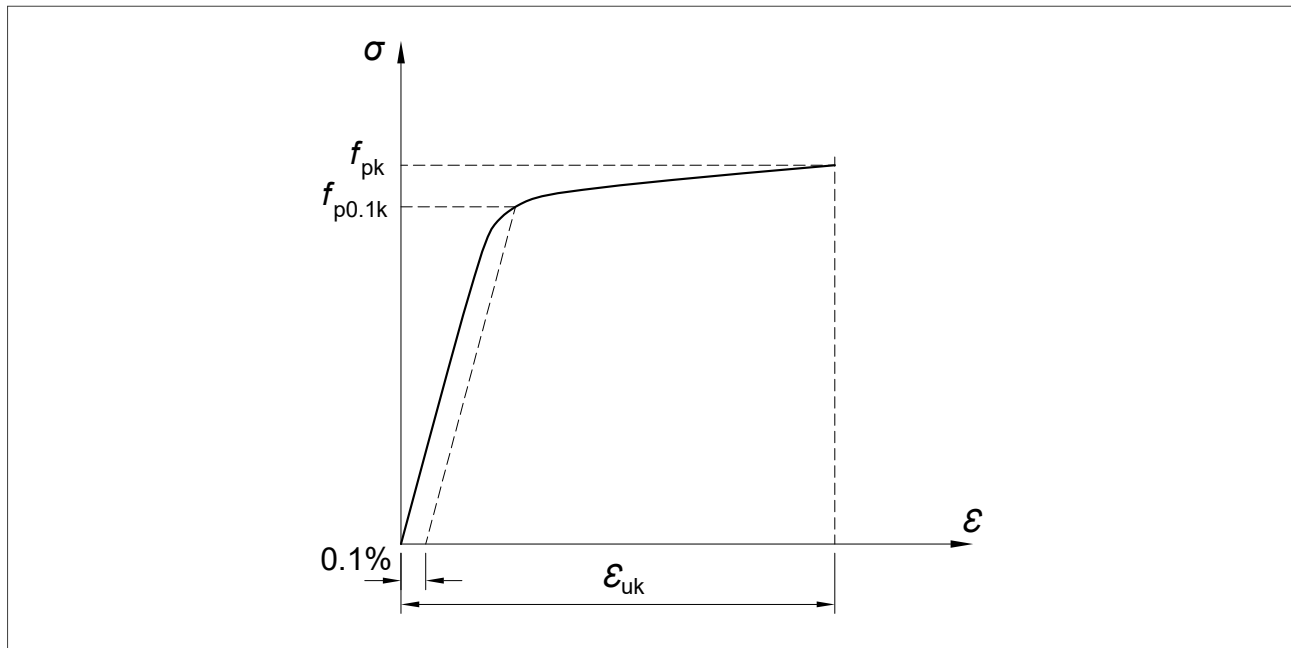
The relaxation losses are calculated for different time intervals/ stages where greater accuracy is required. The long term (final) values of the relaxation losses may be estimated for a time t equal to 500 000 hours (around 57 years).

Relaxation losses are very sensitive to the temperature of the steel. Where this temperature is greater than 50 °C the relaxation losses should be verified.

4.3.3.3 Strength

The 0.1 % proof stress ($f_{p0.1k}$) and the specified value of the tensile strength (f_{pk}) are defined as the characteristic value of the 0.1% proof load and the characteristic maximum load in axial tension respectively, divided by the nominal cross-sectional area as shown in Figure 4.9.

Figure 4.9 Stress-strain Diagram for Typical Prestressing Steel



4.3.3.4 Ductility Characteristics

The prestressing tendons shall have adequate ductility. Adequate ductility in elongation may be assumed if the prestressing tendons obtain the specified value of the elongation at maximum load given in EN 10138.

Adequate ductility in bending may be assumed if the prestressing tendons satisfy the requirements for bendability of EN ISO 15630.

Stress-strain diagrams for the prestressing tendons, based on production data, shall be prepared and made available by the producer.

Adequate ductility in tension may be assumed for the prestressing tendons if $(f_p / f_{p0.1})_k \geq 1.1$.

4.3.3.5 Fatigue

Prestressing tendons shall have adequate fatigue strength. The fatigue stress range for prestressing tendons shall be in accordance with EN 10138.

4.3.3.6 Design Assumptions

Structural analysis is performed on the basis of the nominal cross-section area of the prestressing steel and the characteristic values $f_{p0.1k}$, f_{pk} and ϵ_{uk} .

The design value for the modulus of elasticity E_p for wires and bars may be assumed equal to 205 GPa. The actual value can range from 195 to 210 GPa, depending on the manufacturing process. Producer's Certificates should give the appropriate value.

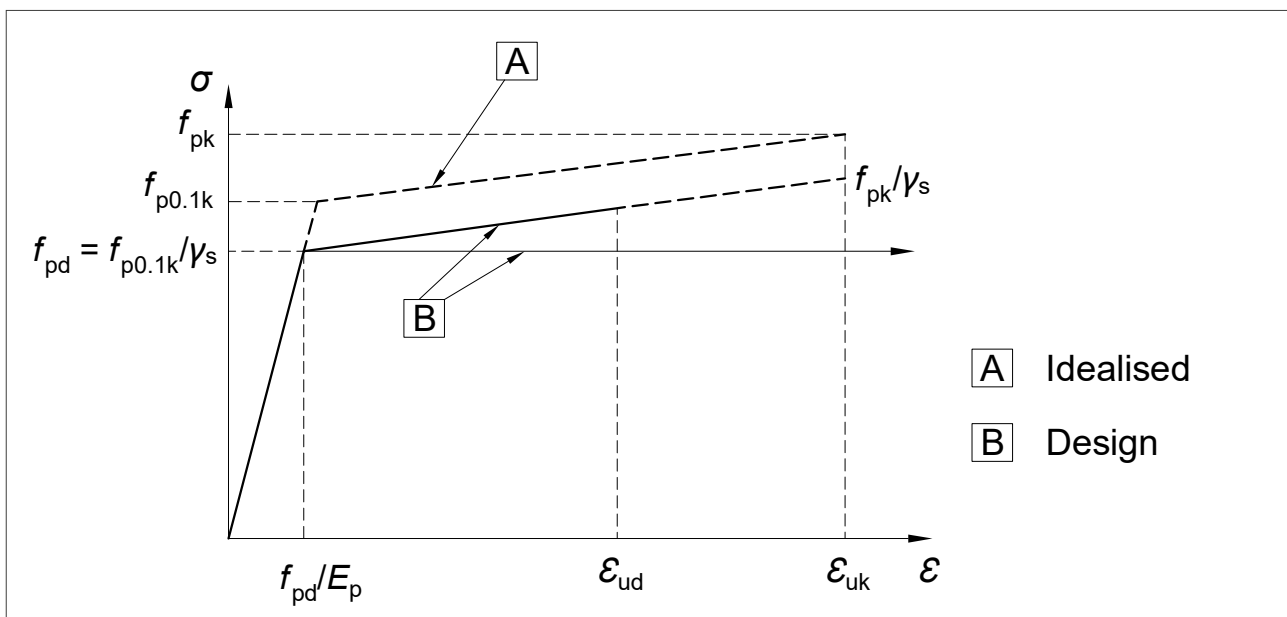
The design value for the modulus of elasticity E_p for strand may be assumed equal to 195 GPa. The actual value can range from 185 GPa to 205 GPa, depending on the manufacturing process. Producer's Certificates should give the appropriate value.

The mean density of prestressing tendons for the purposes of design may normally be taken as 7850 kg/m³.

The values given above may be assumed to be valid within a temperature range between -40 °C and +100 °C for the prestressing steel in the finished structure.

The design value for the steel stress, f_{pd} , is taken as $f_{p0.1k} / \gamma_s$

Figure 4.10 Idealised and Design Stress-strain Diagrams for Prestressing Steel



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For cross-section design, either of the following assumptions may be made:

- an inclined branch, with a strain limit ε_{ud} . The design may also be based on the actual stress/strain relationship, if this is known, with stress above the elastic limit reduced analogously with Figure 4.10, or
- a horizontal top branch without strain limit.

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The recommended value of ε_{ud} is $0.9 \varepsilon_{uk}$. If more accurate values are not known, $\varepsilon_{ud} = 0.02$ and $f_{p0.1k}/f_{pk} = 0.9$ may be taken.

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4.3.3.7 Prestressing Tendons in Sheaths

Prestressing tendons in sheaths (e.g. bonded tendons in ducts, unbonded tendons etc.) shall be adequately and permanently protected against corrosion. Prestressing tendons in sheaths shall be adequately protected against the effects of fire.

4.3.4 Prestressing Devices

4.3.4.1 Anchorages and Couplers

The anchoring devices (anchorages) and coupling devices (couplers) are used for application in post-tensioned construction. The anchorages are used to transmit the forces in tendons to the concrete in the anchorage zone. The couplers are used to connect individual lengths of tendon to make continuous tendons.

4.3.4.1.1 Anchored Tendons

Prestressing tendon anchorage assemblies and prestressing tendon coupler assemblies shall have strength, elongation and fatigue characteristics sufficient to meet the requirements of the design.

4.3.4.1.2 Anchorage Devices and Anchorage Zones

The strength of the anchorage devices and zones shall be sufficient for the transfer of the tendon force to the concrete and the formation of cracks in the anchorage zone shall not impair the function of the anchorage.

4.3.4.2 External Non-bonded Tendons

An external non-bonded tendon is a tendon situated outside the original concrete section and is connected to the structure by anchorages and deviators only.

4.4 Durability and Cover to Reinforcement

4.4.1 General

A durable structure shall meet the requirements of serviceability, strength and stability throughout its design working life, without significant loss of utility or excessive unforeseen maintenance. The required protection of the structure shall be established by considering its intended use, design working life, maintenance programme and actions.

Corrosion protection of steel reinforcement depends on density, quality and thickness of concrete cover and cracking. The cover, density and quality are achieved by controlling the maximum water/cement ratio and minimum cement content and may be related to a minimum strength class of concrete.

Where metal fastenings are inspectable and replaceable, they may be used with protective coatings in exposed situations. Otherwise, they should be of corrosion resistant material.

4.4.2 Environmental Conditions

Exposure conditions are chemical and physical conditions to which the structure is exposed in addition to the mechanical actions. Environmental conditions are classified according to Table 4.8.

Table 4.8 Exposure Classes Related to Environmental Conditions

Class Designation	Description of Environment	Informative examples where exposure classes may occur
1. No risk of corrosion or attack (X0 class)		
X0	- For concrete without reinforcement or embedded metal: all exposures except where there is freeze/thaw, abrasion or chemical attack. - For concrete with reinforcement or embedded metal: very dry	- Unreinforced concrete surfaces inside structures - Unreinforced concrete completely buried in soil classed as AC-1 and with a hydraulic gradient not greater than 5. - Unreinforced concrete permanently submerged in non-aggressive water - Unreinforced concrete surfaces in cyclic wet and dry conditions not subject to abrasion, freezing or chemical attack - Reinforced concrete surfaces exposed to very dry conditions
2. Corrosion induced by carbonation (XC classes) (Where concrete containing reinforcement or other embedded metal is exposed to air and moisture)		
XC1	Dry or permanently wet	- Reinforced and prestressed concrete surfaces inside enclosed structures except voided superstructures and areas of structures with high humidity - Reinforced and prestressed concrete surfaces permanently submerged in non-aggressive water
XC2	Wet, rarely dry	Reinforced and prestressed concrete surfaces permanently in contact with soil not containing chlorides
XC3	Moderate humidity	- External reinforced and prestressed concrete surfaces sheltered from, or exposed to, direct rain - Reinforced and prestressed concrete surfaces subject to high humidity (e.g. poorly ventilated bathrooms, kitchens) - Reinforced and prestressed concrete surfaces exposed to alternate wetting and drying - Interior concrete surfaces of pedestrian subways not subject to de-icing salts, voided superstructures or cellular abutments - Reinforced or prestressed concrete surfaces protected by waterproofing
XC4	Cyclic wet and dry	
3. Corrosion induced by chlorides other than from sea water (XD classes) (Where concrete containing reinforcement or other embedded metal is subject to contact with water containing chlorides, including de-icing salts, from sources other than from sea water)		
XD1	Moderate humidity	- Concrete surfaces exposed to airborne chlorides - Reinforced and prestressed concrete wall and structure supports more than 10 m horizontally from a carriageway - Bridge deck soffits more than 5 m vertically above the carriageway - Parts of structures exposed to occasional or slight chloride conditions
XD2	Wet, rarely dry	- Reinforced and prestressed concrete surfaces totally immersed in water containing chlorides - Buried highway structures more than 1 m below adjacent carriageway
XD3	Cyclic wet and dry	- Reinforced and prestressed concrete walls and structure supports within 10 m of a carriageway - Bridge parapet edge beams - Buried highway structures less than 1 m below carriageway level - Reinforced pavements and car park slabs

Table continues on the next page...

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Class Designation	Description of Environment	Informative examples where exposure classes may occur
4. Corrosion induced by chlorides from sea water (XS classes) (Where concrete containing reinforcement or other embedded metal is subject to contact with sea water or airborne salt originating from sea water)		
XS1	Exposed to airborne salt but not in direct contact with sea water	External reinforced and prestressed concrete surfaces in coastal areas
XS2	Permanently submerged	Reinforced and prestressed concrete surfaces completely submerged and remaining saturated, e.g. concrete below mid-tide level
XS3	Tidal, splash and spray zones	Reinforced and prestressed concrete surfaces in the upper tidal zones and the splash and spray zones including exposed soffits above sea water
5. Freeze/Thaw attack (XF classes)		
XF1	Moderate water saturation, without de-icing agent	- Vertical concrete surfaces as facades and columns exposed to rain and freezing - Non-vertical concrete surfaces not highly saturated, but exposed to freezing and to rain or water
XF2	Moderate water saturation, with de-icing agent	Concrete surfaces such as parts of bridges, which would otherwise be classified as XF1, but which are exposed to de-icing salts either directly or as spray or run-off
XF3	High water saturation, without de-icing agent	- Horizontal or nearly horizontal concrete surfaces, which are exposed to freezing whilst wet - Concrete surfaces subjected to frequent splashing with water and exposed to freezing
XF4	High water saturation, with de-icing agent or sea water	- Horizontal concrete surfaces, such as roads and pavements, exposed to freezing and to de-icing salts either directly or as spray or run-off - Concrete surfaces subjected to frequent splashing with water containing de-icing agents and exposed to freezing
6. Chemical attack (XA classes) Where concrete is exposed to chemical attack (refer BS 8500-1)		
XA1	Slightly aggressive chemical environment	Concrete exposed to natural and ground water according to BS EN 206. These European classes are not used in UK, instead Table A.2 of BS 8500-1 is used to determine the aggressive chemical environment for concrete (ACEC) exposure class
XA2	Moderate aggressive chemical environment	
XA3	Highly aggressive chemical environment	

In addition to the conditions in Table 4.8, particular forms of aggressive or indirect action should be considered including:

Chemical attack, arising from e.g.

- The use of the structure (storage of liquids, etc)
- Solutions of acids or sulphate salts
- Chlorides contained in the concrete
- Alkali-aggregate reactions

Physical attack, arising from e.g.

- Temperature change
- Abrasion
- Water penetration

The composition of the concrete affects both the protection of the reinforcement and the resistance of the concrete to attack. This may lead to the choice of higher strength classes than required for the structural design. In such cases the value of f_{ctm} should be associated with the higher strength in the calculation of minimum reinforcement and crack width control.

There are four environmental classes related to corrosion of reinforcement: X0, XC, XD and XS according to Table 4.8. For reinforced concrete, one of these exposure classes shall be identified. For reinforced concrete at least XC1 class shall be used. Water penetration or the possibility of leakage from the carriageway into the inside of voided structures should be considered in the design. For a concrete surface protected by waterproofing, the exposure class can be considered as XC3.

4.4.3 Requirements for Durability

In order to achieve the required design working life of the structure, adequate measures shall be taken to protect each structural element against the relevant environmental actions. Following provisions shall be considered to meet the requirements for durability.

- Structural conception.
- Material selection.
- Construction details.
- Execution.
- Quality Control.
- Inspection.
- Verifications.
- Special measures (e.g. use of stainless steel, coatings, cathodic protection).

4.4.4 Concrete Cover

The concrete cover is the distance between the surface of the reinforcement closest to the nearest concrete surface (including links and stirrups and surface reinforcement where relevant) and the nearest concrete surface.

4.4.4.1 Nominal Cover

The nominal cover c_{nom} shall be specified on drawings. It is given as minimum cover c_{min} plus allowance in design for deviation Δc_{dev} as follows.

$$c_{nom} = c_{min} + \Delta c_{dev}$$

The nominal cover should be at least 40 mm for concrete cast against prepared ground (including blinding) and 75 mm for concrete cast directly against soil.

4.4.4.2 Minimum Cover

The minimum cover shall be provided in order to ensure safe transmission of bond forces and protection of the steel against corrosion. The minimum cover shall be greater of the minimum cover required for bond strength ($c_{min,b}$), environmental conditions ($c_{min,dur}$), 10 mm.

Minimum cover for bond requirement of reinforcement and prestressing steel shall not be less than $c_{min,b}$ is provided in following Table.

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Table 4.9 Minimum Cover, $c_{min,b}$, for Bond Requirement of Reinforcing Steel

Arrangement and Type of Reinforcement	Minimum cover for nominal maximum aggregate size	
	$\leq 32\text{mm}$	$> 32\text{mm}$
Individual bars	Diameter of bar	Diameter of bar + 5 mm
Bundled bars	Equivalent diameter	Equivalent diameter + 5 mm
Post-tensioned circular ducts	Diameter of duct but not more than 80mm	
Post-tensioned rectangular ducts	Greater of the smaller dimension or half the greater dimension but not more than 80mm	
Pre-tensioned strand or plain wire	1.5 x diameter of strand or plain wire	
Pre-tensioned indented wire	2.5 x diameter of indented wire	

For durability requirements, the minimum cover values for reinforcement and prestressing tendons in normal weight concrete taking account of the exposure classes and the structural classes is given by $c_{min,dur}$. The concrete composition and the recommended values for minimum cover required for durability in different environmental conditions are provided in

Table 4.10 for 50 years design working life and Table 4.11 for 100 years design working life. These tables give nominal cover to reinforcement in the form of minimum cover for durability plus allowance in design for deviation i.e. $c_{min,dur} + \Delta c_{dev}$.

Table 4.10 Durability Recommendations for Reinforced or Prestressed Concrete Elements with an Intended Working Life of at Least 50 Years

Nominal cover mm	Compressive strength class, maximum water-cement ratio and minimum cement or combination content for normal-weight concrete with 20 mm maximum aggregate size										Cement/ Combination Type*
	15 + Δc_{dev}	20 + Δc_{dev}	25 + Δc_{dev}	30 + Δc_{dev}	35 + Δc_{dev}	40 + Δc_{dev}	45 + Δc_{dev}	50 + Δc_{dev}	60 + Δc_{dev}	70 + Δc_{dev}	
Corrosion induced by carbonation (XC exposure classes)											
XC ₁	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	All
	—	—	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	All
XC ₂	—	—	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	All
	—	C40/50 0.45 340	C32/40 0.55 300	C28/35 0.60 280	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	All except IVB P, IVB Q, IVB V
XC _{3/4}	—	—	C40/50 0.45 340	C32/40 0.55 300	C28/35 0.60 280	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	IVB P, IVB Q, IVB V
	Corrosion induced by chlorides other than sea water (XD exposure classes) adequate for any associated carbonation induced corrosion (XC classes)										
XD ₁	—	—	C40/50 0.45 360	C32/40 0.55 320	C28/35 0.60 300	C28/35 0.60 300	C28/35 0.60 300	C28/35 0.60 300	C28/35 0.60 300	C28/35 0.60 300	All
	—	—	—	C40/50 0.40 380	C32/40 0.50 340	C28/35 0.60 300	C28/35 0.60 300	C28/35 0.60 300	C28/35 0.60 300	C28/35 0.60 300	CEM I, IIA, IIB M, IIB S, CEM I SR0, CEM I SR3
XD ₂	—	—	—	C35/45 0.40 380	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIA
	—	—	—	C32/40 0.40 380	C25/30 0.50 340	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	IIIB, IVB P, IVB Q, IVB V
XD ₃	—	—	—	—	—	C45/55 0.35 380	C40/50 0.40 380	C35/45 0.45 360	C35/45 0.45 360	C35/45 0.45 360	CEM I, IIA, IIB M, IIB S, CEM I SR0, CEM I SR3
	—	—	—	—	—	C35/45 0.40 380	C32/40 0.45 360	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIA
	—	—	—	—	—	C32/40 0.40 380	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIA
	—	—	—	—	—	C32/40 0.40 380	C28/35 0.45 360	C25/30 0.50 340	C25/30 0.55 340	C25/30 0.50 340	IIIB, IVB P, IVB Q, IVB V

Table continues on next page...

Notes: A dash (—) indicates that greater cover is recommended.

* For Cement/ combination types, refer BS 8500-1:2015+A2:2019, Table A.6

Table 4.10 Durability Recommendations for Reinforced or Prestressed Concrete Elements with an Intended Working Life of at Least 50 Years (*continued*)

Nominal cover mm	Compressive strength class, maximum water-cement ratio and minimum cement or combination content for normal-weight concrete with 20 mm maximum aggregate size											Cement/ Combination Type*
	30 + Δc_{lev}	35 + Δc_{lev}	40 + Δc_{lev}	45 + Δc_{lev}	50 + Δc_{lev}	55 + Δc_{lev}	60 + Δc_{lev}	65 + Δc_{lev}	70 + Δc_{lev}	75 + Δc_{lev}	80 + Δc_{lev}	
Corrosion induced by chlorides from sea water (XS exposure classes) adequate for any associated carbonation induced corrosion (XC)												
XS ₁	—	—	—	C45/55 0.35 380	C40/50 0.40 380	C35/45 0.45 360	C32/40 0.50 340	C28/35 0.55 320	C28/35 0.55 320	C28/35 0.55 320	C28/35 0.55 320	CEM I, IIA, IIB M, IIB S
	C40/50 0.35 380	C32/40 0.45 360	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIA
	C35/45 0.40 380	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P or Q ≥25% pozzolana, IIB V ≥25% fly ash, IIIA ≥46% ggbs
	C32/40 0.40 380	C25/30 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IVB P, IVB Q, IVB V, IIIB
XS ₂	—	—	—	—	—	—	C45/55 0.35 380	C40/50 0.40 380	C40/50 0.40 380	C35/45 0.45 360	C32/40 0.50 340	CEM I, IIA, IIB M, IIB S
	—	—	C35/45 0.40 380	C32/40 0.45 360	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIA
	—	C40/50 0.35 380	C32/40 0.45 360	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P or Q ≥25% pozzolana, IIB V ≥25% fly ash, IIIA ≥46% ggbs
	—	C35/45 0.35 380	C28/35 0.45 360	C25/30 0.50 340	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	IVB P, IVB Q, IVB V, IIIB
XS ₃	—	—	—	—	—	—	—	—	—	—	C45/55 0.35 380	CEM I, IIA, IIB M, IIB S
	—	—	—	—	C40/50 0.35 380	C35/45 0.40 380	C32/40 0.45 360	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIA
	—	—	—	C40/50 0.35 380	C32/40 0.45 360	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P or Q ≥25% pozzolana, IIB V ≥25% fly ash, IIIA ≥46% ggbs
	—	—	—	C35/45 0.35 380	C28/35 0.45 360	C25/30 0.50 340	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	IVB P, IVB Q, IVB V, IIIB

Notes: A dash (—) indicates that greater cover is recommended.

* For Cement/ combination types, refer BS 8500-1:2015+A2:2019, Table A.6

Table 4.11 Durability Recommendations for Reinforced or Prestressed Concrete Elements with an Intended Working Life of at Least 100 Years

Nominal cover mm	Compressive strength class, maximum w/c ratio and minimum cement or combination content for normal weight concrete with 20 mm maximum aggregate size												Cement/ Combination Types*
	15 + $\Delta f_{c,dev}$	20 + $\Delta f_{c,dev}$	25 + $\Delta f_{c,dev}$	30 + $\Delta f_{c,dev}$	35 + $\Delta f_{c,dev}$	40 + $\Delta f_{c,dev}$	45 + $\Delta f_{c,dev}$	50 + $\Delta f_{c,dev}$	55 + $\Delta f_{c,dev}$	60 + $\Delta f_{c,dev}$	70 + $\Delta f_{c,dev}$	80 + $\Delta f_{c,dev}$	
Corrosion induced by carbonation (XC exposure classes)													
XC ₁	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	All
XC ₂	—	—	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	All
XC _{3/4}	—	—	—	C40/50 0.45 340	C35/45 0.50 320	C32/40 0.55 300	C28/35 0.60 280	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	All except I/B P, I/B Q, I/B V
	—	—	—	C40/50 0.45 340	C35/45 0.50 320	C32/40 0.55 300	C28/35 0.60 280	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	C25/30 0.65 260	I/B P, I/B Q, I/B V
Corrosion induced by chlorides other than sea water (XD exposure classes) adequate for any associated carbonation induced corrosion (XC)													
XD ₁	—	—	—	C45/55 0.40 380	C40/50 0.45 360	C35/45 0.50 340	C32/40 0.55 320	C28/35 0.60 300	C28/35 0.60 300	C28/35 0.60 300	C28/35 0.60 300	C28/35 0.60 300	All
XD ₂	—	—	—	—	—	C35/45 0.45 360	C32/40 0.50 340	C28/35 0.55 320	C28/35 0.55 320	C28/35 0.55 320	C28/35 0.55 320	C28/35 0.55 320	CEM I, IIA, IIB M, IIB S, CEM I SR0, CEM I SR3
	—	—	—	—	—	C32/40 0.45 360	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIA
XD ₃	—	—	—	—	—	C28/35 0.45 360	C25/30 0.50 340	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	III B, I/B P, I/B Q, I/B V
	—	—	—	—	—	—	—	—	—	—	—	—	CEM I, IIA, IIB M, IIB S, CEM I SR0, CEM I SR3
XD ₃	—	—	—	—	—	—	—	—	—	—	—	—	IIB P, IIB Q, IIB V, IIIA
	—	—	—	—	—	—	—	—	—	—	—	—	IIB P, IIB Q, IIB V, IIIA
XD ₃	—	—	—	—	—	—	—	—	—	—	—	—	III B, I/B P, I/B Q, I/B V
	—	—	—	—	—	—	—	—	—	—	—	—	III B, I/B P, I/B Q, I/B V

Table 4.11 Durability Recommendations for Reinforced or Prestressed Concrete Elements with an Intended Working Life of at Least 100 Years (*continued*)

Nominal cover mm		Compressive strength class, maximum w/c ratio and minimum cement or combination content for normal weight concrete with 20 mm maximum aggregate size												Cement/ Combination Types*	
		25 + ΔC_{lev}	30 + ΔC_{lev}	35 + ΔC_{lev}	40 + ΔC_{lev}	45 + ΔC_{lev}	50 + ΔC_{lev}	55 + ΔC_{lev}	60 + ΔC_{lev}	65 + ΔC_{lev}	70 + ΔC_{lev}	75 + ΔC_{lev}	80 + ΔC_{lev}		
Corrosion induced by chlorides from sea water (XS exposure classes) adequate for any associated carbonation induced corrosion (XC)															
XD ₁	—	—	—	—	—	—	—	—	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	C20/25 0.70 240	CEM I, IIA, IIB M, IIB S		
	—	—	C40/50 0.35 380	C40/50 0.35 380	C35/45 0.40 380	C32/40 0.45 360	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIA		
	—	—	—	C35/45 0.40 380	C32/40 0.45 360	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P or Q $\geq 25\%$ pozzolana, IIB V $\geq 25\%$ fly ash, IIIA $\geq 46\%$ ggbs		
	—	—	—	C32/40 0.45 380	C28/35 0.45 360	C25/30 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIIB		
XD ₂	—	—	—	C35/45 0.40 380	C30/37 0.45 360	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	C25/30 0.55 320	IIB P, IIB Q, IIB V		
	—	—	—	—	—	—	—	—	—	—	—	—	CEM I, IIA, IIB M, IIB S		
	—	—	—	—	—	C40/50 0.35 380	C35/45 0.40 380	C32/40 0.45 360	C28/35 0.50 340	C28/35 0.50 340	C25/30 0.55 320	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIA		
	—	—	—	—	—	C40/50 0.35 380	C35/45 0.40 380	C32/40 0.45 360	C28/35 0.50 340	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	IIB P or Q $\geq 25\%$ pozzolana, IIB V $\geq 25\%$ fly ash, IIIA $\geq 46\%$ ggbs		
XD ₃	—	—	—	—	—	C35/45 0.35 380	C32/40 0.40 380	C28/35 0.45 360	C25/30 0.50 340	C20/25 0.55 320	C20/25 0.55 320	C20/25 0.55 320	IIB P, IIB Q, IIB V, IIIB		
	—	—	—	—	—	—	—	—	—	—	—	—	CEM I, IIA, IIB M, IIB S		
	—	—	—	—	—	—	—	—	—	—	—	—	IIB P, IIB Q, IIB V, IIIA		
	—	—	—	—	—	—	—	—	—	—	—	—	IIB P or Q $\geq 25\%$ pozzolana, IIB V $\geq 25\%$ fly ash, IIIA $\geq 46\%$ ggbs		
XD ₃	—	—	—	—	—	—	—	—	C35/45 0.35 380	C32/40 0.40 380	C28/35 0.45 360	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIB		
	—	—	—	—	—	—	—	—	C35/45 0.35 380	C32/40 0.40 380	C28/35 0.45 360	C25/30 0.55 320	IIB P or Q $\geq 25\%$ pozzolana, IIB V $\geq 25\%$ fly ash, IIIA $\geq 46\%$ ggbs		
	—	—	—	—	—	—	—	—	C35/45 0.35 380	C32/40 0.40 380	C28/35 0.45 360	C25/30 0.55 320	IIB P, IIB Q, IIB V, IIIB		
	—	—	—	—	—	—	—	—	C35/45 0.35 380	C32/40 0.40 380	C28/35 0.45 360	C25/30 0.55 320	IIB P or Q $\geq 25\%$ pozzolana, IIB V $\geq 25\%$ fly ash, IIIA $\geq 46\%$ ggbs		

Notes: A dash (—) indicates that greater cover is recommended.

* For Cement/ combination types, refer BS 8500-1:2015+A2:2019, Table A.6

Where in-situ concrete is placed against an existing concrete surface (precast or in-situ) the requirements for cover to the reinforcement from the interface may be modified. If following conditions are met, the minimum cover satisfying the bond requirements can be provided.

- The existing concrete surface has not been subject to an outdoor environment for more than 28 days.
- The existing concrete surface is rough.
- The strength class of the existing concrete is at least C25/30.

For uneven surfaces (e.g. exposed aggregate) the minimum cover should be increased by at least 5 mm. Where chemical attack on concrete (Classes XF and XA) is expected, special attention should be given to the concrete composition. For the concrete members subject to abrasion, the minimum cover should be increased as below.

Table 4.12 Increase on Minimum Cover Due to Abrasion

Abrasion Class	Increase on Minimum Cover
XM1 (moderate abrasion like for members of industrial sites frequented by vehicles with air tyres)	5 mm
XM2 (heavy abrasion like for members of industrial sites frequented by fork lifts with air or solid rubber tyres)	10 mm
XM3 (extreme abrasion like for members industrial sites frequented by fork lifts with elastomer or steel tyres or track vehicles)	15 mm

The bare concrete decks of road bridges, without waterproofing or surfacing, should be classified as Abrasion Class XM2. Where a concrete surface is subject to abrasion caused by solid transportation in running water, the cover should be increased by a minimum of 10 mm.

4.4.4.3 Allowance in Design for Deviation

To calculate the nominal cover, c_{nom} , an addition to the minimum cover shall be made in design to allow for the deviation (Δc_{dev}). The value of allowance in design for deviation is in the range from 5 mm to 15 mm and may be taken as 10 mm. Where construction is subjected to quality assurance system in which the monitoring studies includes measurement of the concrete cover, the allowance in design for deviation may be reduced as $10 \text{ mm} \geq \Delta c_{dev} \geq 5 \text{ mm}$.

For concrete cast against uneven surfaces, the nominal cover should be increased complying with the difference cause by unevenness by allowing larger deviations.

4.5 Structural Analysis

4.5.1 General

Structural analysis is carried out to establish the distribution of either internal forces and moments, or stresses, strains and displacements, over the whole or part of a structure. Additional local analysis shall be carried out where necessary. When designing the bridge deck slab, the local effects due to wheel loads shall be considered in addition to global effects.

4.5.2 Geometric Imperfections

The unfavourable effects of possible deviations in the geometry of the structure and the position of loads shall be considered in the analysis of members and structures. Imperfections shall be considered in ultimate limit states in persistent and accidental design situations. Imperfections need not be considered for serviceability limit states.

1

The following provisions apply to members with axial compression and structures with vertical load. Imperfections may be represented by an inclination θ_i given by

$$\theta_i = \theta_0 \alpha_h$$

2

Where,

θ_0 = The basis value, recommended as 1/200

α_h = The reduction factor for length or height, $\alpha_h = 2/\sqrt{l}$; $\alpha_h \leq 1$

3

l = The length or height (m)

4

For isolated members, the effect of imperfections may be taken into account in two alternative ways a) or b):

a. As an eccentricity, e_i , given by

$$e_i = \theta_i l_0 / 2$$

Where, l_0 is the effective length

For walls and isolated columns in braced systems, $e_i = l_0 / 400$ may always be used as a simplification, corresponding to $\alpha_h = 1$.

b. As a transverse force, H_i in the position that gives maximum moment:

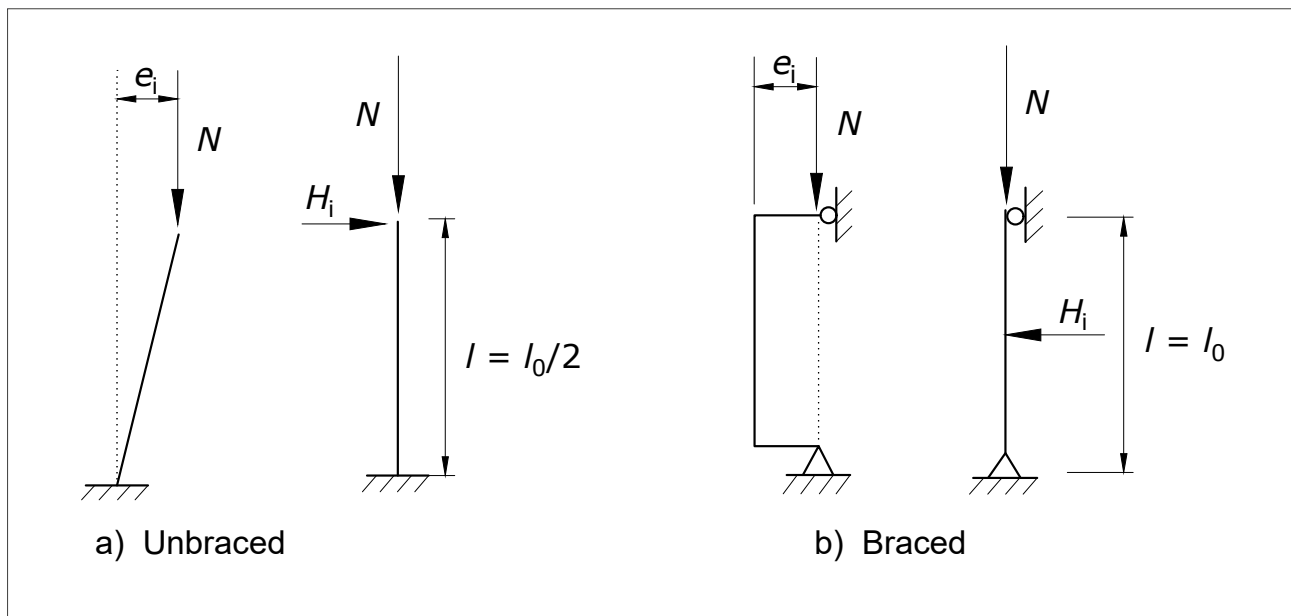
For unbraced members: $H_i = \theta_i N$

For braced members: $H_i = 2 \theta_i N$

Where, N is the axial load.

Eccentricity is suitable for statically determinate members, whereas transverse load can be used for both determinate and indeterminate members.

Figure 4.11 Geometric Imperfections for Isolated Members



4.5.3 Idealisation of the Structure

4.5.3.1 Structural Models

The elements of a structure are classified, by consideration of their nature and function, as beams, columns, slabs, walls, etc.

Beam: A beam is a member for which the span is not less than 3 times the overall section depth. Otherwise, it should be considered as a deep beam.

Slab: A slab is a member for which the minimum panel dimension is not less than 5 times the overall slab thickness. A slab subjected to dominantly uniformly distributed loads may be considered one way spanning if either:

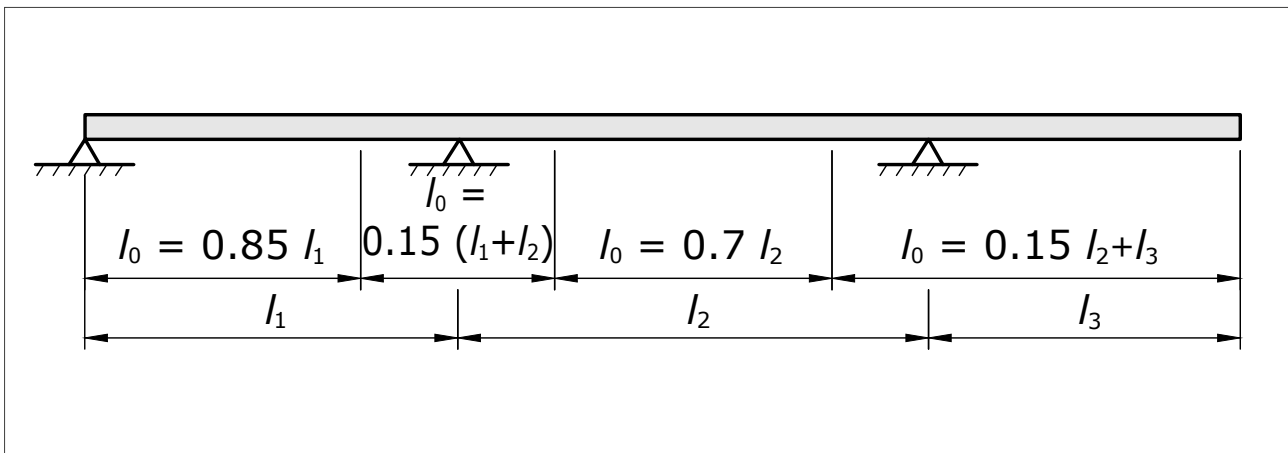
- it possesses two free (unsupported) and sensibly parallel edges, or
- it is the central part of a sensibly rectangular slab supported on four edges with a ratio of the longer to shorter span greater than 2.

Column: A column is a member for which the section depth does not exceed 4 times its width and the height is at least 3 times the section depth. Otherwise, it should be considered as a wall.

4.5.3.2 Effective Width of Flanges

In T beams, the effective flange width over which uniform conditions of stress can be assumed, depends on the web and flange dimensions, the type of loading, the span, the support conditions and the transverse reinforcement. The effective width of flange should be based on the distance l_0 between points of zero moment.

Figure 4.12 Definition of l_0 , for Calculation of Effective Flange Width



The length of the cantilever, l_3 , should be less than half the adjacent span and the ratio of adjacent spans should lie between 2/3 and 1.5.

The effective flange width b_{eff} for a T beam or L beam may be derived as:

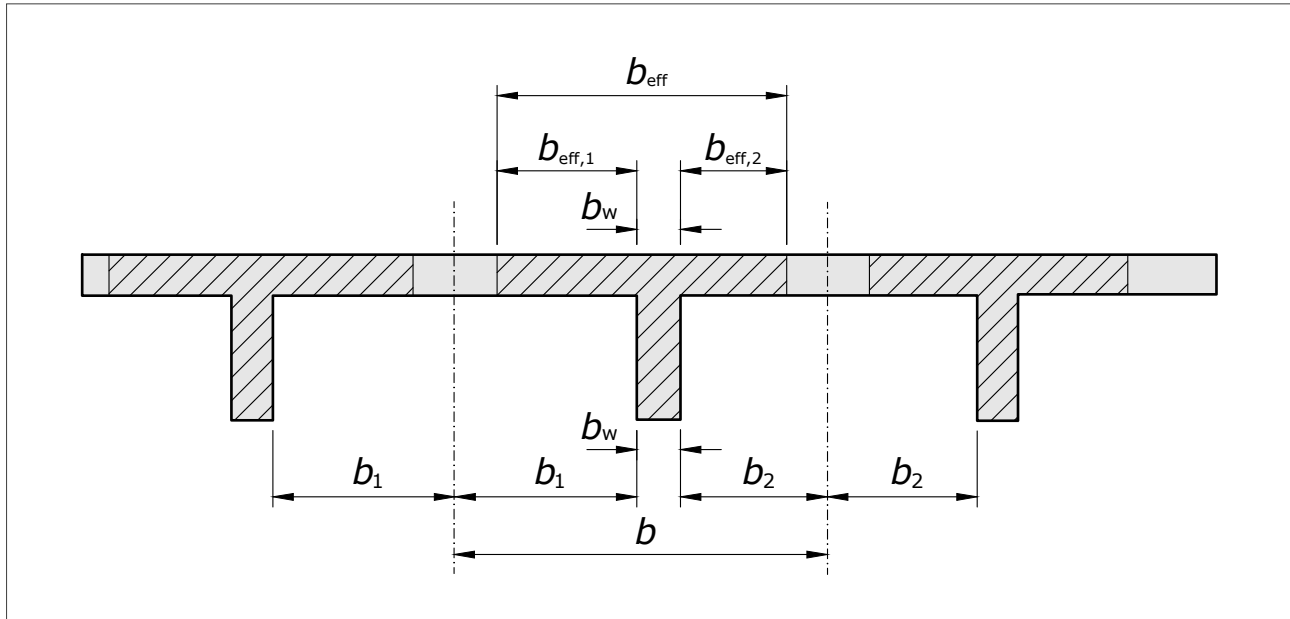
$$b_{eff} = b_w + b_{eff,1} + b_{eff,2} \leq b$$

Where,

$$b_{eff,1} = (0.2 b_1 + 0.1 l_0) \leq 0.2 l_0 \text{ and } \leq b_1$$

$$b_{eff,2} = (0.2 b_2 + 0.1 l_0) \leq 0.2 l_0 \text{ and } \leq b_2$$

Figure 4.13 Effective Flange Width Parameters



For structural analysis, where a great accuracy is not required, a constant width may be assumed over the whole span. The value applicable to the span section should be adopted.

4.5.3.3 Effective Span of Beams and Slabs

The effective span, l_{eff} of a member should be calculated as follows:

$$l_{eff} = l_n + a_1 + a_2$$

Where,

l_n = The clear distance between the faces of the supports.

Values for $a_1 + a_2$ at each end of the span, may be determined from the appropriate a_i values in Figure 4.14 where t is the width of the supporting element as shown.

Continuous slabs and beams may generally be analysed on the assumption that the supports provide no rotational restraint. Where a beam or slab is monolithic with its supports, the critical design moment at the support should be taken as that at the face of the support. The design moment and reaction transferred to the supporting element (e.g. column, wall, etc.) should be generally taken as the greater of the elastic or redistributed values. The moment at the face of the support should not be less than 0.65 that of the full fixed end moment.

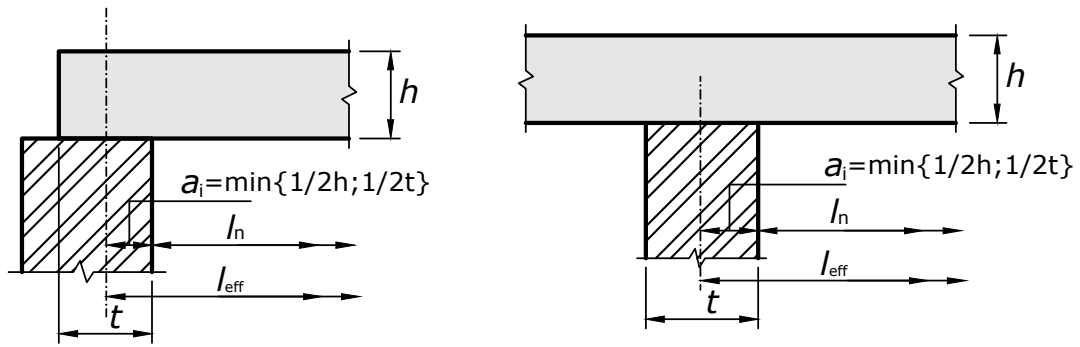
Where a beam or slab is continuous over a support which may be considered to provide no restraint to rotation (e.g. over walls) and the analysis assumes point support, the design support moment, calculated on the basis of a span equal to the centre-centre distance between supports, may be reduced by an amount ΔM_{Ed} as follows:

$$\Delta M_{Ed} = F_{Ed,sup} t / 8$$

Where,

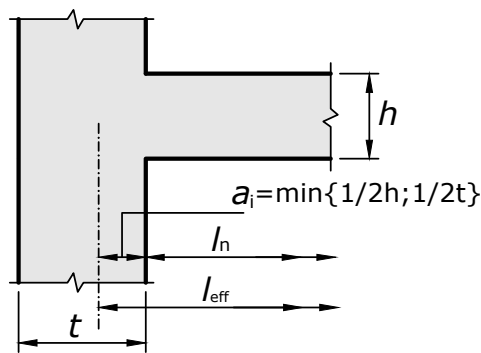
$F_{Ed,sup}$ = The design support reaction.

t = The breadth of the bearing.

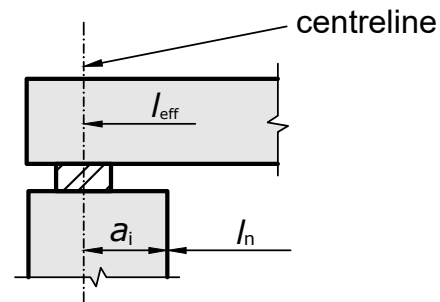
Figure 4.14 Effective Span (l_{eff}) For Different Support Conditions

(a) Non-continuous members

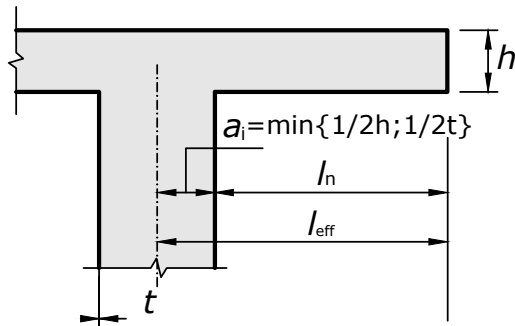
(b) Continuous members



(c) Supports considered fully restrained



(d) Bearing provided



(e) Cantilever

4.5.4 Linear Elastic Analysis

Linear analysis of elements based on the theory of elasticity may be used for both the serviceability and ultimate limit states. For the determination of the action effects, linear analysis may be carried out assuming:

- Uncracked cross sections.
- Linear stress-strain relationships.
- Mean value of the modulus of elasticity.

1

2

3

4

Concrete Structures

For thermal deformation, settlement and shrinkage effects at the ultimate limit state (ULS), a reduced stiffness corresponding to the cracked sections, neglecting tension stiffening but including the effects of creep, may be assumed. For the serviceability limit state (SLS) a gradual evolution of cracking should be considered.

4.5.5 Linear Elastic Analysis with Limited Redistribution

Linear analysis with limited redistribution may be applied to the analysis of structural members for the verification of ULS. The moments at ULS calculated using a linear elastic analysis may be redistributed, provided that the resulting distribution of moments remains in equilibrium with the applied loads.

In continuous beams or slabs which:

- a. Are predominantly subject to flexure; and
- b. Have the ratio of the lengths of adjacent spans in the range of 0.5 to 2.

Redistribution of bending moments may be carried out without explicit check on the rotation capacity, provided that:

$$\delta \geq 0.44 + 1.25 (0.6 + 0.0014 / \varepsilon_{cu2}) x_u / d \quad \text{for } f_{ck} \leq 50 \text{ MPa}$$

$$\delta \geq 0.54 + 1.25 (0.6 + 0.0014 / \varepsilon_{cu2}) x_u / d \quad \text{for } f_{ck} > 50 \text{ MPa}$$

$$\delta \geq 0.85 \text{ where Class B and Class C reinforcement is used}$$

No redistribution is allowed for Class A steel

Where,

δ = The ratio of the redistributed moment to the elastic bending moment.

x_u = The depth of the neutral axis at the ultimate limit state after redistribution.

d = The effective depth of the section.

Redistribution should not be carried out in circumstances where the rotation capacity cannot be defined with confidence (e.g. in curved and or skewed bridges). For the design of columns, the elastic moments from frame action should be used without any redistribution.

If a linear elastic analysis with limited redistribution is undertaken, the shears and reactions used in the design should be taken greater of those either prior to redistribution or after redistribution.

4.5.6 Plastic Analysis

4.5.6.1 General

Methods based on plastic analysis should not be used in design of bridges except for the case of analysing response to earthquake at ULS.

The plastic analysis may be based on the lower bound (static) method or upper bound (kinematic) methods. The ductility of the critical sections shall be sufficient for the envisaged mechanism to be formed.

4.5.6.2 Plastic Analysis for Beams, Frames and Slabs

Plastic analysis without any direct check of rotation capacity may be used for the ultimate limit state if the following conditions of required ductility are met. The required ductility may be deemed to be satisfied if all the following are fulfilled:

- i. The area of tensile reinforcement is limited such that, at any section:
 - $x_u/d \leq 0.15$ for concrete strength classes $\leq C50/60$.
 - $x_u/d \leq 0.10$ for concrete strength classes $\geq C55/67$.
- ii. Reinforcement class is either Class B or C.
- iii. The ratio of the moments at intermediate supports to the moments in the span is between 0.5 and 2.

Columns should be checked for the maximum plastic moments which can be transmitted by connecting members. For connections to flat slabs this moment should be included in the punching shear calculation.

When plastic analysis of slabs is carried out, account should be taken of any non-uniform reinforcement, corner tie down forces, and torsion at free edges.

Plastic methods may be extended to non-solid slabs (ribbed, hollow, waffle slabs) if their response is similar to that of a solid slab, particularly with regard to the torsional effects.

4.5.6.3 Analysis with Strut and Tie Models

Strut and tie models may be used for design in ULS of continuity regions (cracked state of beams and slabs) and for the design in ULS and detailing of discontinuity regions. In general, these extend up to a distance h (section depth of member) from the discontinuity. Strut and tie models may also be used for members where a linear distribution within the cross section is assumed, e.g. plane strain.

Verifications in SLS may also be carried out using strut-and-tie models, e.g. verification of steel stresses and crack width control, if approximate compatibility for strut-and-tie models is ensured (in particular the position and direction of important struts should be oriented according to linear elasticity theory).

Strut-and-tie models consist of struts representing compressive stress fields, of ties representing the reinforcement, and of the connecting nodes. The forces in the elements of a strut-and-tie model should be determined by maintaining the equilibrium with the applied loads in the ultimate limit state. The ties of a strut-and-tie model should coincide in position and direction with the corresponding reinforcement.

Possible means for developing suitable strut-and-tie models include the adoption of stress trajectories and distributions from linear-elastic theory or the load path method. All strut-and-tie models may be optimised by energy criteria.

4.5.7 Non-linear Analysis

Non-linear methods of analysis may be used for both ULS and SLS, provided that equilibrium and compatibility are satisfied and an adequate non-linear behaviour for materials is assumed. The analysis may be first or second order.

At the ultimate limit state, the ability of local critical sections to withstand any inelastic deformations implied by the analysis should be checked, taking appropriate account of uncertainties.

For structures predominantly subjected to static loads, the effects of previous applications of loading may generally be ignored, and a monotonic increase of the intensity of the actions may be assumed.

The use of material characteristics which represent the stiffness in a realistic way but take account of the uncertainties of failure shall be used when using non-linear analysis. Only those design formats which are valid within the relevant fields of application shall be used.

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Concrete Structures

Non-linear analysis may be used provided that the model can appropriately cover all failure modes (e.g. bending, axial force, shear, compression failure affected by reduced effective concrete strength, etc.) and that the concrete tensile strength is not utilised as a primary load resisting mechanism. If one analysis is not sufficient to verify all the failure mechanisms, separate additional analyses should be carried out.

When using non-linear analysis, the following assumptions should be made:

- For reinforcing steel, the stress-strain diagram to be used should be based on Figure 4.8, curve A. In this diagram, f_{yk} and $k f_{yk}$ should be replaced by $1.1 f_{yk}$ and $1.1 k f_{yk}$.
- For prestressing steel, the idealised stress-strain diagram Figure 4.10, curve A should be used. In this diagram f_{pk} should be replaced with $1.1 f_{pk}$.
- For concrete, the stress-strain diagram should be based on Equation 4.14 in Clause 4.3.1.4. In this Expression, and in the k -value, f_{cm} should be replaced by $\gamma_{cf} f_{ck}$ with $\gamma_{cf} = 1.1 \gamma_s / \gamma_c$.

The following design format should be used:

- The resistance should be evaluated for different levels of appropriate actions which should be increased from their serviceability values by incremental steps, such that the value of $\gamma_G \cdot G_k$ and $\gamma_Q \cdot Q_k$ are reached in the same step. The incrementing process should be continued until one region of the structure attains the ultimate strength, evaluated taking account of α_{cc} or there is global failure of the structure. The corresponding load is referred to as q_{ud} .
- Apply an overall safety factor γ_o and obtain the corresponding strength $R(q_{ud}/\gamma_o)$.
- One of the following inequalities should be satisfied:

$$\gamma_{Rd} E (\gamma_G G + \gamma_Q Q) \leq R \left(\frac{q_{ud}}{\gamma_o} \right) \quad \text{Equation 4.27}$$

$$\begin{aligned} &\text{or} \\ &E (\gamma_G G + \gamma_Q Q) \leq R \left(\frac{q_{ud}}{\gamma_{Rd} \gamma_o} \right) \\ &\text{i.e. } R \left(\frac{q_{ud}}{\gamma_o} \right) \end{aligned} \quad \text{Equation 4.28}$$

$$\begin{aligned} &\text{or} \\ &\gamma_{Rd} \gamma_{Sd} E (\gamma_G G + \gamma_Q Q) \leq R \left(\frac{q_{ud}}{\gamma_o} \right) \end{aligned} \quad \text{Equation 4.29}$$

Where,

- γ_{Rd} = The partial factor for model uncertainty for resistance, $\gamma_{Rd} = 1.06$.
- γ_{Sd} = The partial factor for model uncertainty for action/action effort, $\gamma_{Sd} = 1.15$.
- γ_o = The overall safety factor, $\gamma_o = 1.20$.

When model uncertainties γ_{Rd} and γ_{Sd} are not considered explicitly in the analysis (i.e. $\gamma_{Rd} = \gamma_{Sd} = 1$), $\gamma_o = 1.27$ should be used.

If design properties of materials are used for non-linear analysis particular care should be exercised to allow for the effects of indirect actions (e.g. imposed deformations).

4.5.8 Analysis of Second Order Effects with Axial Load

4.5.8.1 Definitions

Some definitions are as under:

Biaxial bending	Simultaneous bending about two principal axes.
Braced members or systems	Structural members or subsystems, which in analysis and design are assumed not to contribute to the overall horizontal stability of a structure.
Bracing members or systems	Structural members or subsystems, which in analysis and design are assumed to contribute to the overall horizontal stability of a structure.
Buckling	Failure due to instability of a member or structure under perfectly axial compression and without transverse load.
Buckling load	The load at which buckling occurs; for isolated elastic members it is synonymous with the Euler load.
Effective length	A length used to account for the shape of the deflection curve; it can also be defined as buckling length, i.e. The length of a pin-ended column with constant normal force, having the same cross section and buckling load as the actual member.
First order effects	Action effects calculated without consideration of the effect of structural deformations, but including geometric imperfections.
Isolated members	Members that are isolated, or members in a structure that for design purposes may be treated as being isolated; examples of isolated members with different boundary conditions are shown in Figure 4.15.
Nominal second order moment	A second order moment used in certain design methods, giving a total moment compatible with the ultimate cross section resistance.
Second order effects	Additional action effects caused by structural deformations.

4.5.8.2 General

When second order effects are taken into account, equilibrium and resistance shall be verified in the deformed state. Deformations shall be calculated taking into account the relevant effects of cracking, non-linear material properties and creep. Where relevant, analysis shall include the effect of flexibility of adjacent members and foundations (soil-structure interaction).

The structural behaviour shall be considered in the direction in which deformations can occur, and biaxial bending shall be taken into account when necessary. Uncertainties in geometry and position of axial loads shall be taken into account as additional first order effects based on geometric imperfections.

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Second order effects may be ignored if they are less than 10 % of the corresponding first order effects. Simplified criteria are given below for isolated members.

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4.5.8.3 Simplified Criteria For Second Order Effects

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4.5.8.3.1 Slenderness Criterion For Isolated Members

The second order effects may be ignored if the slenderness λ is below a certain value λ_{lim} , calculated as below.

$$\lambda_{lim} = 20 A B C / \sqrt{n}$$

4

Where,

A = $1 / (1 + 0.2 \varphi_{ef})$ (if φ_{ef} is not known, A may be taken as 0.7).

B = $\sqrt{1 + 2\omega}$ (if ω is not known, B may be taken as 1.1).

C = $1.7 - r_m$ (if r_m is not known, C may be taken as 0.7).

φ_{ef} = Effective creep ratio.

ω = $A_s f_{yd} / (A_c f_{cd})$; mechanical reinforcement ratio.

A_s = Total area of longitudinal reinforcement.

n = $N_{Ed} / (A_c f_{cd})$; relative normal force.

r_m = M_{01} / M_{02} ; moment ratio.

M_{01}, M_{02} = first order end moments, $|M_{02}| \geq |M_{01}|$.

If the end moments M_{01} and M_{02} give tension on the same side, r_m should be taken positive (i.e. $C \leq 1.7$), otherwise negative (i.e. $C > 1.7$).

In the following cases, r_m should be taken as 1.0 (i.e. $C = 0.7$):

- For braced members in which the first order moments arise only from or predominantly due to imperfections or transverse loading.
- For unbraced members in general.

In cases with biaxial bending, the slenderness criterion may be checked separately for each direction. Depending on the outcome of this check, second order effects (a) may be ignored in both directions, (b) should be taken into account in one direction, or (c) should be taken into account in both directions.

4.5.8.3.2 Slenderness and Effective Length of Isolated Members

The slenderness ratio is defined as follows:

$$\lambda = \frac{l_0}{i}$$

Equation 4.30

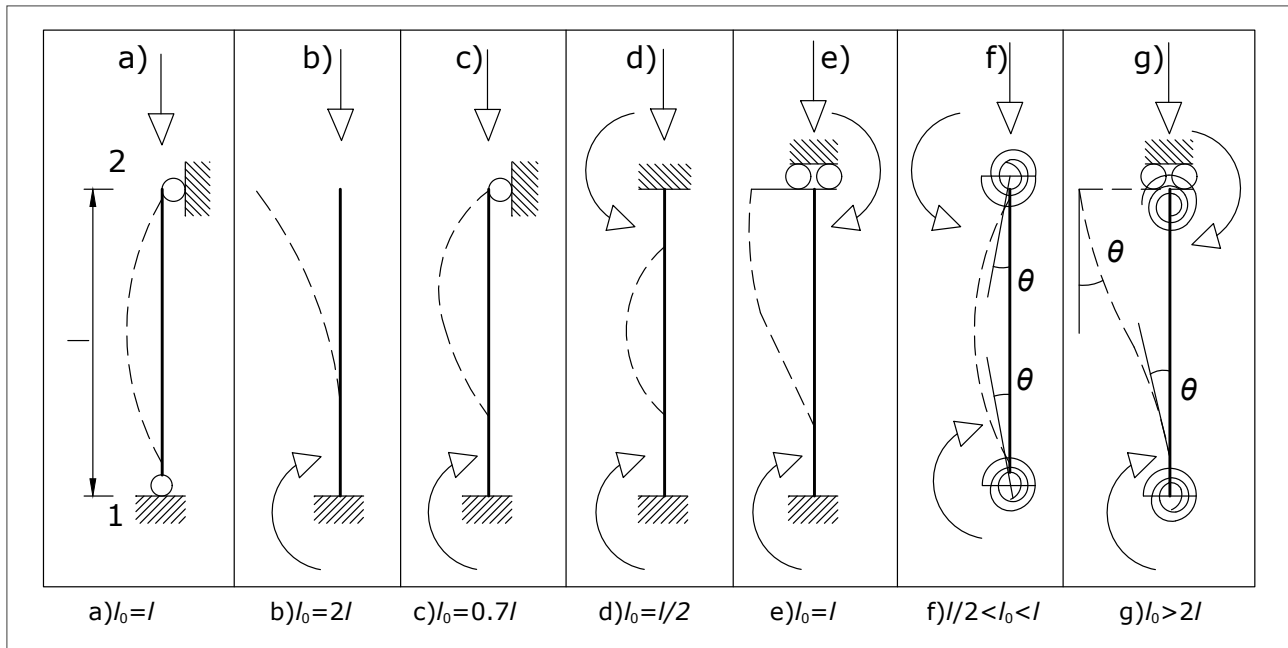
Where,

l_0 = Effective length.

i = Radius of gyration of the uncracked concrete section.

Effective length for isolated members with constant cross section are given in Figure 4.15.

Figure 4.15 Buckling Modes and Effective Lengths for Isolated Members



For compression members in regular frames, the slenderness criterion should be checked with an effective length l_0 determined in the following way:

Braced members (refer Figure 4.15-f)

$$l_0 = 0.5 l \sqrt{\left(1 + \frac{k_1}{0.45 + k_1}\right) \left(1 + \frac{k_2}{0.45 + k_2}\right)}$$

Equation 4.31

Unbraced members (refer Figure 4.15-g)

$$l_0 = l \cdot \max \left\{ \sqrt{1 + 10 \cdot \frac{k_1 \cdot k_2}{k_1 + k_2}}; \left(1 + \frac{k_1}{1 + k_2}\right) \left(1 + \frac{k_2}{1 + k_2}\right) \right\}$$

Equation 4.32

Where,

k_1, k_2 = are the relative flexibilities of rotational restraints at ends 1 and 2 respectively.

$k = (\theta / M) (EI / l)$

θ = rotation of restraining members for bending moment M

EI = bending stiffness of compression member

l = clear height of compression member between end restraints

Note: $k = 0$ is the theoretical limit for rigid rotational restraint, and $k = \infty$ represents the limit for no restraint at all. Since fully rigid restraint is rare in practice, a minimum value of 0.1 is recommended for k_1 and k_2 .

If an adjacent compression member (column) in a node is likely to contribute to the rotation at buckling, then (EI/l) in the definition of k should be replaced by $[(EI/l)_a + (EI/l)_b]$, a and b representing the compression member (column) above and below the node.

In the definition of effective lengths, the stiffness of restraining members should include the effect of cracking, unless they can be shown to be uncracked in ULS.

1

For other cases than those described above, e.g. members with varying normal force and/or cross section, the criterion of slenderness should be checked with an effective length based on the buckling load (calculated e.g. by a numerical method).

2

$$l_0 = \pi \sqrt{\frac{EI}{N_B}}$$

Where,

3

EI = representative bending stiffness

N_B = buckling load expressed in terms of this EI (In Equation 4.30, i should also correspond to this EI)

4

4.5.8.4 Creep

The effect of creep shall be taken into account in second order analysis, with due consideration of both the general conditions for creep and the duration of different loads in the load combination considered. The duration of loads may be taken into account in a simplified way by means of an effective creep ratio, φ_{ef} , which, used together with the design load, gives a creep deformation (curvature) corresponding to the quasi-permanent load.

$$\varphi_{ef} = \varphi_{(\infty, t0)} \cdot \frac{M_{0Eqp}}{M_{0Ed}}$$

Where,

$\varphi_{(\infty, t0)}$ = Final creep coefficient.

M_{0Eqp} = First order bending moment in quasi-permanent load combination (SLS).

M_{0Ed} = The first order bending moment in design load combination (ULS).

It is also possible to base φ_{ef} on total bending moments M_{Eqp} and M_{Ed} , but this requires iteration and a verification of stability under quasi-permanent load with $\varphi_{ef} = \varphi_{(\infty, t0)}$.

If M_{0Eqp} / M_{0Ed} varies in a member or structure, the ratio may be calculated for the section with maximum moment, or a representative mean value may be used.

The effect of creep may be ignored, i.e. $\varphi_{ef} = 0$ may be assumed, if the following three conditions are met:

- $\varphi_{(\infty, t0)} \leq 2$
- $\lambda \leq 75$
- $M_{0Ed} / N_{Ed} \geq h$
 - M_{0Ed} first order moment
 - h cross section depth in corresponding direction

4.5.8.5 Methods of Analysis

The methods of analysis include a general method, based on non-linear second order analysis, the following two simplified methods:

- a. Method based on nominal stiffness.
- b. Method based on nominal curvature.

Method (a) may be used for both isolated members and whole structures, if nominal stiffness values are estimated appropriately. Method (b) is mainly suitable for isolated members.

4.5.8.6 General Method

The general method is based on non-linear analysis, including geometric non-linearity i.e. second order effects.

Stress-strain curves for concrete and steel suitable for overall analysis shall be used. The effect of creep shall be taken into account.

Stress-strain relationships for concrete and steel given in Clause 4.3.1.4, Equation 4.14 and Clause 4.3.2.7 (Figure 4.8) may be used. With stress-strain diagrams based on design values, a design value of the ultimate load is obtained directly from the analysis. In Equation 4.14, and in the k value, f_{cm} is then substituted by the design compressive strength f_{cd} and E_{cm} is substituted by:

$$E_{cd} = E_{cm} / \gamma_{CE}, \text{ where } \gamma_{CE} = 1.2$$

In the absence of more refined models, creep may be taken into account by multiplying all strain values in the concrete stress-strain diagram with a factor $(1 + \varphi_{ef})$, where φ_{ef} is the effective creep ratio.

Normally, conditions of equilibrium and strain compatibility are satisfied in a number of cross sections. A simplified alternative is to consider only the critical cross section(s), and to assume a relevant variation of the curvature in between, e.g. similar to the first order moment or simplified in another appropriate way.

4.5.8.7 Method Based on Nominal Stiffness

4.5.8.7.1 General

In a second order analysis based on stiffness, nominal values of the flexural stiffness should be used, taking into account the effects of cracking, material non-linearity and creep on the overall behaviour. This also applies to adjacent members involved in the analysis, e.g. beams, slabs or foundations. Where relevant, soil-structure interaction should be taken into account. The resulting design moment is used for the design of cross sections.

4.5.8.7.2 Nominal Stiffness

The following model may be used to estimate the nominal stiffness of slender compression members with arbitrary cross section:

$$EI = K_c E_{cd} I_c + K_s E_s I_s$$

Equation 4.33

Where,

E_{cd} = Design value of the modulus of elasticity of concrete.

I_c = Moment of inertia of concrete cross section.

E_s = Design value of the modulus of elasticity of reinforcement.

I_s = Second moment of area of reinforcement, about the centre of area of the concrete.

K_c = Factor for effects of cracking, creep etc.

K_s = Factor for contribution of reinforcement.

The following factors may be used in above Equation, provided $\rho \geq 0.002$.

$$K_s = 1$$

$$K_c = \frac{k_1 k_2}{1 + \varphi_{ef}}$$

Equation 4.34

1

ρ = Geometric reinforcement ratio, A_s/A_c .

A_s = Total area of reinforcement.

A_c = Area of concrete section.

2

φ_{ef} = Effective creep ratio.

k_1 = Factor which depends on concrete strength class.

k_2 = Factor which depends on axial force and slenderness.

3

$$k_1 = \sqrt{\left(\frac{f_{ck}}{20}\right)} \quad \text{where } f_{ck} \text{ in MPa}$$

Equation 4.35

4

$$k_2 = n \left(\frac{\lambda}{170} \right) \leq 0.20$$

Equation 4.36

n = Relative axial force, $N_{Ed} / (A_c f_{cd})$.

λ = Slenderness ratio.

If slenderness ratio λ is not defined, k_2 may be taken as

$$k_2 = 0.30 n \leq 0.20$$

Equation 4.37

As a simplified alternative, provided $\rho \geq 0.01$, the following factors may be used to estimate nominal stiffness;

$$K_s = 0$$

$$K_c = \frac{0.3}{1 + 0.5 \varphi_{ef}}$$

Equation 4.38

In statically indeterminate structures, unfavourable effects of cracking in adjacent members should be taken into account. Equations 4.33 to 4.38 are not generally applicable to such members. Partial cracking and tension stiffening may be taken into account. However, as a simplification, fully cracked sections may be assumed. The stiffness should be based on an effective concrete modulus.

$$E_{cd,eff} = \frac{E_{cd}}{1 + \varphi_{ef}}$$

Where,

E_{cd} = Design value of the modulus of elasticity.

φ_{ef} = Effective creep ratio.

4.5.8.7.3 Moment Magnification Factor

The total design moment, including second order moment, may be expressed as a magnification of the bending moments resulting from a first order analysis, namely:

$$M_{Ed} = M_{0Ed} \left[1 + \frac{\beta}{(N_B/N_{Ed}) - 1} \right]$$

Equation 4.39

Where,

M_{0Ed} = first order moment

β = a factor which depends on distribution of 1st and 2nd order moments

N_{Ed} = design value of axial load

N_B = buckling load based on nominal stiffness

For isolated members with constant cross section and axial load, the second order moment may normally be assumed to have a sine-shaped distribution. Then:

$$\beta = \frac{\pi^2}{c_0}$$

Equation 4.40

Where,

- c_0 = coefficient which depends on the distribution of first order moment
- = 8 for a constant first order moment
- = 9.6 for a parabolic distribution
- = 12 for a symmetric triangular distribution

For members without transverse load, differing first order end moments M_{01} and M_{02} may be replaced by an equivalent constant first order moment M_{0e} according to Equation 4.42. Consistent with the assumption of a constant first order moment, $c_0 = 8$ should be used. The value of $c_0 = 8$ also applies to members bent in double curvature.

Where above is not applicable, $\beta = 1$ is normally a reasonable simplification. Equation 4.39 can then be reduced to

$$M_{Ed} = \frac{M_{0Ed}}{1 - (N_{ED}/N_B)}$$

Equation 4.41

4.5.8.8 Method Based on Nominal Curvature

4.5.8.8.1 General

This method is primarily suitable for isolated members with constant normal force and a defined effective length l_0 . The method gives a nominal second order moment based on a deflection, which in turn is based on the effective length and an estimated maximum curvature.

The resulting design moment is used for the design of cross sections with respect to bending moment and axial force.

4.5.8.8.1 Bending Moments

The total design moment is given by:

$$M_{Ed} = M_{0Ed} + M_2$$

Where,

- M_{0Ed} = 1st order moment, including the effect of imperfections
- M_2 = nominal second order moment

The maximum value of M_{Ed} is given by the distributions of M_{0Ed} and M_2 ; the latter may be taken as parabolic or sinusoidal over the effective length.

For statically indeterminate members, M_{0Ed} is determined for the actual boundary conditions, whereas M_2 will depend on boundary conditions via the effective length.

For members without loads applied between their ends, differing first order end moments M_{01} and M_{02} may be replaced by an equivalent first order end moment M_{0e}

$$M_{0e} = 0.6 M_{02} + 0.4 M_{01} \geq 0.4 M_{02}$$

Equation 4.42

1

M_{01} and M_{02} should have the same sign if they give tension on the same side, otherwise opposite signs. Furthermore, $|M_{02}| \geq |M_{01}|$

2

The nominal second order moment M_2 in above Equation is:

$$M_2 = N_{Ed} e_2$$

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Where,

N_{Ed} = The design value of axial force

e_2 = Deflection = $(1/r) l_0^2 / c$

$1/r$ = Curvature,

l_0 = Effective length,

c = A factor depending on the total curvature distribution,

4

For constant cross section, $c = 10 (\approx \pi^2)$ is normally used. If the first order moment is constant, a lower value should be considered (8 is a lower limit, corresponding to constant total moment). The value π^2 corresponds to a sinusoidal curvature distribution. The value for constant curvature is 8.

4.5.8.8.3 Curvature

For members with constant symmetrical cross sections (incl. reinforcement), the following may be used:

$$1/r = K_r K_\varphi (1/r_0)$$

Where,

K_r = Correction factor depending on axial load,

K_φ = Factor for taking account of creep,

$1/r_0 = \varepsilon_{yd} / (0.45 d)$

$\varepsilon_{yd} = f_{yd} / E_s$

d = Effective depth

If all reinforcement is not concentrated on opposite sides, but part of it is distributed parallel to the plane of bending, d is defined as

$$d = \left(\frac{h}{2} \right) + i_s$$

Where i_s is the radius of gyration of the total reinforcement area

K_r in above Equation should be taken as:

$$K_r = \frac{n_u - n}{n_u - n_{bal}} \leq 1$$

Where,

$n = N_{Ed} / (A_c f_{cd})$, relative axial force

N_{Ed} = design value of axial force

$n_u = 1 + \omega$

n_{bal} = value of n at maximum moment resistance; the value 0.4 may be used

$\omega = A_s f_{yd} / (A_c f_{cd})$

A_s = total area of reinforcement

A_c = area of concrete cross section

The effect of creep should be taken into account by the following factor

$$K_\varphi = 1 + \beta \varphi_{ef} \geq 1$$

Where,

φ_{ef} = Effective creep ratio.

β = $0.35 + f_{ck}/200 - \lambda/150$.

λ = The slenderness ratio.

4.5.8.9 Biaxial Bending

Separate design in each principal direction, disregarding biaxial bending, may be made as a first step. Imperfections need to be taken into account only in the direction where they will have the most unfavourable effect. No further check is necessary if the slenderness ratios satisfy the following two conditions:

$$\frac{\lambda_y}{\lambda_z} \leq 2 \text{ and } \frac{\lambda_z}{\lambda_y} \leq 2$$

Equation 4.43

And if the relative eccentricities e_y/h_{eq} and e_z/b_{eq} (refer Figure 4.16) satisfy one the following conditions:

$$\frac{e_y/h_{eq}}{e_z/b_{eq}} \leq 0.2 \text{ or } \frac{e_z/b_{eq}}{e_y/h_{eq}} \leq 0.2$$

Equation 4.44

Where,

b, h = Width and depth of the section.

$b_{eq} = i_y \sqrt{12}$ and $h_{eq} = i_z \sqrt{12}$ for an equivalent rectangular section.

λ_y, λ_z = The slenderness ratios l_0/i with respect to y- and z-axis respectively.

i_y, i_z = Radii of gyration with respect to y- and z-axis respectively.

e_z = M_{Edy} / N_{Ed} ; eccentricity along z-axis.

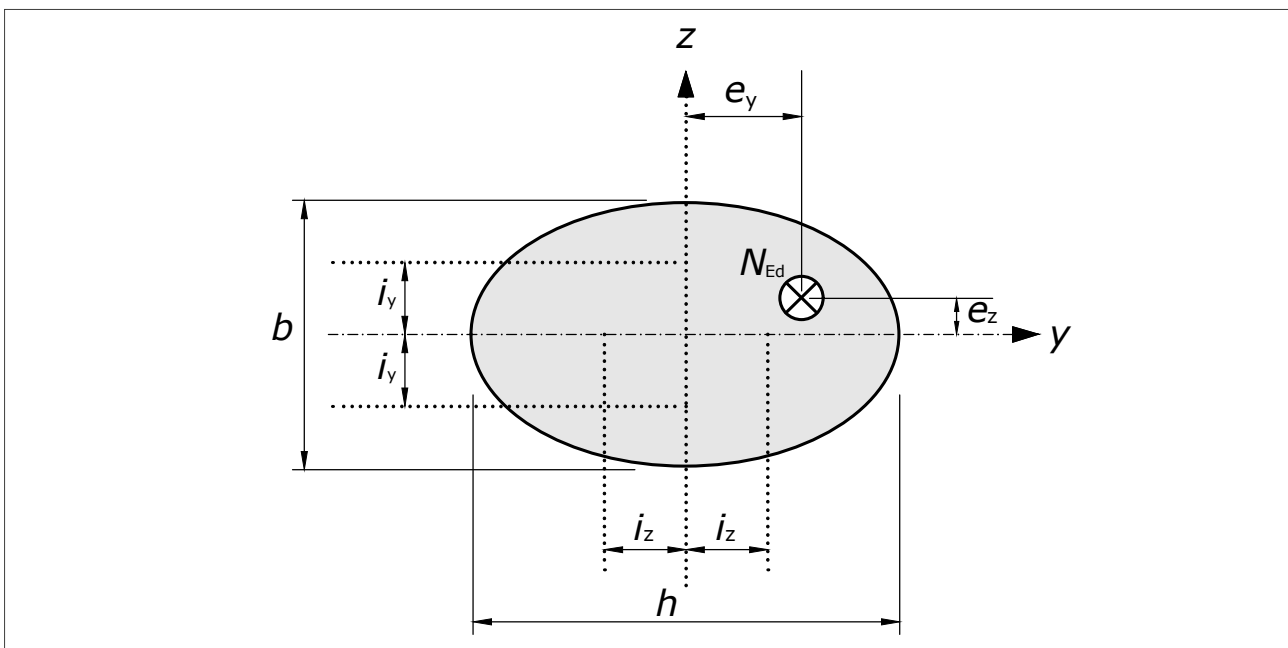
e_y = M_{Edz} / N_{Ed} ; eccentricity along y-axis.

M_{Edy} = Design moment about y-axis, including second order moment.

M_{Edz} = Design moment about z-axis, including second order moment.

N_{Ed} = Design value of axial load in the respective load combination.

Figure 4.16 Definition of eccentricities e_y and e_z



If the above conditions of Equation 4.43 and 4.44 is not fulfilled, biaxial bending should be taken into account including the second order effects in each direction unless they may be ignored. In the absence of an accurate cross section design for biaxial bending, the following simplified criterion may be used.

$$\left(\frac{M_{Edz}}{M_{Rdz}} \right)^a + \left(\frac{M_{Edy}}{M_{Rdy}} \right)^a \leq 1.0$$

Equation 4.45

Where,

M_{Edz}, M_{Edy} = Design moment around the respective axis, including a second order moment.

M_{Rdz}, M_{Rdy} = Moment resistance in the respective direction.

a = The exponent.

= For circular and elliptical cross sections: $a = 2$.

For rectangular cross sections;

N_{Ed} / N_{Rd}	0.1	0.7	1.0
$a =$	1.0	1.5	2.0

With linear interpolation for intermediate values:

N_{Ed} = Design value of axial force.

N_{Rd} = $A_c f_{cd} + A_s f_{yd}$, design axial resistance of section.

A_c = Is the gross area of the concrete section.

A_s = Is the area of longitudinal reinforcement.

4.5.9 Lateral Instability of Slender Beams

Lateral instability of slender beams shall be taken into account where necessary, e.g. for precast beams during transport and erection, for beams without sufficient lateral bracing in the finished structure etc. Geometric imperfections shall be taken into account.

A lateral deflection of $l / 300$ should be assumed as a geometric imperfection in the verification of beams in unbraced conditions, with l = total length of beam. In finished structures, bracing from connected members may be taken into account.

Second order effects in connection with lateral instability may be ignored if the following conditions are fulfilled:

$$\text{For persistent situations} \quad \frac{l_{0t}}{b} \leq \frac{50}{(h/b)^{1/3}} \quad \text{and} \quad h / b \leq 2.5$$

Equation 4.46

$$\text{For transient situations} \quad \frac{l_{0t}}{b} \leq \frac{70}{(h/b)^{1/3}} \quad \text{and} \quad h / b \leq 3.5$$

Equation 4.47

Where,

l_{0t} = Distance between torsional restraints.

h = Total depth of beam in central part of l_{0t} .

b = Width of compression flange.

Torsion associated with lateral instability should be taken into account in the design of supporting structures.

4.5.10 Prestressed Members and Structures

4.5.10.1 General

The prestress is that applied to the concrete by stressed tendons. The effects of pre-stressing may be considered as an action or a resistance caused by pre-strain and pre-curvature. The bearing capacity should be calculated accordingly.

In general prestress is introduced in the action combinations as part of the loading cases and its effects should be included in the applied internal moment and axial force. Following this assumption, the contribution of the prestressing tendons to the resistance of the section should be limited to their additional strength beyond prestressing. This may be calculated assuming that the origin of the stress/strain relationship of the tendons is displaced by the effects of prestressing.

Brittle failure of the member caused by failure of prestressing tendons shall be avoided (refer Clause 4.6.1).

4.5.10.2 Prestressing force During Tensioning

4.5.10.2.1 Maximum Stressing Force

The force applied to a tendon, P_{max} (i.e. the force at the active end during tensioning) shall not exceed the following value:

$$P_{max} = A_p \sigma_{p,max}$$

Where,

A_p = Cross-sectional area of the tendon.

$\sigma_{p,max}$ = Maximum stress applied to the tendon.

$$= \text{Min} \{ k_1 f_{pk} ; k_2 f_{p0.1k} \}.$$

The recommended values are $k_1 = 0.8$ and $k_2 = 0.9$

Overstressing is permitted if the force in the jack can be measured to an accuracy of $\pm 5\%$ of the final value of the prestressing force. In such cases the maximum prestressing force P_{max} may be increased to $k_3 f_{p0.1k} A_p$ (e.g. for the occurrence of an unexpected high friction in long-line pre-tensioning). The recommended value of k_3 is 0.95.

4.5.10.2.2 Limitation of Concrete Stress

The concrete compressive stress in the structure resulting from the prestressing force and other loads acting at the time of tensioning or release of prestress, should be limited to:

$$\sigma_c \leq 0.6 f_{ck}(t)$$

Where $f_{ck}(t)$ is the characteristic compressive strength of the concrete at time t when it is subjected to the prestressing force.

For pretensioned elements the stress at the time of transfer of prestress may be increased to $0.7 f_{ck}(t)$, if it can be justified by tests or experience that longitudinal cracking is prevented. If the compressive stress permanently exceeds $0.45 f_{ck}(t)$ the non-linearity of creep should be taken into account.

4.5.10.2.3 Measurements

In post-tensioning the prestressing force and the related elongation of the tendon shall be checked by measurements and the actual losses due to friction shall be controlled.

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4.5.10.3 Prestress Force

At a given time t and distance x (or arc length) from the active end of the tendon the mean prestress force $P_{m,t}(x)$ is equal to the maximum force P_{max} imposed at the active end, minus the immediate losses and the time dependent losses. Absolute values are considered for all the losses.

The value of the initial prestress force $P_{m0}(x)$ (at time $t = t_0$) applied to the concrete immediately after tensioning and anchoring (post-tensioning) or after transfer of prestressing (pre-tensioning) is obtained by subtracting from the force at tensioning P_{max} the immediate losses $\Delta P_i(x)$ and should not exceed the following value:

$$P_{m0}(x) = A_p \sigma_{pm0}(x)$$

Where,

$\sigma_{pm0}(x)$ = is the stress in the tendon immediately after tensioning or transfer

$$= \min \{ 0.75 f_{pk} ; 0.85 f_{p0.1k} \}$$

When determining the immediate losses $\Delta P_i(x)$ the following immediate influences should be considered for pre-tensioning and post-tensioning where relevant:

- Losses due to elastic deformation of concrete ΔP_{el}
- Losses due to short term relaxation ΔP_r
- Losses due to friction $\Delta P_\mu(x)$
- Losses due to anchorage slip ΔP_{sl}

The mean value of the prestress force $P_{m,t}(x)$ at the time $t > t_0$ should be determined with respect to the prestressing method. In addition to the immediate losses given above, the time dependent losses of prestress $\Delta P_{c+s+r}(x)$ as a result of creep and shrinkage of the concrete and the long term relaxation of the prestressing steel should be considered and $P_{m,t}(x) = P_{m0}(x) - \Delta P_{c+s+r}(x)$.

4.5.10.4 Immediate Losses of Prestress for Pre-tensioning

The following losses occurring during pre-tensioning should be considered:

- i. During the stressing process: loss due to friction at the bends (in the case of curved wires or strands) and losses due to wedge draw-in of the anchorage devices.
- ii. Before the transfer of prestress to concrete: loss due to relaxation of the pre-tensioning tendons during the period which elapses between the tensioning of the tendons and prestressing of the concrete.
- iii. In case of heat curing, losses due to shrinkage and relaxation are modified and should be assessed accordingly; direct thermal effect should also be considered.
- iv. At the transfer of prestress to concrete: loss due to elastic deformation of concrete as the result of the action of pre-tensioned tendons when they are released from the anchorages.

4.5.10.5 Immediate Losses of Prestress for Post-tensioning

4.5.10.5.1 Losses Due to the Instantaneous Deformation of Concrete

Account should be taken of the loss in tendon force corresponding to the deformation of concrete, taking account the order in which the tendons are stressed.

This loss, ΔP_{el} , may be assumed as a mean loss in each tendon as follows:

$$\Delta P_{el} = A_p \cdot E_p \cdot \Sigma \left[\frac{j \cdot \Delta \sigma_c(t)}{E_{cm}(t)} \right]$$

Equation 4.48

Where,

$\Delta\sigma_c(t)$ = Variation of stress at the centre of gravity of the tendons applied at time t .

j = Coefficient equal to

= $(n - 1)/2n$ where n is the number of identical tendons successively prestressed. As an approximation j may be taken as $(1/2)$

= 1 for the variations due to permanent actions applied after prestressing

4.5.10.5.1 Losses Due to Friction

The losses due to friction $\Delta P_\mu(x)$ in post-tensioned tendons may be estimated from:

$$\Delta P_\mu(x) = P_{max} (1 - e^{-\mu(\theta + kx)})$$

Equation 4.49

Where,

θ = is the sum of the angular displacements over a distance x (irrespective of direction or sign)

μ = is the coefficient of friction between the tendon and its duct

k = is an unintentional angular displacement for internal tendons (per unit length)

x = is the distance along the tendon from the point where the prestressing force is equal to P_{max} (the force at the active end during tensioning)

The value μ depends on the surface characteristics of the tendons and the duct, on the presence of rust, on the elongation of the tendon and on the tendon profile. In absence of more exact data, the values for μ given in Table 4.13 may be assumed.

The value k for unintentional angular displacement depends on the quality of workmanship, on the distance between tendon supports, on the type of duct or sheath employed, and on the degree of vibration used in placing the concrete. In absence of more exact data, values for unintended regular displacements for internal tendons will generally be in the range $0.005 < k < 0.01$ per metre. For external tendons, the losses of prestress due to unintentional angles may be ignored.

Table 4.13 Coefficients of Friction μ of Post-tensioned Tendons

Type	Internal Tendons	External Unbonded Tendons			
		Steel duct/non lubricated	HDPE duct/non lubricated	Steel duct/lubricated	HDPE duct/lubricated
Cold drawn wire	0.17	0.25	0.14	0.18	0.12
Strand	0.19	0.24	0.12	0.16	0.10
Deformed bar	0.65	–	–	–	–
Smooth round bar	0.33	–	–	–	–
HDPE - High density polyethylene					

4.5.10.5.3 Losses at Anchorage

Account should be taken of the losses due to wedge draw-in of the anchorage devices, during the operation of anchoring after tensioning, and due to the deformation of the anchorage itself.

Values of the wedge draw-in are normally given by manufacturer. In absence of more exact data, a value of 6mm may be assumed.

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4.5.10.6 Time Dependent Losses of Prestress for Pre- and Post-tensioning

The time dependent losses may be calculated by considering the following two reductions of stress:

- Due to the reduction of strain, caused by the deformation of concrete due to creep and shrinkage, under the permanent loads.
- The reduction of stress in the steel due to the relaxation under tension.

The relaxation of steel depends on the concrete deformation due to creep and shrinkage. This interaction can generally and approximately be taken into account by a reduction factor 0.8.

A simplified method to evaluate time dependent losses at location x under the permanent loads is given by following Equation:

$$\Delta P_{c+s+r} = A_p \Delta \sigma_{p,c+s+r} = A_p \frac{\varepsilon_{cs} E_p + 0.8 \Delta \sigma_{pr} + \frac{E_p}{E_{cm}} \varphi(t, t_0) \sigma_{c,QP}}{1 + \frac{E_p A_p}{E_{cm} A_c} \left(1 + \frac{A_c}{I_c} Z_{cp}^2\right) [1 + 0.8 \varphi(t, t_0)]} \quad \text{Equation 4.50}$$

Where,

$\Delta \sigma_{p,c+s+r}$ = Absolute value of the variation of stress in the tendons due to creep, shrinkage and relaxation at location x , at time t .

ε_{cs} = Estimated shrinkage strain according to Clause 4.3.1.3.2 in absolute value.

E_p = Modulus of elasticity for the prestressing steel.

E_{cm} = Modulus of elasticity for the concrete Table 4.2.

$\Delta \sigma_{pr}$ = Absolute value of the variation of stress in the tendons at location x , at time t , due to the relaxation of the prestressing steel. It is determined for a stress of $\sigma_p = \sigma_p (G + P_{m0} + \psi_2 Q)$

where, $\sigma_p = \sigma_p (G + P_{m0} + \psi_2 Q)$ is the initial stress in the tendons due to initial prestress and quasi-permanent actions.

$\varphi(t, t_0)$ = creep coefficient at a time t and load application at time t_0 .

$\sigma_{c,QP}$ = stress in the concrete adjacent to the tendons, due to self-weight and initial prestress and other quasi-permanent actions where relevant. The value of $\sigma_{c,QP}$ may be the effect of part of self-weight and initial prestress or the effect of a full quasi-permanent combination of action ($\sigma_c (G + P_{m0} + \psi_2 Q)$), depending on the stage of construction considered.

A_p = Area of all the prestressing tendons at the location x .

A_c = Area of the concrete section.

I_c = Second moment of area of the concrete section.

Z_{cp} = Distance between the centre of gravity of the concrete section and the tendons.

Compressive stresses and the corresponding strains given in Equation 4.50 should be used with a positive sign. This Equation applies for bonded tendons when local values of stresses are used and for unbonded tendons when mean values of stresses are used. The mean values should be calculated between straight sections limited by the idealised deviation points for external tendons or along the entire length in case of internal tendons.

4.5.10.7 Consideration of Prestress in Analysis

Second order moments can arise from prestressing with external tendons. Moments from secondary effects of prestressing arise only in statically indeterminate structures.

For linear analysis both the primary and secondary effects of prestressing should be applied before any redistribution of forces and moments is considered.

In plastic and non-linear analysis, the secondary effect of prestress may be treated as additional plastic rotations which should then be included in the check of rotation capacity.

Rigid bond between steel and concrete may be assumed after grouting of post-tensioned tendons. However, before grouting the tendons should be considered as unbonded.

External tendons may be assumed to be straight between deviators.

4.5.10.8 Effects of Prestressing at Ultimate Limit State

In general the design value of the prestressing force may be determined by $P_{d,t}(x) = \gamma_P P_{m,t}(x)$.

For prestressed members with permanently unbonded tendons, it is generally necessary to take the deformation of the whole member into account when calculating the increase of the stress in the prestressing steel. If no detailed calculation is made, it may be assumed that the increase of the stress from the effective prestress to the stress in the ultimate limit state is $\Delta\sigma_{p,ULS}$.

The recommended value of $\Delta\sigma_{p,ULS}$ is 100 MPa.

If the stress increase in external tendons is calculated using the deformation state of the overall member, non-linear analysis should be used.

4.5.10.9 Effects of Prestressing at Serviceability Limit State and Limit State of Fatigue

For serviceability and fatigue calculations allowance shall be made for possible variations in prestress. Two characteristic values of the prestressing force at the serviceability limit state are estimated from:

$$P_{k,sup} = r_{sup} P_{m,t}(x)$$

$$P_{k,inf} = r_{inf} P_{m,t}(x)$$

Where,

$P_{k,sup}$ = The upper characteristic value.

$P_{k,inf}$ = The lower characteristic value.

The recommended values are $r_{sup} = 1.0$ and $r_{inf} = 1.0$.

4.6 Ultimate Limit States (ULS)

4.6.1 Bending With or Without Axial Force

1. This section applies to undisturbed regions of beams, slabs and similar types of members for which sections remain approximately plane before and after loading. The discontinuity regions of beams and other members in which plane sections do not remain plane may be designed and detailed according to Clause 4.6.5.
2. When determining the ultimate moment resistance of reinforced or prestressed concrete cross-sections, the following assumptions are made:
 - Plane sections remain plane.
 - The strain in bonded reinforcement or bonded prestressing tendons, whether in tension or in compression, is the same as that in the surrounding concrete.
 - The tensile strength of the concrete is ignored.
 - The stresses in the concrete in compression are derived from the design stress/strain relationship given in Clause 4.3.1.6.
 - The stresses in the reinforcing or prestressing steel are derived from the design curves in Clause 4.3.2 (Figure 4.8) and Clause 4.3.3 (Figure 4.10).
 - The initial strain in prestressing tendons is taken into account when assessing the stresses in the tendons.

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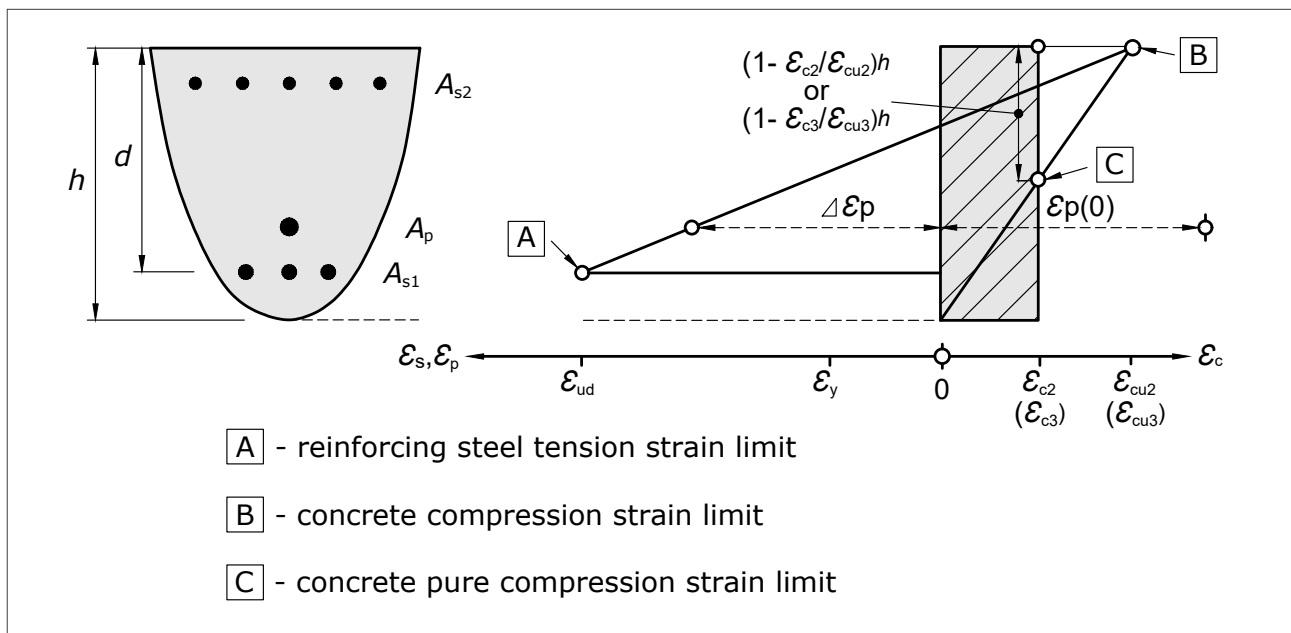
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3. The compressive strain in the concrete shall be limited to ε_{cu2} , or ε_{cu3} , depending on the stress-strain diagram used, refer Clause 4.3.1.6 and Table 4.2. The strains in the reinforcing steel and the prestressing steel shall be limited to ε_{ud} (where applicable); refer Clause 4.3.2.7 and 4.3.3.6 respectively.
4. For cross-sections loaded by the compression force it is necessary to assume the minimum eccentricity, $e_0 = h/30$ but not less than 20 mm where h is the depth of the section.
5. In parts of cross-sections which are subjected to approximately concentric loading ($e_d/h \leq 0.1$), such as compression flanges of box girders, the mean compressive strain in that part of the section should be limited to ε_{c2} if parabola-rectangle relation of Figure 4.3 is used or ε_{c3} if the bi-linear relation of Figure 4.4 is used.
6. The possible range of strain distributions in ultimate limit state is shown in Figure 4.17.
7. For prestressed members with permanently unbonded tendons, refer Clause 4.5.10.8.

Figure 4.17 Possible Strain Distributions in the Ultimate Limit State



8. For external prestressing tendons, the strain in the prestressing steel between two consecutive fixed points is assumed to be constant. The strain in the prestressing steel is then equal to the remaining strain, after losses, increased by the strain resulting from the structural deformation between the fixed points considered.

9. For prestressed structures, brittle failure by failure of individual tendons shall be avoided. This may be satisfied by any of the following methods (a or b):

a. Verification

Verifying the load capacity using a reduced area of prestress. This verification should be undertaken as follows:

- i. Calculate the applied bending moment due to the frequent combination of actions M_{freq} .
- ii. Determine the reduced area of prestress steel $A_{p,Red}$, that results in the tensile stress reaching f_{ctm} at the extreme tension fibre when the section is subject to the bending moment calculated as above.

$$A_{p,Red} = \frac{\frac{M_{freq}}{Z} - f_{ctm}}{\sigma_p \left(\frac{1}{A_c} + \frac{e}{Z} \right)}$$

Where,

Z = Section modulus.

f_{ctm} = Mean value of axial tensile strength of concrete.

σ_p = Stress in tendons just prior to cracking.

A_c = Cross-sectional area of concrete.

e = Eccentricity of the tendons.

- iii. Using this reduced area of prestress, calculate the ultimate flexural capacity. It should be ensured that this ultimate flexural capacity exceeds the bending moment due to the frequent combination. Redistribution of internal actions within the structure may be taken into account for this verification and the ultimate bending resistance should be calculated using the material partial safety factors for accidental design situations given in Table 4.1.

Assuming there is no conventional reinforcement then,

$$M_{freq} \geq \left[\frac{f_{ctm} z_s}{\frac{z_s}{Z} - \frac{\sigma_p \left(\frac{1}{A_c} + \frac{e}{Z} \right)}{f_{p,0.1k}}} \right]$$

Where,

z_s = Lever arm for the prestress, where $A_{p,Red}$ has been used to determine the accidental ultimate limit state.

f_{ctm} = Mean value of concrete cylinder compressive strength.

σ_p = Stress in tendons just prior to cracking.

A_c = Cross-sectional area of concrete.

e = Eccentricity of the tendons.

b. Minimum reinforcement provision

Providing a minimum reinforcing steel area according to the Expression below. Reinforcing steel provided for other purposes may be included in $A_{s,min}$.

$$A_{s,min} = \frac{M_{rep}}{z_s f_{yk}} \quad \text{Equation 4.51}$$

Where,

M_{rep} = Cracking bending moment calculated using an appropriate tensile strength, f_{ctx} at the extreme tension fibre of the section, ignoring any effect of prestressing. At the joint of segmental precast elements M_{rep} should be assumed to be zero.

z_s = The lever arm at the ultimate limit state related to the reinforcing steel.

The recommended value for $f_{ctx} = f_{ctk,0.05}$.

10. In cases where method b) above is chosen, the following rules apply:

- a. The minimum reinforcement steel area should be provided in regions where tensile stresses occur in the concrete under the characteristic combination of actions. In this check the parasitic effects of prestressing should be considered and the primary effects should be ignored.
- b. For pretensioned members, Equation 4.51 should be applied using one of the alternative approaches i. or ii. described below:
 - i. Tendons with concrete cover at least k_{cm} times the minimum specified in Clause 4.4.4.2 are considered as effective in $A_{s,min}$. The value of z_s based on the effective strands is used in the Equation and f_{yk} is replaced with $f_{p0.1k}$.
 - ii. Tendons subject to stresses lower than $0.6 f_{pk}$ after losses under the characteristic combination of actions are considered as fully active regardless of cover. In this case Equation 4.51 is replaced by:

$$A_{s,min} f_{yk} + A_p \Delta \sigma_p \geq \frac{M_{rep}}{z} \quad \text{Equation 4.52}$$

Where,

$\Delta \sigma_p$ = the smaller of $0.4 f_{ptk}$ and 500 MPa

The recommended value of k_{cm} is 1.0.

- c. To ensure adequate ductility, the minimum reinforcing steel area $A_{s,min}$ defined in Equations 4.50 and 4.52, in continuous beams should extend to the intermediate support of the span considered.

However, this extension is not necessary if, at the ultimate limit state, the resisting tensile capacity provided by reinforcing and prestressing steel above the supports, calculated with the characteristic strength f_{yk} and $f_{p0.1k}$ respectively, is less than the resisting compressive capacity of the bottom flange, which means that the failure of the compressive zone is not likely to occur:

$$A_s f_{yk} + k_p A_p f_{p0.1k} < t_{inf} b_0 \alpha_{cc} f_{ck} \quad \text{Equation 4.53}$$

Where,

t_{inf}, b_0 = Respectively, the thickness and the width of the bottom flange of the section. In case of T sections, t_{inf} is taken as equal to b_0

A_s, A_p = Denote respectively the area of reinforced and prestressing steel in the tensile zone at the ultimate limit state.

The recommended value for k_p is 1.0.

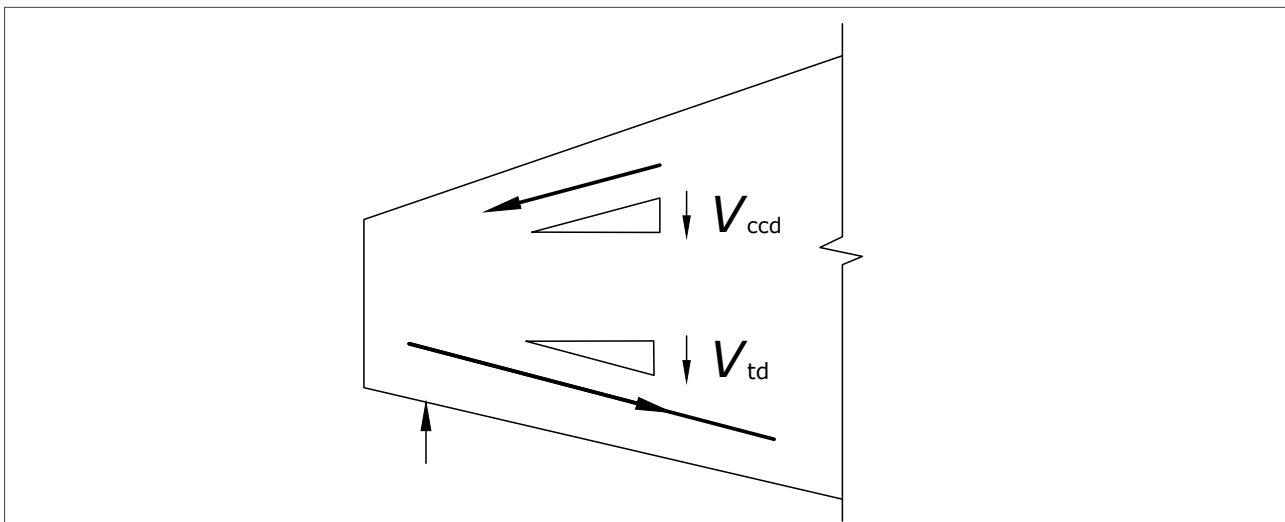
4.6.2 Shear

4.6.2.1 General Verification Procedure

1. For the verification of the shear resistance the following symbols are defined:

- $V_{Rd,c}$ = The design shear resistance of the member without shear reinforcement.
- $V_{Rd,s}$ = The design value of the shear force which can be sustained by the yielding shear reinforcement
- $V_{Rd,max}$ = The design value of the maximum shear force which can be sustained by the member, limited by crushing of the compression struts
- V_{Ed} = The design shear force in the section considered resulting from external loading and prestressing (bonded or unbonded). In members with inclined chords the following additional values are defined (refer Figure 4.18):
- V_{ccd} = The design value of the shear component of the force in the compression area, in the case of an inclined compression chord
- V_{td} = The design value of the shear component of the force in the tensile reinforcement, in the case of an inclined tensile chord

Figure 4.18 Shear Component for Members with Inclined Chords



2. The shear resistance of a member with shear reinforcement is equal to:

$$V_{Rd} = V_{Rd,s} + V_{ccd} + V_{td}$$

Equation 4.54

3. In regions of the member where $V_{Ed} \leq V_{Rd,c}$ no calculated shear reinforcement is necessary.
4. When, on the basis of the design shear calculation, no shear reinforcement is required, minimum shear reinforcement should be provided according to Clause 4.9.2.2. The minimum shear reinforcement may be omitted in members such as slabs (solid, ribbed or hollow core slabs) where transverse redistribution of loads is possible. Minimum reinforcement may also be omitted in members of minor importance (e.g. lintels with span ≤ 2 m) which do not contribute significantly to the overall resistance and stability of the structure.
5. In regions where $V_{Ed} \leq V_{Rd,c}$ according to Equation 4.55 and 4.56, sufficient shear reinforcement should be provided in order that $V_{Ed} \leq V_{Rd,c}$.
6. The sum of the design shear force and the contributions of the flanges, $V_{Ed} - V_{ccd} - V_{td}$, should not exceed the permitted maximum value $V_{Rd,max}$ anywhere in the member.

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7. The longitudinal tension reinforcement should be able to resist the additional tensile force caused by shear (refer Clause 4.6.2.3 (7))
8. For members subject to predominantly uniformly distributed loading, the design shear force need not to be checked at a distance less than d from the face of the support. Any shear reinforcement required should continue to the support. In addition it should be verified that the shear at the support does not exceed $V_{Rd,max}$ (refer Clause 4.6.2.2(5) and 4.6.2.3(8)).
9. Where a load is applied near the bottom of a section, sufficient vertical reinforcement to carry the load to the top of the section should be provided in addition to any reinforcement required to resist shear.

4.6.2.2 Members Not Requiring Design Shear Reinforcement

1. The design value for the shear resistance $V_{Rd,c}$ is given by:

$$V_{Rd,c} = [C_{Rd,c} k(100 \rho_l f_{ck})^{1/3} + k_1 \sigma_{cp}] b_w d$$

Equation 4.55

With a minimum of

$$V_{Rd,c} = (v_{min} + k_1 \sigma_{cp}) b_w d$$

Equation 4.56

Where,

f_{ck} = In MPa.

k = $1 + \sqrt{(200/d)}$ ≤ 2.0 with d in mm.

ρ_l = $A_{sl} / (b_w d)$ ≤ 0.02 .

A_{sl} = The area of the tensile reinforcement, which extends $\geq (l_{bd} + d)$ beyond the section considered (refer Figure 4.19): the area of bonded prestressing steel may be included in the calculation of A_{sl} . In this case a weighted mean value of d may be used.

b_w = The smallest width of the cross-section in the tensile area [mm].

σ_{cp} = $N_{Ed} / A_c < 0.2 f_{cd}$ [MPa].

N_{Ed} = The axial force in the cross-section due to loading or to the acting effect of prestressing in Newtons ($N_{Ed} > 0$ for compression). The influence of imposed deformations on N_{Ed} may be ignored.

A_c = The area of concrete cross section [mm²].

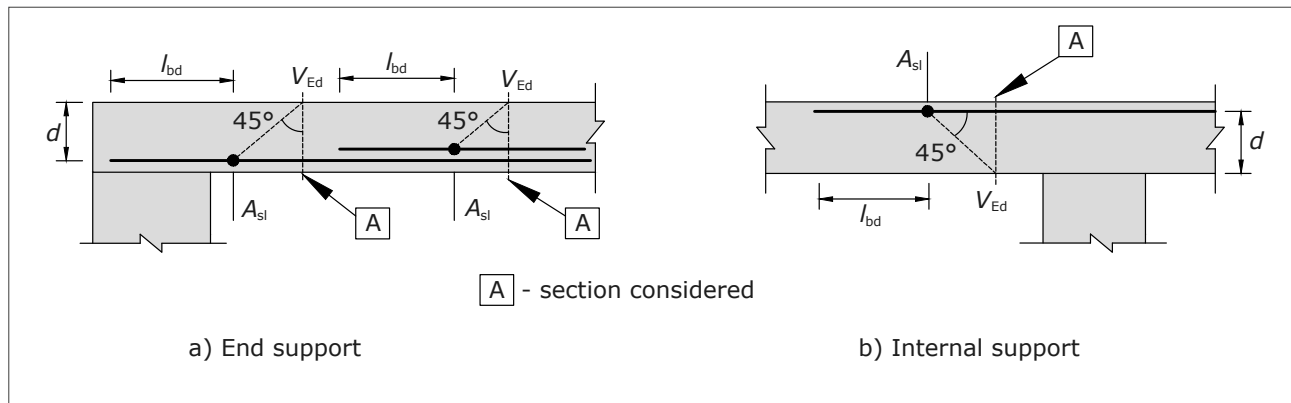
$V_{Rd,c}$ = in Newtons.

The recommended value of $C_{Rd,c}$ is $0.18 / \gamma_c$, that for v_{min} is given by Equation 4.57 and k_1 is 0.15.

$$v_{min} = 0.035 k^{3/2} f_{ck}^{1/2}$$

Equation 4.57

Figure 4.19 Definition of A_{sl} in Equation 4.55 and 4.56



2. In prestressed single span members without shear reinforcement, the shear resistance of the regions cracked in bending may be calculated using Equation 4.55. In regions uncracked in bending (where the flexural tensile stress is smaller than $f_{ctk,0.05}/\gamma_c$) the shear resistance should be limited by the tensile strength of the concrete. In these regions the shear resistance is given by:

$$V_{Rd,c} = \frac{I b_w}{S} \sqrt{(f_{ctd})^2 + \alpha_l \sigma_{cp} f_{ctd}}$$

Equation 4.58

Where,

I = The second moment of area.

b_w = The width of the cross-section at the centroidal axis, allowing for the presence of ducts in accordance with Equations 4.72 and 4.73

S = The first moment of area above and about the centroidal axis.

$\alpha_l = l_x / l_{pt2} \leq 1.0$ for pretensioned tendons.
 $= 1.0$ for other types of prestressing.

l_x = The distance of section considered from the starting point of the transmission length.

l_{pt2} = The upper bound value of the transmission length of the prestressing element as defined in Clause 4.8.10.2.2.

σ_{cp} is the concrete compressive stress at the centroidal axis due to axial loading and/or prestressing ($\sigma_{cp} = N_{Ed} / A_c$ in MPa, $N_{Ed} > 0$ in compression).

For cross-sections where the width varies over the height, the maximum principal stress may occur on an axis other than the centroidal axis. In such a case the minimum value of the shear resistance should be found by calculating $V_{Rd,c}$ at various axes in the cross-section.

3. The calculation of the shear resistance according to Equation 4.58 is not required for cross-sections that are nearer to the support than the point which is the intersection of the elastic centroidal axis and a line inclined from the inner edge of the support at an angle of 45° .
4. For the design of the longitudinal reinforcement, in the region cracked in flexure, the M_{Ed} - line should be shifted over a distance $a_l = d$ in the unfavourable direction (refer Clause 4.9.2.1.3 (2)).
5. For members with loads applied on the upper side within a distance $0.5d \leq a_v \leq 2d$ from the edge of a support (or centre of bearing where flexible bearings are used), the contribution of this load to the shear force V_{Ed} may be multiplied by $\beta = a_v/2d$. This reduction may be applied for checking $V_{Rd,c}$ in Equation 4.55. This is only valid provided that the longitudinal reinforcement is fully anchored at the support. For $a_v \leq 0.5d$ the value $a_v = 0.5d$ should be used.

The shear force V_{Ed} , calculated without reduction by β , should however always satisfy the condition:

$$V_{Ed} \leq 0.5 b_w d v f_{cd}$$

Equation 4.59

Where,

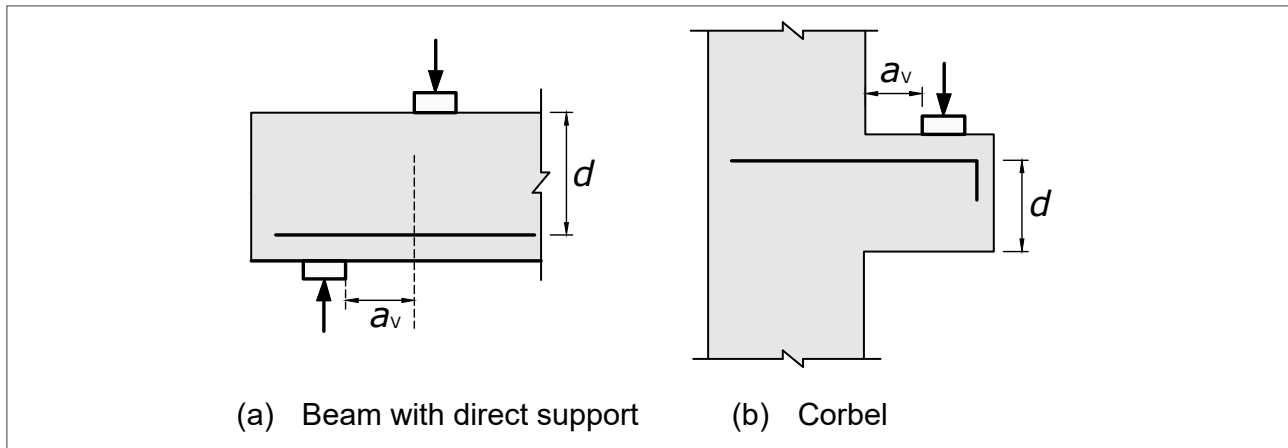
v = A strength reduction factor for concrete cracked in shear.

The recommended value of v follows from:

$$v = 0.6 \left[1 - \frac{f_{ck}}{250} \right] \quad \text{where } f_{ck} \text{ in MPa}$$

Equation 4.60

Figure 4.20 Loads Near Supports



6. Beams with loads near to supports and corbels may alternatively be designed with strut and tie models.

4.6.2.3 Members Requiring Design Shear Reinforcement

1. The design of members with shear reinforcement is based on a truss model (Figure 4.21). The following notations are shown in this figure:

α = The angle between shear reinforcement and the beam axis perpendicular to the shear force (measured positive as shown in Figure 4.21)

θ = The angle between the concrete compression strut and the beam axis perpendicular to the shear force

F_{td} = The design value of the tensile force in the longitudinal reinforcement

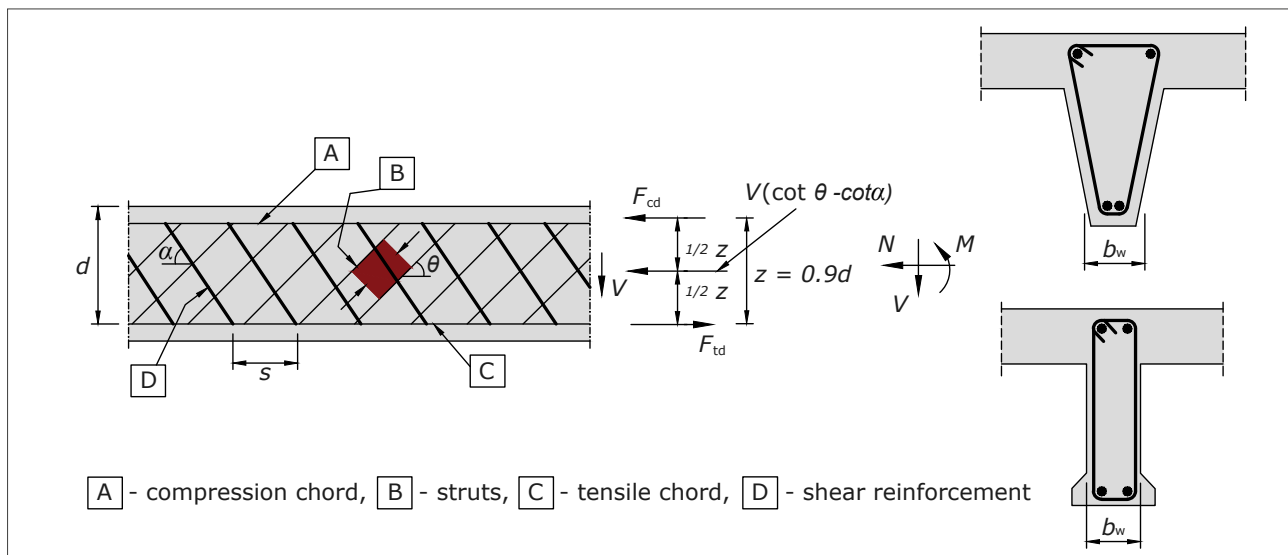
F_{cd} = The design value of the concrete compression force in the direction of the longitudinal member axis

b_w = The minimum width between tension and compression chords

z = The inner lever arm, for a member with constant depth, corresponding to the bending moment in the element under consideration. In the shear analysis of reinforced concrete without axial force, the approximate value $z = 0.9d$ may normally be used

In elements with inclined prestressing tendons, longitudinal reinforcement at the tensile chord should be provided to carry the longitudinal tensile force due to shear defined in para (7).

Figure 4.21 Truss Model and Notation for Shear Reinforcement Members



2. The angle θ should be limited.

The recommended limiting values of $\cot\theta$ are given in following Equation.

$$1 \leq \cot\theta \leq 2.5$$

3. For members with vertical shear reinforcement, the shear resistance, V_{Rd} is the smaller value of:

$$V_{Rd,s} = \frac{A_{sw}}{s} z f_{ywd} \cot\theta$$

Equation 4.61

Note 1: If Equation 4.63 -4.64 is used, the value of f_{ywd} should be reduced to $0.8 f_{yk}$ in Equation 4.61.

and

$$V_{Rd,max} = \frac{\alpha_{cw} b_w z v_1 f_{cd}}{\cot\theta + \tan\theta}$$

Equation 4.62

Where,

A_{sw} = The cross-sectional area of the shear reinforcement

s = The spacing of the stirrups

f_{ywd} = The design yield strength of the shear reinforcement

v_1 = A strength reduction factor for concrete cracked in shear

α_{cw} = A coefficient taking account of the state of the stress in the compression chord

Note 2: The recommended value of v_1 is $v_1 = v (1 - 0.5 \cos \alpha)$ where v is given by Equation 4.60.

Note 3: If the design stress of the shear reinforcement is below 80% of the characteristic yield stress f_{yk} , v_1 may be taken as:

$$v_1 = 0.54(1 - 0.5 \cos \alpha) \text{ for } f_{ck} \leq 60 \text{ MPa}$$

Equation 4.63

$$v_1 = (0.84 - \frac{f_{ck}}{200}) (1 - 0.5 \cos \alpha) > 0.5 \text{ for } f_{ck} \geq 60 \text{ MPa}$$

Equation 4.64

Note 4: The recommended value of α_{cw} is as follows:

$\alpha_{cw} = 1$ for non-prestressed structures

$$= (1 + \sigma_{cp} / f_{cd}) \quad \text{for } 0 < \sigma_{cp} \leq 0.25 f_{cd}$$

Equation 4.65

$$= 1.25 \quad \text{for } 0.25 f_{cd} < \sigma_{cp} \leq 0.5 f_{cd}$$

Equation 4.66

$$= 2.5 (1 - \sigma_{cp} / f_{cd}) \quad \text{for } 0.5 f_{cd} < \sigma_{cp} \leq 1.0 f_{cd}$$

Equation 4.67

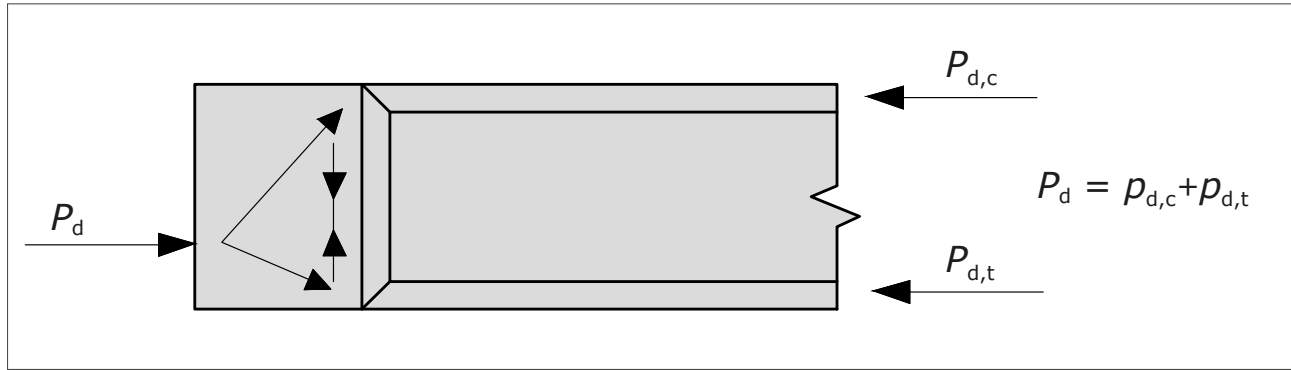
Where,

σ_{cp} is the mean compressive stress, measured positive, in the concrete due to the design axial force. This should be obtained by averaging it over the concrete section taking account of the reinforcement. The value of σ_{cp} need not be calculated at a distance less than $0.5d \cot\theta$ from the edge of the support.

Note: The value of v_1 and α_{cw} should not give rise to a value of $V_{Rd,max}$ greater than $200 b_w^2$ at sections more than a distance d from the edge of a support. For this purpose, the value of b_w does not need to be reduced for ducts.

In the case of straight tendons, a high level of prestress ($\sigma_{cp} / f_{cd} > 0.5$) and thin webs, if the tension and the compression chords are able to carry the whole prestressing force and blocks are provided at the extremity of beams to disperse the prestressing force (refer Figure 4.22), it may be assumed that the prestressing force is distributed between the chords. In these circumstances, the compression field due to shear only should be considered in the web ($\alpha_{cw} = 1$).

Figure 4.22 Dispersion of Prestressing by End Blocks within the Chords



Note 5: The maximum effective cross-sectional area of the shear reinforcement, $A_{sw,max}$, for $\cot\theta = 1$ is given by:

$$\frac{A_{sw,max} f_{ywd}}{b_w s} \leq \frac{1}{2} \alpha_{cw} v_1 f_{cd} \quad \text{Equation 4.68}$$

4. For members with inclined shear reinforcement, the shear resistance is the smaller value of:

$$V_{Rd,s} = \frac{A_{sw}}{s} z f_{ywd} (\cot\theta + \cot\alpha) \sin\alpha \quad \text{Equation 4.69}$$

and

$$V_{Rd,max} = \alpha_{cw} b_w z v_1 f_{cd} \frac{\cot\theta + \cot\alpha}{1 + \cot^2 \theta} \quad \text{Equation 4.70}$$

Note: The maximum effective shear reinforcement, $A_{sw,max}$ for $\cot\theta = 1$ follows from:

$$\frac{A_{sw,max} f_{ywd}}{b_w s} \leq \frac{1/2 \alpha_{cw} v_1 f_{cd}}{\sin\alpha} \quad \text{Equation 4.71}$$

5. In regions where there is no discontinuity of V_{Ed} (e.g. for uniformly distributed loading using the applied at the top) the shear reinforcement in any length increment $l = z (\cot\theta)$ may be calculated using the smallest value of V_{Ed} in the increment.
6. Where the web contains grouted metal ducts with a diameter $\phi > b_w/8$, the shear resistance $V_{Rd,max}$ should be calculated on the basis of a nominal web thickness given by:

$$b_{w,nom} = b_w - 0.5 \Sigma\phi \quad \text{Equation 4.72}$$

Where ϕ is the outer diameter of the duct and $\Sigma\phi$ is determined for the most unfavourable level.

For grouted metal ducts with $\phi \leq b_w/8$, the value of $b_{w,nom} = b_w$

For non-grouted ducts, grouted plastic ducts and unbonded tendons the nominal web thickness is:

$$b_{w,nom} = b_w - 1.2 \Sigma\phi \quad \text{Equation 4.73}$$

The value 1.2 in Equation 4.73 is introduced to take account of splitting of the concrete struts due to transverse tension. If adequate transverse reinforcement is provided this value may be reduced to 1.0.

7. The additional tensile force, ΔF_{td} , in the longitudinal reinforcement due to shear V_{Ed} may be calculated from:

$$\Delta F_{td} = 0.5 V_{Ed} (\cot\theta - \cot\alpha)$$

Equation 4.74

$(M_{Ed}/z) + \Delta F_{td}$, should be taken not greater than $M_{Ed,max}/z$.

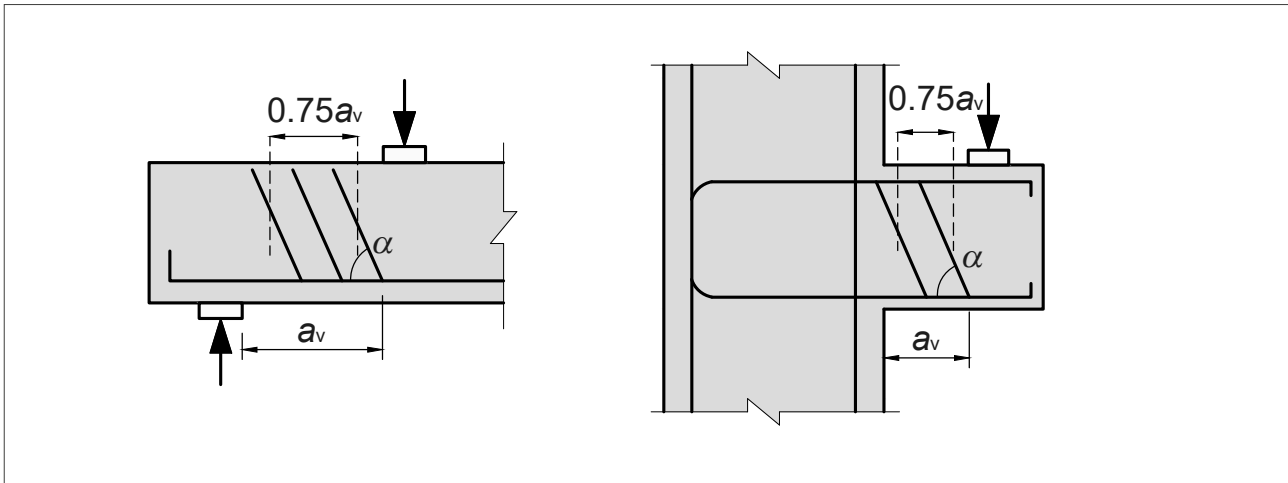
8. For members with loads applied on the upper side within a distance $0.5d \leq a_v \leq 2.0d$ the contribution of this load to the shear force V_{Ed} may be reduced by $\beta = a_v / 2d$. The shear force V_{Ed} , calculated in this way, should satisfy the condition.

$$V_{Ed} \leq A_{sw} f_{ywd} \sin\alpha$$

Equation 4.75

Where $A_{sw}f_{ywd}$ is the resistance of the shear reinforcement crossing the inclined shear crack between the loaded areas (see Figure 4.23). Only the shear reinforcement within the central $0.75 a_v$ should be taken into account. The reduction by β should only be applied for calculating the shear reinforcement. It is only valid provided that the longitudinal reinforcement is fully anchored at the support.

Figure 4.23 Shear Reinforcement in Short Shear Spans with Direct Strut Action



For $a_v < 0.5d$ the value $a_v = 0.5d$ should be used. The value V_{Ed} calculated without reduction by β , should however always be less than $V_{Rd,max}$, see Equation 4.62.

4.6.2.4 Shear Between Web And Flanges

1. The shear strength of the flange may be calculated by considering the flange as a system of compressive struts combined with ties in the form of tensile reinforcement.
2. A minimum amount of longitudinal reinforcement should be provided, as specified in Clause 4.9.3.1.
3. The longitudinal shear stress, V_{Ed} at the junction between one side of a flange and the web is determined by the change of the normal (longitudinal) force in the part of the flange considered, according to:

$$V_{Ed} = \frac{\Delta F_d}{h_f \Delta x}$$

Equation 4.76

Where,

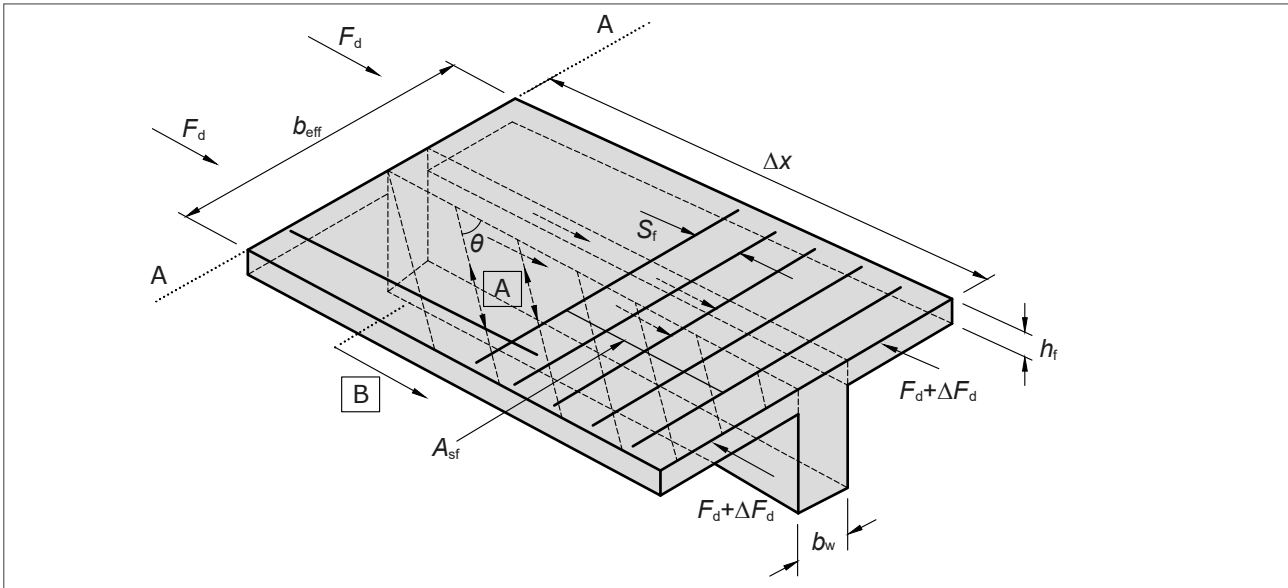
h_f = The thickness of flange at the junctions

Δx = The length under consideration, refer Figure 4.24

ΔF_d = The change of the normal force in the flange over the length Δx

The maximum value that may be assumed for Δ_x is half the distance between the section where the moment is 0 and the section where the moment is maximum. Where point loads are applied the length Δ_x should not exceed the distance between point loads.

Figure 4.24 Notations for the Connection Between Flange and Web



4. The transverse reinforcement per unit length A_{sf}/s_f may be determined as follows:

$$\frac{A_{sf} f_{yd}}{s_f} \geq \frac{v_{Ed} h_f}{\cot \theta_f}$$

Equation 4.77

To prevent crushing of the compression struts in the flange, the following condition should be satisfied:

$$v_{Ed} \leq v f_{cd} \sin \theta_f \cos \theta_f$$

Equation 4.78

Note: The recommended values for $\cot \theta_f$ in the absence of more rigorous calculation are:

$1.0 \leq \cot \theta_f \leq 2.0$ for compression flanges ($45^\circ \geq \theta_f \geq 26.5^\circ$)

$1.0 \leq \cot \theta_f \leq 1.25$ for tension flanges ($45^\circ \geq \theta_f \geq 38.6^\circ$)

v is a strength reduction factor for concrete cracked in shear = $0.6 (1 - f_{ck} / 250)$.

5. In the case of combined shear between the flange and the web, and transverse bending, the area of steel should be the greater of that given by Equation 4.77 or half that given by Equation 4.77 plus that required for transverse bending.

For the verification of concrete compression crushing according to Equation 4.78, h_f should be reduced by the depth of compression considered in the bending assessment.

6. If v_{Ed} is less than or equal to $0.4 f_{cd}$, no extra reinforcement above that for flexure is required.
7. Longitudinal tension reinforcement in the flange should be anchored beyond the strut required to transmit the force back to the web at the section where this reinforcement is required (See Section (A - A) of Figure 4.24).

4.6.2.5 Shear at the Interface Between Concrete Cast at Different Times

1. In addition to the requirements of Clause 4.6.2.1 - 4.6.2.4, the shear stress at the interface between concrete cast at different times should also satisfy the following:

$$v_{Edi} \leq v_{Rdi}$$

Equation 4.79

v_{Edi} is the design value of the shear stress in the interface and is given by:

$$v_{Edi} = \beta V_{Ed} / (z b_i)$$

Where,

β = The ratio of the longitudinal force in the new concrete area and the total longitudinal force either in the compression or tension zone, both calculated for the section considered.

V_{Ed} = The transverse shear force.

z = The lever arm of composite section.

b_i = The width of the interface (see Figure 4.25).

v_{Rdi} = The design shear resistance at the interface and is given by:

$$v_{Rdi} = c f_{ctd} + \mu \sigma_n + \rho f_{yd} (\mu \sin \alpha + \cos \alpha) \leq 0.5 v f_{cd}$$

Equation 4.80

Where,

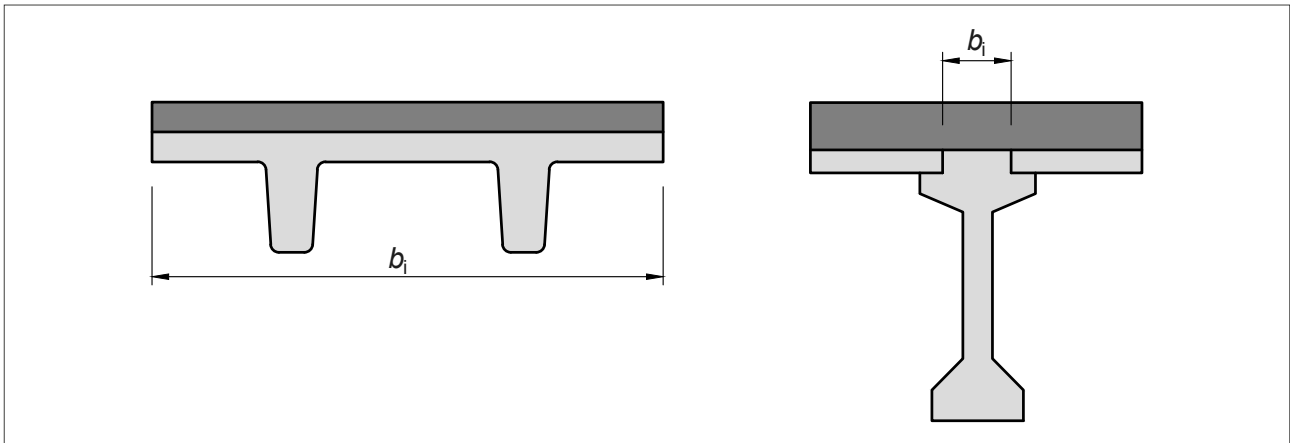
c and μ are factors which depend on the roughness of the interface

f_{ctd} = As defined in Clause 4.3.1.5.

σ_n = Stress per unit area caused by the minimum external normal force across the interface that can act simultaneously with the shear force, positive for compression, such that $\sigma_n < 0.6 f_{cd}$, and negative for tension. When σ_n is tensile $c f_{ctd}$ should be taken as 0.

$$\rho = A_s / A_i$$

Figure 4.25 Examples of Interfaces



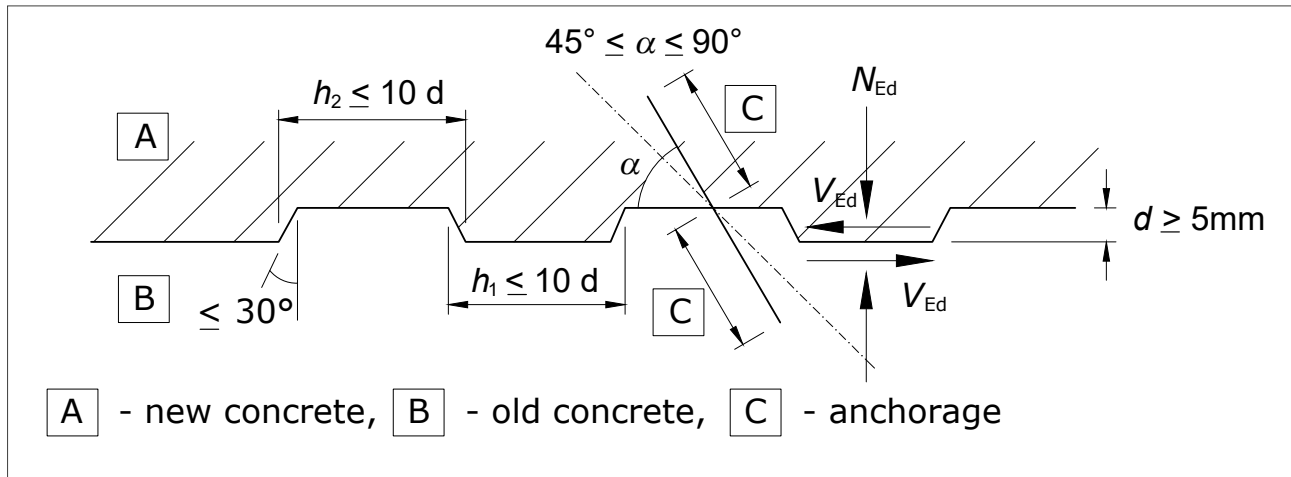
A_s = The area of reinforcement crossing the interface, including ordinary shear reinforcement (if any), with adequate anchorage at both sides of the interface.

A_i = The area of the joint.

α = Defined in Figure 4.26, and should be limited by $45^\circ \leq \alpha \leq 90^\circ$.

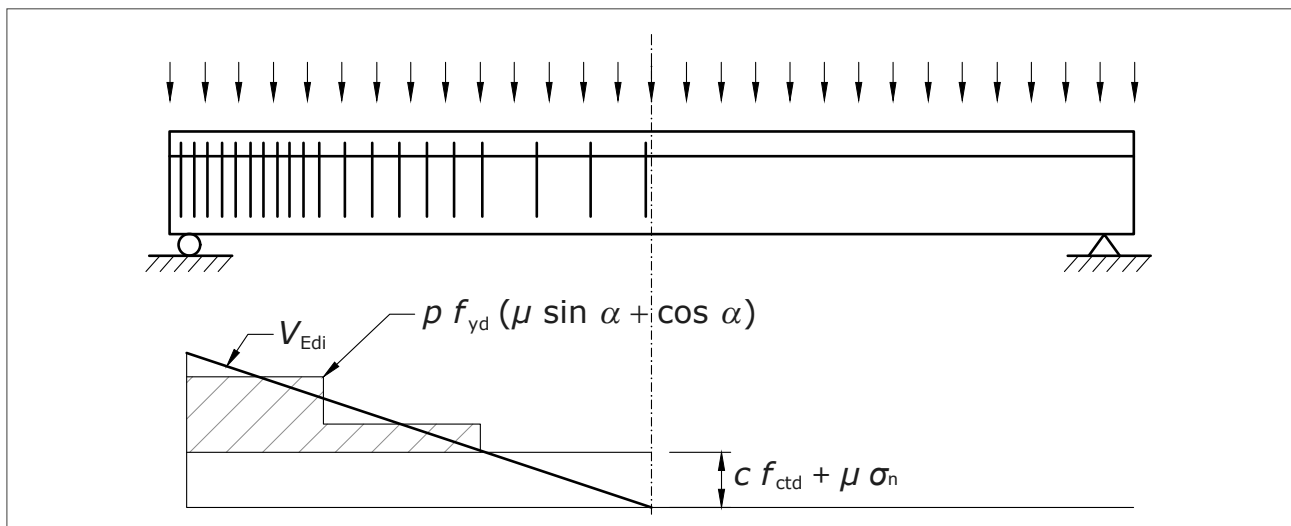
v = A strength reduction factor, recommended as $0.6 (1 - f_{ck} / 250)$.

Figure 4.26 Indented Construction Joint



2. In the absence of more detailed information, surfaces may be classified as very smooth, smooth, rough or indented, with the following examples:
 - Very smooth: a surface cast against steel, plastic or specially prepared wooden moulds: $c = 0.025$ to 0.10 and $\mu = 0.5$.
 - Smooth: a slip-formed or extruded surface, or a free surface left without further treatment after vibration: $c = 0.20$ and $\mu = 0.6$.
 - Rough: a surface with at least 3 mm roughness at about 40 mm spacing, achieved by raking, exposing of aggregate or other methods giving an equivalent behaviour: $c = 0.40$ and $\mu = 0.7$.
 - Indented: a surface with indentations complying with Figure 4.26: $c = 0.50$ and $\mu = 0.9$.
3. A stepped distribution of the transverse reinforcement may be used, as indicated in Figure 4.27. Where the connection between the two different concretes is ensured by reinforcement, the steel contribution to v_{Rdi} may be taken as the resultant of the forces taken from each of the diagonals provided that $45^\circ \leq \alpha \leq 135^\circ$.
4. The longitudinal shear resistance of grouted joints between slab or wall elements may be calculated according to 4.6.2.5 (1). However, in cases where the joint can be significantly cracked, c should be taken as 0 for smooth and rough joints and 0.5 for indented joints.
5. For fatigue or dynamic verifications, the values for c in para (1) should be taken as zero.

Figure 4.27 Shear Diagram Representing the Required Interface Reinforcement



4.6.2.6 Shear and Transverse Bending

1. Due to the presence of compressive stress fields arising from shear and bending, the interaction between longitudinal shear and transverse bending in the webs of box girder sections should be considered in the design.
2. When $V_{Ed}/V_{Rd,max} < 0.2$ or $M_{Ed}/M_{Rd,max} < 0.1$ this interaction can be disregarded; where $V_{Rd,max}$ and $M_{Rd,max}$ represent respectively the maximum web capacity for longitudinal shear and transverse bending.

4.6.3 Torsion

4.6.3.1 General

1. Where the static equilibrium of a structure depends on the torsional resistance of elements of the structure, a full torsional design covering both ultimate and serviceability limit states shall be made.
2. Where, in statically indeterminate structures, torsion arises from consideration of compatibility only, and the structure is not dependent on the torsional resistance for its stability, then it will normally be unnecessary to consider torsion at the ultimate limit state. In such cases a minimum reinforcement, given in Clause 4.7.3 and 4.9.2, in the form of stirrups and longitudinal bars should be provided in order to prevent excessive cracking.
3. The torsional resistance of a section may be calculated on the basis of a thin-walled closed section, in which equilibrium is satisfied by a closed shear flow. Solid sections may be modelled by equivalent thin-walled sections. Complex shapes, such as T-sections, may be divided into a series of sub-sections, each of which is modelled as an equivalent thin-walled section, and the total torsional resistance taken as the sum of the capacities of the individual elements.
4. The distribution of the acting torsional moments over the sub-sections should be in proportion to their uncracked torsional stiffnesses. For non-solid sections the equivalent wall thickness should not exceed the actual wall thickness.
5. Each sub-section may be designed separately.

4.6.3.2 Design Procedure

1. The shear stress in a wall of a section subject to a pure torsional moment may be calculated from:

$$\tau_{t,i} t_{ef,i} = \frac{T_{Ed}}{2 A_k}$$

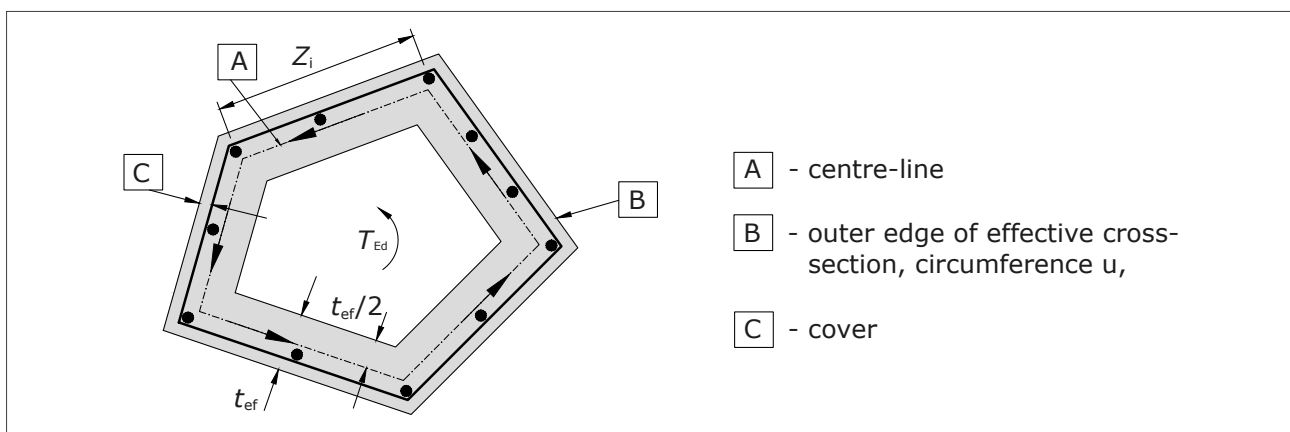
Equation 4.81

The shear force $V_{Ed,i}$ in a wall i due to torsion is given by,

$$V_{Ed,i} = \tau_{t,i} t_{ef,i} z_i$$

Equation 4.82

Figure 4.28 Notations and Definitions Used for Torsion



Where,

T_{Ed} = The applied design torsion (refer Figure 4.28).

A_k = The area enclosed by the centre-lines of the connecting walls, including inner hollow areas.

$\tau_{t,i}$ = The torsional shear stress in wall i .

$t_{ef,i}$ = The effective wall thickness. It may be taken as A/u , but should not be taken as less than twice the distance between edge and centre of the longitudinal reinforcement. For hollow sections the real thickness is an upper limit.

A = The total area of the cross-section within the outer circumference, including inner hollow areas.

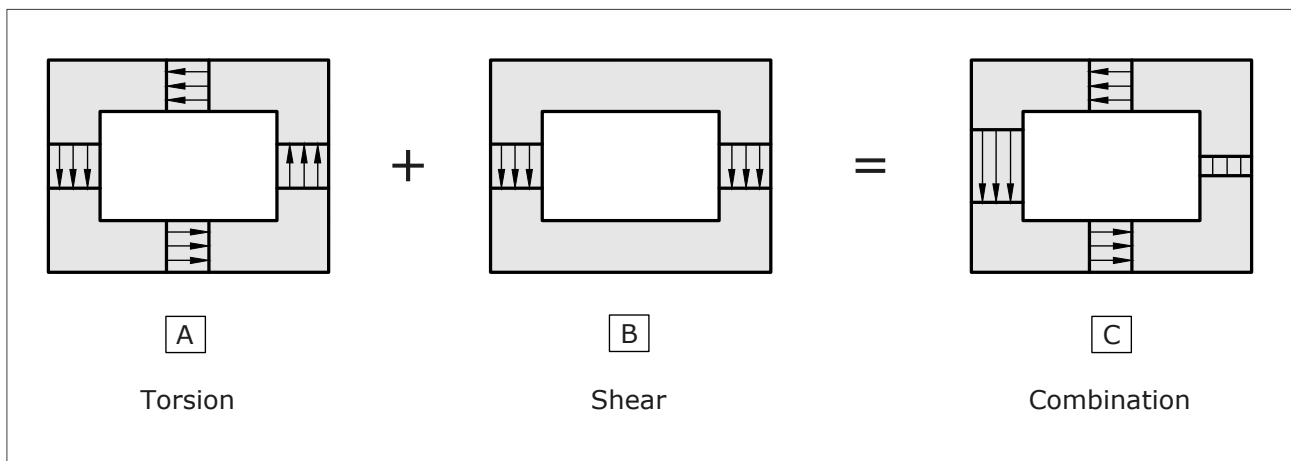
u = The outer circumference of the cross-section.

z_i = The side length of wall i defined by the distance between the intersection points with the adjacent walls.

- The effects of torsion and shear for both hollow and solid members may be superimposed, assuming the same value for the strut inclination θ . The limits for θ given in Clause 4.6.2.3 (2) are also fully applicable for the case of combined shear and torsion. The maximum bearing capacity of a member loaded in shear and torsion follows from Clause 4.6.3.2 (4).

For box sections, each wall should be verified separately, for the combination of shear forces derived from shear and torsion (Figure 4.29).

Figure 4.29 Internal Actions Combination within the Different Walls of a Box Section



- The required area of the longitudinal reinforcement for torsion ΣA_{sl} may be calculated from following Equation 4.83:

$$\frac{\Sigma A_{sl} f_{yd}}{u_k} = \frac{T_{Ed}}{2A_k} \cot \theta$$

Equation 4.83

Where,

u_k = The perimeter of the area A_k .

f_{yd} = The design yield stress of the longitudinal reinforcement A_{sl} .

θ = The angle of compression struts (see Figure 4.21).

In compressive chords, the longitudinal reinforcement may be reduced in proportion to the available compressive force. In tensile chords the longitudinal reinforcement for torsion should be added to the other reinforcement. The longitudinal reinforcement should generally be distributed over the length of side, z_i but for smaller sections it may be concentrated at the ends of this length.

Bonded prestressing tendons can be taken into account limiting their stress increase to $\Delta \sigma_p \leq 500$ MPa. In that case, $\Sigma A_{sl} f_{yd}$ in Equation 4.83 is replaced by $\Sigma A_{sl} f_{yd} + A_p \Delta \sigma_p$.

4. The maximum resistance of a member subjected to torsion and shear is limited by the capacity of the concrete struts. In order not to exceed this resistance the following condition should be satisfied:

For solid cross-sections:

$$\frac{T_{Ed}}{T_{RD,max}} + \frac{V_{Ed}}{V_{RD,max}} \leq 1.0 \quad \text{Equation 4.84}$$

Where,

T_{Ed} = The design torsional moment.

V_{Ed} = The design transverse force.

$T_{RD,max}$ = The design torsional resistance moment according to.

$$T_{RD,max} = 2 \nu \alpha_{cw} f_{cd} A_k t_{ef,i} \sin\theta \cos\theta \quad \text{Equation 4.85}$$

Where ν follows from Equation 4.60 and α_{cw} from Equation 4.62.

$V_{RD,max}$ is the maximum design shear resistance according to Equation 4.62 or 4.70. In solid cross-sections the full width of the web may be used to determine $V_{RD,max}$.

For box sections:

Each wall should be designed separately for combined effects of shear and torsion. The ultimate limit state for concrete should be checked with reference to the design shear resistance $V_{RD,max}$.

5. For approximately rectangular solid sections only minimum reinforcement is required (refer Clause 4.9.2.1.1) provided that the following condition is satisfied:

$$\frac{T_{Ed}}{T_{RD,c}} + \frac{V_{Ed}}{V_{RD,c}} \leq 1.0 \quad \text{Equation 4.86}$$

Where,

$T_{RD,c}$ = The torsional cracking moment, which may be determined by setting $\tau_{t,i} = f_{ctd}$.

$V_{RD,c}$ = Follows from Equation 4.55 to 4.56.

4.6.3.3 Warping Torsion

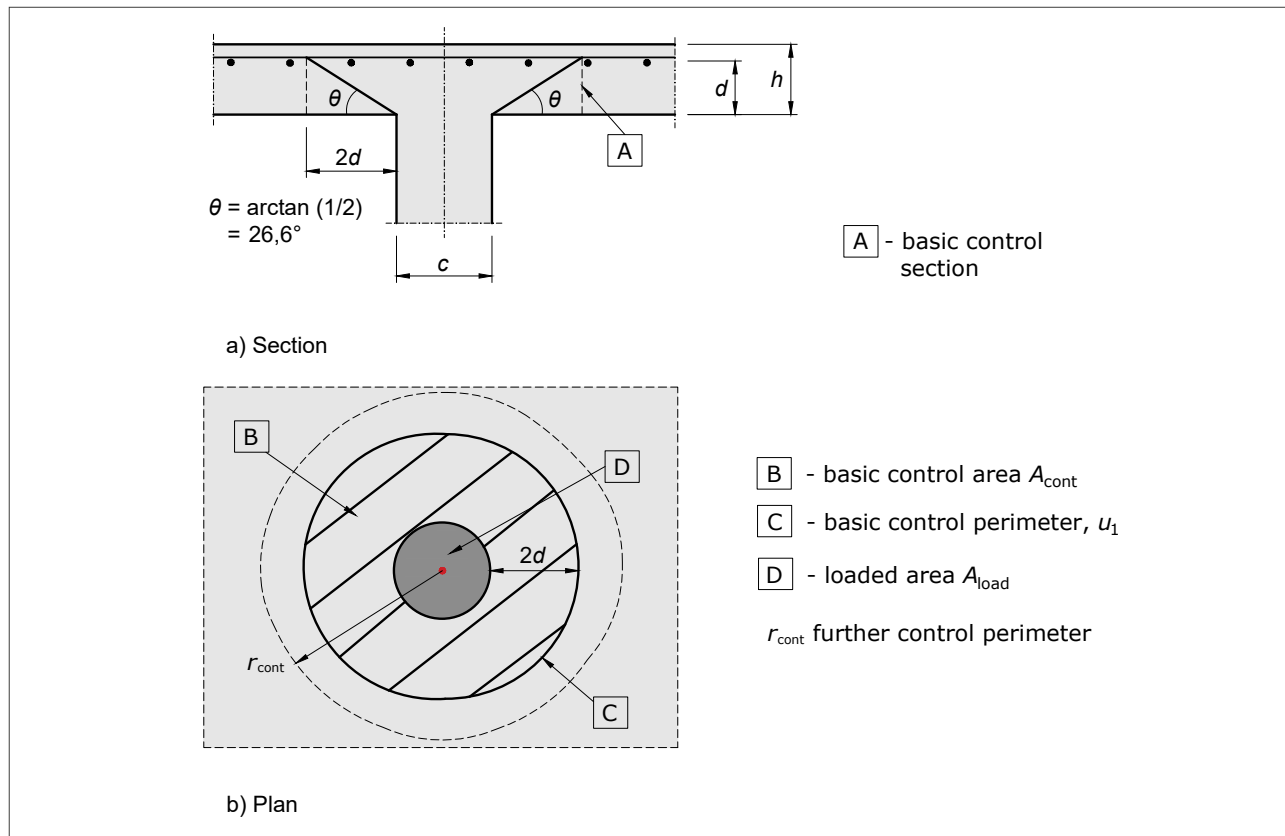
1. For closed thin-walled sections and solid sections, warping torsion may normally be ignored.
2. In open thin-walled members, it may be necessary to consider warping torsion. For very slender cross-sections the calculation should be carried out on the basis of a beam-grid model and for other cases on the basis of a truss model. In all cases the design should be carried out according to the design rules for bending and longitudinal normal force, and for shear.

4.6.4 Punching

4.6.4.1 General

1. The punching shear is a local shear in solid slabs and can be applied for the design of deck slabs, pile caps, pad footings, slabs supported over columns.
2. Punching shear can result from a concentrated load or reaction acting on a relatively small area, called the loaded area A_{load} of a slab or a foundation.
3. An appropriate verification model for checking punching failure at the ultimate limit state is shown in Figure 4.30.

Figure 4.30 Verification Model for Punching Shear at the Ultimate Limit State



4. The shear resistance should be checked at the face of the column and at the basic control perimeter u_1 . If shear reinforcement is required a further perimeter $u_{out,ef}$ should be found where shear reinforcement is no longer required.
5. The guidelines given for punching shear are principally formulated for the case of uniformly distributed loading. In special cases, such as footings, the load within the control perimeter adds to the resistance of the structural system, and may be subtracted when determining the design punching shear stress.

4.6.4.2 Load Distribution and Basic Control Perimeter

1. The basic control perimeter u_1 may normally be taken to be at a distance $2.0d$ from the loaded area and should be constructed so as to minimise its length (see Figure 4.31). The effective depth of the slab is assumed constant and may normally be taken as:

$$d_{eff} = \frac{d_y + d_z}{2}$$

Equation 4.87

Where d_y and d_z are the effective depths of the reinforcement in two orthogonal directions.

2. Control perimeters at a distance less than $2d$ should be considered where the concentrated force is opposed by a high pressure (e.g. soil pressure on a base), or by the effects of a load or reaction within a distance $2d$ of the periphery of area of application of the force.
3. For loaded areas situated near openings, if the shortest distance between the perimeter of the loaded area and the edge of the opening does not exceed $6d$, that part of the control perimeter contained between two tangents drawn to the outline of the opening from the centre of the loaded area is considered to be ineffective (see Figure 4.32).

Figure 4.31 Control Perimeters Near an Opening

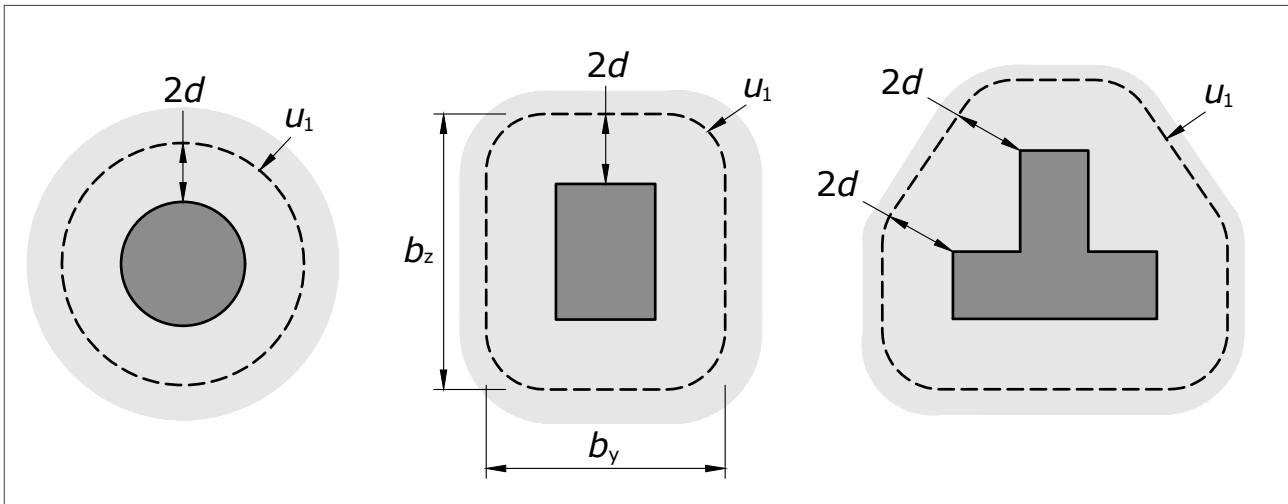
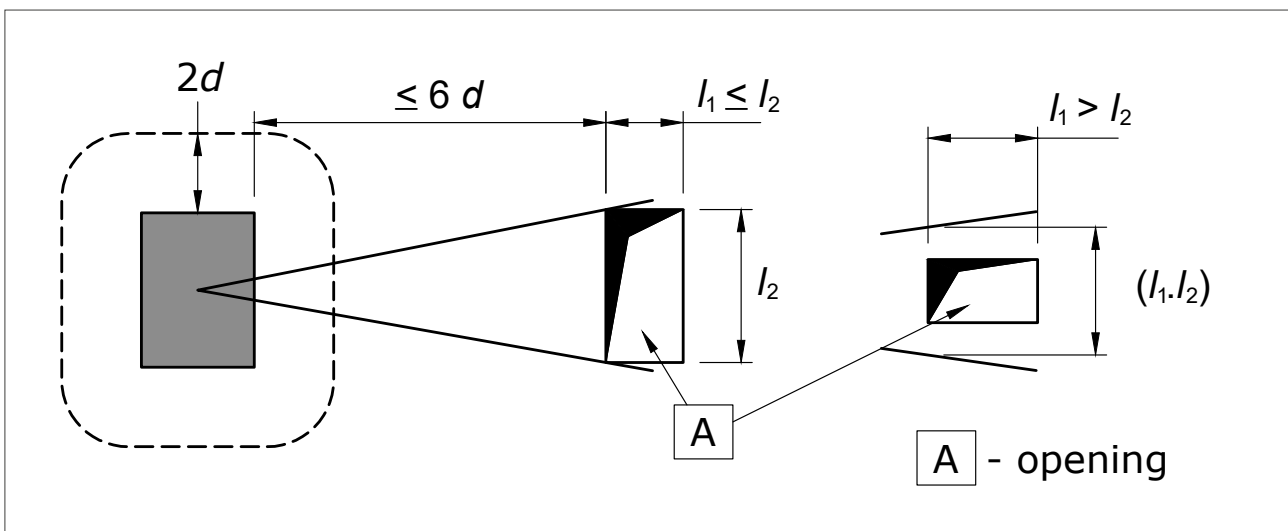
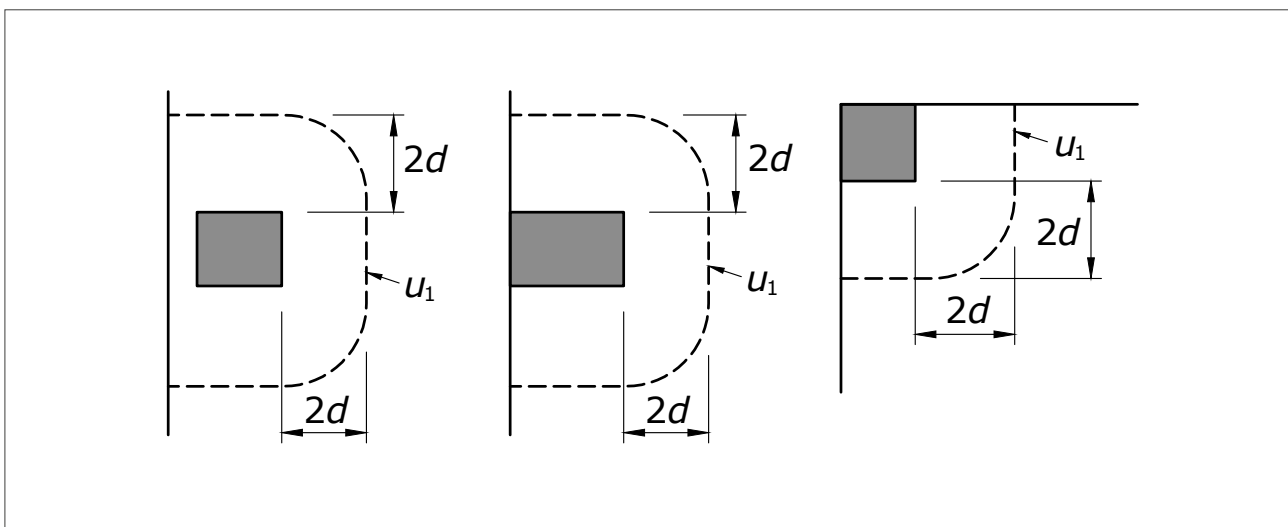


Figure 4.32 Typical Basic Control Perimeters Around Corner



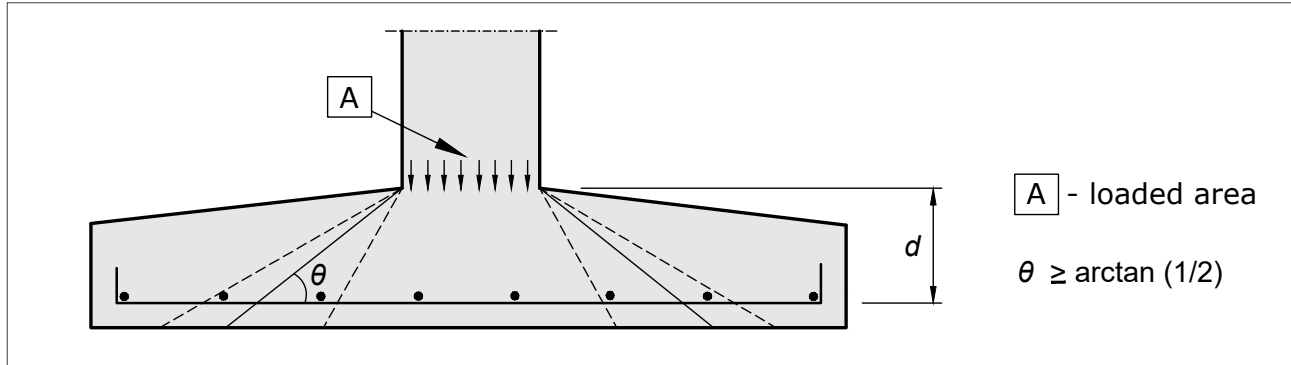
4. For a loaded area situated near an edge or a corner, the control perimeter should be taken as shown in Figure 4.33, if this gives a perimeter (excluding the unsupported edges) smaller than that obtained from paragraph (1) and (2) above.

Figure 4.33 Basic Control Parameters for Loaded Areas Close to or at Edge or Corner



5. For loaded areas situated near an edge or corner, i.e. at a distance smaller than d , special edge reinforcement should always be provided, refer Clause 4.9.3.1.4.
6. The control section is that which follows the control perimeter and extends over the effective depth d . For slabs of constant depth, the control section is perpendicular to the middle plane of the slab. For slabs or footings of variable depth other than step footings, the effective depth may be assumed to be the depth at the perimeter of the loaded area as shown in Figure 4.34.

Figure 4.34 Depth of Control Section in a Footing with Variable Depth



7. Further perimeters, u_i , inside and outside the basic control area should have the same shape as the basic control perimeter.
8. For slabs with circular column heads for which $l_H < 2h_H$ (see Figure 4.35) a check of the punching shear stresses according to Clause 4.6.4.3 is only required on the control section outside the column head. The distance of this section from the centroid of the column r_{cont} may be taken as:

$$r_{cont} = 2d + l_H + 0.5c$$

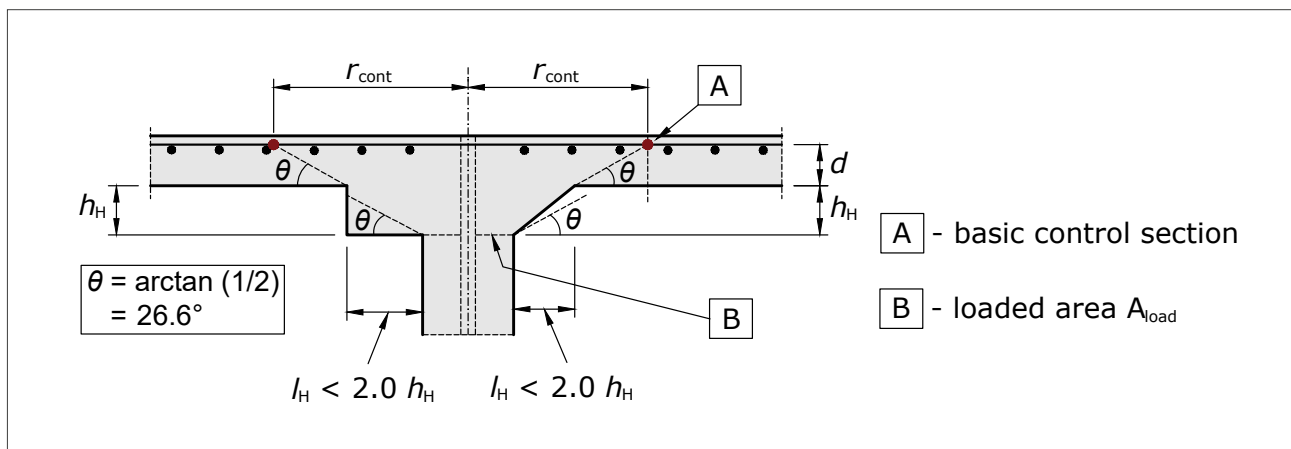
Equation 4.88

Where,

l_H = The distance from the column face to the edge of the column head.

c = The diameter of a circular column.

Figure 4.35 Slab with Enlarged Column Head Where $l_H < 2.0 h_H$



For a rectangular column with a rectangular head with $l_H < 2.0 h_H$ (refer Figure 4.35) and overall dimensions l_1 and l_2 ($l_1 = c_1 + 2l_{H1}$, $l_2 = c_2 + 2l_{H2}$, $l_1 \leq l_2$), the value r_{cont} may be taken as the lesser of:

$$r_{cont} = 2d + 0.56\sqrt{l_1 l_2}$$

Equation 4.89

and

$$r_{cont} = 2d + 0.69l_1$$

Equation 4.90

9. For slabs with enlarged column heads where $l_H > 2h_H$ (see Figure 4.36) control sections both within the head and in the slab should be checked.
10. The provisions of Clause 4.6.4.2 and 4.6.4.3 also apply for checks within the column head with d taken as d_H according to Figure 4.36.
11. For circular columns the distances from the centroid of the column to the control sections in Figure 4.36 may be taken as:

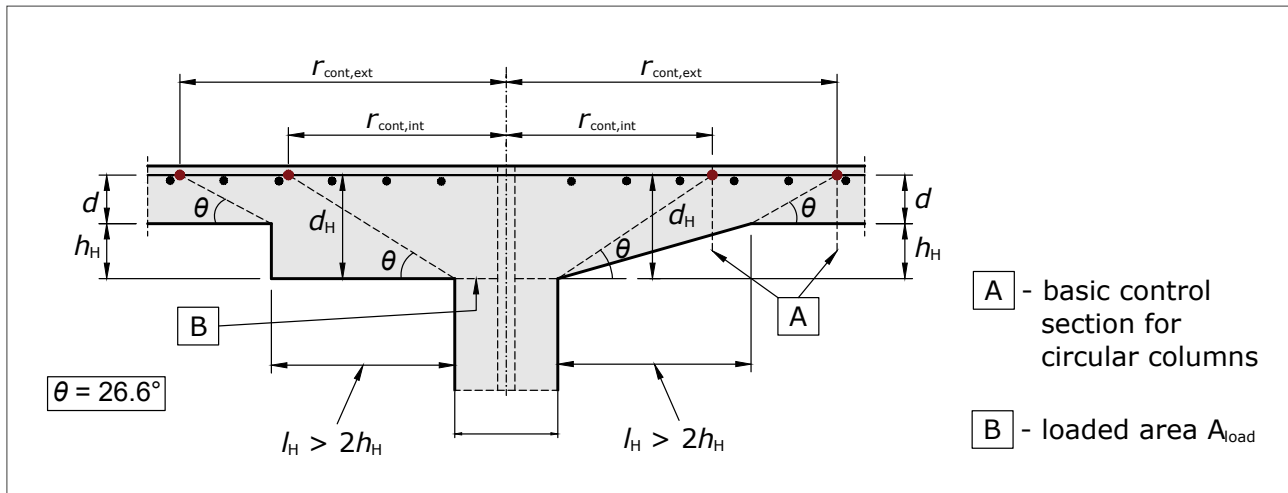
$$r_{cont,ext} = l_H + 2d + 0.5c$$

Equation 4.91

$$r_{cont,ext} = 2(d + h_H) + 0.5c$$

Equation 4.92

Figure 4.36 Slab with Enlarged Column Head Where $l_H > 2.0h_H$



4.6.4.3 Punching Shear Calculation

1. The design procedure for punching shear is based on checks at the face of the column and at the basic control perimeter u_1 . If shear reinforcement is required a further perimeter $u_{out,ef}$ (refer Figure 4.40) should be found where shear reinforcement is no longer required. The following design shear stresses (MPa) along the control sections, are defined:
 - $v_{Rd,c}$ = The design value of the punching shear resistance of a slab without punching shear reinforcement along the control section considered.
 - $v_{Rd,cs}$ = The design value of the punching shear resistance of a slab with punching shear reinforcement along the control section considered.
 - $v_{Rd,max}$ = The design value of the maximum punching shear resistance along the control section considered.
2. The following checks should be carried out:
 - a. At the column perimeter, or the perimeter of the loaded area, the maximum punching shear stress should not be exceeded:

$$v_{Ed} \leq v_{Rd,max}$$

b. Punching shear reinforcement is not necessary if:

$$v_{Ed} \leq v_{Rd,c}$$

Where v_{Ed} exceeds the value $v_{Rd,c}$ for the control section considered, punching shear reinforcement should be provided according to Clause 4.6.4.5.

3. Where the support reaction is eccentric with regard to the control perimeter, the maximum shear stress should be taken as:

$$v_{Ed} = \frac{\beta V_{Ed}}{u_i d}$$

Equation 4.93

Where,

d = The mean effective depth of the slab, which may be taken as $(d_y + d_z)/2$ where:

d_y, d_z = The effective depths in the y- and z- directions of the control section.

u_i = The length of the control perimeter being considered.

β = given by:

$$\beta = 1 + k \frac{M_{Ed}}{V_{Ed}} \frac{u_1}{W_1}$$

Equation 4.94

Where,

u_1 = The length of the basic control perimeter

k = A coefficient dependent on the ratio between the column dimensions c_1 and c_2 : its value is a function of the proportions of the unbalanced moment transmitted by uneven shear and by bending and torsion (see Table 4.14)

W_1 = Corresponds to a distribution of shear as illustrated in Figure 4.37 and is a function of the basic control perimeter u_1 :

$$W_1 = \int_0^{u_1} |e| dl$$

Equation 4.95

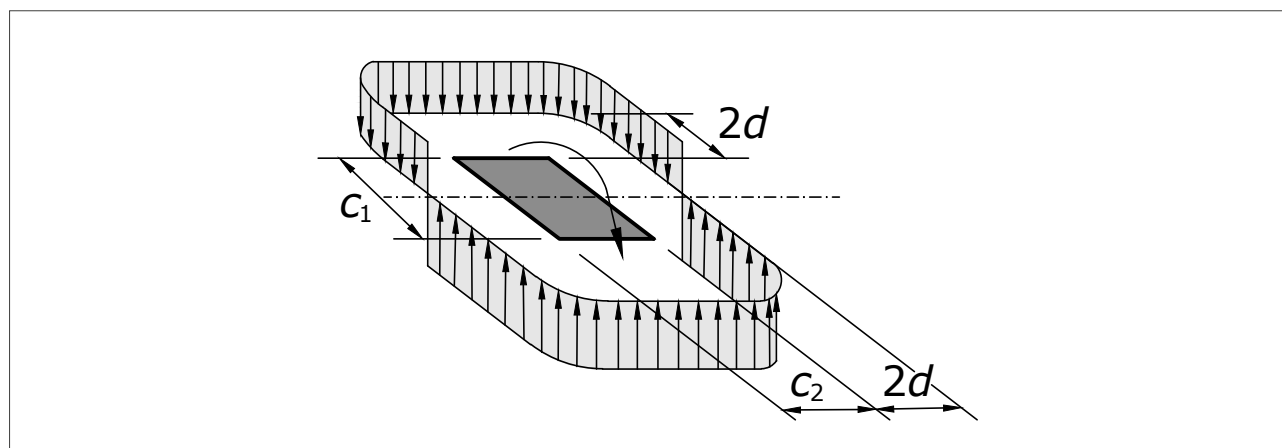
dl = A length increment of the perimeter

e = The distance of dl from the axis about which the moment M_{Ed} acts

Table 4.14 Values of k for Rectangular Loaded Areas

c_1/c_2	≤ 0.5	1.0	2.0	≥ 3.0
k	0.45	0.60	0.70	0.80

Figure 4.37 Shear Distribution Due to an Unbalanced Moment at a Slab-Internal Column Connection



For a rectangular column:

$$W_1 = \frac{c_1^2}{2} + c_1 c_2 + 4 c_2 d + 16 d^2 + 2\pi d c_1$$

Equation 4.96

Where,

c_1 = The column dimension parallel to the eccentricity of the load.

c_2 = The column dimension perpendicular to the eccentricity of the load.

For internal circular columns β follows from:

$$\beta = 1 + 0.6 \pi \frac{e}{D + 4 d}$$

Equation 4.97

Where,

D = The diameter of the circular column

e = The eccentricity of the applied load $e = M_{Ed} / V_{Ed}$

For an internal rectangular column where the loading is eccentric to both axes, the following approximate Equation for β may be used:

$$\beta = 1 + 1.8 \sqrt{\left(\frac{e_y}{b_z}\right)^2 + \left(\frac{e_z}{b_y}\right)^2}$$

Equation 4.98

Where,

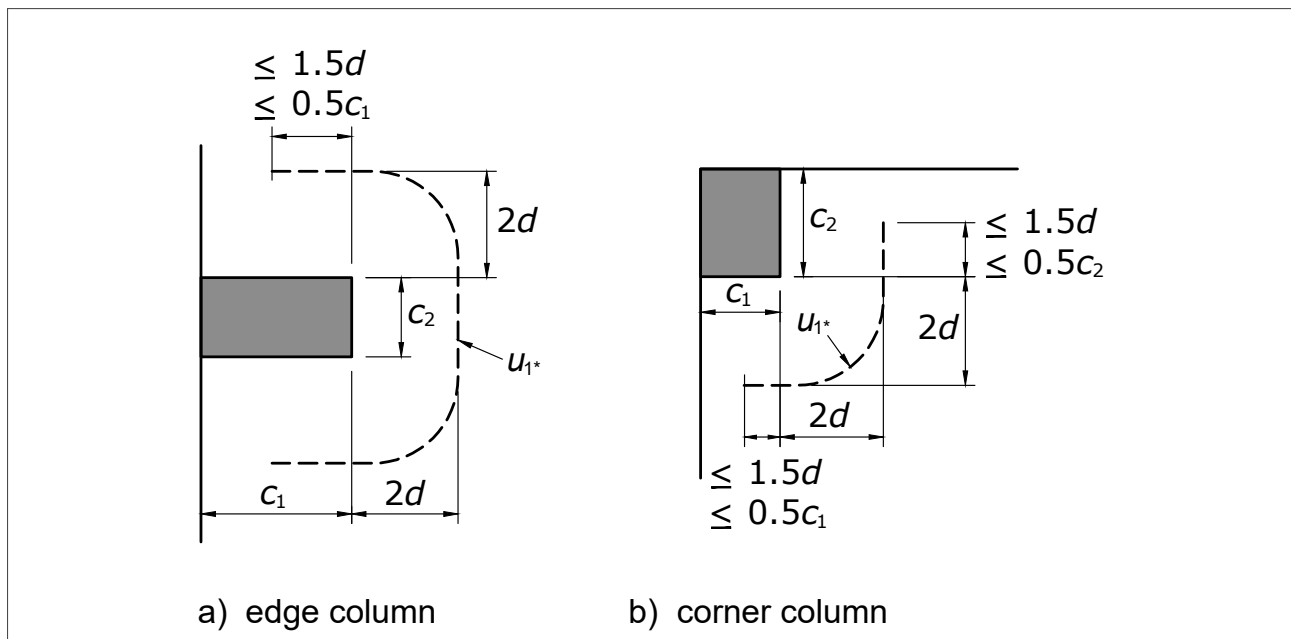
e_y and e_z = The eccentricities M_{Ed}/V_{Ed} along y and z axes respectively

b_y and b_z = The dimensions of the control perimeter (see Figure 4.31)

Note: e_y results from a moment about the z axis and e_z from a moment about the y axis

4. For edge column connections, where the eccentricity perpendicular to the slab edge (resulting from a moment about an axis parallel to the slab edge) is toward the interior and there is no eccentricity parallel to the edge, the punching force may be considered to be uniformly distributed along the control perimeter u_1^* as shown in Figure 4.38(a).

Figure 4.38 Reduced Basic Control Perimeter u_1



Where there are eccentricities in both orthogonal directions, β may be determined using the following Equation:

$$\beta = \frac{u_1}{u_{1*}} + \frac{u_1}{W_1} e_{par}$$

Equation 4.99

Where,

u_1 = The basic control perimeter (see Figure 4.33)

u_{1*} = The reduced basic control perimeter (see Figure 4.38(a))

e_{par} = The eccentricity parallel to the slab edge resulting from a moment about an axis perpendicular to the slab edge

k may be determined from Table 6.1 with the ratio c_1/c_2 replaced by $c_1/2c_2$

W_1 is calculated for the basic control perimeter u_1 (see Figure 4.31)

For a rectangular column as shown in Figure 4.38(a):

$$W_1 = \frac{c_2^2}{4} + c_1 c_2 + 4c_1 d + 8d^2 + \pi d c_2$$

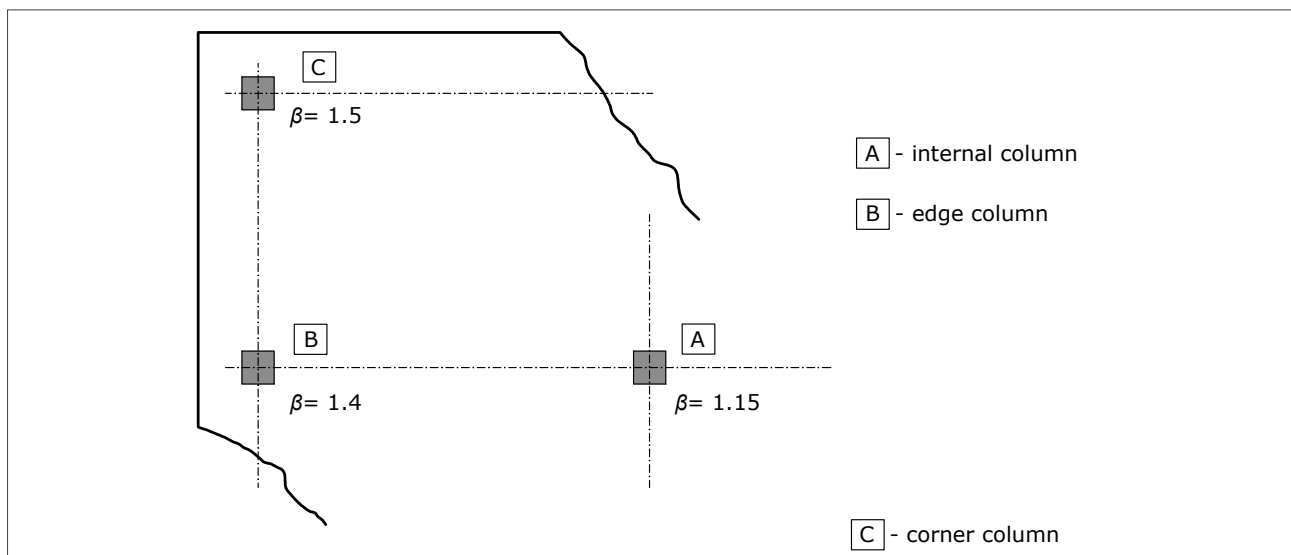
Equation 4.100

If the eccentricity perpendicular to the slab edge is not toward the interior, Equation 4.94 applies. When calculating W_1 the distance e should be measured from the centroid axis of the control perimeter.

5. For corner column connections, where the eccentricity is toward the interior of the slab, it is assumed that the punching force is uniformly distributed along the reduced control perimeter u_{1*} , as defined in Figure 4.38(b). The β -value may then be considered as $\beta = u_1 / u_{1*}$. If the eccentricity is toward the exterior, Equation 4.94 applies.
6. For structures where the lateral stability does not depend on frame action between the slabs and the columns, and where the adjacent spans do not differ in length by more than 25%, approximate values for β may be used.

Note: Recommended values of β are given in Figure 4.39.

Figure 4.39 Recommended Value for β



7. Where a concentrated load is applied close to a flat slab column support, the shear force reduction according to Clause 4.6.2.2(5) and 4.6.2.3(8) respectively is not valid and should not be included.
8. The punching shear force V_{Ed} in a foundation slab may be reduced due to the favourable action of the soil pressure.
9. The vertical component V_{pd} resulting from inclined prestressing tendons crossing the control section may be taken into account as a favourable action where relevant.

4.6.4.4 Punching Shear Resistance Of Slabs And Column Bases Without Shear Reinforcement

1. The punching shear resistance of a slab should be assessed for the basic control section according to Clause 4.6.4.2. The design punching shear resistance [MPa] may be calculated as follows:

$$V_{Rd,c} = C_{Rd,c} k (100 \rho_l f_{ck})^{1/3} + k_1 \sigma_{cp} \geq (V_{min} + k_1 \sigma_{cp})$$

Equation 4.101

Where,

f_{ck} = In MPa.

$k = 1 + \sqrt{(200/d)} \leq 2.0$ d in mm.

$\rho_l = \sqrt{(\rho_{ly} \rho_{lz})} \leq 0.02$.

ρ_{ly}, ρ_{lz} = relate to the bonded tension steel in y- and z- directions respectively. The values value ρ_{ly} and ρ_{lz} should be calculated as mean values taking into account a slab width equal to the column width plus $3d$ each side.

$\sigma_{cp} = (\sigma_{cy} + \sigma_{cz})/2$.

Where,

σ_{cy}, σ_{cz} = the normal concrete stresses in the critical section in y- and z directions (MPa, positive if compression).

$\sigma_{cy} = N_{Ed,y} / A_{cy}$ and $\sigma_{cz} = N_{Ed,z} / A_{cz}$.

$N_{Ed,y}, N_{Ed,z}$ = the longitudinal forces across the full bay for internal columns and the longitudinal force across the control section for edge columns. The force may be from a load or prestressing action.

A_c is the area of concrete according to the definition of N_{Ed} .

Note: The recommended value for $C_{Rd,c}$ is $0.18/\gamma_c$, for v_{min} is given by Equation 4.57 and that for k_1 is 0.1.

2. The punching resistance of column bases should be verified at control perimeters within $2d$ from the periphery of the column.

For concentric loading the net applied force is given by:

$$V_{Ed,red} = V_{Ed} - \Delta V_{Ed}$$

Equation 4.102

Where,

V_{Ed} = The applied shear force.

ΔV_{Ed} = The net upward force within the control perimeter considered i.e. upward pressure from soil minus self weight of base.

$$V_{Ed} = \frac{V_{Ed,red}}{u d}$$

Equation 4.103

$$V_{Rd} = C_{Rd,c} k (100 \rho_l f_{ck})^{1/3} 2 \frac{d}{a} \geq v_{min} \frac{2d}{a}$$

Equation 4.104

Where,

a = The distance from the periphery of the column to the control perimeter considered

$C_{Rd,c}, v_{min}$ and k are defined in paragraph (1).

For eccentric loading:

$$V_{Ed} = \frac{V_{Ed,red}}{u d} \left[1 + k \frac{M_{Ed} u}{V_{Ed,red} W} \right]$$

Equation 4.105

Where k is defined in Clause 4.6.4.3 (3) or 4.6.4.3(4) as appropriate and W is similar to W_1 but for perimeter u .

4.6.4.5 Punching Shear Resistance of Slabs and Column Bases with Shear Reinforcement

1. Where shear reinforcement is required, it should be calculated in accordance with following Equation:

$$v_{Rd,cs} = 0.75 v_{Rd,c} + 1.5 \frac{d}{s_r} A_{sw} f_{ywd,ef} \frac{1}{u_1 d} \sin \alpha \leq k_{max} v_{Rd,c}$$

Equation 4.106

Where,

 A_{sw} = The area of one perimeter of shear reinforcement around the column [mm²]. s_r = The radial spacing of perimeters of shear reinforcement [mm]. $f_{ywd,ef}$ = The effective design strength of the punching shear reinforcement, according to:

$$f_{ywd,ef} = 250 + 0.25 d \leq f_{ywd} \text{ [MPa]}$$

 d = The mean of the effective depths in the orthogonal directions [mm]. α = The angle between the shear reinforcement and the plane of the slab. $v_{Rd,c}$ = given in Clause 4.6.4.4. k_{max} = The factor limiting the maximum capacity that can be achieved by application of shear reinforcement. Recommended value of k_{max} is 1.5.

If a single line of bent-down bars is provided, then the ratio d/s_r in Equation 4.106 may be given the value 0.67.

2. Detailing requirements for punching shear reinforcement are given in Clause 4.9.4.3.
3. Adjacent to the column the punching shear resistance is limited to a maximum of:

$$v_{Ed} = \frac{\beta V_{Ed}}{u_0 d} \leq v_{Rd,max}$$

Equation 4.107

Where,

 u_o for an interior column, u_o = enclosing minimum periphery [mm]for an edge column, $u_o = c_2 + 3d \leq c_2 + 2c_1$ [mm]for a corner column, $u_o = 3d \leq c_1 + c_2$ [mm] c_1, c_2 are the column dimensions as shown in Figure 4.38 β refer section 4.6.4.3(3), (4) and (5)

Note: The recommended value $v_{Rd,max}$ is $0.5 v_{fcd}$ where v is given in Expression (4.60). In addition, v_{Ed} should be limited to $2 v_{rdc}$ at the first control perimeter.

4. The control perimeter at which shear reinforcement is not required, u_{out} (or $u_{out,ef}$ refer Figure 4.40) should be calculated from following Equation:

$$u_{out,ef} = \frac{\beta V_{Ed}}{v_{Rdc} d}$$

Equation 4.108

The outermost perimeter of shear reinforcement should be placed at a distance not greater than kd within u_{out} (or $u_{out,ef}$ refer Figure 4.40). The recommended value of k is 1.5.

1

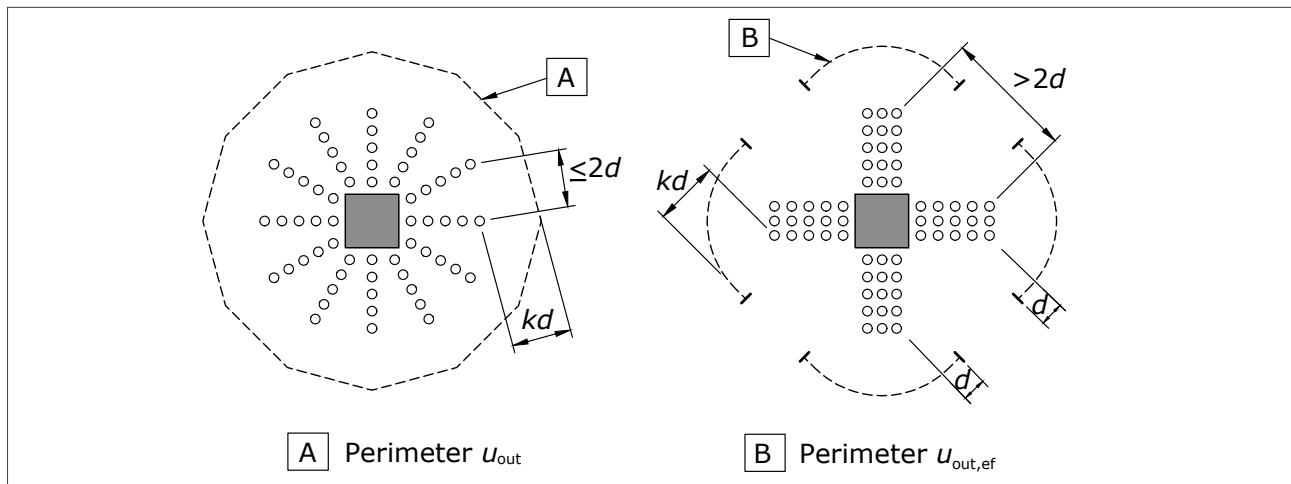
2

3

4

Concrete Structures

Figure 4.40 Control Perimeters at Internal Columns



4.6.5 Design with Strut and Tie Models

4.6.5.1 General

Where a non-linear strain distribution exists (e.g. supports, near concentrated loads or plain stress) strut-and-tie models may be used.

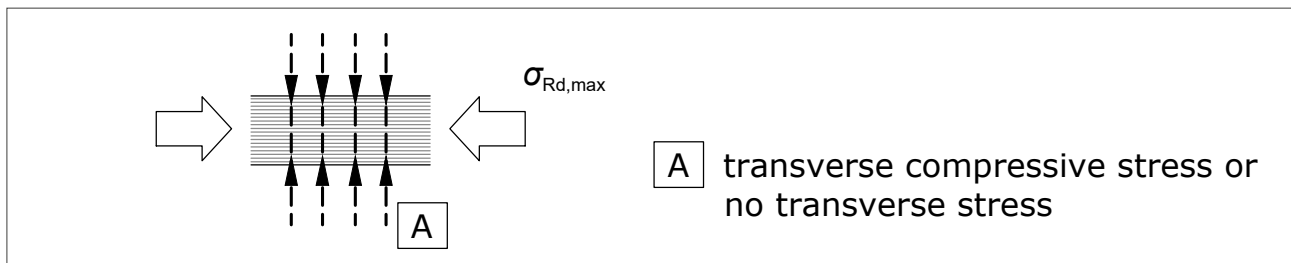
4.6.5.2 Struts

The design strength for a concrete strut in a region with transverse compressive stress or no transverse stress may be calculated from following Equation.

$$\sigma_{Rd,max} = f_{cd}$$

Equation 4.109

Figure 4.41 Design Strength of Concrete Struts Without Transverse Tension

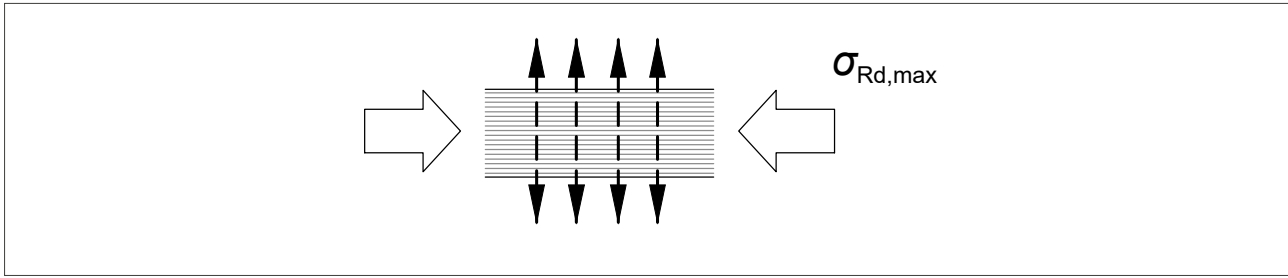


It may be appropriate to assume a higher design strength in regions where multi-axial compression exists.

1. The design strength for concrete struts should be reduced in cracked compression zones and, unless a more rigorous approach is used, may be calculated from following Equation.

$$\sigma_{Rd,max} = 0.6 v' f_{cd}$$

Equation 4.110

Figure 4.42 Design Strength Of Concrete Struts With Transverse Tension

Note: The recommended value of v' is given by following Expression.

$$v' = 1 - \frac{f_{ck}}{250}$$

Equation 4.111

- For struts between directly loaded areas, such as corbels or short deep beams, alternative calculation methods are given in Clause 4.6.2.2 and 4.6.2.3.

4.6.5.3 Ties

- The design strength of transverse ties and reinforcement should be limited in accordance with Clause 4.3.2 and 4.3.3.
- Reinforcement should be adequately anchored in the nodes.
- Reinforcement required to resist the forces at the concentrated nodes may be smeared over a length (see Figure 4.43). When the reinforcement in the node area extends over a considerable length of an element, the reinforcement should be distributed over the length where the compression trajectories are curved (ties and struts). The tensile force T may be obtained by:

- For partial discontinuity regions ($b \leq H/2$),

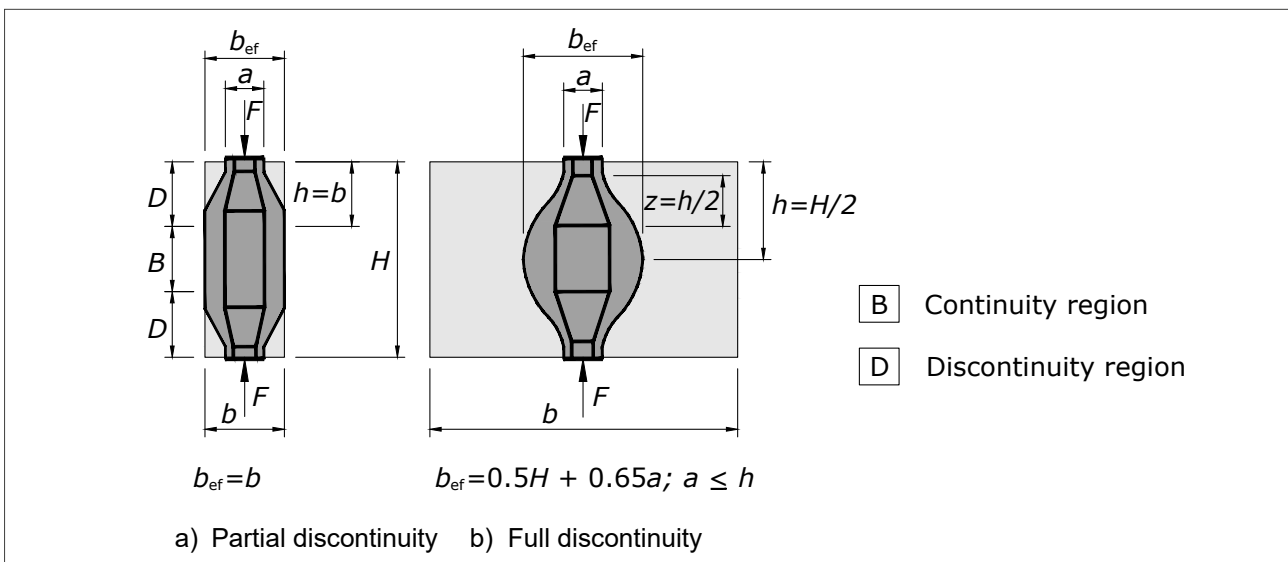
$$T = \frac{1}{4} \frac{b-a}{b} F$$

Equation 4.112

- For full discontinuity regions ($b > H/2$),

$$T = \frac{1}{4} \left(1 - 0.7 \frac{a}{h}\right) F$$

Equation 4.113

Figure 4.43 Parameters For The Determination Of Transverse Tensile Forces In A Compression Field With Smeared Reinforcement

4.6.5.4 Nodes

1. The rules for nodes also apply to regions where concentrated forces are transferred in a member and which are not designed by the strut-and-tie method.
2. The forces acting at nodes shall be in equilibrium. Transverse tensile forces perpendicular to an in-plane node shall be considered.
3. The dimensioning and detailing of concentrated nodes are critical in determining their loadbearing resistance. Concentrated nodes may develop, e.g. where point loads are applied, at supports, in anchorage zones with concentration of reinforcement or prestressing tendons, at bends in reinforcing bars, and at connections and corners of members.
4. The design values for the compressive stresses within nodes may be determined by:
 - a. In compression nodes where no ties are anchored at the node:

$$\sigma_{Rd,max} = 1.0 \left(1 - \frac{f_{ck}}{250} \right) f_{cd}$$

Equation 4.114

Where $\sigma_{Rd,max}$ is the maximum stress which can be applied at the edges of the node.

Figure 4.44 Compression Node Without Ties

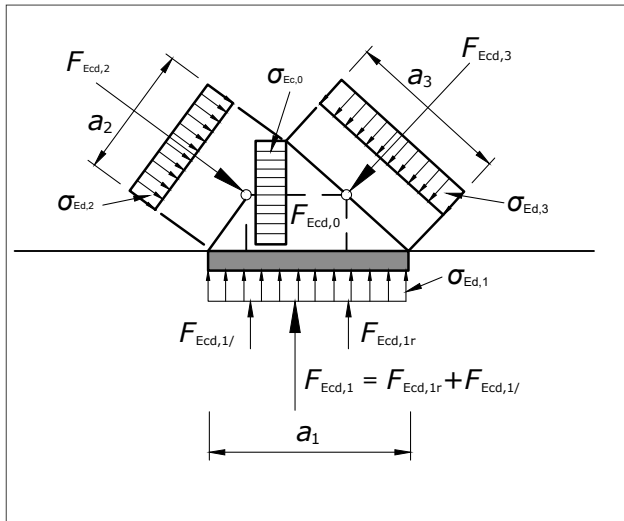
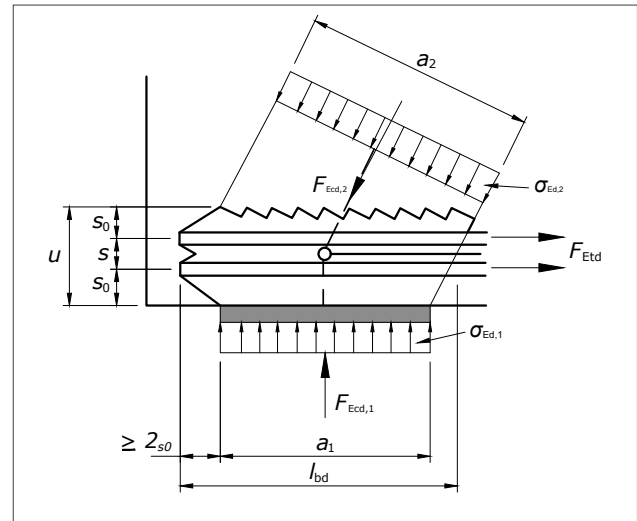


Figure 4.45 Compression Tension Node With Reinforcement Provided in One Direction



- b. In compression - tension nodes with anchored ties provided in one direction:

$$\sigma_{Rd,max} = 0.85 \left(1 - \frac{f_{ck}}{250} \right) f_{cd}$$

Equation 4.115

Where,

$\sigma_{Rd,max}$ is the maximum of $\sigma_{Ed,1}$ and $\sigma_{Ed,2}$.

- c. In compression - tension nodes with anchored ties provided in more than one direction:

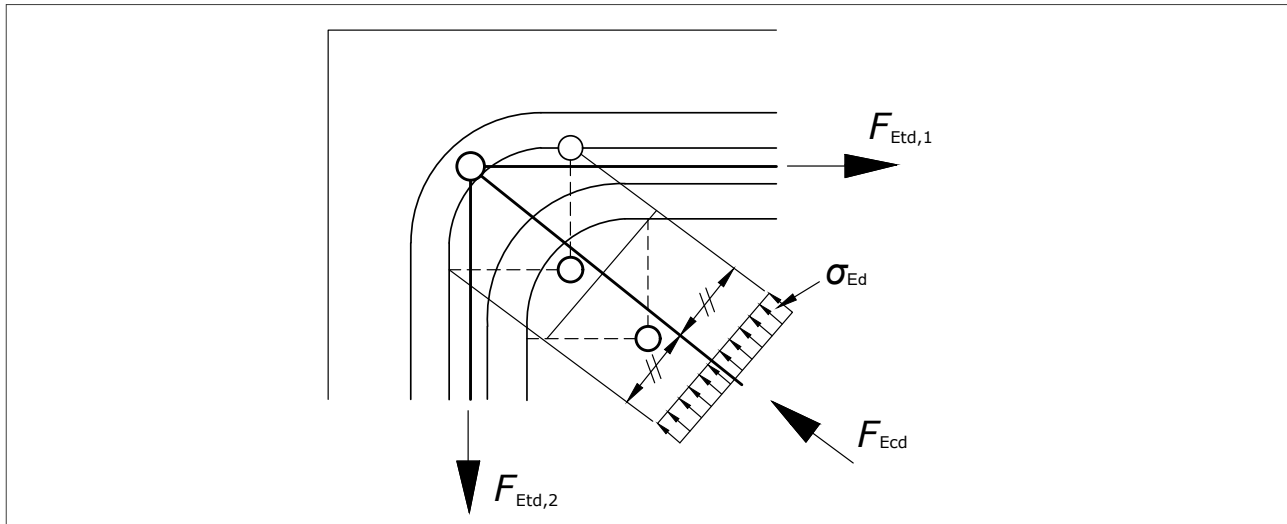
$$\sigma_{Rd,max} = 0.75 \left(1 - \frac{f_{ck}}{250} \right) f_{cd}$$

Equation 4.116

Where,

$\sigma_{Rd,max}$ is the maximum compressive stress which can be applied at the edges of the nodes.

Figure 4.46 Compression Tension Node With Reinforcement Provided In Two Directions



5. Under the conditions listed below, the design compressive stress values given Clause 4.6.5.4(4) may be increased by up to 10% where at least one of the following applies:
 - Triaxial compression is assured
 - All angles between struts and ties are $\geq 55^\circ$
 - The stresses applied at supports or at point loads are uniform, and the node is confined by stirrups
 - The reinforcement is arranged in multiple layers
 - The node is reliably confined by means of bearing arrangement or friction

6. Triaxially compressed nodes may be checked according to Equation 4.20 and 4.21 with an upper limit $\sigma_{Ed,max} \leq k_4 v' f_{cd}$ if for all three directions of the struts the distribution of load is known.

Note: The recommended value of k_4 is 3.0.

7. The anchorage of the reinforcement in compression-tension nodes starts at the beginning of the node, e.g. in case of a support anchorage starting at its inner face (see Figure 4.45). The anchorage length should extend over the entire node length. In certain cases, the reinforcement may also be anchored behind the node.
8. In-plane compression nodes at the junction of three struts may be verified in accordance with Figure 4.44. The maximum average principal node stresses (σ_{c0} , σ_{c1} , σ_{c2} , σ_{c3}) should be checked in accordance with Clause 4.6.5.4 (4) a). Normally the following may be assumed:

$$F_{cd,1} / a_1 = F_{cd,2} / a_2 = F_{cd,3} / a_3 \text{ resulting in } \sigma_{cd,1} = \sigma_{cd,2} = \sigma_{cd,3} = \sigma_{cd,0}.$$

9. Nodes at reinforcement bends may be analysed in accordance with Figure 4.46. The average stresses in the struts should be checked in accordance with Clause 4.6.5.4 (5). The diameter of the mandrel should be checked in accordance with Clause 4.8.3.

4.6.6 Anchorages and Laps

1. The design bond stress is limited to a value depending on the surface characteristics of the reinforcement, the tensile strength of the concrete and confinement of surrounding concrete. This depends on cover, transverse reinforcement and transverse pressure.
2. The length necessary for developing the required tensile force in an anchorage or lap is calculated on the basis of a constant bond stress.
3. Application rules for the design and detailing of anchorages and laps are given in 4.8.4 to 4.8.8.

4.6.7 Partially Loaded Areas

1. For partially loaded areas, local crushing and transverse tension forces (refer section 4.6.5) shall be considered.
2. For a uniform distribution of load on an area A_{c0} (see Figure 4.47) the concentrated resistance force may be determined as follows:

$$F_{Rdu} = A_{c0} f_{cd} \sqrt{\frac{A_{c1}}{A_{c0}}} \leq 3.0 f_{cd} A_{c0}$$

Equation 4.117

Where,

A_{c0} = The loaded area,

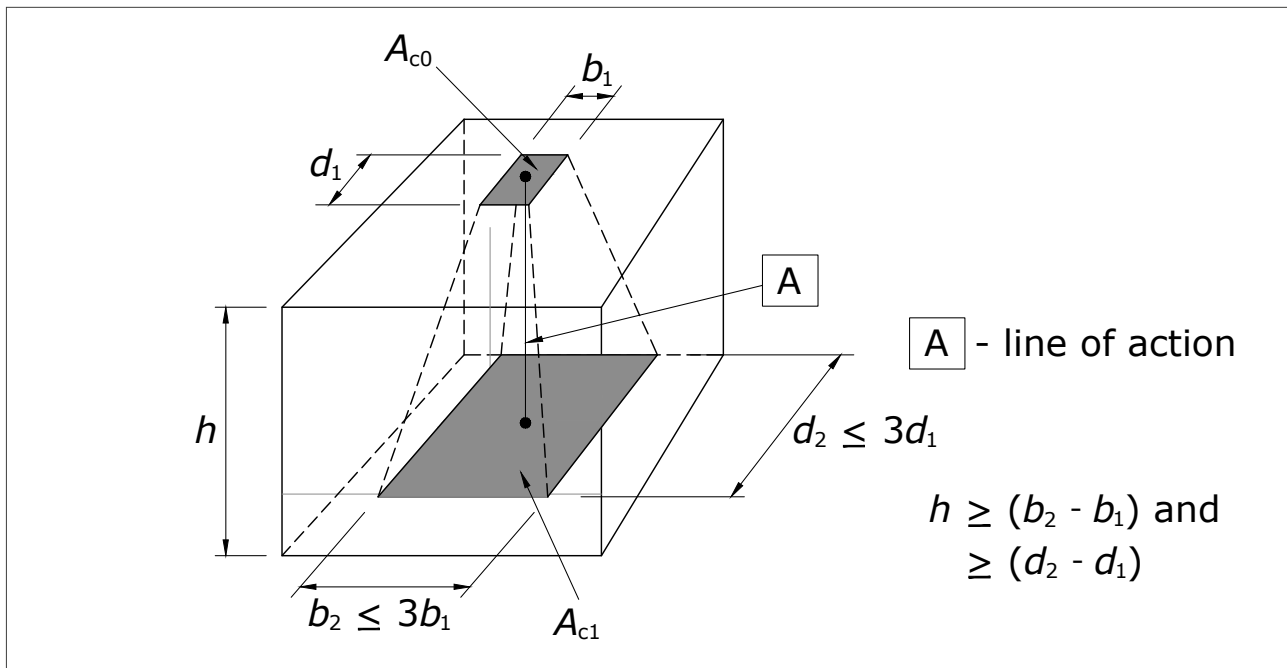
A_{c1} = The maximum design distribution area with a similar shape to A_{c0}

3. The design distribution area A_{c1} required for the resistance force F_{Rdu} should correspond to the following conditions:
 - The height for the load distribution in the load direction should correspond to the conditions given in Figure 4.47.
 - the centre of the design distribution area A_{c1} should be on the line of action passing through the centre of the load area A_{c0} .
 - If there is more than one compression force acting on the concrete cross section, the designed distribution areas should not overlap.

The value of F_{Rdu} should be reduced if the load is not uniformly distributed on the area A_{c0} or if high shear forces exist.

4. Reinforcement should be provided for the tensile force due to the effect of the action.
5. The design of bearing zones for bridges should be carried out using recognised methods.

Figure 4.47 Design Distribution For Partially Loaded Areas



4.6.8 Bearing Zones

The bearing zones of bridges should be in accordance with the guidelines provided in Clause 4.6.7 and following additional provisions.

The distance of the free edge of the concrete section from the edge of the loaded area should not be less than 1/6 of the corresponding dimension of the loaded area measured in the same direction. In no case should the distance to the free edge be taken as less than 50 mm.

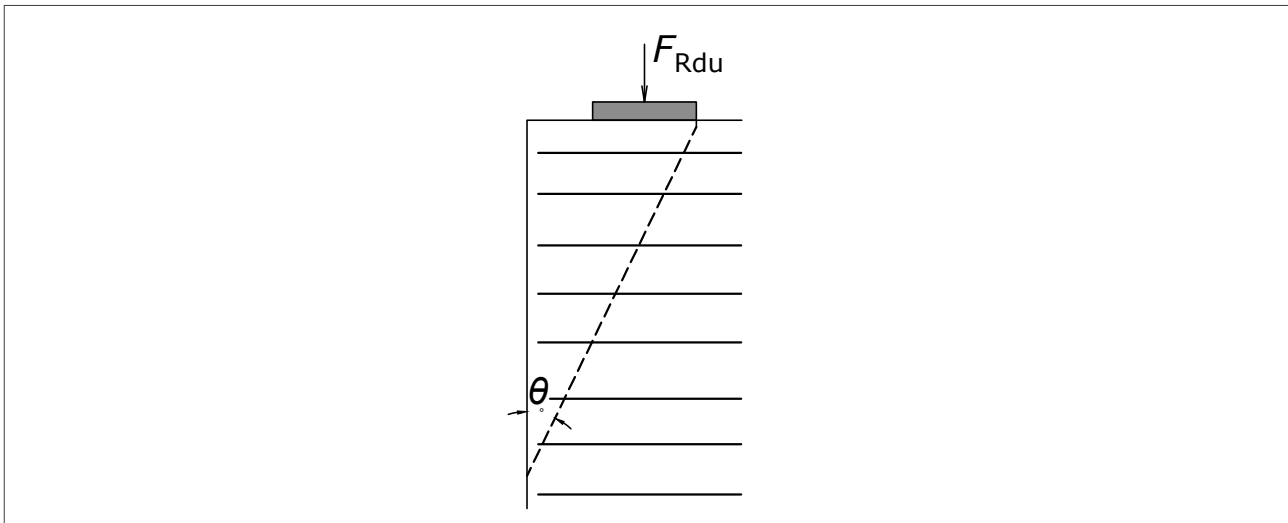
For concrete classes equal to or higher than C55/67, the concentrated resistance force may be determined as follows:

$$F_{Rdu} = A_{co} f_{cd} \frac{0.46 f_{ck}^{2/3}}{1 + 0.1 f_{ck}} \sqrt{\frac{A_{c1}}{A_{c0}}} \leq 3.0 A_{c0} f_{cd} \frac{0.46 f_{ck}^{2/3}}{1 + 0.1 f_{ck}}$$

Equation 4.118

To avoid edge sliding, uniformly distributed reinforcement parallel to the loaded face should be provided to the point at which local compressive stresses are dispersed. This point is determined by drawing a line inclined at an angle $\theta = 30^\circ$ to the direction of load application to intersect with the opposite edge of the loaded surface, as shown in following Figure. The reinforcement provided to avoid edge sliding should be adequately anchored.

Figure 4.48 Edge Sliding Mechanism



The reinforcement provided in order to avoid edge sliding (A_r) should be calculated in accordance with the Expression $A_{r,fyd} \geq F_{Rdu}/2$.

4.6.9 Fatigue

4.6.9.1 Verification Conditions

1. The resistance of structures to fatigue shall be verified in special cases. This verification shall be performed separately for concrete and steel.
2. A fatigue verification should be carried out for structures and structural components which are subjected to regular load cycles.

Note: A fatigue verification is generally not necessary for the following structures and structural elements:

- a. Footbridges, with the exception of structural components very sensitive to wind action.
- b. Buried arch and frame structures with a minimum earth cover of 1.00 m for road bridges.
- c. Foundations.

- d. Piers and columns which are not rigidly connected to superstructures.
- e. Retaining walls of embankments for roads.
- f. Abutments of road bridges which are not rigidly connected to superstructures, abutments; except the slabs of hollow abutments.
- g. Prestressing and reinforcing steel, in regions where, under the frequent combination of actions and P_k , only compressive stresses occur at the extreme concrete fibres.

Fatigue verification for road bridges is not necessary for local effects of wheel loads applied directly to a slab spanning between beams or webs provided that:

- The slab does not contain welded reinforcement, or reinforcement couplers.
- The clear span to overall depth ratio of the slab does not exceed 18.
- The slab acts compositely with its supporting beams or webs.
- Either:
 - i. the slab also acts compositely with transverse diaphragms; or
 - ii. the width of the slab perpendicular to its span exceeds three times its clear span

4.7 Serviceability Limit State (SLS)

4.7.1 General

1. This section covers the common serviceability limit states. These are:
 - Stress limitation
 - Crack control
 - Deflection control
2. In the calculation of stresses and deflections, cross-sections should be assumed to be uncracked provided that the flexural tensile stress does not exceed $f_{ct,eff}$. The value of $f_{ct,eff}$ may be taken as f_{ctm} or $f_{ctm,fl}$ provided that the calculation for minimum tension reinforcement is also based on the same value. For the purposes of calculating crack widths and tension stiffening f_{ctm} should be used.

4.7.2 Stress Limitation

1. The compressive stress in the concrete shall be limited in order to avoid longitudinal cracks, micro-cracks or high levels of creep, where they could result in unacceptable effects on the function of the structure.
2. Longitudinal cracks may occur if the stress level under the characteristic combination of loads exceeds a critical value. Such cracking may lead to a reduction of durability. In the absence of other measures, such as an increase in the cover to reinforcement in the compressive zone or confinement by transverse reinforcement, it may be appropriate to limit the compressive stress to a value $0.6 f_{ck}$ in areas exposed to environments of exposure classes XD, XF and XS (Table 4.8). The maximum stress limit in the presence of confinement may be taken as $0.66 f_{ck}$.
3. If the stress in the concrete under the quasi-permanent loads is less than $0.45 f_{ck}$, linear creep may be assumed. If the stress in concrete exceeds $0.45 f_{ck}$, non-linear creep should be considered.
4. Tensile stresses in the reinforcement shall be limited in order to avoid inelastic strain, unacceptable cracking or deformation.
5. For the appearance unacceptable cracking or deformation may be assumed to be avoided if, under the characteristic combination of loads, the tensile strength in the reinforcement does not exceed $0.8 f_{yk}$. Where the stress is caused by an imposed deformation, the tensile strength should not exceed f_{yk} . The mean value of the stress in prestressing tendons should not exceed $0.75 f_{yk}$.

4.7.3 Crack Control

4.7.3.1 General Considerations

1. Cracking shall be limited to an extent that will not impair the proper functioning or durability of the structure or cause its appearance to be unacceptable.
2. Cracking is normal in reinforced concrete structures subject to bending, shear, torsion or tension resulting from either direct loading or restraint or imposed deformations.
3. Cracks may also arise from other causes such as plastic shrinkage or expansive chemical reactions within the hardened concrete. Such cracks may be unacceptably large but their avoidance and control lie outside the scope of this Section.
4. Cracks may be permitted to form without any attempt to control their width, provided they do not impair the functioning of the structure.
5. The value of w_{max} is given in Table 4.15. The decompression limit requires that all concrete within a certain distance of bonded tendons or their ducts should remain in compression under the specified loading. The distance within which all concrete should remain in compression should be taken as the value of $c_{min,dur}$. Where the most tensile face of a section is not subject to XD or XS exposure but another face is, the decompression limit should require all tendons within 100 mm of a surface subject to XD or XS exposure to have a depth $c_{min,dur}$ of concrete in compression between them and surfaces subject to XD or XS exposure.

The recommended value for w_{max} and the application of the decompression limit are given in Table 4.15.

Table 4.15 Recommended Values of w_{max} in mm and Relevant Combination Rules

Exposure Class ^a	Reinforced members and prestressed members without bonded tendons	Prestressed members with bonded tendons
	Quasi-permanent load combination ^b	Frequent load combination ^b
X0, XC1	0.3 ^c	0.2
XC2, XC3, XC4	0.3	0.2 ^d
XD1, XD2, XD3 XS1, XS2, XS3	0.3	0.2 ^e and decompression

a The exposure class considered, including at transfer, applies to the most severe exposure, the surface will be subjected to in service.
b For the crack width checks under combinations which include temperature distribution, the resulting member forces should be calculated using gross section concrete properties and self-equilibrating thermal stresses within a section may be ignored.
c For X0, XC1 exposure classes, crack width has no influence on durability and this limit is set to guarantee acceptable appearance. In the absence of appearance conditions this limit may be relaxed.
d For these exposure classes, in addition, decompression should be checked under the quasi-permanent combination of loads.
e 0.2 applies to the parts of the member that do not have to be checked for decompression.

6. For members with only unbonded tendons, the requirements for reinforced concrete elements apply. For members with a combination of bonded and unbonded tendons requirements for prestressed concrete members with bonded tendons apply.
7. Special measures may be necessary for members subjected to exposure class XD3. The choice of appropriate measures will depend upon the nature of the aggressive agent involved.
8. When using strut-and-tie models with the struts oriented according to the compressive stress trajectories in the uncracked state, it is possible to use the forces in the ties to obtain the corresponding steel stresses to estimate the crack width (refer Clause 4.5.6.3).
9. Crack widths may be calculated according to Clause 4.7.3.4. A simplified alternative is to limit the bar size or spacing according to Clause 4.7.3.3.
10. In some cases, it may be necessary to check and control shear cracking in webs.

4.7.3.2 Minimum Reinforcement Areas

1. If crack control is required, a minimum amount of bonded reinforcement is required to control cracking in areas where tension is expected. The amount may be estimated from equilibrium between the tensile force in concrete just before cracking and the tensile force in reinforcement at yielding or at a lower stress if necessary to limit the crack width.
2. Unless a more rigorous calculation shows lesser areas to be adequate, the required minimum areas of reinforcement may be calculated as follows. In profiled cross sections like T-beams and box girders, minimum reinforcement should be determined for the individual parts of the section (webs, flanges).

$$A_{s,min} \sigma_s = k_c k f_{ct,eff} A_{ct}$$

Equation 4.119

Where,

$A_{s,min}$ = The minimum area of reinforcing steel within the tensile zone

A_{ct} = The area of concrete within tensile zone. The tensile zone is that part of the section which is calculated to be in tension just before formation of the first crack. In flanged cross sections such as T-beams and box girders the division into parts should be as indicated in Figure 4.49.

σ_s = The absolute value of the maximum stress permitted in the reinforcement immediately after formation of the crack. This may be taken as the yield strength of the reinforcement, f_{yk} . A lower value may, however, be needed to satisfy the crack width limits according to the maximum bar size or the maximum bar spacing (refer Clause 4.7.3.3)

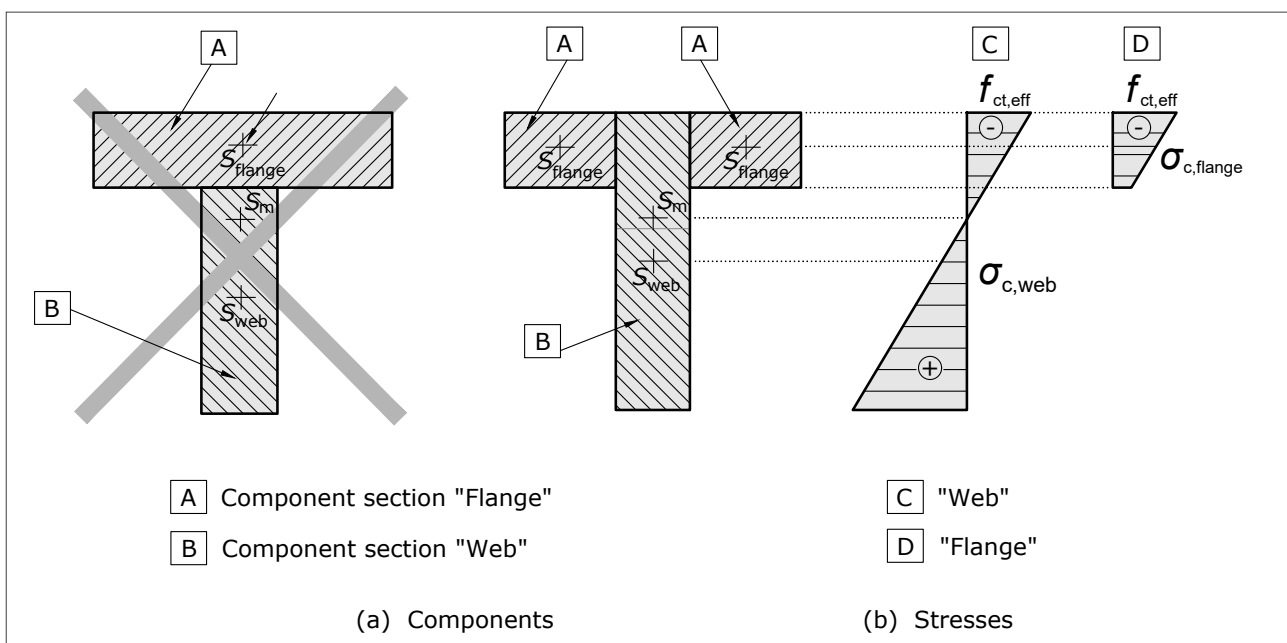
$f_{ct,eff}$ = The mean value of the tensile strength of the concrete effective at the time when the cracks may first be expected to occur

$f_{ct,eff}$ = f_{ctm} or lower ($f_{ctm}(t)$), if cracking is expected earlier than 28 days

k = The coefficient which allows for the effect of non-uniform self-equilibrating stresses, which lead to a reduction of restraint forces
 = 1.0 for webs with $h \leq 300$ mm or flanges with widths less than 300 mm
 = 0.65 for webs with $h \geq 800$ mm or flanges with widths greater than 800 mm (Intermediate values may be interpolated)

k_c = A coefficient which takes account of the stress distribution within the section immediately prior to cracking and of the change of the lever arm

Figure 4.49 Example for a Division of a Flanged Cross-Section for Analysis of Cracking



For pure tension $k_c = 1.0$

For bending or bending combined with axial forces:

- For rectangular sections and webs of box sections and T-sections:

$$k_c = 0.4 \left[1 - \frac{\sigma_c}{k_1 (h/h^*) f_{ct,eff}} \right] \leq 1 \quad \text{Equation 4.120}$$

- For flanges of box sections and T-sections:

$$k_c = 0.9 \frac{F_{cr}}{A_{ct} f_{ct,eff}} \geq 0.5 \quad \text{Equation 4.121}$$

Where,

σ_c = The mean stress of the concrete acting on the part of the section under consideration:

$$\sigma_c = \frac{N_{Ed}}{bh} \quad \text{Equation 4.122}$$

N_{Ed} = The axial force at the serviceability limit state acting on the part of the cross-section under consideration (compressive force positive). N_{Ed} should be determined considering the characteristic values of prestress and axial forces under the relevant combination of actions

h^* = h for $h < 1.0\text{m}$

h^* = 1.0 for $h \geq 1.0\text{m}$

k_1 = a coefficient considering the effects of axial forces on the stress distribution:

k_1 = 1.5 if N_{Ed} is a compressive force

k_1 = $(2 h^*) / (3 h)$ if N_{Ed} is a tensile force

F_{cr} = The absolute value of the tensile force within the flange immediately prior to cracking due to the cracking moment calculated with $f_{ct,eff}$

- Bonded tendons in the tension zone may be assumed to contribute to crack control within a distance ≤ 150 mm from the centre of the tendon. This may be taken into account by adding the term $\xi_1 A_p' \Delta\sigma_p$ to the left hand side of Equation 4.119 as under:

$$A_{s,min} \sigma_s + \xi_1 A_p' \Delta\sigma_p = k_c k f_{ct,eff} A_{ct}$$

Where,

A_p' = The area of pre or post-tensioned tendons within $A_{c,eff}$

$A_{c,eff}$ = The effective area of concrete in tension surrounding the reinforcement or prestressing tendons of depth, $h_{c,eff}$, where $h_{c,eff}$ is the lesser of $2.5(h-d)$, $(h-x)/3$ or $h/2$ (see Figure 4.50)

ξ_1 = The adjusted ratio of bond strength taking into account the different diameters of prestressing and reinforcing steel:

$$= \sqrt{\xi \cdot \frac{\phi_s}{\phi_p}} \quad \text{Equation 4.123}$$

ξ = The ratio of bond strength of prestressing and reinforcing steel, according to Table 4.16

ϕ_s = The largest bar diameter of reinforcing steel.

ϕ_p = The equivalent diameter of prestressing tendon.

$\phi_p = 1.6 \sqrt{A_p}$ for bundles.

$\phi_p = 1.75 \phi_{wire}$ for single 7 wire strands where ϕ_{wire} is the wire diameter.

$\phi_p = 1.20 \phi_{wire}$ for single 3 wire strands where ϕ_{wire} is the wire diameter.

If only prestressing steel is used to control cracking, $\xi_1 = \sqrt{\xi}$.

$\Delta\sigma_p$ = Stress variation in prestressing tendons from the state of zero strain of the concrete at the same level.

1

2

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4

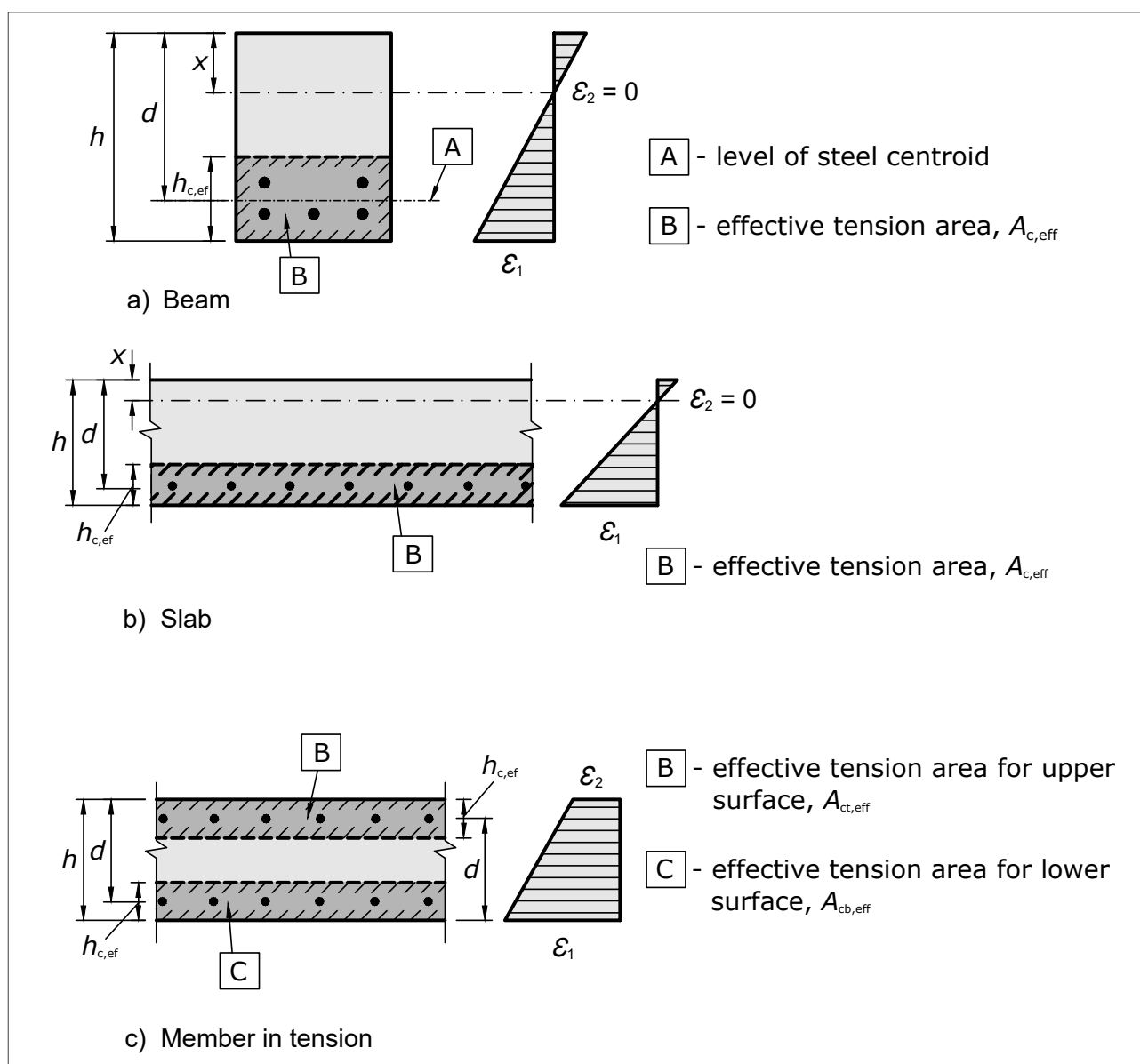
Concrete Structures

Table 4.16 Ratio of Bond Strength, ξ , Between Tendons and Reinforcing Steel

Prestressing Steel	Value of ξ		
	Pre-tensioned	Bonded, post-tensioned	
		$\leq \text{C50/60}$	$\geq \text{C70/85}$
Smooth bars and wires	Not applicable	0.3	0.15
Strands	0.6	0.5	0.25
Indented wires	0.7	0.6	0.30
Ribbed bars	0.8	0.7	0.35

4. In prestressed members no minimum reinforcement is required in sections where, under the characteristic combination of loads and the characteristic value of prestress, the concrete is compressed or the absolute value of the tensile stress in the concrete is below $\sigma_{ct,p}$, which may be taken as $f_{ct,eff}$.

Figure 4.50 Effective Tension Area (typical classes)



5. For bridges, in calculating the minimum reinforcement to cater for shrinkage, $f_{ct,eff}$ in Equation 4.119 should be taken as the greater of 2.9 MPa or $f_{ctm}(t)$.

4.7.3.3 Control of Cracking Without Direct Calculation

1. Where the minimum reinforcement given by Clause 4.7.3.2 is provided, crack widths are unlikely to be excessive if:

- For cracking caused dominantly by restraint, the bar sizes given in Table 4.17 are not exceeded where the steel stress is the value obtained immediately after cracking (i.e. σ_s in Equation 4.119).
- For cracks caused mainly by loading, either the provisions of Table 4.17 or the provisions of Table 4.18 are complied with. The steel stress should be calculated on the basis of a cracked section under the relevant combination of actions.

For pre-tensioned concrete, where crack control is mainly provided by tendons with direct bond, Table 4.17 and Table 4.18 may be used with a stress equal to the total stress minus prestress. For post-tensioned concrete, where crack control is provided mainly by ordinary reinforcement, the tables may be used with the stress in this reinforcement calculated with the effect of prestressing forces included.

Table 4.17 Maximum Bar Diameters Φ_s^* for Crack Control¹

Steel stress ² MPa	Maximum bar size mm		
	$W_k = 0.4$ mm	$W_k = 0.3$ mm	$W_k = 0.2$ mm
160	40	32	25
200	32	25	16
240	20	16	12
280	16	12	8
320	12	10	6
360	10	8	5
400	8	6	4
450	6	5	-

Notes: 1. The values in the table are based on the following assumptions: $c = 25\text{mm}$, $f_{ct,eff} = 2.9\text{MPa}$, $h_{cr} = 0.5h$, $(h-d) = 0.1h$, $k_1 = 0.8$, $k_2 = 0.5$, $k_c = 0.4$, $k = 1.0$, $k_3 = 0.4$ and $k_4 = 1.0$

2. Under the relevant combinations of actions.

Table 4.18 Maximum Bar Spacing for Crack Control¹

Steel stress ² MPa	Maximum bar size mm		
	$W_k = 0.4$ mm	$W_k = 0.3$ mm	$W_k = 0.2$ mm
160	40	32	25
200	32	25	16
240	20	16	12
280	16	12	8
320	12	10	6
360	10	8	5
400	8	6	4
450	6	5	-

Notes: Same as for Table 4.17

2. The maximum bar diameter should be modified as follows:

- Bending (at least part of section in compression):

$$\phi_s = \phi_s^* \frac{f_{ct,eff}}{2.9} \frac{k_c h_{cr}}{2(h-d)}$$

Equation 4.124

- Tension (uniform axial tension):

$$\phi_s = \phi_s^* \frac{f_{ct,eff}}{2.9} \frac{h_{cr}}{8(h-d)}$$

Equation 4.125

Where,

ϕ_s = The adjusted maximum bar diameter.

ϕ_s^* = The maximum bar size given in Table 4.17.

h = The overall depth of the section.

h_{cr} = The depth of the tensile zone immediately prior to cracking, considering the characteristic values of prestress and axial forces under the quasi-permanent combination of actions.

d = The effective depth to the centroid of the outer layer of reinforcement.

Where all the section is under tension $h - d$ is the minimum distance from the centroid of the layer of reinforcement to the face of the concrete (consider each face where the bar is not placed symmetrically).

3. Beams with a total depth of 1000 mm or more, where the main reinforcement is concentrated in only a small proportion of the depth, should be provided with additional skin reinforcement to control cracking on the side faces of the beam. This reinforcement should be evenly distributed between the level of the tension steel and the neutral axis and should be located within the links. The area of the skin reinforcement should not be less than the amount obtained from Clause 4.7.3.2(2) taking k as 0.5 and σ_s as f_{yk} . The spacing and size of suitable bars may be obtained from Clause 4.7.3.4 or a suitable simplification assuming pure tension and a steel stress of half the value assessed for the main tension reinforcement.
4. It should be noted that there are particular risks of large cracks occurring in sections where there are sudden changes of stress, e.g.
- At changes of section.
 - Near concentrated loads.
 - Positions where bars are curtailed.
 - Areas of high bond stress, particularly at the ends of laps.

Care should be taken at such areas to minimise the stress changes wherever possible. However, the requirements for crack control given above will normally ensure adequate control at these points provided that the provisions for detailing reinforcement given in sections 4.8 and 4.9 are applied.

4.7.3.4 Calculation of Crack Widths

1. The crack width, w_k , may be calculated from following Equation:

$$w_k = s_{r,max} (\epsilon_{sm} - \epsilon_{cm})$$

Equation 4.126

Where,

$s_{r,max}$ = The maximum crack spacing.

ε_{sm} = The mean strain in the reinforcement under the relevant combination of loads, including the effect of imposed deformations and taking into account the effects of tension stiffening. Only the additional tensile strain beyond the state of zero strain of the concrete at the same level is considered.

ε_{cm} = The mean strain in the concrete between cracks.

2. The $\varepsilon_{sm} - \varepsilon_{cm}$ may be calculated from the Equation:

$$\varepsilon_{sm} - \varepsilon_{cm} = \frac{\sigma_s - k_t \frac{f_{ct,eff}}{\rho_{p,eff}} (1 + \alpha_e \rho_{p,eff})}{E_s} \geq 0.6 \sigma_s / E_s$$

Equation 4.127

Where,

σ_s = The stress in the tension reinforcement assuming a cracked section. For pretensioned members, σ_s may be replaced by $\Delta\sigma_p$ the stress variation in prestressing tendons from the state of zero strain of the concrete at the same level.

α_e = The ratio E_s/E_{cm} .

$\rho_{p,eff} = (A_s + \xi 1 A'_p) / A_{c,eff}$

A'_p and $A_{c,eff}$ are as defined in Clause 4.7.3.2.

$\xi 1$ = According to Equation 4.123.

k_t = A factor dependent on the duration of the load.

$k_t = 0.6$ for short term loading.

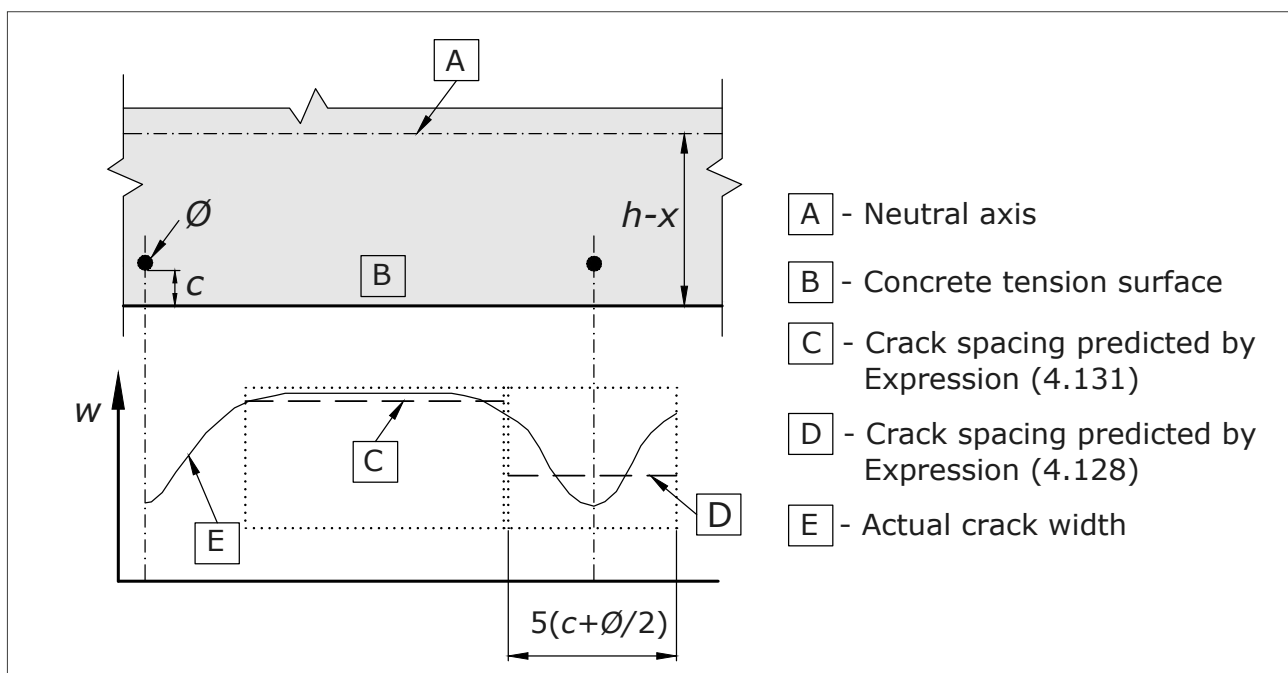
$k_t = 0.4$ for long term loading.

3. In situations where bonded reinforcement is fixed at reasonably close centres within the tension zone (spacing $\leq 5(c + \phi/2)$), the maximum final crack spacing may be calculated from Equation 4.128 (refer Figure 4.51):

$$s_{r,max} = k_3 c + \frac{k_1 k_2 k_4 \phi}{\rho_{p,eff}}$$

Equation 4.128

Figure 4.51 Crack Width, w , at Concrete Surface Relative to Distance From Bar



Where,

ϕ = The bar diameter. Where a mixture of bar diameters is used in a section, an equivalent diameter, ϕ_{eq} , should be used. For a section with n_1 bars of diameter ϕ_1 and n_2 bars of diameter ϕ_2 , the following Equation should be used

$$\phi_{eq} = \frac{n_1 \phi_1^2 + n_2 \phi_2^2}{n_1 \phi_1 + n_2 \phi_2}$$

Equation 4.129

c = The cover to the longitudinal reinforcement, shall be taken as c_{nom}

k_1 = A coefficient which takes account of the bond properties of the bonded reinforcement:

= 0.8 for high bond bars

= 1.6 for bars with an effectively plain surface (e.g. prestressing tendons)

k_2 = A coefficient which takes account of the distribution of strain:

= 0.5 for bending

= 1.0 for pure tension

For cases of eccentric tension or for local areas, intermediate values of k_2 should be used which may be calculated from the relation:

$$k_2 = \frac{\varepsilon_1 + \varepsilon_2}{2 \varepsilon_1}$$

Equation 4.130

Where ε_1 is the greater and ε_2 is the lesser tensile strain at the boundaries of the section considered, assessed on the basis of a cracked section.

k_3 , k_4 recommended values are 3.4 and 0.425 respectively.

Where the spacing of the bonded reinforcement exceeds $5(c + \phi/2)$ (refer Figure 4.51) or where there is no bonded reinforcement within the tension zone, an upper bound to the crack width may be found by assuming a maximum crack spacing:

$$s_{r,max} = 1.3(h - x)$$

Equation 4.131

4. Where the angle between the axes of principal stress and the direction of the reinforcement, for members reinforced in two orthogonal directions, is significant ($>15^\circ$), then the crack spacing $s_{r,max}$ may be calculated from the following Equation:

$$s_{r,max} = \frac{1}{\frac{\cos \theta}{s_{r,max,y}} + \frac{\sin \theta}{s_{r,max,z}}}$$

Equation 4.132

Where,

θ = The angle between the reinforcement in the y direction and the direction of the principal tensile stress

$s_{r,max,y}$, $s_{r,max,z}$ are the crack spacings calculated in the y and z directions respectively, according to Clause 4.7.3.4 (3)

5. For walls subjected to early thermal contraction where the horizontal steel area, A_s does not fulfil the requirements of Clause 4.7.3.2 and where the bottom of the wall is restrained by a previously cast base, $s_{r,max}$ may be assumed to be equal to 1.3 times the height of the wall.

4.7.4 Deflection Control

4.7.4.1 General Considerations

1. The deformation of a member or structure shall not be such that it adversely affects its proper functioning or appearance.
2. Appropriate limiting values of deflection taking into account the nature of the structure, of the finishes, partitions and fixings and upon the function of the structure should be established.

4.7.4.2 Checking Deflections by Calculation

1. Where a calculation is deemed necessary, the deformations shall be calculated under load conditions which are appropriate to the purpose of the check.
2. The calculation method adopted shall represent the true behaviour of the structure under relevant actions to an accuracy appropriate to the objectives of the calculation.
3. Members which are not expected to be loaded above the level which would cause the tensile strength of the concrete to be exceeded anywhere within the member should be considered to be uncracked. Members which are expected to crack, but may not be fully cracked, will behave in a manner intermediate between the uncracked and fully cracked conditions and, for members subjected mainly to flexure, an adequate prediction of behaviour is given by following Expression:

$$\alpha = \zeta \alpha_{II} + (1 - \zeta) \alpha_I$$

Equation 4.133

Where,

α = The deformation parameter considered.

α_I, α_{II} = The values of the parameter calculated for the uncracked and fully cracked conditions respectively.

ζ = A distribution coefficient (allowing for tensioning stiffening at a section) given by as follows:

$$\zeta = 1 - \beta \left(\frac{\sigma_{sr}}{\sigma_s} \right)^2$$

Equation 4.134

ζ = 0 for uncracked sections.

β = A coefficient taking account of the influence of the duration of the loading or of repeated loading on the average strain.

= 1.0 for a single short-term loading.

= 0.5 for sustained loads or many cycles of repeated loading.

σ_s = The stress in the tension reinforcement calculated on the basis of a cracked section.

σ_{sr} = The stress in the tension reinforcement calculated on the basis of a cracked section under the loading conditions causing first cracking.

Note: σ_{sr} / σ_s may be replaced by M_{cr}/M for flexure or N_{cr}/N for pure tension, where M_{cr} is the cracking moment and N_{cr} is the cracking force.

4. Deformations due to loading may be assessed using the tensile strength and the effective modulus of elasticity of the concrete.

Table 4.2 indicates the range of likely values for tensile strength. In general, the best estimate of the behaviour will be obtained if f_{ctm} is used. Where it can be shown that there are no axial tensile stresses (e.g. those caused by shrinkage or thermal effects) the flexural tensile strength, $f_{ctm,fl}$, may be used.

5. For loads with a duration causing creep, the total deformation including creep may be calculated by using an effective modulus of elasticity for concrete according to following Equation:

$$E_{c,eff} = \frac{E_{cm}}{1 + \varphi(\infty, t_0)}$$

Equation 4.135

Where,

$\varphi(\infty, t_0)$ = The creep coefficient relevant for the load and time interval

6. Shrinkage curvatures may be assessed using following Equation:

$$\frac{1}{r_{cs}} = \varepsilon_{cs} \alpha_e \frac{S}{I}$$

Equation 4.136

Where,

$1 / r_{cs}$ = the curvature due to shrinkage.

ε_{cs} = the free shrinkage strain.

S = the first moment of area of the reinforcement about the centroid of the section.

I = the second moment of area of the section.

α_e = the effective modular ratio, $\alpha_e = E_s / E_{c,eff}$.

S and I should be calculated for the uncracked condition and the fully cracked condition, the final curvature being assessed by use of Equation 4.133.

7. The most rigorous method of assessing deflections using the method given in para (3) above is to compute the curvatures at frequent sections along the member and then calculate the deflection by numerical integration. In most cases it will be acceptable to compute the deflection twice, assuming the whole member to be in the uncracked and fully cracked condition in turn, and then interpolate using Equation 4.133.

4.8 Detailing of Reinforcement and Prestressing Tendons

4.8.1 General

The guidelines given in this Section apply to reinforcement and prestressing tendons subjected predominantly to static loading. They are applicable for normal bridges. They may not be sufficient for:

- Elements subjected to dynamic loading caused by seismic effects or machine vibration, impact loading.
- To elements incorporating specially painted, epoxy or zinc coated bars.
- Additional guidelines are provided for large diameter bars.

The requirements concerning minimum concrete cover shall be satisfied.

4.8.2 Spacing of Bars

1. The spacing of bars shall be such that the concrete can be placed and compacted satisfactorily for the development of adequate bond.
2. The clear distance (horizontal and vertical) between individual parallel bars or horizontal layers of parallel bars should be not less than the maximum of (a) largest bar diameter, (b) $(d_g + 5 \text{ mm})$ where d_g is the maximum size of aggregate, (c) 20 mm.
3. Where bars are positioned in separate horizontal layers, the bars in each layer should be located vertically above each other. There should be sufficient space between the resulting columns of bars to allow access for vibrators and good compaction of the concrete.
4. Lapped bars may be allowed to touch one another within the lap length.

4.8.3 Mandrel Diameters for Bent Bars

1. The minimum diameter to which a bar is bent shall be such as to avoid bending cracks in the bar, and to avoid failure of the concrete inside the bend of the bar.
2. In order to avoid damage to the reinforcement the diameter to which the bar is bent (Mandrel diameter) should not be less than $\phi_{m,min}$, as provided in Table 4.19.

Table 4.19 Minimum Mandrel Diameter to Avoid Damage to Reinforcement

a) For bars and wire	
Bar diameter ϕ	Minimum mandrel diameter for bends, hooks and loops (Refer Figure 4.52)
$\phi \leq 16\text{mm}$	4ϕ
$\phi > 16\text{mm}$	7ϕ
b) For welded reinforcement and fabrics bent after welding	
Location of transverse bar defined as a multiple of the bar diameter, bar diameter ϕ	Minimum mandrel diameter $\phi_{m,min}$ in mm
Transverse bar inside or outside a bend or centre of a transverse bar $\leq 4 \phi$ from a bend	20ϕ
Centre of transverse bar $> 4 \phi$ from a bend	4ϕ for $\phi \leq 16$ 7ϕ for $\phi \leq 20$

3. The mandrel diameter need not be checked to avoid concrete failure if the following conditions exist:
 - Either the anchorage of the bar does not require a length more than 5ϕ past the end of the bend or the bar is not positioned at the edge (plane of bend close to concrete face) and there is a cross bar diameter $\geq \phi$ inside the bend.
 - The mandrel diameter is at least equal to the recommended values given in Table 4.19.

Otherwise the mandrel diameter, $\phi_{m,min}$, should be increased in accordance with following Equation:

$$\phi_{m,min} \geq \frac{F_{bt} \left(\frac{1}{a_b} + \frac{1}{2\phi} \right)}{f_{cd}} \quad \text{Equation 4.137}$$

Where,

F_{bt} = The tensile force from ultimate loads in a bar or group of bars in contact at the start of a bend

a_b = A given bar (or group of bars in contact) is half of the centre-to-centre distance between bars (or groups of bars) perpendicular to the plane of the bend. For a bar or group of bars adjacent to the face of the member, a_b should be taken as the cover plus $\phi/2$

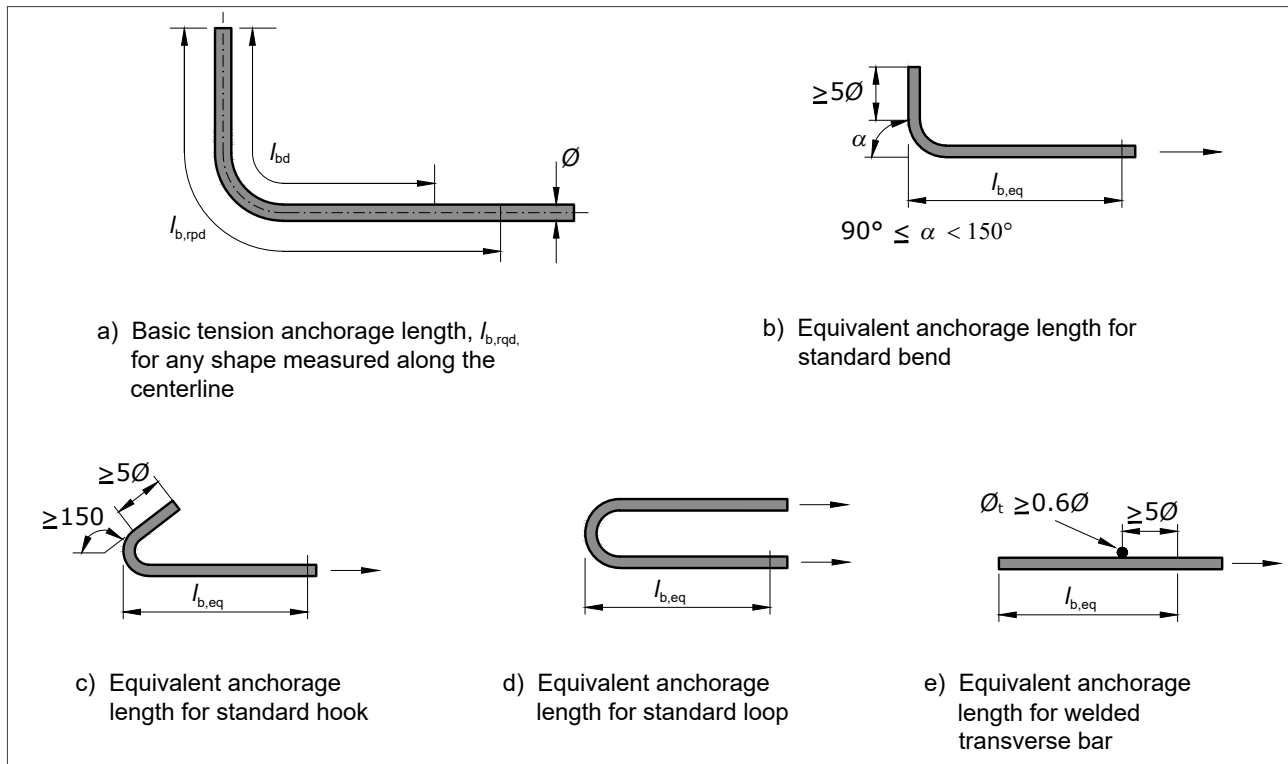
The value of f_{cd} should not be taken greater than that for concrete class C55/67.

4.8.4 Anchorage of Longitudinal Reinforcement

4.8.4.1 General

1. Reinforcing bars, wires or welded mesh fabrics shall be so anchored that the bond forces are safely transmitted to the concrete avoiding longitudinal cracking or spalling. Transverse reinforcement shall be provided if necessary.
2. Methods of anchorage are shown in Figure 4.52.

Figure 4.52 Methods of Anchorage Other than by a Straight Bar



3. Bends and hooks do not contribute to compression anchorages.
4. Concrete failure inside bends should be prevented by complying with Clause 4.8.3 (3).
5. Where mechanical devices are used, the test requirements should be in accordance with the relevant product standard.
6. For the transmission of prestressing forces to the concrete, refer Clause 4.8.10.

4.8.4.2 Ultimate Bond Stress

1. The ultimate bond strength shall be sufficient to prevent bond failure.
2. The design value of the ultimate bond stress, f_{bd} , for ribbed bars may be taken as:

$$f_{bd} = 2.25 \eta_1 \eta_2 f_{ctd}$$

Equation 4.138

Where,

f_{ctd} = The design value of concrete tensile strength according to Clause 4.3.1.5. Due to the increasing brittleness of higher strength concrete, $f_{ctk,0.05}$ should be limited here to the value for C60/75, unless it can be verified that the average bond strength increases above this limit.

η_1 = A coefficient related to the quality of the bond condition and the position of the bar during concreting (refer Figure 4.53):

η_1 = 1.0 when 'good' conditions are obtained.

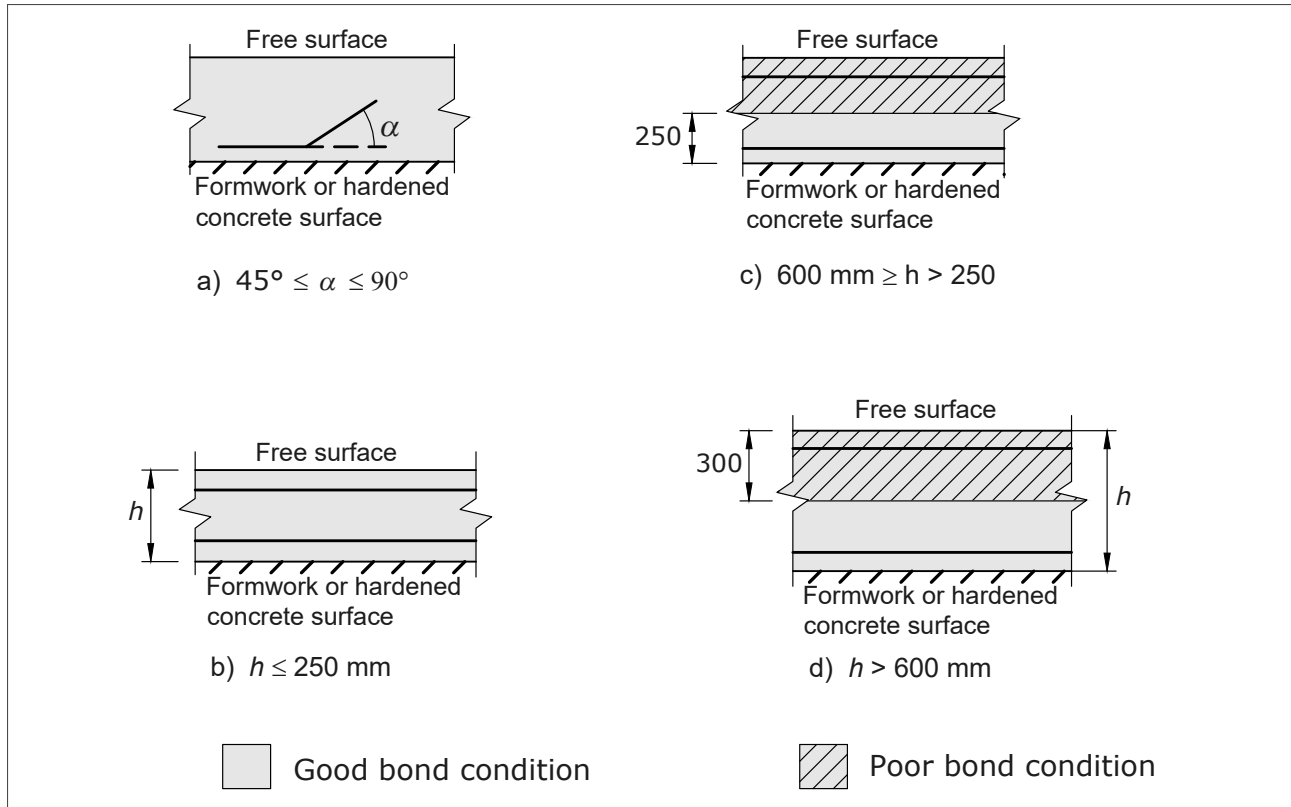
η_1 = 0.7 for all the cases and for bars in structural elements built with slip-forms, unless it can be shown that 'good' bond condition exist.

η_2 = Related to the bar diameter:

η_2 = 1.0 for $\phi \leq 32$ mm.

η_2 = $(132 - \phi)/100$ for $\phi > 32$ mm.

Figure 4.53 Description of Bond Conditions



4.8.4.3 Basic Anchorage Length

1. The calculation of the required anchorage length shall take into consideration the type of steel and bond properties of the bars.
2. The basic required anchorage length, $l_{b,rqd}$, for anchoring the force $A_s \sigma_{sd}$ in a straight bar assuming constant bond stress equal to f_{bd} follows from:

$$l_{b,rqd} = \frac{\phi}{4} \frac{\sigma_{sd}}{f_{bd}}$$

Equation 4.139

Where σ_{sd} is the design stress of the bar at the position from where the anchorage is measured from. Values for f_{bd} are given in Clause 4.8.4.2.

3. For bent bars the basic required anchorage length, $l_{b,rqd}$, and the design length, l_{bd} , should be measured along the centre-line of the bar (refer Figure 4.52a).
4. Where pairs of wires/bars form welded fabrics the diameter, ϕ , in Equation 4.139 should be replaced by the equivalent diameter $\phi_n = \phi\sqrt{2}$.

4.8.4.4 Design Anchorage Length

1. The design anchorage length, l_{bd} , is:

$$l_{bd} = \alpha_1 \alpha_2 \alpha_3 \alpha_4 \alpha_5 l_{b,rqd} \geq l_{b,min}$$

Equation 4.140

Where,

$\alpha_1, \alpha_2, \alpha_3, \alpha_4$ and α_5 are coefficients given in Table 4.20.

α_1 = for the effect of the form of the bars assuming adequate cover (refer Figure 4.52).

α_2 = for the effect of concrete minimum cover (refer Figure 4.54).

α_3 = for the effect of confinement by transverse reinforcement.

α_4 = for the influence of one or more welded transverse bars ($\phi_t > 0.6 \phi$) along the design anchorage length l_{bd} .

α_5 = for the effect of the pressure transverse to the plane of splitting along the design anchorage length.

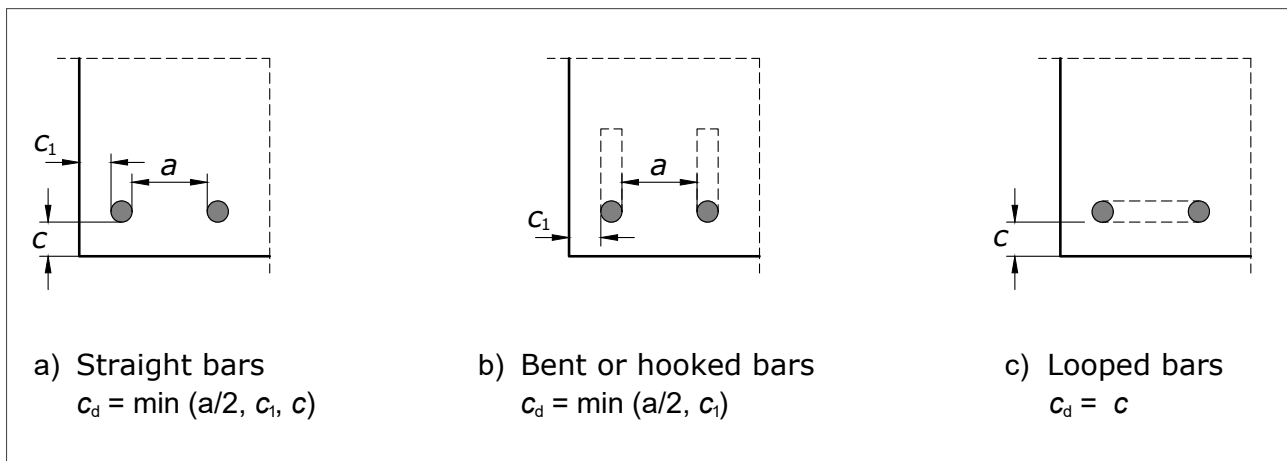
The product ($\alpha_2 \alpha_3 \alpha_5$) ≥ 0.7 .

$l_{b,rqd}$ = taken from Equation 4.139.

$l_{b,min}$ = the minimum anchorage length if no other limitation is applied:

- for anchorages in tension: $l_{b,min} \geq \max \{0.3 l_{b,rqd}; 10 \phi; 100 \text{ mm}\}$.
- for anchorages in compression: $l_{b,min} \geq \max \{0.6 l_{b,rqd}; 10 \phi; 100 \text{ mm}\}$.

Figure 4.54 Values of c_d for Beams and Slabs



2. As a simplified alternative to above, the tension anchorage of certain shapes shown in Figure 4.52 may be provided as an equivalent anchorage length, $l_{b,eq}$. The $l_{b,eq}$ is defined in this figure and may be taken as:

- $\alpha_1 l_{b,rqd}$ for shapes shown in Figure 4.52 b to d.
- $\alpha_4 l_{b,rqd}$ for shapes shown in Figure 4.52 e.

Where,

α_1 and α_4 are defined above

$l_{b,rqd}$ is calculated from Equation 4.139.

Table 4.20 Values of α_1 , α_2 , α_3 , α_4 and α_5 coefficients

Influence Factor	Shape of Anchorage	Reinforcement Bar	
		In Tension	In Compression
Shape of bars	Straight	$\alpha_1 = 1.0$	$\alpha_1 = 1.0$
	Other than straight See Figure 4.52 (b), (c) and (d)	$\alpha_1 = 0.7$ if $c_d > 3\phi$ $\alpha_1 = 1.0$ for other cases (refer Figure 4.54 for values of c_d)	$\alpha_1 = 1.0$
Concrete cover	Straight	$\alpha_2 = 1 - 0.15 (c_d - \phi) / \phi$ ≥ 0.7 ≤ 1.0	$\alpha_2 = 1.0$
	Other than straight See Figure 4.52 (b), (c) and (d)	$\alpha_2 = 1 - 0.15 (cd - 3\phi) / \phi$ ≥ 0.7 ≤ 1.0 (refer Figure 4.54 for c_d)	$\alpha_2 = 1.0$
Confinement by transverse reinforcement not welded to main reinforcement.	All types	$\alpha_3 = 1 - k\lambda$ ≥ 0.7 ≤ 1.0	$\alpha_3 = 1.0$
Confinement by welded transverse reinforcement*	All types, position and size as specified in Figure 4.52 (e)	$\alpha_4 = 0.7$	$\alpha_4 = 0.7$
Confinement by transverse pressure	All types	$\alpha_5 = 1 - 0.04 p$ ≥ 0.7 ≤ 1.0	

Where:

$$\lambda = (\sum A_{st} - \sum A_{st,min}) / A_s$$

$\sum A_{st}$ cross-sectional area of the transverse reinforcement along design anchorage length l_{bd} .

$\sum A_{st,min}$ cross-sectional area of the minimum transverse reinforcement

A_s area of a single anchored bar with maximum bar diameter

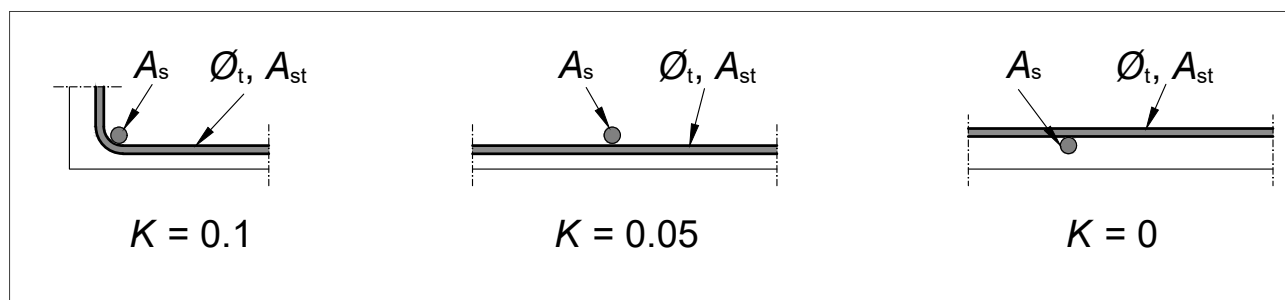
K values shown in Figure 4.55

P transverse pressure [MPa] at ultimate limit state along l_{bd}

***Also refer Clause 4.8.6:**

For direct supports l_{bd} may be taken as less than $l_{b,min}$ provided that there is at least one transverse wire welded within the support.

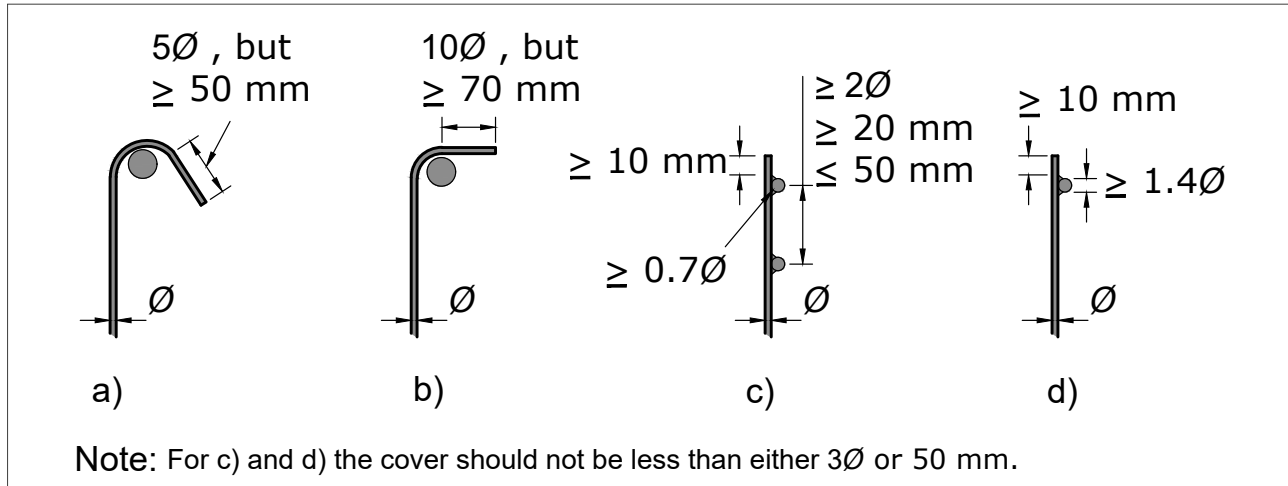
This should be at least 15mm from the face of support.

Figure 4.55 Values of K for Beams and Slabs

4.8.5 Anchorage of Links and Shear Reinforcement

1. The anchorage of links and shear reinforcement should normally be provided by means of bends and hooks, or by welded transverse reinforcement. A bar should be provided inside a hook or bend.
2. The anchorage should comply with Figure 4.56. Welding should be carried out in accordance with EN ISO 17660 and have a welding capacity in accordance with Clause 4.8.6(2).

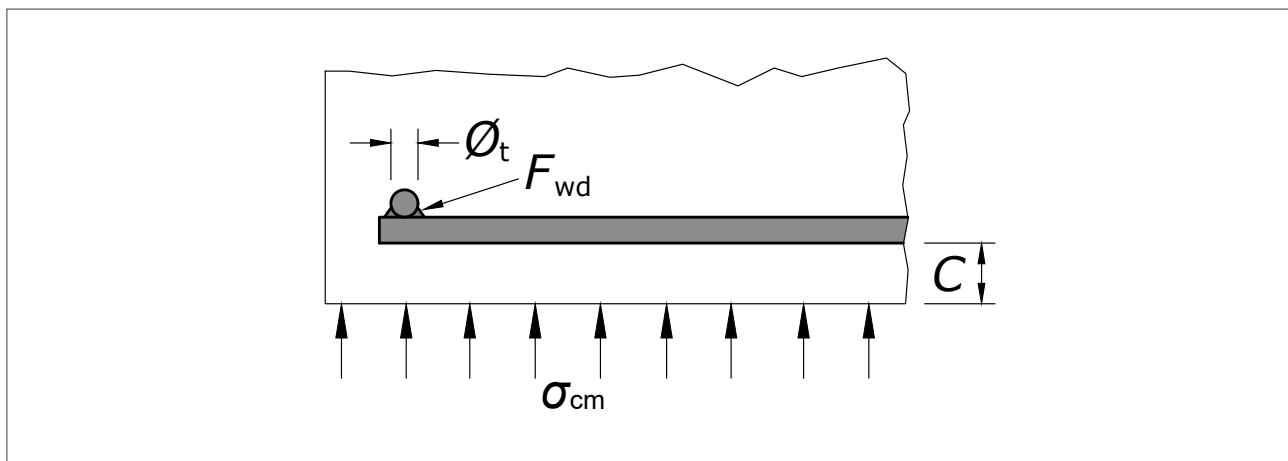
Figure 4.56 Anchorage of Links



4.8.6 Anchorage by Welded Bars

1. Additional anchorage to that of Clause 4.8.4 and 4.8.5 may be obtained by transverse welded bars (refer Figure 4.57) bearing on the concrete. The quality of the welded joints should be shown to be adequate.

Figure 4.57 Welded Transverse Bar as Anchoring Device



2. The anchorage capacity of one welded transverse bar (diameter 14 mm- 32 mm), welded on the inside of the main bar, is F_{btd} . The σ_{sd} in Equation 4.139 may then be reduced by F_{btd} / A_s in where A_s is the area of the bar.

Note: The recommended value of F_{btd} is determined from following Expression:

$$F_{btd} = l_{td} \phi_t \sigma_{td} \quad \text{but not greater than } F_{wd}$$

Equation 4.141

Where,

F_{wd} = The design shear strength of weld (specified as a factor times $A_s f_{yd}$, say $0.5 A_s f_{yd}$ where A_s is the cross-section of the anchored bar and f_{yd} is its design yield strength).

l_{td} = The design length of transverse bar: $l_{td} = 1.16 \phi_t (f_{yd} / \sigma_{td})^{0.5} \leq l_t$

l_t = The length of transverse bar, but not more than the spacing of bars to be anchored.

ϕ_t = The diameter of transverse bar.

σ_{td} = The concrete stress: $\sigma_{td} = (f_{ctd} + \sigma_{cm}) / \gamma \leq f_{cd}$

σ_{cm} = The compression in the concrete perpendicular to both bars (mean value, positive for compression).

γ = A function: $\gamma = 0.015 + 0.14 e^{(-0.18\gamma)}$.

x = A function accounting for the geometry: $x = 2 (c / \phi_t) + 1$.

c = The concrete cover perpendicular to both bars.

3. If two bars of the same size are welded on opposite sides of the bar to be anchored, the capacity calculated from Clause 4.8.6 (2) may be doubled provided that the cover to the outer bar is in accordance with the requirements.
4. If two bars are welded to the same side with a minimum spacing of 3ϕ , the capacity should be multiplied by a factor of 1.41.
5. For nominal bar diameters of 12 mm and less, the anchorage capacity of a welded cross bar is mainly dependent on the design strength of the welded joint. It may be calculated as follows:

$$F_{btd} = F_{wd} \leq \frac{16 A_s f_{cd} \phi_t}{\phi_l} \quad \text{Equation 4.142}$$

Where,

F_{wd} = Design shear strength of weld.

ϕ_t = Nominal diameter of transverse bar: $\phi_t \leq 12$ mm.

ϕ_l = Nominal diameter of bar to anchor: $\phi_l \leq 12$ mm.

If two welded cross bars with a minimum spacing of ϕ_t are used, the anchorage capacity given by Equation 4.142 should be multiplied by a factor of 1.41.

4.8.7 Laps and Mechanical Couplers

4.8.7.1 General

Forces are transmitted from one bar to another by:

- Lapping of bars, with or without bends or hooks.
- Welding.
- Mechanical devices assuring load transfer in tension-compression or in compression only.

4.8.7.1 Laps

1. The detailing of laps between bars shall be such that:

- The transmission of the forces from one bar to the next is assured.
- Spalling of the concrete in the neighbourhood of the joints does not occur.
- Large cracks which affect the performance of the structure do not occur.

2. The Laps:

- Between bars should normally be staggered and not located in areas of high moments / forces (e.g. plastic hinges).
- At any section should normally be arranged symmetrically.

3. The arrangement of lapped bars should comply with Figure 4.58.

- The clear distance between lapped bars should not be greater than 4ϕ or 50 mm,
- otherwise, the lap length should be increased by a length equal to the clear space where it exceeds 4ϕ or 50 mm.
- The longitudinal distance between two adjacent laps should not be less than 0.3 times the lap length, l_0 .
- In case of adjacent laps, the clear distance between adjacent bars should not be less than 2ϕ or 20 mm.

4. When the provisions comply with above, the permissible percentage of lapped bars in tension may be 100 % where the bars are all in one layer. Where the bars are in several layers the percentage should be reduced to 50 %.

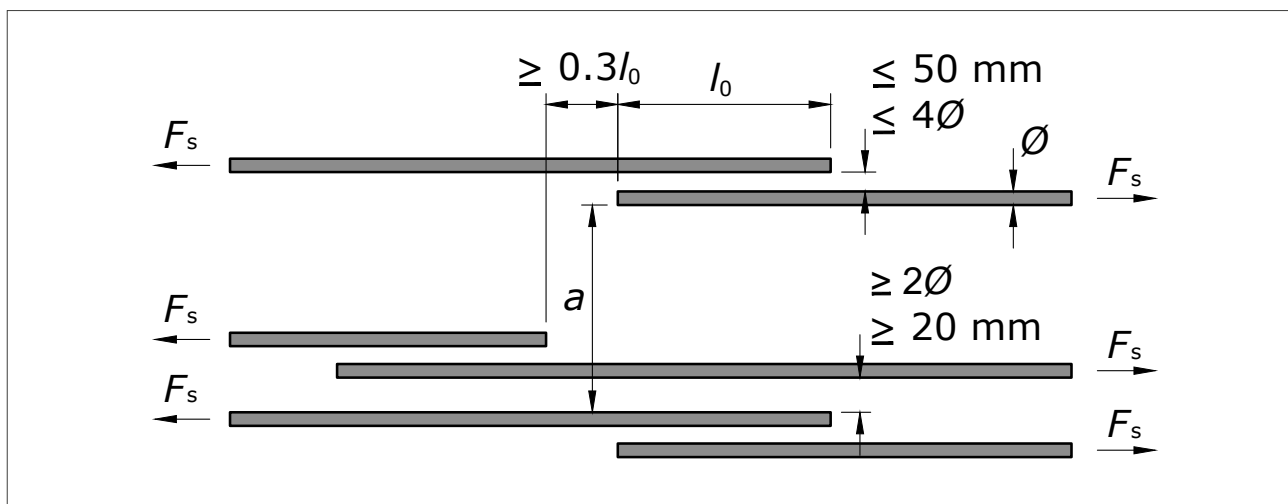
All bars in compression and secondary (distribution) reinforcement may be lapped in one section.

4.8.7.3 Lap Length

The design lap length is:

$$l_0 = \alpha_1 \alpha_2 \alpha_3 \alpha_5 \alpha_6 l_{b,req} \geq l_{0,min}$$

Figure 4.58 Adjacent Laps



Equation 4.143

Where,

$l_{b,req}$ = Calculated from Equation 4.139.

$l_{0,min}$ $\geq \max\{0.3 \alpha_6 l_{b,req}; 15\phi; 200 \text{ mm}\}$

Values of α_1 , α_2 , α_3 and α_5 may be taken from Table 4.20; however, for the calculation of α_3 , $\Sigma A_{st,min}$ should be taken as $1.0 A_s (\sigma_{sd} / f_{yd})$, with A_s = area of one lapped bar.

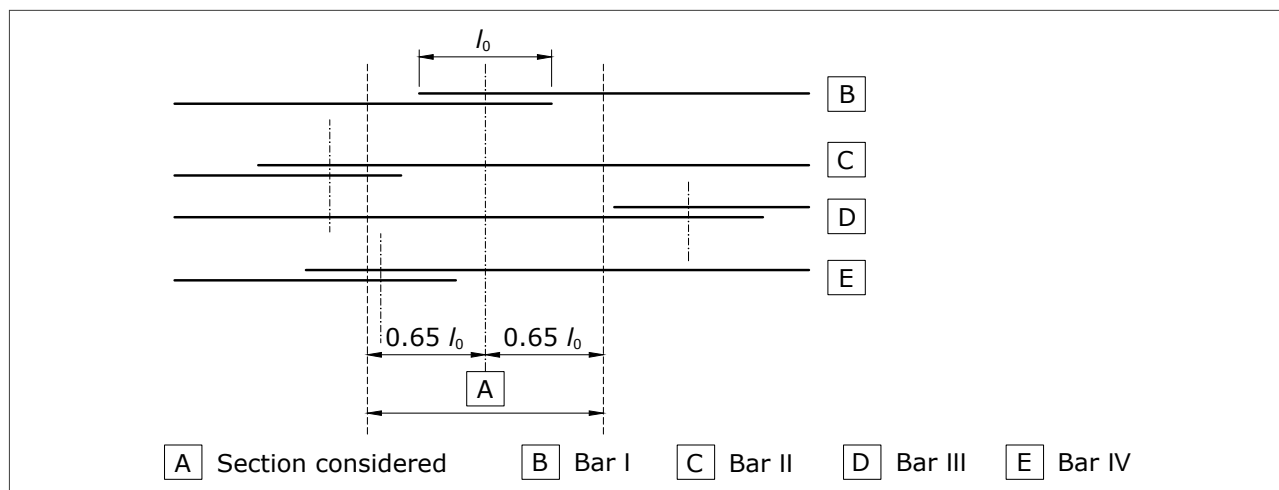
α_6 = $(\rho_1 / 25)^{0.5}$ but not exceeding 1.5 nor less than 1.0, where ρ_1 is the percentage of reinforcement lapped within $0.65 l_0$ from the centre of the lap length considered (refer Figure 4.59). Values of α_6 are given in Table 4.21.

Table 4.21 Values of the coefficient α_6

Percentage of lapped bars relative to the total cross-section area	<25%	33%	50%	>50%
α_6	1	1.15	1.4	1.5

Note: Intermediate values may be determined by interpolation.

Figure 4.59 Percentage of Lapped Bars in One Lapped Direction



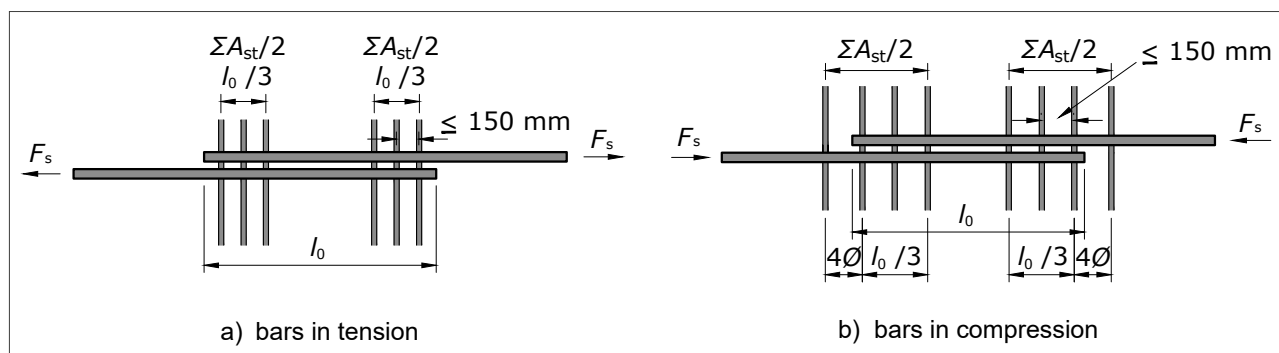
4.8.7.4 Transverse Reinforcement in the Lap Zone

4.8.7.4.1 Transverse Reinforcement for Bars in Tension

1. Transverse reinforcement is required in the lap zone to resist transverse tension forces.
2. Where the diameter, ϕ , of the lapped bars is less than 20 mm, or the percentage of lapped bars in any section is less than 25%, then any transverse reinforcement or links necessary for other reasons may be assumed sufficient for the transverse tensile forces without further justification.
3. Where the diameter, ϕ , of the lapped bars is greater than or equal to 20 mm, the transverse reinforcement should have a total area, ΣA_{st} (sum of all legs parallel to the layer of the spliced reinforcement) of not less than the area A_s of one lapped bar ($\Sigma A_{st} \geq 1.0 A_s$). The transverse bar should be placed perpendicular to the direction of the lapped reinforcement.
4. If more than 50% of the reinforcement is lapped at one point and the distance, a , between adjacent laps at a section is $\leq 10\phi$ (refer Figure 4.58) transverse reinforcement should be formed by links or U bars anchored into the body of the section.

The transverse reinforcement provided for above, should be positioned at the outer sections of the lap as shown in Figure 4.60(a).

Figure 4.60 Transverse Reinforcement for Lapped Splices



4.8.7.4.2 Transverse Reinforcement for Bars Permanently in Compression

In addition to the rules for bars in tension one bar of the transverse reinforcement should be placed outside each end of the lap length and within 4ϕ of the ends of the lap length (Figure 4.60b).

4.8.7.5 Laps for Welded Mesh Fabrics Made of Ribbed Wires

4.8.7.5.1 Laps of the Main Reinforcement

1. Laps may be made either by intermeshing or by layering of the fabrics (Figure 4.61).
2. Where fatigue loads occur, intermeshing should be adopted.
3. For intermeshed fabric, the lapping arrangements for the main longitudinal bars should conform with Clause 4.8.7.2. Any favourable effects of the transverse bars should be ignored, thus taking $\alpha_3 = 1.0$.
4. For layered fabric, the laps of the main reinforcement should generally be situated in zones where the calculated stress in the reinforcement at ultimate limit state is not more than 80% of the design strength.
5. Where above condition is not fulfilled, the effective depth of the steel for the calculation of bending resistance should apply to the layer furthest from the tension face. In addition, when carrying out a crack-verification next to the end of the lap, the steel stress used in Table 4.17 and Table 4.18 should be increased by 25% due to the discontinuity at the ends of the laps.
6. The percentage of the main reinforcement, which may be lapped in anyone section, should comply with the following:

For intermeshed fabric, the values given in Table 4.21 are applicable.

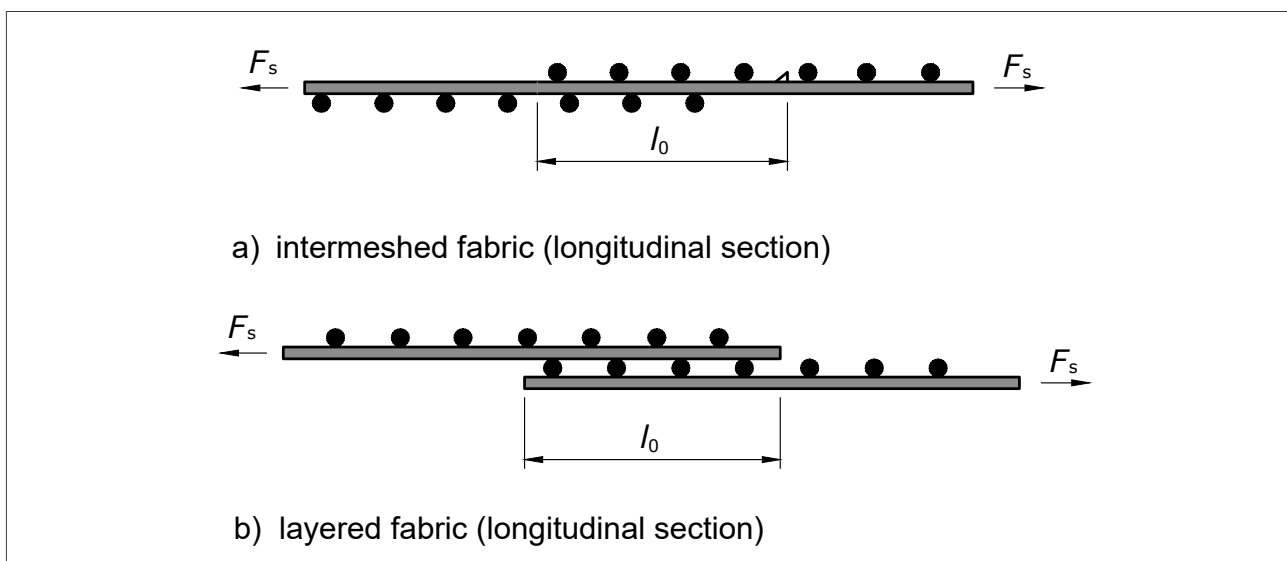
For layered fabric the permissible percentage of the main reinforcement that may be spliced by lapping in any section, depends on the specific cross-section area of the welded fabric provided $(A_s/s)_{prov}$, where s is the spacing of the wires:

- 100% if $(A_s/s)_{prov} \leq 1200 \text{ mm}^2/\text{m}$
- 60% if $(A_s/s)_{prov} > 1200 \text{ mm}^2/\text{m}$.

The joints of the multiple layers should be staggered by at least $1.3 l_0$ (l_0 is determined from Clause 4.8.7.3).

7. Additional transverse reinforcement is not necessary in the lapping zone.

Figure 4.61 Lapping of Welded Fabric



4.8.7.5.2 Laps of Secondary or Distribution Reinforcement

All secondary reinforcement may be lapped at the same location. The minimum values of the lap length l_0 are given in Table 4.22; the lap length of two secondary bars should cover two main bars.

Table 4.22 Required Lap Lengths for Secondary Wires of Fabrics

Diameter of Secondary Wires (mm)	Lap Lengths
$\phi \leq 6$	$\geq 150\text{mm}$: at least 1 wire pitch within the lap length.
$6 < \phi \leq 8.5$	$\geq 250\text{mm}$: at least 2 wire pitches
$8.5 < \phi \leq 12$	$\geq 350\text{mm}$: at least 3 wire pitches

4.8.8 Additional Rules for Large Diameter Bars

- For bars with a diameter larger than ϕ_{large} , the following rules supplement those given in Clause 4.8.4 and 4.8.7.

Note: The recommended value of ϕ_{large} is 40 mm.

- When such large diameter bars are used, crack control may be achieved either by using surface reinforcement (Clause 4.9.2.4) or by calculation (Clause 4.7.3.4).
- Splitting forces are higher and dowel action is greater with the use of large diameter bars. Such bars should be anchored with mechanical devices. As an alternative they may be anchored as straight bars, but links should be provided as confining reinforcement.
- Generally large diameter bars should not be lapped. Exceptions include sections with a minimum dimension 1.0 m or where the reinforcement stress is not greater than 80 % of the design ultimate strength.
- Transverse reinforcement, additional to that for shear, should be provided in the anchorage zones where transverse compression is not present.
- For straight anchorage lengths (refer Figure 4.62 for the notation used), the additional reinforcement referred to in above should not be less than the following:

- In the direction parallel to the tension face:

$$A_{sh} = 0.25 A_s n_1$$

- In the direction perpendicular to the tension face:

$$A_{sv} = 0.25 A_s n_2$$

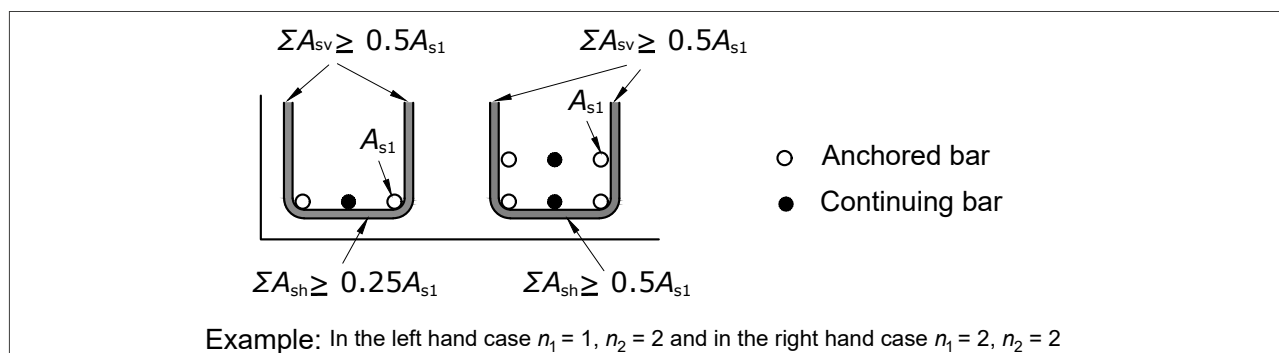
Where,

A_s = The cross-sectional area of an anchored bar.

n_1 = The number of layers with bars anchored at the same point in the member.

n_2 = The number of bars anchored in each layer.

Figure 4.62 Additional Reinforcement in an Anchorage for Large Diameter Bars



7. The additional transverse reinforcement should be uniformly distributed in the anchorage zone and the spacing of bars should not exceed 5 times the diameter of the longitudinal reinforcement.
8. For surface reinforcement, Clause 4.9.2.4 applies, but the area of surface reinforcement should not be less than $0.01 A_{ct,ext}$ in the direction perpendicular to large diameter bars, and $0.02 A_{ct,ext}$ parallel to those bars.

4.8.9 Bundled Bars

4.8.9.1 General

1. Unless otherwise stated, the rules for individual bars also apply for bundles of bars. In a bundle, all the bars should be of the same characteristics (type and grade). Bars of different sizes may be bundled provided that the ratio of diameters does not exceed 1.7.
2. In design, the bundle is replaced by a notional bar having the same sectional area and the same centre of gravity as the bundle. The equivalent diameter, ϕ_n of this notional bar is such that:

$$\phi_n = \phi \sqrt{n_b} \leq 55 \text{ mm}$$

Where,

n_b = The number of bars in the bundle, which is limited to:

$n_b \leq 4$ for vertical bars in compression and for bars in a lapped joint.

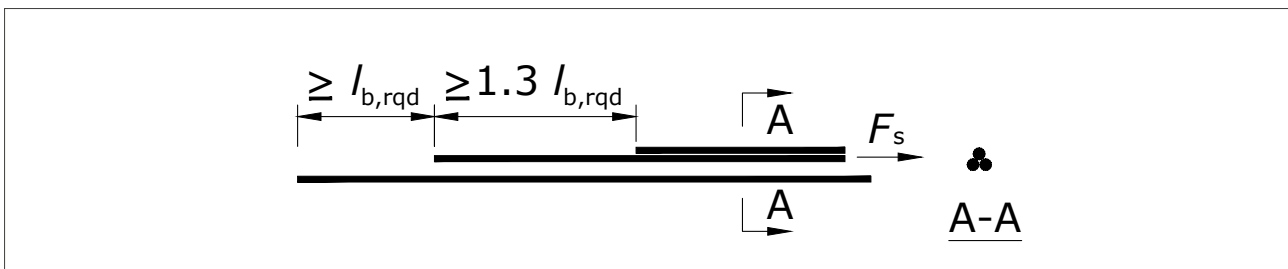
$n_b \leq 3$ for all other cases.

3. For a bundle, the rules given in Clause 4.8.2 for spacing of bars apply. The equivalent diameter, ϕ_n , should be used but the clear distance between bundles should be measured from the actual external contour of the bundle of bars. The concrete cover should be measured from the actual external contour of the bundles and should not be less than ϕ_n .
4. Where two touching bars are positioned one above the other, and where the bond conditions are good, such bars need not be treated as a bundle.

4.8.9.2 Anchorage of Bundles of Bars

1. Bundles of bars in tension may be curtailed over end and intermediate supports. Bundles with an equivalent diameter < 32 mm may be curtailed near a support without the need for staggering bars. Bundles with an equivalent diameter ≥ 32 mm which are anchored near a support should be staggered in the longitudinal direction as shown in Figure 4.63.
2. Where individual bars are anchored with a staggered distance greater than $1.3 l_{b,rqd}$ (where $l_{b,rqd}$ is based on the bar diameter), the diameter of the bar may be used in assessing l_{bd} (refer Figure 4.63). Otherwise, the equivalent diameter of the bundle, ϕ_n , should be used.
3. For compression anchorages, bundled bars need not be staggered. For bundles with an equivalent diameter ≥ 32 mm, at least four links having a diameter ≥ 12 mm should be provided at the ends of the bundle. A further link should be provided just beyond the end of the curtailed bar.

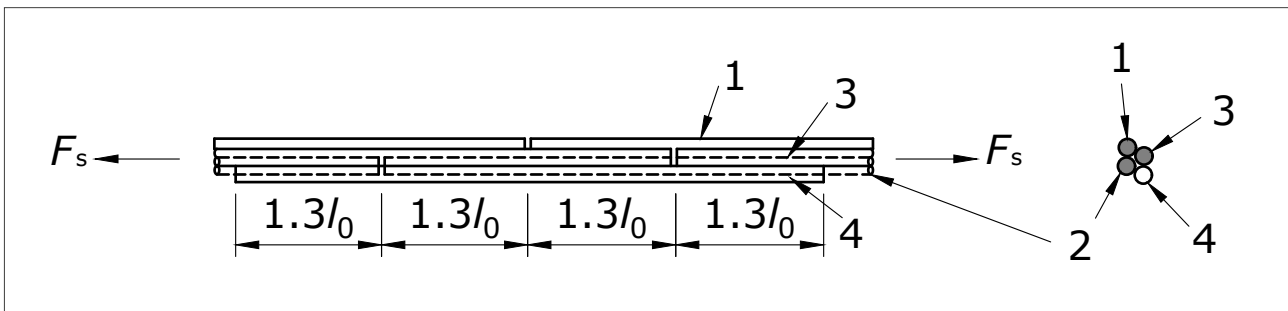
Figure 4.63 Anchorage of Widely Staggered Bars in a Bundle



4.8.9.3 Lapping Bundles of Bars

1. The lap length should be calculated in accordance with Clause 4.8.7.3 using ϕ_n as the equivalent diameter of bar.
2. For bundles which consist of two bars with an equivalent diameter < 32 mm the bars may be lapped without staggering individual bars. In this case the equivalent bar size should be used to calculate l_0 .
3. For bundles which consist of two bars with an equivalent diameter ≥ 32 mm or of three bars, individual bars should be staggered in the longitudinal direction by at least $1.3l_0$ as shown in Figure 4.64, where l_0 is based on a single bar. For this case bar No.4 is used as the lapping bar. Care should be taken to ensure that there are not more than four bars in any lap cross section. Bundles of more than three bars should not be lapped.

Figure 4.64 Lap Joint in Tension Including a Fourth Bar



4.8.10 Prestressing Tendons

4.8.10.1 Arrangement of Prestressing Tendons and Ducts

4.8.10.1.1 General

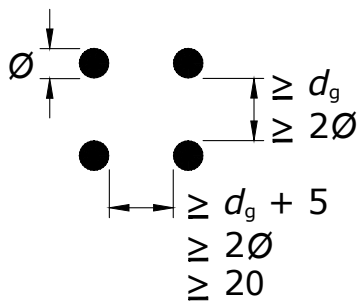
The spacing of ducts or of pre-tensioned tendons shall be such as to ensure that placing and compacting of the concrete can be carried out satisfactorily and that sufficient bond can be attained between the concrete and the tendons.

4.8.10.1.2 Pre-tensioned Tendons

1. The minimum clear horizontal and vertical spacing of individual pre-tensioned tendons should be in accordance with that shown in Figure 4.65. Other layouts may be used provided that test results show satisfactory ultimate behaviour with respect to:
 - The concrete in compression at the anchorage
 - The spalling of concrete
 - The anchorage of pre-tensioned tendons
 - The placing of the concrete between the tendons

Consideration should also be given to durability and the danger of corrosion of the tendon at the end of elements.

Figure 4.65 Minimum Clear Spacing Between Pre-Tensioned Tendons



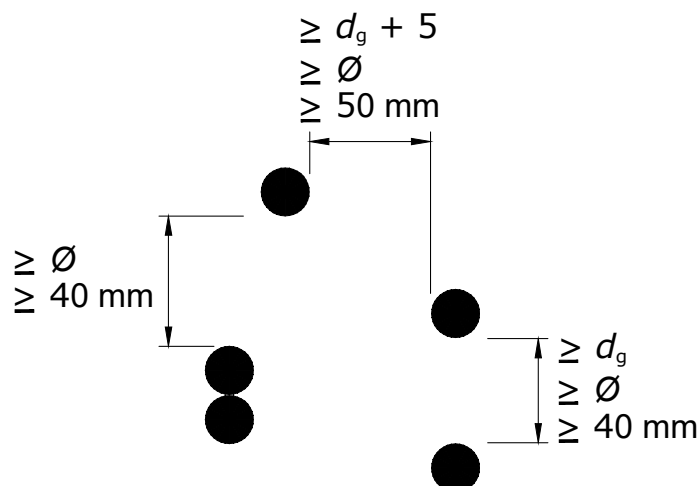
Note: Where $\emptyset$ is the diameter of pre-tensioned tendon and d_g is the maximum size of aggregate.

2. Bundling of tendons should not occur in the anchorage zones, unless placing and compacting of the concrete can be carried out satisfactorily and sufficient bond can be attained between the concrete and the tendons.

4.8.10.1.3 Post-tension Ducts

1. The ducts for post-tensioned tendons shall be located and constructed so that:
 - The concrete can be safely placed without damaging the ducts
 - The concrete can resist the forces from the ducts in the curved parts during and after stressing.
 - No grout will leak into other ducts during grouting process.
2. Ducts for post-tensioned members, should not normally be bundled except in the case of a pair of ducts placed vertically one above the other.
3. The minimum clear spacing between ducts should be in accordance with that shown in Figure 4.66.

Figure 4.66 Minimum Clear Spacing Between Post-tension Ducts



Note: Where $\emptyset$ is the diameter of post-tension duct and d_g is the maximum size of aggregate.

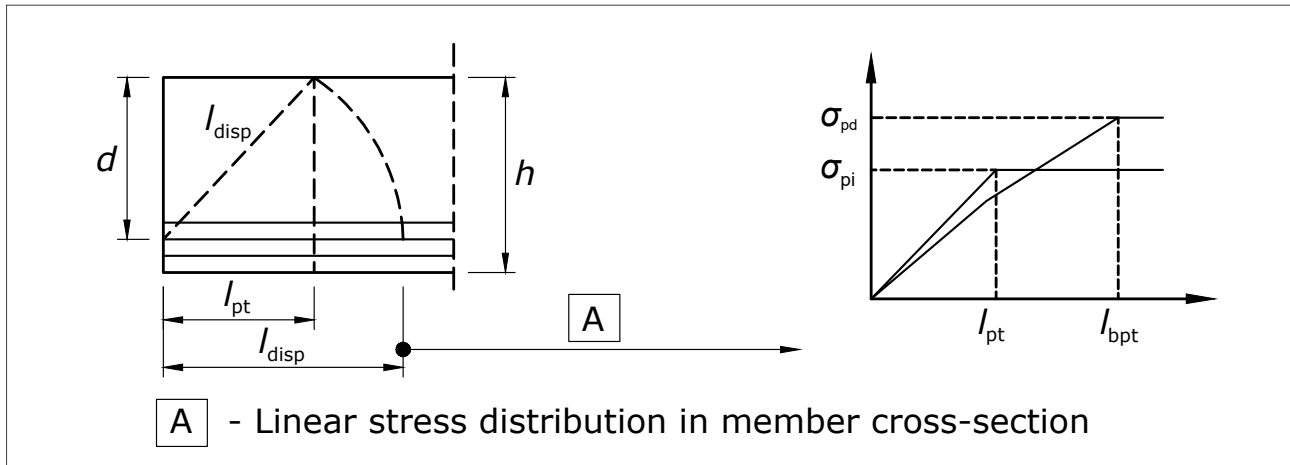
4.8.10.2 Anchorage of Pre-tensioned Tendons

4.8.10.2.1 General

In anchorage regions for pre-tensioned tendons, the following length parameters should be considered, refer Figure 4.67:

- Transmission length, l_{pt} , over which the prestressing force (P_0) is fully transmitted to the concrete.
- Dispersion length, l_{disp} over which the concrete stresses gradually disperse to a linear distribution across the concrete section.
- Anchorage length, l_{bpd} , over which the tendon force F_{pd} in the ultimate limit state is fully anchored in the concrete.

Figure 4.67 Transfer of Prestress in Pretensioned Elements, Length Parameters



4.8.10.2.2 Transfer of Prestress

- At release of tendons, the prestress may be assumed to be transferred to the concrete by a constant bond stress f_{bpt} , where:

$$f_{bpt} = \eta_{p1} \eta_1 f_{ctd}(t)$$

Where,

η_{p1} = a coefficient that takes into account the type of tendon and the bond situation at release

$\eta_{p1} = 2.7$ for indented wires

$\eta_{p1} = 3.2$ for 3 and 7-wire strands

$\eta_1 = 1.0$ for good bond conditions

= 0.7 otherwise

$f_{ctd}(t)$ is the design tensile value of strength at time of release

$$f_{ctd}(t) = \alpha_{ct} 0.7 f_{ctm} / \gamma_c$$

- The basic value of the transmission length, l_{pt} , is given by:

$$l_{pt} = \frac{\alpha_1 \alpha_2 \phi \sigma_{pm0}}{f_{bpt}}$$

Where,

α_1 = 1.0 for gradual release

= 1.25 for sudden release

α_2 = 0.25 for tendons with circular cross section

= 0.19 for 3 and 7-wire strands

ϕ = the nominal diameter of tendon

σ_{pm0} = the tendon stress just after release

- The design value of the transmission length should be taken as the less favourable of two values, depending on the design situation:

$$l_{pt1} = 0.8 l_{pt} \text{ or } l_{pt2} = 1.2 l_{pt}$$

Note: Normally the lower value is used for verifications of local stresses at release, the higher value for ultimate limit states (shear, anchorage etc.).

- Concrete stresses may be assumed to have a linear distribution outside the dispersion length, refer Figure 4.67.

$$l_{disp} = \sqrt{l_{pt}^2 + d^2}$$

4.8.10.2.3 Anchorage of Tendons for the Ultimate Limit State

- The anchorage of tendons should be checked in sections where the concrete tensile stress exceeds $f_{ctk,0.05}$. The tendon force should be calculated for a cracked section, including the effect of shear according to Clause 4.6.2.3(7). Where the concrete tensile stress is less than $f_{ctk,0.05}$, no anchorage check is necessary.
- The bond strength for anchorage in the ultimate limit state is:

$$f_{bpt} = \eta_{p2} \eta_1 f_{ctd}$$

Where,

η_{p2} = A coefficient that takes into account the type of tendon and the bond situation at anchorage.

= 1.4 for indented wires, or

= 1.2 for 7-wire strands.

η_1 = as defined in Clause 4.8.10.2.2 (1).

- Due to increasing brittleness with higher concrete strength, $f_{ctk,0.05}$ should here be limited to the value for C60/75.
- The total anchorage length for anchoring a tendon with stress σ_{pd} is:

$$l_{bpd} = l_{pt2} + \frac{\alpha_2 \phi (\sigma_{pd} - \sigma_{pm\infty})}{f_{bpd}}$$

Where,

l_{pt2} = The upper design value of transmission length

α_2 = Defined earlier

σ_{pd} = The tendon stress corresponding to the force described in paragraph (1)

$\sigma_{pm\infty}$ = The prestress after all losses

- Tendon stresses in the anchorage zone are illustrated in Figure 4.68
- In case of combined ordinary and pre-tensioned reinforcement, the anchorage capacities of each may be summed.

1

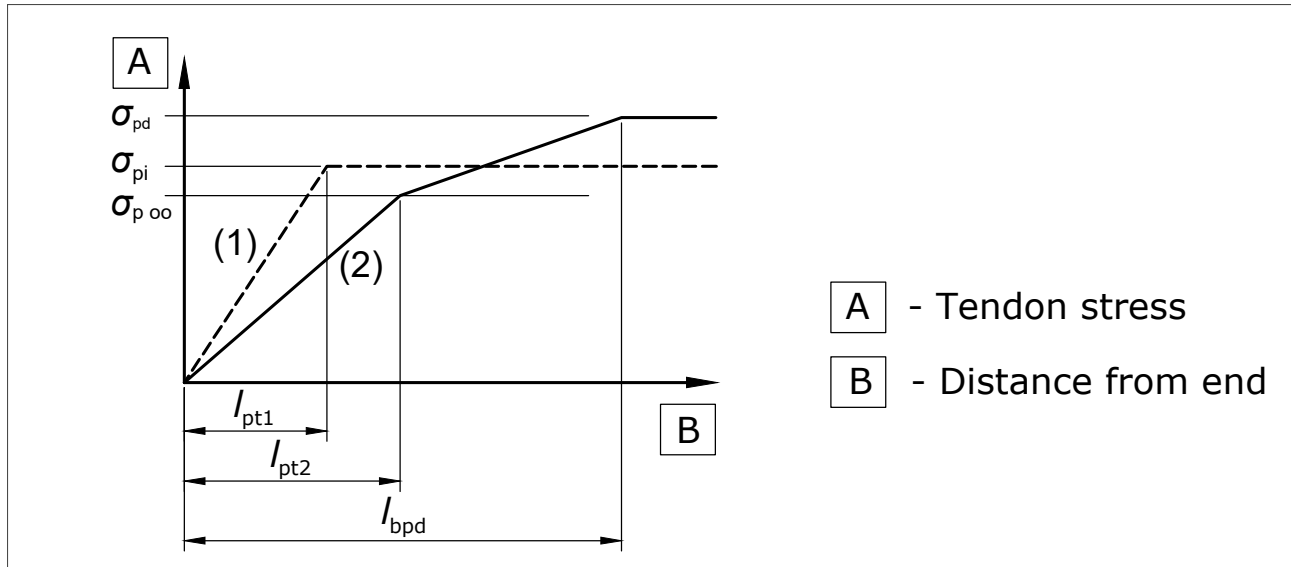
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Concrete Structures

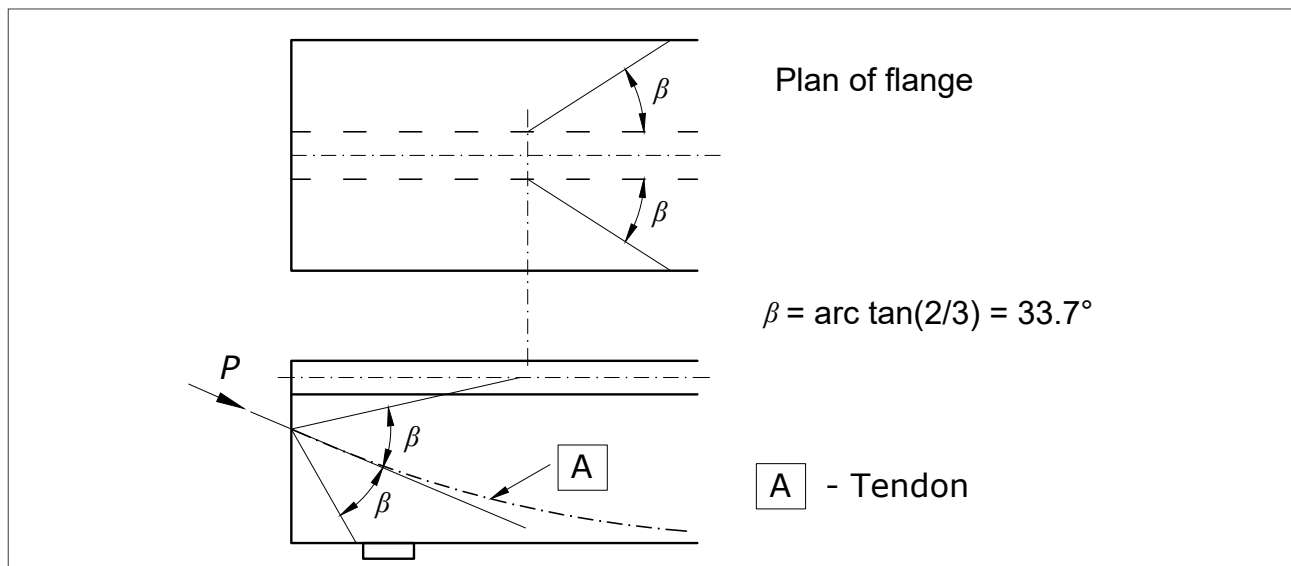
Figure 4.68 Stresses in the Anchorage Zone of Pre-Tensioned Members: (1) At release of tendons, (2) at ultimate state



4.8.10.3 Anchorage Zones of Post-tensioned Members

1. The design of anchorage zones should be in accordance with the guidelines given in this Clause and those in Clause 4.6.5.3.
2. When considering the effects of the prestress as a concentrated force on the anchorage zone, the design value of the prestressing tendons should be in accordance with Clause 4.2.4.2.2 and the lower characteristic tensile strength of the concrete should be used.
3. Tensile forces due to concentrated forces should be assessed by a strut and tie model, or other appropriate representation. Reinforcement should be detailed assuming that it acts at its design strength. If the stress in this reinforcement is limited to 250 MPa, no check of crack widths is necessary.
4. As a simplification, the prestressing force may be assumed to disperse at an angle of spread 2β (see Figure 4.69), starting at the end of the anchorage device, where β may be assumed to be $\arctan(2/3)$.
5. Particular consideration should be given to the design of anchorage zones where two or more tendons are anchored.

Figure 4.69 Dispersion of Prestress



4.8.10.4 Anchorages and Couplers for Prestressing Tendons

1. The anchorage devices used for post-tensioned tendons shall be in accordance with those specified for the prestressing system, and the anchorage lengths in the case of pre-tensioned tendons shall be such as to enable the full design strength of the tendons to be developed, taking account of any repeated, rapidly changing action effects.
2. Where couplers are used, they shall be in accordance with those specified for the prestressing system and shall be so placed - taking account of the interference caused by these devices - that they do not affect the bearing capacity of the member and that any temporary anchorage which may be needed during construction can be introduced in a satisfactory manner.
3. Calculations for local effects in the concrete and for the transverse reinforcement should be made in accordance with Clause 4.6.5 and 4.8.10.3.
4. In general, couplers should be located away from intermediate supports.
5. The placing of couplers on more than 50 % of the tendons at one cross-section should be avoided unless:
 - Continuous minimum reinforcement according to Equation 4.119 is provided, or
 - There is a minimum residual compressive stress of 3 MPa at the cross-section under the characteristic combination of actions.

Note: The maximum percentage of tendons to be coupled at a section should not exceed 67%.

Where a proportion of tendons are joined with couplers at a particular cross section, remaining tendons may not be joined with couplers within distance 'a' of the that cross section.

Note: The recommended value of a is given in Table 4.23.

Table 4.23 Minimum Distance Between Sections at Which Tendons are Joined with Couplers

Construction Depth h (m)	Distance a (m)
≤ 1.5	1.5
$1.5 < h < 3.0$	$a = h$
≥ 3.0	3.0

6. If slabs are transversely prestressed, special consideration should be given to the arrangement of prestressing, to achieve a reasonably uniform distribution of prestress.
7. In an aggressive environment openings and pockets which are necessary to apply the prestress to the tendons should be avoided on the upper side of carriageway slabs. Where, in exceptional circumstances, openings and pockets are provided on the upper side of carriageway slabs appropriate precautions should be taken to ensure durability.
8. If tendons are anchored at a construction joint or within a concrete member (whether on an external rib, within a pocket or entirely inside the member), it should be checked that a minimum residual compressive stress of at least 3 MPa is present in the direction of the anchored prestressing force, under the frequent load combination. If the minimum residual stress is not present, reinforcement should be provided to cater for the local tension behind the anchor. The check for residual stress is not required if the tendon is coupled at the anchorage considered.

1

2

3

4

Concrete Structures

4.8.10.5 Deviators

1. A deviator shall satisfy the following requirements:
 - Withstand both longitudinal and transverse forces that the tendon applies to it and transmit these forces to the structure.
 - Ensure that the radius of curvature of the prestressing tendon does not cause any overstressing or damage to it
2. In the deviation zones the tubes forming the sheaths shall be able to sustain the radial pressure and longitudinal movement of the prestressing tendon, without damage and without impairing its proper functioning.
3. The radius of curvature of the tendon in a deviation zone shall be in accordance with EN 10138.
4. Designed tendon deviations up to an angle of 0.01 radians may be permitted without using a deviator.

4.9 Detailing of Members and Particular Requirements

4.9.1 General

1. The requirements for safety, serviceability and durability are satisfied by following the rules given in this section in addition to the general rules given elsewhere.
2. The detailing of members should be consistent with the design models adopted.
3. Minimum areas of reinforcement are given in order to prevent a brittle failure, wide cracks and also to resist forces arising from restrained actions.

4.9.2 Beams

4.9.2.1 Longitudinal Reinforcement

4.9.2.1.1 Minimum and Maximum Reinforcement Areas

1. The area of longitudinal tension reinforcement should not be taken as less than $A_{s,min}$.

Note 1: Also refer Clause 4.7.3 for area of longitudinal tension reinforcement to control cracking.

Note 2: The recommended value of $A_{s,min}$ for beams is given in the following:

$$A_{s,min} = 0.26 \frac{f_{cm}}{f_{yk}} b_t d \geq 0.0013 b_t d$$

Where,

b_t denotes the mean width of the tension zone; for a T-beam with the flange in compression, only the width of the web is taken into account in calculating the value of b_t .

f_{ctm} should be determined with respect to the relevant strength class according to Table 4.2.

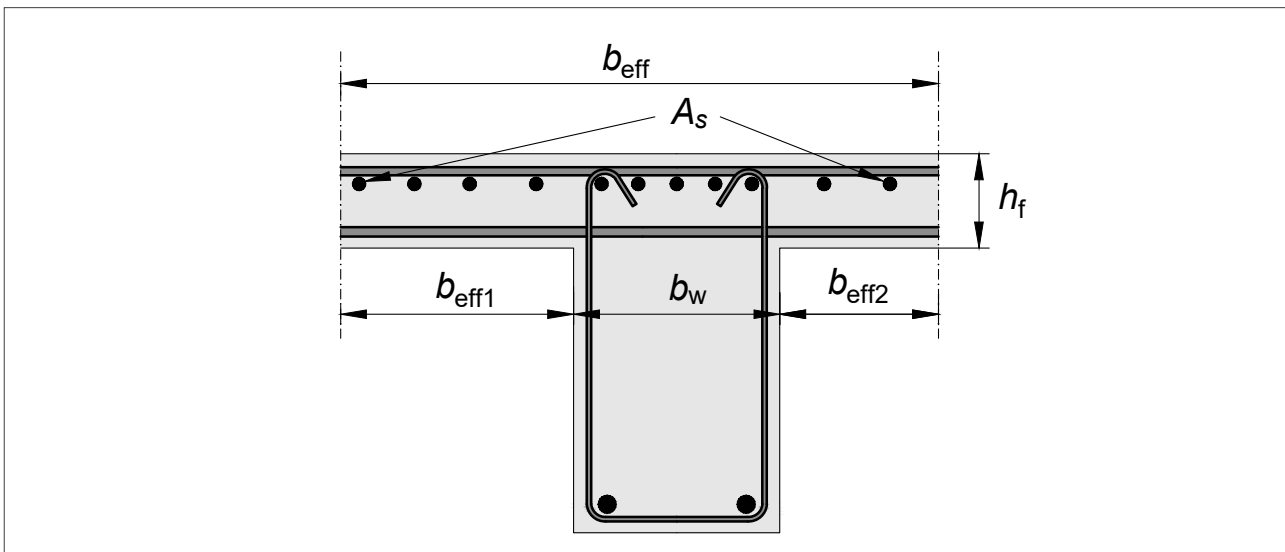
Alternatively, for secondary elements, where some risk of brittle failure may be accepted, $A_{s,min}$ may be taken as 1.2 times the area required in ULS verification.

2. Sections containing less reinforcement than $A_{s,min}$ should be considered as unreinforced.
3. The cross-sectional area of tension or compression reinforcement should not exceed $A_{s,max}$ outside lap locations. The value of $A_{s,max}$ for beams is $0.04 A_c$.
4. For members prestressed with permanently unbonded tendons or with external prestressing cables, it should be verified that the ultimate bending capacity is larger than the flexural cracking moment. A capacity of 1.15 times the cracking moment is sufficient.

4.9.2.1.2 Other Detailing Arrangements

1. In monolithic construction, even when simple supports have been assumed in design, the section at supports should be designed for a bending moment arising from partial fixity of at least 25% of the maximum bending moment in the span. The minimum area of longitudinal reinforcement shall be as defined above.
2. At intermediate supports of continuous beams, the total area of tension reinforcement A_s of a flanged cross-section should be spread over the effective width of flange. Part of it may be concentrated over the web width (refer Figure 4.70).
3. Any compression longitudinal reinforcement (diameter ϕ) which is included in the resistance calculation should be held by transverse reinforcement with spacing not greater than 15ϕ .

Figure 4.70 Placing Of Tension Reinforcement In Flanged Cross-Section



4.9.2.1.3 Curtailment of Longitudinal Tension Reinforcement

1. Sufficient reinforcement should be provided at all sections to resist the envelope of the acting tensile force, including the effect of inclined cracks in webs and flanges.
2. For members with shear reinforcement the additional tensile force, ΔF_{td} , should be calculated according to Clause 4.6.2.3 (7). For members without shear reinforcement ΔF_{td} may be estimated by shifting the moment curve a distance $a_l = d$ according to Clause 4.6.2.2 (4). This "shift rule" may also be used as an alternative for members with shear reinforcement, where:

$$a_l = z (\cot \theta - \cot \alpha) / 2$$

The symbols are defined in Clause 4.6.2.3.

The additional tensile force is illustrated in Figure 4.71.

3. The resistance of bars within their anchorage lengths may be taken into account, assuming a linear variation of force. As a conservative simplification this contribution may be ignored.
4. The anchorage length of a bent-up bar which contributes to the resistance to shear should be not less than $1.3 l_{bd}$ in the tension zone and $0.7 l_{bd}$ in the compression zone. It is measured from the point of intersection of the axes of the bent-up bar and the longitudinal reinforcement.

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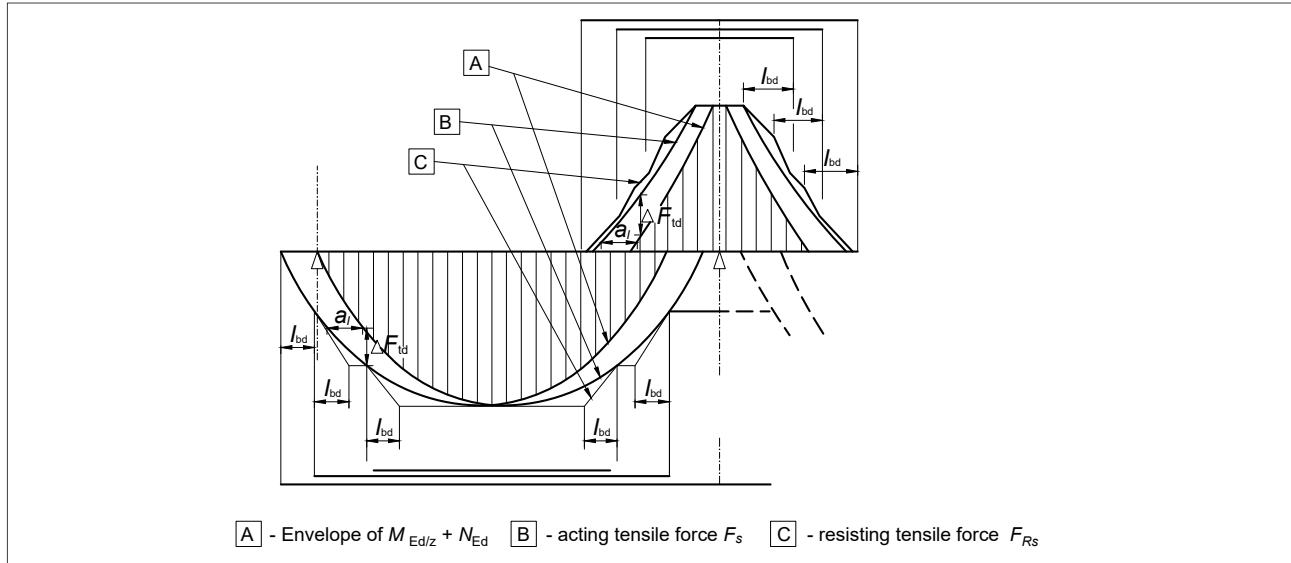
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Figure 4.71 Illustration of the Curtailment of Longitudinal Reinforcement



4.9.2.1.4 Anchorage of Bottom Reinforcement at an End Supports

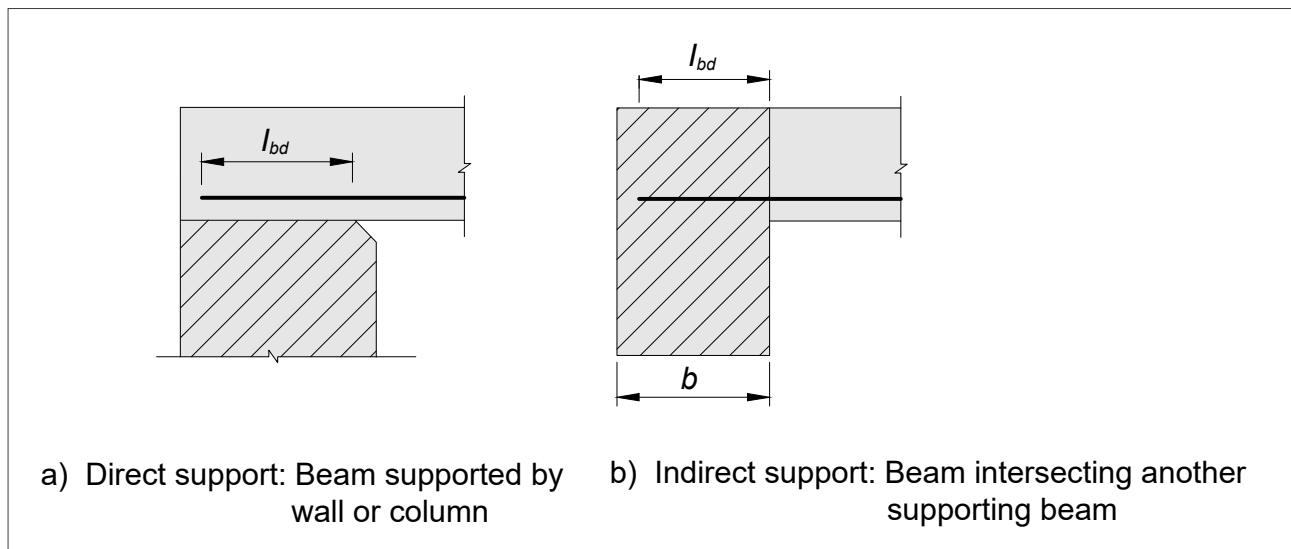
1. The area of bottom reinforcement provided at end supports with little or no end fixity assumed in design, should be at least 25 % of the area of steel provided in the span.
2. The tensile force to be anchored may be determined according to Clause 4.6.2.3(7) (members with shear reinforcement) including the contribution of the axial force if any, or according to the shift rule:

$$F_{Ed} = \frac{|V_{Ed}| a_l}{z} + N_{Ed}$$

Where, N_{Ed} is the axial force, to be added to or subtracted from the tensile force; a_l see Clause 4.9.2.1.3 (2).

3. The anchorage length is l_{bd} according to Clause 4.8.4.4, measured from the line of contact between beam and support. Transverse pressure may be taken into account for direct support. Refer Figure 4.72.

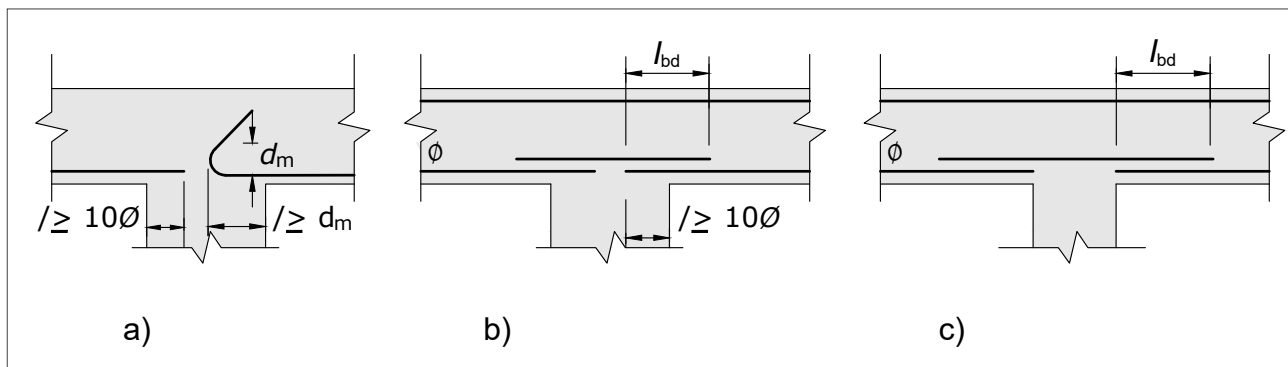
Figure 4.72 Anchorage of Bottom Reinforcement at End Supports



4.9.2.1.5 Anchorage of Bottom Reinforcement at Intermediate Supports

1. The area of reinforcement given in Clause 4.9.2.1.4 (1) applies.
2. The anchorage length should not be less than 10ϕ (for straight bars) or not less than the diameter of the mandrel (for hooks and bends with bar diameters at least equal to 16 mm) or twice the diameter of the mandrel (in other cases) (see Figure 4.73 (a)). These minimum values are normally valid but a more refined analysis may be carried out in accordance with Clause 4.6.6.
3. The reinforcement required to resist possible positive moments (e.g. settlement of the support, explosion, etc.) should be specified in contract documents. This reinforcement should be continuous which may be achieved by means of lapped bars (see Figure 4.73 (b) or (c))

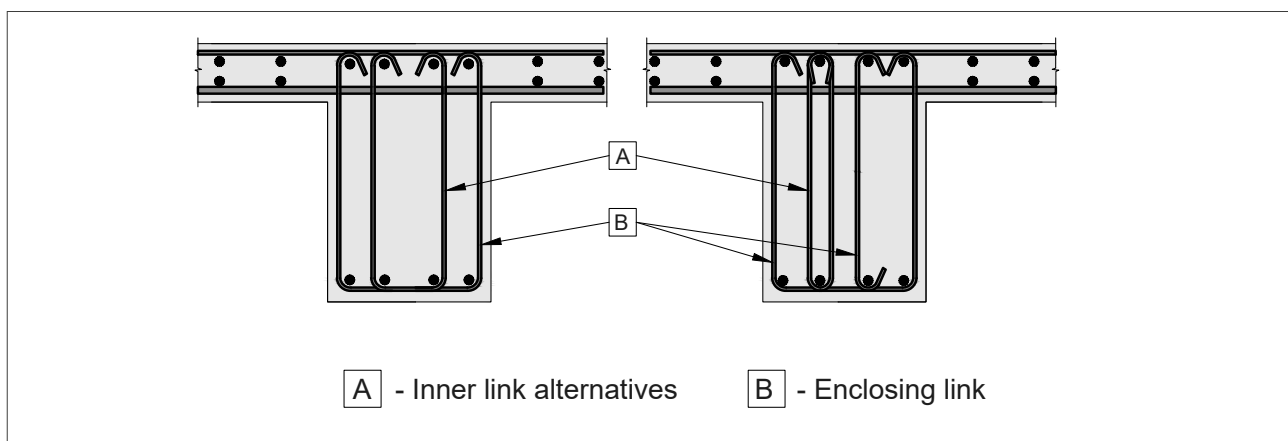
Figure 4.73 Anchorage at Intermediate Supports



4.9.2.2 Shear Reinforcement

1. The shear reinforcement should form an angle α of between 45° and 90° to the longitudinal axis of the structural element. The shear reinforcement shall be provided in the form of:
 - Links enclosing the longitudinal tension reinforcement and the compression zone (refer Figure 4.74).
 - Bent-up bars.
 - Or a combination of the two.

Figure 4.74 Examples of Shear Reinforcement



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2. Links should be effectively anchored. A lap joint on the leg near the surface of the web is permitted provided that the link is not required to resist torsion.
3. At least 50% of the necessary shear reinforcement should be in the form of links.
4. The ratio of shear reinforcement is given by following Equation:

$$\rho_w = A_{sw} / (s b_w \sin \alpha)$$

Where,

ρ_w = The shear reinforcement ratio, should not be less than $\rho_{w,min}$ given as

$$\rho_{w,min} = (0.08 \sqrt{f_{ck}}) / f_{yk}$$

A_{sw} = The area of shear reinforcement within length s

s = The spacing of the shear reinforcement measured along the longitudinal axis of the member

b_w = The breadth of the web of the member

α = The angle between shear reinforcement and the longitudinal axis

The $\rho_{w,min}$ for beams is given in following Equation.

$$\rho_{w,min} = (0.08 \sqrt{f_{ck}}) / f_{yk}$$

5. The maximum longitudinal spacing between shear assemblies should not exceed $s_{l,max}$ given by:

$$s_{l,max} = 0.75 d (1 + \cot \alpha)$$

where α is the inclination of the shear reinforcement to the longitudinal axis of the beam.

6. The maximum longitudinal spacing of bent-up bars should not exceed $s_{b,max}$ given by following Equation:

$$s_{b,max} = 0.6 d (1 + \cot \alpha)$$

7. The transverse spacing of the legs in a series of shear links should not exceed $s_{t,max}$, given by Equation:

$$s_{t,max} = 0.75 d \leq 600 \text{ mm}$$

4.9.2.3 Torsion Reinforcement

1. The torsion links should be closed and be anchored by means of laps or hooked ends, see Figure 4.75, and should form an angle of 90° with the axis of the structural element.
2. The provisions of Clause 4.9.2.2 (4) and (5) are generally sufficient to provide the minimum torsion links required.
3. The maximum longitudinal spacing of the torsion links $s_{l,max}$ should comply the following:

$$s_{l,max} \leq \min \left[\frac{u}{8}; 0.75 d (1 + \cot \alpha); h, b \right]$$

Where,

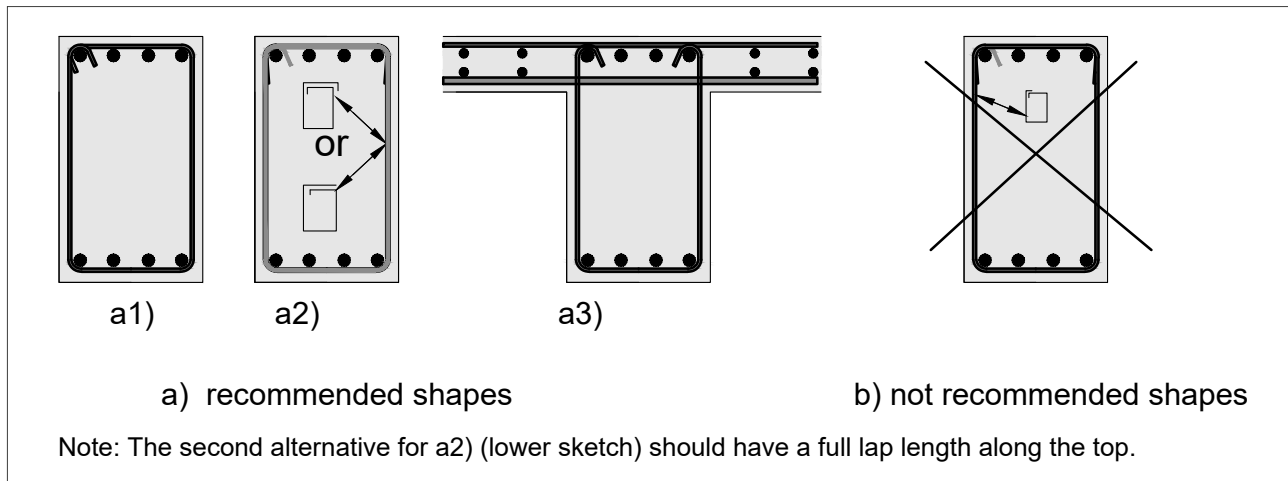
u = Circumference of outer edge of effective cross-section as defined in Figure 4.28.

d = Effective depth of beam.

h, b = Depth and width of beam.

4. The longitudinal bars should be so arranged that there is at least one bar at each corner, the others being distributed uniformly around the inner periphery of the links, with a spacing not greater than 350 mm.

Figure 4.75 Examples of Shapes for Torsion Links



4.9.2.4 Surface Reinforcement

It may be necessary to provide surface reinforcement either to control cracking or to ensure adequate resistance to spalling of the cover. Surface reinforcement should be used where the main reinforcement is made up of:

- Bars with diameter greater than 32 mm or.
- Bundled bars with equivalent diameter greater than 32 mm.

The surface reinforcement should consist of wire mesh or small diameter bars, and be placed outside the links as indicated in Figure 4.76.

The area of surface reinforcement $A_{s,surf}$ should be not less than $(0.01 A_{ct,ext})$ in the two directions parallel and orthogonal to the tension reinforcement in the beam. $A_{ct,ext}$ is the area of the tensile concrete external to the links. Where the cover to reinforcement is greater than 70 mm, for enhanced durability similar surface reinforcement should be used, with an area of $0.005 A_{ct,ext}$ in each direction. The minimum cover needed for the surface reinforcement shall be as given in Clause 4.4.4.2.

Figure 4.76 Surface Reinforcement

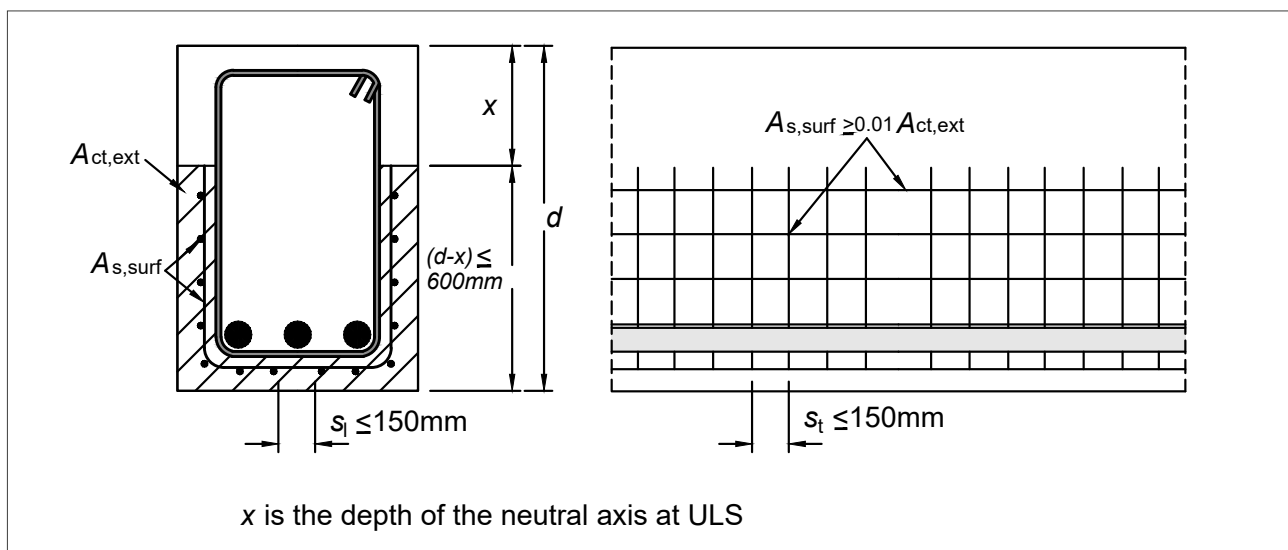
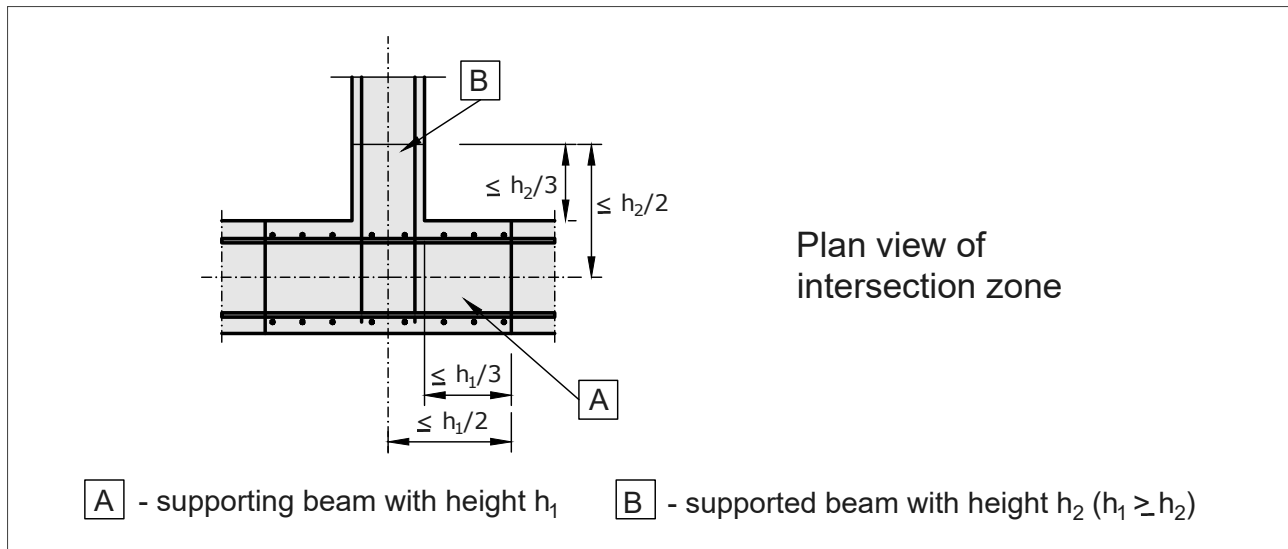


Figure 4.77 Placing Of Supporting Reinforcement in the Intersection Zone of Two Beams (Plan View)



4.9.2.5 Indirect Supports

1. Where a beam is supported by a beam instead of a wall or column, reinforcement should be provided and designed to resist the mutual reaction. This reinforcement is in addition to that required for other reasons. This rule also applies to a slab not supported at the top of a beam.
2. The supporting reinforcement between two beams should consist of links surrounding the principal reinforcement of the supporting member. Some of these links may be distributed outside the volume of the concrete, which is common to the two beams, (refer Figure 4.77)

4.9.3 Solid Slabs

This section applies to one-way and two-way solid slabs for which b and l_{eff} are not less than $5h$.

4.9.3.1 Flexural Reinforcement

4.9.3.1.1 General

1. For the minimum and the maximum steel percentages in the main direction, Clause 4.9.2.1.1 apply.
2. Secondary transverse reinforcement of not less than 20% of the principal reinforcement should be provided in one-way slabs. In areas near supports transverse reinforcement to principal top bars is not necessary where there is no transverse bending moment.
3. The spacing of bars should not exceed $s_{max,slabs}$ given as:
 - For the principal reinforcement, $3h \leq 400$ mm, where h is the total depth of the slab
 - For the secondary reinforcement, $3.5h \leq 450$ mm

In areas with concentrated loads or areas of maximum moment those provisions become respectively:

- For the principal reinforcement, $2h \leq 250$ mm
- For the secondary reinforcement, $3h \leq 400$ mm

4.9.3.1.2 Reinforcement in Slabs Near Supports

1. In simply supported slabs, half the calculated span reinforcement should continue up to the support and be anchored therein in accordance with Clause 4.8.4.4.
2. Where partial fixity occurs along an edge of a slab, but is not taken into account in the analysis, the top reinforcement should be capable of resisting at least 25 % of the maximum moment in the adjacent span. This reinforcement should extend at least 0.2 times the length of the adjacent span, measured from the face of the support. It should be continuous across internal supports and anchored at end supports. At an end support the moment to be resisted may be reduced to 15 % of the maximum moment in the adjacent span.

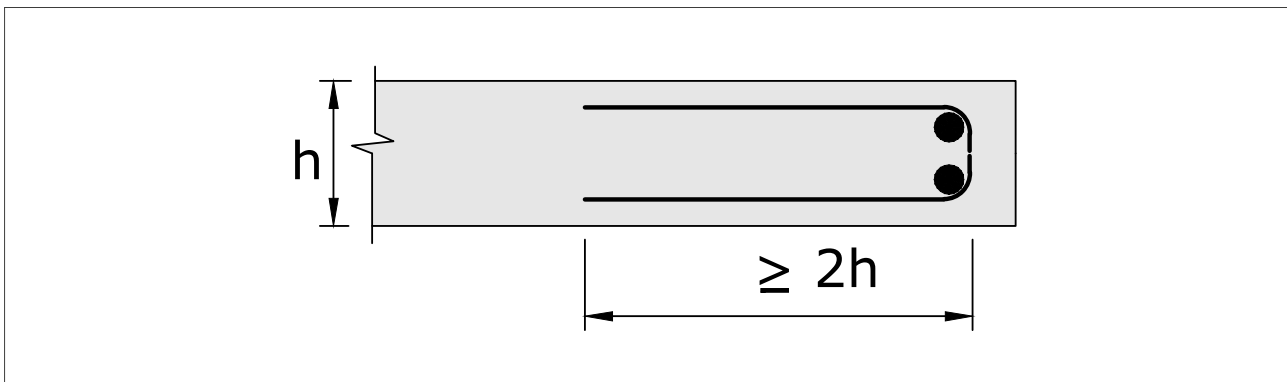
4.9.3.1.3 Corner Reinforcement

If the detailing arrangements at a support are such that lifting of the slab at a corner is restrained, suitable reinforcement should be provided.

4.9.3.1.4 Reinforcement at the Free Edges

1. Along a free (unsupported) edge, a slab should normally contain longitudinal and transverse reinforcement, generally arranged as shown in Figure 4.78.
2. The normal reinforcement provided for a slab may act as edge reinforcement.

Figure 4.78 Edge Reinforcement for Slab



4.9.3.2 Shear Reinforcement

1. A slab in which shear reinforcement is provided should have a depth of at least 200 mm.
2. In detailing the shear reinforcement, the minimum value and definition of reinforcement ratio in Clause 4.9.2.2 apply, unless modified by the following.
3. In slabs, if $|V_{Ed}| \leq 1/3 V_{Rd,max}$, (refer Clause 4.6.2), the shear reinforcement may consist entirely of bent-up bars or of shear reinforcement assemblies.
4. The maximum longitudinal spacing of successive series of links is given by:

$$s_{max} = 0.75 d (1 + \cot \alpha)$$

where α is the inclination of the shear reinforcement.

The maximum longitudinal spacing of bent-up bars is given by:

$$s_{max} = d$$

5. The maximum transverse spacing of shear reinforcement should not exceed $1.5d$.

4.9.4 Flat Slabs

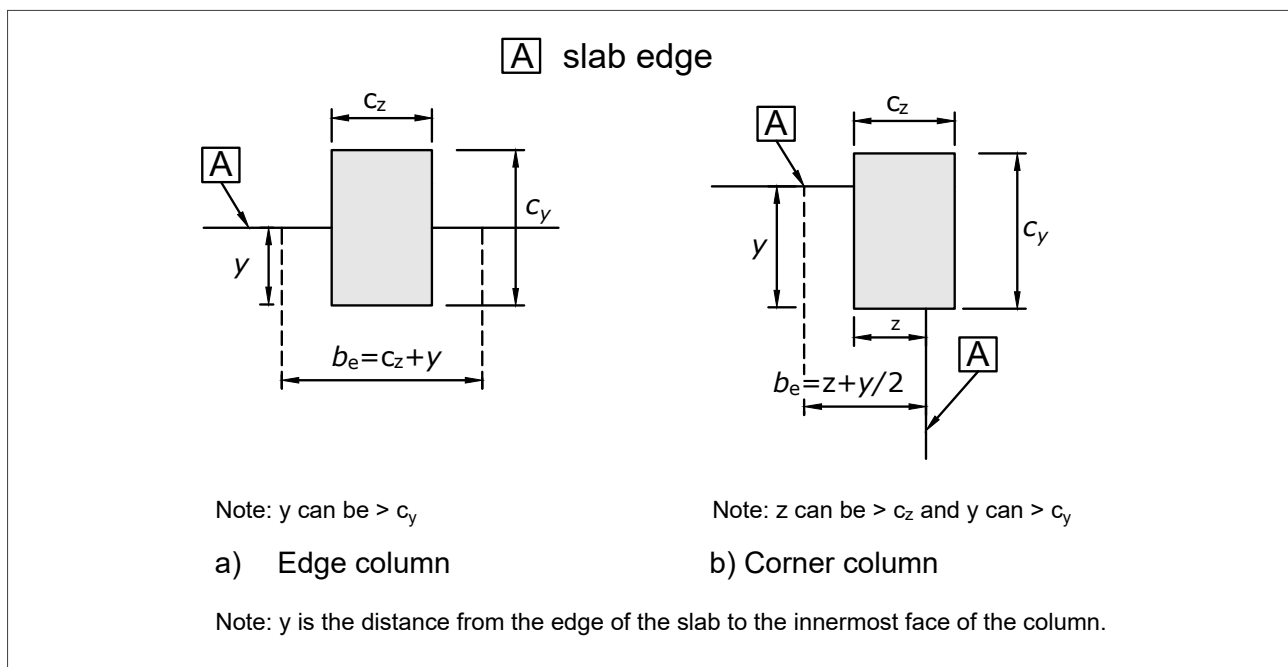
4.9.4.1 Slab at internal columns

1. The arrangement of reinforcement in flat slab construction should reflect the behaviour under working conditions. In general, this will result in a concentration of reinforcement over the columns.
2. At internal columns, unless rigorous serviceability calculations are carried out, top reinforcement of area $0.5 A_t$ should be placed in a width equal to the sum of 0.125 times the panel width on either side of the column. A_t represents the area of reinforcement required to resist the full negative moment from the sum of the two half panels each side of the column.
3. Bottom reinforcement (≥ 2 bars) in each orthogonal direction should be provided at internal columns and this reinforcement should pass through the column.

4.9.4.2 Slab at Edge and Corner Columns

Reinforcement perpendicular to a free edge required to transmit bending moments from the slab to an edge or corner column should be placed within the effective width be shown in Figure 4.79.

Figure 4.79 Effective Width, b_e , of a Flat Slab



4.9.4.3 Punching Shear Reinforcement

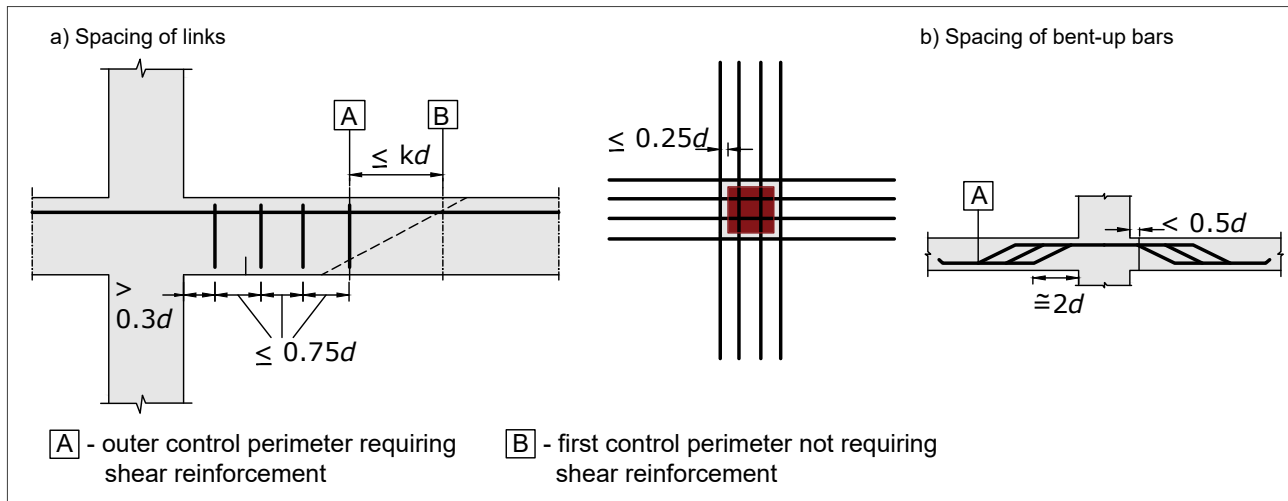
1. Where punching shear reinforcement is required (see 6.4) it should be placed between the loaded area/column and kd inside the control perimeter at which shear reinforcement is no longer required. It should be provided in at least two perimeters of link legs (refer Figure 4.80). The spacing of the link leg perimeters should not exceed $0.75d$.

The spacing of link legs around a perimeter should not exceed $1.5d$ within the first control perimeter ($2d$ from loaded area), and should not exceed $2d$ for perimeters outside the first control perimeter where that part of the perimeter is assumed to contribute to the shear capacity (see Figure 4.40).

For bent down bars as arranged in Figure 4.80 b) one perimeter of link legs may be considered sufficient.

Note: See Clause 4.6.4.5 (4) for the value of k

Figure 4.80 Punching Shear Reinforcement



2. Where shear reinforcement is required the area of a link leg (or equivalent), $A_{sw,min}$, is given by following Expression.

$$A_{sw,min} \frac{1.5 \sin \alpha + \cos \alpha}{s_r s_t} \geq 0.08 \frac{\sqrt{f_{ck}}}{f} \quad \text{Equation 4.144}$$

Where,

α = The angle between the shear reinforcement and the main steel (i.e. for vertical links $\alpha = 90^\circ$ and $\sin \alpha = 1$).

s_r = The spacing of shear links in the radial direction.

s_t = The spacing of shear links in the tangential direction.

f_{ck} = in MPa.

3. Bent-up bars passing through the loaded area or at a distance not exceeding $0.25d$ from this area may be used as punching shear reinforcement (see Figure 4.80 b), top).
4. The distance between the face of a support, or the circumference of a loaded area, and the nearest shear reinforcement taken into account in the design should not exceed $d/2$. This distance should be taken at the level of the tensile reinforcement. If only a single line of bent-up bars is provided, their slope may be reduced to 30° .

4.9.5 Columns

4.9.5.1 General

This Clause deals with columns for which the larger dimension h is not greater than 4 times the smaller dimension b .

4.9.5.2 Longitudinal Reinforcement

1. Longitudinal bars should have a diameter of not less than ϕ_{min} .

Note: The recommended value of ϕ_{min} is 12 mm.

2. The total amount of longitudinal reinforcement should not be less than $A_{s,min}$ given by following Equation:

$$A_{s,min} = \frac{0.10 N_{ED}}{f_{yd}} \quad \text{or} \quad 0.002 A_c \quad \text{whichever is the greater}$$

Where,

f_{yd} = The design yield strength of the reinforcement

N_{Ed} = The design axial compression force

3. The area of longitudinal reinforcement should not exceed $A_{s,max}$. The value of $A_{s,max}$ may be taken as $0.04 A_c$ outside lap locations. This limit should be increased to $0.08 A_c$ at laps.
4. For columns having a polygonal cross-section, at least one bar should be placed at each corner. The number of longitudinal bars in a circular column should not be less than four.

4.9.5.3 Transverse Reinforcement

1. The diameter of the transverse reinforcement (links, loops or helical spiral reinforcement) should not be less than 6 mm or one quarter of the maximum diameter of the longitudinal bars, whichever is the greater. The diameter of the wires of welded mesh fabric for transverse reinforcement should not be less than 5 mm.
2. The transverse reinforcement should be anchored adequately.
3. The spacing of the transverse reinforcement along the column should not exceed the following:
 - a. For concrete class $\leq$ C50/60
 - 20 times the minimum diameter of the longitudinal bars
 - the lesser dimension of the column
 - 400 mm
 - b. For concrete class $>$ C50/60

The transverse reinforcement should satisfy the following condition

$$\alpha_n \alpha_s \omega_{wd} \geq 0.04$$

Where,

ω_{wd} = (Volume of confining hoops $\times f_{yd}$ / volume of concrete $\times f_{cd}$)

For rectangular columns:

$$\alpha_n = 1 - \sum (b_i^2 / 6) / b_0 h_0$$

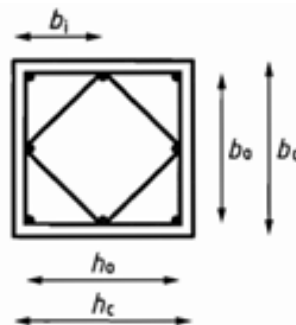
$$\alpha_s = [1 - (s/2b_0)] [1 - (s/2h_0)]$$

n is the number of longitudinal bars laterally restrained by links or ties

b_i is the distance between consecutive bars that are laterally restrained

s is the longitudinal spacing of links

b_0 & h_0 are the widths of confinement in the two directions (dimensions to centre of links)



For circular columns with circular links: $\alpha_n = 1$, $\alpha_s = (1 - (s/2D_0))^2$

For circular columns with spiral links: $\alpha_n = 1$, $\alpha_s = (1 - (s/2D_0))$

D_0 is the diameter of the confined core (to the centre line of links).

4. The maximum spacing required as above should be reduced by a factor 0.6:
 - i. In sections within a distance equal to the larger dimension of the column cross-section above or below a beam or slab
 - ii. Near lapped joints, if the maximum diameter of the longitudinal bars is greater than 14 mm. A minimum of 3 bars evenly placed in the lap length is required
5. Where the direction of the longitudinal bars changes, (e.g. at changes in column size), the spacing of transverse reinforcement should be calculated, taking account of the lateral forces involved. These effects may be ignored if the change of direction is less than or equal to 1 in 12.
6. Every longitudinal bar or bundle of bars placed in a corner should be held by transverse reinforcement. No bar within a compression zone should be further than 150 mm from a restrained bar.

4.9.6 Walls

4.9.6.1 General

This Clause refers to reinforced concrete walls with a length to thickness ratio of 4 or more and in which the reinforcement is taken into account in the strength analysis.

4.9.6.2 Vertical Reinforcement

1. The area of the vertical reinforcement should lie between $A_{s,min} = 0.002 A_c$ and $A_{s,max} = 0.04 A_c$ outside lap locations. This limit may be doubled at laps.
2. Where the minimum area of reinforcement, $A_{s,min}$, controls in design, half of this area should be located at each face.
3. The distance between two adjacent vertical bars shall not exceed 3 times the wall thickness or 400 mm whichever is the lesser.

4.9.6.3 Horizontal Reinforcement

1. Horizontal reinforcement running parallel to the faces of the wall (and to the free edges) should be provided at each surface. It should not be less than $A_{s,hmin}$, maybe taken as either 25% of the vertical reinforcement or $0.001 A_c$, whichever is greater.
2. The spacing between two adjacent horizontal bars should not be greater than 400 mm.

4.9.6.4 Transverse Reinforcement

1. In any part of a wall where the total area of the vertical reinforcement in the two faces exceeds $0.02 A_c$, transverse reinforcement in the form of links should be provided in accordance with the requirements for columns. The large dimension referred to in section need not be taken greater than 4 times thickness of wall.
2. Where the main reinforcement is placed nearest to the wall faces, transverse reinforcement should also be provided in the form of links with at least of 4 per m² of wall area. Transverse reinforcement need not be provided where welded wire mesh and bars of diameter $\phi \leq 16$ mm are used with concrete cover larger than 2ϕ .

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4.9.7 Deep Beams

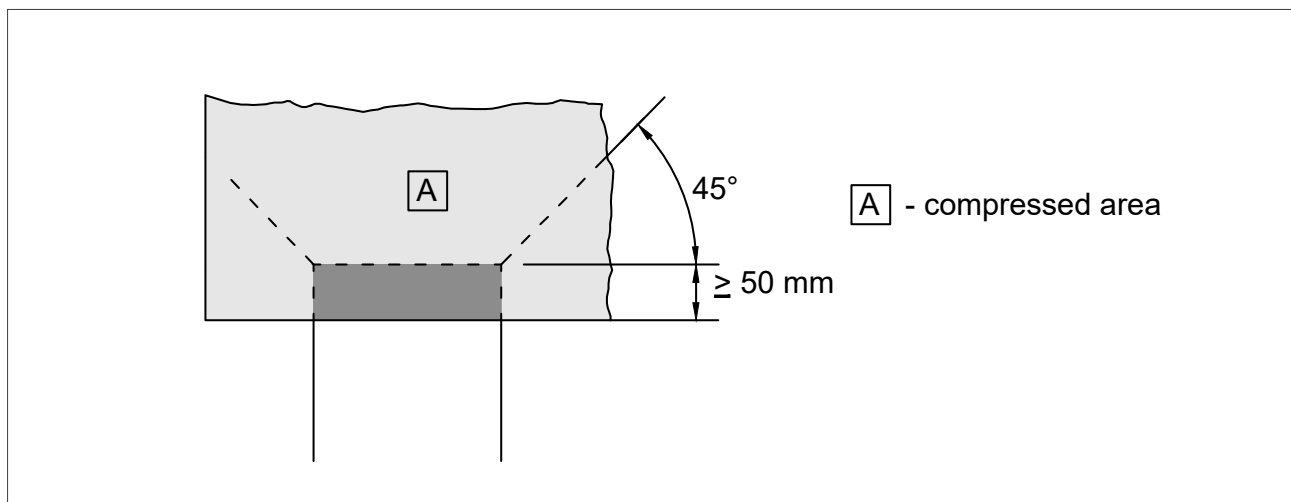
1. Deep beams (for definition see Clause 4.5.3.1) should normally be provided with an orthogonal reinforcement mesh near each face, with a minimum of $A_{s,dbmin}$ which may be taken as $0.002 A_c$ but not less than $150 \text{ mm}^2/\text{m}$ in each face and each direction.
2. The distance between two adjacent bars of the mesh should not exceed S_{mesh} . The recommended value of S_{mesh} is the lesser of the web thickness or 300 mm.
3. Reinforcement, corresponding to the ties considered in the design model, should be fully anchored for equilibrium in the node, see Clause 4.6.5.4, by bending the bars, by using U-hoops or by anchorage devices, unless a sufficient length is available between the node and the end of the beam permitting an anchorage length of l_{bd} .

4.9.8 Foundations

4.9.8.1 Pile caps

1. The distance from the outer edge of the pile to the edge of the pile cap should be such that the tie forces in the pile cap can be properly anchored. The expected deviation of the pile on site should be taken into account.
2. Reinforcement in a pile cap should be calculated either by using strut-and-tie or flexural methods as appropriate.
3. The main tensile reinforcement to resist the action effects should be concentrated in the stress zones between the tops of the piles. A minimum bar diameter 12 mm should be provided. If the area of this reinforcement is at least equal to the minimum reinforcement, evenly distributed bars along the bottom surface of the member may be omitted.
4. Welded transverse bars may be used for the anchorage of the tension reinforcement. In this case the transverse bar may be considered to be part of the transverse reinforcement in the anchorage zone of the reinforcement bar considered.
5. (The compression caused by the support reaction from the pile may be assumed to spread at 45° angles from the edge of the pile (see Figure 4.81). This compression may be taken into account when calculating the anchorage length.

Figure 4.81 Compressed Area Increasing The Anchorage Capacity

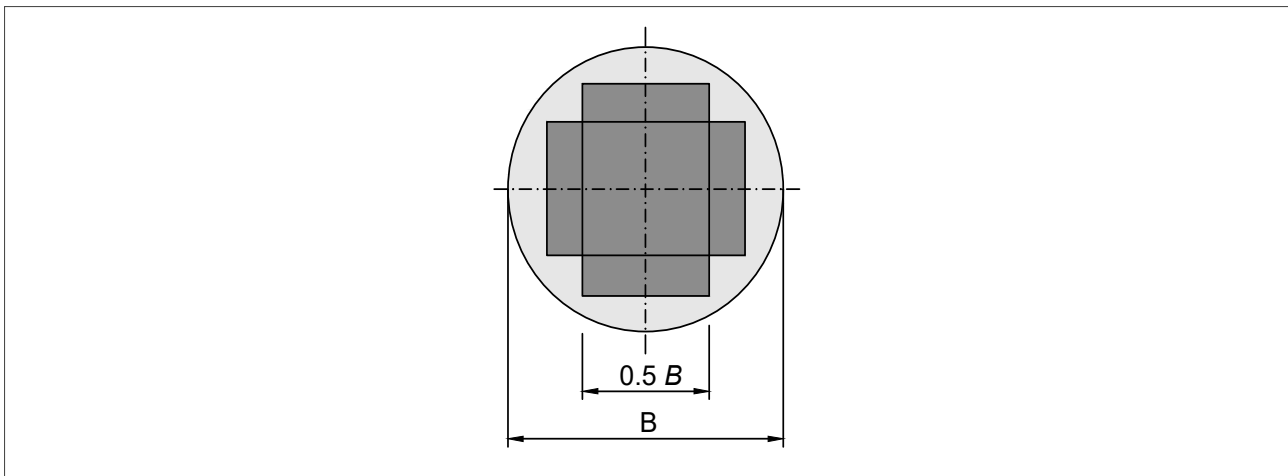


4.9.8.2 Column and Wall Footings

4.9.8.2.1 General

1. The main reinforcement should be anchored in accordance with the requirements of Clause 4.8.4 and 4.8.5. A minimum bar diameter ϕ_{min} should be provided. In footings the design model shown in Clause 4.9.8.2.2 may be used. The recommended value of ϕ_{min} is 12 mm.
2. The main reinforcement of circular footings may be orthogonal and concentrated in the middle of the footing for a width of $50\% \pm 10\%$ of the diameter of the footing, see Figure 4.82. In this case the unreinforced parts of the element should be considered as plain concrete for design purposes.
3. If the action effects cause tension at the upper surface of the footing, the resulting tensile stresses should be checked and reinforced as necessary.

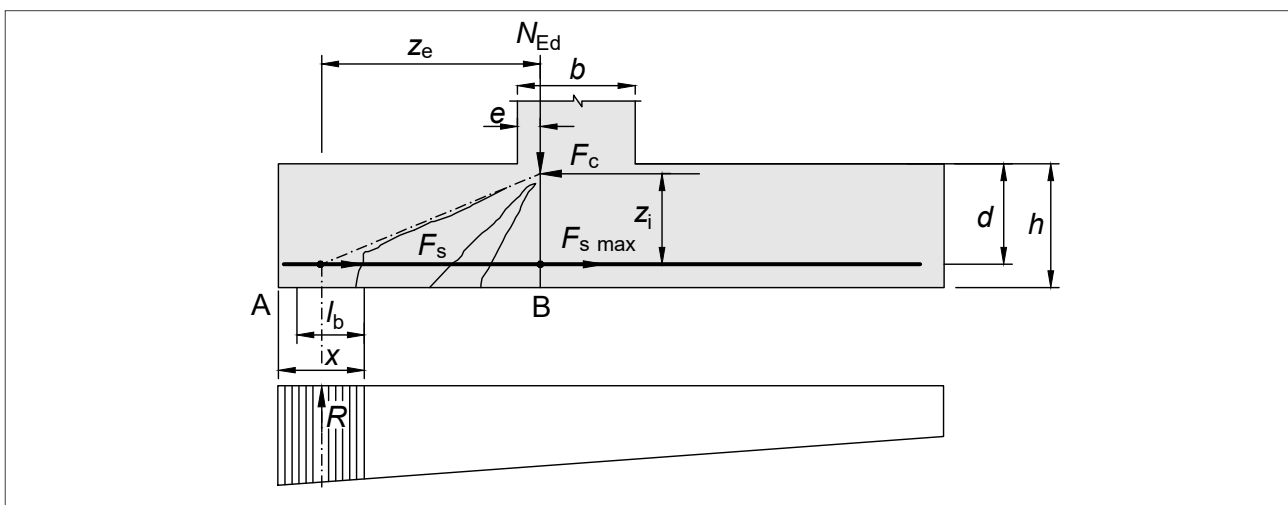
Figure 4.82 Orthogonal Reinforcement in Circular Spread Footing on Soil



4.9.8.2.2 Anchorage of Bars

1. The tensile force in the reinforcement is determined from equilibrium conditions, taking into account the effect of inclined cracks, see Figure 4.83. The tensile force F_s at a location x should be anchored in the concrete within the same distance x from the edge of the footing.

Figure 4.83 Model for Tensile Force With Regard to Inclined Cracks



2. The tensile force to be anchored is given by:

$$F_s = \frac{R z_e}{z_i}$$

Where,

R = The resultant of ground pressure within distance x .

Z_e = The external lever arm, i.e. distance between R and the vertical force.

N_{Ed} = The vertical force corresponding to total ground pressure between sections A and B.

z_i = The internal lever arm, i.e. distance between the reinforcement and the horizontal force F_c .

F_c = The compressive force corresponding to maximum tensile force $F_{s,max}$.

3. Lever arms z_e and z_i may be determined with regard to the necessary compression zones for N_{Ed} and F_c respectively. As simplifications, z_e may be determined assuming $e = 0.15b$, see Figure 4.83 and z_i may be taken as $0.9d$.
4. The available anchorage length for straight bars is denoted l_b in Figure 4.83. If this length is not sufficient to anchor F_s , bars may either be bent up to increase the available length or be provided with end anchorage devices.
5. For straight bars without end anchorage the minimum value of x is the most critical. As a simplification $x_{min} = h/2$ may be assumed. For other types of anchorage, higher values of x may be more critical.

4.9.8.3 Tie Beams

1. Tie beams may be used to eliminate the eccentricity of loading of the foundations. The beams should be designed to resist the resulting bending moments and shear forces. A minimum bar diameter ϕ_{min} for the reinforcement resisting bending moments should be provided. The recommended value of ϕ_{min} is 12 mm.
2. Tie beams should also be designed for a minimum downward load of q_1 if the action of compaction machinery can cause effects to the tie beams. The recommended value of q_1 is 10 kN/m.

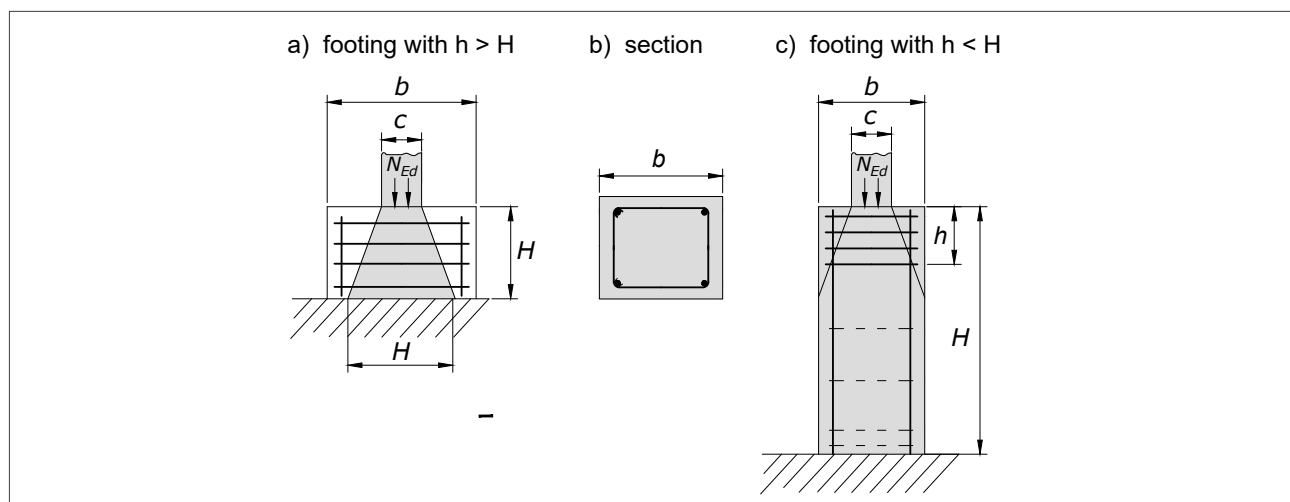
4.9.8.4 Column Footing On Rock

1. Adequate transverse reinforcement should be provided to resist the splitting forces in the footing, when the ground pressure in the ultimate states exceeds q_2 . This reinforcement may be distributed uniformly in the direction of the splitting force over the height h (see Figure 4.84). A minimum bar diameter, ϕ_{min} , should be provided. The recommended values of q_2 is 5 MPa and of ϕ_{min} is 8 mm.
2. The splitting force, F_s , may be calculated as follows (see Figure 4.84) :

$$F_s = 0.25 \left(1 - \frac{c}{h} \right) N_{Ed}$$

Where h is the lesser of b and H

Figure 4.84 Splitting Reinforcement In Footing on Rock



4.9.8.5 Bored Piles

1. The following Clauses apply for reinforced bored piles.
2. In order to allow the free flow of concrete around the reinforcement it is of primary importance that reinforcement, reinforcement cages and any attached inserts are detailed such that the flow of concrete is not adversely affected.
3. Bored piles should be provided with a minimum longitudinal reinforcement $A_{s,bpmin}$ related to pile cross section A_c .

Note: The recommended values of $A_{s,bpmin}$ are given in Table 4.24. This reinforcement should be distributed along the periphery of the section

Table 4.24 Recommended Minimum Longitudinal Reinforcement Area in Cast-in-place Bored Piles

Pile Cross-section: A_c (m ²)	Minimum area of Longitudinal Reinforcement: $A_{s,bpmin}$
$A_c \leq 0.5$	$A_s \geq 0.005 A_c$
$0.5 < A_c \leq 1.0$	$A_s \geq 25 \text{ cm}^2$
$A_c > 1.0$	$A_s \geq 0.0025 A_c$

The minimum diameter for the longitudinal bars should not be less than 16 mm. Piles should have at least 6 longitudinal bars. The clear distance between bars should not exceed 200 mm measured along the periphery of the pile.

4. For the detailing of longitudinal and transverse reinforcement in bored piles, refer EN 1536.

4.9.10 Regions With Discontinuity in Geometry or Action

1. D-regions should normally be designed with strut-and-tie models according to section 4.6.5 and detailed according to the rules given in Section 4.8.
2. The reinforcement, corresponding to the ties, shall be fully anchored by an anchorage of l_{bd} according to 4.8.4.

4.9.10 Bearings

Bearings shall be designed and detailed to ensure correct positioning taking into account the production and assembling deviations. The nominal length, a , of a simple bearing as shown in Figure 4.85 may be calculated as:

$$a = a_1 + a_2 + a_3 + \sqrt{(\Delta a_2^2 + \Delta a_3^2)}$$

Equation 4.145

Where,

a_1 = The net bearing length with regard to bearing stress, $a_1 = F_{Ed} / (b_1 f_{Rd})$, but not less than minimum values in Table 4.25.

F_{Ed} = Design value of support reaction.

b_1 = Net bearing width.

f_{Rd} = Design value of bearing strength.

a_2 = Distance assumed ineffective beyond outer end of supporting member, see Figure 4.85 and Table 4.26.

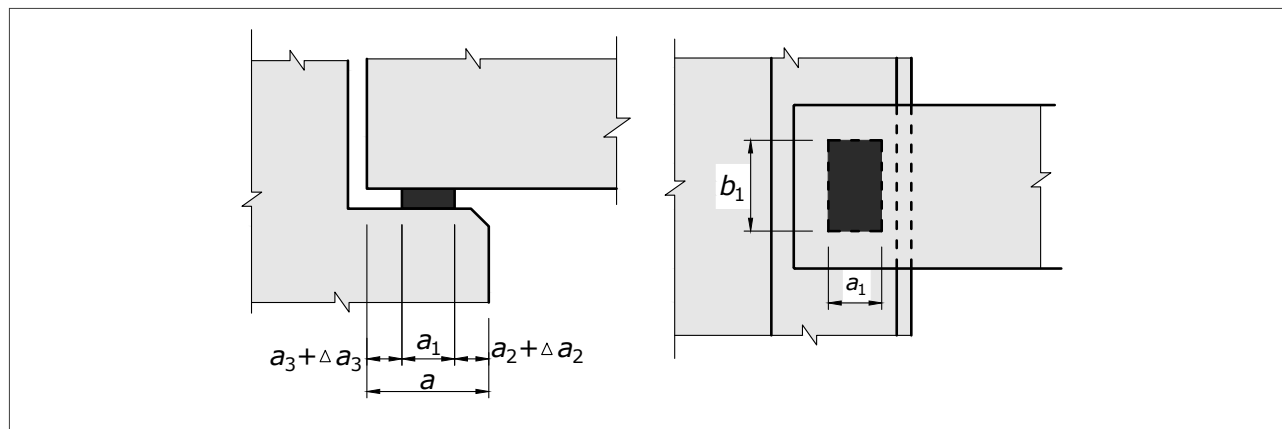
a_3 = Similar distance for supported member, see Figure 4.85 and Table 4.27.

Δa_2 = Allowance for deviations for the distance between supporting members, see Table 4.28.

Δa_3 = Allowance for deviations for the length of supporting member.
= $l_n / 2500$.

l_n = Length of member.

Figure 4.85 Example of Bearing

Table 4.25 Minimum Value of a_1 in mm

Relative Bearing Stress, σ_{Ed} / f_{cd}	≤ 0.15	$0.15 - 0.4$	> 0.4
Line supports (floors, roofs)	25	30	40
Ribbed floors and purlins	55	70	80
Concentrated supports (beams)	90	110	140

Table 4.26 Distance a_2 in mm

Support Material and Type		Relative Bearing Stress, σ_{Ed} / f_{cd}		
		≤ 0.15	$0.15 - 0.4$	> 0.4
Steel	Line	0	0	10
	Concentrated	5	10	15
Reinforced concrete $\geq C30/37$	Line	0	10	15
	Concentrated	10	15	25
Plain concrete and reinforced concrete $< C30/37$	Line	10	15	25
	Concentrated	15	25	35
Brickwork	Line	10	15	*
	Concentrated	20	25	*

*Notes: * Concrete padstone should be used in these cases*

Table 4.27 Distance a_3 in mm

Detailing of Reinforcement	Support	
	Line	Concentrated
Continuous bars over support (restrained or not)	0	0
Straight bars, horizontal, close to end of member	5	15, but not less than end cover
Tendons or straight bars exposed at end of member	5	15
Vertical loop reinforcement	15	End cover + inner radius of bending

Table 4.28 Allowance Δa_2

Support Material	Δa_2 (mm)
Steel or precast concrete	$10 \leq l/1200 \leq 30$ mm
Brickwork or cast in-situ concrete	$15 \leq l/1200 + 5 \leq 40$ mm
<i>Note: l = Span length</i>	

In the absence of other specifications, the following values can be used for the bearing strength:

$$f_{Rd} = 0.4 f_{cd} \text{ for dry connections (without bedding material)}$$

$$f_{Rd} = f_{bed} \leq 0.85 f_{cd} \text{ for all other cases}$$

Where,

f_{cd} = The lower of the design strengths for supported and supporting member

f_{bed} = The design strength of bedding material

If measures are taken to obtain a uniform distribution of the bearing pressure, e.g. with mortar, neoprene or similar pads, the design bearing width b_1 may be taken as the actual width of the bearing. Otherwise, and in the absence of a more accurate analysis, b_1 should not be greater than to 600 mm.

4.10 Detailing for Seismic Resistance

This Clause applies only to bridges designed for ductile behaviour and aim to ensure a minimum level of curvature/rotation ductility at the plastic hinges. In general, plastic hinge formation is not allowed in the superstructure. Therefore, there is no need for the application of special detailing guidelines for the superstructure due to seismic actions. Bridge foundation system shall be designed, as far as practicable, to remain elastic under design seismic action.

4.10.1 Concrete Piers

4.10.1.1 General Requirements for Confining Reinforcement

Ductile behaviour of the compression zone of concrete shall be ensured by providing confinement of concrete in the potential plastic hinge regions. Where normalised axial force exceeds the following limit in the regions of potential plastic hinge, confinement of compression zone shall be provided.

$$\eta_k = \frac{N_{Ed}}{A_c f_{ck}} > 0.08$$

Equation 4.146

The confinement is provided through rectangular hoops and/or cross-ties for rectangular sections or through circular hoops or spirals for circular sections. The quantity of confining reinforcement is defined through the mechanical reinforcement ratio:

$$\omega_{wd} = \rho_w \frac{f_{yd}}{f_{cd}}$$

Equation 4.147

The ρ_w is defined as below:

a. For rectangular sections

$$\rho_w = \frac{A_{sw}}{s_L b}$$

Equation 4.148

Where,

- 1 A_{sw} = The total area of hoops or ties in the one direction of confinement.
- 2 s_L = The spacing of hoops or ties in the longitudinal direction.
- 3 b = The dimension of the concrete core perpendicular to the direction of the confinement under consideration, measured to the outside of the perimeter hoop.

Bars inclined at an angle $\alpha > 0$ to the transverse direction in which ρ_w refers to shall be assumed to contribute to the total area A_{sw} by their area multiplied by $\cos\alpha$.

b. For circular sections

$$\rho_w = \frac{4 A_{sp}}{D_{sp} s_L}$$

Equation 4.149

Where,

- A_{sp} = The area of the spiral or hoop bar
- D_{sp} = The diameter of the spiral or hoop
- s_L = The spacing of hoops or ties in the longitudinal direction

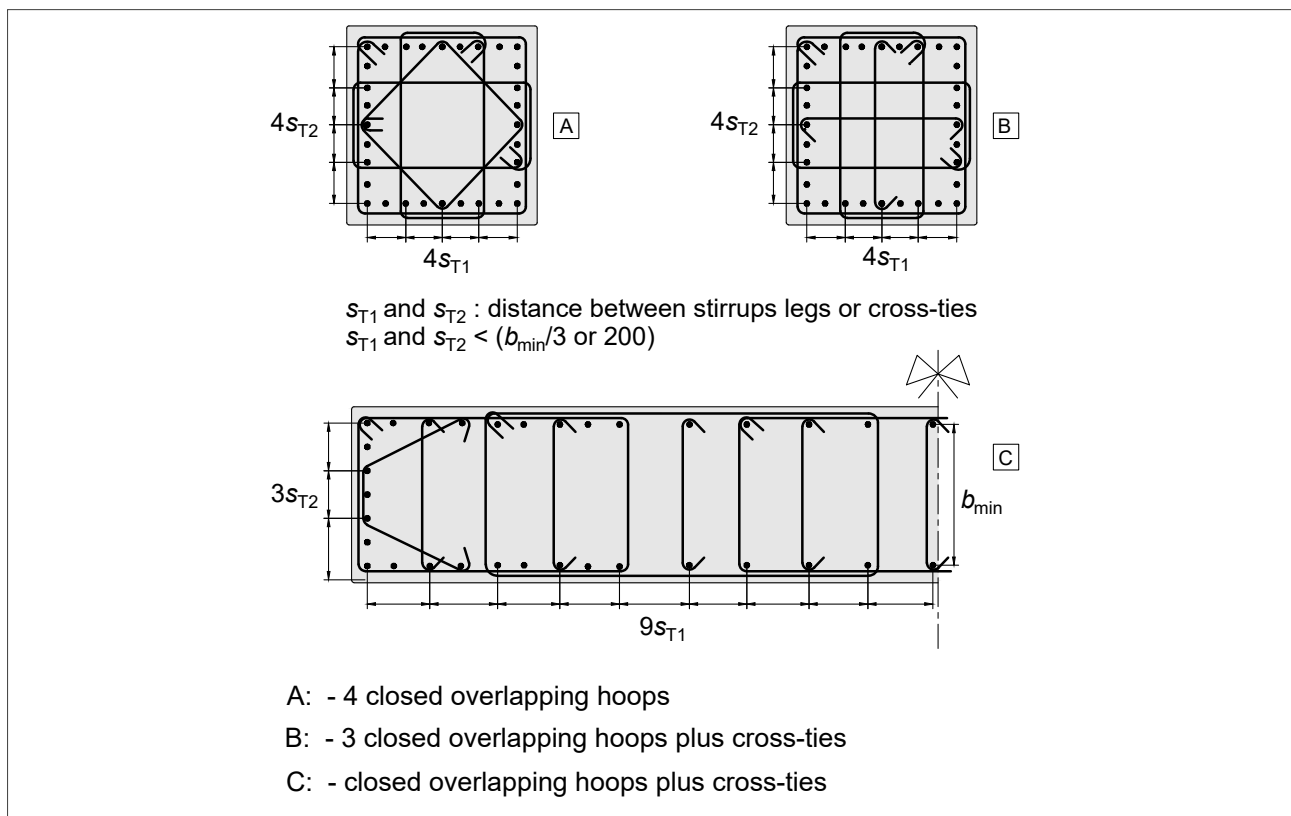
4.10.1.2 Spacing of Confining Reinforcement

The spacing of hoops or ties in the longitudinal direction, s_L , shall satisfy both of the following conditions:

- $s_L \leq 6$ times the longitudinal bar diameter, d_{bL}
- $s_L \leq 1/5$ of the smallest dimension of the confined concrete core, to the hoop centre line

For rectangular sections, the transverse distance s_T between hoop legs or supplementary cross-ties shall not exceed $1/3$ of the smallest dimension b_{min} of the concrete core to the hoop centre line, nor 200mm.

Figure 4.86 Typical Confinement Details in Concrete Piers



4.10.1.3 Minimum Confining Reinforcement

For rectangular stirrups and cross-ties, the minimum amount of confining reinforcement is the greater of two values given as follow:

$$\omega_{wd,r} \geq \max(\omega_{w,req}, \left(\frac{2}{3}\right) \omega_{w,min})$$

Equation 4.150

The $\omega_{w,req}$ is given by:

$$\omega_{w,req} = \frac{A_c}{A_{cc}} \lambda_{\eta k} + 0.13 \frac{f_{yd}}{f_{cd}} (\rho_L - 0.01)$$

Equation 4.151

Where,

A_c = The area of the gross concrete section.

A_{cc} = The confined (core) concrete area of the to the section hoop centreline.

$\omega_{w,min}$, λ = Are the factors given below.

Seismic Behaviour	λ	$\omega_{w,min}$
Ductile	0.37	0.18
Limited ductile	0.28	0.12

For circular hoops and spirals, the minimum amount of confining reinforcement is the greater of two values given as follow:

$$\omega_{wd,c} \geq \max(1.4\omega_{w,req}, \omega_{w,min})$$

Equation 4.152

When rectangular hoops and cross-ties are used, the minimum reinforcement condition shall be satisfied in both transverse directions.

4.10.1.4 Extent of Confinement - Length of Potential Plastic Hinges

When $\eta_k = N_{Ed}/A_c f_{ck} \leq 0.3$, the design length L_h of potential plastic hinges shall be estimated as the largest of the following values:

- The depth of the pier section within the plane of bending (perpendicular to the axis of rotation of the hinge).
- The distance from the point of maximum moment to the point where the design moment is less than 80% of the value of the maximum moment.

When $0.6 \geq \eta_k > 0.3$, the design length of the potential plastic hinges as determined above shall be increased by 50%.

The design length of plastic hinges (L_h) defined above should be used exclusively for detailing the reinforcement of the plastic hinge. It should not be used for estimating the plastic hinge rotation.

When confining reinforcement is required, the amount specified in Clause 4.10.1.3 shall be provided over the entire length of the plastic hinge. Outside the length of the hinge, the transverse reinforcement may be gradually reduced to the amount required by other criteria. The amount of transverse reinforcement provided over an additional length L_h adjacent to the theoretical end of the plastic hinge shall not be less than 50% of the amount of the confining reinforcement required in the plastic hinge.

4.10.1.5 Buckling of Longitudinal Compression Reinforcement

Buckling of longitudinal reinforcement shall be avoided along the length of the potential hinge areas. To meet this requirement, all main longitudinal bars should be restrained against out ward buckling by transverse reinforcement (hoops or cross-ties for rectangular section and though circular hoops or spirals for circular section) perpendicular to the longitudinal bars at a longitudinal spacing s_L not exceeding five times d_{bL} , the diameter of the smallest longitudinal bars. The coefficient δ depends on the ratio f_t/f_y of the tensile strength f_{tk} to the yield strength f_{yk} of the transverse reinforcement in terms of characteristic values, in accordance with the following relation:

$$5 \leq \delta = 2.5 \frac{f_{tk}}{f_{yk}} + 2.25 \leq 6$$

Along straight section boundaries, restraining of longitudinal bars should be achieved in either one of the following ways:

- Through a perimeter tie engaged by intermediate cross-ties at alternate locations of longitudinal bars, at transverse (horizontal) spacing s_T not exceeding 200 mm. The cross-ties shall have 135° hooks at one end, and 135° or 90° hook at the other end. 90° hooks are not permitted if η_k is greater than 0.30. Cross ties having 135° hooks on both ends may consist of two lapped spliced pieces. In sections of large dimensions, the perimeter tie may be spliced using appropriate lapping length combined with hooks; or
- Through overlapping closed ties arranged so that every corner bar and at least every alternate internal longitudinal bar is engaged by a tie leg. The transverse (horizontal) spacing s_T of the tie legs should not exceed 200 mm.

The minimum transverse ties shall be determined as follows:

$$\min \frac{A_t}{s_L} = \frac{\Sigma A_s f_{ys}}{1.6 f_{yt}} \text{ (mm}^2\text{/m)}$$

Equation 4.153

Where,

- A_t = The area of one tie leg, in mm².
- s_L = The spacing of the legs along the axis of the member in m.
- ΣA_s = The sum of the areas of the longitudinal bars restrained by the tie, in mm².
- f_{yt} = The yield strength of the tie.
- f_{ys} = The yield strength of the longitudinal reinforcement.

4.10.1.6 Other Requirements

Due to the possibility of loss of concrete cover in the plastic hinge region, the confining reinforcement shall be anchored by 135° hooks (unless a 90° hook is used in accordance with Clause 4.10.1.5) around a longitudinal bar plus adequate extension (minimum 10 diameters) into the core concrete. Where 90° hook is used, the bar shall extend by minimum 10 diameters into the core concrete.

- Similar anchoring or a full-strength weld is required for the lapping of spirals or hoops within potential plastic hinge regions. In this case laps of successive spirals or hoops, when located along the perimeter of the member, should be staggered.
- Splicing by lapping or welding of longitudinal reinforcement is not allowed within the plastic hinge region.

4.10.2 Foundations

4.10.2.1 Spread Foundations

Spread foundations such as footings, rafts, piers etc., shall not enter the plastic range under the design seismic action, and hence do not require special detailing reinforcement.

4.10.2.2 Pile Foundations

When it is not feasible to avoid localised hinge formation in the piles by designing pier to form hinges earlier, integrity and ductile behaviour of piles shall be ensured. The following locations along the pile should be treated as potential plastic hinges.

- a. At the pile heads adjacent to the pile cap, when the rotation of the pile cap about a horizontal axis transverse to the seismic action is restrained by the large stiffness of the pile group.
- b. At location of maximum bending moment in piles taking into account soil-pile interaction, using appropriate pile flexural stiffnesses, lateral soil stiffnesses and rotational stiffness of the pile group at the pile cap.
- c. At the interfaces of soil layers with markedly different shear deformability (e.g. change of strata).

At locations as in (a) above, confining reinforcement of the amount specified in Clause 4.10.1.3 along a vertical length equal to 3 times the pile diameter, shall be provided.

Unless a more accurate analysis is made, longitudinal as well as confining reinforcement of the same amount as that required at the pile head, shall be provided over a length of two pile diameters on each side of the point of maximum moment at location as in (b) above and of each side of the interface at locations as in (c) above.

4.10.3 Structures of Limited Ductile Behaviour

4.10.3.1 Verification of Ductility of Critical Sections

The following rules apply at the critical sections of structures designed for limited ductile behaviour (with $q \leq 1.5$) in cases other than those of low seismicity, to ensure a minimum of limited ductility.

A minimum of limited ductility shall be ensured at critical sections of the structures designed for limited ductile behaviour (with $q < 1.5$). A section is considered to be critical, i.e. location of a potential plastic hinge, when:

$$\frac{M_{Rd}}{M_{Ed}} < 1.30$$

Where,

M_{Ed} = The maximum design moment at the section in the seismic design situation.

M_{Rd} = The minimum flexural resistance of the section in the seismic design situation.

As far as possible, the location of potential plastic hinges should be accessible for inspection.

Confining reinforcement as required by Clause 4.10.1.3 for limited ductility), shall be provided in concrete members. In such cases it is also required to secure the longitudinal reinforcement against buckling in accordance with Clause 4.10.1.5.

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Concrete Structures

5 Steel Structures

5.1 General

5.1.1 Scope

This Chapter deals with the design of steel road bridges. It provides a general basis for the structural design of steel bridges and steel parts of composite bridges complying the requirements of safety and serviceability of structures. The basic design rules are applicable for steel structures with material thicknesses $t \geq 3$ mm. The fabrication and erection shall comply with “EN 1090: Execution of steel structures - Technical requirements”.

Reference documents

Following documents may be referred while using this Manual.

EN 1090	Execution of steel structures - Technical requirements.
EN 10025-1	Hot-rolled products of structural steels – Part 1: General delivery conditions.
EN 10025-2	Hot-rolled products of structural steels – Part 2: Technical delivery conditions for non-alloy structural steels.
EN 10025-3	Hot-rolled products of structural steels – Part 3: Technical delivery conditions for normalised/normalised rolled weldable fine grain structural steels.
EN 10025-4	Hot-rolled products of structural steels – Part 4: Technical delivery conditions for thermomechanical rolled weldable fine grain structural steels.
EN 10025-5	Hot-rolled products of structural steels – Part 5: Technical delivery conditions for structural steels with improved atmospheric corrosion resistance.
EN 10025-6	Hot-rolled products of structural steels – Part 6: Technical delivery conditions for flat products of high yield strength structural steels in the quenched and tempered condition.
EN 14399-1	High strength structural bolting for preloading Part 1: General Requirements.
EN 14399-2	High strength structural bolting for preloading Part 2: Suitability Test for preloading.
EN 14399-3	High strength structural bolting for preloading Part 3: System HR -Hexagon bolt and nut assemblies.
EN 14399-4	High strength structural bolting for preloading - Part 4: System HY -Hexagon bolt and nut assemblies.
EN 14399-5	High strength structural bolting for preloading - Part 5: Plain washers for system HR.
EN 14399-6	High strength structural bolting for preloading - Part 6: Plain chamfered washers for systems HR and HY.
EN ISO 4014	Hexagon head bolts - Product grades A and B (ISO 4014).
EN ISO 4016	Hexagon head bolts - Product grade C (ISO 4016).
EN ISO 4017	Hexagon head screws - Product grades A and B (ISO 4017).
EN ISO 4018	Hexagon head screws - Product grades C (ISO 4018).
EN ISO 4032	Hexagon nuts, style 1 - Product grades A and B (ISO 4032).
EN ISO 4033	Hexagon nuts, style 2 - Product grades A and B (ISO 4033).
EN ISO 4034	Hexagon nuts - Product grades C (ISO 4034).
KS 572	Hot-rolled structural steel sections - Specification.

5.1.3 Definitions

For general definitions, refer initial sections of this Manual. Other definitions are as under.

Capacity design

Design method for achieving the plastic deformation capacity of a member by providing additional strength in its connections and in other parts connected to it.

Effective cross-section and effective width

The gross cross-section or width reduced for the effects of plate buckling or shear lag or both; to distinguish between their effects the word "effective" is clarified as follows:

- "effective^p" denotes effects of plate buckling.
- "effective^s" denotes effects of shear lag.
- "effective" denotes effects of plate buckling and shear lag.

Plated structure

A structure built up from nominally flat plates which are connected together; the plates may be stiffened or unstiffened.

Rotational capacity

The angle through which the joint can rotate for a given resistance level without failing.

Rotational stiffness

The moment required to produce unit rotation in a joint.

Shear lag effect

Non-uniform stress distribution in wide flanges due to shear deformation; it is taken into account by using a reduced "effective" flange width in safety assessments.

Stiffened plate

Plate with transverse or longitudinal stiffeners or both.

Stiffeners

A plate or section attached to a plate to resist buckling or to strengthen the plate; a stiffener is denoted:

- Longitudinal if its direction is parallel to the member;
- Transverse if its direction is perpendicular to the member.

Subpanel

Unstiffened plate portion surrounded by flanges and/or stiffeners.

5.1.4 Symbols

The general symbols used in this Chapter are as under. Additional symbols are also defined where they appear. Symbols may have various meanings.

$x-x$	Axis along a member.
$y-y$	Axis along a cross-section.
$z-z$	Axis along a cross-section.
$u-u$	Major principal axis (where this does not coincide with the $y-y$ axis).
$v-v$	Minor principal axis (where this does not coincide with the $z-z$ axis).
b	Width of a cross section.
h	Depth of cross section.
d	Depth of straight portion of web / length of a diagonal of a built-up column.
t_w	Web thickness.
t_f	Thickness of flange.
r	Radius of root fillet.
r_1	Radius of root fillet.
r_2	Toe radius.
t	Thickness.
G_k	Nominal value of the effect of permanent actions.

R_d	Design value of resistance.
γ_M	General partial factor.
f_y	Yield strength.
f_u	Ultimate strength.
A_0	Original cross-section area.
ε_y	Yield strain.
ε_u	Ultimate strain.
E	Modulus of elasticity.
G	Shear modulus.
ν	Poisson's ratio in elastic stage.
α_{cr}	Factor by which the design loads would have to be increased to cause elastic instability in a global mode.
F_{Ed}	Design loading on the structure.
F_{cr}	Elastic critical buckling load for global instability mode based on initial elastic stiffnesses.
$\bar{\lambda}$	Non dimensional slenderness.
N_{Ed}	Design value of the axial force
ϕ	Global initial sway imperfection.
ϕ_0	Basic value of global initial sway imperfection.
α_h	Reduction factor for height h applicable to columns.
α_m	Reduction factor for the number of columns in a row.
m	Number of columns in a row.
e_0	Maximum amplitude of a member imperfection.
L	Member length.
$e_{0,d}$	Design value of maximum amplitude of an imperfection.
α	Imperfection factor / portion of a part of a cross section in compression / parameter introducing the effect of biaxial bending / angle between axes of chord and lacings.
$EI \eta_{cr}$	Bending moment due to η_{cr} at the critical cross section.
α_{cr}	Minimum force amplifier to reach the elastic critical buckling load.
M_{Ed}	Design bending moment.
k	Factor for $e_{0,d}$.
ε	Strain / factor depending on f_y .
c	Width or depth of a part of a cross-section
ψ	Stress or strain ratio.
γ_{M0}	Partial factor of resistance of cross-section whatever the class is.
γ_{M1}	Partial factor of resistance of members to instability assessed by member checks.
γ_{M2}	Partial factor of resistance of cross-sections in tension to fracture.
$\sigma_{x,Ed}$	Design value of the local longitudinal stress.
$\sigma_{z,Ed}$	Design value of the local transverse stress.
τ_{Ed}	Design value of the local shear stress.
N_{Ed}	Design normal force.
$M_{y,Ed}$	Design bending moment, y-y axis.
$M_{z,Ed}$	Design bending moment, z-z axis.
N_{Rd}	Design values of the resistance to normal forces.
$M_{y,Rd}$	Design values of resistance to bending moments, y-y-axis.

1

 $M_{z,Rd}$

Design values of resistance to bending moments, z-z-axis.

 s

Staggered pitch, the spacing of the centres of two consecutive holes in the chain measured parallel to the member axis.

2

 p

Spacing of the centres of the same two holes measured perpendicular to the member axis.

3

 n

Number of holes extending in any diagonal or zig-zag line progressively across the member or part of the member / ratio of design normal force to design plastic resistance to normal forces of the gross cross-section / number of planes of lacings or battens.

4

 d_0

Diameter of hole.

 e_N Shift of the centroid of the effective area A_{eff} relative to the centre of gravity of the cross section.

5

 ΔM_{Ed} Additional moment from shift of the centroid of the effective area A_{eff} relative to the centre of gravity of the gross cross section. A_{eff}

Effective area of a cross-section.

 $N_{t,Rd}$

Design values of the resistance to tension forces.

 $N_{pl,Rd}$

Design plastic resistance to normal forces of the gross cross-section.

 $N_{u,Rd}$

Design ultimate resistance to normal forces of the net cross-section at holes for fasteners.

 A_{net}

Net area of a cross-section.

 $N_{net,Rd}$

Design plastic resistance to normal forces of the net cross-section.

 $N_{c,Rd}$

Design resistance to normal forces of the cross-section for uniform compression.

 $M_{c,Rd}$

Design resistance of bending about one principal axis of a cross-section.

 W_{pl}

Plastic section modulus.

 $W_{el,min}$

Minimum elastic section modulus.

 $W_{eff,min}$

Minimum effective section modulus.

 A_f

Area of the tension flange.

 $A_{f,net}$

Net area of the tension flange.

 V_{Ed}

Design shear force.

 $V_{c,Rd}$

Design shear resistance.

 $V_{pl,Rd}$

Design plastic shear resistance.

 A_v

Shear area.

 η

Factor for shear area.

 S

First moment of area.

 I

Second moment of area.

 A

Cross-sectional area.

 A_w

Area of a web.

 A_f

Area of one flange.

 T_{Ed}

Design value of total torsional moments.

 T_{Rd}

Design resistance to torsional moments.

 $T_{t,Ed}$

Design value of internal St. Venant torsional moment.

 $T_{w,Ed}$

Design value of internal warping torsional moment.

 $\tau_{t,Ed}$

Design shear stresses due to St. Venant torsion.

 $\tau_{w,Ed}$

Design shear stresses due to warping torsion.

 $\sigma_{w,Ed}$

Design direct stresses due to the bi-moment bed.

 B_{Ed}

Design value of the bi-moment.

$V_{pl,T,Rd}$	Reduced design plastic shear resistance making allowance for the presence of a torsional moment.
ρ	Reduction factor to determine reduced design values of the resistance to bending moments making allowance for the presence of shear forces.
$M_{V,Rd}$	Reduced design values of the resistance to bending moments making allowance for the presence of shear forces.
$M_{N,Rd}$	Reduced design values of the resistance to bending moments making allowance for the presence of normal forces.
a	Ratio of web area to gross area / distance between restraints of chords.
β	Parameter introducing the effect of biaxial bending / correction factor for the lateral torsional buckling curves for rolled and welded sections.
$e_{N,y}$	Shift of the centroid of the effective area A_{eff} relative to the centre of gravity of the gross cross section (y-y axis).
$e_{N,z}$	Shift of the centroid of the effective area A_{eff} relative to the centre of gravity of the gross cross section (z-z axis).
$N_{b,Rd}$	Design buckling resistance of a compression member.
χ	Reduction factor for the relevant buckling curve.
Φ	Value to determine the reduction factor χ .
a_0, a, b, c, d	Class indexes for buckling curves.
N_{cr}	Elastic critical force for the relevant buckling mode based on the gross cross-sectional properties.
i	Radius of gyration about the relevant axis, determined using the properties of the gross cross-section.
$\bar{\lambda}_I$	Slenderness value to determine the relative slenderness.
$\bar{\lambda}_T$	Relative slenderness for torsional or torsional-flexural buckling.
$N_{cr,TF}$	Elastic torsional-flexural buckling force.
$N_{cr,T}$	Elastic torsional buckling force.
$M_{b,Rd}$	Design buckling resistance moment.
χ_{LT}	Reduction factor for lateral-torsional buckling.
Φ_{LT}	Value to determine the reduction factor χ_{LT} .
α_{LT}	Imperfection factor.
$\bar{\lambda}_{LT}$	Non dimensional slenderness for lateral torsional buckling.
M_{cr}	Elastic critical moment for lateral-torsional buckling.
$\bar{\lambda}_{LT0}$	Plateau length of the lateral torsional buckling curves for rolled and welded sections.
$\chi_{LT,mod}$	Modified reduction factor for lateral-torsional buckling.
F	Modification factor for χ_{LT} .
k_c	Correction factor for moment distribution.
ψ	Ratio of moments in segment.
L_c	Length between lateral restraints.
$\bar{\lambda}_f$	Equivalent compression flange slenderness.
$i_{f,z}$	Radius of gyration of compression flange about the minor axis of the section.
$I_{eff,f}$	Effective second moment of area of compression flange about the minor axis of the section.
$A_{eff,f}$	Effective area of compression flange.

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Steel Structures

1

 $A_{eff,w,c}$

Effective area of compressed part of web.

 $\lambda_{\bar{c}\bar{0}}$

Slenderness parameter.

 k_f

Modification factor.

2

 $\Delta M_{y,Ed}$

Moments due to the shift of the centroidal y-y axis.

 $\Delta M_{z,Ed}$

Moments due to the shift of the centroidal z-z axis.

3

 χ_y

Reduction factor due to flexural buckling (y-y axis).

 χ_z

Reduction factor due to flexural buckling (z-z axis).

 $\bar{\lambda}_{op}$

Global non dimensional slenderness of a structural component for out-of-plane buckling

4

 χ_{op} Reduction factor for the non-dimensional slenderness $\bar{\lambda}_{op}$. $\alpha_{ult,k}$

Minimum load amplifier of the design loads to reach the elastic critical buckling load with regards to lateral or lateral torsional buckling.

5

 N_{Rk}

Characteristic value of resistance to compression.

 $M_{y,Rk}$

Characteristic value of resistance to bending moments about y-y axis.

 $M_{z,Rk}$

Characteristic value of resistance to bending moments about z-z axis.

 Q_m

Local force applied at each stabilised member at the plastic hinge locations.

 L_{ch}

Buckling length of chord.

 h_0

Distance of centrelines of chords of a built-up section.

 i_{min}

Minimum radius of gyration of single angles.

 A_{ch}

Area of one chord of a built-up column.

 $N_{ch,Ed}$

Design chord force in the middle of a built-up member.

 M_{Ed}^I

Design value of the maximum first order moment in the middle of the built-up member.

 I_{eff}

Effective second moment of area of the built-up member.

 S_v

Shear stiffness of built-up member from the lacings or battened panel.

 A_d

Area of one diagonal of a built-up column.

 A_v

Area of one post (or transverse element) of a built-up column.

 I_{ch}

In plane second moment of area of a chord.

 I_b

In plane second moment of area of a batten.

 μ

Efficiency factor.

 $\sigma_{Ed,ser}$

Nominal stresses from the characteristic load combinations.

 $\tau_{Ed,ser}$ $\lambda, \lambda_1, \lambda_2,$

Damage equivalent factors.

 $\lambda_3, \lambda_4,$ $\lambda_{max}, \lambda_{loc},$ λ_{glo} $\Phi_2, \Phi_{loc},$

Damage equivalent impact factors.

 Φ_{glo} $\Delta\sigma_p, \Delta\sigma_{loc},$ Stress ranges from load p . $\Delta\sigma_{glo}$ μ_k

Characteristic value of friction coefficient.

Stiffener: A_{sf}

Total area of all the longitudinal stiffeners of a stiffened plate.

 A_{st}

Gross cross-sectional area of one transverse stiffener.

 A_{eff}

Effective cross-sectional area.

$A_{c,eff}$	Effective ^p cross sectional area.
$A_{c,eff,loc}$	Effective ^p cross sectional area for local buckling.
a	Length of a stiffened or unstiffened plate.
b	Width of a stiffened or unstiffened plate.
b_w	Clear width between welds for welded sections or between ends of radii for rolled sections.
b_{eff}	Effective ^s width for elastic shear lag.
F_{Ed}	Design transverse force.
h_w	Clear web depth between flanges.
L_{eff}	Effective length for resistance to transverse forces.
$M_{f,Rd}$	Design plastic moment of resistance of a cross-section consisting of the flanges only.
$M_{pl,Rd}$	Design plastic moment of resistance of the cross-section (irrespective of cross-section class).
M_{Ed}	Design bending moment.
N_{Ed}	Design axial force.
t	Thickness of the plate.
V_{Ed}	Design shear force including shear from torque.
W_{eff}	Effective elastic section modulus.
β	Effective ^s width factor for elastic shear lag.

Joint and Connections:

d	The nominal bolt diameter, the diameter of the pin or the diameter of the fastener.
d_0	The hole diameter for a bolt, a rivet or a pin.
$d_{0,t}$	The hole size for the tension face, generally the hole diameter, but for a slotted holes perpendicular to the tension face the slot length should be used.
f_{ur}	The specified ultimate tensile strength of the rivet.
e_1	The end distance from the centre of a fastener hole to the adjacent end of any part, measured in the direction of load transfer.
e_2	The edge distance from the centre of a fastener hole to the adjacent edge of any part, measured at right angles to the direction of load transfer.
e_3	The distance from the axis of a slotted hole to the adjacent end or edge of any part.
e_4	The distance from the centre of the end radius of a slotted hole to the adjacent end or edge of any part.
l_{eff}	The effective length of fillet weld.
n	The number of the friction surfaces or the number of fastener holes on the shear face.
p_1	The spacing between centres of fasteners in a line in the direction of load transfer.
$p_{1,0}$	The spacing between centres of fasteners in an outer line in the direction of load transfer.
$p_{1,i}$	The spacing between centres of fasteners in an inner line in the direction of load transfer.
p_2	The spacing measured perpendicular to the load transfer direction between adjacent lines of fasteners.
t_p	The thickness of the plate under the bolt or the nut.
t_w	The thickness of the web or bracket.

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 A

The gross cross-section area of bolt.

 A_s

The tensile stress area of the bolt or of the anchor bolt.

 $B_{p,Rd}$

The design punching shear resistance of the bolt head and the nut.

2

 E

The elastic modulus.

 $F_{p,cd}$

The design preload force.

 $F_{t,Ed}$

The design tensile force per bolt for the ultimate limit state.

3

 $F_{t,Rd}$

The design tension resistance per bolt.

 $F_{v,Rd}$

The design shear resistance per bolt.

4

 $F_{b,Rd}$

The design bearing resistance per bolt.

 $F_{s,Rd,ser}$

The design slip resistance per bolt at the serviceability limit state.

 $F_{s,Rd}$

The design slip resistance per bolt at the ultimate limit state.

5

 $F_{v,Ed,ser}$

The design shear force per bolt for the serviceability limit state.

 $F_{v,Ed}$

The design shear force per bolt for the ultimate limit state.

 z

The lever arm.

 μ

The slip factor.

 ϕ

The rotation of a joint.

Fatigue: $\Delta\sigma$

Stress range (direct stress).

 $\Delta\tau$

Stress range (shear stress).

 $\Delta\sigma_E, \Delta\tau_E$ Equivalent constant amplitude stress range related to n_{max} . $\Delta\sigma_{E,2}, \Delta\tau_{E,2}$

Equivalent constant amplitude stress range related to 2 million cycles.

 $\Delta\sigma_c, \Delta\tau_c$ Reference value of the fatigue strength at $n_c = 2$ million cycles. γ_{Ff} Partial factor for equivalent constant amplitude stress ranges $\Delta\sigma_e, \Delta\tau_e$. γ_{Mf} Partial factor for fatigue strength $\Delta\sigma_c, \Delta\tau_c$. m

Slope of fatigue strength curve.

 λi

Damage equivalent factors.

 ψ_I

Factor for frequent value of a variable action.

 Q_k

Characteristic value of a single variable action.

 k_s

Reduction factor for fatigue stress to account for size effects.

 k_I

Magnification factor for nominal stress ranges to account for secondary bending moments in trusses.

 k_f

Stress concentration factor.

5.1.5 Convention for Member Axis

Following axes conventions are used.

1. The convention for axes is:

x-x along the member.

y-y axis of the cross-section.

z-z axis of the cross-section.

2. For steel members, the c conventions used for cross-section axes are:

For general cross sections:

y-y cross-section axis parallel to the flanges.

z-z cross-section axis perpendicular to the flanges.

For angle sections:

y-y cross-section axis parallel to the smaller leg.

z-z cross-section axis perpendicular to the smaller leg.

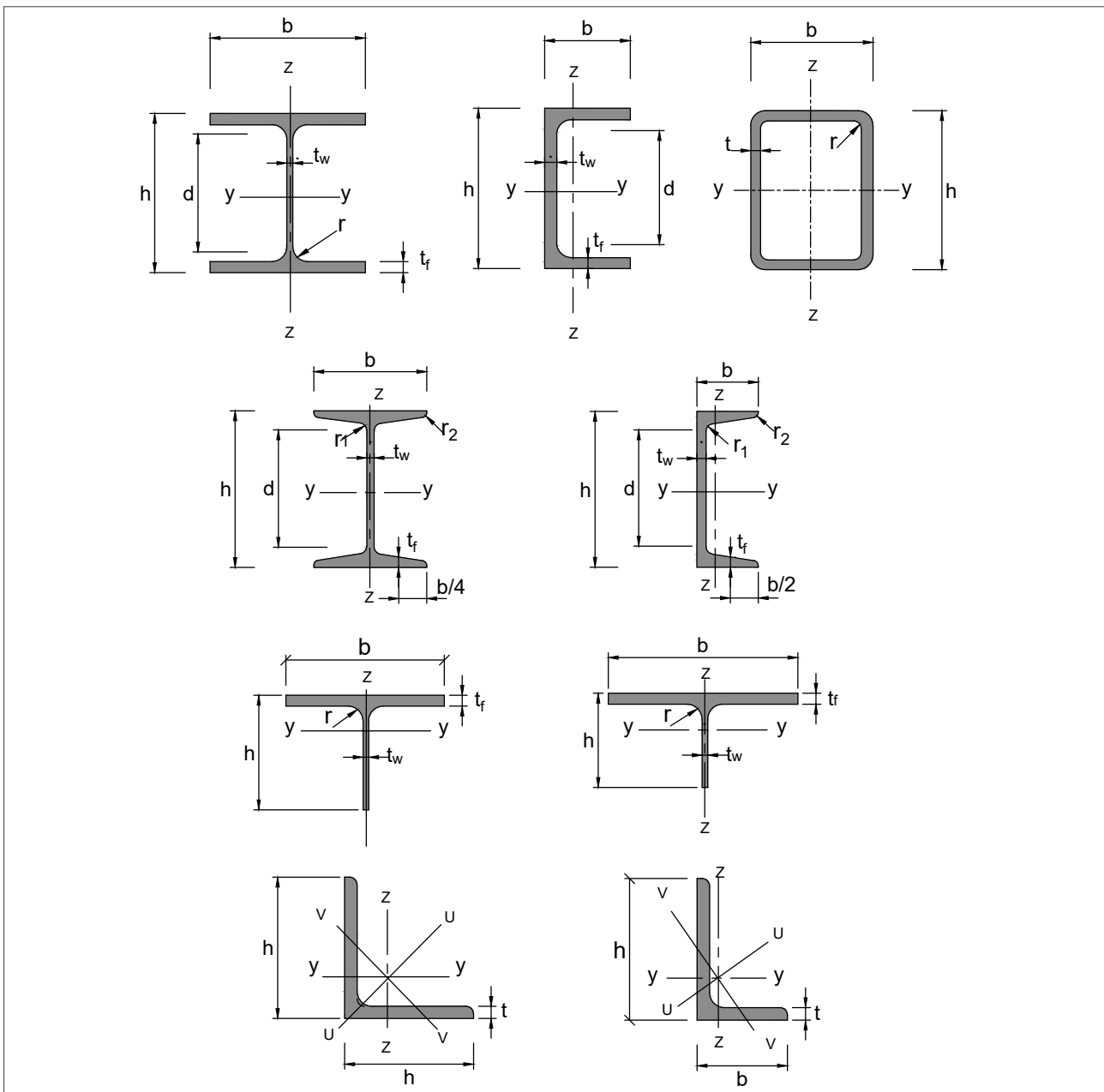
Principal axes, if different from y-y, z-z axis:

u-u major principal axis (where this does not coincide with the yy axis).

v-v minor principal axis (where this does not coincide with the zz axis).

3. The symbols used for dimensions and axes of rolled steel sections are indicated in Figure 5.1. The convention used for subscripts that indicate axes for moments is: "Use the axis about which the moment acts."

Figure 5.1 Dimensions and Axes of Sections



5.2 Basis of Design

Steel bridge structures are designed and constructed to satisfy the design requirements for the resistance, serviceability and durability of steel structures. The design shall be based on the characteristic values of material strengths and the loads applied to the structure. The design values are derived from the characteristic values by using partial safety factors both for material strengths and for loads. The reliability of design is ensured when:

$$E_d \leq R_d$$

Where,

E_d = The design value of the effect of actions

R_d = The design value of the corresponding resistance (strength)

The steel structures are designed for following limit states for the duration of their design working life:

- Ultimate Limit State.
- Serviceability Limit State.
- Limit state of fatigue.

5.2.1 Design Actions

The actions and combination of actions for the design of bridges shall be considered as specified in Chapter 3 along with the partial factors for actions.

5.2.2 Design Resistance

The design resistance is obtained from characteristic value of the resistance and partial factor for material.

$$R_d = \frac{R_k}{\gamma_M}$$

Where,

R_d = The design resistance.

R_k = The characteristic value of the particular resistance determined with characteristic or nominal values for the material properties and dimensions.

γ_M = Partial factor for material.

5.3 Material

5.3.1 General

The nominal values of material properties given in this section should be adopted as characteristic values in design calculations.

This Chapter covers the design of steel structures fabricated from steel material conforming to the steel grades listed in Table 5.1.

5.3.2 Material Properties

The nominal values of the yield strength f_y and the ultimate strength f_u for structural steel should be obtained either by adopting the values directly from the product standard or by using the simplification given in Table 5.1.

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5.3.3 Ductility Requirements

In elastic global analysis, a minimum ductility for structural steels is required that should be expressed in terms of limits for following parameters:

- The ratio f_u/f_y of the specified minimum ultimate tensile strength f_u to the specified minimum yield strength f_y . This ratio $f_u/f_y \geq 1.10$.
- The elongation at failure on a gauge length of $5.65 \sqrt{A_0}$ (where A_0 is the original cross-sectional area). The elongation at failure not less than 15%.
- The ultimate strain ε_u , where ε_u corresponds to the ultimate strength f_u . The ultimate strain $\varepsilon_u \geq 15 \varepsilon_y$, where ε_y is the yield strain ($\varepsilon_y = f_y/E$).

Plastic global analysis should not be used for bridges. Steel conforming with one of the steel grades listed in Table 5.1 should be accepted as satisfying these requirements.

Table 5.1 Nominal Values of Yield Strength f_y and Ultimate Tensile Strength f_u for Hot Rolled Structural Steel

Standard and steel grade		Nominal thickness of the element t (mm)			
		$t \leq 40$ mm		$40\text{mm} < t \leq 80$ mm	
		f_y (N/mm ²)	f_u (N/mm ²)	f_y (N/mm ²)	f_u (N/mm ²)
EN 10025-2	S 235	235	360	215	360
	S 275	275	430	255	410
	S 355	355	490	335	470
	S 450	440	550	410	550
EN 10025-3	S 275 N/NL	275	390	255	370
	S 355 N/NL	355	490	335	470
	S 420 N/NL	420	520	390	520
	S 460 N/NL	460	540	430	540
EN 10025-4	S 275 M/ML	275	370	255	360
	S 355 M/ML	355	470	335	450
	S 420 M/ML	420	520	390	500
	S 460 M/ML	460	540	430	530
EN 10025-5	S 235 W	235	360	215	340
	S 355 W	355	490	335	490
EN 10025-6	S 460 Q/QL/QL1	460	570	440	550

5.3.4 Tolerances

The dimensional and mass tolerances of rolled steel sections and plates should confirm with the relevant product standard, unless more severe tolerances are specified. For welded components the tolerances in EN 1090 should be applied.

For structural analysis and design, the nominal values of dimensions should be used.

5.3.5 Design values of Material Coefficients

The material coefficients to be adopted in calculations for the structural steels shall be taken as follows :

- Modulus of elasticity $E = 210\,000$ N/mm².
- Shear modulus $G = E / (2(1 + \nu)) = 81\,000$ N/mm².
- Poisson's ratio in elastic stage $\nu = 0.3$.
- The coefficient of linear thermal expansion $\alpha = 12 \times 10^{-6}$ per K (for $T \leq 100^\circ\text{C}$).

Note: For calculating the structural effects of unequal temperatures in composite concrete-steel structures, the coefficient of linear thermal expansion is taken as $\alpha = 10 \times 10^{-6}$ per K .

5.3.6 Connecting Devices

Bolts, nuts and washers should conform to the Reference Standards given in Clause 5.1.2. The bolts grade as given in Table 5.2 shall be applicable. The nominal values of the yield strength f_{yb} and the ultimate tensile strength f_{ub} are given in Table 5.2 and they should be adopted as characteristic values in calculations.

Table 5.2 Nominal Values of Yield Strength f_{yb} and Ultimate Tensile Strength f_{ub} for Bolts

Bolt Grade	4.6	5.6	6.8	8.8	10.9
f_{yb} (N/mm ²)	240	300	480	640	900
f_{ub} (N/mm ²)	400	500	600	800	1000

The High strength structural bolts of bolt grades 8.8 and 10.9 which conform to the Reference standards given in Clause 5.1.2 may be used as preloaded bolts.

The rivets, anchor bolts welding consumables shall be in accordance with the Reference standards given in Clause 5.1.2 and the relevant product standards. The nominal yield strength for anchor bolts should not exceed 640 N/mm².

5.4 Durability

The structure shall be designed such that deterioration over its design working life does not impair the performance of the structure below that intended, having due regard to its environment and the anticipated level of maintenance.

Following standards may be referred for execution and corrosion protection measures for steel structures.

EN 1090: Execution of steel structures
Technical requirements.

EN ISO 12944: Paints and varnishes
Corrosion protection of steel structures by protective paint systems.

EN ISO 1461: Hot dip galvanized coatings on fabricated iron and steel articles
Specifications and test methods.

The means of executing the protective treatment undertaken off-site and on-site shall be in accordance with EN 1090.

Parts susceptible to corrosion, mechanical wear or fatigue should be designed such that inspection maintenance and reconstruction can be carried out satisfactorily and access is available for in-service inspection and maintenance. For elements that cannot be inspected, fatigue checks should be carried out and appropriate corrosion allowances should be provided.

The inaccessible surfaces that have not been effectively sealed against the effects of air, moisture and contaminants should be provided with an additional sacrificial thickness appropriate to the environment as shown in Table 5.3.

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Table 5.3 Additional Sacrificial Thickness

Atmospheric corrosion classification (ISO 9223)	Additional sacrificial thickness (mm)	
	Structural steel	Weathering steel
C1: Very low	Not applicable	Not applicable
C2: Low	2	0.5
C3: Medium	4	1.0
C4: High	6	2.5
C5: Very high	Not applicable	Not applicable
Note 1: C1 relates to dry heated interiors, which are seldom applicable to bridges. Note 2: ISO 9223 and BS EN ISO 12944-2 may be referred for Atmospheric corrosion classification C1 to C5. Note 3: The sacrificial thickness given above assume free drainage of the steel surfaces. Where free drainage cannot be guaranteed, unprotected steel surfaces are not recommended.		

The required fatigue life of the structure and its components should be attained by the required fatigue design of details and with serviceability checks.

Components that cannot be designed with sufficient reliability to achieve the total design working life of the bridge should be replaceable.

5.5 Structural Analysis

5.5.1 Structural Modelling for Analysis

Structural analysis shall be based upon calculation models of the structure that are appropriate for the limit state under consideration. The calculation model and basic assumptions for the calculations should reflect the structural behaviour at the relevant limit state with appropriate accuracy and reflect the anticipated type of behaviour of the cross sections, members, joints and bearings. The method used for the analysis shall be consistent with the design assumptions.

The effects of the behaviour of the joints on the distribution of internal forces and moments within a structure, and on the overall deformations of the structure, may generally be neglected, but where such effects are significant (such as in the case of semi-continuous joints) they should be taken into account.

To identify whether the effects of joint behaviour on the analysis need be taken into account, a distinction may be made between three joint models as follows:

- Simple, in which the joint may be assumed not to transmit bending moments.
- Continuous, in which the behaviour of the joint may be assumed to have no effect on the analysis.
- Semi-continuous, in which the behaviour of the joint needs to be taken into account in the analysis.

For bridges, the type of joint and its modelling should be chosen to ensure that the required fatigue life can be attained.

Account should be taken of the deformation characteristics of the supports, where significant.

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5.2.2 Global Analysis

5.5.2.1 Effects of Deformed Geometry of the Structure

The internal forces and moments may generally be determined using either:

- First-order analysis, using the initial geometry of the structure; or
- Second-order analysis, taking into account the influence of the deformation of the structure.

The effects of the deformed geometry (second-order effects) should be considered if they increase the action effects significantly or modify significantly the structural behaviour.

First order analysis may be used for the structure, if the increase of the relevant internal forces or moments or any other change of structural behaviour caused deformations can be neglected. This condition may be assumed to be fulfilled, if the following criterion is satisfied:

$$\alpha_{cr} = F_{cr}/F_{Ed} \geq 10 \quad \text{for elastic analysis}$$

$$\alpha_{cr} = F_{cr}/F_{Ed} \geq 15 \quad \text{for plastic analysis}$$

Equation 5.1

Where,

α_{cr} = The factor by which the design loading would have to be increased to cause elastic instability in a global mode.

F_{Ed} = The design loading on the structure.

F_{cr} = The elastic critical buckling load for global instability mode based on initial elastic stiffnesses.

Bridges may be checked with first order theory if $\alpha_{cr} \geq 10$.

The effects of shear lag and of local buckling on the stiffness should be taken into account if this significantly influences the global analysis. For rolled sections and welded sections with similar dimensions shear lag effects may be neglected.

For global analysis, the effect of shear lag may be taken by using effective^S width of the flange elements and may be assumed to be uniform over the length of the span. The effective^S width of flanges for each span should be taken as the lesser of the full width and $L/8$ per side of the web. The L is taken as span length or twice the distance from the support to the end of a cantilever whichever is appropriate.

The effects of plate buckling for elastic global analysis, may be taken by effective^P cross sectional areas of the elements in compression as defined in Clause 5.6.8.3. For global analysis, the effect of plate buckling on the stiffness may be ignored if the effective^P cross-sectional area of an element in compression is larger than 0.5 times the gross cross-sectional area of the same element.

The effects on the global analysis of the slip in bolt holes and similar deformations of connection devices like studs and anchor bolts on action effects should be taken into account, where relevant and significant.

5.5.2.2 Structural Stability of Frames

The structural stability shall be verified considering deformation of the structure (imperfection and second order effects), if required in accordance with Clause 5.5.2.1. The second order effects and imperfections may be accounted by one of the following methods:

- Both totally by the global analysis.
- Partially by the global analysis and partially through individual stability check of the members according to Clause 5.6.18.
- For basic cases by individual stability checks of equivalent members according to Clause 5.6.18 using appropriate buckling lengths.

Second order effects may be calculated by using an analysis appropriate to the structure.

Where the behaviour of a bridge or its components is governed by the first buckling mode (single degree of freedom) the second order effects M_{II} may be calculated by applying a factor to the bending moments M_I as follows:-

$$M_{II} = M_I \frac{1}{1 - \frac{1}{\alpha_{cr}}} \quad \text{Equation 5.2}$$

Where $\alpha_{cr} \geq 3$

The stability of individual members should be checked according to the following:

- a. If second order effects in individual members and relevant member imperfections are totally accounted for in the global analysis of the structure, no individual stability check for the members according to Clause 5.6.18 is necessary.
- b. If second order effects in individual members or certain individual member imperfections are not totally accounted for in the global analysis, the individual stability of members should be checked according to the relevant criteria in Clause 5.6.18 for the effects not included in the global analysis.

Where the stability of a frame is assessed by a check with the equivalent column method according to Clause 5.6.18, the buckling length values should be based on a global buckling mode of the frame accounting for the stiffness behaviour of members and joints. In this case internal forces to be used in resistance checks are calculated according to first order theory without considering imperfections.

The effective lengths for isolated members for different buckling modes can be determined in accordance with the details given in Chapter 4.

5.5.3 Imperfections

5.5.3.1 Basis

Appropriate allowances should be incorporated in the structural analysis to cover the effects of imperfections, including residual stresses and geometrical imperfections such as lack of verticality, lack of straightness, lack of flatness, lack of fit eccentricities greater than the essential tolerances give in EN 1090-2 present in joints of the unloaded structure

The following imperfections should be taken into account:

- a. Global imperfections for frames and bracing systems.
- b. Local imperfections for individual members.

Global imperfections may be taken into account by modelling the frame out-of-plumb, or by a series of equivalent horizontal forces applied to a frame modelled vertically. The latter approach is recommended. Imperfections in individual members may be modelled, or members may be modelled as straight and imperfections automatically allowed for by verifying member resistances in accordance with Clause 5.6.18. The latter approach is recommended.

5.5.3.2 Imperfection for Global Analysis of Frames

The assumed shape of global imperfections and local imperfections may be derived from the elastic buckling mode of a structure in the plane of buckling considered. Both in and out of plane buckling with symmetric and asymmetric buckling shapes should be taken into account in the most unfavourable direction and form. For frames sensitive to buckling in a sway mode the effect of imperfections should be allowed for in frame analysis by means of an equivalent imperfection in the form of an initial sway imperfection and individual bow imperfections of members. The imperfections may be determined from:

a. Global initial sway imperfections is given as:

$$\phi = \phi_0 \alpha_h \alpha_m$$

Where,

ϕ_0 = The basic value: $\phi_0 = 1/200$

α_h = The reduction factor for height h applicable to columns:

$$\alpha_h = \frac{2}{\sqrt{h}} \text{ but } \frac{2}{3} \leq \alpha_h \leq 1.0$$

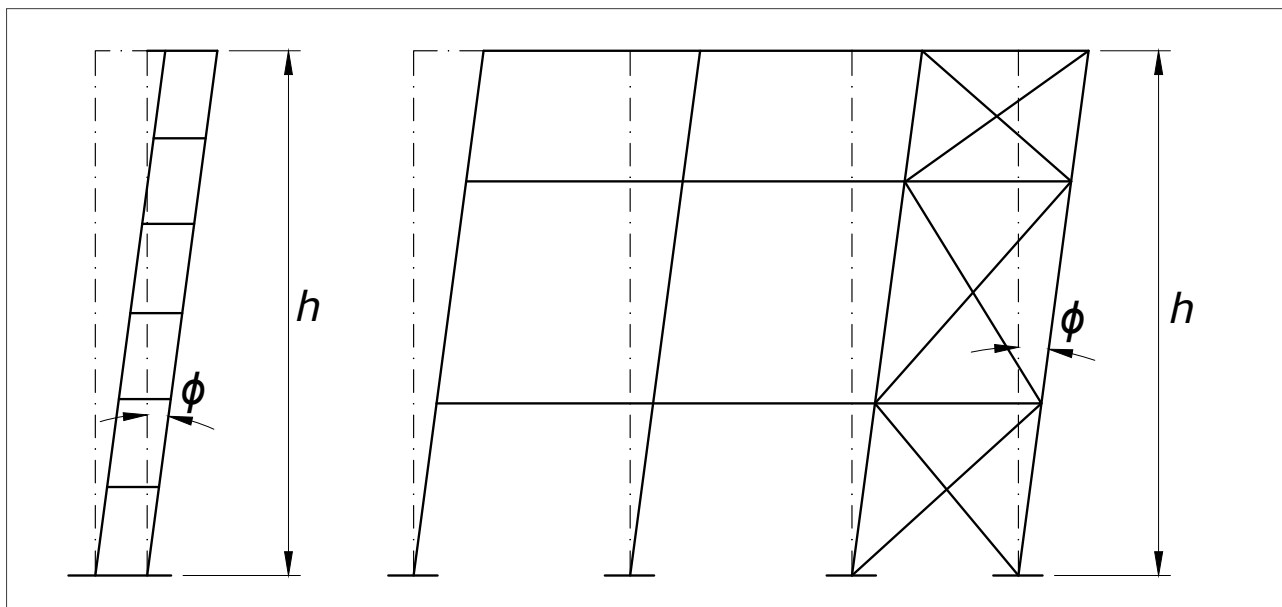
h = The height of the structure in meters

α_m = The reduction factor for the number of columns in a row: $\alpha_m = \sqrt{0.5(1 + 1/m)}$

For piers, α_m would be applicable, if cumulative effects from contributions of various piers occur (e.g. piers forming a frame with superstructure).

m = The number of columns in a row that are capable of resisting the sway and which carry a vertical load N_{Ed} not less than 50% of the average value of the column load in the vertical plane considered

Figure 5.2 Equivalent Sway Imperfections



b. relative initial local bow imperfections of members for flexural buckling is given as:

$$\frac{e_0}{L}$$

where L is the member length. The recommended values of e_0/L are given in below table.

Table 5.4 Design Value of Initial Local Bow Imperfection e_0/L for Members

Buckling curve as per Table 5.12	Elastic analysis	Plastic analysis
	e_0/L	e_0/L
a_0	1/350	1/300
a	1/300	1/250
b	1/250	1/200
c	1/200	1/150
d	1/150	1/100

When performing the global analysis for determining end forces and end moments to be used in member checks according to Clause 5.6.18, local bow imperfections may be neglected. However, for frames sensitive to second order effects local bow imperfections of members additionally to global sway imperfections should be introduced in the structural analysis of the frame for each compressed member where the following conditions are met.

- At least one moment resistant joint at one member end.

- $\bar{\lambda} > 0.5 \sqrt{\frac{A f_y}{N_{Ed}}}$

Where,

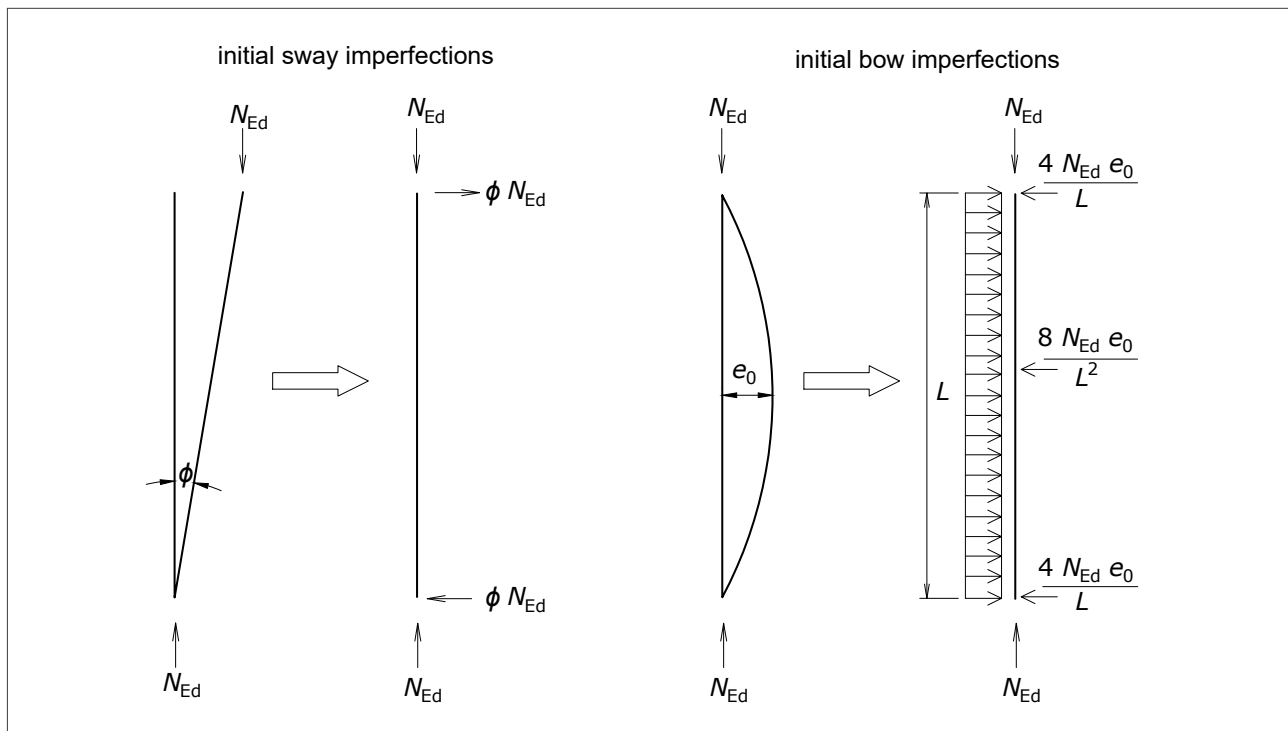
N_{Ed} = The design value of the compression force.

$\bar{\lambda}$ = The in-plane non-dimensional slenderness calculated for the member considered as hinged at its ends.

= Local bow imperfections are taken into account in member checks.

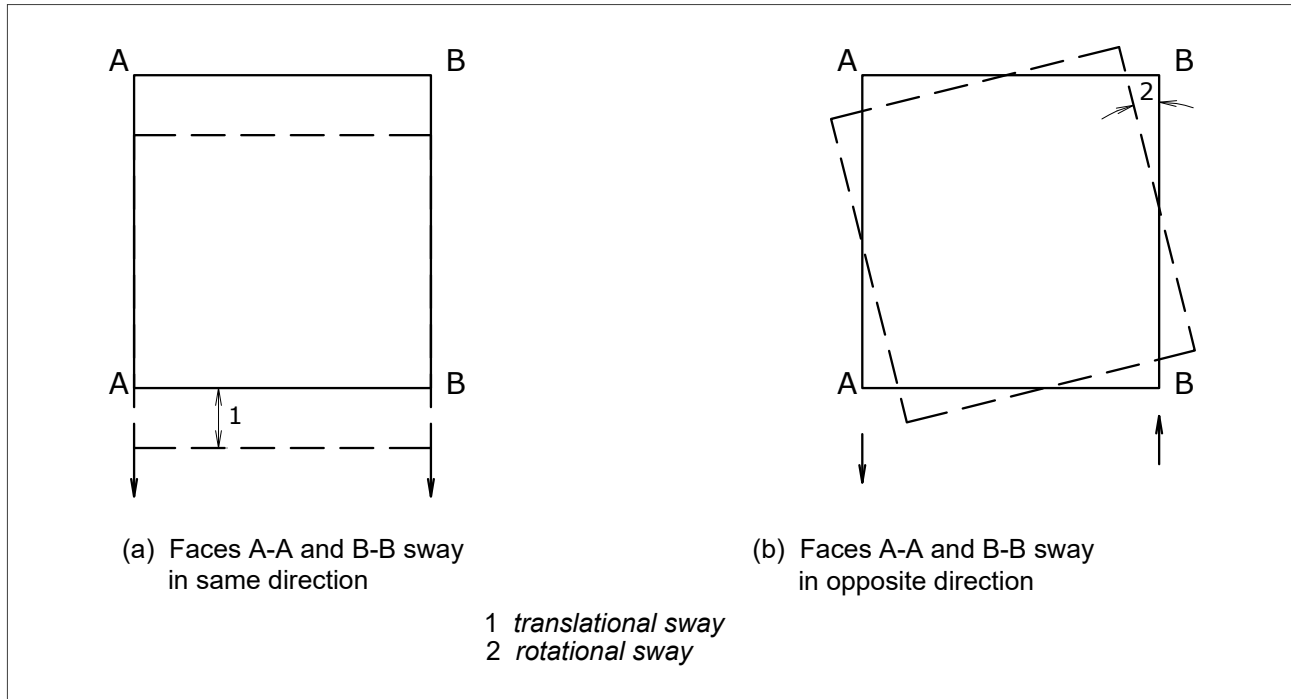
The effects of initial sway imperfection and local bow imperfections may be replaced by systems of equivalent horizontal forces, introduced for each column as shown in Figure 5.3. These initial sway imperfections shall be applied in all relevant directions but only in one direction at a time.

Figure 5.3 Replacement of Initial Imperfections by Equivalent Horizontal Forces



The possible torsional effects on a structure caused by anti-symmetric sways at the two opposite faces, should also be considered as shown in Figure 5.4.

Figure 5.4 Translational and Torsional Effects



5.5.3.3 Imperfection for Analysis of Bracing Systems

In the analysis of bracing systems which are required to provide lateral stability within the length of beams or compression members, the effects of imperfections should be included by means of an equivalent geometric imperfection of the members to be restrained, in the form of an initial bow imperfection.

$$e_0 = \alpha_m \frac{L}{500}$$

Equation 5.3

Where,

L = The span of the bracing system

$\alpha_m = \sqrt{0.5(1 + (1/m))}$

m = The number of members to be restrained

For convenience, the effects of the initial bow imperfections of the members to be restrained by a bracing system, may be replaced by the equivalent stabilising force as shown in Figure 5.5:

$$q_d = \sum N_{Ed} \ 8 \frac{e_0 + \delta_q}{L^2}$$

Equation 5.4

Where δ_q is the in-plane deflection of the bracing system due to q plus any external loads calculated from first order analysis. δ_q may be taken as 0 if second order analysis is used.

Where the bracing system is required to stabilise the compression flange of a beam of constant height, the force N_{Ed} in Figure 5.5 may be obtained from:

$$N_{Ed} = \frac{M_{Ed}}{h}$$

Equation 5.5

Where,

M_{Ed} = The maximum moment in the beam.

h = The overall depth of the beam.

Note: Where a beam is subjected to external compression, the N_{Ed} should include a part of the compression force carried by the flange.

At points where beams or compression members are spliced, it should also be verified that the bracing system is able to resist a local force equal to $(\alpha_m N_{Ed} / 100)$ applied to it by each beam or compression member which is spliced at that point, and to transmit this force to the adjacent points at which that beam or compression member is restrained, see Figure 5.6.

For checking for the local force according to the preceding requirement (points where members are spliced), any external loads acting on bracing systems should also be included, but the forces arising from the imperfection in the bracing system may be omitted.

Figure 5.5 Equivalent Stabilising Force

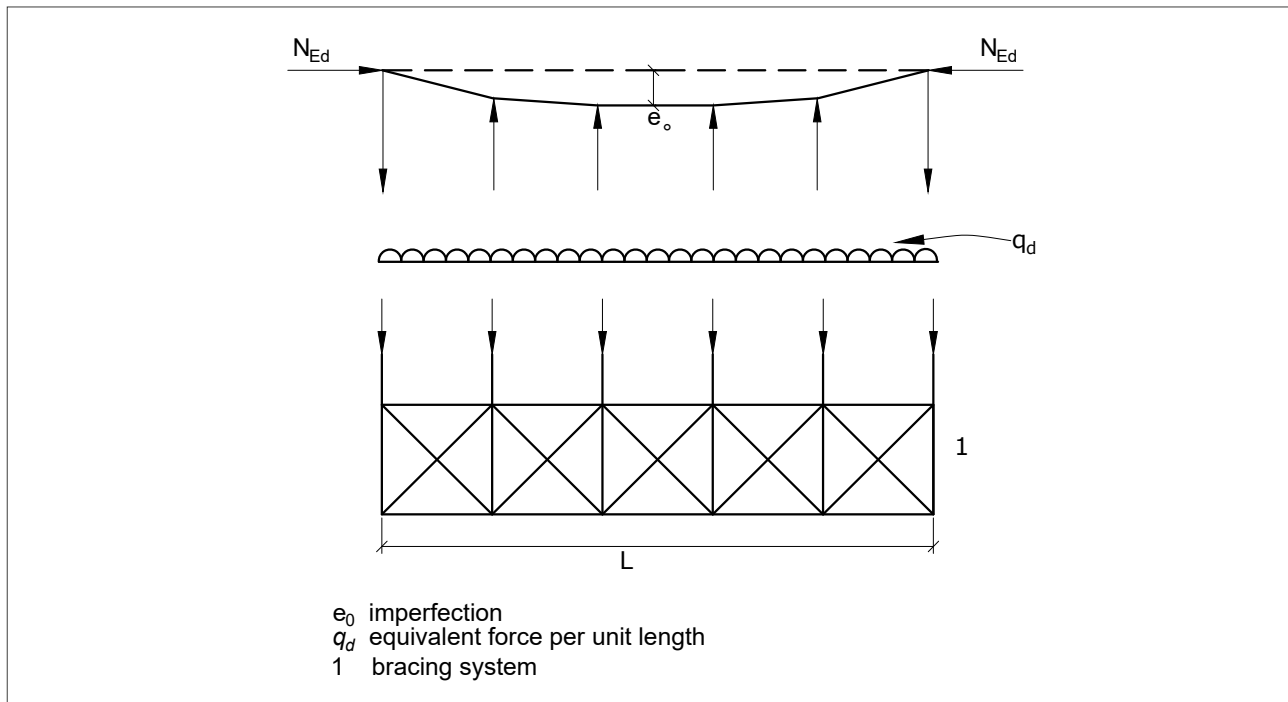
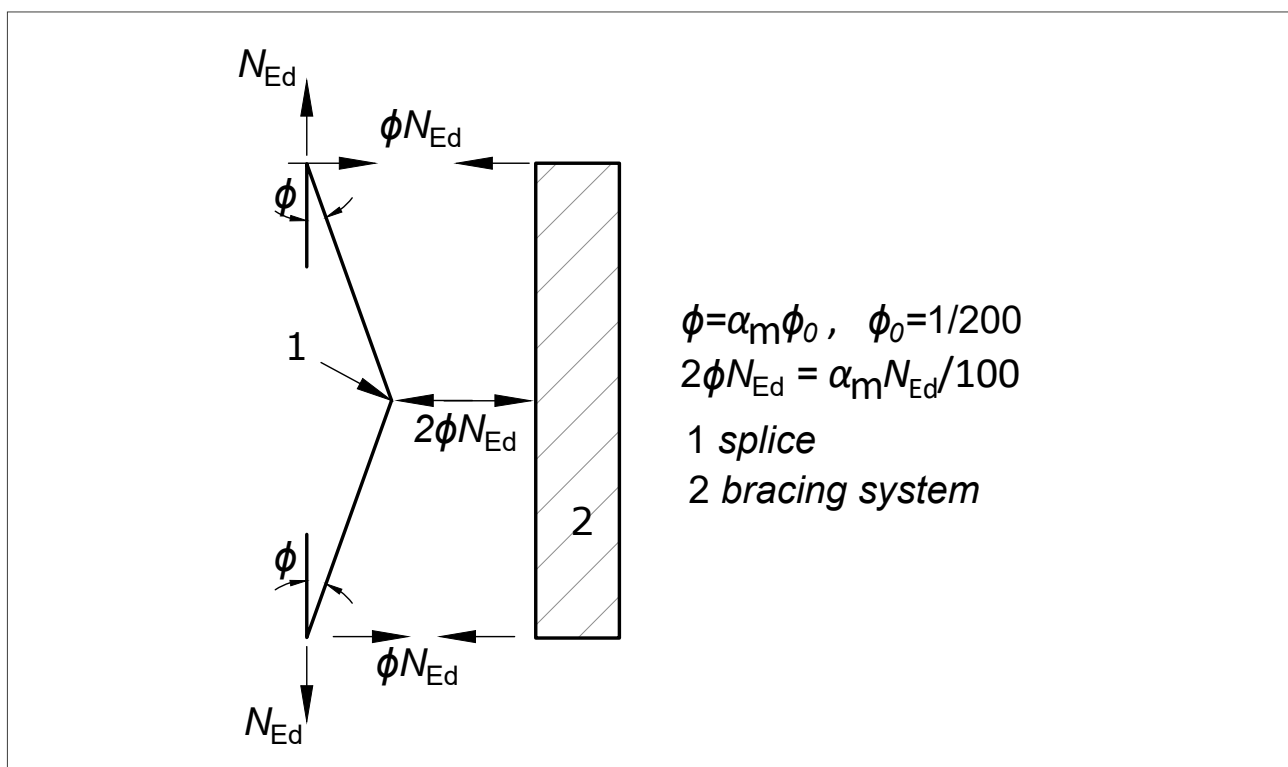


Figure 5.6 Bracing Forces at Splices in Compression Elements



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5.5.3.4 Member Imperfections

The effects of local bow imperfections of members are incorporated within the formulas given for buckling resistance for members together with the effective lengths of members, refer Clause 5.6.18. Where the stability of members is accounted for second order analysis, the compression member imperfection shall be considered as in Clause 5.5.3.2.

For a second order analysis taking account of lateral torsional buckling of a member in bending, the imperfections may be adopted as $k e_0/d$, where e_0 is the equivalent initial bow imperfection of the weak axis of the profile considered. In general, an additional torsional imperfection need not to be allowed for. The value of k may be taken as 1.0.

5.5.4 Methods of Analysis Considering Material Non-linearities

Elastic analysis should be used to determine the internal forces and moments for all persistent and transient design situations in road bridges.

Elastic global analysis should be based on the assumption that the stress-strain behaviour of the material is linear at all levels of stress. Internal forces and moments may be calculated according to elastic global analysis even if the resistance of a cross section is based on its plastic resistance. Elastic global analysis may also be used for cross sections the resistance of which are limited by local buckling. When global analysis is used for a continuous beam, the sagging moment at a Class 1 or 2 section should not exceed 90% of its plastic bending resistance when the adjacent hogging sections are in Class 3 or 4.

If all sections are class 1, the effects of differential temperature, shrinkage and settlement at the ultimate limit state may be ignored.

5.5.5 Classification of Cross-sections

5.5.5.2 Basis

The local buckling of plate elements of a cross section before the limit state is achieved can be avoided by limiting the width to depth ratio of each element of a cross section subjected to axial or bending compression. When plastic analysis is used, the member shall be capable of forming plastic hinge with sufficient rotation capacity without local buckling. When elastic analysis is used, the member shall be capable of developing yield stress under compression without local buckling.

The purpose of cross section classification is to identify the extent to which the resistance and rotation capacity of cross sections is limited by its local buckling resistance.

5.5.5.2 Classification

Four classes of cross-sections are defined, as follows:

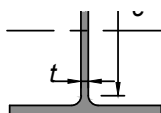
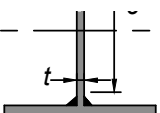
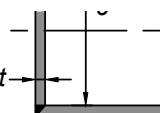
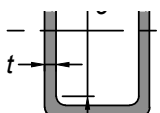
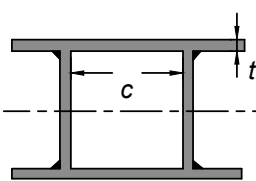
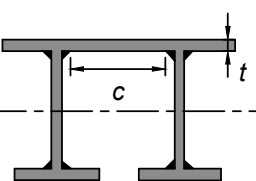
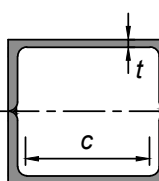
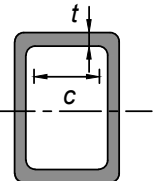
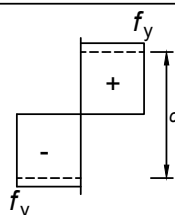
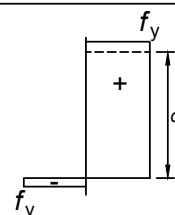
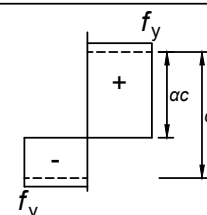
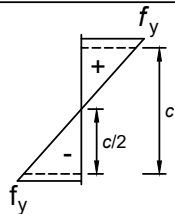
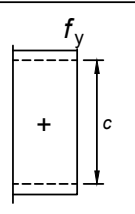
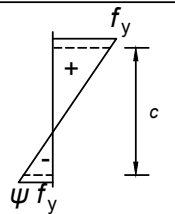
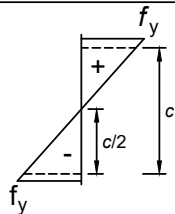
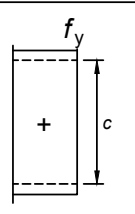
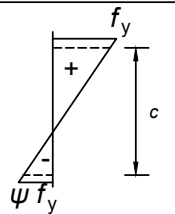
- Class 1 cross-sections are those which can form a plastic hinge with the rotation capacity required from plastic analysis without reduction of the resistance. The width to thickness ratio of the plate elements shall be less than that specified for Class 1 in Table 5.5.
- Class 2 cross-sections are those which can develop their plastic moment resistance, but have limited rotation capacity because of local buckling. The width to thickness ratio of the plate elements shall be less than that specified for Class 2 but greater than that specified for Class 1 in Table 5.5.
- Class 3 cross-sections are those in which the stress in the extreme compression fibre of the steel member assuming an elastic distribution of stresses can reach the yield strength, but can not develop the plastic moment resistance due to local buckling. The width to thickness ratio of the plate elements shall be less than that specified for Class 3 but greater than that specified for Class 2 in Table 5.5.
- Class 4 cross-sections are those in which local buckling will occur before the attainment of yield stress in one or more parts of the cross-section. The width to thickness ratio of the plate elements shall be greater than that specified for Class 3 in Table 5.5. For such cross sections, effective widths may be used to make the necessary allowances for reductions in resistance due to the effects of local buckling.

The classification of a cross-section depends on the width to thickness ratio of the parts subject to compression. Compression parts include every part of a cross-section which is either totally or partially in compression under the load combination considered. The various compression parts in a cross-section (such as a web or flange) can, in general, be in different classes.

When different parts of a cross section fall under different classes, the cross-section is classified according to the highest (least favourable) class of its compression parts. Alternatively, the classification of a cross-section may be defined by quoting both the flange classification and the web classification.

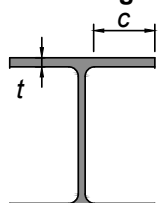
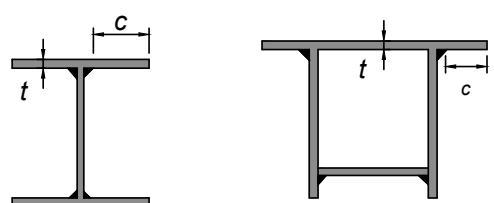
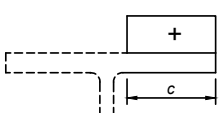
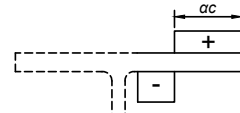
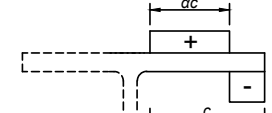
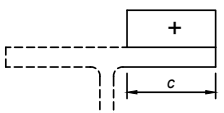
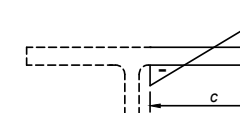
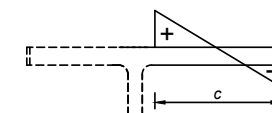
The limiting proportions for Class 1, 2, and 3 compression parts should be obtained from Table 5.5. A part which fails to satisfy the limits for Class 3 should be taken as Class 4. The Class 4 sections may be treated as Class 3 sections if the width to thickness ratios are less than the limiting proportions for Class 3 obtained from Table 5.5 when ε is increased by $\sqrt{((f_y/\gamma_{M0})/\sigma_{com,Ed})}$, where $\sigma_{com,Ed}$ is the maximum design compressive stress in the part taken from first order or where necessary second order analysis. However, when verifying the design buckling resistance of a member using Clause 5.6.18, the limiting proportions for Class 3 should always be obtained from Table 5.5.

Table 5.5 Maximum Width To Thickness Ratios (C/T) for Compression Parts

								Axis of bending
								Axis of bending
Class	Part subject to bending		Part subject to compression		Part subject to bending and compression			
1								
	$c / t \leq 72\varepsilon$		$c / t \leq 33\varepsilon$		when $\alpha > 0.5$: $c / t \leq \frac{396\varepsilon}{13\alpha - 1}$ when $\alpha \leq 0.5$: $c / t \leq \frac{36\varepsilon}{\alpha}$			
2								
	$c / t \leq 83\varepsilon$		$c / t \leq 38\varepsilon$		when $\alpha > 0.5$: $c / t \leq \frac{456\varepsilon}{13\alpha - 1}$ when $\alpha \leq 0.5$: $c / t \leq \frac{41.5\varepsilon}{\alpha}$			
3								
	$c / t \leq 124\varepsilon$		$c / t \leq 42\varepsilon$		when $\psi > -1$: $c / t \leq \frac{42\varepsilon}{0.67 + 0.33\psi}$ when $\psi \leq -1^*$: $c / t \leq 62\varepsilon(1-\psi)\sqrt{(1-\psi)}$			
$\varepsilon = \sqrt{235/f_y}$		f_y	235	275	355	420	460	
		ε	1.00	0.92	0.81	0.75	0.71	

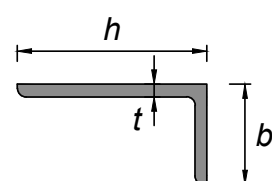
* $\psi \leq -1$ applies where either the compression stress $\sigma \leq f_y$ or the tensile strain $\varepsilon_y > f_y/E$

Table 5.5 Continued...

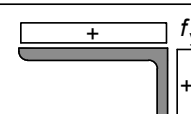
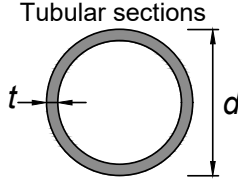
Outstand flanges						
						
Rolled sections			Welded sections			
Class	Part subject to compression	Part subject to bending and compression				
		Tip in compression		Tip in tension		
Stress distribution in parts (compression positive)						
1	$c / t \leq 9 \epsilon$	$c / t \leq \frac{9 \epsilon}{\alpha}$		$c / t \leq \frac{9 \epsilon}{\alpha \sqrt{\alpha}}$		
2	$c / t \leq 10 \epsilon$	$c / t \leq \frac{10 \epsilon}{\alpha}$		$c / t \leq \frac{10 \epsilon}{\alpha \sqrt{\alpha}}$		
Stress distribution in parts (compression positive)						
3	$c / t \leq 14 \epsilon$	$c / t \leq 21 \epsilon \sqrt{k_a}$				
$\epsilon = \sqrt{235 / f_y}$	f_y	235	275	355	420	460
	ϵ	1.00	0.92	0.81	0.75	0.71

Angles

Refer also to "Outstand flanges" (see sheet 2 of 3)



Does not apply to angles in continuous contact with other components.

Class		Section in compression				
Stress distribution across section (compression positive)						
3		$h / t \leq 15 \epsilon$ and $\frac{b+h}{2 t} \leq 11.5 \epsilon$				
		Tubular sections 				
Class		Section in bending and/or compression				
1		$d / t \leq 50 \epsilon^2$				
2		$d / t \leq 70 \epsilon^2$				
3		$d / t \leq 90 \epsilon^2$				
$\epsilon = \sqrt{235 / f_y}$	f_y	235	275	355	420	460
	ϵ	1.00	0.92	0.81	0.75	0.71
	ϵ^2	1.00	0.85	0.66	0.56	0.51

5.6 Ultimate Limit State

5.6.1 Partial Factors for Resistance

The partial factors γ_M shall be applied to the various characteristic values of resistance in this section. Recommended values of γ_M are given in Table 5.6.

Table 5.6 Partial Factors

Resistance		Value of Partial Factors	
a) Resistance of members and cross section:			
Resistance of cross sections to excessive yielding including local buckling		γ_{M0}	1.00
Resistance of members to instability assessed by member checks		γ_{M1}	1.10
Resistance of cross sections in tension to fracture		γ_{M2}	1.25
b) Resistance of joints			
Resistance of bolts		γ_{M2}	1.25
Resistance of rivets			
Resistance of pins			
Resistance of welds			
Resistance of plates in bearing			
Slip resistance	At ultimate limit state (Category C)	γ_{M3}	1.25
	At serviceability limit state	$\gamma_{M2,ser}$	1.10
Bearing resistance of an injection bolt		γ_{M4}	1.10
Resistance of joints in hollow section lattice girders		γ_{M5}	1.10
Resistance of pins at serviceability limit state		$\gamma_{M6,ser}$	1.00
Preload of high strength bolts		γ_{M7}	1.10

5.6.2 Resistance of Cross Sections

The design value of an action effect in each cross section shall not exceed the corresponding design resistance and if several action effects act simultaneously the combined effect shall not exceed the resistance for that combination. Shear lag effects and local buckling effects should be included by an effective width (refer Clause 5.6.4). Shear buckling effects should also be considered.

The design values of resistance should depend on the classification of the cross-section. Elastic verification according to the elastic resistance may be carried out for all cross-sectional classes provided the effective cross-sectional properties are used for the verification of class 4 cross sections.

For the elastic verification, where local longitudinal, transverse and shear stresses coexist at the critical point of the cross section, the following yield criterion may be used unless other interaction formulae apply.

$$\left(\frac{\sigma_{x,Ed}}{f_y/\gamma_{M0}}\right)^2 + \left(\frac{\sigma_{z,Ed}}{f_y/\gamma_{M0}}\right)^2 - \left(\frac{\sigma_{x,Ed}}{f_y/\gamma_{M0}}\right)\left(\frac{\sigma_{z,Ed}}{f_y/\gamma_{M0}}\right) + 3\left(\frac{\tau_{Ed}}{f_y/\gamma_{M0}}\right)^2 \leq 1$$

Equation 5.6

Where,

$\sigma_{x,Ed}$ = The design value of the longitudinal stress at the point of consideration.

$\sigma_{z,Ed}$ = The design value of the transverse stress at the point of consideration.

τ_{Ed} = The design value of the shear stress at the point of consideration.

The plastic resistance of cross sections should be verified by finding a stress distribution which is in equilibrium with the internal forces and moments without exceeding the yield strength. This stress distribution should be compatible with the associated plastic deformations.

As a conservative approximation for all cross-section classes a linear summation of the utilisation ratios for each stress resultant may be used. For class 1, class 2 or class 3 cross-sections subjected to the combination of N_{Ed} , $M_{y,Ed}$ and $M_{z,Ed}$ this method may be applied by using the following criteria:

$$\frac{N_{Ed}}{N_{Rd}} + \frac{M_{y,Ed}}{M_{y,Rd}} + \frac{M_{z,Ed}}{M_{z,Rd}} \leq 1 \quad \text{Equation 5.7}$$

Where N_{Rd} , $M_{y,Rd}$ and $M_{z,Rd}$ are the design values of the resistance depending on the cross sectional classification and including any reduction that may be caused by shear effects.

Where all the compression parts of a cross-section are Class 1 or Class 2, the cross-section may be taken as capable of developing its full plastic resistance in bending. Where all the compression parts of a cross-section are Class 3, its resistance should be based on an elastic distribution of strains across the cross-section. Compressive stresses should be limited to the yield strength at the extreme fibres. The extreme fibres may be assumed at the midplane of the flanges for ULS checks.

Where yielding first occurs on the tension side of the cross section, the plastic reserves of the tension zone may be utilised by accounting for partial plastification when determining the resistance of a Class 3 cross-section.

5.6.3 Section Properties

5.6.3.1 Gross Cross-section

The properties of the gross cross-section should be determined using the nominal dimensions. Holes for fasteners need not be deducted, but allowance should be made for larger openings. Splice materials should not be included.

5.6.3.1 Net Area

The net area of a cross-section should be taken as its gross area less appropriate deductions for all holes and other openings. For calculating net section properties, the deduction for a single fastener hole should be the gross cross-sectional area of the hole in the plane of its axis. For countersunk holes, appropriate allowance should be made for the countersunk portion.

Provided that the fastener holes are not staggered, the total area to be deducted for fastener holes shall be the maximum sum of the sectional areas of the holes in any cross-section perpendicular to the member axis (as failure plane ② in Figure 5.7). Where the fastener holes are staggered, the total area to be deducted for fasteners should be greater of:

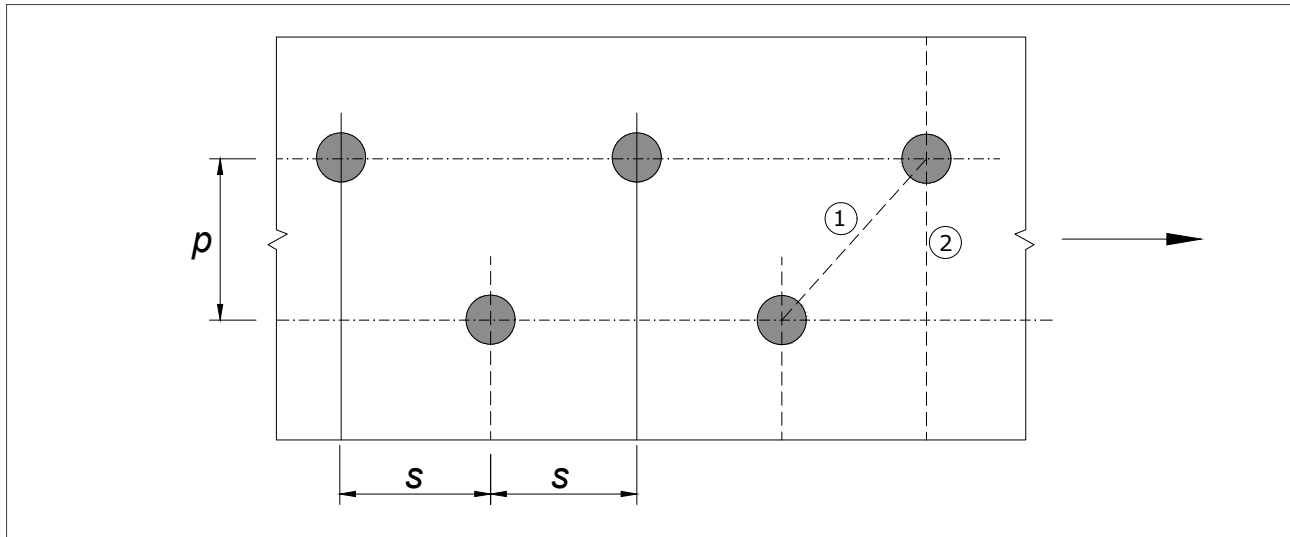
- The deduction for non-staggered holes shall be the maximum sum of the sectional areas of the holes in any cross-section perpendicular to the member axis (as failure plane ② in Figure 5.7).
- The deduction on staggered holes, as failure plane ① in Figure 5.7, shall be equal to

$$t \left(n d_0 - \sum s^2 \frac{s^2}{4p} \right) \quad \text{Equation 5.8}$$

Where,

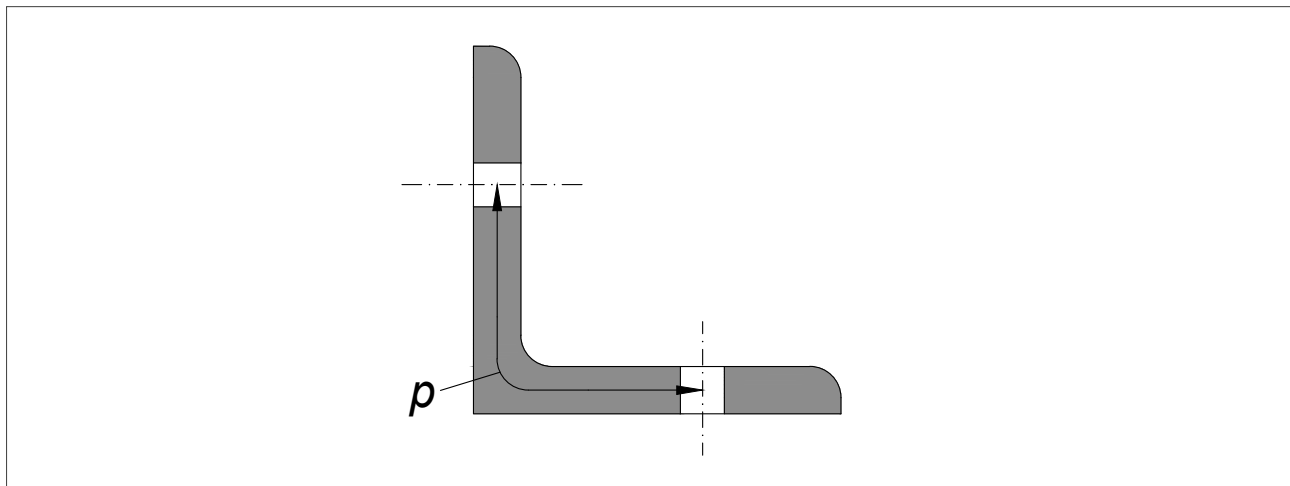
- s = The staggered pitch, the spacing of the centres of two consecutive holes in the chain measured parallel to the member axis.
- p = The spacing of the centres of the same two holes measured perpendicular to the member axis
- t = The thickness.
- n = The number of holes extending in any diagonal or zig-zag line progressively across the member or part of the member, refer Figure 5.7.
- d_0 = the diameter of hole.

Figure 5.7 Staggered Holes and Critical Fracture Lines 1 and 2



In an angle or other member with holes in more than one plane, the spacing p should be measured along the centre of thickness of the material (as in Figure 5.8).

Figure 5.8 Angles with Holes in Both Legs



5.6.4 Shear Lag Effects

5.6.4.1 General

In wide flanges, in-plane shear flexibility leads to a non-uniform distribution of bending stress across the flange width. This effect is known as shear lag and shall be accounted in the design. Shear lag in flanges may be neglected if $b_0 < L_e/50$ where b_0 is taken as the flange outstand or half the width of an internal element and L_e is the length between points of zero bending moment. Where this limit for b_0 is exceeded, the effects due to shear lag in flanges should be considered at serviceability and fatigue limit state verifications by the use of an effective^s width as in Clause 5.6.4.2.1 and a stress distribution as in Clause 5.6.4.2.2.

For the ultimate limit state verification, an effective area according to Clause 5.6.4.3 may be used. The stresses due to patch loading at the flange level shall be determined as provided in Clause 5.6.4.2.3.

5.6.4.2 Effective^s Width for Elastic Shear Lag

5.6.4.2.1 Effective^s Width

The effective^s width b_{eff} for shear lag under elastic conditions should be determined from:

$$b_{eff} = \beta b_0$$

Equation 5.9

Where the effective^s factor β is given in Table 5.7.

This effective^s width may be relevant for serviceability and fatigue limit states.

Table 5.7 Effective^s width factor β

κ	Verification	β - Value
$\kappa \leq 0.02$		$\beta = 1.0$
$0.02 < \kappa \leq 0.70$	Sagging bending	$\beta = \beta_1 = 1/(1 + 6.4\kappa^2)$
	Hogging bending	$\beta = \beta_2 = 1/(1 + 6.0(\kappa - 1/(2500\kappa)) + 1.6\kappa^2)$
$\kappa > 0.70$	Sagging bending	$\beta = \beta_1 = 1/(5.9\kappa)$
	Hogging bending	$\beta = \beta_2 = 1/(8.6\kappa)$
All κ	End support	$\beta_0 = (0.55 + 0.025/\kappa)\beta_1$, but $\beta_0 < \beta_1$
	Cantilever	$\beta = \beta_2$ at support and at the end

$\kappa = \alpha_0 b_0 / L_e$ with $\alpha_0 = \sqrt{(1 + A_{sl}/(b_0 t))}$
in which A_{sl} is the area of all longitudinal stiffeners within the width b_0 and other symbols are as defined in Figure 5.9 and Figure 5.10.

Provided adjacent spans do not differ more than 50% and any cantilever span is not larger than half the adjacent span, the effective lengths L_e may be determined from Figure 5.9. For all other cases L_e should be taken as the distance between adjacent points of zero bending moment.

Figure 5.9 Effective Length L_e and Effective^s Width for Continuous Beams

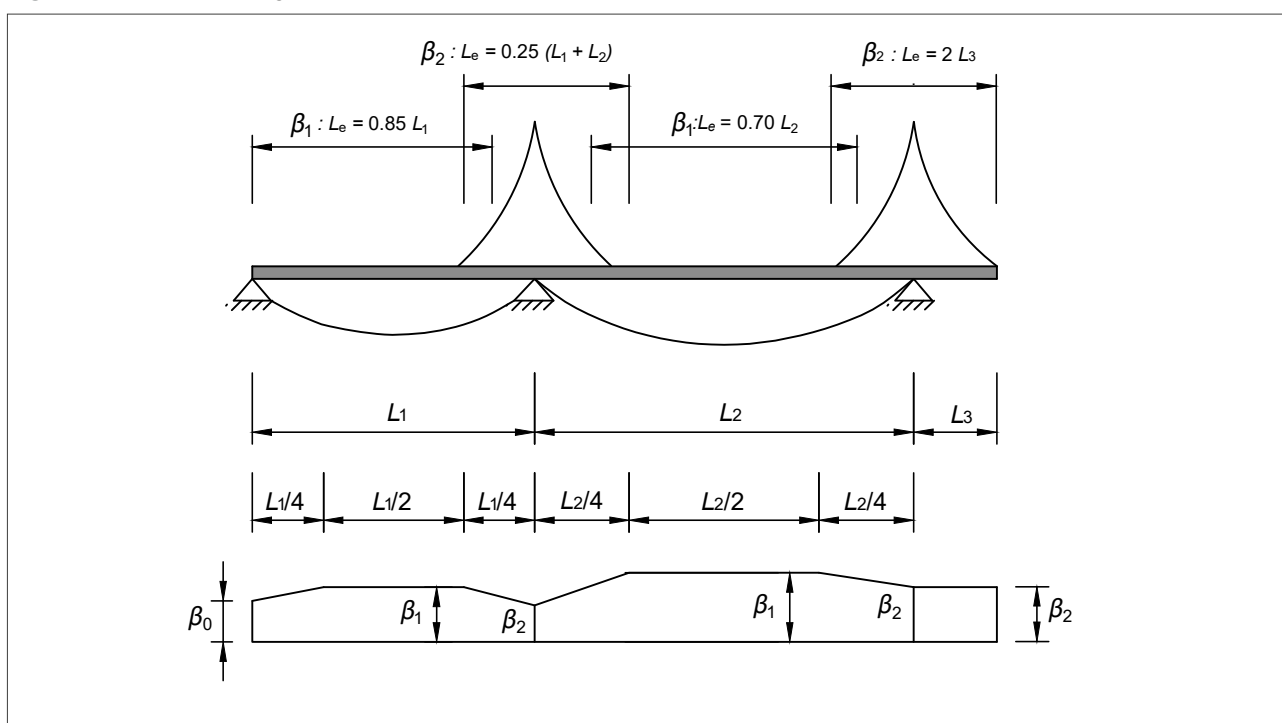
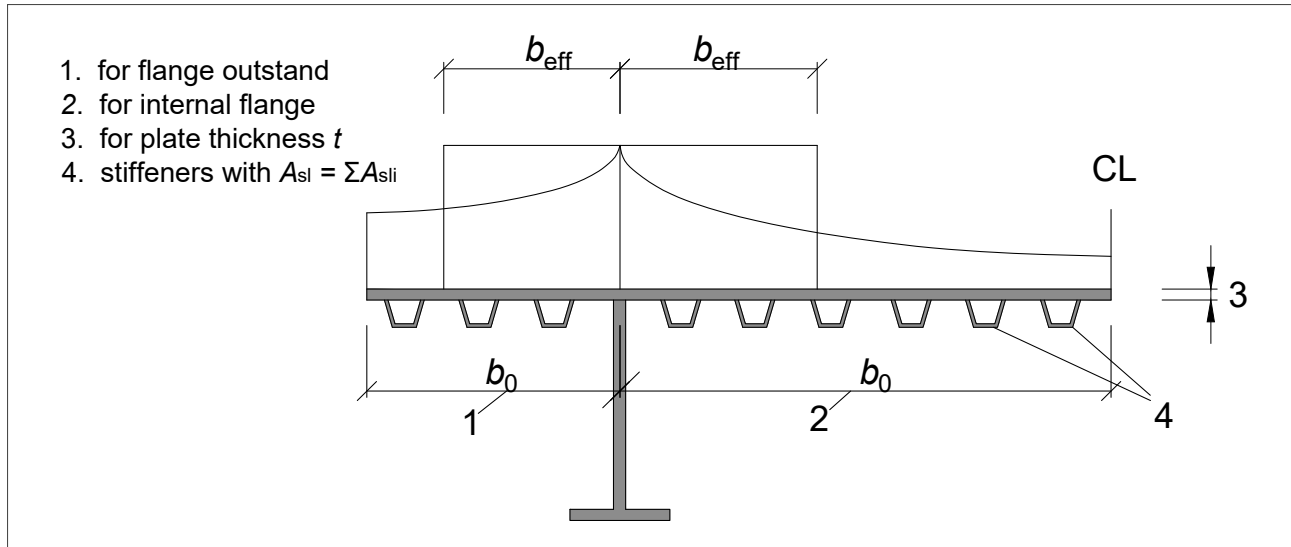


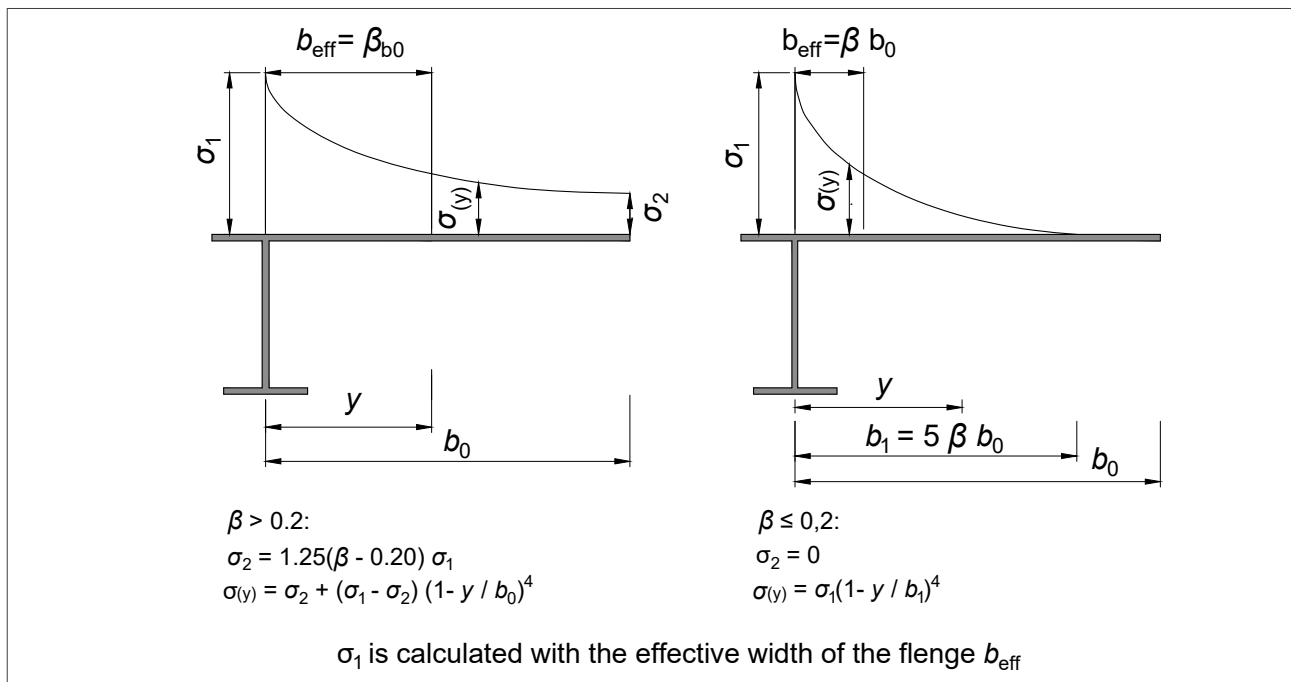
Figure 5.10 Notations for Shear Lag



5.6.4.2.2 Stress Distribution Due to Shear Lag

The distribution of longitudinal stresses across the flange plate due to shear lag should be obtained from Figure 5.11.

Figure 5.11 Distribution of Stresses Due to Shear Lag



5.6.4.2.3 In-plane Load Effects

The elastic stress distribution in a stiffened or unstiffened plate due to the local introduction of in-plane forces (patch loads), as shown in Figure 5.12, should be determined from:

$$\sigma_{z,Ed} = \frac{F_{Ed}}{b_{eff}(t_w + a_{st,1})}$$

Equation 5.10

Where,

$$b_{eff} = s_e \sqrt{1 + (z/(s_e n))^2}$$

$$n = 0.636 \sqrt{1 + (0.878 a_{st,1}/t_w)}$$

$$s_e = s_s + 2 t_f$$

Where,

$a_{st,1}$ = The gross cross-sectional area of the directly loaded stiffeners divided over the length s_e . This may be taken as the area of a stiffener smeared over the length of the spacing s_{st}

t_w = The web thickness

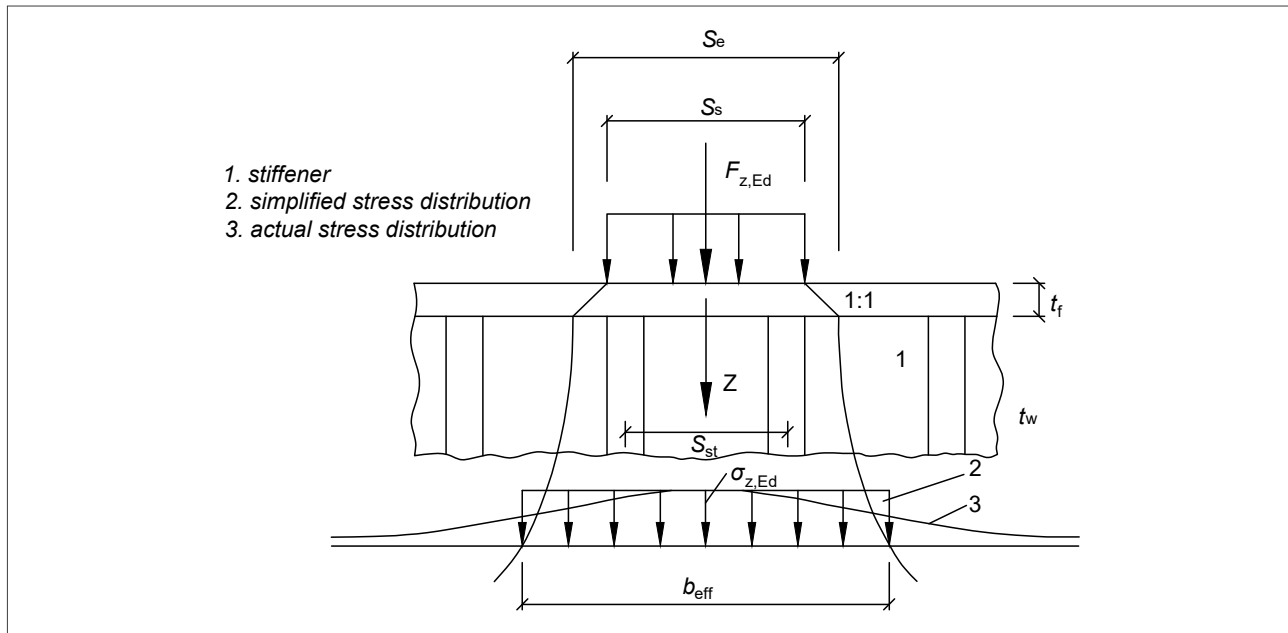
z = The distance to flange

s_e = The length of stiff bearing

s_{st} = The spacing of stiffeners

The above Expression is valid when $s_{st}/s_e \leq 0.5$; otherwise, the contribution of stiffeners should be neglected.

Figure 5.12 In-plane Load Introduction



The above stress distribution may also be used for the fatigue verification.

For an unstiffened flange, with a load applied through the flange, the spread width simplifies to:

$$b_{eff} = s_e \sqrt{1 + \left(\frac{z}{s_e \cdot 0.636} \right)^2}$$

and the design transverse stress at depth z below the loaded flange is:

$$\sigma_{z,Ed} = \frac{F_{Ed}}{b_{eff} t_w}$$

Equation 5.11

5.6.4.3 Shear Lag at Ultimate Limit State

Shear lag effects at the ultimate limit state, may be determined as follows:

- Elastic shear lag effects as determined for serviceability and fatigue limit states;
- Combined effects of shear lag and of plate buckling;
- Elastic-plastic shear lag effects allowing for limited plastic strains.

Note 1: the method of elastic-plastic shear lag effects allowing for limited plastic strains as provided in Note 3, is recommended.

Note 2: The combined effects of shear lag and plate buckling may be taken into account by using A_{eff} as given by:

$$A_{eff} = A_{c,eff} \beta_{ult}$$

Equation 5.12

Where,

$A_{c,eff}$ = The effective^p area of the compression flange due to plate buckling (refer Clause 5.6.8.4 and 5.6.8.5)

β_{ult} = The effective^s width factor for the effect of shear lag at the ultimate limit state, which may be taken as β determined from Table 5.7 with α_0 replaced by

$$\alpha_0^* = \sqrt{\frac{A_{c,eff}}{b_0 t_f}}$$

Equation 5.13

t_f = The flange thickness

Note 3: Elastic-plastic shear lag effects allowing for limited plastic strains may be taken into account using A_{eff} as follows:

$$A_{eff} = A_{c,eff} \beta^k \geq A_{c,eff} \beta$$

Equation 5.14

Where, β and κ are taken from Table 5.7.

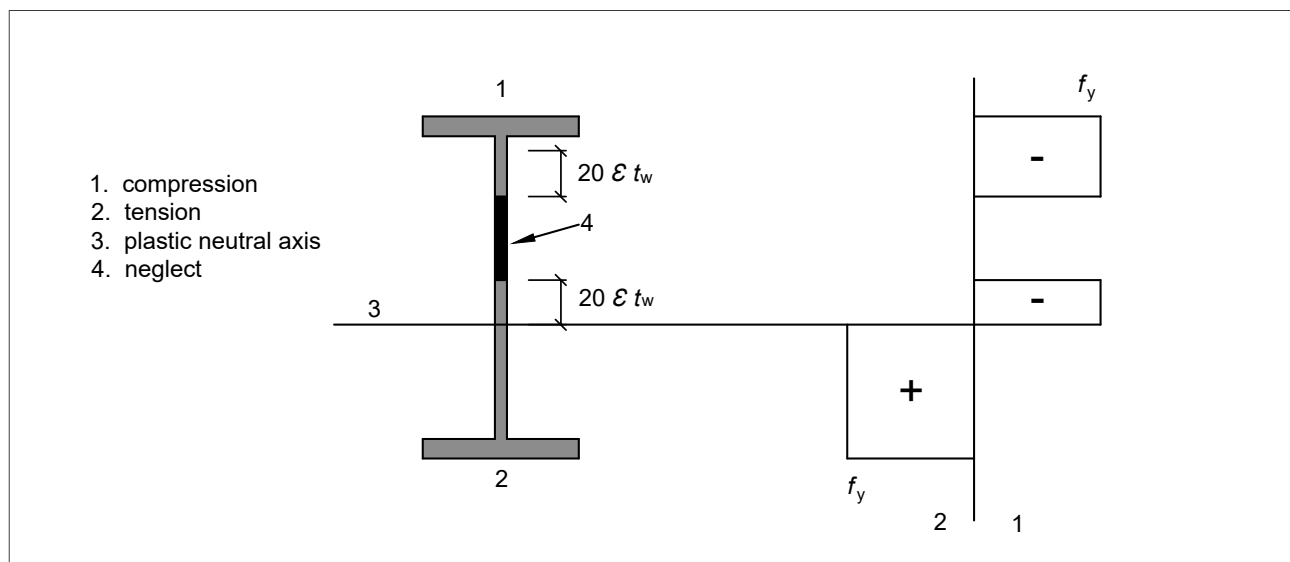
The Equations in Note 2 and Note 3 may also be applied for flanges in tension in which case A_{eff} should be replaced by the gross area of the tension flange.

The recommended method in Note 3 shall be used.

5.6.5 Effective Properties of Cross Section with Class 3 Webs and Class 1 or 2 Flanges

Where cross-sections with a class 3 web and class 1 or 2 flanges are classified as effective Class 2 cross-sections, the proportion of the web in compression should be replaced by a part of $20\epsilon t_w$ adjacent to the compression flange, with another part of $20\epsilon t_w$ adjacent to the plastic neutral axis of the effective cross-section in accordance with Figure 5.13.

Figure 5.13 Effective Class 2 Web



5.6.6 Effects of Local Buckling for Class 4 Cross Sections

The effects of local buckling should be considered using one of the following two methods:

- Effective cross section properties of class 4 sections, as given in Clause 5.6.8.3.
- Limiting the stress levels to achieve cross section properties, as given in Clause 5.6.20.2.

5.6.7 Effective Cross Section Properties of Class 4 Sections

The effective cross-section properties of Class 4 cross-sections should be based on the effective widths of the compression parts. Where a class 4 cross section is subjected to an axial compression force, the method given in Clause 5.6.8.3 be used to determine the possible shift e_N of the centroid of the effective area A_{eff} relative to the centre of gravity of the gross cross section and the resulting additional moment:

$$\Delta M_{Ed} = N_{Ed} e_N$$

Equation 5.15

5.6.8 Plate Buckling Effects Due to Direct Stresses at the Ultimate Limit State

5.6.8.1 General

This section gives guidelines to account for plate buckling effects from direct stresses at the ultimate limit state when the following criteria are met:

- The panels are rectangular and flanges are parallel or nearly parallel.
- Stiffeners, if any, are provided in the longitudinal or transverse direction or both.
- Open holes and cut outs are small.
- Members are of uniform cross section.
- No flange induced web buckling occurs.

5.6.8.2 Resistance to Direct Stresses

The resistance of plated members may be determined using the effective^p areas of plate elements in compression for class 4 sections using cross sectional data (A_{eff} , I_{eff} , W_{eff}) for cross sectional verifications and member verifications for column buckling and lateral torsional buckling.

Effective^p areas should be determined on the basis of the linear strain distributions with the attainment of yield strain in the mid plane of the compression plate.

5.6.8.3 Effective Cross Section

In calculating longitudinal stresses, account should be taken of the combined effect of shear lag and plate buckling using the effective areas given in Clause 5.6.4.3.

The effective cross-sectional properties of members should be based on the effective areas of the compression elements and on the effective^s area of the tension elements due to shear lag. The effective area A_{eff} should be determined assuming that the cross section is subject only to stresses due to uniform axial compression. For non-symmetrical cross sections the possible shift e_N of the centroid of the effective area A_{eff} relative to the centre of gravity of the gross cross-section, as in Figure 5.14, gives an additional moment which should be taken into account in the cross section verification, refer Clause 5.6.8.5.5.

The effective section modulus W_{eff} should be determined assuming the cross section is subject only to bending stresses, refer Figure 5.15. For biaxial bending, effective section moduli should be determined about both main axes. The stress in a flange should be calculated using the elastic section modulus with reference to the midplane of the flange.

Figure 5.14 Class 4 Cross Sections - Axial Force

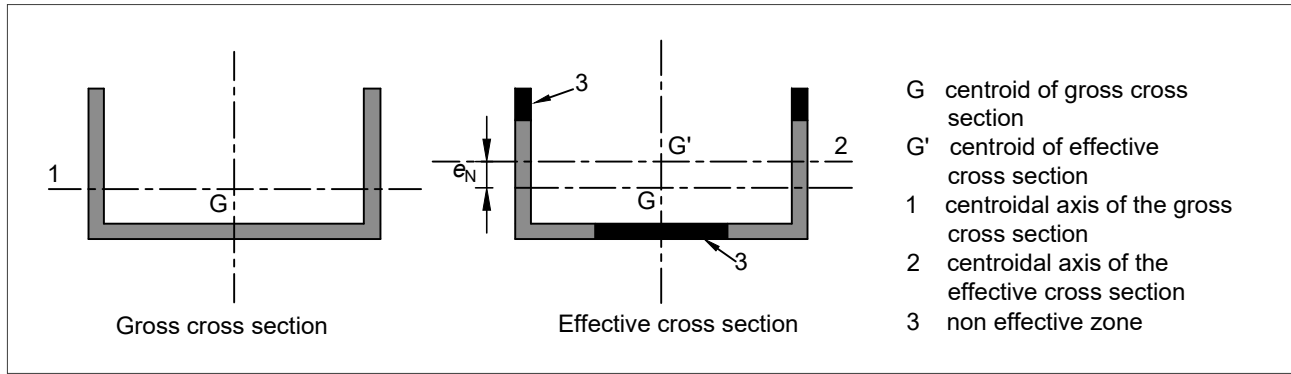
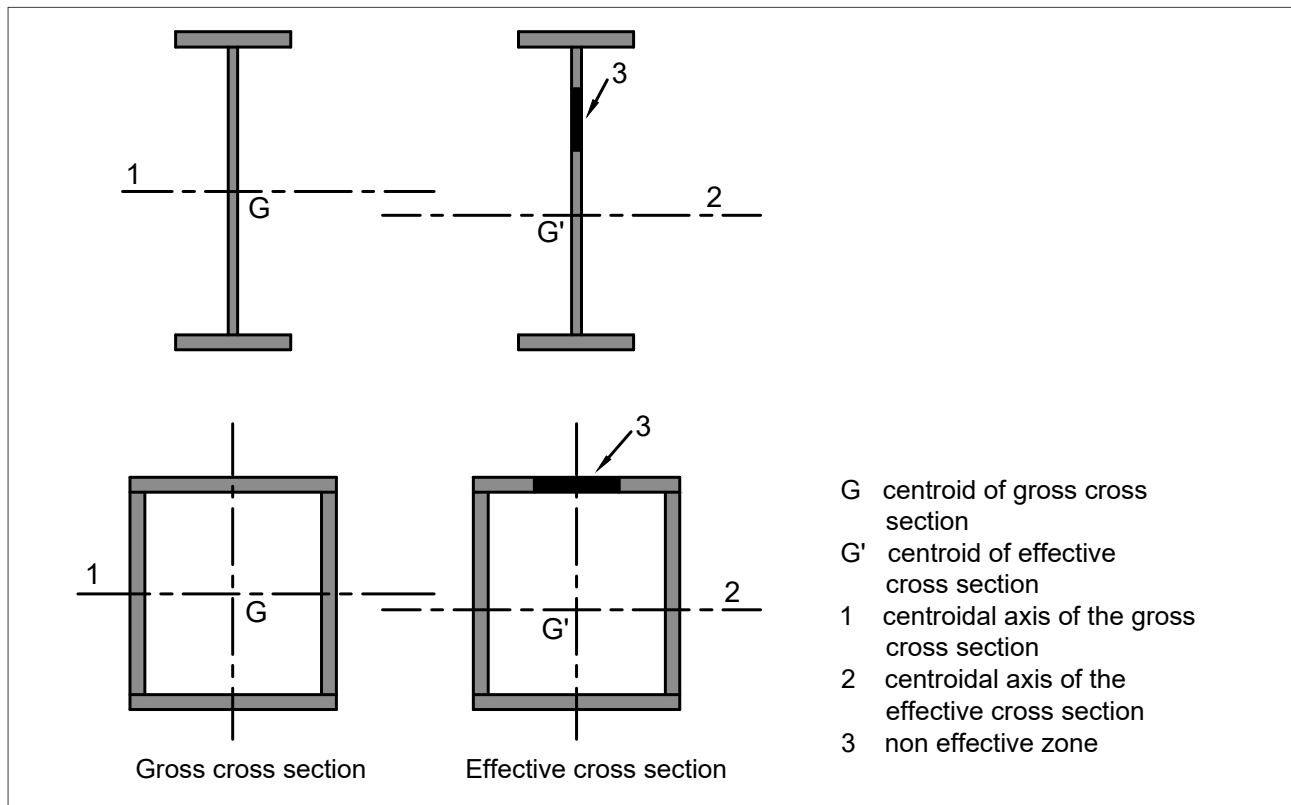


Figure 5.15 Class 4 Cross Sections - Bending Moment



5.6.8.4 Plate Elements Without Longitudinal Stiffeners

The effective^p areas of flat compression elements should be obtained using Table 5.8 for internal elements and Table 5.9 for outstand elements. The effective^p area of the compression zone of a plate with the gross cross-sectional area A_c should be obtained from:

$$A_{c,eff} = \rho A_c$$

Equation 5.16

Where ρ is the reduction factor for plate buckling.

The reduction factor ρ may be taken as follows:

- Internal compression elements:

$$\rho = 1.0 \quad \text{for } \bar{\lambda}_p \leq 0.5 + \sqrt{(0.085 - 0.055 \psi)}$$

$$\rho = \frac{\bar{\lambda}_p - 0.055(3 + \psi)}{\bar{\lambda}_p^2} \leq 1.0 \quad \text{for } \bar{\lambda}_p > 0.5 + \sqrt{(0.085 - 0.055 \psi)}$$

Equation 5.17

- Outstand compression elements:

$$\rho = 1.0 \quad \text{for } \bar{\lambda}_p \leq 0.748$$

$$\rho = \frac{\bar{\lambda}_p - 0.188}{\bar{\lambda}_p^2} \leq 1.0 \quad \text{for } \bar{\lambda}_p > 0.748$$

Equation 5.18

Where,

$$\bar{\lambda}_p = \sqrt{\frac{f_y}{\sigma_{cr}}} = \frac{\bar{b}/t}{28.4 \varepsilon \sqrt{k_\sigma}}$$

ψ = The stress ratio determined in accordance with this section (next paragraph)

$\bar{b}$ = The appropriate width to be taken as follows (for definitions, see Table 5.5)

b_w = For webs

b = For internal flange elements (except rectangular hollow section)

$b - 3t$ = For flanges of rectangular hollow section

c = For outstand flanges

h = For equal-leg angles

h = For unequal-leg angles

k_σ = The buckling factor corresponding to the stress ratio ψ and boundary conditions. For long plates k_σ is given in Table 5.8 or Table 5.9 as appropriate

t = The thickness

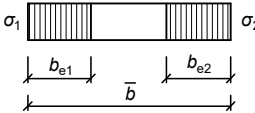
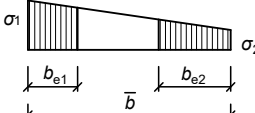
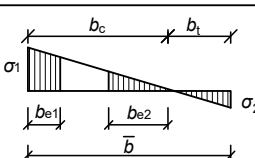
σ_{cr} = The elastic critical plate buckling stress

ε = $\sqrt{(235/f_y)}$ using f_y is in N/mm^2

For flange elements of *I*-sections and box girders the stress ratio ψ used in Table 5.8 and Table 5.9 should be based on the properties of the gross cross-sectional area, due allowance being made for shear lag in the flanges if relevant. For web elements the stress ratio used in Table 5.8 should be obtained using a stress distribution based on the effective area of the compression flange and the gross area of the web.

If the stress distribution results from different stages of construction (as e.g. in a composite bridge) the stresses from the various stages may first be calculated with a cross section consisting of effective flanges and gross web and these stresses are added together. This resulting stress distribution determines an effective web section that can be used for all stages to calculate the final stress distribution for stress analysis.

Table 5.8 Internal Compression Elements

Stress distribution (compression positive)				Effective ^p width b_{eff}		
				$\psi = 1:$ $b_{eff} = \rho \bar{b}$ $b_{e1} = 0.5 b_{eff} \quad b_{e2} = 0.5 b_{eff}$		
				$1 > \psi \geq 0:$ $b_{eff} = \rho \bar{b}$ $b_{e1} = \frac{2}{5 - \psi} b_{eff} \quad b_{e2} = b_{eff} - b_{e1}$		
				$\psi < 0:$ $b_{eff} = \rho b_c = \rho \bar{b} / (1 - \psi)$ $b_{e1} = 0.4 b_{eff} \quad b_{e2} = 0.6 b_{eff}$		
$\psi = \sigma_2 / \sigma_1$	1	$1 > \psi > 0$	0	$0 > \psi > -1$	-1	$1 > \psi \geq -3$
Buckling factor k_σ	4.0	$8.2 / (1.05 + \psi)$	7.81	$7.81 - 6.29\psi + 9.78\psi^2$	23.9	$5.98(1 - \psi)^2$

Except as given below, the plate slenderness $\bar{\lambda}_p$ of an element may be replaced by:

$$\bar{\lambda}_{p,red} = \bar{\lambda}_p \sqrt{\frac{\sigma_{com,Ed}}{f_y \gamma_{M0}}}$$

Equation 5.19

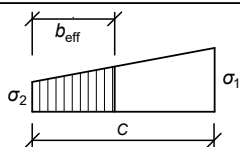
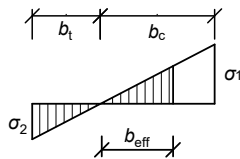
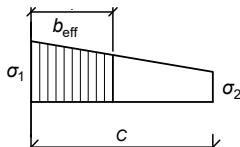
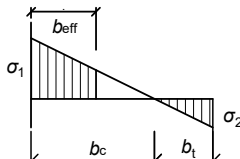
Where $\sigma_{com,Ed}$ is the maximum design compressive stress in the element determined using the effective area of the section caused by all simultaneous actions.

Note 1: The above procedure is conservative and requires an iterative calculation in which the stress ratio ψ (see Table 5.8 and Table 5.9) is determined at each step from the stresses calculated on the effective^P cross section defined at the end of the previous step.

For the verification of the design buckling resistance of a class 4 member using Clause 5.6.18.1 or 5.6.18.2 or 5.6.18.4, either the plate slenderness $\bar{\lambda}_p$ or $\bar{\lambda}_{p,red}$ with $\sigma_{com,Ed}$ based on second order analysis with global imperfections should be used.

For aspect ratios $a/b < 1$, a column type of buckling may occur and the check should be performed according to Clause 5.6.8.5.4 using the reduction factor ρ_c .

Table 5.9 Outstand Compression Elements

Stress distribution (compression positive)		Effective ^p width b_{eff}			
		$1 > \psi \geq 0:$ $b_{\text{eff}} = \rho c$			
		$\psi < 0:$ $b_{\text{eff}} = \rho b_c = \rho c / (1 - \psi)$			
$\psi = \sigma_2/\sigma_1$	1	0	-1	$1 \geq \psi \geq -3$	
Buckling factor k_σ	0.43	0.57	0.85	$0.57 - 0.21\psi + 0.07\psi^2$	
		$1 > \psi \geq 0:$ $b_{\text{eff}} = \rho c$			
		$\psi < 0:$ $b_{\text{eff}} = \rho b_c = \rho c / (1 - \psi)$			
$\psi = \sigma_2/\sigma_1$	1	$1 > \psi > 0$	0	$0 > \psi > -1$	-1
Buckling factor k_σ	0.43	$0.578 / (\psi + 0.34)$	1.70	$1.7 - 5\psi + 17.1\psi^2$	23.8

5.6.8.5 Stiffened Plate Elements With Longitudinal Stiffeners

5.6.8.5.1 General

For plates with longitudinal stiffeners the effective^P areas from local buckling of the various subpanels between the stiffeners and the effective^P areas from the global buckling of the stiffened panel should be accounted for.

The effective^P section area of each subpanel should be determined by a reduction factor in accordance section with Clause 5.6.8.4 to account for local plate buckling. The stiffened plate with effective^P section areas for the stiffeners should be checked for global plate buckling (by modelling it as an equivalent orthotropic plate) and a reduction factor ρ_c should be determined for overall plate buckling.

The effective^P area of the compression zone of the stiffened plate should be taken as:

$$A_{c,eff} = \rho_c A_{c,eff,loc} + \sum b_{edge,eff} t$$

Equation 5.20

Where $A_{c,eff,loc}$ is the effective^P section area of all the stiffeners and subpanels that are fully or partially in the compression zone except the effective parts supported by an adjacent plate element with the width $b_{edge,eff}$ (refer Figure 5.16)

Area $A_{c,eff,loc}$ should be obtained from:

$$A_{c,eff,loc} = A_{sl,eff} + \sum_c \rho_{loc} b_{c,loc} t$$

Equation 5.21

Where,

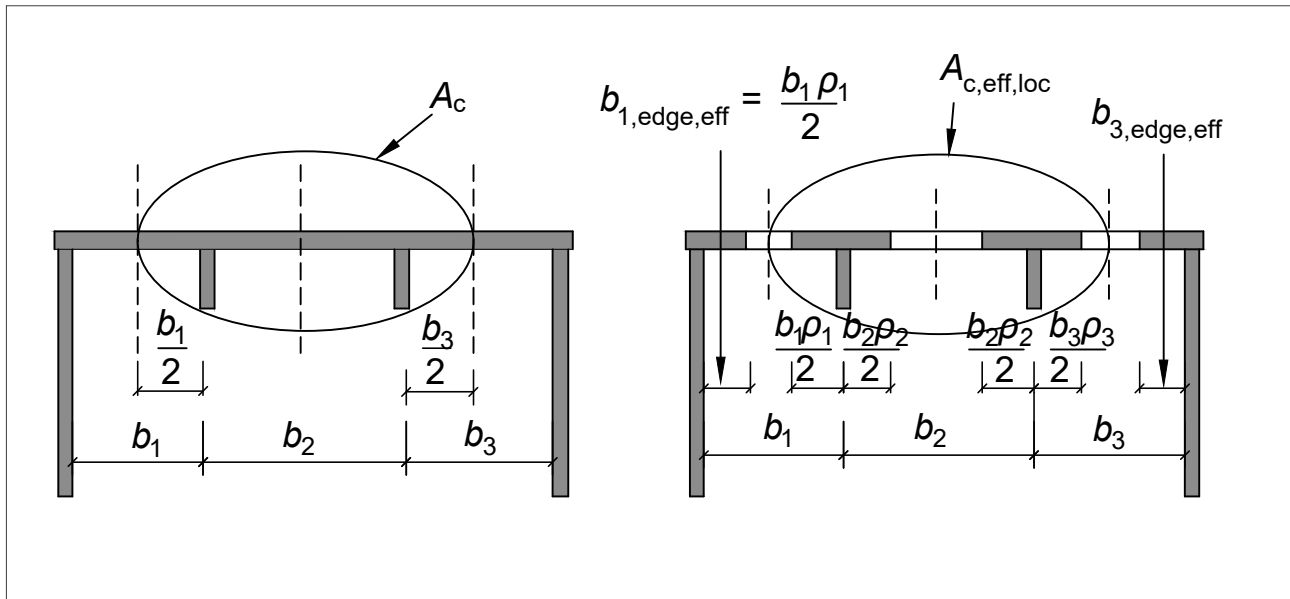
$\sum_c$ = Applies to the part of the stiffened panel width that is in compression except the parts $b_{edge,eff}$.

$A_{sl,eff}$ = The sum of the effective^P sections according Clause 5.6.8.4 of all longitudinal stiffeners with gross area A_{sl} located in the compression zone.

$b_{c,loc}$ = The width of the compressed part of each subpanel.

ρ_{loc} = The reduction factor from Clause 5.6.8.4 for each subpanel.

Figure 5.16 Stiffened Plate Under Uniform Compression



In determining the reduction factor ρ_c for overall buckling, the reduction factor for column-type buckling, which is more severe than the reduction factor for plate buckling, should be considered. Interpolation should be carried out in accordance with Clause 5.6.8.5.4 – between the reduction factor ρ for plate buckling and the reduction factor χ_c for column buckling to determine ρ_c , refer Clause 5.6.8.5.4. The reduction of the compressed area $A_{c,eff,loc}$ through ρ_c may be taken as a uniform reduction across the whole cross section.

If shear lag is relevant, Clause 5.6.4.3, the effective cross-sectional area $A_{c,eff}$ of the compression zone of the stiffened plate should then be taken as $A_{c,eff}^*$ accounting not only for local plate buckling effects but also for shear lag effects. The effective cross-sectional area of the tension zone of the stiffened plate should be taken as the gross area of the tension zone reduced for shear lag if relevant, refer Clause 5.6.4.3.

The effective section modulus W_{eff} should be taken as the second moment of area of the effective cross section divided by the distance from its centroid to the mid depth of the flange plate.

5.6.8.5.2 Plate Type Behaviour

The relative plate slenderness $\bar{\lambda}_p$ of the equivalent plate is defined as:

$$\bar{\lambda}_p = \sqrt{\frac{\beta_{A,c} f_y}{\sigma_{cr,p}}}$$

Equation 5.22

$$\text{with } \beta_{A,c} = \frac{A_{c,eff,loc}}{A_c}$$

Where,

A_c = The gross area of the compression zone of the stiffened plate except the parts of subpanels supported by an adjacent plate, see Figure 5.16 (to be multiplied by the shear lag factor if shear lag is relevant, Clause 5.6.4.3).

$A_{c,eff,loc}$ = The effective area of the same part of the plate (including shear lag effect, if relevant) with due allowance made for possible plate buckling of subpanels and/or stiffeners.

The reduction factor ρ for the equivalent orthotropic plate is obtained from 5.6.8.4 provided $\bar{\lambda}_p$ is calculated from above Equation.

5.6.8.5.3 Column Type Buckling Behaviour

The elastic critical column buckling stress $\sigma_{cr,c}$ of an unstiffened or stiffened plate should be taken as the buckling stress with the supports along the longitudinal edges removed.

For an unstiffened plate, the elastic critical column buckling stress $\sigma_{cr,c}$ may be obtained from:

$$\sigma_{cr,c} = \frac{\pi^2 E t^2}{12(1 - \nu^2) a^2}$$

Equation 5.23

For a stiffened plate, $\sigma_{cr,c}$ may be determined from the elastic critical column buckling stress $\sigma_{cr,sl}$ of the stiffener closest to the panel edge with the highest compressive stress as follows:

$$\sigma_{cr,sl} = \frac{\pi^2 EI_{sl,1}}{A_{sl,1} a^2}$$

Equation 5.24

Where,

$I_{sl,1}$ = The second moment of area of the gross cross section of the stiffener and the adjacent parts of the plate, relative to the out-of-plane bending of the plate.

$A_{sl,1}$ = The gross sectional area of the stiffener and the adjacent parts of the plate

The relative column slenderness $\bar{\lambda}_c$ is defined as follows:

$$\bar{\lambda}_c = \sqrt{\frac{f_y}{\sigma_{cr,c}}} \text{ for unstiffened plates}$$

$$\bar{\lambda}_c = \sqrt{\frac{\beta_{A,c} f_y}{\sigma_{cr,c}}} \text{ for stiffened plates}$$

$$\text{with } \beta_{A,c} = \frac{A_{sl,1,eff}}{A_{sl,1}}$$

$A_{sl,1,eff}$ = The effective cross sectional area of the stiffener and the adjacent parts of the plate with due allowance for plate buckling.

For unstiffened plates $\alpha = 0.21$ corresponding to buckling curve “a” should be used. For stiffened plates its value should be increased to:

$$\alpha_e = \alpha + \frac{0.09}{i/e}$$

Equation 5.25

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$$\text{with } i = \sqrt{\frac{I_{sl,1}}{A_{sl,1}}}$$

e = max (e_1, e_2) is the largest distance from the respective centroids of the plating and the one sided stiffener (or of the centroids of either set of stiffeners when present on both sides) to the neutral axis of the effective column.

α = 0.34 (curve b) for closed section stiffeners.

= 0.49 (curve c) for open section stiffeners.

5.6.8.5.4 Interaction Between Plate and Column Buckling

The final reduction factor ρ_c should be obtained by interpolation between χ_c and ρ as follows:

$$\rho_c = (\rho - \chi_c) \zeta (2 - \zeta) + \chi_c$$

Where,

ζ = $\sigma_{cr,p}/\sigma_{cr,c} - 1$ but $0 \leq \zeta \leq 1$

$\sigma_{cr,p}$ = The elastic critical plate buckling stress.

$\sigma_{cr,c}$ = The elastic critical column buckling stress.

χ_c = The reduction factor due to column buckling.

P = The reduction factor due to plate buckling.

5.6.8.5.5 Verification

Member verification for compression and uniaxial bending should be performed as follows:

$$\eta_1 = \frac{N_{Ed}}{f_y A_{eff}} + \frac{M_{Ed} + N_{Ed} e_N}{f_y W_{eff}} \leq 1.0$$

Equation 5.26

Where,

A_{eff} = The effective cross-section area.

e_N = The shift in the position of neutral axis.

M_{Ed} = The design bending moment.

N_{Ed} = The design axial force.

W_{eff} = The effective elastic section modulus.

γ_{M0} = The partial factor.

For members subject to compression and biaxial bending the above Equations may be modified as follows:

$$\eta_1 = \frac{N_{Ed}}{f_y A_{eff}} + \frac{M_{y,Ed} + N_{Ed} e_{yN}}{f_y W_{y,eff}} + \frac{M_{z,Ed} + N_{Ed} e_{zN}}{f_y W_{z,eff}} \leq 1.0$$

Equation 5.27

$M_{y,Ed}, M_{z,Ed}$ = The design bending moments with respect to y-y and z-z axes respectively.

e_{yN}, e_{zN} = The eccentricities with respect to neutral axis

Action effects M_{Ed} and N_{Ed} should include global second order effects where relevant.

The plate buckling verification of the panel should be carried out for the stress resultants at a distance 0.4a or 0.5b, whichever is the smallest, from the panel end where the stresses are the greater. In this case the gross sectional resistance needs to be checked at the end of the panel.

5.6.9 Tension

The design value of the tension force N_{Ed} at each cross section shall be less than the design tension resistance $N_{t,Rd}$. It shall satisfy:

$$\frac{N_{Ed}}{N_{t,Rd}} \leq 1.0$$

Equation 5.28

The design resistance of a member under axial tension is the lower of the design resistance due to yielding of gross section or by rupture of critical section. For sections with holes the design tension resistance $N_{t,Rd}$ should be taken as the smaller of:

- a. The design plastic resistance under axial tension, as governed by yielding of gross section, is given by:

$$N_{pl,Rd} = \frac{A f_y}{\gamma_{M0}}$$

Equation 5.29

- b. The design ultimate resistance of the net cross-section at holes for fasteners, as governed by rupture of net cross section, is given by

$$N_{u,Rd} = \frac{0.9 A_{net} f_u}{\gamma_{M2}}$$

Equation 5.30

Where capacity design is requested, the design plastic resistance $N_{pl,Rd}$ should be less than the design ultimate resistance of the net section at fasteners holes $N_{u,Rd}$.

In category C connections (the design tension resistance $N_{t,Rd}$ of the net section at holes for fasteners should be taken as $N_{net,Rd}$, where:

$$N_{net,Rd} = \frac{A_{net} f_y}{\gamma_{M0}}$$

Equation 5.31

For angles connected through one leg, Clause 5.8.1.7.3 shall be considered. Similar consideration should also be given to other type of sections connected through outstands.

5.6.10 Compression

The design value of the compression force N_{Ed} at each cross-section shall be less than design compression resistance $N_{c,Rd}$. It shall satisfy:

$$\frac{N_{Ed}}{N_{c,Rd}} \leq 1.0$$

Equation 5.32

The design resistance of cross sections for uniform compression $N_{c,Rd}$ should be determined as follows:

- a. Without local buckling: the cross-sections in Class 1, 2 or 3 will be able to achieve the yield stress in compression without local buckling occurring and the cross-section compression resistance is as under:

$$N_{c,Rd} = \frac{A f_y}{\gamma_{M0}} \text{ for class 1,2 or 3 cross sections}$$

Equation 5.33

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- b. With local buckling: the cross-sections in Class 4 are susceptible to local buckling at a stress lower than the yield stress and the cross-section compression resistance can be given by either of the two methods. The first one is to use the effective section method and other is to use the limiting stress method as under.

$$N_{c,Rd} = \frac{A_{eff} f_y}{\gamma_{M0}} \quad \text{for class 4 cross sections or} \quad \text{Equation 5.34}$$

$$N_{c,Rd} = \frac{A \sigma_{limit}}{\gamma_{M0}} \quad \text{for stress limits} \quad \text{Equation 5.35}$$

Where $\sigma_{limit} = \rho_{x,fy}$ is the limiting stress of the weakest part of the cross section in compression.

Fastener holes except for oversize and slotted holes as defined in EN 1090 need not be allowed for in compression members, provided that they are filled by fasteners.

In the case of unsymmetrical Class 4 sections, the method given in section Clause 5.6.16.3 should be used to allow for the additional moment ΔM_{Ed} due to the eccentricity e_N of the centroidal axis of the effective section.

5.6.11 Bending Moment

The design value of the bending moment M_{Ed} at each cross-section shall be less than the design bending resistance $M_{c,Rd}$. It shall satisfy:

$$\frac{M_{Ed}}{M_{c,Rd}} \leq 1.0 \quad \text{Equation 5.36}$$

Where $M_{c,Rd}$ is determined considering fastener holes.

The design resistance for bending about the major axis should be determined as follows:

- a. Without local buckling

$$M_{c,Rd} = \frac{W_{pl} f_y}{\gamma_{M0}} \quad \text{for class 1 and 2 cross sections} \quad \text{Equation 5.37}$$

$$M_{c,Rd} = \frac{W_{el,min} f_y}{\gamma_{M0}} \quad \text{for class 3 cross sections} \quad \text{Equation 5.38}$$

- b. With local buckling:

$$M_{c,Rd} = \frac{W_{eff,min} f_y}{\gamma_{M0}} \quad \text{for class 4 cross sections or} \quad \text{Equation 5.39}$$

$$M_{c,Rd} = \frac{W_{el,min} \sigma_{limit}}{\gamma_{M0}} \quad \text{for stress limits} \quad \text{Equation 5.40}$$

Where,

$W_{el,min}$ and $W_{eff,min}$ are the elastic moduli which correspond to the fibre with the maximum elastic stress.
 σ_{limit} = The limiting stress of the weakest part of the cross section in compression.

For bending about both axes, the methods given in Clause 5.6.16 should be used.

1

Fastener holes in the tension flange may be ignored provided that for the tension flange:

$$\frac{A_{f,net}}{\gamma_{M2}} \frac{0.9 f_u}{\gamma_{M0}} \geq \frac{A_f f_y}{\gamma_{M0}}$$

Equation 5.41

2

Where, A_f is the area of the tension flange.

3

Fastener holes in tension zone of the web need not be allowed for, provided that the limit given above is satisfied for the complete tension zone comprising the tension flange plus the tension zone of the web. Fastener holes except for oversize and slotted holes in compression zone of the cross-section need not be allowed for, provided that they are filled by fasteners.

4

5.6.12 Shear

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The design value of the shear force V_{Ed} at each cross section shall be less than the design shear resistance. It shall satisfy:

$$\frac{V_{Ed}}{V_{c,Rd}} \leq 1.0$$

Equation 5.42

Where $V_{c,Rd}$ is the design shear resistance. For elastic design $V_{c,Rd}$ is the design elastic shear resistance. For plastic design, $V_{c,Rd}$ is the design plastic shear resistance $V_{pl,Rd}$. The plastic and elastic shear resistance shall be calculated as given below.

5.6.12.1 Shear Resistance Without Shear Buckling

In the absence of torsion, the design plastic shear resistance is given by:

$$V_{pl,Rd} = \frac{A_v (f_y / \sqrt{3})}{\gamma_{M0}}$$

Equation 5.43

Where A_v is the shear area, which may be taken as follows:

- a. Rolled I and H sections, load parallel to web,
 $A - 2bt_f + (t_w + 2r) t_f$ but not less than $\eta h_w t_w$
- b. Rolled channel sections, load parallel to web
 $A - 2bt_f + (t_w + r) t_f$
- c. Rolled T-section, load parallel to web
 - For rolled T-sections: $A_v = A - bt_f + (t_w + 2r) (t_f/2)$
 - For welded T-sections: $A_v = t_w (h - (t_f/2))$
- d. Welded I , H and box sections, load parallel to web: $\eta \Sigma (h_w t_w)$
- e. Welded I , H , channel and box sections, load parallel to flanges: $A - \Sigma (h_w t_w)$
- f. Rolled rectangular hollow sections of uniform thickness:
 - Load parallel to depth: $A h / (b + h)$
 - Load parallel to width: $A b / (b + h)$
- g. Circular hollow sections and tubes of uniform thickness: $2 A / \pi$

Where,

A = The cross-sectional area.

b = The overall breadth.

h = The overall depth.

h_w = The depth of the web.

r = The root radius.

t_f = The flange thickness.

t_w = The web thickness (If the web thickness is not constant, t_w should be taken as the minimum thickness).

η = May be conservatively taken equal 1.0.

For verifying the design elastic shear resistance $V_{c,Rd}$ the following criterion for a critical point of the cross section may be used unless the buckling verification applies:

$$\frac{\tau_{Ed}}{f_y/(\sqrt{3} \gamma_{M0})} \leq 1.0 \quad \text{Equation 5.44}$$

τ_{Ed} is the elastic shear stress and may be obtained from:

$$\tau_{Ed} = \frac{V_{Ed} S}{I t} \quad \text{Equation 5.45}$$

Where,

V_{Ed} = The design value of the shear force.

S = The first moment of area about the centroidal axis of that portion of the cross-section between the point at which the shear is required and the boundary of the cross-section.

I = Second moment of area of the whole cross section.

t = The thickness at the examined point.

The verification according to above is conservative as it excludes partial plastic shear distribution, which is permitted in elastic design. Therefore, it should only be carried out where the verification on the basis of $V_{c,Rd}$ according to Equation 5.42 cannot be performed.

For I - or H -sections the shear stress in the web may be taken as:

$$\tau_{Ed} = \frac{V_{Ed}}{A_w} \quad \text{if} \quad \frac{A_f}{A_w} \geq 0.6 \quad \text{Equation 5.46}$$

Where,

A_f = The area of one flange

A_w = The area of the web: $A_w = h_w t_w$

Fastener holes need not be allowed for in the shear verification except in verifying the design shear resistance at connection zones.

Where the shear force is combined with a torsional moment, the plastic shear resistance $V_{pl,Rd}$ should be reduced to $V_{pl,T,Rd}$ to account for torsional effects.

5.6.12.2 Shear Resistance Considering Shear Buckling

The sections shall be checked for resistance to shear buckling, if $h_w/t_w > (72/\eta)\varepsilon$ for unstiffened web or $(31/\eta)\varepsilon\sqrt{k_\tau}$ for stiffened web and should be provided with transverse stiffeners at the supports. The value of ε is given by, $\varepsilon = \sqrt{(235/f_y)}$. The f_y is in N/mm^2 .

η may be conservatively taken equal 1.0.

h_w is clear web depth between flanges, k_τ is the shear buckling coefficient as given in Clause 5.6.12.2.1.

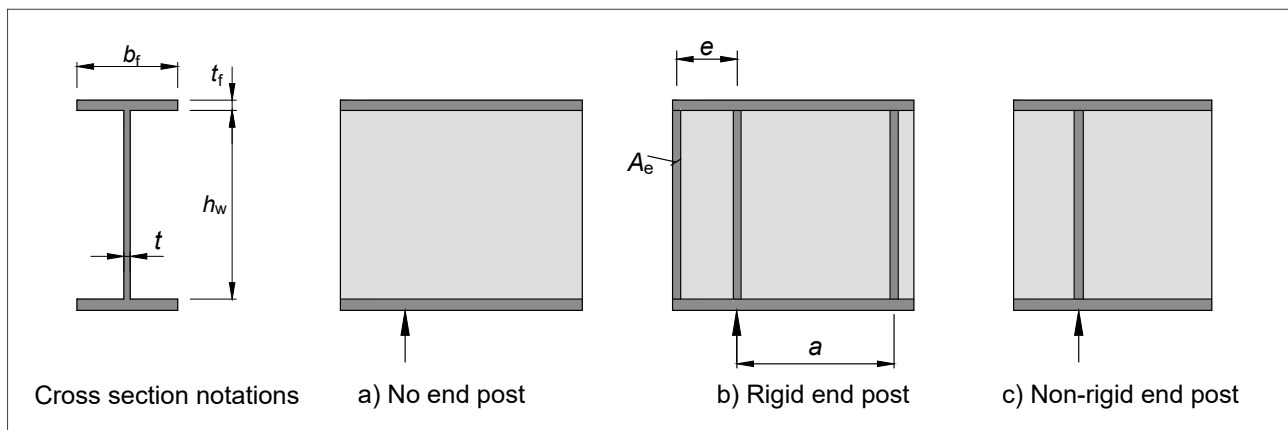
The design shear resistance for unstiffened or stiffened webs should be taken as:

$$V_{b,Rd} = V_{bw,Rd} + V_{bf,Rd} \leq \frac{\eta f_{yw} h_w t}{\sqrt{3} \gamma_{M1}} \quad \text{Equation 5.47}$$

Where $V_{bw,Rd}$ is the contribution from the web and $V_{bf,Rd}$ is the contribution from the flanges.

Stiffeners and weld should comply with the requirements in Clause 5.6.21.

Figure 5.17 End Supports



5.6.12.2.1 Contribution from the Web

The contribution from web is given by:

$$V_{bw,Rd} = \frac{\chi_w f_{yw} h_w t}{\sqrt{3} \gamma_{M1}} \quad \text{Equation 5.48}$$

For webs with transverse stiffeners at supports only and for webs with either intermediate transverse stiffeners or longitudinal stiffeners or both, the factor χ_w for the contribution of the web to the shear buckling resistance should be obtained from Table 5.10 or Figure 5.18. The type of end supports for girders are shown in Figure 5.17.

- No end post
- Rigid end post
- Non-rigid end post

Table 5.10 Contribution from Web χ_w to Shear Buckling Resistance

	Rigid End Post	Non-rigid End Post
$\bar{\lambda} < 0.83 / \eta$	η	η
$0.83 / \eta \leq \bar{\lambda} < 1.08$	$0.83 / \bar{\lambda}$	$0.83 / \bar{\lambda}$
$\bar{\lambda} \geq 1.08$	$1.37 / (0.7 + \bar{\lambda})$	$0.83 / \bar{\lambda}$

The modified slenderness $\bar{\lambda}_w$ in Table 5.10 and Figure 5.18 should be taken as:

$$\bar{\lambda}_w = 0.76 \sqrt{\frac{f_{yw}}{\tau_{cr}}}$$

Equation 5.49

Where, $\tau_{cr} = k_\tau \sigma_E$

Note 1: Values for k_τ and σ_E may be taken from Appendix C.

Note 2: The modified slenderness $\bar{\lambda}_w$ may be taken as follows

a. Transverse stiffeners at supports only:

$$\bar{\lambda}_w = \frac{h_w}{86.4 t \varepsilon}$$

Equation 5.50

b. Transverse stiffeners at supports and intermediate transverse or longitudinal stiffeners or both:

$$\bar{\lambda}_w = \frac{h_w}{37.4 t \varepsilon \sqrt{k_\tau}}$$

Equation 5.51

In which k_τ is the minimum shear buckling coefficient for the web panel.

Note 3: Where non-rigid transverse stiffeners are also used in addition to rigid transverse stiffeners k_τ is taken as the minimum of the values from the web panels between any two transverse stiffeners (e.g. $a_2 h_w$ and $a_3 h_w$) and that between two rigid stiffeners containing non-rigid transverse stiffeners (e.g. $a_4 h_w$).

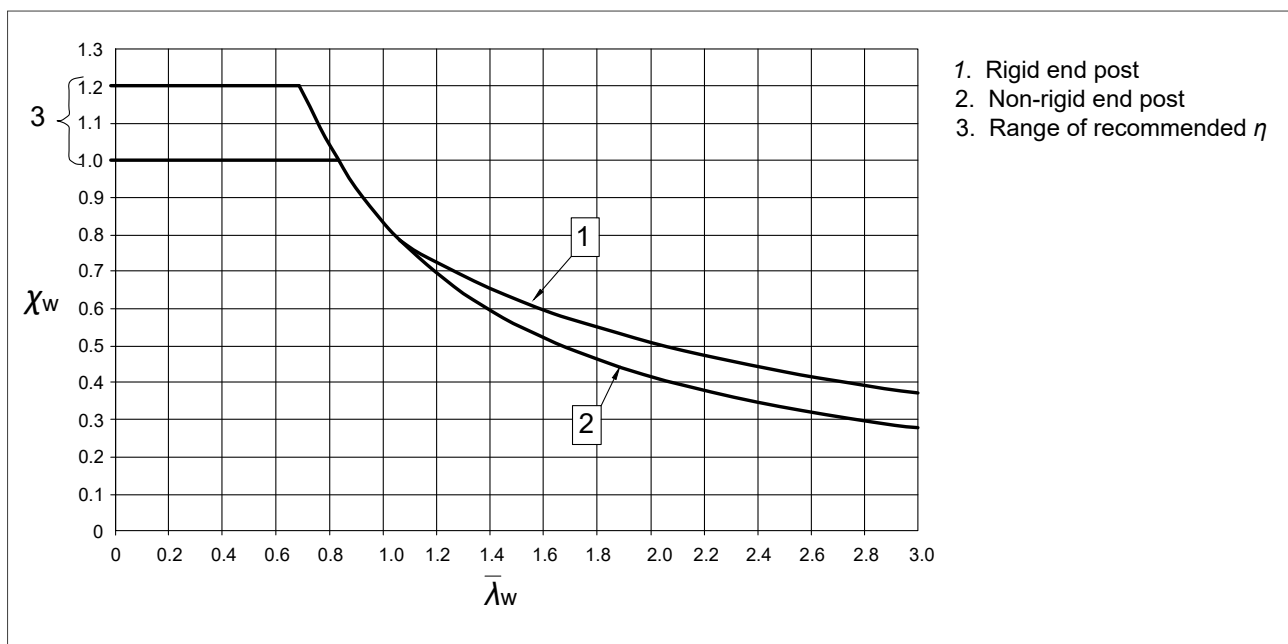
Note 4: Rigid boundaries may be assumed for panels bordered by flanges and rigid transverse stiffeners. The web buckling analysis can then be based on the panels between two adjacent transverse stiffeners (e.g. $a_1 h_w$).

Note 5: For non-rigid transverse stiffeners, the minimum value k_τ may be obtained from the buckling analysis of the following:

1. A combination of two adjacent web panels with one flexible transverse stiffener.
2. A combination of three adjacent web panels with two flexible transverse stiffeners. For procedure to determine k_τ , refer Appendix C.

The second moment of area of a longitudinal stiffener should be reduced to 1/3 of its actual value when calculating k_τ . Formulae for k_τ taking this reduction into account in Appendix C may be used.

Figure 5.18 Shear Buckling Factor χ_w



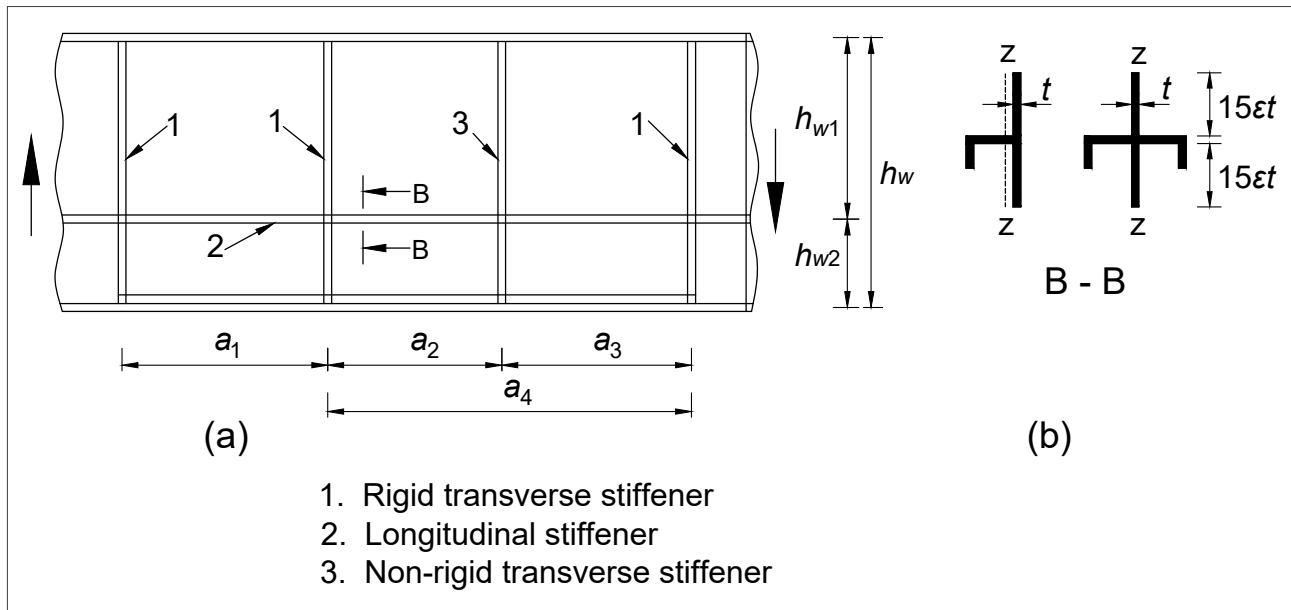
For webs with longitudinal stiffeners, the modified slenderness $\bar{\lambda}_w$ may should not be taken as less than:

$$\bar{\lambda}_w = \frac{h_{wi}}{37.4 \text{ t } \varepsilon \sqrt{k_{\tau i}}} \quad \text{Equation 5.52}$$

Where h_{wi} and k_{ti} refer to the subpanel with the largest modified slenderness $\bar{\lambda}_w$ of all subpanels within the web panel under consideration.

Note: To calculate k_{ti} , the Expression given in Appendix C may be used with $k_{tst} = 0$.

Figure 5.19 Web With Transverse and Longitudinal Stiffeners



5.6.12.2.2 Contribution from the Flange

When the flange resistance is not completely utilised in resisting the bending moment ($M_{Ed} < M_{f,Rd}$) the contribution from the flanges should be obtained as follows:

$$V_{bf,Rd} = \frac{b_f t_f^2 f_{yf}}{c \gamma_{M1}} \left(1 - \left(\frac{M_{Ed}}{M_{f,Rd}} \right)^2 \right)$$

Equation 5.53

b_f and t_f are taken for the flange which provides the least axial resistance.

b_f being taken as not larger than $(15 \varepsilon t_f)$ on each side of the web.

$M_{f,Rd} = M_{f,k} / \gamma_M$ is the moment of resistance of the cross section consisting of the effective area of the flanges only.

$$c = a \left(0.25 + \frac{1.6 b_f t_f^2 f_{yf}}{t h_w^2 f_{vw}} \right)$$

When an axial force N_{Ed} is present, the value of $N_{f,Rd}$ should be reduced by multiplying it by the following factor:

$$1 - \frac{N_{Ed}}{(A_{f1} + A_{f2})f_{yf}\gamma_{M0}} \quad \text{Equation 5.54}$$

Where A_{f1} and A_{f2} are the areas of the top and bottom flanges respectively.

5.6.12.2.3 Verification

The verification should be performed as follows:

$$\eta_3 = \frac{V_{Ed}}{V_{b,Rd}} \leq 1.0$$

Equation 5.55

Where V_{Ed} is the design shear force including shear from torque.

5.6.13 Torsion

5.6.13.1 General

Torsional and distortional effects should be taken into account for members subject to torsion. The effects of transverse stiffness in the cross section, and/or of diaphragms that are built in to reduce distortional deformations, may be taken into account by considering an appropriate elastic model which is subject to the combined effect of bending, torsion and distortion.

Distortional effects in the members may be disregarded where the effects from distortion, due to the transverse bending stiffness in the cross section and/or diaphragm action, do not exceed 10% of the bending effects.

Diaphragms Should be designed to take into account the action effects resulting from their load distributing effect.

5.6.13.2 Torsion for Which Distortional Effects May Be Neglected

For members subject to torsion for which distortional deformations may be disregarded the design value of the torsional moment T_{Ed} at each cross-section should satisfy:

$$\frac{T_{Ed}}{T_{Rd}} \leq 1.0$$

Equation 5.56

Where T_{Rd} is the design torsional resistance of the cross section.

The total torsional moment at any cross-section should be considered as the sum of two internal effects:

$$T_{Ed} = T_{t,Ed} + T_{w,Ed}$$

Equation 5.57

Where,

$T_{t,Ed}$ = The design value of the internal St. Venant torsion moment.

$T_{w,Ed}$ = The design value of the internal warping torsional moment.

The values of $T_{t,Ed}$ and $T_{w,Ed}$ at any cross-section may be determined from T_{Ed} by elastic analysis, taking account of the section properties of the member, the conditions of restraint at the supports and the distribution of the actions along the member.

The following stresses due to torsion should be taken into account:

- The shear stresses $\tau_{t,Ed}$ due to St. Venant torsion $T_{t,Ed}$.
- The direct stresses $\sigma_{w,Ed}$ due to the bi-moment B_{Ed} and shear stresses $\tau_{w,Ed}$ due to warping torsion $T_{w,Ed}$.

For the elastic verification, the yield criterion given in Clause 5.6.2 may be applied. For determining the plastic moment resistance of a cross section due to bending and torsion, only torsion effects B_{Ed} should be derived from elastic analysis.

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As a simplification, in the case of a member with a closed hollow cross-section, such as a structural hollow section, it may be assumed that the effects of torsional warping can be neglected. Also, as a simplification, in the case of a member with open cross section, such as I or H , it may be assumed that the effects of St. Venant torsion can be neglected. For the calculation of the resistance T_{Rd} of closed hollow sections the design shear strength of the individual parts of the cross section should be taken into account.

For combined shear force and torsional moment the plastic shear resistance accounting for torsional effects should be reduced from $V_{pl,Rd}$ to $V_{pl,T,Rd}$ and the design shear force should satisfy:

$$\frac{V_{Ed}}{V_{pl,T,Rd}} \leq 1.0 \quad \text{Equation 5.58}$$

In which $V_{pl,T,Rd}$ may be derived as follows:

- for an I or H section:

$$V_{pl,T,Rd} = \sqrt{1 - \frac{\tau_{t,Ed}}{1.25(f_y I \sqrt{3})/\gamma_{M0}}} V_{pl,Rd} \quad \text{Equation 5.59}$$

- for a channel section:

$$V_{pl,T,Rd} = \left[\sqrt{1 - \frac{\tau_{t,Ed}}{1.25(f_y I \sqrt{3})/\gamma_{M0}}} - \frac{\tau_{w,Ed}}{1.25(f_y I \sqrt{3})/\gamma_{M0}} \right] V_{pl,Rd} \quad \text{Equation 5.60}$$

- for a structural hollow section:

$$V_{pl,T,Rd} = \left[1 - \frac{\tau_{t,Ed}}{(f_y I \sqrt{3})/\gamma_{M0}} \right] V_{pl,Rd} \quad \text{Equation 5.61}$$

Where $V_{pl,Rd}$ is given 5.6.12.

5.6.14 Bending, Axial Load, Shear and Transverse Loads

The interaction between bending, axial load, shear and transverse loads may be determined using one of the following two methods:

- Interaction methods given in Clause 5.6.15 to 5.6.16.
- Interaction of stresses using the yielding criterion given in Clause 5.6.2.

For the local buckling effects due to transverse loads, Eurocode EN 1993-1-5 should be referred.

5.6.15 Bending and Shear

Where the shear force is present, allowance should be made for its effect on the moment resistance. Where the shear force is less than half the plastic shear resistance ($V_{Ed} < 0.5 V_{pl,Rd}$), its effect on the moment resistance may be neglected except where shear buckling reduces the section resistance.

Where ($V_{Ed} \geq 0.5 V_{pl,Rd}$), the reduced moment resistance should be taken as the design resistance of the cross-section, calculated using a reduced yield strength for the shear area.

$$(1-\rho)f_y$$

Where $\rho = \left(\frac{2 V_{Ed}}{V_{pl,Rd}} - 1 \right)^2$ and $V_{pl,Rd}$ is obtained from Clause 5.6.12.

When torsion is present, ρ should be obtained from $\rho = \left(\frac{2 V_{Ed}}{V_{pl,T,Rd}} - 1 \right)^2$, but should be taken as 0 for $V_{Ed} \leq 0.5 V_{pl,T,Rd}$.

The reduced design plastic resistance moment allowing for the shear force may alternatively be obtained for I -cross-sections with equal flanges and bending about the major axis as follows:

$$M_{y,V,Rd} = \frac{\left[W_{pl,y} - \frac{\rho A_w^2}{4 t_w} \right] f_y}{\gamma_{M0}} \quad \text{but } M_{y,V,Rd} \leq M_{y,c,Rd} \quad \text{Equation 5.62}$$

Where, $M_{y,c,Rd}$ is obtained from Clause 5.6.11 and $A_w = h_w t_w$

For the interaction of bending, shear and transverse loads, refer Clause 5.6.20.3 and 5.6.20.4.

5.6.16 Bending and Axial Force

5.6.16.1 Class 1 and Class 2 Cross Sections

Where an axial force is present, allowance should be made for its effect on the plastic moment resistance.

For class 1 and 2 cross sections, the following criterion shall be satisfied:

$$\frac{M_{Ed}}{M_{N,Rd}} \leq 1.0$$

Where $M_{N,Rd}$ is the design plastic moment resistance reduced due to the axial force N_{Ed} .

For a rectangular solid section without fastener holes $M_{N,Rd}$ should be taken as:

$$M_{N,Rd} = M_{pl,Rd} \left[1 - \left(\frac{N_{Ed}}{N_{pl,Rd}} \right)^2 \right] \quad \text{Equation 5.63}$$

For doubly symmetrical I - and H -sections or other flanges sections, allowance need not be made for the effect of the axial force on the plastic resistance moment about the y - y axis when both the following criteria are satisfied:

$$N_{Ed} \leq 0.25 N_{pl,Rd} \quad \text{Equation 5.64}$$

$$N_{Ed} \leq \frac{0.5 h_w t_w f_y}{\gamma_{M0}} \quad \text{Equation 5.65}$$

For doubly symmetrical I - and H -sections, allowance need not be made for the effect of the axial force on the plastic resistance moment about the z - z axis when:

$$N_{Ed} \leq \frac{h_w t_w f_y}{\gamma_{M0}} \quad \text{Equation 5.66}$$

For cross-sections where fastener holes are not to be accounted for, the following approximations may be used for standard rolled I or H sections and for welded I or H sections with equal flanges:

$$M_{N,y,Rd} = M_{pl,y,Rd} \frac{1-n}{1-0.5a} \quad \text{but } M_{N,y,Rd} \leq M_{pl,y,Rd} \quad \text{Equation 5.67}$$

For $n \leq a$: $M_{N,z,Rd} = M_{pl,z,Rd}$

For $n > a$: $M_{N,z,Rd} = M_{pl,z,Rd} \left[1 - \left(\frac{n-a}{1-a} \right)^2 \right]$

Where,

$$n = N_{Ed} / N_{pl,Rd}$$

$$a = (A - 2 b t_f) / A \quad \text{but } a \leq 0.5$$

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For cross-sections where fastener holes are not to be accounted for, the following approximations may be used for rectangular structural hollow sections of uniform thickness and for welded box sections with equal flanges and equal webs:

$$M_{N,y,Rd} = M_{pl,y,Rd} \frac{1-n}{1-0.5 a_w} \text{ but } M_{N,y,Rd} \leq M_{pl,y,Rd}$$

Equation 5.68

$$M_{N,z,Rd} = M_{pl,z,Rd} \frac{1-n}{1-0.5 a_f} \text{ but } M_{N,z,Rd} \leq M_{pl,z,Rd}$$

Equation 5.69

Where,

$a_w = (A - 2 b t) / A$ but $a_w \leq 0.5$ for hollow sections

$a_w = (A - 2 b t_f) / A$ but $a_w \leq 0.5$ for welded box sections

$a_f = (A - 2 h t) / A$ but $a_f \leq 0.5$ for hollow sections

$a_f = (A - 2 h t_w) / A$ but $a_f \leq 0.5$ for welded box sections

For bi-axial bending the following criterion may be used:

$$\left[\frac{M_{y,Ed}}{M_{N,y,Rd}} \right]^\alpha + \left[\frac{M_{z,Ed}}{M_{N,z,Rd}} \right]^\beta \leq 1$$

Equation 5.70

Where, α and β are constants, which may conservatively be taken as unity, otherwise as follows:

- I and H sections:

$$\alpha = 2 ; \quad \beta = 5n \text{ but } \beta \geq 1$$

- Rectangular hollow sections:

$$\alpha = \beta = 1.66 / (1 - 1.13 n^2) \text{ but } \alpha = \beta \leq 6$$

Where, $n = N_{Ed} / N_{pl,Rd}$.

5.6.16.2 Class 3 Cross-sections

In the absence of shear force, for Class 3 cross-sections the maximum longitudinal stress shall satisfy the criterion:

$$\sigma_{x,Ed} \leq \frac{f_y}{\gamma_{M0}}$$

Equation 5.71

Where $\sigma_{x,Ed}$ is the design value of the local longitudinal stress due to moment and axial force taking account of fastener holes where relevant.

The following should be met for local buckling consideration, when the limit stress method is used:

$$\sigma_{x,Ed} \leq \frac{\sigma_{limit}}{\gamma_{M0}} \leq \frac{f_y}{\gamma_{M0}}$$

Equation 5.72

Where σ_{limit} should be determined from Clause 5.6.20.2.

5.6.16.3 Class 4 Cross-sections

In the absence of shear force, for Class 4 cross-sections the maximum longitudinal stress $\sigma_{x,Ed}$ calculated using the effective cross sections shall satisfy the criterion:

$$\sigma_{x,Ed} \leq \frac{f_y}{\gamma_{M0}}$$

Equation 5.73

Where $\sigma_{x,ED}$ is the design value of the longitudinal stress due to moment and axial force taking account of fastener holes where relevant.

As an alternative to above criterion, the following simplified criterion may be used:

$$\frac{N_{Ed}}{A_{eff} f_y / \gamma_{M0}} + \frac{M_{y,Ed} + N_{Ed} e_{Ny}}{W_{eff,y,min} f_y / \gamma_{M0}} + \frac{M_{z,Ed} + N_{Ed} e_{Nz}}{W_{eff,z,min} f_y / \gamma_{M0}} \leq 1$$

Equation 5.74

Where,

A_{eff} = The effective area of the cross-section when subjected to uniform compression.

$W_{eff,min}$ = The effective section modulus (corresponding to the fibre with the maximum elastic stress) of the cross-section when subjected only to moment about the relevant axis.

e_N = The shift of the relevant centroidal axis when the cross-section is subjected to compression only.

Note: The signs of N_{Ed} , $M_{y,Ed}$, $M_{z,Ed}$ and $\Delta M_i = N_{Ed} e_{Ni}$ depends on the combination of the respective direct stresses.

5.6.17 Bending, Shear and Axial Force

Where shear and axial force are present, allowance should be made for the effect of both shear force and axial force on the resistance moment. Provided that the value of the shear force V_{Ed} does not exceed 50 % of the design plastic shear resistance $V_{pl,Rd}$, no reduction of the resistances defined for bending and axial force in 5.6.16 need be made, except where shear buckling reduces the section resistance.

Where V_{Ed} exceeds 50% of $V_{pl,Rd}$, the design resistance of the cross-section to combinations of moment and axial force should be calculated using a reduced yield strength as follows for the shear area:

$$\text{Where } \rho = \left(\frac{V_{Ed}}{V_{pl,Rd}} - 1 \right)^2 \quad (1 - \rho) f_y$$

5.6.18 Buckling Resistance of Members

5.6.18.1 Uniform Members in Compression

5.6.18.1.1 Buckling Resistance

A compression member should be verified against buckling as follows:

$$\frac{N_{Ed}}{N_{b,Rd}} \leq 1.0$$

Equation 5.75

Where,

N_{Ed} = The design value of the compression force

$N_{b,Rd}$ = The design buckling resistance of the compression member.

For members with non-symmetric Class 4 sections allowance should be made for the additional moment ΔM_{Ed} due to the eccentricity of the centroidal axis of the effective section and the interaction should be carried out according to Clauses 5.6.18.3 or 5.6.18.4.

The design buckling resistance of a compression member should be taken as:

$$N_{b,Rd} = \frac{\chi A f_y}{\gamma_{M1}} \quad \text{for Class 1,2 and 3 cross-sections}$$

$$N_{b,Rd} = \frac{\chi A_{eff} f_y}{\gamma_{M1}} \quad \text{for class 4 cross-sections}$$

Where χ is the reduction factor for the relevant buckling mode.

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In determining A and A_{eff} , holes for fasteners at the column ends need not to be taken into account.

5.6.18.1.2 Buckling Curves

For axial compression in members, the value of χ for the appropriate non-dimensional slenderness should be determined from the relevant buckling curve according to:

$$\chi = \frac{1}{\phi + \sqrt{\phi^2 - \bar{\lambda}^2}} \quad \text{but } \chi \leq 1.0$$

Where,

$$\phi = 0.5 [1 + \alpha(\bar{\lambda} - 0.2) + \bar{\lambda}^2]$$

$$\bar{\lambda} = \sqrt{A f_y / N_{cr}} \quad \text{for class 1, 2 and 3 cross-sections}$$

$$\bar{\lambda} = \sqrt{A_{eff} f_y / N_{cr}} \quad \text{for class 4 cross-sections}$$

α = An imperfection factor

N_{cr} = The elastic critical force for the relevant buckling mode based on the gross cross-sectional properties.

The imperfection factor α corresponding to the appropriate buckling curve should be obtained from Table 5.11 and Table 5.12.

Table 5.11 Imperfection Factors for Buckling Curves

Buckling curve	a_0	a	b	c	d
Imperfection factor α	0.13	0.21	0.34	0.49	0.76

Values of the reduction factor χ for the appropriate non-dimensional slenderness $\bar{\lambda}$ may be obtained from Figure 5.20.

For slenderness $\bar{\lambda} \leq 0.2$ or for $N_{Ed} / N_{cr} \leq 0.04$ the buckling effects may be ignored and only cross-sectional checks apply.

5.6.18.1.3 Slenderness for Flexural Buckling

The non-dimensional slenderness $\bar{\lambda}$ is given by:

$$\bar{\lambda} = \sqrt{\frac{A f_y}{N_{cr}}} = \frac{L_{cr}}{i} \frac{1}{\lambda_1} \quad \text{for Class 1, 2 and 3 cross-sections} \quad \text{Equation 5.76}$$

$$\bar{\lambda} = \sqrt{\frac{A_{eff} f_y}{N_{cr}}} = \frac{L_{cr}}{i} \sqrt{\frac{A_{eff}}{A}} \frac{1}{\lambda_1} \quad \text{for Class 4 cross-sections} \quad \text{Equation 5.77}$$

Where,

L_{cr} = The buckling length in the buckling plane considered.

i = The radius of gyration about the relevant axis, determined using the properties of the gross cross-section.

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = 93.9 \varepsilon$$

$$\varepsilon = \sqrt{\frac{235}{f_y}} \quad \text{taking } f_y \text{ in N/mm}^2$$

For flexural buckling the appropriate buckling curve should be determined from Table 5.12.

Table 5.12 Selection of Buckling Curves in a Cross Section

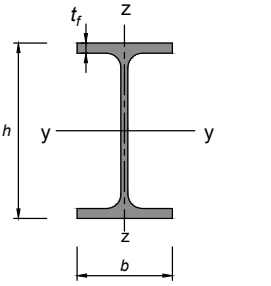
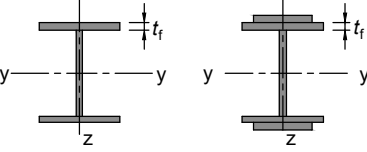
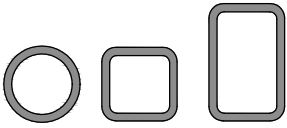
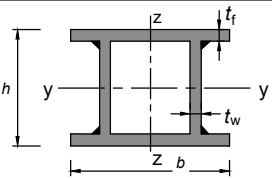
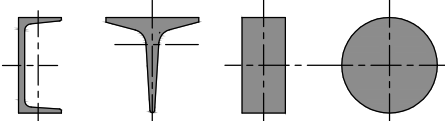
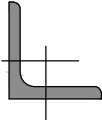
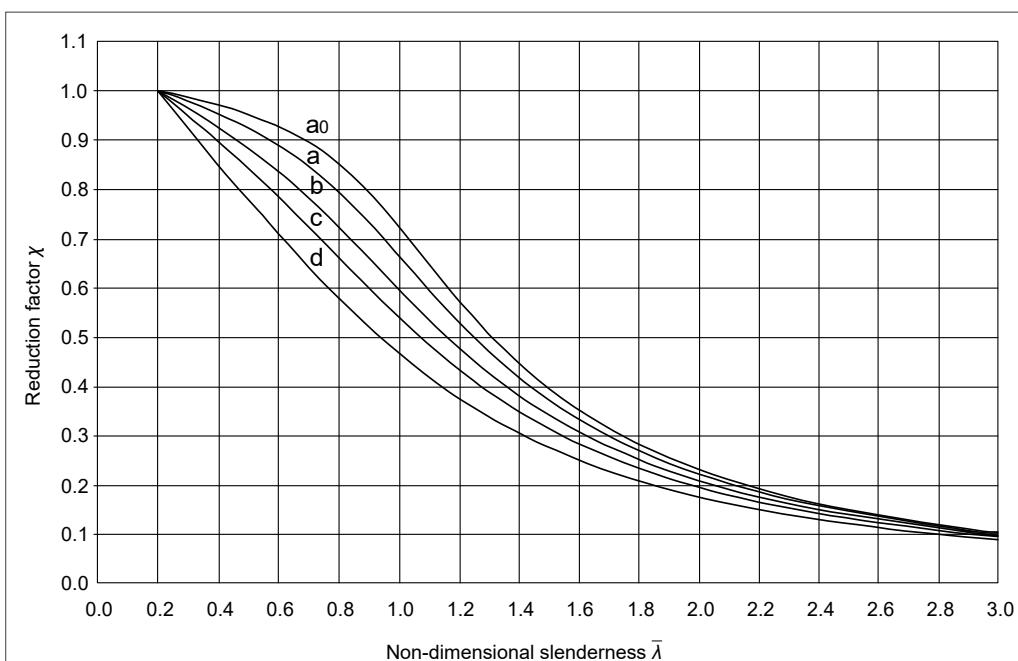
Cross section		Limits		Buckling about axis	Buckling curve	
					S235 S275 S355 S420	S 460
Rolled sections		$h/b > 1.2$	$t_f \leq 40$ mm	y - y z - z	a b	a ₀ a ₀
			$40 \text{ mm} < t_f \leq 100$	y - y z - z	b c	a a
		$h/b \leq 1.2$	$t_f \leq 100$ mm	y - y z - z	b c	a a
			$t_f > 100$ mm	y - y z - z	d d	c c
Welded I-sections		$t_f \leq 40$ mm		y - y z - z	b c	b c
		$t_f > 40$ mm		y - y z - z	c d	c d
Hollow sections		hot finished		any	a	a ₀
		cold formed		any	c	c
Welded box sections		generally (except as below)		any	b	b
		thick welds: $a > 0.5t_f$ $b/t_f < 30$ $h/t_w < 30$		any	c	c
U-, T- and solid sections				any	c	c
L-sections				any	b	b

Figure 5.20 Buckling Curves



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5.6.18.1.4 Slenderness for Torsional and Torsional Flexural Buckling

For members with open cross-sections, account should be taken of the possibility that the of the member to either torsional or torsional-flexural buckling could be less than its resistance to flexural buckling.

The non-dimensional slenderness $\bar{\lambda}_T$ for torsional or torsional-flexural buckling should be taken as:

$$\bar{\lambda}_T = \sqrt{\frac{A f_y}{N_{cr}}} \quad \text{for Class 1, 2 and 3 cross-sections}$$

$$\bar{\lambda}_T = \sqrt{\frac{A_{eff} f_y}{N_{cr}}} \quad \text{for Class 4 cross-sections}$$

Where,

$$N_{cr} = N_{cr,TF} \quad \text{but } N_{cr} < N_{cr,T}$$

$N_{cr,TF}$ = The elastic torsional-flexural buckling force

$N_{cr,T}$ = The elastic torsional buckling force

For torsional or torsional-flexural buckling the appropriate buckling curve may be determined from Table 5.12 considering the one related to the z-axis.

5.6.18.2 Uniform Members in Bending

5.6.18.2.1 Buckling Resistance

A laterally unrestrained member subject to major axis bending should be verified against lateral-torsional buckling as follows:

$$\frac{M_{Ed}}{M_{b,Rd}} \leq 1.0$$

Equation 5.78

Where,

M_{Ed} = The design value of the moment

$M_{b,Rd}$ = The design buckling resistance moment

Beams with sufficient restraint to the compression flange are not susceptible to lateral-torsional buckling. In addition, beams with certain types of cross-sections, such as square or circular hollow sections, fabricated circular tubes or square box sections are not susceptible to lateral-torsional buckling.

The design buckling resistance moment of a laterally unrestrained beam should be taken as:

$$M_{b,Rd} = \chi_{LT} W_y \frac{f_y}{\gamma_{MI}}$$

Equation 5.79

Where,

W_y = The appropriate section modulus as follows:

$W_y = W_{pl,y}$ for Class 1 or 2 cross sections

$W_y = W_{el,y}$ for Class 3 cross sections

$W_y = W_{eff,y}$ for Class 4 cross sections

χ_{LT} = The reduction factor for lateral-torsional buckling

In determining W_y holes for fasteners at the beam end need not to be taken into account.

5.6.18.2.2 Lateral Torsional Buckling Curves - General Case

Unless otherwise specified, for bending members of constant cross-section, the value of χ_{LT} for the appropriate non-dimensional slenderness $\bar{\lambda}$ should be determined from:

$$\chi_{LT} = \frac{1}{\phi_{LT} + \sqrt{\phi_{LT}^2 - \bar{\lambda}_{LT}^2}} \quad \text{but } \chi_{LT} \leq 1.0 \quad \text{Equation 5.80}$$

Where,

$$\phi_{LT} = 0.5 [1 + \alpha_{LT} (\bar{\lambda}_{LT} - 0.2) + \bar{\lambda}_{LT}^2]$$

α_{LT} = An imperfection factor.

$$\bar{\lambda}_{LT} = \sqrt{(W_y f_y / M_{cr})}$$

M_{cr} = The elastic critical moment for lateral-torsional buckling.

M_{cr} = Based on gross cross-sectional properties and takes into account the loading conditions, the real moment distribution and the lateral restraints.

The recommended values α_{LT} are in Table 5.13.

Table 5.13 Imperfection Factors for Buckling Curves

Buckling curve	a	b	c	d
Imperfection factor α_{LT}	0.21	0.34	0.49	0.76

The recommendations for buckling curves are given in Table 5.14.

Table 5.14 Lateral Torsional Buckling Curves for General Case

Cross-section	Limits	Buckling curve
Rolled <i>I</i> -sections	$h/b \leq 2$	a
	$h/b > 2$	b
Welded <i>I</i> -sections	$h/b \leq 2$	c
	$h/b > 2$	d
Other cross-sections	—	d

Values of the reduction factor χ_{LT} for the appropriate non-dimensional slenderness $\bar{\lambda}$ may be obtained from Figure 5.20. Lateral torsional buckling effects may be ignored if the slenderness parameter $\bar{\lambda} \leq 0.2$ or $(M_{ED} / M_{crit}) \leq 0.04$.

5.6.18.2.3 Lateral Torsional Buckling Curves for Rolled or Equivalent Welded Sections

For rolled or equivalent welded sections in bending the values of χ_{LT} for the appropriate non-dimensional slenderness may be determined from:

$$\chi_{LT} = \frac{1}{\phi_{LT} + \sqrt{\phi_{LT}^2 - \beta \bar{\lambda}_{LT}^2}} \quad \text{but } \begin{cases} \chi_{LT} \leq 1.0 \\ \chi_{LT} \leq \frac{1}{\bar{\lambda}_{LT}^2} \end{cases} \quad \text{Equation 5.81}$$

$$\phi_{LT} = 0.5 [1 + \alpha_{LT} (\bar{\lambda}_{LT} - \bar{\lambda}_{LT,0}) + \beta \bar{\lambda}_{LT}^2] \quad \text{Equation 5.82}$$

Note: The following values of parameters $\bar{\lambda}$ and β and any limitation of validity concerning the beam depth or h/b ratio, are recommended

For rolled sections and hot-finished and cold formed hollow sections $\bar{\lambda}_{LT,0} = 0.4$ $\beta = 0.75$

For welded sections $\bar{\lambda}_{LT,0} = 0.2$ $\beta = 1.0$

The recommendations for buckling curves are given in Table 5.15. For taking into account the moment

Table 5.15 Selection of Buckling Curves for Cross Sections

Cross-section	Limits	Buckling curve
Rolled doubly symmetric <i>I</i> and <i>H</i> sections and hot-finished hollow sections	$h/b \leq 2$	<i>b</i>
	$2.0 < h/b \leq 3.1$	<i>c</i>
	$h/b > 3.1$	<i>d</i>
Angles (for moments in the major principal plane)		<i>d</i>
All other hot-rolled sections		<i>d</i>
Welded doubly symmetric sections and cold-formed hollow sections	$h/b \leq 2$	<i>c</i>
	$2 < h/b < 3.1$	<i>d</i>

distribution between the lateral restraints of members the reduction factor χ_{LT} may be modified as follows:

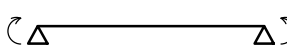
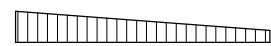
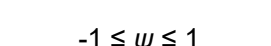
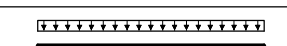
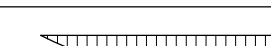
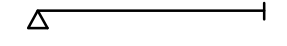
$$\chi_{LT,mod} = \frac{\chi_{LT}}{f} \quad \text{but} \quad \begin{cases} \chi_{LT,mod} \leq 1 \\ \chi_{LT,mod} \leq \frac{1}{\bar{\lambda}_{LT}^2} \end{cases} \quad \text{Equation 5.83}$$

Note: The following minimum values of f are recommended:

$$f = 1 - 0.5(1 - k_c)[1 - 2.0(\bar{\lambda}_{LT} - 0.8)^2] \quad \text{but } f \leq 1.0 \quad \text{Equation 5.84}$$

k_c is a correction factor according to Table 5.16

Table 5.16 Correction Factors k_c

Loading		Moment distribution	k_c
End moment loading		 $\psi = 1$	1.0
		 M  ψM $-1 \leq \psi \leq 1$	$\frac{1}{1.33 - 0.33\psi}$
Intermediate transverse loading			0.94
			0.90
			0.91
			0.86
			0.77
			0.82

5.6.18.3 Uniform Members in Bending and Axial Compression

Unless second order analysis is carried out using the imperfections given in section 5.5.3.2, the stability of uniform members subject to axial compression and bending in the plane of buckling should be checked in accordance with the details given below.

For members of structural systems, the resistance check may be carried out on the basis of the individual single span members regarded as cut out of the system. Second order effects of the sway system ($P-\Delta$ effects) have to be taken into account, either by the end moments of the member or by means appropriate buckling lengths.

Members which are subjected to combined bending and axial compression should satisfy:

$$\frac{\frac{N_{Ed}}{\chi_y N_{Rk}}}{\gamma_{M1}} + \frac{C_{mi,0} (M_{y,Ed} + \Delta M_{y,Ed})}{\frac{M_{y,Rk}}{\gamma_{M1}}} \leq 0.9 \quad \text{Equation 5.85}$$

Where,

N_{Ed} = The design values of the compression force.

$M_{y,Ed}$ = The design values of the maximum moments about the y-y axis of the member obtained from first order analysis without considering imperfections.

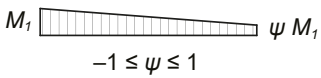
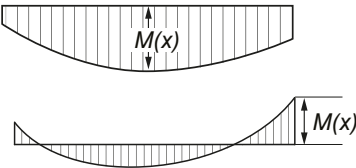
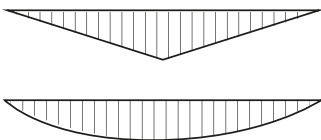
$\Delta M_{y,Ed}$ = The moments due to the shift of the centroidal axis, refer section 5.6.16.3 for class 4 sections, given as:

$\Delta M_{y,Ed} = e_{N,y} N_{Ed}$

$C_{mi,0}$ = The equivalent moment factor given in Table 5.17.

χ_y = The reduction factor due to flexural buckling as in section 5.6.18.1.

Table 5.17 Equivalent Uniform Moment Factors $C_{mi,0}$

Moment diagram	$C_{mi,0}$
 M_1 ψM_1 $-1 \leq \psi \leq 1$	$C_{mi,0} = 0.79 + 0.21 \psi_i + 0.36 (\psi_i - 0.33) \frac{N_{Ed}}{N_{cr,i}}$
 $M(x)$ $M(x)$	$C_{mi,0} = 1 + \left(\frac{\pi^2 EI_i \delta_x }{L^2 M_{i,Ed}(x) } - 1 \right) \frac{N_{Ed}}{N_{cr,i}}$ $M_{i,Ed}(x)$ is the maximum moment $M_{y,Ed}$ or $M_{z,Ed}$ according to the first order analysis. δ_x is the maximum member deflection along the member.
	$C_{mi,0} = 1 - 0.18 \frac{N_{Ed}}{N_{cr,i}}$ $C_{mi,0} = 1 + 0.03 \frac{N_{Ed}}{N_{cr,i}}$

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5.6.18.4 General Method for Lateral and Lateral Torsional Buckling of Structural Components

5.6.18.4.1 General Method

The following method allows the verification of the resistance to lateral and lateral torsional buckling for structural components such as:

- Single members with mono symmetric cross sections, built-up or not, uniform or not, with complex support conditions or not; or
- Plane frames or subframes composed of such members;

Which are subject to compression and/or mono-axial bending in the plane, but which do not contain rotative plastic hinges.

This method is valid only for nominally straight components subject to in-plane mono-axial bending and / or compression.

Overall resistance to out-of-plane buckling for any structural component conforming to the scope in above can be verified by ensuring that:

$$\frac{\chi_{op} \alpha_{ult,k}}{\gamma_{M1}} \geq 1.0$$

Equation 5.86

Where,

$\alpha_{ult,k}$ = The minimum load amplifier of the design loads to reach the characteristic resistance of the most critical cross section of the structural component considering its in plane behaviour without taking lateral or lateral torsional buckling into account however accounting for all effects due to in plane geometrical deformation and imperfections, global and local, where relevant.

χ_{op} = The reduction factor for the non-dimensional slenderness $\bar{\lambda}_{op}$, take account of lateral and lateral torsional buckling. χ_{op} should be taken as the minimum value of χ determined for lateral buckling and χ_{LT} determined for lateral torsional buckling.

The global non dimensional slenderness $\bar{\lambda}_{op}$ for the structural component should be determined from:

$$\bar{\lambda}_{op} = \sqrt{\frac{\alpha_{ult,k}}{\alpha_{cr,op}}}$$

Equation 5.87

Where,

$\alpha_{ult,k}$ = Defined above.

$\alpha_{cr,op}$ = The minimum amplifier for the in-plane design loads to reach the elastic critical load of the structural component with regards to lateral or lateral torsional buckling without accounting for in plane flexural buckling.

- The reduction factor χ_{op} may be determined from either of the following methods:

a. The minimum value of:

χ for lateral buckling

χ_{LT} for lateral torsional buckling

each calculated for the global non dimensional slenderness $\bar{\lambda}_{op}$.

Note: For example where $\alpha_{ult,k}$ is determined by the cross-section check $1 / \alpha_{ult,k} = N_{Ed} / N_{Rk} + M_{y,Ed} / M_{y,Rk}$ this method leads to:

$$\frac{N_{Ed}}{N_{Rk} / \gamma_{M1}} + \frac{M_{y,Ed}}{M_{y,Rk} / \gamma_{M1}} \leq \chi_{op}$$

Equation 5.88

- b. A value interpolated between the values χ and χ_{LT} as determined in a) by using the formula for $\alpha_{ult,k}$ corresponding to critical cross section.

Note: For example where $\alpha_{ult,k}$ is determined by the cross-section check $1/\alpha_{ult,k} = N_{Ed}/N_{Rk} + M_{y,Ed}/M_{y,Rk}$ this method leads to:

$$\frac{N_{Ed}}{\chi N_{Rk} / \gamma_{M1}} + \frac{M_{y,Ed}}{\chi_{LT} M_{y,Rk} / \gamma_{M1}} \leq 1$$

Equation 5.89

5.6.18.4.2 Simplified Method

Members with discrete lateral restraint to the compression flange are not susceptible to lateral-torsional buckling if the length L_c between restraints or the resulting slenderness $\bar{\lambda}_f$ of the equivalent compression flange satisfies:

$$\bar{\lambda}_f = \frac{k_c L_c}{i_{f,z} \lambda_1} \leq \bar{\lambda}_{c0} \frac{M_{c,Rd}}{M_{y,Ed}}$$

Equation 5.90

Where,

$M_{y,Ed}$ = The maximum value of the bending moment within the restraint spacing.

$$M_{c,Rd} = W_y \frac{f_y}{\gamma_{M1}}$$

W_y = The appropriate section modulus corresponding to the compression flange.

k_c = A slenderness correction factor for moment distribution between restraints, as in Table 5.16.

$i_{f,z}$ = The radius of gyration of the equivalent compression flange composed of the compression flange plus 1/3 of the compressed part of the web area, about the minor axis of the section.

$\bar{\lambda}_{c0}$ = A slenderness limit of the equivalent compression flange defined above, may be taken as:

$$\bar{\lambda}_{c0} = \lambda_{LT,0} + 0.1$$

$$\lambda_1 = \pi \sqrt{\frac{E}{f_y}} = 93.9 \varepsilon$$

$$\varepsilon = \sqrt{\frac{235}{f_y}} \text{ where } f_y \text{ in } \text{N/mm}^2$$

For class 4 sections, $i_{f,z}$ may be taken as

$$i_{f,z} = \sqrt{\frac{I_{efff}}{A_{efff} + (1/3) A_{eff,e,e}}}$$

Where,

i_{efff} = The effective second moment of area of the compression flange about the minor axis of the section.

A_{efff} = The effective area of the compression flange.

$A_{eff,e,e}$ = The effective area of the compressed part of the web.

Truss chords and flanges in compression that are subject to lateral buckling may be verified by modelling the elements as a column subject to the compression force N_{Ed} and supported by continuous or discrete elastic restraint modelled as springs.

The buckling mode and the elastic critical buckling load N_{cr} may be determined from an elastic critical buckling analysis. If continuous springs are used to represent the restraints which are basically discrete, the critical buckling load should not be taken as being larger than that corresponding to the buckling with nodes at the locations of the restraints.

The safety verification may be carried out in accordance with 5.6.18.2 using:

$$\bar{\lambda}_{LT} = \sqrt{\frac{A_{eff} f_y}{N_{crit}}}$$

Equation 5.91

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Where,

 A_{eff} = The effective area of the chord N_{crit} = The elastic critical load determined with A_{gross} .

2

For chords in compression or the bottom flanges of continuous girders between rigid supports, the effect of initial imperfections and second order effects on a supporting spring may be taken into account by applying an additional lateral force F_{ED} at the connection of the chord to the spring such that:

3

$$F_{Ed} = \frac{N_{Ed}}{100} \text{ if } l_k \leq 1.2 l$$

Equation 5.92

4

$$F_{Ed} = \frac{l}{l_k} \frac{N_{Ed}}{80} \frac{1}{1 - \frac{N_{Ed}}{N_{crit}}} \text{ if } l_k > 1.2 l$$

Equation 5.93

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Where,

$$l_k = \pi \sqrt{EI N_{crit}}$$

 l = The distance between the springs.

If the design value of the compression force N_{Ed} is constant over the length of the chord, the critical axial load N_{crit} may be calculated from:

$$N_{crit} = m N_E$$

Where,

$$N_E = \pi^2 EI / L^2$$

$$m = (2/\pi^2) \sqrt{\gamma} \text{ but not less than 1.0.}$$

$$\gamma = (c L^4) / (E I)$$

$$c = C_d / l$$

Where,

 L = The span length between the rigid supports l = The distance between the springs C_d = The spring stiffness

A lateral support to a compressed flange may be assumed to be rigid if its stiffness C_d satisfies:

$$C_d > \frac{4N_E}{L}$$

Where, N_E is the elastic critical load which is determined assuming hinged ends.

The procedure given above may also be applied to the flanges of girders in compression when A_{eff} in Equation 5.91 is substituted by:

$$A_{eff} + \frac{A_{wc}}{3}$$

Where, A_{wc} is the area of the compression zone of the web. In the case of a class 4 section the areas should be taken as the effective areas.

5.6.19 Uniform Built-up Compression Members

5.6.19.1 General

Uniform built-up compression members with hinged ends that are laterally supported should be designed with the following model, see Figure 5.21.

- Member may be considered as a column with a bow imperfection $e_0 = L/500$.
- The elastic deformations of lacing or battens, may be considered by a continuous shear stiffness S_v of the column.

For other end conditions, appropriate modifications may be performed. The model of a uniform built-up compression member applies when:

- The lacings or battens consist of equal modules with parallel chords
- The minimum numbers of modules in a member is three.

The design procedure is applicable to built-up members with lacings in two planes, refer Figure 5.22. The chords may be solid members or may themselves be laced or battened in the perpendicular plane.

Figure 5.21 Uniform Built-up Columns with Lacings and Battens

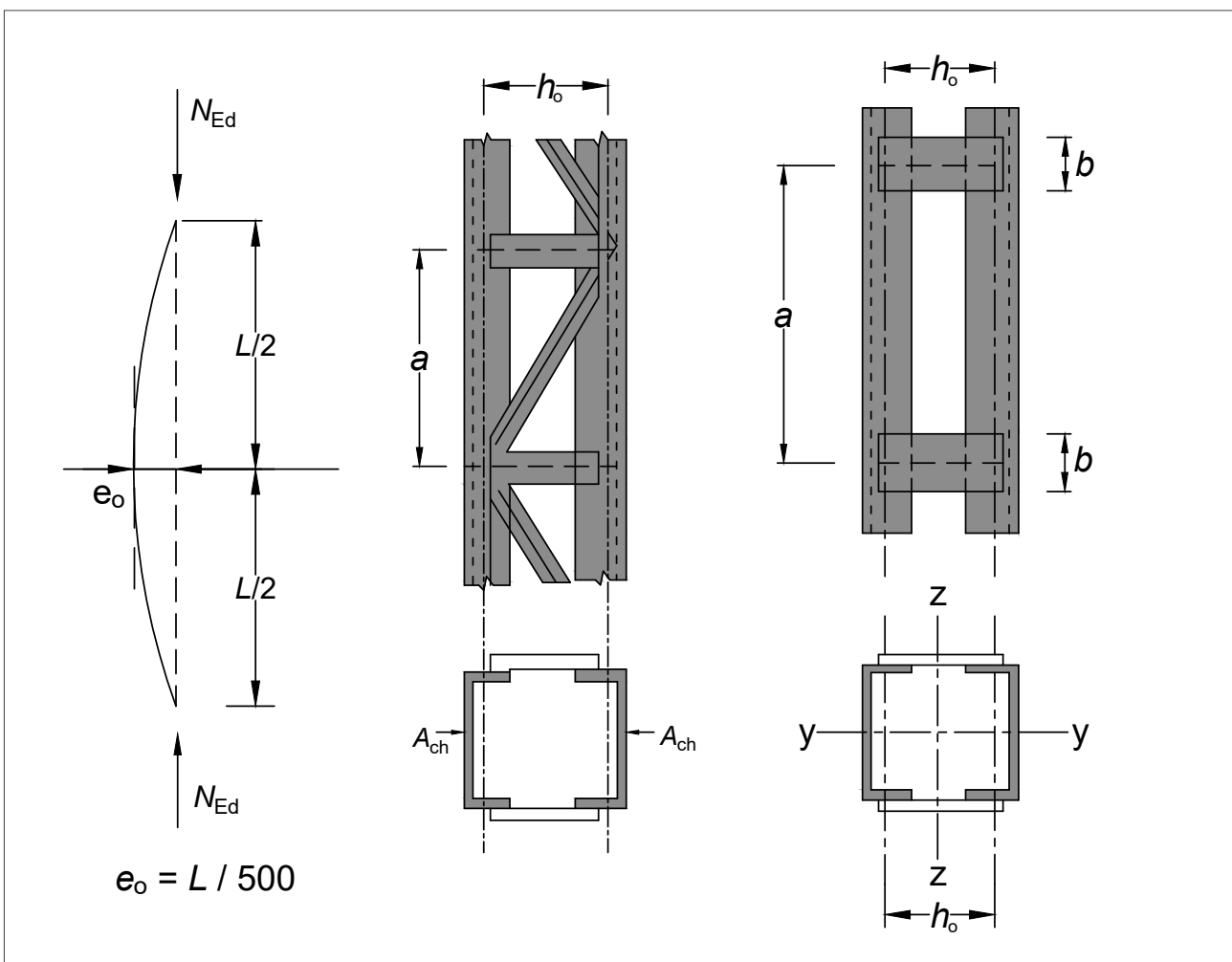
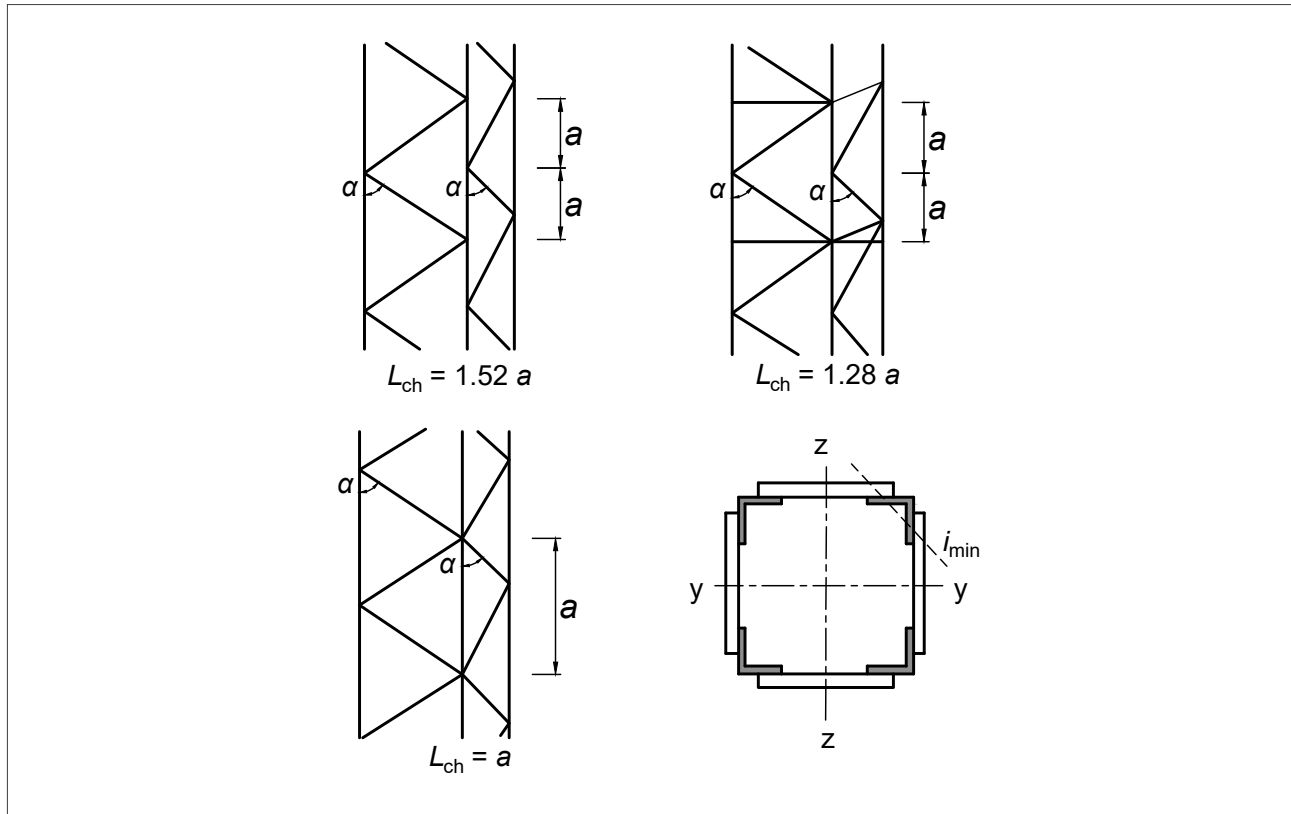


Figure 5.22 Lacings on Four Sides and Buckling Length L_{ch} of Chords

Checks should be performed for chords using the design chord forces $N_{ch,Ed}$ from compression forces N_{Ed} and moments M_{Ed} at mid span of the built-up member.

For a member with two identical chords the design force $N_{ch,Ed}$ should be determined from:

$$N_{ch,Ed} = 0.5N_{Ed} + \frac{M_{Ed} h_0 A_{ch}}{2I_{eff}}$$

Equation 5.94

Where,

$$M_{Ed} = \frac{N_{Ed} e_0 + M_{Ed}^I}{1 - \frac{N_{Ed}}{N_{cr}} - \frac{N_{Ed}}{s_v}}$$

Equation 5.95

$N_{cr} = \pi^2 E I_{eff} / L^2$ is the effective critical force of the built-up member

N_{Ed} = The design value of the compression force to the built-up member

M_{Ed} = The design value of the maximum moment in the middle of the built-up member considering second order effects

M_{Ed}^I = The design value of the maximum moment in the middle of the built-up member without second order effects

h_0 = The distance between the centroids of chords

A_{ch} = The cross-sectional area of one chord

I_{eff} = The effective second moment of area of the built-up member

S_v = The shear stiffness of the lacings or battened panel

The checks for the lacings of laced built-up members or for the frame moments and shear forces of the battened panels of battened built-up members should be performed for the end panel taking account of the shear force in the built-up member

$$V_{Ed} = \pi \frac{M_{Ed}}{L}$$

Equation 5.96

5.6.19.2 Laced Compression Members

5.6.19.2.1 Resistance of Components of Laced Compression Members

The chords and diagonal lacings subject to compression should be designed for buckling. For chords, the buckling verification should be performed as follows:

$$\frac{N_{ch,Ed}}{N_{b,Rd}} \leq 1.0$$

Equation 5.97

Where,

$N_{ch,Ed}$ = The design compression force in the chord at mid-length of the built-up member

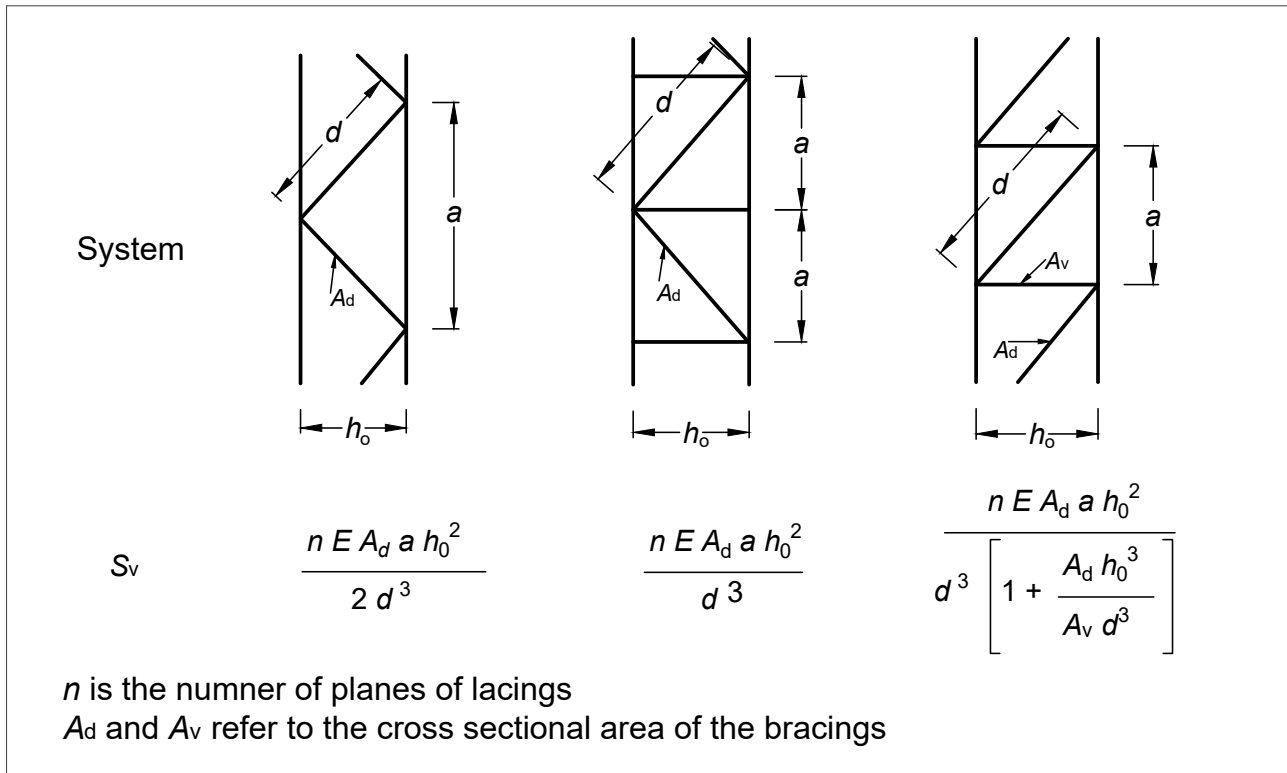
$N_{b,Rd}$ = The design value of the buckling resistance of the chord taking the buckling length L_{ch} from Figure 5.22

The shear stiffness S_v of the lacings should be taken from Figure 5.23.

The effective second order moment of area of laced built-up members may be taken as:

$$I_{eff} = 0.5h_0^2 A_{ch}$$

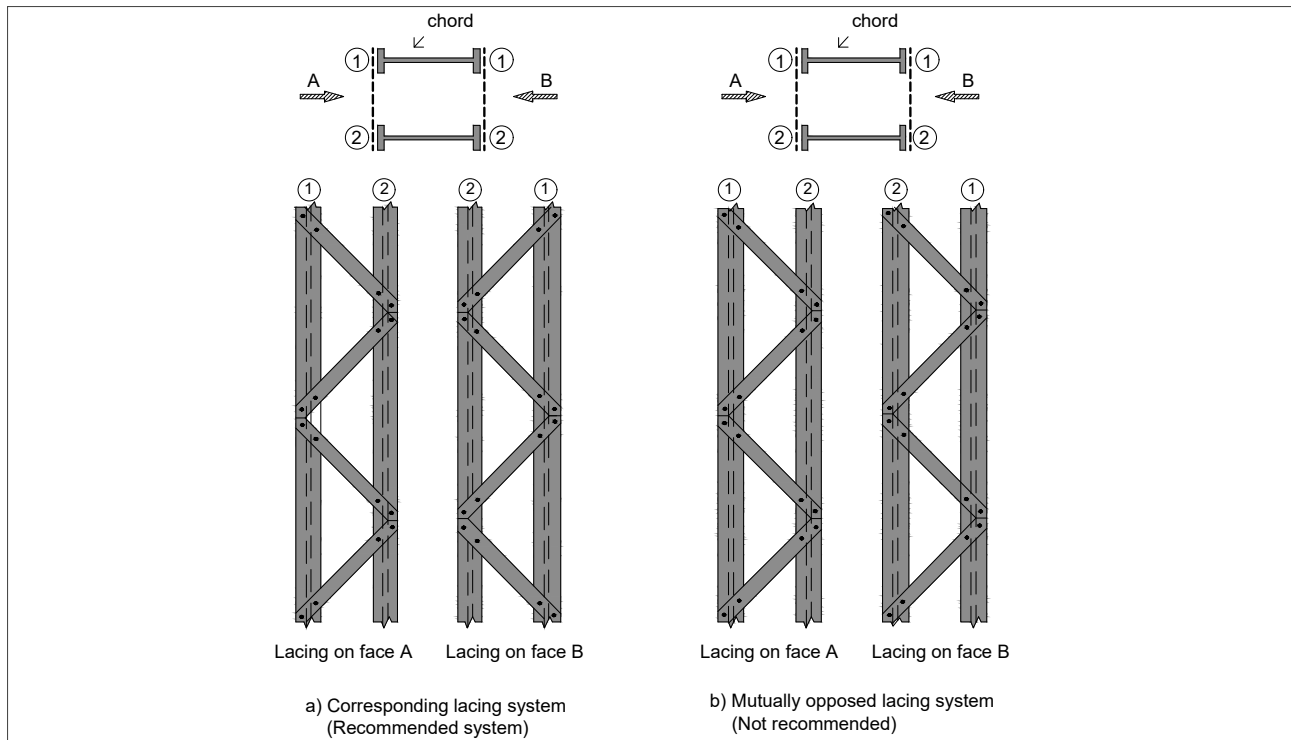
Figure 5.23 Shear Stiffness of Lacings of Built-Up Members



5.6.19.2.2 Constructional Details

Single lacing systems in opposite faces of the built-up member with two parallel laced planes should be corresponding systems as shown in Figure 5.24(a), arranged so that one is the shadow of the other. When the single lacing systems on opposite faces of a built-up member with two parallel laced planes are mutually opposed in direction as shown in Figure 5.24(b), the resulting torsional effects in the member should be taken into account. Tie panels should be provided at the ends of lacing systems, at points where the lacing is interrupted and at joints with other members.

Figure 5.24 Single Lacing System on Opposite Faces of a Built-Up Member With Two Parallel Laced Planes



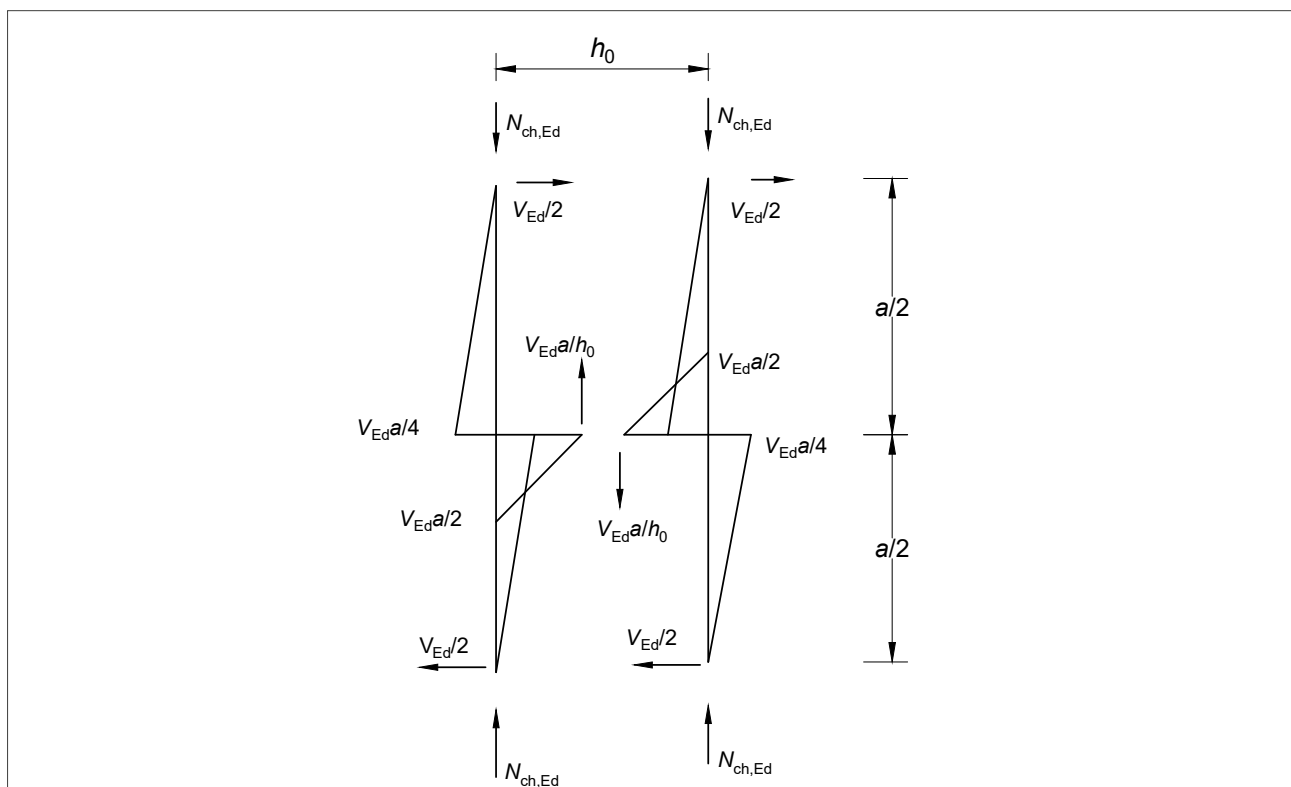
5.6.19.3 Battened Compression Members

5.6.19.3.1 Resistance of Components of Battened Compression Members

The chords and the battens and their joints to the chords should be checked for the actual moments and forces in an end panel and at mid-span as indicated in Figure 5.25.

Note: For simplicity the maximum chord forces $N_{ch,Rd}$ may be combined with the maximum shear force V_{Ed} .

Figure 5.25 Moments and Forces in an End Panel of a Battened Built-up Member



The shear stiffness S_v should be taken as follows:

$$S_v = \frac{24EI_{ch}}{a^2 \left[1 + \frac{2I_{ch}}{nI_b} \frac{h_0}{a} \right]} \leq \frac{2\pi^2 EI_{ch}}{a^2} \quad \text{Equation 5.98}$$

The effective second moments of area of battened built-up members may be taken as:

$$I_{eff} = 0.5h_0^2 A_{ch} + 2\mu I_{ch} \quad \text{Equation 5.99}$$

Where,

I_{ch} = in plane second moment of area of one chord.

I_b = in plane second moment of area of one batten.

μ = efficiency factor from Table 5.18.

n = number of planes of battens.

Table 5.18 Efficiency Factor μ

Criterion	Efficiency factor μ
$\lambda \geq 150$	0
$75 < \lambda < 150$	$\mu = 2 - \lambda/75$
$\lambda \leq 75$	1.0
Where, $\lambda = \frac{L}{i_0}$; $i_0 = \sqrt{I_1/2A_{ch}}$; $I_1 = 0.5h_0^2 A_{ch} + 2I_{ch}$	

5.6.19.3.2 Design Details

Battens should be provided at each end of a member

Where parallel planes of battens are provided, the battens in each plane should be arranged opposite each other.

Battens should also be provided at intermediate points where loads are applied or lateral restraint is supplied.

5.6.19.4 Closely Spaced Built-up Members

Built-up compression members with chords in contact or closely spaced and connected through packing plates, see Figure 5.26, or star battened angle members connected by pairs of battens in two perpendicular planes, see Figure 5.27, should be checked for buckling as a single integral member ignoring the effect of shear stiffness ($S_v = \infty$), when the conditions in Table 5.19 are met.

Figure 5.26 Closely Spaced Built-Up Members

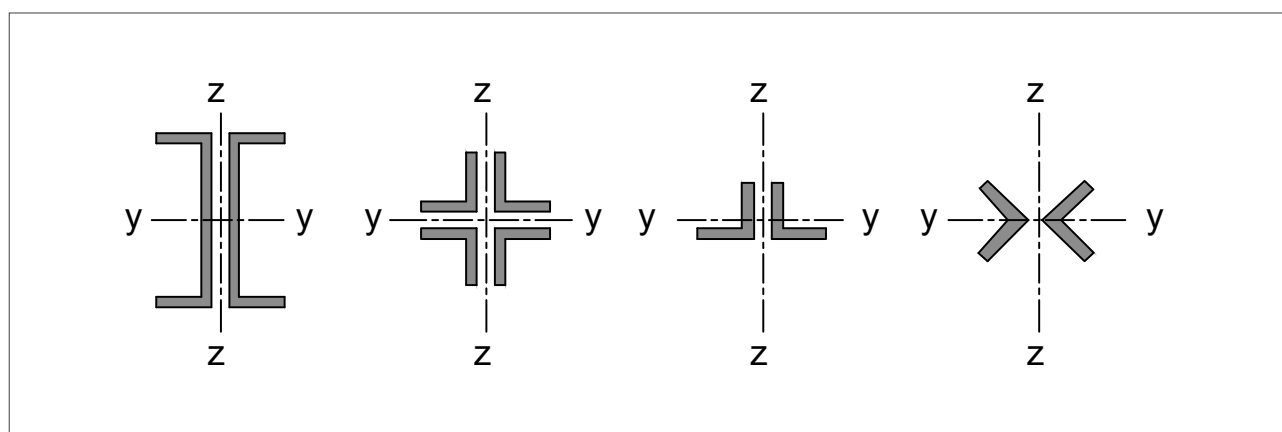


Table 5.19 Maximum Spacings for Interconnections in Closely Spaced Built-Up Members

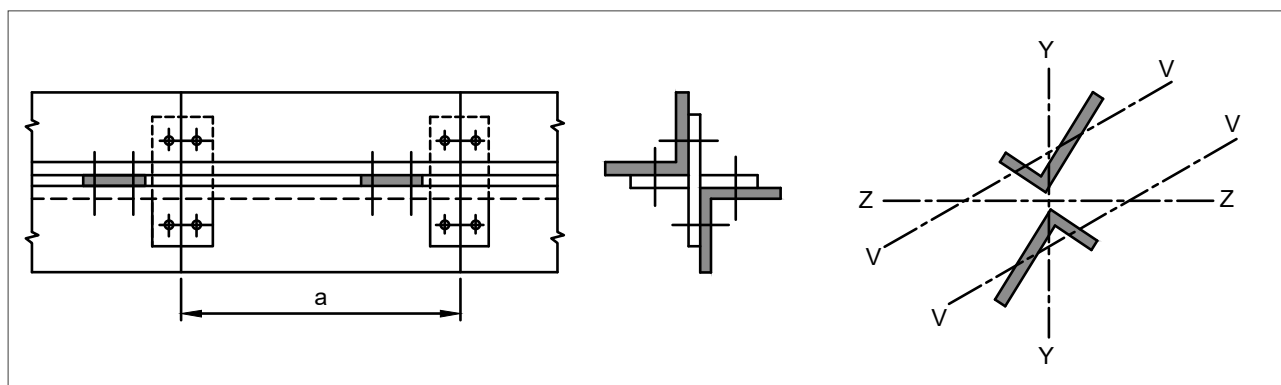
Type of built-up member	Maximum spacing between interconnections *
Members according to Figure 5.26 connected by bolts or welds	$15 i_{min}$
Members according to Figure 5.27 connected by pair of battens	$70 i_{min}$
*Centre-to-centre distance of interconnections i_{min} is the minimum radius of gyration of one chord or one angle	

The shear forces to be transmitted by the battens should be determined from Clause 5.6.19.3.1. In the case of unequal-leg angles, refer Figure 5.27, buckling about the y-y axis may be verified with:

$$i_y = \frac{i_0}{1.15}$$

Equation 5.100

Where, i_0 is the minimum radius of gyration of the built-up member.

Figure 5.27 Star Battened Angle Members

5.6.20 Buckling of Plates

5.6.20.1 General

The plate buckling verification of members at the ultimate limit state should be carried out using either a) or b) as follows:

- Direct stresses, shear stresses and transverse forces verification as per the provisions for the plate elements in Clause 5.4, 5.5 and 5.6. Additionally, the interaction criteria in Clause 5.6.20.3 should also be met.
- Reduced stress method on the basis of stress limits governed by local buckling according to Clause 5.6.20.2.

For web stiffeners or stiffened deck plates which are subjected to compression and additional bending moments from loads transverse to the plane of the stiffened plate, the stability may be verified in accordance with Clause 5.6.18.2.3.

5.6.20.2 Reduced Stress Method

The reduced stress method may be used to determine the stress limits for stiffened or unstiffened plates.

This method is an alternative to the effective width method in respect of the following:

- $\sigma_{x,Ed}$, $\sigma_{z,Ed}$ and τ_{Ed} are considered as acting together.
- The stress limits of the weakest part of the cross section may govern the resistance of the full cross section.

For unstiffened or stiffened panels subjected to combined stresses $\sigma_{x,Ed}$, $\sigma_{z,Ed}$ and τ_{Ed} , class 3 section properties may be assumed, where:

$$\frac{\rho \alpha_{ult,k}}{\gamma_{M1}} \geq 1 \quad \text{Equation 5.101}$$

Where,

$\alpha_{ult,k}$ = The minimum load amplifier for the design loads to reach the characteristic value of resistance of the most critical point of the plate.

ρ = The reduction factor depending on the plate slenderness $\bar{\lambda}_p$ to take account of plate buckling

γ_{M1} = The partial factor applied to this method.

The modified plate slenderness $\bar{\lambda}_p$ should be taken from:

$$\bar{\lambda}_p = \sqrt{\frac{\alpha_{ult,k}}{\alpha_{cr}}} \quad \text{Equation 5.102}$$

Where,

α_{cr} = The minimum load amplifier for the design loads to reach the elastic critical load of the plate under the complete stress field.

Note 1: For calculating α_{cr} for the complete stress field, the stiffened plate may be modelled without reduction of the second moment of area of longitudinal stiffeners.

Note 2: When α_{cr} cannot be determined for the panel and its subpanels as a whole, separate checks for the subpanel and the full panel may be applied.

In determining $\alpha_{ult,k}$, the yield criterion may be used for resistance:

$$\frac{1}{\alpha_{ult,k}^2} = \left(\frac{\sigma_{x,Ed}}{f_y} \right)^2 + \left(\frac{\sigma_{z,Ed}}{f_y} \right)^2 - \left(\frac{\sigma_{x,Ed}}{f_y} \right) \left(\frac{\sigma_{z,Ed}}{f_y} \right) + 3 \left(\frac{\tau_{Ed}}{f_y} \right)^2 \quad \text{Equation 5.103}$$

Where, $\sigma_{x,Ed}$, $\sigma_{z,Ed}$ and τ_{Ed} are the components (longitudinal direction, the direct stress in the transverse direction and the shear stress) of the stress field in the ultimate limit state.

Note: By using the above Expression, it is assumed that the resistance is reached when yielding occurs without plate buckling.

The reduction factor ρ may be determined using either of the following methods:

a. The minimum value of the following reduction factors:

ρ_x for longitudinal stresses taking into account column-like behaviour where relevant

ρ_z for transverse stresses taking into account column-like behaviour where relevant

χ_w for shear

Each calculated for the modified plate slenderness $\bar{\lambda}_p$.

Note: This method leads to the verification formula:

$$\left(\frac{\sigma_{x,Ed}}{f_y / \gamma_{M1}} \right)^2 + \left(\frac{\sigma_{z,Ed}}{\rho_z f_y / \gamma_{M1}} \right)^2 - \left(\frac{\sigma_{x,Ed}}{f_y / \gamma_{M1}} \right) \left(\frac{\sigma_{z,Ed}}{\rho_z f_y / \gamma_{M1}} \right) + 3 \left(\frac{\tau_{Ed}}{\chi_w f_y / \gamma_{M1}} \right)^2 \leq \rho^2 \quad \text{Equation 5.104}$$

b. A value interpolated between the values of ρ_x , ρ_z and χ_w as determined in a) by using the formula for $\alpha_{ult,k}$ as interpolation function. This method leads to verification of following:

$$\left(\frac{\sigma_{x,Ed}}{\rho_x f_y / \gamma_{M1}} \right)^2 + \left(\frac{\sigma_{z,Ed}}{\rho_z f_y / \gamma_{M1}} \right)^2 - \left(\frac{\sigma_{x,Ed}}{\rho_x f_y / \gamma_{M1}} \right) \left(\frac{\sigma_{z,Ed}}{\rho_z f_y / \gamma_{M1}} \right) + 3 \left(\frac{\tau_{Ed}}{\chi_w f_y / \gamma_{M1}} \right)^2 \leq 1 \quad \text{Equation 5.105}$$

1

For panels with tension and compression, the Equations 5.104 and 5.105 shall be applied only for compression parts.

2

Where α_{cr} values for the complete stress field are not available and only $\alpha_{cr,i}$ values for the various components of the stress field $\sigma_{x,Ed}$, $\sigma_{z,Ed}$ and τ_{Ed} can be used, the α_{cr} value may be determined from:

3

$$\frac{1}{\alpha_{cr}} = \frac{1 + \psi_x}{4 \alpha_{cr,x}} + \frac{1 + \psi_z}{4 \alpha_{cr,z}} + \left[\left(\frac{1 + \psi_x}{4 \alpha_{cr,x}} + \frac{1 + \psi_z}{4 \alpha_{cr,z}} \right)^2 + \frac{1 - \psi_x}{2 \alpha_{cr,x}^2} + \frac{1 - \psi_z}{2 \alpha_{cr,z}^2} + \frac{1}{\alpha_{cr,\tau}^2} \right]^{1/2}$$

Equation 5.106

Where,

4

$$\alpha_{cr,x} = \frac{\sigma_{cr,x}}{\sigma_{x,Ed}}$$

Equation 5.107

$$\alpha_{cr,z} = \frac{\sigma_{cr,z}}{\sigma_{z,Ed}}$$

Equation 5.108

5

$$\alpha_{cr,\tau} = \frac{\tau_{cr}}{\tau_{Ed}}$$

Equation 5.109

Stiffeners and detailing of plate panels should be designed according to section 5.6.21.

5.6.20.3 Interaction Between Shear Force, Bending Moment and Axial Force

Provided that $\bar{\eta}_3$ (see below) does not exceed 0.5, the design resistance to bending moment and axial force need not be reduced to allow for the shear force. If $\bar{\eta}_3$ is more than 0.5, the combined effects of bending and shear in the web of an I or box girder should satisfy:

$$\bar{\eta}_1 + \left(1 - \frac{M_{f,Rd}}{M_{pl,Rd}} \right) (2\bar{\eta}_3 - 1)^2 \leq 1.0 \quad \text{for } \bar{\eta}_1 \geq \frac{M_{f,Rd}}{M_{pl,Rd}}$$

Equation 5.110

Where,

$M_{f,Rd}$ = The design plastic moment of resistance of the section consisting of the effective area of the flanges.

$M_{pl,Rd}$ = The design plastic resistance of the cross section consisting of the effective area of the flanges and the fully effective web irrespective of its section class.

$$\bar{\eta}_1 = \frac{M_{Ed}}{M_{pl,Rd}}$$

Equation 5.111

$$\bar{\eta}_3 = \frac{V_{Ed}}{V_{bw,Rd}} \quad \text{for } V_{bw,Rd} \quad \text{Refer to Equation 5.48.}$$

Equation 5.112

In addition, the requirements in Clause 5.6.8.5.5. and 5.6.12.2.3 should be met. Action effects should include global second order effects of members where relevant.

The criterion given above should be verified at all sections other than those located at a distance less than $h_w/2$ from a support with vertical stiffeners.

The plastic moment of resistance $M_{f,Rd}$ may be taken as the product of the yield strength, the effective area of the flange with the smallest value of $A_f f_y / \gamma_{M0}$ and the distance between the centroids of the flanges. If an axial force N_{Ed} is present, $M_{pl,Rd}$ and $M_{f,Rd}$ should be reduced as in Clause 5.6.16 and 5.6.12.2.2 respectively.

When axial force is large such that the whole web section is in compression, following provision shall be applied, a flange in a box girder should be verified taking $M_{f,Rd} = 0$. The τ_{Ed} shall be taken as the average shear stress in the flange which should not be less than half the maximum shear stress in the flange and $\bar{\eta}_1$ is taken as η_1 . In addition, the subpanels should be checked using the average

shear stress within the subpanel and χ_w determined for shear buckling of the subpanel according to Clause 5.6.12.2.1 assuming the longitudinal stiffeners to be rigid.

5.6.20.4 Interaction Between Transverse Force, Bending Moment and Axial Force

If the girder is subjected to a concentrated transverse force acting on the compression flange in conjunction with bending and axial force, the resistance should be verified using the following interaction Equation:

$$\eta_2 + 0.8 \eta_1 \leq 1.4$$

Where,

η_1 = According to Clause 5.6.8.5.5.

$\eta_2 = \frac{F_{Ed}}{f_{yw} L_{eff} t_w I \gamma_{M1}}$ According to Clause 5.6.8.5.5.

F_{Ed} = Is the design transverse force.

L_{eff} = Is the effective length for resistance.

t_w = Is the thickness of the plate.

5.6.21 Stiffeners and Detailing

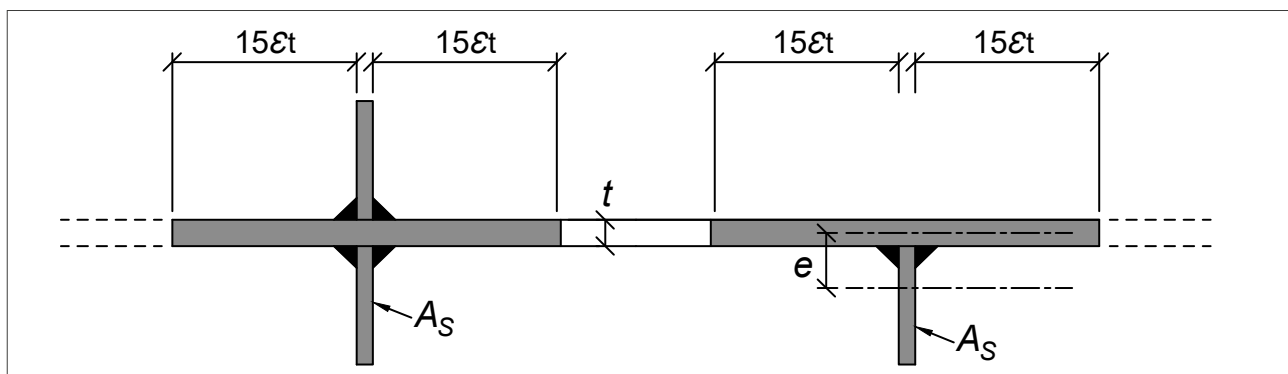
5.6.21.1 General

This section gives design rules for stiffeners in plated structures.

Where the web of a beam is subjected to a net compressive force, whether due to externally applied compressive force or due to the flexural neutral axis not being at the mid-height of the web, the effect of the net compressive force should be considered in the design of rigid transverse stiffeners resisting shear.

When checking the buckling resistance, the section of a stiffener may be taken as the gross area comprising the stiffener plus a width of plate equal to $15\epsilon t$ but not more than the actual dimension available, on each side of the stiffener avoiding any overlap of contributing parts to adjacent stiffeners, see Figure 5.28. The axial force in a transverse stiffener should be taken as the sum of the force resulting from shear and any external loads.

Figure 5.28 Effective Cross-section of Stiffener



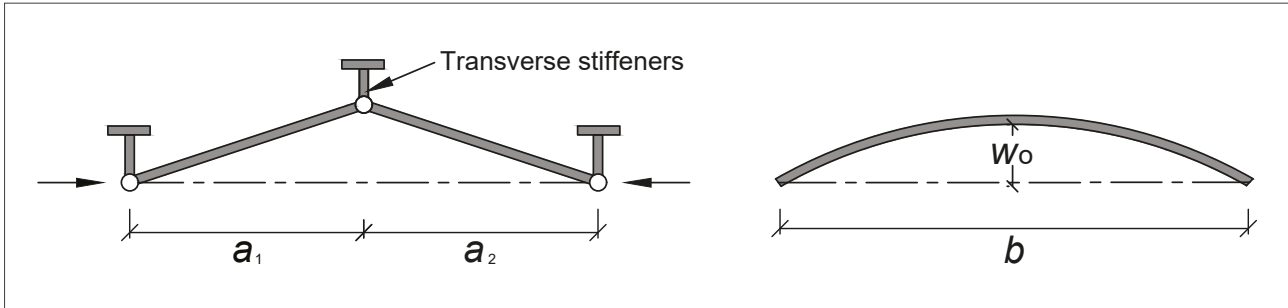
5.6.21.2 Direct Stresses

5.6.21.2.1 Minimum Requirements For Transverse Stiffeners

In order to provide a rigid support for a plate with or without longitudinal stiffeners, intermediate transverse stiffeners should satisfy the criteria that follows.

The transverse stiffener should be treated as a simply supported member subject to lateral loading with an initial sinusoidal imperfection w_0 equal to $s/300$, where s is the smallest of a_1 , a_2 or b , refer Figure 5.29, where a_1 and a_2 are the lengths of the panels adjacent to the transverse stiffener under consideration and b is the height between the centroids of the flanges or span of the transverse stiffener. Eccentricities should be accounted for.

Figure 5.29 Transverse Stiffener



The transverse stiffener should carry the deviation forces from the adjacent compressed panels under the assumption that both adjacent transverse stiffeners are rigid and straight together with any external load and axial force. The compressed panels and the longitudinal stiffeners are considered to be simply supported at the transverse stiffeners.

It should be verified that using a second order elastic method analysis both the following criteria are satisfied at the ultimate limit state:

- That the maximum stress in the stiffener should not exceed f_y/γ_{M1} .
- That the additional deflection should not exceed $b/300$.

In the absence of an axial force in the transverse stiffener both the criteria in above may be assumed to be satisfied provided that the second moment of area I_{st} of the transverse stiffeners is not less than:

$$I_{st} = \frac{\sigma_m}{E} \left(\frac{b}{\pi} \right)^4 (1 + w_0 \frac{300}{b} u)$$

Equation 5.113

Where,

$$\sigma_m = \frac{\sigma_{cr,c}}{\sigma_{cr,p}} \frac{N_{Ed}}{b} \left(\frac{1}{a_1} + \frac{1}{a_2} \right)$$

$$u = \frac{\pi^2 E e_{max}}{f_y 300 b / \gamma_{M1}} \geq 1.0$$

e_{max} = The maximum distance from the extreme fibre of the stiffener to the centroid of the stiffener.

N_{Ed} = The maximum compressive force of the adjacent panels but not less than the maximum compressive stress times half the effective^P compression area of the panel including stiffeners.

When applying this Clause, the $\sigma_{cr,c}$ and $\sigma_{cr,p}$ should generally be calculated for a panel length of $a_1 + a_2$ with w_0 taken as the lesser of $b/300$, $a_1/300$ and $a_2/300$. However, to safeguard against a mode of buckling with adjacent transverse stiffeners bowing in alternate directions, it should also be applied with $\sigma_{cr,c}$ and $\sigma_{cr,p}$ calculated for a panel length of $0.5(a_1 + a_2)$. In this case, w_0 should be taken as the lesser of $b/150$, $a_1/150$ and $a_2/150$.

For panels with longitudinal stiffeners, the ratio $\sigma_{cr,c} / \sigma_{cr,p}$ should be taken to be equal to 1

If the stiffener carries axial compression this should be increased by $\Delta N_{st} = \sigma_m b^2 / \pi^2$ in order to account for deviation forces.

Unless a more advanced method of analysis is carried out in order to prevent torsional buckling of stiffeners with open cross-sections, the following criterion should be satisfied:

$$\frac{I_T}{I_p} \geq 5.3 \frac{f_y}{E}$$

Equation 5.114

Where,

I_p = The polar second moment of area of the stiffener alone around the edge fixed to the plate

I = The St. Venant torsional constant for the stiffener alone

Where warping stiffness is considered, stiffeners should criterion

$$\sigma_{cr} \geq \theta f_y$$

Where,

σ_{cr} = The elastic critical stress for torsional buckling not considering rotational restraint from the plate.

θ = A parameter to ensure class 3 behaviour, may be taken as $\theta = 2$.

5.6.21.2.2 Minimum Requirements for Longitudinal Stiffeners

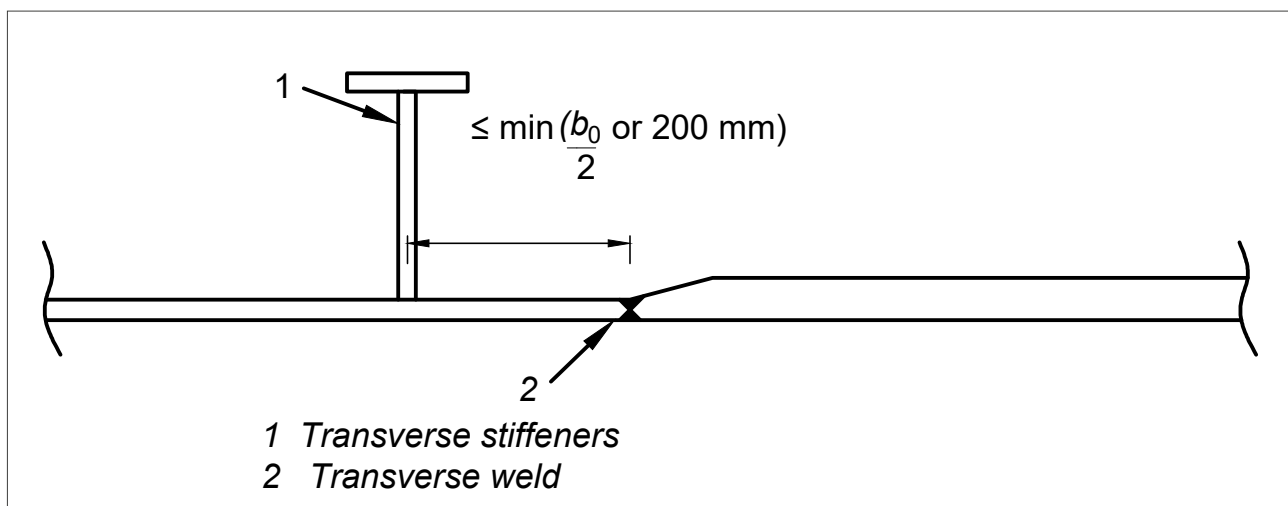
Discontinuous longitudinal stiffeners that do not pass through openings made in the transverse stiffeners or are not connected to either side of the transverse stiffeners should be:

- Used only for webs (i.e., not allowed in flanges)
- Neglected in global analysis
- Neglected in the calculation of stresses
- Considered in the calculation of the effective^P widths of web sub-panels
- Considered in the calculation of the elastic critical stresses.
- Strength assessments for stiffeners should be performed according to Clause 5.6.8.5.3 and 5.6.8.5.5.

5.6.21.2.3 Welded Plates

Plates with changes in plate thickness should be welded adjacent to the transverse stiffener, see Figure 5.30. The effects of eccentricity need not be taken into account unless the distance to the stiffener from the welded junction exceeds $b_0/2$ or 200 mm whichever is the smallest, where b_0 is the width of the plate between longitudinal stiffeners.

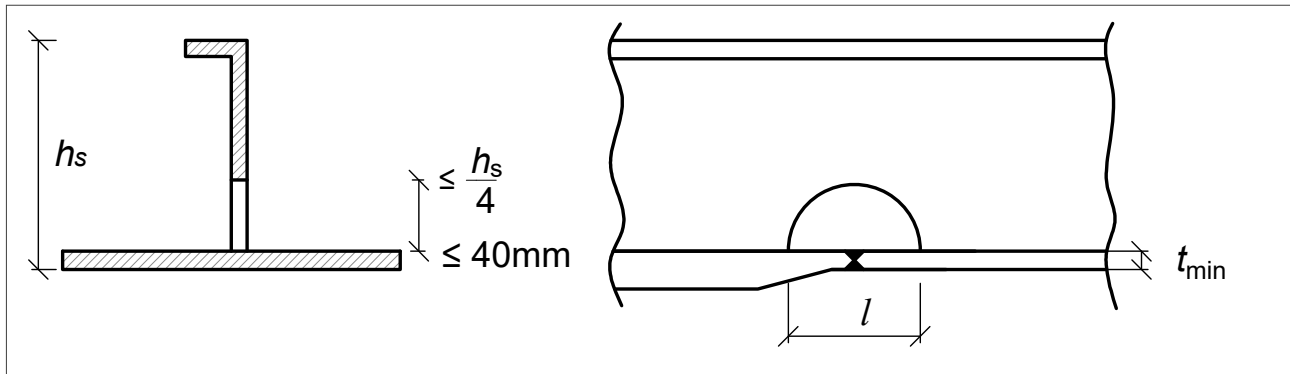
Figure 5.30 Welded Plates



5.6.21.2.4 Cut Outs in Stiffeners

The dimensions of cut outs in longitudinal stiffeners should be as shown in Figure 5.31.

Figure 5.31 Cut Outs in Longitudinal Stiffeners



The length l should not exceed:

$l \leq 6 t_{min}$ for flat stiffeners in compression.

$l \leq 8 t_{min}$ for other stiffeners in compression.

$l \leq 15 t_{min}$ for stiffeners without compression.

Where,

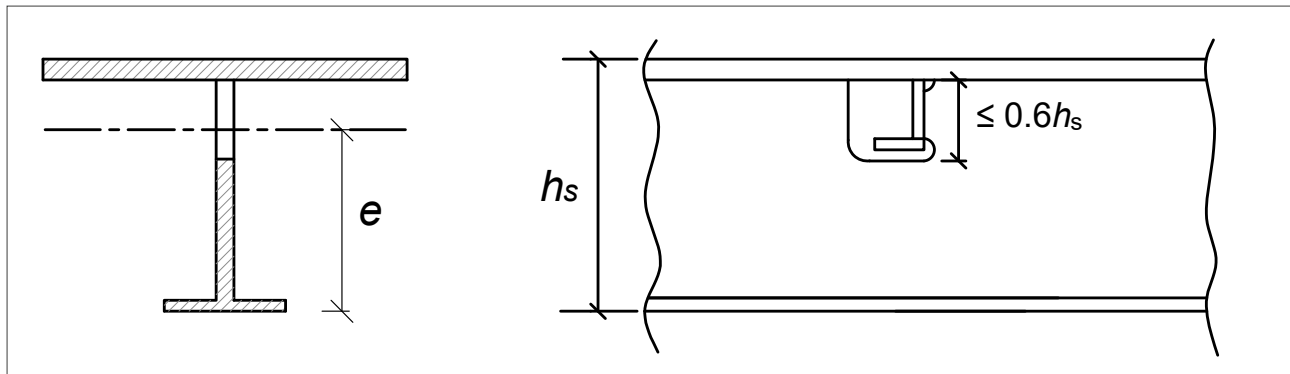
t_{min} is the lesser of the plate thicknesses

The limiting values l in above point for stiffeners in compression may be increased by $\sqrt{\frac{\sigma_{x,Rd}}{\sigma_{x,Ed}}}$ when $\sigma_{x,Ed} \leq \sigma_{x,Rd}$ and $l \leq 15t_{min}$.

$\sigma_{x,Ed}$ is the compression stress at the location of the cut-out.

The dimensions of cut outs in transverse stiffeners should be as shown in Figure 5.32.

Figure 5.32 Cut Outs in Traverse Stiffeners



The gross web adjacent to the cut out should resist a shear force V_{Ed} , where:

$$V_{Ed} = \frac{I_{net}}{e} \frac{f_{yk}}{\gamma_{M0}} \frac{\pi}{b_G}$$

Equation 5.115

Where,

I_{net} = The second moment of area for the net section of the transverse stiffener

e = The maximum distance from the underside of the flange plate to the neutral axis of net section

b_G = The length of the transverse stiffener between the flanges.

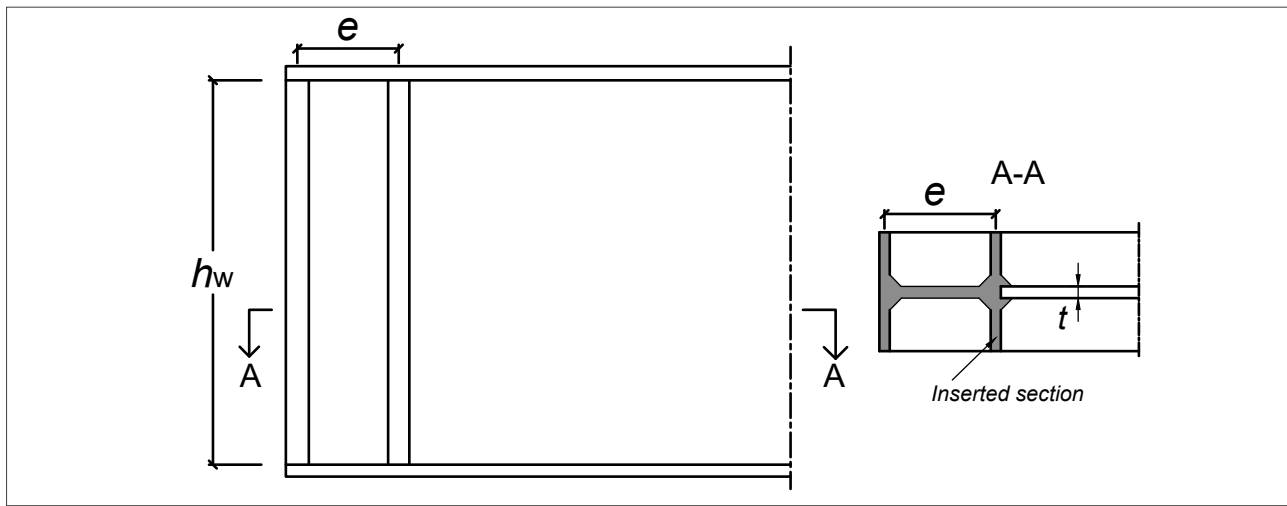
5.6.21.3 Shear

5.6.21.3.1 Rigid End Post

The rigid end post should act as a bearing stiffener resisting the reaction from the support, and should be designed as a short beam resisting the longitudinal membrane stresses in the plane of the web.

A rigid end post should comprise of two double-sided transverse stiffeners that form the flanges of a short beam of length h_w , the strip of web plate between the stiffeners forms the web of the short beam. Alternatively, a rigid end post may be in the form of a rolled section, connected to the end of the web plate as shown in Figure 5.33.

Figure 5.33 Rolled Section Forming an End Post



Each double-sided stiffener consisting of flats should have a cross sectional area of at least $4h_w t^2 / e$, where e is the centre-to-centre distance between the stiffeners and $e > 0.1 h_w$. Where a rolled section other than flats is used for the end-post its section modulus should be not less than $4h_w t^2$ for bending around a horizontal axis perpendicular to the web.

As an alternative the girder end may be provided with a single double-sided stiffener and a vertical stiffener adjacent to the support so that the subpanel resists the maximum shear when designed with a nonrigid end post.

5.6.21.3.2 Stiffeners Acting as Non-rigid End Post

A non-rigid end post may be a single double-sided. It may act as a bearing stiffener resisting the reaction at the girder support.

5.6.21.3.3 Intermediate Transverse Stiffeners

Intermediate stiffeners that act as rigid supports to interior panels of the web should be designed for strength and stiffness.

When flexible intermediate transverse stiffeners are used, their stiffness should be considered in the calculation of k_τ .

The effective section of intermediate stiffeners acting as rigid supports for web panels should have a minimum second moment of area I_{st} :

$$\text{if } a / h_w < \sqrt{2} : I_{st} \geq 1.5 h_w^3 t^3 / a^2$$

$$\text{if } a / h_w \geq \sqrt{2} : I_{st} \geq 0.75 h_w^3 t^3$$

1

Intermediate rigid stiffeners may be designed for an axial force ρ_{Ed} acting in the mid-plane of the web in addition to any externally applied loads, as below.

$$\rho_{Ed} = V_{Ed} - h_w t 0.8 \tau \sqrt{1 - \frac{\sigma_{x,Ed}}{0.8 \sigma_{cr,x}}} \quad \text{for } a \geq h_w$$

Equation 5.116

2

$$\rho_{Ed} = \left[V_{Ed} - h_w t 0.8 \tau \sqrt{1 - \frac{\sigma_{x,Ed}}{0.8 \sigma_{cr,x}}} \right] \frac{a}{h_w} \quad \text{for } a < h_w$$

Equation 5.117

3

Where,

τ_{cr} = The elastic critical shear buckling stress determined as in 5.6.12.2.1.

$\sigma_{cr,x}$ = The elastic critical plate buckling stress for direct stress determined as in 5.6.8.4 or 5.6.8.5 for sub-panel buckling and overall stiffened plate buckling respectively.

$\sigma_{x,Ed}$ = The greatest longitudinal compressive stress in the web panel under consideration. It is taken as positive when compressive and should not be taken as being greater than $0.8 \sigma_{cr,x}$.

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5.6.21.3.4 Longitudinal Stiffeners

If longitudinal stiffeners are taken into account in the stress analysis, they should be checked for direct stresses for the cross-sectional resistance.

5.6.21.3.5 Welds

The web to flange welds may be designed for the nominal shear flow V_{Ed}/h_w if V_{Ed} does not exceed $\chi_w f_{yw} h_w t / (\sqrt{3} \gamma_{M1})$. For larger values V_{Ed} the weld between flanges and webs should be designed for the shear flow $\eta f_{yw} t / (\sqrt{3} \gamma_{M1})$.

In all other cases welds should be designed to transfer forces along and across welds making up sections taking into account analysis method (elastic/plastic) and second order effects.

5.6.21.3.6 Transverse Loads

If the design resistance of an unstiffened web is insufficient, transverse stiffeners should be provided.

The out-of-plane buckling resistance of the transverse stiffener under transverse loads and shear force should be determined using buckling curve *c*. When both ends are assumed to be fixed laterally a buckling length *l* of not less than $0.75 h_w$ should be used. A larger value of *l* should be used for conditions that provide less end restraint. If the stiffeners have cut outs at the loaded end, the cross-sectional resistance should be checked at this end.

Where single sided or other asymmetric stiffeners are used, the resulting eccentricity should be allowed. If the stiffeners are assumed to provide lateral restraint to the compression flange, they should comply with the stiffness and strength criteria in the design for lateral torsional buckling.

5.7 Serviceability Limit States

5.7.1 General

A steel structure should be designed and constructed such that all relevant serviceability criteria are satisfied.

Any serviceability limit state and the associated loading and analysis model should be specified for a project.

The following serviceability criteria should be met:

- a. Restriction to elastic behaviour in order to limit:
 - Excessive yielding
 - Deviations from the intended geometry by residual deflections
 - Excessive deformations
- b. Limitation of deflections and curvature in order to prevent:
 - Unwanted dynamic impacts due to traffic (combination of detection and natural frequency limitations)
 - Infringement of required clearances
 - Cracking of surfacing layers
 - Damage of drainage
- c. Limitation of natural frequencies, in order to:
 - Exclude vibrations due to traffic or wind which are unacceptable to pedestrians or passengers in cars using the bridge
 - Limit fatigue damages caused by resonance
 - Limit excessive noise emission
- d. Restriction of plate slenderness, in order to limit:
 - Excessive rippling of plates
 - Breathing of plates
 - Reduction of stiffness due to plate buckling, resulting in an increase of deflection
- e. Improved durability by appropriate detailing to reduce corrosion and excessive wear
- f. Ease of maintenance and repair, to ensure:
 - Accessibility of structural parts for maintenance and inspection, renewal of corrosion protection and asphaltic pavements
 - Replacement of bearings, anchors, cables, expansion joints with minimum disruption to the use of the structure

In most situations serviceability aspects should be dealt with in the conceptual design of the bridge, or by suitable detailing. However, in appropriate cases, serviceability limit states may be verified by numerical assessment, e.g. for calculating deflections or eigen frequencies.

5.7.2 Calculation Model

Stresses at serviceability limit states should be determined from a linear elastic analysis, using the appropriate section properties.

In modelling the structure, the non-uniform distribution of loads and stiffness resulting from the changes in plate thickness, stiffening etc. should be taken into account.

Deflections should be determined by linear elastic analysis using the appropriate section properties.

5.7.3 Limitations for Stress

The nominal stresses $\sigma_{Ed,ser}$ and $\tau_{Ed,ser}$ resulting from the characteristic load combinations calculated making due allowance for the effects of shear lag in flanges and the secondary effects caused by deflections (e.g. secondary moments in trusses), should be limited as follows:

$$\sigma_{Ed,ser} \leq \frac{f_y}{\gamma_{M,ser}} \quad \text{Equation 5.118}$$

$$\tau_{Ed,ser} \leq \frac{f_y}{\sqrt{3} \gamma_{M,ser}} \quad \text{Equation 5.119}$$

$$\sqrt{\sigma_{Ed,ser}^2 + 3\tau_{Ed,ser}^2} \leq \frac{f_y}{\gamma_{M,ser}} \quad \text{Equation 5.120}$$

Note 1: Where relevant the above checks should include stresses σ_z from transverse loads.

Note 2: The $\gamma_{M,ser} = 1.00$ is recommended.

Note 3: Plate buckling effects may be ignored for global analysis when the effective cross-sectional area of an element in compression is larger than ρ_{lim} times the gross cross-sectional area of the same element. Value of $\rho_{lim} = 0.5$ is recommended.

The nominal stress range $\Delta\sigma_{fre}$, due to the frequent load combination should be limited to $1.5f_y / \gamma_{M,ser}$.

For non-preloaded bolted connections subject to shear, the bolt forces due to the characteristic load combination should be limited to:

$$F_{b,Rd,ser} \leq 0.7 F_{b,Rd}$$

Where $F_{b,Rd}$ is the bearing resistance for ultimate limit states verifications.

For slip-resistant preloaded bolted connections category B (slip resistant at serviceability), the assessment for serviceability should be carried out using the characteristic load combination.

5.7.4 Limitation of Web Breathing

The slenderness of web plates should be limited to avoid excessive breathing that might result in fatigue at or adjacent to the web-to-flange connections.

Web breathing may be neglected for web panels without longitudinal stiffeners or for subpanels of stiffened webs, where the following criteria are met:

$$\frac{b}{t} \leq 30 + 4.0 L \leq 300$$

Where L is the span length in m, but not less than 20 m

If the above provision is not satisfied web breathing should be checked as follows:

$$\sqrt{\left(\frac{\sigma_{x,Ed,ser}}{k_\sigma \sigma_E}\right)^2 + \left(\frac{1.1 \tau_{x,Ed,ser}}{k_\tau \sigma_E}\right)^2} \leq 1.1 \quad \text{Equation 5.121}$$

Where,

$\sigma_{x,Ed,ser}$, $\tau_{Ed,ser}$ are the stresses for the frequent load combination. If the stresses are not uniform along the length of the panel, the verification shall be carried out for the stress resultant at a distance of $0.4a$ or $0.5b$ whichever is smallest, from the panel end where the stresses are the greater.

$$\sigma E = 190000 (t / b_p)^2 \text{ in } N/mm^2$$

b_p is the smaller of a and b .

5.7.5 Limits for Clearance Gauges

Specified clearance gauges should be maintained without encroachment by any part of the structure under the effects of the characteristic load combination.

5.7.6 Limits for Visual Impression

To achieve a satisfactory appearance of the bridge, consideration should be given to pre-cambering.

In calculating camber, the effects of shear deformation and slip in riveted or bolted connections should be considered.

For connections with rivets or fitted bolts, a fastener slip of 0.2 mm should be assumed. For preloaded bolts, slip does not need to be considered.

5.7.7 Performance Criteria for Road Bridges

5.7.7.1 General

Excessive deformations should be avoided where it could:

- Endanger traffic by excessive transverse slope when the surface is iced
- Affect the dynamic load on the bridge by impact from wheels
- Affect the dynamic behaviour causing discomfort to users
- Lead to cracks in asphaltic surfacing
- Adversely affect the drainage of water from the bridge deck

Deformations should be calculated using the frequent load combination.

The natural frequency of vibrations and deflections of the bridge should be limited to avoid discomfort to users.

5.7.7.2 Deflection Limits to Avoid Excessive Impact from Traffic

The deck structure should be designed to ensure that its deflection along the length is uniform and that there is no abrupt change in cross section giving rise to impact. Sudden changes in the slope of the deck and changes of level at the expansion joints should be eliminated. Any transverse girders at the end of the bridge should be designed to ensure that the deflection does not exceed:

- The limit specified for the proper functioning of the expansion joint
- 5 mm under frequent loads unless other limits are specified for the particular type of expansion joint.

Where the deck structure is irregularly supported (e.g. by additional bracings at intermediate bridge piers), the deck area adjacent to these additional deck supports should be designed for the enhanced impact factors given in Chapter 3 for the area close to the expansion joints.

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5.7.7.3 Resonance Effects

Mechanical resonance should be taken into account when relevant. Where light bracing members, cable stays or similar components have natural frequencies that are close to the frequency of any mechanical excitation due to regular passage of vehicles over deck joints, consideration should be given to either increasing the stiffness or providing artificial dampers, i.e. oscillation dampers.

5.7.8 Performance Criteria for Pedestrian Bridges

For footbridges and cycle bridges with excessive vibrations could cause discomfort to users, measures should be taken to minimise such vibrations by designing the bridge with appropriate natural frequency or by providing suitable damping devices. Refer Clause 9.8.7 and 3.13.5 for pedestrian traffic class and comfort criteria for design of footbridges.

5.7.9 Performance Criteria for the Effect of Wind

Vibrations of slender members induced by vortex excitation should be minimised to prevent repetitive stresses of sufficient magnitude that could cause fatigue. Reference can be made to literature specific to this effect.

5.7.10 Accessibility of Joint Details and Surfaces

All steel work should be designed and detailed to minimise the risk of corrosion and to permit inspection and maintenance.

All parts should normally be designed to be accessible for inspection, cleaning and painting. Where such access is not possible, all inaccessible parts should either be effectively sealed against corrosion (e.g. the interior of boxes or hollow portions) or they should be constructed in steel with improved atmospheric corrosion resistance. Where the environment or access provisions are such that corrosion can occur during the life of the bridge, a suitable allowance for this should be made in the proportioning of the parts.

5.7.11 Drainage

All decks should be waterproofed and the surfaces of carriageways and walkways should be sealed to prevent the ingress of water. The layout of the drainage should take into account the slope of the bridge deck as well as the location, diameter and slope of the pipes. Free fall drains should carry water to a point clear of the underside of the structure to prevent water entering into the structure.

Drainage pipes should be designed so that they can be cleaned easily. The distance between centres of the cleaning openings should be shown on drawings. Where drainage pipes are used inside box girder bridges, provisions should be made to prevent accumulation of water during leaks or breakage of pipes.

Provision should be made for the drainage of all closed cross sections, unless these are fully sealed by welding.

5.8 Fasteners, Welds, Connections and Joints

5.8.1 Connections Made of Bolts, Rivets and Pins

5.8.1.1 Categories of Bolted Connections

5.8.1.1.1 Shear Connections

Bolted connections loaded in shear should be designed as one of the following:

a. Category A: Bearing type

In this category bolts from class 4.6 up to and including class 10.9 should be used. No preloading and special provisions for contact surfaces are required. The design ultimate shear load should not exceed the design shear resistance, obtained from Clause 5.8.1.3, nor the design bearing resistance, obtained from Clause 5.8.1.3 and 5.8.1.4.

b. Category B: Slip-resistant at serviceability limit state

In this category preloaded bolts should be used. Slip should not occur at the serviceability limit state. The design serviceability shear load should not exceed the design slip resistance, obtained from Clause 5.8.1.6. The design ultimate shear load should not exceed the design shear resistance, obtained from Clause 5.8.1.3, nor the design bearing resistance, obtained from Clause 5.8.1.3 and 5.8.1.4.

c. Category C: Slip-resistant at ultimate limit state

In this category preloaded bolts should be used. Slip should not occur at the ultimate limit state. The design ultimate shear load should not exceed the design slip resistance obtained from Clause 5.8.1.6 nor the design bearing resistance, obtained from Clause 5.8.1.3 and 5.8.1.4. In addition for a connection in tension, the design plastic resistance of the net cross-section at bolt holes $N_{net,Rd}$, should be checked, at the ultimate limit state.

The design checks for these connections are summarised in Table 5.20.

5.8.1.1.2 Tension Connections

Bolted connection loaded in tension should be designed as one of the following:

a. Category D: non-preloaded

In this category bolts from class 4.6 up to and including class 10.9 should be used. No preloading is required. This category should not be used where the connections are frequently subjected to variations of tensile loading. However, they may be used in connections designed to resist normal wind loads.

b. Category E: preloaded

In this category preloaded 8.8 and 10.9 bolts with controlled tightening should be used. The checks for these connections are summarised in Table 5.20.

Table 5.20 Category of Bolted Connections

Category	Criteria	Remarks
Shear Connections		
A bearing type	$F_{v,Ed} \leq F_{v,Rd}$	No preloading required. Bolt classes from 4.6 to 10.0 may be used.
	$F_{v,Ed} \leq F_{b,Rd}$	
B slip-resistant at serviceability	$F_{v,Ed,ser} \leq F_{s,Rd,ser}$	Preloaded 8.8 or 10.9 bolts should be used.
	$F_{v,E} \leq F_{v,Rd}$	
	$F_{v,E} \leq F_{b,Rd}$	
C slip-resistant at ultimate	$F_{v,Ed} \leq F_{s,Rd}$	Preloaded 8.8 or 10.9 bolts should be used.
	$F_{v,Ed} \leq F_{b,Rd}$	
	$\sum F_{v,Ed} \leq N_{net,Rd}$	
Tension Connections		
D Non-preloaded	$F_{t,Ed} \leq F_{t,Rd}$	No preloading required. Bolt classes from 4.6 to 10.9 may be used. $B_{p,Rd}$, refer Table 5.22.
	$F_{t,Ed} \leq B_{p,Rd}$	
E Preloaded	$F_{t,Ed} \leq F_{t,Rd}$	Preloaded 8.8 or 10.9 bolts should be used. $B_{p,Rd}$, refer Table 5.22.
	$F_{t,Ed} \leq B_{p,Rd}$	

The design tensile force $F_{t,Ed}$ should include any force due to prying action. Bolts subjected to both shear force and tensile force should also satisfy the criteria given in Table 5.22.

5.8.1.2 Positioning of Holes for Bolts and Rivets

Minimum and maximum spacing and end and edge distances for bolts and rivets are given in Table 5.21.

Figure 5.34 Symbols For End and Edge Distance and Spacing of Fasteners

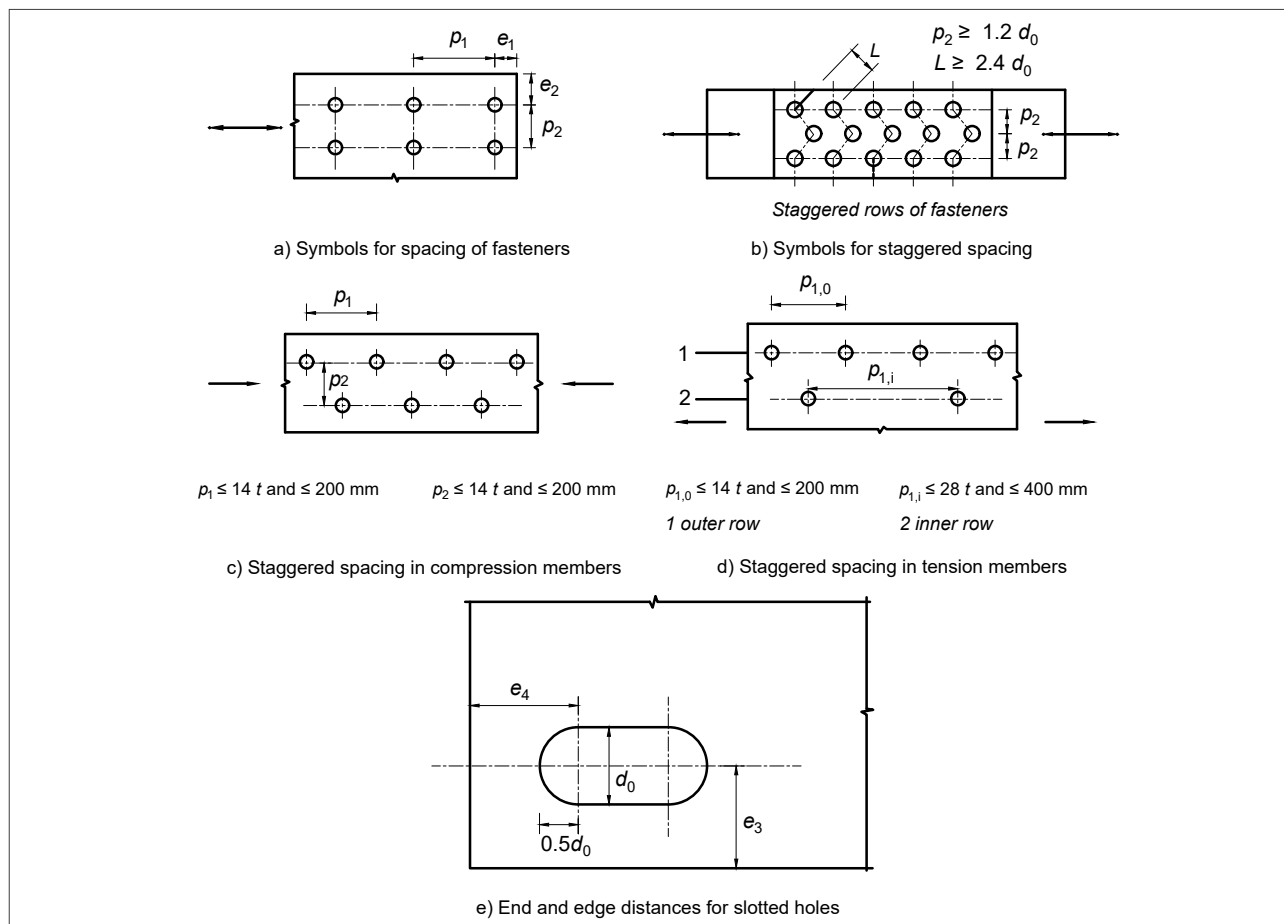


Table 5.21 Minimum and Maximum Spacing, End and Edge Distances

Distances and spacings see Figure 5.34	Minimum	Maximum ^{1,2,3}		
		Structures made from steels conforming to EN 10025 except steels conforming to EN 10025-5		Structures made from steels conforming to EN 10025-5
		Steel exposed to the weather or other corrosive influences	Steel not exposed to the weather or other corrosive influences	Steel used unprotected
End distance e_1	$1.2 d_0$	$4t + 40$ mm		The larger of $8t$ or 125mm
Edge distance e_2	$1.2 d_0$	$4t + 40$ mm		The larger of $8t$ or 125mm
Distance e_3 in slotted holes	$1.5 d_0^4$			
Distance e_4 in slotted holes	$1.5 d_0^4$			
Spacing p_1	$2.2 d_0$	The smaller of $14t$ or 200 mm	The smaller of $14t$ or 200 mm	The smaller of $14t_{min}$ or 175 mm
Spacing $p_{1,0}$		The smaller of $14t$ or 200 mm		
Spacing $p_{1,i}$		The smaller of $18t$ or 400 mm		
Spacing $p_{2,5}$	$2.4 d_0$	The smaller of $14t_i$ or 200 mm	The smaller of $14t$ or 200 mm	The smaller of $14t_{min}$ or 175 mm
1 Maximum values for spacings, edge and end distances are unlimited, except in the following cases: - For compression members in order to avoid local buckling and to prevent corrosion in exposed members (the limiting values are given in the table); and - For exposed tension members to prevent corrosion (the limiting values are given in the table) 2 The local buckling resistance of the plate in compression between the fasteners should be calculated using $0.6 p_1$ as buckling length. Local buckling between the fasteners need not to be checked if p_1/t is smaller than 9ε. The edge distance should not exceed the local buckling requirements for an outstand element in the compression members. The end distance is not affected by this requirement. 3 t is the thickness of the thinner outer connected part. 4 The dimensional limits for the slotted holes are given in EN 1090-2. 5 For staggered rows of fasteners, a minimum line spacing of $p_2 = 1.2 d_0$ may be used, provided that the minimum distance, L, between any two fasteners is greater or equal to $2.4 d_0$, (refer Figure 5.34b).				

5.8.1.3 Design Resistance of Individual Fasteners

5.8.1.3.1 Bolts and Rivets

The design resistance for an individual fastener subjected to shear and/or tension is given in Table 5.22. For preloaded bolts, the design preload, $F_{p,Cd}$, to be used in design calculations should be taken as:

$$F_{p,Cd} = \frac{0.7 f_{ub} A_s}{\gamma_{M7}}$$

Equation 5.122

The resistances for tension and for shear through the threaded portion of a bolt given in Table 5.22 should only be used for bolts manufactured in conformity with Clause 5.1.2. For bolts with cut threads, such as anchor bolts or tie rods fabricated from round steel bars where the threads comply with EN 1090, the relevant values from Table 5.22 should be used. For bolts with cut threads where the threads do not comply with EN 1090 the relevant values from Table 5.22 should be multiplied by a factor of 0.85. The design shear resistance $F_{v,Rd}$ given in Table 5.22 should only be used where the bolts are used in holes with nominal clearances not exceeding those for normal holes as specified in EN 1090-2.

1

M12 and M14 bolts may also be used in 2 mm clearance holes provided that the design resistance of the bolt group based on bearing is less than or equal to the design resistance of the bolt group based on bolt shear. In addition for class 5.8, 6.8, 8.8 and 10.9 bolts, the design shear resistance $F_{v,Rd}$ should be taken as 0.85 times the value given in Table 5.22.

2

Fit bolts should be designed using the method for bolts in normal holes. The thread of a fit bolt should not be included in the shear plane. The length of the threaded portion of a fit bolt included in the bearing length should not exceed $1/3$ of the thickness of the plate, see Figure 5.35. The hole tolerance used for fit bolts should be in accordance with EN 1090-2.

3

In single lap joints with only one bolt row, see Figure 5.36, the bolts should be provided with washers under both the head and the nut. The design bearing resistance $F_{b,Rd}$ for each bolt should be limited to:

4

$$F_{b,Rd} \leq \frac{1.5f_u d t}{\gamma_{M2}}$$

Equation 5.123

5

Note: Single rivets should not be used in single lap joints.

In the case of class 8.8 or 10.9 bolts, hardened washers should be used for single lap joints with only one bolt or one row of bolts.

Where bolts or rivets transmitting load in shear and bearing pass through packing of total thickness t_p greater than one-third of the nominal diameter d , see Figure 5.37, the design shear resistance $F_{v,Rd}$ calculated as specified in Table 5.22, should be multiplying by a reduction factor β_p given by:

$$\beta_p = \frac{9d}{8d + 3t_p} \text{ but } \beta_p \leq 1$$

For double shear connections with packing on both sides of the splice, t_p should be taken as the thickness of the thicker packing.

Riveted connections should be designed to transfer shear forces. If tension is present the design tensile force $F_{t,Ed}$ should not exceed the design tension resistance $F_{t,Rd}$ in Table 5.22. for grade S 235 steel the "as driven" value of f_{ur} may be taken as 400 N/mm². As a general rule, the grip length of a rivet should not exceed 4.5d for hammer riveting and 6.5d for press riveting.

Figure 5.35 Threaded Portion of The Shank In Bearing Length for Fit Bolts

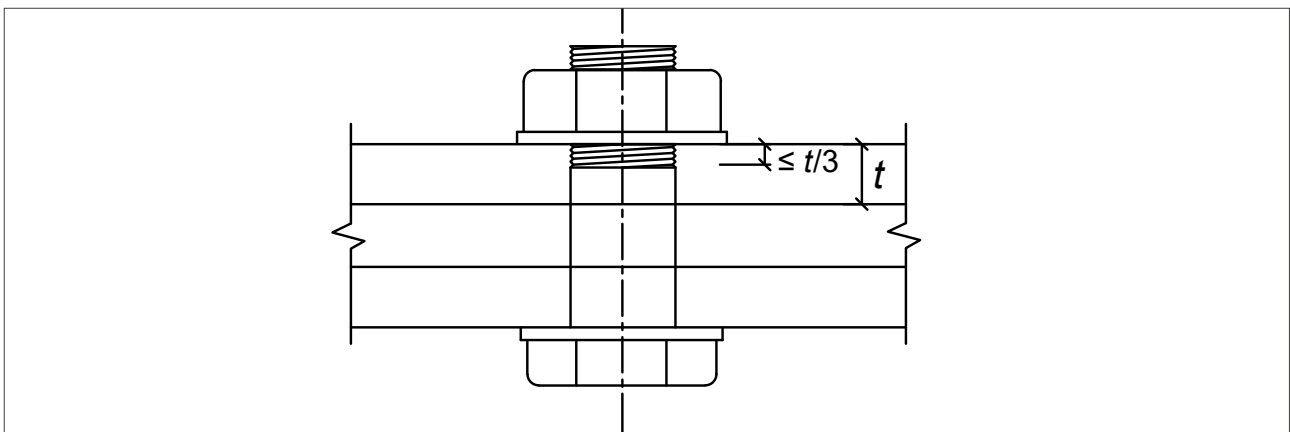


Figure 5.36 Single Lap Joint With One Row Of Bolts

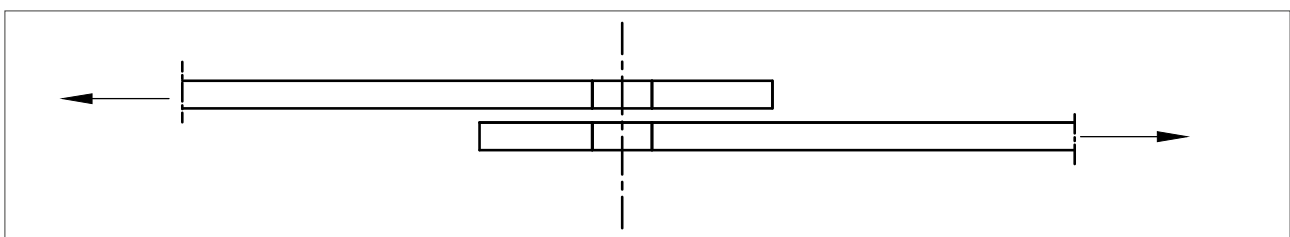


Figure 5.37 Fasteners Through Packings

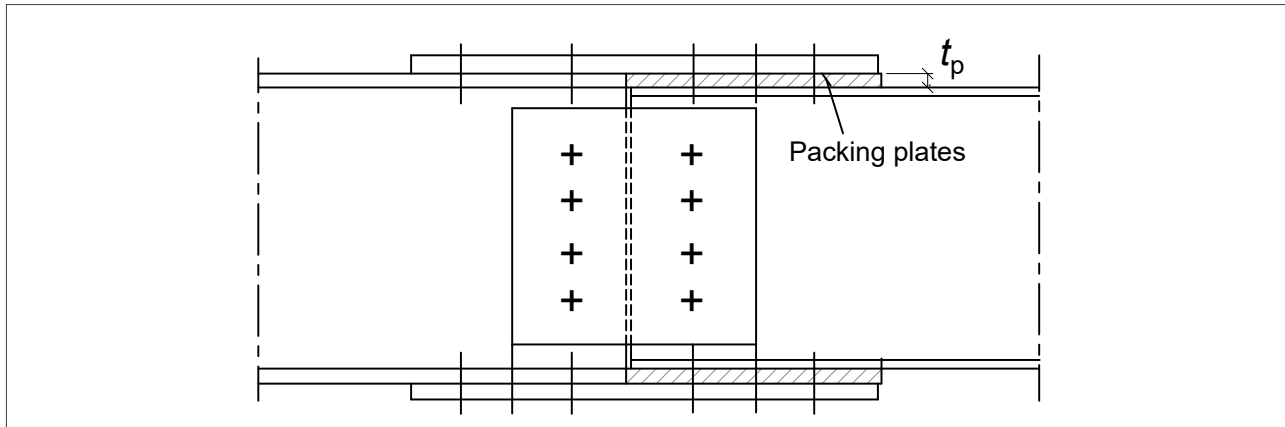


Table 5.22 Design Resistance for Individual Fasteners Subjected to Shear and/or Tension

Failure Mode	Bolts	Rivets
Shear resistance per shear plane	$F_{v,Rd} = \frac{\alpha_v f_{ub} A}{\gamma_{M2}}$ Where, the shear plane passes through the threaded portion of the bolt (<i>A is the tensile stress area of the bolt As</i>): - For classes 4.6, 5.6 and 8.8: $\alpha_v = 0.6$ - For classes 4.8, 5.8, 6.8 and 10.9: $\alpha_v = 0.5$ Where, the shear plane passes through the unthreaded portion of the bolt (<i>A is the gross cross section of the bolt</i>): $\alpha_v = 0.6$	$F_{v,Rd} = \frac{0.6 f_{ur} A_0}{\gamma_{M2}}$
Bearing resistance ^{1, 2, 3}	$F_{b,Rd} = \frac{k_1 \alpha_b f_u d t}{\gamma_{M2}}$ Where, α_b is the smallest of α_d; f_{ub}/f_u or 1.0; in the direction of load transfer: - For end bolts: $\alpha_d = \frac{e_1}{3d_0}$; for inner bolts: $\alpha_d = \frac{p_1}{3d_0} - 1/4$ perpendicular to the direction of load transfer: - For edge bolts: k_1 is the smallest of $2.8 \frac{e_2}{d_0} - 1.7$, $1.4 \frac{p_2}{d_0} - 1.7$ and 2.5 - For inner bolts: k_1 is the smallest of $1.4 \frac{p_2}{d_0} - 1.7$ or 2.5	
Tension resistance ²	$F_{t,Rd} = \frac{k_2 f_{ub} A_s}{\gamma_{M2}}$ - Where, $k_2 = 0.63$ for countersunk bolt, - Otherwise $k_2 = 0.9$	$F_{t,Rd} = \frac{0.6 f_{ur} A_0}{\gamma_{M2}}$
Punching shear resistance	$B_{p,Rd} = \frac{0.6 \pi d_m t_p f_u}{\gamma_{M2}}$	no check needed
Combined shear and tension	$\frac{F_{v,Ed}}{F_{v,Rd}} + \frac{F_{t,Ed}}{1.4 F_{t,Rd}} \leq 1.0$	
1: The bearing resistance $F_{b,Rd}$ for bolts: In oversized holes is 0.8 times the bearing resistance for bolts in normal holes; In slotted holes, where the longitudinal axis of the slotted hole is perpendicular to the direction of the force transfer, is 0.6 times the bearing resistance for bolts in round, normal holes. 2: For countersunk bolts: The bearing resistance $F_{b,Rd}$ should be based on a plate thickness t equal to the thickness of the connected plate minus half the depth of the countersinking; For the determination of the tension resistance $F_{t,Rd}$, the angle and depth of countersinking should conform with section 5.1.2, otherwise the tension resistance $F_{t,Rd}$ should be adjusted accordingly. 3 When the load on a bolt is not parallel to the edge, the bearing resistance may be verified separately for the bolt load components parallel and normal to the end.		

5.8.1.3.2 Injection Bolts

Injection bolts may be used as an alternative to ordinary bolts and rivets for category A, B and C connections specified in Clause 5.8.1.1. Fabrication and erection details for injection bolts are given in EN 1090-2. The design ultimate shear load of any bolt in a Category A connection, should not exceed the smaller of the following:

- The design shear resistance of the bolt or a group of bolts
- The design bearing resistance of the resin

Preloaded injection bolts should be used for category B and C connections, for which preloaded bolt assemblies should be used.

The design serviceability shear load of any bolt in a category B connection and the design ultimate shear load of any bolt in a category C connection should not exceed the design slip resistance of the bolt at the relevant limit state plus the design bearing resistance of the resin at the relevant limit state. In addition, the design ultimate shear load of a bolt in a category B or C connection should not exceed either the design shear resistance of the bolt, nor the bearing resistance of the bolt.

The bearing resistance of the resin, $F_{b,Rd,resin}$, may be determined according to the following Equation:

$$F_{b,Rd,resin} = \frac{k_t k_s d t_{b,resin} \beta f_{b,resin}}{\gamma_{M4}} \quad \text{Equation 5.124}$$

Where,

$F_{b,Rd,resin}$ = The strength of an injection bolt.

β = A coefficient depending of the thickness ratio of the connected plates as given in Table 5.23 and Figure 5.38.

$F_{b,resin}$ = The bearing strength of the resin to be determined according to the EN 1090-2.

$t_{b,resin}$ = The effective bearing thickness of the resin, given in Table 5.23.

k_t = 1.0 for serviceability limit state (long duration).

= 1.2 for ultimate limit state

k_s = Taken as 1.0 for holes with normal clearance or (1.0 – 0.1 m) for oversized holes

m = The difference (in mm) between the normal and oversized hole dimensions. In the case of short slotted holes, $m = 0.5$ (the difference (in mm) between the hole length and width).

When calculating the bearing resistance of a bolt with a clamping length exceeding $3d$, a value of not more than $3d$ should be taken to determine the effective bearing thickness $t_{b,resin}$ (see Figure 5.39).

Figure 5.38 Factor β As A Function of the Thickness Ratio of the Connected Plates

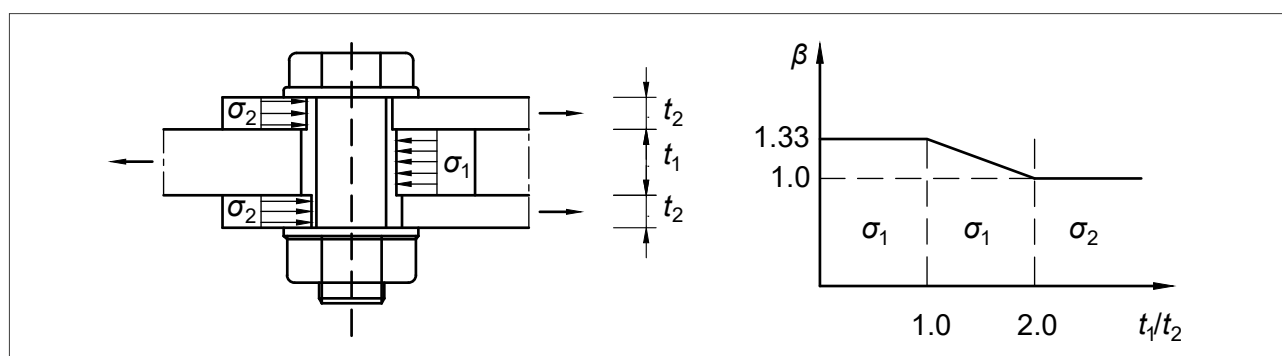
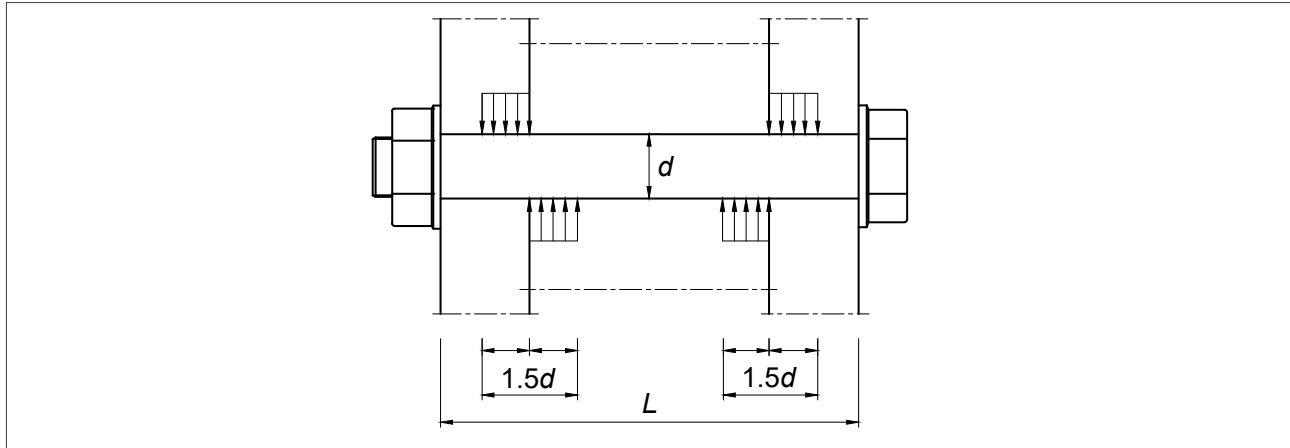


Table 5.23 Values of β and $t_{b,resin}$

t_1 / t_2	β	$t_{b,resin}$
≥ 2.0	1.0	$2 t_2 \leq 1.5 d$
$1.0 < t_1 / t_2 < 2.0$	$1.66 - 0.33 (t_1 / t_2)$	$t_1 \leq 1.5 d$
≤ 1.0	1.33	$t_1 \leq 1.5 d$

Figure 5.39 Limiting Effective Length for Long Injection Bolts



5.8.1.4 Groups of Fasteners

The design resistance of a group of fasteners may be taken as the sum of the design bearing resistances $F_{b,Rd}$ of the individual fasteners provided that the shear resistance $F_{v,Rd}$ of each individual fastener is greater than or equal to the design bearing resistance $F_{b,Rd}$. Otherwise, the design resistance of a group of fasteners should be taken as the number of fasteners multiplied by the smallest design resistance of any of the individual fasteners.

5.8.1.5 Long Joints

Where the distance L_j between the centres of the end fasteners in a joint, measured in the direction measured in the direction of force transfer (refer Figure 5.40), is more than $15d$, the design shear resistance $F_{v,Rd}$ of all the fasteners calculated according to Table 5.22 should be reduced by multiplying it by a reduction factor β_{Lf} , given by:

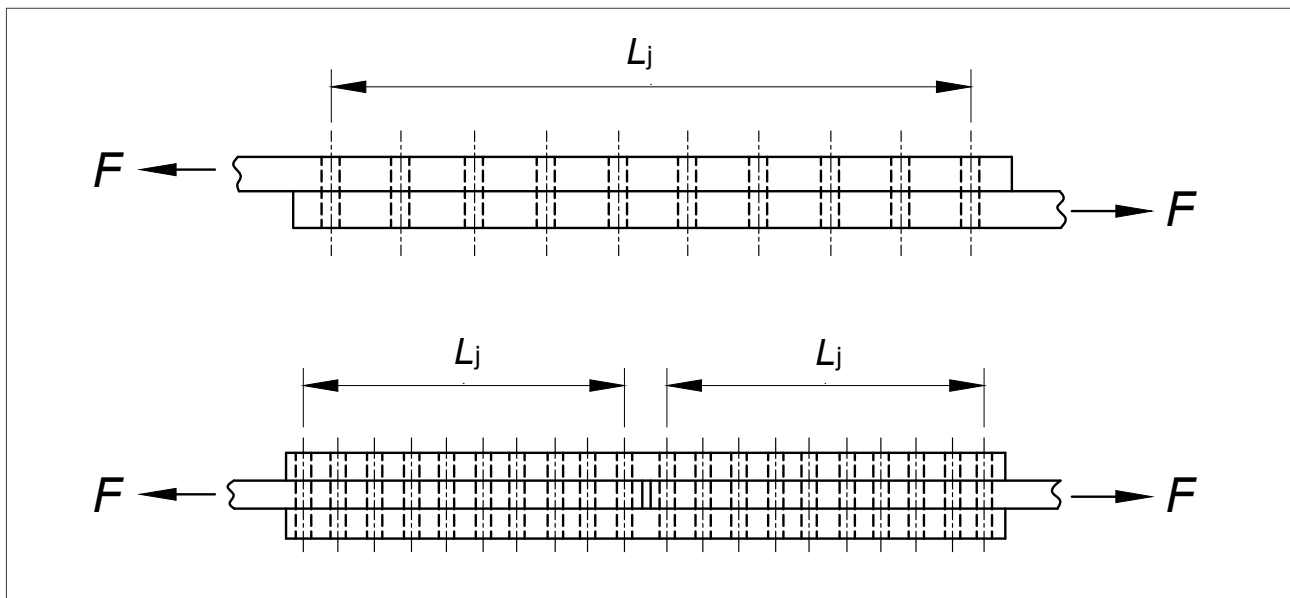
$$\beta_{Lf} = 1 - \frac{L_j - 15d}{200d}$$

Equation 5.125

$$\beta_{Lf} \leq 1.0 \text{ and } \beta_{Lf} \geq 0.75$$

The provision given above does not apply where there is a uniform distribution of force transfer over the length of the joint, e.g. the transfer of shear force between the web and the flange of a section.

Figure 5.40 Long Joints



5.8.1.6 Slip Resistant Connections using 8.8 and 10.9 Bolts

Slip Resistance

The design slip resistance of a preloaded class 8.8 or 10.9 bolt should be taken as:

$$F_{s,Rd} = \frac{k_s n \mu}{\gamma_{M3}} F_{p,C}$$

Equation 5.126

$$F_{s,Rd,ser} = \frac{k_s n \mu}{\gamma_{M3,ser}} F_{p,C}$$

Equation 5.127

Where,

k_s = Given in Table 5.24.

n = The number of friction planes

μ = The slip factor obtained either by specific tests for the friction surface in accordance with EN 1090-2 or when relevant as given in Table 5.25.

For class 8.8 and 10.9 bolts conforming with Clause 5.1.2, with controlled tightening in conformity with EN 1090-2, the preloading force $F_{p,c}$ should be taken as:

$$F_{p,C} = 0.7 f_{ub} A_s$$

Equation 5.128

Table 5.24 Values of k_s

Description	K_s
Bolts in normal holes.	1.0
Bolts in either oversized holes or short slotted holes with the axis of the slot perpendicular to the direction of load transfer.	0.85
Bolts in long slotted holes with the axis of the slot perpendicular to the direction of load transfer.	0.7
bolts in short slotted holes with the axis of the slot parallel to the direction of load transfer.	0.76
Bolts in long slotted holes with the axis of the slot parallel to the direction of load transfer.	0.63

Table 5.25 Slip Factor, μ , for Pre-loaded Bolts

Class of friction surfaces (refer EN 1090-2)	Slip factor μ
A	0.5
B	0.4
C	0.3
D	0.2

Note 1: The requirements for testing and inspection are given in EN 1090-2

Note 2: The classification of any other surface treatment should be based on test specimen representatives of the surfaces used in the structure using the procedure set out in EN 1090-2.

Note 3: The definitions of the class of friction surface are given in EN 1090-2.

Note 4: With painted surface treatments a loss of pre-load may occur over time.

Combined Tension and Shear

If a slip-resistant connection is subjected to an applied tensile force, or in addition to the shear force, $F_{v,Ed}$ or $F_{v,Ed,ser}$ tending to produce slip, the design slip resistance per bolt should be taken as follows:

- For a category B connection: $F_{s,Rd,ser} = \frac{k_s n \mu F_{p,C} - 0.8 F_{t,Ed,ser}}{\gamma_{M3,ser}}$

Equation 5.129

- For a category C connection: $F_{s,Rd} = \frac{k_s n \mu F_{p,C} - 0.8 F_{t,Ed}}{\gamma_{M3}}$

Equation 5.130

If in a moment connection, a contact force on the compression side counter balances the applied tensile force no reduction in slip resistance is required.

5.8.1.7 Deductions for Fastener Holes

5.8.1.7.1 General

Deduction for holes in the member design should be made according to Clause 5.6.3.2.

5.8.1.7.2 Design for Block Tearing

Block tearing consists of failure in shear at the row of bolts along the shear face of the hole group accompanied by tensile rupture along the line of bolt holes on the tension face of the bolt group. Figure 5.41 shows block tearing.

For a symmetric bolt group subject to concentric loading the design block tearing resistance, $V_{eff,1,Rd}$ is given by:

$$V_{eff,1,Rd} = \frac{f_u A_{nt}}{\gamma_{M2}} + \frac{1}{\sqrt{3}} \frac{f_y A_{nv}}{\gamma_{M0}}$$

Equation 5.131

Where,

A_{nt} = Net area subjected to tension.

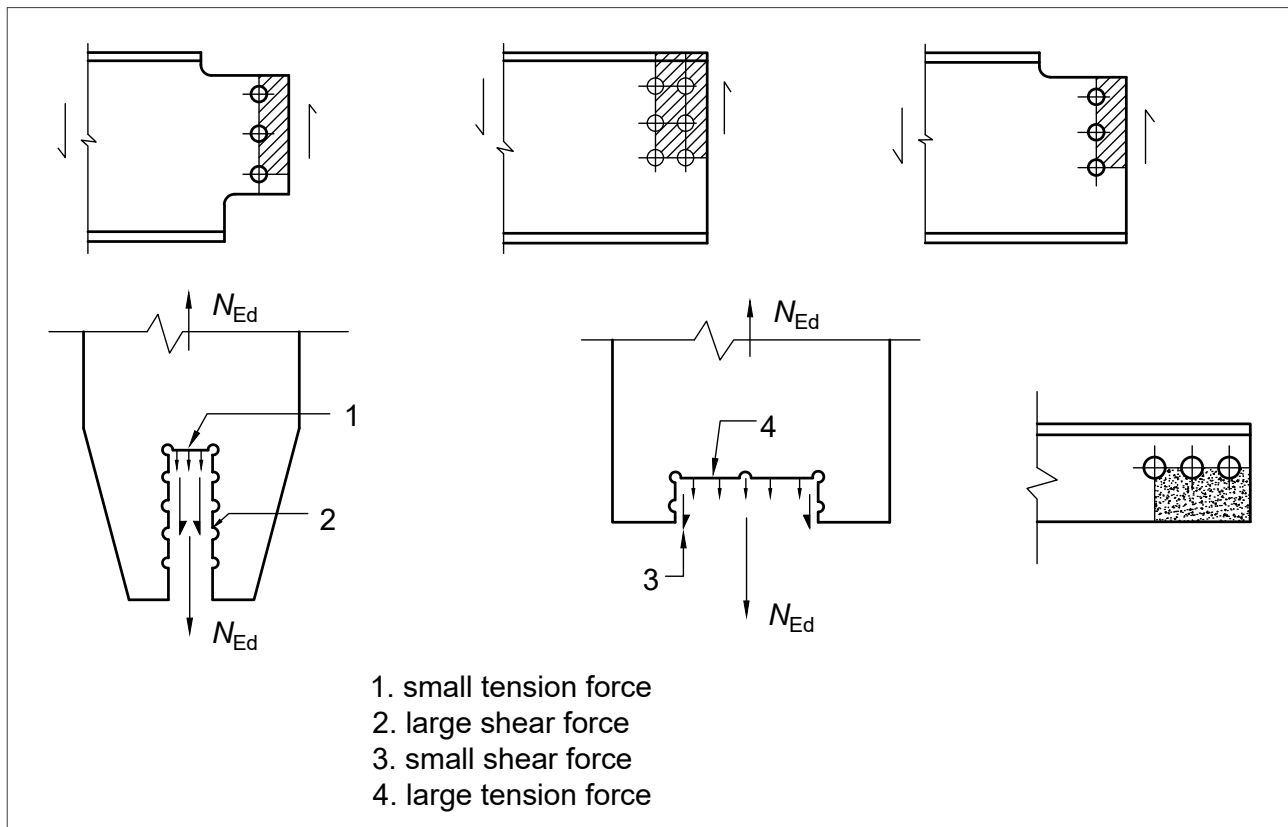
A_{nv} = Net area subjected to shear.

For a bolt group subject to eccentric loading the design block shear tearing resistance $V_{eff,2,Rd}$ is given by:

$$V_{eff,2,Rd} = \frac{0.5 f_u A_{nt}}{\gamma_{M2}} + \frac{1}{\sqrt{3}} \frac{f_y A_{nv}}{\gamma_{M0}}$$

Equation 5.132

Figure 5.41 Block Tearing



5.8.1.7.3 Angles Connected by One Leg and Other Unsymmetrically Connected Members in Tension

The eccentricity in joints and the effects of the spacing and edge distance of the bolts, should be taken into account in determining the design resistance of:

- Unsymmetrical members.
- Symmetrical members that are connected unsymmetrically, such as angles connected by one leg.

A single angle in tension connected by a row of bolts in one leg, see Figure 5.42, may be treated as concentrically loaded over an effective net section for which the design ultimate resistance should be determined as follows:

• With 1 bolt:
$$N_{u,Rd} = \frac{2.0 (e_2 - 0.5 d_0) t f_u}{\gamma_{M2}}$$

Equation 5.133

• With 2 bolts:
$$N_{u,Rd} = \frac{\beta_2 A_{net} f_u}{\gamma_{M2}}$$

Equation 5.134

• With 3 or more bolts:
$$N_{u,Rd} = \frac{\beta_3 A_{net} f_u}{\gamma_{M2}}$$

Equation 5.134

Where,

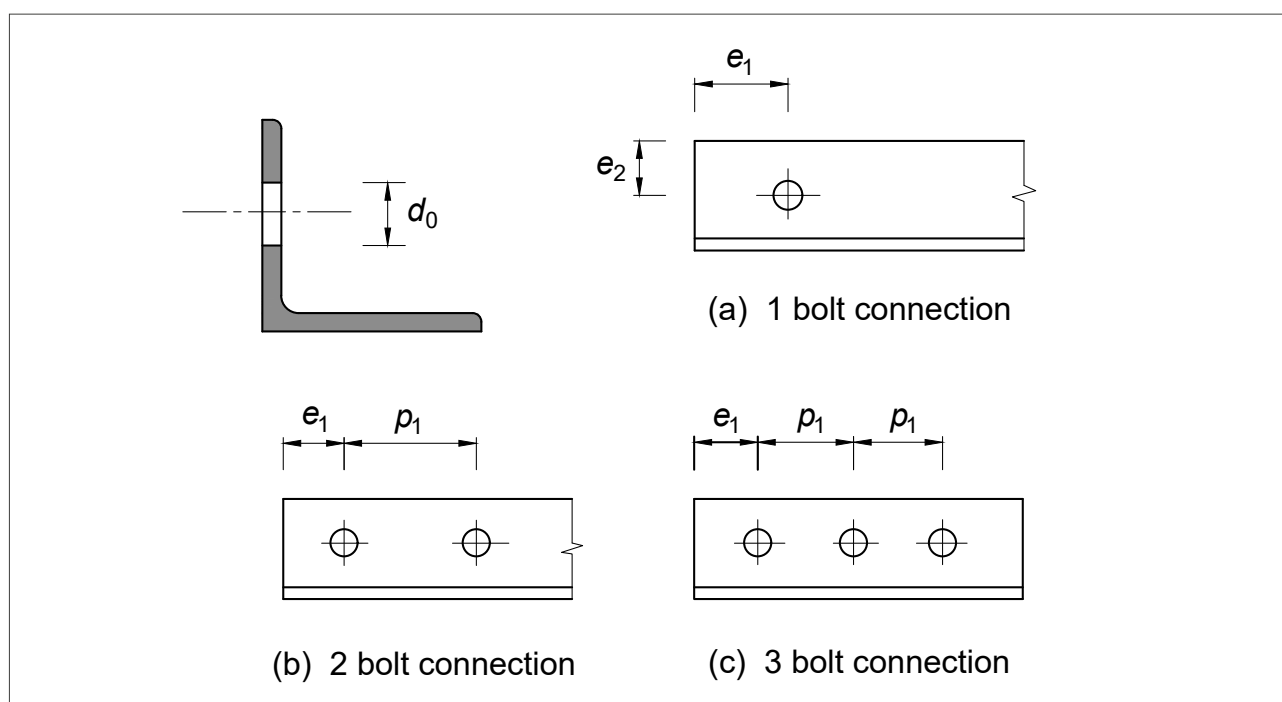
β_2, β_3 = reduction factors dependent on the pitch p_1 as given in Table 5.26. For intermediate values of p_1 , the value of β may be determined by linear interpolation.

A_{net} = The net area of the angle. For an unequal-leg angle connected by its smaller leg, A_{net} should be taken as equal to the net section area of an equivalent equal-leg angle of leg size equal to that of the smaller leg.

Table 5.26 Reduction Factors, β_1 and β_3

Pitch	p_1	$\leq 2.5 d_0$	$\geq 5.0 d_0$
2 Bolts	β_2	0.4	0.7
3 Bolts or more	β_3	0.5	0.7

Figure 5.42 Angles Connected by One Leg

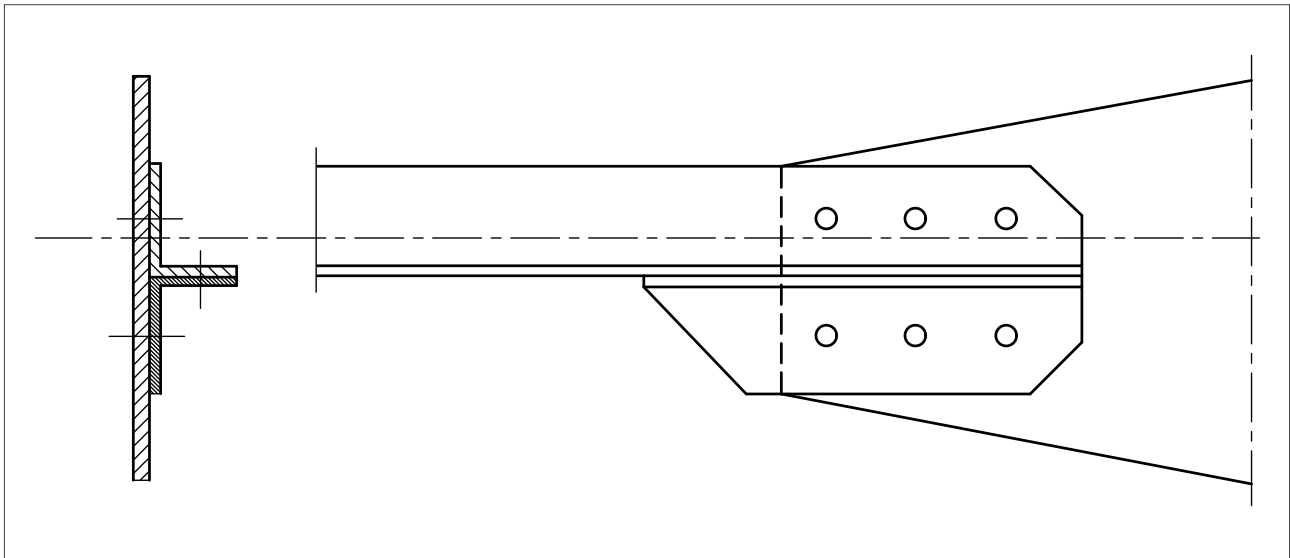


5.8.1.7.4 Lug Angles

The Lug angle shown in Figure 5.43 connects angle members and their fasteners to a gusset or other supporting part and should be designed to transmit a force 1.2 times the force in the outstand of the angle connected. The fasteners connecting the lug angle to the outstand of the angle member should be designed to transmit a force 1.4 times the force in the outstand of the angle member. Connecting a channel or a similar member should be designed to transmit a force 1.1 times the force in the channel flanges to which they are attached. The fasteners connecting the lug angle to the channel or similar member should be designed to transmit a force 1.2 times the force in the channel which they connect.

In no case should less than two bolts or rivets be used to attach a lug angle to a gusset or other supporting part. The connection of a lug angle to a gusset plate or other supporting part should terminate at the end of the member connected. The connection of the lug angle to the member should run from the end of the member to a point beyond the direct connection of the member to the gusset or other supporting part.

Figure 5.43 Lug Angles



5.8.1.8 Distribution of Forces Between Fasteners at the Ultimate Limit State

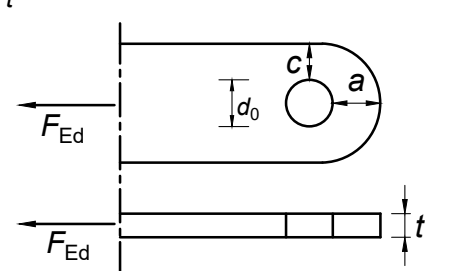
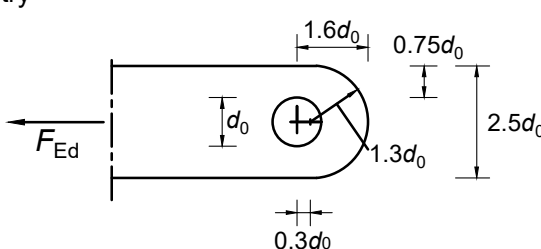
If a moment is applied to a joint, the distribution of internal forces should be linearly proportional to the distance from the centre of rotation. When a joint is loaded by a concentric shear only, the load may be assumed to be uniformly distributed amongst the fasteners, provided that the size and the class of fasteners is the same.

5.8.1.9 Connections Made With Pins

5.8.1.9.1 General

Wherever there is a risk of pins becoming loose, they should be secured. Pin connections in which no rotation is required may be designed as bolted connections, provided that the length of the pin is less than 3 times the diameter of the pin. For all other cases the method given in Clause 5.8.1.9.2 should be followed. In pin-connected members the geometry of the unstiffened element that contains a hole for the pin should satisfy the dimensional requirements given in Table 5.27.

Table 5.27 Geometrical Requirements for Pin Ended Members

Type A:	Given thickness t  $a \geq \frac{F_{Ed} \gamma_{M0}}{2 t f_y} + \frac{2 d_0}{3} : c \geq \frac{F_{Ed} \gamma_{M0}}{2 t f_y} + \frac{d_0}{3}$
Type B:	Given geometry  $t \geq 0.7 \sqrt{\frac{F_{Ed} \gamma_{M0}}{f_y}} : d_0 \leq 2.5 t$

Pin connected members should be arranged such to avoid eccentricity and should be of sufficient size to distribute the load from the area of the member with the pin hole into the member away from the pin.

5.8.1.9.2 Design of Pins

The requirements for solid circular pins are given in Table 5.28. The moments in a pin should be calculated on the basis that the connected parts form simple supports. It should be generally assumed that the reactions between the pin and the connected parts are uniformly distributed along the in contact on each part as indicated in Figure 5.44.

If the pin is intended to be replaceable, in addition to the provisions given above, the contact bearing stress should satisfy:

$$\sigma_{h,Ed} \leq f_{h,Rd} \quad \text{Equation 5.136}$$

Where,

$$\sigma_{h,Ed} = 0.591 \sqrt{\frac{E F_{b,Ed,ser} (d_0 - d)}{d^2 t}} \quad \text{Equation 5.137}$$

$$f_{h,Rd} = \frac{2.5 f_y}{\gamma_{M6,ser}} \quad \text{Equation 5.138}$$

Where,

d = The diameter of the pin

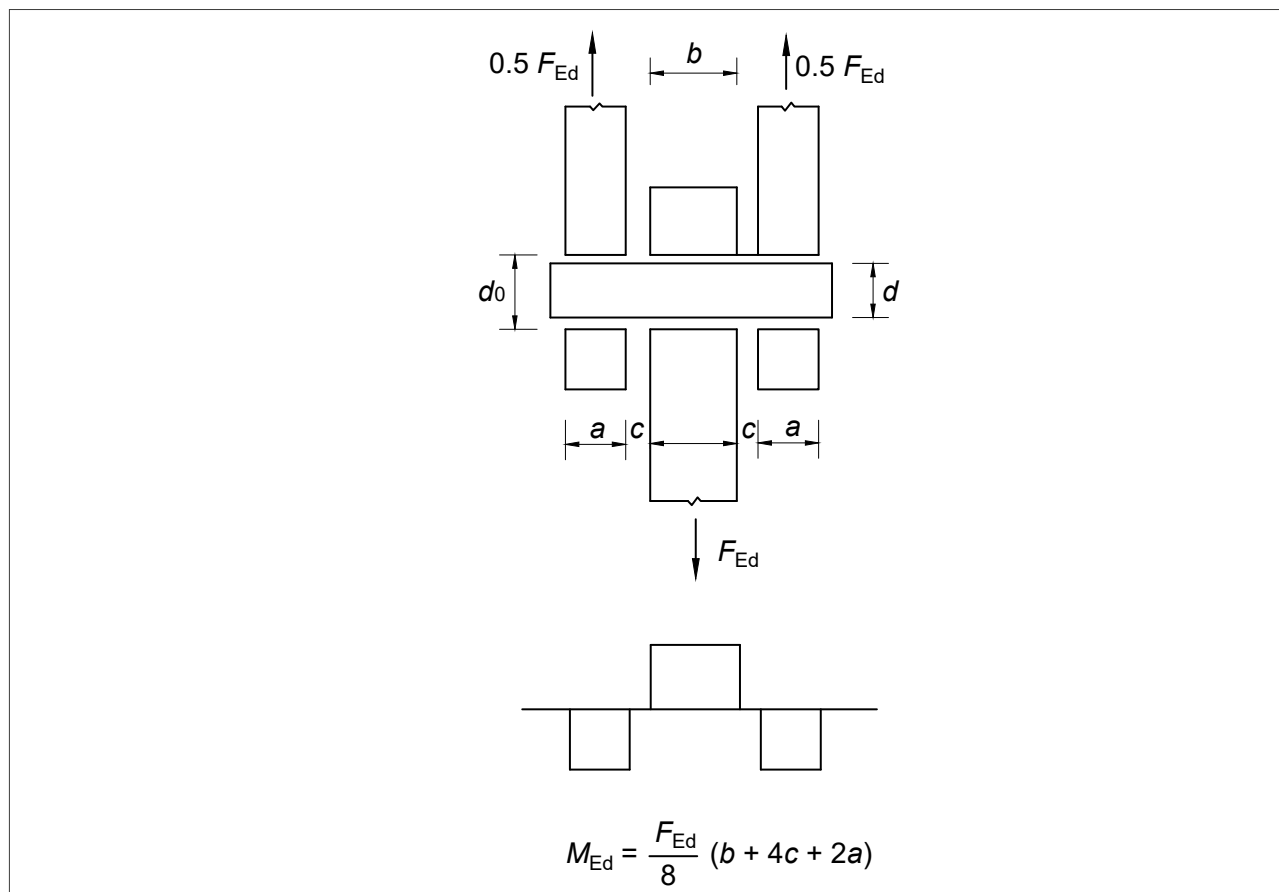
d_0 = The diameter of the pin hole

$F_{b,Ed,ser}$ = The design value of the force to be transferred in bearing, under the characteristic load combination for serviceability limit states

Table 5.28 Design Criteria for Pin Connections

Failure Mode	Design Requirements
Shear resistance of the pin	$F_{v,Rd} = \frac{0.6 A f_{up}}{\gamma_{M2}} \geq F_{v,Ed}$
Bearing resistance of the plate and the pin	$F_{b,Rd} = \frac{1.5 t d f_y}{\gamma_{M0}} \geq F_{b,Ed}$
<i>If the pin is intended to be replaceable this requirement should also be satisfied.</i>	$F_{b,Rd,ser} = \frac{0.6 t d f_y}{\gamma_{M6,ser}} \geq F_{b,Ed,ser}$
Bending resistance of the pin	$M_{Rd} = \frac{1.5 W_{cf} f_{yp}}{\gamma_{M0}} \geq M_{Ed}$
<i>If the pin is intended to be replaceable this requirement should also be satisfied.</i>	$M_{Rd,ser} = \frac{0.8 W_{cf} f_{yp}}{\gamma_{M6,ser}} \geq M_{Ed,ser}$
Combined shear and bending resistance of the pin	$\left[\frac{M_{Ed}}{M_{Rd}} \right]^2 + \left[\frac{F_{v,Ed}}{F_{v,Rd}} \right]^2 \leq 1$
Notes: d = The diameter of the pin. f_y = The lower of the yield strengths of the pin and the connected part. f_{up} = The ultimate tensile strength of the pin. f_{yp} = The yield strength of the pin. t = The thickness of the connected part. A = The cross-sectional area of a pin.	

Figure 5.44 Bending Moment in Pin



5.8.2 Welded Connections

5.8.2.1 Geometry and Dimensions

5.8.2.1.1 Type of Weld

This Standard covers the design of fillet welds, fillet welds all round, butt welds, plug welds and flare groove welds. Butt welds may be either full penetration butt welds or partial penetration butt welds. Both, fillet welds all round and plug welds may be either in circular holes or in elongated holes. The most common types of joints and welds are illustrated in EN 12345.

5.8.2.1.2 Fillet Welds

Fillet welds may be used for connecting parts where the fusion faces form an angle of between 60° and 120° . Angles smaller than 60° are also permitted. However, in such cases the weld should be considered to be a partial penetration butt weld. For angles greater than 120° the resistance of fillet welds should be determined by testing.

Fillet welds finishing at the ends or sides of parts should be returned continuously, full size, around the corner for a distance of at least twice the leg length of the weld, unless access or the configuration of the joint renders this impracticable. In the case of intermittent welds this rule applies only to the last intermittent fillet weld at corners. End returns should be indicated on the drawings.

For eccentricity of single-sided fillet welds, refer Clause 5.8.2.10.

Intermittent fillet weld should not be used at locations, where they could result in the possible formation of rust pockets. Where the connection is protected from weather, e.g. in the interior of box sections, intermittent fillet welds are permitted.

5.8.2.1.3 Fillet Welds all Round

Fillet welds all round, comprising fillet welds in circular or elongated holes, may be used only to transmit shear or to prevent the buckling or separation of lapped parts.

The diameter of a circular hole, or width of an elongated hole, for a fillet weld all round should not be less than four times the thickness of the part containing it. The ends of elongated holes should be semi-circular, except for those ends which extend to the edge of the part concerned. The centre to centre spacing of fillet welds all round should not exceed the value necessary to prevent local buckling, refer Table 5.21.

5.8.2.1.4 Butt Welds

A full penetration butt weld is defined as a weld that has complete penetration and fusion of weld and parent metal throughout the thickness of the joint. A partial penetration butt weld is defined as a weld that has joint penetration which is less than the full thickness of the parent material. Intermittent butt welds should not be used.

5.8.2.1.5 Plug Welds

Plug welds may be used:

- To transmit shear
- To prevent the buckling or separation of lapped parts, and
- To inter-connect the components of built-up members

But should not be used to resist externally applied tension.

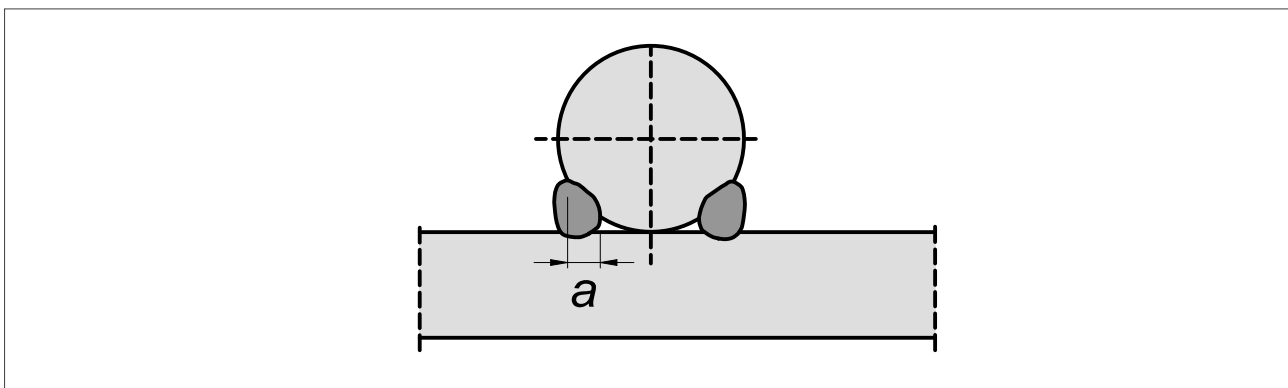
The diameter of a circular hole, or width of an elongated hole, for a plug weld should be at least 8 mm more than the thickness of the part containing it. The ends of elongated holes should either be semi-circular or else should have corners which are rounded to a radius of not less than the thickness of the part containing the slot, except for those ends which extend to the edge of the part concerned.

The thickness of a plug weld in parent material up to 16 mm thick should be equal to the thickness of the parent material. The thickness of a plug weld in parent material over 16 mm thick should be at least half the thickness of the parent material and not less than 16 mm. The centre to centre spacing of plug welds should not exceed the value necessary to prevent local buckling, refer Table 5.21.

5.8.2.1.6 Flare Groove Welds

For solid bars the design effective throat thickness of flare groove welds when fitted flush to the surface of the solid section of the bars is defined in Figure 5.45.

Figure 5.45 Effective Throat Thickness of Flare Groove Welds in Solid Sections



5.8.2.2 Welds with Packings

In the case of welds with packing, the packing should be trimmed flush with the edge of the part that is to be welded.

Where two parts connected by welding are separated by packing having a thickness less than the leg length of weld necessary to transmit the force, the required leg length should be increased by the thickness of the packing. Where two parts connected by welding are separated by packing having a thickness equal to or greater than the leg length of weld necessary to transmit the force, each of the parts should be connected to the packing by a weld capable of transmitting the design force.

5.8.2.3 Design Resistance of a Fillet Weld

5.8.2.3.1 Length of Welds

The effective length of a fillet weld l_{eff} should be taken as the length over which the fillet is full-size. This may be taken as the overall length of the weld reduced by twice the effective throat thickness a . Provided that the weld is full size throughout its length including starts and terminations, no reduction in effective length need be made for either the start or the termination of the weld.

A fillet weld with an effective length less than 30 mm or less than 6 times its throat thickness, whichever is larger, should not be designed to carry load.

5.8.2.3.2 Effective Throat Thickness

The effective throat thickness, a , of a fillet weld should be taken as the height of the largest triangle (with equal or unequal legs) that can be inscribed within the fusion faces and the weld surface, measured perpendicular to the outer side of this triangle, refer Figure 5.46. The effective throat thickness of a fillet weld should not be less than 3 mm.

In determining the design resistance of a deep penetration fillet weld, account may be taken of its additional throat thickness, see Figure 5.47, provided that preliminary tests show that the required penetration can consistently be achieved.

Figure 5.46 Throat Thickness of a Fillet Weld

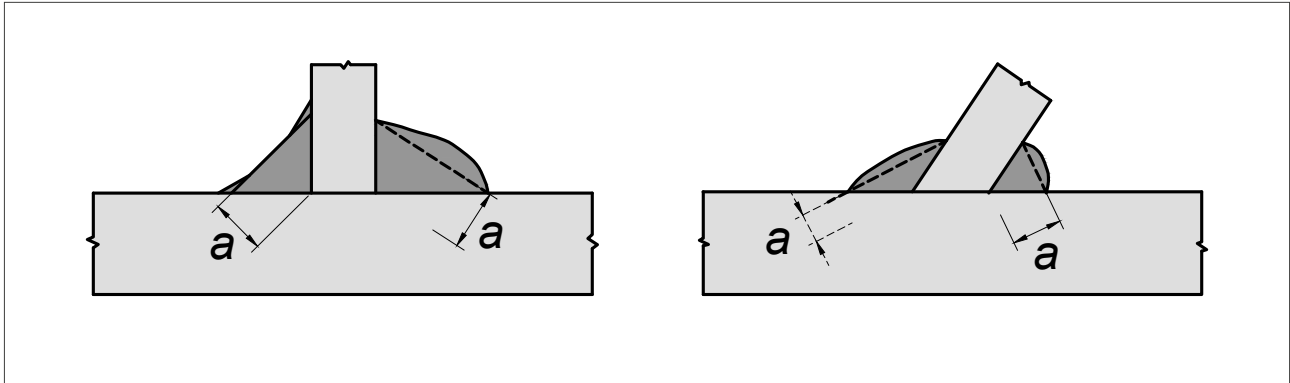
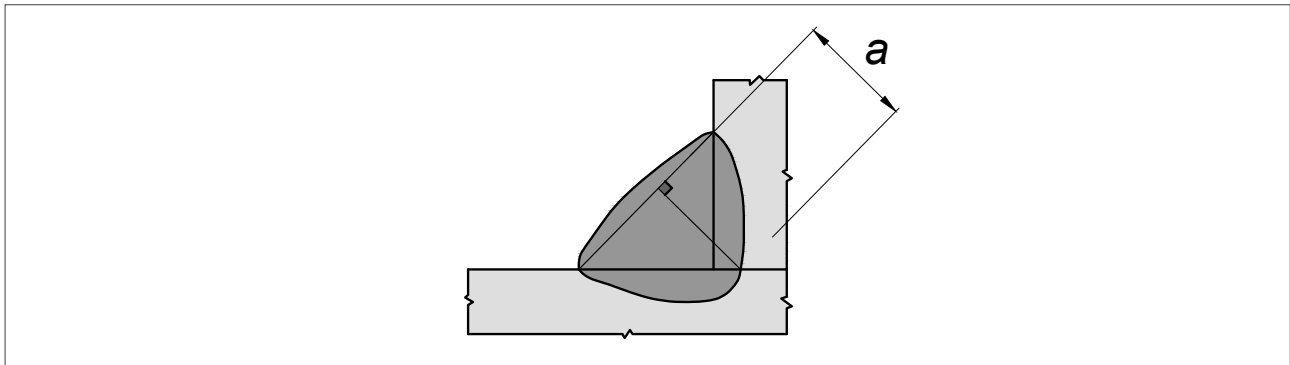


Figure 5.47 Throat Thickness of a Deep Penetration Fillet Weld



5.8.2.3.3 Design Resistance of Fillet Welds

The design resistance of a fillet weld should be determined using either the Directional method or the Simplified method.

Directional Method

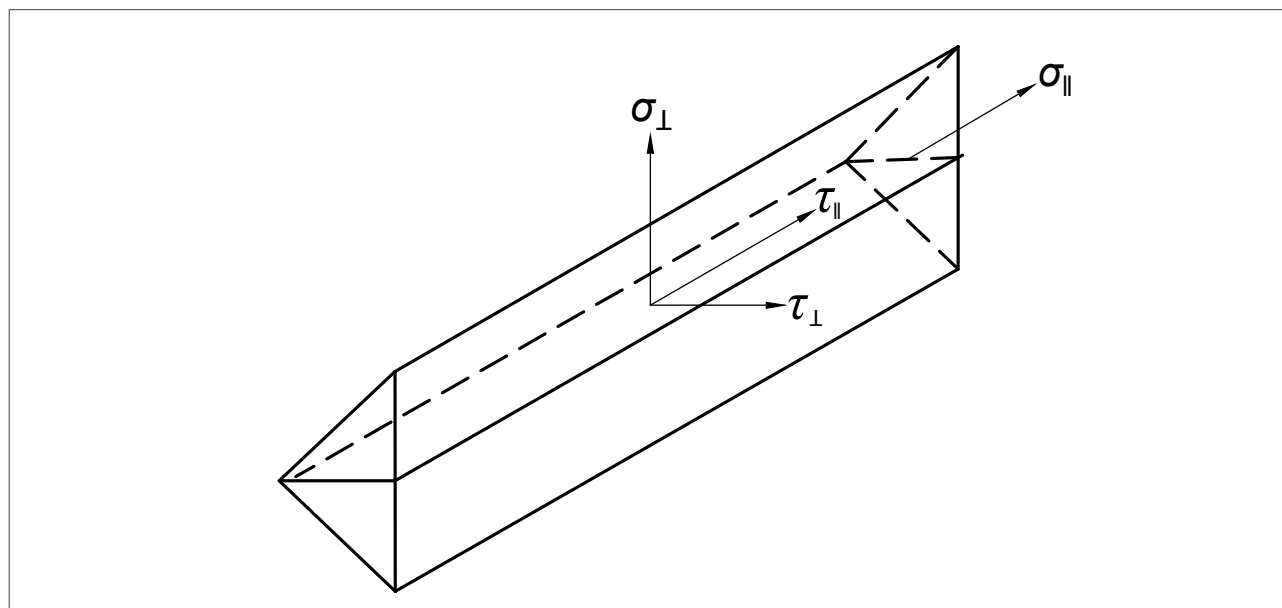
In this method, the forces transmitted by a unit length of weld are resolved into components parallel and transverse to the longitudinal axis of the weld and normal and transverse to the plane of its throat.

The design throat area A_w should be taken as $A_w = \sum a l_{eff}$. The location of the design throat area should be assumed to be concentrated in the root.

A uniform distribution of stress is assumed on the throat section of the weld, leading to the normal stresses and shear stresses shown in Figure 5.48, as follows:

- $\sigma_{\perp}$ = The normal stress perpendicular to the throat
- $\sigma_{\parallel}$ = The normal stress parallel to the axis of the weld
- $\tau_{\perp}$ = The shear stress (in the plane of the throat) perpendicular to the axis of the weld
- $\tau_{\parallel}$ = The shear stress (in the plane of the throat) parallel to the axis of the weld.

Figure 5.48 Stresses on the Throat Section of a Fillet Weld



The normal stress $\sigma_{\parallel}$ parallel to the axis is not considered when verifying the design resistance of the weld.

The design resistance of the fillet weld will be sufficient if the following both conditions are satisfied:

$$[\sigma_{\perp}^2 + 3(\tau_{\perp}^2 + \tau_{\parallel}^2)]^{0.5} \leq \frac{f_u}{\beta_w \gamma_{M2}} \quad \text{and} \quad \sigma_{\perp} \leq 0.9 \frac{f_u}{\gamma_{M2}}$$

Equation 5.139

Where,

f_u is the nominal ultimate tensile strength of the weaker part joined.

β_w is the appropriate correlation factor taken from Table 5.29.

Welds between parts with different material strength grades should be designed using the properties of the material with the lower strength grade.

Table 5.29 Correlation Factor β_w for Fillet Weld

Standard and Steel Grade			Correction Factor β_w
EN 10025	EN 10201	EN 10219	
S 235 S 235W	S 235 H	S 235 H	0.8
S 275 S 275 N/NL S 275 M/ML	S 275 H S 275 NH/NLH	S 275 H S 275 NH/NLH S 275 MH/MLH	0.85
S 335 S 335 N/NL S 335 M/ML S 335 W	S 355 H S 355 NH/NLH	S 355 H S 355 NH/NLH S355 MH/MLH	0.9
S 420 N/NL S 420 M/ML		S 420 MH/MLH	1.0
S 460 N/NL S 460 M/ML S 460 Q/QL/QL1	S 460 NH/NLH	S 460 NH/NLH S 460 MH/MLH	1.0

1

Simplified Method for Design Resistance of Fillet Weld

Alternatively, the design resistance of a fillet weld may be assumed to be adequate if, at every point along its length, the resultant of all the forces per unit length transmitted by the weld satisfy the following criterion:

$$F_{w,Ed} \leq F_{w,Rd}$$

Equation 5.140

2

3

Where,

$F_{w,Ed}$ is the design value of the weld force per unit length.

$F_{w,Rd}$ is the weld resistance per unit length.

4

Independent of the orientation of the weld throat plane to the applied force, the design resistance per unit length $F_{w,Rd}$ should be determined from:

$$F_{w,Rd} = f_{vw,d} a$$

Equation 5.141

Where, $f_{vw,d}$ is the design shear strength of the weld.

The design shear strength $f_{vw,d}$ of the weld should be determined from:

$$f_{vw,d} = \frac{f_u / \sqrt{3}}{\beta_w \gamma_{M2}}$$

Equation 5.142

Where, f_u and β_w are defined above,

5.8.2.4 Design Resistance of Fillet Welds All Round

The design resistance of a fillet weld all round should be determined using one of the methods given in Clause 5.8.2.3.

5.8.2.5 Design Resistance of Butt Welds**5.8.2.5.1 Full Penetration Butt Welds**

The design resistance of a full penetration butt weld should be taken as equal to the design resistance of the weaker of the parts connected, provided that the weld is made with a suitable consumable which will produce all-weld tensile specimens having both a minimum yield strength and a minimum tensile strength not less than those specified for the parent metal.

5.8.2.5.2 Partial Penetration Butt Welds

The design resistance of a partial penetration butt weld should be determined using the method for a deep penetration fillet weld given in Clause 5.8.2.3.2. The throat thickness of a partial penetration butt weld should not be greater than the depth of penetration that can be consistently achieved, refer Clause 5.8.2.3.2.

5.8.2.5.3 T-butt Joints

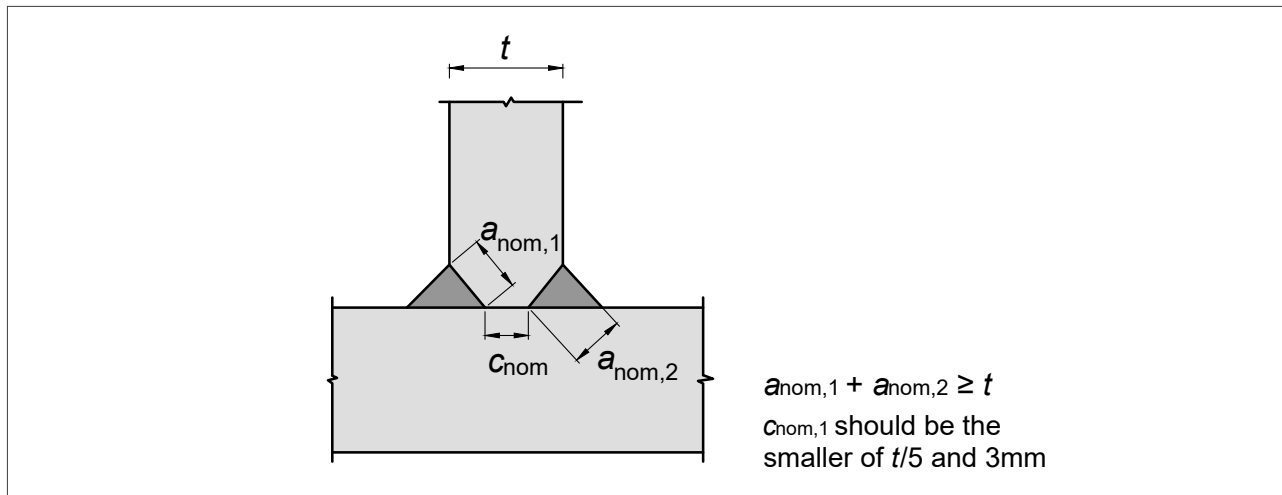
The design resistance of a T-butt joint, consisting of a pair of partial penetration butt welds reinforced by superimposed fillet welds, may be determined as for a full penetration butt weld (see **Clause 4.7.1**) if the total nominal throat thickness, exclusive of the unwelded gap, is not less than the thickness t of the part forming the stem of the tee joint, provided that the unwelded gap is not more than $(t/5)$ or 3 mm, whichever is less, see Figure 5.49.

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The design resistance of a T-butt joint which does not meet the requirements given in above should be determined using the method for a fillet weld or a deep penetration fillet weld in Clause 5.8.2.3 depending on the amount of penetration. The throat thickness should be determined in conformity with the provisions for fillet welds (refer Clause 5.8.2.3.2) or partial penetration butt welds (refer Clause 5.8.2.5.2) as relevant.

Figure 5.49 Effective Full Penetration of T-butt Welds



5.8.2.6 Design Resistance of Plug Welds

The design resistance $F_{w,Rd}$ of a plug weld should be taken as:

$$F_{w,Rd} = f_{vw,d} A_w$$

Equation 5.143

Where,

$F_{vw,d}$ = The design shear strength of a weld;

A_w = The design throat area and should be taken as the area of the hole.

5.8.2.7 Distribution of Forces

The distribution of forces in a welded connection may be calculated on the assumption of either elastic or plastic behaviour. It is acceptable to assume a simplified load distribution within the welds.

Residual stresses and stresses not subjected to transfer of load need not be included when checking the resistance of a weld. This applies specifically to the normal stress parallel to the axis of a weld.

Welded joints should be designed to have adequate deformation capacity. However, ductility of the welds should not be relied upon. In joints where plastic hinges may form, the welds should be designed to provide at least the same design resistance as the weakest of the connected parts.

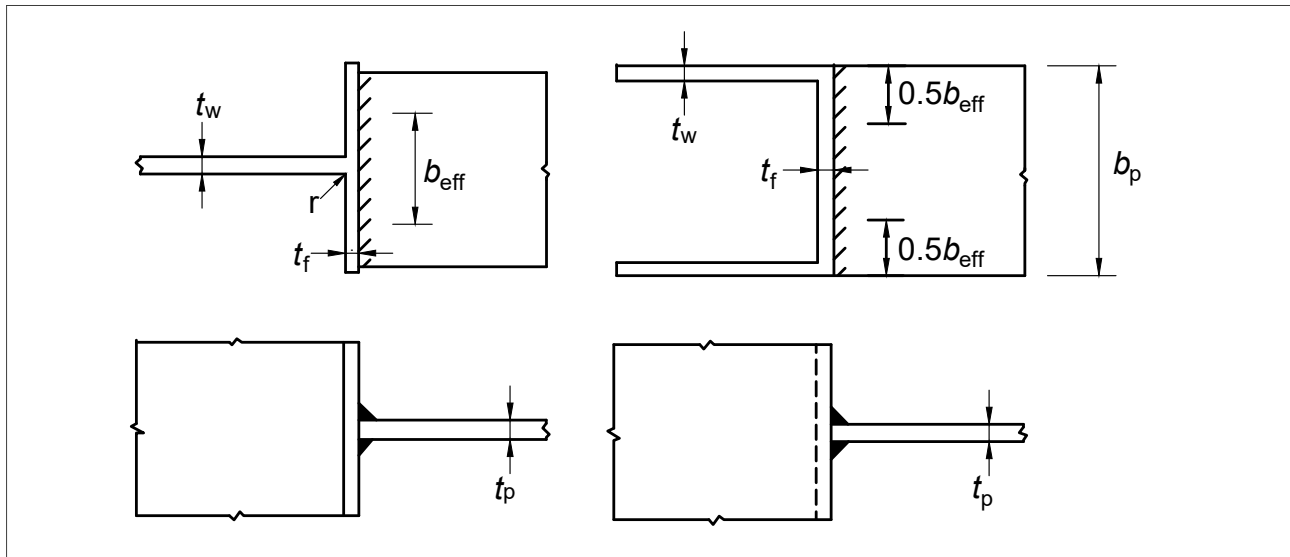
In other joints where deformation capacity for joint rotation is required due to the possibility of excessive straining, the welds sufficient strength not to rupture before general yielding in the adjacent parent material.

5.8.2.8 Connections to Unstiffened Flanges

Where a transverse plate (or beam flange) is welded to a supporting unstiffened flange of I , H or other sections, refer Figure 5.50, the applied force perpendicular to the unstiffened flange should not exceed any of the relevant design resistances as follows:

- That of the web of the supporting member of I or H sections
- Those for a transverse plate on a RHS member
- That of the supporting flange calculated assuming the applied force is concentrated over an effective width, b_{eff} , of the flange

Figure 5.50 Effective Width of an Unstiffened T-joint



For an unstiffened I or H section the effective width b_{eff} should be obtained from:

$$b_{eff} = t_w + 2s + 7k t_f$$

Equation 5.144

Where,

$$k = \frac{t_f}{t_p} \frac{f_{yf}}{f_{yp}} \quad \text{but } k \leq 1$$

Equation 5.145

- f_{yf} = The yield strength of the flange of the I or H section.
 f_{yp} = The yield strength of the plate welded to the I or H section.

The dimension s should be obtained from:

- For a rolled I or H section: $s = r$
- For a welded I or H section: $s = \sqrt{2} a$

For an unstiffened flange of an I or H section, the following criterion should be satisfied:

$$b_{eff} \geq \frac{f_{yf}}{f_{up}} b_p$$

Equation 5.146

Where,

- f_{up} is the ultimate strength of the plate of the I or H section.
 b_p is the width of the plate welded to the I or H section.
 Otherwise, the joint should be stiffened.

For other sections such as box sections or channel sections where the width of the connected plate is similar to the width of the flange, the effective width b_{eff} should be obtained from:

$$b_{eff} = 2t_w + 5t_f \quad \text{but} \quad b_{eff} \leq 2t_w + 5k t_f$$

Equation 5.147

Even if, $b_{eff} \leq b_p$, the welds connecting the plate to the flange need to be designed to transmit the design resistance of the plate $b_p t_p f_{y,P} / \gamma_{M0}$ assuming a uniform stress distribution.

5.8.2.9 Long Joints

In lap joints the resistance of a fillet weld should be reduced by multiplying it by a reduction factor β_{Lw} to allow for the effects of non-uniform distribution of stress along its length.

The provisions in this Clause do not apply when the stress distribution along the weld corresponds to the stress distribution in the adjacent base metal, as, for example, in the case of a weld connecting the flange and the web of a plate girder.

In lap joints longer than $150a$ the reduction factor β_{Lw} should be taken as $\beta_{Lw,1}$ given by:

$$\beta_{Lw,1} = 1.2 - \frac{0.2L_j}{150a} \quad \text{but} \quad \beta_{Lw,1} \leq 1.0$$

Equation 5.148

Where; L_j is the overall length of the lap in the direction of the force transfer.

fillet welds longer than 1.7 metres connecting transverse stiffeners in plated members, the reduction factor β_{Lw} may be taken as $\beta_{Lw,2}$ given by:

$$\beta_{Lw,2} = 1.2 - \frac{L_w}{17} \quad \text{but} \quad \beta_{Lw,2} \leq 1.0 \quad \text{and} \quad \beta_{Lw,2} \geq 0.6$$

Equation 5.149

Where, L_w is the length of the weld in metres.

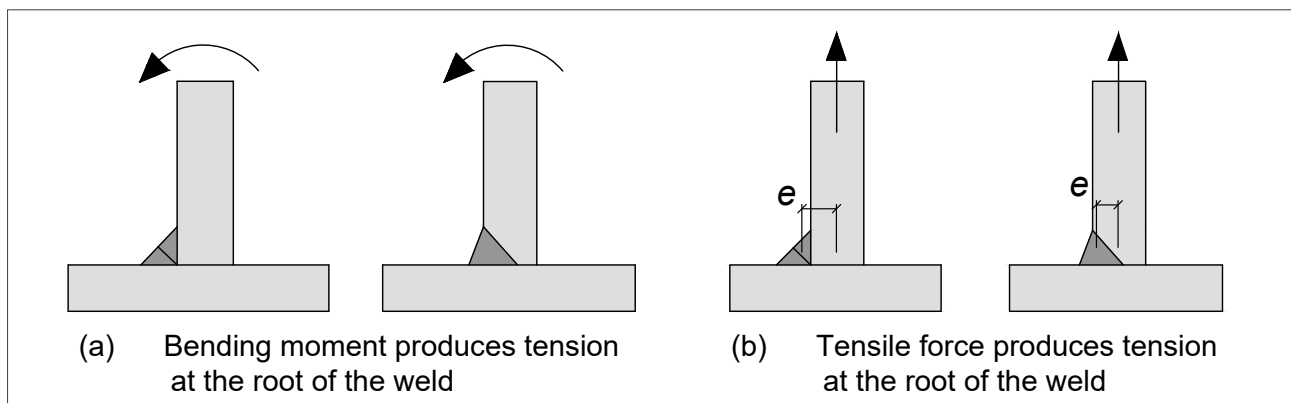
5.8.2.10 Eccentrically Loaded Single Fillet or Single-Sided Partial Penetration Butt Welds

Local eccentricity should be avoided whenever it is possible. Local eccentricity (relative to line of action of the force to be resisted) should be taken into account in the following cases:

- Where a bending moment transmitted about the longitudinal axis of the weld produces tension at the root of the weld, refer Figure 5.51 (a).
- Where a tensile force transmitted perpendicular to the longitudinal axis of the weld produces a bending moment, resulting in a tension force at the root of the weld, refer Figure 5.51 (b).

Eccentrically loaded single fillet welds or single-sided partial penetration butt welds should not be used to transmit a bending moment about the longitudinal axis of the weld.

Figure 5.51 Single Fillet Weld and Single Sided Partial Penetration Butt Welds



5.8.2.11 Angles Connected by One Leg

In angles connected by one leg, the eccentricity of welded lap joint end connections may be allowed for by adopting an effective cross-sectional area and then treating the member as concentrically loaded.

For an equal-leg angle, or an unequal-leg angle connected by its larger leg, the effective area may be taken as equal to the gross area.

For an unequal-leg angle connected by its smaller leg, the effective area should be taken as equal to the gross cross-sectional area of an equivalent equal-leg angle of leg size equal to that of the smaller leg, when determining the design resistance of the cross-section. However, when determining the design buckling resistance of a compression member, the actual gross cross-sectional area should be used.

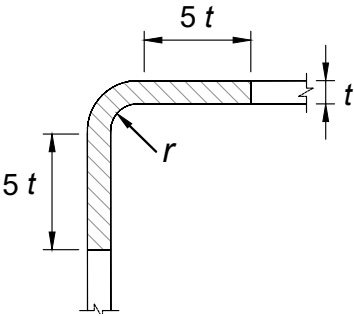
5.8.2.12 Welding in Cold-formed Zones

Welding may be carried out within a length $5t$ either side of a cold-formed zone, refer Table 5.30, provided that one of the following conditions is fulfilled:

- The cold-formed zones are normalised after cold-forming but before welding
- The r/t -ratio satisfy the relevant value obtained from Table 5.30.

Table 5.30 Conditions for Welding Cold-formed Zones and Adjacent Material

r/t	Strain due to cold forming (%)	Maximum thickness (mm)		
		Predominantly static loading	Where fatigue predominates	Fully killed aluminium-killed steel ($Al \geq 0.02\%$)
≥ 25	≤ 2	any	any	any
≥ 10	≤ 5	any	16	any
≥ 3.0	≤ 14	24	12	24
≥ 2.0	≤ 20	12	10	12
≥ 1.5	≤ 25	8	8	10
≥ 1.0	≤ 33	4	4	6



Note: Cold formed hollow sections according to EN 10219 which do not satisfy the limits given in Table 4.2 can be assumed to satisfy these limits if these sections have a thickness not exceeding 12.5 mm and are Al-killed with a quality J2H, K2H, MH, MLH, NH or NLH and further satisfy $C \leq 0.18\%$, $P \leq 0.020\%$ and $S \leq 0.012\%$.

In other cases, welding is only permitted within a distance of $5t$ from the corners if it can be shown by tests that welding is permitted for that particular application

5.9 Fatigue Assessment

5.9.1 General Requirements

Fatigue assessment is not applicable to bridges that are predominantly statically loaded e.g. pedestrian bridges and the parts of the bridges that are not stressed by traffic loads, nor likely to be excited by wind loads. Fatigue assessments should be carried out for all components of the road bridges using the procedures given in this section.

5.9.2 Fatigue Loading

For the fatigue assessment of road bridges, the fatigue load model in accordance with Chapter 3 of this Manual should be applied

5.9.3 Partial Factors for Fatigue Verifications

The partial factor for fatigue loads shall be taken as γ_{Ff} . Recommended value of $\gamma_{Ff} = 1.0$.

The partial factor for fatigue resistance shall be taken as γ_{Mf} . Recommended value of $\gamma_{Mf} = 1.1$ should be used

5.9.4 Fatigue Stress Range

5.9.4.1 General

The following procedure may be used to determine the design stress range for the fatigue loads.

The maximum stress $\sigma_{p,max}$ and the minimum stress $\sigma_{p,min}$ should be determined by evaluating influence areas. The reference stress range $\Delta\sigma_p$ for determining the damage effects of the stress range spectrum should be obtained from:

$$\Delta\sigma_p = | \sigma_{p,max} - \sigma_{p,min} |$$

The damage effects of the stress range spectrum may be represented by the damage equivalent stress range related to 2×10^6 cycles:

$$\Delta\sigma_{E,2} = \lambda \Phi_2 \Delta\sigma_p$$

Equation 5.150

Where,

λ = The damage equivalent factor.

Φ_2 = The damage equivalent impact factor.

For road bridges, Φ_2 may be taken as 1.0.

5.9.4.2 Analysis for Fatigue

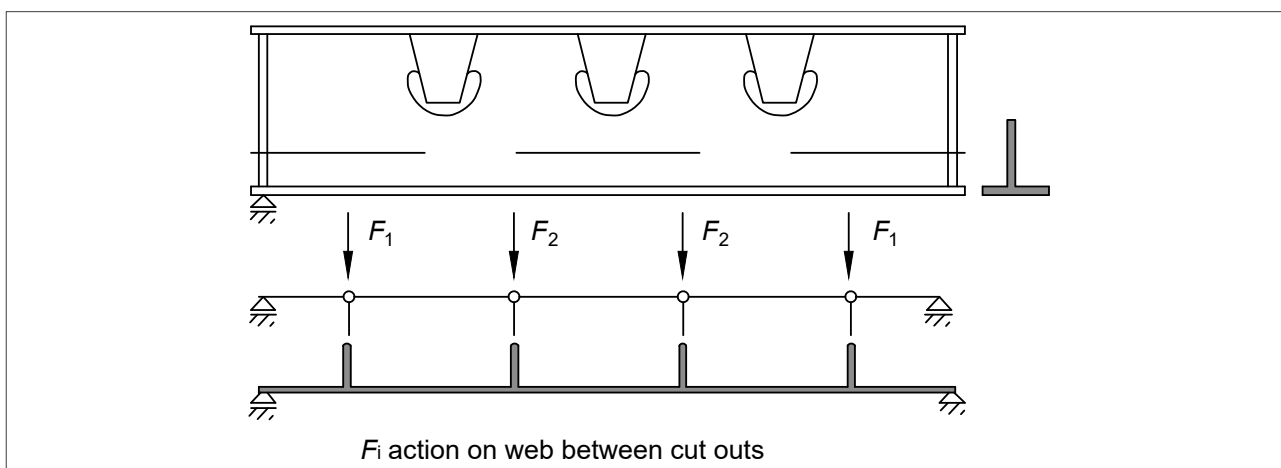
5.9.4.2.1 Longitudinal Stiffeners

Longitudinal stiffeners should be analysed using a model for the integral structure or for simplicity, as continuous beams on elastic supports.

5.9.4.2.2 Cross Beams

The influence of the cut outs should be taken into account in the analysis for crossbeams. Where crossbeams are provided with cut outs as given in Figure 5.52, the action effects may be determined with a Vierendeel-model (where the deck plate and a part of the cross beam below the cut outs are the flanges and the areas between the cut outs are the posts).

Figure 5.52 Vierendeel-model for Cross Beam



1

2

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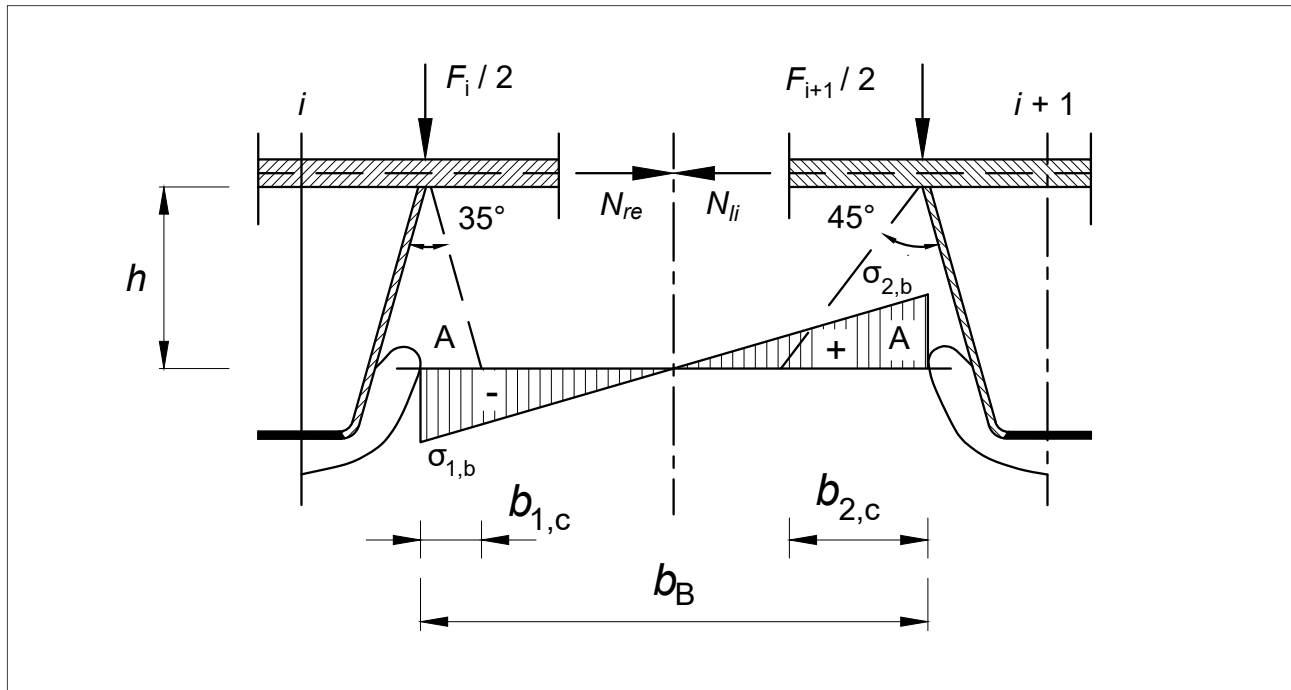
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In the analysis of the model for a crossbeam the following should be taken into account:

- The connections of the crossbeam to the transverse stiffeners of the webs of main should form a continuous transverse frame
- The contributions of the deformations of components of the Vierendeel-beams due to bending moments, axial forces and shear forces to the overall deformation
- The effects of shear between the deck plate and the web of the cross beam on the direct stresses and shear stresses at the critical section in Figure 5.53.
- The effects of local introduction of loads from the stiffeners into the web
- The shear stresses from the horizontal and vertical shear in the critical section in Figure 5.53.

Figure 5.53 Stress Distribution at Cope Hole



The direct stresses in the critical section in Figure 5.53 may be determined as follows:

$$\sigma_1 = \sigma_{1b} + \sigma_{1c}$$

Equation 5.151

$$\sigma_2 = \sigma_{2b} + \sigma_{2c}$$

Equation 5.152

Where, $-\sigma_{1b} = +\sigma_{2b} = \frac{M_{Ed}}{W}$ are the stresses due to bending.

Equation 5.153

$$\sigma_{1c} = -\frac{F_i}{2A_{1c}} \quad \text{and} \quad \sigma_{2c} = -\frac{F_{i+1}}{2A_{2c}}$$

are the compressive stresses due to local load from stiffeners.

Equation 5.154

$$W = \frac{1}{6} t b_B^2$$

$$A_{1c} = b_{1c} t$$

$$A_{2c} = b_{2c} t$$

$$V_{Ed} = \text{The horizontal shear force.}$$

$$M_{Ed} = V_{Ed} h \text{ is the bending moment in the critical section.}$$

$$F_i, F_{i+1} = \text{The loads introduced from the stiffeners.}$$

$$t = \text{The plate thickness of the web.}$$

Where no cope holes are provided the stresses at the critical section may be determined using flanges from the webs of the stiffeners with an effective width $b_{eff} = 5 t_{w,st}$, where $t_{w,st}$ is the plate thickness of the stiffeners.

5.9.5 Fatigue Assessment Procedures

5.9.5.1 Fatigue Assessment

The fatigue assessment should be carried out as follows:

$$\gamma_{Ff} \Delta\sigma_{E2} \leq \frac{\Delta\sigma_c}{\gamma_{Mf}} \quad \text{Equation 5.155}$$

and

$$\gamma_{Ff} \Delta\tau_{E2} \leq \frac{\Delta\tau_c}{\gamma_{Mf}} \quad \text{Equation 5.156}$$

5.9.5.2 Damage Equivalence Factors λ for Road Bridges

The damage equivalence factor λ for road bridges up to 80m span should be obtained from:

$$\lambda = \lambda_1 \times \lambda_2 \times \lambda_3 \times \lambda_4 \quad \text{but } \lambda \leq \lambda_{max} \quad \text{Equation 5.157}$$

Where,

λ_1 = The factor for the damage effect of traffic and depends on the length of the critical influence line or area

λ_2 = The factor for the traffic volume

λ_3 = The factor for the design life of the bridge

λ_4 = The factor for the traffic on other lanes

λ_{max} = The maximum λ -value taking account of the fatigue limit

In determining λ_1 the critical length of the influence line or area may be taken as follows:

a. For moments:

- for a simply supported span, the span length L_i .
- for continuous spans in midspan sections, refer Figure 5.56, the span length L_i of the span under consideration.
- for continuous spans in support sections, refer Figure 5.56, the mean of the two spans L_i and L_j adjacent to that support.
- for cross girders supporting stringers, the sum of the two adjacent spans of the stiffeners carried by the cross girder.

b. For shear for a simply supported span and a continuous span:

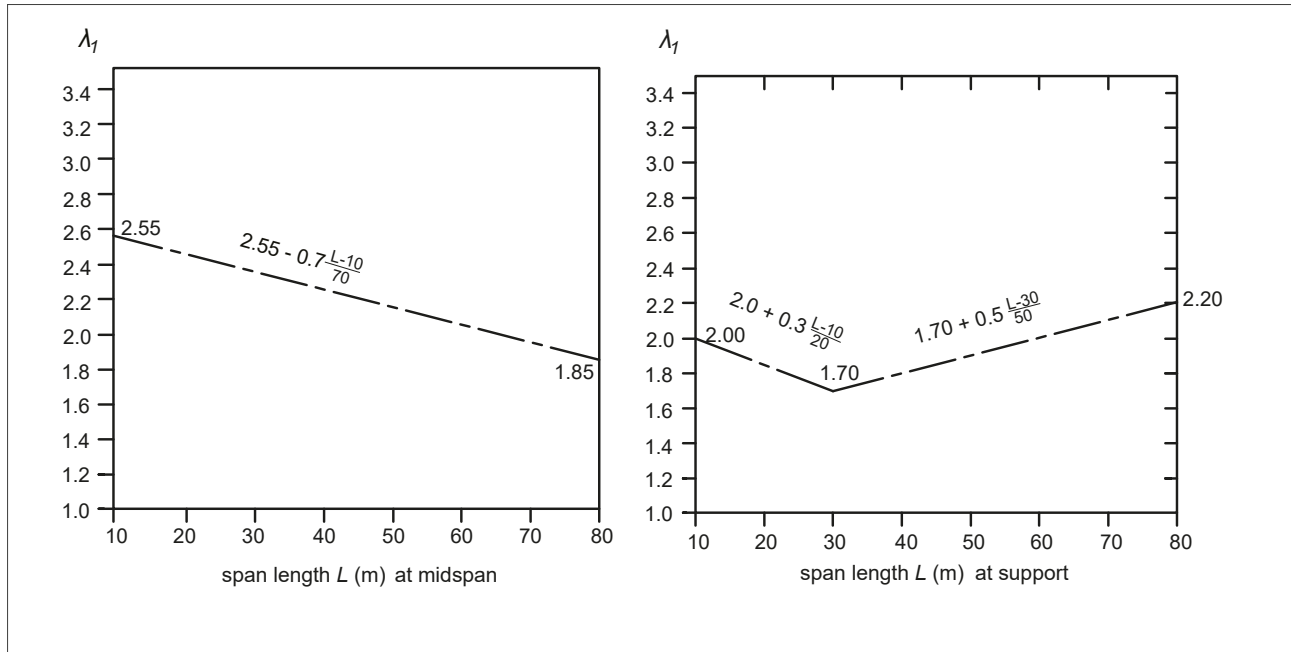
- for the support section, refer Figure 5.56, the span under consideration L_i
- for the midspan section, see refer Figure 5.56, $0.4 \times$ the span under consideration L_i

c) for reactions:

- for end support, the span under consideration L_i
- for intermediate supports, the sum of the two adjacent spans $L_i + L_j$

The recommended value of λ_1 is in Figure 5.54

Figure 5.54 λ_1 for Moments for Road Bridges



λ_2 should be calculated as follows:

$$\lambda_2 = \frac{Q_{m1}}{Q_0} \left(\frac{N_{Obs}}{N_0} \right)^{1/5}$$

Equation 5.158

Where Q_{m1} is the average gross weight (kN) of the lorries in the slow lane obtained from:

$$Q_{m1} = \left(\frac{\sum n_i Q_i^5}{\sum n_i} \right)^{1/5}$$

Q_0 = 480 kN.

N_0 = 0.5×10^6 .

N_{Obs} = The total number of lorries per year in the slow lane.

Q_i = The gross weight in kN of the lorry i in the slow lane as specified by the competent authority.

n_i = The number of lorries of gross weight Q_i in the slow lane as specified by the competent authority.

Note 1: UK National Annexe specifies a value of Q_{m1} as 260 kN for UK Roads that are not subject to weight restrictions.

For given values of Q_{m1} and N_{Obs} , λ_2 may be obtained from Table 5.31.

Table 5.31 λ_2

Q_{m1}	N_{Obs}							
	0.25×10^6	0.50×10^6	0.75×10^6	1.00×10^6	1.25×10^6	1.50×10^6	1.75×10^6	2.00×10^6
200	0.362	0.417	0.452	0.479	0.500	0.519	0.535	0.550
300	0.544	0.625	0.678	0.712	0.751	0.779	0.803	0.825
400	0.725	0.833	0.904	0.957	1.001	1.0381	1.071	1.100
500	0.907	1.042	1.130	1.197	1.251	1.298	1.338	1.374
600	1.088	1.250	1.356	1.436	1.501	1.557	1.606	1.649

The factor λ_3 should be calculated as follows:

$$\lambda_3 = \left(\frac{t_{Ld}}{100} \right)^{1/5}$$

Equation 5.159

Where t_{Ld} is the design life of the bridge in years.

Table 5.32 λ_3

Design life in years	50	60	70	80	90	100	120
Factor λ_3	0.871	0.903	0.931	0.956	0.979	1.00	1.037

The design life of the bridge t_{Ld} may be taken as 100 years.

The factor λ_4 should be calculated as follows:

$$\lambda_4 = \left[1 + \frac{N_2}{N_1} \left(\frac{\eta_2 Q_{m2}}{\eta_1 Q_{m1}} \right)^5 + \frac{N_3}{N_1} \left(\frac{\eta_3 Q_{m3}}{\eta_1 Q_{m1}} \right)^5 + \dots + \frac{N_k}{N_1} \left(\frac{\eta_k Q_{mk}}{\eta_1 Q_{m1}} \right)^5 \right]^{1/5}$$

Equation 5.160

Where,

k = The number of lanes with heavy traffic.

N_j = The number of lorries per year in lane j .

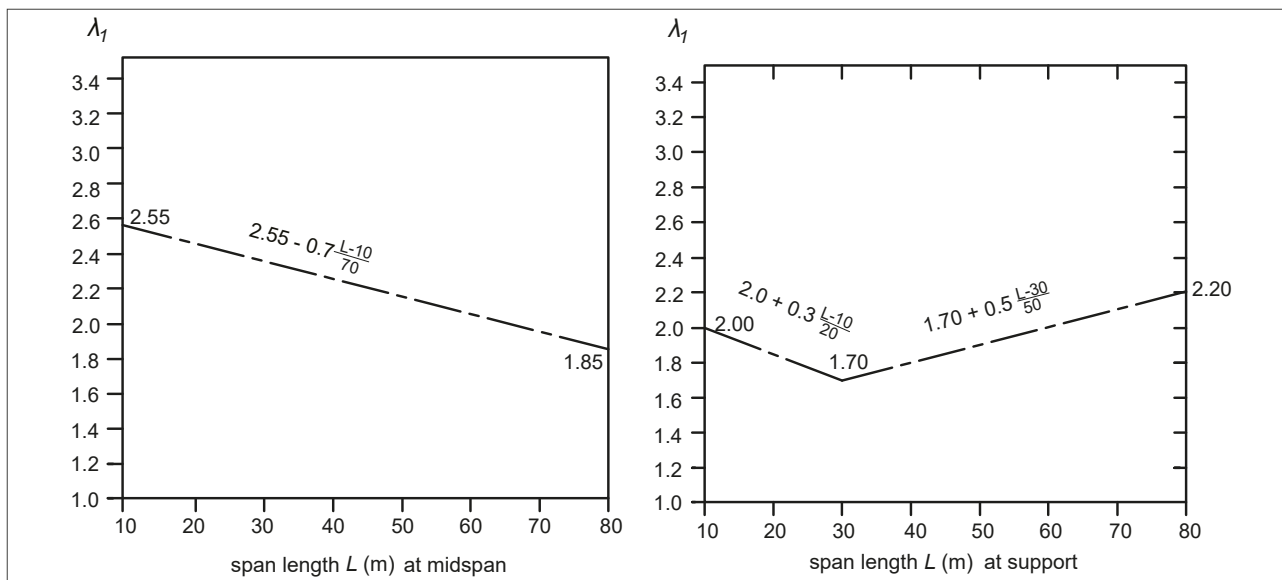
Q_{mj} = The average gross weight of the lorries in lane j .

η_j = The value of the influence line for the internal force that produces the stress range in the middle of lane j to be inserted in Equation 5.160 with positive sign.

Note 2: Q_{m1} may be taken as 260 kN for all traffic lanes in deriving λ_4 .

The factor λ_{max} should be obtained from the relevant stress -range spectrum.

Note 3: The use of the factors λ_{max} in Figure 5.55 is recommended.

Figure 5.54 λ_{max} for Moments for Road Bridges

5.9.5.3 Combination of Damage From Local and Global Stress Ranges

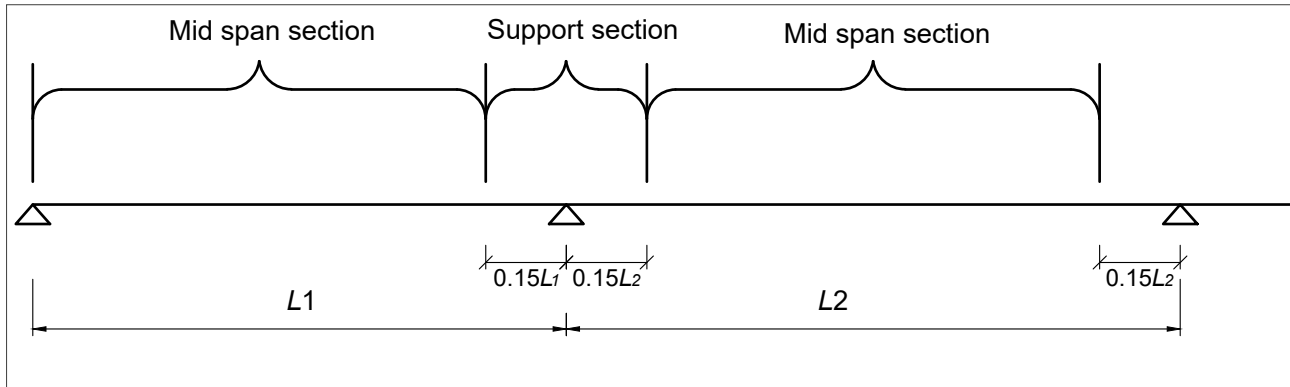
Where the stress verification in a member is due to the combined effects of flexure of the bridge (global effects) and flexure of the internal elements (local effects), the combined effects $\Delta\sigma_{E2}$ should be as follows:

$$\Delta\sigma_{E2} = \lambda_{loc} \times \Phi_{loc} \times \Delta\sigma_{loc} + \lambda_{glo} \times \Phi_{glo} \times \Delta\sigma_{glo}$$

Equation 5.161

Where the suffix "loc" refers to local effects and "glo" refers to global effects.

Figure 5.56 Location of Midspan or Support Section



5.9.6 Fatigue Strength

5.9.6.1 General

The fatigue strength for nominal stress ranges is represented by a series of $(\log \Delta\sigma_R) - (\log N)$ curves and $(\log \Delta\tau_R) - (\log N)$ curves ($S-N$ -curves), which correspond to typical detail categories. Each detail category is designated by a number which represents, in N/mm^2 , the reference value $\Delta\sigma_c$ and $\Delta\tau_c$ for the fatigue strength at 2 million cycles.

For the fatigue strength assessment of bridges, Eurocode EN 1993-1-9 and other special literature should be used.

6 Composite Structures

6.1 General

6.1.1 Scope

This Chapter deals with the design of steel-concrete composite road bridges. In composite construction, steel girders are used as primary members and cast-in-situ reinforced and/or pre-cast concrete slab with necessary in-situ concrete as deck slab. It provides a general basis for the structural design of steel-concrete composite bridges complying the requirements of resistance, durability, safety and serviceability of structures.

6.1.2 References Documents

Following reference documents may be referred while using this Manual.

EN 1090-2	Execution of steel structures and aluminium Structures-Part 2: Technical requirements for execution of steel structures
EN 10025-1	Hot-rolled products of structural steels – Part 1: General delivery conditions
EN 10025-2	Hot-rolled products of structural steels – Part 2: Technical delivery conditions for non-alloy structural steels
EN 10025-3	Hot-rolled products of structural steels – Part 3: Technical delivery conditions for normalised/normalised rolled weldable fine grain structural steels
EN 10025-4	Hot-rolled products of structural steels – Part 4: Technical delivery conditions for thermomechanical rolled weldable fine grain structural steels
EN 10025-5	Hot-rolled products of structural steels – Part 5: Technical delivery conditions for structural steels with improved atmospheric corrosion resistance
EN 10025-6	Hot-rolled products of structural steels – Part 6: Technical delivery conditions for flat products of high yield strength structural steels in the quenched and tempered condition
EN 10326	Continuously hot-dip coated strip and sheet of structural steel – Technical delivery conditions
EN 10149-2	Hot-rolled flat products made of high yield strength steels for cold-forming – Part 2: Delivery conditions for thermo-mechanically rolled steels
EN 10149-3	Hot-rolled flat products made of high yield strength steels for cold-forming – Part 3: Delivery conditions for normalised or normalised rolled steels
EN ISO 13918	Studs and ceramic ferrules for arc stud welding
EN ISO 14555	Arc stud welding of metallic materials

6.1.3 Definitions

For general definitions, refer initial sections of this Manual. Other definitions are as under.

Composite member	A structural member with components of concrete and of structural or cold-formed steel, interconnected by shear connection so as to limit the longitudinal slip between concrete and steel and the separation of one component from the other.
Shear connection	An interconnection between the concrete and steel components of a composite member that has sufficient strength and stiffness to enable the two components to be designed as parts of a single structural member.
Composite behaviour	Behaviour which occurs after the shear connection has become effect due to hardening of concrete.

1

Composite beam

A composite member subjected mainly to bending.

2

Composite column

A composite member subjected mainly to compression or to compression and bending.

3

Composite slab

a slab in which profiled steel sheets are used initially as permanent shuttering subsequently combine structurally with the hardened concrete and act as tensile reinforcement in the finished floor.

4

Composite joint

A joint between a composite member and another composite, steel or reinforced concrete member, in which reinforcement is taken into account in design for the resistance and the stiffness of the joint.

5

Propped structure or member

A structure or member where the weight of concrete elements is applied to the steel elements which are supported in the span, or is carried independently until the concrete elements are able to resist stresses.

6

Un-propped structure or member

A structure or member in which the weight of concrete elements is applied to steel elements which are unsupported in the span.

Un-cracked flexural stiffnessThe stiffness E_{a1} of a cross-section of a composite member where I_1 is the second moment of area of the effective equivalent steel section calculated assuming that concrete in tension is un-cracked.**Cracked flexural stiffness**The stiffness E_{a2} of a cross-section of a composite member where I_2 is the second moment of area of the effective equivalent steel section calculated neglecting concrete in tension but including reinforcement.**Filler beam deck**

A deck consisting of a reinforced concrete slab and partially concrete-encased rolled or welded steel beams, having their bottom flange on the level of the slab bottom.

Composite plate

Composite member consisting of a flat bottom steel plate connected to a concrete slab, in which both the length and width are much larger than the thickness of the composite plate.

6.1.4 Symbols

The general symbols used in this Chapter are as under. Additional symbols are also defined where they appear. Symbols may have various meanings.

A	Cross-sectional area of the effective composite section neglecting concrete in tension.
A_a	Cross-sectional area of the structural steel section.
A_b	Cross-sectional area of bottom transverse reinforcement.
A_{bh}	Cross-sectional area of bottom transverse reinforcement in a haunch.
A_c	Cross-sectional area of concrete.
A_{ct}	Cross-sectional area of the tensile zone of the concrete.
A_{fc}	Cross-sectional area of the compression flange.
A_s	Cross-sectional area of reinforcement.
A_{sf}	Cross-sectional area of transverse reinforcement.
A_t	Cross-sectional area of top transverse reinforcement.
A_v	Shear area of a structural steel section.
A_l	Loaded area under the gusset plate.
E_a	Modulus of elasticity of structural steel.
$E_{c,eff}$	Effective modulus of elasticity for concrete.
E_{cm}	Secant modulus of elasticity of concrete.
E_s	Design value of modulus of elasticity of reinforcing steel.

$(EA)_{eff}$	Effective longitudinal stiffness of cracked concrete.
$(EI)_{eff}$	Effective flexural stiffness for calculation of relative slenderness.
$(EI)_{eff,II}$	Effective flexural stiffness for use in second-order analysis.
$(EI)_2$	Cracked flexural stiffness per unit width of the concrete or composite slab.
F_d	Component in the direction of the steel beam of the design force of a bonded or unbonded tendon applied after the shear connection has become effective.
F_{ten}	Design tensile force per stud.
G_a	Shear modulus of structural steel.
G_c	Shear modulus of concrete.
I	Second moment of area of the effective composite section neglecting concrete in tension.
I_a	Second moment of area of the structural steel section.
I_{at}	St. Venant torsion constant of the structural steel section.
I_c	Second moment of area of the un-cracked concrete section.
I_{eff}	Effective second moment of area of filler beams.
I_s	Second moment of area of the steel reinforcement.
I_1	Second moment of area of the effective equivalent steel section assuming that the concrete in tension is un-cracked.
I_2	Second moment of area of the effective equivalent steel section neglecting concrete in tension but including reinforcement.
$K_c, K_{c,II}$	Correction factors to be used in the design of composite columns.
K_0	Calibration factor to be used in the design of composite columns.
L	Length; span; effective span.
L_e	Equivalent span.
L_i	Span.
L_{A-B}	Length of inelastic region, between points A and B, corresponding to $M_{el,Rd}$ and $M_{Ed,max}$, respectively.
L_v	Length of shear connection.
M	Bending moment.
$M_{a,Ed}$	Design bending moment applied to the structural steel section.
$M_{b,Rd}$	Design value of the buckling resistance moment of a composite beam.
$M_{c,Ed}$	The part of the design bending moment acting on the composite section.
M_{cr}	Elastic critical moment for lateral-torsional buckling of a composite beam.
M_{Ed}	Design bending moment.
$M_{Ed,max}$	Total design bending moment applied to the steel and composite member.
$M_{Ed,max,f}$	Maximum bending moment or internal force due to fatigue loading.
$M_{Ed,min,f}$	Minimum bending moment due to fatigue loading.
$M_{f,Rd}$	Design resistance moment.
$M_{pl,a,Rd}$	Design value of the plastic resistance moment of the structural steel section.
$M_{pl,N,Rd}$	Design value of the plastic resistance moment of the composite section taking into account the compressive normal force.
$M_{pl,Rd}$	Design value of the plastic resistance moment of the composite section with full shear connection.
$M_{pl,y,Rd}$	Design value of the plastic resistance moment about the y-y axis of the composite section with full shear connection.

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$M_{pl,z,Rd}$ Design value of the plastic resistance moment about the z-z axis of the composite section with full shear connection.

2

M_{Rd} Design value of the resistance moment of a composite section.

M_{Rk} Characteristic value of the resistance moment of a composite section or joint.

$M_{y,Ed}$ Design bending moment applied to the composite section about the y-y axis.

$M_{z,Ed}$ Design bending moment applied to the composite section about the z-z axis.

3

N Compressive normal force; number of stress range cycles; number of shear connectors.

N_c Design value of the compressive normal force in the concrete flange.

4

N_{cd} Design compressive force in concrete slab corresponding to $M_{Ed,max}$.

$N_{c,f}$ Design value of the compressive normal force in the concrete flange with full shear connection.

5

$N_{c,el}$ Compressive normal force in the concrete flange corresponding to $M_{el,Rd}$.

$N_{cr,eff}$ Elastic critical load of a composite column corresponding to an effective flexural stiffness.

N_{cr} Elastic critical normal force.

6

N_{Ed} Design value of the compressive normal force.

$N_{Ed,serv}$ Normal force of concrete tension member for SLS.

$N_{Ed,ult}$ Normal force of concrete tension member for ULS.

$N_{G,Ed}$ Design value of the part of the compressive normal force that is permanent.

$N_{pl,Rd}$ Design value of the plastic resistance of the composite section to compressive normal force.

$N_{pl,Rk}$ Characteristic value of the plastic resistance of the composite section to compressive normal force.

$N_{pm,Rd}$ Design value of the resistance of the concrete to compressive normal force.

N_R Number of stress-range cycles.

P_{Ed} Longitudinal force on a connector at distance x from the nearest web.

P_{Rd} Design value of the shear resistance of a single connector.

$V_{a,Ed}$ Design value of the shear force acting on the structural steel section.

$V_{b,Ed}$ Design value of the shear buckling resistance of a steel web.

$V_{c,Ed}$ Design value of the shear force acting on the reinforced concrete cross-section of a filler beam.

V_{Ed} Design value of the shear force acting on the composite section.

V_L Longitudinal shear force, acting along the steel-concrete flange interface.

$V_{L,Ed}$ Longitudinal shear force acting on length L_{A-B} of the inelastic region.

$V_{pl,Rd}$ Design value of the plastic resistance of the composite section to vertical shear.

$V_{pl,a,Rd}$ Design value of the plastic resistance of the structural steel section to vertical shear.

V_{Rd} Design value of the resistance of the composite section to vertical shear.

a Spacing between parallel beams; diameter or width; distance.

a_w Steel flange projection outside the web of the beam.

b Width of the flange of a steel section; width of slab, half the distance between adjacent webs, or the distance between the web and the free edge of the flange.

b_{eff} Total effective width.

$b_{eff,1}$ Effective width at mid-span for a span supported at both ends.

$b_{eff,2}$ Effective width at an internal support.

b_{ei} Effective width of the concrete flange on each side of the web, effective width of composite bottom flange of a box section.

b_f Width of the flange of a steel section.

b_i	Geometric width of the concrete flange on each side of the web.
b_0	Distance between the centres of the outstand shear connectors; mean width of a concrete rib (minimum width for re-entrant sheeting profiles); width of haunch.
c	Width of the outstand of a steel flange; effective perimeter of reinforcing bar.
c_{st}	Concrete cover above the steel beams of filler beam decks.
c_y, c_z	Thickness of concrete cover.
d	Clear depth of the web of the structural steel section; diameter of the shank of a stud connector; overall diameter of circular hollow steel section; minimum transverse dimension of a column.
d_{d0}	Diameter of the weld collar to a stud connector.
e_D	Edge distance.
e_d	Either of $2e_h$ or $2e_v$.
e_h	Lateral distance from the point of application of force F_d to the relevant steel web, if F_d is applied to the concrete slab.
e_v	Vertical distance from the point of application of force F_d to the plane of shear connection concerned, if F_d is applied to the steel element.
f_{cd}	Design value of the cylinder compressive strength of concrete.
f_{ck}	Characteristic value of the cylinder compressive strength of concrete at 28 days.
f_{cm}	Mean value of the measured cylinder compressive strength of concrete.
$f_{ct,eff}$	Mean value of the effective tensile strength of the concrete.
f_{ctm}	Mean value of the axial tensile strength of concrete.
$f_{ct,0}$	Reference strength for concrete in tension.
f_{lctm}	Mean value of the axial tensile strength of lightweight concrete.
f_{pd}	Limiting stress of prestressing tendons.
f_{sd}	Design value of the yield strength of reinforcing steel.
f_{sk}	Characteristic value of the yield strength of reinforcing steel.
f_u	Specified ultimate tensile strength.
f_y	Nominal value of the yield strength of structural steel.
f_{yd}	Design value of the yield strength of structural steel.
h	Overall depth; thickness.
h_c	Thickness of the concrete flange.
h_s	Depth between the centroids of the flanges of the structural steel section.
h_{sc}	Overall nominal height of a stud connector.
k	Amplification factor for second-order effects; coefficient; empirical factor for design shear resistance.
k_c	Coefficient.
k_s	Reduction factor for shear resistance of stud connector.
k_ϕ	Parameter.
k_1	Flexural stiffness of the cracked concrete slab.
k_2	Flexural stiffness of the web.
m	Slope of fatigue strength curve; empirical factor for design shear resistance.
n_L	Modular ratio depending on the type of loading.
n_0	Modular ratio for short-term loading.
n_{0G}	Modular ratio (shear moduli) for short term loading.
n_{LG}	Modular ratio (shear moduli) for short term loading.

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S_f Clear distance between the upper flanges of the steel beams of filler beam decks.

2

S_t Transverse spacing centre-to-centre of the stud shear connectors.

3

S_w Spacing of webs of steel beams of filler beam decks.

4

t Age; thickness.

5

t_w Thickness of the web of the structural steel section.

6

t_f Thickness of the steel flange of the steel beams of filler beam decks.

t_0 Age at loading.

v_{Ed} Design longitudinal shear stress.

$v_{L,Ed}$ Design longitudinal shear force per unit length at the interface between steel and concrete.

$v_{L,Ed,max}$ Maximum design longitudinal shear force per unit length at the interface between steel and concrete.

w_k Design value of crack width.

x_{pl} Distance between the plastic neutral axis and the extreme fibre of the concrete slab in compression.

z_0 Vertical distance.

$\Delta\sigma$ Stress range.

$\Delta\sigma_c$ Reference value of the fatigue strength at 2 million cycles.

$\Delta\sigma_E$ Equivalent constant amplitude stress range.

$\Delta\sigma_{E,glob}$ Equivalent constant amplitude stress range due to global effects.

$\Delta\sigma_{E,loc}$ Equivalent constant amplitude stress range due to local effects.

$\Delta\sigma_{E,2}$ Equivalent constant amplitude stress range related to 2 million cycles.

$\Delta\sigma_s$ Increase of stress in steel reinforcement due to tension stiffening of concrete.

$\Delta\sigma_{s,equ}$ Damage equivalent stress range.

$\Delta\tau$ Range of shear stress for fatigue loading.

$\Delta\tau_c$ Reference value of the fatigue strength at 2 million cycles.

$\Delta\tau_E$ Equivalent constant amplitude stress range.

$\Delta\tau_{E,2}$ Equivalent constant amplitude range of shear stress related to 2 million cycles.

$\Delta\tau_R$ Fatigue shear strength.

α_{cr} Factor by which the design loads would have to be increased to cause elastic instability.

α_M Coefficient related to bending of a composite column.

$\alpha_{M,y}$, $\alpha_{M,z}$ Coefficient related to bending of a composite column about the y-y axis and the z-z axis respectively.

β Factor; transformation parameter, Half of the angle of spread of longitudinal shear force V_I into the concrete slab.

γ_C Partial factor for concrete.

γ_F Partial factor for actions, also accounting for model uncertainties and dimensional variations.

γ_{Ff} Partial factor for equivalent constant amplitude stress range

γ_M Partial factor for a material property, also accounting for model uncertainties and dimensional variations.

γ_{M0} Partial factor for structural steel applied to resistance of cross-sections,

γ_{M1} Partial factor for structural steel applied to resistance of members to instability assessed by member checks.

γ_{Mf} Partial factor for fatigue strength.

$\gamma_{Mf,s}$ Partial factor for fatigue strength of studs in shear.

γ_S	Partial factor for reinforcing steel.
γ_V	Partial factor for design shear resistance of a headed stud.
δ	Factor; steel contribution ratio; central deflection.
ε	$\sqrt{(235/f_y)}$, where f_y is in N/mm ² .
η_a, η_{ao}	Factors related to the confinement of concrete.
η_c, η_{co}	Factors related to the confinement of concrete.
η_{cL}	
θ	Angle.
λ, λ_v	Damage equivalent factors.
$\lambda_{v,l}$	Factor to be used for the determination of the damage equivalent factor λ_v for headed studs in shear.
$\lambda_{glob}, \lambda_{loc}$	Damage equivalent factors for global effects and local effects, respectively.
$\bar{\lambda}$	Relative slenderness.
$\bar{\lambda}_{LT}$	Relative slenderness for lateral-torsional buckling.
μ	Coefficient of friction; nominal factor.
μ_d	Factor related to design for compression and uniaxial bending.
μ_{dy}, μ_{dz}	Factor μ_d related to plane of bending.
ν_a	Poisson's ratio for structural steel.
ρ	Parameter related to reduced design bending resistance accounting for vertical shear.
ρ_s	Parameter; reinforcement ratio.
$\sigma_{c,Rd}$	Local design strength of concrete.
σ_{ct}	Extreme fibre tensile stress in the concrete.
$\sigma_{max,f}$	Maximum stress due to fatigue loading.
$\sigma_{min,f}$	Minimum stress due to fatigue loading.
$\sigma_{s,max,f}$	Stress in the reinforcement due to the bending moment $M_{Ed,max,f}$.
$\sigma_{s,min,f}$	Stress in the reinforcement due to the bending moment $M_{Ed,min,f}$.
σ_s	Stress in the tension reinforcement.
$\sigma_{s,max}$	Stress in the reinforcement due to the bending moment M_{max} .
$\sigma_{s,max,0}$	Stress in the reinforcement due to the bending moment M_{max} , neglecting concrete in tension.
$\sigma_{s,0}$	Stress in the tension reinforcement neglecting tension stiffening of concrete.
τ_{Rd}	Design shear strength.
φ	Diameter (size) of a steel reinforcing bar; damage equivalent impact factor.
φ^*	Diameter (size) of a steel reinforcing bar.
φt	Creep coefficient.
$\varphi(t, t_0)$	Creep coefficient, defining creep between times t and t_0 , related to elastic deformation at 28 days.
χ	Reduction factor for flexural buckling.
χ_{LT}	Reduction factor for lateral-torsional buckling.
ψ_L	Creep multiplier.

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Composite Structures

6.2 Basis of Design

6.2.1 General

Steel concrete composite bridge structures are designed and constructed to satisfy the design requirements for the resistance, serviceability and durability of composite structures. The design shall be based on the characteristic values of material strengths and the loads applied to the structure. The design values are derived from the characteristic values by using partial safety factors both for material strengths and for loads.

6.2.2 Design Actions

The actions and combination of actions for the design of bridges shall be considered as given in Chapter 3 along with the partial factors for actions. Appropriate load combinations with corresponding partial factors shall be used to determine the maximum values of action effects.

6.2.3 Design Resistance

The design values of the resistance shall be obtained by using the characteristic or other representative values applying with partial factor and other factors as provided in relevant Chapters of this Manual.

The design value of material strengths can be derived by applying partial factors to lower characteristic or nominal strengths unless upper estimate of strength is required.

For concrete, a partial factor γ_c shall be applied. The design compressive strength shall be given by

$$f_{cd} = \frac{f_{ck}}{\gamma_c}$$

The recommended value of partial factor for different materials can be taken as below.

- For concrete, γ_c , refer Chapter 4.
- For steel reinforcement, γ_s , refer Chapter 4.
- For structural steel, γ_M , refer Chapter 5.
- For shear connection, headed stud in shear, partial factor $\gamma_v = 1.25$
- For fatigue verification of headed studs, $\gamma_{Mf} =$ refer Chapter 5 and $\gamma_{Mf,s} = 1.0$

The geometric data for cross sections (dimensions, thickness, etc.) may be taken from product standards or drawings for execution.

6.3 Materials

The properties of different materials can be obtained as described below.

6.3.1 Concrete

The properties of concrete shall be obtained as given Chapter 4, unless given in this Chapter. The grade of concrete to be used in composite construction shall not be lower than C20/25 and higher than C60/75.

6.3.2 Reinforcing Steel for Bridges

The properties of reinforcing steel shall be obtained as given Chapter 4. For composite structures, the design value of the modulus of elasticity E_s may be taken as equal to the value of structural steel given in Chapter 5. Ductility characteristics should comply with the provisions given in Chapter 4.

6.3.3 Structural Steel for Bridges

The properties of structural steel shall be obtained as given Chapter 5. The yield strength of structural steel shall not be more than 460 N/mm².

6.3.4 Connecting Devices

6.3.4.1 General

The properties of connecting devices such as fasteners and welding shall be obtained as given Chapter 5.

6.3.4.2 Headed Stud Shear Connectors

Reference should be made to EN 13918.

6.3.4.3 Prestressing Steel & Devices

Reference for same should be made to Chapter 4.

6.4 Durability

6.4.1 General

Relevant provisions given in Chapter 4 and Chapter 5 should be followed.

Detailing of shear connection should be in accordance with Clause 6.6.6.5.

6.4.2 Corrosion Protection at the Steel-concrete Interface in Bridges

The corrosion protection of the steel flange should extend into the steel-concrete interface at least 50mm.

6.5 Structural Analysis

6.5.1 Structural Modelling for Analysis

The structural model shall be adopted such that it reflects the anticipated behaviour of the cross-sections, members, joints and bearings. The effects of the behaviour of the joints on the distribution of internal forces and moments within a structure, and on the overall deformations of the structure, may generally be neglected, but where such effects are significant (such as in the case of semi-continuous joints) they should be taken into account. Semi-continuous joints shall not be used in bridge structures.

Deformation characteristic of the supports where significant, shall be accounted for ground-structure interaction. Effects due to settlements may generally be neglected in ultimate limit states other than fatigue for composite members where all cross sections are in class 1 or 2 and bending resistance is not reduced by lateral torsional buckling.

6.5.2 Structural Stability

The effects of actions may generally be determined using first-order analysis with initial geometry of the structure or second-order analysis considering the deformation of the structure. The effect of deformations (second-order analysis) shall be considered if they increase the action effects significantly or modify significantly the structural behaviour.

First-order analysis may be used if the increase of the relevant internal forces or moments caused by the deformations given by first-order analysis is less than 10%. This condition may be assumed

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to be fulfilled if the following criterion is satisfied:

$$\alpha_{cr} = \frac{F_{cr}}{F_{Ed}} \geq 10$$

Equation 6.1

2

Where,

α_{cr} = The factor by which the design loading would have to be increased to cause elastic instability

F_{Ed} = The design loading on the structure

3

F_{cr} = The elastic critical buckling load for elastic instability

4

Appropriate allowances shall be made for cracking and creep of concrete and for the behaviour of the joints while determining the stiffness of the structure,

5

The methods of analysis for bridge structures shall be in accordance with the provisions given under section "Structural stability of frames" in Chapter 5.

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6.5.3 Imperfections

Structural analysis shall consider appropriate allowances for the effects of imperfections, including residual stresses and geometrical imperfections such as lack of verticality, lack of straightness, lack of flatness, lack of fit and the unavoidable minor eccentricities present in joints of the unloaded structure.

Appropriate value of equivalent geometric imperfections should be used for system imperfections and also member imperfections unless these effects are included in the resistance formulae.

The imperfections and design transverse forces for stabilising transverse frames should be calculated in accordance with Chapter 5. For composite columns and composite compression members, the member imperfections should always be considered when verifying stability within a member's length in accordance with Clause 6.6.7.3.6 or 6.6.7.3.7. Design values of equivalent initial bow imperfection should be taken from Table 6.4. Imperfections within steel compression members should be considered in accordance with Chapter 5.

6.5.4 Calculation of Action Effects

6.5.4.1 Methods of Global Analysis

6.5.4.1.1 General

Action effects may be calculated by elastic global analysis, even where the resistance of a cross-section is based on its plastic or non-linear resistance. Elastic global analysis should be used for serviceability limit states, with appropriate corrections for non-linear effects such as cracking of concrete. Elastic global analysis should be used for verifications of the limit state of fatigue.

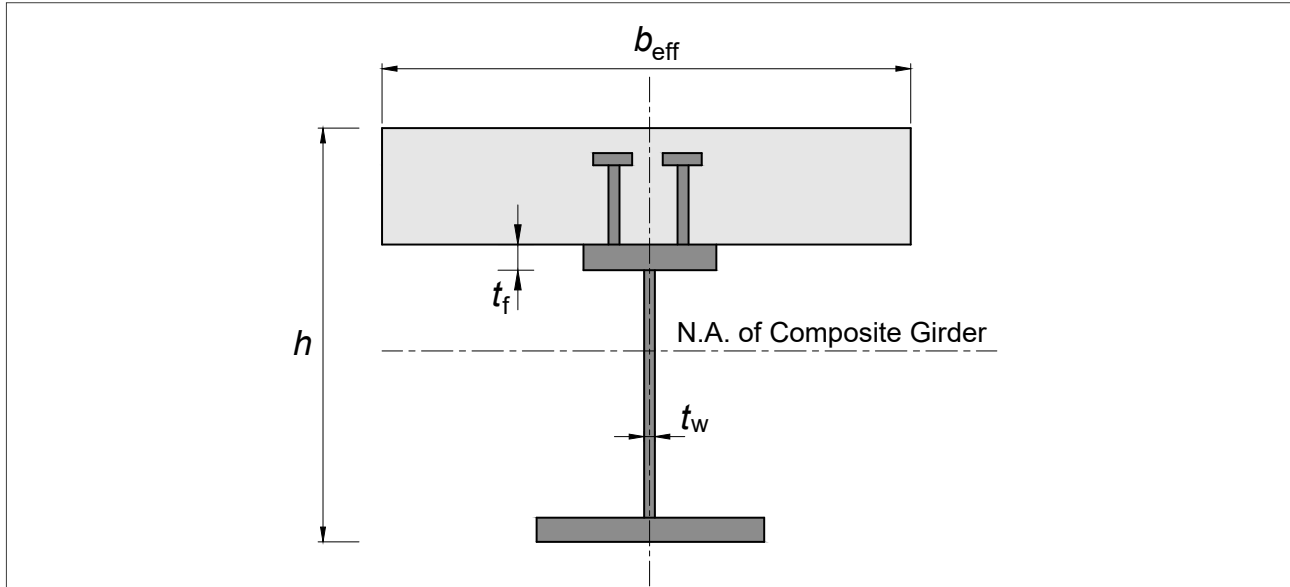
The effects of shear lag and of local buckling shall be taken into account if these significantly influence the global analysis. The effects of local buckling of steel elements on the choice of method of analysis may be taken into account by classifying cross-sections. The effects of local buckling of steel elements on stiffness may be ignored in normal composite sections. For cross-sections of Class 4, refer Chapter 5.

The effects on the global analysis of slip in bolt holes and similar deformations of connecting devices should be considered. Unless non-linear analysis is used, the effects of slip and separation on calculation of internal forces and moments may be neglected at interfaces between steel and concrete where shear connection is provided.

For transient design situations during erection stages, uncracked global analysis and the distribution of effective width according to Clause 6.5.4.1.2 may be used. A constant effective width may be assumed over the whole of each span.

A typical composite girder system is shown in Figure 6.1.

Figure 6.1 Typical Section for Composite Girder



6.5.4.1.2 Effective Width of Flange for Shear Lag

Allowance shall be made for the flexibility of steel or concrete flanges affected by shear in their plane (shear lag) by using an effective width of flange. The effective width of plate elements shall be determined as per Chapter 5. The effective width of concrete flange shall be determined with the following provisions.

When elastic global analysis is used, a constant effective width may be assumed over the whole of each span. This value may be taken as the value b_{eff1} at mid-span for a span supported at both ends, or the value b_{eff2} at the support for a cantilever. At mid-span or an internal support, the total effective width b_{eff} may be determined as follows.

$$b_{eff} = b_0 + \sum b_{ei}$$

Where,

b_0 = The distance between the centres of the outstand shear connectors;

b_{ei} = The effective width of concrete flange on each side of the web
 $= L_e/8$ but not greater than b_i

b_i = The distance from the outstand shear connector to a point mid-way between adjacent webs. At free edge of outer girder, the distance shall be up to free edge.

L_e = The approximate distance between points of zero bending moment

= May be assumed as shown in Figure 6.2.

The effective width at an end support may be determined as:

$$b_{eff} = b_0 + \sum \beta_i b_{ei}$$

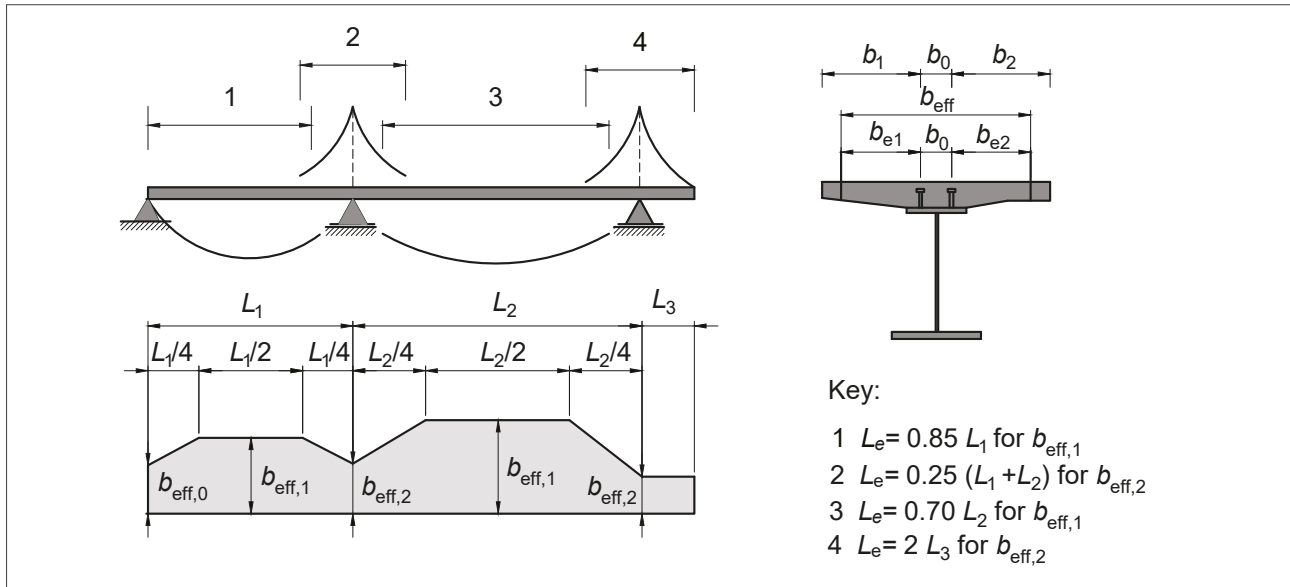
With,

$$\beta_i = \left(0.55 + \frac{0.025L_e}{b_{ei}} \right) \leq 1.0$$

Where,

b_{ei} = The effective width of the end span at mid-span and L_e is the equivalent span of the end span according to Figure 6.2.

Figure 6.2 Effective Width of Concrete Flange



The transverse distribution of stresses due to shear lag may be taken in accordance with Chapter 5 for both concrete and steel flanges.

For cross-sections with bending moments resulting from the main-girder system and from a local system (for example in composite trusses with direct actions on the chord between nodes), the relevant effective widths for the main girder system and the local system should be used for the relevant bending moments.

6.5.4.2 Linear Elastic Analysis

6.5.4.2.1 General

Allowance should be made for the effects of cracking of concrete, creep and shrinkage of concrete, sequence of construction and pre-stressing.

6.5.4.2.2 Creep and Shrinkage

Appropriate allowance shall be made for the effects of creep and shrinkage of concrete. Except for members with both flanges composite, the effects of creep may be taken into account by using modular ratios n_L for the concrete. The modular ratios depending on the type of loading (subscript L) are given by:

$$n_L = n_0 (1 + \psi_L \varphi_t)$$

Where,

n_0 = The modular ratio, E_a/E_{cm} for short-term loading

E_a = The modulus of elasticity of the structural steel

E_{cm} = The secant modulus of elasticity of the concrete for short-term loading

φ_t = The creep coefficient $\varphi(t, t_0)$ depending on the age (t) of concrete at the time of moment considered and the age (t_0) at loading

ψ_L = The creep multiplier depending on the type of loading, which should be taken as 1.1 for permanent loads, 0.55 for primary and secondary effects of shrinkage and 1.5 for pre-stressing by imposed deformations

For permanent loads on composite structures cast in several stages one mean value to may be used for the determination of the creep coefficient. This assumption may also be used for prestressing by imposed deformations, if the age of all of the concrete in the relevant spans at the time of pre-stressing is more than 14 days.

For shrinkage, the age at loading should generally be assumed to be one day. Where prefabricated slabs are used or when pre-stressing of the concrete slab is carried out before the shear connection has become effective, the creep coefficient and the shrinkage values from the time when the composite action becomes effective should be used.

Where in bridges the bending moment distribution at t_0 is significantly changed by creep, for example in continuous beams of mixed structures with both composite and non-composite spans, the time-dependent secondary effects due to creep should be considered, except in global analysis for the ultimate limit state for members where all cross-sections are in Class 1 or 2 and in which no allowance for lateral torsional buckling is necessary. For the time-dependent secondary effects the modular ratio may be determined with a creep multiplier ψ_L of 0.55.

Appropriate account should be taken of the primary and secondary effects caused by shrinkage and creep of the concrete flange. The effects of creep and shrinkage of concrete may be neglected in analysis for verifications of ultimate limit states other than fatigue, for composite members with all cross-sections in Class 1 or 2 and in which no allowance for lateral-torsional buckling is necessary; for serviceability limit states, see Section 6.7.

In regions where the concrete slab is assumed to be cracked, the primary effects due to shrinkage may be neglected in the calculation of secondary effects.

In composite columns and compression members, account should be taken of the effects of creep in accordance with Clause 6.6.7.3.4. For double composite action with both flanges un-cracked (e.g., in case of pre-stressing) the effects of creep and shrinkage should be determined by more accurate methods.

The St. Venant torsional stiffness of box girders should be calculated for a transformed cross section in which the concrete slab thickness is reduced by the modular ratio $n_{0G} = G_a/G_c$ where G_a and G_c are the elastic shear moduli of structural steel and concrete respectively. The effects of creep should be taken into account with the modular ratio $n_{LG} = n_{0G} (1 + \psi_L \varphi_t)$.

6.5.4.3 Effects of Cracking of Concrete

Appropriate allowance shall be made for the effects of cracking of concrete. The following method may be used for the determination of the effects of cracking in composite beams with concrete flanges. First the envelope of the internal forces and moments for the characteristic combinations, including long-term effects should be calculated using the flexural stiffness $E_a I_1$ of the un-cracked sections. This is defined as “uncracked analysis”.

In regions where the extreme fibre tensile stress in the concrete due to the envelope of global effects exceeds twice the strength f_{ctm} , the stiffness should be reduced to $E_a I_2$. This distribution of stiffness may be used for ultimate limit states and for serviceability limit states. A new distribution of internal forces and moments, and deformation if appropriate, is then determined by re-analysis. This is defined as “cracked analysis”.

For continuous composite beams with the concrete flanges above the steel section and not prestressed, including beams in frames that resist horizontal forces by bracing, the following simplified method may be used. Where all the ratios of the length of adjacent continuous spans (shorter / longer) between supports are at least 0.6, the effect of cracking may be taken into account by using the flexural stiffness $E_a I_2$ over 15% of the span on each side of each internal support, and as the uncracked values $E_a I_1$ elsewhere.

The effect of cracking of concrete on the flexural stiffness of composite columns and compression members should be determined in accordance with Clause 6.6.7.3.4. Unless a more precise method is used, in multiple beam decks where transverse composite members are not subjected to tensile forces, it may be assumed that the transverse members are uncracked throughout.

1

The torsional stiffness of box girders should be calculated for a transformed cross section. In areas where the concrete slab is assumed to be cracked due to bending, the calculation should be performed considering a slab thickness reduced to one half, unless the effect of cracking is considered in a more precise way.

2

For ultimate limit states the effects of cracking on the longitudinal shear forces at the interface between the steel and concrete section should be taken into account according to Clause 6.6.6.2. For serviceability limit states the longitudinal shear forces at the interface between the steel and concrete section should be calculated by uncracked analysis. If alternatively, the effects of cracking are taken into account, tension stiffening and over-strength of concrete in tension should be considered.

3

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6.5.4.4 Stages and Sequence of Construction

Appropriate analysis shall be made to cover the effects of staged construction including where necessary separate effects of actions applied to structural steel and to wholly or partially composite members.

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The effects of sequence of construction may be neglected in analysis for ultimate limit states other than fatigue, for composite members where all cross-sections are in Class 1 or 2 and in which no allowance for lateral-torsional buckling is necessary.

6.5.4.5 Temperature effects

Account should be taken of effects due to temperature in accordance with Chapter 3.

Temperature effects may normally be neglected in analysis for the ultimate limit states other than fatigue, for composite members where all cross-sections are in Class 1 or Class 2 and in which no allowance for lateral-torsional buckling is necessary.

For simplification in global analysis and for the determination of stresses for composite structures, the value of the coefficient of linear thermal expansion for structural steel may be taken as 10×10^{-6} per °C. For calculation of change in length of the bridge, the coefficient of thermal expansion should be taken as 12×10^{-6} per °C for all structural materials.

6.5.4.6 Tension Members in Composite Bridges

1. In this Clause, concrete tension member means either:

- a. An isolated reinforced concrete tension member acting together with a tension member of structural steel, with shear connection only at the ends of the member, which causes a global tensile force in the concrete tension member; or
- b. The reinforced concrete part of a composite member with shear connection over the member length (a composite tension member) subjected to longitudinal tension.

Typical examples occur in bowstring arches and trusses where the concrete or composite members act as tension members in the main composite system.

2. The internal forces and moments in bowstring arches with isolated reinforced concrete tension members with shear connection only at the ends of the member may be determined as follows:

- Determination of the internal forces of the steel structure with an effective longitudinal stiffness $(EA_s)_{eff}$ of the cracked concrete tension member.

$$(EA_s)_{eff} = \frac{E_s A_s}{1 - 0.35/(1 + n_0 \rho_s)}$$

Equation 6.2

Where,

n_0 = The modular ratio for short term loading.

A_s = The longitudinal reinforcement of the concrete tension member within the effective width and

ρ_s = The reinforcement ratio.

$\rho_s = A_s/A_c$ determined with the effective concrete cross-section area A_c ,

- The normal forces of the concrete tension member $N_{Ed, serv}$ for the serviceability limit state and $N_{Ed, ult}$ for the ultimate limit state are given by:

$$N_{Ed, serv.} = 1.15 A_c f_{ct, eff} (1 + n_0 \rho_s)$$

Equation 6.3

$$N_{Ed, ult.} = 1.45 A_c f_{ct, eff} (1 + n_0 \rho_s)$$

Equation 6.4

Where $f_{ct, eff}$ is the effective tensile strength of concrete.

Unless verified by more accurate methods, the effective tensile strength may be assumed as $f_{ct, eff} = 0.7 f_{ctm}$ where the concrete tension member is simultaneously acting as a deck and is subjected to combined global and local effects.

- For composite tension members subjected to normal forces and bending moments, the cross-section properties of the cracked section and the normal force of the reinforced concrete part of the composite member should be determined with the effective longitudinal stiffness of the reinforcement according to Equation 6.2. if the normal forces of reinforced concrete part of the member do not exceed the values given by the Equation 6.3 and 6.4, these values should be used for the design. Stresses in reinforcement should be determined with these forces but taking into account the actual cross-section area A_s of reinforcement.

6.5.4.7 Filler Beam Decks for Bridges

Where the detailing is in accordance with Clause 6.6.3, in longitudinal bending the effects of slip between the concrete and the steel beams and effects of shear lag may be neglected. The contribution of formwork supported from the steel beams, which becomes part of the permanent construction, should be neglected.

Where the distribution of loads applied after hardening of concrete is not uniform in the direction transverse to the span of the filler beams, the analysis should take account of the transverse distribution of forces due to the difference between the deformation of adjacent filler beams and of the flexural stiffness transverse to the filler beam, unless it is verified that sufficient accuracy is obtained by a simplified analysis assuming rigid behaviour in the transverse direction.

Account may be taken of the effects described above by using one of the following methods of analysis:

- Modelling by an orthotropic slab by smearing of the steel beams.
- Considering the concrete as discontinuous so as to have a plane grid with members having flexural and torsional stiffness where the torsional stiffness of the steel section may be neglected. For the determination of internal forces in the transverse direction, the flexural and torsional stiffness of the transverse concrete members may be assumed to be 50 % of the uncracked stiffness.

The nominal value of Poisson's ratio of concrete may be assumed to be zero for ultimate limit states and 0.2 for serviceability limit states.

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Composite Structures

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Internal forces and moments should be determined by elastic analysis, neglecting redistribution of moments and internal forces due to cracking of concrete.

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Hogging bending moments of continuous filler beams with Class 1 cross-sections at internal supports may be redistributed for ultimate limit states other than fatigue by amounts not exceeding 15% to take into account inelastic behaviour of materials. For each load case the internal forces and moments after redistribution should be in equilibrium with the loads.

3

Effects of creep on deformations may be taken into account according to Clause 6.5.4.2.2. The effects of shrinkage of concrete may be neglected.

4

For the determination of deflections and precamber for the serviceability limit state as well as for dynamic analysis, the effective flexural stiffness of filler beam decks may be taken as

$$E_a I_{eff} = 0.5(E_a I_1 + E_a I_2)$$

5

Where I_1 and I_2 are the uncracked and the cracked values of second moment of area of the composite cross-section subjected to sagging bending. The second moment of area I_2 should be determined with the effective cross-section of structural steel, reinforcement and concrete in compression. The area of concrete in compression may be determined from the plastic stress distribution.

6

6.5.5 Combination of Global and Local Action Effects

To determine the design values, global and local action effects should be added taking into account a combination factor as follows.

$$\sigma_{Ed} = \sigma_{loc,d} + \psi \sigma_{glob,d}$$

Equation 6.5

$$\sigma_{Ed} = \psi \sigma_{loc,d} + \sigma_{glob,d}$$

Equation 6.6

Where,

- σ_{Ed} = The design value of stress in the stringer due to combined effects of local load $\sigma_{loc,Ed}$ and global load $\sigma_{glob,Ed}$.
- $\sigma_{loc,d}$ = The design value of stress in the stringer due to local wheel or tyre load from a single heavy vehicle.
- $\sigma_{glob,d}$ = The design value of stress in the stringer due to bridge loads comprising one or more heavy vehicle.
- ψ = The combination factor, should be taken as 1.0 for bridges other than road bridges. For road bridges, it shall be taken as 1.0 for spans length less than 20 m, 0.7 for spans length greater than 40 m. Interpolated values to be used for span length between 20 m and 40 m.

6.5.6 Classification of Cross-sections

6.5.6.1 General

The classification system for steel cross sections as defined in Chapter 5, shall be applicable to cross sections of composite beams. The composite cross section shall be classified according to the least favourable class of its steel elements in compression. The class of a composite section normally depends on the direction of the bending moment at that section.

For classification, the plastic stress distribution should be used except at the boundary between Classes 3 and 4, where the elastic stress distribution should be used taking into account sequence of construction and the effects of creep and shrinkage. For classification, design values of strengths of materials should be used. Concrete in tension should be neglected. The distribution of the stresses should be determined for the gross cross-section of the steel web and the effective flanges.

For cross-sections in Class 1 and 2 with bars in tension, reinforcement used within the effective width should have a ductility Class B or C. Additionally for a section whose resistance moment is

determined by Clause 6.6.2.1.2, 6.6.2.1.3 or 6.6.2.1.4, a minimum area of reinforcement within the effective width of the concrete flange should be provided to satisfy the following condition:

$$A_s \geq \rho_s A_c$$

Equation 6.7

With,

$$\rho_s = \delta \frac{f_y}{235} \frac{f_{ctm}}{f_{sk}} \sqrt{k_c}$$

Equation 6.8

Where,

A_c = The effective area of the concrete flange;

f_y = The nominal value of the yield strength of the structural steel in N/mm^2

f_{sk} = The characteristic yield strength of the reinforcement

f_{ctm} = The mean tensile strength of the concrete

k_c = A coefficient given in Clause 6.7.2.

δ = Equal to 1.0 for Class 2 cross-sections, and equal to 1.1 for Class 1 cross-sections at which plastic hinge rotation is required.

Welded mesh should not be included in the effective section unless it has been shown to have sufficient ductility, when built into a concrete slab, to ensure that it will not fracture.

In global analysis for stages in construction, account should be taken of the class of the steel section at the stage considered.

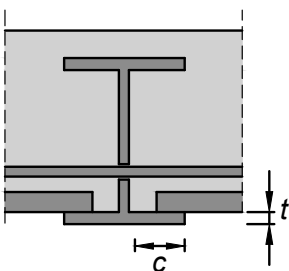
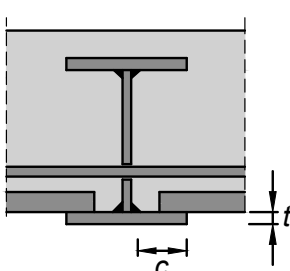
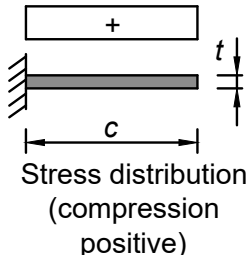
6.5.6.2 Classification of Composite Sections Without Concrete Encasement

A steel compression flange that is restrained from buckling by effective attachment to a concrete flange by shear connectors may be assumed to be in Class 1. The other steel flanges and webs in compression in composite beams without concrete encasement should be classified on the basis of width to thickness ratios in accordance with Chapter 5. An element that fails to satisfy the limits for Class 3 should be taken as Class 4. Cross-sections with webs in Class 3 and flanges in Classes 1 or 2 may be treated as an effective cross-section in Class 2 with an effective web in accordance with Chapter 5.

6.5.6.3 Classification of Sections of Filler Beam Decks for Bridges

A steel outstand of a composite section should be classified in accordance with Table 6.1. A web in Class 3 that is encased in concrete may be represented by an effective web of the same cross-section in Class 2.

Table 6.1 Maximum Values c/t for Steel Flanges in Filler Beams

 rolled section  welded section			 Stress distribution (compression positive)
Class	Type	Limit max (c/t)	
1	Rolled or welded	$c/t \leq 9\epsilon$	$\epsilon = \sqrt{\frac{235}{f_y}}$ with f_y in N/mm^2
2		$c/t \leq 14\epsilon$	
3		$c/t \leq 20\epsilon$	

1

6.6 Ultimate Limit States

2

6.6.1 Composite Beams

The Composite beams should be checked for:

- Resistance of cross-sections.
- Resistance to lateral-torsional buckling.
- Resistance to shear buckling and in-plane applied to webs.
- Resistance to longitudinal shear.
- Resistance to fatigue.

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The effective width of the concrete flange for verification of cross-sections should be determined in accordance with Clause 6.5.4.1.2 taking into account the distribution of effective width between supports and mid-span regions.

6

6.6.2 Resistance of cross-sections of beams

6.6.2.1 Resistance to Bending

6.6.2.1.1 General

The design bending resistance shall be determined by rigid-plastic theory only where the effective composite cross-section is in Class 1 or Class 2.

Elastic analysis and non-linear theory for bending resistance may be applied to cross-sections of any class. For elastic analysis and non-linear theory, it may be assumed that the composite cross-section remains plane if the shear connection and the transverse reinforcement are designed in accordance with Clause 6.6.6, considering appropriate distributions of design longitudinal shear force. The tensile strength of concrete shall be neglected.

Where the steel section of a composite member is curved in plan, the effects of curvature should be taken into account.

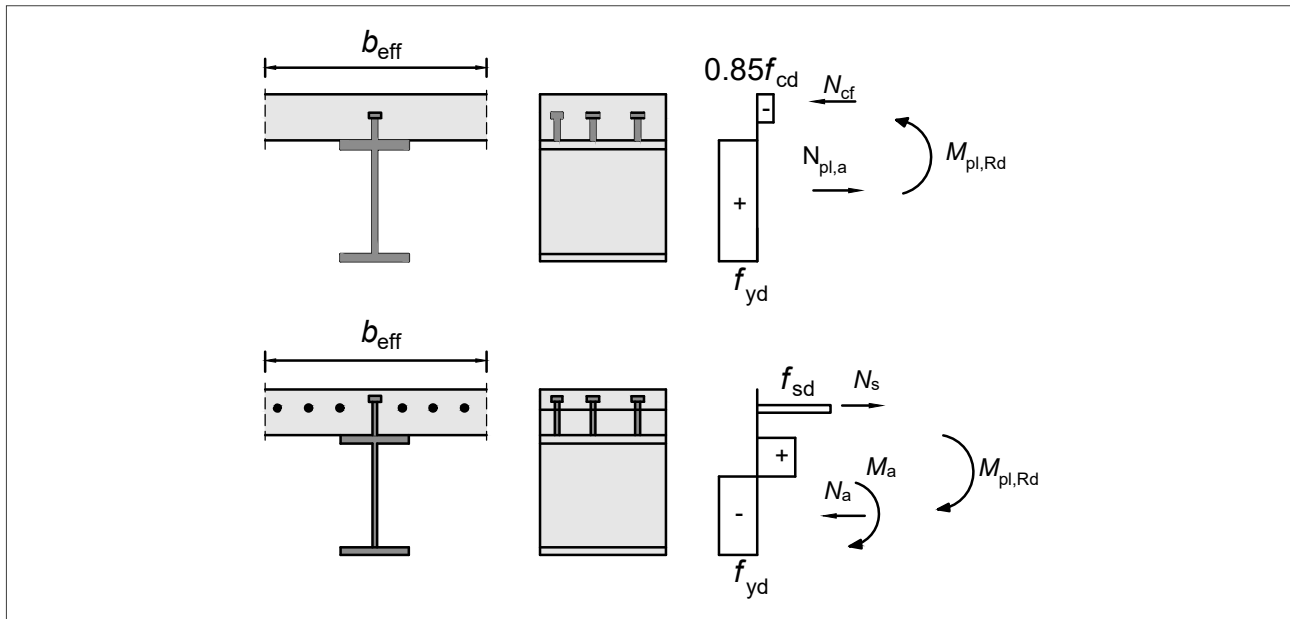
6.6.2.1.2 Plastic Resistance Moment $M_{pl,Rd}$

Following assumptions are made in the calculation of plastic resistance moment of a composite cross-section:

- a. There is full interaction between structural steel, reinforcement, and concrete
- b. The effective area of the structural steel member is stressed to its design yield strength f_{yd} in tension or compression
- c. The effective areas of longitudinal reinforcement in tension and in compression are stressed to their design yield strength f_{sd} in tension or compression. Alternatively, reinforcement in compression in a concrete slab may be neglected
- d. The effective area of concrete in compression resists a stress of $0.85 f_{cd}$, constant over the whole depth between the plastic neutral axis and the most compressed fibre of the concrete, where f_{cd} is the design cylinder compressive strength of concrete

Typical plastic stress distributions for a composite beam with a solid slab and full shear connection in sagging and hogging bending are shown in:

Figure 6.3 Typical Plastic Stress Distribution in Sagging and Hogging Bending

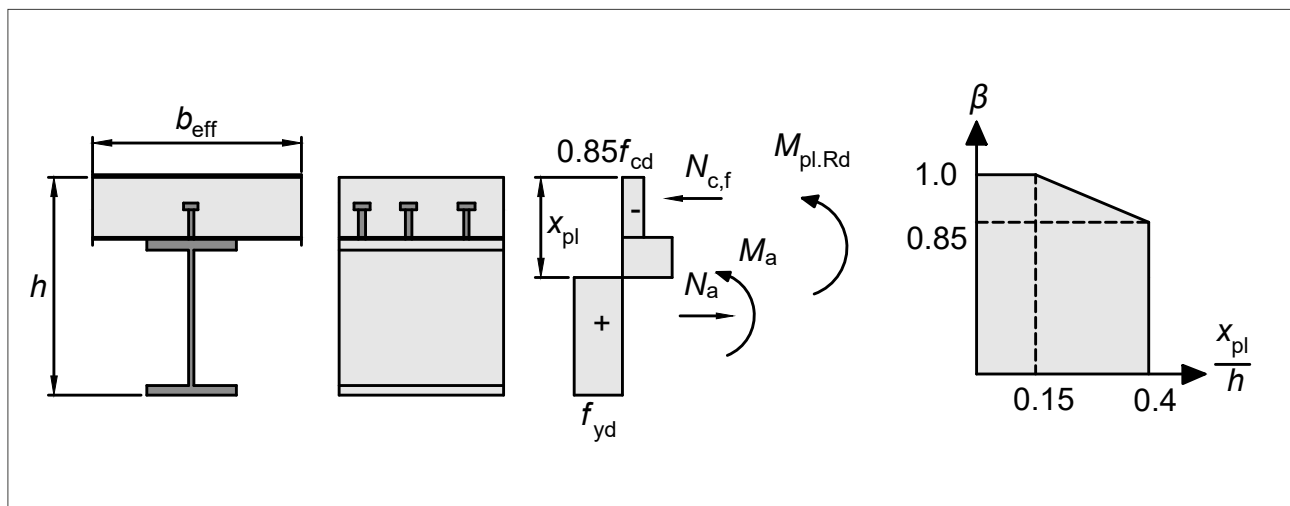


For composite cross-sections with structural steel grade S420 or S460, where the distance x_{pl} between the plastic neutral axis and the extreme fibre of the concrete slab in compression exceeds 15% of the overall depth h of the member, the design resistance moment M_{Rd} should be taken as $\beta M_{pl,Rd}$ where β is the reduction factor given in Figure 6.4. For values of x_{pl} / h greater than 0.4, the resistance to bending should be determined from Clause 6.6.2.1.3 or 6.6.2.1.4.

$$\beta = 1.0, \text{ if } x_{pl} / h \leq 0.15$$

$$\beta = 0.85, \text{ if } x_{pl} / h = 0.40$$

The value of β to be interpolated for intermediate values of x_{pl} / h .

Figure 6.4 Reduction Factor β for $M_{pl,Rd}$ 

Where plastic theory is used and reinforcement is in tension, that reinforcement should be in accordance with Clause 6.5.6.

1

Where a composite beam is subjected to biaxial bending, combined bending and torsion, or combined global and local effects, following yield criterion may be used for the elastic verification:

$$\left(\frac{\sigma_{x,Ed}}{f_y/\gamma_{M0}}\right)^2 + \left(\frac{\sigma_{z,Ed}}{f_y/\gamma_{M0}}\right)^2 - \left(\frac{\sigma_{x,Ed}}{f_y/\gamma_{M0}}\right)\left(\frac{\sigma_{z,Ed}}{f_y/\gamma_{M0}}\right) + 3\left(\frac{\tau_{Ed}}{f_y/\gamma_{M0}}\right)^2 \leq 1$$

Equation 6.9

2

Where,

$\sigma_{x,Ed}$ = The design value of the longitudinal stress at the point of consideration

$\sigma_{z,Ed}$ = The design value of the transverse stress at the point of consideration

τ_{Ed} = The design value of the shear stress at the point of consideration

3

4

Where elastic global analysis is used for a continuous beam, M_{Ed} should not exceed $0.9 M_{pl,Rd}$ at any cross-section in Class 1 or 2 in sagging bending with the concrete slab in compression where both:

5

- The cross-section in hogging bending at or near an adjacent support is in Class 3 or 4, and
- The ratio of lengths of the spans adjacent to that support (shorter/longer) is less than 0.6.

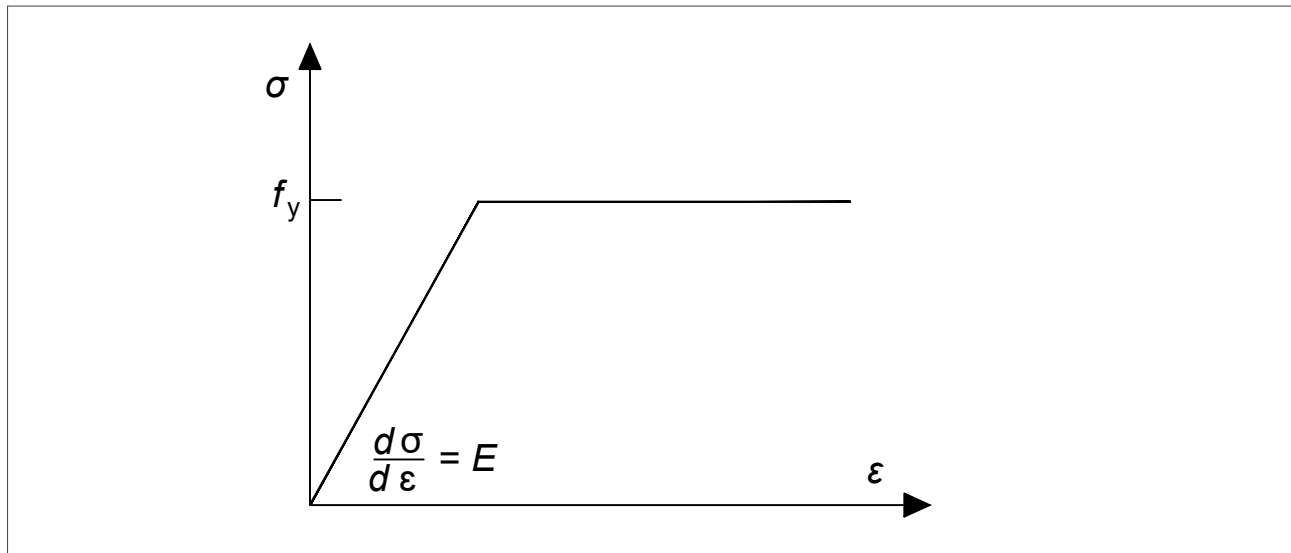
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6.6.2.1.3 Non-linear Resistance to Bending

Where the bending resistance of a composite cross-section is determined by non-linear theory, the stress-strain relationships of the materials shall be taken into account. It should be assumed that the composite cross-section remains plane and that the strain in bonded reinforcement, whether in tension or compression, is the same as the mean strain in the surrounding concrete.

The stresses in the concrete in compression should be derived from the stress-strain curves given in Chapter 4. The stresses in the reinforcement should be derived from the bi-linear diagrams given in Chapter 4. The stresses in structural steel in compression or tension should be derived from the bi-linear diagram given in Figure 6.5 and should take account of the effects of the method of construction (e.g. propped or un-propped).

Figure 6.5 Bi-linear Stress-strain Relationship



For Class 1 and Class 2 composite cross-sections with the concrete flange in compression, the non-linear resistance to bending M_{Rd} may be determined as a function of the compressive force in the concrete N_c as shown in Figure 6.6:

$$M_{Rd} = M_{a,Ed} + (M_{cl,Rd} - M_{a,Ed}) \frac{N_c}{N_{c,el}} \quad \text{for } N_c \leq N_{c,el}$$

Equation 6.10

$$M_{Rd} = M_{el,Rd} + (M_{pl,Rd} - M_{el,Rd}) \frac{N_c - N_{c,el}}{N_{c,f} - N_{c,el}} \quad \text{for } N_{c,el} \leq N_c \leq N_{c,f}$$

Equation 6.11

With,

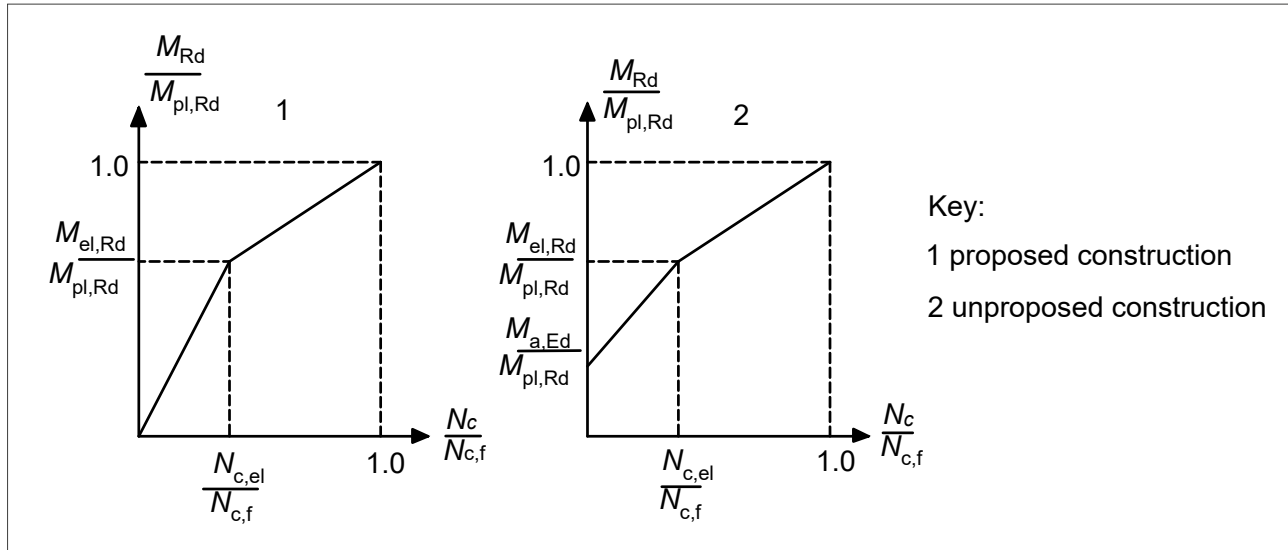
$$M_{el,Rd} = M_{a,Ed} + k M_{c,Ed}$$

Equation 6.12

Where,

 $M_{a,Ed}$ = The design bending moment applied to structural steel section before composite behaviour. $M_{c,Ed}$ = The part of the design bending moment acting on the composite section. k = The lowest factor such that a stress limit is reached; where un-propped construction is used, the sequence of construction should be taken into account. $N_{c,el}$ = The compressive force in the concrete flange corresponding to moment $M_{el,Rd}$. $M_{el,Rd}$ = Design elastic resistance moment of the composite section.

Reduced value $\beta M_{pl,Rd}$ instead of $M_{pl,Rd}$ shall be used in Equation 6.11 and Figure 6.6 for the cross sections where reduction is applicable as in Clause 6.6.2.1.2.

Figure 6.6 Simplified Relationship Between M_{Rd} and N_c 

Where the bending resistance of a composite cross-section is determined by non-linear theory, the stresses in prestressing steel should be derived from the appropriate design curves given in Chapter 4. The design initial pre-strain in prestressing tendons should be taken into account when assessing the stresses in the tendons.

6.6.2.1.4 Elastic Resistance to Bending

Stresses should be calculated by elastic theory, using an effective width of the concrete flange in accordance with Clause 6.6.1. For cross-sections in class 4, the effective structural steel section should be determined as given in Chapter 5. In the calculation of the elastic resistance to bending based on the effective cross-section, the limiting stresses should be taken as:

- f_{cd} in concrete in compression
- f_{yd} in structural steel in tension or compression
- f_{sd} in reinforcement in tension or compression. Alternatively, reinforcement in compression in a concrete slab may be neglected.

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Composite Structures

Stresses due to actions on the structural steelwork alone shall be added to stresses due to actions on the composite member. Unless a more precise method is used, the effect of creep should be taken into account by use of a modular ratio according to Clause 6.5.4.2.2. In cross-sections with concrete in tension and assumed to be cracked, the stresses due to primary (isostatic) effects of shrinkage may be neglected. Compression flanges should be checked for lateral torsional buckling in accordance with Clause 6.6.4.

For composite bridges with cross-sections in Class 4 designed considering plate buckling effects as in Chapter 5, the sum of stresses from different stages of construction and use, calculated on gross sections, should be used for calculating the effective steel cross-section at the time considered. These effective cross-sections should be used for checking stresses in the composite section at the different stages of construction and use.

In the calculation of the elastic resistance to bending based on the effective cross-section, the limiting stress in prestressing tendons should be taken as f_{pd} as given in Chapter 4.

As an alternative to above, for cross-sections in Class 4, the reduced stress method given Clause 5.6.20.2 may also be used.

6.6.2.2 Resistance to Vertical Shear

This Clause is applicable for composite beams with rolled or welded structural steel sections with a solid web, which may be stiffened.

6.6.2.2.1 Plastic Resistance to Vertical Shear

The plastic resistance to vertical shear $V_{pl,Rd}$ should be taken as the plastic resistance of the structural steel section $V_{pl,a,Rd}$. The design plastic shear resistance $V_{pl,a,Rd}$ of the structural steel section shall be determined as given in Chapter 5.

6.6.2.2.2 Shear Buckling Resistance

The shear buckling resistance $V_{b,Rd}$ of an uncased steel web should be determined in accordance with Chapter 5 neglecting the contribution from concrete slab.

6.6.2.2.3 Bending and Vertical Shear

Where the vertical shear force V_{Ed} exceeds half the shear resistance V_{Rd} given by $V_{pl,Rd}$ or $V_{b,Rd}$, whichever is the smaller, allowance should be made for its effect on the resistance moment. For cross-sections in Class 1 or 2, the influence of the vertical shear on the resistance to bending may be taken into account by a reduced design steel strength $(1 - \rho) f_{yd}$ in the shear area as shown in Figure 6.7 where:

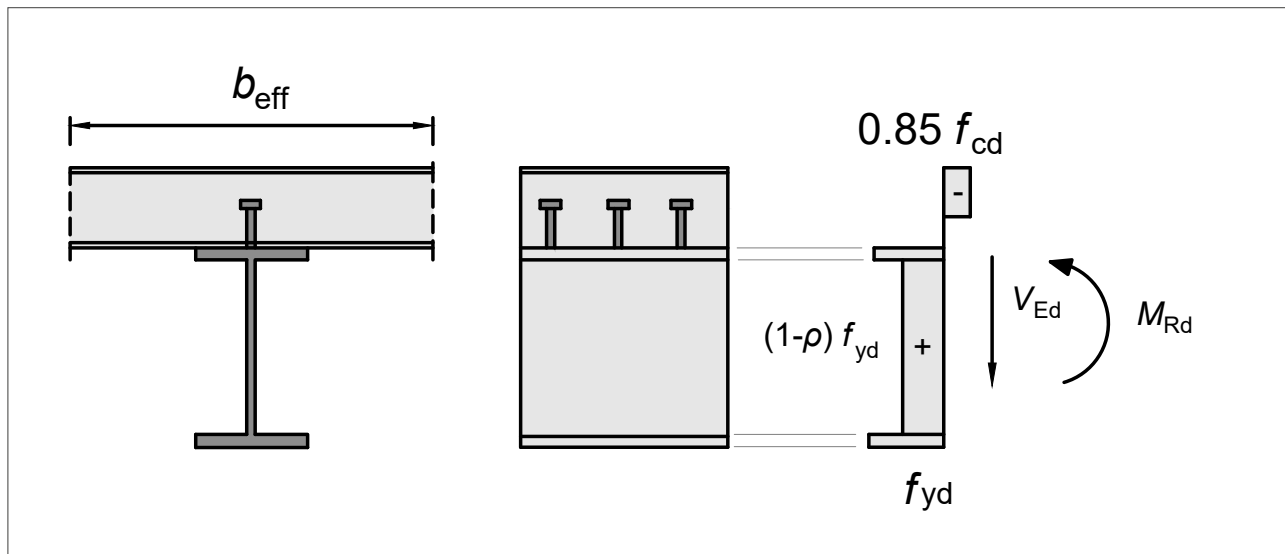
$$\rho = \left(\frac{2 V_{Ed}}{V_{Rd}} - 1 \right)^2$$

Equation 6.13

V_{Rd} = The appropriate resistance to vertical shear $V_{pl,Rd}$ or $V_{b,Rd}$.

For cross-sections in Classes 3 and 4, the section for "Interaction between shear force, bending moment and axial force" in Chapter 5 shall be applicable using as M_{Ed} the total bending moment in the considered cross section and both $M_{pl,Rd}$ and $M_{f,Rd}$ for the composite cross section. No account should be taken of the change in the position of the plastic neutral axis of the cross-section caused by the reduced yield strength $(1 - \rho) f_{yd}$ when classifying the web in accordance with Clause 6.5.6.

Figure 6.7 Plastic Stress Distribution Modified by Effect of Vertical Shear



When calculating the shear resistance contribution from flange, as in Chapter 5 (Clause 5.6.12.2.2), for a beam with one flange composite, the dimension of the non-composite flange may be used even if that is the larger steel flange. The axial normal force N_{Ed} should be taken as the axial force acting on the composite section. For composite flanges, the effective area should be used.

For the calculation of $M_{f,Rd}$ in applying interaction between shear force, bending moment and axial force, as in Chapter 5 (Clause 5.6.20.3), the design plastic resistance to bending of the effective composite section excluding the steel web should be used.

For vertical shear in a concrete flange of a composite member reference shall be made to Chapter 4 considering member not requiring design shear reinforcement.

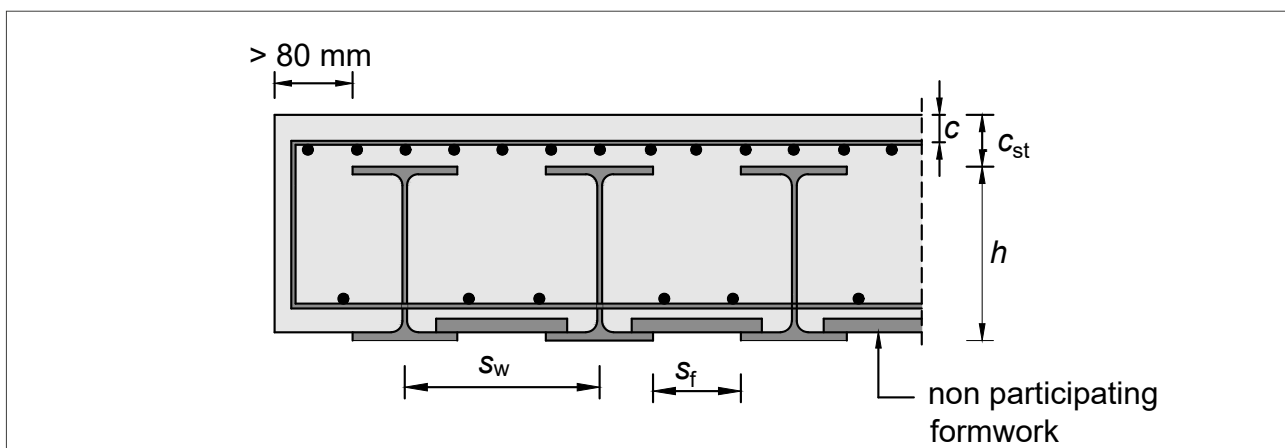
Note: While determining design shear resistance of concrete flange in tension, the value for k_1 should take into account specific aspects of composite action. The recommended values are $C_{Rd,c} = 0.15/\gamma_c$ and $k_1 = 0.12$. Also, where the stress σ_{cp} is tensile (that is, $\sigma_{cp} < 0$) and $\sigma_{cp} > \sigma_{cp,0}$, then σ_{cp} should be replaced by $\sigma_{cp,0}$ with the recommended value $\sigma_{cp,0} = -1.85 \text{ N/mm}^2$.

6.6.3 Filler Beam Decks

6.6.3.1 General

Filler beam deck consists of a reinforced concrete slab and partially concrete-encased rolled or welded steel beams, having their bottom flange on the level of the slab bottom. A typical cross-section of a filler beam deck with non-participating permanent formwork is shown in Figure 6.8.

Figure 6.8 Typical Cross-section of a Filler Beam Deck



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Steel beams may be rolled sections, or welded sections with a uniform cross-section. For welded sections, both the width of the flanges and the depth of the web should be within the ranges that are available for rolled H - or I - sections. Spans may be simply supported or continuous. Supports may be square or skew.

2

Filler-beam decks should comply with the following:

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- The steel beams are not curved in plan.
- The skew angle θ should not be greater than 30° (for non-skew deck the value $\theta = 0$).
- The nominal depth h of the steel beams complies with: $210 \leq h \leq 1100$ mm.
- The spacing s_w of webs of the steel beams should not exceed the lesser of $h/3 + 600$ mm and 750 mm, where h is the nominal depth of the steel beams in mm.
- The concrete cover c_{st} above the steel beams satisfies the conditions:

5

$$c_{st} \geq 70 \text{ mm}, \quad c_{st} \leq 150 \text{ mm}, \quad c_{st} \leq h/3, \quad c_{st} \leq x_{pl} - t_f$$

6

Where x_{pl} is the distance between the plastic neutral axis for sagging bending and the extreme fibre of the concrete in compression, and t_f is the thickness of the steel flange.

- The concrete cover to the side of an encased steel flange is not less than 80 mm.
- The clear distance s_f between the upper flanges of the steel beams is not less than 150 mm, so as to allow pouring and compaction of concrete.
- The soffit of the lower flange of the steel beams is not encased.
- A bottom layer of transverse reinforcement passes through the webs of the beams, and is anchored beyond the end steel beams, and at each end of each bar, so as to develop its yield strength; ribbed bars are used; their diameter is not less than 16 mm and their spacing is not more than 300 mm.
- Normal-density concrete is used.
- The surface of the steel beams should be descaled. The soffit, the upper surfaces and the edges of the lower flange of the steel beam should be protected against corrosion.
- For road bridges, the holes in the webs of the steel section should be drilled.

Filler beam decks should be designed for ultimate limit states and for the serviceability limit state. Steel beams with bolted connections and/or welding should be checked against fatigue. Composite cross-sections should be classified according to Clause 6.5.6.3. Mechanical shear connection need not be provided.

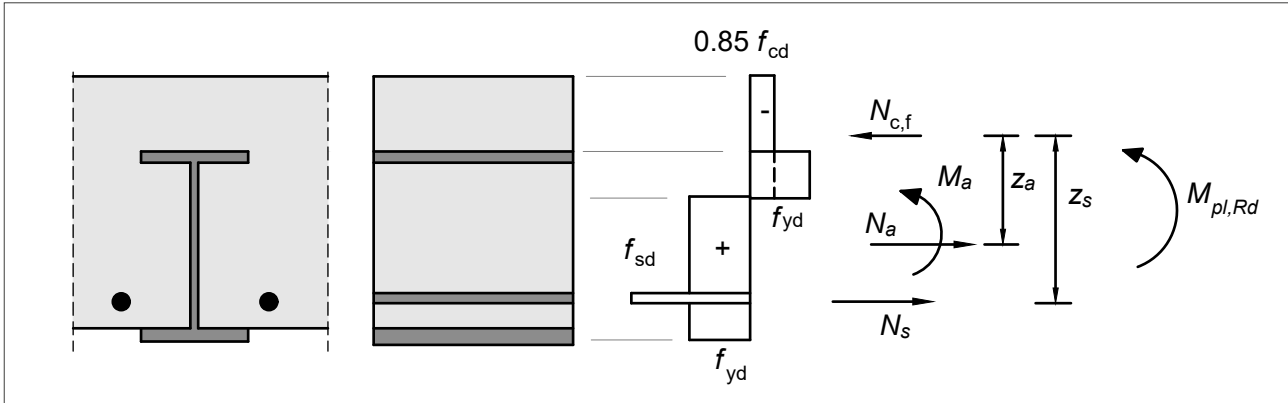
6.6.3.2 Resistance in Bending

The design resistance of composite cross-sections to bending moments should be determined according to Clause 6.6.2.1. Where the vertical shear force $V_{a,Ed}$ on the steel section exceeds half of the shear resistance by Clause 6.6.3.3, allowance should be made for its effect on the resistance moment in accordance with Clause 6.6.2.2.3. The design resistance of reinforced concrete sections to transverse bending moments should be determined according to Chapter 4.

6.6.3.3 Vertical Shear

The resistance of the composite cross-section to vertical shear should be taken as the resistance of the structural steel section $V_{pl,a,Rd}$ unless the value of a contribution from the reinforced concrete part has been established in accordance with Chapter 4. Unless a more accurate analysis is used, the part $V_{c,Ed}$ of the total vertical shear V_{Ed} acting on the reinforced concrete part may be taken as $V_{c,Ed} = V_{Ed} (M_{s,Rd} / M_{pl,Rd})$, with $M_{s,Rd} = N_s z_s = A_s f_{sd} z_s$. The lever arm z_s is shown in Figure 6.9 for a filler-beam deck in Class 1 or 2.

Figure 6.9 Stress Distribution for Filler-beam Deck in Class 1 or 2



The design resistance to vertical shear of reinforced concrete sections between filler beams should be verified according to Chapter 4. The resistance and stability of steel beams during execution before hardening of concrete should be verified according to provisions given in Chapter 5.

6.6.4 Lateral-torsional Buckling of Composite Beams

6.6.4.1 General

A steel flange that is attached to a concrete or composite slab by shear connection may be assumed to be laterally stable, provided that lateral instability of the concrete slab is prevented. All other steel flanges in compression should be checked for lateral stability.

The methods given for determining the resistance of uniform members in bending and general method for lateral and lateral torsional buckling of structural components in Chapter 5 are applicable to the steel section on the basis of the cross-sectional forces on the composite section, taking into account effects of sequence of construction in accordance with Clause 6.5.4.4. The lateral and elastic torsional restraint at the level of the shear connection to the concrete slab may be taken into account.

6.6.4.2 Beams in Bridges with Uniform cross-sections in Class 1, 2 or 3

For beams with a uniform steel cross-section in Class 1, 2, or 3, restrained by reinforced concrete slab, the design buckling resistance moment should be taken as:

$$M_{b,Rd} = \chi_{LT} M_{Rd}$$

Equation 6.14

Where,

χ_{LT} = The reduction factor for lateral-torsional buckling corresponding to the relative slenderness $\bar{\lambda}_{LT}$, may be obtained as given in Chapter 5.

M_{Rd} = The design resistance moment at the relevant cross-section.

For cross-sections in Class 1 or 2, $M_{Rd} = M_{pl,Rd}$, determined according to Clause 6.6.2.1.2. For cross-sections in Class 3, M_{Rd} should be taken as $M_{el,Rd}$ given by Equation 6.12, but as the design bending moment that causes either a tensile stress f_{sd} in the reinforcement or a stress f_{yd} in an extreme fibre of the steel section, whichever is the smaller.

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Composite Structures

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The relative slenderness $\bar{\lambda}_{LT}$ may be calculated from:

$$\bar{\lambda}_{LT} = \sqrt{\frac{M_{Rk}}{M_{cr}}}$$

Equation 6.15

2

Where,

M_{Rk} = The resistance moment of the composite section using the characteristic material properties and the method specified for M_{Rd} .

3

M_{cr} = The elastic critical moment for lateral-torsional buckling determined at the relevant cross-section.

4

Where the slab is attached to one or more supporting steel members which are approximately parallel to the composite beam considered and the conditions (a) and (b) below are satisfied, the calculation of the elastic critical moment, M_{cr} , may be based on the "continuous inverted-U frame" model. This model takes into account the lateral displacement of the bottom flange causing bending of the steel web, and the rotation of the top flange as shown in Figure 6.10.

5

a. The top flange of the steel member is attached to a reinforced concrete slab by shear connectors in accordance with Clause 6.6.6.

b. At each support of the steel member, the bottom flange is laterally restrained and the web is stiffened. Elsewhere, the web is un-stiffened.

6

At the level of the top steel flange, a rotational stiffness k_s per unit length of steel beam may be adopted to represent the U-frame model by a beam alone:

$$k_s = \frac{k_1 k_2}{k_1 + k_2}$$

Equation 6.16

Where,

k_1 = The flexural stiffness of the cracked concrete slab in the direction transverse to the steel beam, which may be taken as:

$$k_1 = \frac{\alpha E_a I_2}{a}$$

Equation 6.17

Where,

α = 2 for k_1 for an edge beam, with or without a cantilever, and $\alpha = 3$ for an inner beam. For inner beams in a bridge deck with four or more similar beams, $\alpha = 4$ may be used.

a = The spacing between the parallel beams

$E_a I_2$ = The "cracked" flexural stiffness per unit width of the concrete or composite slab, where I_2 should be taken as the lowest of the value at midspan, for sagging bending, and the values at the supporting steel members, for hogging bending

k_2 = the flexural stiffness of the steel web, to be taken as:

$$k_2 = \frac{E_a t_w^3}{4(1 - \nu_a^2) h_s}$$

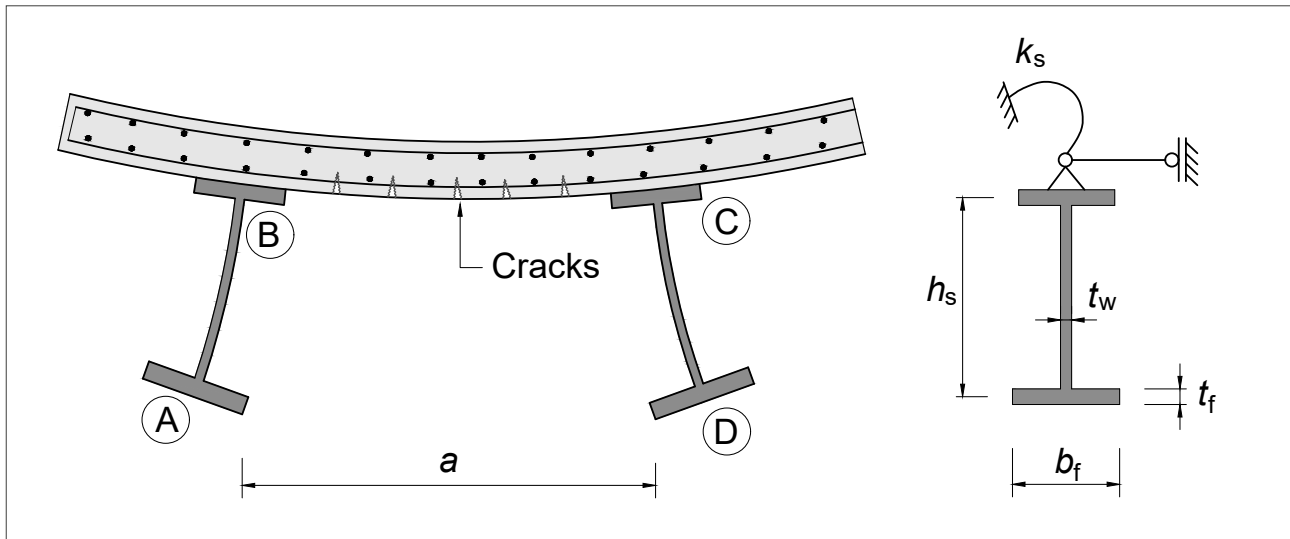
Equation 6.18

Where,

ν_a = Poisson's ratio for steel and h_s and t_w are defined in Figure 6.10.

In the U-frame model, the favourable effect of the St. Venant torsional stiffness, $G_a I_{at}$, of the steel section may be taken into account for the calculation of M_{cr} .

Figure 6.10 U-frame Model



6.6.5 Transverse Forces on Webs

The rules given in Chapter 5 to determine the design resistance of an unstiffened or stiffened web to transverse forces applied through a flange are applicable to the non-composite steel flange of a composite beam, and to the adjacent part of the web.

If the transverse force acts in combination with bending and axial force, the resistance should be verified according to Chapter 5, Clause 5.6.20.4.

6.6.6 Shear Connection

6.6.6.1 General

Shear connection and transverse reinforcement shall be provided to transmit the longitudinal shear force between the concrete and the structural steel element, ignoring the effect of natural bond between the two.

Shear connectors may be either of mild steel or high tensile steel irrespective of the grade of steel used in the parent girder. Flexible shear connectors are preferred because of their better performance. Channel, Angle and Tee shear connectors may be of mild steel, whereas, the shear studs may be made of high tensile steel.

Shear connector shall be generally designed for full transfer of longitudinal shear. Shear connector strength and spacing are to be checked separately for all the limit states using appropriate factored load combinations and factored strength.

Shear connectors shall be capable of preventing separation of the concrete element from the steel element. Headed stud shear connectors may be assumed to provide sufficient resistance to uplift, unless the shear connection is subjected to direct tension.

For verification for ultimate limit states, the shear connectors provided in terms of size and shape may be kept constant over any length where the design longitudinal shear per unit length does not exceed the design shear resistance by more than 10 percent. Over every such length, the total design longitudinal shear force should not exceed the total design shear resistance.

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6.6.6.2 Longitudinal Shear Force in Beams

6.6.6.2.1 Beams in Which Elastic Or Non-Linear Theory Is Used For Resistances Of Cross-Sections

For any load combination and arrangement of design actions, the longitudinal shear per unit length at the interface between steel and concrete in a composite member, $v_{L,Ed}$, should be determined from the rate of change of the longitudinal force in either the steel or the concrete element of the composite section. Where elastic theory is used for calculating resistances of sections, the envelope of transverse shear force in the relevant direction may be used.

In general, the elastic properties of the uncracked section should be used for the determination of the longitudinal shear force, even where cracking of concrete is assumed in global analysis. The effects of cracking of concrete on the longitudinal shear force may be taken into account, if in global analysis and for the determination of the longitudinal shear force account is taken of the effects of tension stiffening and possible over-strength of concrete.

Where concentrated longitudinal shear forces occur, account should be taken of the local effects of longitudinal slip, as provided in Clause 6.6.6.2.3 and 6.6.6.2.4. Otherwise, the effects of longitudinal slip may be neglected.

For composite box girders, the longitudinal shear force on the connectors should include the effects of bending and torsion, and also of distortion if appropriate. For box girders with a flange designed as a composite plate, refer Clause 6.9.4.

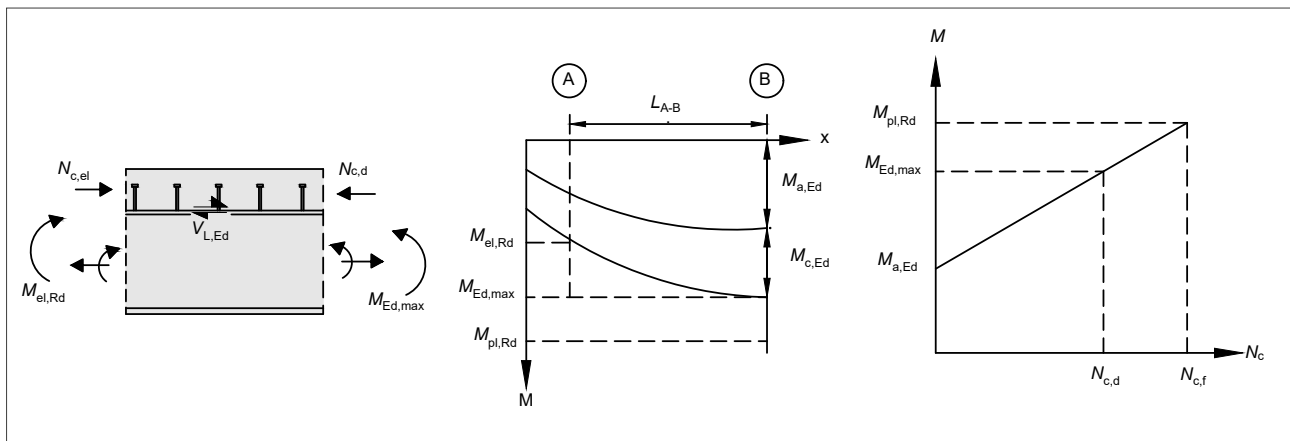
6.6.6.2.2 Beams in Bridges With Cross-Sections in Class 1 or 2

In members with cross-sections in Class 1 or 2, if the total design bending moment $M_{Ed,max} = M_{a,Ed} + M_{c,Ed}$ exceeds the elastic bending resistance $M_{el,Rd}$, account should be taken of the non-linear relationship between transverse shear and longitudinal shear within the inelastic lengths of the member. $M_{a,Ed}$ and $M_{c,Ed}$ are defined in Clause 6.6.2.1.3.

This paragraph applies to regions where the concrete slab is in compression, as shown in Figure 6.11. Shear connectors should be provided within the inelastic length L_{A-B} to resist the longitudinal shear force $V_{L,Ed}$, resulting from the difference between the normal forces N_{cd} and $N_{c,el}$ in the concrete slab at the cross-sections B and A, respectively. The bending resistance $M_{el,Rd}$ is defined in Clause 6.6.2.1.3. If the maximum bending moment $M_{Ed,max}$ at section B is smaller than the plastic bending resistance $M_{pl,Rd}$, the normal force N_{cd} at section B may be determined according to Clause 6.6.2.1.3 and Figure 6.6, or alternatively using the simplified linear relationship according to Figure 6.11.

The longitudinal shear forces should be determined by elastic analysis with the cross-section properties of the uncracked section taking into account effects of sequence of construction.

Figure 6.11 Determination of Longitudinal Shear in Beams With Inelastic Behaviour of Cross Sections



6.6.6.2.3 Local Effects of Concentrated Longitudinal Shear Force due to Introduction of Longitudinal Forces

Where a force F_{Ed} parallel to the longitudinal axis of the composite beam is applied to the concrete or steel element by a bonded or unbonded tendon, the distribution of the concentrated longitudinal shear force $V_{L,Ed}$ along the interface between steel and concrete, should be determined as the details given in this section. The distribution of $V_{L,Ed}$ caused by several forces F_{Ed} should be obtained by summation.

The force $V_{L,Ed}$ may be assumed to be distributed along a length L_v of shear connection with a maximum shear force per unit length given by Equation (6.19) and (Figure 6.12a) for load introduction within a length of a concrete flange and by Equation (6.20) and (Figure 6.12b) at an end of a concrete flange.

$$v_{L,Ed,max} = \frac{V_{L,Ed}}{e_d + \frac{b_{eff}}{2}} \quad \text{Equation 6.19}$$

$$v_{L,Ed,max} = \frac{V_{L,Ed}}{e_d + \frac{b_{eff}}{2}} \quad \text{Equation 6.20}$$

Where,

- b_{eff} = The effective width for global analysis, given by Clause 6.5.4.1.2,
- e_d = Either $2e_h$ or $2e_v$ (the length over which the force F_{Ed} is applied may be added to e_d)
- e_h = The lateral distance from the point of application of force F_{Ed} to the relevant steel web, if it is applied to the slab,
- e_v = The vertical distance from the point of application of force F_{Ed} to the plane of the shear connection concerned, if it is applied to the steel element.

Where stud shear connectors are used, at ultimate limit states a rectangular distribution of shear force per unit length may be assumed within the length L_v , so that within a length of concrete flange,

$$v_{L,Ed,max} = \frac{V_{L,Ed}}{e_d + b_{eff}} \quad \text{Equation 6.21}$$

and at an end of a flange,

$$v_{L,Ed,max} = \frac{2V_{L,Ed}}{e_d + b_{eff}} \quad \text{Equation 6.22}$$

In the absence of a more precise determination, the force $F_{Ed} - V_{L,Ed}$ may be assumed to disperse into the concrete or steel element at an angle of spread 2β , where $\beta = \arctan 2/3$.

6.6.6.2.4 Local Effects of Concentrated Longitudinal Shear Forces at Sudden Change of Cross-Sections

Concentrated longitudinal shear at the end of the concrete slab, e.g., due to the primary effects of shrinkage and thermal actions (see Figure 6.12c), and taken into account where appropriate. This applies also for intermediate stages of construction of a concrete slab (Figure 6.12d).

Concentrated longitudinal shear at a sudden change of cross-sections, e.g. change from steel to composite section according to (Figure 6.12d), should be taken into account.

Where the primary effects of temperature and shrinkage cause a design longitudinal shear force to be transferred across the interface between steel and concrete at each free end of the member considered, its distribution may be assumed to be triangular, with a maximum shear force per unit length (Figure 6.12c and d).

$$v_{L,Ed,max} = \frac{2V_{L,Ed}}{b_{eff}} \quad \text{Equation 6.23}$$

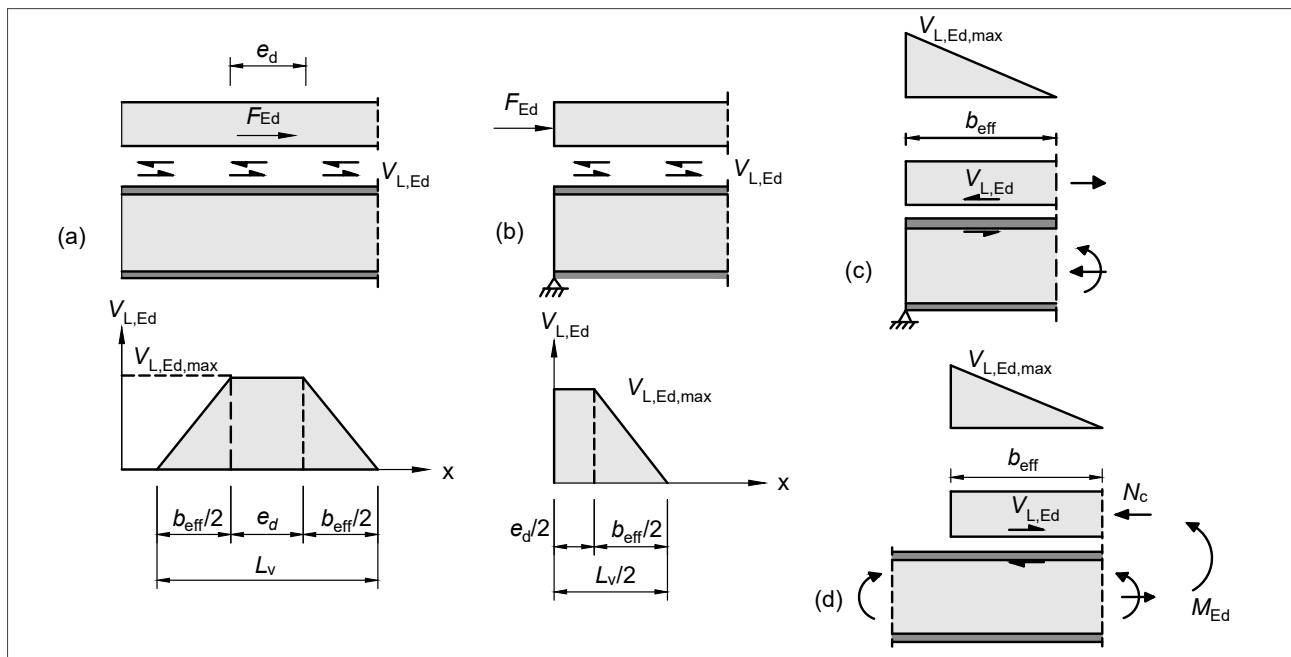
at the free end of the slab, where b_{eff} is the effective width for global analysis, given by Clause 6.5.4.1.2. Where stud shear connectors are used, for the ultimate limit state the distribution may alternatively be assumed to be rectangular along a length b_{eff} adjacent to the free end of the slab.

For calculating the primary effects of shrinkage at intermediate stages of the construction of a concrete slab, the equivalent span for the determination of the width b_{eff} should be taken as the continuous length of concrete slab where the shear connection is effective, within the span considered.

Where at a sudden change of cross-section according to Figure 6.12d the concentrated longitudinal shear force results from the force N_c due to bending, the triangular distribution with maximum shear force per unit length given by Equation 6.23 may be used.

The forces transferred by shear connectors should be assumed to disperse into the concrete slab at an angle of spread 2β , where $\beta = \arctan 2/3$.

Figure 6.12 Distribution of Longitudinal Shear Force Along the Interface



6.6.6.3 Headed Stud Connectors in Solid Slabs and Concrete Encasement

6.6.6.3.1 Design Resistance

The design shear resistance of a headed stud automatically welded in accordance with EN 14555 should be determined from:

$$P_{Rd} = \frac{0.8 f_u \pi d^2 / 4}{\gamma_V} \quad \text{Equation 6.24}$$

or:

$$P_{Rd} = \frac{0.29 \alpha d^2 \sqrt{f_{ck} E_{cm}}}{\gamma_V} \quad \text{Equation 6.25}$$

Whichever is smaller, with:

$$P_{Rd} = 0.2 \left(\frac{h_{sc}}{d} + 1 \right) \quad \text{for } 3 \leq \frac{h_{sc}}{d} \leq 4 \quad \text{Equation 6.26}$$

$$\alpha = 1 \quad \text{for } \frac{h_{sc}}{d} > 4 \quad \text{Equation 6.27}$$

Where,

γ_V = The partial factor (recommended value is 1.25).

d = The diameter of the shank of the stud, $16 \text{ mm} \leq d \leq 25 \text{ mm}$.

f_u = The specified ultimate tensile strength of the material of the stud but not greater than 500 N/mm^2 .

f_{ck} = The characteristic cylinder compressive strength of the concrete at the age considered, of density not less than 1750 kg/m^3 .

h_{sc} = The overall nominal height of the stud.

The weld collars should comply with the requirements of EN 13918. Where studs are arranged in such a way that splitting forces occur in the direction of the slab thickness, the above Equations are not applicable for design shear resistance. Studs of diameter up to 25 mm shall be used. For studs of diameter greater than 25 mm, verification by tests will also be required before being used.

6.6.6.3.2 Influence of Tension on Shear Resistance

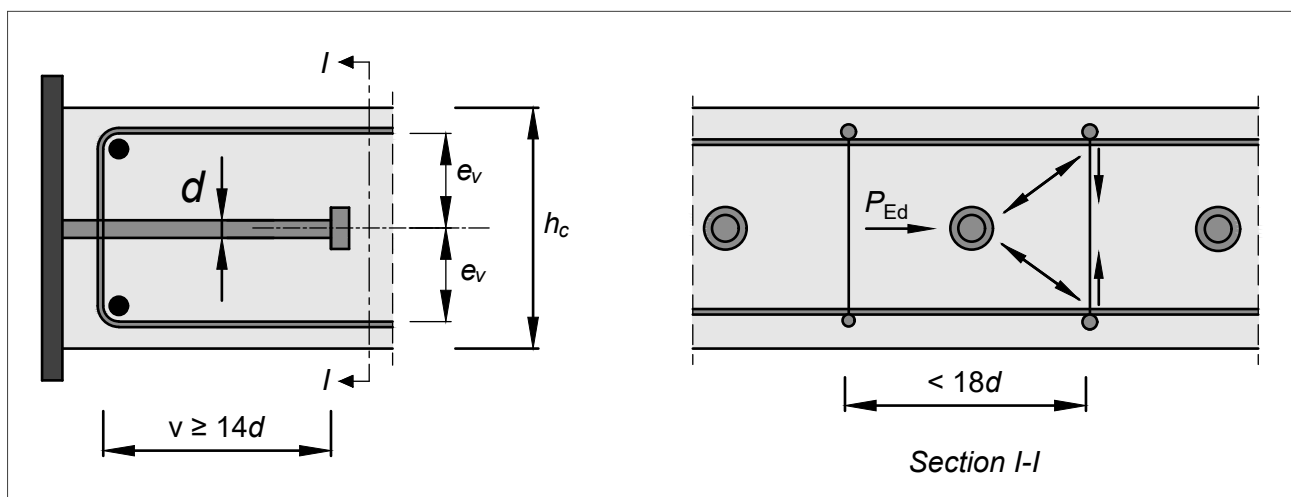
Where headed stud connectors are subjected to direct tensile force in addition to shear, the design tensile force per stud F_{ten} should be calculated. If $F_{ten} \leq 0.1 P_{Rd}$, where P_{Rd} is the design shear resistance, the tensile force may be neglected. If $F_{ten} > 0.1 P_{Rd}$, the connection is not within the scope of this Manual.

6.6.6.4 Headed Studs that Cause Splitting in the Direction of the Slab Thickness

Where, in bridges, headed stud connectors are arranged in such a way that splitting forces can occur in the direction of the slab thickness (refer Figure 6.13) and where there is no transverse shear, the design resistance to longitudinal shear may be determined according to Clause 6.6.6.3.1, provided that following two conditions are fulfilled.

- Transverse reinforcement should be provided, as shown in Figure 6.13, such that $e_v \geq 6d$, and the anchoring length v should be greater than or equal to $14d$.
- The splitting force should be resisted by stirrups which should be designed for a tensile force $0.3P_{Rd}$ per stud connector. The spacing of these stirrups should not exceed the smaller of $18d$ and the longitudinal spacing of the connectors.

Figure 6.13 Local Reinforcement for Splitting Forces



6.6.6.5 Detailing of the Shear Connection

The surface of a connector that resists separation forces (for example, the underside of the head of a stud) should extend not less than 30 mm clear above the bottom reinforcement, see Figure 6.14. Cover over shear connectors should not be less than that required for reinforcement adjacent to the same surface of concrete.

In execution, the rate and sequence of concreting should be required to be such that partly matured concrete is not damaged as a result limited composite action occurring from deformation of the steel beam under subsequent concreting operations. Wherever possible, deformation should not be imposed on a shear connection until the concrete has reached a cylinder strength of at least 20 N/mm^2 .

Where the shear connection is adjacent to a longitudinal edge of a concrete slab, transverse reinforcement provided in accordance with Clause 6.6.6.6 should be fully anchored in the concrete between the edge of the slab and the adjacent row of connectors.

To prevent longitudinal splitting of the concrete flange caused by the shear connectors, the following additional recommendations should be applied where the distance from the edge of the concrete flange to the centreline of the nearest row of shear connectors is less than 300 mm:

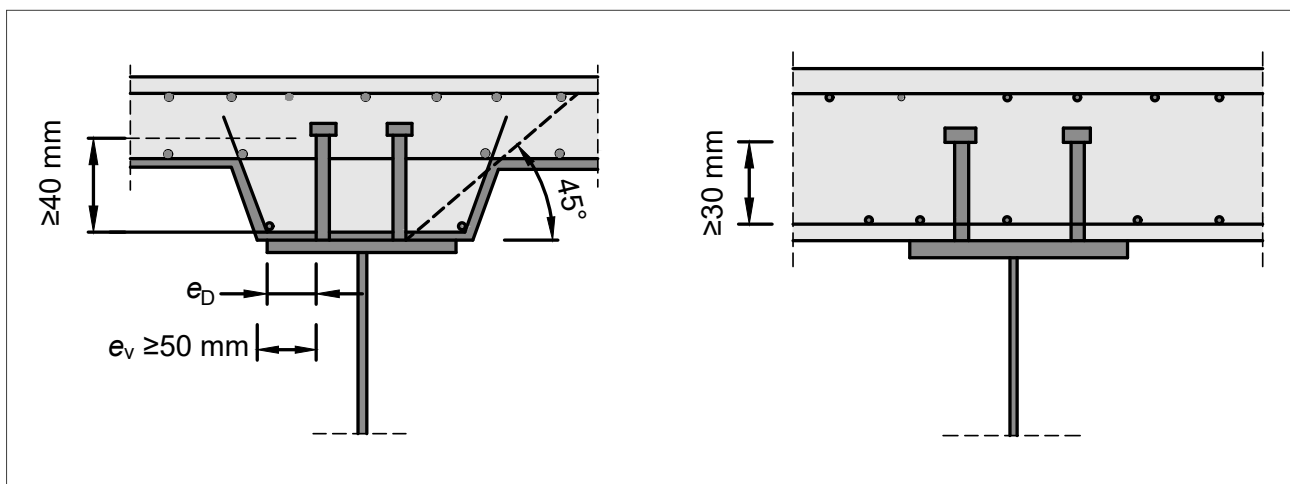
- Transverse reinforcement should be supplied by U-bars passing around the shear connectors,
- Where headed studs are used as shear connectors, the distance from the edge of the concrete flange to the centre of the nearest stud should not be less than $6d$, where d is the nominal diameter of the stud, and the U-bars should be not less than $0.5d$ in diameter and
- The U-bars should be placed as low as possible while still providing sufficient bottom cover.

At the end of a composite cantilever, sufficient local reinforcement shall be provided to transfer forces from the shear connectors to the longitudinal reinforcement.

Where a concrete haunch is used between the steel section and the soffit of the concrete slab, the sides of the haunch should lie outside a line drawn at 45° from the outside edge of the connector, see Figure 6.14. The nominal concrete cover from the side of the haunch to the connector should be not less than 50mm.

Transverse reinforcing bars sufficient to satisfy the requirements of Clause 6.6.6.6 should be provided in the haunch at not less than 40 mm clear below the surface of the connector that resists uplift.

Figure 6.14 Connection Detailing



Where a steel compression flange that would otherwise be in Class 3 or Class 4 is assumed to be in Class 1 or Class 2 because of restraint from shear connectors, the centre-to-centre spacing of the shear connectors in the direction of compression should not be greater than the following limits:

- Where the slab is in contact over the full length (e.g. solid slab): $22 t_f \sqrt{(235 / f_y)}$.
- Where the slab is not in contact over the full length (e.g. slab with ribs transverse to the beam): $15 t_f \sqrt{235 f_y}$.

Where,

t_f = The thickness of the flange.

f_y = The nominal yield strength of the flange in N/mm^2 .

In addition, the clear distance from the edge of a compression flange to the nearest line of shear connectors should be not greater than $9 t_f \sqrt{235 f_y}$.

The maximum longitudinal centre-to-centre spacing of individual shear connectors should not exceed the lesser of four times the slab thickness and 800 mm.

The thickness of the steel plate or flange to which a connector is welded shall be sufficient to allow proper welding and proper transfer of load from the connector to the plate without local failure or excessive deformation.

The distance e_D between the edge of a connector and the edge of the flange of the beam to which it is welded, see Figure 6.14, should not be less than 25 mm.

The overall height of a stud should be not less than $3d$, where d is the diameter of the shank. The head should have a diameter of not less than $1.5d$ and a depth of not less than $0.4d$. For elements in tension and subjected to fatigue loading, the diameter of a welded stud should not exceed 1.5 times the thickness of the flange to which it is welded. This applies also to studs directly over a web.

The spacing of studs in the direction of the shear force should be not less than $5d$; the spacing in the direction transverse to the shear force should be not less than $2.5d$ in solid slabs and $4d$ in other cases. Except when the studs are located directly over the web, the diameter of a welded stud should be not greater than 2.5 times the thickness of that part to which it is welded.

6.6.6.6 Longitudinal Shear in Concrete Slabs

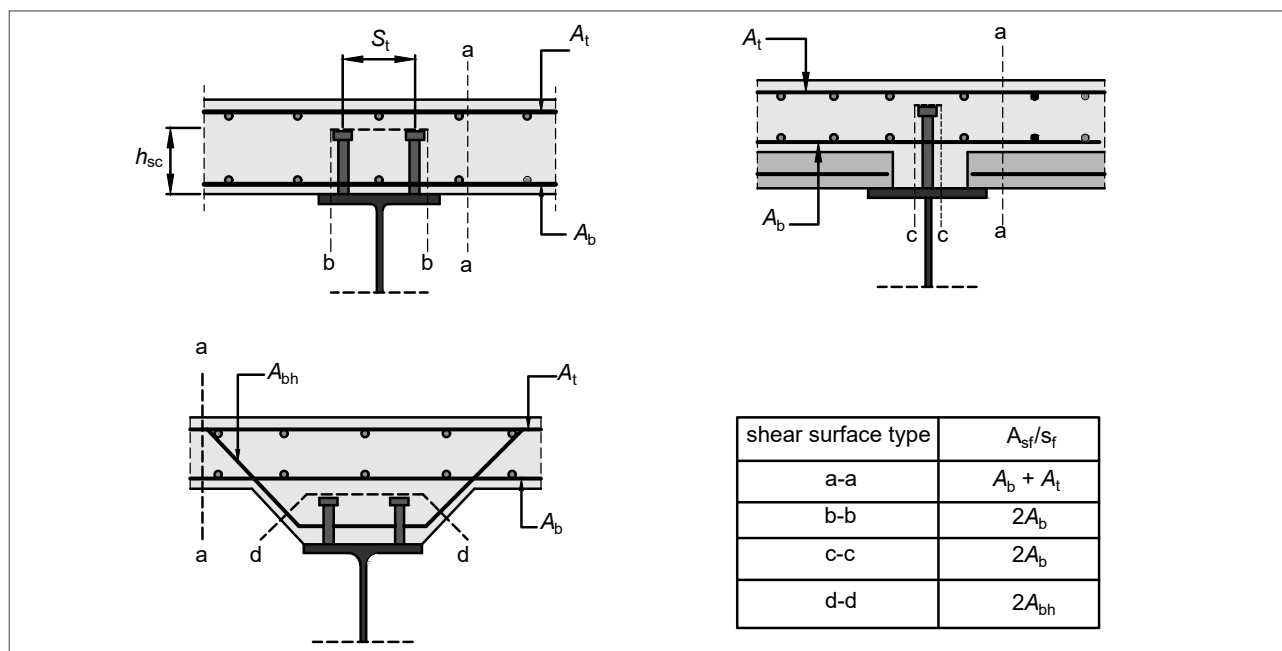
6.6.6.6.1 General

Transverse reinforcement in the slab shall be designed for the ultimate limit state so that premature longitudinal shear failure or longitudinal splitting shall be prevented. The design longitudinal shear stress for any potential surface of longitudinal shear failure within the slab shall not exceed the design longitudinal shear strength of the shear surface considered.

The length of the shear surface b-b shown in Figure 6.15 should be taken as equal to $(2h_{sc} + s_t)$ plus the head diameter for stud shear connectors arranged in pairs and shear surface c-c equal to $2h_{sc}$ plus the head diameter for a single row of stud shear connectors or staggered stud connectors, where h_{sc} is the height of the studs and s_t is the transverse spacing centre-to-centre of the studs.

The design longitudinal shear per unit length of beam on a shear surface should be determined in accordance with Clause 6.6.6.2. For each type of shear surface considered, the design longitudinal shear stress v_{Ed} should be determined from the design longitudinal shear per unit length of beam, taking account of the number of shear planes and the length of shear surface.

Figure 6.15 Typical Potential Surfaces of Shear Failure



6.6.6.6.2 Design Resistance to Longitudinal Shear

The design shear strength of the concrete flange (shear planes a-a in Figure 6.15) should be determined in accordance with Chapter 4.

For a shear surface passing around the shear connectors (e.g., shear surface b-b in Figure 6.15), the dimension h_f should be taken as the length of the shear surface.

The effective transverse reinforcement per unit length, A_{sf}/s_f , should be as shown in Figure 6.15, in which A_b , A_t and A_{bh} are areas of reinforcement per unit length of beam.

Where a combination of pre-cast elements and in-situ concrete is used, the resistance to longitudinal shear should be determined in accordance with Chapter 4.

6.6.6.6.3 Minimum Transverse Reinforcement

The minimum area of reinforcement should be determined in accordance with requirements given in Chapter 4.

6.6.7 Composite Columns

6.6.7.1 General

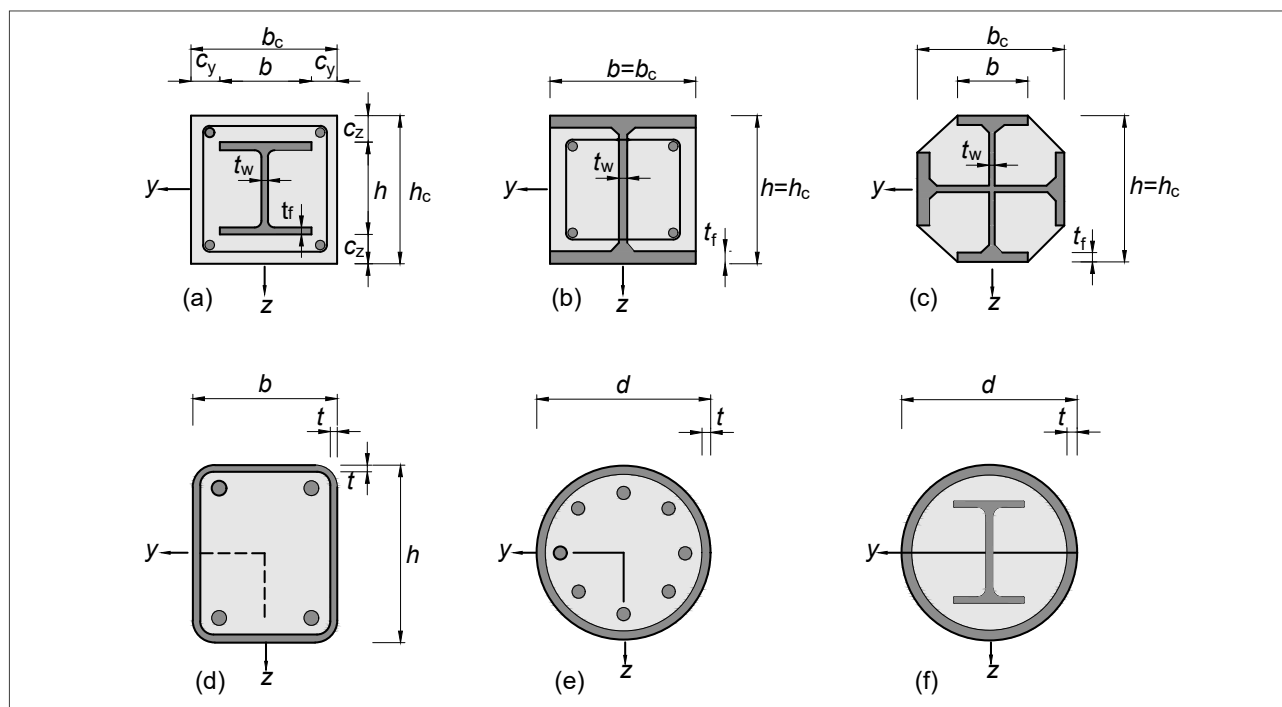
This Clause is applicable for the design of composite columns and composite compression members with concrete encased sections, partially encased sections and concrete filled rectangular and circular tubes, Figure 6.16. The steel grades shall be S235 to S460 and normal weight concrete of strength classes C20/25 to C50/60.

The steel contribution ratio δ , as defined in Clause 6.6.7.3.3, should be ≥ 0.2 and ≤ 0.9 .

Composite columns or compression members of any cross-section should be checked for:

- Resistance of the member.
- Resistance to local buckling.
- Introduction of loads.
- Resistance to shear between steel and concrete elements.

Figure 6.16 Typical Cross-sections of Composite Columns

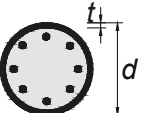
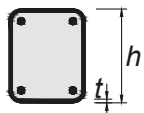
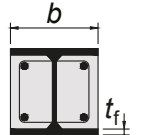


Two methods of design are given: a) a general method in Clause 6.6.7.2 whose scope includes members with non-symmetrical or non-uniform cross-sections over the column length and b) a simplified method in Clause 6.6.7.3 for members of doubly symmetrical and uniform cross section over the member length.

For composite compression members subjected to bending moments and normal forces resulting from independent actions, the partial factor γ_F for those internal forces that lead to an increase of resistance should be reduced by 20%.

The influence of local buckling of the steel section on the resistance shall be considered in design. The effects of local buckling may be neglected for a steel section fully encased in accordance with Clause 6.6.7.5 and for other types of cross-sections provided the maximum values of parameters in Table 6.2 are not exceeded.

Table 6.2 Maximum Values of d/t , h/t and b/t_f with f_y in N/mm^2

Cross-section	max (d/t), max (h/t) and max (b/t_f)
Circular hollow steel sections 	$\max (d/t) = 90 \frac{235}{f_y}$
Rectangular hollow steel sections 	$\max (h/t) = 52 \sqrt{\frac{235}{f_y}}$
Partially encased I-sections 	$\max (b/t_f) = 44 \sqrt{\frac{235}{f_y}}$

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Composite Structures

6.6.7.2 General Method of Design

This method is applicable for members with non-symmetrical or non-uniform cross-sections over the column length. Design for structural stability shall take account of second-order effects including residual stresses, geometrical imperfections, local instability, cracking of concrete, creep and shrinkage of concrete and yielding of structural steel and of reinforcement. The design shall ensure that instability does not occur for the most unfavourable combination of actions at the ultimate limit state and that the resistance of individual cross-sections subjected to bending, longitudinal force and shear is not exceeded.

Second-order effects shall be considered in any direction in which failure might occur, if they affect the structural stability significantly.

Internal forces shall be determined by elasto-plastic analysis. Plane sections may be assumed to remain plane. Full composite action up to failure may be assumed between the steel and concrete components of the member.

The tensile strength of concrete shall be neglected. The influence of tension stiffening of concrete between cracks on the flexural stiffness may be taken into account.

Shrinkage and creep effects shall be considered if they are likely to reduce the structural stability significantly. For simplification, creep and shrinkage effects may be ignored if the increase in the first-order bending moments due to creep deformations and longitudinal force resulting from permanent loads is not greater than 10%.

The following stress-strain relationships should be used in the non-linear analysis:

- For concrete in compression as given in Chapter 4.
- For reinforcing steel as given in Chapter 4.
- For structural steel as given in Chapter 5.

For simplification, instead of the effect of residual stresses and geometrical imperfections, the equivalent initial bow imperfections for members may be used as given in Table 6.4.

6.6.7.3 Simplified Method of Design

6.6.7.3.1 General

This method is applicable for members of doubly symmetrical and uniform cross-section over the member length with rolled, cold-formed or welded steel sections. The simplified method is not applicable if the structural steel component consists of two or more unconnected sections. The relative slenderness $\bar{\lambda}$, as defined in Clause 6.6.7.3.3, should be ≤ 2.0 .

For a fully encased steel section, see Figure 6.16a, limits to the maximum thickness of concrete cover that may be used in calculation are: $\max c_z = 0.3 h$ and $\max c_y = 0.4 b$.

The longitudinal reinforcement that may be used in calculation, should not exceed 6% of the concrete area.

The ratio of the cross-section's depth to width of the composite section should be within the limits 0.2 and 5.0.

6.6.7.3.2 Resistance of Cross-sections

The plastic resistance to compression $N_{pl,Rd}$ of a composite cross-section should be calculated by adding the plastic resistances of its components:

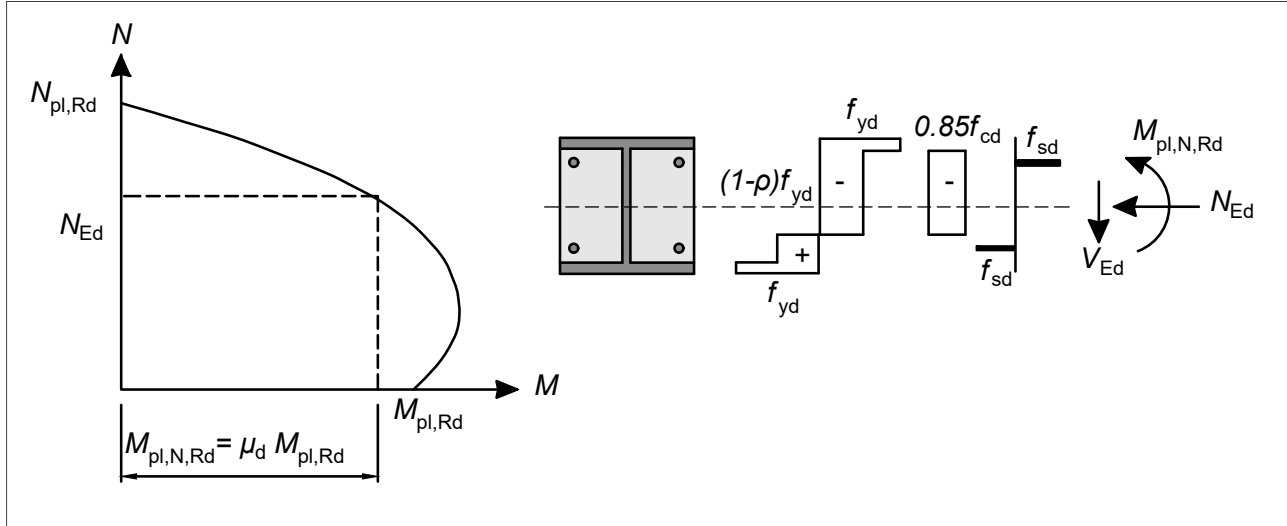
$$N_{pl,Rd} = A_a f_{yd} + 0.85 A_c f_{cd} + A_s f_{sd}$$

Equation 6.28

This Equation applies for concrete encased and partially concrete encased steel sections. For concrete filled sections the coefficient 0.85 may be replaced by 1.0.

The resistance of a cross-section to combined compression and bending and the corresponding interaction curve may be calculated assuming rectangular stress blocks as shown in Figure 6.17, taking account of the design shear force V_{Ed} as below. The tensile strength of the concrete should be neglected.

Figure 6.17 Interaction Curve for Combined Compression and Uniaxial Bending



The influence of transverse shear forces on the resistance to bending and normal force should be considered when determining the interaction curve, if the shear force $V_{a,Ed}$ on the steel section exceeds 50% of the design shear resistance $V_{pl,a,Rd}$ of the steel section.

Where $V_{a,Ed} > 0.5 V_{pl,a,Rd}$, the influence of the transverse shear on the resistance in combined bending and compression should be taken into account by a reduced design steel strength $(1 - \rho) f_{yd}$ in the shear area A_v in accordance with Clause 6.6.2.2.3 and Figure 6.17.

The shear force $V_{a,Ed}$ should not exceed the resistance to shear of the steel section determined according to Clause 6.6.2.2. The resistance to shear $V_{c,Ed}$ of the reinforced concrete part should be verified as given in Chapter 4.

The design shear force V_{Ed} may be distributed into $V_{a,Ed}$ acting on the structural steel and $V_{c,Ed}$ acting on reinforced concrete section by:

$$V_{a,Ed} = V_{Ed} \frac{M_{pl,a,Rd}}{M_{pl,Rd}} \quad \text{Equation 6.29}$$

$$V_{c,Ed} = V_{Ed} - V_{a,Ed} \quad \text{Equation 6.30}$$

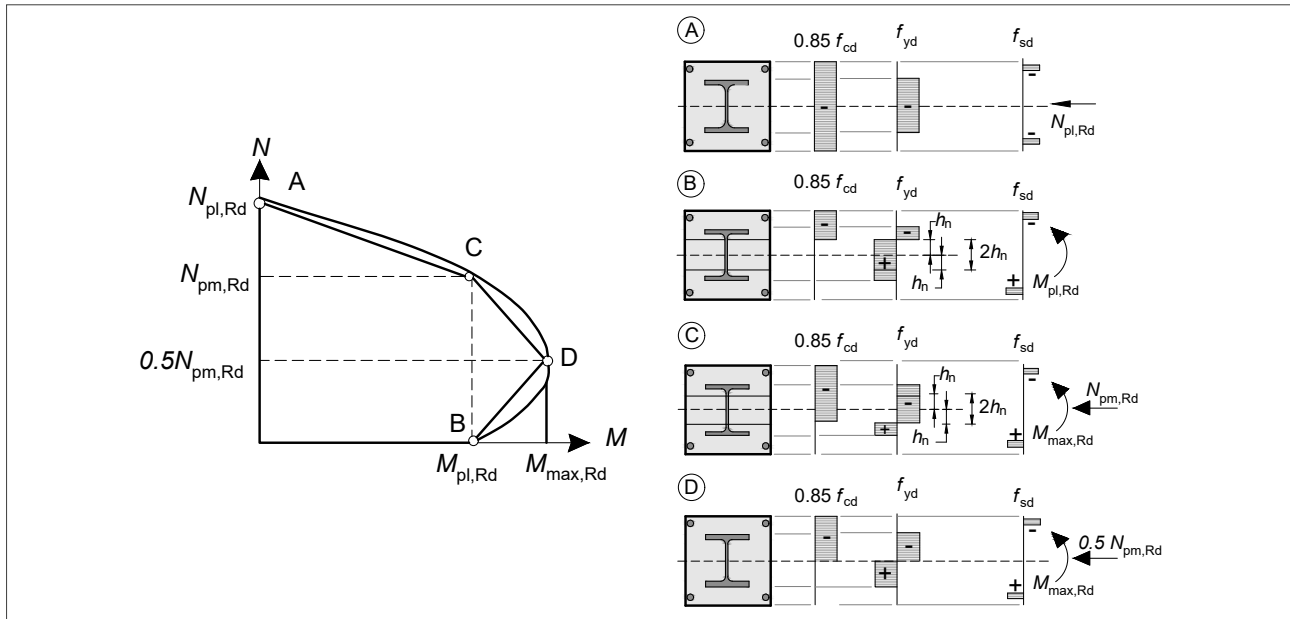
Where,

$M_{pl,a,Rd}$ = The plastic resistance moment of the steel section

$M_{pl,Rd}$ = The plastic resistance moment of the steel section

As a simplification, the interaction curve may be replaced by a polygonal diagram (the dashed line in Figure 6.18). The Figure 6.18 shows as an example the plastic stress distribution of a fully encased cross section for the points A to D. $N_{pm,Rd}$ should be taken as $0.85 f_{cd} A_c$ for concrete encased and partially concrete encased sections, as in Figure 6.16a-c, and as $f_{cd} A_c$ for concrete filled sections, as in Figure 6.16d - f.

Figure 6.18 Simplified Interaction Curve and Stress Distributions



For concrete filled tubes of circular cross-section, account may be taken of increase in strength of concrete caused by confinement provided that the relative slenderness $\bar{\lambda}$ defined in Clause 6.6.7.3.3 does not exceed 0.5 and $e/d < 0.1$, where e is the eccentricity of loading given by M_{Ed} / N_{Ed} and d is the external diameter of the column. The plastic resistance to compression may then be calculated from the following Equation:

$$N_{pl,Rd} = \eta_a A_a f_{yd} + A_c f_{cd} \left(1 + \eta_c \frac{t}{d} \frac{f_y}{f_{ck}} \right) + A_s f_{sd}$$

Equation 6.31

Where,

t = The wall thickness of the steel tube.

For members with $e = 0$ the values $\eta_a = \eta_{a0}$ and $\eta_c = \eta_{c0}$ are given by the following Equations:

$$\eta_{a0} = 0.25(3 + 2\bar{\lambda}) \quad (\text{but } \leq 1.0)$$

$$\eta_{c0} = 4.9 - 18.5\bar{\lambda} + 17\bar{\lambda}^2 \quad (\text{but } \geq 0)$$

For members in combined compression and bending with $0 < e/d \leq 0.1$, the value of η_a and η_c should be determined from Equations below, where η_{a0} and η_{c0} are given by Equations above:

$$\eta_a = \eta_{a0} + (1 - \eta_{a0})(10 e/d)$$

$$\eta_c = \eta_{c0} (1 - 10 e/d)$$

For $e/d > 0.1$, $\eta_a = 1.0$ and $\eta_c = 0$.

6.6.7.3.3 Effective Flexural Stiffness, Steel Contribution Ratio and Relative Slenderness

The steel contribution ratio δ is defined as:

$$\delta = \frac{A_a f_{yd}}{N_{pl,Rd}}$$

Equation 6.32

Where,

$N_{pl,Rd}$ is the plastic resistance to compression as given in Clause 6.6.7.3.2.

The relative slenderness $\bar{\lambda}$ for the plane of bending being considered is given by:

$$\bar{\lambda} = \sqrt{\frac{N_{pl,Rd}}{N_{cr}}}$$

Equation 6.33

Where,

$N_{pl,Rk}$ = The characteristic value of the plastic resistance to compression as given by Equation 6.28 if, instead of the design strengths, the characteristic values are used

N_{cr} = The elastic critical normal force for the relevant buckling mode, calculated with the effective flexural stiffness $(EI)_{eff}$ determined as below.

The characteristic value of the effective flexural stiffness $(EI)_{eff}$ of a cross section of a composite column should be calculated from:

$$(EI)_{eff} = E_a I_a + E_s I_s + K_c E_{cm} I_c$$

Equation 6.34

Where,

K_c = A correction factor that should be taken as 0.6.

I_a, I_c, I_s = The second moments of area of the structural steel section, the un-cracked concrete section and the reinforcement for the bending plane being considered.

Account should be taken to the influence of long-term effects on the effective elastic flexural stiffness. The modulus of elasticity of concrete E_{cm} should be reduced to the value $E_{c,eff}$ in accordance with the following Equation:

$$E_{c,eff} = E_{cm} \frac{1}{1 + (N_{G,Ed} / N_{Ed}) \varphi_t}$$

Equation 6.35

Where,

φ_t = The creep coefficient.

N_{cr} = The total design normal force.

$N_{G,Ed}$ = The part of this normal force that is permanent.

6.6.7.3.4 Methods of Analysis and Member Imperfections

For member verification, analysis should be based on second-order linear elastic analysis. For the determination of the internal forces the design value of effective flexural stiffness $(EI)_{eff,II}$ should be determined from the following Equation:

$$(EI)_{eff,II} = K_0 (E_a I_a + E_s I_s + K_{c,II} E_{cm} I_c)$$

Equation 6.36

Where,

$K_{c,II}$ = A correction factor which should be taken as 0.5

K_0 = A calibration factor which should be taken as 0.9.

Long-term effects on the effective elastic flexural stiffness by reducing E_{cm} to $E_{c,eff}$ should be taken into account in accordance with Clause 6.6.7.3.3.

Second-order effects need not to be considered where Clause 6.5.2 applies and the elastic critical load is determined with the flexural stiffness $(EI)_{eff,II}$ in accordance with Equation (6.36).

The influence of geometrical and structural imperfections may be taken into account by equivalent geometrical imperfections. Equivalent member imperfections for composite columns are given in Table 6.4, where L is the column length.

Within the column length, second-order effects may be allowed for by multiplying the greatest first-

order design bending moment M_{Ed} , by a factor k given by:

$$k = \frac{\beta}{1 - N_{Ed}/N_{cr,eff}} \geq 1.0$$

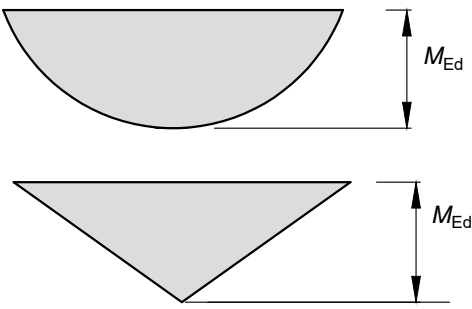
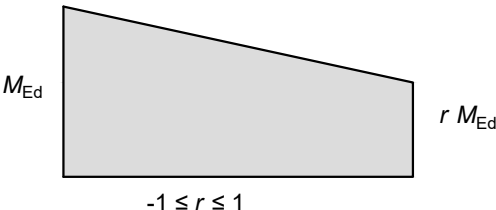
Equation 6.37

Where,

$N_{cr,eff}$ = The critical normal force for the relevant axis and corresponding to the effective flexural stiffness given as above in this Clause with the effective length taken as the column length,

β = An equivalent moment factor given in Table 6.3

Table 6.3 Factors β for the Determination of Moments to Second Order Theory

Moment distribution	Moment factors β	Comment
	First-order bending moments from member imperfection or lateral load: $\beta = 1.0$	M_{Ed} is the maximum bending moment within the column length ignoring second-order effects
	End moments: $\beta = 0.66 + 0.44 r$ but $\beta \geq 0.44$	M_{Ed} and $r M_{Ed}$ are the end moments from first-order or second-order global analysis

6.6.7.3.5 Resistance of Members in Axial Compression

Members may be verified using second order analysis according to Clause 6.6.7.3.6 taking into account member imperfections.

For simplification for members in axial compression, the design value of the normal force N_{Ed} should satisfy:

$$\frac{N_{Ed}}{\chi N_{pl,Rd}} \leq 1.0$$

Equation 6.38

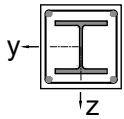
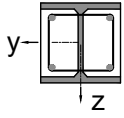
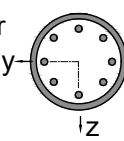
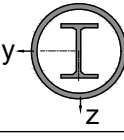
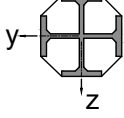
Where,

$N_{pl,Rd}$ = The plastic resistance of the composite section determined according Clause 6.6.7.3.2, but with f_{yd} determined using the partial factor γ_{M1} given in Chapter 5.

χ = the reduction factor for the relevant buckling mode given in Chapter 5 in terms of the relevant relative slenderness $\bar{\lambda}$

The relevant buckling curves for cross-sections of composite columns are given in Table 6.4, where ρ_s is the reinforcement ratio A_s / A_c .

Table 6.4 Buckling Curves and Member Imperfections for Composite Columns

Cross-section	Limits	Axis of buckling	Buckling curve	Member imperfection
concrete encased section 		y-y z-z	b c	$L / 200$ $L / 150$
partially concrete encased section 		y-y z-z	b c	$L / 200$ $L / 150$
circular and rectangular hollow steel section 	$\rho_s \leq 3\%$ $3\% < \rho_s \leq 6\%$	any any	a b	$L / 300$ $L / 200$
circular hollow steel sections with additional I-section 		y-y z-z	b b	$L / 200$ $L / 200$
partially concrete encased section with crossed I-sections 		any	b	$L / 200$

6.6.7.3.6 Resistance of Members in Combined Compression and Uniaxial Bending

The following Equation, based on the interaction curve determined according to Clause 6.6.7.3.2, should be satisfied:

$$\frac{M_{Ed}}{M_{pl,N,Rd}} = \frac{M_{Ed}}{\mu_d M_{pl,Rd}} \leq \alpha_M$$

Equation 6.39

Where,

M_{Ed} = The greatest of the end moments and the maximum bending moment within the column length, including imperfections and second order effects if necessary.

$M_{pl,N,Rd}$ = The plastic bending resistance taking into account the normal force N_{Ed} , given by $\mu_d M_{pl,Rd}$, refer Figure 6.17.

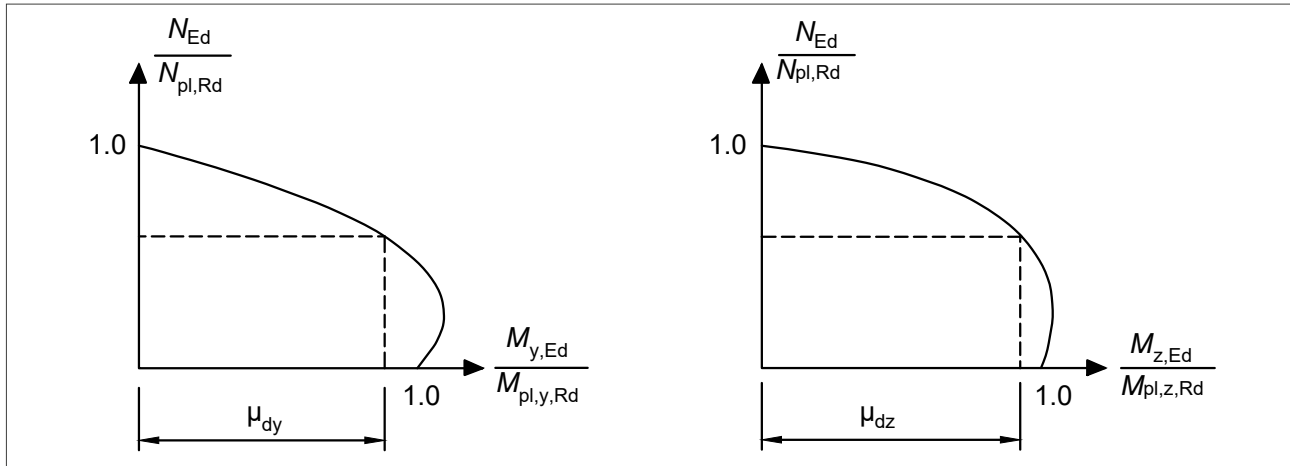
$M_{pl,Rd}$ = The plastic bending resistance, given by point B in Figure 6.18.

α_M = 0.9 for steel grade from S235 to S355.

= 0.8 for steel grade S420 and S460.

The value $\mu_d = \mu_{dy}$ or μ_{dz} , as shown in Figure 6.19, refers to the design plastic resistance moment $M_{pl,Rd}$ for the plane of bending being considered. Values μ_d greater than 1.0 should only be used where the bending moment M_{Ed} depends directly on the action of the normal force N_{Ed} , for example where the moment M_{Ed} results from an eccentricity of the normal force N_{Ed} .

Figure 6.19 Design for Compression and Biaxial Bending



6.6.7.3.7 Combined Compression and Biaxial Bending

For composite columns and compression members with biaxial bending the values μ_{dy} and μ_{dz} in Figure 6.19 may be calculated separately for each axis. Imperfections should be considered only in the plane in which failure is expected to occur. If it is not evident which plane is the more critical, checks should be made for both planes.

For combined compression and biaxial bending the following conditions should be satisfied for the stability check within the column length and for the check at the end:

$$\frac{M_{y,Ed}}{\mu_{dy} M_{pl,y,Rd}} \leq \alpha_{M,y} \quad \frac{M_{z,Ed}}{\mu_{dz} M_{pl,z,Rd}} \leq \alpha_{M,z} \quad \text{Equation 6.40}$$

$$\frac{M_{y,Ed}}{\mu_{dy} M_{pl,y,Rd}} + \frac{M_{z,Ed}}{\mu_{dz} M_{pl,z,Rd}} \leq 1.0 \quad \text{Equation 6.41}$$

Where,

$M_{pl,y,Rd}$ and $M_{pl,z,Rd}$ are the plastic bending resistance of the relevant plane of bending

$M_{y,Ed}$ and $M_{z,Ed}$ are the design bending including second order effects and imperfections according to Clause 6.6.7.3.4

μ_{dy} or μ_{dz} are defined in Clause 6.6.7.3.6

$\alpha_M = \alpha_{M,y}$ and $\alpha_M = \alpha_{M,z}$ are given in Clause 6.6.7.3.6

6.6.7.4 Shear Connection and Load Introduction

6.6.7.4.1 General

Provision shall be made in regions of load introduction for internal forces and moments applied from members connected to the ends and for loads applied within the length to be distributed between the steel and concrete components, considering the shear resistance at the interface between steel and concrete. A clearly defined load path shall be provided that does not involve an amount of slip at this interface that would invalidate the assumptions made in design.

Where composite columns and compression members are subjected to significant transverse shear, provision shall be made for the transfer of the corresponding longitudinal shear stress at the interface between steel and concrete.

For axially loaded columns and compression members, longitudinal shear outside the areas of load introduction need not be considered.

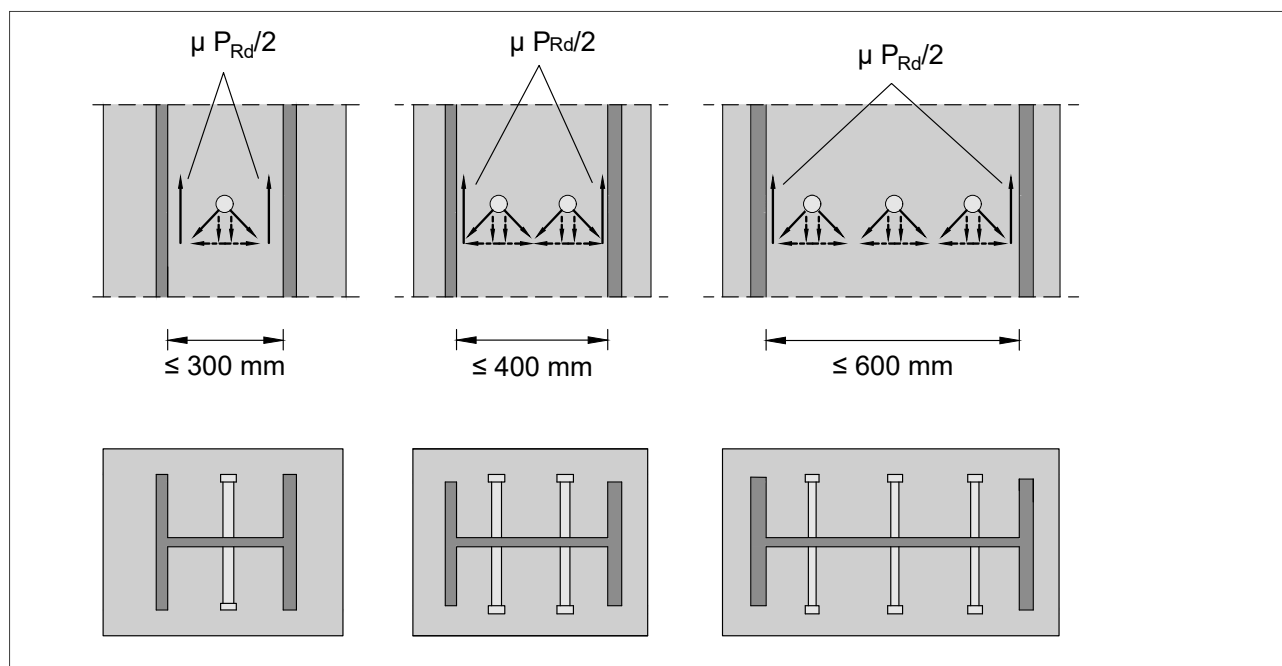
6.6.7.4.2 Load Introduction

Shear connectors should be provided in the load introduction area and in areas with change of cross section, if the design shear strength τ_{Rd} , is exceeded at the interface between steel and concrete. The shear forces should be determined from the change of sectional forces of the steel or reinforced concrete section within the introduction length. If the loads are introduced into the concrete cross section only, the values resulting from an elastic analysis considering creep and shrinkage should be taken into account. Otherwise, the forces at the interface should be determined by elastic theory or plastic theory, to determine the more severe case.

In absence of a more accurate method, the introduction length should not exceed $2d$ or $L/3$, where d is the minimum transverse dimension of the column and L is the column length.

For composite columns and compression members no shear connection need, be provided for load introduction by endplates if the full interface between the concrete section and endplate is permanently in compression, taking account of and shrinkage. Otherwise, the load introduction should be verified according to details given below for cross-sections partially loaded. For concrete filled tubes of circular cross-section the effect caused by the confinement may be taken into account if the conditions given in Clause 6.6.7.3.2 are satisfied using the values η_a and η_c for $\bar{\lambda}$ equal to zero.

Figure 6.20 Additional Frictional Forces in Composite Columns by Use of Headed Studs



Where stud connectors are attached to the web of a fully or partially concrete encased steel I-section or a similar section, account may be taken of the frictional forces that develop from the prevention of lateral expansion of the concrete by the adjacent steel flanges. This resistance may be added to the calculated resistance of the shear connectors. The additional resistance may be assumed to be $(\mu P_{Rd})/2$ on each flange and each horizontal row of studs, as shown in Figure 6.20, where μ is the relevant coefficient of friction that may be assumed. For steel sections without painting, μ may be taken as 0.5. P_{Rd} is the resistance of a single stud in accordance with Clause 6.6.6.3.1. In absence of better information from tests, the clear distance between the flanges should not exceed the values given in Figure 6.20.

If the cross-section is partially loaded (as, for example, Figure 6.21a), the loads may be distributed with a ratio of 1:2.5 over the thickness t_c of the end plate. Concrete stresses should then be limited in the area of the effective load introduction, for concrete filled hollow sections in accordance with the provision given below and for all other type of cross sections in accordance with Chapter 4.

If the concrete in a filled circular hollow section or a square hollow section is only partially loaded, for example by gusset plates through the profile or by stiffeners as shown in Figure 6.21, the local design strength of concrete, $\sigma_{c,Rd}$ under the gusset plate or stiffener resulting from the sectional forces of the concrete section should be determined by:

$$\sigma_{c,Rd} = f_{cd} \left(1 + \eta_{cL} \frac{t}{a} \frac{f_y}{f_{ck}} \right) \sqrt{\frac{A_c}{A_1}} \leq \frac{A_c f_{cd}}{A_1} \leq f_{yd}$$

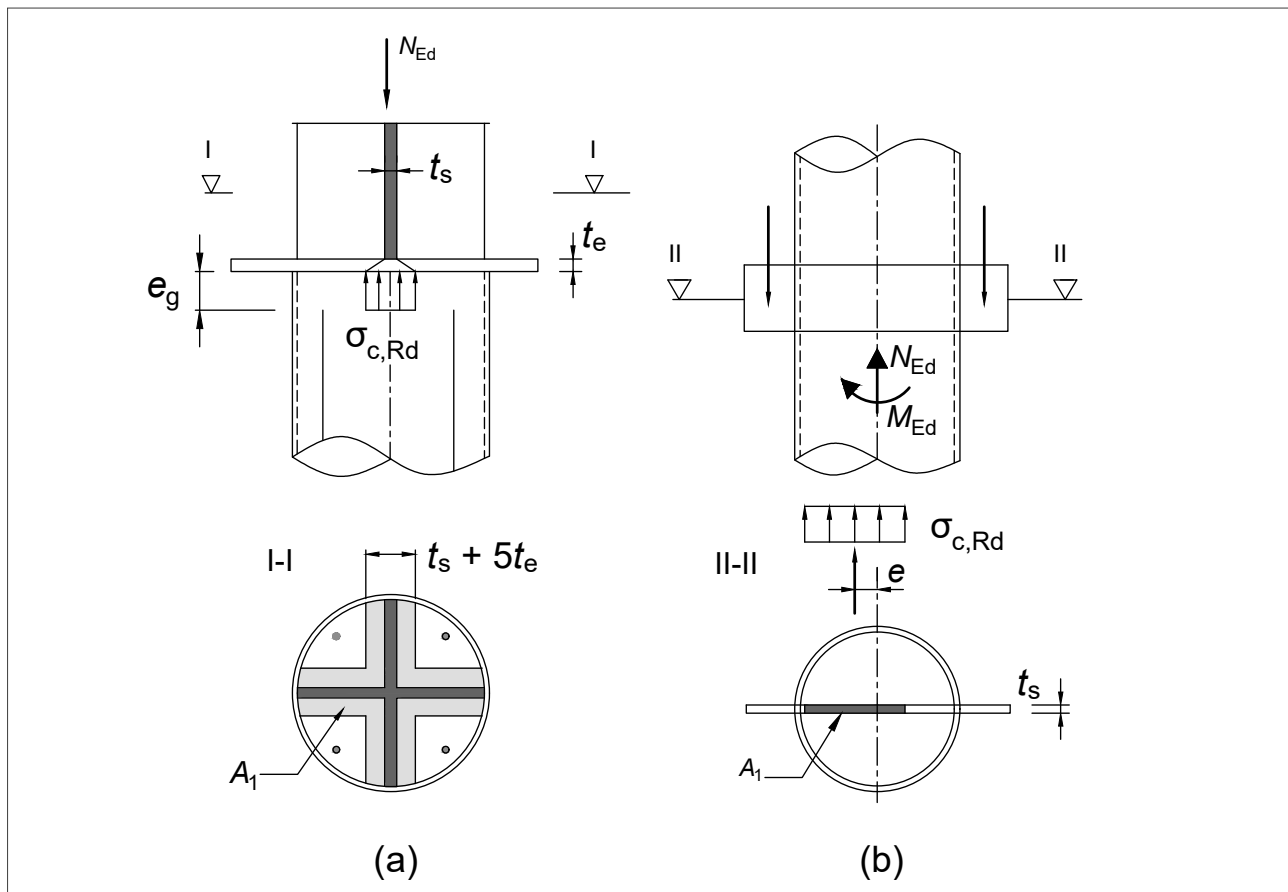
Equation 6.42

Where,

- t = The wall thickness of the steel tube.
- a = The diameter of the tube or the width of the square section.
- A_c = The cross-sectional area of the concrete section of the column.
- A_1 = The loaded area under the gusset plate, see Figure 6.21.
- η_{cL} = 4.9 for circular steel tube and 3.5 for square sections.

The ratio A_c/A_1 should not exceed the value 20.

Figure 6.21 Partially Loaded Circular Concrete Filled Hollow Section



For concrete filled circular hollow sections, longitudinal reinforcement may be taken into account for the resistance of the column, even where the reinforcement is not welded to the end plates or in direct contact with the endplates provided that:

- Verification for fatigue is not required
- The gap e_g , between the reinforcement and the end plate does not exceed 30 mm, see Figure 6.21a

Transverse reinforcement shall be in accordance with Chapter 4. In case of partially encased steel sections, concrete should be held in place by transverse reinforcement arranged as shown in Figure 6.22.

Figure 6.22 Arrangement of Stirrups

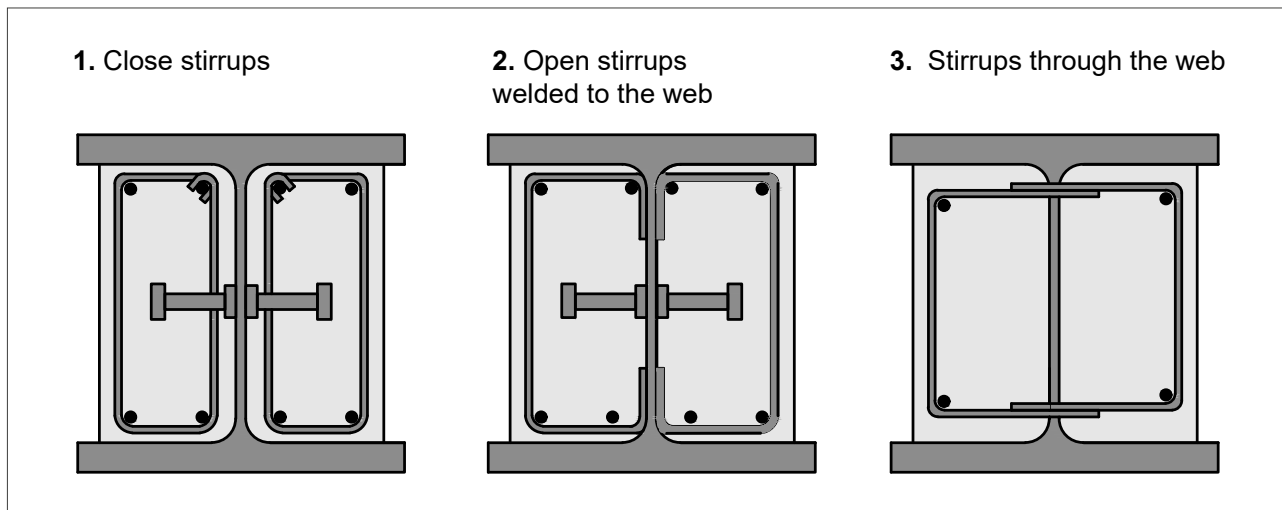
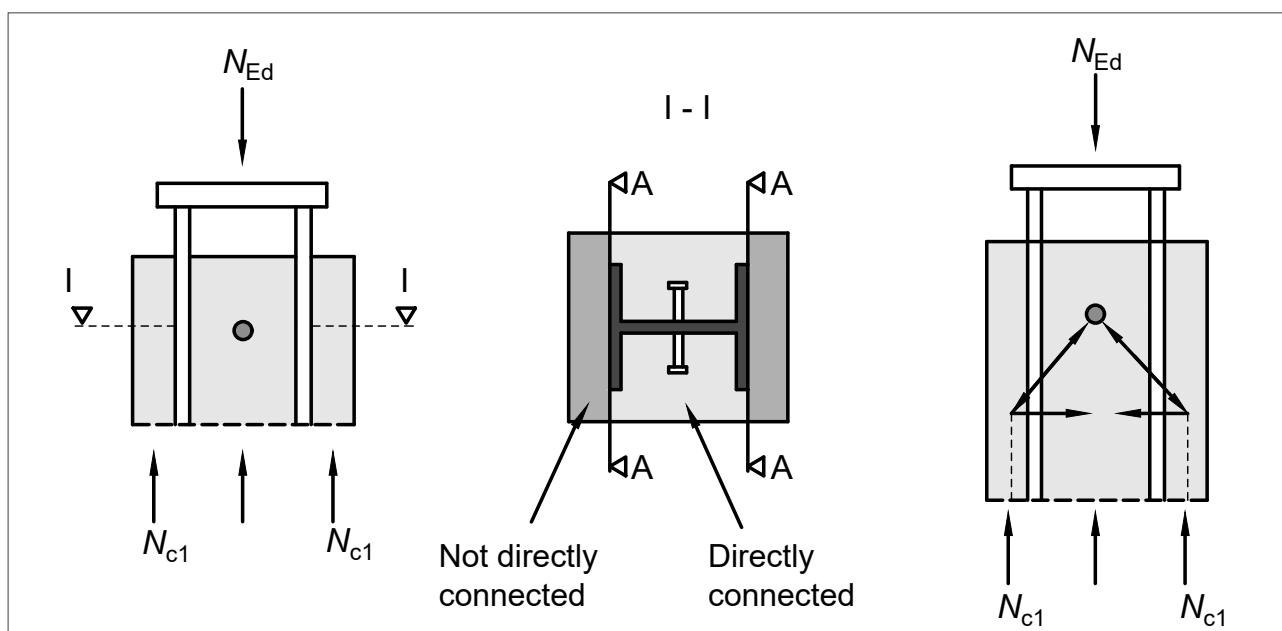


Figure 6.23 Directly and Not Directly Connected Concrete Areas for the Design of Transverse Reinforcement



In the case of load introduction through only the steel section or the concrete section, for fully encased steel sections the transverse reinforcement should be designed for the longitudinal shear that results from the transmission of normal force (N_{c1} in Figure 6.23) from the parts of concrete directly connected by shear connectors into the parts of the concrete without direct shear connection (see Figure 6.23, section A-A; the hatched area outside the flanges of Figure 6.23 should be considered as not directly connected). The design and arrangement of transverse reinforcement should be based on a truss model assuming an angle of 45° between concrete compression struts and the member axis.

6.6.7.4.3 Longitudinal Shear Outside the Areas of Load Introduction

Outside the area of load introduction, longitudinal shear at the interface between concrete and steel should be verified where it is caused by transverse loads and /or end moments. Shear connectors should be provided, based on the distribution of the design value of longitudinal shear, where this exceeds the design shear strength τ_{Rd} .

In absence of a more accurate method, elastic analysis, considering long term effects and cracking of concrete, may be used to determine the longitudinal shear at the interface.

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Provided that the surface of the steel section in contact with the concrete is unpainted and free from oil, grease and loose scale or rust, the values given in Table 6.5 may be assumed for τ_{Rd} .

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Table 6.5 Design Shear Strength τ_{Rd}

Type of Cross Section	τ_{Rd} (N/mm ²)
Completely concrete encased steel sections	0.30
Concrete filled circular hollow sections	0.55
Concrete filled rectangular hollow sections	0.40
Flanges of partially encased sections	0.20
Webs of partially encased sections	0.00

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The value of τ_{Rd} given in Table 6.5 for completely concrete encased steel sections applies to sections with a minimum concrete cover of 40 mm and transverse and longitudinal reinforcement in accordance with Clause 6.6.7.5.2. For greater concrete cover and adequate reinforcement, higher values of τ_{Rd} may be used. Unless verified by tests, for completely encased sections the increased value $\beta_c \tau_{Rd}$ may be used, with β_c given by:

$$\beta_c = 1 + 0.02c_z \left(1 - \frac{c_{z,min}}{c_z}\right) \leq 2.5$$

Equation 6.43

Where,

c_z = The nominal value of concrete cover in mm, see Figure 6.16a.

$c_{z,min}$ = 40 mm is the minimum concrete cover.

Unless otherwise verified, for partially encased *I*-sections with transverse shear due to bending about the weak axis due to lateral loading or end moments, shear connectors should always be provided. If the resistance to transverse shear is not be taken as only the resistance of the structural steel, then the required transverse reinforcement for the shear force $V_{c,Ed}$ according to Clause 6.6.7.3.2, should be welded to the web of the steel section or should pass through the web of the steel section.

6.6.7.5 Detailing Provisions

6.6.7.5.1 Concrete Cover of Steel Profiles and Reinforcement

For fully encased steel sections at least a minimum cover of reinforced concrete shall be provided to ensure the safe transmission of bond forces, the protection of the steel against corrosion and spalling of concrete.

The concrete cover to a flange of a fully encased steel section should be not less than 40 mm, nor less than one-sixth of the breadth *b* of the flange. The concrete cover for steel reinforcement shall be as per the requirements given in Chapter 4.

6.6.7.5.2 Longitudinal and Transverse Reinforcement

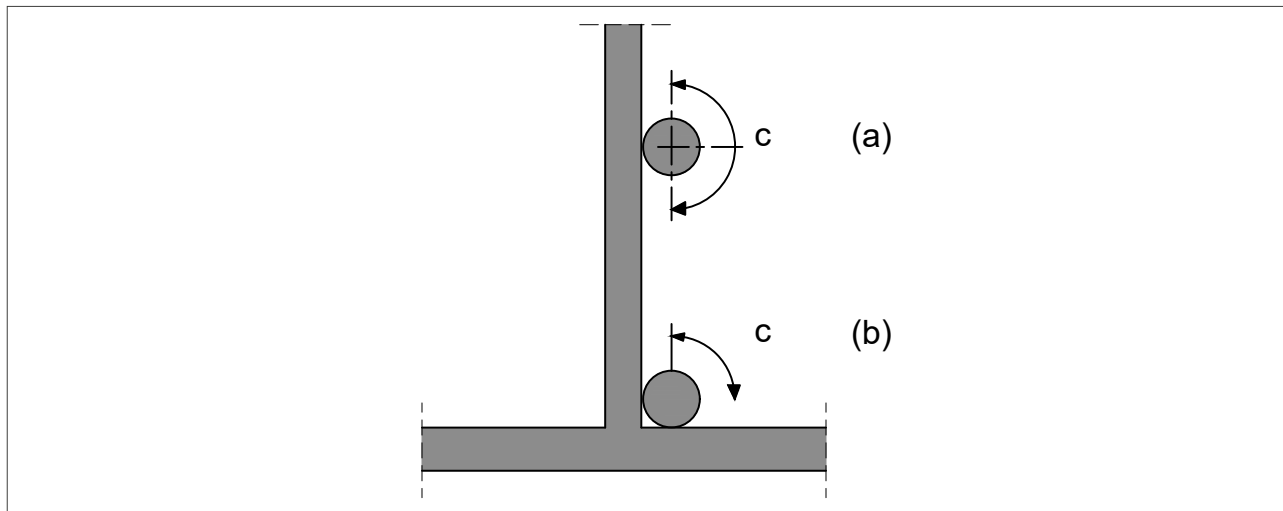
The longitudinal reinforcement in concrete-encased columns which is allowed for in the resistance of the cross-section should be not less than 0.3 % of the cross-section of the concrete. In concrete filled hollow sections normally no longitudinal reinforcement is necessary, if design for fire resistance is not required.

The transverse and longitudinal reinforcement in fully or partially concrete encased columns should be designed and detailed in accordance with Chapter 4.

The clear distance between longitudinal reinforcing bars and the structural steel section may be smaller than required as in Chapter 4, even zero. In this case, for bond the effective perimeter *c* of the reinforcing bar should be taken as half or one quarter of its perimeter, as shown in Figure 6.24 at (a) and (b) respectively.

For fully or partially encased members, where environmental conditions (refer Chapter 4) are class X_0 and longitudinal reinforcement is neglected in design, a minimum longitudinal reinforcement of diameter 8 mm and 250 mm spacing and a transverse reinforcement of diameter 6 mm and 200 mm spacing should be provided. Alternatively welded mesh reinforcement of diameter 4 mm may be used.

Figure 6.24 Effective Perimeter c of Reinforcing Bar



6.6.8 Fatigue

The resistance of composite structures to fatigue shall be verified where the structures are subjected to repeated fluctuations of stresses.

For headed stud shear connectors in bridges, under characteristic combination of actions, the maximum longitudinal shear force per connector should not exceed $k_s P_{Rd}$ where P_{Rd} is determined according to Clause 6.6.6.3.1. The recommended value of reduction factor k_s is 0.75.

For structural steel, no fatigue assessment is required where Clause 5.9.1 of Chapter 5 applies. For concrete and reinforcement, no fatigue assessment is required when the exceptions listed in Clause 4.6.9.1 of Chapter 5 apply.

Partial factors γ_{Mf} for fatigue strength are given in Chapter 5 for structural steel elements and in Chapter 4 for concrete and reinforcement. For headed studs in shear, a partial factor $\gamma_{Mf,s} = 1.0$ should be applied. Partial factors for fatigue loading γ_{Ff} should be applied as in Chapter 5.

6.6.8.1 Fatigue Strength

The fatigue strength for structural steel and for welds should be taken from Chapter 5. The fatigue strength of reinforcing steel and pre-stressing steel should be taken from Chapter 4.

The fatigue strength curve of an automatically welded headed stud in accordance with Clause 6.6.6.3.1 is shown in Figure 6.25 and given for normal weight concrete by:

$$(\Delta\tau_R)^m N_R = (\Delta\tau_c)^m N_c$$

Where,

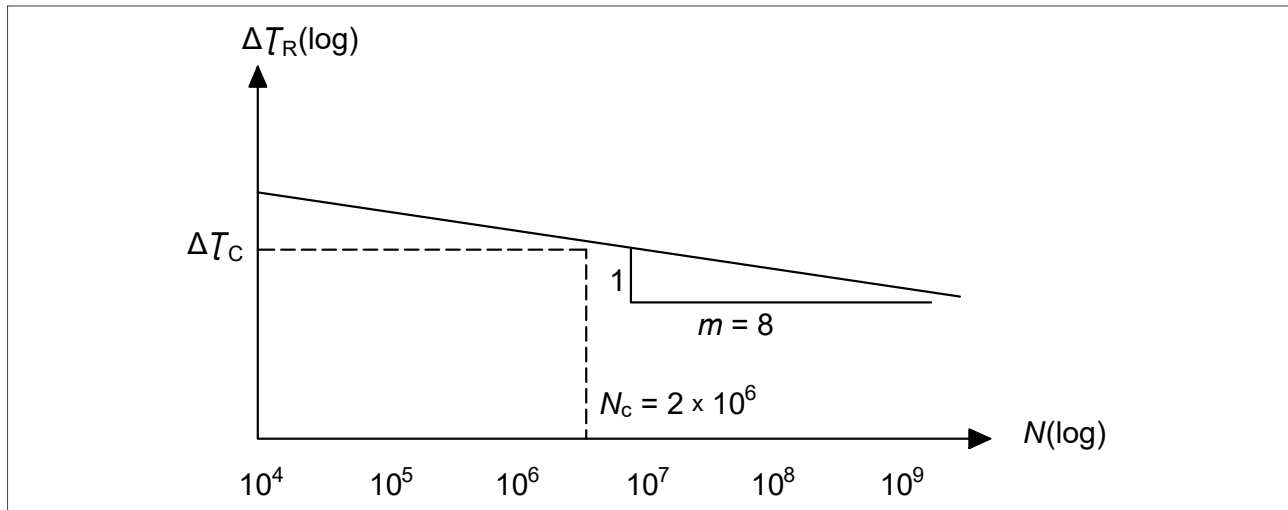
$\Delta\tau_R$ = The fatigue shear strength related to the cross-sectional area of the shank of the stud, using the nominal diameter d of the shank

$\Delta\tau_c$ = The reference value at $N_c = 2 \times 10^6$ cycles with $\Delta\tau_c$ equal to 90 N/mm²

m = The slope of the fatigue strength curve with the value $m = 8$

N_R = The number of stress cycles

Figure 6.25 Fatigue Strength Curve for Headed Studs in Solid Slab



6.6.8.2 Internal Forces and Fatigue Loadings

Internal forces and moments should be determined by elastic global analysis of the structure in accordance with Clause 6.5.4.1 and 6.5.4.2. The maximum and minimum internal bending moments and/or internal forces resulting from the load combination are defined as $M_{Ed,max,f}$ and $M_{Ed,min,f}$.

6.6.8.3 Stresses

6.6.8.3.1 General

The calculation of stresses should be based on Clause 6.7.1. For the determination of stresses in cracked regions the effect of tension stiffening of concrete on the stresses in reinforcement shall be taken into account.

The effect of tension stiffening on the stresses in reinforcement may be taken into account according to Clause 6.6.8.3.3. For the determination of stresses in structural steel the effect of tension stiffening may be neglected. For the determination of stresses in concrete elements, refer Chapter 4.

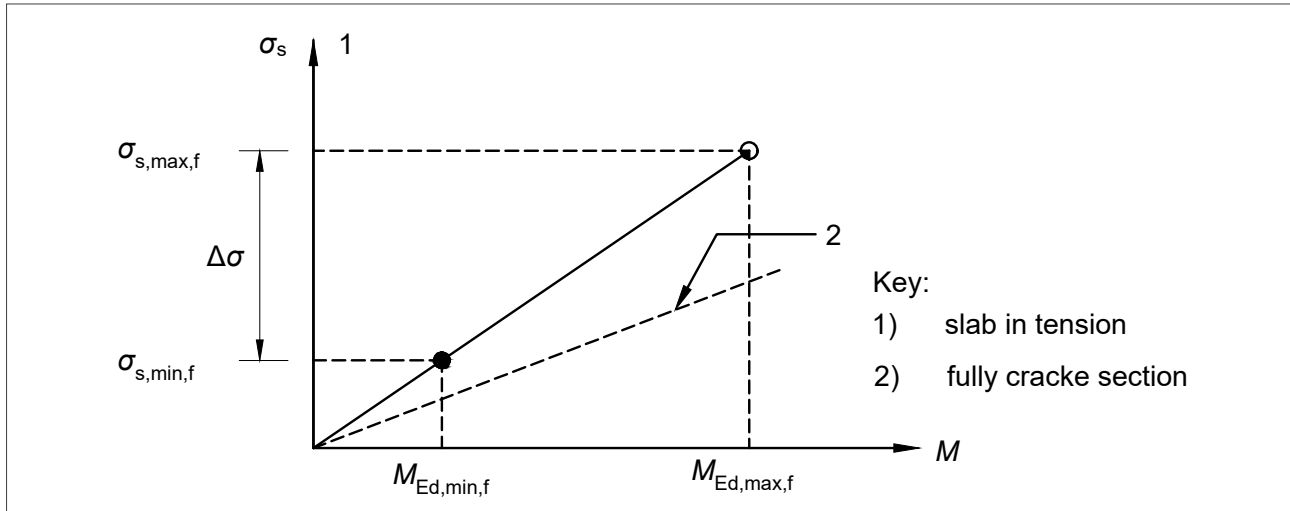
6.6.8.3.2 Concrete and Structural Steel

Where the bending moments $M_{Ed,max,f}$ and $M_{Ed,min,f}$ cause tensile stresses in the concrete slab, the stresses in structural steel for these bending moments may be determined based on the second moment of area I_2 .

Where $M_{Ed,min,f}$ and $M_{Ed,max,f}$ or only $M_{Ed,min,f}$ cause compression in the concrete slab, the stresses in structural steel for these bending moments should be determined with the cross-section properties of the un-cracked section.

6.6.8.3.3 Reinforcement

Where the bending moment $M_{Ed,max,f}$ causes tensile stresses in the concrete slab and where no more accurate method is used, the effects of tension stiffening of concrete on the stress $\sigma_{s,max,f}$ in reinforcement due to $M_{Ed,max,f}$ should be determined from the Equations 6.52 to 6.54 given in Clause 6.7.2, also a factor 0.2 should be used, in place of the factor 0.4 in Equation 6.53.

Figure 6.26 Determination of Stresses $\sigma_{s,max,f}$ and $\sigma_{s,min,f}$ in Cracked Regions

Where also the bending moment $M_{Ed,min,f}$ causes tensile stress in the concrete slab the stress range $\Delta\sigma$ is given by Figure 6.26 and the stress $\sigma_{s,max,f}$ in reinforcement due to $M_{Ed,min,f}$ can be determined from:

$$\sigma_{s,min,f} = \sigma_{s,max,f} \frac{M_{Ed,min,f}}{M_{Ed,max,f}} \quad \text{Equation 6.44}$$

Where $M_{Ed,min,f}$ and $M_{Ed,max,f}$ or only $M_{Ed,min,f}$ cause compression in the concrete slab, the stresses in reinforcement for these bending moments should be determined with the cross-section properties of the un-cracked section.

6.6.8.3.4 Shear Connection

The longitudinal shear per unit length shall be calculated by elastic analysis. The longitudinal shear forces at the interface between structural steel and concrete may be determined by using the properties of the uncracked section.

6.6.8.4 Stress Ranges

6.6.8.4.1 Structural Steel and Reinforcement

The stress ranges should be determined from the stresses determined in accordance with Clause 6.6.8.3. Where the verification for fatigue is based on damage equivalent stress ranges, in general a range should be determined from:

$$\Delta\sigma_E = \lambda \phi [\sigma_{max,f} - \sigma_{min,f}]$$

Where:

$\sigma_{max,f}$ and $\sigma_{min,f}$ are the maximum and minimum stresses.

λ = A damage equivalent factor.

ϕ = A damage equivalent impact factor.

Where a member is subjected to combined global and local effects the separate effects should be considered. The equivalent constant amplitude stress due to global effects and local effects should be combined using:

$$\Delta\sigma_E = \lambda_{glob} \phi_{glob} \Delta\sigma_{E,glob} + \lambda_{loc} \phi_{loc} \Delta\sigma_{E,loc}$$

$$\text{Equation 6.45}$$

The subscripts "glob" and "loc" refer to global and local effects, respectively.

1

The damage equivalent factor λ depends on the loading spectrum and the slope of the fatigue strength curve. The factor λ for structural steel elements is given in Chapter 5 for road bridges. For road bridges the damage equivalent impact factor may be taken as equal to 1.0.

2

6.6.8.4.2 Shear Connection

For verification of stud shear connectors based on nominal stress ranges, the equivalent constant range of shear stress $\Delta\tau_{E,2}$ for 2 million cycles is given by:

$$\Delta\tau_{E,2} = \lambda_v \Delta\tau$$

Where,

λ_v = The damage equivalent factor depending on the spectra and the slope m of the fatigue strength curve

$\Delta\tau$ = The range of shear stress due to fatigue loading, related to the cross-sectional area of the shank of the stud using the nominal diameter d of the shank

For bridges, the damage equivalent factor λ_v for headed studs in shear should be determined from $\lambda_v = \lambda_{v,1} \lambda_{v,2} \lambda_{v,3} \lambda_{v,4}$ where the factors $\lambda_{v,1}$ to $\lambda_{v,4}$ are defined below.

For road bridges of span up to 100 m, the factor $\lambda_{v,1} = 1.55$ should be used. The factors $\lambda_{v,2}$ to $\lambda_{v,4}$ should be determined in accordance with Clause 5.9.5.2 of Chapter 5 but using exponents 8 and 1/8 in place of those given, to allow for the relevant slope $m = 8$ of the fatigue strength curve for headed studs, given in Clause 6.6.8.1.

6.6.8.5 Fatigue Assessment Based on Nominal Stress Ranges

6.6.8.5.1 Structural Steel, Reinforcement and Concrete

The fatigue assessment for reinforcement should follow Chapter 4. The verification for concrete in compression should follow Chapter 4. The fatigue assessment for structural steel should comply with Chapter 5.

6.6.8.5.2 Shear Connection

For stud connectors welded to a steel flange that is always in compression under the relevant combination of actions, the fatigue assessment should be made by checking the criterion:

$$\gamma_{Ff} \Delta\tau_{E,2} \leq \frac{\Delta\tau_c}{\gamma_{Mf,s}} \quad \text{Equation 6.46}$$

Where,

$\Delta\tau_{E,2}$ = Defined in Clause 6.6.8.4.2

$\Delta\tau_c$ = The reference value of fatigue strength at 2 million cycles determined in accordance with Clause 6.6.8.1.

The stress range $\Delta\tau$ in the stud should be determined with the cross-sectional area of the shank of the stud using the nominal diameter d of the shank.

Where the maximum stress in the steel flange to which stud connectors are welded is tensile under the relevant combination, the interaction at any cross-section between shear stress range $\Delta\tau_E$ in the weld of stud connectors and the normal stress range $\Delta\sigma_E$ in the steel flange should be verified using the following interaction Equations.

$$\frac{\gamma_{Ff} \Delta\sigma_{E,2}}{\Delta\sigma_c / \gamma_{Mf}} + \frac{\gamma_{Ff} \Delta\tau_{E,2}}{\Delta\tau_c / \gamma_{Mf,s}} \leq 1.3 \quad \text{Equation 6.47}$$

$$\frac{\gamma_{Ff} \Delta\sigma_{E,2}}{\Delta\sigma_c / \gamma_{Mf}} \leq 1.0 \quad \frac{\gamma_{Ff} \Delta\tau_{E,2}}{\Delta\tau_c / \gamma_{Mf,s}} \leq 1.0$$

$$\text{Equation 6.48}$$

Where,

$\Delta\sigma_{E,2}$ = The stress range in the flange

$\Delta\sigma_c$ = The reference value of fatigue strength by applying category 80

$\Delta\tau_{E,2}$ and $\Delta\tau_c$ are the stress range defined above

6.7 Serviceability Limit States

6.7.1 General

Calculation of stresses for beams at the serviceability limit state shall take into account the following effects, where relevant:

- Shear lag.
- Creep and shrinkage of concrete.
- Cracking of concrete and tension stiffening of concrete.
- Sequence of construction.
- Increased flexibility resulting from significant incomplete interaction due to slip of shear connection.
- Inelastic behaviour of steel and reinforcement, if any.
- Torsional and distortional warping, if any.

Shear lag may be taken into account according to Clause 6.5.4.1.2. Creep and shrinkage may be taken into account by use of modular ratios according to Clause 6.5.4.2.2. In cracked sections the primary effects of shrinkage may be neglected when verifying stresses.

In section analysis the tensile strength of concrete shall be neglected. The influence of tension stiffening of concrete between cracks on stresses in reinforcement and pre-stressing steel should be taken into account. The influences of tension stiffening on stresses in structural steel may be neglected.

Stresses in the concrete slab and its reinforcement caused by simultaneous global and local actions should be added.

Excessive creep and microcracking shall be avoided by limiting the compressive stress in concrete.

Stress limitation for concrete to the value $k_i f_{ck}$ should be in accordance with Chapter 4. Under the characteristic combination of actions, the stresses should be limited to $k_i f_{sk}$ in reinforcing steel. Refer the section Serviceability Limit state of Chapter 4 for values of k_i .

The stresses in structural steel should be in accordance with Chapter 5.

For serviceability limit states the longitudinal shear force per connector should be limited according to Clause 6.6.8

The slenderness of unstiffened or stiffened web plates of composite girders should be limited according to Chapter 5.

The limitation of crack width depends on the exposure classes according to Chapter 4. An estimation of crack width can be obtained from Chapter 4, where the stress σ_s should be calculated by taking into account the effects of tension stiffening. The σ_s may be determined as per the Equation 6.52.

As a simplified and conservative alternative, crack width limitation to acceptable width can be achieved by ensuring a minimum reinforcement defined in Clause 6.7.2, and bar spacing or diameters not exceeding the limits.

6.7.2 Minimum Reinforcement

In all sections subjected to significant tension due to restraint of imposed deformations (e.g. primary and secondary effects of shrinkage), in combination or not with effects of direct loading the required minimum reinforcement area A_s for the slabs of composite beams is given by:

$$A_s = \frac{k_s k_c k f_{ct,eff} A_{ct}}{\sigma_s}$$

Equation 6.49

Where,

$f_{ct,eff}$ = The mean value of the tensile strength of the concrete effective at the time when cracks may first be expected to occur. Values of $f_{ct,eff}$ may be taken as those f_{ctm} , taking as the class the strength at the time cracking is expected to occur. When the age of the concrete at cracking cannot be established with confidence as being less than 28 days, a minimum tensile strength of 3 N/mm² may be adopted

k = A coefficient which allows for the effect of non-uniform self-equilibrating stresses which may be taken as 0.8

k_s = A coefficient which allows for the effect of the reduction of the normal force of concrete slab due to initial cracking and local slip of the shear connection, which may be taken as 0.9

k_c = A coefficient which takes account of the stress distribution within the section immediately prior to cracking and is given by:

$$k_c = \frac{1}{1 + h_c / (2z_0)} + 0.3 \leq 1.0$$

Equation 6.50

h_c = The thickness of the concrete flange, excluding any haunch or ribs.

z_0 = The vertical distance between the centroids of the un-cracked concrete flange and the uncracked composite section, calculated using the modular ratio n_0 for short-term loading.

σ_s = The maximum stress permitted in the reinforcement immediately after cracking. This may be taken as its characteristic yield strength f_{sk} . A lower value, depending on the bar size, may however be needed to satisfy the required crack width limits. This value is given in Table 6.6.

A_{ct} = The area of the tensile zone (caused by direct loading and primary effects of shrinkage) immediately prior to cracking of the cross section. For simplicity the area of the concrete section within the effective width may be used.

Table 6.6 Maximum Bar Diameter for High Bond Bars

Steel stress σ_s (N/mm ²)	Maximum bar diameter ϕ^* (mm) for design crack width w_k		
	$w_k = 0.4$ mm	$w_k = 0.3$ mm	$w_k = 0.2$ mm
160	40	32	25
200	32	25	16
240	20	16	12
280	16	12	8
320	12	10	6
360	10	8	5
400	8	6	4
450	6	5	-

The maximum bar diameter for the minimum reinforcement may be modified to a value ϕ given by:

$$\phi = \phi^* \frac{f_{ct,eff}}{f_{ct,0}}$$

Equation 6.51

Where,

ϕ^* = The maximum bar size given in Table 6.6

$f_{ct,0}$ = A reference strength of 2.9 N/mm²

At least half of the required minimum reinforcement should be placed between mid-depth of the slab and the face subjected to the greater tensile strain. The minimum reinforcement should be placed where the stresses in concrete are tensile under the characteristic combination of actions.

Where at least the minimum reinforcement determined above is provided, the limitation of crack widths to acceptable values may generally be achieved by limiting bar spacing or bar diameters. Maximum bar diameter and maximum bar spacing depend on the stress σ_s in the reinforcement and the design crack width. Maximum bar diameters are given in Table 6.6 and maximum bar spacing in Table 6.7.

Table 6.7 Maximum Bar Spacing for High Bond Bars

Steel stress σ_s (N/mm ²)	Maximum bar spacing mm) for design crack width w_k		
	$w_k = 0.4$ mm	$w_k = 0.3$ mm	$w_k = 0.2$ mm
160	300	300	200
200	300	250	150
240	250	200	100
280	200	150	50
320	150	100	-
360	100	50	-

In composite beams where the concrete slab is assumed to be cracked, stresses in reinforcement increase due to the effects of tension stiffening of concrete between cracks compared with the stresses based on a composite section neglecting concrete. The tensile stress in reinforcement σ_s due to direct loading may be calculated from:

$$\sigma_s = \sigma_{s,0} + \Delta\sigma_s$$

Equation 6.52

With,

$$\Delta\sigma_s = \frac{0.4 f_{ctm}}{\alpha_{st} \rho_s}$$

Equation 6.53

$$\alpha_{st} = \frac{A I}{A_a I_a}$$

Equation 6.54

Where,

$\sigma_{s,0}$ = The stress in the reinforcement caused by the internal forces acting on the composite section, calculated neglecting concrete in tension.

f_{ctm} = The mean tensile strength of the concrete, for normal concrete taken as f_{ctm} .

ρ_s = The reinforcement ratio, given by $\rho_s = (A_s / A_{ct})$.

A_{ct} = The effective area of the concrete flange within the tensile zone; for simplicity the area of the concrete section within the effective width should be used.

A_s = The total area of all layers of longitudinal reinforcement within the effective area A_{ct} .

A, I = Are area and second moment of area, respectively, of the effective composite section neglecting concrete in tension.

A_a, I_a are the corresponding properties of the structural steel section.

6.7.3 Filler Beam Decks

The action effects for the serviceability limit states should be determined according to Clause 6.5.4.7. The minimum longitudinal top reinforcement $A_{s,min}$ per filler beam should be determined as follows:

$$A_{s,min} \geq 0.01 A_{c,eff}$$

Equation 6.55

Where,

$A_{c,eff}$ = The effective area of concrete given by $A_{c,eff} = s_w c_{st} \leq s_w d_{eff}$

d_{eff} = The effective thickness of the concrete given by $d_{eff} = c + 7.5 \phi_s$

ϕ_s = The diameter of the longitudinal reinforcement in mm within the range $10\text{mm} \leq \phi_s \leq 16\text{mm}$

c_{st} = The concrete cover of the longitudinal reinforcement and the structural steel section (see Figure 6.8)

s_w is defined in Figure 6.8

The bar spacing s of the longitudinal reinforcement shall be $100\text{ mm} \leq s \leq 150\text{ mm}$

6.8 Precast Concrete Slabs in Composite Bridges

This Clause deals with reinforced or prestressed precast concrete slabs, used either as full depth flanges of bridge decks or as partial depth slabs acting with in-situ concrete.

Precast bridge slabs should be designed in accordance with Chapter 4 and also for composite action with the steel beam. Precast elements acting as permanent formwork shall be designed for the intended construction loads taking into account the construction method.

Where it is assumed that the precast slab acts with in-situ concrete, they should be designed as continuous in both the longitudinal and the transverse directions. The joints between slabs should be designed to transmit in-plane forces as well as bending moments and shears. Compression perpendicular to the joint may be assumed to be transmitted by contact pressure if the joint is filled with mortar or glue or if it is shown by tests that the meeting surfaces are in sufficiently close contact.

For the use of stud connectors in groups, see Clause 6.6.6.5. The shear connection between steel beam and concrete slab and the transverse reinforcement in concrete slab shall be designed in accordance with the relevant Clauses.

If shear connectors welded to the steel beam project into recesses within slabs or joints between slabs, which are filled with concrete after erection, the detailing and the properties of the concrete (e.g. size of the aggregate) should be such that it can be cast properly. The clear distance between the shear connectors and the precast element should be sufficient in all directions to allow for full compaction of the infill concrete.

6.9 Composite Plates in Bridges

6.9.1 General

This Clause is valid for composite plates consisting of a nominally flat plate of structural steel connected to a site cast concrete layer by headed studs for use as a flange in a bridge deck carrying transverse loads as well as in-plane forces, or as a bottom flange in a box girder.

The steel plate should be supported during casting either permanently or by temporary, supports in order to limit its deflection to less than 0.05 times the thickness of the concrete layer unless the additional weight of concrete due to the deflection of the plate is taken into account in the design of the steel plate.

The effective width should be determined according to Clause 6.5.4.1.2, where b_0 should be taken as $2a_w$ with a_w defined in Figure 6.27.

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Composite Structures

6.9.2 Design for Local Effects

Local effects are bending moments and shears caused by transverse loads on the composite plate acting as a one- or two-way slab. For the purpose of analysis of local action effects, the composite plate may be assumed to be elastic and uncracked. A top flange of an *I*-girder need not be designed as composite in the transverse direction.

The concrete and the steel plate may be assumed to act compositely without slip.

The resistance to bending and vertical shear force may be verified as for a reinforced concrete slab where the steel plate is considered as reinforcement. The design resistance for vertical shear in Clause 6.6.2.2 is applicable, where the distance, in longitudinal and transverse direction, between shear connectors does not exceed three times the thickness of the composite plate.

6.9.3 Design for Global Effects

The composite plate shall be designed to resist all forces from axial loads and global bending and torsion of all longitudinal girders or cross-girders of which it forms a part.

The design resistance to in-plane compression may be taken as the sum of the design resistances of the concrete and the steel plate within the effective width. Reduction in strength due to second order effects should be considered.

The design resistance for in-plane tension should be taken as the sum of the design resistances of the steel plate and the reinforcement within the effective width.

Interaction with local load effects should be considered for the shear connectors. Otherwise it need not be considered. Connectors designed for shear forces in both the longitudinal and transverse directions should be verified for the vector sum of the simultaneous forces on the connector.

6.9.4 Design of Shear Connectors

Resistance to fatigue and requirements for serviceability limit states shall be verified for the combined local and simultaneous global effect.

The design strength of stud connectors in Clause 6.6.6.3 and 6.6.8.1 may be used provided that the concrete slab has bottom reinforcement with area not less than 0.002 times the concrete area in each of two perpendicular directions.

The detailing rules of Clause 6.6.6.5 are applicable.

For wide girder flanges the distribution of longitudinal shear due to global effects for serviceability and fatigue limit states may be determined as follows in order to account for slip and shear lag. The longitudinal force P_{Ed} on a connector at distance x from the nearest web may be taken as

$$P_{Ed} = \frac{v_{L,Ed}}{n_{tot}} \left[\left(3.85 \left(\frac{n_w}{n_{tot}} \right)^{-0.17} - 3 \right) \left(1 - \frac{x}{b} \right)^2 + 0.15 \right]$$

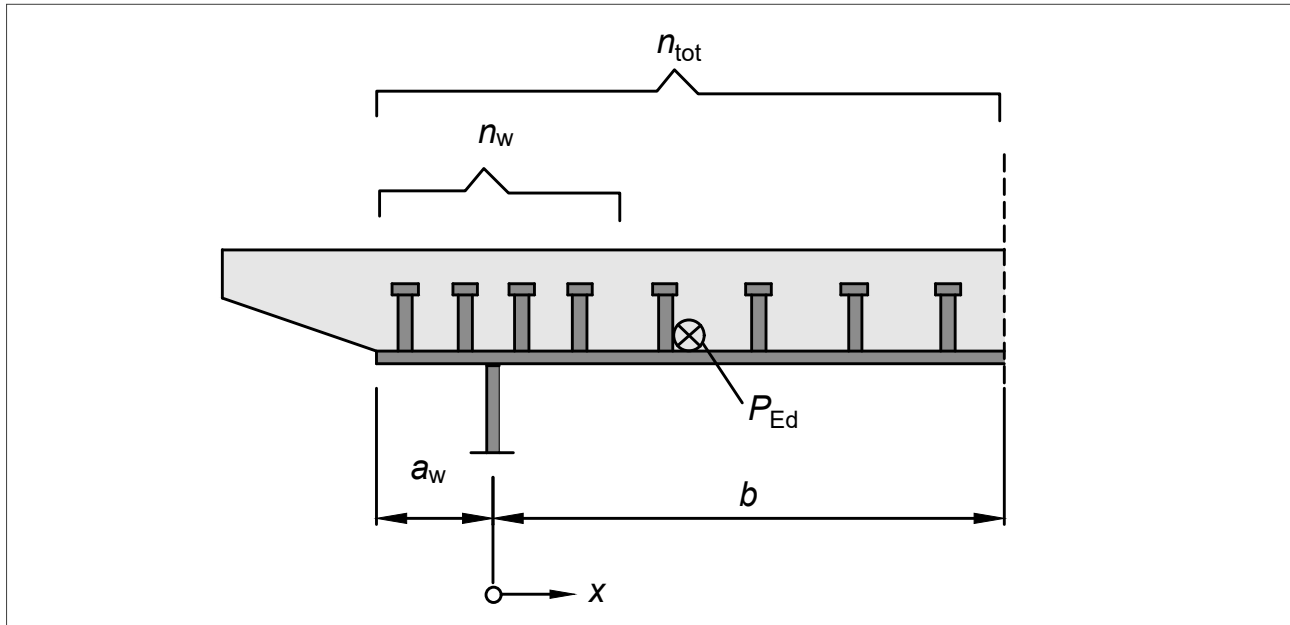
Equation 6.56

Where,

- $v_{L,Ed}$ = The design longitudinal shear per unit length in the concrete slab due to global effects for the web considered, determined using effective widths for shear lag.
- n_{tot} = The total number of connectors of the same size per unit length of girder as shown in Figure 6.27, provided that the number of connectors per unit area does not increase with x .
- n_w is the number of connectors per unit length placed within a distance from the web equal to the larger of $10 t_f$ and 200 mm, where t_f is the thickness of the steel plate. For these connectors x should be taken as 0.
- b = Equal to half the distance between adjacent webs or the distance between the web and the free edge of the flange.

In case of a flange projecting distance a_w outside the web according to Figure 6.27, the number of connectors n_{tot} and n_w may include connectors placed on this flange. Shear connectors should be concentrated in the region for n_w according to Figure 6.27. The spacing of the connectors should fulfil the conditions given in Table 6.8 to avoid premature local buckling of the plate.

Figure 6.27 Notations in Composite Plates



A more accurate determination of the distribution of longitudinal shear forces in composite bottom flanges of box sections is not required, if the arrangement of the shear connectors is based on the following rules:

- Shear connectors should be concentrated in the corners of the box girder
- At least 50% of the total amount of shear connectors, which are responsible for the transfer of the longitudinal shear force from a web to the bottom concrete flange should be attached to the web and within the width b_f of the steel bottom flange. The width b_f of the steel bottom flange should be taken as the largest of:

$$b_f = 20 t_f \quad b_f = 0.2 b_{ci} \quad \text{and} \quad b_f = 400 \text{ mm}$$

Where b_{ci} is the effective width of the lower flange according to Clause 6.5.4.1.2 and t_f is the thickness of the steel bottom flange.

For ultimate limit states it may be assumed that all connectors within the effective width carry the same longitudinal force.

Where restraint from shear connectors is relied upon to prevent local buckling of the element of a composite plate in compression, the centre-to-centre spacings of the connectors should not exceed the limits in Table 6.8.

Table 6.8 Maximum Spacing of Shear Connectors for Composite Plate in Compression

		Class 2	Class 3
Transverse to the direction of compressive stress	Outstand flange:	$14 t_\varepsilon$	$20 t_\varepsilon$
	Interior flange:	$45 t_\varepsilon$	$50 t_\varepsilon$
In the direction of compressive stress	Outstand and interior flanges:	$22 t_\varepsilon$	$25 t_\varepsilon$
$\varepsilon = \sqrt{\left(\frac{235}{f_y}\right)}, \text{ with } f_y \text{ in } \text{N/mm}^2 \quad t - \text{thickness of the steel flange}$			

7 Masonry and Timber Structures

This Chapter cover the guidelines on timber and masonry structures. The details are provided in different sections.

7.1 Masonry Structures

7.1.1 General

7.1.1.1 Scope

Scope of this section covers general guidelines on masonry works. It does not cover masonry bridges. The masonry works can be adopted for gravity retaining walls.

7.1.1.2 Reference Standards

EN 771-1	Specification for masonry units- Part-1: Clay masonry units
EN 771-2	Specification for masonry units- Part-2: Calcium silicate masonry units
EN 771-3	Specification for masonry units- Part-3: Aggregate Concrete Masonry units
EN 771-4	Specification for masonry units- Part-4: Autoclaved aerated Concrete Masonry units
EN 771-5	Specification for masonry units- Part-5: Manufactured stone masonry units
EN 771-6	Specification for masonry units- Part-6: Natural stone masonry units
EN-998-1	Specifications for mortar for masonry- Part-1: Rendering and plastering mortar
EN-998-2	Specifications for mortar for masonry- Part-2: Masonry mortar

7.1.1.3 Definitions

For general definitions, refer initial sections of this Manual. Other definitions are as under.

Masonry	An assemblage of masonry units laid in a specified pattern and joined together with mortar.
Masonry unit	A preformed component, intended for use in masonry construction.
Groups 1, 2,3 and 4 masonry units	Group designations for masonry units, according to the percentage size and orientation of holes in the units when laid.
Bed face	The top or bottom surface of a masonry unit when laid as intended
Frog	A depression formed during manufacturing, in one or both bed faces of a masonry unit.
Hole	A formed void which may or may not pass completely through a masonry unit.
Griphole	A formed void in masonry unit to enable it to be more readily grasped and lifted with one or both hands or by machine.
Web	The solid material between the holes in masonry unit.
Shell	The peripheral material between the hole and face of a masonry unit.
Bed joint	A mortar layer between the bed faces of masonry units.
Perpend joint	A mortar joint perpendicular to the bed joint and to the face of wall.
Longitudinal joint	A vertical mortar joint within the thickness of a wall, parallel to the face of the wall.
Thin layer joint	A joint made with thin layer mortar.

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Masonry and Timber Structures

7.1.1.4 Symbols

f_b	Normalised mean compressive strength of a masonry unit.
f_d	Design compressive strength of a masonry.
f_k	Characteristic compressive strength of masonry.
f_m	Compressive strength of mortar.
K	Constant used in the calculation of the compressive strength of masonry.
N_{Ed}	Design value of the vertical load.
N_{Rd}	Design value of the vertical resistance.

7.1.2 Basis of Design

7.1.2.1 Principles of Limit State Design

For masonry structures, the ultimate limit state and serviceability limit state shall be considered for all aspects of the structure.

Actions shall be obtained as per the general guidelines provided in Chapter 3 and other relevant standards.

Properties of materials and construction products and geometrical data to be used for design should be those specified in the relevant standards. The design value for a material property is obtained by dividing its characteristic value by the relevant partial factor for materials, γ_M .

7.1.2.2 Ultimate Limit States

The partial factor for materials γ_M shall be used for the ultimate limit state for ordinary and accidental situations. It depends upon the category of masonry unit and class of execution control.

When considering the effect of misuse or accident, the values provided in this table may be halved.

Table 7.1 Geometrical Requirements for Grouping of Masonry Units

Material Masonry	γ_M for Execution Class	
	1	2
1. When in a start of direct or flexural compression		
i. Unreinforced masonry made with:		
Units of category I	2.3	2.7
Units of category II	2.6	3.0
ii. Reinforced masonry made with:		
Units of category I	2.0	
Units of category II	2.3	
2. When in a start of flexural tension		
Units of category I and II	2.3	2.7
3. When in a start of shear		
i. Unreinforced masonry made with:		
Units of category I and II	2.5	2.5
ii. Reinforced masonry made with:		
Units of category I and II	2.0	

The class of execution are defined as:

Class 1: Execution control corresponds the special category of construction control and workmanship including appropriate supervision and inspection. Mortar confirming to the strength requirements and regular testing used on site.

Class 2: Execution control corresponds the normal category of construction control and workmanship including appropriate supervision.

7.1.2.3 Serviceability Limit States

Detailed calculations are not required when simplified rules given in the relevant Clauses are followed. When needed, the partial factor for materials properties γ_M shall be 1.0 for serviceability limit state.

7.1.3 Materials

7.1.3.1 Masonry units

The masonry units shall comply the relevant standards given in Clause 7.1.1.2. The mean compressive strength of a product must be declared by the manufacturer. Masonry units may be of Category I or Category II.

Category I: The probability of failing to achieve the declared strength does not exceed 5 %)

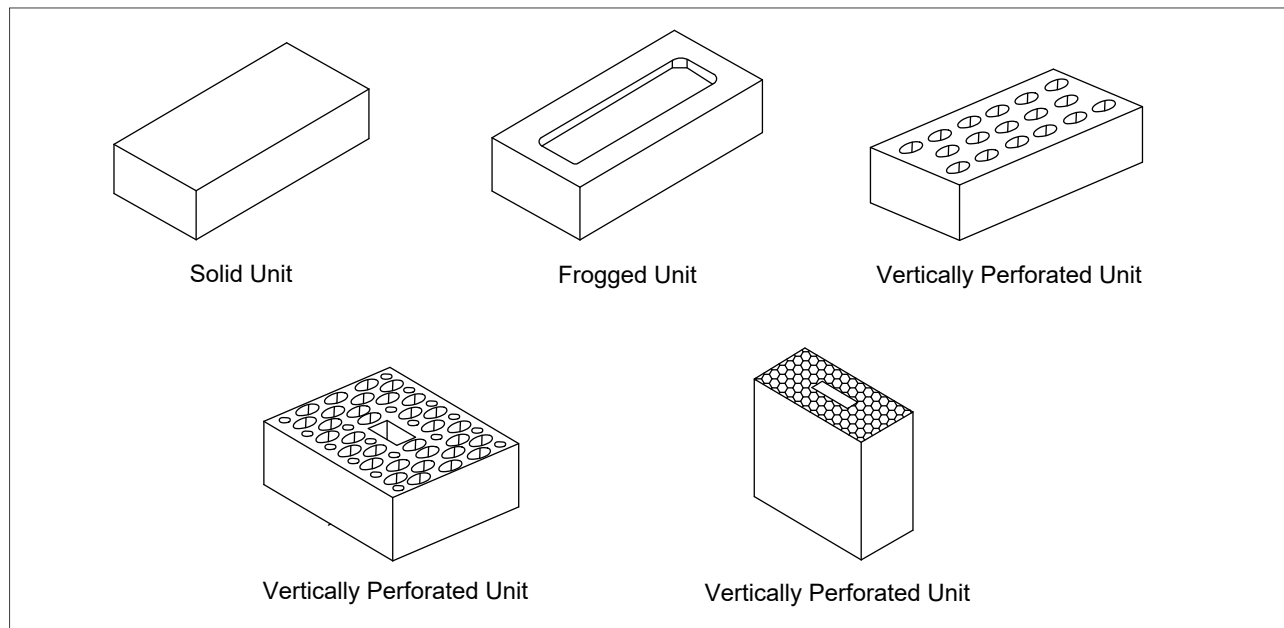
Category II: if the criteria for Category I is not complied.

The masonry units can be classified into four groups depending upon the percentage, size and orientation of voids or perforations.

Table 7.2 Geometrical Requirements for Grouping of Masonry Units

Parameter	Units	Group 1	Group 2	Group 3	Group 4
			Vertical holes		Horizontal holes
Volume of all holes as % of gross volume	Clay	≤ 25	$> 25; \leq 55$	$> 25; \leq 70$	$> 25; \leq 70$
	Concrete	≤ 25	$> 25; \leq 60$	$> 25; \leq 70$	$> 25; \leq 50$
Volume of any hole as % of gross volume	Clay	≤ 12.5	Each of multiple hole ≤ 2 Grip holes up to a total of 12.5	Each of multiple hole ≤ 2 Grip holes up to a total of 12.5	Each of multiple hole ≤ 30
	Concrete	≤ 12.5	Each of multiple hole ≤ 30 Grip holes up to a total of 30	Each of multiple hole ≤ 30 Grip holes up to a total of 30	Each of multiple hole ≤ 25
Thickness of web & shell (mm)	Clay	No requirement	Web ≥ 5 Shell ≥ 8	Web ≥ 3 Shell ≥ 6	Web ≥ 5 Shell ≥ 6
	Concrete	No requirement	Web ≥ 15 Shell ≥ 18	Web ≥ 15 Shell ≥ 15	Web ≥ 20 Shell ≥ 20
Combined thickness of webs and shells as % of overall width	Clay	No requirement	≥ 16	≥ 12	≥ 12
	Concrete	No requirement	≥ 18	≥ 15	≥ 45

Figure 7.1 Examples of Masonry Units



The compressive strength of masonry units, to be used in design, shall be the normalised mean compressive strength, f_b .

7.1.3.2 Mortar

Masonry mortars are considered as designed or prescribed mortars according to the method of defining their composition. The masonry mortars may be pre-mixed (factory made) or site-made, according to the method of manufacture.

Mortars should be classified according to their compressive strength, expressed as the letter M followed by the compressive strength in N/mm^2 , for example, $M5$. Prescribed masonry mortars, additionally to the M number, will be described by their prescribed constituents, e. g. 1: 1: 5 cement: lime: sand by volume.

7.1.3.3 Characteristic Compressive Strength of Masonry

The characteristic compressive strength of masonry, f_k , shall be determined from the results of tests on masonry. The compressive strength of masonry mainly depends on strength of the masonry units and compressive strength of mortar. Additional parameters that may affect the masonry strength are masonry bond, thickness of bed joints, number of bed joints over the height of wall, etc.

The characteristic compressive strength of masonry, may be calculated using following Expression:

$$f_k = K f_b^\alpha f_m^\beta \quad \text{in } N/mm^2$$

Equation 7.1

Where,

- f_b = Normalised mean compressive strength of a masonry unit, in N/mm^2 .
- f_m = Compressive strength of mortar, in N/mm^2 .
- K = A constant given in Table 7.3.

Table 7.3 Value of Constant *K*

Masonry Unit		Value of <i>K</i>	
		General purpose mortar	Thin layer mortar, bed joints 0.5mm to 3mm
Clay	Group 1	0.50	0.75
	Group 2	0.40	0.70
	Group 3	-	-
	Group 4	-	-
Dimensioned natural stone	Group 1	0.45	-*

* Natural stone unit not used with thin layer mortar

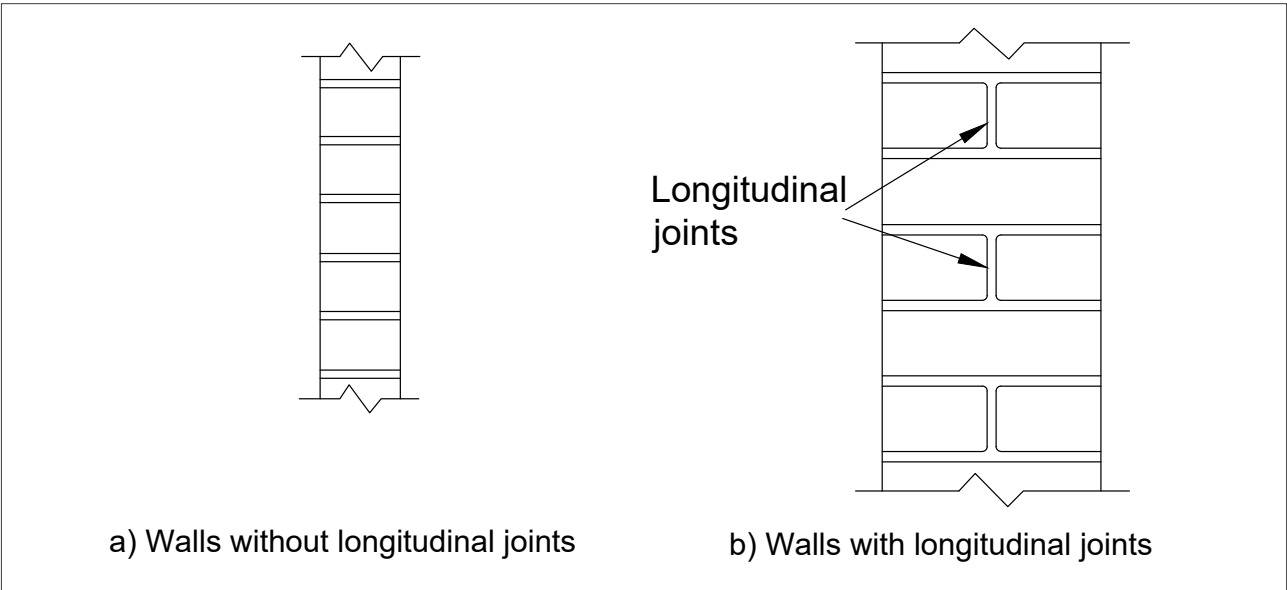
α & β are constants given as follows:

- For general purpose mortar, $\alpha = 0.7$ and $\beta = 0.3$
- For thin layer mortar (bed joints 0.5 to 3 mm),
 - using clay units of Group 1, $\alpha = 0.85$ and $\beta = 0$
 - using clay units of Group 2, $\alpha = 0.7$ and $\beta = 0$

The above Equation is applicable provided that following conditions are satisfied.

- Masonry with all joints to be considered as filled
- f_b is not taken to be greater than 110 N/mm² when units are laid in general purpose mortar and 50 N/mm² when units are laid in thin layer mortar.
- f_b is not taken to be greater than 50 N/mm² when units are laid in general purpose mortar.
- f_m is not taken to be greater than 12 N/mm² nor greater than $2 f_b$ when units are laid in general purpose mortar
- For masonry made with general purpose mortar and the thickness of the masonry is equal to the width or length of the unit, so that there is no mortar joint parallel to the face of the wall through all or any part of the length of the wall (refer Figure 7.2(a)); *K* shall be obtained by Table 7.3.
- For masonry made with general purpose mortar where there is a mortar joint parallel to the face of the wall through all or any part of the length of the wall (refer Figure 7.2(b)), the values of *K* can be obtained by multiplying the values given in Table 7.3 by 0.8.

Figure 7.2 Masonry Walls with and Without Longitudinal Joints



7.1.3.4 Characteristic Shear Strength of Masonry

The characteristic shear strength of masonry, f_{vk} , shall be determined from the results of tests on masonry. The characteristic initial shear strength of masonry, f_{vko} , should be determined from tests.

The characteristic shear strength of masonry, f_{vk} , using general purpose mortar or thin layer mortar in beds of thickness 0.5 mm to 3.0 mm, with all joints considered as filled, may be taken from following Equation.

$$F_{vk} = f_{vko} + 0.4 \sigma_d$$

Equation 7.2

But not greater than $0.065 f_b$

Where,

f_{vko} = The characteristic initial shear strength, under zero compressive stress

σ_d = The design compressive stress perpendicular to the shear

f_b = The normalised compressive strength of the masonry units,

The initial shear strength of the masonry, f_{vko} shall be determined from the database on the results of the tests on the initial shear strength of masonry. In the absence of such database, the values given in following table may be taken.

Table 7.4 Value of Initial Shear Strength of Masonry f_{vko}

Masonry Units		f_{vko} (in N/mm ²)	
		General purpose mortar of strength class	Thin layer mortar, bed joints 0.5mm to 3mm
Clay	M12	0.30	0.30
	M4 & M6	0.20	
	M2	0.10	
Dimensioned natural stone	M1 – M2	0.10	0.30

7.1.3.5 Design Strength of Masonry

The design strength of masonry is given as characteristic strength divided by appropriate partial safety factor.

In compression, $f_d = f_k / \gamma_M$

In shear, $f_{vd} = f_{vk} / \gamma_M$

7.1.4 Ultimate Limit State

The resistance of masonry walls, subjected mainly to vertical loading, shall be based on the geometry of the wall, the effect of the applied eccentricities and the material properties of the masonry. It shall be ensured that there is no tensile stress perpendicular to the bed joints.

At the ultimate limit state, the design value of the vertical load applied to a masonry wall, N_{Ed} , shall be less than or equal to the design value of the vertical resistance of wall, N_{Rd} ,

$$N_{Ed} \leq N_{Rd}$$

7.1.5 Serviceability Limit State

A masonry structure shall be designed and constructed so as not to exceed the Serviceability Limit State. The deflections/ settlements should be checked for verification of the serviceability limit state.

7.2 Timber Structures

7.2.1 General

7.2.1.1 Scope

This section gives general design guidelines for the structural parts of bridges made of timber or other wood-based materials, either singly or compositely with concrete, steel or other materials.

7.2.1.2 Reference Standards

Following reference documents may be referred while using this Manual.

EN 10138-1 Prestressing steels – Part 1: General requirements

EN 10138-4 Prestressing steels – Part 4: Bars

EN 912 Timber fasteners – specifications for connectors for timber

7.2.1.3 Definitions

For general definitions, refer initial sections of this Manual. Other definitions are as under.

Grooved connection	Shear connection consisting of the integral part of one member embedded in the contact face of the other member. The contacted parts are normally held together by mechanical fasteners.
Laminated deck plates	Deck plates made of laminations, arranged edgewise or flatwise, held together by mechanical fasteners or gluing.
Stress-laminated deck plates	Laminated deck plates made of edgewise arranged laminations with surfaces either sawn or planed, held together by pre-stressing.
Cross-laminated deck plates	Laminated deck plates made of laminations in layers of different grain direction (crosswise or at different angles). The layers are glued together or connected using mechanical fasteners.

7.2.1.4 Symbols

$E_{0,mean}$	Mean modulus of elasticity parallel to grain
$E_{90,mean}$	Mean modulus of elasticity perpendicular to grain
F	Force
$G_{0,mean}$	Mean shear modulus parallel to grain
$G_{90,mean}$	Mean shear modulus perpendicular to grain (rolling shear)
b_{ef}	Effective width
$b_{ef,c}$	Total Effective width of concrete slab
$b_{ef,1}; b_{ef,2}$	Effective width of concrete slab
b_{lam}	Width of the lamination
b_w	Width of the loaded area of the contact surface of deck plate
$b_{w,middle}$	Width of the loaded in the middle of the deck plate
d	Diameter; outer diameter of rod; distance
h	Depth of beam; thickness of plate
$f_{c,90,d}$	Design compressive strength perpendicular to grain
f_k	Characteristic strength
$f_{m,d,deck}$	Design bending strength of deck plate

1	$f_{v,d,deck}$	Design shear strength of deck plate
	$f_{m,d,lam}$	Design bending strength of laminations
2	$f_{v,d,lam}$	Design shear strength of laminations
	$k_{c,90}$	Factor for compressive strength perpendicular to grain
	k_{sys}	System strength factor
3	l_1	Distance
	n	Number of loaded laminations; number of pedestrians
4	t	Time; thickness of lamination
	β	Factor based on the damage consequence; angle of stress dispersion
	γ_M	Partial factor for timber material properties
5	$\gamma_{M,c}$	Partial factor for concrete material properties, also accounting for model uncertainties and dimensional variations
	$\gamma_{M,s}$	Partial factor for steel material properties, also accounting for model uncertainties and dimensional variations
6	$\gamma_{M,v}$	Partial factor for shear connectors, also accounting for model uncertainties and dimensional variations
7	$\gamma_{M,fat}$	Partial safety factor for fatigue verification of materials, also accounting for model uncertainties and dimensional variations
	μ_d	Design coefficient of friction
	$\sigma_{p,min}$	Minimum long-term residual compressive stress due to prestressing

7.2.2 Basis of Design

7.2.2.1 Principles of Limit State Design

7.2.2.1.1 General

The design models for the different limit states shall, as appropriate, take into account the following:

- Different material properties (e.g., strength and stiffness)
- Different time-dependent behaviour of the materials (duration of load, creep)
- Different climatic conditions (temperature, moisture variations)
- Different design situations (stages of construction, change of support conditions)

7.2.2.1.2 Ultimate Limit States

The analysis of structures shall be carried out using the following values for stiffness properties:

- For a first order linear elastic analysis of a structure, whose distribution of internal forces is not affected by the stiffness distribution within the structure (e.g. all members have the same time-dependent properties), mean values shall be used.
- For a first order linear elastic analysis of a structure, whose distribution of internal forces is affected by the stiffness distribution within the structure (e.g. composite members containing materials having different time-dependent properties), final mean values adjusted to the load component causing the largest stress in relation to strength shall be used.
- For a second order linear elastic analysis of a structure, design values, not adjusted for duration of load, shall be used.

The slip modulus of a connection for the ultimate limit state, K_u should be taken as:

$$K_u = \frac{2}{3} K_{ser}$$

Equation 7.3

Where, K_{ser} is the slip modulus, refer Table 7.5.

ρ_m is the mean density of wood member in kg/m³ and d or d_c connector diameter in mm.

Table 7.5 Values of K_{ser} for Fasteners and Connectors in N/mm In Timber to Timber and Wood-Based Panel to Timber Connections

Fastener Type		K_{ser}
Dowels		$\rho_m^{1.5} d / 23$
Bolts with or without clearance ^a		
Screws		
Nails (with pre-drilling)		
Nails (without pre-drilling)		$\rho_m^{1.5} d^{0.8} / 30$
Staples		$\rho_m^{1.5} d^{0.8} / 80$
Split-ring connectors type A according to EN 912		$\rho_m d_c / 2$
Shear-plate connectors type B according to EN 912		
Toothed-plate connectors:	Connectors type C1 to C9 according to EN 912	$1.5 \rho_m d_c / 4$
	Connectors type C10 and C11 according to EN 912	$\rho_m d_c / 2$

^a The clearance should be added separately to the deformation.

7.2.2.1.3 Serviceability Limit States

The deformation of a structure which results from the effects of actions (such as axial and shear forces, bending moments and joint slip) and from moisture shall remain within appropriate limits, having regard to the possibility of damage to surfacing materials, ceilings, floors, partitions and finishes, and to the functional needs as well as any appearance requirements.

The instantaneous deformation, u_{inst} , (refer Figure 7.3), should be calculated for the characteristic combination of actions, using mean values of the appropriate moduli of elasticity, shear moduli and slip moduli.

The final deformation, u_{fin} , should be calculated for the quasi-permanent combination of actions.

If the structure consists of members or components having different creep behaviour, the final deformation should be calculated using final mean values of the appropriate moduli of elasticity, shear moduli and slip moduli.

For structures consisting of members, components and connections with the same creep behaviour and under the assumption of a linear relationship between the actions and the corresponding deformations, the final deformation, u_{fin} may be taken as:

$$u_{fin} = u_{fin,G} + u_{fin,Q_1} + \sum u_{fin,Q_i} \quad \text{Equation 7.4}$$

Where,

$$u_{fin,G} = u_{inst,G} (1 + k_{def}) \text{ for a permanent action, } G \quad \text{Equation 7.5}$$

$$u_{fin,Q_1} = u_{inst,Q_1} (1 + \psi_{2,1} k_{def}) \text{ for the leading variable action, } Q_1 \quad \text{Equation 7.6}$$

$$u_{fin,Q_i} = u_{inst,Q_i} (\psi_{0,i} + \psi_{2,i} k_{def}) \text{ for accompanying variable actions, } Q_i (i > 1) \quad \text{Equation 7.7}$$

$u_{inst,G}$, u_{inst,Q_1} , u_{inst,Q_i} are the instantaneous deformations for action G , Q_1 , Q_i respectively.

$\psi_{2,l}$, $\psi_{2,i}$ are the factors for the quasi-permanent value of variable actions.

$\psi_{0,i}$ are the factors for combination value of variable actions.

k_{def} is deformation factor given in Table 7.6 for timber and wood-based materials

For serviceability limit states with respect to vibrations, mean values of the appropriate stiffness moduli should be used.

Following notations are used in Figure 7.3.

w_c = Precamber, if applied.

w_{inst} = Instantaneous deflection due to permanent and variable actions.

w_{creep} = Creep deflection due to permanent and variable actions.

w_{fin} = Final deflection due to permanent and variable actions = $w_{inst} + w_{creep}$.

$w_{net,fin}$ = Net final deflection due to permanent and variable actions = $w_{fin} - w_c$.

Figure 7.3 Components of Deflection

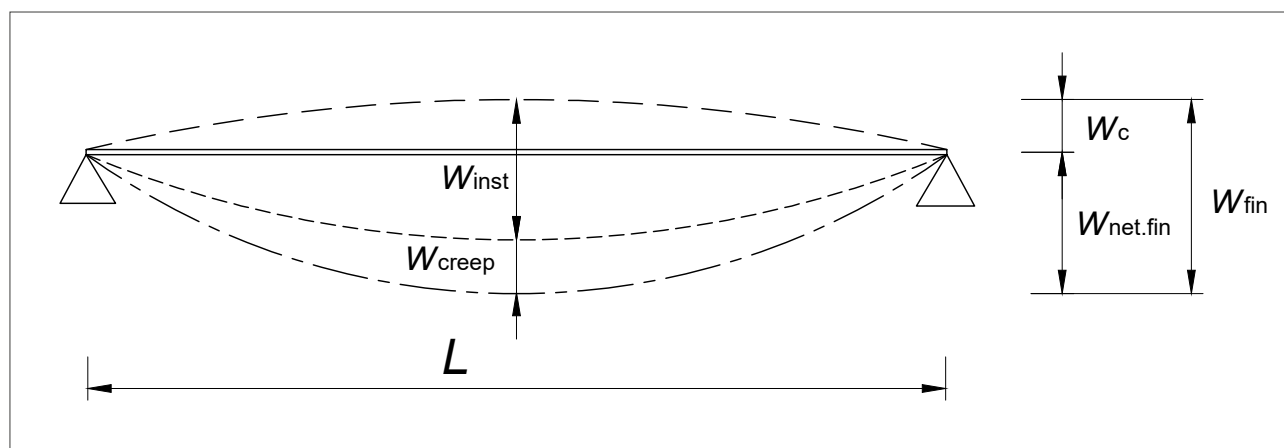


Table 7.6 Values of K_{def} for Timber and Wood-Based Materials

Material	Standard	Service Class*		
		1	2	3
Solid timber	EN 14081-1	0.60	0.80	2.00
Glued laminated timber	EN 14374, EN 14279	0.60	0.80	2.00

*The service class shall be as defined in Table 7.8.

7.2.2.2 Basic Variables

The main variables are the actions, the material properties and the geometrical data.

7.2.2.2.1 General

Actions to be used in design of bridges may be obtained from Chapter 3.

7.2.2.2.2 Load Duration Classes

Duration of load and moisture content affect the strength and stiffness properties of timber elements and shall be taken into account in the design for mechanical resistance and serviceability.

Variable actions due to the passage of vehicular and pedestrian traffic should be regarded as short-term actions.

Initial pre-stressing forces perpendicular to the grain should be regarded as short-term actions.

Actions occurring during erection should be regarded as short-term actions.

Actions shall be assigned to one of the load-duration classes given in following for strength and stiffness calculations.

Table 7.7 Load-duration Classes

Load-duration Class	Order of Accumulated Duration of Characteristic Load	Examples of Loading
Permanent	More than 10 years	Self-weight
Long-term	6 Months to 10 years	Storage loading
Medium-term	1 Week to 6 months	Imposed floor loading
Short-term	Less than one week	Maintenance or man loading on roof
Instantaneous	–	Wind, impact loading, accidental load

7.2.2.2.3 Service Classes

Service class is characterised by a moisture content in the materials corresponding to a temperature and relative humidity of the surrounding air. The structures shall be assigned to one of the service classes given below:

Table 7.8 Service Class

Service Class	Service Condition	Moisture Content in Wood
1	Temperature 20°C Relative humidity of surround air exceeding 65% for a few weeks per year	≤ 12%
2	Temperature 20°C Relative humidity of surround air exceeding 85% for a few weeks per year	≤ 20%
3	Climatic condition leading to a higher moisture condition than in service class 2	> 20%

7.2.2.3 Verification by the Partial Factor Method

7.2.2.3.1 Design Value of Material Property

The design value R_d of a resistance (load carrying capacity) shall be calculated as:

$$R_d = k_{mod} \frac{R_k}{\gamma_M}$$

Where,

R_k = The characteristic value of the load carrying capacity.

γ_M = The partial factor for a material property.

k_{mod} = The modification factor taking into account the duration of the load and moisture content.

Table 7.9 Value of Modification Factor k_{mod}

Material	Standard	Service Class	Load Duration Class				
			Permanent Actions	Long Term Actions	Medium Term Actions	Short Term Actions	Instantaneous Action
Solid timber	EN 14081-1	1	0.60	0.70	0.80	0.90	1.10
		2	0.60	0.70	0.80	0.90	1.10
		3	0.50	0.55	0.65	0.70	0.90
Glued laminate timber	EN 14080	1	0.60	0.70	0.80	0.90	1.10
		2	0.60	0.70	0.80	0.90	1.10
		3	0.50	0.55	0.65	0.70	0.90

For fundamental combinations, the recommended partial factors (γ_M) for material properties, are given in Table 7.10. For accidental combinations, the recommended value of partial factor is 1.0.

Table 7.10 Recommended Partial Factors γ_M for Material Properties

Material Property	Symbol	Value
1. Fundamental combinations		
• Solid timber – untreated	γ_M	1.3
• Solid timber – preservatively treated	γ_M	1.3
• Glued laminated timber	γ_M	1.25
• LVL, plywood, OSB	γ_M	1.2
• Connections	γ_M	1.3
• Shear connectors between timber and concrete in composite members	$\gamma_{M,v}$	1.25
• Steel used in composite members	$\gamma_{M,s}$	1.15
• Prestressing steel elements	$\gamma_{M,s}$	1.15
• Concrete used in composite members	$\gamma_{M,c}$	1.5
2. Fatigue combinations		
• Timber and wood-based materials	$\gamma_{M,fat}$	1.0
• Connections	$\gamma_{M,fat}$	1.0
• Shear connectors between timber and concrete in composite members	$\gamma_{M,v,fat}$	1.0
3. Accidental combinations		
• All Materials	$\gamma_{M,acc}$	1.0

7.2.3 Material Properties

Strength and stiffness parameters shall be determined on the basis of tests for the types of action effects to which the material will be subjected in the structure, or on the basis of comparisons with similar timber species and grades or wood-based materials, or on well-established relations between the different properties.

The characteristic values are determined on the assumption of a linear relation between stress and strain until failure. The strength verification of individual members shall also be based on such a linear relation.

Strength modification factors for service classes and load duration classes, deformation modification factors for service classes shall be applied to determine the design values of material properties.

Following standards shall be referred for material properties.

- Solid timber members shall comply with EN 14081-1.
- Strength class for timber are given in EN 338.
- Glued laminated timber members shall comply with EN 14080.
- Glued laminated timber, strength and stiffness properties given in EN 1194.
- Metal fasteners and connections shall comply with EN 14592, EN 14545.
- Pre-Stressing steels shall comply with EN10138-1 & EN10138-4.

7.2.4 Durability

7.2.4.1 Timber

The effect of precipitation, wind and solar radiation should be taken into account.

The effect of direct weathering by precipitation or solar radiation of structural timber members can be reduced by constructional preservation measures, or by using timber with sufficient natural durability, or timber treated with preservative against biological attacks.

- Where a partial or complete covering of the main structural elements is not practical, durability can be improved by one or more of the following measures.
- Limiting standing water on timber surfaces through appropriate inclination of surfaces; limiting openings, slots, etc., where water may accumulate or infiltrate.
- Limiting direct absorption of water (e.g. capillary absorption from concrete foundation) through use of appropriate barriers.
- Limiting fissures and delamination, especially at locations where the end grain would be exposed, by appropriate sealing or cover plates.
- Limiting swelling and shrinking movements by ensuring an appropriate initial moisture content and by reducing in-service moisture changes through adequate surface protection.
- Choosing a geometry for the structure that ensures natural ventilation of all timber parts.

The risk of increased moisture content near the ground, e.g. due to insufficient ventilation due to vegetation between the timber and the ground, or splashing water, can be reduced by one or more of the following measures:

- Covering of the ground by coarse gravel or similar to limit vegetation.
- Use of an increased distance between the timber parts and the ground level.

Where structural timber members are exposed to abrasion by traffic, the depth used in the design shall be the minimum permitted before replacement.

7.2.4.2 Resistance to Corrosion

Metal fasteners and other structural connections shall, where necessary, either be inherently corrosion-resistant or be protected against corrosion. The possibility of stress corrosion shall be taken into account. Steel parts encased in concrete, such as reinforcing and pre-stressing steel, should be protected by providing adequate cover in accordance with Chapter 4 of this Manual. The effect of chemical treatment of timber, or timber with high acidic content, on the corrosion protection of fasteners should be taken into account.

7.2.4.3 Protection of Timber Decks from Water by Sealing

The elasticity of the seal layers shall be sufficient to follow the movement of the timber deck.

7.2.5 Basis of Structural Analysis

7.2.5.1 Laminated Deck Plates

7.2.5.1.1 General

The analysis of laminated timber deck plates should be based upon one of the following:

- The orthotropic plate theory.
- Modelling the deck plate by a grid.
- A simplified method as given in Clause 7.2.5.1.3.

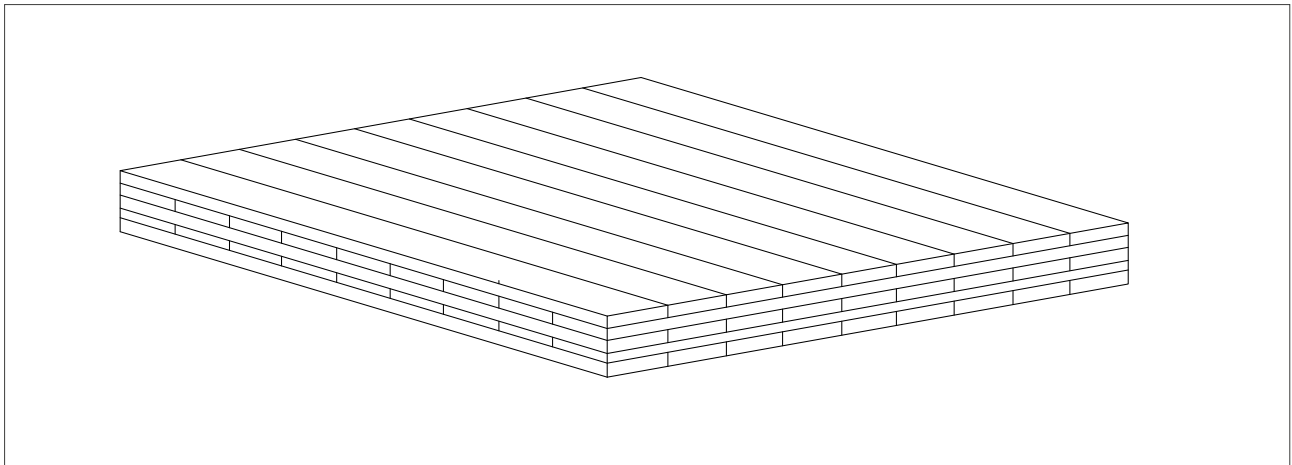
In an advanced analysis, for deck plates made of softwood laminations, the relationships for the system properties should be taken from Table 7.11. The Poisson ratio ν may be taken as zero.

Table 7.11 System Properties of Laminated Deck Plates

Type of Deck Plate		$E_{90, \text{mean}} / E_{0, \text{mean}}$	$G_{0, \text{mean}} / E_{0, \text{mean}}$	$G_{90, \text{mean}} / G_{0, \text{mean}}$
Nail-laminated		0	0.06	0.05
Stress-laminated	Sawn	0.015	0.06	0.08
	Planed	0.020	0.06	0.10
Glued-laminated		0.030	0.06	0.15

For cross-laminated deck plates, refer Figure 7.4, shear deformations should be taken into account.

Figure 7.4 Example of Cross-laminated Deck Plate



7.2.5.1.2 Concentrated Vertical Loads

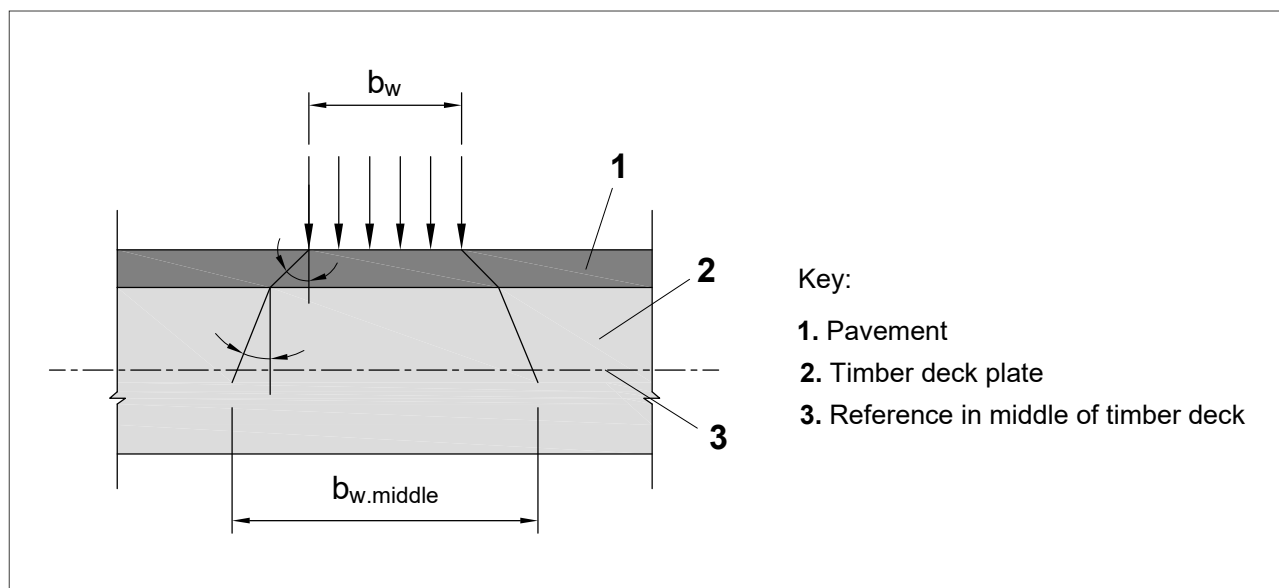
Loads should be considered at a reference plane in the middle of the deck plate. For concentrated loads an effective load area with respect to the middle plane of the deck plate should be assumed, refer Figure 7.5.

Where,

b_w = The width of the loaded area on the contact surface of the pavement.

$b_{w, \text{middle}}$ = The width of the loaded area at the reference plane in the middle of the deck plate.

β = The angle of dispersion according to Table 7.12.

Figure 7.5 Dispersion of Concentrated Loads From Contact Area Width b_w Table 7.12 Dispersion Angle β of Concentrated Loads for Various Materials

Material		Dispersion Angle
Pavement		45°
Boards and planks		45°
Laminated timber deck plates	In the direction of the grain	45°
	Perpendicular to the grain	15°
Plywood and cross laminated deck plates		45°

7.2.5.1.3 Simplified Analysis

The deck plate may be replaced by one or several beams in the direction of the laminations with the effective width b_{ef} calculated as

$$b_{ef} = b_{w,middle} + a$$

Where,

$b_{w,middle}$ should be calculated according to Clause 7.2.5.1.2.

a should be taken from Table 7.13.

Table 7.13 Width a in m for Determination of Effective Width of Beam

Deck Plate System	a m
Nail-laminated deck plate	0.1
Stress-laminated or glue laminated	0.3
Cross-laminated timber	0.5
Composite concrete/timber deck structure	0.6

7.2.5.2 Composite Members

For composite action of deck plate systems, the influence of joint slip shall be taken into account.

7.2.5.3 Timber-concrete Composite Members

The steel fasteners and the grooved connections should be designed to transmit all forces due to composite action. Friction and adhesion between wood and concrete should not be taken into account.

The effective width of the concrete plate of composite timber beam/concrete deck structures should be determined as:

$$b_{ef,c} = b + b_{ef,1} + b_{ef,2}$$

Equation 7.8

Where,

b = The width of the timber beam

$b_{ef,1}$, $b_{ef,2}$ = Are the effective widths of the concrete flanges, as determined for a concrete T-section according to Chapter 4.

For verification at ultimate limit state, cracks in the concrete plate shall be taken into account.

The effect of concrete tension stiffening may be included. As a simple approach the stiffness of the cracked part of the concrete cross-section may be taken as 40 % of the stiffness in uncracked condition. In such areas the need for an adequate crack distributing reinforcement should be observed.

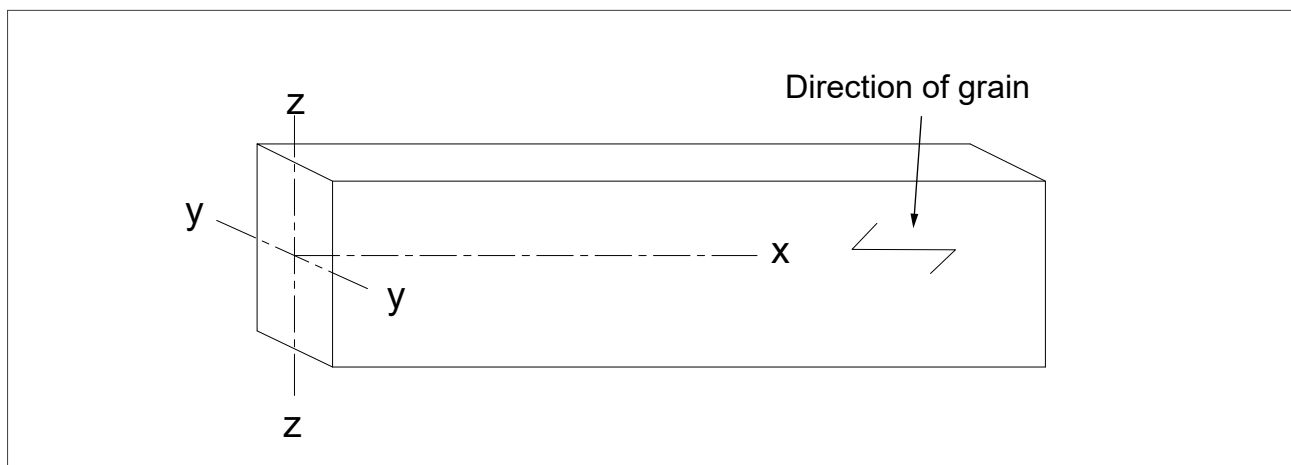
7.2.6 Ultimate Limit States

Timber structures are analysed using elastic structural analysis. This is quite appropriate for the serviceability limit state and the ultimate limit state, which models the failure of structural element under an extreme loading condition, can also be reasonably modelled using an elastic analysis.

7.2.6.1 Design of Cross-sections Subjected to Stress in One Principal Direction

This section deals with the design of members subjected to stress in one principal direction and applies to straight solid timber, glued laminated timber or wood-based structural products of constant cross-section, whose grain runs essentially parallel to the length of the member.

Figure 7.6 Member Axes



7.2.6.1.1 Tension Parallel to the Grain

For members subjected to tension parallel to the grain, following Expression shall be satisfied.

$$\sigma_{t,0,d} \leq f_{t,0,d}$$

Equation 7.9

Where,

$\sigma_{t,0,d}$ = The design tensile stress along the grain.

$f_{t,0,d}$ = The design tensile strength along the grain.

7.2.6.1.2 Compression Parallel to the Grain

For members subjected to compression parallel to the grain, following Equation shall be satisfied.

$$\sigma_{c,0,d} \leq f_{c,0,d}$$

Equation 7.10

Where,

$\sigma_{c,0,d}$ = The design compressive stress along the grain.

$f_{c,0,d}$ = The design compressive strength along the grain.

7.2.6.1.2 Compression Perpendicular to the Grain

For members subjected to compression perpendicular to the grain, following Equation shall be satisfied.

$$\sigma_{c,90,d} \leq k_{c,90} f_{c,90,d}$$

Equation 7.11

With,

$$\sigma_{c,90,d} = \frac{F_{c,90,d}}{A_{ef}}$$

Equation 7.12

Where,

$\sigma_{c,90,d}$ = The design compressive stress in the effective contact area perpendicular to the grain.

$F_{c,90,d}$ = The design compressive load perpendicular to the grain.

A_{ef} = The effective contact area in compression perpendicular to the grain.

$f_{c,90,d}$ = The design compressive strength perpendicular to the grain.

$k_{c,90}$ = A factor taking into account the load configuration, possibility of splitting and degree of compressive deformation.

The effective contact area perpendicular to the grain, A_{ef} , should be determined taking into account an effective contact length parallel to the grain, where the actual contact length, l , at each side is increased by 30 mm, but not more than a , l or $l/2$, refer Figure 7.7.

The value of $k_{c,90}$ should be taken as 1.0 unless the following member support conditions exists. In these cases, the higher value of $k_{c,90}$ specified may be taken, with a limiting value of $k_{c,90} = 1.75$.

- For a beam member resting on continuous supports provided that $l/1 \geq 2h$, refer Figure 7.7(a), the value of $k_{c,90}$ should be taken as:

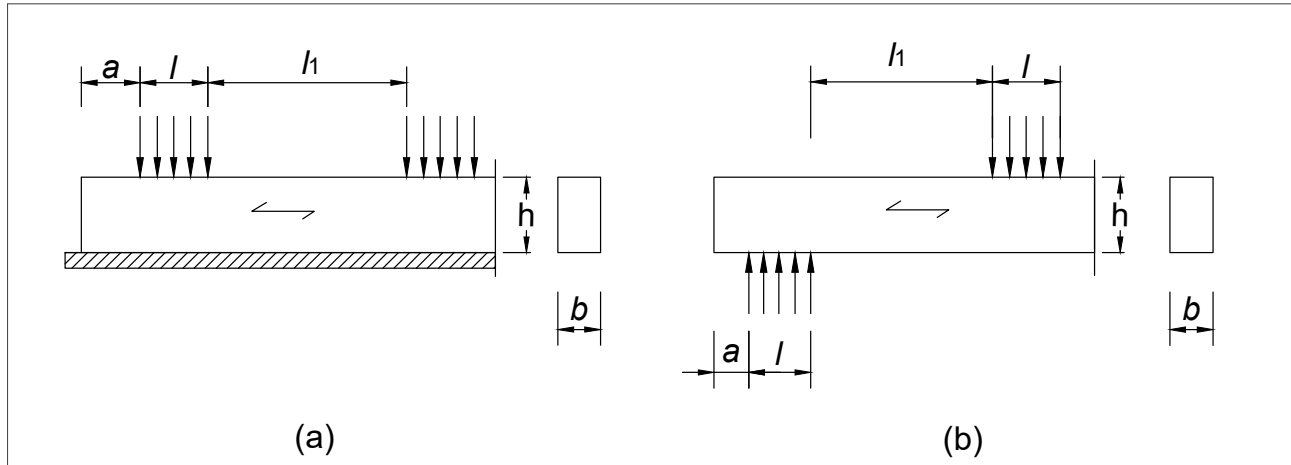
$k_{c,90} = 1.25$ for solid softwood timber

$k_{c,90} = 1.50$ for glued laminated softwood timber

Where h is the depth of the member and l is the contact length.

- For beam members on discrete supports, provided that $l/l \geq 2h$, refer Figure 7.7(b), the value of $k_{c,90}$ be taken as:
 $k_{c,90} = 1.50$ for solid softwood timber.
 $k_{c,90} = 1.75$ for glued laminated softwood timber, provided that $l \leq 400$ mm.

Figure 7.7 Member on Continuous and Discrete Supports



7.2.6.1.4 Members in Bending

For members subjected to bending, following Equation shall be satisfied.

$$\frac{\sigma_{m,y,d}}{f_{m,y,d}} + k_m \frac{\sigma_{m,z,d}}{f_{m,z,d}} \leq 1 \quad \text{Equation 7.13}$$

$$k_m \frac{\sigma_{m,y,d}}{f_{m,y,d}} + \frac{\sigma_{m,z,d}}{f_{m,z,d}} \leq 1 \quad \text{Equation 7.14}$$

Where,

$\sigma_{m,y,d}$ and $\sigma_{m,z,d}$ are the design bending stresses about the principal y-axis and z-axis.
 $f_{m,y,d}$ and $f_{m,z,d}$ are the corresponding design bending strengths.

The factor k_m makes allowance for re-distribution of stresses and the effect of inhomogeneities of the material in a cross-section.

The value of the factor should be taken as follows:

- For solid timber, glued laminated timber and *LVL*:
for rectangular or square sections: $k_m = 0.7$
for other cross-sections: $k_m = 1.0$
- For other wood-based structural products, for all cross-sections: $k_m = 1.0$

For members with rectangular cross-sections:

$$\sigma_{m,y,d} = \frac{M_{y,d}}{W_y} = \frac{M_{y,d}}{b h^2/6}$$

$$\sigma_{m,z,d} = \frac{M_{z,d}}{W_z} = \frac{M_{z,d}}{h b^2/6}$$

Where,

$M_{y,d}$ and $M_{z,d}$ are design bending moments about y-y and z-z axis.

W_y and W_z are the values of elastic modulus of section about y-y and z-z axis.

b is the width of section and h is the depth of section.

7.2.6.1.5 Members in Shear

Shear stresses will be developed due to transverse loading. Shear stresses transverse to the beam axis will always be accompanied by shear stresses parallel to the beam axis due to bending.

For members in shear with a stress component parallel to the grain, as in Figure 7.8(a), as well as for shear with both stress components perpendicular to the grain, as in Figure 7.8(b), the following Equation shall be satisfied:

$$\tau_d \leq f_{v,d}$$

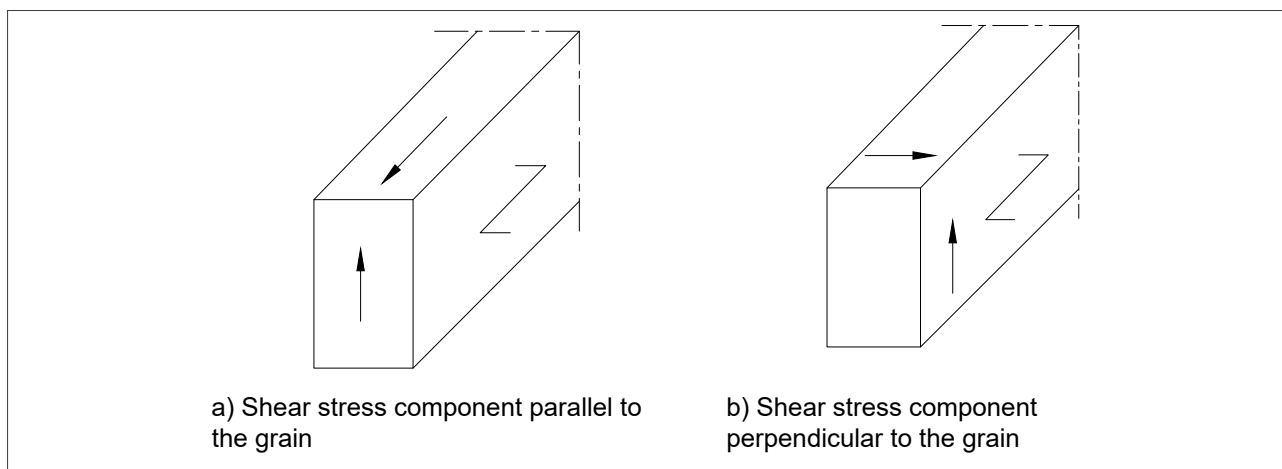
Equation 7.15

Where,

τ_d = The design shear stress.

$f_{v,d}$ = The design shear strength for the actual condition.

Figure 7.8 Members with Shear Stress



For the verification of shear resistance of members in bending, the influence of cracks should be taken into account using an effective width of the member given as:

$$b_{ef} = k_{cr} b$$

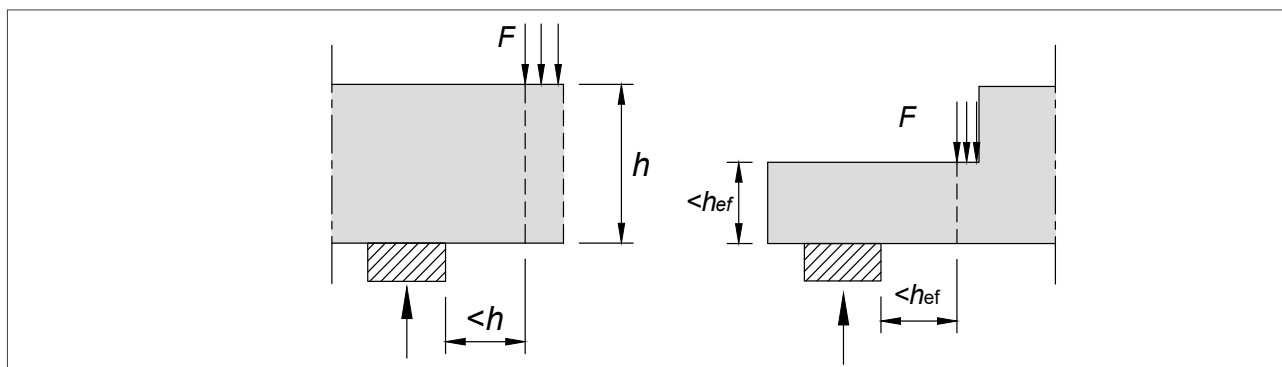
Equation 7.16

Where; b is the width of the relevant section of the member.

The value for k_{cr} may be taken as 0.67 for solid timber and glued laminated timber and 1.0 for other wood products.

At supports, the contribution to the total shear force of a concentrated load F acting on the top side of the beam and within a distance h or h_{ef} from the edge of the support may be disregarded (refer Figure 7.9). For beams with a notch at the support this reduction in the shear force applies only when the notch is on the opposite side to the support.

Figure 7.9 Conditions at Support for Shear Force Due to Concentrated Force



7.2.6.1.6 Members in Torsion

Torsional stresses are introduced when the applied load tends to twist a member. This will occur when a beam supports a load which is applied eccentric to the principal cross sectional axis. For members subjected to torsion, following Expression shall be satisfied

$$\tau_{tor,d} \leq k_{shape} f_{v,d}$$

Equation 7.17

With,

k_{shape} = 1.2 for a circular cross section

k_{shape} = Minimum of $\{(1.0+0.15 h/b) \text{ and } 2.0\}$ for a rectangular cross section

Where,

$\tau_{tor,d}$ = The design torsional stress

$f_{v,d}$ = The design shear strength

k_{shape} = A factor depending on the shape of the cross-section

h = The larger cross-sectional dimension

b = The smaller cross-sectional dimension

7.2.6.2 Design of Cross-sections Subjected to Combined Stresses

This section applies to straight solid timber, glued laminated timber or wood-based structural products of constant cross-section, whose grain runs essentially parallel to the length of the member. The member is assumed to be subjected to stresses from combined actions or to stresses acting in two or three of its principal axes

7.2.6.2.1 Combined Bending and Axial Tension

For members with combined bending and axial tension, following Equation shall be satisfied.

$$\frac{\sigma_{t,0,d}}{f_{t,0,d}} + \frac{\sigma_{m,y,d}}{f_{m,y,d}} + k_m \frac{\sigma_{m,z,d}}{f_{m,z,d}} \leq 1$$

Equation 7.18

$$\frac{\sigma_{t,0,d}}{f_{t,0,d}} + k_m \frac{\sigma_{m,y,d}}{f_{m,y,d}} + \frac{\sigma_{m,z,d}}{f_{m,z,d}} \leq 1$$

Equation 7.19

The values of k_m shall be taken as given earlier.

7.2.6.2.2 Combined Bending and Axial Compression

For members with combined bending and axial compression, following Equation shall be satisfied.

$$\left(\frac{\sigma_{c,0,d}}{f_{c,0,d}} \right)^2 + \frac{\sigma_{m,y,d}}{f_{m,y,d}} + k_m \frac{\sigma_{m,z,d}}{f_{m,z,d}} \leq 1$$

Equation 7.20

$$\left(\frac{\sigma_{c,0,d}}{f_{c,0,d}} \right)^2 + k_m \frac{\sigma_{m,y,d}}{f_{m,y,d}} + \frac{\sigma_{m,z,d}}{f_{m,z,d}} \leq 1$$

Equation 7.21

The values of k_m shall be taken as given earlier.

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Masonry and Timber Structures

7.2.6.3 Deck Plates

7.2.6.3.1 System Strength

When several equally spaced similar members, components or assemblies are laterally connected by a continuous load distribution system, the member strength properties may be multiplied by a system strength factor k_{sys} .

The design bending and shear strength of the deck plate should be calculated as:

$$f_{m,d,deck} = k_{sys} f_{m,d,lam}$$

Equation 7.22

$$f_{v,d,deck} = k_{ys} f_{v,d,lam}$$

Equation 7.23

Where,

$f_{m,d,lam}$ = The design bending strength of the laminations.

$f_{v,d,lam}$ = The design shear strength of the laminations.

k_{sys} = The system strength factor.

For the calculation of k_{sys} , the number of loaded laminations should be taken as:

$$n = \frac{b_{ef}}{b_{lam}}$$

Equation 7.24

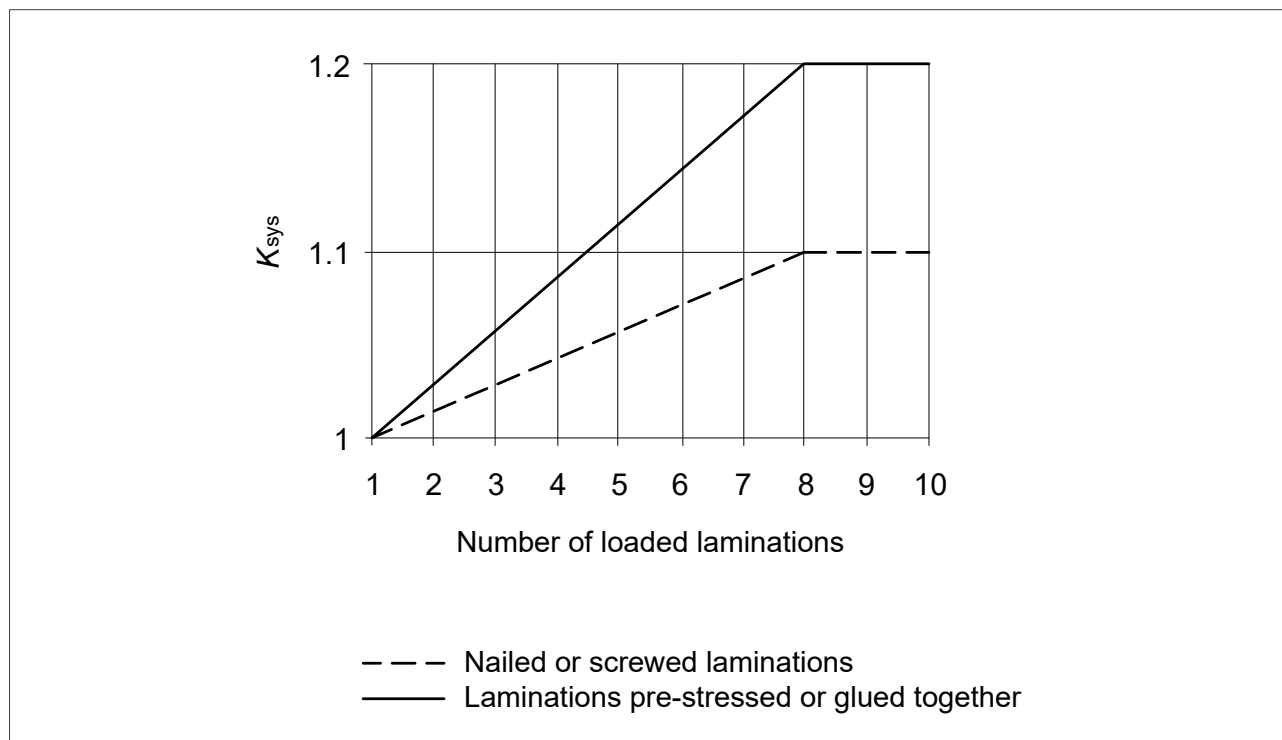
With,

b_{ef} is the effective width

b_{lam} is the width of the laminations.

The system strength factor k_{sys} as given in following figure shall be used.

Figure 7.10 System Strength factor k_{sys} for Laminated Deck Plates



7.2.6.3.2 Stress-laminated Deck Plates

The long-term pre-stressing forces shall be such that no inter-laminar slip occurs. The following requirement should be satisfied:

$$F_{v,Ed} \leq \mu_d \sigma_{p,min} h$$

Equation 7.25

Where,

$F_{v,Ed}$ = The design shear force per unit length, caused by vertical and horizontal actions.

μ_d = The design value of coefficient of friction.

$\sigma_{p,min}$ = The minimum long-term residual compressive stress due to pre-stressing.

h = The thickness of the plate.

The coefficient of friction should take into account the wood species; roughness of contact surface; treatment of the timber; residual stress level between laminations.

Unless other values have been verified, the design static friction coefficients, μ_d between softwood timber laminations, and between softwood timber laminations and concrete, should be taken from Table 7.14. For moisture contents between 12 and 16 %, the values may be obtained by linear interpolation.

In areas subjected to concentrated loads, the minimum long-term residual compressive stress, $\sigma_{p,min}$, due to pre-stressing between laminations should be not less than 0.35 N/mm^2 .

The long-term residual pre-stressing stress may normally be assumed to be greater than 0.35 N/mm_2 , provided that:

- The initial pre-stress is at least 1.0 N/mm^2
- The moisture content of the laminations at the time of pre-stressing is not more than 16%
- The variation of in-service moisture content in the deck plate is limited by adequate protection, e.g., a sealing layer

Table 7.14 Design Values of Coefficient of Friction μ_d

Lamination Surface Roughness	Perpendicular to Grain		Parallel to Grain	
	Moisture Content $\leq 12 \%$	Moisture Content $\geq 16 \%$	Moisture Content $\leq 12 \%$	Moisture Content $\geq 16 \%$
Sawn timber to sawn timber	0.30	0.45	0.23	0.35
Planed timber to planed timber	0.20	0.40	0.17	0.30
Sawn timber to planed timber	0.30	0.45	0.23	0.35
Timber to concrete	0.40	0.40	0.40	0.40

The resulting pre-stressing forces should act centrally on the timber cross-section. The compressive stress perpendicular to the grain during pre-stressing in the contact area of the anchorage plate shall be verified. The factor $k_{c,90}$ may be taken as 1.3.

Not more than one butt joint should occur in any four adjacent laminations within a distance l_1 given as

$$l_1 = \min \begin{cases} 2 d \\ 30 t \\ 1.2 m \end{cases}$$

Equation 7.26

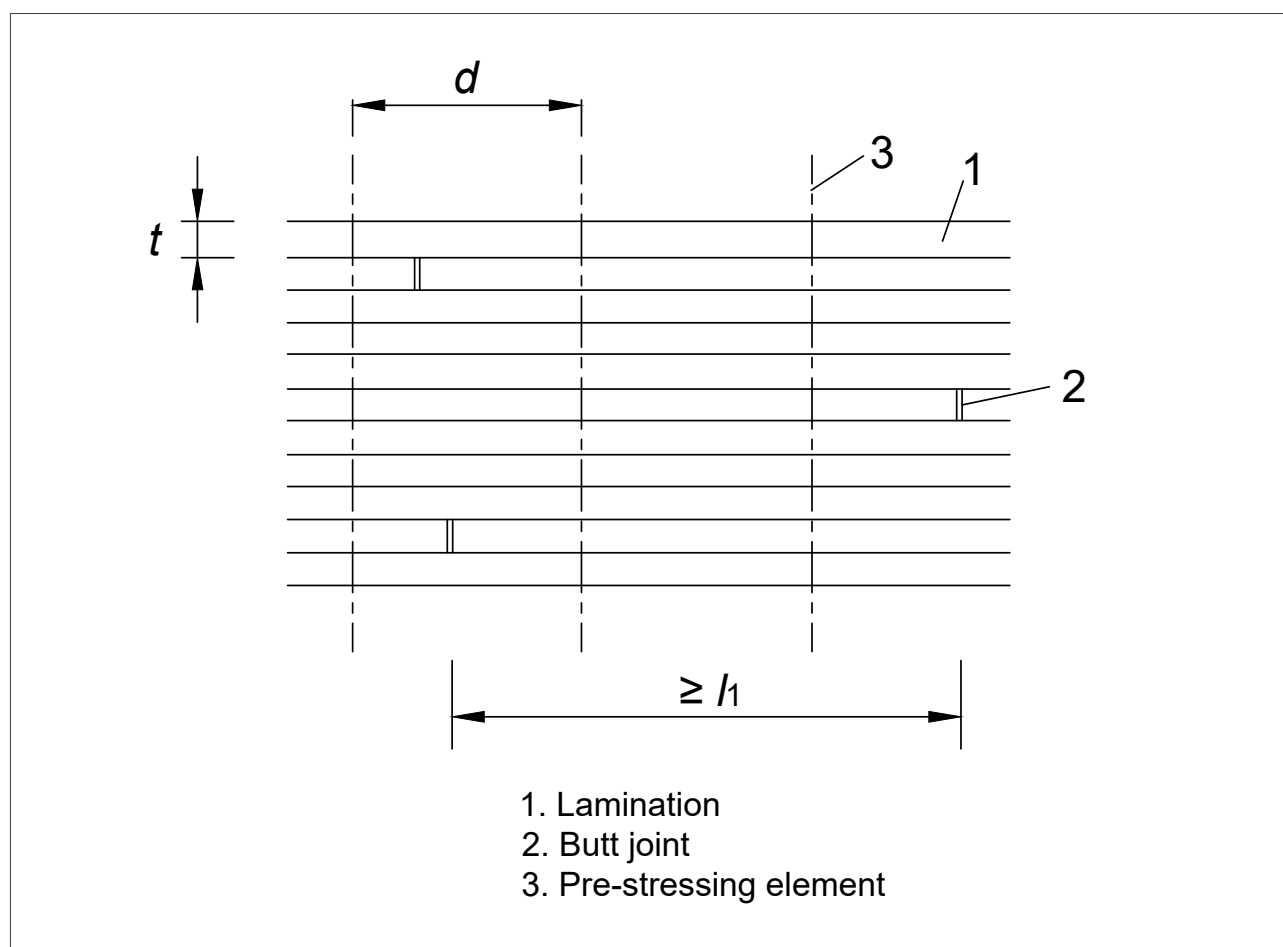
Where,

d = The distance between the pre-stressing elements.

t = The thickness of the laminations in the direction of pre-stressing.

In calculating the longitudinal strength of stress-laminated deck plates, the section should be reduced in proportion to the number of butt joints within a distance of 4 times the thickness of laminations in the direction of pre-stressing.

Figure 7.11 Butt Joints in Stress-laminated Deck Plates



7.2.7 Serviceability Limit States

7.2.7.1 General

In the calculations, mean values of density should be used.

7.2.7.2 Limiting Values for Deflections

The limiting values for deflections due to traffic load only, for beams and trusses with span l given in Table 7.15.

Table 7.15 Limiting Values for Deflections for Beams and Trusses

Action	Limiting Values
Characteristic traffic load	$l / 400$
Pedestrian load and low traffic load	$l / 200$

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Masonry and Timber Structures

8 Expansion Joints and Bearings

This Chapter deals in expansion joints and bearings for bridges.

8.1 Expansion Joints

8.1.1 General

Expansion joints in bridges are usually provided to allow for thermal expansion and contraction of the bridge deck, and to also allow for movement due to traffic actions on the bridges. They shall provide a level and skid resistant surface capable of supporting traffic loads and managing the drainage of deck surface water to avoid leakage through the joint. Expansion joints can be installed at the transverse joints between the end of the bridge deck and abutment ballast wall or between adjacent decks in non-continuous multi-span bridges

8.1.2 Scope

The scope of this section covers the requirements and characteristics of different types of expansion joints for road bridge.

8.1.3 Design Principles

Expansion joints shall be designed in accordance with the requirements provided in this Manual. Expansion joints shall not cause hazard to road users. Expansion joints in highway bridge decks shall be capable to accommodate design movements without damage to the surfacing or supporting structures during service.

The same joint system, seal or sealant shall continue across the full width of the deck including walkway, verge, hard shoulder and central reserve. The expansion joint shall have the ease of replacement. The expansion joint shall not permit water to enter the structure, nor permit it to accumulate within the joint or structure.

8.1.4 Selection of Joint Type

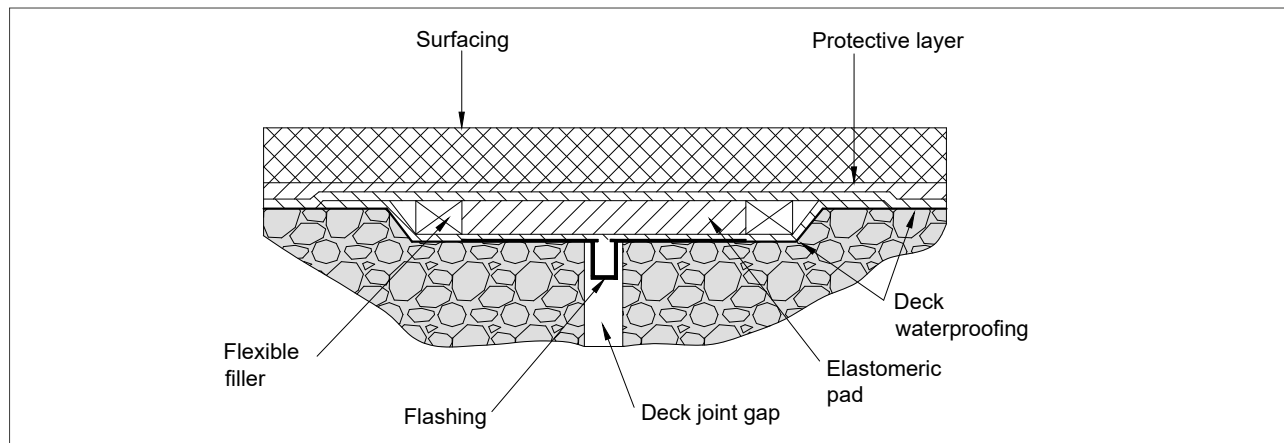
Joints shall be selected such that it will accommodate all the horizontal, vertical and rotational movement likely to be encountered in service. Following are different category of expansion joints:

- Buried expansion joints
- Flexible plug expansion joints
- Nosing expansion joints
- Mat expansion joints
- Cantilever expansion joints
- Supported expansion joints
- Modular expansion joints

8.1.4.1 Buried Expansion Joint

This expansion joint is formed in situ using components such as waterproofing membranes or an elastomeric pad, to distribute the deformations to a greater width and to support the surfacing which is continuous over the deck joint gap. The components of the expansion joint are situated under the surfacing.

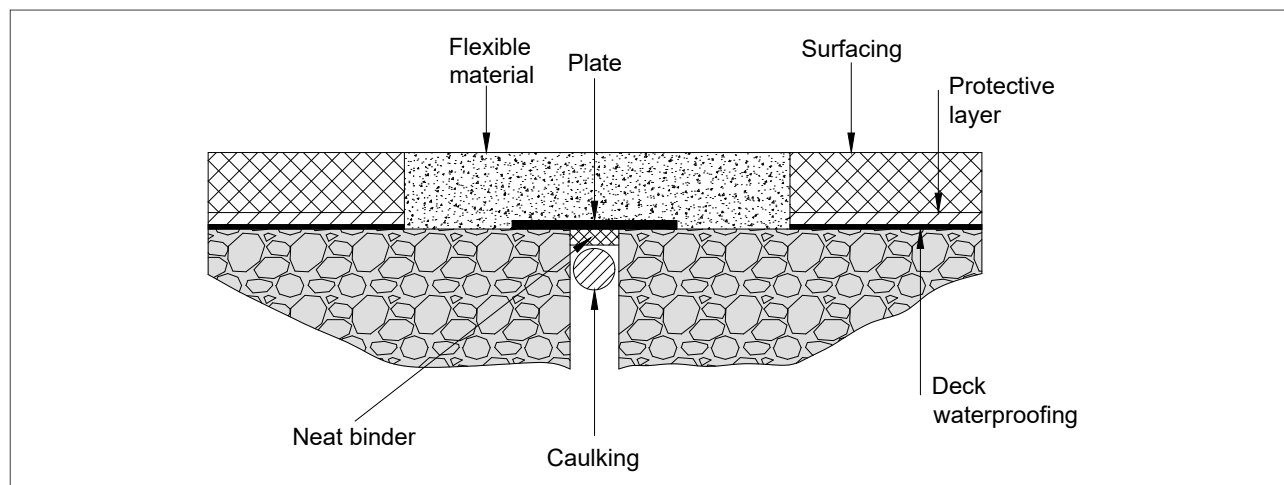
Figure 8.1 Buried Expansion Joint



8.1.4.2 Flexible Plug Expansion Joint

An in-situ poured joint comprising a band of specially formulated flexible material (binder and aggregates), which also forms the surfacing, supported over the deck joint gap by thin metal plates or other suitable components. The joint material is flush with the running surface.

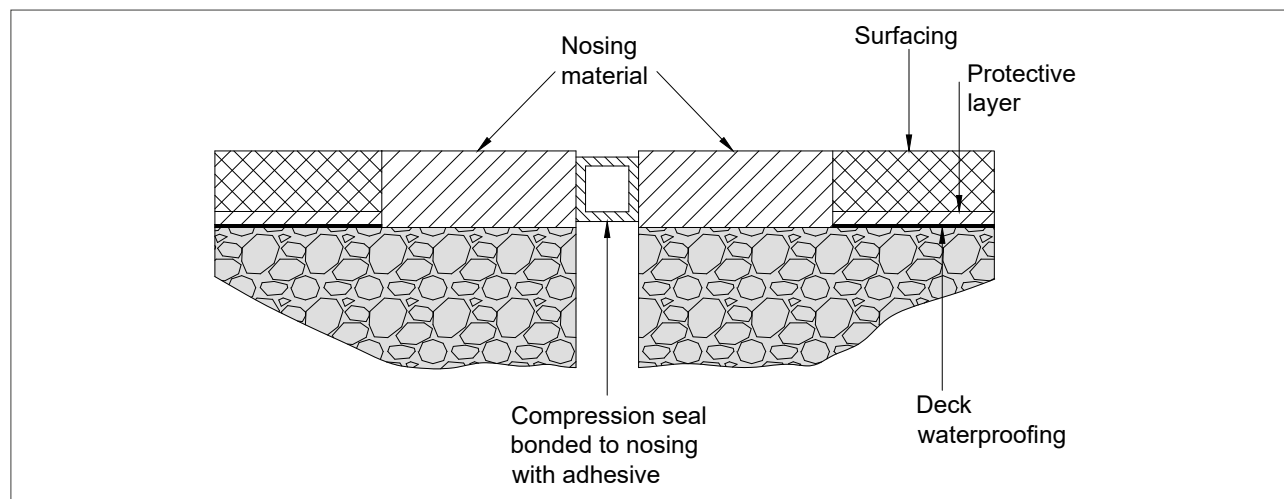
Figure 8.2 Flexible Plug Expansion Joint



8.1.4.3 Nosing Expansion Joint

This expansion joint has edges prepared with concrete, resin mortar or elastomer. The gap between the edges is filled by a flexible profile, which is not traffic load carrying.

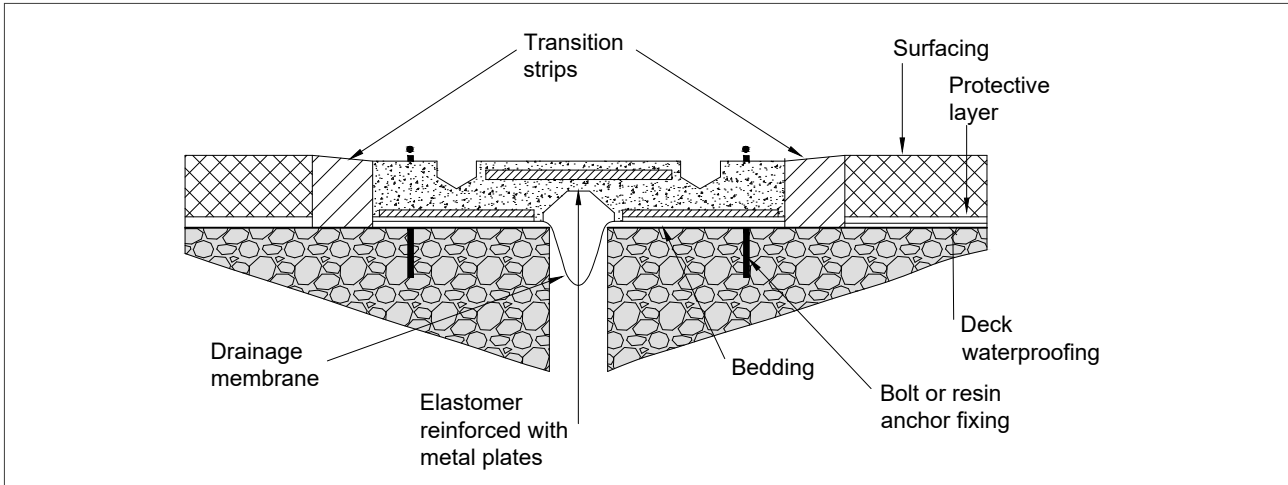
Figure 8.3 Nosing Expansion Joint



8.1.4.4 Mat Expansion Joint

This expansion joint uses the elastic properties of a prefabricated elastomeric strip or pad to allow the expected movements of the structure. The strip is fixed by e.g. bolts to the structure. The joint subcomponent flush with the running surface.

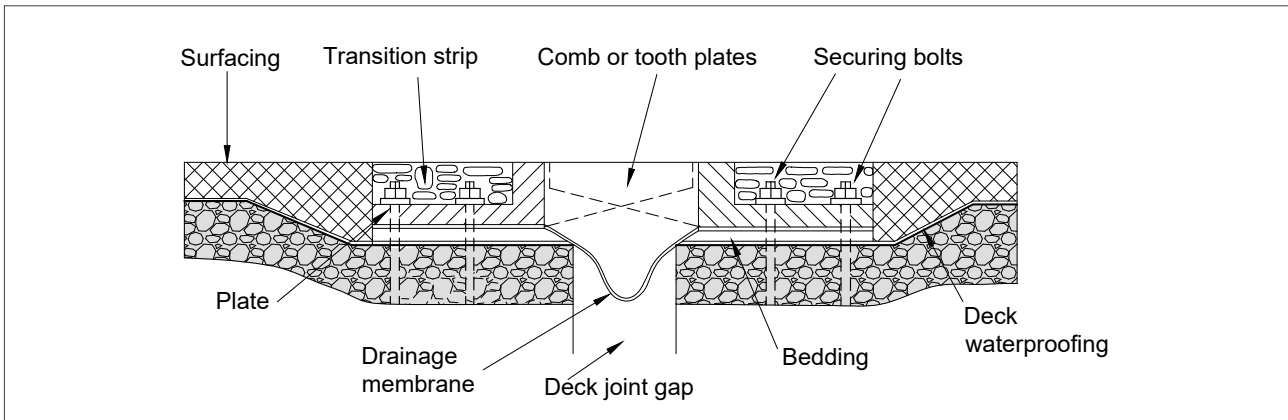
Figure 8.4 Mat Expansion Joint



8.1.4.5 Cantilever Expansion Joint

This expansion joint consists of cantilever symmetrical and non-symmetrical sub-components (such as comb or saw-tooth plates), which are anchored on one side of the deck joint gap and interpenetrated to bridge the deck joint gap. The sub-components are flush with the running surface.

Figure 8.5 Cantilever Expansion Joint



8.1.4.6 Supported Expansion Joint

This expansion joint consists of one sub-component flushed with the running surface, which is fixed by hinges on one side and sliding supports on the other side (by a second element), and which spans the deck joint gap. The expected structure movement is allowed through sliding on the non-fixed side of the hinged sub-component, i.e. on the supporting element that is anchored to the substructure.

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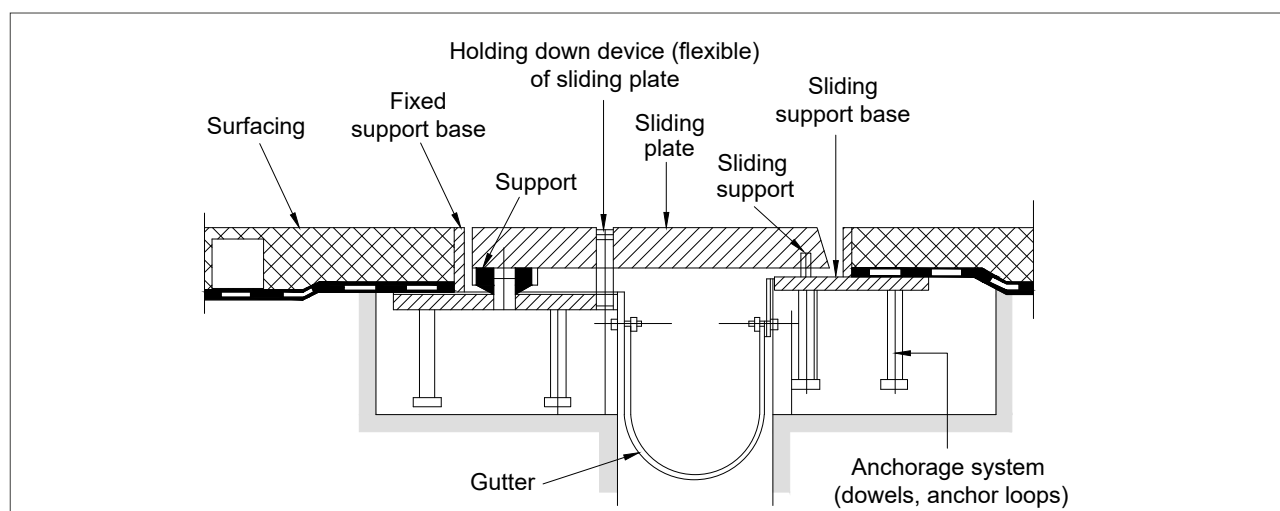
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Expansion Joints and Bearings

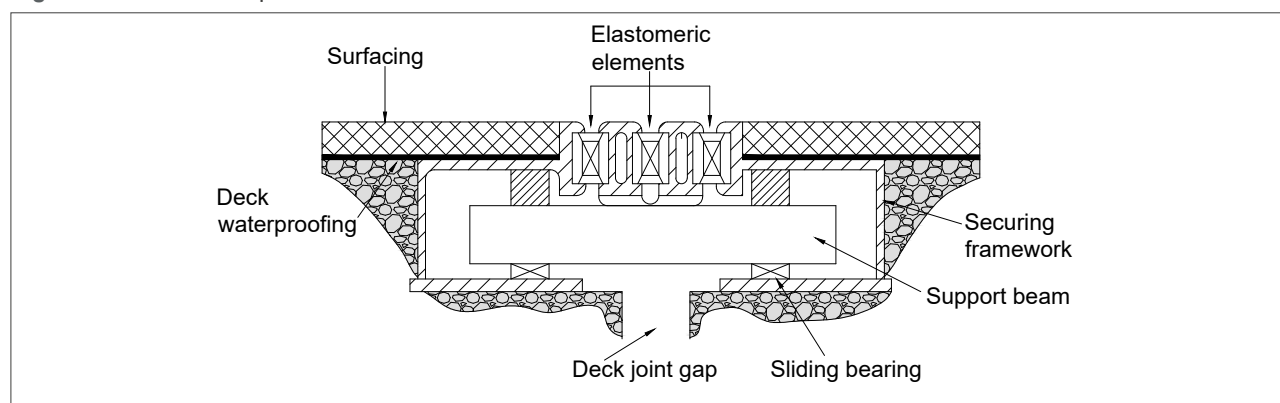
Figure 8.6 Supported Expansion Joint



8.1.4.7 Modular Expansion Joint

This expansion joint consists of a succession of watertight sub-components (in the traffic direction) comprising movement-controlled metal beams supported by moveable substructures bridging the structural gap (i.e. cross beams, cantilevers and pantographs). The metal beams are flush with the running surface.

Figure 8.7 Modular Expansion Joint



8.1.4.8 Movement Range of Expansion Joints

Indicative ranges of movement for various types of expansion joint are shown in following Table.

Table 8.1 Indicative Ranges of Movement for Various Types of Expansion Joints at SLS

Type of Expansion Joint	Total longitudinal movement (mm)		Maximum vertical movement between sides of joint (mm)
	Minimum**	Maximum	
Buried expansion joint	5	20	1.3
Flexible plug joint	5	40	3.0
Nosing expansion joint, with poured sealant	5	12	3.0
Nosing expansion joint, with pre-formed compression seal	5	40	3.0
Mat expansion joint	5	Varies*	3.0
Cantilever expansion joint	25	Varies*	3.0
Supported expansion joint	150	Varies*	3.0
Modular expansion joint	25	Varies*	3.0

* Maximum value varies according to manufacturer or type.

** Minimum of the range is given to indicate when the type of joint may not be economical.

8.1.5 Design Requirements

8.1.5.1 General

At the preliminary stage, the basic requirements to specify the expansion joint shall be established from the following:

- Required movement capacity
- User category, i.e. vehicular, pedestrian or other use
- Any special needs for the site, e.g. limited depth of surfacing, high skew.

Expansion joints shall not cause a hazard to any class of road user, including motorcyclists, cyclists, pedestrians and animals where they have access. Expansion joints should be installed in a straight line. The deck waterproofing system shall be compatible with the expansion joint system.

When the total design horizontal movement is less than 5 mm and vertical movement is less than 0.5 mm, there should be no need to provide an expansion joint.

8.1.5.2 Movement

The joint shall be designed for the relative movements that could occur at the relevant limit states. Movements can arise from the effects of actions, including thermal actions, shrinkage, creep, traffic and settlement. Relative movements can occur due to compression of bearings, structural deflections due to traffic on one side of the joint, or rotation of the deck.

Bridge movements that can occur simultaneously shall be combined. Partial factors and combinations of actions for the calculation of bridge movements shall be in accordance with Chapter 3.

The longitudinal movement due to changes in temperature shall be calculated in accordance with Chapter 3 considering the installation temperature and the extreme temperature range. The longitudinal movement for concrete bridge shall include the estimated change in deck dimensions due to shrinkage and creep, if significant.

In wide, curved or skew bridges, the joint requirements shall be determined according to the combination of the transverse and longitudinal movement of the deck.

Any longitudinal and vertical movement at joints arising from the predicted differential settlement between the bridge supports and the tilt of the bridge supports, such as abutment tilt, shall be included in the calculations of movement.

8.1.5.3 Joint Gaps, Groves and Cover Plates

Where there is an expansion joint gap it shall be of a uniform width, and never less than 10% of the specified range of movement of the joint system or 6mm, whichever is the greater.

Where pedestrians have access, all expansion joints with an open joint gap at surface level in the walkway shall be closed with either by a cover plate or a load bearing seal 5 ± 2 mm below the top surface of the joint.

Where the expansion joint in the carriageway consists of toothed, finger or comb plates and where motorcyclists, cyclists, wheelchair users, pedestrians or animals have access, the spaces between the mating teeth or fingers shall not exceed 150 mm in length or 20 mm in width, unless compatible cover plates are provided over the expansion joint.

Steel cover plates subject to traffic (including accidental wheel loading on walkways) should be at least 12 mm thick.

8.1.5.4 Transition Strip

Where a joint system requires a transition strip (to infill the gap between the joint and the adjacent surfacing), the width of the strip shall be kept as narrow as possible with the edge of the strip adjacent to the joint chamfered to assist in the dissipation of wheel impact loads. The width of the transition strip should be no wider than 400 mm.

8.1.5.5 Skid Resistance

The skid resistance of an expansion joint shall be at least equal to the minimum requirement for the adjacent carriageway surfacing throughout its working life.

The upper surfaces of expansion joints or compatible cover plates in walkways and cycleways shall have a slip-resistant finish, which, when new, has a mean corrected Pendulum Test Value (PTV) of not less than 65 under wet conditions.

The finish of the upper surfaces of expansion joints or compatible in walkways and cycleways cover plates shall have an effective life of at least five years and also retain a PTV of not less than 45 under wet conditions throughout this period.

8.1.5.6 Drainage

Water is not permitted to accumulate on joints. Joints shall be protected from surface water and ponding through incorporating suitable falls on the carriageway around joints and the use of appropriate drainage systems.

Joints provided with a compression seal, elastomeric element or sealant at surfacing level (such as nosing joints), shall be designed to be watertight.

For joints which are not sealed at surfacing level (such as modular and comb joints), an effective secondary drainage system shall be provided immediately beneath the expansion joint with access for inspection and maintenance.

Water from the drainage system shall be discharged away from the structure into a suitable road drainage system or soakaway.

8.2 Bearings

8.2.1 Scope

This section gives guidance on requirements/ characteristics for structural bearings used in bridges. The various bearings covered are elastomeric bearings, pot bearings, spherical and cylindrical bearings. The following bearings are not covered:

- a. Bearings that transmit moments as a primary function
- b. Bearings that resist uplift
- c. Bearings for moving bridges
- d. Concrete hinges
- e. Seismic devices

Fixed bearings prevent movements but other bearings such as guided bearings allow movements in one direction while free bearings allow movements in all directions.

The documents BS EN 1337 may be referred while using this Manual.

8.2.2 Definitions

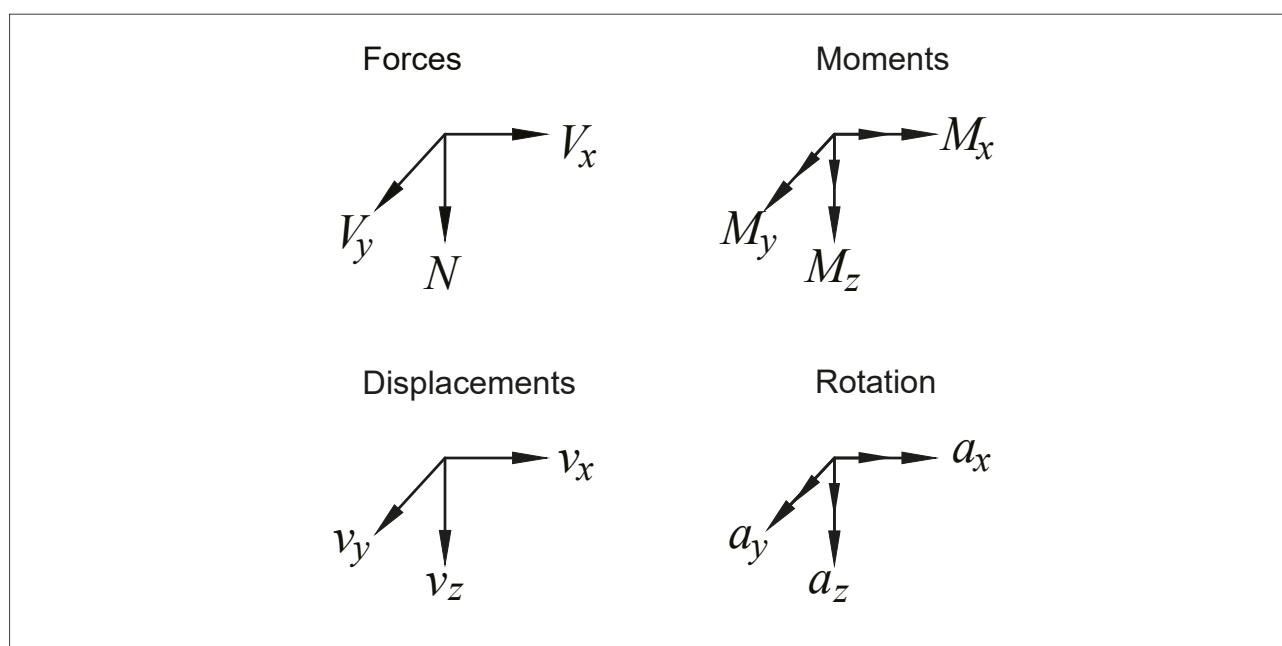
Bearings: Bearings are elements allowing rotation between two members of a structure and transmitting the loads defined in the relevant requirements as well as preventing displacements (fixed bearings), allowing displacements in only one direction (guided bearings) or in all directions of a plane (free bearings) as required.

Bearing systems: The bearing system for a structure is the combination of bearings which together provide for the movements and transmission of forces.

8.2.3 Types of Bearings

Common type of bearings used in bridges are given in Table 8.2.

The notations used for forces and displacements in different directions are as under.

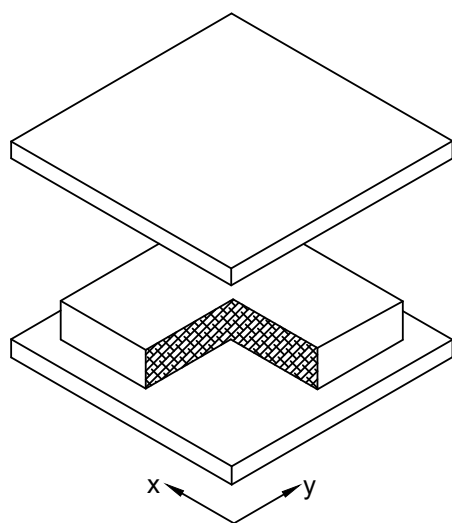


In general, x is the main direction of movement for bridges and z is the direction of the forces due to vertical loads.

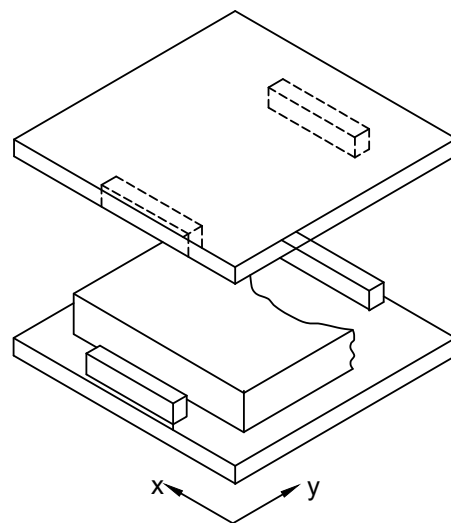
Table 8.2 Common Type of Bearings

NS	Type of Bearing	Symbol in Plan	Displacements			Rotations			Reactions			
			v_x	v_y	v_z	α_x	α_y	α_z	Forces		Mom-ent	
1	Elastomeric bearing		Deforming	Deforming	Small	Deforming	Deforming	Deforming	V_x	V_y	N	
2	Elastomeric bearing with restraints for one direction		Deforming	None	Small	Deforming	Deforming	Deforming	V_x	V_y	N	
3	Elastomeric bearing with sliding in one direction and restraints for other direction		Sliding and deforming	None	Small	Deforming	Deforming	Deforming		V_y	N	
4	Elastomeric bearing with multi-directional sliding		Sliding and deforming	Sliding and deforming	Small	Deforming	Deforming	Deforming			N	
5	Pot bearing		None	None	Very small	Deforming	Deforming	Deforming	V_x	V_y	N	
6	Pot bearing with uni-directional sliding		Sliding	None	Very small	Deforming	Deforming	Deforming		V_y	N	
7	Pot bearing with multi-directional sliding		Sliding	Sliding	Very small	Deforming	Deforming	Sliding & deforming			N	
8	Spherical bearing with restraints beyond the rotating part		None	None	None	Sliding	Sliding	Sliding	V_x	V_y	N	
9	Spherical bearing with uni-directional sliding		Sliding	None	None	Sliding	Sliding	Sliding		V_y	N	
10	Spherical bearing with multi-directional sliding		Sliding	Sliding	None	Sliding	Sliding	Sliding			N	
11	Fixed cylindrical bearing		None	None	Almost none	None	Sliding	None	V_x	V_y	N	M_x
12	Guided cylindrical bearing, moveable in y-direction		None	Sliding	Almost none	None	Sliding	None	V_x		N	M_x
13	Guided cylindrical bearing, moveable in x-direction		Sliding	None	Almost none	None	Sliding	None		V_y	N	M_x
14	Free cylindrical bearing, moveable in x and y-direction		Sliding	Sliding	Almost none	None	Sliding	Sliding			N	M_x
15	Guided bearing with restraints for two axes – thrust bearing		None	None	Sliding	Sliding or deforming	Sliding or deforming	Sliding or deforming	V_x	V_y		

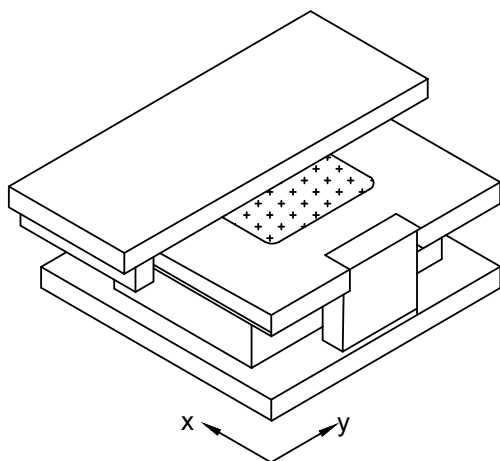
Figure 8.8 Bearings



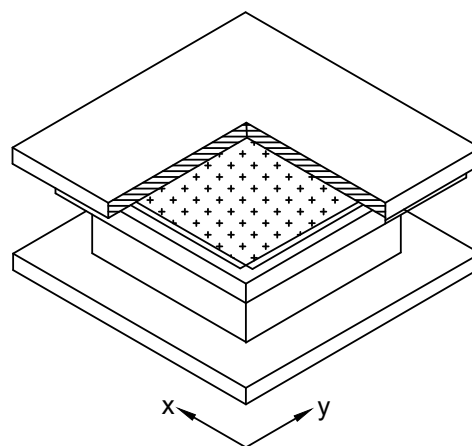
a. Elastomeric bearing



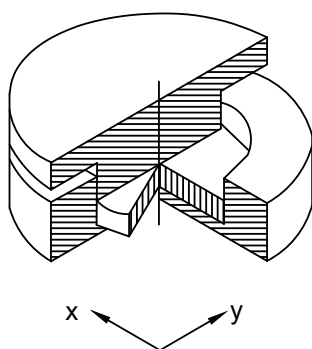
a. Elastomeric bearing with restraints for one direction



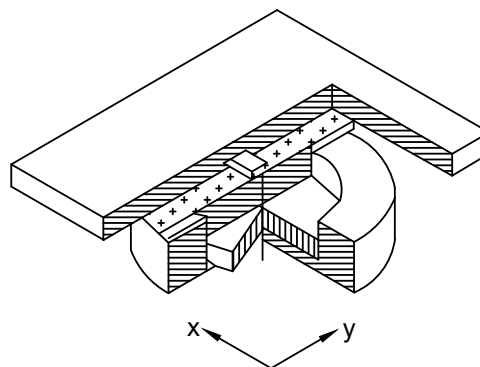
c. Elastomeric bearing with sliding in one direction and restraints for other direction



a. Elastomeric bearing with multi-directional sliding



e. Pot bearing



f. Pot bearing with uni-directional sliding

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8.2.4 Bearing Layout

Bearing layout should be designed to permit the required movements of the structure with minimum possible resistance to these movements. The arrangement of bearings for a structure should be considered together with the design of the structure as a whole. The bearing forces and movements should be given to the manufacturer to ensure that bearing meet the requirements.

The drawing showing bearing layout shall include the following:

General arrangement of the bridge showing the bearing layout in plan.

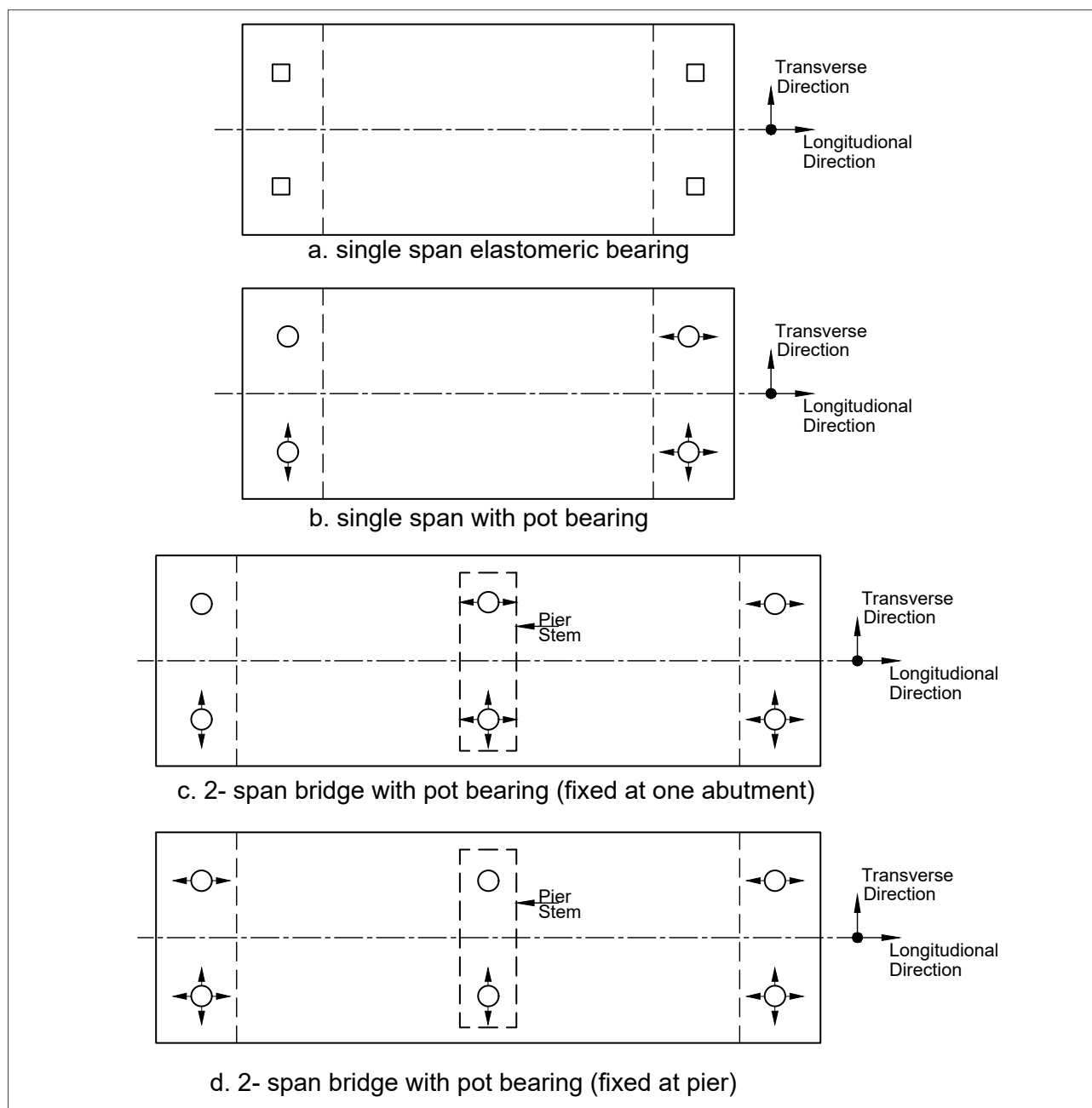
Details at the bearing location (bearing seat, recess, etc.)

Indication for type of bearing at each location.

A bearing schedule giving detail requirements for each bearing.

Examples for typical bearing layout plans are shown in Figure 8.9.

Figure 8.9 Example of Typical Layout Plans for Bearings



8.2.5 Bearing Schedule

The bearing schedule is prepared to list all the information normally required for the design of bearings for a particular structure. A drawing shall be prepared showing bearing layout plan with identification marks for each bearing and typical cross section of the bridge. The bearing functions shall be indicated on the drawings by the bearing symbols. The information will enable the manufacturers to design and construct the bearings.

A typical schedule of bearings is shown in Table 8.3. Whenever required, additional information should be added.

Table 8.3 Typical Schedule of Bearings

SN	Bearing Information				Value	
1	Structure name or reference					
2	Bearing Identification Mark					
3	Type of Bearing					
4	Number of bearings	Seating Material*		Upper Surface		
				Lower Surface		
5	Design Load Effects (kN)	Serviceability Limit State		Vertical, N	Maximum	
					Permanent	
					Minimum	
				Transverse, $V_{y,sd}$		
		Longitudinal, $V_{x,sd}$				
		Ultimate Limit State		Vertical, N		
				Transverse, $V_{y,ud}$		
				Longitudinal, $V_{x,ud}$		
6	Displacement (mm)			Serviceability Limit State	Irreversible	Transverse, $v_{y,sdi}$
		Longitudinal, $v_{x,sdi}$				
		Reversible	Transverse, $v_{y,sdr}$			
			Longitudinal, $v_{x,sdr}$			
		Ultimate Limit State	Irreversible	Transverse, $v_{y,udi}$		
				Longitudinal, $v_{x,udi}$		
			Reversible	Transverse, $v_{y,udr}$		
				Longitudinal, $v_{x,udr}$		
7	Rotation (Radians)	Serviceability Limit State	Irreversible	Transverse, $\alpha_{y,sdi}$		
				Longitudinal, $\alpha_{x,sdi}$		
			Reversible	Transverse, $\alpha_{y,sdr}$		
				Longitudinal, $\alpha_{x,sdr}$		
8	Maximum Bearing Dimensions (mm)	Upper Surface		Transverse		
				Longitudinal		
		Lower Surface		Transverse		
				Longitudinal		
9	Special requirements					

* Seating material shall be specified such as cement mortar, epoxy mortar, in situ concrete, precast concrete, steel, etc.

8.2.6 Design Principles

8.2.6.1 General

Bearings and supports shall be designed so that bearings can be inspected, maintained and replaced, if necessary, in order to enable them to fulfil their functions throughout the intended life of structure.

Bearings shall be designed to permit the specified movement with minimum possible resisting force against permitted movements.

Pre-setting shall be avoided as far as possible. If necessary, the required pre-setting shall be carried out at the factory. If adjustment on site cannot be avoided, it shall be carried out by the manufacturer of the bearing or under his supervision.

Because the movement restriction imposed by a bearing, it must be compatible with the movements allowed by the adjacent expansion joint. The expansion joints and bearings must be designed interdependently and in conjunction with the anticipated behaviour of the overall structure.

The design of bearing shall be based on serviceability and/or ultimate state depending on the safety classification of the limit state in consideration.

8.2.6.2 Safety Against Sliding

Where the position of bearing or part of bearing is maintained completely or partially by friction, its safety against sliding shall be checked at ULS in accordance with the following:

$$V_{Sd} \leq V_{Rd}$$

Equation 8.1

Where,

V_{Sd} = The design shear force resulting from the actions.

V_{Rd} = The design value of the shear resistance.

The design value of shear resistance V_{Rd} is given as under:

$$V_{Rd} = \frac{\mu_k}{\gamma_\mu} N_{Sd} + V_{pd}$$

Equation 8.2

Where,

N_{Sd} = The minimum design force acting normal to the joint in conjunction with V_{Sd} .

V_{pd} = The design strength of any fixing device.

μ_k = The characteristic value of the friction coefficient.

= 0.4 for steel on steel

= 0.6 for steel on concrete

γ_μ = The partial safety factor for friction

= 2.0 for steel on steel

= 1.2 for steel on concrete

The values of μ_k and γ_μ mentioned above shall be used provided that prior to installation or assembly of the surfaces of steel components are un-coated and free from grease or coated with a fully hardened zinc silicate. In case of dynamic loads where extreme load fluctuations can occur, e.g. seismic condition, the horizontal forces shall not be resisted by friction and the value of μ_k shall be taken as zero.

8.2.6.3 Increased Movements

In order to ensure that the bearings function as they intended with sufficient reliability, the design movements shall be increased by following values.

a. Rotation

± 0.005 radians or $\pm 10/r$ radians whichever is greater (r is measured in mm)

Where,

r = radius of pot or lid for pot bearings.

= radius of curvature for dimensioning the rocker of rocker bearings.

= radius of recess for dimensioning the limit of stop of the outer bearing plate of the rocker bearings.

= radius of curvature of the sphere for dimensioning the sliding surface of spherical bearings.

= half of the inside diameter of the guide edge for dimensioning the limit of stop.

b. Translation

± 20 mm in both directions of movement with a minimum movement of ± 50 mm in the direction of maximum movement and ± 20 mm transversely unless the bearing is mechanically restrained.

These requirements apply for design of movement capacities of bearings. They shall not be used in stress calculations. These values shall not be applied to elastomeric bearings.

8.2.6.4 Minimum Movements for Strength Analysis

For the purpose of strength analysis of bearings, the resultant rotational movement shall not be taken less than ± 0.003 radians and the resultant translational movement not less than ± 20 mm or ± 10 mm for elastomeric bearings.

If a bearing cannot rotate about one axis, a minimum eccentricity of $l/10$ perpendicular to that axis shall be assumed, where l is the total length of the bearing perpendicular to that axis.

8.2.7 Bearing Resistances

8.2.7.1 General Specifications

The values to be used for calculating the resistance to movement of various types of bearings are given in subsequent Clauses of this Chapter. For more details, refer relevant part of EN 1337.

8.2.7.2 Reactions to Rolling or Sliding of a Set of Bearings

Where a number of bearings are so arranged that the adverse forces, resulting from reaction to movement by some, are partly relieved by the forces resulting from the reactions to movement by others, the respective coefficient of friction μ_a and μ_r shall be calculated as following:

$$\mu_a = 0.5 \mu_{max} (1 + \alpha)$$

Equation 8.3

$$\mu_r = 0.5 \mu_{max} (1 - \alpha)$$

Equation 8.4

Where,

μ_a = The adverse coefficient of friction.

μ_r = The relieving coefficient of friction.

μ_{max} = The maximum coefficient of friction for the bearing.

α = The factor dependent on the type and number of bearings which are exerting either an adverse or relieving force as appropriate and may be taken as following.

n (number of bearings)	α
≤ 4	1
$4 < n < 10$	$(16 - n)/12$
≥ 10	0.5

8.2.8 General Design Guidance

Forces from the deck are transferred to sub-structure through bearings. The maximum and minimum vertical forces, horizontal forces and rotations for bearing design are obtained from the deck analysis. The type of bearings will depend upon the vertical forces and types of restraints required to support the deck.

Horizontal forces on bearings from deck are produced by temperature actions, wind actions, creep movement, braking and acceleration effects, centrifugal forces if deck is in curve, and seismic actions where appropriate.

Vertical forces on bearings from deck are produced by self-weight of the structure, superimposed loads and traffic loads.

Translational and rotational movements can arise from the effects of actions, including thermal actions, shrinkage, creep, traffic and settlement.

Longitudinal forces from temperature effects in the deck will depend upon the type of bearings used. The elastomeric bearings undergo shear deformation when the deck expands or contracts. The longitudinal force produced by this shear deformation is proportional to the shear stiffness of the bearing and the magnitude of the movement. Sliding bearings produce longitudinal force which is proportional to the reaction due to permanent actions and the coefficient of friction between the sliding surfaces. The coefficient of friction depends upon the type of bearing surfaces and the bearing stress.

The longitudinal forces from the temperature effect will act equally on both supports of a span. If sliding bearings are used then the load transmitted is equal to the friction at the bearing under permanent actions. If elastomeric bearings are used then the load transmitted is equal to the force required to distort the bearing by the distance by which the deck expands or contracts.

The longitudinal forces due to braking and acceleration are assumed to be transmitted to the fixed support only. The frictional resistance from sliding bearings at the free end of the deck would produce a relieving effect on the fixed bearing and should therefore be ignored when designing the fixed bearing. If only elastomeric bearings are used, i.e. there is no fixed support, then the loads due to braking and acceleration are shared between the both side supports.

8.2.9 Design Requirements of Sliding Bearings

8.2.9.1 Materials

Following materials are generally used for sliding bearings:

- PTFE sheets
- Composite material CM1
- Composite material CM2
- Austenitic steel sheet
- Hard chromium plated surfaces
- Ferrous materials for backing plates
- Aluminium alloy
- Lubricant
- Adhesive for bonding austenitic sheets

The material shall comply the requirements specified in BS EN 1337-2.

8.2.9.2 Combination of Sliding Material

The sliding material shall be combined as shown in Table 8.4. Only one combination shall be used in a sliding bearing.

Table 8.4 Permissible Combination of Materials for Sliding Surfaces

Plane Surface		Curved Surface		Guides	
Dimples PTFE	Austenitic	Dimples PTFE	Austenitic steel	Undimpled PTFE	Austenitic steel
			Hard chromium	CM1	
			Aluminium	CM2	

8.2.9.3 Characteristic Compressive Strength of Sliding Materials

The characteristic compressive strengths for sliding materials are given in Table 8.5. These values are valid for effective bearing temperature up to 30 °C. If bearings are exposed to a maximum effective bearing temperature > 30 °C and up to 48 °C, the values provided in this table shall be reduced by 2% per degree above 30 °C.

Table 8.5 Characteristic Compressive Strength for Sliding Materials

Material	Action	f_{ck} (in MPa)
PTFE for main bearing surfaces	Permanent and variable loads	90
	Variable load	90
PTFE for guides	Temperature, shrinkage and creep	30
	Permanent loads	10
CM1	Permanent and variable horizontal loads	200
CM2	Permanent and variable horizontal loads	120

8.2.9.4 Coefficient of Friction

The maximum coefficient of friction (μ_{max}) as given in Table 8.6 may be used for verification of the bearings and the structure where it is installed. Linear interpolation may be used for intermediate values. These values are not used in the presence of high dynamic forces which may occur in seismic zones. The effects of friction shall not be used to relieve the effects of externally applied horizontal loads. The values shown in this table are applicable only for the dimpled lubricated PTFE.

Table 8.6 Coefficient of Friction μ_{max}

Contact Pressure σp (MPa)	≤ 5	10	20	≥ 30
PTFE dimpled / austenitic steel or hard chromium plating	0.08	0.06	0.04	0.03 (0.025) ^a
PTFE dimples / aluminium alloy anodised	0.12	0.09	0.06	0.045 (0.038) ^a

^a These values are applicable to the frictional resistance of curved sliding surfaces.

In the zones where, the bearing temperature does not fall below -5 °C, the values in this table may be multiplied by a factor 2/3.

For guides with combination of sliding materials given in third column of Table 8.4, the coefficient of friction shall be considered as independent of contact pressure and the μ_{max} may be taken as 0.08 for PTFE and 0.20 for composite materials.

8.2.9.5 Design Verification of Sliding Surfaces

All the internal forces and moments due to actions and frictional resistance shall be considered in dimensioning and designing the sliding surfaces. Design values of the actions and resistances shall be determined in accordance with the provisions given in earlier sections of this Chapter. Deformation of sliding material shall not be used to accommodate rotations.

Lack of confinement of PTFE may lead to separation of sliding surface. For serviceability limit state, with the exception of guides, it shall be verified that compressive stress on sliding surfaces $\sigma_p > 0$ under the characteristic combination of actions.

Excessive pressure may cause loss of the sliding function and may lead to structural failure. Following condition shall be verified at ultimate limit state for combination of materials according to Table 8.4.

$$N_{Sd} \leq \frac{f_k}{\gamma_m} A_r$$

Equation 8.5

Where,

N_{Sd} = The design value of the axial force due to the design value of actions

f_k = The characteristic compressive strength as give in Table 8.5.

A_r = The reduced contact area of sliding surface whose centroid is the point through which N_{Sd} acts with the eccentricity e , which is caused by both mechanical and geometrical effects (refer Annex A of EN 1337-2 for details). Eccentricity can be neglected for guides.

γ_m = The partial safety factor, recommended value is 1.4.

8.2.10 Design Requirements of Elastomeric Bearings

8.2.10.1 Material

The elastomer used in the manufacture of Elastomeric Bearings should be either natural rubber (NR) or chloroprene rubber (CR) as the raw polymer. Blending up to 5% of another polymer to aid processing is permitted. No reclaimed or ground vulcanised rubber shall be used.

The inner and outer reinforcing plates shall be of steel grade S 235 according to EN 10025 or steel with a minimum equivalent elongation at break.

The properties of elastomer and steel plates shall comply the requirements given in BS EN 1337-3. Materials used in sliding elastomeric bearings shall comply the requirements given in BS EN 1337-3 and BS EN 1337-2.

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Expansion Joints and Bearings

The shear modulus (G_g) is the apparent "conventional shear modulus" of elastomeric bearings determined by testing. At a nominal temperature of $23\text{ }^{\circ}\text{C} \pm 2\text{ }^{\circ}\text{C}$, the value G_g of the conventional shear modulus shall comply with one of the values given in Table 8.7.

Table 8.7 Shear Modulus of Elastomeric Bearing at Nominal Temperature

SN	Hardness (IRHD)	Gg (MPa)	Tolerance of Gg (MPa)
1*	50 ± 5	0.7	± 0.10
2	60 ± 5	0.5	± 0.15
3*	70 ± 5	1.15	± 0.20

* The values at row 1 and 3 shall be used only if specified by the structural designer.

8.2.10.2 Design Requirements

Elastomeric bearings shall be designed at ultimate limit state to ensure that the strength and stability of bearings are adequate to withstand the ultimate design loads and movements of the structure.

Elastomeric bearings shall be designed such that the design value of actions (S_d) does not exceed the design value of resistance (R_d) considering all action effects and the relative movements.

8.2.10.3 Laminated Elastomeric Bearings

8.2.10.3.1 Size and Shape

Laminated elastomeric bearings are rectangular or circular. For particular applications, elliptical or octagonal (approximating to elliptical) shapes may be accepted.

A particular bearing shall have internal rubber layers of the same thickness between 5 mm and 25 mm each. The minimum thickness of top and bottom elastomer cover layers shall not be less than 2.5 mm and minimum thickness of side cover shall not be less than 4 mm. Design calculations shall not be applied to the external top and bottom layer when their thickness is less or equal to 2.5mm.

The minimum thickness of reinforcing plates shall not be less than 2 mm for inner plates. For elastomeric bearing type C with elastomer layers less than or equal to 8 mm thick, the minimum thickness of outer steel plate shall not be less than 15 mm. For thicker elastomer layers, the minimum thickness of outer steel plate shall not be less than 18mm.

The elastomeric bearing types or a combination of the types are given in Table 8.8. A typical cross section of an elastomeric bearing type B showing different symbols used is shown in Figure 8.10. The recommended standard sizes in accordance with BS EN 1337-3 are given in Table 8.9.

Figure 8.10 Typical Cross Section of an Elastomeric Bearing Type B

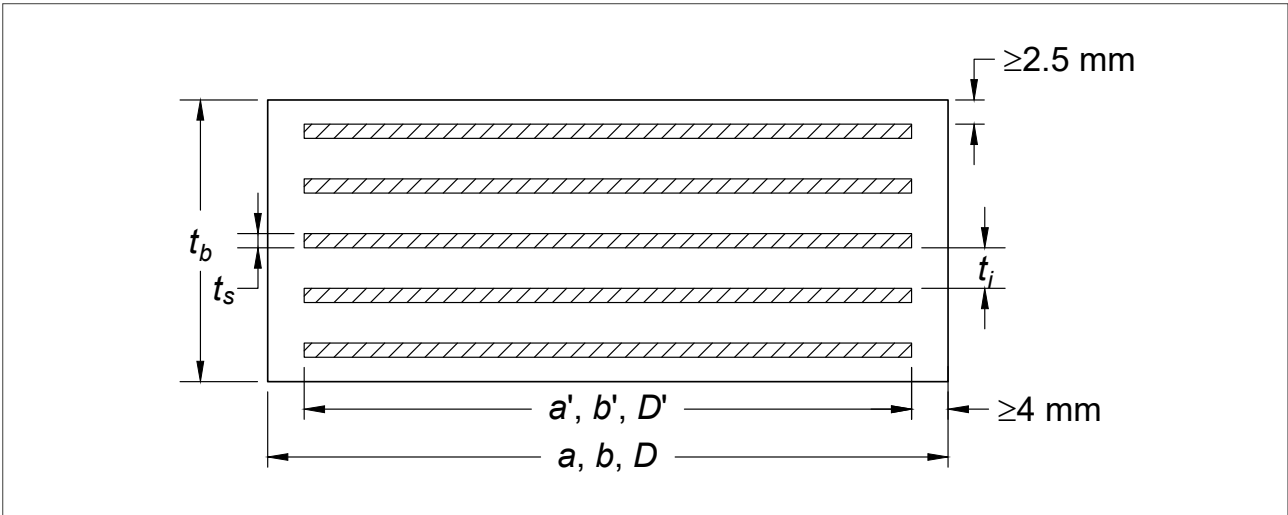
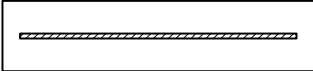
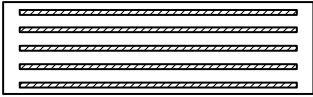
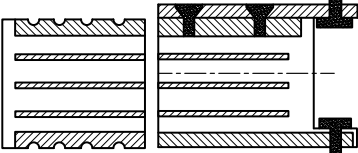
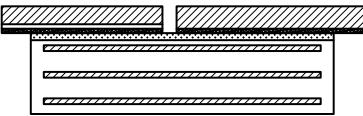
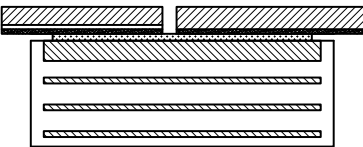


Table 8.8 Different Types of Bearing Cross Sections

Type of Bearing	Typical Cross Section
Type A: Laminated bearing fully covered with elastomer comprising only one steel reinforcing plate	
Type B: Laminated bearing fully covered with elastomer comprising at least two steel reinforcing plates	
Type C: Laminated bearing with outer steel plates (profiled or allowing fixing). NOTE The sketch shows examples of a few fixing methods; other methods can be used by agreement	
Type D: Type B with PTFE sheet bonded to the elastomer	
Type E: Type C with one outer plate bonded to the elastomer and PTFE sheet recessed in the steel	
Type F: Plain pad bearings and strip bearings	

8.2.10.3.2 Basis of Design

All designed bearings including standard sizes given in Table 8.9 shall meet the following requirements:

a. Maximum design strain

At any point of bearings, the sum of the strains ($\epsilon_{t,d}$) due to the design load effects (E_d) is given by the following expression:

$$\epsilon_{t,d} = K_L (\epsilon_{c,d} + \epsilon_{q,d} + \epsilon_{a,d})$$

Equation 8.6

Where,

$\epsilon_{c,d}$ = The design strain due to compressive loads.

$\epsilon_{q,d}$ = The design shear strain due to design translatory movements.

$\epsilon_{a,d}$ = The design strain due to the design angular rotation.

K_L = The type-loading factor and is equal to 1.0.

If requested by the structural designer, K_L may be taken as 1.5 for live load effects and 1.0 for all other effects including wind and temperature.

The sum of strains $\epsilon_{t,d}$ shall not exceed the maximum value of $\epsilon_{u,d}$ given by following expression:

$$\epsilon_{u,d} = \frac{\epsilon_{u,k}}{\gamma_m}$$

Equation 8.7

Where,

$\epsilon_{u,k}$ = The maximum permissible value of 7 for ULS.

γ_m = The partial safety factor, recommended value is 1.0.

Table 8.9 Standard Sizes for Elastomeric Bearings Type B

Dimensions <i>a</i> x <i>b</i> (mm) or <i>D</i>	Thickness in mm						Number of layers <i>n</i>	
	Unloaded bearing		Elastomer (total ^{a)})		Elastomer layers	Reinforcing plates		
	Min.	Max.	Min.	Max.			Min.	Max.
100 x 150	30	41	16	24	8	3	2	3
100 x 200	30	41	16	24	8	3	2	3
150 x 200	30	52	16	32	8	3	2	4
ϕ 200	30	52	16	32	8	3	2	4
150 x 250	30	52	16	32	8	3	2	4
150 x 300	30	52	16	32	8	3	2	4
ϕ 250	30	52	16	32	8	3	2	4
200 x 250	41	74	24	48	8	3	3	6
200 x 300	41	74	24	48	8	3	3	6
200 x 350	41	74	24	48	8	3	3	6
ϕ 300	41	74	24	48	8	3	3	6
200 x 400	41	74	24	48	8	3	3	6
250 x 300	41	85	24	56	8	3	3	7
ϕ 350	41	85	24	56	8	3	3	7
250 x 400	41	85	24	56	8	3	3	7
300 x 400	57	105	36	72	12	4	3	6
ϕ 400	57	105	36	72	12	4	3	6
300 x 500	57	105	36	72	12	4	3	6
ϕ 450	57	105	36	72	12	4	3	6
300 x 600	57	105	36	72	12	4	3	6
350 x 450	57	121	36	84	12	4	3	7
ϕ 500	57	121	36	84	12	4	3	7
400 x 500	73	137	48	96	12	4	4	8
ϕ 550	73	137	48	96	12	4	4	8
400 x 600	73	137	48	96	12	4	4	8
450 x 600	73	153	48	108	12	4	4	9
ϕ 600	73	153	48	108	12	4	4	9
500 x 600	73	169	48	120	12	4	4	10
ϕ 650	73	169	48	120	12	4	4	10
600 x 600	94	199	64	144	16	5	4	9
ϕ 700	94	199	64	144	16	5	4	9
600 x 700	94	199	64	144	16	5	4	9
ϕ 750	94	199	64	144	16	5	4	9
700 x 700	94	220	64	160	16	5	4	10
ϕ 800	94	220	64	160	16	5	4	10
700 x 800	94	220	64	160	16	5	4	10
ϕ 850	94	220	64	160	16	5	4	10
800 x 800	110	285	80	220	20	5	4	10
ϕ 900	110	285	80	220	20	5	4	10
900 x 900	110	285	80	220	20	5	4	11

^a Total thickness without 2.5 mm top and bottom cover.

^a Total thickness without 2.5 mm top and bottom cover.

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- b. Maximum tensile stresses in reinforcing plates.

The reinforcing plates shall be designed for ULS as defined in Clause 8.2.10.3.7.

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- c. Stability criteria.

Stability shall be evaluated according to Clause 8.2.10.3.8 considering the following:

- Stability regarding rotation.
- Stability regarding buckling.
- Stability regarding sliding.

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- d. Forces, moments, and deformations exerted on the structure.

Forces, moments and deformations shall be evaluated according to Clause 8.2.10.3.9 taking into account the following:

- The pressure at the contact surfaces between the bearing and the structure.
- The force exerted on the structure by the bearing resisting translatory movement.
- The restoring moment due to the bearing resisting rotational movement.
- Vertical deflection due to the vertical load.

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8.2.10.3.3 Shape Factor

The effect of shape of the elastomer in strength and deflection calculations is taken in the form of a shape factor S . It is the ratio of the effective plan area of an elastomeric slab to its force-free surface area, including holes

For laminated bearings the shape factor S for each individual elastomer layer is given by the equation:

$$S = \frac{A_1}{l_p t_e} \quad \text{Equation 8.8}$$

For plain pad bearings the shape factor S is given by the equation.

$$S = \frac{A}{l_p t_e} \quad \text{Equation 8.9}$$

For strip bearings the shape factor S is given by the equation.

$$S = \frac{a}{2 t_e} \quad \text{Equation 8.10}$$

Where,

A_1 = The effective plan area of the bearing, i.e. the plan area common to elastomer and steel plate, excluding the area of any holes if these are not later effectively plugged. For rectangular bearing without holes, $A_1 = a' \cdot b'$

A = The overall plan area of the elastomeric bearing

a = The overall width of the strip bearing

l_p = The force-free perimeter of the bearing including that of any holes if these are not later effectively plugged. For rectangular bearing without holes, $l_p = 2(a' + b')$

t_e = The effective thickness of an individual elastomer layer in compression.

- a. For in laminated bearings, it is taken as actual thickness t_i for inner layers and $1.4 t_i$ for outer layers with a thickness ≥ 3 mm;

- b. For plain pad and strip bearings, it is taken as $1.8 t_i$ (where t_i is the thickness of an individual elastomer layer)

a' is the effective width of the bearing (i.e. the width of reinforcing plates)

b' is the effective length of the bearing (i.e. the length of reinforcing plates)

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Expansion Joints and Bearings

8.2.10.3.4 Design Strain Due to Compressive Load

For calculation purpose, the shear modulus G shall be taken as defined in Table 8.7. The design strain due to compression loads ($\varepsilon_{c,d}$) is given by following equation.

$$\varepsilon_{c,d} = \frac{1.5 F_{z,d}}{G A_r S} \quad \text{Equation 8.11}$$

Where,

$F_{z,d}$ = The vertical design force.

G = The nominal value of conventional shear modulus of elastomeric bearing, according to Table 8.7

A_r = The reduced effective plan area due to the loading effects

S = The shape factor as determined in Clause 8.2.10.3.3.

The reduced effective plan area due to the loading effects is given by the following equation:

$$A_r = A_1 \left(1 - \frac{v_{x,d}}{a'} - \frac{v_{y,d}}{b'} \right) \quad \text{Equation 8.12}$$

Where,

$v_{x,d}$ = The maximum horizontal relative displacement of parts of the bearing in the direction of dimension, a , of the bearing due to all design load effects.

$v_{y,d}$ = The maximum horizontal relative displacement of parts of the bearing in the direction of dimension, b , of the bearing due to all design load effects.

8.2.10.3.5 Shear Strain

The shear strain ($\varepsilon_{q,d}$) of the elastomer due to translatory movement shall not exceed 1.0 and is given by the expression:

$$\varepsilon_{q,d} = \frac{v_{xy,d}}{T_q} \quad \text{Equation 8.13}$$

Where,

$v_{xy,d}$ = The maximum resultant horizontal relative displacement of parts of the bearing obtained by vectorial addition of $v_{x,d}$ and $v_{y,d}$.

T_q = The total thickness of elastomer in shear including the top and bottom cover, unless relative movement between the outer plates of the bearing and the structure is restrained by doweling or other means.

8.2.10.3.6 Design Strain Due to Angular Rotation

The nominal strain due to angular rotation is given by the equation:

$$\varepsilon_{a,d} = \frac{(a'^2 \alpha_{a,d} + b'^2 \alpha_{b,d}) t_i}{2 \sum (t_i^3)} \quad \text{Equation 8.14}$$

Where,

$\alpha_{a,d}$ = The angle of rotation across the width, a , of the bearing.

$\alpha_{b,d}$ = The angle of rotation across the width, b , of the bearing.

t_i = The thickness of an individual layer of elastomer.

8.2.10.3.7 Reinforcing Plate Thickness

The minimum thickness of the steel plates in a laminated bearing to resist induced tensile stresses under load is given by the equation:

$$t_s = \frac{K_p F_{z,d} (t_1 + t_2) K_h \gamma_m}{A_r f_y} \quad \text{and} \quad t_s \geq 2 \text{ mm} \quad \text{Equation 8.15}$$

Where,

t_1 and t_2 = The thickness of elastomer on either side of the plate.

f_y = The yield stress of the steel.

K_h = A factor for induced tensile stresses in reinforcing plate.

= 1 for plates without holes.

= 2 for plates with holes.

γ_m = The partial safety factor, recommended value is 1.0.

K_p = A stress correction factor = 1.3.

8.2.10.3.8 Limiting Conditions

a. Rotational limitation condition

For laminated bearings, the rotational limitation shall be satisfied when the total vertical deflection $\Sigma v_{z,d}$ (calculated as in Clause 8.2.10.3.9) complies with following equation.

For rectangular bearings:

$$\Sigma v_{z,d} - \frac{a' \alpha_{a,d} + b' \alpha_{b,d}}{K_{r,d}} \geq 0 \quad \text{Equation 8.16}$$

For circular bearings:

$$\Sigma v_{z,d} - \frac{D' \alpha_d}{K_{r,d}} \geq 0 \quad \text{Equation 8.17}$$

Where,

D' = The effective diameter of the bearing.

$K_{r,d}$ = A rotation limitation factor, may be taken as 3.0.

$v_{z,d}$ = The total vertical deflection producing α_a and α_b .

b. Buckling stability

For laminated bearings, the pressure, $(F_{z,d} / A_r)$ shall satisfy the following expression

For rectangular bearings

$$\frac{F_z}{A_r} < \frac{2 a' G S_1}{3 T_e} \quad \text{Equation 8.18}$$

For circular bearings, a' shall be the diameter

c. Non-sliding condition

For non-anchored bearings, the following expression shall be satisfied:

$$F_{xy,d} \leq \mu_e F_{z,d \min} \quad \text{Equation 8.19}$$

And under permanent loads:

$$\sigma_{c,d \min} = \frac{F_{z,d \min}}{A_r} \geq 3 \quad (\text{in N/mm}^2) \quad \text{Equation 8.20}$$

Where,

$F_{xy,d}$ = The resultant of all the horizontal forces.

$F_{z,d min}$ = The minimum vertical design force coexisting with $F_{xy,d}$.

μ_e = The friction coefficient given by the equation.

$$\mu_e = 0.1 + (1.5 K_f) / \sigma_m$$

K_f = 0.6 for concrete.

= 0.2 for all other surfaces including bedding resin mortars.

σ_m = The average of the compressive stress in megapascals from $F_{z,d min}$.

Where the bearing fail to satisfy the requirements for stability against sliding, alternative means shall be provided to resist the whole of the horizontal forces.

Forces, moments, and deformations exerted on the structure

a. Pressure on the contact surfaces

Elastomeric bearing exerts a non-uniform pressure on the contact surface with the structure. It is sufficient to ensure that mean pressure does not exceed the strength of the supporting material.

b. Force exerted on the structure by the bearing resisting translatory movement

The force R_{xy} exerted on the structure by the bearing resisting translatory movement is given by the expression -

$$R_{xy} = \frac{A G v_{xy}}{T_e}$$

Equation 8.21

Where,

R_{xy} = The resultant of the forces resisting to translatory movement

A = The total plan area of the bearing

G = The shear modulus of the bearing

T_e = The total thickness of elastomer in shear

The force R_{xy} shall not exceed the value specified.

c. Resistance to rotation

The design value of the restoring moment due to rotation about an axis through the centre of the bearing, parallel to the length (b direction), is given by the following equations:

For rectangular bearings:

$$M = G \frac{\alpha a'^5 b'}{n t_i^3 K_s}$$

Equation 8.22

For circular bearings:

$$M = G \frac{\alpha \pi D^6}{512 n t_i^3}$$

Equation 8.23

The value of restoring moment factor K_s may be taken from Table 8.10.

Table 8.10 Restoring Moment Factor K_s

b/a	0.5	0.75	1	1.2	1.25	1.3	1.4	1.5
K_s	137	100	86.2	80.4	79.3	78.4	76.7	75.3
b/a	1.6	1.7	1.8	1.9	2	2.5	10	∞
K_s	74.1	73.1	72.2	71.5	70.8	68.3	61.9	60

Note: If $b < a$, the formula is still applicable for rotation about the axis parallel to b , but in this case, b is the shorter dimension and a is the longer dimension.

d. Vertical deflection

The total vertical deflection v_c of a laminated bearing is the sum of the vertical deflection of the individual layers given by the expression -

$$v_c = \sum \frac{F_z t_i}{A'} \left(\frac{1}{5 G S_1^2} + \frac{1}{E_b} \right)$$

Equation 8.24

The value of the bulk modulus $E_b = 2000$ MPa may be used.

8.2.10.4 Plain Pad Bearings

This type of bearing, consisting of a solid block of elastomer without reinforcing plates, is not generally used for bridge structures. These bearings are only suitable for low pressure and predominantly static actions as indicated below:

8.2.10.4.1 Geometry

Plain pad bearings are generally of square, rectangular or circular plan area. The thickness shall not be less than 8 mm.

8.2.10.4.2 Loads

The mean design pressure, σ_{cd} , on a plain pad bearing is defined by the following equation:

$$\sigma_{cd} = \frac{F_{z,d}}{A}$$

Equation 8.25

Where,

$F_{z,d}$ = The vertical design load effect

A = The overall plan area of the plain pad bearing

The mean design pressure, σ_{cd} , shall not exceed $(1.4 G_d S)$ or $(7 G_d)$ whichever is the lesser where G_d is the design shear modulus of the elastomer and S is the shape factor for the elastomer pad.

8.2.10.4.3 Shear Strain

The provisions of Clause 8.2.10.3.5 shall apply for plain pad bearings also.

8.2.10.4.4 Stability Criteria**a. Rotation**

The provisions of Clause 8.2.10.3.8 shall apply.

b. Buckling

Thickness shall be less than 1/4 of minimum lateral dimension

c. Sliding

The provisions of 8.2.10.3.8 shall apply for all loads and following equation for permanent loads.

$$\frac{F_{z,d}}{A_r} > 1 + \frac{a}{b}$$

Equation 8.26

8.2.10.4.5 Deformations and Forces Exerted on the Structure

a. Vertical deflection

The deflection is given by the equation for a single layer in Clause 8.2.10.3.9 (ignoring term involving the bulk modulus).

b. Mean Pressure

The mean pressure is given by $F_{z,d} / A$.

c. Translatory

The translatory force arising from the shear strain given in Clause 8.2.10.3.9 shall be applicable.

8.2.10.5 Strip Bearings

This type of bearing consisting of a solid strip of elastomer without reinforcing plates is not generally used for bridge structures.

8.2.10.5.1 Geometry

The thickness of strip bearings shall not be less than 8 mm.

8.2.10.5.2 Loads

The mean design pressure, σ_{cd} , on a plain pad bearing is defined by the following equation:

$$\sigma_{cd} = \frac{F_{z,d}}{A}$$

Equation 8.27

Where,

$F_{z,d}$ = The vertical design load effect

A = The overall plan area of the strip bearing

The mean design pressure, σ_{cd} , shall not exceed the value of $(1.4 G S)$ or $(7 G)$ whichever is the lesser, where G is the nominal shear modulus of the elastomer and S is the shape factor of the elastomer strip.

8.2.10.5.3 Shear Strain

The calculation for determining $\epsilon_{q,d}$ described in Clause 8.2.10.3.5 shall apply. The shear strain $\epsilon_{q,d}$ shall be ≤ 0.3 .

8.2.10.5.4 Stability Criteria

a. Rotation

It should be ensured that: $\sum \delta > \frac{a \alpha_a}{3}$

Equation 8.28

b. Buckling

Thickness of the strip pad shall be less than 1/4 of width

c. Sliding

The provisions of Clause 8.2.10.3.8 for all loads shall be applied and following expression for permanent loads.

$$\frac{F_{z,d}}{A_1} > 1 + \frac{a}{b}$$

Equation 8.29

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*8.2.10.5.5 Deformations and Forces Exerted on the Structure***a. Vertical deflection**

The deflection is given by the equation for a single layer in Clause 8.2.10.3.9 (ignoring term involving the bulk modulus).

2

b. Mean Pressure

The mean pressure ($F_{z,d} / A$) $< G_d S$ or $5 G_d$ whichever is the lesser.

3

c. Translatory

The force arising from the shear strain shall be as given in Clause 8.2.10.3.9.

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8.2.10.5.6 Sliding Elastomeric Bearings

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Bearings of type D and E in Table 8.8 shall conform to the design requirements for laminated bearings (refer Cause 8.2.10.3.).

6

The maximum frictional force $F_{xy,d}$, when calculated in accordance with the provisions in Clause 8.2.9 (sliding bearings) shall comply with ($F_{xy,d} \leq R_d$) and ($R_d = A G$).

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9 Footbridges

9.1 General

A footbridges/ pedestrian bridges are intended to be used by pedestrians and/or cyclists. Where required, footbridges shall be provided to cross the physical obstacle such as over rivers or to avoid pedestrian – vehicle conflicts such as over roads to cross. This Chapter provides guidelines on general requirements, load models other design requirements of footbridges.

9.2 Layouts

Elegant proportions, appropriate structural form, landscaping and attractive details are important for any good footbridge design. The advice of an architect may be sought for good aesthetics of the footbridge. Adverse environmental and social impacts shall be minimised.

The location and arrangement of the footbridge shall be selected so as to maximise the use of local topography, placing the abutments at or close to adjacent ground level, where possible. This will minimise the need for stairs and ramps. The access ramps and stairs shall be simple, short and direct wherever possible avoiding long detours or unnecessary climbing.

Location of footbridges shall be carefully decided based on the aspects of planning considering the pedestrian volumes, type and need of the pedestrian facilities, environmental, social and other considerations, e.g. near schools, hospitals or crossing over a busy expressway, etc. Where a footbridge crosses a road, the design shall be such as to encourage pedestrians to use the footbridge rather than to cross the road at grade. Pedestrians can be encouraged to use the footbridge by providing suitable guardrails, fencing or appropriate planting to act as a physical barrier alongside the road.

Access to the deck of footbridge shall be provided by both ramps and stairs. The stairs may be omitted where ramps alone provide the most direct route to the deck.

Footbridges shall be designed with full or partial enclosure when there is a particularly high risk of the following:

- a. Objects being dropped or thrown from the footbridge; or
- b. Persons jumping onto the carriageway from the footbridge.

Some typical details and sections of a footbridge are shown in Figure 9.1.

Figure 9.1 Typical Details for Footbridges a) Typical Long Section of a Footbridge Over Road

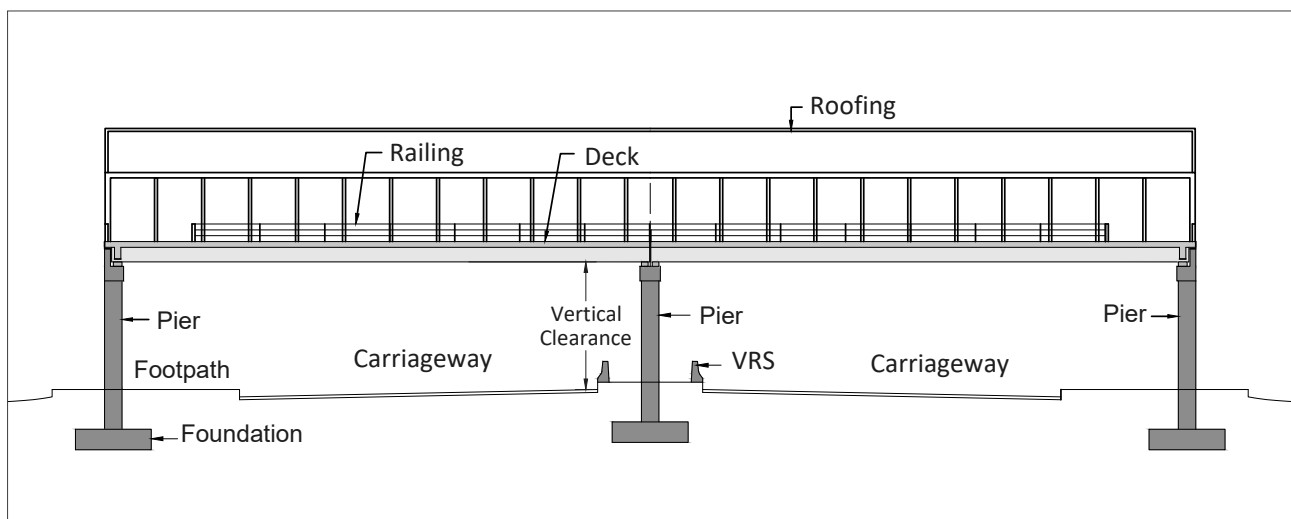


Figure 9.1 Typical Details for Footbridges b) Typical Plan of a Footbridge Over Road

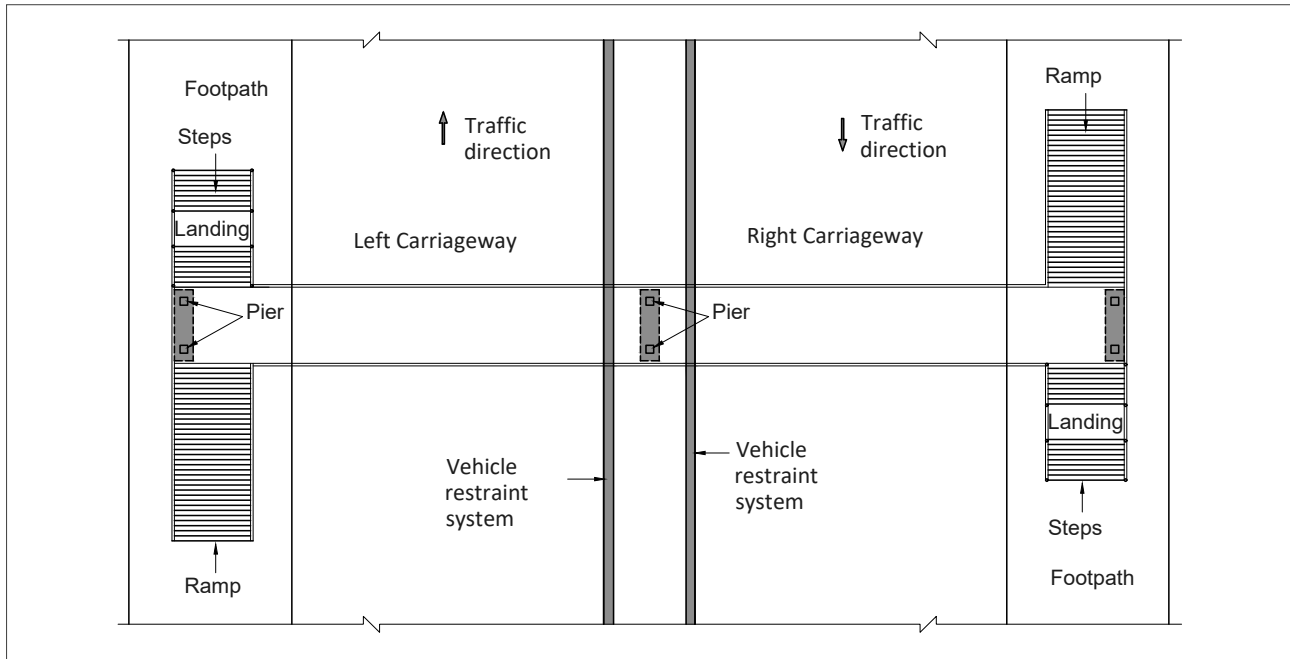


Figure 9.1 Typical Details for Footbridges c) Typical Arrangement of Ramp/stairs for a Footbridge

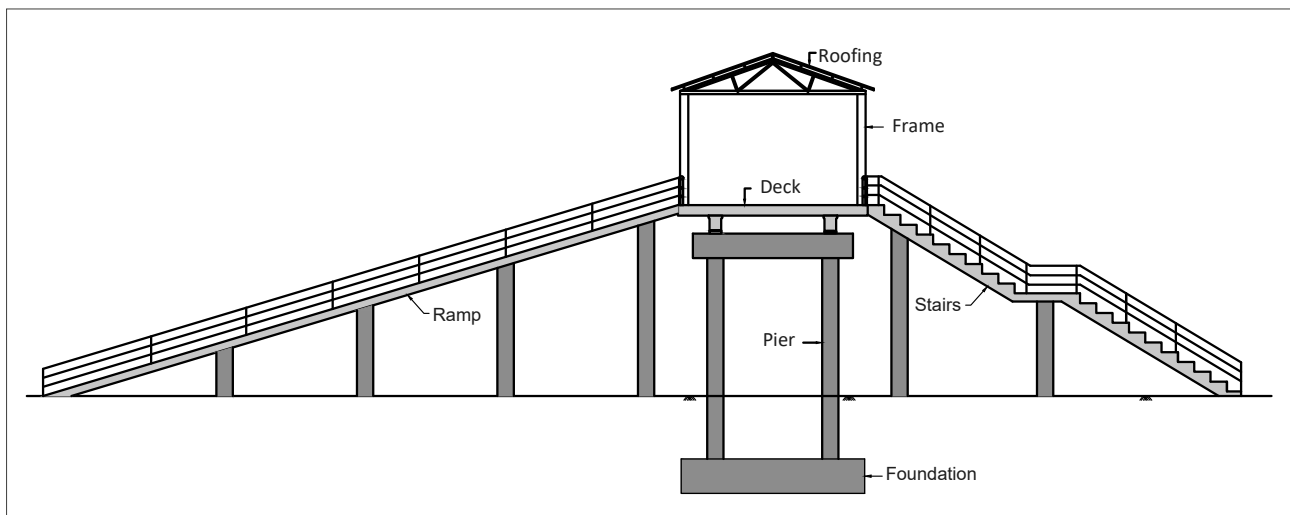
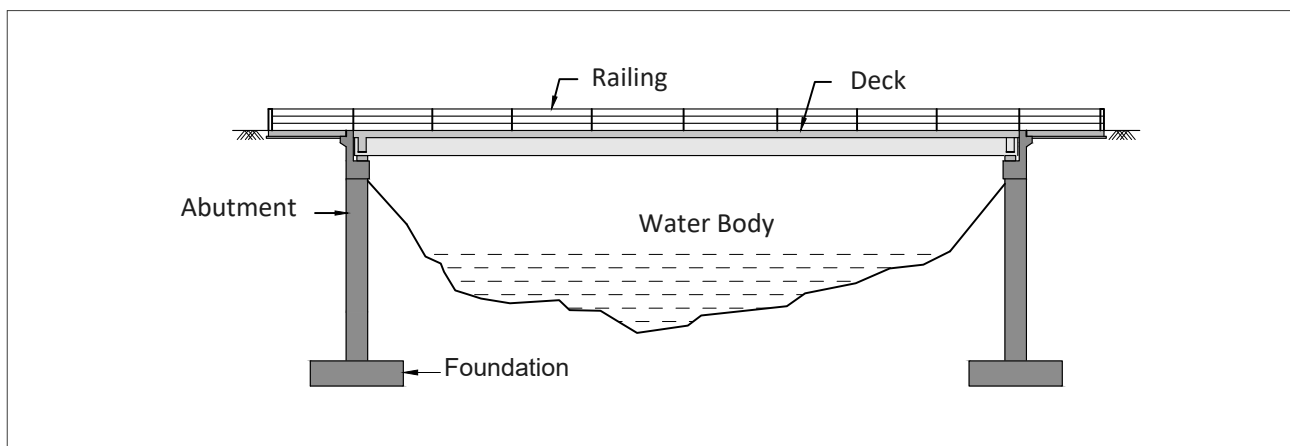


Figure 9.1 Typical Details for Footbridges d) Typical Long Section of a Footbridge Over Water Body



9.3 Dimensional Standards

9.3.1 Clearances

The vertical clearances over highway under a footbridge after the maximum deflection of the superstructure at the serviceability limit state shall be greater than the minimum requirement of 5.70 m. When the road passing under the footbridge is on sag curve, the vertical clearance shall be increased to 5.80 m if the sag radius is less than 6000 m. The lateral clearance from the edge of the carriageway of highway to the supports of footbridge shall be a minimum of 4.5 m. The vertical and horizontal clearances to railways, canals and watercourses should be agreed with the Approving Authority

The minimum vertical clearance on the footbridge to the underside of any overhead obstacles such as roof enclosures, measured vertically from the walkway surface shall be as provided in Table 9.1.

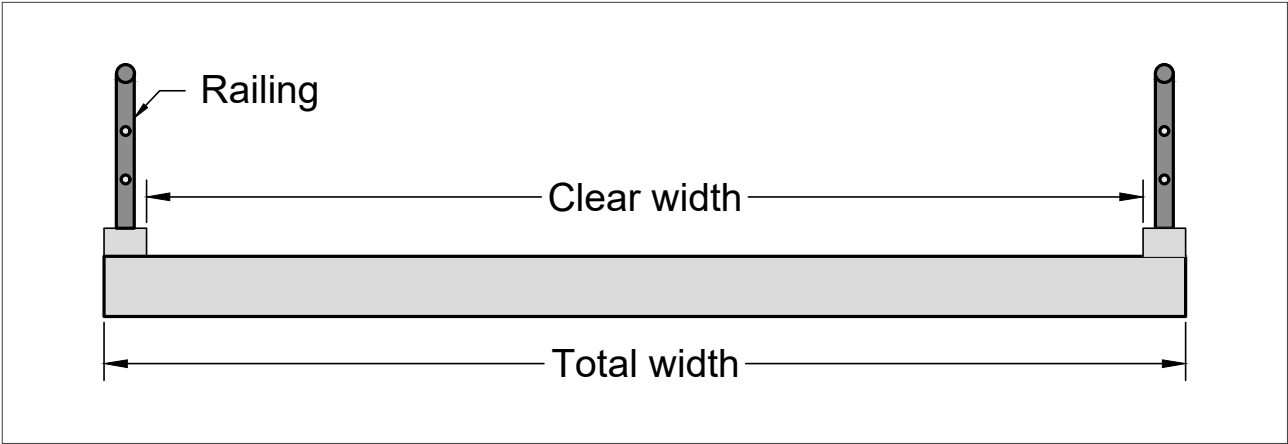
Table 9.1 Minimum Vertical Clearance on Footbridges

Type of Use	Minimum Clearance
For use by pedestrians only	≥ 2.3 m
For use by pedestrians and cyclists	≥ 2.4 m

9.3.2 Width of Footbridges

Pedestrian facility should provide enough space for safe and comfortable walking. The width should be provided based on the current and expected pedestrians. Footbridge may have segregated provision for pedestrians and cyclists, in which case it will need to be wider. Parapets are provided for the safety of the pedestrians. Continuity of pedestrian facilities shall be maintained on approaches to footbridge.

Figure 9.2 Typical Cross Section for Footbridge



The width of footbridge shall be in accordance with the provision in “RDM 1.3”. The minimum clear width of a pedestrian bridge, ramps and stairs should be 2.0 m. Where the pedestrian volume is more, the clear width of pedestrian bridge shall be greater of 2.0 m or as determined from the Table 9.2 for the predicted peak pedestrian traffic.

Footbridges should normally be designed for pedestrian Level of Service (LOS) B. This will provide pedestrian facilities for pleasant and comfortable walking. Under resource constraint, Level of Service C can be adopted in deciding the width of pedestrian facilities.

Table 9.2 Minimum Clear Width of Footbridges for Pedestrian Traffic

Width of Walkway (m)	Design Flow in Number of Persons Per Hour			
	In Both Directions		All in One Direction	
	LOS B	LOS C	LOS B	LOS C
1.8	1350	1890	2025	2835
2.0	1800	2520	2700	3780
2.5	2250	3150	3375	4725
3.0	2700	3780	4050	5670
3.5	3150	4410	4725	6615
4.0	3600	5040	5400	7560

The footbridges designed for the combined use by pedestrians and cyclists may be either segregated or unsegregated. The minimum clear usable widths for the walkway and cycle path on shared use of footbridges and ramps shall be as provided in Table 9.3.

Table 9.3 Minimum Width of Footbridges for Shared Use by Pedestrians and Cyclists

Footbridge Configuration	Minimum Width for		
	Walkway	Cyclepath	Total
When segregated by a kerb not less than 50mm high	2.0 m	2.7 m	4.7 m
When segregated by a physical barrier not less than 900mm high	2.0 m	3.0 m	5.0 m
When segregated by a white line and/or contrasting surface colours or textures	1.5 m	2.5 m	4.0 m
Unsegregated	–	–	3.5 m

When service vehicles are to be carried on a pedestrian bridge, the width shall not be less than 3.0m. This vehicle may be a vehicle for maintenance, emergencies, ambulance, or other services.

9.3.3 Maximum Gradients

The maximum gradient on the footbridge and approach ramps shall not be steeper than 1 in 20 unless special circumstances apply. Where special circumstances apply, a relaxation in ramp gradient to 1 in 15 may be permitted, or even to 1 in 12 in cases of extreme difficulty. Gradients steeper than 1 in 12 shall not be permitted.

9.3.4 Landings on Ramps

The intermediate horizontal landings shall be provided on ramps as follows:

- Intermediate landings are not required if the gradients are flatter than 1 in 22
- If the gradients are between 1 in 20 to 1 in 22, intermediate landings shall be provided at equal vertical rise interval not exceeding 2.5 m.
- For gradients steeper than 1 in 20, intermediate landings shall be provided at vertical rise interval not exceeding 0.65 m.

The length of a landing shall not be less than 2.0m along centre centreline for straight ramps or at a distance of 0.90m from inside handrail for curved ramps.

When gradient is steeper than 1 in 20, there shall be change in alignment at intervals corresponding to a vertical rise of 3.5 m. This change may be achieved either by change in direction not less than 30° angle or by an offset in the alignment not less than width of the walkway. This change may be exempted if no other arrangement of ramps is possible on site.

9.3.5 Spiral and Curved Ramps

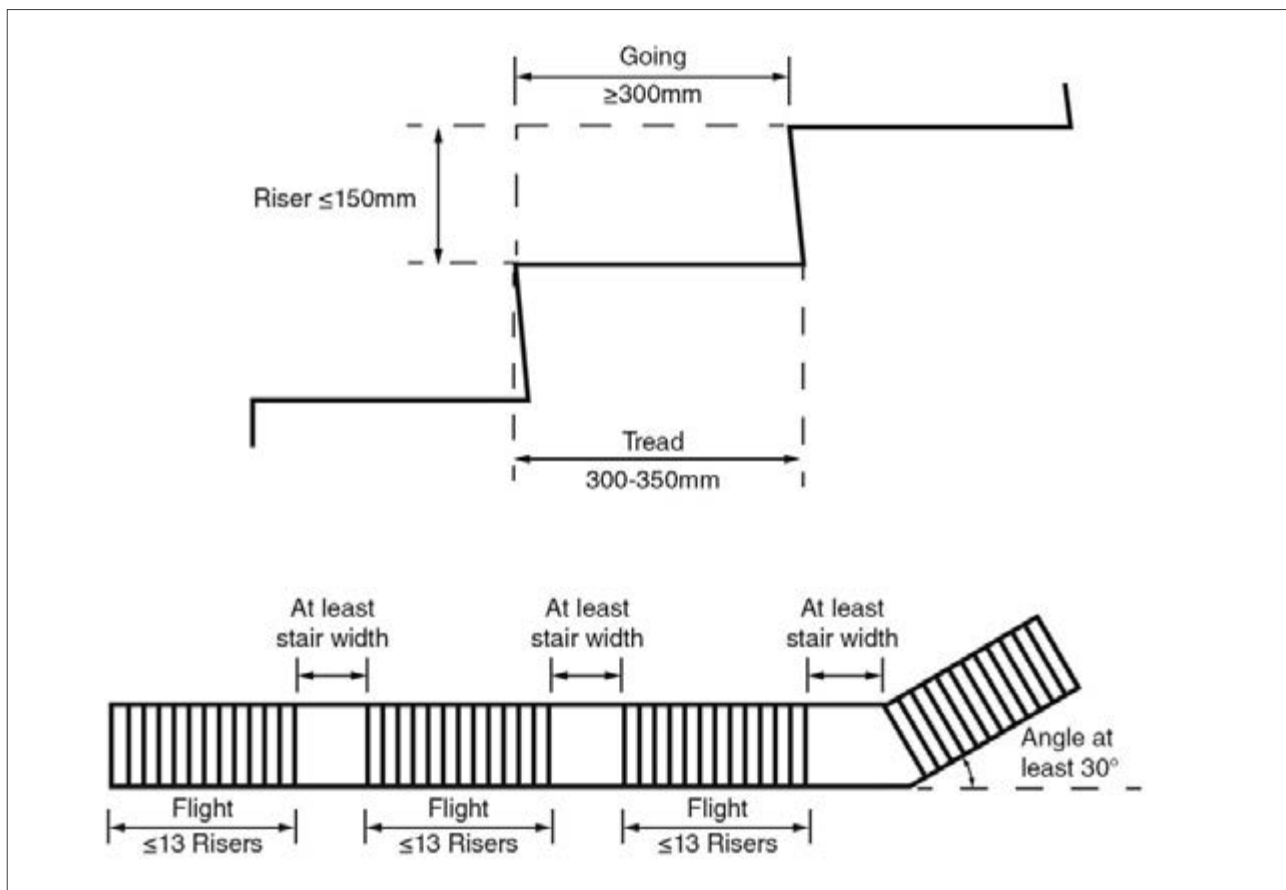
The governing dimensions and effective gradients for spiral and curved ramps shall be measured at 0.90 m away from inner edge of walkway on inside of the curve for the purpose of assessing the maximum gradient requirements. The minimum inside radius of the walkway for spiral and curved ramps shall be 5.5 m. A larger curve radius shall be provided if the bridges are to be used by cyclists.

9.3.6 Stairs

Stairs shall be designed with following requirements.

- a. Number of risers in a single flight shall be ≤ 13 .
- b. Riser dimension shall be ≤ 150 mm and be constant over the width of the stairs.
- c. Going dimensions ≥ 300 mm.
- d. The risers and goings of each step in a flight of stairs shall be uniform.
- e. After having maximum three successive flight, there shall be change in direction of at least 30° angle.
- f. The length of landing shall not be less than the width of the stairs or 2 metres, whichever is the greater, when measured along the centre line of the stairs.
- g. Risers may be solid or perforated but shall not be completely open.
- h. If perforated risers are provided, the openings shall meet the following requirements:
 - i. Maximum principal dimension of the perforation = 50 mm
 - ii. Maximum ratio of the open area to the total area of the riser = 0.4

Figure 9.3 Plan and Section of Public Stairs for Footbridges



9.3.7 Parapets

All footbridges, ramps and stairs shall be provided with parapets. Parapets provide safety and sense of security to users. It also affects the visual appearance and landscape of the bridge. The parapets shall be robust and durable.

The parapets shall meet the requirements provided in “RDM 1.3”.

The height of the edge upstand under the parapet for plain or spiral ramps shall be not less than 25 mm and no more than 50 mm. An upstand is not required under the parapet on stairs. Where a stair parapet is provided with a bottom rail, the clearance from the rail to the nose of the stairs shall be at least 50 mm but not greater than 100 mm.

Handrails shall be provided on both sides of stairs and the walkway on bridge decks and ramps. The top of the handrail be between 0.9m and 1.0 m above the walkway surface measured vertically. The handrails of circular section shall have a diameter between 40 mm and 50 mm. The non-circular handrail section should not be less than 50 mm wide by 38 mm deep with rounded edges.

The handrail and its fixings shall be designed to resist a uniformly distributed load of 0.7 kN/m applied separately in the horizontal and vertical directions such that the system is designed for the most severe effects.

9.4 Lighting

Footbridges shall be illuminated where they are located in area where public lighting is provided. The lighting shall conform the requirements of BS 5489-1. The design shall take into account the level and nature of illumination for safety reasons and also provide visual impact and aesthetics.

Lighting design is a specialist field and the services of a specialist lighting designer shall be obtained to achieve the quality of lighting demanded by bridge and its environment.

The design shall avoid unnecessarily excessive illumination and any direct glare in the eyes of bridge users. The illumination shall be consistent to ensure that all walkway surfaces, stairs and handrails are visible.

Footbridges may be illuminated by means of existing adjacent road or walkway lighting. Where it is not possible to illuminate a footbridge by means of existing lighting from the adjacent road or walkway, the footbridge may be illuminated by:

- Light fittings incorporated into the parapet or handrail; or,
- Other light fittings mounted on the bridge.

Footbridge lighting shall be on a circuit that can be isolated. All components of the lighting system, including all fittings, connections, wiring and switchgear, shall be robust and tamper proof. Parapet members should not be used as cable ducts. Wherever possible, luminaires shall be accessible by authorised personnel for maintenance without the need to provide special access facilities.

9.5 Drainage

Adequate provisions shall be made for drainage of water from the footbridges and from its roof in case of enclosed footbridges. The walkway surfaces, steps, ramps and roof shall have adequate gradients/ falls and suitable detailing to allow water to drain off in the intended manner. Water from the footbridge shall be carried away to a drainage system in the area or to a soakway.

Drainage shall be provided beneath all deck movement joints and be directed away from bearings and bearing shelves. Steel decks shall be carefully detailed and fabricated such that it should not lead to water ponding on the deck. Deck distortions can be avoided by the use of 8 mm deck plate spanning no more than 800 mm. The deck plate can be cambered to provide a transverse cross fall to the edges of the deck to ensure surface water is shed to the sides and directed to a drainage system.

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Footbridges

The longitudinal profile of the bridge shall be designed with pre-camber to overcome deflections due to self-weight and live action. The pre-camber should be sufficient to ensure positive fall to the drains under SLS load conditions.

The design and detailing shall ensure that there is no possibility of water ponding, particularly at corners of landings. Steps and nosing of stairs shall be detailed to prevent water getting trapped on the steps.

Erosion protection at drainage discharge points shall be provided to prevent damage to structure foundations and adjacent soft surfaced areas. The provision of access to drainage system shall be made to allow safe periodic/routine maintenance operations to be undertaken.

9.6 Surfacing

The walkway surfacing system is an important part of the overall appearance, performance and durability of the bridge, and shall be assessed as an integral part of the design to suit the particular application. The suitability of the surfacing system shall be assessed in line with all performance requirements such as corrosion resistance, slip resistance, environmental deterioration, noise transmission, free drainage and durability.

The upper surface of bridge decks, ramps and stairs shall be waterproofed or otherwise protected against deterioration due to water or surface contaminants.

All traversed surfaces shall be designed to have minimum slip resistance for the life of the walkway surfacing equivalent to a mean corrected pendulum test value of 45 units using a standard skid resistance pendulum test in accordance with BS EN 13036-4.

Cover plates to expansion joints at deck level shall be flush with the walkway surface and provided with a suitable slip resistant coating or be profiled to provide a non-slip finish. The width of any exposed gap in the walkway surface, such as at movement joints, shall not exceed 12 mm, taking into account movements due to temperature

Where the bridge is designated for use by cyclists, any gap, joint or discontinuity in the bridge deck surface that is aligned within a plan angle of less than 30° to the normal line of travel shall be flush with the surface and such that a bicycle wheel cannot be caught or deflected when it passes over.

9.7 Roofing on Footbridges

Footbridges shall be designed with full enclosure (from sides and the roof) or partial enclosure (from sides) where it is assessed that there is a particularly high risk of the following:

- Objects being dropped or thrown from the footbridge, or
- Persons jumping onto the carriageway from footbridge.

Where a footbridge is exposed to very adverse weather or strong winds, such as where it is unusually high above the surrounding ground level, or where it is so high above the road that pedestrians can feel insecure, then full enclosure may be required. Depending on the issues relevant at a particular site, a high parapet with an inward canted top may suffice in place of full enclosure. The minimum clearance below the roof shall comply with Clause 9.3.1.

Where full or partial bridge enclosure is proposed, the aerodynamic and other effects of wind both on the structure and on the comfort of its users shall be investigated and analysed. Specialist advice may be necessary to assess the aerodynamic and other effects of wind. Wind tunnel testing shall be considered where reliable design guidance is not available for the form of bridge design.

9.8 Actions on Footbridges

The load models defined in this section are applicable for footbridges used for pedestrians and cycle tracks. For the design of footbridges, load models for uniformly distributed loads, concentrated loads and service vehicle loads if applicable shall be considered. This section provides the details on load models for pedestrians/ cyclists traffic, actions for accidental design situations, dynamic load models. For other actions and combination of actions, refer Chapter 3.

9.8.1 Static models for vertical loads

Three models, mutually exclusive, shall be taken into account where relevant. The characteristic values of these load models shall be used for both persistent and transient design situations.

9.8.1.1 Uniformly Distributed Load

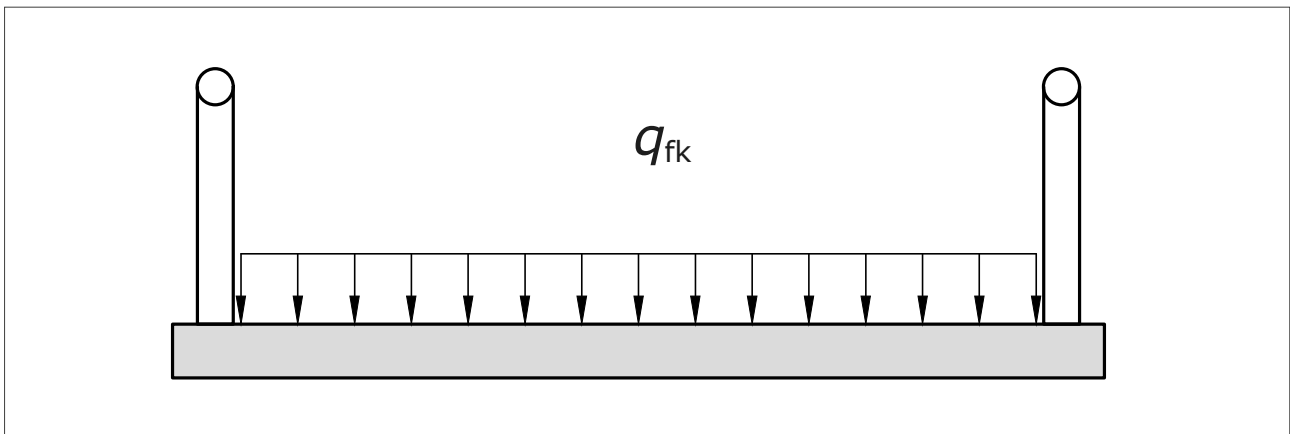
For design of footbridges, where the risk of continuous dense crowd exists, a uniformly distributed load (Load Model 4 defined in Clause 3.3.3.4) with a characteristic value $q_{fk} = 5 \text{ kN/m}^2$ shall be used. For other cases, the uniformly distributed load q_{fk} shall be taken as under.

$$q_{fk} = 2.0 + 120 / (L + 10) \quad \text{kN/m}^2, \text{ where } L \text{ is loaded length}$$

$$2.5 \text{ kN/m}^2 \leq q_{fk} \leq 5.0 \text{ kN/m}^2$$

For the design of footbridges, the load should be applied longitudinally and transversally on an area that produce most unfavourable effect.

Figure 9.4 Live Load on Footbridges



9.8.1.2 Concentrated Load Model

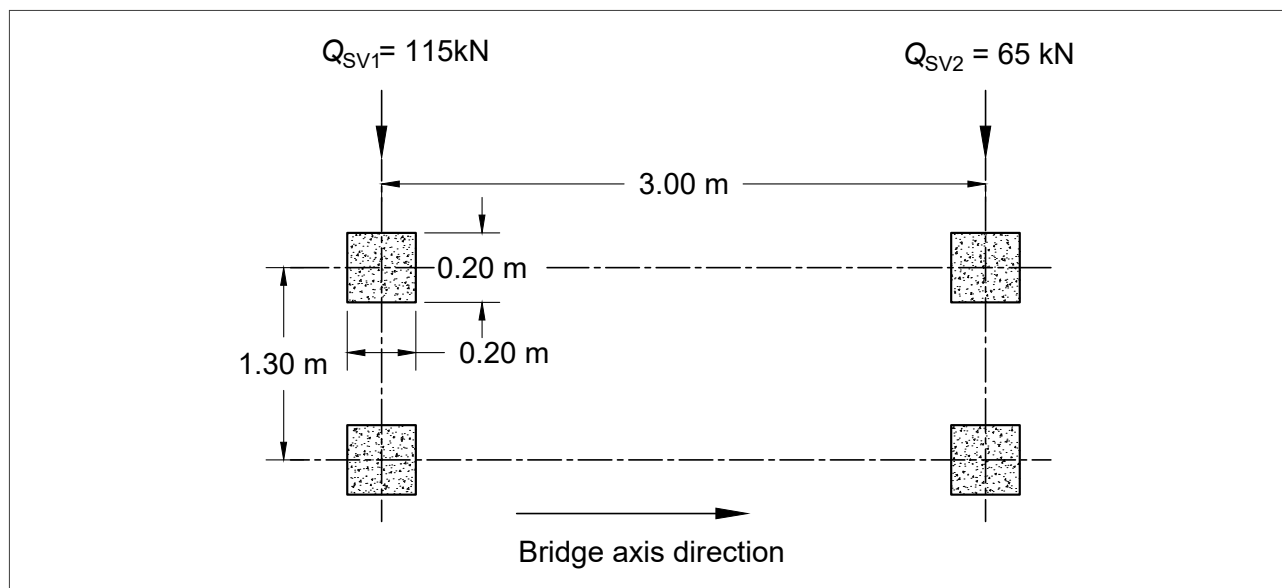
This load model consists of a concentrated load with a characteristic value of $Q_{fwk} = 10 \text{ kN}$ acting on a square surface of sides 0.10 m. This shall be considered for verification of local effects. For footbridges, if a service vehicle is considered, this model should not be considered.

9.8.1.3 Service Vehicle Load

When service vehicles are to be carried on a footbridge, one service vehicle Q_{serv} shall be taken into account. This vehicle may be a vehicle for maintenance, emergencies, ambulance, or other services.

The characteristics of the service vehicle (axle weight and spacing, contact area of wheels, etc.), the dynamic amplification and all other appropriate loading rules may be defined depending upon the particular service allowed for the footbridge. Where footbridges do not have permanent provision to prevent the entry of vehicles on the footbridges, the vehicle model given in Figure 9.5 shall be used.

Figure 9.5 Service Vehicle Load



Service vehicle needs not be considered if permanent provisions are made to prevent access of all vehicles to the footbridge.

9.8.2 Static Model for Horizontal Forces

For footbridges, a horizontal force Q_{flk} should be taken into account, acting along the bridge deck axis at the pavement level. The characteristic value of the horizontal force should be taken equal to the greater of the following two value

10% of the total load corresponding to the uniformly distributed load,

60% of the total weight of the service vehicle, if relevant

The horizontal force is considered as acting simultaneously with the corresponding vertical load, and in no case with the concentrated load Q_{fwk} .

9.8.3 Groups of Traffic Loads on Footbridges

The vertical loads and horizontal forces due to traffic should be taken into account by considering groups of loads defined in Table 9.4. Each of these groups of loads, which are mutually exclusive, should be considered as defining a characteristic action for combination with non-traffic loads.

Table 9.4 Group of Traffic Loads on Footbridges

Load Type		Vertical forces		Horizontal forces
Load System		Uniformly distributed load	Service vehicle	
Group of loads	gr1	q_{fk}	0	Q_{flk}
	gr2	0	Q_{serv}	Q_{flk}

9.8.4 Actions for Accidental Design Situations for Footbridge

9.8.4.1 General

The accidental actions may be due to collision of road traffic under the bridge or accidental presence of heavy vehicle on the footbridge. Measures to protect footbridge from collision shall be adopted by:

- Providing restraint systems/ safety barriers at appropriate distance before the piers or by
- Providing a higher clearance than on the neighbouring road bridge over the same road.

9.8.4.2 Collision Forces on Piers of Footbridges

When vehicular collision forces are to be taken into account, for stiff piers, a minimum impact force of 1000 kN in the direction of vehicle travel or 500 kN perpendicular to that direction at a height of 1.25 m above the adjacent level shall be considered in the design. These recommended values are as per the Eurocode. However, according to the UK National Annex, reference shall be made to EN 1991-1-7 with its UK National Annex and PD 6688-1-7 for details on collision forces on piers of footbridges and collision forces on decks of footbridges.

9.8.4.3 Collision Forces on Decks of Footbridges

The vehicle collision loads on superstructure of footbridges are not applicable as the footbridges are required to have vertical clearance exceeding the applicable limit and less likely to have impact. However, adequate restraint on the deck of footbridge shall be provided to prevent the deck being removed from the support under the action of the vehicle collision forces.

9.8.4.4 Accidental Presence of Heavy Vehicles

The vehicle load model to determine the actions due to accidental presence of heavy vehicle on the footbridges, where no permanent obstacle is provided, shall be taken as defined in Clause 9.8.1.3.

9.8.5 Action on Parapets

For footbridges, the pedestrian parapets shall be designed in accordance with the provisions given in Clause 3.5.7.

9.8.6 Load Model for Abutments and Walls Adjacent to Footbridges

The area external to a carriageway and located behind abutments, wing walls, side walls and other parts of the footbridge in contact with earth, should be loaded with a uniformly distributed vertical load of 5 kN/m².

9.8.7 Dynamic Models of Pedestrian Loads

9.8.7.1 General

The modern lightweight footbridges can be susceptible to vibrations caused by users. Depending on the dynamic characteristics of the structure, the relevant natural frequencies corresponding to vertical, horizontal, torsional vibrations of the main structure of the bridge deck should be determined from an appropriate structural model.

Forces exerted by pedestrians with a frequency identical to one of the natural frequencies of the bridge can result into resonance and need to be taken into account for limit state verifications in relation with vibrations.

Note: The dynamic response of footbridge structures is a relatively specialist area and specialist advice where necessary to derive an appropriate prediction of the structural response and pedestrian comfort shall be obtained.

In the absence of significant response of the bridge, a pedestrian normally walking exerts on it the following simultaneous periodic forces:

- In the vertical direction, with a frequency range of between 1 and 3 Hz, and
- In the horizontal direction, with a frequency range of between 0.5 and 1.5 Hz.
- Groups of joggers may cross a footbridge with a frequency of 3 Hz.

9.8.7.2 Assessment of Pedestrian Traffic Classes

The assessment of pedestrian traffic classes and the corresponding pedestrian densities are to be made according to following Table (based on JRC Scientific and Technical Report - EUR 23984 EN).

Table 9.5 Pedestrian Traffic Classes and Densities

Traffic Class	Density (d) Pedestrian/m ²	Description	Characteristics
TC 1	Group of 15 pedestrian $d = 15 Pl (B L)$	Very weak traffic	B = width of deck L = length of deck
TC 2	$d = 0.2$	Weak traffic	Comfortable and free walking, overtaking is possible, single pedestrians can freely choose pace
TC 3	$d = 0.5$	Dense traffic	Still unrestricted walking, overtaking can intermittently be inhibited
TC 4	$d = 1.0$	Very dense traffic	Freedom of movement is restricted, obstructed walking, overtaking is no longer possible
TC 5	$d = 1.5$	Exceptionally dense traffic	Unpleasant walking, crowding begins, one can no longer freely choose pace

The traffic class to be adopted for design shall be agreed for individual project.

9.8.7.3 Assessment of Pedestrian Comfort Class

Pedestrian comforts are commonly represented as a limiting acceleration for the footbridge. The recommended comfort class are as under:

Table 9.6 Comfort Classes and Common Acceleration Ranges

Comfort Class	Degree of Comfort		Vertical Acceleration Limit (m/s ²) a_{Limit}	Lateral Acceleration Limit (m/s ²) a_{Limit}
CL1	Maximum	Acceleration undergone by the structure are practically imperceptible to users	< 0.5	< 0.1
CL2	Medium	Acceleration undergone by the structure are merely perceptible to users	0.5 – 1.0	0.1 – 0.3
CL3	Minimum	Acceleration undergone by the structure are perceived by the users, but do not become intolerable	1.0 – 2.5	0.3 – 0.8
CL4	Unacceptable discomfort	Uncomfortable acceleration levels that are not acceptable	> 2.5	> 0.8

The comfort criteria should be defined as appropriate for the individual project in terms of the maximum acceptable acceleration of any part of the deck.

9.9 Structural Design of Footbridges

The footbridges are subjected to various actions as detailed in this Chapter and the different elements of footbridges are analysed to determine the action effects.

The minimum thickness of structural elements of footbridges shall be as in Table 9.7.

Table 9.7 Minimum Thickness of Structural Elements for Footbridges

Element	Dimension
Concrete deck slab thickness	≥ 150 mm
Thickness of steel plates and sections other than hollow sections effectively sealed by welding	≥ 6 mm
Thickness of steel hollow sections effectively sealed by welding	≥ 5 mm

Structural design of concrete footbridges shall be done in accordance with the guidelines provided in Chapter 4. The guidelines provided in Chapter 5 and Chapter 6 shall be used for structural design of steel and composite footbridges. The verification of structural elements shall be made under Ultimate Limit State and Serviceability Limit State.

Modern lightweight footbridges can be susceptible to vibrations caused by users. Footbridges with vertical modes of vibration less than 5 Hz and/or lateral modes of vibration less than 2.5 Hz can be particularly susceptible, and the resulting motions can cause discomfort to bridge users and even lead to structural damage in extreme cases. Footbridges shall be checked for comfort class.

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Footbridges

10 Culverts

10.1 General

10.1.1 Scope

Culverts are small cross drainage structures used to carry the storm water across any road embankment. The location, size and other details of culverts shall be decided judiciously to cater for the requirements of design discharge and balancing of water levels on either side of the road embankment. This Chapter provides guidelines on the design of reinforced concrete box culverts and circular pipe culverts which are most commonly-used type of culverts. The concrete box culverts can be pre-cast in segments or cast-in-situ.

10.1.2 Reference Documents

- BS 5911-1** Concrete pipes and ancillary concrete products (unreinforced and reinforced concrete pipes).
- BS 5911-5** Concrete pipes and ancillary concrete products (pre-stressed non-pressure concrete pipes).

10.1.3 Definitions

For general definitions, refer initial sections of this Manual. Other definitions are as under

Buried pipe system	Composite structure comprising the pipe and the surrounding soil embedment
Earth cover	Height of fill measured between road surface level and the top of the roof of a buried structure.
Embedment	Arrangement and type of materials around a buried pipe which contribute to its structural performance
K_{max} pressure	Maximum pressure applied behind the abutment of a buried structure.
Traffic surcharge	Traffic applied behind bridge abutment and retaining walls which generates horizontal pressure on the walls

10.1.4 Symbols

The general symbols used in this Chapter are as under. Additional symbols are also defined where they appear. Symbols may have various meanings.

B_c	Outside diameter of pipe
B_d	Effective width of trench
C_c	Soil load coefficient in embankment or wide trench condition
C_d	Soil load coefficient in narrow trench condition
C_w	Water load coefficient
D	Mean diameter of pipe
F_m	Bedding factor
F_{se}	Factor of safety for rigid pipe material in external load design
H	Depth of cover to top of pipe
H_c	Height of earth cover between road surface level and the top surface of a buried structure
H_e	Height of plane of equal settlement above top of pipe
K	Coefficient of lateral earth pressure

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 K_a Coefficient of active earth pressure

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 K_d Design value of K based on K_0 K_{max} Coefficient of earth pressure applied to buried structures which takes account of pressure increases caused by expansion of the structure

3

 K_{min} Coefficient of minimum earth pressure applied when earth pressure is favourable K_p Coefficient of passive earth pressure

4

 K_r Coefficient of earth pressure resisting overturning or sliding P_s Vehicle surcharge pressure

5

 S_f Settlement of pipe invert S_g Settlement of natural ground S_m Settlement of fill

6

 t Pipe wall thickness W'_c Soil load per unit length of pipe in embankment or wide trench conditions W_c Soil load per unit length of pipe in narrow trench conditions

7

 W_{csu} Concentrated surcharge load per unit length of pipe W_e Total external load per unit length of pipe $W_{e,Ed}$ External load per unit length of pipe

8

 W_p Proof load for pipes W_s Safe supporting strength of the pipe/bed combination W_t Crushing strength of pipe

9

 W_w Weight of water inside of pipe r_{sd} Settlement deflection ratio γ Unit weight of soil

10

 γ_G Partial factor for permanent actions $\gamma_{G,intw}$ Partial factor for internal water weight γ_Q Partial factor for traffic/surcharge $\gamma_{Sd,K}$ Model factor for to be applied to earth pressure at ULS $\gamma_{Sd,ec}$ Model factor to be applied to the weight of earth cover at SLS and ULS γ_w Unit weight of water θ Skew angle of bridge/ culvert μ Coefficient of sliding friction within a mass of soil μ' Coefficient of sliding friction of backfill against trench sides

10.2 Design Standards

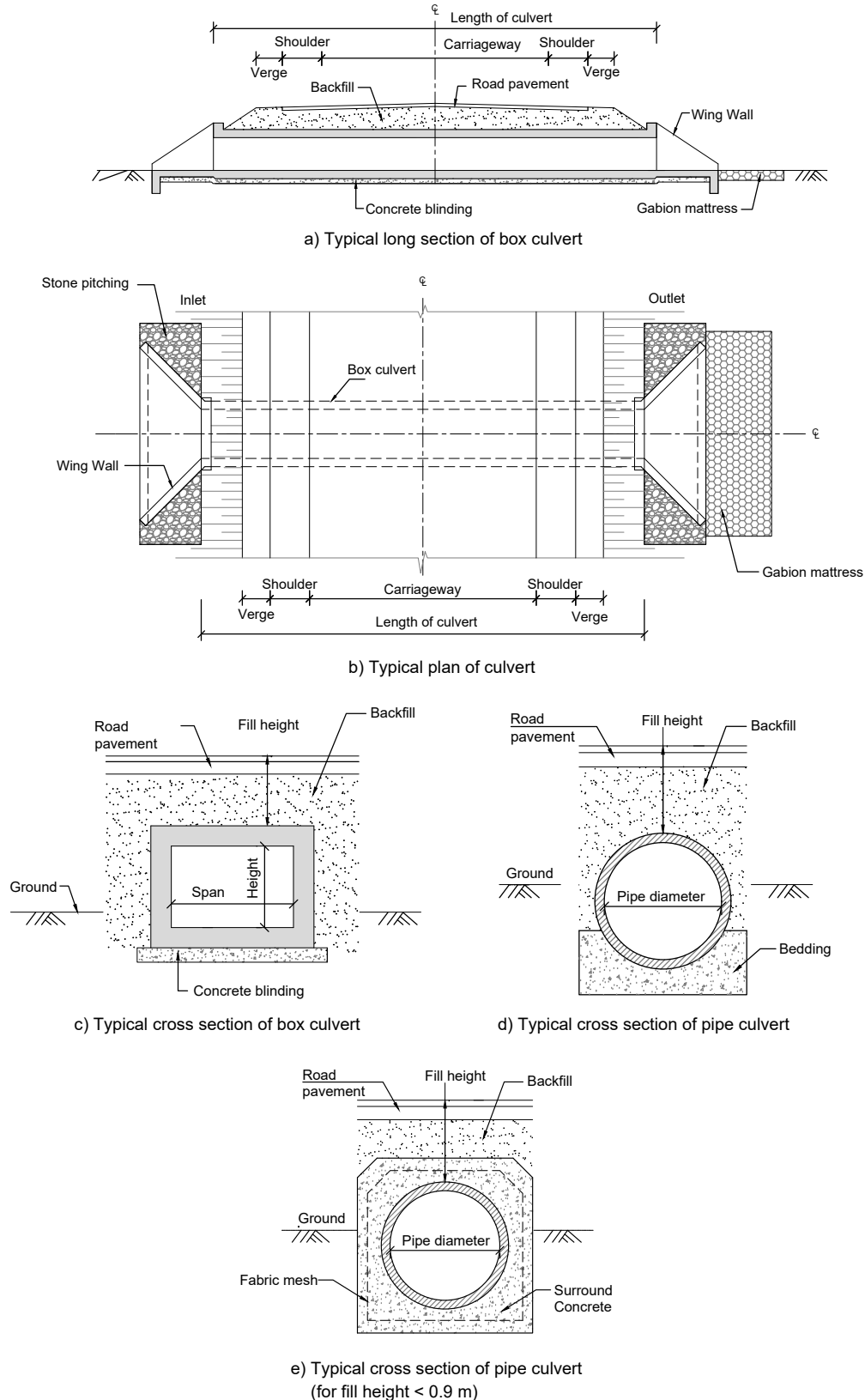
Sizing of the culvert shall be assessed with the hydraulic design guidelines provided in "RDM Vol 2" for the estimated design flood flows. The minimum culvert diameter shall not be less than 450mm for small culverts (e.g. on minor roads) as small sizes are prone to blockage. Where a circular culvert is longer than 12 metre, the diameter shall not be less than 1.2 metres to facilitate access for maintenance. The minimum size for rectangular culvert shall not be less than 500mm span and 450 mm height. Where a box culvert is longer than 12 metres, the span and height shall not be less than 1.2 metres to facilitate access for maintenance. The fill height over top of pipe up to road level shall not be less than 0.90 m when granular bedding is used. When this height is not achieved due to alignment or other constraints, the pipe shall be surrounded all around with concrete. The fill height is the total height from top of pipe to road surfacing including the thickness of pavement layers. Some

typical sections of culverts are shown in Figure 10.1.

10.3 Layout of Culverts

Culverts beneath highways have to accommodate restrictions imposed by carriageway alignment

Figure 10.1 Typical Sections for Culverts



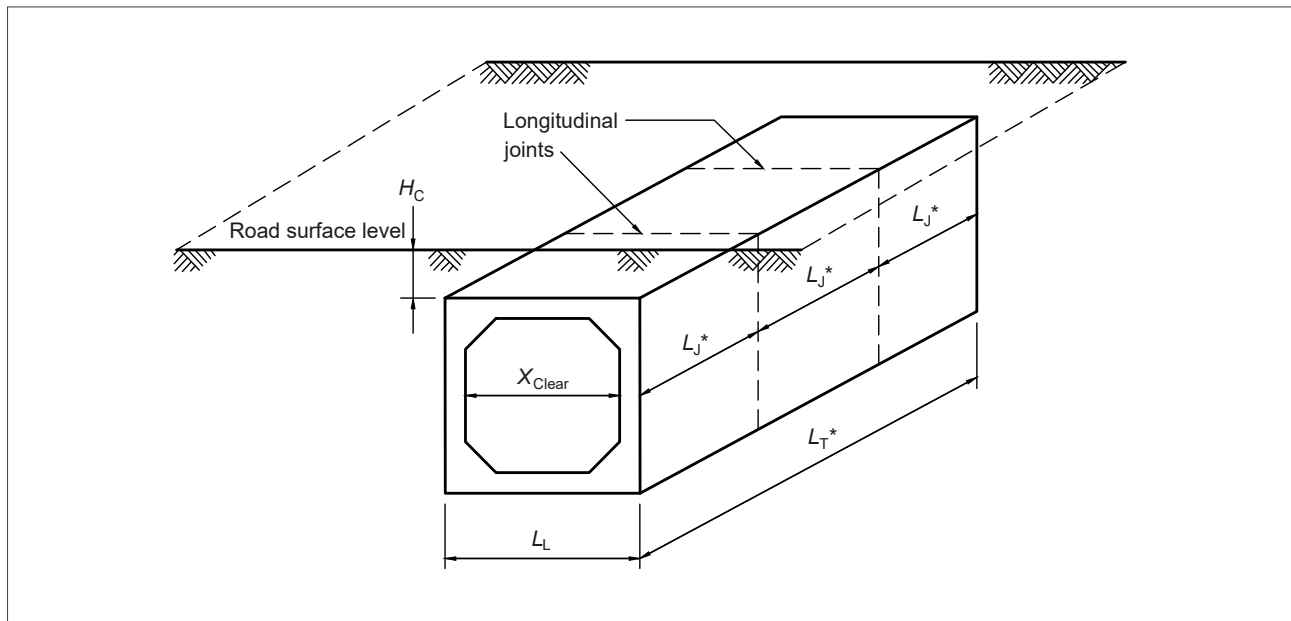
and road construction, as these affect the cover to the top of the culvert. Some typical sections of culverts are shown Figure 10.1.

Bedding is required to provide support to the pipe culvert. The bedding material can be compacted earth, granular material or lean concrete depending upon the nature of the ground and the shape and material of the culvert pipe. Foundation shall be excavated to the extent required to place the bedding material. Any weak soils found should be removed and replaced with compacted backfill.

10.4 Box Culverts

10.4.1 General

The guidelines provided in this Clause are applicable for design of buried concrete box and portal frame structures, where the overall longitudinal length (L_L) does not exceed 15 m and the fill height over culvert up to road surface level (H_C) does not exceed 11 m. The symbols used for a typical



buried box culvert are shown in Figure 10.2.

10.4.2 Design Principles for Actions

The buried concrete structures (box and portal frame structures) with a height of earth cover less than 0.6m shall be treated as normal bridge structure and be designed considering traffic actions as per Clause 3.3, thermal actions as per Clause 3.10 and earth pressures as per Clause 3.7 and as described in this section. The buried concrete structures with more than 0.6m earth cover may also be designed with these guidelines considering following additional provisions:

- Dispersion of the vertical traffic actions through the fill in accordance with Clause 10.4.7.
- Reduction of the braking and acceleration forces applied to the structure (refer Clause 10.4.8.2).
- Modification in thermal actions as described in Clause 3.10 and 10.4.9.

10.4.3 Superimposed Permanent Load

The superimposed permanent load over buried structure consists of weight of the earth cover and the road construction materials above the structure. The effect of superimposed permanent load over buried structure may be influenced by positive and negative arching action. The possible effects of positive arching, which has reducing effect to this load, should be ignored.

When the consolidation or settlement of the fill adjacent to a buried structure takes place, it causes negative arching of the fill above the roof and the increased loading will be generated on the roof slab. The downdrag effects can be greater in situations where the structure is founded on hard material or piles so that the backfill settles more than the structure.

In the absence of a more rigorous analysis, the effects of differential settlement between the structure and the adjacent ground should be taken into account by applying a supplementary model factor $\gamma_{Sd,ec}$ to the effects of the weight of the earth cover and the road construction materials. The $\gamma_{Sd,ec}$ is given as under.

$\gamma_{Sd,ec} = 1.5 + 0.5 (H_c - 8) / 3$ but not less than 1.5 for structural foundations on hard material or piles

$\gamma_{Sd,ec} = 1.15 + 0.35 (H_c - 8) / 3$ but not less than 1.15 for other foundation material.

10.4.4 Horizontal Earth Pressure

The earth pressure on walls of buried structures shall be determined in accordance with “RDM 4.3” and the guidelines provided in this section. The effect of wall friction, thermal expansion and contraction shall be incorporated in calculating the horizontal earth pressure applied to the walls.

The simplified method as given in Appendix D.1, may be used to determine the horizontal earth pressure on walls of buried structure backfilled with free draining granular material, properly compacted and when there is uniformity of ground conditions. This method gives directly determined values of earth pressure coefficients that take into account the effects of wall friction and thermal expansion and contraction.

10.4.5 Vertical Road Traffic Actions

Vertical traffic actions, walkway and cycle track actions and accidental traffic loading on edge members should be applied in accordance with relevant Clauses of the Chapter 3. The wheel loads are applied on culvert structure by considering the dispersion through the fill as described in Clause 10.4.7. On dual carriageways, the portion of structure supporting the central reservation, shall be designed for full carriageway loading.

10.4.6 Construction Traffic

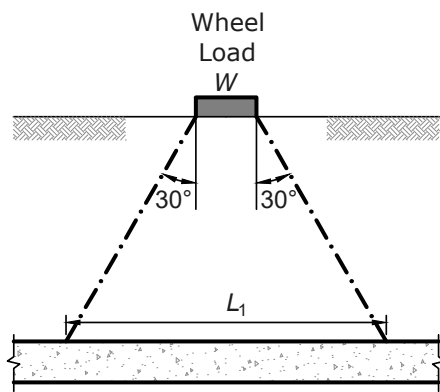
During construction stage and under low earth cover condition, the structure can be subjected to load conditions that are more severe than those experienced in normal service condition. The type of construction traffic likely to be relevant at different stages should be identified and be taken into account at design stage.

10.4.7 Dispersion of Vertical Loads Through the Fill

In the absence of more rigorous analysis, for dispersal of wheel loads through backfills or earth properly consolidated, the dispersal angle, transversely and longitudinally, from vertical equal to 30° may be considered as shown in Figure 10.3(a).

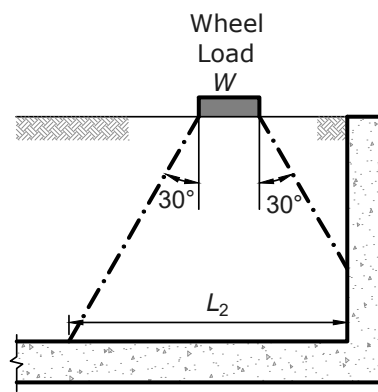
When the wheel is near to a vertical wall, such as wing wall or head wall, if the dispersal zone is curtailed by the wall, the pressure distribution may be based on the wheel load dispersion over the curtailed dispersion zone as shown in Figure 10.3(b).

Figure 10.3 Dispersal Wheel Load Through Earth Fill



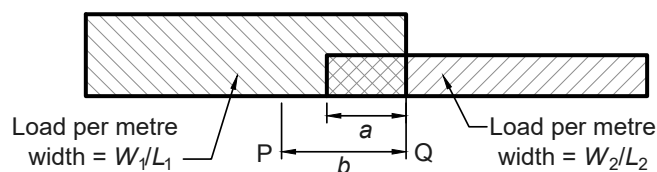
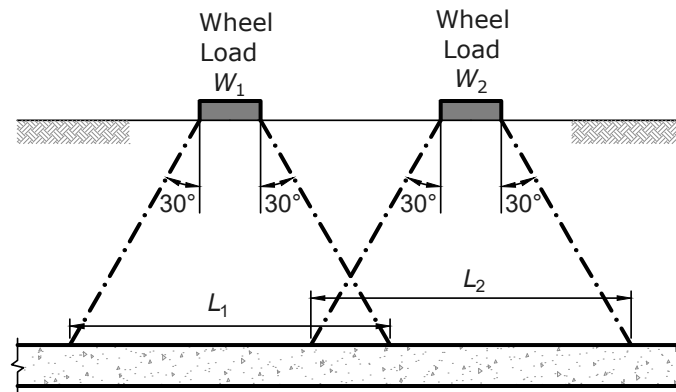
(a) Load dispersion in normal zone

Note: Load dispersion in longitudinal and transverse direction over a length L_1



(b) Load dispersion in curtailed zone

Note: Load dispersion in transverse direction over a curtailed length L_2 (curtailed by wing wall). Dispersion in longitudinal direction over length L_1 unless curtailed by transverse wall or similar



(c) Load dispersion in overlapping zone

Note 1: For a two-wheel overlap, the load on most heavily loaded strip PQ of width b may be taken as $(b W_1 / L_1 + a W_2 / L_2)$, where W_1 is larger wheel load, W_2 is other wheel load and a is the length of overlap. For a metre-width design, b to be taken as 1 m. For narrow segmental width, unit b may be taken as the segment width

Note 2: When the dispersion zones of more than two wheel overlap, the local pressure may be taken based on the most heavily loaded segment or metre-width as appropriate

When the dispersal zone of various wheel loads overlap, the following assumptions may be made.

- a. For in-situ structures which are to be analysed using a three-dimensional model (3-D), the dispersed patch loads may be superimposed
- b. For in-situ structures which are to be analysed using a two-dimensional model (2-D) or on a metre-strip analysis basis, the load per metre width in the transverse direction may be based on the total load on the most heavily loaded metre-strip as shown in Figure 10.3(c).
- c. For segmental units, the total load on the most heavily loaded segment may also be determined as in Figure 10.3(c). However, where the earth cover is small, the effect of applying concentrated wheel loads close to the edge of a segment should also be taken into account.
- d. A uniformly distributed traffic load may be considered to be dispersed at 30° to the vertical from each edge. Where the dispersal zone is curtailed by a wing wall or any similar member, the load may be considered to be dispersed over the curtailed dispersal zone.
- e. Where the transverse moments in an in-situ structure are not analysed using a 3-D computer model or where the width of a segmental unit (L_j) is more than twice the depth of earth cover (H_c), the transverse bending moment per metre may be taken as one half of the longitudinal moment per metre due to the vertical traffic load.

10.4.8. Longitudinal Road Traffic Actions

10.4.8.1 Traffic Surcharge

The areas of carriageway on the backfill beyond the ends of the structure should be loaded with the traffic surcharge load as specified in Clause 3.5.8. The resulting horizontal pressures on the abutments can be calculated in accordance with Appendix A.1 and Table A.2.

10.4.8.2 Braking and Acceleration Forces

For a buried structure with depth of earth cover (H_c) greater than overall length of the structure (L_L), the effects of braking and acceleration on the structure may be ignored.

For buried structures with less than 0.6 m depth of earth cover (H_c), the full value of the braking or acceleration action described in **Clause 3.3.5.1**, should be applied. Where the structure is skew to the direction of traffic, the combined effect of the longitudinal and transverse components of the horizontal actions should be assessed.

For buried structures where the depth of earth cover (H_c) is greater than 0.6 m and less than the overall length of the structure (L_L), the horizontal force may be assumed to reduce linearly to zero, as H_c increases from 0.6 m to L_L . The reduction factor, η , to be applied to the horizontal force in this range is therefore given by $\eta = (L_L - H_c)/(L_L - 0.6)$.

The braking or acceleration force should be applied directly to the top of the roof of the structure over a notional lane width without lateral dispersion. However, the in-plane rigidity of the roof and walls of in-situ box structures provides a robust resistance to distortional warping, and provided account is taken of the tendency for plan rotation, the effects of the braking and acceleration forces may be assumed to be distributed structurally over a length of L_j .

The total braking or acceleration force applied to the top of the roof of a buried structure need not be taken as greater than the friction force that can be generated between the earth and the roof, taking the weight of the vehicle into account.

10.4.9 Thermal Expansion and Contraction

The effects of thermal expansion and contraction due to uniform temperature component in buried concrete structures should be identified in accordance with Clause 3.10. If the overall length of the structure L_L is less than 3 m, such effects may be ignored.

The effect of the thermal expansion and contraction of the roof slab, ignoring any thermal movements of the base of the structure, should be accounted for in the analysis of the structure. Interaction between the backfill and structure due to thermal effects may be ignored if the directly determined earth pressure coefficients as given in Appendix D.1 are used, because any increase in earth pressure due to expansion is deemed to be covered by the specified K_{max} pressure coefficient. K_{max} represents the maximum pressure coefficients to be applied taking into account the pressure increase caused by expansion of the structure.

10.4.10 Temperature Difference

The effect of temperature difference shall be assessed in accordance with Clause 3.10. Temperature gradient within the walls and base slab of buried structures may be ignored, but the effect on these members of roof flexure induced by temperature difference should be considered in the design.

The heating and cooling temperature difference may be ignored for buried concrete box and portal frame structures if the total filling height from top of roof slab to top of the surfacing is more than 0.50 m.

10.4.11 Design of Foundations

10.4.11.1 Rotational Stiffness of Foundations

The bearing capacity of buried box structures shall be assessed based on the geotechnical investigations carried out in accordance with "RDM 3.1". The following guidelines may be used while analysing the foundation for box structure.

- a. Where a box structure is founded on hard material, the footings may be assumed to be rotationally rigid.
- b. Where a box structure is founded on softer material, it may be assumed that the pressure across the base slab is trapezoidal.
- c. Where a portal frame is founded on material other than hard material, it may be assumed that the base slab is rotationally flexible and the portal frame can therefore be considered to be pinned at the base of the legs.
- d. For portal frames, where the structures or elements of the structure are particularly sensitive to the rotational stiffness of the base, an analysis of both the rigid and flexible (fixed and pinned) conditions should be carried out and the more severe effects shall be used in the design.

10.4.11.2 Bearing Pressure

For verifying the bearing pressure under the base slab of a buried structure which is subject to braking or acceleration forces and traffic surcharge, it may initially be assumed that the pressure on the passive wall is the K_{max} pressure as given in Table D.4 of Appendix D.1. If this loading results in excessive bearing pressure, the assumed pressure on the passive wall may be increased, provided that the component members of the structures are designed for this increased pressure and the rotations required to mobilise this pressure are acceptable at the relevant limit states.

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Culverts

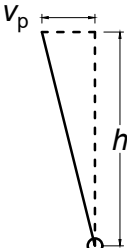
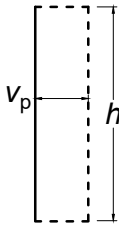
10.4.11.3 Movements Required to Generate Passive Pressure

The mobilisation of full passive pressure can involve unacceptably large horizontal movements, especially at SLS. The movements in terms of ratio v/h (where v is the wall displacement and h is the height of wall) required to mobilise a proportion of passive pressure can be found from the Table 10.1. The table shows the range of values of ratio v/h needed to mobilise full passive pressure (v_p/h) and half passive pressure (termed here as v_2/h), the latter being given in brackets beside the value of v_p/h . For intermediate values, it may be assumed that the coefficient of earth pressure (K) increases linearly from K_0 to $0.5K_p$ as the movement (v/h) increases from zero to v_2/h . For values of K greater than $0.5K_p$, the relationship between K and v can be found from the following equation.

$$K = K_p \left[1 - 0.5 \left\{ \frac{(v_p/h) - (v/h)}{(v_p/h) - (v_2/h)} \right\}^3 \right]$$

Equation 10.1

Table 10.1 Ratio of v_p/h for σ_p and v_2/h for $0.5 \sigma_p$ for Non-Cohesive Soils

Mode of Wall Movement		v_p/h (v_2/h for $0.5 \sigma_p$) in %	v_p/h (v_2/h for $0.5 \sigma_p$) in %
		Loose Soil	Dense Soil
a		7 (1.5) to 25 (4.0)	5 (1.1) to 10 (2.0)
b		4 (0.9) to 10 (1.5)	3 (0.5) to 6 (1.0)
c		6 (1.0) to 15 (1.5)	5 (0.5) to 6 (1.3)
Where; v = The wall displacement. v_p = The wall displacement to mobilise the full passive earth pressure. v_2 = The wall displacement to mobilise the half passive earth pressure. h = The height of the wall. σ_p = The fully mobilised passive earth pressure.			

10.4.11.4 Sliding and Overturning

Where sliding or overturning of a box structure is initiated by braking or acceleration, the earth pressure on the passive face may initially be based on the pressure coefficient K_{max} . If, however, this pressure is insufficient to prevent sliding or overturning, the earth pressure on the passive face may be increased until equilibrium is achieved, provided that the component members of the structures are designed for this increased pressure and the movements required to mobilise this pressure are acceptable at the relevant limit states.

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The movements required to mobilise a proportion of passive earth pressure can be found in accordance with Clause 10.4.11.3.

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Culverts

10.4.11.5 Sliding of Portal Legs

Under symmetrical earth pressure loading, both legs of a portal structure tend to slide inwards. If this tendency can be fully resisted by friction under the base slabs, then the portal can be considered to be restrained against horizontal movement at the base of the legs. If the tendency to slide cannot be fully resisted by base friction, the legs flex inwards until equilibrium is achieved, producing additional flexural moments and shears in the structure which should be assessed in the structural design. If this leg displacement is likely to be irreversible, the consequences of the legs being permanently displaced inwards should be identified and accounted for in all relevant design situations.

10.4.12 Skew Culverts

10.4.12.1 Analysis Method

The in-situ skewed structures which are transversely long relative to their span, may be designed on the basis of either the square or the skew span. However, for structures designed on the basis of the square span, the structural elements within a width of $X_{clear} \sin \theta$ from the edge of the structure should be designed on the basis of the skew span. Alternatively, the skewed in-situ structures may be analysed using more rigorous methods such as a 3-D computer analysis. Skewed pre-cast units shall generally be avoided.

10.4.12.2 Effect of Offset Pressure

With skewed in-situ structures, the line of thrust of the horizontal earth pressure forces on one abutment is offset laterally from the line of thrust of the earth pressure forces on the opposite abutment. This results in a plan torque which for box structures is resisted by torsional friction on the base.

If the line of thrust of the earth pressure forces on one wall passes within the middle third of the other wall (that is, if $L_L \tan \theta < L_j/6$), this plan twisting effect may be ignored.

If $L_L \tan \theta \geq L_j/6$ and the applied torque is greater than the rotational frictional resistance of the base, the unbalanced torque may be resisted by increasing the horizontal earth pressure on the walls towards the obtuse corner provided the structural elements are designed to resist the increased pressure.

The value of passive pressure that can be mobilised near the obtuse corners should be limited by the value of movement and plan rotation that is acceptable at the relevant limit state, but it should not exceed the lower-bound value of passive pressure based on zero wall friction.

The movements required to mobilise a proportion of passive earth pressure can be found in accordance with Clause 10.4.11.3.

10.4.13 Stages to be Analysed

Following three stages shall generally be analysed in the design of box culverts:

- a. The completed structure and backfilled up to the top of the box roof.
- b. The structure backfilled to an intermediate level between box roof level and finished road surface level. This corresponds to a stage at which it is proposed to use the structure for construction traffic.
- c. The structure is fully backfilled and is in service.

Each element of the structure should be designed for stages a) and b) taking into account the following:

- i. Permanent loads with maximum or minimum superimposed dead loads (refer Clause 10.4.3) (excluding differential settlement)
- ii. Relevant traffic or construction loads (Refer Clause 10.4.6)
- iii. Maximum or minimum horizontal earth pressures
- iv. Thermal actions

For the in-service stage c), the box culvert shall be designed for the most adverse action effects for various load cases together with the requirements in the relevant Clauses of this Chapter. Some important load cases are shown in Appendix D.1.

10.5 Pipe Culverts

10.5.1 General

The guidelines provided in this Clause are applicable for design of buried concrete pipe culverts under different conditions of loading. The buried pipe system comprises the pipe and the surrounding soil embedment.

10.5.2 Pipes

The precast pipes shall meet the requirements of following standards

BS 5911-1: Concrete pipes and ancillary concrete products (unreinforced and reinforced concrete pipes).

BS 5911-5: Concrete pipes and ancillary concrete products (pre-stressed non-pressure concrete pipes).

The concrete pipes are classified as rigid pipes. Circular concrete pipes are manufactured to Strength Class 120. This defines the minimum crushing load per meter length of pipe under test conditions and is equal to 120 multiplied by the nominal diameter (DN) of a pipe in metres.

10.5.3 Design Considerations

The magnitude and distribution of earth loads on culverts depend upon the relative stiffness of culvert and the soil. The concrete pipe culverts are treated as rigid culverts. The soil surrounding the culvert and the method of installation of pipe affects the load transmission to the buried pipe. Based on these factors, the culverts can be categorised as narrow trench condition and wide trench or embankment condition. The wide trench and embankment conditions can be further subdivided into positive projecting or negative projecting. Most highway culverts fall under the wide trench and embankment condition.

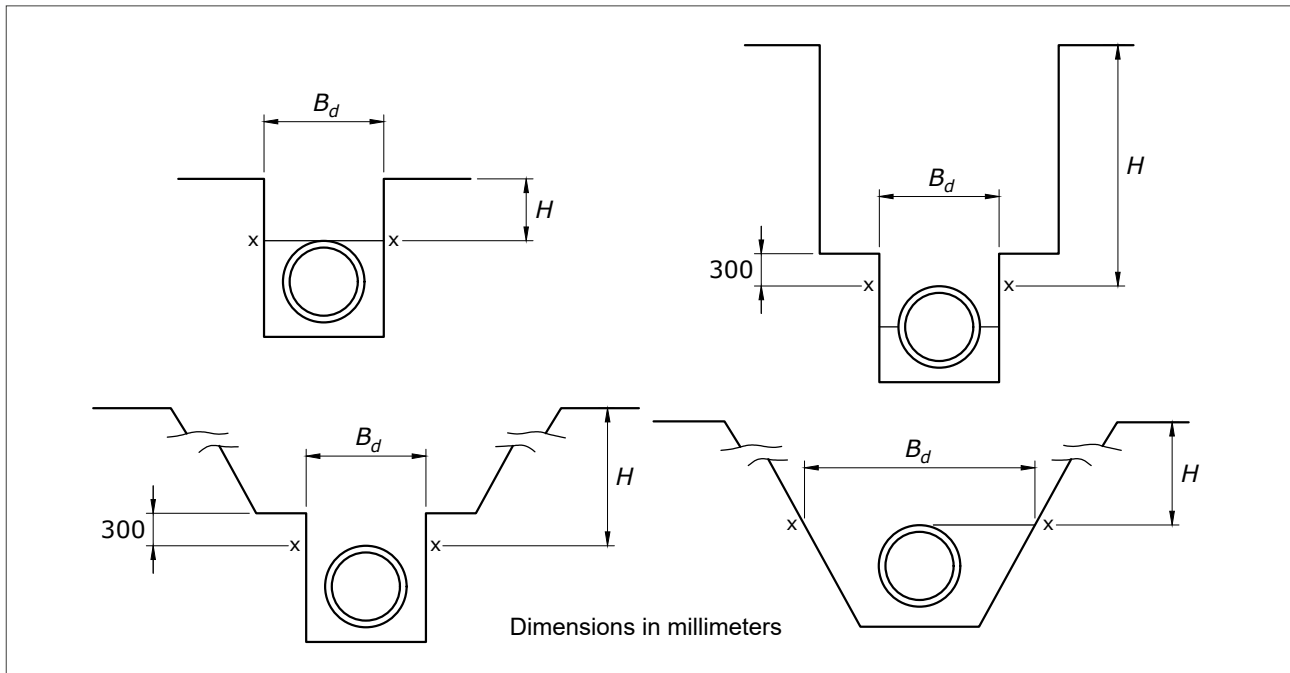
The total vertical earth load on circular pipe depends upon the unit weight of fill material, trench width or width of the pipe and load coefficient. The load coefficient depends on the geometry of the soil-culvert system and physical properties of the fill and culvert materials and is usually expressed as a function of the height of fill above pipe, horizontal width of the culvert or trench, the coefficient of internal friction of the soil, the projection ratio and the settlement ratio. In general, the total load on pipe is equal to the weight of the prism of soil above the pipe plus or minus the shearing resistance developed due to relative deformation of this soil prism and the adjacent column of soil.

The various design parameters which are important in design of pipe culverts are given in this section.

10.5.3.1 Trench Width

The zone of soil which has a structural influence on the buried pipe typically ranges between one and two diameters from the pipe wall in all directions. This zone typically includes the trench base, pipe embedment material and the native soil of the trench wall. The shape and width of trench are important parameters and has an influence on design loadings. The Figure 10.4 defines the effective trench width for different shapes of excavations.

Figure 10.4 Effective Trench Width



10.5.3.2 Wide and Narrow Trenches

There are two installation conditions for the pipe culverts:

- a. Pipes installed in wide trenches or embankments; and
- b. Pipes installed in narrow trenches.

For all pipe installations the loads arising from both these conditions should be calculated for the selected maximum trench width and the lower of the two loads, W_c or W'_c , used in the design specifying the installation condition. In the absence of specific information, it is safer to assume the trench is wide.

A separate calculation of vertical backfill load on the pipe is generally conducted for each condition and the results are compared. Depending on the assumed trench geometry, either calculated load might be larger.

10.5.3.3 Complete and Incomplete Projection

The wide trench design condition can further be sub divided into complete and incomplete projection. These refer to a line known as the “plane of equal settlement”. If this line falls above ground level for complete projection or below for incomplete projection. The plane of equal settlement refers to the point at which the settlements of soil to the sides of the pipe and over the pipe are equal.

Loads should be calculated using both complete and incomplete projection formulae and the lesser of the two calculated loads shall be used for design. Generally, the incomplete projection case is the more common situation, except for pipes installed at a shallow depth of cover.

Figure 10.5 Complete Projection

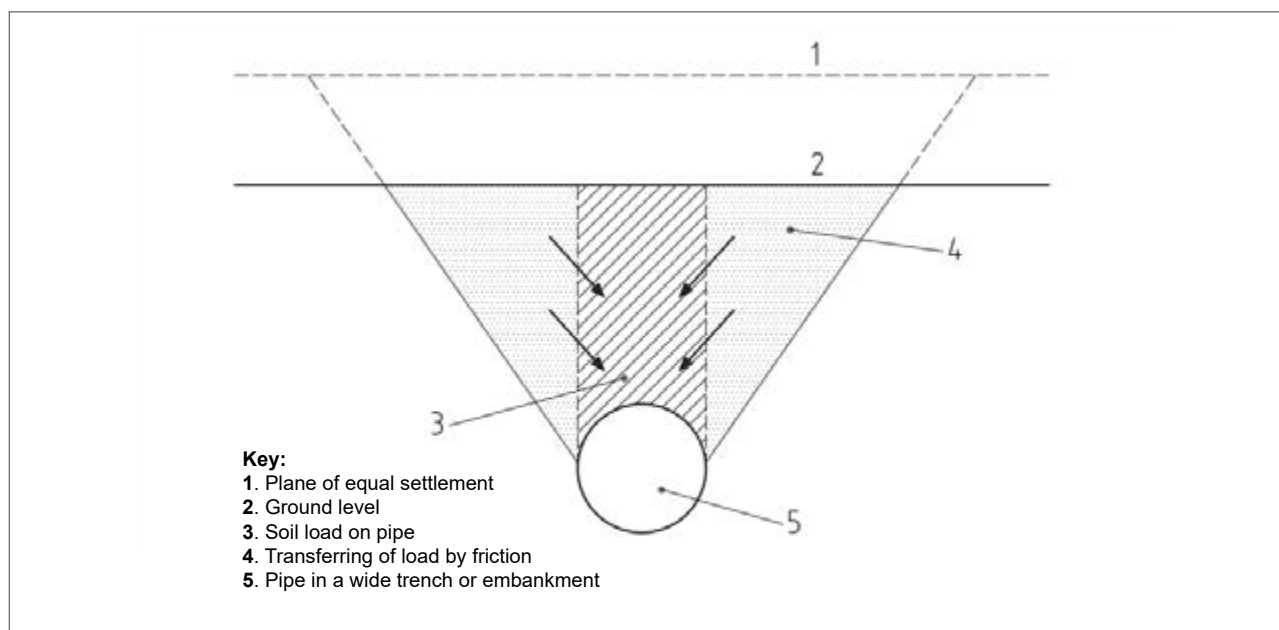
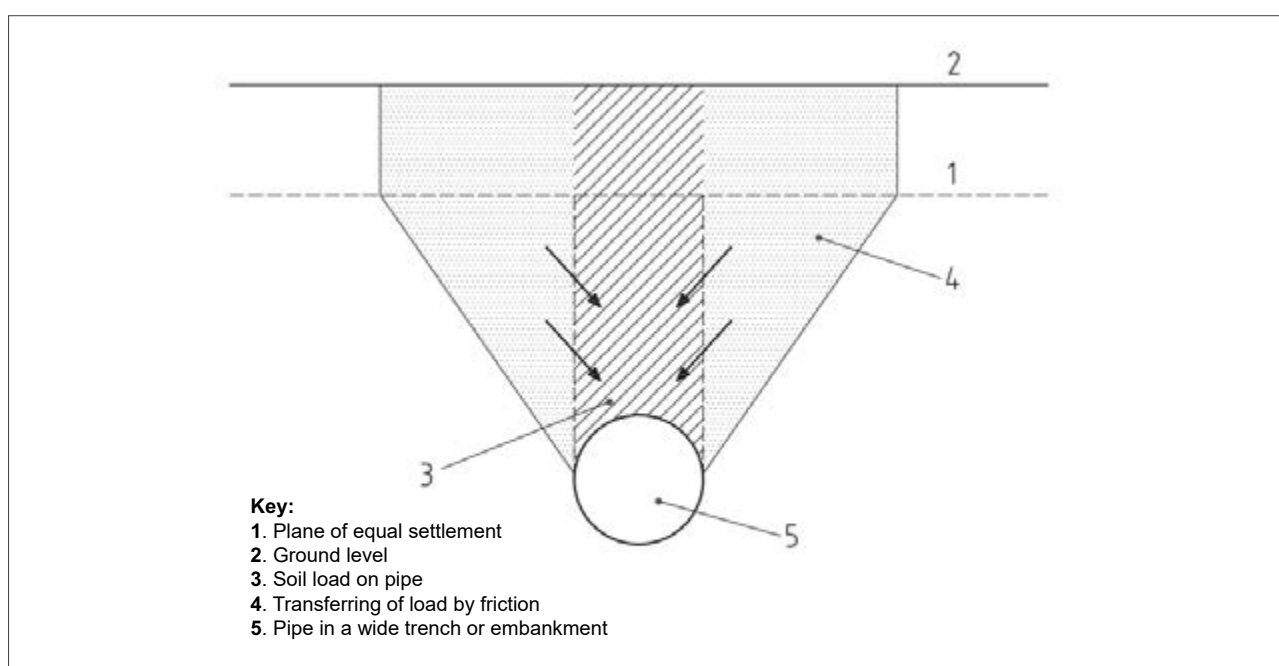


Figure 10.6 Incomplete Projection



10.5.3.4 Settlement Deflection Ratio

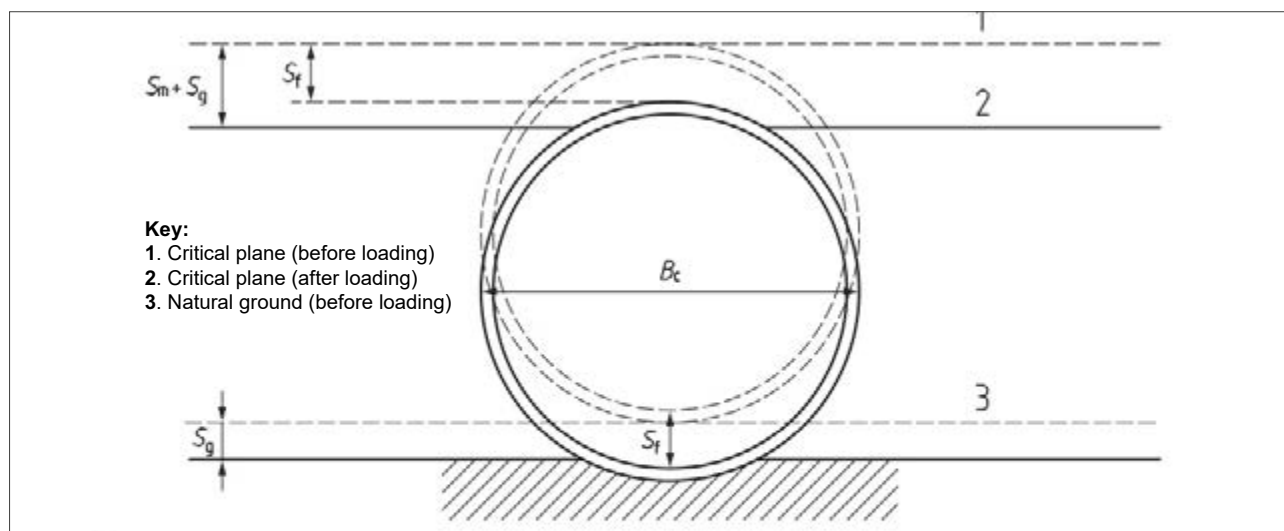
The settlement deflection ratio, r_{sd} , in a wide trench or embankment, is given by:

$$r_{sd} = \frac{(S_m + S_g) - S_f}{S_m}$$

Equation 10.2

The S_m , S_g and S_f are shown in Figure 10.7. The combined value of the settlement deflection and projection ratios are given in Clause 10.5.3.5.

Figure 10.7 Settlement Deflection Ratio



10.5.3.5 Projection Ratio

The projection ratio, ρ , is the proportion of the pipe external diameter that is above firm bedding level or the natural ground level. For classes D, N and F beddings, the value of ρ is unity. For other granular and concrete beddings, ρ is 0.7.

In calculations, the settlement deflection and projection ratios are used in combination. The design values of their product, $r_{sd} \rho$, are often taken as 0.7 for bedding classes D, N and F and 0.5 for bedding classes B and S.

10.5.3.6 Soil Properties

The detail soil properties shall be determined in accordance with RDM 3.1. In the absence of data on the soil type supporting the use of other values, it is normal practice to assume the following.

$$K\mu = 0.19$$

$$K\mu' = 0.13$$

$$\gamma = 19.6 \text{ kN/m}^3$$

The combination $K\mu$ and $K\mu'$ is generally used as single design parameters, rather than determining values of K , μ and μ' separately. Following values may be taken if the soils are properly identified.

Table 10.2 Values of $K\mu$ or $K\mu'$ for Specific Soil Types

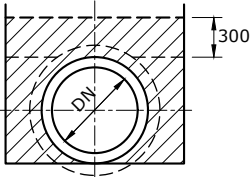
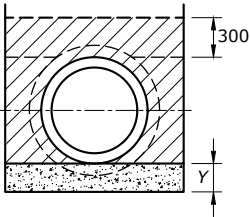
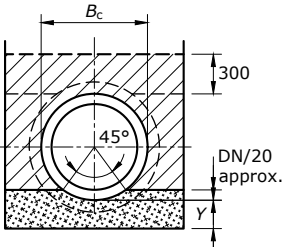
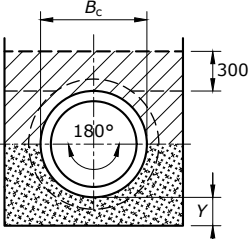
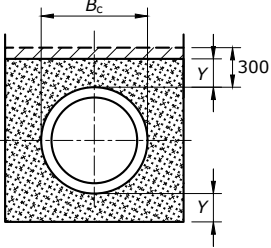
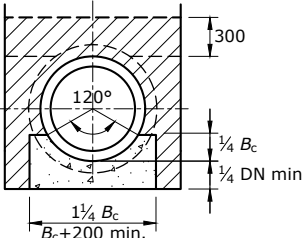
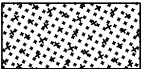
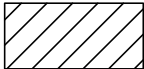
Type of Soil	$K\mu$ or $K\mu'$
Granular soils without cohesion	0.190
Maximum for sand and gravel	0.165
Saturated top soil	0.150
Maximum for ordinary clay	0.130
Maximum for saturated clay	0.110

The value of $K\mu'$ is taken as the lower of the values for the backfill material and the native soil in the trench sides.

10.5.3.7 Bedding Factors

The load bearing capacity of concrete pipe when buried in ground is the crushing strength of the pipe multiplied by the bedding factor. The crushing strength is determined by line load testing (in accordance with BS EN 1916) which produce the most onerous pipe wall circumferential bending moment. The Table 10.3 shows the bedding factors for different class of beddings.

Table 10.3 Bedding Factors

Bedding Details $Y = (1/6) B_c$	Bedding Class	Bedding Factors F_m	
		For Narrow Trench	For Wide Trench/ Embankment*
	D	1.1	1.1 to 1.3
	N	1.1	1.1 to 1.3
	F	1.5	1.9
	B	1.9	2.5
	S	2.2	2.5
	A Unreinforced	2.6	Wide trench: 2.6 Embankment: 2.6 to 3.7
	A Unreinforced	3.4	Wide trench: 3.4 Embankment: 3.4 to 5.7
Legend:  <i>In situ concrete</i>  <i>Granular bedding material</i>  <i>Selected backfill material</i>  <i>All in granular material</i> * Values applicable for pipe less than 900mm outside diameter			

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Culverts

Table 10.4 Bedding Factors for Pipes Outside Diameter $\geq 900\text{mm}$

Laid in Wide Trench and Embankment Condition Classes B and S								
H/B_c	0.5	1	1.5	2	3	5	10	15
Bedding Factors F_m	2.8	2.58	2.44	2.39	2.36	2.35	2.33	2.33

Note: Assumption is made $\rho = 0.7$, $r_{sd} \rho = 0.5$

10.5.4 Design Loads for Concrete Pipes

The design for pipes in pipe culverts shall be based on more conservative wide trench model. Following are the different load to be considered in design of buried pipes.

- Loads from backfill.
- Vehicle surcharge loads.
- Internal water loads.

10.5.4.1 Load From Backfill in Wide Trench or Embankment Conditions

Load from the backfill for wide trench or embankment installation is given by the following expression.

$$W'_c = C_c \gamma B_c^2$$

Equation 10.3

Where,

B_c = Outside diameter of pipe.

C_c = Soil load coefficient in wide trench and embankment conditions.

W'_c = Soil load per unit length of pipe in embankment or wide trench conditions.

γ = Unit weight of soil.

The value of soil load coefficient C_c shall be taken as lower of the values calculated from following equations:

$$C_c = \frac{e^{2K\mu H/B_c} - 1}{2K\mu}$$

Equation 10.4

$$C_c = \left\{ \frac{e^{2K\mu H_e/B_c} - 1}{2K\mu} + \left(\frac{H}{B_c} - \frac{H_e}{B_c} \right) e^{2K\mu H_e/B_c} \right\}$$

Equation 10.5

If the lower value is given by Equation (10.4), then complete projection conditions exist. If the lower value is given by Equation (10.5), the incomplete projection conditions apply.

H_e is the height of the plane of equal settlement above the top of the pipe. Derivation of H_e is complex. The Table 10.5 provides simplified equations for determining C_c assuming $K\mu = 0.19$.

Table 10.5 Soil Load Coefficients

$r_{sd}\rho$	Equation for C_c
0	$C_c = H/B_c$
+0.1	$C_c = 1.23 (H/B_c) - 0.02$
+0.3	$C_c = 1.39 (H/B_c) - 0.05$
+0.5	$C_c = 1.50 (H/B_c) - 0.07$
+0.7	$C_c = 1.59 (H/B_c) - 0.09$
+1.0	$C_c = 1.69 (H/B_c) - 0.12$
+2.0	$C_c = 1.93 (H/B_c) - 0.17$
+3.0	$C_c = 2.08 (H/B_c) - 0.20$
+4.0	$C_c = 2.19 (H/B_c) - 0.21$
+5.0	$C_c = 2.28 (H/B_c) - 0.22$

10.5.4.2 Load from Backfill in Narrow Trench Conditions

Load from the backfill for narrow trench installation is given by the following equation.

$$W_c = C_d \gamma B_d^2$$

Equation 10.6

Where,

B_d = Effective trench width

C_d = Soil load coefficient in narrow trench conditions

The value of soil load coefficient C_d shall be calculated from following equations:

$$C_d = \frac{1 - e^{-2K\mu'H/B_d}}{2K\mu'}$$

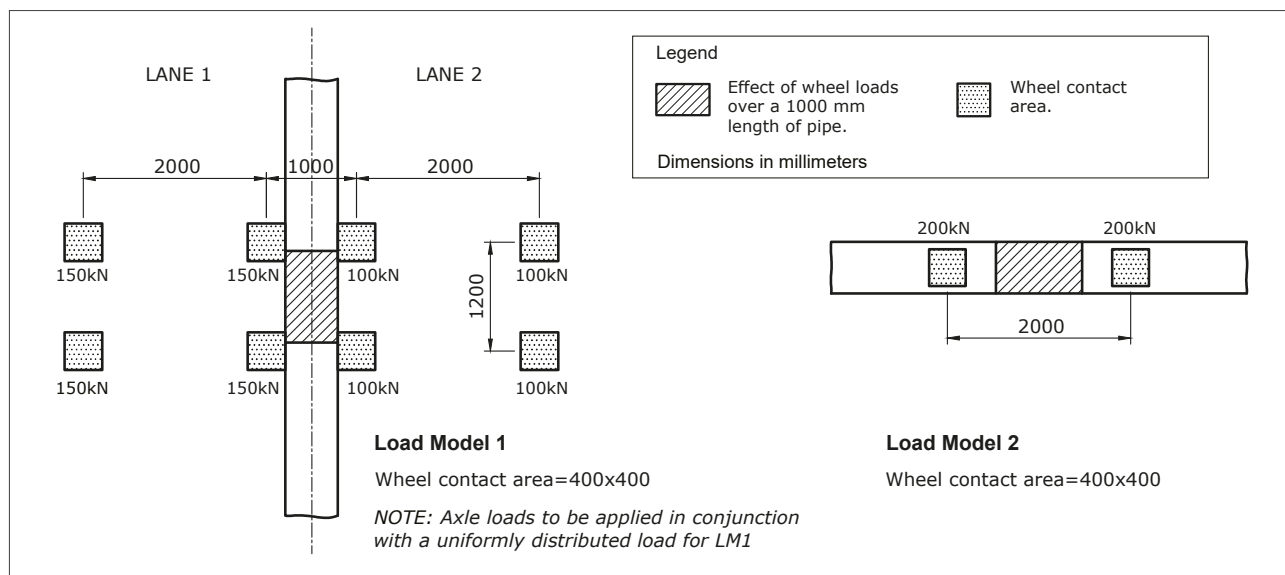
Equation 10.7

For pipes in trenches, the backfill load is calculated for wide trench and narrow trench conditions and the lower value is used in design.

10.5.4.3 Traffic Surcharge Load

Vehicle loads are due to the traffic on road. The maximum load on a pipe depends upon its depth and diameter. The traffic load decreases with depth of cover whereas backfill load increases. The vehicle surcharge loading corresponds to the worst case of either Load Model 1 or Load Model 2. The vehicle load configurations for Model 1 or Load Model 2 are shown in Figure 10.8.

Figure 10.8 Traffic Load Configurations



The surcharge pressure due to road vehicles for different depths of cover over the pipe crown are given in Table 10.6. These values are based on the higher value of the pressures for Load Model 1 and Load Model 2.

Table 10.6 Road Traffic Surcharge Pressure for Different Depths of Cover

Depth of cover over pipe crown	Vehicle surcharge pressure P_s (kN/m ²)	Depth of cover over pipe crown H (m)	Vehicle surcharge pressure P_s (kN/m ²)	Depth of cover over pipe crown H (m)	Vehicle surcharge pressure P_s (kN/m ²)
0.5	—	1.7	45	3.8	22
0.6	144	1.8	43	4.0	21
0.7	120	1.9	41	4.5	19
0.8	101	2.0	39	5.0	17
0.9	86	2.2	35	5.5	16
1.0	74	2.4	33	6.0	15
1.1	67	2.6	31	6.5	14
1.2	62	2.8	29	7.0	13
1.3	58	3.0	27	8.0	12
1.4	54	3.2	26	9.0	11
1.5	51	3.4	24	10.0	10
1.6	48	3.6	23	—	—

The values of surcharge pressure given in the table can also be determined approximately using following equation.

$$P_s = 76.46 H^{0.923}$$

Equation 10.8

Where,

P_s = The surcharge pressure in kN/m².

H = The depth of cover over pipe crown in meters.

The concentrated surcharge load due to traffic is given by following equation.

$$W_{csu} = P_s B_c$$

Equation 10.9

10.5.4.4 Internal Water Load

The equivalent load due to weight of water in pipe W_w is given by following equation.

$$W_w = \frac{C_w \gamma_w \pi (D - t)^2}{4}$$

Equation 10.10

Where the value of C_w is normally taken as 0.75.

10.5.4.5 Total Design External Load

The total design external load is given by following equation.

$$W_e = W'_c \text{ (or } W_c) + W_{csu} + W_w$$

Equation 10.11

Use W'_c for wide trench or embankment condition and W_c for narrow trench condition.

10.5.5 Supporting Strength of Concrete Pipes

The safe supporting strength of the pipe/bed combination (W_s) is given by following equation.

$$W_s = \frac{W_t F_m}{F_{se}}$$

Equation 10.12

Where,

W_t = Crushing strength of pipe, obtained from product standards.

F_m = Bedding factor.

F_{se} = Factor of safety.

The factor of safety F_{se} for concrete pipes is 1.25. For reinforced pipes, it shall not be less than W_t/W_p which may increase up to 1.50. Confirmation shall be obtained from manufactures or conservatively, the factor of safety may be taken as 1.50. Pipes with diameter greater than 600 mm are reinforced and diameter less than or equal to 600 mm may be unreinforced.

The safe supporting strength of pipe (W_s) shall be greater or equal to the total design external load (W_e). The minimum recommended bedding factor F_m is given by following equation.

$$F_m \geq \frac{W_e F_{se}}{W_t}$$

Equation 10.13

10.5.6 Limit State Design

10.5.6.1 General

The traditional approach for design of concrete pipes is to combine the load in accordance with Clause 10.5.4.5 and determine the minimum bedding factor using factor of safety given in Clause 10.5.5. The limit state method can be used with verification by partial factor method as an alternative to global factor of safety.

The pipe crushing strength W_t is used to assess the ultimate limit state (ULS). The product standards specify crushing strength as the absolute minimum and can be used in place of characteristic strength for the purpose of ULS design check.

The serviceability limit state (SLS) is assessed using the proof load W_p which is applicable for only reinforced products where a maximum crack width is specified in product standards.

10.5.6.2 Design Loads

Following actions are determined as per the methodology given in Clause 10.5.4.

- Backfill load W'_c for wide trench and embankment, W_c for narrow trench condition.
- Concentrated surcharge load due to traffic, W_{csu} .
- Internal water load, W_w .

10.5.6.3 Partial Factors

The total design load is calculated using partial factors to respective actions as in Table 10.7.

Table 10.7 Partial Factors of Actions

Action	Type		Partial Factors in	
			ULS	SLS
Permanent	Soil	γ_G	1.35	1.0
Variable	Traffic/Surcharge	γ_Q	1.35	1.0
Permanent	Internal water	$\gamma_{G,intw}$	1.2	1.0

10.5.6.4 Design Value of Actions

The total design value of actions is calculated by following equation.

$$W_{e,Ed} = \gamma_G W_c' + \gamma_Q W_{csu} + \gamma_{G,intw} W_w$$

Equation 10.14

Substitute W_c in place of W_c' for narrow trench condition.

10.5.6.5 Minimum Recommended Bedding Factor

The minimum recommended bedding factor shall be calculated with following equation for ULS and SLS conditions.

$$F_m \geq \frac{W_{e,Ed}}{W_t} \quad \text{for ULS}$$

Equation 10.15

$$F_m \geq \frac{W_{e,Ed}}{W_p} \quad \text{for SLS}$$

Equation 10.16

For reinforced pipes, minimum bedding factor F_m required for both ULS and SLS shall be calculated and the critical or higher value used to specify the bedding class. For unreinforced pipes, only the ULS is applicable and the resulting minimum F_m for ULS shall be used.

10.5.7 Simplified Bedding Table for Concrete Pipes

The Table 10.8 gives the applicable range of depth of cover for different pipes sizes and type of bedding considering main road traffic load conditions for concrete pipes of strength class 120 with a maximum depth up to 6.0m. Where depths can be exceeded, the value 6.0+ is shown.

Table 10.8 Simplified Bedding and Fill Height for Concrete Pipes

DN	Trench width condition	Bedding class	Bedding factor	Pipe strength class	Crushing strength kN/m	Fill height m
450	Narrow	F	1.5	120	54	–
		B	1.9			0.9 – 5.7
		S	2.2			0.9 – 6.0+
	Wide	B/S	2.5	120	54	0.9 – 5.9
600	Narrow	F	1.5	120	72	–
		B	1.9			0.9 – 6.0
		S	2.2			0.9 – 6.0+
	Wide	B/S	2.5	120	72	0.9 – 5.8
900	Narrow	F	1.5	120	108	–
		B	1.9			0.9 – 4.7
		S	2.2			0.9 – 6.0
	Wide	B/S	As per Table 10.4	120	108	0.9 – 4.6
1050	Narrow	F	1.5	120	126	–
		B	1.9			0.9 – 3.8
		S	2.2			0.9 – 6.0
	Wide	B/S	As per Table 10.4	120	126	0.9 – 5.2
1200	Narrow	F	1.5	120	144	–
		B	1.9			0.9 – 3.9
		S	2.2			0.9 – 5.2
	Wide	B/S	As per Table 10.4	120	144	0.9 – 4.6
1500	Narrow	B	1.9	120	180	0.9 – 3.9-
		S	2.2			0.9 – 5.3
	Wide	B/s	As per Table 10.4	120	180	0.9 – 5.3

Note 1: The values provided above are based on BS EN 9295 considering main road traffic loading.

Note 2: The depths of cover are calculated using $W_t/W_p = 0.67$.

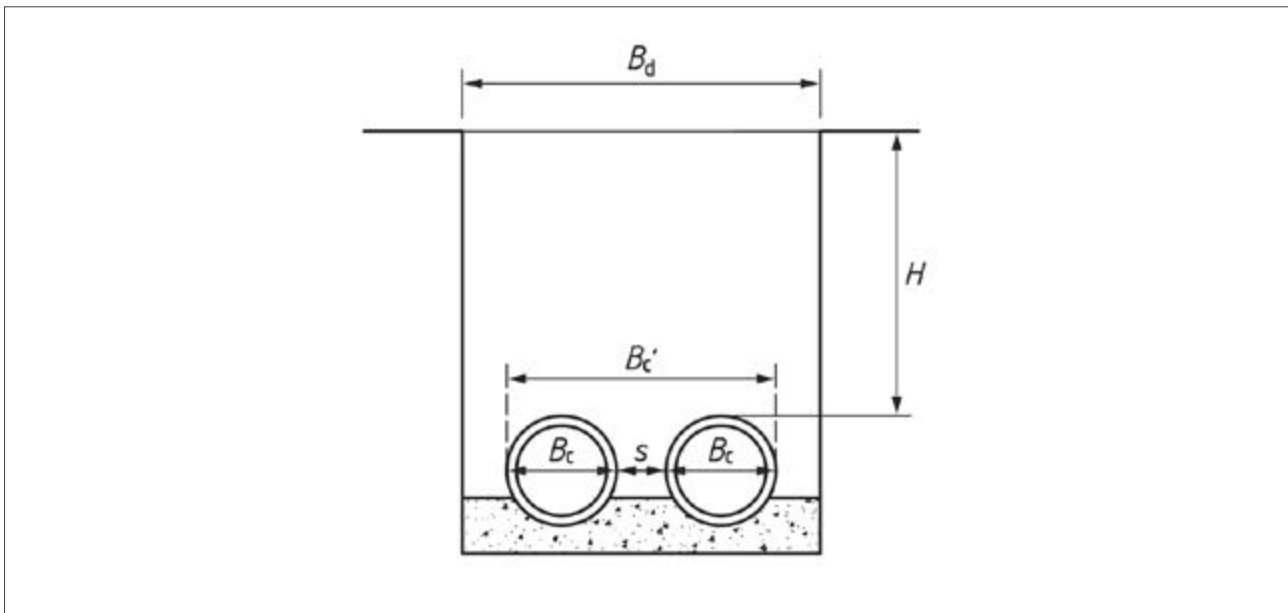
For depths greater than 6.0m, the advice shall be obtained from manufacturer and an experienced structural design engineer.

Pipes up to and including DN 600 mm are assumed to be unreinforced and greater than DN 600 mm are assumed as reinforced. Confirmation shall be obtained from the manufacturer whether the pipe is reinforced or unreinforced.

10.5.8 Design for Multiple Pipes in Single Trench

Based on the requirements, two or more pipes may be laid in a single trench. The soil load for rigid pipes can be estimated by treating the multiple pipes as equivalent to a single pipe of outside width as shown in Figure 10.9. For two pipes, the equivalent width will be $B'_c = 2 B_c + s$.

Figure 10.9 Equivalent Width of Two Pipes in One Trench



10.5.9 Concrete Pipes with Concrete Beds or Surround

When the fill height over the top of concrete pipe is less than 0.9m, the pipe need to be protected by surrounding it by concrete. In such situations, it is important to ensure that the bed formation below concrete bed or surround shall provide a firm foundation by using a blinding concrete or granular material. Any soft patches shall be excavated and replaced with some more suitable material such as granular bedding material and compact.

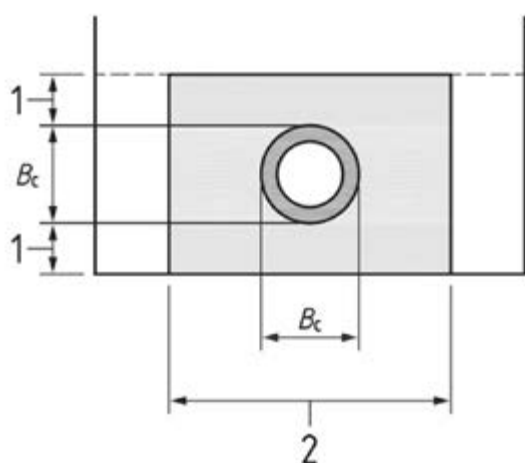
The dimensions for concrete bedding or surround shall be sufficient to ensure that the specified bedding factors are achieved. The concrete bed or surround shall extend at least 150mm on either side of pipe. The depth of concrete below pipe and above pipe for concrete surround shall be at least 150 mm or one-fourth of the outside diameter, which ever is greater.

The concrete for pipe bedding or surround shall be of structural grade C20/25 as a minimum and should be thoroughly compacted into place. Care should be taken in placing concrete so as not to move pipes or construction joints.

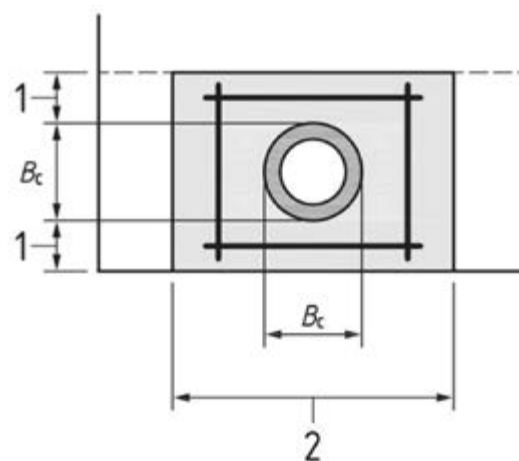
The details of plain and reinforced concrete surround are are shown in Figure 10.10. In the design verificatiuon of pipe-bedding, the width of the top of concrete instead of out side diameter is considered to calculate the load on pipe.

The area of transverse steel reinforcement for reinforced concrete beds shall not be less than 0.4 % of the area of concrete in longitudinal section. If the area of tranverse steel is increased to 1 % of the concrete area in longitudinal section in concrete bed or surround both above and below the pipe, the bedding factor may be increased up to 4.8.

Figure 10.10 Plain and Reinforced Concrete Surrounds

**Bedding Factor 4.5**

Maximum diameter 600 mm.

a) Plain concrete surround.**Bedding Factor 4.8**

Steel reinforcement 4 x 1 %.

b) Reinforced concrete surround.

Key: 1. The greater of $\frac{1}{4} B_c$ or 150 mm 2. The greater of $1\frac{1}{2} B_c$ or $B_c + 300$ mm

11 Foundations

11.1 General

11.1.1 Scope

This Chapter deals with the design of foundations which includes spread foundations and pile foundations. The foundations shall be provided such that it transmits the loads of the structure to ground safely. Different sections of this Chapter will provide guidelines on verification requirements to ensure that the foundation will be safe for various limit state conditions under different design situations.

11.1.2 Symbols

Following symbols are used in this section. Additional symbols are also defined where they appear. Symbols may have various meanings.

Latin Letters

A'	Effective base area ($A' = B' \times L'$).
A_b	Base area under pile.
A_c	Total base area under compression.
$A_{s,i}$	Pile shaft surface area in layer i .
B	Width of a foundation.
C_d	Limiting design value of the relevant serviceability criterion.
c	Cohesion intercept.
c'	Cohesion intercept in terms of effective stress.
c_u	Undrained shear strength.
$c_{u,d}$	Design value of undrained shear strength.
E_d	Design value of the effect of actions.
$E_{stb,d}$	Design value of the effect of stabilising actions.
$E_{dst,d}$	Design value of the effect of destabilising actions.
$F_{c,d}$	Design axial compression load on a pile or a group of piles.
F_d	Design value of an action.
F_k	Characteristic value of an action.
F_{rep}	Representative value of an action.
$G_{dst,d}$	Design value of the destabilising permanent actions for uplift verification.
$G_{stb,d}$	Design value of the stabilising permanent vertical actions for uplift verification.
$G'_{stb,d}$	Design value of the stabilising permanent vertical actions for heave verification (submerged weight).
H	Horizontal load, or component of total action acting parallel to the foundation base.
H_d	Design value of H .
n	Number of e.g. piles or test profiles.
$q_{b,k}$	Characteristic value of unit base resistance.
$q_{s,i,k}$	Characteristic value of unit shaft resistance in stratum i .
q_u	Unconfined compressive strength.
$R_{b,k}$	Characteristic value of the base resistance of a pile.
R_c	Compressive resistance of the ground against a pile, at the ultimate limit state.

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Foundations

1

 $R_{c,d}$ Design value of R_c . $R_{c,m}$ Measured value of R_c . R_d

Design value of the resistance to an action.

2

 $R_{p,d}$

Design value of the resisting force caused by earth pressure on the side of a foundation.

3

 $R_{s,d}$

Design value of the shaft resistance of a pile.

 $R_{s,cal}$

Ultimate shaft friction, calculated using ground parameters from test results.

 $R_{s,k}$

Characteristic value of the shaft resistance of a pile.

4

 $R_{t,d}$

Design value of the tensile resistance of a pile or a group of piles.

 $R_{t,k}$

Characteristic value of the tensile resistance of a pile or a group of piles.

5

 $R_{t,m}$

Measured tensile resistance of an isolated pile in one or several pile load tests.

 T_d

Design value of total shearing resistance that develops around a block of ground in which a group of tension piles is placed, or on the part of the structure in contact with the ground.

6

 V

Vertical load, or component of the total action acting normal to the foundation base.

 V_d Design value of V .

7

 V'_d

Design value of the effective vertical action or component of the total action acting normal to the foundation base.

8

 $V_{dst,d}$

Design value of the destabilising vertical action on a structure.

 $V_{dst,k}$

Characteristic value of the destabilising vertical action on a structure.

9

 X_d

Design value of a material property.

 X_k

Characteristic value of a material property.

Greek Letters

10

 δ

Structure-ground interface friction angle.

 δ_d Design value of δ .

11

 γ_b

Partial factor for the base resistance of a pile.

 γ_c

Partial factor for the effective cohesion.

 γ_{cu}

Partial factor for the undrained shear strength.

 γ_E

Partial factor for the effect of an action.

 γ_F

Partial factor for an action.

 γ_G

Partial factor for a permanent action.

 $\gamma_{G,dst}$

Partial factor for a permanent destabilising action.

 $\gamma_{G,stb}$

Partial factor for a permanent stabilising action.

 γ_m

Partial factor for a soil parameter (material property).

 $\gamma_{m,i}$ Partial factor for a soil parameter in stratum i . γ_M

Partial factor for a soil parameter (material property), also accounting for model uncertainties.

 γ_Q

Partial factor for a variable action.

 γ_{qu}

Partial factor for unconfined strength.

 γ_R

Partial factor for a resistance.

 $\gamma_{R,d}$

Partial factor for uncertainty in a resistance model.

 $\gamma_{R,e}$

Partial factor for passive earth resistance.

 $\gamma_{R,h}$

Partial factor for sliding resistance.

$\gamma_{R,v}$	Partial factor for bearing resistance.
γ_s	Partial factor for shaft resistance of a pile.
$\gamma_{S,d}$	Partial factor for uncertainties in modelling the effects of actions.
$\gamma_{Q,dst}$	Partial factor for a variable destabilising action.
$\gamma_{Q,stb}$	Partial factor for a variable stabilising action.
$\gamma_{s,t}$	Partial factor for tensile resistance of a pile.
γ_t	Partial factor for total resistance of a pile.
γ_w	Weight density of water.
$\gamma_{\varphi'}$	Partial factor for the angle of shearing resistance ($\tan \varphi'$).
γ_γ	Partial factor for weight density.
ξ	Correlation factor depending on the number of piles tested or of profiles of tests.
$\xi_1 ; \xi_2$	Correlation factors to evaluate the results of static pile load tests.
$\xi_3 ; \xi_4$	Correlation factors to derive the pile resistance from ground investigation results, not being pile load tests.
$\xi_5 ; \xi_6$	Correlation factors to derive the pile resistance from dynamic impact tests.
ψ	Factor for converting the characteristic value to the representative value.
$\sigma_{stb,d}$	Design value of stabilising total vertical stress.
φ'	Angle of shearing resistance in terms of effective stress.
φ_{cv}	Critical state angle of shearing resistance.
$\varphi_{cv,d}$	Design value of φ_{cv} .
φ'_d	Design value of φ' .

11.2 Basis of Geotechnical Design

11.2.1 Design Requirements

For each geotechnical design situation, it shall be verified that no relevant limit state is exceeded. Limit states can occur either in the ground or in the structure or by combined failure in the structure and the ground. The verification is carried out under Ultimate Limit state (ULS) and Serviceability Limit state (SLS).

Under the ultimate limit state, the loss of equilibrium of the structure, excessive deformation of the ground and loss of equilibrium of the structure or the ground due to uplift by buoyancy, and adequacy of capacities of various elements etc. are verified.

Under serviceability limit state, the bearing pressure under spread foundations and settlement for the foundations are to be investigated.

11.2.2 Geotechnical Design

Geotechnical design shall be done by calculating the effect of different actions and the resistance offered by the ground against those action effects.

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11.2.2.1 Actions

In geotechnical design, the following should be considered for inclusion as actions:

- The weight of soil, rock and water.
- Stresses in the ground.
- Earth pressures.
- Free water pressures, including wave pressures.
- Ground-water pressures.
- Dead and imposed loads from structures.
- Surcharges.
- Traffic loads.
- Downdrag.

The characteristic values of relevant actions shall be taken in accordance with Chapter 3. The design values of actions (F_d) shall be derived using following Equation.

$$F_d = \gamma_F F_{rep}$$

Equation 11.1

With,

$$F_{rep} = \psi F_k$$

Equation 11.2

Where,

F_{rep} = The representative value of an action.

γ_F = The partial factor for the action.

The ψ factors shall be taken from Chapter 3. The partial factors γ_F on actions for persistent and transient situation shall be taken in accordance with the provisions in this Chapter.

11.2.2.2 Ground Properties

Properties of soil and rock masses, as quantified for design calculations by geotechnical parameters, shall be obtained from test results, either directly or through correlation and from other relevant data.

The characteristic values for geotechnical parameters shall be based on results and derived values from laboratory and field tests, complemented by well-established experience.

Design values of geotechnical parameters (X_d) shall either be derived from characteristic values using the following Equation:

$$X_d = \frac{X_k}{\gamma_M}$$

Equation 11.3

Or, shall be assessed directly.

Where,

X_k = The characteristic value of a geotechnical parameter.

γ_M = The partial factor for the geotechnical parameter.

The partial factors γ_M on geotechnical parameters for persistent and transient situation shall be taken in accordance with the provisions in this Chapter.

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11.2.2.3 Structural Properties

The design strength properties of structural materials and the design resistances of structural elements shall be calculated in accordance with Chapter 4, Chapter 5 and Chapter 6 for concrete, steel and composite elements respectively.

11.2.2.4 Permanent Actions Arising from a Single Source

Unfavourable (or destabilising) and favourable (or stabilising) permanent actions may in some situations be considered as coming from a single source. If they are considered so, a single partial factor may be applied to the sum of these actions or to the sum of their effects. Design values of geotechnical actions are dependent upon both γ_F and γ_M and this “single source principle” may therefore be applied to both γ_F and γ_M for geotechnical actions. However, as in most cases the destabilising action is dependent on active or at rest pressure and the stabilizing action is dependent on passive pressure, a single value of γ_M normally produces the most unfavourable effect.

For bridges and earth retaining structures, the “single source principle” can be applied to the following for the structural ultimate limit state (STR) and the geotechnical ultimate limit state (GEO):

- a. The soil either side of the embedded length of piles of an embedded abutment.
- b. Natural ground above excavation level on an embedded wall, if this is similar to the soil in the embedded section below excavation level.
- c. Earth pressure on different bays of a horizontal spanning counterfort or buttress wall.
- d. Backfill from the same source and compacted to the same specification installed behind propped abutments or the abutments of integral bridges or buried structures.
- e. Fill placed on different spans of a multi-span buried structure.

The “single source principle” is not applied at the equilibrium limit state (EQU).

The “single source principle” should not normally be applied to the fill placed in front of a retaining structure (i.e. if the earth pressure arising from the fill is treated as an action, it should be taken as favourable), since it might not be compacted to the same quality as the fill behind the retaining structure or it might subsequently be altered to accommodate road widening, installation of services, etc.

Vertical and horizontal components of a single action should be subject to the same partial factor.

11.3 Ultimate Limit State

11.3.1 General

Where relevant, verification shall be carried out for not to exceed the following limit states:

- Loss of equilibrium of the structure or the ground (EQU)
- Internal failure or excessive deformation of the structure or structural elements (STR)
- Failure or excessive deformation of the ground (GEO)
- Loss of equilibrium of the structure or the ground due to uplift by water pressure (buoyancy) or other vertical actions (UPL)

The partial factors in persistent and transient design situations shall be as taken as defined in this Chapter. The partial factors for actions or action effects in accidental situations shall be taken as 1.0. More severe values than those recommended in this Chapter, in agreement with the authority, shall be used in case of abnormal risk or unusual or exceptionally difficult ground or loading conditions

When calculating the design value of the resistance, (R_d), or the design value of the effect of actions, (E_d), the model factors shall be applied to partial factors to ensure that the results of the design calculation are either accurate or on the safe side.

For bridges and other structures subject to highway loading, an additional model factor may be introduced for evaluation of the earth pressure coefficient K (e.g. for abutments and walls). The model factors may be applied where the method of analysis of a bridge is innovative or where the results of calculations are of uncertain reliability. A model factor of 1.2 shall be applied to the earth pressure coefficients at active and at rest condition (K_a and K_0) for unfavourable pressure at all ultimate limit states for the design approach combinations 1 and 2 (refer Clause 11.3.3.2). The model factor shall not be applied to earth pressure in the design of embedded walls nor to the traffic surcharge, nor for serviceability limit states. Model factors required in pile design shall be as provided in Clause 11.3.3.4.2.

11.3.2 Verification of Static Equilibrium

For verification of static equilibrium (EQU), following condition shall be complied.

$$E_{dst,d} \leq E_{stb,d} + T_d$$

Equation 11.4

Where,

$E_{dst,d}$ = The design value of the effect of destabilising actions.

$E_{stb,d}$ = The design value of the effect of stabilising actions.

T_d = The design value of any stabilising shear resistance of the ground or of structural elements.

For the verification of equilibrium (EQU) limit state under ULS in persistent and transient design situations, the partial factors on actions and soil parameters shall be applied as in Table 11.1 and Table 11.2 respectively. Design value of earth pressures should be based on the design value of the actions giving rise to the earth pressure.

Table 11.1 Partial Factors of Actions (γ_F) for Verification of Equilibrium (EQU) Limit States.

Action	Value
Actions on bridges	Values given in Table 3.31. The term $\gamma_{G\ sup}$ and $\gamma_{G\ inf}$ correspond with $\gamma_{G\ dst}$ and $\gamma_{G\ stb}$

Table 11.2 Partial Factors on Soil Parameters (γ_M) for Verification of Equilibrium (EQU) Limit States

Action	Symbol	Value
Tangent of the angle of shearing resistance (i.e. $\tan \varphi'$ and $\tan \varphi'_{cv}$)	$\gamma_{\varphi'}$	1.1
Effective Cohesion	$\gamma_{c'}$	1.1
Undrained Shear Strength	γ_{cu}	1.2
Unconfined Strength	γ_{qu}	1.2

11.3.3 Verification of Resistance

11.3.3.1 General

The verification of resistance for structural (STR) and ground limit states (GEO) in persistent and transient design situations shall be carried out and it shall be verified that:

$$E_d \leq R_d$$

Equation 11.5

Where,

E_d = The design value of the effects of actions

R_d = The design value of the corresponding resistance to the actions

11.3.3.2 Design Approach

For verification of resistance of foundations other than axially loaded piles and anchors, it shall be verified that a limit state of rupture or excessive deformation will not occur with either of the following combinations of sets of partial factors:

Combination 1: $A1$ "+" $M1$ "+" $R1$

Combination 2: $A2$ "+" $M2$ "+" $R1$

Where, "+" implies: "to be combined with".

In these combinations 1 and 2, partial factors are applied to actions and to ground strength parameters. A denotes actions or action effects, M denotes for soil parameters and R denotes for resistances.

For the design of axially loaded piles and anchors, it shall be verified that a limit state of rupture or excessive deformation will not occur with either of the following combinations of sets of partial factors:

Combination: 1: $A1$ "+" $M1$ "+" $R1$

Combination 2: $A2$ "+" ($M1$ or $M2$) "+" $R4$

In combination 1, partial factors are applied to actions and to ground strength parameters. In combination 2, partial factors are applied to actions, to ground resistances and sometimes to ground strength parameters.

In combination 2, set $M1$ is used for calculating resistances of piles or anchors and set $M2$ for calculating unfavourable actions on piles e.g. to negative friction or transverse loading.

If one of the two combinations governs the design, calculating for the other combination need not be carried out. However, different combinations may be critical to different aspects of the same design.

11.3.3.3 Design Actions

In calculating the design value of action effects, the partial factor may directly be applied to actions themselves or to action effects.

For the verification of structural (STR) and geotechnical (GEO) limit states, set $A1$ or set $A2$ of the partial factors on actions (γ_F) or the effects of actions (γ_E) shall be applied as in Table 11.3. Design value of earth pressures should be based on the design value of the actions giving rise to the earth pressure.

Table 11.3 Partial Factors on Actions (γ_F) or the Effect of Actions (γ_E) for Structural (STR) and Geotechnical (GEO) Limit States

Action	Symbol	
	$A1$	$A2$
Actions on bridges	Values given in Table 3.32	Values given in Table 3.33

For the verification of structural (STR) and geotechnical (GEO) limit states, set $M1$ or set $M2$ of the partial factors on soil parameters (γ_M) shall be applied as in Table 11.4.

Table 11.4 Partial Factors for Soil Parameters (γ_M) for Structural (STR) and Geotechnical (GEO) Limit States

Soil Parameter	Symbol	Value	
		M1	M2
Tangent of the angle of shearing resistance (i.e. $\tan \phi'$ and $\tan \phi'_{cv}$)	$\gamma_{\phi'}$	1.0	1.25
Effective Cohesion	$\gamma_{c'}$	1.0	1.25
Undrained Shear Strength	γ_{cu}	1.0	1.4
Unconfined Strength	γ_{qu}	1.0	1.4

11.3.3.4 Design Resistances

In calculating the design value of resistances, the partial factor may directly be applied to ground properties or resistances or to both. In design procedures where the effect of actions are factored, the partial factor for actions $\gamma_F = 1.0$ shall be adopted.

11.3.3.4.1 Spread Foundations

For verifications of structural (STR) and geotechnical (GO) limit states of spread foundations, set R1 of the partial factors ($\gamma_{R,v}$) on bearing resistance and ($\gamma_{R,h}$) on sliding resistance shall be applied as in Table 11.5.

Table 11.5 Partial Resistance Factors (γ_R) for Spread Foundations for Structural (STR) and Geotechnical (GEO) Limit States

Resistance	Symbol	Set R1
Bearing	$\gamma_{R,v}$	1.0
Sliding	$\gamma_{R,h}$	1.0

11.3.3.4.2 Pile Foundations

For verifications of structural (STR) and geotechnical (GEO) limit states of pile foundations, the partial factors on resistance (γ_R) shall be applied for driven and bored piles as in Table 11.6 and Table 11.7.

These values are used to convert characteristic values to design values for ultimate limit state calculation. They apply irrespective of the process by which the characteristic resistances are derived. Characteristic resistances may be derived from static load testing, dynamic load testing or ground test results.

The characteristic resistance may be derived from static load tests (using Clause 11.6.3.3 for compressive loading and 11.6.4.3 for tensile loading) or from ground test results (using Equations 11.17 or 11.18 for compressive loading and 11.27 or 11.28 for tensile loading). When Equation 11.18 or 11.28 is used to derive the characteristic resistance, a model factor shall be applied to the shaft and base resistance calculated using characteristic value of soil properties. The value of the model factor shall be 1.4 except that it may be reduced to 1.2 if the resistance is verified by a maintained load test taken to the calculated unfactored ultimate resistance.

Table 11.6 Partial Resistance Factors for Driven Piles (γ_R) for Structural (STR) and Geotechnical (GEO) Limit States

Resistance	Symbol	Set R1	Set R4 without explicit verification of SLS	Set R4 with explicit verification of SLS
Base	γ_b	1.0	1.7	1.5
Shaft (compression)	γ_s	1.0	1.5	1.3
Total/ combined (compression)	γ_t	1.0	1.7	1.5
Shaft in tension	$\gamma_{s,t}$	1.0	2.0	1.7

Table 11.7 Partial Resistance Factors for Bored Piles (γ_R) for Structural (STR) and Geotechnical (GEO) Limit States

Resistance	Symbol	Set R1	Set R4 without explicit verification of SLS	Set R4 with explicit verification of SLS
Base	γ_b	1.0	2.0	1.7
Shaft (compression)	γ_s	1.0	1.6	1.4
Total/ combined (compression)	γ_t	1.0	2.0	1.7
Shaft in tension	$\gamma_{s,t}$	1.0	2.0	1.7

11.3.3.4.3 Retaining Structures

For verifications of structural (STR) and geotechnical (GEO) limit states of retaining structure, set R1 of the partial factors on resistance (γ_R) shall be applied as in Table 11.8.

Table 11.8 Partial Resistance Factors for Retaining Structure (γ_R) for Structural (STR) and Geotechnical (GEO) Limit States

Resistance	Symbol	Set R1
Bearing capacity	$\gamma_{R,v}$	1.0
Sliding resistance	$\gamma_{R,h}$	1.0
Earth resistance	$\gamma_{R,e}$	1.0

11.3.4 Verification for Uplift

Verification for uplift shall be carried out by checking that the design value of the combination of destabilising permanent and variable vertical actions ($V_{dst,d}$) is less than or equal to the sum of the design value of the stabilising permanent vertical actions ($G_{stb,d}$) and of the design value of any additional resistance to uplift (R_d):

$$V_{dst,d} \leq G_{stb,d} + R_d$$

Equation 11.6

Where,

$$V_{dst,d} = G_{dst,d} + Q_{dst,d}$$

Equation 11.7

For the verification of uplift limit state in persistent and transient design situations, the following partial factors on actions (γ_F) shall be applied for $G_{dst,d}$, $Q_{dst,d}$, $G_{stb,d}$ and R_d for persistent and transient situations as in Table 11.9 and Table 11.10.

Table 11.9 Partial Factors on Actions (γ_R) for Verification of Uplift Limit State

Action	Symbol	Value
Permanent actions:		
Destabilising (unfavourable)	$\gamma_{G,dst}$	1.0
Stabilising (favourable)	$\gamma_{G,stb}$	0.9
Variable actions:		
Destabilising (unfavourable)	$\gamma_{Q,dst}$	1.5
Stabilising (favourable)	$\gamma_{Q,stb}$	0

Table 11.10 Partial Factors on Soil Parameters (γ_M) and Resistances (γ_R) for Verification of Uplift

Soil Parameter	Symbol	Value
Tangent of the angle of shearing resistance (i.e. $\tan \phi'$ and $\tan \phi'_{cv}$)	$\gamma_{\phi'}$	1.25
Effective cohesion	$\gamma_{c'}$	1.25
Undrained shear strength	γ_{cu}	1.40
Resistance	Symbol	Value
Tensile pile resistance	$\gamma_{s,t}$	Ref note below
Anchorage resistance	γ_a	1.40
Note: Pile design shall comply with the provisions in Clause 11.3.3.4.2		

11.4 Serviceability Limit State

Verification for serviceability limit states in the ground or in a structural section, element or connection, shall either require that:

$$E_d \leq C_d$$

Equation 11.8

Where,

E_d = Design value of the effect of actions.

C_d = Limiting design value of the relevant serviceability criterion.

Values of partial factors for serviceability limit states should normally be taken equal to 1.0.

In foundation design, limiting values shall be established for the foundation movements. Any differential movements of foundations leading to deformation in the supported structure shall be limited to ensure that they do not lead to a limit state in the supported structure.

For normal structures with isolated foundations, total settlements up to 50 mm are often acceptable. The differential settlement shall be checked between adjacent foundations. The angular distortion of superstructure resulting on account of differential settlement shall not exceed 1 in 400.

11.5 Spread Foundations

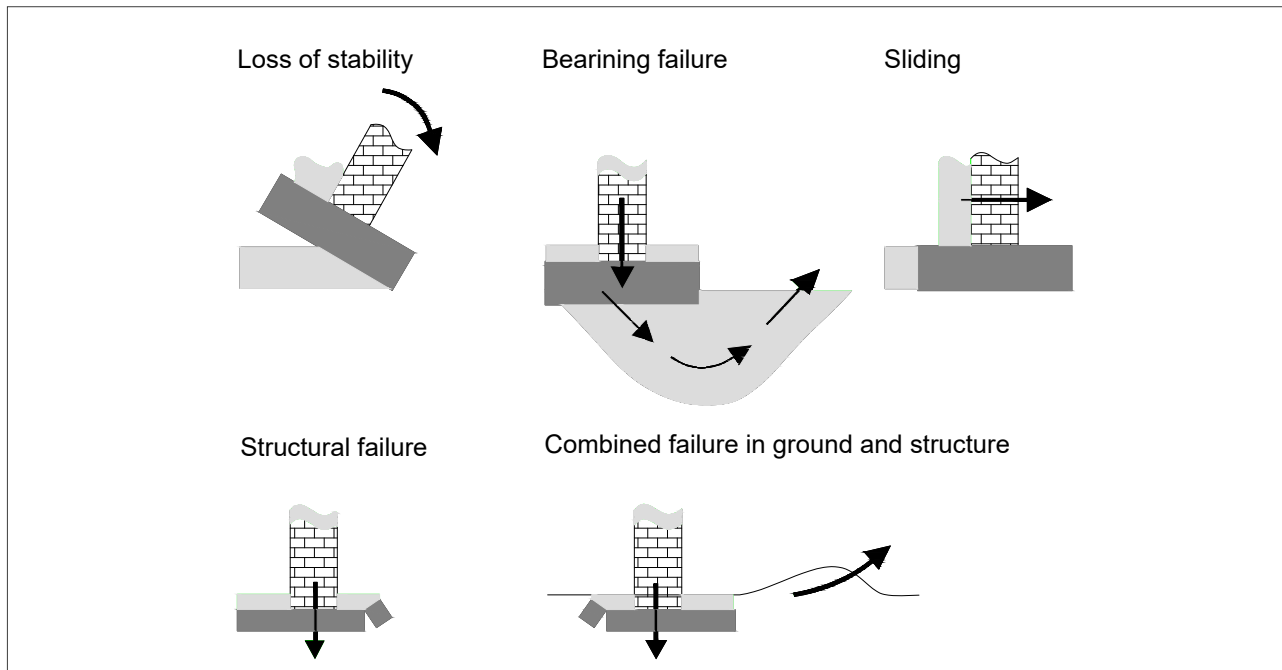
11.5.1 General

The provisions of this Section apply to spread foundations including pads, strips and rafts. The following limit states shall be considered for verification of foundations:

- Loss of equilibrium.
- Bearing resistance failure.
- Failure by sliding.
- Combined failure in the ground and in the structure.
- Structural failure due to foundation movement.
- Excessive settlements.
- Excessive heave due to swelling and other causes.

The different states of failure of spread foundations are shown in Figure 11.1.

Figure 11.1 Failure of Shallow Foundations



Some of above are ultimate limit states and some are serviceability limit states. The design philosophy is that both types of limit states need to be considered as possibly critical in selecting the dimensions of the foundations.

For the earth pressures to be used in designing the spread foundations, it can be assumed that fully active pressure is applied to the earth retaining element except when the structure is propped or tend to rotate backwards. Appropriate value of the model factor (refer Clause 11.3.1) shall be applied to unfavourable active and at rest earth pressures.

11.5.2 Ultimate Limit State Design

Verification shall be carried out for the following:

11.5.2.1 Verification for Equilibrium

Verification against overturning for all structures and for foundations supporting the retaining structure or abutments shall be carried out.

11.5.2.2 Verification for Bearing Resistance

The following inequality shall be satisfied for all ultimate limit states:

$$V_d \leq R_d$$

Equation 11.9

Where,

V_d = The design value of the vertical load or component of the total action acting normal to the foundation base including weight of foundation, weight of backfill material and earth pressures, water pressures.

R_d = The design value of the bearing resistance.

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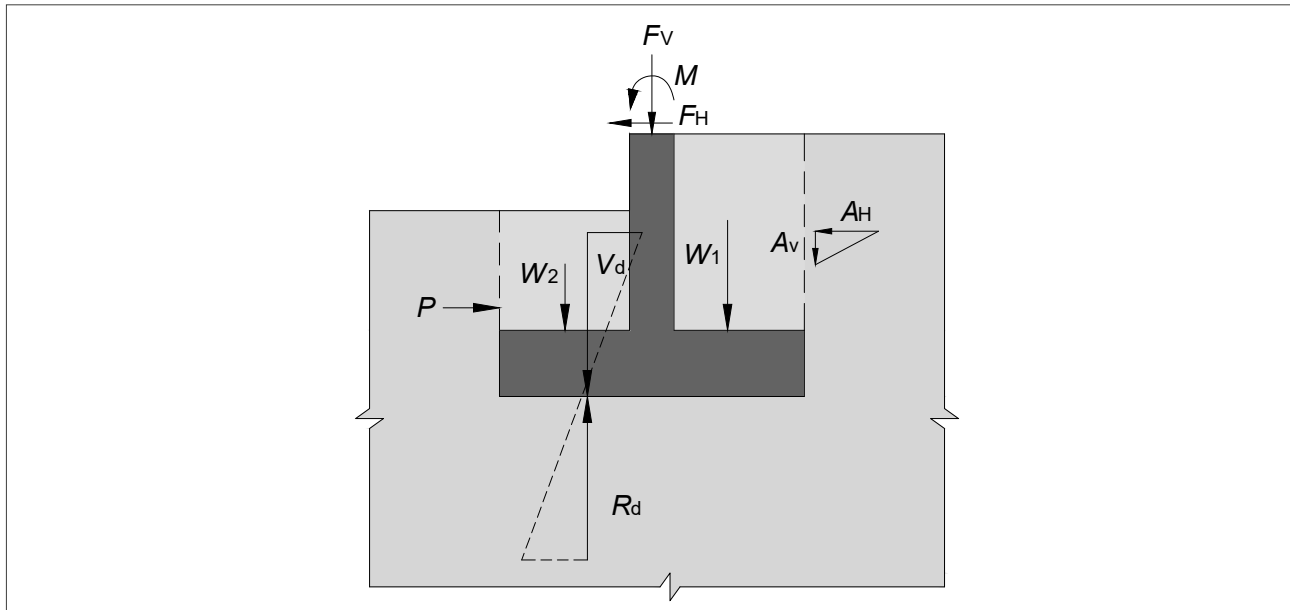
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The design value of actions V_d , includes the variable and permanent vertical loads. The permanent loads may include the following

- Supported permanent loads
- Weight of foundation
- Weight of the back fill
- Loads from water pressure
- Uplift

Figure 11.2 Action on Spread Foundation



The bearing strength (strength per unit area) at the founding level can be determined by commonly recognised analytical method of geotechnical engineering and is a part of Geotechnical investigations Manual. A sample method for spread foundations is given in Appendix E.

11.5.2.3 Verification for Sliding Resistance

Where the loading is not normal to the foundation base, foundations shall be checked against failure by sliding on the base. The following condition shall be satisfied.

$$H_d \leq R_d + R_{p,d}$$

Equation 11.10

Where:

H_d = The design value of the horizontal load or component of the total action acting parallel to the foundation base. It shall include the design value of any active earth forces imposed on foundations.

R_d = The design value of the resistance.

$R_{p,d}$ = The design value of the resisting force caused by earth pressure on the side of a foundation

The possibility that the soil in front of the foundation may be removed by erosion or human activity shall be considered.

a. Design shear resistance for drained conditions.

For drained conditions, the design shear resistance, R_d , shall be calculated either by factoring the ground properties or the ground resistance as follows:

$$R_d = V'_d \tan \delta_d$$

Equation 11.11

or

$$R_d = \frac{V'_d \tan \delta_k}{\gamma_{R,h}}$$

Equation 11.12

Where,

V'_d = The design value of the effective vertical action or component of the total action acting normal to the foundation base.

δ_d = The design value of structure-ground interface friction angle.

δ_k = The characteristic value of structure-ground interface friction angle.

$\gamma_{R,h}$ = The partial factor for sliding resistance.

For the design approach proposed in this Manual, only Equation 11.11 shall be used.

In design procedures where the effects of actions are factored, the partial factor for the actions (γ_F) is 1.0 and $V'_d = V_k$ in Equation 11.12.

Where V_k is the characteristic value of the effective vertical action or component of the total action acting normal to the foundation base.

The design friction angle δ_d may be assumed equal to the design value of the effective critical state angle of shearing resistance, $\phi'_{cv,d}$, for cast-in-situ concrete foundations and equal to $2/3 \phi'_{cv,d}$ for smooth precast foundations. Any effective cohesion c' should be neglected

Where , $\phi'_{cv,d}$ is the design value of critical state angle of shearing resistance.

b. Design shear resistance for undrained conditions.

For undrained conditions, the design shearing resistance, R_d , shall be calculated either by factoring the ground properties or the ground resistance as follows:

$$R_d = A c_{u,d}$$

Equation 11.13

or

$$R_d = \frac{A c_{u,k}}{\gamma_{R,h}}$$

Equation 11.14

Where,

A = The base area of foundation ($A = B \times L$).

$c_{u,d}$ = The design value of undrained shear strength.

$c_{u,k}$ = The characteristic value of undrained shear strength.

For the design approach proposed in this Manual, only Equation 11.13 shall be used.

If it is possible for water or air to reach the interface between a foundation and an undrained clay subgrade, the following check shall be made:

$$R_d \leq 0.4 V_d$$

Equation 11.15

11.5.2.4 Loads with Large Eccentricities

Special precautions shall be taken where the eccentricity of loading exceeds 1/3 of the width of a rectangular footing or 0.6 of the radius of a circular footing. Such precautions include:

- Careful review of the design values of actions.
- Designing the location of the foundation edge by taking into account 0.10m construction tolerances.

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11.5.2.5 Structural Failure Due to Foundation Movement

Differential vertical and horizontal foundation displacements shall be considered to ensure that they do not lead to an ultimate limit state occurring in the supported structure.

11.5.3 Serviceability Limit State Design

11.5.3.1 General

Account shall be taken of displacements caused by actions on the foundation. For soft clays, settlement calculations shall always be carried out. For spread foundations on stiff and firm clays, calculations of vertical displacement (settlement) should usually be undertaken. The serviceability limit state design loads shall be used when calculating foundation displacements for comparison with serviceability criteria. Foundation displacements shall be considered both in terms of displacement of the entire foundation and differential displacements of parts of the foundation.

When determining the dimensions of the foundations, it is prudent to ensure that no uplift occurs anywhere under the foundations at SLS under the characteristic combination of actions, because, once uplift has started to occur, the foundation rotation becomes increasingly sensitive to variations in the eccentricity of the applied load.

11.5.3.2 Settlement

Calculations of settlements shall include both immediate and delayed settlement. Following three components of settlement should be considered for partially or fully saturated soils:

- Immediate settlement; for fully-saturated soil due to shear deformation at constant volume, and for partially-saturated soil due to both shear deformation and volume reduction.
- Settlement caused by consolidation.

The depth of the compressible soil layer to be considered when calculating settlement may normally be taken as the depth at which the effective vertical stress due to the foundation load is 20 % of the effective overburden stress. For many cases this depth may also be roughly estimated as 1 to 2 times the foundation width, but may be reduced for lightly-loaded, wider foundation rafts.

To ensure the avoidance of a serviceability limit state, assessment of differential settlements and relative rotations shall take account of both the distribution of loads and the possible variability of the ground. An analysis of ground-structure interaction may be used to justify realistic values of differential settlements. For spread foundations on natural ground, it should be taken into account that some differential settlement normally occurs even if the calculation predicts uniform settlement only.

The tilting of an eccentrically loaded foundation should be estimated by assuming a linear bearing pressure distribution and then calculating the settlement at the corner points of the foundation, using the vertical stress distribution in the ground beneath each corner point.

11.5.4 Foundations on Rock - Additional Design Considerations

The design of spread foundations on rock shall take account of the following features:

- The deformability and strength of the rock mass and the permissible settlement of the supported structure.
- The presence of any weak layers, for example solution features or fault zones, beneath the foundation.
- The presence of bedding joints and other discontinuities and their characteristics (for example filling, continuity, width, spacing).
- The state of weathering, decomposition and fracturing of the rock.

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Spread foundations on rock may normally be designed using the method of presumed bearing pressures. For strong intact igneous rocks, gneissic rocks, limestones and sandstones, the presumed bearing pressure is limited by the compressive strength of the concrete foundation.

11.5.5 Structural Design of Spread Foundations

The bearing pressure beneath a stiff foundation may be assumed to be distributed linearly. A more detailed analysis of soil-structure interaction may be used to justify a more economic design.

The distribution of bearing pressure beneath a flexible foundation may be derived by modelling the foundation as a beam or raft resting on a deforming continuum or series of springs, with appropriate stiffness and strength

The serviceability of strip and raft foundations shall be checked assuming serviceability limit state loading and a distribution of bearing pressure corresponding to the deformation of the foundation and the ground

For design situations with concentrated loads acting on a strip or raft foundation, forces and bending moments in the foundation may be derived from a subgrade reaction model of the ground, using linear elasticity.

Total and differential settlements of the structure as a whole should be calculated in accordance with Clause 11.5.3.2.

11.6 Pile Foundations

11.6.1 Scope

This section describes the provisions for the design of pile foundations. These provisions apply to end-bearing piles, friction piles, tension piles and transversely loaded piles. They apply to piles installed by driving or boring.

11.6.2 Axially Loaded Piles

The axially loaded pile foundations shall be verified for following limit states.

- Ultimate limit states of compressive or tensile resistance failure of a single pile.
- Ultimate limit states of compressive or tensile resistance failure of the pile foundation as a whole (group of piles).
- Ultimate limit states of collapse or severe damage to a supported structure caused by excessive displacement or differential displacements of the pile foundation.
- Serviceability limit states in the supported structure caused by displacement of the piles.

Normally the design should consider the margin of safety with respect to compressive or tensile resistance failure. The failure in resistance is the state in which the pile foundation displaces significantly downwards or upwards with negligible increase or decrease of resistance.

For piles in compression, it is often difficult to define an ultimate limit state from a load settlement plot showing a continuous curvature. In these cases, settlement of the pile top equal to 10 % of the pile base diameter should be adopted as the "failure" criterion.

For piles that undergo significant settlements, ultimate limit states may occur in supported structures before the resistance of the piles is fully mobilised. In these cases, a cautious estimate of the possible range of the settlements shall be adopted in design.

11.6.3 Ground Resistance of Piles in Axial Compression

11.6.3.1 General

To verify that the pile foundation will support the design load with adequate safety against compressive failure, the following inequality shall be satisfied for all ultimate limit state load cases and load combinations:

$$F_{c,d} \leq R_{c,d}$$

Equation 11.16

In principle, $F_{c,d}$ should include the weight of the pile itself and $R_{c,d}$ should include the overburden pressure of the soil at the foundation base. However, these two items may be disregarded if they cancel approximately. They need not cancel if:

- Downdrag is significant.
- The soil is very light.
- The pile extends above the surface of the ground.

For piles in groups, two failure mechanisms shall be taken into account:

- Compressive resistance failure of the piles individually.
- Compressive resistance failure of the piles and the soil contained between them acting as a block.
- The design resistance shall be taken as the lower value caused by these two mechanisms.

The compressive resistance of the pile group acting as a block may be calculated by treating the block as a single pile of large diameter.

If the layer in which the piles bear overlies a layer of weak soil, the effect of the weak layer on the compressive resistance of the foundation shall be considered.

11.6.3.2 Ultimate Compressive Resistance from Ground Test Results

The characteristic values of pile resistance $R_{b,k}$ and $R_{s,k}$ from values of ground parameters may be obtained either by calculating from:

$$R_{c,k} = R_{b,k} + R_{s,k}$$

$$= \frac{R_{b,cal} + R_{s,cal}}{\zeta} = \frac{R_{c,cal}}{\zeta} = \text{Min} \left\{ \frac{(R_{c,cal})_{mean}}{\zeta_3}, \frac{(R_{c,cal})_{min}}{\zeta_4} \right\}$$

Equation 11.17

Where ζ_3 and ζ_4 are correlation factors related to the number of profiles tests, n , and are applied respectively to:

$$\text{mean values } (R_{c,cal})_{mean} = (R_{b,cal} + R_{s,cal})_{mean} = (R_{b,cal})_{mean} + (R_{s,cal})_{mean}$$

$$\text{and the lowest value } (R_{c,cal})_{min} = (R_{b,cal} + R_{s,cal})_{min}$$

or by

$$R_{c,k} = R_{b,k} + R_{s,k}$$

And $R_{b,k}$, $R_{s,k}$, are given as

$$R_{b,k} = A_b q_{b,k} \text{ and } R_{s,k} = \sum_i A_{s,i} q_{s,i,k}$$

Equation 11.18

Where $q_{b,k}$ and $q_{s,i,k}$ are characteristic values of base resistance and shaft friction in the various strata, obtained from values of ground parameters.

For verification of structural (STR) and geotechnical (GEO) limit states, the recommended values of correlation factors ξ_3 and ξ_4 to be applied to derive the characteristic value of resistance of axially loaded piles from ground test results are given in Table 11.11.

Table 11.11 Correlation Factors ξ_3 and ξ_4 for Resistance of Axially Loaded Piles from Ground Test Results

n	1	2	3	4	5	7	10
ξ_3	1.55	1.47	1.42	1.38	1.36	1.33	1.30
ξ_4	1.55	1.39	1.33	1.29	1.26	1.20	1.15

Note: For structures having sufficient stiffness and strength to transfer loads from weak to strong piles, the values of ξ_3 and ξ_4 may be divided by 1.1 provided that ξ_3 is never less than 1.0

The design compressive resistance of a pile ($R_{c,d}$), shall be derived from:

$$R_{c,d} = R_{b,d} + R_{s,d} = \frac{R_{b,k}}{\gamma_b} + \frac{R_{s,k}}{\gamma_s} \quad \text{Equation 11.19}$$

The recommended values of the partial factors for persistent and transient situations are given in Table 11.6 and Table 11.7 for driven and bored piles.

11.6.3.3 Ultimate Compressive Resistance from Static Load Tests

The ultimate compressive resistance of piles can be derived from static pile load tests. When deriving the ultimate characteristic compressive resistance $R_{c,k}$ from values $R_{c,m}$ measured in one or several static pile load tests, an allowance shall be made for the variability of the ground and the variability of the effect of pile installation. It can be derived from following Equation:

$$R_{c,k} = \text{Min} \left\{ \frac{(R_{c,m})_{\text{mean}}}{\xi_1}, \frac{(R_{c,m})_{\text{min}}}{\xi_2} \right\} \quad \text{Equation 11.20}$$

Where ξ_1 and ξ_2 are correlation factors related to the number of piles tested, n , and are applied to the mean value $(R_{c,m})_{\text{mean}}$ and the lowest value $(R_{c,m})_{\text{min}}$ of $R_{c,m}$ respectively.

For verification of structural (STR) and geotechnical (GEO) limit states, the recommended values of correlation factors ξ_1 and ξ_2 to be applied to derive the characteristic value of resistance of axially loaded piles from static pile load tests are given in Table 11.12.

Table 11.12 Correlation Factors ξ_1 and ξ_2 for Resistance of Axially Loaded Piles from Static Pile Load Tests

n (number of tested piles)	1	2	3	4	≥ 5
ξ_1	1.55	1.47	1.42	1.38	1.35
ξ_2	1.55	1.35	1.23	1.15	1.08

Note: For structures having sufficient stiffness and strength to transfer loads from weak to strong piles, the values of ξ_1 and ξ_2 may be divided by 1.1 provided that ξ_1 is never less than 1.0.

When it is possible to measure the base resistance and shaft resistance separately from static load test on piles, the characteristic compressive resistance of the ground, $R_{c,k}$, may be derived from the characteristic values of the base resistance, $R_{b,k}$, and of the shaft resistance, $R_{s,k}$, such that:

$$R_{c,k} = R_{b,k} + R_{s,k} \quad \text{Equation 11.21}$$

These components may be derived directly from static load test results, or estimated on the basis of ground test results or dynamic load tests.

The design resistance, shall be derived from either:

$$R_{c,d} = \frac{R_{c,k}}{\gamma_t}$$

Equation 11.22

or

$$R_{c,d} = \frac{R_{b,k}}{\gamma_b} + \frac{R_{s,k}}{\gamma_s}$$

Equation 11.23

The recommended values of the partial factors for persistent and transient situations are given in Table 11.6 and Table 11.7 for driven and bored piles.

11.6.3.4 Ultimate Compressive Resistance from Dynamic Impact Tests

Where a dynamic impact (hammer blow) pile test, which measures strain and acceleration versus time during the impact event, is used to assess the resistance of individual compression piles, the validity of the result shall be demonstrated by previous evidence of acceptable performance in static load tests on the same pile type of similar length and cross-section and in similar ground conditions.

When using a dynamic impact load test, the driving resistance of the pile should be measured directly on the site in question (working pile).

The design value of the compressive resistance of the pile, $R_{c,d}$ shall be derived from:

$$R_{c,k} = \text{Min} \left\{ \frac{(R_{c,m})_{\text{mean}}}{\xi_5}, \frac{(R_{c,m})_{\text{min}}}{\xi_6} \right\}$$

Equation 11.24

and

$$R_{c,d} = \frac{R_{c,k}}{\gamma_t}$$

Equation 11.25

Where, ξ_5 and ξ_6 are correlation factors related to the number of piles tested, n , and are applied to the mean $(R_{c,m})_{\text{mean}}$ and the lowest $(R_{c,m})_{\text{min}}$ value of $R_{c,m}$ respectively.

For verification of structural (STR) and geotechnical (GEO) limit states, the recommended values of correlation factors ξ_5 and ξ_6 to be applied to derive the characteristic value of resistance of axially loaded pile from dynamic impact tests are given in Table 11.13.

Table 11.13 Correlation Factors ξ_5 and ξ_6 for Resistance of Axially Loaded Piles from Dynamic Impact Tests

ξ for n (n = number of tested piles)	≥ 2	≥ 5	≥ 10	≥ 15	≥ 20
ξ_5	1.94	1.85	1.83	1.82	1.81
ξ_6	1.90	1.76	1.70	1.67	1.66

Notes:

a) The ξ values in the table are valid for dynamic impact tests.

b) The ξ -values may be multiplied with a model factor of 0.85 when using dynamic impact tests with Signal matching.

c) The ξ - values should be multiplied with a model factor of 1.10 when using a pile driving formula with measurement of the quasi-elastic pile head displacement during the impact.

d) The ξ -values shall be multiplied with a model factor of 1.20 when using a pile driving formula without measurement of the quasi-elastic pile head displacement during the impact.

e) If different piles exist in the foundation, groups of similar piles should be considered separately when selecting the number 'n' of test piles.

11.6.4 Ground Resistance of Piles in Tension

11.6.4.1 General

To verify that the pile foundation will support the design load with adequate safety against failure in tension, the following inequality shall be satisfied for all ultimate limit state load cases and load combinations:

$$F_{t,d} \leq R_{t,d}$$

Equation 11.26

For the verification of piles in tension, following two failure mechanisms shall be considered:

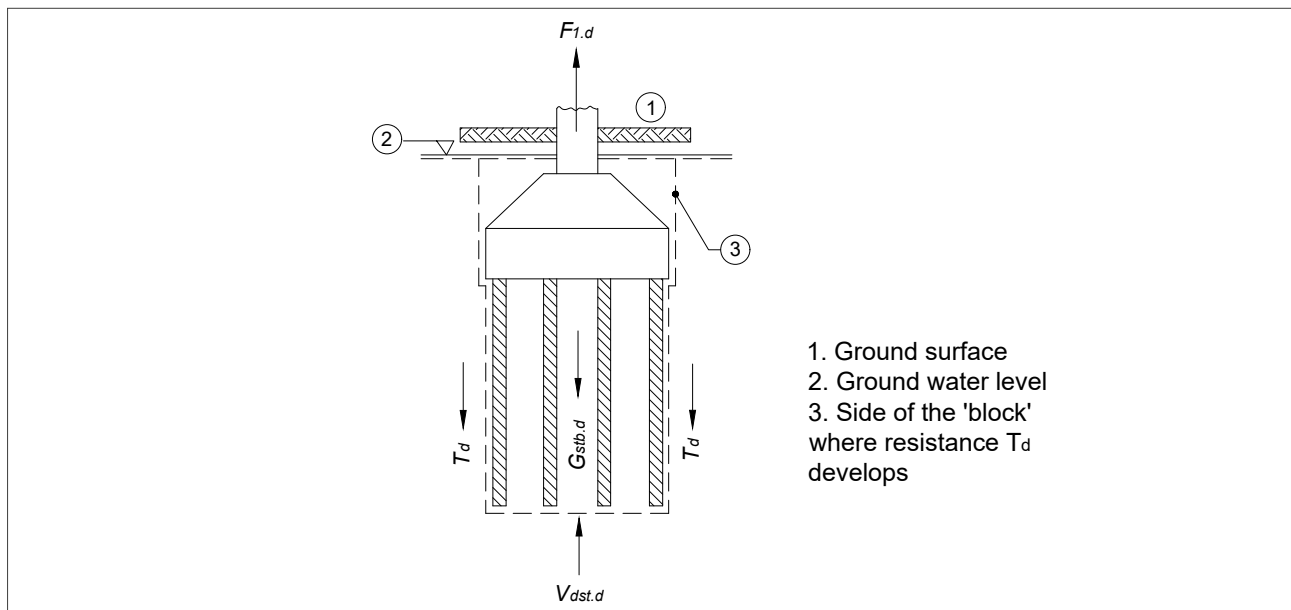
- Pull-out of the piles from the ground mass
- Uplift of the block of ground containing the piles

When considering the uplift of the block of ground containing the piles, the shear resistance T_d along the sides of the block may be added to the resisting forces shown in Figure 10.3.

Normally the block effect will govern the design tensile resistance if the distance between the piles is equal to or less than the square root of the product of the pile diameter and the pile penetration into the main resisting stratum.

The group effect, which may reduce the effective vertical stresses in the soil and hence the shaft resistances of individual piles in the group, shall be considered when assessing the tensile resistance of a group of piles.

Figure 11.3 Examples of Uplift of a Group of Piles



11.6.4.2 Ultimate Tensile Resistance from Ground Test Results

The characteristic value of tensile resistance from ground test results may be obtained either by calculating from:

$$R_{t,k} = \text{Min} \left\{ \frac{(R_{s,cal})_{mean}}{\xi_3} ; \frac{(R_{s,cal})_{min}}{\xi_4} \right\}$$

Equation 11.27

Where ξ_3 and ξ_4 are correlation factors that depends upon the number of profiles tests, n , and are applied respectively to mean $(R_{s,cal})_{mean}$ and to the lowest value $(R_{s,cal})_{min}$. The recommended values of ξ_3 and ξ_4 are in Table 11.11.

Or by

$$R_{t,k} = \sum_i A_{s,i} \cdot q_{s,i,k}$$

Equation 11.28

Where $q_{s,i,k}$ are characteristic values of shaft friction in the various strata obtained from values of ground properties.

The design resistance, shall be derived from:

$$R_{t,d} = \frac{R_{t,k}}{\gamma_t}$$

Equation 11.29

Where, $R_{t,k} = R_{s,k}$

The recommended values of the partial factors for persistent and transient situations are given in Table 11.6 and Table 11.7 for driven and bored piles.

11.6.4.3 Ultimate Tensile Resistance from Pile Load Tests

The design tensile resistance, $R_{t,d}$, shall be derived from:

$$R_{t,d} = \frac{R_{t,k}}{\gamma_{s,t}}$$

Equation 11.30

The recommended values of the partial factors for persistent and transient situations are given in Table 11.6 and Table 11.7 for driven and bored piles.

Normally when piles are to be loaded in tension, more than one pile should be tested. In the case of a large number of tension piles, at least 2 % should be tested.

The characteristic value of the pile tensile resistance shall be determined by:

$$R_{t,k} = \text{Min} \left\{ \frac{(R_{t,m})_{\text{mean}}}{\xi_1} ; \frac{(R_{t,m})_{\text{min}}}{\xi_2} \right\}$$

Equation 11.31

Where ξ_1 and ξ_2 are correlation factors related to the number of piles tested, n , and are applied respectively to the mean $(R_{t,m})_{\text{mean}}$ and the lowest $(R_{t,m})_{\text{min}}$ value of the measured tensile resistances. The recommended values of the correlation factors ξ_1 and ξ_2 are given in Table 11.12.

11.6.5 Serviceability Limit State

11.6.5.1 General

Vertical displacement of pile foundations is required for verification of the serviceability limit state of the supported structure. It shall be assessed and verified that the required serviceability criteria for displacement is met.

11.6.5.2 Pile foundations in Compression

The occurrence of a serviceability limit state in the supported structure due to pile settlements shall be checked, taking into account downdrag, where probable.

When the pile toe is placed in a medium-dense or firm layer overlying rock or very hard soil, the partial factors for ultimate limit state conditions are normally sufficient to satisfy serviceability limit state conditions.

Assessment of settlements shall include both the settlement of individual piles and the settlement due to group action. The settlement analysis should include an estimate of the differential settlements that may occur.

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When no load test results are available for an analysis of the interaction of the pile foundation with the superstructure, the load-settlement performance of individual piles should be assessed on the basis of empirically established safe assumptions.

11.6.5.3 *Pile Foundations in Tension*

The assessment of upward displacements of the pile foundation shall be done for verification of the required serviceability criteria.

For piles bearing in medium-to-dense soils and for tension piles, the requirements for the ultimate limit state design are normally sufficient to prevent a serviceability limit state in the supported structure.

Particular attention should be paid to the elongation of the pile material.

When very severe criteria are set for the serviceability limit state, a separate check of the upward displacements shall be carried out.

11.6.6 *Transversely Loaded Piles*

11.6.6.1 *General*

For assessing the resistance under transverse loads on piles, one of the following failure mechanisms should be considered:

- For short piles, rotation or translation as a rigid body.
- For long slender piles, bending failure of pile, accompanied by local yielding and displacement of the soil near the top of the pile.

The group effect shall be considered when assessing the resistance of transversely loaded piles.

It should be considered that a transverse load applied to a group of piles may result in a combination of compression, tension and transverse forces in the individual piles.

11.6.6.2 *Transverse Load Resistance from Pile Load Tests*

Transverse pile load tests shall be carried out to assess the resistance under transverse loading. The tests on transversely loaded piles need not normally be continued to a state of failure. The magnitude and line of action of the test load should simulate the design loading of the pile.

An allowance shall be made for the variability of the ground, particularly over the top few metres of the pile, when choosing the number of piles for testing and when deriving the design transverse resistance from load test results.

Any deviation from the normal construction conditions should be accounted for in the interpretation of the pile load test results. For pile groups, the effects of interaction and head fixity should be accounted for when deriving the transverse resistance from the results of load tests on individual test piles.

11.6.6.3 *Transverse Load Resistance from Ground Test Results*

The calculation of the transverse resistance of a long slender pile may be carried out using the theory of a beam loaded at the top and supported by a deformable medium characterised by a horizontal modulus of subgrade reaction.

The analysis of a transversely loaded pile shall include the possibility of structural failure of the pile in the ground.

The degree of freedom of rotation of the piles at the connection with the structure shall be taken into account when assessing the foundation's transverse resistance.

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11.6.6.4 Transverse Displacement

The assessment of the transverse displacement of a pile foundation shall take into account:

- The stiffness of the ground and its variation with strain level.
- The flexural stiffness of the individual piles.
- The moment fixity of the piles at the connection with the structure.
- The group effect.

11.6.7 Structural Design of Piles

Piles shall be verified against structural failure. The design strength properties of materials and design resistance shall be taken in accordance with the design guidelines provided in Chapter 4 for concrete and Chapter 5 for structural steel elements. The structure of piles shall be designed to accommodate all the situations to which the piles will be subjected.

Slender piles passing through water or thick deposits of extremely low strength fine soil shall be checked against buckling. A check for buckling is not required when the piles are contained by soils with a characteristic shear strength, c_u , exceeding 10 kPa.

11.6.8 Pile Load Tests

Pile load tests are used to assess the pile resistance and overall judgement of the representative pile under applied loads. If one pile load test is carried out, it shall normally be located where the most adverse ground conditions are believed to occur. If load tests are carried out on two or more test piles, the test locations shall be representative of the site of the pile foundation and one of the test piles shall be located where the most adverse ground conditions are believed to occur. Before carrying out the tests on piles, adequate time shall be allowed to ensure that the required strength of the pile material is achieved. Pile load tests can be static or dynamic.

11.6.8.1 Static Load Tests

The static load can be performed on trial piles or working piles and shall be carried out in accordance with the standard procedures.

The pile load test procedures, number of loading steps and duration shall be such that conclusions can be drawn about the deformation and rebound of a piled foundation from the measurements on the pile. For trial piles, the loading shall be applied to determine the ultimate failure load.

The number of trial piles for tests depends on the ground conditions and their variability across the site, total number and types of pile in the foundation design.

The number of working pile load tests shall be selected on the basis of the findings during installation and the total number of piles. The test load on working piles shall be at least equal to the design load for the pile but not more than 1.5 times the design load for the pile.

11.6.8.2 Dynamic Load Tests

Dynamic load tests are carried out to estimate the compressive resistance of the piles. Adequate site investigation shall be carried out and the method shall be calibrated against static load tests on the same type of pile, of similar length and cross-section, and in comparable soil conditions. Dynamic load tests may also be used as an indicator of the consistency of the piles and to detect weak piles.

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11.6.8.3 Load Test Report

A detail load test report shall be prepared for all load tests conducted at site. In general, a test report may include:

- A description of the site.
- The ground conditions with reference to ground investigations.
- The pile type.
- Description of the pile installation and of any problems encountered during the works.
- A description of the loading and measuring apparatus and the reaction system.
- Calibration documents for the load cells, the jacks and the gauges.
- The installation records of the test piles.
- Photographic records of the pile and the test site.
- Test results in numerical form.
- Time-displacement plots for each applied load when a step loading procedure is used.
- The measured load-displacement behaviour.
- Reasons for any departures from the above requirements.

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Appendices

A Actions and Combination of Actions

A.1 Horizontal Earth Pressure From Traffic Loads On Abutments And Walls Adjacent To Bridges

The traffic load models to determine the horizontal earth pressure on abutment walls and retaining/wing walls shall be in accordance with the Clause 3.5.8.1.

A.1.1 Simplified Models To Represent The Effects Of Traffic Loads

As an alternative to the detailed methods, simplified models that reflect the effects of traffic loads on abutments and walls may be used as follows:

- For walls and other retaining structures adjacent to the carriageway, where the traffic does not cross over the structure, the simplified traffic surcharge models given in Table A.1 may be used.
- For abutments, where the traffic cross over the structure, the simplified traffic surcharge model given in Table A.2 and Figure A.1 may be used.

Table A.1 Simplified Traffic Surcharge Model for Retaining Structures (where traffic does not cross over the structure)

Traffic Type	Characteristic vertical UDL (kN/m ²) applied at ground level to the area extending from the back face of the wall to a distance at least 1.5H from the structure, where H is the total height of the structure						
Distance from the centreline of the closest wheel to the back face of the wall, <i>a</i>	<i>a</i> < 1 m	<i>a</i> = 1 m	<i>a</i> = 2 m	<i>a</i> = 3 m	<i>a</i> = 4 m	<i>a</i> = 5 m	<i>a</i> ≥ 6 m
Normal highway traffic	Simplified approach not valid	25	15	10	10	5	5
SV80 + accompanying normal highway traffic		30	15	10	10	10	5
SV100 + accompanying normal highway traffic		40	20	15	10	10	10
SV 196 + accompanying normal highway traffic		45	30	25	20	20	15
SOV + accompanying normal highway traffic		45	35	25	20	20	15
Walkways and cycle tracks	5						
Note 1: Horizontal effects can be determined by multiplying the vertical UDL by <i>k_d</i> , which is the design value of <i>k_a</i> or <i>k_o</i> (as appropriate) based on <i>φ_d</i> .							
Note 2: The alternative detailed approaches are described in clause A.1.2.							
Note 3: For the design of walls subject to highway traffic with <i>a</i> < 1, can be modelled with a detailed approach as described in Clause A.1.2.							
Note 4: The surcharge due to construction plant can exceed the surcharge due to traffic.							

Table A.2 Simplified Traffic Surcharge Model for Abutments (where traffic crosses over the structure)

Category of Traffic Areas	Characteristic horizontal UDL acting on abutment wall, σ_h (kN/m ²), applied to the lane width	Characteristic horizontal line loads acting on wall, F (kN), applied to both edges of the lane width
Walkways, cycle tracks and footbridges	$5 K_d$	0
Normal highway traffic	$20 K_d$	$330 K_d$
Special vehicle traffic	$45 K_d$ (SOV) $30 K_d$ (SV196, SV100) $20 K_d$ (SV80)	$330 K_d$

Note 1: For the effects of load models not included in this table, the surcharge loads should be determined for the specific project or derived using the detailed analysis.

Note 2: For normal highway traffic, the loads should be applied to up to three lanes simultaneously, and the loads on the third lane may be reduced by a factor of 0.5.

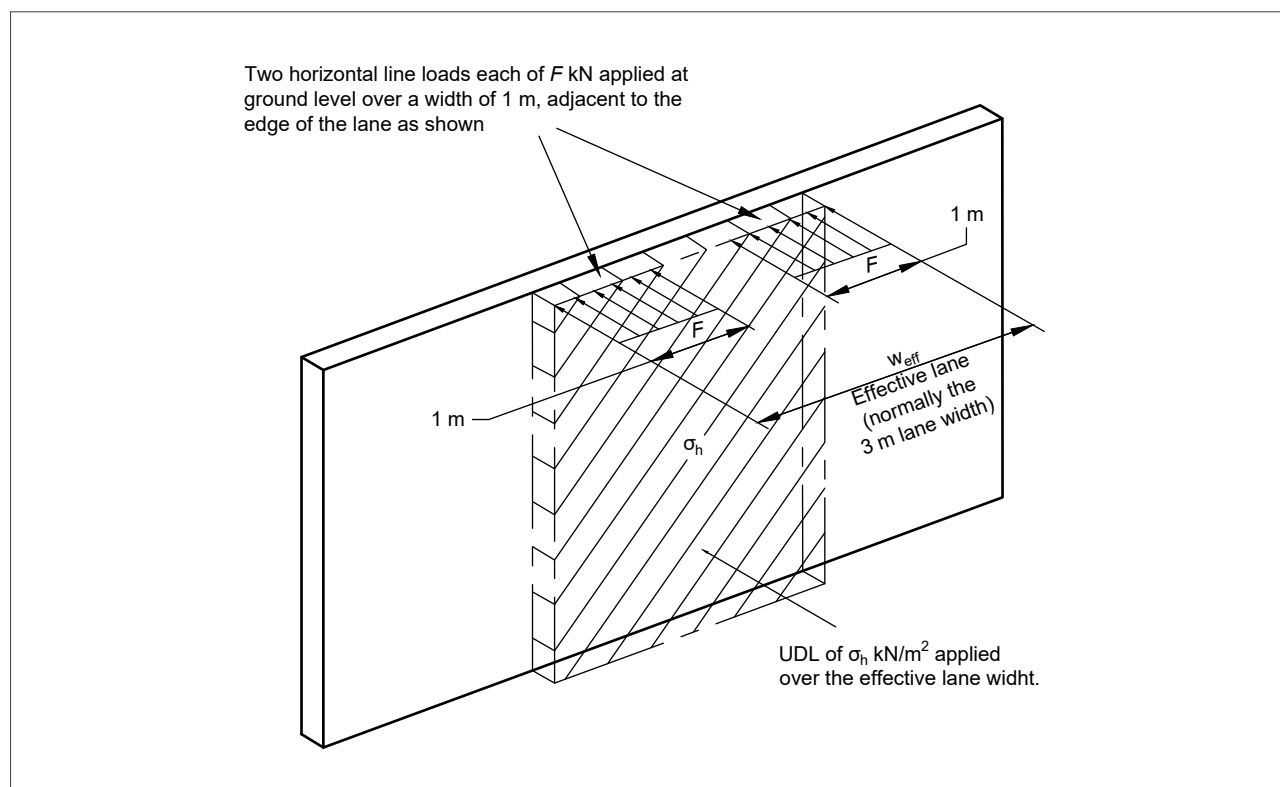
Note 3: For special vehicle traffic loading, one lane should be loaded with the special vehicle loading, and up to two adjacent lanes loaded with the normal traffic loading, with the normal traffic loading reduced by a factor of 0.75.

Note 4: K_d is the design value of K_a or K_o (as appropriate), based on ϕ'_d .

Note 5: For buried structures (culverts) with less than 2 m of fill above the top of the structure, the horizontal line loads F , should be applied at the top of the structure and reduced by the factor $(1 - H_c/2)^2$, where H_c is the depth of fill measured in metres from the surface of the carriageway to the top of the structure. For cover depths in excess of 2 m, the horizontal line loads may be ignored.

Note 6: For abutments with a height, H , less than 2 m, the horizontal line loads F may be reduced by the factor $1 - [1 - (H/2)]^2$.

Figure A.1 Horizontal Surcharge Load for Abutments



Note 1: The figure shows the surcharge model for one lane of effective width w_{eff} . The models for adjacent lanes are similar. For most situations, w_{eff} will be equal to 3.0 m. However, for carriageways between 5.4 m and 6.0 m in width subject solely to normal traffic, the effective lane width, w_{eff} , will be equal to the notional lane width.

Note 2: For carriageways subject to SV or SOV (LM3) loading which encroaches on the adjacent notional lane, the effective width of that lane available for carrying normal traffic should be taken as the distance from the edge of the LM3 vehicle to the far edge of the notional lane, except that if that width is less than 2.5 m, no normal traffic need be applied adjacent to the LM3 vehicle in that lane.

Note 3: The effective width of a lane carrying LM3 loading used in applying the surcharge model is always 3.0 m.

A.1.2 Detailed Analysis of Horizontal Pressure From Traffic Loads

The horizontal pressures from traffic loads may be analysed based on either of the following approaches:

- a. A rigorous analysis of the horizontal effects of the applied actions in the backfill, considering the strength of the backfill, the movement of the wall, the proximity of wheel loads to the wall and the high-pressure concentrations that develop just below the surface when a heavy wheel load is applied immediately behind the wall.
- b. The general model presented below for vertically spanning earth retaining structures as presented below.

The horizontal effects of vertical traffic loads on vertically spanning earth retaining structures (including wing walls and abutments) may be determined using the model shown in Figure A.2. The distance 'a' given in this figure is measured from the centreline of the load to the back of the wall.

For determining local concentrated pressures on a wall, the pressures arising from all the wheels affecting the structure or relevant part of the structure should be superimposed. However, for global effects on walls, each lane of traffic may be modelled as two parallel, uniformly distributed line loads, except for the SOV model vehicle, which can be modelled as four parallel, uniformly-distributed line loads.

The model in Figure A.2 can also be used for determining the horizontal effects of vertical loads other than traffic loads. Patch loads or strip loads of significant width can be simulated by superimposing the effects of a number of parallel line loads.

Figure A.2 Lateral and Vertical Dispersion of Finite Line Loads for Horizontal Pressure

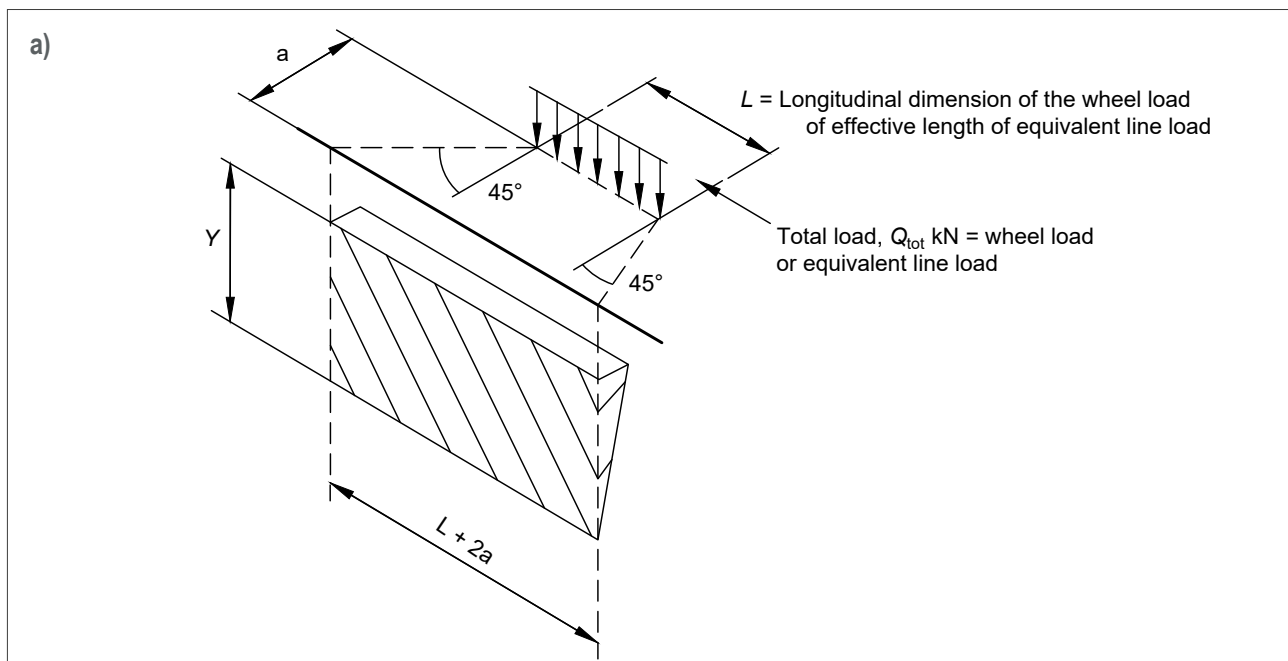


Figure A.2 Lateral and Vertical Dispersion of Finite Line Loads for Horizontal Pressure

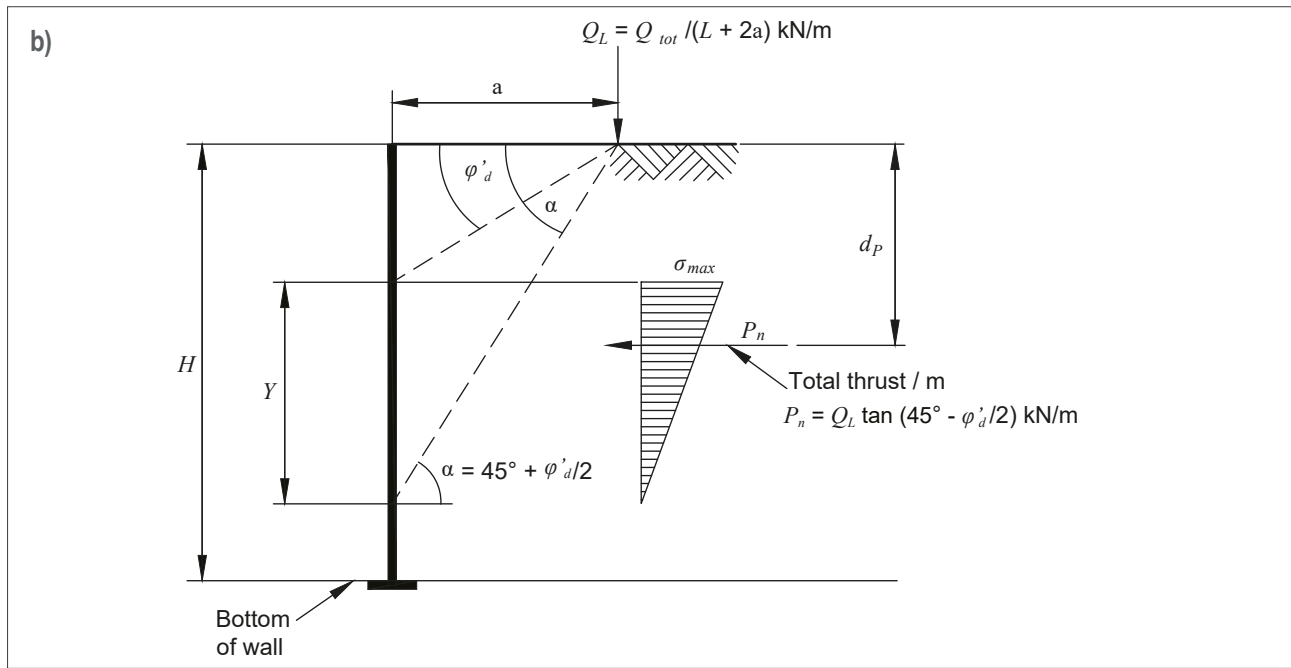
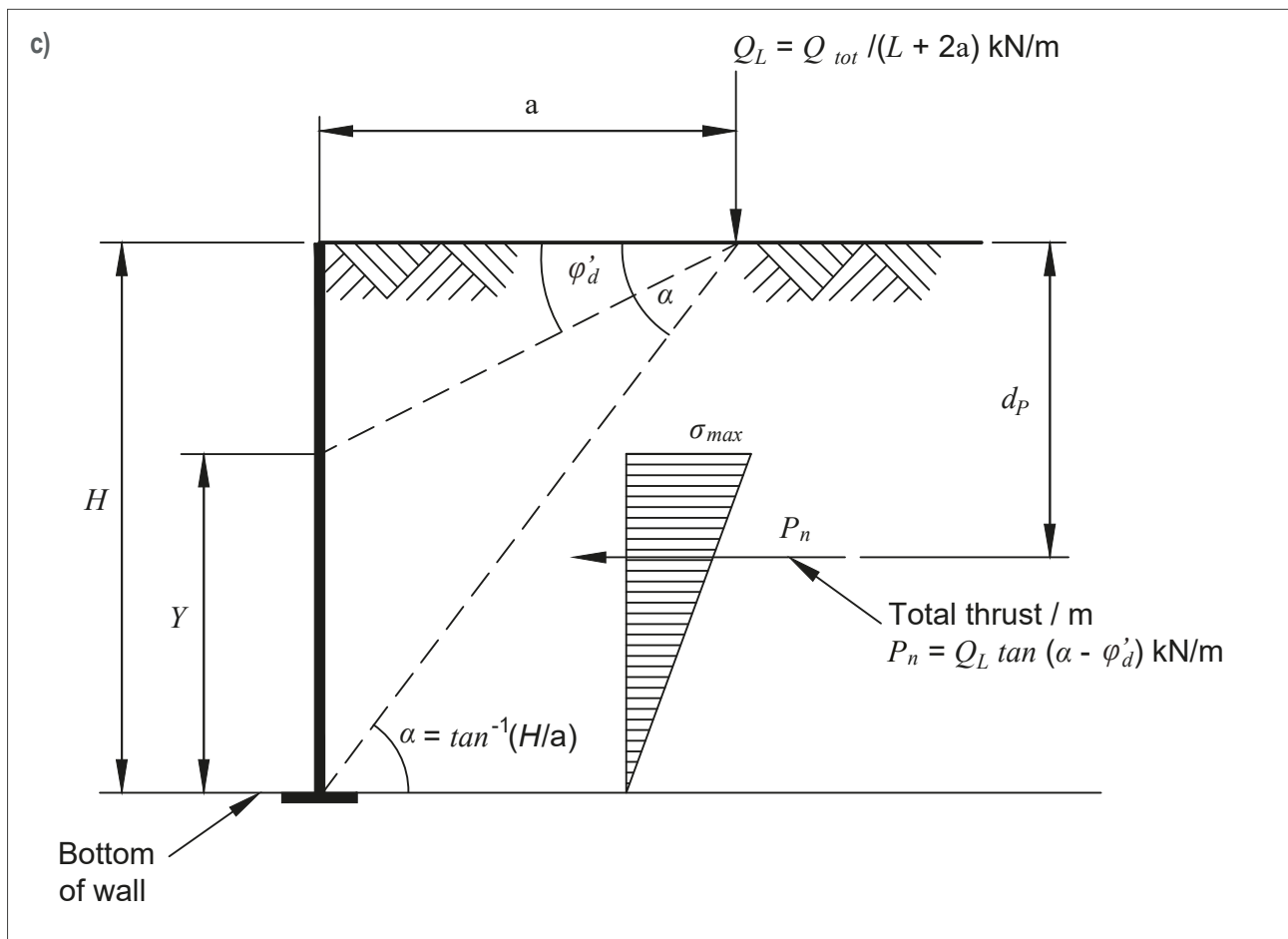


Figure A.2 Lateral and Vertical Dispersion of Finite Line Loads for Horizontal Pressure



A.2 Temperature Difference Component – Approach 2

This approach considers non-linear effects of vertical temperature gradients in reference to Note under Clause 3.10.3. The recommended values of vertical temperature difference for bridge decks to be used are given in Figure A.3 - Figure A.5. These values are valid for 40 mm surfacing depth for deck type 1 and 100 mm for deck type 2 and 3. The recommended values for other depths of surfacing are given in Table A.3 - Table A.5. In these figures, the “heating” refers to conditions such that solar radiation and other effects cause a gain in heat through the top surface of the bridge deck. The “cooling” refers to conditions such that heat is lost from top surface of the bridge deck as a result of re-radiation and other effects.

Figure A.3 Temperature Differences for Bridge Decks – Type 1: Steel Decks

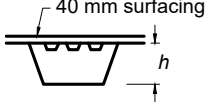
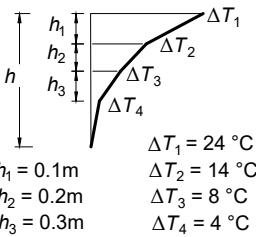
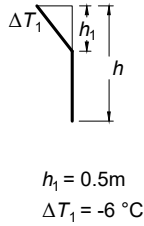
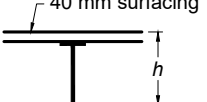
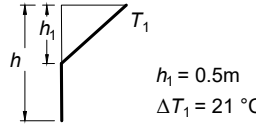
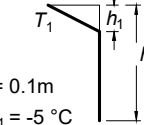
Type of Construction	Temperature Difference (ΔT)	
	(a) Heating	(b) Cooling
1a. Steel deck on steel box girders 	 $h_1 = 0.1\text{m}$ $h_2 = 0.2\text{m}$ $h_3 = 0.3\text{m}$ $\Delta T_1 = 24\text{ }^{\circ}\text{C}$ $\Delta T_2 = 14\text{ }^{\circ}\text{C}$ $\Delta T_3 = 8\text{ }^{\circ}\text{C}$ $\Delta T_4 = 4\text{ }^{\circ}\text{C}$	 $h_1 = 0.5\text{m}$ $\Delta T_1 = -6\text{ }^{\circ}\text{C}$
1b. Steel deck on steel truss or plate girders 	 $h_1 = 0.5\text{m}$ $\Delta T_1 = 21\text{ }^{\circ}\text{C}$	 $h_1 = 0.1\text{m}$ $\Delta T_1 = -5\text{ }^{\circ}\text{C}$

Figure A.4 Temperature Differences For Bridge Decks – Type 2: Composite Decks

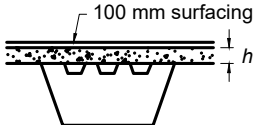
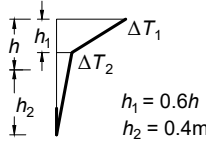
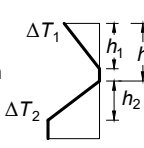
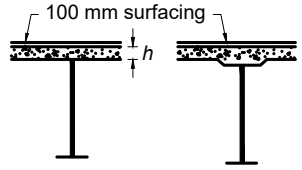
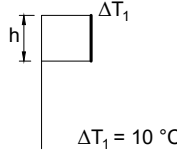
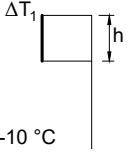
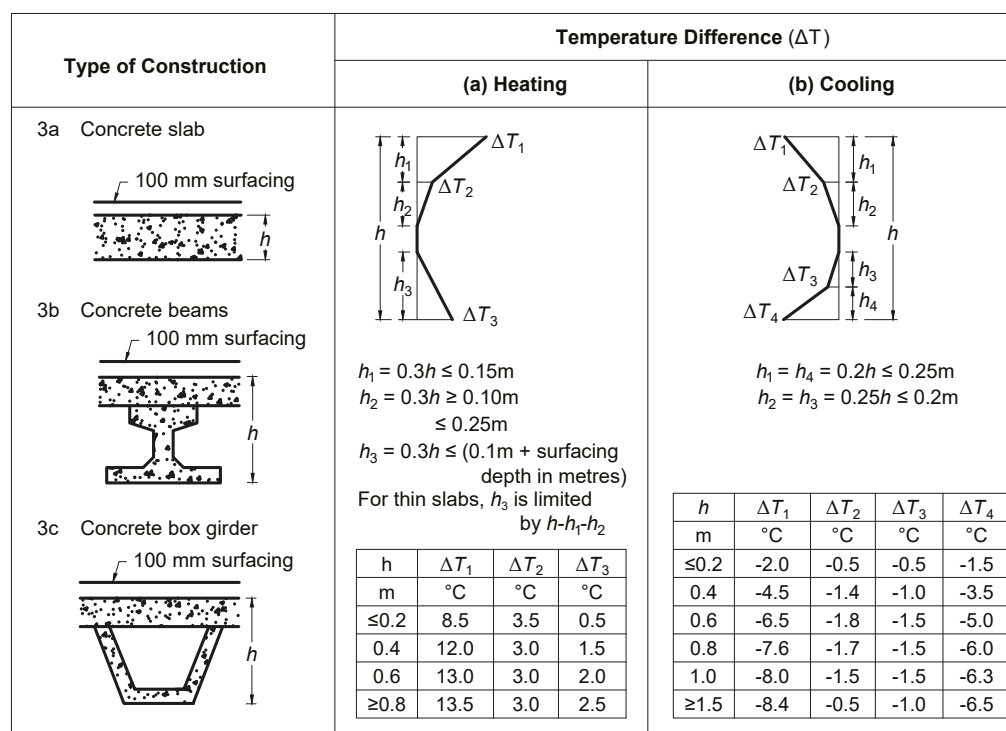
Type of Construction	Temperature Difference (ΔT)																			
	(a) Heating	(b) Cooling																		
2. Concrete deck on steel box, truss or plate girders 	 $h_1 = 0.6h$ $h_2 = 0.4\text{m}$ <table border="1"> <thead> <tr> <th>h m</th><th>ΔT_1 °C</th><th>ΔT_2 °C</th></tr> </thead> <tbody> <tr> <td>0.2</td><td>13.0</td><td>4.0</td></tr> <tr> <td>0.3</td><td>16.0</td><td>4.0</td></tr> </tbody> </table>	h m	ΔT_1 °C	ΔT_2 °C	0.2	13.0	4.0	0.3	16.0	4.0	 $h_1 = 0.6h$ $h_2 = 0.4\text{m}$ <table border="1"> <thead> <tr> <th>h m</th><th>ΔT_1 °C</th><th>ΔT_2 °C</th></tr> </thead> <tbody> <tr> <td>0.2</td><td>-3.5</td><td>-8.0</td></tr> <tr> <td>0.3</td><td>-5.0</td><td>-8.0</td></tr> </tbody> </table>	h m	ΔT_1 °C	ΔT_2 °C	0.2	-3.5	-8.0	0.3	-5.0	-8.0
h m	ΔT_1 °C	ΔT_2 °C																		
0.2	13.0	4.0																		
0.3	16.0	4.0																		
h m	ΔT_1 °C	ΔT_2 °C																		
0.2	-3.5	-8.0																		
0.3	-5.0	-8.0																		
	 $\Delta T_1 = 10\text{ }^{\circ}\text{C}$	 $\Delta T_1 = -10\text{ }^{\circ}\text{C}$																		

Figure A.5 Temperature Differences for Bridge Decks – Type 3: Concrete Decks

Table A.3 Recommended Values of ΔT for Deck Type 1

Surfacing Thickness (mm)	Temperature Difference				
	Heating				Cooling
	ΔT_1 (°C)	ΔT_2 (°C)	ΔT_3 (°C)	ΔT_4 (°C)	ΔT_1 (°C)
Unsurfaced	30	16	6	3	8
20	27	15	9	5	6
40	24	14	8	4	6

Table A.4 Recommended Values of ΔT for Deck Type 2

Depth of Slab (h) (m)	Surfacing Thickness (mm)	Temperature Difference	
		Heating	Cooling
		ΔT_1 (°C)	ΔT_1 (°C)
0.20	Unsurfaced	16.5	5.9
	Waterproofed*	23.0	5.9
	50	18.0	4.4
	100	13.0	3.5
	150	10.5	2.3
	200	8.5	1.6
0.30	Unsurfaced	18.5	9.0
	Waterproofed*	26.5	9.0
	50	20.5	6.8
	100	16.0	5.0
	150	12.5	3.7
	200	10.0	2.7

*These values represent upper bound values for dark colour

Table A.5 Recommended Values of ΔT for Deck Type 3

Depth of Slab (h) (m)	Surfacing Thickness (mm)	Temperature Difference						
		Heating			Cooling			
		ΔT_1 (°C)	ΔT_2 (°C)	ΔT_3 (°C)	ΔT_1 (°C)	ΔT_2 (°C)	ΔT_3 (°C)	ΔT_4 (°C)
≤ 0.20	Unsurfaced	12.0	5.0	0.1	4.7	1.7	0.0	0.7
	Waterproofed*	19.5	8.5	0.0	4.7	1.7	0.0	0.7
	50	13.2	4.9	0.3	3.1	1.0	0.2	1.2
	100	8.5	3.5	0.5	2.0	0.5	0.5	1.5
	150	5.6	2.5	0.2	1.1	0.3	0.7	1.7
	200	3.7	2.0	0.5	0.5	0.2	1.0	1.8
0.40	Unsurfaced	15.2	4.4	1.2	9.0	3.5	0.4	4.6
	Waterproofed*	23.6	6.5	1.0	9.0	3.5	0.4	4.6
	50	17.2	4.6	1.4	6.4	2.3	0.6	4.9
	100	12.0	3.0	1.5	4.5	1.4	1.0	5.0
	150	8.5	2.0	1.2	3.2	0.9	1.4	5.1
	200	6.2	1.3	1.0	2.2	0.5	1.9	5.1
0.60	Unsurfaced	15.2	4.0	1.4	11.8	4.0	0.9	4.6
	Waterproofed*	23.6	6.0	1.4	11.8	4.0	0.9	4.6
	50	17.6	4.0	1.8	8.7	2.7	1.2	4.9
	100	13.3	3.0	2.0	6.5	1.8	1.5	5.0
	150	9.7	2.2	1.7	4.9	1.1	1.7	5.1
	200	7.2	1.5	1.5	3.6	0.6	1.9	5.1
0.8	Unsurfaced	15.4	4.0	2.0	12.8	3.3	0.9	5.6
	Waterproofed*	23.6	5.0	1.4	12.8	3.3	0.9	5.6
	50	17.8	4.0	2.1	9.8	2.4	1.2	5.8
	100	13.5	3.0	2.5	7.6	1.7	1.5	6.0
	150	10.0	2.5	2.0	5.8	1.3	1.7	6.2
	200	7.5	2.1	1.5	4.5	1.0	1.9	6.0
1.0	Unsurfaced	15.4	4.0	2.0	13.4	3.0	0.9	6.4
	Waterproofed*	23.6	5.0	1.4	13.4	3.0	0.9	6.4
	50	17.8	4.0	2.1	10.3	2.1	1.2	6.3
	100	13.5	3.0	2.5	8.0	1.5	1.5	6.3
	150	10.0	2.5	2.0	6.2	1.1	1.7	6.2
	200	7.5	2.1	1.5	4.8	0.9	1.9	5.8
≥ 1.5	Unsurfaced	15.4	4.5	2.0	13.7	1.0	0.6	6.7
	Waterproofed*	23.6	5.0	1.4	13.7	1.0	0.6	6.7
	50	17.8	4.0	2.1	10.6	0.7	0.8	6.6
	100	13.5	3.0	2.5	8.4	0.5	1.0	6.5
	150	10.0	2.5	2.0	6.5	0.4	1.1	6.2
	200	7.5	2.1	1.5	5.0	0.3	1.2	5.6

*These values represent upper bound values for dark colour

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B Concrete Structures

B.1 Basic Equations for Determining the Creep Coefficient

The creep coefficient $\varphi(t, t_0)$ may be calculated from:

$$\varphi(t, t_0) = \varphi_0 \cdot \beta_c(t, t_0)$$

Equation B.1

Where,

φ_0 is the notional creep coefficient and may be estimated from:

$$\varphi_0 = \varphi_{RH} \beta_{f_{cm}} \beta(t_0)$$

Equation B.2

φ_{RH} is a factor to allow for the effect of relative humidity on the notional creep coefficient:

$$\varphi_{RH} = 1 + \frac{1 - RH/100}{0.1 \sqrt[3]{h_0}} \quad \text{for } f_{cm} \leq 35 \text{ MPa}$$

$$\varphi_{RH} = \left[1 + \frac{1 - RH/100}{0.1 \sqrt[3]{h_0}} \alpha_1 \right] \alpha_2 \quad \text{for } f_{cm} > 35 \text{ MPa}$$

Equation B.3

RH = The relative humidity of the ambient environment in %

$\beta(f_{cm})$ = A factor to allow for the effect of concrete strength on the notional creep coefficient:

$$\frac{16.8}{\sqrt{(f_{cm})}}$$

Equation B.4

f_{cm} = The mean compressive strength of concrete in MPa at the age of 28 days

$\beta(t_0)$ = A factor to allow for the effect of concrete age at loading on the notional creep coefficient:

$$\beta(t_0) = \frac{1}{(0.1 + t_0^{0.20})}$$

Equation B.5

h_0 = The notional size of the member in mm where:

$$h_0 = \frac{2 A_c}{u}$$

Equation B.6

A_c = The cross-sectional area

u = The perimeter of the member in contact with the atmosphere

$\beta_c(t, t_0)$ = A coefficient to describe the development of creep with time after loading, and may be estimated using the following Equation:

$$\beta_c(t, t_0) = \left[\frac{(t - t_0)}{(\beta_H + t - t_0)} \right]^{0.3}$$

Equation B.7

t = The age of concrete in days at the moment considered

t_0 = The age of concrete at loading in days

$t - t_0$ = The non-adjusted duration of loading in days

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β_H = A coefficient depending on the relative humidity (RH in %) and the notional member size (h_0 in mm). It may be estimated from:

$$\beta_H = 1.5 [1 + (0.012 RH)^{18}] h_0 + 250 \leq 1500 \quad \text{for } f_{cm} \leq 35$$

$$\beta_H = 1.5 [1 + (0.012 RH)^{18}] h_0 + 250 \alpha_3 \leq 1500 \alpha_3 \quad \text{for } f_{cm} > 35$$

Equation B.8

$\alpha_{1/2/3}$ are coefficients to consider the influence of the concrete strength:

$$\alpha_1 = \left[\frac{35}{f_{cm}} \right]^{0.7} \quad \alpha_2 = \left[\frac{35}{f_{cm}} \right]^{0.2} \quad \alpha_3 = \left[\frac{35}{f_{cm}} \right]^{0.5}$$

Equation B.9

The effect of type of cement on the creep coefficient of concrete may be taken into account by modifying the age of loading t_0 in Equation B.5 according to the following Equation:

$$t_0 = t_{0,T} \left(\frac{9}{2 + t_{0,T}^{1.2}} + 1 \right)^\alpha \geq 0.5$$

Equation B.10

Where,

$t_{0,T}$ = the temperature adjusted age of concrete at loading in days adjusted according to Equation B.10.

α = A power which depends on type of cement

= -1 for cement Class S (slow early strength)

= 0 for cement Class N (normal early strength)

= 1 for cement Class R (high early strength)

The effect of elevated or reduced temperatures within the range 0 – 80°C on the maturity of concrete may be taken into account by adjusting the concrete age according to the following Equation:

$$t_T = \sum_{i=1}^n e^{-(4000/[273 + T(\Delta t_i)] - 13.65)} \Delta t_i$$

Equation B.11

Where,

t_T = The temperature adjusted concrete age which replaces t in the corresponding equations

$T(\Delta t_i)$ = The temperature in °C during the time period Δt_i

Δt_i = The number of days where a temperature T prevails.

The mean coefficient of variation of the above predicted creep data, deduced from a computerised data bank of laboratory test results, is of the order of 20%.

The values of $\varphi(t, t_0)$ given above should be associated with the tangent modulus E_c .

When a less accurate estimate is considered satisfactory, the values given in Chapter 4 may be adopted for creep of concrete at 70 years.

B.2 Basic Equations for Determining the Drying Shrinkage Strain

The basic drying shrinkage strain $\varepsilon_{cd,0}$ is calculated from:

$$\varepsilon_{cd,0} = 0.85 \left[(220 + 110 \alpha_{ds1}) \exp \left(-\alpha_{ds2} \cdot \frac{f_{cm}}{f_{cm0}} \right) \right] 10^{-6} \beta_{RH}$$

Equation B.12

$$\beta_{RH} = 1.55 \left[1 - \left(\frac{RH}{RH_0} \right)^3 \right]$$

Equation B.13

Where,

f_{cm} = The mean compressive strength (MPa)

f_{cm0} = 10 Mpa

α_{ds1} = A coefficient which depends on the type of cement

= 3 for cement Class S

= 4 for cement Class N

= 6 for cement Class R

α_{ds2} = A coefficient which depends on the type of cement

= 0.13 for cement Class S

= 0.12 for cement Class N

= 0.11 for cement Class R

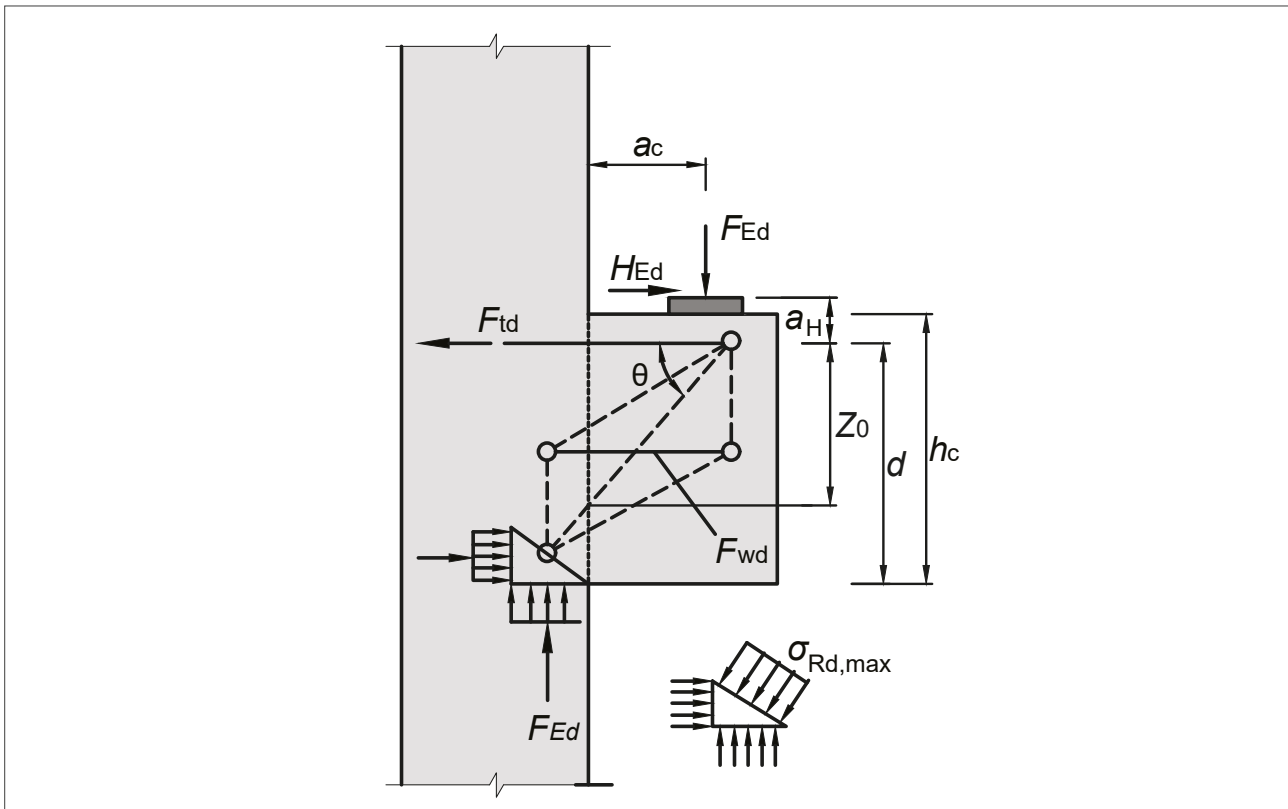
RH = The ambient relative humidity (%)

$RH_0 = 100\%$.

B.3 Corbels

Corbels ($a_c < z_0$) may be designed using strut-and-tie models, refer following figure. The inclination of the strut is limited by $1.0 \leq \tan \theta \leq 2.5$.

Figure B.1 Corbel Strut and Tie Model

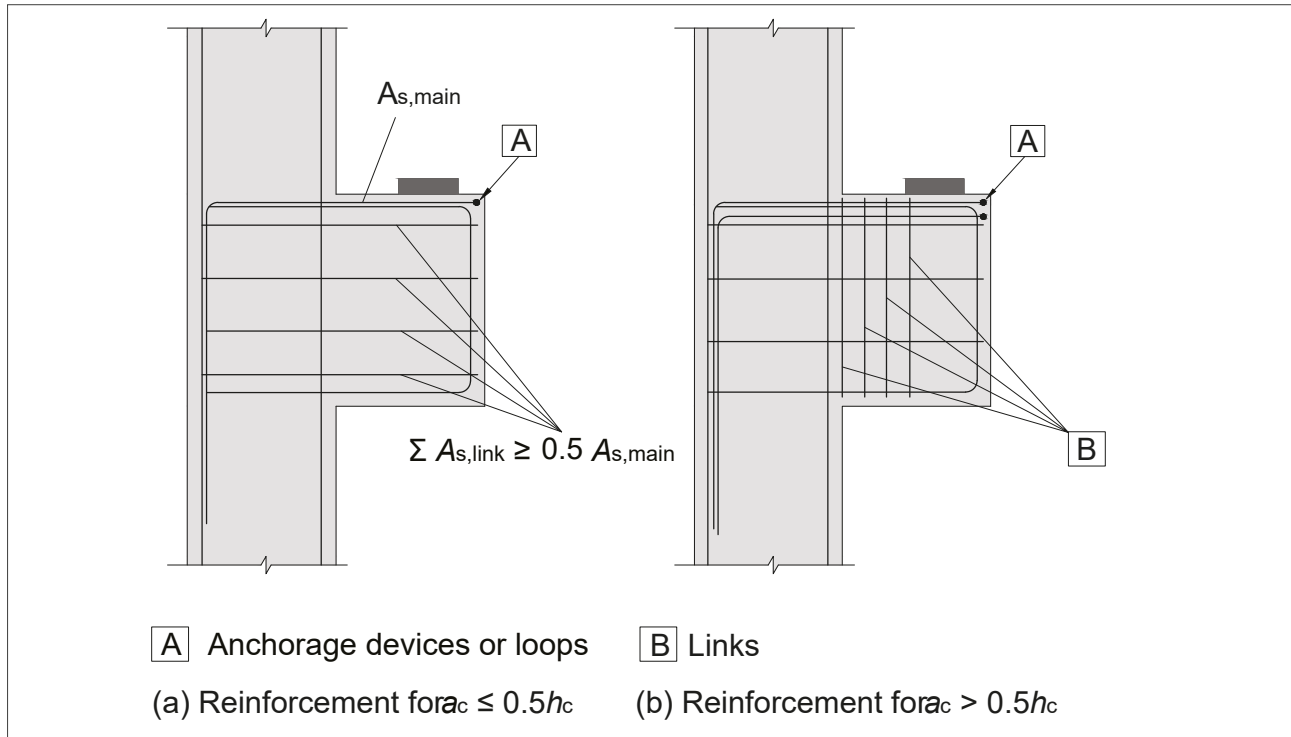


If $a_c \leq 0.5 h_c$ closed horizontal or inclined links with $A_{s,link} \geq 0.5 A_{s,main}$ should be provided in addition to the main tension reinforcement.

If $a_c > 0.5 h_c$ and $F_{Ed} > V_{Rd,c}$, closed vertical links $A_{s,link} \geq 0.5 F_{Ed}/f_{yd}$ should be provided in addition to the main tension reinforcement.

The main tension reinforcement should be anchored at both ends. It should be anchored in the supporting element on the far face and the anchorage length should be measured from the location of the vertical reinforcement in the near face. The reinforcement should be anchored in the corbel and the anchorage length should be measured from the inner face of the loading plate.

Figure B.2 Corbel Detailing



B.4 Anchorage Zones of Post-tensioned Members

The following rules apply in addition to those given in Chapter 4 for the design of anchorage zones where two or more tendons are anchored. The bearing stress behind anchorage plates should be checked as follows:

The reinforcement required to prevent bursting and spalling in anchorage zones is determined in relation to a rectangular prism of concrete, known as the primary regularisation prism, located behind each anchorage. The cross section of the prism associated with each anchorage is known as the associate rectangle. The associate rectangle has the same centre and the same axes of symmetry as the anchorage plate (which should have two axes of symmetry) and should satisfy:

$$\frac{P_{max}}{c c'} \leq 0.6 f_{ck}(t)$$

Equation B.14

Where,

P_{max} = The maximum force applied to the tendon.

c, c' = The dimensions of the associate rectangle.

$f_{ck}(t)$ = The concrete strength at the time of tensioning.

The associate rectangle should have approximately the same aspect ratio as the anchorage plate. This requirement is satisfied if c/a and c'/a' are not greater than $1.25\sqrt{(c/c')/(a/a')}$ where a and a' are the dimensions of the smallest rectangle including the anchorage plate. Rectangles associated with anchorages located in the same cross section should remain inside the concrete and should not overlap.

The "primary regularisation prism" represents very approximately the volume of concrete in which the stresses change from very high values just behind the anchorage plate to a reasonable value for concrete under uniaxial compression. The axis of the prism is taken as the axis of the tendon, its base is the associate rectangle and its depth behind the anchorage is taken as $1.2 \max(c, c')$. The prisms associated with different anchorages may overlap when the tendons are not parallel, but should remain inside the concrete.

Reinforcement to prevent bursting and spalling of the concrete, in each regularisation prism should not be less than: $A_s = 0.15 (P_{max}/f_{yd}) \gamma_{p,unfav}$ with $\gamma_{p,unfav} \geq 1.20$

Where,

P_{max} is the maximum force applied to the tendon and f_{yd} is the design strength of the reinforcing steel.

This reinforcement should be distributed in each direction over the length of the prism. The area of the surface reinforcement at the loaded face should not be less than $0.03 (P_{max}/f_{yd}) \gamma_{p,unfav}$ in each direction.

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C Steel Structures

C.1 Critical stresses for stiffened plates

Plates with at least three longitudinal stiffeners may be treated as equivalent orthotropic plates. The elastic critical plate buckling stress of the equivalent orthotropic plate may be taken as:

$$\sigma_{cr,p} = k_{\sigma,p} \sigma_E$$

Equation C.1

Where,

$$\sigma_E = \frac{\pi^2 E t^2}{12(1 - \nu^2)b^2} = 190000 \left(\frac{t}{b}\right)^2 \text{ in [MPa]}$$

Equation C.2

$k_{\sigma,p}$ = The buckling coefficient according to orthotropic plate theory with the stiffeners smeared over the plate.

b = Defined in Figure C.1.

t = The thickness of the plate.

The $\sigma_{cr,p}$ is the elastic critical plate buckling stress at the edge of the panel where the maximum compression stress occurs, refer Figure C.1. Where a web is of concern, the width b should be replaced by h_w .

For stiffened plates with at least three equally spaced longitudinal stiffeners the plate buckling coefficient $k_{\sigma,p}$ (global buckling of the stiffened panel) may be approximated by:

$$k_{\sigma,p} = \frac{2((1 + \alpha^2)^2 + \gamma - 1)}{\alpha^2(\psi + 1)(1 + \delta)} \quad \text{if } \alpha \leq \sqrt[4]{\gamma}$$

Equation C.3

$$k_{\sigma,p} = \frac{4(1 + \sqrt[4]{\gamma})}{(\psi + 1)(1 + \delta)} \quad \text{if } \alpha > \sqrt[4]{\gamma}$$

Equation C.4

With, $\psi = \frac{\sigma_2}{\sigma_1} \geq 0.5$

$$\gamma = \frac{I_{sl}}{I_p}$$

$$\delta = \frac{A_{sl}}{A_p}$$

$$\alpha = \frac{a}{b} \geq 0.5$$

Where,

I_{sl} = The second moment of area of the whole stiffened plate.

I_p = The second moment of area for bending of plate.

$$= \frac{bt^3}{12(1 - \nu^2)} = \frac{bt^3}{10.92}$$

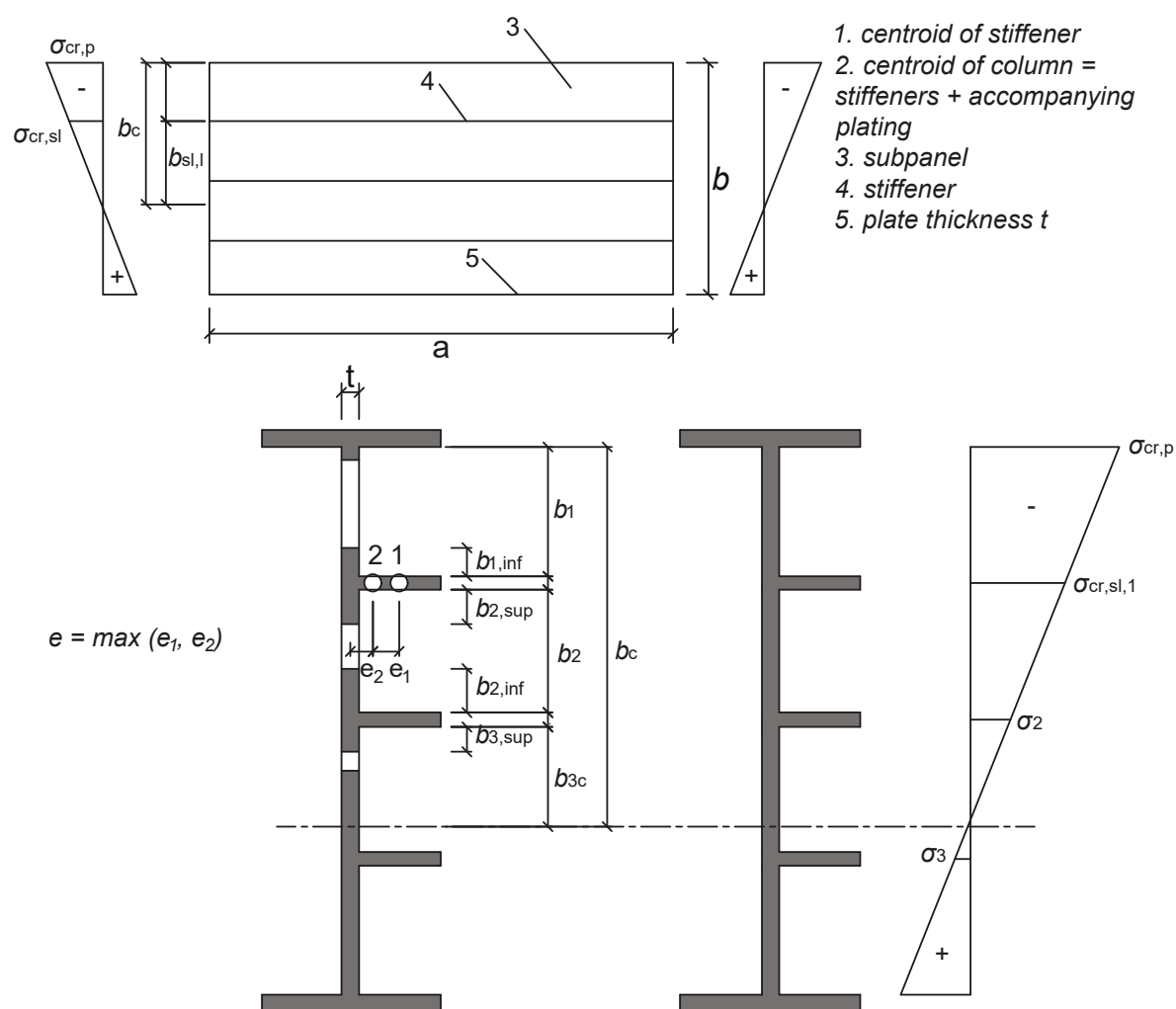
A_{sl} = The sum of the gross areas of the individual longitudinal stiffeners.

A_p = The gross area of the plate = bt .

σ_1 = The larger edge stress.

σ_2 = The smaller edge stress.

Figure C.1 Notations for Longitudinally Stiffened Plates



	Width for gross area	Width for effective area	Condition for ψ_i
$b_{1,inf}$	$\frac{3 - \psi_1}{5 - \psi_1} b_1$	$\frac{3 - \psi_1}{5 - \psi_1} b_{1,eff}$	$\psi_1 = \frac{\sigma_{cr,sl,1}}{\sigma_{cr,p}} > 0$
$b_{2,sup}$	$\frac{2}{5 - \psi_2} b_2$	$\frac{2}{5 - \psi_2} b_{2,eff}$	$\psi_2 = \frac{\sigma_2}{\sigma_{cr,sl,1}} > 0$
$b_{2,inf}$	$\frac{3 - \psi_2}{5 - \psi_2} b_2$	$\frac{3 - \psi_2}{5 - \psi_2} b_{2,eff}$	$\psi_2 > 0$
$b_{3,sup}$	$0.4 b_{3c}$	$0.4 b_{3c,eff}$	$\psi_3 = \frac{\sigma_3}{\sigma_2} < 0$

C.2 Shear Buckling Coefficients

For plates with rigid transverse stiffeners and without longitudinal stiffeners or with more than two longitudinal stiffeners, the shear buckling coefficient k_τ can be obtained as follows:

$$k_\tau = 5.34 + 4.00 \left(\frac{h_w}{a} \right)^2 + k_{\tau sl} \quad \text{when } \frac{a}{h_w} \geq 1$$

Equation C.5

$$k_\tau = 4.00 + 5.34 \left(\frac{h_w}{a} \right)^2 + k_{\tau sl} \quad \text{when } \frac{a}{h_w} < 1$$

Equation C.6

Where,

$$k_{\tau sl} = 9 \left(\frac{h_w}{a} \right)^2 \sqrt[4]{\left(\frac{I_{sl}}{t^3 h_w} \right)^3} \quad \text{but not less than } \frac{2.1^3}{t} \sqrt{\frac{I_{sl}}{h_w}}$$

Equation C.7

a = The distance between transverse stiffeners

I_{sl} = The second moment of area of the longitudinal stiffener about the z-z axis

For webs with longitudinal stiffeners, not necessarily equally spaced, I_{sl} is the sum of the stiffness of the individual stiffeners.

The above equation also applies to plates with one or two longitudinal stiffeners and an aspect ratio $\alpha \geq 3$. For plates with one or two longitudinal stiffeners and an aspect ratio $\alpha < 3$ the shear buckling coefficient should be taken from:

$$k_\tau = 4.1 + \frac{6.3 + 0.18 \frac{I_{sl}}{t^3 h_w}}{\alpha^2} + 2.2 \sqrt[3]{\frac{I_{sl}}{t^3 h_w}}$$

Equation C.8

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D Culverts

D.1 Load Cases To Be Considered For Design Of Buried Concrete Structures

Like the bridges, culverts are designed for Ultimate Limit state, Serviceability Limit state and Accidental Limit state. Culverts need not to be checked for seismic effects. The loading conditions assumed for the design of culvert shall consider the heaviest likely traffic loads and possibly increased dead loads due to future development above the culvert.

Following cases may be considered for determination of earth pressures on buried concrete box and portal structures that may be used with various loading combinations. Assumption is being made that the structure is backfilled with free draining granular material, properly compacted and there is uniformity of ground conditions. For other situations, the earth pressures shall be calculated using geotechnical design guidelines.

Case-1: Maximum vertical load with maximum horizontal load.

Case-2: Minimum vertical load with maximum horizontal load.

Case-3: Maximum vertical load with minimum horizontal load.

The critical condition shall be analysed with:

- a. Water inside the culvert.
- b. No water inside the culvert.

Case-4: Braking and acceleration with maximum vertical load and active pressure.

Case-5: Braking and acceleration with minimum vertical load and active pressure.

Case-6: Sliding.

The different load cases are represented in Figure D.1 to Figure D.6 along with the design values of earth pressure coefficient K in Table D.1 to Table D.6 for corresponding load cases.

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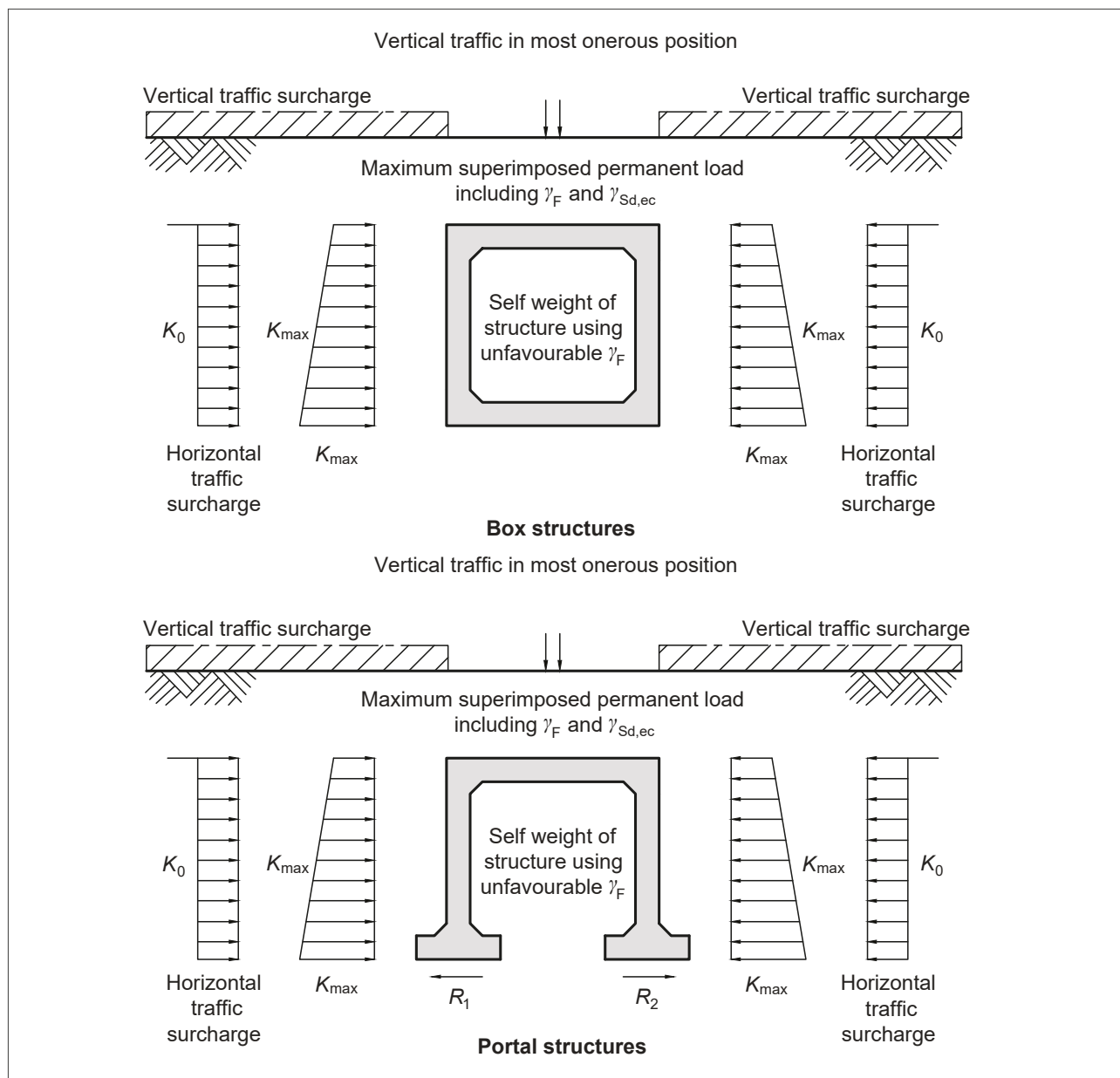
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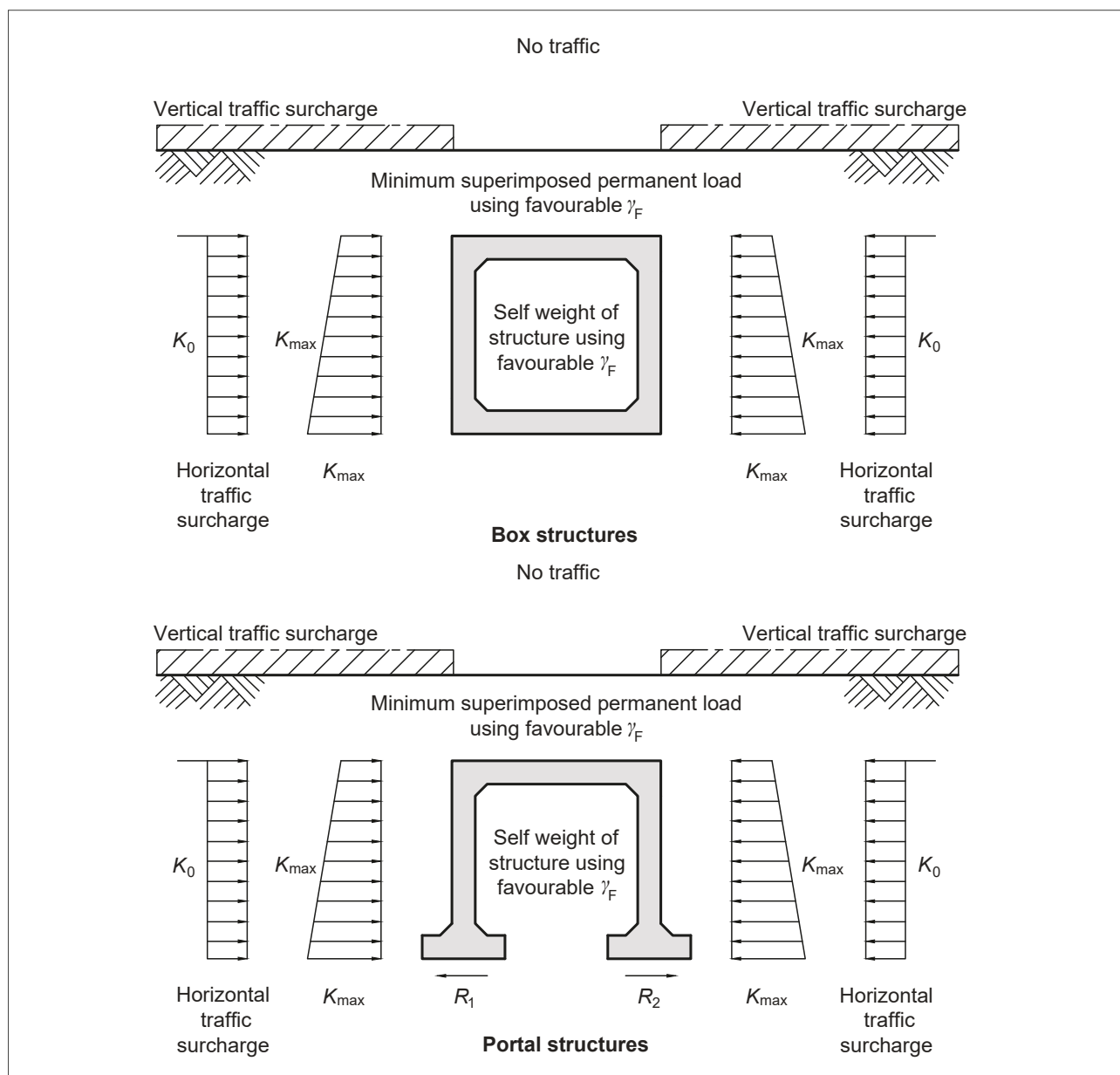
Figure D.1 Maximum Vertical Load with Maximum Horizontal Load

Table D.1 Design Values of the Earth Pressure Coefficient K for Load Case as in Figure D.1

Horizontal traffic surcharge K_0	Earth pressure K_{max}	Limit state	Earth pressure K_{max}	Horizontal traffic surcharge K_0
0.50	0.60	SLS (characteristic)	0.60	0.50
0.54	0.77	EQU	0.77	0.54
0.50	0.72	STR/GEO (Comb 1)	0.72	0.50
0.58	0.84	STR/GEO (Comb 2)	0.84	0.58

Note: The values of K include the partial factor γ_M , and (for permanent earth pressure at ULS) the model factor $\gamma_{Sd,K}$.

Figure D.2 Minimum Vertical Load with Maximum Horizontal Load

Table D.2 Design Values of the Earth Pressure Coefficient K for Load Case as in Figure D.2

Horizontal traffic surcharge K_0	Earth pressure K_{max}	Limit state	Earth pressure K_{max}	Horizontal traffic surcharge K_0
0.5	0.60	SLS (characteristic)	0.60	0.5
0.54	0.77	EQU	0.77	0.54
0.50	0.72	STR/GEO (Comb 1)	0.72	0.50
0.58	0.84	STR/GEO (Comb 2)	0.84	0.58

Note: The values of K include the partial factor γ_M , and (for permanent earth pressure at ULS) the model factor $\gamma_{Sd,K}$

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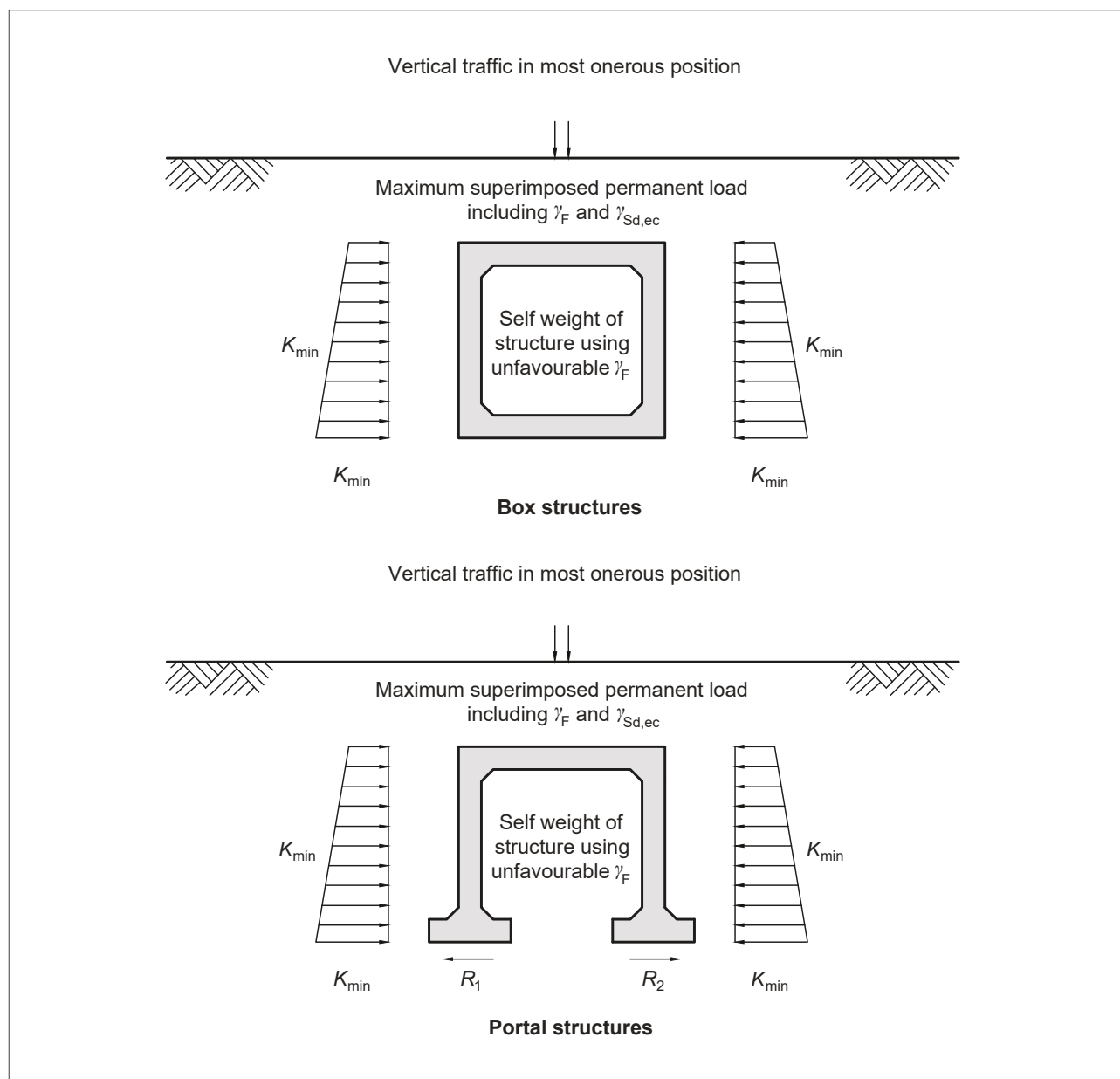
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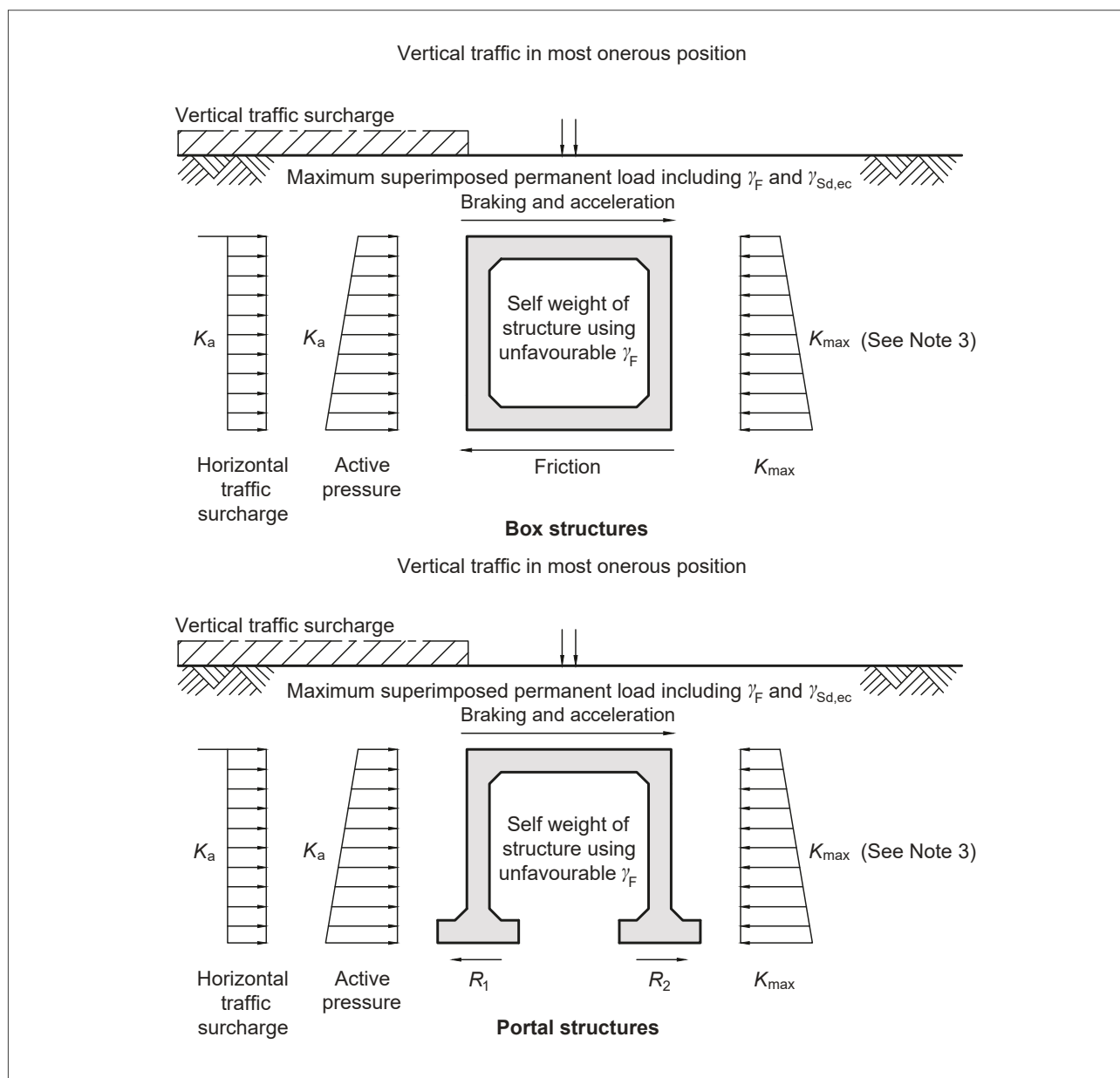
Figure D.3 Maximum Vertical Load with Minimum Horizontal Load

Table D.3 Design Values of the Earth Pressure Coefficient K for Load Case as in Figure D.3

Earth pressure K_{min}	Limit state	Earth pressure K_{min}
0.20	SLS (characteristic)	0.20
0.18	EQU	0.18
0.20	STR/GEO (Comb 1)	0.20
0.16	STR/GEO (Comb 2)	0.16

Note: The values of K include the partial factor γ_M , and (for permanent earth pressure at ULS) the model factor $\gamma_{Sd,K}$.

Figure D.4 Braking and Acceleration With Maximum Vertical Load and Active Pressure

Table D.4 Design Values of the Earth Pressure Coefficient K for Load Case as in Figure D.4

Horizontal traffic surcharge K_a	Earth pressure K_a	Limit state	Earth pressure K_{max}
0.33	0.33	SLS (characteristic)	0.6
0.37	0.44	EQU	0.6
0.33	0.40	STR/GEO (Comb 1)	0.72
0.41	0.49	STR/GEO (Comb 2)	0.84

Note 1: The values of K include the partial factor γ_M , and (for permanent earth pressure at ULS) the model factor $\gamma_{Sd,K}$.

Note 2: If the structure sways towards the active side, this load case may be ignored.

Note 3: The earth pressure coefficient for the passive wall may be taken as greater than K_{max} for bearing, sliding and overturning provided the associated displacements are acceptable at the relevant limit state.

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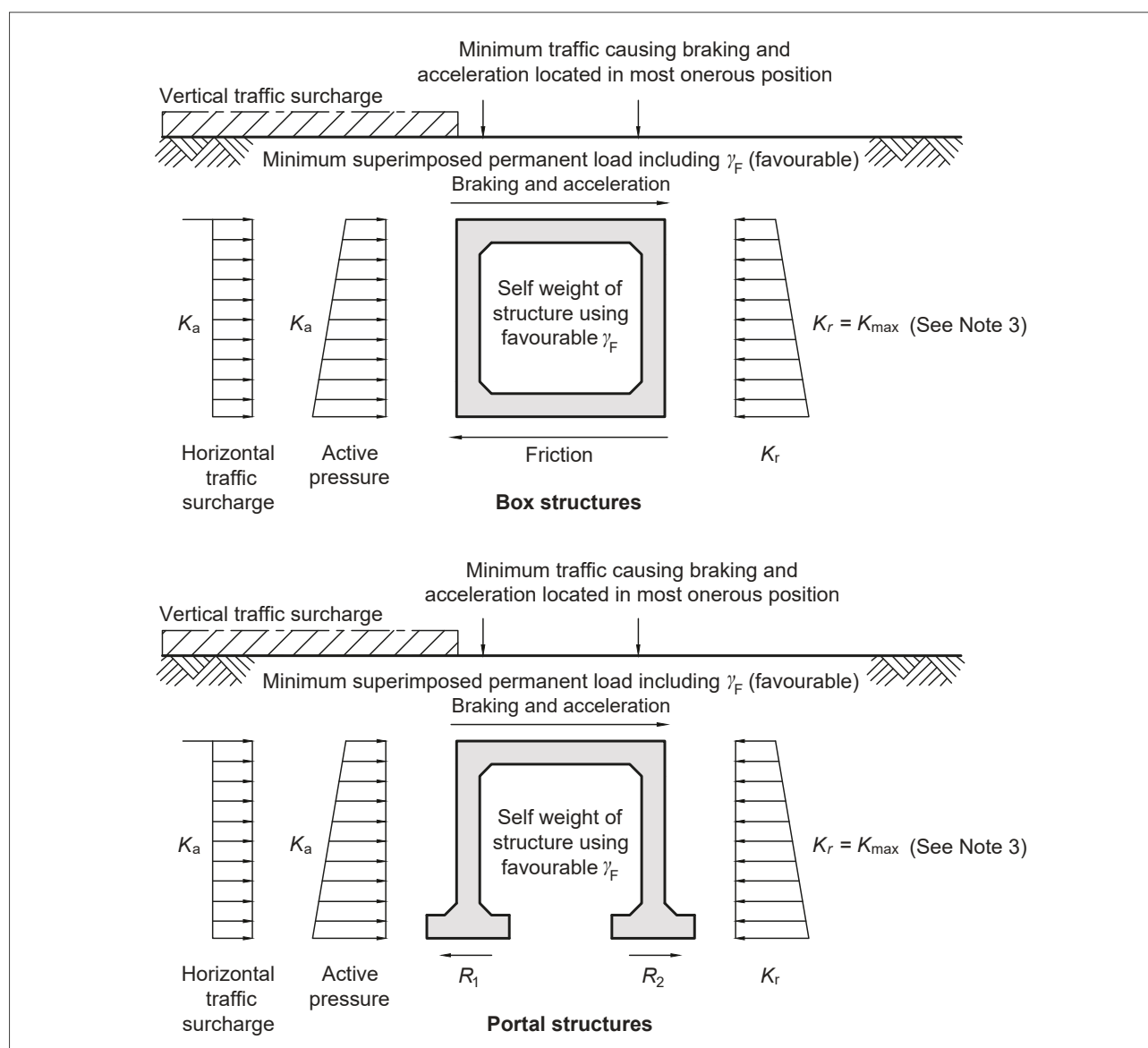
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Appendices

Figure D.5 Braking and Acceleration With Minimum Vertical Load and Active Pressure

Table D.5 Design Values of the Earth Pressure Coefficient K for Load Case as in Figure D.5

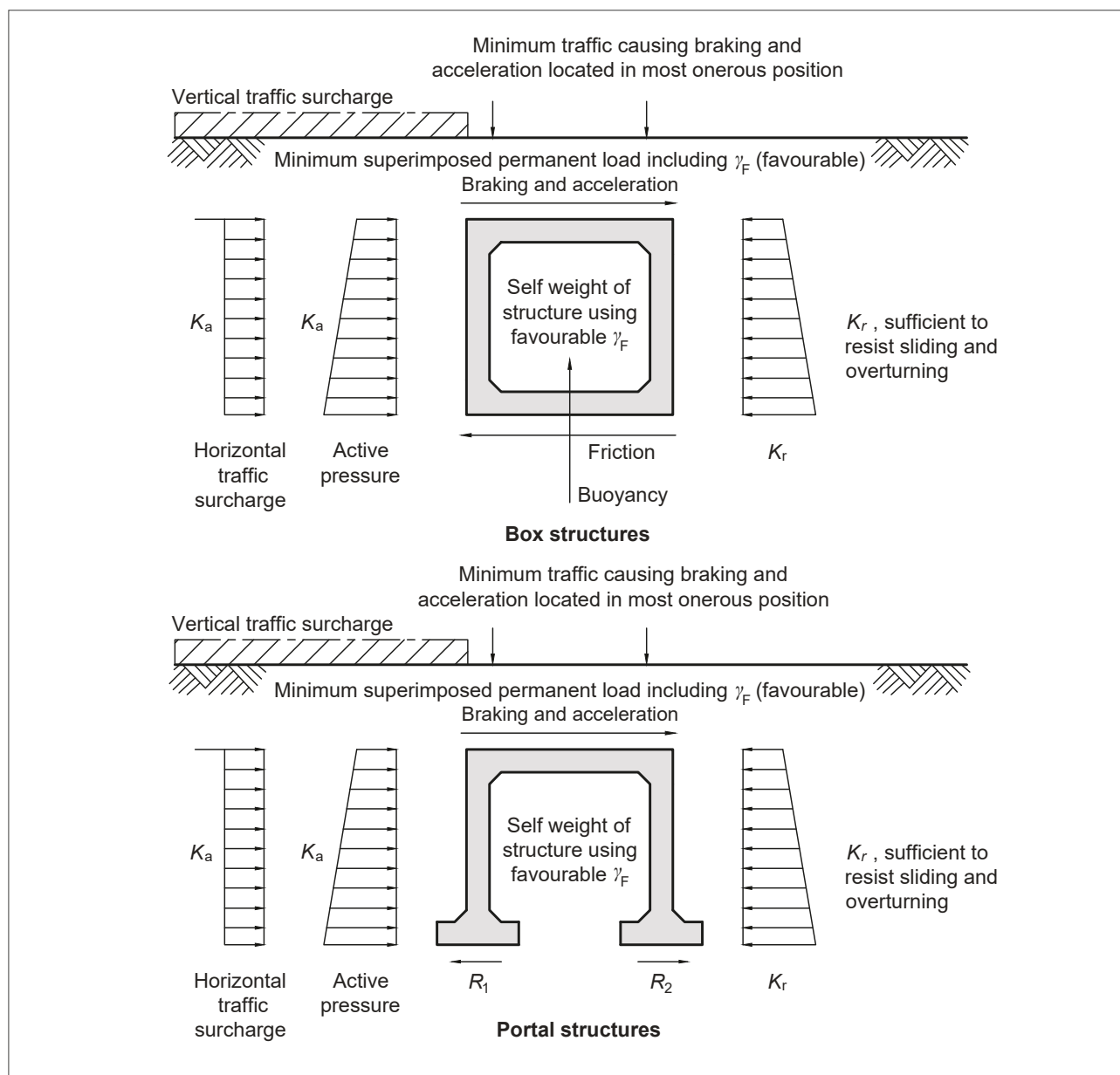
Horizontal traffic surcharge K_a	Earth pressure K_a	Limit state	Earth pressure K_{max}
0.33	0.33	SLS (characteristic)	0.6
0.37	0.44	EQU	0.6
0.33	0.40	STR/GEO (Comb 1)	0.72
0.41	0.49	STR/GEO (Comb 2)	0.84

Note 1: The values of K include the partial factor γ_M , and (for permanent earth pressure at ULS) the model factor $\gamma_{Sd,K}$.

Note 2: If the structure sways towards the active side, this load case may be ignored.

Note 3: The earth pressure coefficient for the passive wall may be taken as greater than K_{max} for bearing, sliding and overturning provided the associated displacements are acceptable at the relevant limit state.

Figure D.6 Sliding

Table D.6 Design Values of the Earth Pressure Coefficient K for Load Case as in Figure D.6

Horizontal traffic surcharge K_a	Earth pressure K_a	Limit state	Earth pressure K_r
0.33	0.33	SLS (characteristic)	K_r
0.37	0.44	EQU	K_r
0.33	0.40	STR/GEO (Comb 1)	K_r
0.41	0.49	STR/GEO (Comb 2)	K_r

Note 1: The values of K include the partial factor γ_M , and (for permanent earth pressure at ULS) the model factor $\gamma_{Sd,K}$.

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E Foundations

E.1 Design Bearing Resistance For Spread Foundations

E.1.1 Undrained conditions

Design bearing resistance = $R / A' = (2 + \pi) c_u b_c s_c i_c + q$.

Where,

b_c = Factor for the inclination of the foundation base.
 $= 1 - 2\alpha / (\pi + 2)$.

α = The inclination of the foundation base to the horizontal .

s_c = Factor for the shape of the foundation.
 $= 1 + 0.2 (B' / L')$ for rectangular foundation.
 $= 1.2$ for square or circular shape.

i_c = Factor for the inclination of the load, caused by a horizontal load H .
 $= 0.5(1 + \sqrt{(1 - H / (A' c_u))})$

H = The horizontal load.
 $\leq A' c_u$.

A' = $B' L'$, the design effective foundation area (reduced area with load acting at centre).

B' = The design effective foundation width ($B - 2e_B$).

L' = The design effective foundation length ($L - 2e_L$).

c_u = Undrained shear strength.

q = Overburden or surcharge pressure at the level of foundation base.

E.1.2 Drained Conditions

Design bearing resistance = $R / A' = c' N_c b_c s_c i_c + q' N_q b_q s_q i_q + (1/2) \gamma' B' N_\gamma b_\gamma s_\gamma i_\gamma$

Where,

c' = Cohesion intercept in terms of effective stress.

q' = The design effective overburden pressure at the level of foundation base.

γ' = The design effective weight density of the soil below foundation level.

N_q, N_c, N_γ are factors for bearing resistance, given as:

$$N_q = e^{\pi \tan \phi'} \tan^2 (45 + \phi'/2)$$

$$N_c = (N_q - 1) \cot \phi'$$

$$N_\gamma = 2 (N_q - 1) \tan \phi'$$

b_c, b_q, b_γ are factors for the inclination of the foundation base, given as:

$$b_c = b_q - (1 - b_q) / (N_c \tan \phi')$$

$$b_q = b_\gamma = (1 - \alpha \tan \phi')^2$$

s_c, s_q, s_γ are factors for the shape of foundation, given as:

$$s_q = 1 + (B' / L') \sin \phi', \text{ for rectangular shape.}$$

$$s_q = 1 + \sin \phi', \text{ for square or circular shape.}$$

$$s_c = (s_q N_q - 1) / (N_q - 1), \text{ for rectangular, square or circular shape.}$$

$$s_\gamma = 1 - 0.30 (B' / L'), \text{ for rectangular shape.}$$

$$s_\gamma = 0.70, \text{ for square or circular shape.}$$

i_c, i_q, i_γ are factors for the inclination of the load, caused by a horizontal load H

$$i_c = i_q - (1 - i_q) / (N_c \tan \phi')$$

$$i_q = (1 - H / (V + A' c' \cot \phi'))^m$$

$$i_\gamma = (1 - H / (V + A' c' \cot \phi'))^{m+1}$$

The m may be calculated as under-

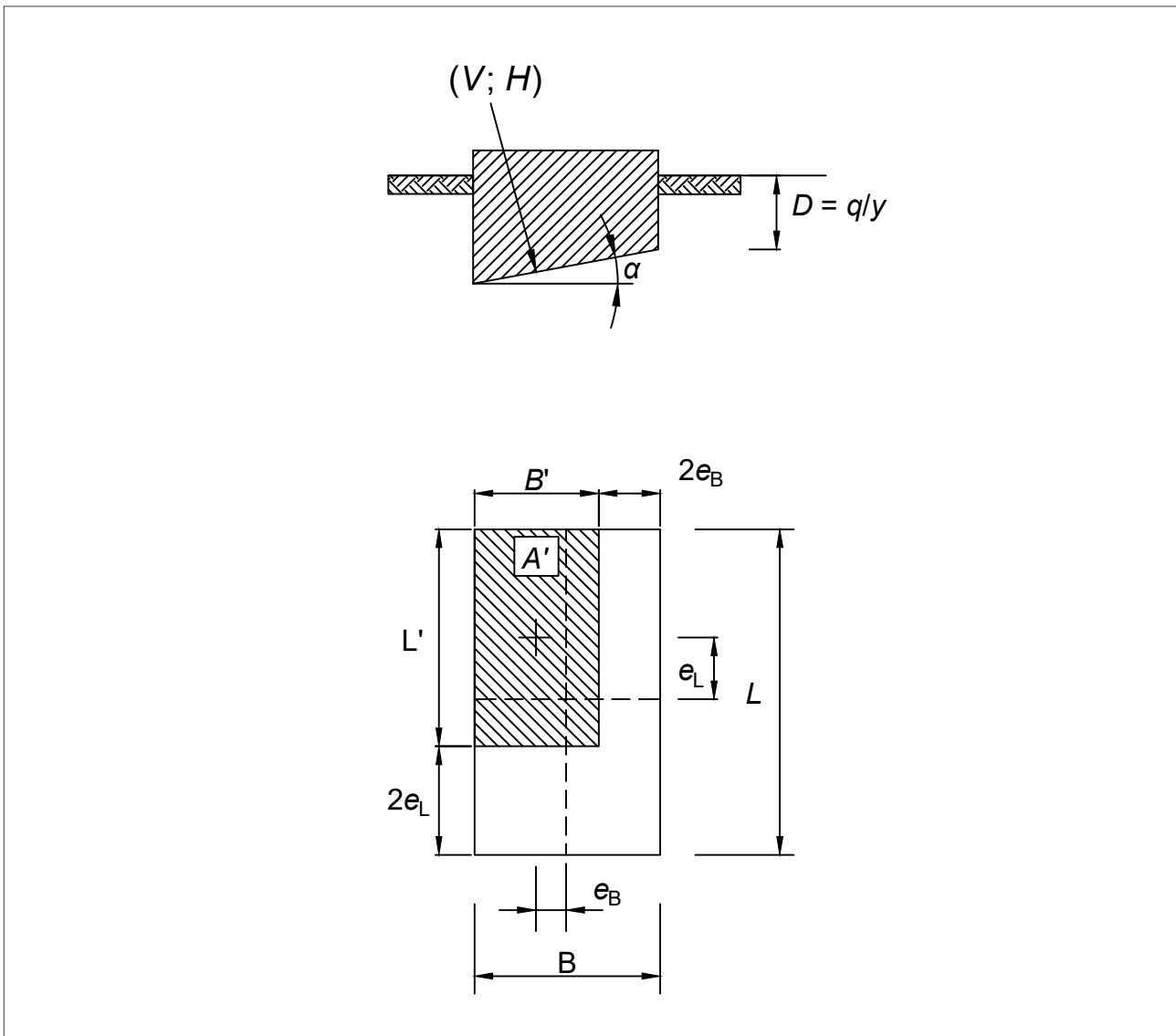
$$m = m_B = [2 + (B'/L')] / [1 + (B'/L')], \text{ when } H \text{ acts in the direction of } B'$$

$$m = m_L = [2 + (L'/B')] / [1 + (L'/B')], \text{ when } H \text{ acts in the direction of } L'$$

In case where the horizontal load acts in a direction forming an angle θ with the direction of L' , the m may be calculated by:

$$m = m_\theta = m_L \cos^2 \theta + m_B \sin^2 \theta$$

Figure E.1 Definition of Effective Area A'



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Appendices

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